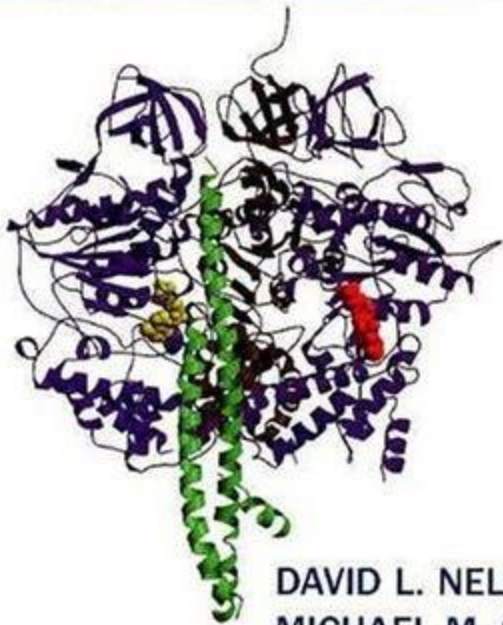


Lehninger

PRINCIPLES OF BIOCHEMISTRY

fourth edition



DAVID L. NELSON
MICHAEL M. COX

Lehninger
PRINCIPLES OF
BIOCHEMISTRY
Fourth Edition

David L. Nelson (University of Wisconsin–Madison)
Michael M. Cox (University of Wisconsin–Madison)

New to This Edition

- **Every chapter fully updated:** Including coverage of the human genome and genomics integrated throughout, and key developments since the publication of the third edition, such as the structure of the ribosome.
- **New treatment of metabolic regulation:** NEW Chapter 15 gives students the most up-to-date picture of how cells maintain biochemical homeostasis by including modern concepts in metabolic regulation.
- **New, earlier coverage of DNA-based information technologies (Chapter 9):** Shows how advances in DNA technology are revolutionizing medicine and biotechnology; examines cloning and genetic engineering, as well as the implications of human gene therapy.
- **Glycolysis and gluconeogenesis** now presented in a single chapter (Chapter 14).
- **Redesigned and Expanded Treatment of Enzyme Mechanisms:** NEW Mechanism Figures designed to lead students through these reactions step by step. The first reaction mechanism treated in the book, chymotrypsin, presents a refresher on how to follow and understand reaction mechanism diagrams. Twelve new mechanisms have been added, including lysozyme.
- **New Medical and Life Sciences Examples:** This edition adds boxed features of biochemical methods, medical applications, and the history of biochemistry, adding to those already present of medicine, biotechnology, and other aspects of daily life.

Web site at: www.whfreeman.com/lehninger4e

For students:

- **Biochemistry in 3D molecular structure tutorials:** Self-paced, interactive tutorials based on the Chemscape Chime molecular visualization browser plug-in.
- **Chime tutorial archive** provides links to some of the best Chime tutorials available on the Web.
- **Online support** for the Biochemistry on the Internet problems in the textbook.
- **Flashcards** on key terms from the text.
- **Online quizzing** for each chapter, a new way for students to review material and prepare for exams.
- **Animated mechanisms** viewed in Flash or PowerPoint formats give students and instructors a way to visualize mechanisms in a two-dimensional format.
- **Living Graphs** illustrate graphed material featured in the text.
- **Bonus Material** from Lehninger, Principles of Biochemistry, Third Edition: fundamental Chapters 1, 2, and 3 from the third edition that instructors find useful for their students as a basis for their biochemistry studies.

For instructors:

- All the figures from the book optimized for projection, available in PowerPoint and JPEG format; also available on the IRCD (see below).

CHIME Student CD, 0-7167-7049-0

This CD allows students to view Chime tutorials without having to install either the older version of Netscape or the Chime plug-in. Available packaged with Lehninger for free, this optional Student CD-ROM also includes the animated mechanisms and living graphs from the Web site.

Instructor's Resource CD-ROM with Test Bank, 0-7167-5953-5

- All the images and tables from the text in JPEG and PowerPoint formats, optimized for projection with enhanced colors, higher resolution and enlarged fonts for easy reading in the lecture hall.
- Animated enzyme mechanisms.
- Living Graphs
- Test Bank organized by chapter in the form of .pdf files and editable Word files.

Supplements

For Instructors

Printed Test Bank, Terry Platt and Eugene Barber, University of Rochester Medical Center), David L. Nelson and Brook Chase Soltvedt, University of Wisconsin-Madison, 0-7167-5952-7

The new Test Bank contains 25% new multiple-choice and short-answer problems and solutions with approximately 50 problems and solutions per chapter. Each problem is keyed to the corresponding chapter of the text and rated by level of difficulty.

Overhead Transparency Set, 0-7167-5956-X

The full-color transparency set contains 150 key illustrations from the text, with enlarged labels that project more clearly for lecture hall presentation.

For Students

The Absolute, Ultimate Guide to Lehninger, Principles of Biochemistry, Fourth Edition: Study Guide and Solutions Manual, Marcy Osgood, University of New Mexico, and Karen Ocorr, University of California, San Diego, 0-7167-5955-1

The *Absolute, Ultimate Guide* combines an innovative study guide with a reliable solutions manual in one convenient volume. A poster-size Cellular Metabolic Map is packaged with the Guide, on which students can draw the reactions and pathways of metabolism in their proper compartments within the cell.

Exploring Genomes, Paul G. Young (Queens University), 0-7167-5738-2

Used in conjunction with the online tutorials found at www.whfreeman.com/young, Exploring Genomes guides students through live searches and analyses on the most commonly used National Center for Biotechnology Information (NCBI) database.

Lecture Notebook, 0-7167-5954-3

Bound volume of black and white reproductions of all the text's line art and tables, allowing students to concentrate on the lecture instead of copying illustrations. Also includes:

- Essential reaction equations and mathematical equations with identifying labels
- Complete pathway diagrams and individual reaction diagrams for all metabolic pathways in the book
- References that key the material in the text to the CD-ROM and Web Site

Lehninger Principles of Biochemistry

Fourth Edition

David L. Nelson (U. of Wisconsin–Madison)

Michael M. Cox (U. of Wisconsin–Madison)

1. The Foundations of Biochemistry

- 1.1 Cellular Foundations
- 1.2 Chemical Foundations
- 1.3 Physical Foundations
- 1.4 Genetic Foundations
- 1.5 Evolutionary Foundations

Distilled and reorganized from Chapters 1–3 of the previous edition, this overview provides a refresher on the cellular, chemical, physical, genetic, and evolutionary background to biochemistry, while orienting students toward what is unique about biochemistry.

PART I. STRUCTURE AND CATALYSIS

2. Water

- 2.1 Weak Interactions in Aqueous Systems
- 2.2 Ionization of Water, Weak Acids, and Weak Bases
- 2.3 Buffering against pH Changes in Biological Systems
- 2.4 Water as a Reactant
- 2.5 The Fitness of the Aqueous Environment for Living Organisms

Includes new coverage of the concept of protein-bound water, illustrated with molecular graphics.

3. Amino Acids, Peptides, and Proteins

- 3.1 Amino Acids
- 3.2 Peptides and Proteins
- 3.3 Working with Proteins
- 3.4 The Covalent Structure of Proteins
- 3.5 Protein Sequences and Evolution

Adds important new material on genomics and proteomics and their implications for the study of protein structure, function, and evolution.

4. The Three-Dimensional Structure of Proteins

- 4.1 Overview of Protein Structure
- 4.2 Protein Secondary Structure
- 4.3 Protein Tertiary and Quaternary Structures
- 4.4 Protein Denaturation and Folding

Adds a new box on scurvy.

5. Protein Function

- 5.1 Reversible Binding of a Protein to a Ligand: Oxygen-Binding Proteins
- 5.2 Complementary Interactions between Proteins and Ligands: The Immune System and Immunoglobulins
- 5.3 Protein Interactions Modulated by Chemical Energy: Actin, Myosin, and Molecular Motors

Adds a new box on carbon monoxide poisoning

6. Enzymes

- 6.1 An Introduction to Enzymes
- 6.2 How Enzymes Work
- 6.3 Enzyme Kinetics as An Approach to Understanding Mechanism
- 6.4 Examples of Enzymatic Reactions
- 6.5 Regulatory Enzymes

Offers a revised presentation of the mechanism of chymotrypsin (the first reaction mechanism in the book), featuring a two-page figure that takes students through this particular mechanism, while serving as a step-by-step guide to interpreting any

reaction mechanism

Features new coverage of the mechanism for lysozyme including the controversial aspects of the mechanism and currently favored resolution based on work published in 2001.

7. Carbohydrates and Glycobiology

7.1 Monosaccharides and Disaccharides

7.2 Polysaccharides

7.3 Glycoconjugates: Proteoglycans, Glycoproteins, and Glycolipids

7.4 Carbohydrates as Informational Molecules: The Sugar Code

7.5 Working with Carbohydrates

Includes new section on polysaccharide conformations.

A striking new discussion of the "sugar code" looks at polysaccharides as informational molecules, with detailed discussions of lectins, selectins, and oligosaccharide-bearing hormones.

Features new material on structural heteropolysaccharides and proteoglycans

Covers recent techniques for carbohydrate analysis.

8. Nucleotides and Nucleic Acids

8.1 Some Basics

8.2 Nucleic Acid Structure

8.3 Nucleic Acid Chemistry

8.4 Other Functions of Nucleotides

9. DNA-Based Information Technologies

9.1 DNA Cloning: The Basics

9.2 From Genes to Genomes

9.3 From Genomes to Proteomes

9.4 Genome Alterations and New Products of Biotechnology

Introduces the human genome. Biochemical insights derived from the human genome are integrated throughout the text.

Tracking the emergence of genomics and proteomics, this chapter establishes DNA technology as a core topic and a path to understanding metabolism, signaling, and other topics covered in the middle chapters of this edition. Includes up-to-date coverage of microarrays, protein chips, comparative genomics, and techniques in cloning and analysis.

10. Lipids

10.1 Storage Lipids

10.2 Structural Lipids in Membranes

10.3 Lipids as Signals, Cofactors, and Pigments

10.4 Working with Lipids

Integrates new topics specific to chloroplasts and archaebacteria

Adds material on lipids as signal molecules.

11. Biological Membranes and Transport

11.1 The Composition and Architecture of Membranes

11.2 Membrane Dynamics

11.3 Solute Transport across Membranes

Includes a description of membrane rafts and microdomains within membranes, and a new box on the use of atomic force microscopy to visualize them.

Looks at the role of caveolins in the formation of membrane caveolae

Covers the investigation of hop diffusion of membrane lipids using FRAP (fluorescence recovery after photobleaching)

Adds new details to the discussion of the mechanism of Ca^{2+} -ATPase (SERCA)

pump), revealed by the recently available high-resolution view of its structure
Explores new facets of the mechanisms of the K⁺ selectivity filter, brought to light
by recent high-resolution structures of the K⁺ channel
Illuminates the structure, role, and mechanism of aquaporins with important new
details
Describes ABC transporters, with particular attention to the multidrug transporter
(MDR1)
Includes the newly solved structure of the lactose transporter of E. coli.

12. Biosignaling

- 12.1 Molecular Mechanisms of Signal Transduction
- 12.2 Gated Ion Channels
- 12.3 Receptor Enzymes
- 12.4 G Protein-Coupled Receptors and Second Messengers
- 12.5 Multivalent Scaffold Proteins and Membrane Rafts
- 12.6 Signaling in Microorganisms and Plants
- 12.7 Sensory Transduction in Vision, Olfaction, and Gustation
- 12.8 Regulation of Transcription by Steroid Hormones
- 12.9 Regulation of the Cell Cycle by Protein Kinases
- 12.10 Oncogenes, Tumor Suppressor Genes, and Programmed Cell Death

Updates the previous edition's groundbreaking chapter to chart the continuing rapid development of signaling research

Includes discussion on general mechanisms for activation of protein kinases in cascades

Now covers the roles of membrane rafts and caveolae in signaling pathways, including the activities of AKAPs (A Kinase Anchoring Proteins) and other scaffold proteins

Examines the nature and conservation of families of multivalent protein binding modules, which combine to create many discrete signaling pathways

Adds a new discussion of signaling in plants and bacteria, with comparison to mammalian signaling pathways

Features a new box on visualizing biochemistry with fluorescence resonance energy transfer (FRET) with green fluorescent protein (GFP)

PART II: BIOENERGETICS AND METABOLISM

13. Principles of Bioenergetics

- 13.1 Bioenergetics and Thermodynamics
 - 13.2 Phosphoryl Group Transfers and ATP
 - 13.3 Biological Oxidation-Reduction Reactions
- Examines the increasing awareness of the multiple roles of polyphosphate
Adds a new discussion of niacin deficiency and pellagra.

14. Glycolysis, Gluconeogenesis, and the Pentose Phosphate Pathway

- 14.1 Glycolysis
- 14.2 Feeder Pathways for Glycolysis
- 14.3 Fates of Pyruvate under Anaerobic Conditions: Fermentation
- 14.4 Gluconeogenesis
- 14.5 Pentose Phosphate Pathway of Glucose Oxidation

Now covers gluconeogenesis immediately after glycolysis, discussing their relatedness, differences, and coordination and setting up the completely new chapter on metabolic regulation that follows

Adds coverage of the mechanisms of phosphohexose isomerase and aldolase

Revises the presentation of the mechanism of glyceraldehyde 3-phosphate dehydrogenase.

New Chapter 15. Principles of Metabolic Regulation, Illustrated with Glucose and Glycogen Metabolism

15.1 The Metabolism of Glycogen in Animals

15.2 Regulation of Metabolic Pathways

15.3 Coordinated Regulation of Glycolysis and Gluconeogenesis

15.4 Coordinated Regulation of Glycogen Synthesis and Breakdown

15.5 Analysis of Metabolic Control

Brings together the concepts and principles of metabolic regulation in one chapter

Concludes with the latest conceptual approaches to the regulation of metabolism, including metabolic control analysis and contemporary methods for studying and predicting the flux through metabolic pathways

16. The Citric Acid Cycle

16.1 Production of Acetyl-CoA (Activated Acetate)

16.2 Reactions of the Citric Acid Cycle

16.3 Regulation of the Citric Acid Cycle

16.4 The Glyoxylate Cycle

Expands and updates the presentation of the mechanism for pyruvate carboxylase.

Adds coverage of the mechanisms of isocitrate dehydrogenase and citrate synthase.

17. Fatty Acid Catabolism

17.1 Digestion, Mobilization, and Transport of Fats

17.2 Oxidation of Fatty Acids

17.3 Ketone Bodies

Updates coverage of trifunctional protein

New section on the role of perilipin phosphorylation in the control of fat mobilization

New discussion of the role of acetyl-CoA in the integration of fatty acid oxidation and synthesis

Updates coverage of the medical consequences of genetic defects in fatty acyl-CoA dehydrogenases

Takes a fresh look at medical issues related to peroxisomes

18. Amino Acid Oxidation and the Production of Urea

18.1 Metabolic Fates of Amino Groups

18.2 Nitrogen Excretion and the Urea Cycle

18.3 Pathways of Amino Acid Degradation

Integrates the latest on regulation of reactions throughout the chapter, with new material on genetic defects in urea cycle enzymes, and updated information on the regulatory function of N-acetylglutamate synthase.

Reorganizes coverage of amino acid degradation to focus on the big picture

Adds new material on the relative importance of several degradative pathways

Includes a new description of the interplay of the pyridoxal phosphate and tetrahydrofolate cofactors in serine and glycine metabolism

19. Oxidative Phosphorylation and Photophosphorylation

Oxidative Phosphorylation

19.1 Electron-Transfer Reactions in Mitochondria

19.2 ATP Synthesis

19.3 Regulation of Oxidative Phosphorylation

19.4 Mitochondrial Genes: Their Origin and the Effects of Mutations

19.5 The Role of Mitochondria in Apoptosis and Oxidative Stress

Photosynthesis: Harvesting Light Energy

19.6 General Features of Photophosphorylation

19.7 Light Absorption

19.8 The Central Photochemical Event: Light-Driven Electron Flow

19.9 ATP Synthesis by Photophosphorylation

Adds a prominent new section on the roles of mitochondria in apoptosis and oxidative stress

Now covers the role of IF1 in the inhibition of ATP synthase during ischemia

Includes revelatory details on the light-dependent pathways of electron transfer in photosynthesis, based on newly available molecular structures

20. Carbohydrate Biosynthesis in Plants and Bacteria

20.1 Photosynthetic Carbohydrate Synthesis

20.2 Photorespiration and the C₄ and CAM Pathways

20.3 Biosynthesis of Starch and Sucrose

20.4 Synthesis of Cell Wall Polysaccharides: Plant Cellulose and Bacterial

Peptidoglycan

20.5 Integration of Carbohydrate Metabolism in the Plant Cell

Reorganizes the coverage of photosynthesis and the C₄ and CAM pathways

Adds a major new section on the synthesis of cellulose and bacterial peptidoglycan

21. Lipid Biosynthesis

21.1 Biosynthesis of Fatty Acids and Eicosanoids

21.2 Biosynthesis of Triacylglycerols

21.3 Biosynthesis of Membrane Phospholipids

21.4 Biosynthesis of Cholesterol, Steroids, and Isoprenoids

Features an important new section on glyceroneogenesis and the triacylglycerol cycle between adipose tissue and liver, including their roles in fatty acid metabolism (especially during starvation) and the emergence of thiazolidinediones as regulators of glyceroneogenesis in the treatment of type II diabetes

Includes a timely new discussion on the regulation of cholesterol metabolism at the genetic level, with consideration of sterol regulatory element-binding proteins (SREBPs).

22. Biosynthesis of Amino Acids, Nucleotides, and Related Molecules

22.1 Overview of Nitrogen Metabolism

22.2 Biosynthesis of Amino Acids

22.3 Molecules Derived from Amino Acids

22.4 Biosynthesis and Degradation of Nucleotides

Adds material on the regulation of nitrogen metabolism at the level of transcription

Significantly expands coverage of synthesis and degradation of heme

23. Integration and Hormonal Regulation of Mammalian Metabolism

23.1 Tissue-Specific Metabolism: The Division of Labor

23.2 Hormonal Regulation of Fuel Metabolism

23.3 Long Term Regulation of Body Mass

23.4 Hormones: Diverse Structures for Diverse Functions

***Reorganized presentation** leads students through the complex interactions of integrated metabolism step by step*

Features extensively revised coverage of insulin and glucagon metabolism that includes the integration of carbohydrate and fat metabolism

New discussion of the role of AMP-dependent protein kinase in metabolic integration

Updates coverage of the fast-moving field of obesity, regulation of body mass, and the leptin and adiponectin regulatory systems

Adds a discussion of Ghrelin and PYY3-36 as regulators of short-term eating behavior

Covers the effects of diet on the regulation of gene expression, considering the role of peroxisome proliferator-activated receptors (PPARs)

PART III. INFORMATION PATHWAYS

24. Genes and Chromosomes

24.1 Chromosomal Elements

24.2 DNA Supercoiling

24.3 The Structure of Chromosomes

Integrates important new material on the structure of chromosomes, including the roles of SMC proteins and cohesins, the features of chromosomal DNA, and the organization of genes in DNA

25. DNA Metabolism

25.1 DNA Replication

25.2 DNA Repair

25.3 DNA Recombination

Adds a section on the "replication factories" of bacterial DNA

Includes latest perspectives on DNA recombination and repair

26. RNA Metabolism

26.1 DNA-Dependent Synthesis of RNA

26.2 RNA Processing

26.3 RNA-Dependent Synthesis of RNA and DNA

Updates coverage on mechanisms of mRNA processing

Adds a subsection on the 5' cap of eukaryotic mRNAs

Adds important new information about the structure of bacterial RNA polymerase and its mechanism of action.

27. Protein Metabolism

27.1 The Genetic Code

27.2 Protein Synthesis

27.3 Protein Targeting and Degradation

Includes a presentation and analysis of the long-awaited structure of the ribosome - one of the most important updates in this new edition

Adds a new box on the evolutionary significance of ribozyme-catalyzed peptide synthesis.

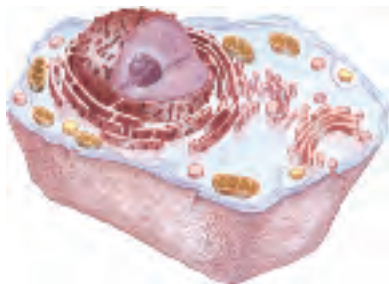
28. Regulation of Gene Expression

28.1 Principles of Gene Regulation

28.2 Regulation of Gene Expression in Prokaryotes

28.3 Regulation of Gene Expression in Eukaryotes

Adds a new section on RNA interference (RNAi), including the medical potential of gene silencing.



THE FOUNDATIONS OF BIOCHEMISTRY

- 1.1 Cellular Foundations 3
- 1.2 Chemical Foundations 12
- 1.3 Physical Foundations 21
- 1.4 Genetic Foundations 28
- 1.5 Evolutionary Foundations 31

With the cell, biology discovered its atom . . . To characterize life, it was henceforth essential to study the cell and analyze its structure: to single out the common denominators, necessary for the life of every cell; alternatively, to identify differences associated with the performance of special functions.

—François Jacob, *La logique du vivant: une histoire de l'hérédité*
(The Logic of Life: A History of Heredity), 1970

We must, however, acknowledge, as it seems to me, that man with all his noble qualities . . . still bears in his bodily frame the indelible stamp of his lowly origin.

—Charles Darwin, *The Descent of Man*, 1871

Fifteen to twenty billion years ago, the universe arose as a cataclysmic eruption of hot, energy-rich subatomic particles. Within seconds, the simplest elements (hydrogen and helium) were formed. As the universe expanded and cooled, material condensed under the influence of gravity to form stars. Some stars became enormous and then exploded as supernovae, releasing the energy needed to fuse simpler atomic nuclei into the more complex elements. Thus were produced, over billions of years, the Earth itself and the chemical elements found on the Earth today. About four billion years ago,

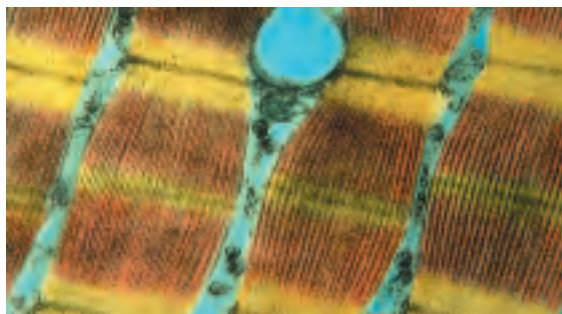
life arose—simple microorganisms with the ability to extract energy from organic compounds or from sunlight, which they used to make a vast array of more complex **biomolecules** from the simple elements and compounds on the Earth's surface.

Biochemistry asks how the remarkable properties of living organisms arise from the thousands of different lifeless biomolecules. When these molecules are isolated and examined individually, they conform to all the physical and chemical laws that describe the behavior of inanimate matter—as do all the processes occurring in living organisms. The study of biochemistry shows how the collections of inanimate molecules that constitute living organisms interact to maintain and perpetuate life animated solely by the physical and chemical laws that govern the nonliving universe.

Yet organisms possess extraordinary attributes, properties that distinguish them from other collections of matter. What are these distinguishing features of living organisms?

A high degree of chemical complexity and microscopic organization. Thousands of different molecules make up a cell's intricate internal structures (Fig. 1–1a). Each has its characteristic sequence of subunits, its unique three-dimensional structure, and its highly specific selection of binding partners in the cell.

Systems for extracting, transforming, and using energy from the environment (Fig. 1–1b), enabling organisms to build and maintain their intricate structures and to do mechanical, chemical, osmotic, and electrical work. Inanimate matter tends, rather, to decay toward a more disordered state, to come to equilibrium with its surroundings.



(a)



(b)



(c)

FIGURE 1-1 Some characteristics of living matter. (a) Microscopic complexity and organization are apparent in this colorized thin section of vertebrate muscle tissue, viewed with the electron microscope. (b) A prairie falcon acquires nutrients by consuming a smaller bird. (c) Biological reproduction occurs with near-perfect fidelity.

A capacity for precise self-replication and self-assembly (Fig. 1-1c). A single bacterial cell placed in a sterile nutrient medium can give rise to a billion identical “daughter” cells in 24 hours. Each cell contains thousands of different molecules, some extremely complex; yet each bacterium is a faithful copy of the original, its construction directed entirely from information contained within the genetic material of the original cell. **Mechanisms for sensing and responding to alterations in their surroundings**, constantly adjusting to these changes by adapting their internal chemistry.

Defined functions for each of their components and regulated interactions among them.

This is true not only of macroscopic structures, such as leaves and stems or hearts and lungs, but also of microscopic intracellular structures and individual chemical compounds. The interplay among the chemical components of a living organism is dynamic; changes in one component cause coordinating or compensating changes in another, with the whole ensemble displaying a character beyond that of its individual parts. The collection of molecules carries out a program, the end result of which is reproduction of the program and self-perpetuation of that collection of molecules—in short, life.

A history of evolutionary change. Organisms change their inherited life strategies to survive in new circumstances. The result of eons of evolution is an enormous diversity of life forms, superficially very different (Fig. 1-2) but fundamentally related through their shared ancestry.

Despite these common properties, and the fundamental unity of life they reveal, very few generalizations about living organisms are absolutely correct for every organism under every condition; there is enormous diversity. The range of habitats in which organisms live, from hot springs to Arctic tundra, from animal intestines to college dormitories, is matched by a correspondingly wide range of specific biochemical adaptations, achieved



FIGURE 1-2 Diverse living organisms share common chemical features. Birds, beasts, plants, and soil microorganisms share with humans the same basic structural units (cells) and the same kinds of macromolecules (DNA, RNA, proteins) made up of the same kinds of monomeric subunits (nucleotides, amino acids). They utilize the same pathways for synthesis of cellular components, share the same genetic code, and derive from the same evolutionary ancestors. Shown here is a detail from “The Garden of Eden,” by Jan van Kessel the Younger (1626–1679).

within a common chemical framework. For the sake of clarity, in this book we sometimes risk certain generalizations, which, though not perfect, remain useful; we also frequently point out the exceptions that illuminate scientific generalizations.

Biochemistry describes in molecular terms the structures, mechanisms, and chemical processes shared by all organisms and provides organizing principles that underlie life in all its diverse forms, principles we refer to collectively as *the molecular logic of life*. Although biochemistry provides important insights and practical applications in medicine, agriculture, nutrition, and industry, its ultimate concern is with the wonder of life itself.

In this introductory chapter, then, we describe (briefly!) the cellular, chemical, physical (thermodynamic), and genetic backgrounds to biochemistry and the overarching principle of evolution—the development over generations of the properties of living cells. As you read through the book, you may find it helpful to refer back to this chapter at intervals to refresh your memory of this background material.

1.1 Cellular Foundations

The unity and diversity of organisms become apparent even at the cellular level. The smallest organisms consist of single cells and are microscopic. Larger, multicellular organisms contain many different types of cells, which vary in size, shape, and specialized function. Despite these obvious differences, all cells of the simplest and most complex organisms share certain fundamental properties, which can be seen at the biochemical level.

Cells Are the Structural and Functional Units of All Living Organisms

Cells of all kinds share certain structural features (Fig. 1–3). The **plasma membrane** defines the periphery of the cell, separating its contents from the surroundings. It is composed of lipid and protein molecules that form a thin, tough, pliable, hydrophobic barrier around the cell. The membrane is a barrier to the free passage of inorganic ions and most other charged or polar compounds. Transport proteins in the plasma membrane allow the passage of certain ions and molecules; receptor proteins transmit signals into the cell; and membrane enzymes participate in some reaction pathways. Because the individual lipids and proteins of the plasma membrane are not covalently linked, the entire structure is remarkably flexible, allowing changes in the shape and size of the cell. As a cell grows, newly made lipid and protein molecules are inserted into its plasma membrane; cell division produces two cells, each with its own membrane. This growth and cell division (fission) occurs without loss of membrane integrity.

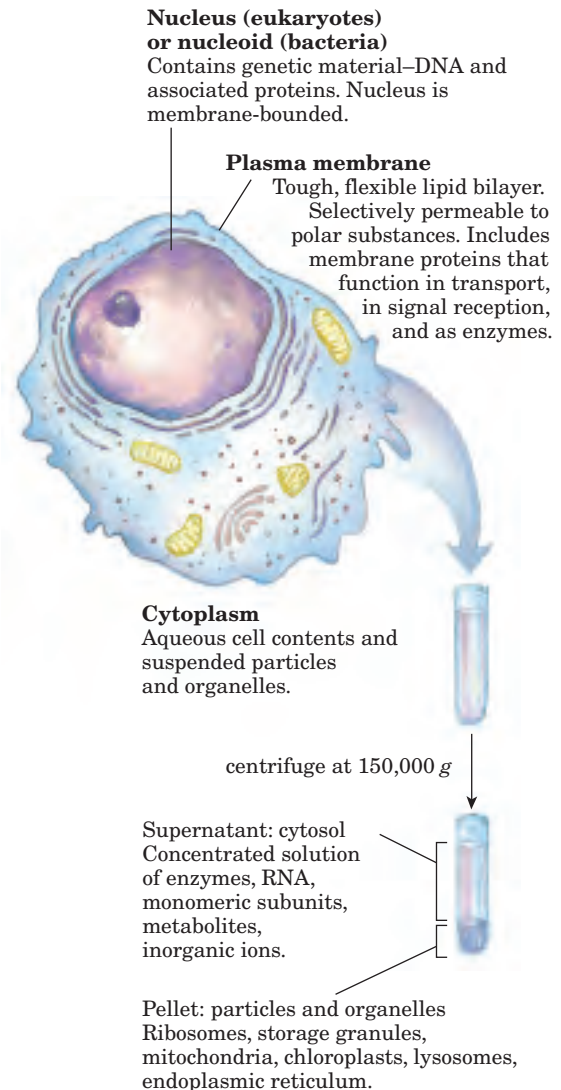


FIGURE 1–3 The universal features of living cells. All cells have a nucleus or nucleoid, a plasma membrane, and cytoplasm. The cytosol is defined as that portion of the cytoplasm that remains in the supernatant after centrifugation of a cell extract at 150,000 g for 1 hour.

The internal volume bounded by the plasma membrane, the **cytoplasm** (Fig. 1–3), is composed of an aqueous solution, the **cytosol**, and a variety of suspended particles with specific functions. The cytosol is a highly concentrated solution containing enzymes and the RNA molecules that encode them; the components (amino acids and nucleotides) from which these macromolecules are assembled; hundreds of small organic molecules called **metabolites**, intermediates in biosynthetic and degradative pathways; **coenzymes**, compounds essential to many enzyme-catalyzed reactions; inorganic ions; and **ribosomes**, small particles (composed of protein and RNA molecules) that are the sites of protein synthesis.

All cells have, for at least some part of their life, either a **nucleus** or a **nucleoid**, in which the **genome**—

the complete set of genes, composed of DNA—is stored and replicated. The nucleoid, in bacteria, is not separated from the cytoplasm by a membrane; the nucleus, in higher organisms, consists of nuclear material enclosed within a double membrane, the nuclear envelope. Cells with nuclear envelopes are called **eukaryotes** (Greek *eu*, “true,” and *karyon*, “nucleus”); those without nuclear envelopes—bacterial cells—are **prokaryotes** (Greek *pro*, “before”).

Cellular Dimensions Are Limited by Oxygen Diffusion

Most cells are microscopic, invisible to the unaided eye. Animal and plant cells are typically 5 to 100 μm in diameter, and many bacteria are only 1 to 2 μm long (see the inside back cover for information on units and their abbreviations). What limits the dimensions of a cell? The lower limit is probably set by the minimum number of each type of biomolecule required by the cell. The smallest cells, certain bacteria known as mycoplasmas, are 300 nm in diameter and have a volume of about 10^{-14} mL. A single bacterial ribosome is about 20 nm in its longest dimension, so a few ribosomes take up a substantial fraction of the volume in a mycoplasmal cell.

The upper limit of cell size is probably set by the rate of diffusion of solute molecules in aqueous systems. For example, a bacterial cell that depends upon oxygen-consuming reactions for energy production must obtain

molecular oxygen by diffusion from the surrounding medium through its plasma membrane. The cell is so small, and the ratio of its surface area to its volume is so large, that every part of its cytoplasm is easily reached by O_2 diffusing into the cell. As cell size increases, however, surface-to-volume ratio decreases, until metabolism consumes O_2 faster than diffusion can supply it. Metabolism that requires O_2 thus becomes impossible as cell size increases beyond a certain point, placing a theoretical upper limit on the size of the cell.

There Are Three Distinct Domains of Life

All living organisms fall into one of three large groups (kingdoms, or domains) that define three branches of evolution from a common progenitor (Fig. 1–4). Two large groups of prokaryotes can be distinguished on biochemical grounds: **archaebacteria** (Greek *archē*, “origin”) and **eubacteria** (again, from Greek *eu*, “true”). Eubacteria inhabit soils, surface waters, and the tissues of other living or decaying organisms. Most of the well-studied bacteria, including *Escherichia coli*, are eubacteria. The archaebacteria, more recently discovered, are less well characterized biochemically; most inhabit extreme environments—salt lakes, hot springs, highly acidic bogs, and the ocean depths. The available evidence suggests that the archaebacteria and eubacteria diverged early in evolution and constitute two separate

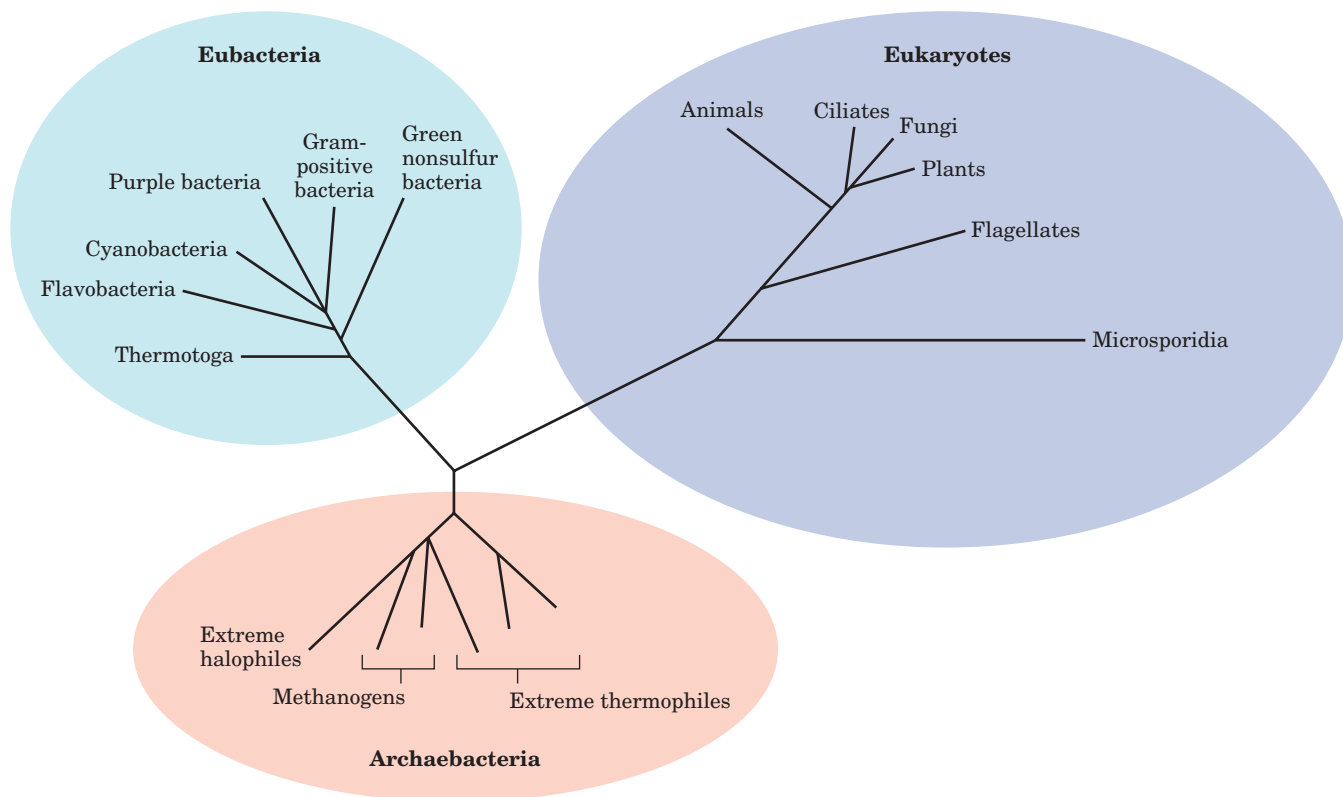


FIGURE 1-4 Phylogeny of the three domains of life. Phylogenetic relationships are often illustrated by a “family tree” of this type. The fewer the branch points between any two organisms, the closer is their evolutionary relationship.

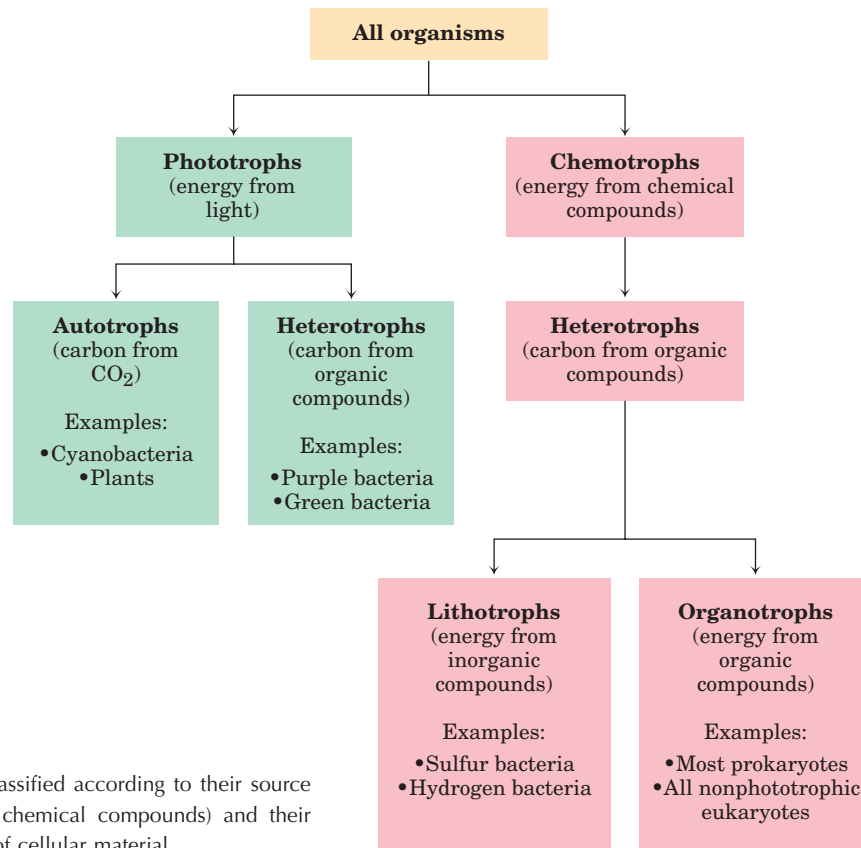


FIGURE 1-5 Organisms can be classified according to their source of energy (sunlight or oxidizable chemical compounds) and their source of carbon for the synthesis of cellular material.

domains, sometimes called Archaea and Bacteria. All eukaryotic organisms, which make up the third domain, Eukarya, evolved from the same branch that gave rise to the Archaea; archaeobacteria are therefore more closely related to eukaryotes than to eubacteria.

Within the domains of Archaea and Bacteria are subgroups distinguished by the habitats in which they live. In **aerobic** habitats with a plentiful supply of oxygen, some resident organisms derive energy from the transfer of electrons from fuel molecules to oxygen. Other environments are **anaerobic**, virtually devoid of oxygen, and microorganisms adapted to these environments obtain energy by transferring electrons to nitrate (forming N_2), sulfate (forming H_2S), or CO_2 (forming CH_4). Many organisms that have evolved in anaerobic environments are *obligate anaerobes*: they die when exposed to oxygen.

We can classify organisms according to how they obtain the energy and carbon they need for synthesizing cellular material (as summarized in Fig. 1–5). There are two broad categories based on energy sources: **phototrophs** (Greek *trophē*, “nourishment”) trap and use sunlight, and **chemotrophs** derive their energy from oxidation of a fuel. All chemotrophs require a source of organic nutrients; they cannot fix CO_2 into organic compounds. The phototrophs can be further divided into those that can obtain all needed carbon from CO_2 (**autotrophs**) and those that require organic nutrients (**heterotrophs**). No chemotroph can get its carbon

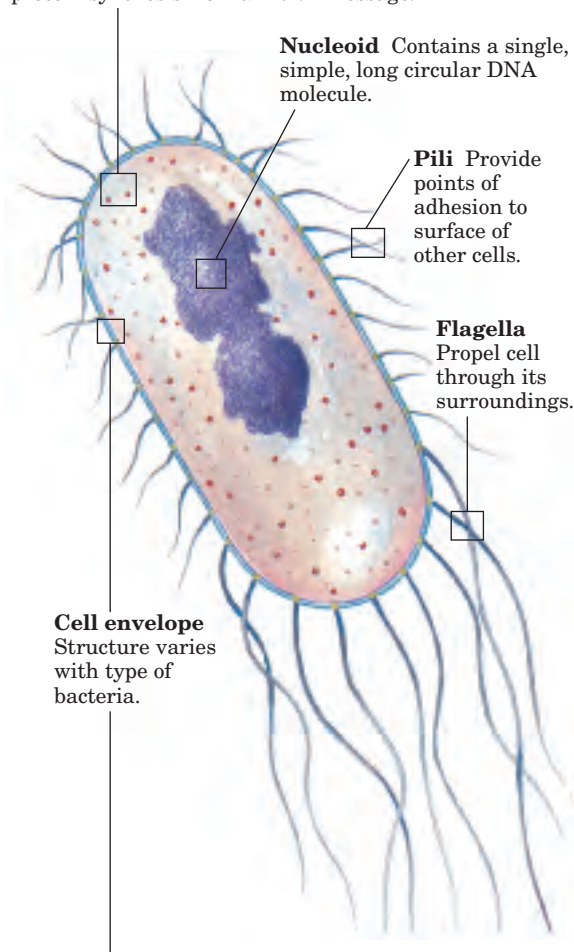
atoms exclusively from CO_2 (that is, no chemotrophs are autotrophs), but the chemotrophs may be further classified according to a different criterion: whether the fuels they oxidize are inorganic (**lithotrophs**) or organic (**organotrophs**).

Most known organisms fall within one of these four broad categories—autotrophs or heterotrophs among the photosynthesizers, lithotrophs or organotrophs among the chemical oxidizers. The prokaryotes have several general modes of obtaining carbon and energy. *Escherichia coli*, for example, is a chemoorganoheterotroph; it requires organic compounds from its environment as fuel and as a source of carbon. Cyanobacteria are photolithoautotrophs; they use sunlight as an energy source and convert CO_2 into biomolecules. We humans, like *E. coli*, are chemoorganoheterotrophs.

Escherichia coli Is the Most-Studied Prokaryotic Cell

Bacterial cells share certain common structural features, but also show group-specific specializations (Fig. 1–6). *E. coli* is a usually harmless inhabitant of the human intestinal tract. The *E. coli* cell is about $2\ \mu\text{m}$ long and a little less than $1\ \mu\text{m}$ in diameter. It has a protective outer membrane and an inner plasma membrane that encloses the cytoplasm and the nucleoid. Between the inner and outer membranes is a thin but strong layer of polymers called peptidoglycans, which gives the cell its shape and rigidity. The plasma membrane and the

Ribosomes Bacterial ribosomes are smaller than eukaryotic ribosomes, but serve the same function—protein synthesis from an RNA message.

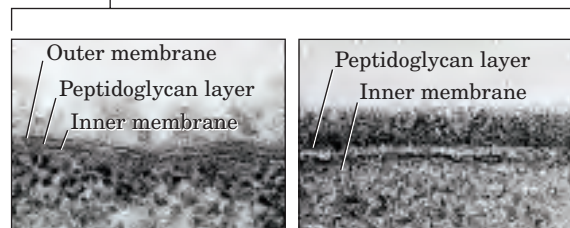


Nucleoid Contains a single, simple, long circular DNA molecule.

Pili Provide points of adhesion to surface of other cells.

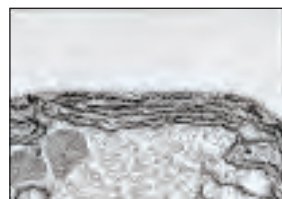
Flagella Propel cell through its surroundings.

Cell envelope
Structure varies with type of bacteria.



Gram-negative bacteria
Outer membrane;
peptidoglycan layer

Gram-positive bacteria
No outer membrane;
thicker peptidoglycan layer



Cyanobacteria
Gram-negative; tougher
peptidoglycan layer;
extensive internal
membrane system with
photosynthetic pigments



Archaeobacteria
No outer membrane;
peptidoglycan layer outside
plasma membrane

FIGURE 1-6 Common structural features of bacterial cells. Because of differences in the cell envelope structure, some eubacteria (gram-positive bacteria) retain Gram's stain, and others (gram-negative bacteria) do not. *E. coli* is gram-negative. Cyanobacteria are also eubacteria but are distinguished by their extensive internal membrane system, in which photosynthetic pigments are localized. Although the cell envelopes of archaeobacteria and gram-positive eubacteria look similar under the electron microscope, the structures of the membrane lipids and the polysaccharides of the cell envelope are distinctly different in these organisms.

layers outside it constitute the **cell envelope**. In the Archaea, rigidity is conferred by a different type of polymer (pseudopeptidoglycan). The plasma membranes of eubacteria consist of a thin bilayer of lipid molecules penetrated by proteins. Archaeobacterial membranes have a similar architecture, although their lipids differ strikingly from those of the eubacteria.

The cytoplasm of *E. coli* contains about 15,000 ribosomes, thousands of copies each of about 1,000 different enzymes, numerous metabolites and cofactors, and a variety of inorganic ions. The nucleoid contains a single, circular molecule of DNA, and the cytoplasm (like that of most bacteria) contains one or more smaller, circular segments of DNA called **plasmids**. In nature, some plasmids confer resistance to toxins and antibiotics in the environment. In the laboratory, these DNA segments are especially amenable to experimental manipulation and are extremely useful to molecular geneticists.

Most bacteria (including *E. coli*) lead existences as individual cells, but in some bacterial species cells tend to associate in clusters or filaments, and a few (the myxobacteria, for example) demonstrate simple social behavior.

Eukaryotic Cells Have a Variety of Membranous Organelles, Which Can Be Isolated for Study

Typical eukaryotic cells (Fig. 1-7) are much larger than prokaryotic cells—commonly 5 to 100 μm in diameter, with cell volumes a thousand to a million times larger than those of bacteria. The distinguishing characteristics of eukaryotes are the nucleus and a variety of membrane-bounded organelles with specific functions: mitochondria, endoplasmic reticulum, Golgi complexes, and lysosomes. Plant cells also contain vacuoles and chloroplasts (Fig. 1-7). Also present in the cytoplasm of many cells are granules or droplets containing stored nutrients such as starch and fat.

In a major advance in biochemistry, Albert Claude, Christian de Duve, and George Palade developed methods for separating organelles from the cytosol and from each other—an essential step in isolating biomolecules and larger cell components and investigating their

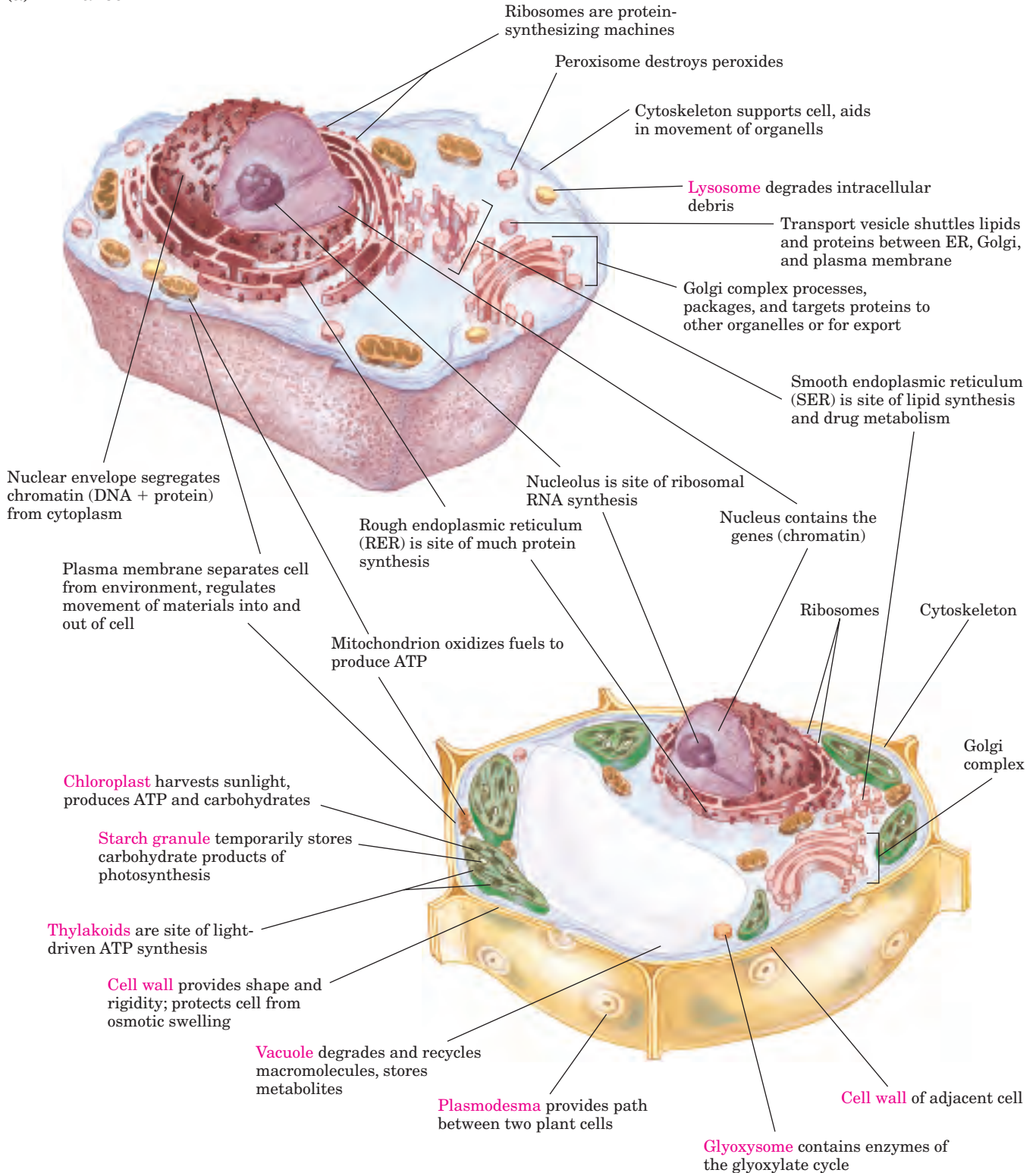
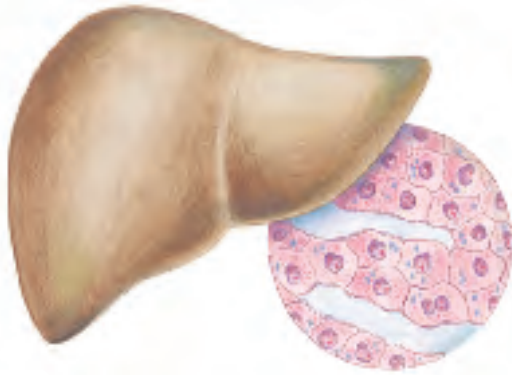
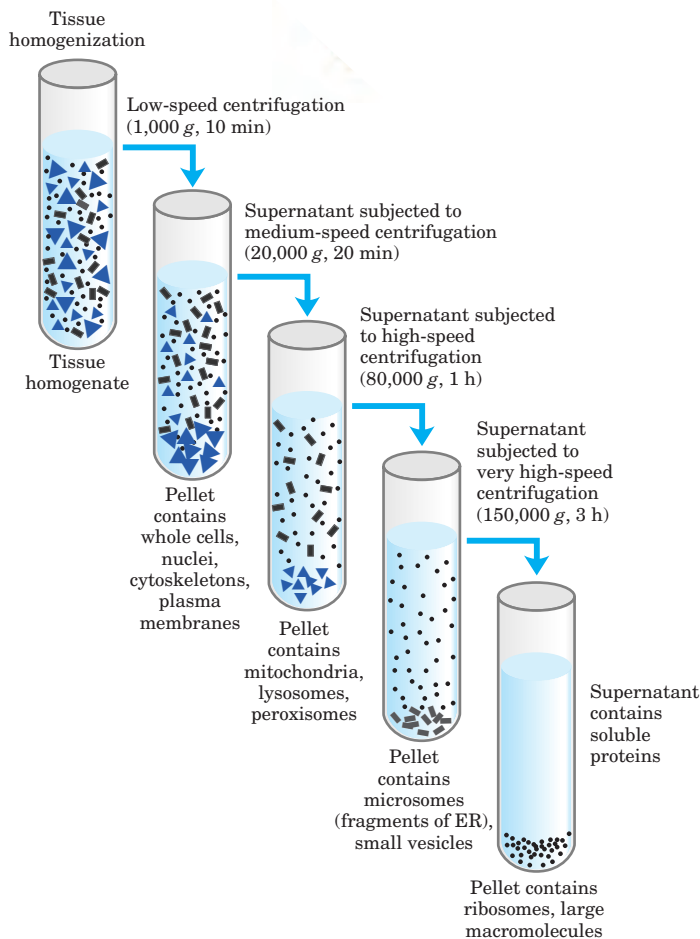
(a) Animal cell**(b) Plant cell**

FIGURE 1-7 Eukaryotic cell structure. Schematic illustrations of the two major types of eukaryotic cell: **(a)** a representative animal cell and **(b)** a representative plant cell. Plant cells are usually 10 to 100 μm in diameter—larger than animal cells, which typically range from 5 to 30 μm . Structures labeled in red are unique to either animal or plant cells.

structures and functions. In a typical cell fractionation (Fig. 1–8), cells or tissues in solution are disrupted by gentle homogenization. This treatment ruptures the plasma membrane but leaves most of the organelles intact. The homogenate is then centrifuged; organelles such as nuclei, mitochondria, and lysosomes differ in size and therefore sediment at different rates. They also differ in specific gravity, and they “float” at different levels in a density gradient.



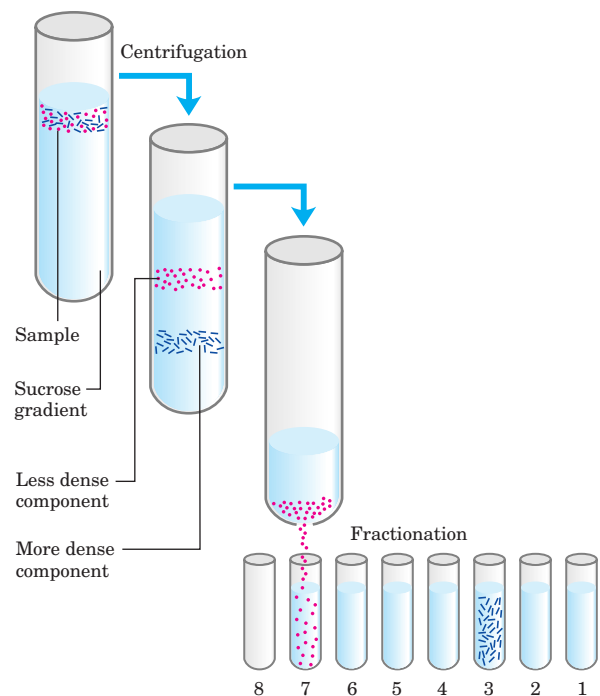
(a) Differential centrifugation



Differential centrifugation results in a rough fractionation of the cytoplasmic contents, which may be further purified by isopycnic (“same density”) centrifugation. In this procedure, organelles of different buoyant densities (the result of different ratios of lipid and protein in each type of organelle) are separated on a density gradient. By carefully removing material from each region of the gradient and observing it with a microscope, the biochemist can establish the sedimentation position of each organelle

FIGURE 1–8 Subcellular fractionation of tissue. A tissue such as liver is first mechanically homogenized to break cells and disperse their contents in an aqueous buffer. The sucrose medium has an osmotic pressure similar to that in organelles, thus preventing diffusion of water into the organelles, which would swell and burst. **(a)** The large and small particles in the suspension can be separated by centrifugation at different speeds, or **(b)** particles of different density can be separated by isopycnic centrifugation. In isopycnic centrifugation, a centrifuge tube is filled with a solution, the density of which increases from top to bottom; a solute such as sucrose is dissolved at different concentrations to produce the density gradient. When a mixture of organelles is layered on top of the density gradient and the tube is centrifuged at high speed, individual organelles sediment until their buoyant density exactly matches that in the gradient. Each layer can be collected separately.

(b) Isopycnic (sucrose-density) centrifugation



and obtain purified organelles for further study. For example, these methods were used to establish that lysosomes contain degradative enzymes, mitochondria contain oxidative enzymes, and chloroplasts contain photosynthetic pigments. The isolation of an organelle enriched in a certain enzyme is often the first step in the purification of that enzyme.

The Cytoplasm Is Organized by the Cytoskeleton and Is Highly Dynamic

Electron microscopy reveals several types of protein filaments crisscrossing the eukaryotic cell, forming an interlocking three-dimensional meshwork, the **cytoskeleton**. There are three general types of cytoplasmic filaments—actin filaments, microtubules, and intermediate filaments (Fig. 1–9)—differing in width (from about 6 to 22 nm), composition, and specific function. All types provide structure and organization to the cytoplasm and shape to the cell. Actin filaments and microtubules also help to produce the motion of organelles or of the whole cell.

Each type of cytoskeletal component is composed of simple protein subunits that polymerize to form filaments of uniform thickness. These filaments are not permanent structures; they undergo constant disassembly

into their protein subunits and reassembly into filaments. Their locations in cells are not rigidly fixed but may change dramatically with mitosis, cytokinesis, amoeboid motion, or changes in cell shape. The assembly, disassembly, and location of all types of filaments are regulated by other proteins, which serve to link or bundle the filaments or to move cytoplasmic organelles along the filaments.

The picture that emerges from this brief survey of cell structure is that of a eukaryotic cell with a meshwork of structural fibers and a complex system of membrane-bounded compartments (Fig. 1–7). The filaments disassemble and then reassemble elsewhere. Membranous vesicles bud from one organelle and fuse with another. Organelles move through the cytoplasm along protein filaments, their motion powered by energy dependent motor proteins. The **endomembrane system** segregates specific metabolic processes and provides surfaces on which certain enzyme-catalyzed reactions occur. **Exocytosis** and **endocytosis**, mechanisms of transport (out of and into cells, respectively) that involve membrane fusion and fission, provide paths between the cytoplasm and surrounding medium, allowing for secretion of substances produced within the cell and uptake of extracellular materials.

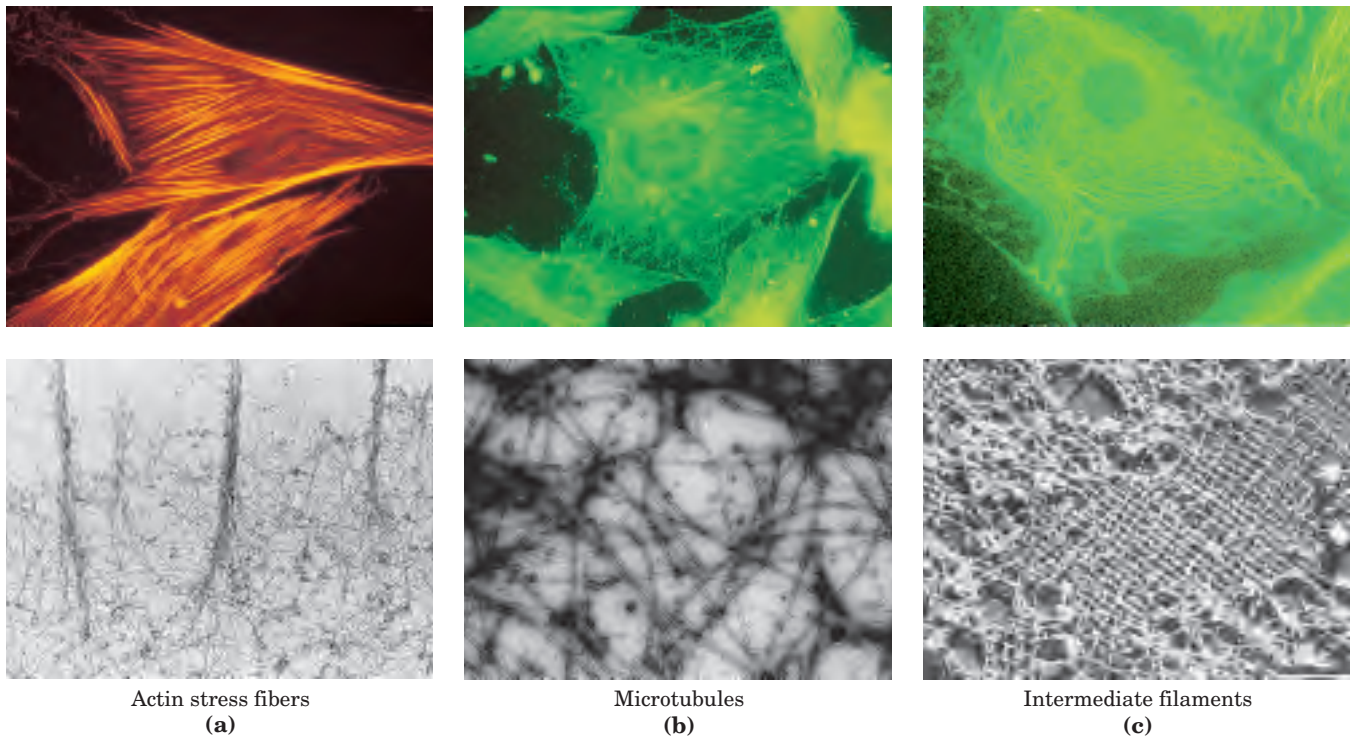


FIGURE 1-9 The three types of cytoskeletal filaments. The upper panels show epithelial cells photographed after treatment with antibodies that bind to and specifically stain (a) actin filaments bundled together to form “stress fibers,” (b) microtubules radiating from the cell center, and (c) intermediate filaments extending throughout the cytoplasm. For these experiments, antibodies that specifically recognize actin, tubu-

lin, or intermediate filament proteins are covalently attached to a fluorescent compound. When the cell is viewed with a fluorescence microscope, only the stained structures are visible. The lower panels show each type of filament as visualized by (a, b) transmission or (c) scanning electron microscopy.

Although complex, this organization of the cytoplasm is far from random. The motion and the positioning of organelles and cytoskeletal elements are under tight regulation, and at certain stages in a eukaryotic cell's life, dramatic, finely orchestrated reorganizations, such as the events of mitosis, occur. The interactions between the cytoskeleton and organelles are noncovalent,

reversible, and subject to regulation in response to various intracellular and extracellular signals.

Cells Build Supramolecular Structures

Macromolecules and their monomeric subunits differ greatly in size (Fig. 1–10). A molecule of alanine is less than 0.5 nm long. Hemoglobin, the oxygen-carrying protein of erythrocytes (red blood cells), consists of nearly 600 amino acid subunits in four long chains, folded into globular shapes and associated in a structure 5.5 nm in diameter. In turn, proteins are much smaller than ribosomes (about 20 nm in diameter), which are in turn much smaller than organelles such as mitochondria, typically 1,000 nm in diameter. It is a long jump from simple biomolecules to cellular structures that can be seen

(a) Some of the amino acids of proteins

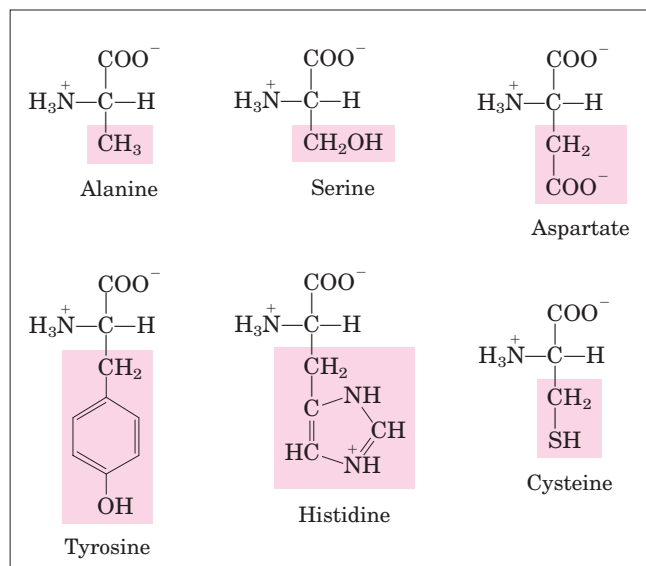
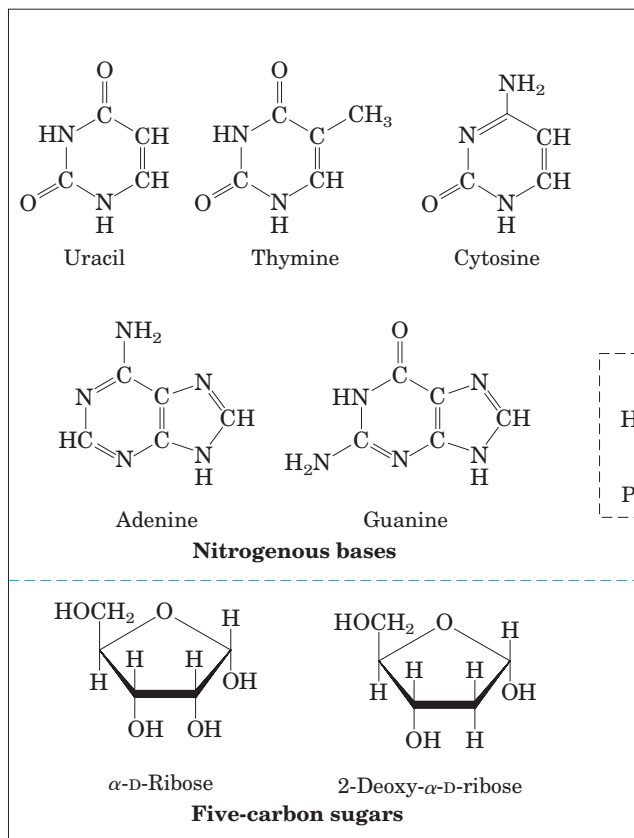
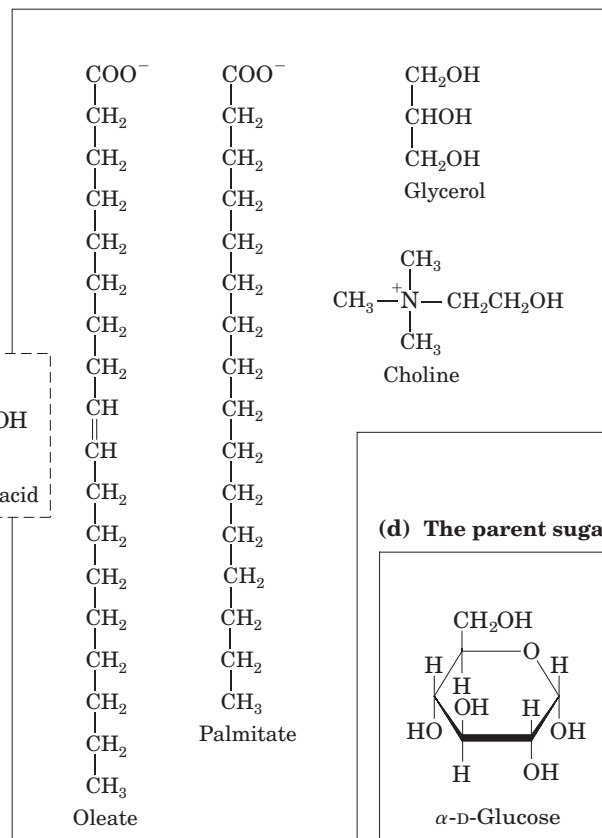


FIGURE 1-10 The organic compounds from which most cellular materials are constructed: the ABCs of biochemistry. Shown here are (a) six of the 20 amino acids from which all proteins are built (the side chains are shaded pink); (b) the five nitrogenous bases, two five-carbon sugars, and phosphoric acid from which all nucleic acids are built; (c) five components of membrane lipids; and (d) D-glucose, the parent sugar from which most carbohydrates are derived. Note that phosphoric acid is a component of both nucleic acids and membrane lipids.

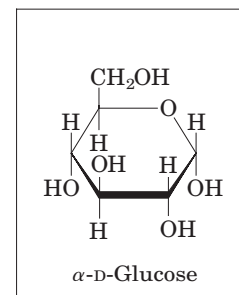
(b) The components of nucleic acids



(c) Some components of lipids



(d) The parent sugar



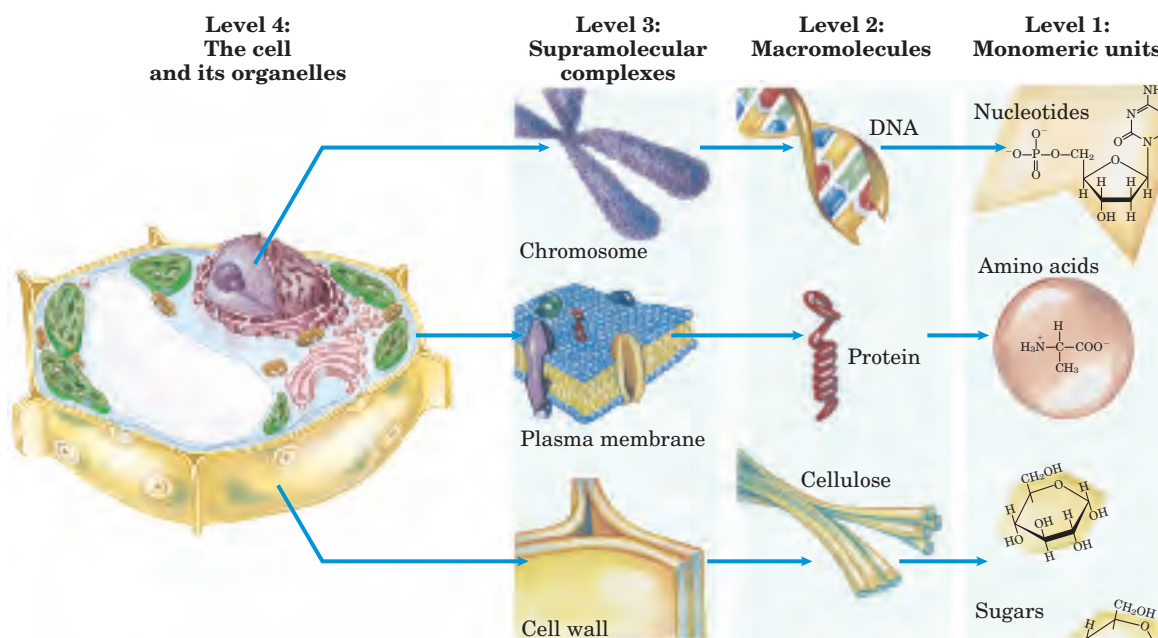


FIGURE 1-11 Structural hierarchy in the molecular organization of cells. In this plant cell, the nucleus is an organelle containing several types of supramolecular complexes, including chromosomes. Chro-

mosomes consist of macromolecules of DNA and many different proteins. Each type of macromolecule is made up of simple subunits—DNA of nucleotides (deoxyribonucleotides), for example.

with the light microscope. Figure 1-11 illustrates the structural hierarchy in cellular organization.

The monomeric subunits in proteins, nucleic acids, and polysaccharides are joined by covalent bonds. In supramolecular complexes, however, macromolecules are held together by noncovalent interactions—much weaker, individually, than covalent bonds. Among these noncovalent interactions are hydrogen bonds (between polar groups), ionic interactions (between charged groups), hydrophobic interactions (among nonpolar groups in aqueous solution), and van der Waals interactions—all of which have energies substantially smaller than those of covalent bonds (Table 1-1). The nature of these noncovalent interactions is described in Chapter 2. The large numbers of weak interactions between macromolecules in supramolecular complexes stabilize these assemblies, producing their unique structures.

In Vitro Studies May Overlook Important Interactions among Molecules

One approach to understanding a biological process is to study purified molecules *in vitro* (“in glass”—in the test tube), without interference from other molecules present in the intact cell—that is, *in vivo* (“in the living”). Although this approach has been remarkably revealing, we must keep in mind that the inside of a cell is quite different from the inside of a test tube. The “interfering” components eliminated by purification may be critical to the biological function or regulation of the molecule purified. For example, *in vitro* studies of pure

enzymes are commonly done at very low enzyme concentrations in thoroughly stirred aqueous solutions. In the cell, an enzyme is dissolved or suspended in a gel-like cytosol with thousands of other proteins, some of which bind to that enzyme and influence its activity.

TABLE 1-1 Strengths of Bonds Common in Biomolecules

Type of bond	Bond dissociation energy* (kJ/mol)	Type of bond	Bond dissociation energy (kJ/mol)
Single bonds		Double bonds	
O—H	470	C=O	712
H—H	435	C=N	615
P—O	419	C=C	611
C—H	414	P=O	502
N—H	389	Triple bonds	
C—O	352	C≡C	816
C—C	348	N≡N	930
S—H	339		
C—N	293		
C—S	260		
N—O	222		
S—S	214		

*The greater the energy required for bond dissociation (breakage), the stronger the bond.

Some enzymes are parts of multienzyme complexes in which reactants are channeled from one enzyme to another without ever entering the bulk solvent. Diffusion is hindered in the gel-like cytosol, and the cytosolic composition varies in different regions of the cell. In short, a given molecule may function quite differently in the cell than in vitro. A central challenge of biochemistry is to understand the influences of cellular organization and macromolecular associations on the function of individual enzymes and other biomolecules—to understand function in vivo as well as in vitro.

SUMMARY 1.1 Cellular Foundations

- All cells are bounded by a plasma membrane; have a cytosol containing metabolites, coenzymes, inorganic ions, and enzymes; and have a set of genes contained within a nucleoid (prokaryotes) or nucleus (eukaryotes).
- Phototrophs use sunlight to do work; chemotrophs oxidize fuels, passing electrons to good electron acceptors: inorganic compounds, organic compounds, or molecular oxygen.
- Bacterial cells contain cytosol, a nucleoid, and plasmids. Eukaryotic cells have a nucleus and are multicompartmented, segregating certain processes in specific organelles, which can be separated and studied in isolation.
- Cytoskeletal proteins assemble into long filaments that give cells shape and rigidity and serve as rails along which cellular organelles move throughout the cell.
- Supramolecular complexes are held together by noncovalent interactions and form a hierarchy of structures, some visible with the light microscope. When individual molecules are removed from these complexes to be studied in vitro, interactions important in the living cell may be lost.

1.2 Chemical Foundations

Biochemistry aims to explain biological form and function in chemical terms. As we noted earlier, one of the most fruitful approaches to understanding biological phenomena has been to purify an individual chemical component, such as a protein, from a living organism and to characterize its structural and chemical characteristics. By the late eighteenth century, chemists had concluded that the composition of living matter is strikingly different from that of the inanimate world. Antoine Lavoisier (1743–1794) noted the relative chemical simplicity of the “mineral world” and contrasted it with the complexity of the “plant and animal worlds”; the latter, he knew, were composed of compounds rich in the elements carbon, oxygen, nitrogen, and phosphorus.

During the first half of the twentieth century, parallel biochemical investigations of glucose breakdown in yeast and in animal muscle cells revealed remarkable chemical similarities in these two apparently very different cell types; the breakdown of glucose in yeast and muscle cells involved the same ten chemical intermediates. Subsequent studies of many other biochemical processes in many different organisms have confirmed the generality of this observation, neatly summarized by Jacques Monod: “What is true of *E. coli* is true of the elephant.” The current understanding that all organisms share a common evolutionary origin is based in part on this observed universality of chemical intermediates and transformations.

Only about 30 of the more than 90 naturally occurring chemical elements are essential to organisms. Most of the elements in living matter have relatively low atomic numbers; only five have atomic numbers above that of selenium, 34 (Fig. 1–12). The four most abundant elements in living organisms, in terms of percentage of total number of atoms, are hydrogen, oxygen, nitrogen, and carbon, which together make up more than 99% of the mass of most cells. They are the lightest elements capable of forming one, two, three, and four bonds, respectively; in general, the lightest elements

1 H																	2 He													
3 Li	4 Be															5 B	6 C	7 N	8 O	9 F	10 Ne									
11 Na	12 Mg															13 Al	14 Si	15 P	16 S	17 Cl	18 Ar									
19 K	20 Ca	21 Sc	22 Ti	23 V	24 Cr	25 Mn	26 Fe	27 Co	28 Ni	29 Cu	30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr													
37 Rb	38 Sr	39 Y	40 Zr	41 Nb	42 Mo	43 Tc	44 Ru	45 Rh	46 Pd	47 Ag	48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe													
55 Cs	56 Ba															72 Hf	73 Ta	74 W	75 Re	76 Os	77 Ir	78 Pt	79 Au	80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
87 Fr	88 Ra																													

Bulk elements

Trace elements

Lanthanides

Actinides

FIGURE 1-12 Elements essential to animal life and health. Bulk elements (shaded orange) are structural components of cells and tissues and are required in the diet in gram quantities daily. For trace elements (shaded bright yellow), the requirements are much smaller: for humans, a few milligrams per day of Fe, Cu, and Zn, even less of the others. The elemental requirements for plants and microorganisms are similar to those shown here; the ways in which they acquire these elements vary.

form the strongest bonds. The trace elements (Fig. 1–12) represent a miniscule fraction of the weight of the human body, but all are essential to life, usually because they are essential to the function of specific proteins, including enzymes. The oxygen-transporting capacity of the hemoglobin molecule, for example, is absolutely dependent on four iron ions that make up only 0.3% of its mass.

Biomolecules Are Compounds of Carbon with a Variety of Functional Groups

The chemistry of living organisms is organized around carbon, which accounts for more than half the dry weight of cells. Carbon can form single bonds with hydrogen atoms, and both single and double bonds with oxygen and nitrogen atoms (Fig. 1–13). Of greatest significance in biology is the ability of carbon atoms to form very stable carbon–carbon single bonds. Each carbon atom can form single bonds with up to four other carbon atoms. Two carbon atoms also can share two (or three) electron pairs, thus forming double (or triple) bonds.

The four single bonds that can be formed by a carbon atom are arranged tetrahedrally, with an angle of

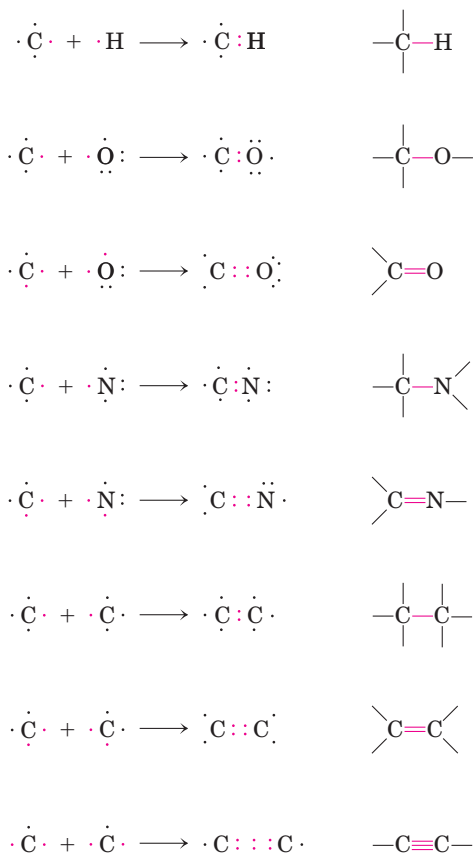


FIGURE 1–13 Versatility of carbon bonding. Carbon can form covalent single, double, and triple bonds (in red), particularly with other carbon atoms. Triple bonds are rare in biomolecules.

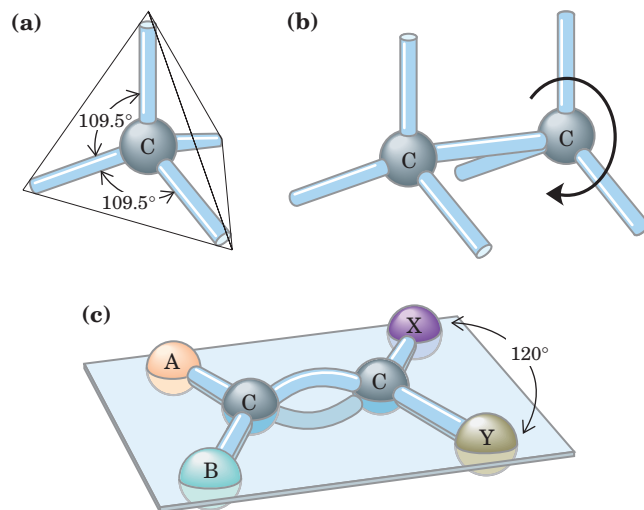


FIGURE 1–14 Geometry of carbon bonding. (a) Carbon atoms have a characteristic tetrahedral arrangement of their four single bonds. (b) Carbon–carbon single bonds have freedom of rotation, as shown for the compound ethane ($\text{CH}_3\text{—CH}_3$). (c) Double bonds are shorter and do not allow free rotation. The two doubly bonded carbons and the atoms designated A, B, X, and Y all lie in the same rigid plane.

about 109.5° between any two bonds (Fig. 1–14) and an average length of 0.154 nm. There is free rotation around each single bond, unless very large or highly charged groups are attached to both carbon atoms, in which case rotation may be restricted. A double bond is shorter (about 0.134 nm) and rigid and allows little rotation about its axis.

Covalently linked carbon atoms in biomolecules can form linear chains, branched chains, and cyclic structures. To these carbon skeletons are added groups of other atoms, called **functional groups**, which confer specific chemical properties on the molecule. It seems likely that the bonding versatility of carbon was a major factor in the selection of carbon compounds for the molecular machinery of cells during the origin and evolution of living organisms. No other chemical element can form molecules of such widely different sizes and shapes or with such a variety of functional groups.

Most biomolecules can be regarded as derivatives of hydrocarbons, with hydrogen atoms replaced by a variety of functional groups to yield different families of organic compounds. Typical of these are alcohols, which have one or more hydroxyl groups; amines, with amino groups; aldehydes and ketones, with carbonyl groups; and carboxylic acids, with carboxyl groups (Fig. 1–15). Many biomolecules are polyfunctional, containing two or more different kinds of functional groups (Fig. 1–16), each with its own chemical characteristics and reactions. The chemical “personality” of a compound is determined by the chemistry of its functional groups and their disposition in three-dimensional space.

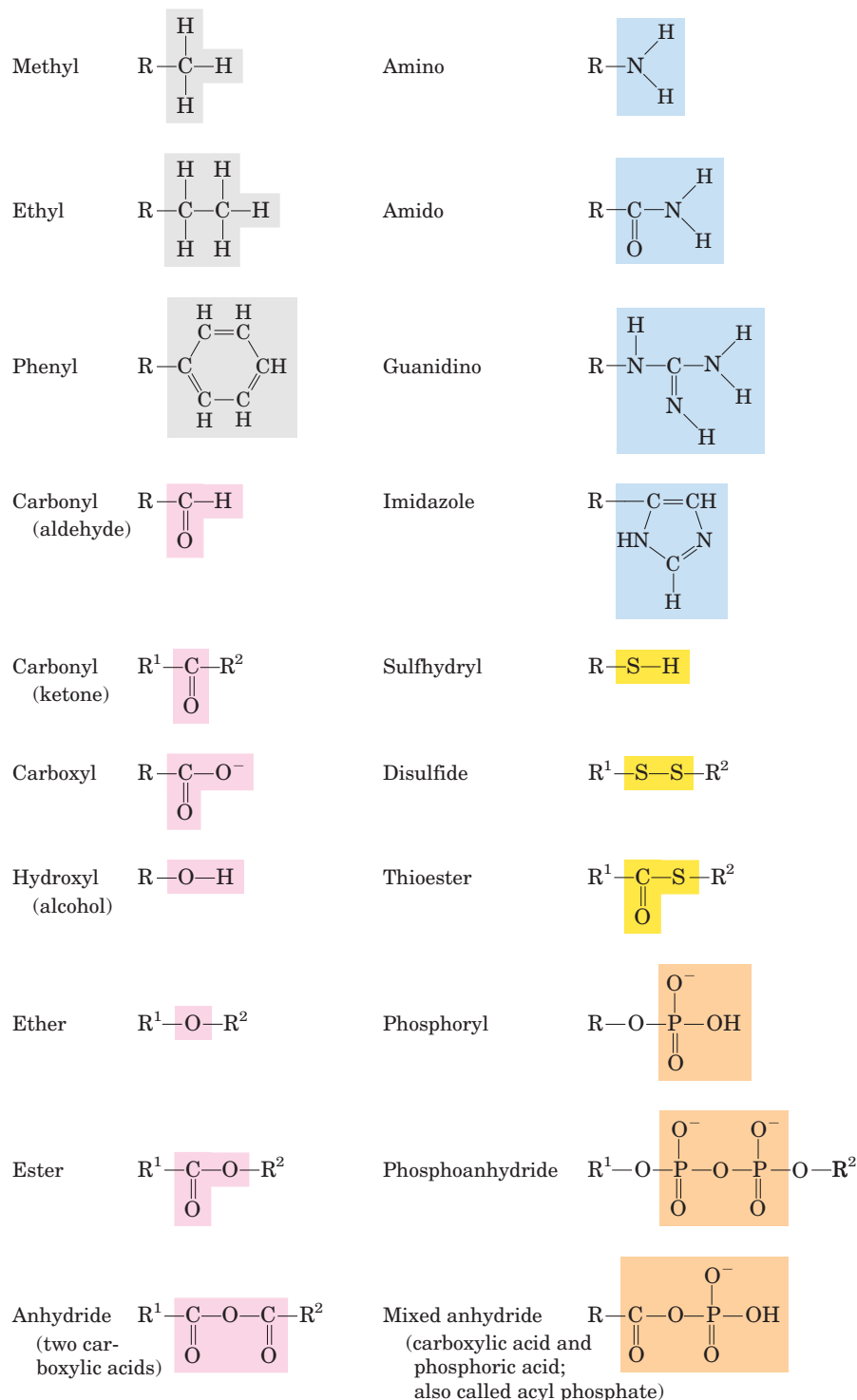


FIGURE 1-15 Some common functional groups of biomolecules. In this figure and throughout the book, we use R to represent “any substituent.” It may be as simple as a hydrogen atom, but typically it is a carbon-containing moiety. When two or more substituents are shown in a molecule, we designate them R¹, R², and so forth.

Cells Contain a Universal Set of Small Molecules

Dissolved in the aqueous phase (cytosol) of all cells is a collection of 100 to 200 different small organic molecules (M_r ~100 to ~500), the central metabolites in the major pathways occurring in nearly every cell—the metabolites and pathways that have been conserved throughout the course of evolution. (See Box 1-1 for an explanation of the various ways of referring to molecu-

lar weight.) This collection of molecules includes the common amino acids, nucleotides, sugars and their phosphorylated derivatives, and a number of mono-, di-, and tricarboxylic acids. The molecules are polar or charged, water soluble, and present in micromolar to millimolar concentrations. They are trapped within the cell because the plasma membrane is impermeable to them—although specific membrane transporters can catalyze the movement of some molecules into and out

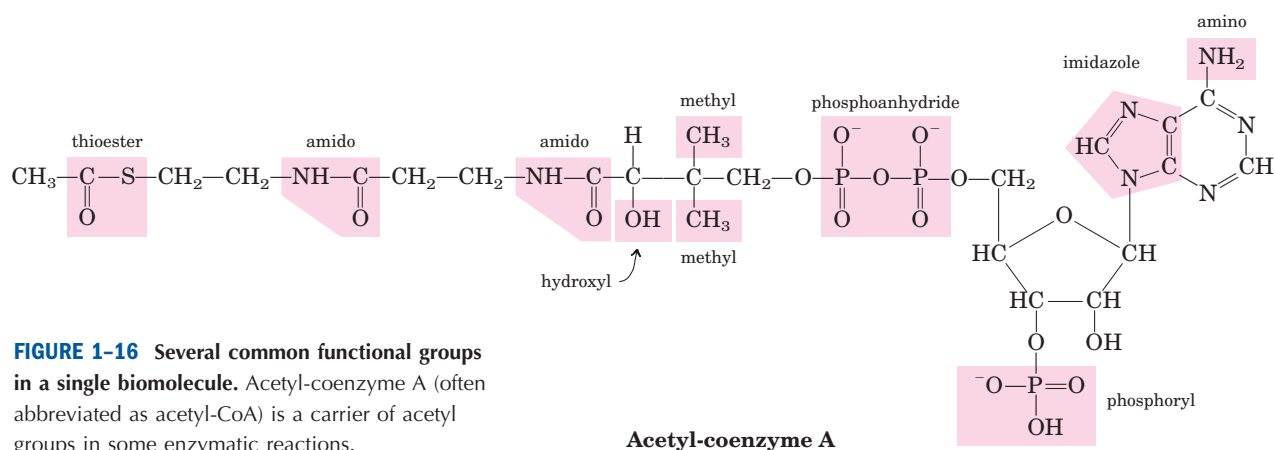


FIGURE 1-16 Several common functional groups in a single biomolecule. Acetyl-coenzyme A (often abbreviated as acetyl-CoA) is a carrier of acetyl groups in some enzymatic reactions.

of the cell or between compartments in eukaryotic cells. The universal occurrence of the same set of compounds in living cells is a manifestation of the universality of metabolic design, reflecting the evolutionary conservation of metabolic pathways that developed in the earliest cells.

There are other small biomolecules, specific to certain types of cells or organisms. For example, vascular plants contain, in addition to the universal set, small molecules called **secondary metabolites**, which play a role specific to plant life. These metabolites include compounds that give plants their characteristic scents, and compounds such as morphine, quinine, nicotine, and caffeine that are valued for their physiological effects on humans but used for other purposes by plants. The entire collection of small molecules in a given cell has been called that cell's **metabolome**, in parallel with the term “genome” (defined earlier and expanded on in

Section 1.4). If we knew the composition of a cell's metabolome, we could predict which enzymes and metabolic pathways were active in that cell.

Macromolecules Are the Major Constituents of Cells

Many biological molecules are macromolecules, polymers of high molecular weight assembled from relatively simple precursors. Proteins, nucleic acids, and polysaccharides are produced by the polymerization of relatively small compounds with molecular weights of 500 or less. The number of polymerized units can range from tens to millions. Synthesis of macromolecules is a major energy-consuming activity of cells. Macromolecules themselves may be further assembled into supramolecular complexes, forming functional units such as ribosomes. Table 1–2 shows the major classes of biomolecules in the bacterium *E. coli*.

BOX 1-1 WORKING IN BIOCHEMISTRY

Molecular Weight, Molecular Mass, and Their Correct Units

There are two common (and equivalent) ways to describe molecular mass; both are used in this text. The first is *molecular weight*, or *relative molecular mass*, denoted M_r . The molecular weight of a substance is defined as the ratio of the mass of a molecule of that substance to one-twelfth the mass of carbon-12 (^{12}C). Since M_r is a ratio, it is dimensionless—it has no associated units. The second is *molecular mass*, denoted m . This is simply the mass of one molecule, or the molar mass divided by Avogadro's number. The molecular mass, m , is expressed in daltons (abbreviated Da). One dalton is equivalent to one-twelfth the mass of carbon-12; a kilodalton (kDa) is 1,000 daltons; a megadalton (MDa) is 1 million daltons.

Consider, for example, a molecule with a mass 1,000 times that of water. We can say of this molecule either $M_r = 18,000$ or $m = 18,000$ daltons. We can also describe it as an “18 kDa molecule.” However, the expression $M_r = 18,000$ daltons is incorrect.

Another convenient unit for describing the mass of a single atom or molecule is the atomic mass unit (formerly amu, now commonly denoted u). One atomic mass unit (1 u) is defined as one-twelfth the mass of an atom of carbon-12. Since the experimentally measured mass of an atom of carbon-12 is 1.9926×10^{-23} g, $1 \text{ u} = 1.6606 \times 10^{-24}$ g. The atomic mass unit is convenient for describing the mass of a peak observed by mass spectrometry (see Box 3–2).

TABLE 1-2 Molecular Components of an *E. coli* Cell

	Percentage of total weight of cell	Approximate number of different molecular species
Water	70	1
Proteins	15	3,000
Nucleic acids		
DNA	1	1
RNA	6	>3,000
Polysaccharides	3	5
Lipids	2	20
Monomeric subunits and intermediates	2	500
Inorganic ions	1	20

Proteins, long polymers of amino acids, constitute the largest fraction (besides water) of cells. Some proteins have catalytic activity and function as enzymes; others serve as structural elements, signal receptors, or transporters that carry specific substances into or out of cells. Proteins are perhaps the most versatile of all biomolecules. The **nucleic acids**, DNA and RNA, are polymers of nucleotides. They store and transmit genetic information, and some RNA molecules have structural and catalytic roles in supramolecular complexes. The **polysaccharides**, polymers of simple sugars such as glucose, have two major functions: as energy-yielding fuel stores and as extracellular structural elements with specific binding sites for particular proteins. Shorter polymers of sugars (oligosaccharides) attached to proteins or lipids at the cell surface serve as specific cellular signals. The **lipids**, greasy or oily hydrocarbon derivatives, serve as structural components of membranes, energy-rich fuel stores, pigments, and intracellular signals. In proteins, nucleotides, polysaccharides, and lipids, the number of monomeric subunits is very large: molecular weights in the range of 5,000 to more than 1 million for proteins, up to several billion for nucleic acids, and in the millions for polysaccharides such as starch. Individual lipid molecules are much smaller (M_r 750 to 1,500) and are not classified as macromolecules. However, large numbers of lipid molecules can associate noncovalently into very large structures. Cellular membranes are built of enormous noncovalent aggregates of lipid and protein molecules.

Proteins and nucleic acids are **informational macromolecules**: each protein and each nucleic acid has a characteristic information-rich subunit sequence. Some oligosaccharides, with six or more different sug-

ars connected in branched chains, also carry information; on the outer surface of cells they serve as highly specific points of recognition in many cellular processes (as described in Chapter 7).

Three-Dimensional Structure Is Described by Configuration and Conformation

The covalent bonds and functional groups of a biomolecule are, of course, central to its function, but so also is the arrangement of the molecule's constituent atoms in three-dimensional space—its stereochemistry. A carbon-containing compound commonly exists as **stereoisomers**, molecules with the same chemical bonds but different stereochemistry—that is, different **configuration**, the fixed spatial arrangement of atoms. Interactions between biomolecules are invariably **stereospecific**, requiring specific stereochemistry in the interacting molecules.

Figure 1-17 shows three ways to illustrate the stereochemical structures of simple molecules. The perspective diagram specifies stereochemistry unambiguously, but bond angles and center-to-center bond lengths are better represented with ball-and-stick models. In space-

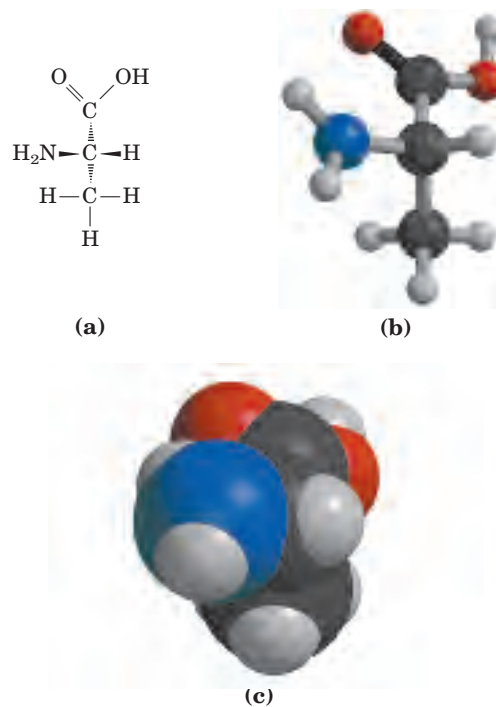


FIGURE 1-17 Representations of molecules. Three ways to represent the structure of the amino acid alanine. (a) Structural formula in perspective form: a solid wedge ($\rightarrow$) represents a bond in which the atom at the wide end projects out of the plane of the paper, toward the reader; a dashed wedge ($\dashrightarrow$) represents a bond extending behind the plane of the paper. (b) Ball-and-stick model, showing relative bond lengths and the bond angles. (c) Space-filling model, in which each atom is shown with its correct relative van der Waals radius.

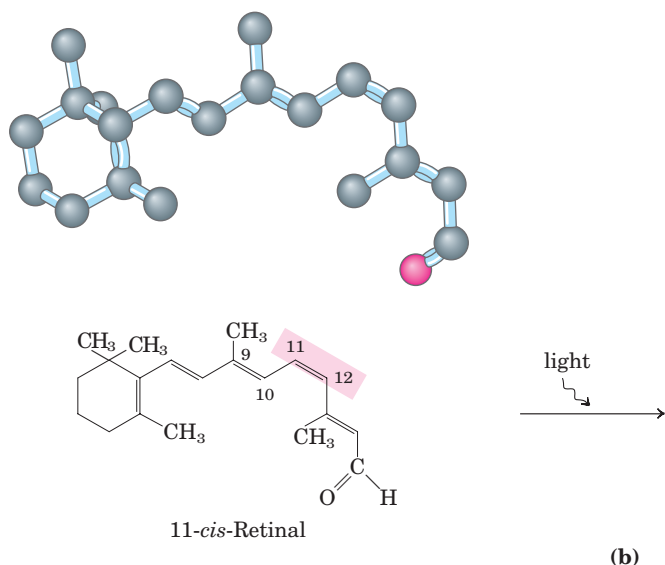
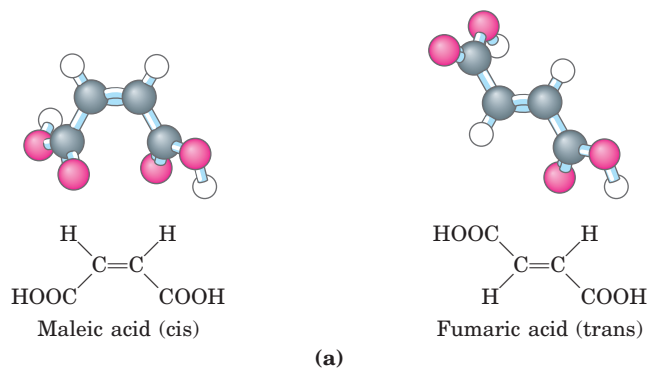


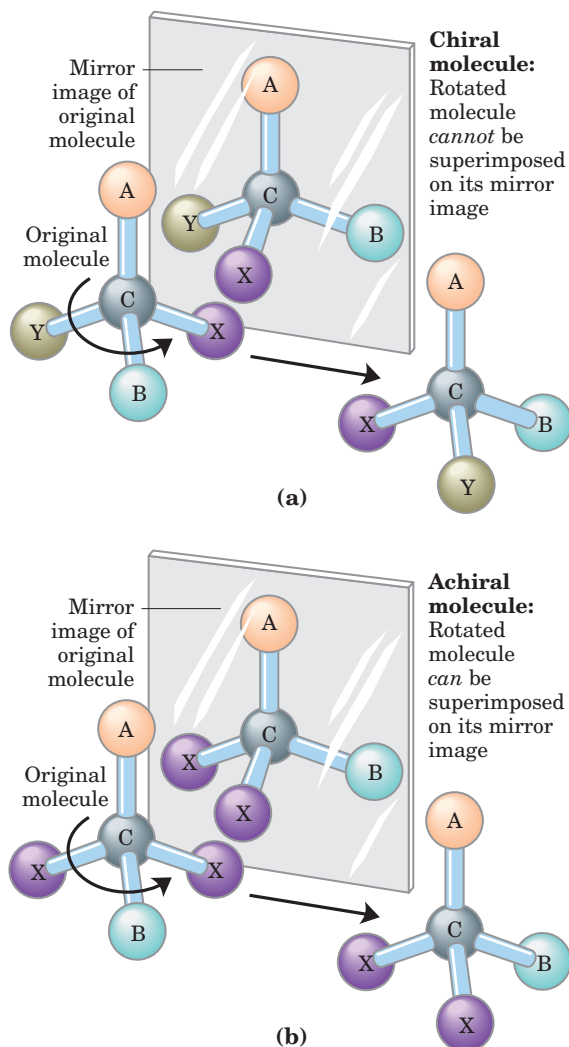
FIGURE 1-18 Configurations of geometric isomers. (a) Isomers such as maleic acid and fumaric acid cannot be interconverted without breaking covalent bonds, which requires the input of much energy. (b) In the vertebrate retina, the initial event in light detection is the absorption of visible light by 11-*cis*-retinal. The energy of the absorbed light (about 250 kJ/mol) converts 11-*cis*-retinal to all-*trans*-retinal, triggering electrical changes in the retinal cell that lead to a nerve impulse. (Note that the hydrogen atoms are omitted from the ball-and-stick models for the retinals.)

filling models, the radius of each atom is proportional to its van der Waals radius, and the contours of the model define the space occupied by the molecule (the volume of space from which atoms of other molecules are excluded).

Configuration is conferred by the presence of either (1) double bonds, around which there is no freedom of rotation, or (2) chiral centers, around which substituent groups are arranged in a specific sequence. The identifying characteristic of configurational isomers is that they cannot be interconverted without temporarily breaking one or more covalent bonds. Figure 1-18a shows the configurations of maleic acid and its isomer, fumaric acid. These compounds are **geometric, or cis-trans, isomers**; they differ in the arrangement of their substituent groups with respect to the nonrotating double bond (Latin *cis*, “on this side”—groups on the same side of the double bond; *trans*, “across”—groups on opposite sides). Maleic acid is the *cis* isomer and fumaric acid the *trans* isomer; each is a well-defined compound that can be separated from the other, and each has its own unique chemical properties. A binding site (on an enzyme, for example) that is complementary to one of these molecules would not be a suitable binding site for the other, which explains why the two compounds have distinct biological roles despite their similar chemistry.

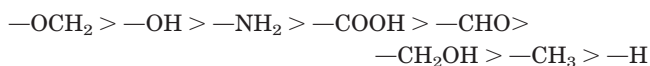
In the second type of configurational isomer, four different substituents bonded to a tetrahedral carbon atom may be arranged two different ways in space—that is, have two configurations (Fig. 1-19)—yielding two stereoisomers with similar or identical chemical properties but differing in certain physical and biological properties. A carbon atom with four different substituents is said to be asymmetric, and asymmetric carbons are called **chiral centers** (Greek *chiros*, “hand”; some stereoisomers are related structurally as the right hand is to the left). A molecule with only one chiral carbon can have two stereoisomers; when two or more (n) chiral carbons are present, there can be 2^n stereoisomers. Some stereoisomers are mirror images of each other; they are called **enantiomers** (Fig. 1-19). Pairs of stereoisomers that are not mirror images of each other are called **diastereomers** (Fig. 1-20).

As Louis Pasteur first observed (Box 1-2), enantiomers have nearly identical chemical properties but differ in a characteristic physical property, their interaction with plane-polarized light. In separate solutions, two enantiomers rotate the plane of plane-polarized light in opposite directions, but an equimolar solution of the two enantiomers (a **racemic mixture**) shows no optical rotation. Compounds without chiral centers do not rotate the plane of plane-polarized light.

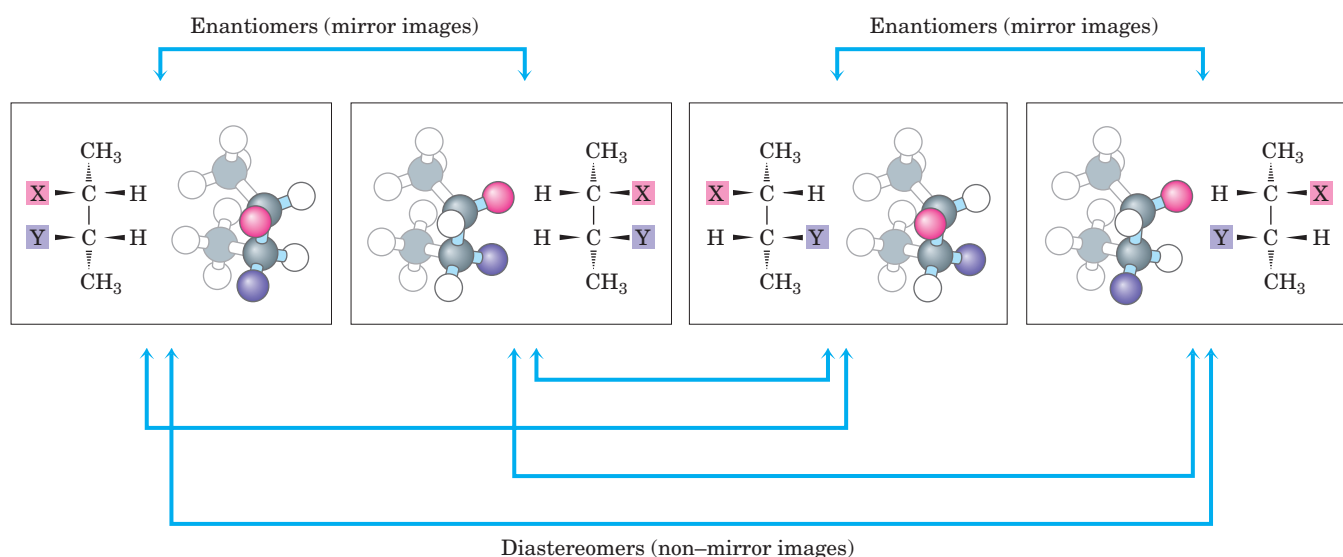
**FIGURE 1-19** Molecular asymmetry: chiral and achiral molecules.

(a) When a carbon atom has four different substituent groups (A, B, X, Y), they can be arranged in two ways that represent nonsuperimposable mirror images of each other (enantiomers). This asymmetric carbon atom is called a chiral atom or chiral center. (b) When a tetrahedral carbon has only three dissimilar groups (i.e., the same group occurs twice), only one configuration is possible and the molecule is symmetric, or achiral. In this case the molecule is superimposable on its mirror image: the molecule on the left can be rotated counter-clockwise (when looking down the vertical bond from A to C) to create the molecule in the mirror.

Given the importance of stereochemistry in reactions between biomolecules (see below), biochemists must name and represent the structure of each biomolecule so that its stereochemistry is unambiguous. For compounds with more than one chiral center, the most useful system of nomenclature is the RS system. In this system, each group attached to a chiral carbon is assigned a *priority*. The priorities of some common substituents are



For naming in the RS system, the chiral atom is viewed with the group of lowest priority (4 in the diagram on the next page) pointing away from the viewer. If the priority of the other three groups (1 to 3) decreases in clockwise order, the configuration is (*R*) (Latin *rectus*, “right”); if in counterclockwise order, the configuration

**FIGURE 1-20** Two types of stereoisomers. There are four different 2,3-disubstituted butanes ($n = 2$ asymmetric carbons, hence $2^n = 4$ stereoisomers). Each is shown in a box as a perspective formula and a ball-and-stick model, which has been rotated to allow the reader to

view all the groups. Some pairs of stereoisomers are mirror images of each other, or enantiomers. Other pairs are not mirror images; these are diastereomers.

BOX 1-2 WORKING IN BIOCHEMISTRY

Louis Pasteur and Optical Activity:
In Vino, Veritas

Louis Pasteur encountered the phenomenon of **optical activity** in 1843, during his investigation of the crystalline sediment that accumulated in wine casks (a form of tartaric acid called paratartaric acid—also called racemic acid, from Latin *racemus*, “bunch of grapes”). He used fine forceps to separate two types of crystals identical in shape but mirror images of each other. Both types proved to have all the chemical properties of tartaric acid, but in solution one type rotated polarized light to the left (levorotatory), the other to the right (dextrorotatory). Pasteur later described the experiment and its interpretation:

In isomeric bodies, the elements and the proportions in which they are combined are the same, only the arrangement of the atoms is different . . . We know, on the one hand, that the molecular arrangements of the two tartaric acids are asymmetric, and, on the other hand, that these arrangements are absolutely identical, excepting that they exhibit asymmetry in opposite directions. Are the atoms of the dextro acid grouped in the form of a right-handed spiral, or are they placed at the apex of an irregular tetrahedron, or are they disposed according to this or that asymmetric arrangement? We do not know.*



Louis Pasteur
1822–1895

Now we do know. X-ray crystallographic studies in 1951 confirmed that the levorotatory and dextrorotatory forms of tartaric acid are mirror images of each other at the molecular level and established the absolute configuration of each (Fig. 1). The same approach has been used to demonstrate that although the amino acid alanine has two stereoisomeric forms (designated D and L), alanine in proteins exists exclusively in one form (the L isomer; see Chapter 3).

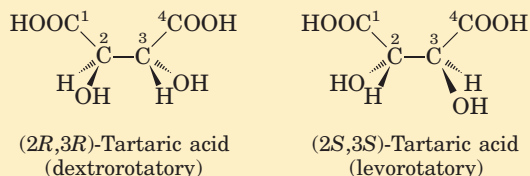
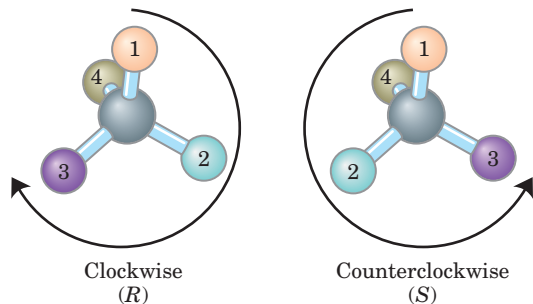


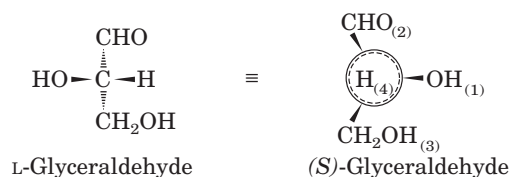
FIGURE 1 Pasteur separated crystals of two stereoisomers of tartaric acid and showed that solutions of the separated forms rotated polarized light to the same extent but in opposite directions. These dextrorotatory and levorotatory forms were later shown to be the (*R,R*) and (*S,S*) isomers represented here. The RS system of nomenclature is explained in the text.

*From Pasteur's lecture to the Société Chimique de Paris in 1883, quoted in DuBos, R. (1976) *Louis Pasteur: Free Lance of Science*, p. 95, Charles Scribner's Sons, New York.

is (*S*) (Latin *sinister*, “left”). In this way each chiral carbon is designated either (*R*) or (*S*), and the inclusion of these designations in the name of the compound provides an unambiguous description of the stereochemistry at each chiral center.



Another naming system for stereoisomers, the D and L system, is described in Chapter 3. A molecule with a single chiral center (glyceraldehydes, for example) can be named unambiguously by either system.



Distinct from configuration is molecular **conformation**, the spatial arrangement of substituent groups that, without breaking any bonds, are free to assume different positions in space because of the freedom of rotation about single bonds. In the simple hydrocarbon ethane, for example, there is nearly complete freedom of rotation around the C—C bond. Many different, interconvertible conformations of ethane are possible, depending on the degree of rotation (Fig. 1–21). Two conformations are of special interest: the staggered, which is more stable than all others and thus predominates, and the eclipsed, which is least stable. We cannot isolate either of these conformational forms, because

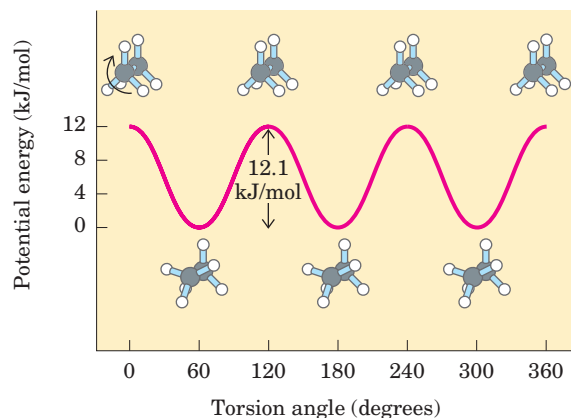


FIGURE 1-21 Conformations. Many conformations of ethane are possible because of freedom of rotation around the C—C bond. In the ball-and-stick model, when the front carbon atom (as viewed by the reader) with its three attached hydrogens is rotated relative to the rear carbon atom, the potential energy of the molecule rises to a maximum in the fully eclipsed conformation (torsion angle 0° , 120° , etc.), then falls to a minimum in the fully staggered conformation (torsion angle 60° , 180° , etc.). Because the energy differences are small enough to allow rapid interconversion of the two forms (millions of times per second), the eclipsed and staggered forms cannot be separately isolated.

they are freely interconvertible. However, when one or more of the hydrogen atoms on each carbon is replaced by a functional group that is either very large or electrically charged, freedom of rotation around the C—C bond is hindered. This limits the number of stable conformations of the ethane derivative.

Interactions between Biomolecules Are Stereospecific

Biological interactions between molecules are stereospecific: the “fit” in such interactions must be stereochemically correct. The three-dimensional structure of biomolecules large and small—the combination of configuration and conformation—is of the utmost importance in their biological interactions: reactant with enzyme, hormone with its receptor on a cell surface, antigen with its specific antibody, for example (Fig. 1-22). The study of biomolecular stereochemistry with precise physical methods is an important part of modern research on cell structure and biochemical function.

In living organisms, chiral molecules are usually present in only one of their chiral forms. For example, the amino acids in proteins occur only as their L isomers; glucose occurs only as its D isomer. (The conventions for naming stereoisomers of the amino acids are described in Chapter 3; those for sugars, in Chapter 7; the RS system, described above, is the most useful for some biomolecules.) In contrast, when a compound with an asymmetric carbon atom is chemically synthesized in the laboratory, the reaction usually pro-

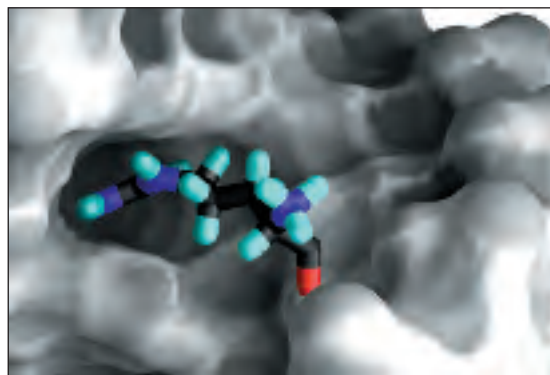


FIGURE 1-22 Complementary fit between a macromolecule and a small molecule. A segment of RNA from the regulatory region TAR of the human immunodeficiency virus genome (gray) with a bound arginamide molecule (colored), representing one residue of a protein that binds to this region. The arginamide fits into a pocket on the RNA surface and is held in this orientation by several noncovalent interactions with the RNA. This representation of the RNA molecule is produced with the computer program GRASP, which can calculate the shape of the outer surface of a macromolecule, defined either by the van der Waals radii of all the atoms in the molecule or by the “solvent exclusion volume,” into which a water molecule cannot penetrate.

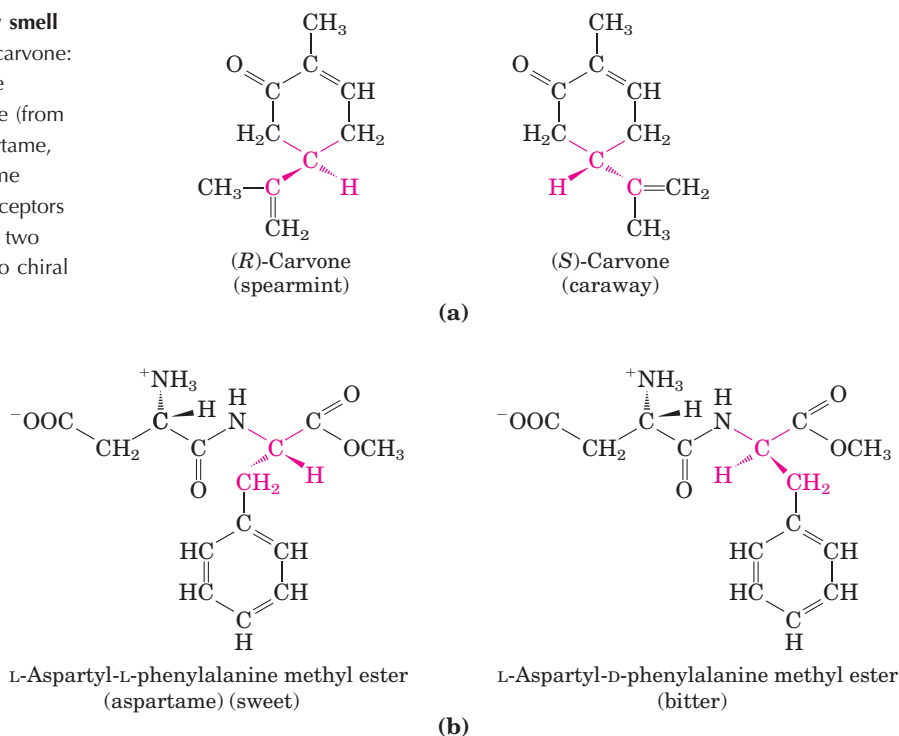
duces all possible chiral forms: a mixture of the D and L forms, for example. Living cells produce only one chiral form of biomolecules because the enzymes that synthesize them are also chiral.

Stereospecificity, the ability to distinguish between stereoisomers, is a property of enzymes and other proteins and a characteristic feature of the molecular logic of living cells. If the binding site on a protein is complementary to one isomer of a chiral compound, it will not be complementary to the other isomer, for the same reason that a left glove does not fit a right hand. Two striking examples of the ability of biological systems to distinguish stereoisomers are shown in Figure 1-23.

SUMMARY 1.2 Chemical Foundations

- Because of its bonding versatility, carbon can produce a broad array of carbon-carbon skeletons with a variety of functional groups; these groups give biomolecules their biological and chemical personalities.
- A nearly universal set of several hundred small molecules is found in living cells; the interconversions of these molecules in the central metabolic pathways have been conserved in evolution.
- Proteins and nucleic acids are linear polymers of simple monomeric subunits; their sequences contain the information that gives each molecule its three-dimensional structure and its biological functions.

FIGURE 1-23 Stereoisomers distinguishable by smell and taste in humans. (a) Two stereoisomers of carvone: (*R*)-carvone (isolated from spearmint oil) has the characteristic fragrance of spearmint; (*S*)-carvone (from caraway seed oil) smells like caraway. (b) Aspartame, the artificial sweetener sold under the trade name NutraSweet, is easily distinguishable by taste receptors from its bitter-tasting stereoisomer, although the two differ only in the configuration at one of the two chiral carbon atoms.



- Molecular configuration can be changed only by breaking covalent bonds. For a carbon atom with four different substituents (a chiral carbon), the substituent groups can be arranged in two different ways, generating stereoisomers with distinct properties. Only one stereoisomer is biologically active. Molecular conformation is the position of atoms in space that can be changed by rotation about single bonds, without breaking covalent bonds.
- Interactions between biological molecules are almost invariably stereospecific: they require a complementary match between the interacting molecules.

1.3 Physical Foundations

Living cells and organisms must perform work to stay alive and to reproduce themselves. The synthetic reactions that occur within cells, like the synthetic processes in any factory, require the input of energy. Energy is also consumed in the motion of a bacterium or an Olympic sprinter, in the flashing of a firefly or the electrical discharge of an eel. And the storage and expression of information require energy, without which structures rich in information inevitably become disordered and meaningless.

In the course of evolution, cells have developed highly efficient mechanisms for coupling the energy obtained from sunlight or fuels to the many energy-consuming processes they must carry out. One goal of

biochemistry is to understand, in quantitative and chemical terms, the means by which energy is extracted, channeled, and consumed in living cells. We can consider cellular energy conversions—like all other energy conversions—in the context of the laws of thermodynamics.

Living Organisms Exist in a Dynamic Steady State, Never at Equilibrium with Their Surroundings

The molecules and ions contained within a living organism differ in kind and in concentration from those in the organism's surroundings. A *Paramecium* in a pond, a shark in the ocean, an erythrocyte in the human bloodstream, an apple tree in an orchard—all are different in composition from their surroundings and, once they have reached maturity, all (to a first approximation) maintain a constant composition in the face of constantly changing surroundings.

Although the characteristic composition of an organism changes little through time, the population of molecules within the organism is far from static. Small molecules, macromolecules, and supramolecular complexes are continuously synthesized and then broken down in chemical reactions that involve a constant flux of mass and energy through the system. The hemoglobin molecules carrying oxygen from your lungs to your brain at this moment were synthesized within the past month; by next month they will have been degraded and entirely replaced by new hemoglobin molecules. The glucose you ingested with your most recent meal is now circulating in your bloodstream; before the day is over these particular glucose molecules will have been

converted into something else—carbon dioxide or fat, perhaps—and will have been replaced with a fresh supply of glucose, so that your blood glucose concentration is more or less constant over the whole day. The amounts of hemoglobin and glucose in the blood remain nearly constant because the rate of synthesis or intake of each just balances the rate of its breakdown, consumption, or conversion into some other product. The constancy of concentration is the result of a *dynamic steady state*, a steady state that is far from equilibrium. Maintaining this steady state requires the constant investment of energy; when the cell can no longer generate energy, it dies and begins to decay toward equilibrium with its surroundings. We consider below exactly what is meant by “steady state” and “equilibrium.”

Organisms Transform Energy and Matter from Their Surroundings

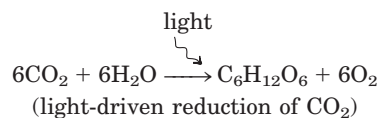
For chemical reactions occurring in solution, we can define a **system** as all the reactants and products present, the solvent that contains them, and the immediate atmosphere—in short, everything within a defined region of space. The system and its surroundings together constitute the **universe**. If the system exchanges neither matter nor energy with its surroundings, it is said to be **isolated**. If the system exchanges energy but not matter with its surroundings, it is a **closed** system; if it exchanges both energy and matter with its surroundings, it is an **open** system.

A living organism is an open system; it exchanges both matter and energy with its surroundings. Living organisms derive energy from their surroundings in two ways: (1) they take up chemical fuels (such as glucose) from the environment and extract energy by oxidizing them (see Box 1–3, Case 2); or (2) they absorb energy from sunlight.

The first law of thermodynamics, developed from physics and chemistry but fully valid for biological systems as well, describes the principle of the conservation of energy: *in any physical or chemical change, the total amount of energy in the universe remains constant, although the form of the energy may change*. Cells are consummate transducers of energy, capable of interconverting chemical, electromagnetic, mechanical, and osmotic energy with great efficiency (Fig. 1–24).

The Flow of Electrons Provides Energy for Organisms

Nearly all living organisms derive their energy, directly or indirectly, from the radiant energy of sunlight, which arises from thermonuclear fusion reactions carried out in the sun. Photosynthetic cells absorb light energy and use it to drive electrons from water to carbon dioxide, forming energy-rich products such as glucose ($\text{C}_6\text{H}_{12}\text{O}_6$), starch, and sucrose and releasing O_2 into the atmosphere:



Nonphotosynthetic cells and organisms obtain the energy they need by oxidizing the energy-rich products of photosynthesis and then passing electrons to atmos-

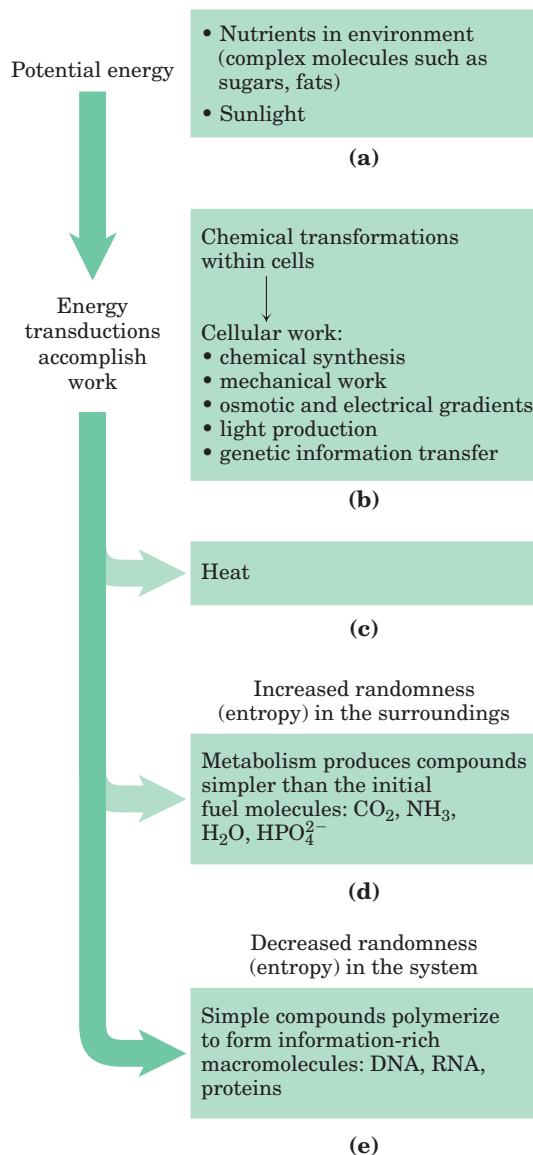
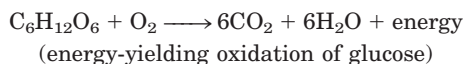


FIGURE 1–24 Some energy interconversion in living organisms. During metabolic energy transductions, the randomness of the system plus surroundings (expressed quantitatively as entropy) increases as the potential energy of complex nutrient molecules decreases. (a) Living organisms extract energy from their surroundings; (b) convert some of it into useful forms of energy to produce work; (c) return some energy to the surroundings as heat; and (d) release end-product molecules that are less well organized than the starting fuel, increasing the entropy of the universe. One effect of all these transformations is (e) increased order (decreased randomness) in the system in the form of complex macromolecules. We return to a quantitative treatment of entropy in Chapter 13.

pheric O_2 to form water, carbon dioxide, and other end products, which are recycled in the environment:



Virtually all energy transductions in cells can be traced to this flow of electrons from one molecule to another, in a “downhill” flow from higher to lower electrochemical potential; as such, this is formally analogous to the flow of electrons in a battery-driven electric circuit. All these reactions involving electron flow are **oxidation-reduction reactions**: one reactant is oxidized (loses electrons) as another is reduced (gains electrons).

Creating and Maintaining Order Requires Work and Energy

DNA, RNA, and proteins are informational macromolecules. In addition to using chemical energy to form the covalent bonds between the subunits in these polymers, the cell must invest energy to order the subunits in their correct sequence. It is extremely improbable that amino acids in a mixture would spontaneously condense into a single type of protein, with a unique sequence. This would represent increased order in a population of molecules; but according to the second law of thermodynamics, the tendency in nature is toward ever-greater disorder in the universe: *the total entropy of the universe is continually increasing*. To bring about the synthesis of macromolecules from their monomeric units, free energy must be supplied to the system (in this case, the cell).

The randomness or disorder of the components of a chemical system is expressed as **entropy, S** (Box 1–3).



J. Willard Gibbs,
1839–1903

Any change in randomness of the system is expressed as entropy change, ΔS , which by convention has a positive value when randomness increases. J. Willard Gibbs, who developed the theory of energy changes during chemical reactions, showed that the **free-energy content, G** , of any closed system can be defined in terms of three quantities: **enthalpy, H** , reflecting the number and kinds of bonds;

entropy, S ; and the absolute temperature, T (in degrees Kelvin). The definition of free energy is $G = H - TS$. When a chemical reaction occurs at constant temperature, the **free-energy change, ΔG** , is determined by the enthalpy change, ΔH , reflecting the kinds and numbers of chemical bonds and noncovalent interactions broken and formed, and the entropy change, ΔS , describing the change in the system's randomness:

$$\Delta G = \Delta H - T \Delta S$$

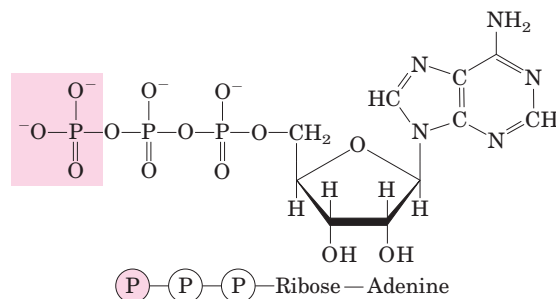


FIGURE 1–25 Adenosine triphosphate (ATP). The removal of the terminal phosphoryl group (shaded pink) of ATP, by breakage of a phosphoanhydride bond, is highly exergonic, and this reaction is coupled to many endergonic reactions in the cell (as in the example in Fig. 1–26b).

A process tends to occur spontaneously only if ΔG is negative. Yet cell function depends largely on molecules, such as proteins and nucleic acids, for which the free energy of formation is positive: the molecules are less stable and more highly ordered than a mixture of their monomeric components. To carry out these thermodynamically unfavorable, energy-requiring (**endergonic**) reactions, cells couple them to other reactions that liberate free energy (**exergonic** reactions), so that the overall process is exergonic: the *sum* of the free-energy changes is negative. The usual source of free energy in coupled biological reactions is the energy released by hydrolysis of phosphoanhydride bonds such as those in adenosine triphosphate (ATP; Fig. 1–25; see also Fig. 1–15). Here, each $\textcircled{P}$ represents a phosphoryl group:

Amino acids $\longrightarrow$ polymer ΔG_1 is positive (endergonic)

$-\textcircled{P}-\textcircled{P} \longrightarrow -\textcircled{P} + \textcircled{P} \quad \Delta G_2$ is negative (exergonic)

When these reactions are coupled, the sum of ΔG_1 and ΔG_2 is negative—the overall process is exergonic. By this coupling strategy, cells are able to synthesize and maintain the information-rich polymers essential to life.

Energy Coupling Links Reactions in Biology

The central issue in *bioenergetics* (the study of energy transformations in living systems) is the means by which energy from fuel metabolism or light capture is coupled to a cell's energy-requiring reactions. In thinking about energy coupling, it is useful to consider a simple mechanical example, as shown in Figure 1–26a. An object at the top of an inclined plane has a certain amount of potential energy as a result of its elevation. It tends to slide down the plane, losing its potential energy of position as it approaches the ground. When an appropriate string-and-pulley device couples the falling object to another, smaller object, the spontaneous downward motion of the larger can lift the smaller, accomplishing a

BOX 1-3 WORKING IN BIOCHEMISTRY

Entropy: The Advantages of Being Disorganized

The term “entropy,” which literally means “a change within,” was first used in 1851 by Rudolf Clausius, one of the formulators of the second law of thermodynamics. A rigorous quantitative definition of entropy involves statistical and probability considerations. However, its nature can be illustrated qualitatively by three simple examples, each demonstrating one aspect of entropy. The key descriptors of entropy are *randomness* and *disorder*, manifested in different ways.

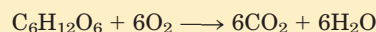
Case 1: The Teakettle and the Randomization of Heat

We know that steam generated from boiling water can do useful work. But suppose we turn off the burner under a teakettle full of water at 100 °C (the “system”) in the kitchen (the “surroundings”) and allow the teakettle to cool. As it cools, no work is done, but heat passes from the teakettle to the surroundings, raising the temperature of the surroundings (the kitchen) by an infinitesimally small amount until complete equilibrium is attained. At this point all parts of the teakettle and the kitchen are at precisely the same temperature. The free energy that was once concentrated in the teakettle of hot water at 100 °C, *potentially* capable of doing work, has disappeared. Its equivalent in heat energy is still present in the teakettle + kitchen (i.e., the “universe”) but has become completely randomized throughout. This energy is no longer available to do work because there is no temperature differential within the kitchen. Moreover, the increase in entropy of the kitchen (the surroundings) is irreversible. We know from everyday experience that heat never spontaneously passes back from the kitchen into the teakettle to raise the temperature of the water to 100 °C again.

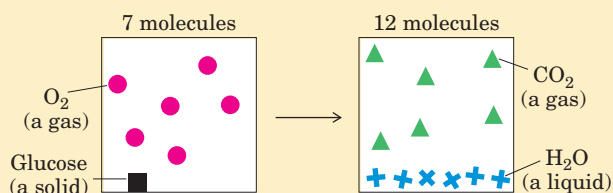
Case 2: The Oxidation of Glucose

Entropy is a state not only of energy but of matter. Aerobic (heterotrophic) organisms extract free en-

ergy from glucose obtained from their surroundings by oxidizing the glucose with O₂, also obtained from the surroundings. The end products of this oxidative metabolism, CO₂ and H₂O, are returned to the surroundings. In this process the surroundings undergo an increase in entropy, whereas the organism itself remains in a steady state and undergoes no change in its internal order. Although some entropy arises from the dissipation of heat, entropy also arises from another kind of disorder, illustrated by the equation for the oxidation of glucose:



We can represent this schematically as



The atoms contained in 1 molecule of glucose plus 6 molecules of oxygen, a total of 7 molecules, are more randomly dispersed by the oxidation reaction and are now present in a total of 12 molecules (6CO₂ + 6H₂O).

Whenever a chemical reaction results in an increase in the number of molecules—or when a solid substance is converted into liquid or gaseous products, which allow more freedom of molecular movement than solids—molecular disorder, and thus entropy, increases.

Case 3: Information and Entropy

The following short passage from *Julius Caesar*, Act IV, Scene 3, is spoken by Brutus, when he realizes that he must face Mark Antony’s army. It is an information-rich nonrandom arrangement of 125 letters of the English alphabet:

certain amount of work. The amount of energy available to do work is the **free-energy change, ΔG** ; this is always somewhat less than the theoretical amount of energy released, because some energy is dissipated as the heat of friction. The greater the elevation of the larger object, the greater the energy released (ΔG) as the object slides downward and the greater the amount of work that can be accomplished.

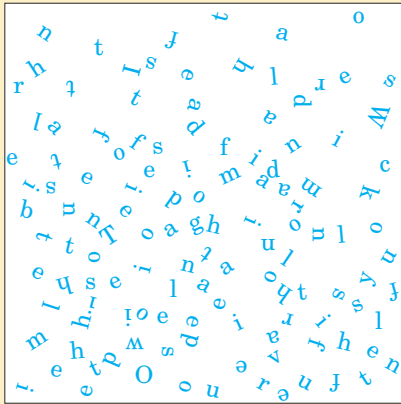
How does this apply in chemical reactions? In closed systems, chemical reactions proceed spontaneously until **equilibrium** is reached. When a system is at equilibrium, the rate of product formation exactly equals the

rate at which product is converted to reactant. Thus there is no net change in the concentration of reactants and products; a *steady state* is achieved. The energy change as the system moves from its initial state to equilibrium, with no changes in temperature or pressure, is given by the free-energy change, ΔG . The magnitude of ΔG depends on the particular chemical reaction and on how far from equilibrium the system is initially. Each compound involved in a chemical reaction contains a certain amount of potential energy, related to the kind and number of its bonds. In reactions that occur spontaneously, the products have less free energy than the re-

There is a tide in the affairs of men,
Which, taken at the flood, leads on to fortune;
Omitted, all the voyage of their life
Is bound in shallows and in miseries.

In addition to what this passage says overtly, it has many hidden meanings. It not only reflects a complex sequence of events in the play, it also echoes the play's ideas on conflict, ambition, and the demands of leadership. Permeated with Shakespeare's understanding of human nature, it is very rich in information.

However, if the 125 letters making up this quotation were allowed to fall into a completely random, chaotic pattern, as shown in the following box, they would have no meaning whatsoever.

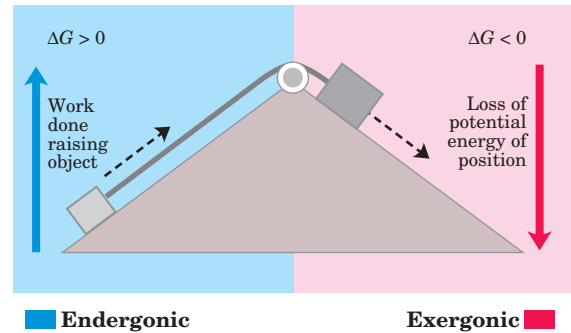


In this form the 125 letters contain little or no information, but they are very rich in entropy. Such considerations have led to the conclusion that information is a form of energy; information has been called “negative entropy.” In fact, the branch of mathematics called information theory, which is basic to the programming logic of computers, is closely related to thermodynamic theory. Living organisms are highly ordered, nonrandom structures, immensely rich in information and thus entropy-poor.

actants, thus the reaction releases free energy, which is then available to do work. Such reactions are exergonic; the decline in free energy from reactants to products is expressed as a negative value. Endergonic reactions require an input of energy, and their ΔG values are positive. As in mechanical processes, only part of the energy released in exergonic chemical reactions can be used to accomplish work. In living systems some energy is dissipated as heat or lost to increasing entropy.

In living organisms, as in the mechanical example in Figure 1–26a, an exergonic reaction can be coupled to an endergonic reaction to drive otherwise unfavorable

(a) Mechanical example



(b) Chemical example

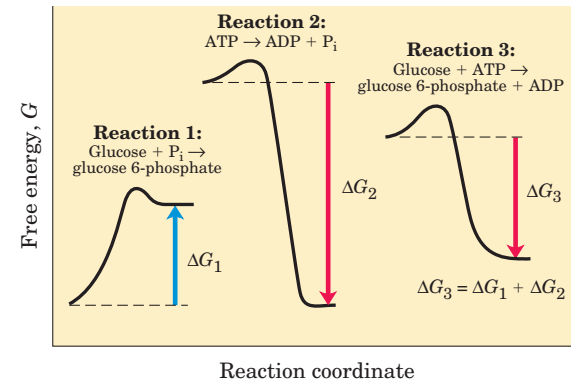
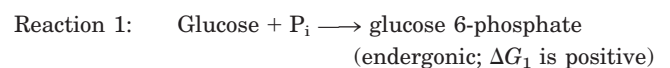


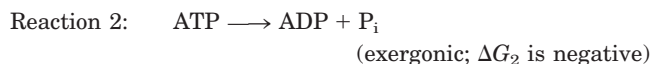
FIGURE 1–26 Energy coupling in mechanical and chemical processes. (a) The downward motion of an object releases potential energy that can do mechanical work. The potential energy made available by spontaneous downward motion, an exergonic process (pink), can be coupled to the endergonic upward movement of another object (blue). (b) In reaction 1, the formation of glucose 6-phosphate from glucose and inorganic phosphate (P_i) yields a product of higher energy than the two reactants. For this endergonic reaction, ΔG is positive. In reaction 2, the exergonic breakdown of adenosine triphosphate (ATP) can drive an endergonic reaction when the two reactions are coupled. The exergonic reaction has a large, negative free-energy change (ΔG_2), and the endergonic reaction has a smaller, positive free-energy change (ΔG_1). The third reaction accomplishes the sum of reactions 1 and 2, and the free-energy change, ΔG_3 , is the arithmetic sum of ΔG_1 and ΔG_2 . Because ΔG_3 is negative, the overall reaction is exergonic and proceeds spontaneously.

reactions. Figure 1–26b (a type of graph called a reaction coordinate diagram) illustrates this principle for the conversion of glucose to glucose 6-phosphate, the first step in the pathway for oxidation of glucose. The simplest way to produce glucose 6-phosphate would be:

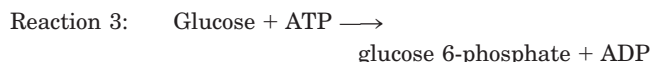


(P_i is an abbreviation for inorganic phosphate, HPO_4^{2-} . Don't be concerned about the structure of these compounds now; we describe them in detail later in the book.) This reaction does not occur spontaneously; ΔG

is positive. A second, very exergonic reaction can occur in all cells:



These two chemical reactions share a common intermediate, P_i , which is consumed in reaction 1 and produced in reaction 2. The two reactions can therefore be coupled in the form of a third reaction, which we can write as the sum of reactions 1 and 2, with the common intermediate, P_i , omitted from both sides of the equation:

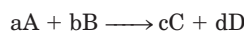


Because more energy is released in reaction 2 than is consumed in reaction 1, the free-energy change for reaction 3, ΔG_3 , is negative, and the synthesis of glucose 6-phosphate can therefore occur by reaction 3.

The coupling of exergonic and endergonic reactions through a shared intermediate is absolutely central to the energy exchanges in living systems. As we shall see, the breakdown of ATP (reaction 2 in Fig. 1–26b) is the exergonic reaction that drives many endergonic processes in cells. In fact, ATP is the major carrier of chemical energy in all cells.

K_{eq} and ΔG° Are Measures of a Reaction's Tendency to Proceed Spontaneously

The tendency of a chemical reaction to go to completion can be expressed as an equilibrium constant. For the reaction



the equilibrium constant, K_{eq} , is given by

$$K_{\text{eq}} = \frac{[\text{C}_{\text{eq}}]^c [\text{D}_{\text{eq}}]^d}{[\text{A}_{\text{eq}}]^a [\text{B}_{\text{eq}}]^b}$$

where $[\text{A}_{\text{eq}}]$ is the concentration of A, $[\text{B}_{\text{eq}}]$ the concentration of B, and so on, *when the system has reached equilibrium*. A large value of K_{eq} means the reaction tends to proceed until the reactants have been almost completely converted into the products.

Gibbs showed that ΔG for any chemical reaction is a function of the **standard free-energy change**, ΔG° —a constant that is characteristic of each specific reaction—and a term that expresses the initial concentrations of reactants and products:

$$\Delta G = \Delta G^\circ + RT \ln \frac{[\text{C}_i]^c [\text{D}_i]^d}{[\text{A}_i]^a [\text{B}_i]^b} \quad (1-1)$$

where $[\text{A}_i]$ is the initial concentration of A, and so forth; R is the gas constant; and T is the absolute temperature.

When a reaction has reached equilibrium, no driving force remains and it can do no work: $\Delta G = 0$. For this special case, $[\text{A}_i] = [\text{A}_{\text{eq}}]$, and so on, for all reactants and products, and

$$\frac{[\text{C}_i]^c [\text{D}_i]^d}{[\text{A}_i]^a [\text{B}_i]^b} = \frac{[\text{C}_{\text{eq}}]^c [\text{D}_{\text{eq}}]^d}{[\text{A}_{\text{eq}}]^a [\text{B}_{\text{eq}}]^b} = K_{\text{eq}}$$

Substituting 0 for ΔG and K_{eq} for $[\text{C}_i]^c [\text{D}_i]^d / [\text{A}_i]^a [\text{B}_i]^b$ in Equation 1–1, we obtain the relationship

$$\Delta G^\circ = -RT \ln K_{\text{eq}}$$

from which we see that ΔG° is simply a second way (besides K_{eq}) of expressing the driving force on a reaction. Because K_{eq} is experimentally measurable, we have a way of determining ΔG° , the thermodynamic constant characteristic of each reaction.

The units of ΔG° and ΔG are joules per mole (or calories per mole). When $K_{\text{eq}} \gg 1$, ΔG° is large and negative; when $K_{\text{eq}} \ll 1$, ΔG° is large and positive. From a table of experimentally determined values of either K_{eq} or ΔG° , we can see at a glance which reactions tend to go to completion and which do not.

One caution about the interpretation of ΔG° : *thermodynamic* constants such as this show where the final equilibrium for a reaction lies but tell us nothing about how fast that equilibrium will be achieved. The rates of reactions are governed by the parameters of *kinetics*, a topic we consider in detail in Chapter 6.

Enzymes Promote Sequences of Chemical Reactions

All biological macromolecules are much less thermodynamically stable than their monomeric subunits, yet they are *kinetically stable*: their *uncatalyzed* breakdown occurs so slowly (over years rather than seconds) that, on a time scale that matters for the organism, these molecules are stable. Virtually every chemical reaction in a cell occurs at a significant rate only because of the presence of **enzymes**—biocatalysts that, like all other catalysts, greatly enhance the rate of specific chemical reactions without being consumed in the process.

The path from reactant(s) to product(s) almost invariably involves an energy barrier, called the activation barrier (Fig. 1–27), that must be surmounted for any reaction to proceed. The breaking of existing bonds and formation of new ones generally requires, first, the distortion of the existing bonds, creating a **transition state** of higher free energy than either reactant or product. The highest point in the reaction coordinate diagram represents the transition state, and the difference in energy between the reactant in its ground state and in its transition state is the **activation energy**, $\Delta G^\ddagger$. An enzyme catalyzes a reaction by providing a more comfortable fit for the transition state: a surface that complements the transition state in stereochemistry, polarity, and charge. The binding of enzyme to the transition state is exergonic, and the energy released by this binding reduces the activation energy for the reaction and greatly increases the reaction rate.

A further contribution to catalysis occurs when two or more reactants bind to the enzyme's surface close to each other and with stereospecific orientations that fa-

vor the reaction. This increases by orders of magnitude the probability of productive collisions between reactants. As a result of these factors and several others, discussed in Chapter 6, enzyme-catalyzed reactions commonly proceed at rates greater than 10^{12} times faster than the uncatalyzed reactions.

Cellular catalysts are, with a few exceptions, proteins. (In some cases, RNA molecules have catalytic roles, as discussed in Chapters 26 and 27.) Again with a few exceptions, each enzyme catalyzes a specific reaction, and each reaction in a cell is catalyzed by a different enzyme. Thousands of different enzymes are therefore required by each cell. The multiplicity of enzymes, their specificity (the ability to discriminate between reactants), and their susceptibility to regulation give cells the capacity to lower activation barriers selectively. This selectivity is crucial for the effective regulation of cellular processes. By allowing specific reactions to proceed at significant rates at particular times, enzymes determine how matter and energy are channeled into cellular activities.

The thousands of enzyme-catalyzed chemical reactions in cells are functionally organized into many sequences of consecutive reactions, called **pathways**, in which the product of one reaction becomes the reactant in the next. Some pathways degrade organic nutrients into simple end products in order to extract chemical energy and convert it into a form useful to the cell; together these degradative, free-energy-yielding reactions are designated **catabolism**. Other pathways start with small precursor molecules and convert them to progressively larger and more complex molecules, including proteins and nucleic acids. Such synthetic pathways,

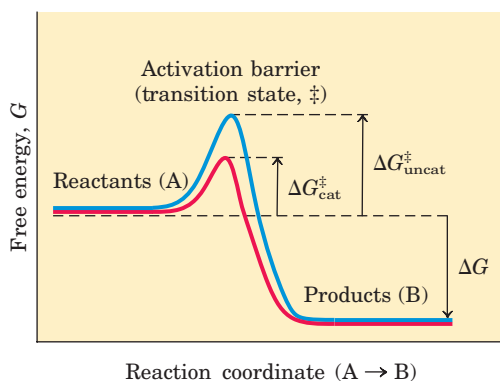


FIGURE 1-27 Energy changes during a chemical reaction. An activation barrier, representing the transition state, must be overcome in the conversion of reactants (A) into products (B), even though the products are more stable than the reactants, as indicated by a large, negative free-energy change (ΔG). The energy required to overcome the activation barrier is the activation energy ($\Delta G^\ddagger$). Enzymes catalyze reactions by lowering the activation barrier. They bind the transition-state intermediates tightly, and the binding energy of this interaction effectively reduces the activation energy from $\Delta G^\ddagger_{\text{uncat}}$ to $\Delta G^\ddagger_{\text{cat}}$. (Note that activation energy is *not* related to free-energy change, ΔG .)

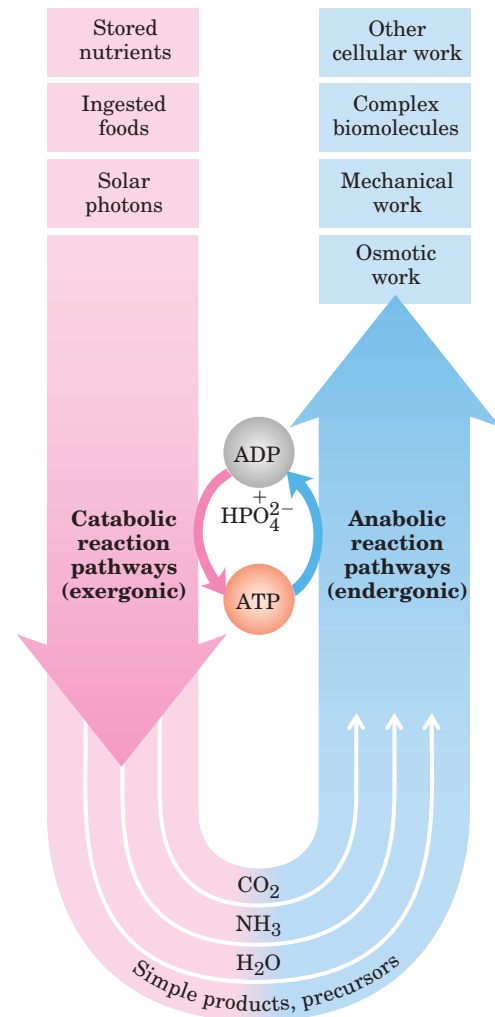


FIGURE 1-28 The central role of ATP in metabolism. ATP is the shared chemical intermediate linking energy-releasing to energy-requiring cell processes. Its role in the cell is analogous to that of money in an economy: it is “earned/produced” in exergonic reactions and “spent/consumed” in endergonic ones.

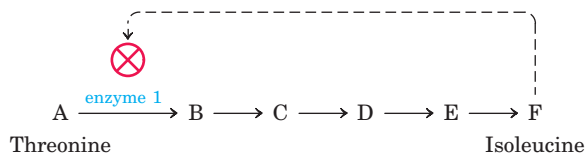
which invariably require the input of energy, are collectively designated **anabolism**. The overall network of enzyme-catalyzed pathways constitutes cellular **metabolism**. ATP is the major connecting link (the shared intermediate) between the catabolic and anabolic components of this network (shown schematically in Fig. 1–28). The pathways of enzyme-catalyzed reactions that act on the main constituents of cells—proteins, fats, sugars, and nucleic acids—are virtually identical in all living organisms.

Metabolism Is Regulated to Achieve Balance and Economy

Not only do living cells simultaneously synthesize thousands of different kinds of carbohydrate, fat, protein, and nucleic acid molecules and their simpler subunits, but they do so in the precise proportions required by

the cell under any given circumstance. For example, during rapid cell growth the precursors of proteins and nucleic acids must be made in large quantities, whereas in nongrowing cells the requirement for these precursors is much reduced. Key enzymes in each metabolic pathway are regulated so that each type of precursor molecule is produced in a quantity appropriate to the current requirements of the cell.

Consider the pathway in *E. coli* that leads to the synthesis of the amino acid isoleucine, a constituent of proteins. The pathway has five steps catalyzed by five different enzymes (A through F represent the intermediates in the pathway):



If a cell begins to produce more isoleucine than is needed for protein synthesis, the unused isoleucine accumulates and the increased concentration inhibits the catalytic activity of the first enzyme in the pathway, immediately slowing the production of isoleucine. Such **feedback inhibition** keeps the production and utilization of each metabolic intermediate in balance.

Although the concept of discrete pathways is an important tool for organizing our understanding of metabolism, it is an oversimplification. There are thousands of metabolic intermediates in a cell, many of which are part of more than one pathway. Metabolism would be better represented as a meshwork of interconnected and interdependent pathways. A change in the concentration of any one metabolite would have an impact on other pathways, starting a ripple effect that would influence the flow of materials through other sectors of the cellular economy. The task of understanding these complex interactions among intermediates and pathways in quantitative terms is daunting, but new approaches, discussed in Chapter 15, have begun to offer important insights into the overall regulation of metabolism.

Cells also regulate the synthesis of their own catalysts, the enzymes, in response to increased or diminished need for a metabolic product; this is the substance of Chapter 28. The expression of genes (the translation of information in DNA to active protein in the cell) and synthesis of enzymes are other layers of metabolic control in the cell. All layers must be taken into account when describing the overall control of cellular metabolism. Assembling the complete picture of how the cell regulates itself is a huge job that has only just begun.

SUMMARY 1.3 Physical Foundations

- Living cells are open systems, exchanging matter and energy with their surroundings, extracting and channeling energy to maintain

themselves in a dynamic steady state distant from equilibrium. Energy is obtained from sunlight or fuels by converting the energy from electron flow into the chemical bonds of ATP.

- The tendency for a chemical reaction to proceed toward equilibrium can be expressed as the free-energy change, ΔG , which has two components: enthalpy change, ΔH , and entropy change, ΔS . These variables are related by the equation $\Delta G = \Delta H - T \Delta S$.
- When ΔG of a reaction is negative, the reaction is exergonic and tends to go toward completion; when ΔG is positive, the reaction is endergonic and tends to go in the reverse direction. When two reactions can be summed to yield a third reaction, the ΔG for this overall reaction is the sum of the ΔG s of the two separate reactions. This provides a way to couple reactions.
- The conversion of ATP to P_i and ADP is highly exergonic (large negative ΔG), and many endergonic cellular reactions are driven by coupling them, through a common intermediate, to this reaction.
- The standard free-energy change for a reaction, ΔG° , is a physical constant that is related to the equilibrium constant by the equation $\Delta G^\circ = -RT \ln K_{eq}$.
- Most exergonic cellular reactions proceed at useful rates only because enzymes are present to catalyze them. Enzymes act in part by stabilizing the transition state, reducing the activation energy, $\Delta G^\ddagger$, and increasing the reaction rate by many orders of magnitude. The catalytic activity of enzymes in cells is regulated.
- Metabolism is the sum of many interconnected reaction sequences that interconvert cellular metabolites. Each sequence is regulated so as to provide what the cell needs at a given time and to expend energy only when necessary.

1.4 Genetic Foundations

Perhaps the most remarkable property of living cells and organisms is their ability to reproduce themselves for countless generations with nearly perfect fidelity. This continuity of inherited traits implies constancy, over millions of years, in the structure of the molecules that contain the genetic information. Very few historical records of civilization, even those etched in copper or carved in stone (Fig. 1–29), have survived for a thousand years. But there is good evidence that the genetic instructions in living organisms have remained nearly unchanged over very much longer periods; many bacteria have nearly

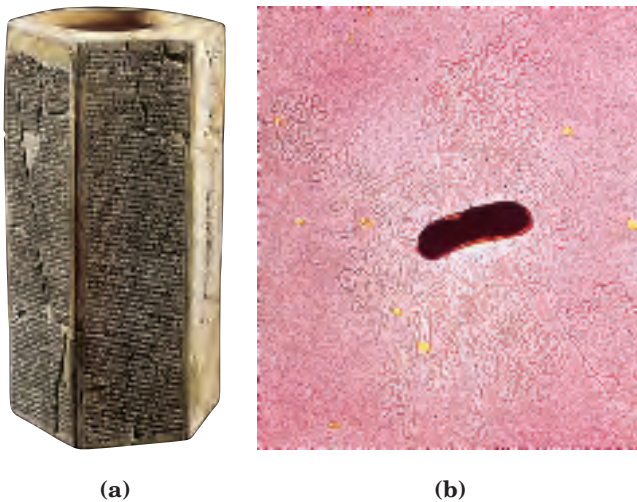


FIGURE 1-29 Two ancient scripts. (a) The Prism of Sennacherib, inscribed in about 700 B.C.E., describes in characters of the Assyrian language some historical events during the reign of King Sennacherib. The Prism contains about 20,000 characters, weighs about 50 kg, and has survived almost intact for about 2,700 years. (b) The single DNA molecule of the bacterium *E. coli*, seen leaking out of a disrupted cell, is hundreds of times longer than the cell itself and contains all the encoded information necessary to specify the cell's structure and functions. The bacterial DNA contains about 10 million characters (nucleotides), weighs less than 10^{-10} g, and has undergone only relatively minor changes during the past several million years. (The yellow spots and dark specks in this colorized electron micrograph are artifacts of the preparation.)

the same size, shape, and internal structure and contain the same kinds of precursor molecules and enzymes as bacteria that lived nearly four billion years ago.

Among the seminal discoveries in biology in the twentieth century were the chemical nature and the three-dimensional structure of the genetic material, **deoxyribonucleic acid, DNA**. The sequence of the monomeric subunits, the nucleotides (strictly, deoxyribonucleotides, as discussed below), in this linear polymer encodes the instructions for forming all other cellular components and provides a template for the production of identical DNA molecules to be distributed to progeny when a cell divides. The continued existence of a biological species requires its genetic information to be maintained in a stable form, expressed accurately in the form of gene products, and reproduced with a minimum of errors. Effective storage, expression, and reproduction of the genetic message defines individual species, distinguishes them from one another, and assures their continuity over successive generations.

Genetic Continuity Is Vested in Single DNA Molecules

DNA is a long, thin organic polymer, the rare molecule that is constructed on the atomic scale in one dimension (width) and the human scale in another (length: a

molecule of DNA can be many centimeters long). A human sperm or egg, carrying the accumulated hereditary information of billions of years of evolution, transmits this inheritance in the form of DNA molecules, in which the linear sequence of covalently linked nucleotide subunits encodes the genetic message.

Usually when we describe the properties of a chemical species, we describe the average behavior of a very large number of identical molecules. While it is difficult to predict the behavior of any single molecule in a collection of, say, a picomole (about 6×10^{11} molecules) of a compound, the *average* behavior of the molecules is predictable because so many molecules enter into the average. Cellular DNA is a remarkable exception. The DNA that is the entire genetic material of *E. coli* is a *single molecule* containing 4.64 million nucleotide pairs. That single molecule must be replicated perfectly in every detail if an *E. coli* cell is to give rise to identical progeny by cell division; there is no room for averaging in this process! The same is true for all cells. A human sperm brings to the egg that it fertilizes just one molecule of DNA in each of its 23 different chromosomes, to combine with just one DNA molecule in each corresponding chromosome in the egg. The result of this union is very highly predictable: an embryo with all of its 35,000 genes, constructed of 3 billion nucleotide pairs, intact. An amazing chemical feat!

The Structure of DNA Allows for Its Replication and Repair with Near-Perfect Fidelity

The capacity of living cells to preserve their genetic material and to duplicate it for the next generation results from the structural complementarity between the two halves of the DNA molecule (Fig. 1-30). The basic unit of DNA is a linear polymer of four different monomeric subunits, **deoxyribonucleotides**, arranged in a precise linear sequence. It is this linear sequence that encodes the genetic information. Two of these polymeric strands are twisted about each other to form the DNA double helix, in which each deoxyribonucleotide in one strand pairs specifically with a complementary deoxyribonucleotide in the opposite strand. Before a cell divides, the two DNA strands separate and each serves as a template for the synthesis of a new complementary strand, generating two identical double-helical molecules, one for each daughter cell. If one strand is damaged, continuity of information is assured by the information present in the other strand, which acts as a template for repair of the damage.

The Linear Sequence in DNA Encodes Proteins with Three-Dimensional Structures

The information in DNA is encoded in its linear (one-dimensional) sequence of deoxyribonucleotide subunits, but the expression of this information results in

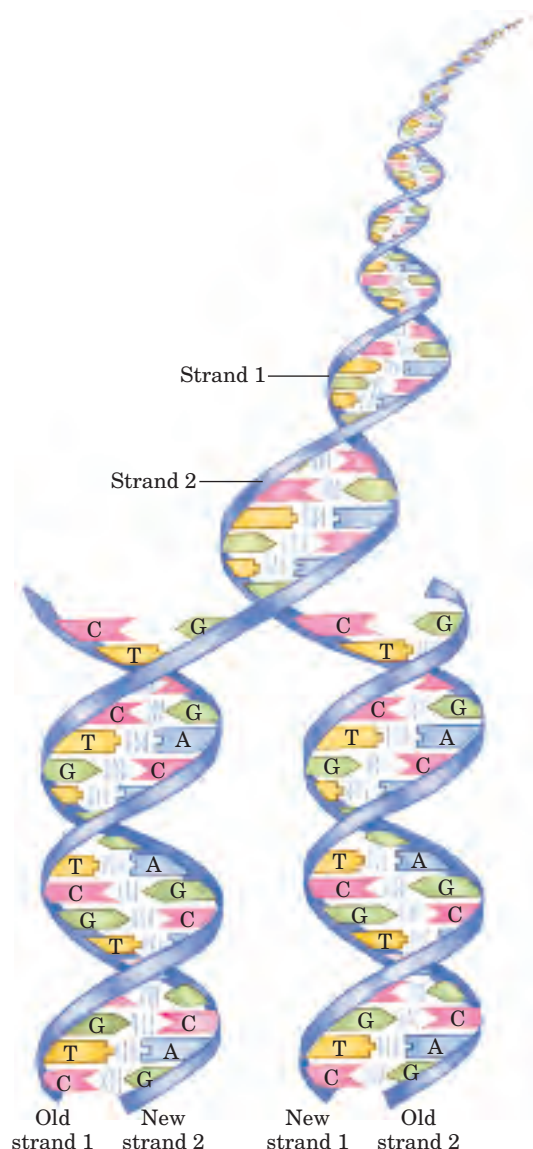


FIGURE 1-30 Complementarity between the two strands of DNA.

DNA is a linear polymer of covalently joined deoxyribonucleotides, of four types: deoxyadenylate (A), deoxyguanylate (G), deoxycytidylate (C), and deoxythymidylate (T). Each nucleotide, with its unique three-dimensional structure, can associate very specifically but noncovalently with one other nucleotide in the complementary chain: A always associates with T, and G with C. Thus, in the double-stranded DNA molecule, the entire sequence of nucleotides in one strand is *complementary* to the sequence in the other. The two strands, held together by hydrogen bonds (represented here by vertical blue lines) between each pair of complementary nucleotides, twist about each other to form the DNA double helix. In DNA replication, the two strands separate and two new strands are synthesized, each with a sequence complementary to one of the original strands. The result is two double-helical molecules, each identical to the original DNA.

a three-dimensional cell. This change from one to three dimensions occurs in two phases. A linear sequence of deoxyribonucleotides in DNA codes (through an intermediary, RNA) for the production of a protein with a corresponding linear sequence of amino acids (Fig. 1-31). The protein folds into a particular three-dimensional

shape, determined by its amino acid sequence and stabilized primarily by noncovalent interactions. Although the final shape of the folded protein is dictated by its amino acid sequence, the folding process is aided by “molecular chaperones,” which catalyze the process by discouraging incorrect folding. The precise three-dimensional structure, or **native conformation**, of the protein is crucial to its function.

Once in its native conformation, a protein may associate noncovalently with other proteins, or with nucleic acids or lipids, to form supramolecular complexes such as chromosomes, ribosomes, and membranes. The individual molecules of these complexes have specific, high-affinity binding sites for each other, and within the cell they spontaneously form functional complexes.

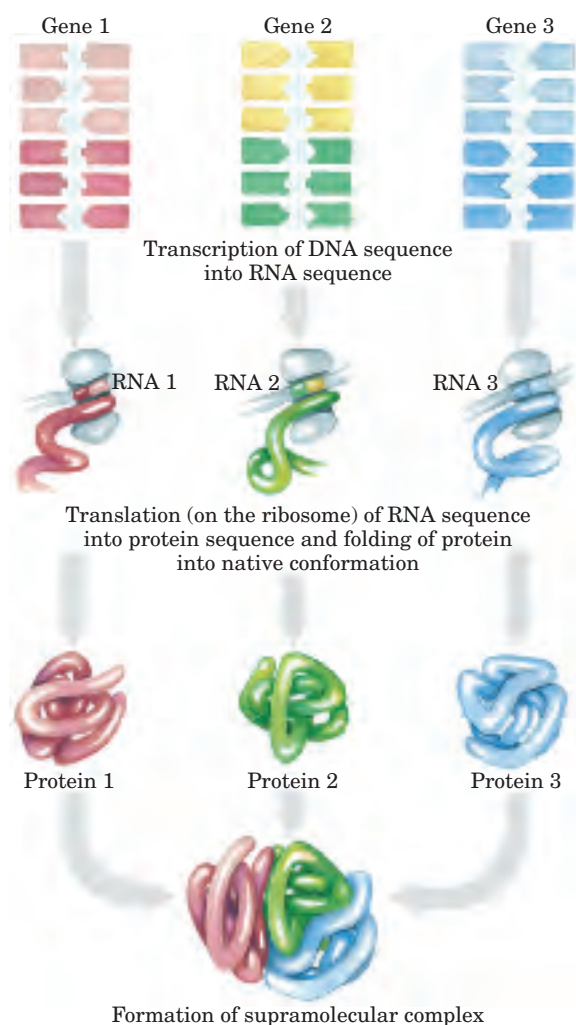


FIGURE 1-31 DNA to RNA to protein. Linear sequences of deoxyribonucleotides in DNA, arranged into units known as genes, are transcribed into ribonucleic acid (RNA) molecules with complementary ribonucleotide sequences. The RNA sequences are then translated into linear protein chains, which fold into their native three-dimensional shapes, often aided by molecular chaperones. Individual proteins commonly associate with other proteins to form supramolecular complexes, stabilized by numerous weak interactions.

Although protein sequences carry all necessary information for the folding into their native conformation, this correct folding requires the right environment—pH, ionic strength, metal ion concentrations, and so forth. Self-assembly therefore requires both information (provided by the DNA sequence) and environment (the interior of a living cell), and in this sense the DNA sequence alone is not enough to dictate the formation of a cell. As Rudolph Virchow, the nineteenth-century Prussian pathologist and researcher, concluded, “*Omnis cellula e cellula*”: every cell comes from another cell.

SUMMARY 1.4 Genetic Foundations

- Genetic information is encoded in the linear sequence of four deoxyribonucleotides in DNA.
- The double-helical DNA molecule contains an internal template for its own replication and repair.
- The linear sequence of amino acids in a protein, which is encoded in the DNA of the gene for that protein, produces a protein's unique three-dimensional structure.
- Individual macromolecules with specific affinity for other macromolecules self-assemble into supramolecular complexes.

1.5 Evolutionary Foundations

Nothing in biology makes sense except in the light of evolution.

—*Theodosius Dobzhansky*, *The American Biology Teacher*,
March 1973

Great progress in biochemistry and molecular biology during the decades since Dobzhansky made this striking generalization has amply confirmed its validity. The remarkable similarity of metabolic pathways and gene sequences in organisms across the phyla argues strongly that all modern organisms share a common evolutionary progenitor and were derived from it by a series of small changes (mutations), each of which conferred a selective advantage to some organism in some ecological niche.

Changes in the Hereditary Instructions Allow Evolution

Despite the near-perfect fidelity of genetic replication, infrequent, unrepaired mistakes in the DNA replication process lead to changes in the nucleotide sequence of DNA, producing a genetic **mutation** (Fig. 1–32) and changing the instructions for some cellular component. Incorrectly repaired damage to one of the DNA strands has the same effect. Mutations in the DNA handed down

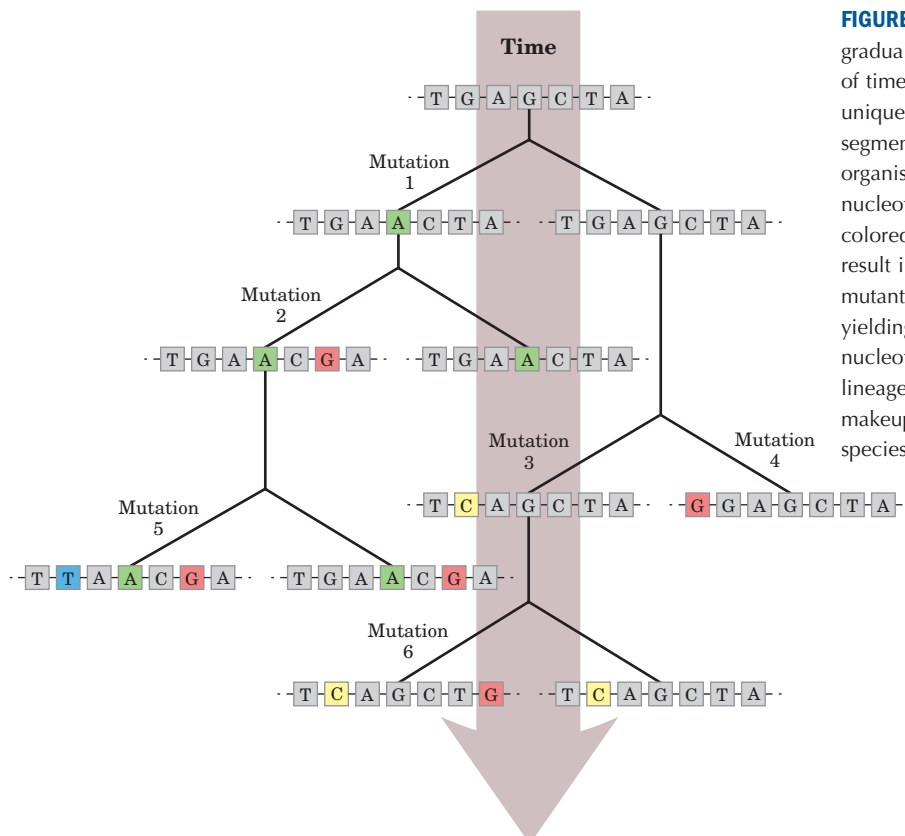


FIGURE 1–32 Role of mutation in evolution. The gradual accumulation of mutations over long periods of time results in new biological species, each with a unique DNA sequence. At the top is shown a short segment of a gene in a hypothetical progenitor organism. With the passage of time, changes in nucleotide sequence (mutations, indicated here by colored boxes), occurring one nucleotide at a time, result in progeny with different DNA sequences. These mutant progeny also undergo occasional mutations, yielding their own progeny that differ by two or more nucleotides from the progenitor sequence. When two lineages have diverged so much in their genetic makeup that they can no longer interbreed, a new species has been created.

to offspring—that is, mutations that are carried in the reproductive cells—may be harmful or even lethal to the organism; they may, for example, cause the synthesis of a defective enzyme that is not able to catalyze an essential metabolic reaction. Occasionally, however, a mutation better equips an organism or cell to survive in its environment. The mutant enzyme might have acquired a slightly different specificity, for example, so that it is now able to use some compound that the cell was previously unable to metabolize. If a population of cells were to find itself in an environment where that compound was the only or the most abundant available source of fuel, the mutant cell would have a selective advantage over the other, unmutated (**wild-type**) cells in the population. The mutant cell and its progeny would survive and prosper in the new environment, whereas wild-type cells would starve and be eliminated. This is what Darwin meant by “survival of the fittest under selective pressure.”

Occasionally, a whole gene is duplicated. The second copy is superfluous, and mutations in this gene will not be deleterious; it becomes a means by which the cell may evolve: by producing a new gene with a new function while retaining the original gene and gene function. Seen in this light, the DNA molecules of modern organisms are historical documents, records of the long journey from the earliest cells to modern organisms. The historical accounts in DNA are not complete; in the course of evolution, many mutations must have been erased or written over. But DNA molecules are the best source of biological history that we have.

Several billion years of adaptive selection have refined cellular systems to take maximum advantage of the chemical and physical properties of the molecular raw materials for carrying out the basic energy-transforming and self-replicating activities of a living cell. Chance genetic variations in individuals in a population, combined with natural selection (survival and reproduction of the fittest individuals in a challenging or changing environment), have resulted in the evolution of an enormous variety of organisms, each adapted to life in its particular ecological niche.

Biomolecules First Arose by Chemical Evolution

In our account thus far we have passed over the first chapter of the story of evolution: the appearance of the first living cell. Apart from their occurrence in living organisms, organic compounds, including the basic biomolecules such as amino acids and carbohydrates, are found in only trace amounts in the earth's crust, the sea, and the atmosphere. How did the first living organisms acquire their characteristic organic building blocks? In 1922, the biochemist Aleksandr I. Oparin proposed a theory for the origin of life early in the history of Earth, postulating that the atmosphere was very different from that of today. Rich in methane, ammonia, and water, and

essentially devoid of oxygen, it was a reducing atmosphere, in contrast to the oxidizing environment of our era. In Oparin's theory, electrical energy from lightning discharges or heat energy from volcanoes caused ammonia, methane, water vapor, and other components of the primitive atmosphere to react, forming simple organic compounds. These compounds then dissolved in the ancient seas, which over many millennia became enriched with a large variety of simple organic substances. In this warm solution (the “primordial soup”), some organic molecules had a greater tendency than others to associate into larger complexes. Over millions of years, these in turn assembled spontaneously to form membranes and catalysts (enzymes), which came together to become precursors of the earliest cells. Oparin's views remained speculative for many years and appeared untestable—until a surprising experiment was conducted using simple equipment on a desktop.

Chemical Evolution Can Be Simulated in the Laboratory

The classic experiment on the abiotic (nonbiological) origin of organic biomolecules was carried out in 1953 by Stanley Miller in the laboratory of Harold Urey. Miller subjected gaseous mixtures of NH_3 , CH_4 , H_2O , and H_2 to electrical sparks produced across a pair of electrodes (to simulate lightning) for periods of a week or more, then analyzed the contents of the closed reaction vessel (Fig. 1–33). The gas phase of the resulting mixture contained CO and CO_2 , as well as the starting materials. The water phase contained a variety of organic compounds, including some amino acids, hydroxy acids, aldehydes, and hydrogen cyanide (HCN). This experiment established the possibility of abiotic production of biomolecules in relatively short times under relatively mild conditions.

More refined laboratory experiments have provided good evidence that many of the chemical components of living cells, including polypeptides and RNA-like molecules, can form under these conditions. Polymers of RNA can act as catalysts in biologically significant reactions (as we discuss in Chapters 26 and 27), and RNA probably played a crucial role in prebiotic evolution, both as catalyst and as information repository.

RNA or Related Precursors May Have Been the First Genes and Catalysts

In modern organisms, nucleic acids encode the genetic information that specifies the structure of enzymes, and enzymes catalyze the replication and repair of nucleic acids. The mutual dependence of these two classes of biomolecules brings up the perplexing question: which came first, DNA or protein?

The answer may be: neither. The discovery that RNA molecules can act as catalysts in their own forma-

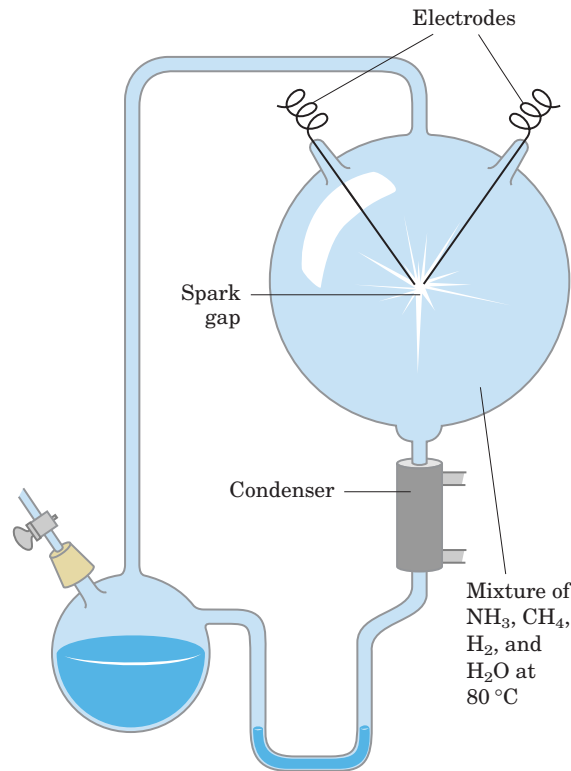


FIGURE 1-33 Abiotic production of biomolecules. Spark-discharge apparatus of the type used by Miller and Urey in experiments demonstrating abiotic formation of organic compounds under primitive atmospheric conditions. After subjection of the gaseous contents of the system to electrical sparks, products were collected by condensation. Biomolecules such as amino acids were among the products.

tion suggests that RNA or a similar molecule may have been the first gene *and* the first catalyst. According to this scenario (Fig. 1-34), one of the earliest stages of biological evolution was the chance formation, in the primordial soup, of an RNA molecule that could catalyze the formation of other RNA molecules of the same sequence—a self-replicating, self-perpetuating RNA. The concentration of a self-replicating RNA molecule would increase exponentially, as one molecule formed two, two formed four, and so on. The fidelity of self-replication was presumably less than perfect, so the process would generate variants of the RNA, some of which might be even better able to self-replicate. In the competition for nucleotides, the most efficient of the self-replicating sequences would win, and less efficient replicators would fade from the population.

The division of function between DNA (genetic information storage) and protein (catalysis) was, according to the “RNA world” hypothesis, a later development. New variants of self-replicating RNA molecules developed, with the additional ability to catalyze the condensation of amino acids into peptides. Occasionally, the peptide(s) thus formed would reinforce the self-replicating ability of the RNA, and the pair—RNA

molecule and helping peptide—could undergo further modifications in sequence, generating even more efficient self-replicating systems. The recent, remarkable discovery that, in the protein-synthesizing machinery of modern cells (ribosomes), RNA molecules, not proteins, catalyze the formation of peptide bonds is certainly consistent with the RNA world hypothesis.

Some time after the evolution of this primitive protein-synthesizing system, there was a further development: DNA molecules with sequences complementary to the self-replicating RNA molecules took over the function of conserving the “genetic” information, and RNA molecules evolved to play roles in protein synthesis. (We explain in Chapter 8 why DNA is a more stable molecule than RNA and thus a better repository of inheritable information.) Proteins proved to be versatile catalysts and, over time, took over that function. Lipidlike compounds in the primordial soup formed relatively impermeable layers around self-replicating collections of molecules. The concentration of proteins and nucleic acids within these lipid enclosures favored the molecular interactions required in self-replication.

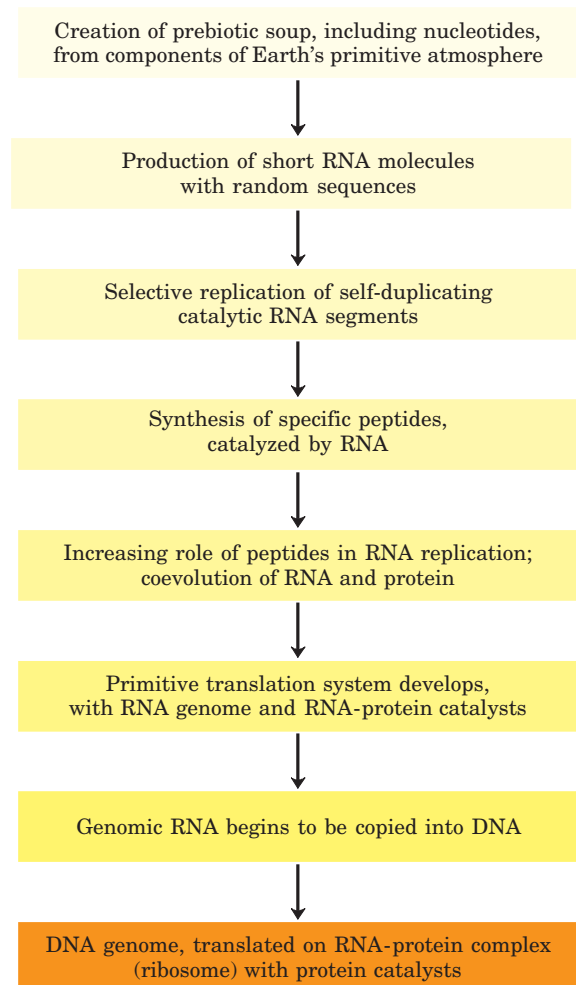


FIGURE 1-34 A possible “RNA world” scenario.

Biological Evolution Began More Than Three and a Half Billion Years Ago

Earth was formed about 4.5 billion years ago, and the first evidence of life dates to more than 3.5 billion years ago. In 1996, scientists working in Greenland found not fossil remains but chemical evidence of life from as far back as 3.85 billion years ago, forms of carbon embedded in rock that appear to have a distinctly biological origin. Somewhere on Earth during its first billion years there arose the first simple organism, capable of replicating its own structure from a template (RNA?) that was the first genetic material. Because the terrestrial atmosphere at the dawn of life was nearly devoid of oxygen, and because there were few microorganisms to scavenge organic compounds formed by natural processes, these compounds were relatively stable. Given this stability and eons of time, the improbable became inevitable: the organic compounds were incorporated into evolving cells to produce increasingly effective self-reproducing catalysts. The process of biological evolution had begun.

The First Cell Was Probably a Chemoheterotroph

The earliest cells that arose in the rich mixture of organic compounds, the primordial soup of prebiotic times, were almost certainly chemoheterotrophs (Fig. 1–5). The organic compounds they required were originally synthesized from components of the early atmosphere—CO, CO₂, N₂, CH₄, and such—by the nonbiological actions of volcanic heat and lightning. Early heterotrophs gradually acquired the ability to derive energy from compounds in their environment and to use that energy to synthesize more of their own precursor molecules, thereby becoming less dependent on outside sources. A very significant evolutionary event was the development of pigments capable of capturing the energy of light from the sun, which could be used to reduce, or “fix,” CO₂ to form more complex, organic compounds. The original electron donor for these **photosynthetic** processes was probably H₂S, yielding elemental sulfur or sulfate (SO₄²⁻) as the by-product, but later cells developed the enzymatic capacity to use H₂O as the electron donor in photosynthetic reactions, eliminating O₂ as waste. Cyanobacteria are the modern descendants of these early photosynthetic oxygen-producers.

Because the atmosphere of Earth in the earliest stages of biological evolution was nearly devoid of oxygen, the earliest cells were anaerobic. Under these conditions, chemoheterotrophs could oxidize organic compounds to CO₂ by passing electrons not to O₂ but to acceptors such as SO₄²⁻, yielding H₂S as the product. With the rise of O₂-producing photosynthetic bacteria, the atmosphere became progressively richer in oxygen—a powerful oxidant and deadly poison to anaerobes. Responding to the evolutionary pressure of the “oxygen holocaust,” some lineages of microorganisms

gave rise to aerobes that obtained energy by passing electrons from fuel molecules to oxygen. Because the transfer of electrons from organic molecules to O₂ releases a great deal of energy, aerobic organisms had an energetic advantage over their anaerobic counterparts when both competed in an environment containing oxygen. This advantage translated into the predominance of aerobic organisms in O₂-rich environments.

Modern bacteria inhabit almost every ecological niche in the biosphere, and there are bacteria capable of using virtually every type of organic compound as a source of carbon and energy. Photosynthetic bacteria in both fresh and marine waters trap solar energy and use it to generate carbohydrates and all other cell constituents, which are in turn used as food by other forms of life. The process of evolution continues—and in rapidly reproducing bacterial cells, on a time scale that allows us to witness it in the laboratory.

Eukaryotic Cells Evolved from Prokaryotes in Several Stages

Starting about 1.5 billion years ago, the fossil record begins to show evidence of larger and more complex organisms, probably the earliest eukaryotic cells (Fig. 1–35).

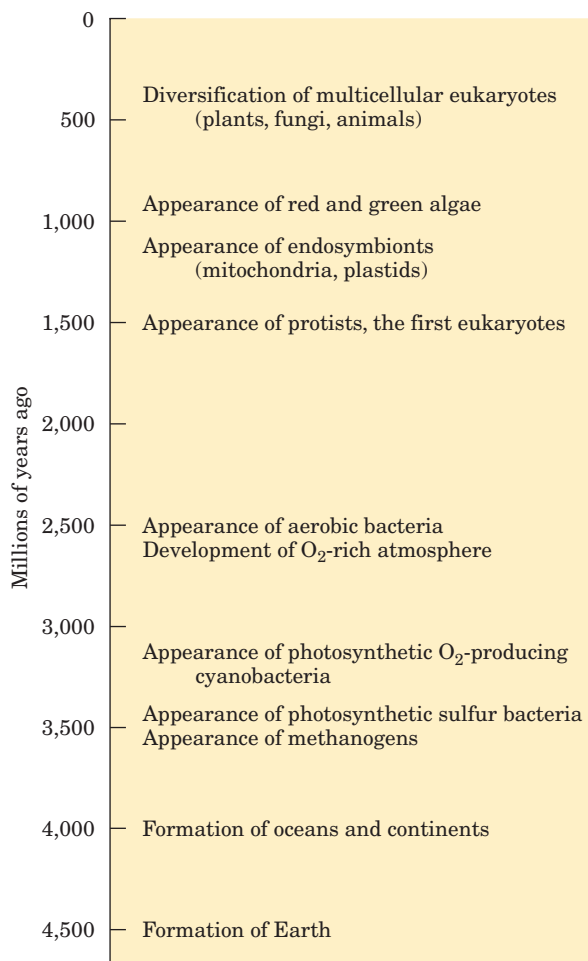


FIGURE 1–35 Landmarks in the evolution of life on Earth.

Details of the evolutionary path from prokaryotes to eukaryotes cannot be deduced from the fossil record alone, but morphological and biochemical comparisons of modern organisms have suggested a sequence of events consistent with the fossil evidence.

Three major changes must have occurred as prokaryotes gave rise to eukaryotes. First, as cells acquired more DNA, the mechanisms required to fold it compactly into discrete complexes with specific proteins and to divide it equally between daughter cells at cell division became more elaborate. For this, specialized proteins were required to stabilize folded DNA and to pull the resulting DNA-protein complexes (*chromosomes*) apart during cell division. Second, as cells became larger, a system of intracellular membranes developed, including a double membrane surrounding the DNA. This membrane segregated the nuclear process of RNA synthesis on a DNA template from the cytoplasmic process of protein synthesis on ribosomes. Finally, early eukaryotic cells, which were incapable of photosynthesis or aerobic metabolism, enveloped aerobic bacteria or photosynthetic bacteria to form **endosymbiotic** associations that became permanent (Fig. 1–36). Some aerobic bacteria evolved into the mitochondria of modern eukaryotes, and some photosynthetic cyanobacteria

became the plastids, such as the chloroplasts of green algae, the likely ancestors of modern plant cells. Prokaryotic and eukaryotic cells are compared in Table 1–3.

At some later stage of evolution, unicellular organisms found it advantageous to cluster together, thereby acquiring greater motility, efficiency, or reproductive success than their free-living single-celled competitors. Further evolution of such clustered organisms led to permanent associations among individual cells and eventually to specialization within the colony—to cellular differentiation.

The advantages of cellular specialization led to the evolution of ever more complex and highly differentiated organisms, in which some cells carried out the sensory functions, others the digestive, photosynthetic, or reproductive functions, and so forth. Many modern multicellular organisms contain hundreds of different cell types, each specialized for some function that supports the entire organism. Fundamental mechanisms that evolved early have been further refined and embellished through evolution. The same basic structures and mechanisms that underlie the beating motion of cilia in *Paramecium* and of flagella in *Chlamydomonas* are employed by the highly differentiated vertebrate sperm cell.

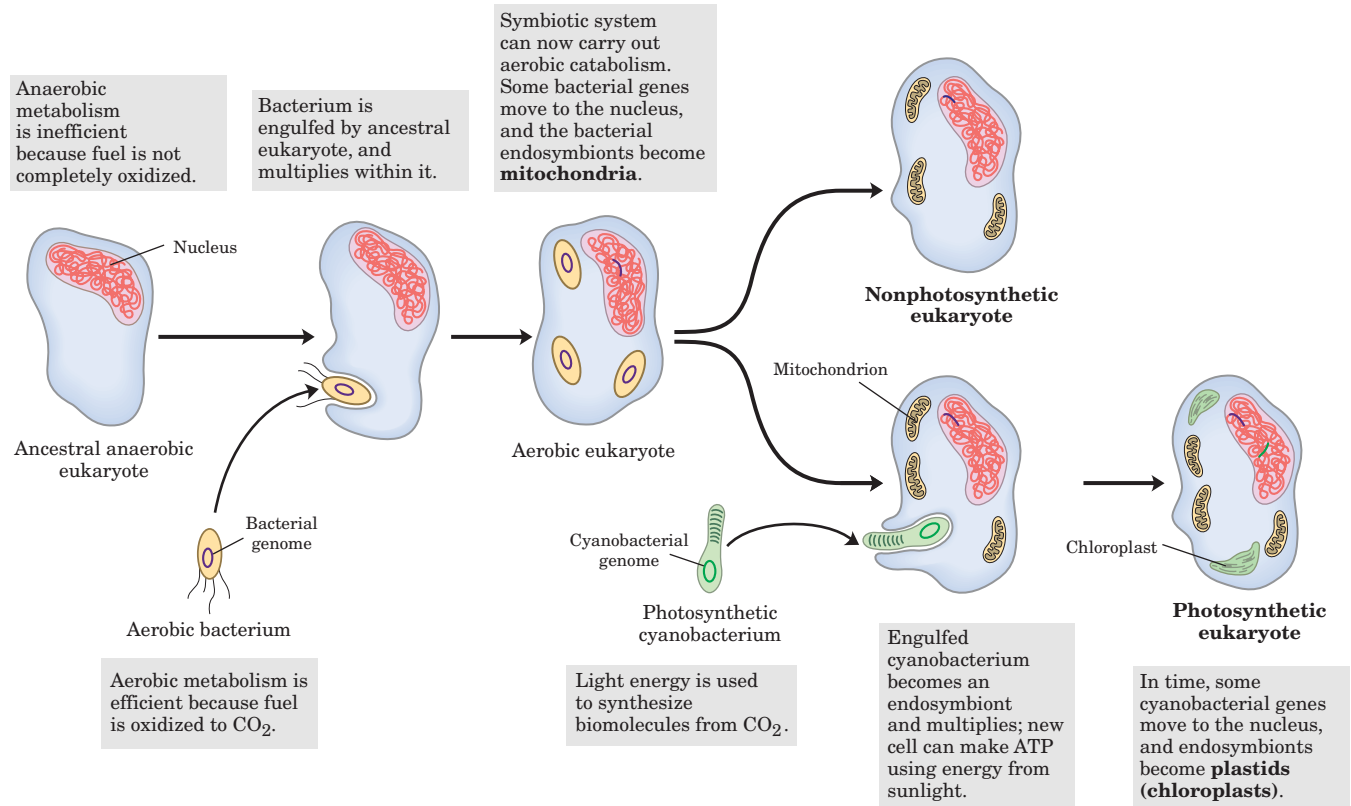


FIGURE 1–36 Evolution of eukaryotes through endosymbiosis. The earliest eukaryote, an anaerobe, acquired endosymbiotic purple bacteria (yellow), which carried with them their capacity for aerobic catabolism and became, over time, mitochondria. When photosynthetic

cyanobacteria (green) subsequently became endosymbionts of some aerobic eukaryotes, these cells became the photosynthetic precursors of modern green algae and plants.

TABLE 1-3 Comparison of Prokaryotic and Eukaryotic Cells

Characteristic	Prokaryotic cell	Eukaryotic cell
Size	Generally small (1–10 μm)	Generally large (5–100 μm)
Genome	DNA with nonhistone protein; genome in nucleoid, not surrounded by membrane	DNA complexed with histone and nonhistone proteins in chromosomes; chromosomes in nucleus with membranous envelope
Cell division	Fission or budding; no mitosis	Mitosis, including mitotic spindle; centrioles in many species
Membrane-bounded organelles	Absent	Mitochondria, chloroplasts (in plants, some algae), endoplasmic reticulum, Golgi complexes, lysosomes (in animals), etc.
Nutrition	Absorption; some photosynthesis	Absorption, ingestion; photosynthesis in some species
Energy metabolism	No mitochondria; oxidative enzymes bound to plasma membrane; great variation in metabolic pattern	Oxidative enzymes packaged in mitochondria; more unified pattern of oxidative metabolism
Cytoskeleton	None	Complex, with microtubules, intermediate filaments, actin filaments
Intracellular movement	None	Cytoplasmic streaming, endocytosis, phagocytosis, mitosis, vesicle transport

Source: Modified from Hickman, C.P., Roberts, L.S., & Hickman, F.M. (1990) *Biology of Animals*, 5th edn, p. 30, Mosby-Yearbook, Inc., St. Louis, MO.

Molecular Anatomy Reveals Evolutionary Relationships

The eighteenth-century naturalist Carolus Linnaeus recognized the anatomic similarities and differences among living organisms and used them to provide a framework for assessing the relatedness of species. Charles Darwin, in the nineteenth century, gave us a unifying hypothesis to explain the phylogeny of modern organisms—the origin of different species from a common ancestor. Biochemical research in the twentieth century revealed the molecular anatomy of cells of different species—the monomeric subunit sequences and the three-dimensional structures of individual nucleic acids and proteins. Biochemists now have an enormously rich and increasing treasury of evidence that can be used to analyze evolutionary relationships and to refine evolutionary theory.

The sequence of the **genome** (the complete genetic endowment of an organism) has been entirely determined for numerous eubacteria and for several archaeobacteria; for the eukaryotic microorganisms *Saccharomyces cerevisiae* and *Plasmodium* sp.; for the plants *Arabidopsis thaliana* and rice; and for the multicellular animals *Caenorhabditis elegans* (a roundworm), *Drosophila melanogaster* (the fruit fly), mice, rats, and *Homo sapiens* (you) (Table 1-4). More sequences are being added to this list regularly. With such sequences in hand, detailed and quantitative comparisons among species can provide deep insight into the evolutionary process. Thus far, the molecular phylogeny derived from gene sequences is

consistent with, but in many cases more precise than, the classical phylogeny based on macroscopic structures. Although organisms have continuously diverged at the level of gross anatomy, at the molecular level the basic unity of life is readily apparent; molecular structures and mechanisms are remarkably similar from the simplest to the most complex organisms. These similarities are most easily seen at the level of sequences, either the DNA sequences that encode proteins or the protein sequences themselves.

When two genes share readily detectable sequence similarities (nucleotide sequence in DNA or amino acid sequence in the proteins they encode), their sequences



Carolus Linnaeus,
1701–1778



Charles Darwin,
1809–1882

TABLE 1-4 Some Organisms Whose Genomes Have Been Completely Sequenced

Organism	Genome size (millions of nucleotide pairs)	Biological interest
<i>Mycoplasma pneumoniae</i>	0.8	Causes pneumonia
<i>Treponema pallidum</i>	1.1	Causes syphilis
<i>Borrelia burgdorferi</i>	1.3	Causes Lyme disease
<i>Helicobacter pylori</i>	1.7	Causes gastric ulcers
<i>Methanococcus jannaschii</i>	1.7	Grows at 85 °C!
<i>Haemophilus influenzae</i>	1.8	Causes bacterial influenza
<i>Methanobacterium thermoautotrophicum</i>	1.8	Member of the Archaea
<i>Archaeoglobus fulgidus</i>	2.2	High-temperature methanogen
<i>Synechocystis</i> sp.	3.6	Cyanobacterium
<i>Bacillus subtilis</i>	4.2	Common soil bacterium
<i>Escherichia coli</i>	4.6	Some strains cause toxic shock syndrome
<i>Saccharomyces cerevisiae</i>	12.1	Unicellular eukaryote
<i>Plasmodium falciparum</i>	23	Causes human malaria
<i>Caenorhabditis elegans</i>	97.1	Multicellular roundworm
<i>Anopheles gambiae</i>	278	Malaria vector
<i>Mus musculus domesticus</i>	2.5×10^3	Laboratory mouse
<i>Homo sapiens</i>	2.9×10^3	Human

are said to be homologous and the proteins they encode are **homologs**. If two homologous genes occur in the *same* species, they are said to be paralogous and their protein products are **paralogs**. Paralogous genes are presumed to have been derived by gene duplication followed by gradual changes in the sequences of both copies (Fig. 1-37). Typically, paralogous proteins are similar not only in sequence but also in three-dimensional structure, although they commonly have acquired different functions during their evolution.

Two homologous genes (or proteins) found in *different* species are said to be orthologous, and their protein products are **orthologs**. Orthologs are commonly found to have the same function in both organisms, and when a newly sequenced gene in one species is found to be strongly orthologous with a gene in another, this gene is presumed to encode a protein with the same function in both species. By this means, the function of gene products can be deduced from the genomic sequence, without any biochemical characterization of the gene product. An **annotated genome** includes, in addition to the DNA sequence itself, a description of the likely function of each gene product, deduced from comparisons with other genomic sequences and established protein functions. In principle, by identifying the pathways (sets of enzymes) encoded in a genome, we can deduce from the genomic sequence alone the organism's metabolic capabilities.

The sequence differences between homologous genes may be taken as a rough measure of the degree to which the two species have diverged during evolution—of how long ago their common evolutionary precursor gave rise to two lines with different evolutionary fates. The larger the number of sequence differences, the earlier the divergence in evolutionary history. One can construct a phylogeny (family tree) in which the evolutionary distance between any two species is represented by their proximity on the tree (Fig. 1-4 is an example).

As evolution advances, new structures, processes, or regulatory mechanisms are acquired, reflections of the changing genomes of the evolving organisms. The genome of a simple eukaryote such as yeast should have genes related to formation of the nuclear membrane, genes not present in prokaryotes. The genome of an insect should contain genes that encode proteins involved in specifying the characteristic insect segmented body plan, genes not present in yeast. The genomes of all vertebrate animals should share genes that specify the development of a spinal column, and those of mammals should have unique genes necessary for the development of the placenta, a characteristic of mammals—and so on. Comparisons of the whole genomes of species in each phylum may lead to the identification of genes critical to fundamental evolutionary changes in body plan and development.

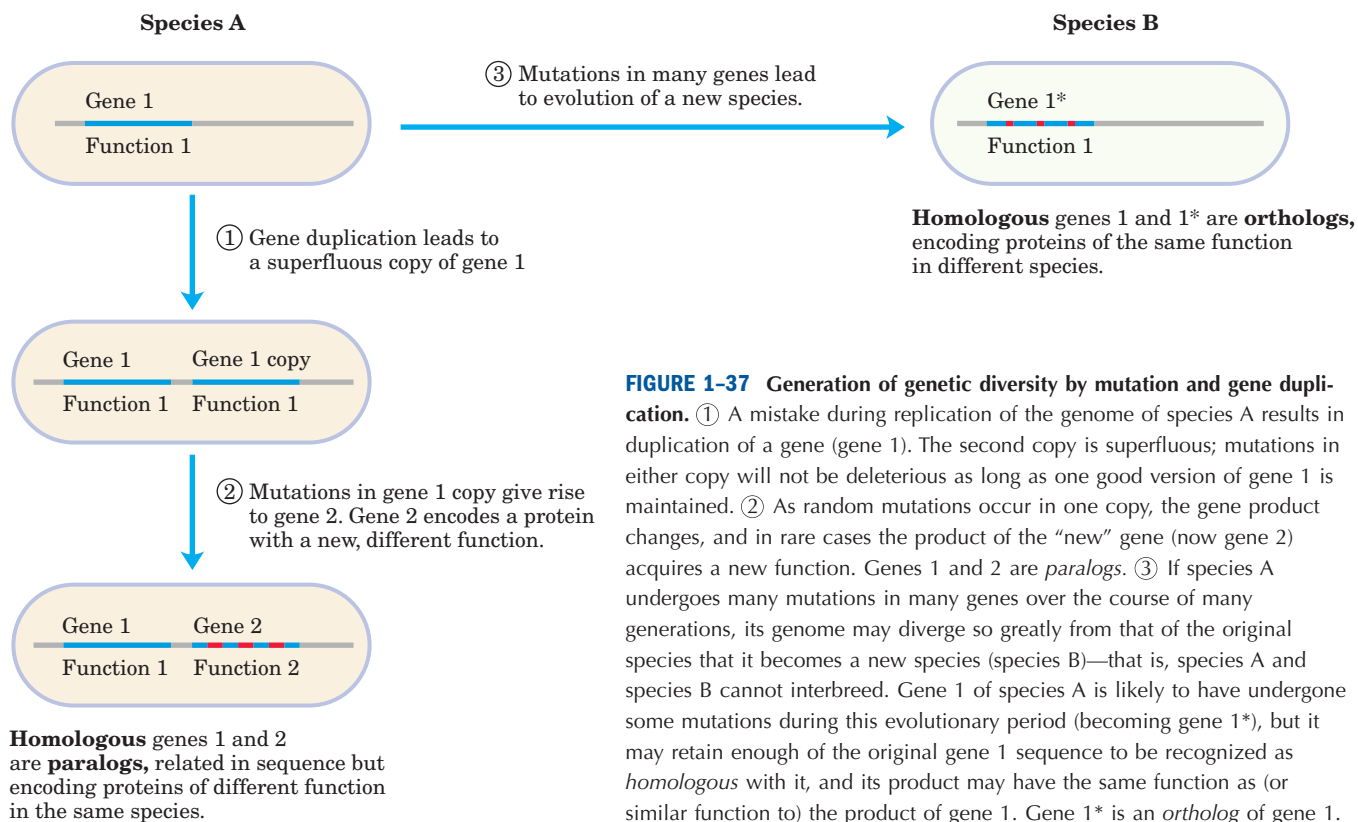


FIGURE 1-37 Generation of genetic diversity by mutation and gene duplication. ① A mistake during replication of the genome of species A results in duplication of a gene (gene 1). The second copy is superfluous; mutations in either copy will not be deleterious as long as one good version of gene 1 is maintained. ② As random mutations occur in one copy, the gene product changes, and in rare cases the product of the “new” gene (now gene 2) acquires a new function. Genes 1 and 2 are *paralogs*. ③ If species A undergoes many mutations in many genes over the course of many generations, its genome may diverge so greatly from that of the original species that it becomes a new species (species B)—that is, species A and species B cannot interbreed. Gene 1 of species A is likely to have undergone some mutations during this evolutionary period (becoming gene 1*), but it may retain enough of the original gene 1 sequence to be recognized as *homologous* with it, and its product may have the same function as (or similar function to) the product of gene 1. Gene 1* is an *ortholog* of gene 1.

Functional Genomics Shows the Allocations of Genes to Specific Cellular Processes

When the sequence of a genome is fully determined and each gene is annotated (that is, assigned a function), molecular geneticists can group genes according to the processes (DNA synthesis, protein synthesis, generation of ATP, and so forth) in which they function and thus find what fraction of the genome is allocated to each of a cell’s activities. The largest category of genes in *E. coli*, *A. thaliana*, and *H. sapiens* consists of genes of as yet unknown function, which make up more than 40% of the genes in each species. The transporters that move ions and small molecules across plasma membranes take up a significant proportion of the genes in all three species, more in the bacterium and plant than in the mammal (10% of the 4,269 genes of *E. coli*, ~8% of the 25,706 genes of *A. thaliana*, and ~4% of the ~35,000 genes of *H. sapiens*). Genes that encode the proteins and RNA required for protein synthesis make up 3% to 4% of the *E. coli* genome, but in the more complex cells of *A. thaliana*, more genes are needed for targeting proteins to their final location in the cell than are needed to synthesize those proteins (about 6% and 2%, respectively). In general, the more complex the organism, the greater the proportion of its genome that encodes genes involved in the *regulation* of cellular processes and the smaller the proportion dedicated to the basic processes themselves, such as ATP generation and protein synthesis.

Genomic Comparisons Will Have Increasing Importance in Human Biology and Medicine



The genomes of chimpanzees and humans are 99.9% identical, yet the differences between the two species are vast. The relatively few differences in genetic endowment must explain the possession of language by humans, the extraordinary athleticism of chimpanzees, and myriad other differences. Genomic comparison will allow researchers to identify candidate genes linked to divergences in the developmental programs of humans and the other primates and to the emergence of complex functions such as language. The picture will become clearer only as more primate genomes become available for comparison with the human genome.

Similarly, the differences in genetic endowment among humans are vanishingly small compared with the differences between humans and chimpanzees, yet these differences account for the variety among us—including differences in health and in susceptibility to chronic diseases. We have much to learn about the variability in sequence among humans, and during the next decade the availability of genomic information will almost certainly transform medical diagnosis and treatment. We may expect that for some genetic diseases, palliatives will be replaced by cures; and that for disease susceptibilities associated with particular genetic markers, forewarning and perhaps increased preventive measures will prevail. Today’s “medical history” may be replaced by a “medical forecast.” ■

SUMMARY 1.5 Evolutionary Foundations

- Occasional inheritable mutations yield an organism that is better suited for survival in an ecological niche and progeny that are preferentially selected. This process of mutation and selection is the basis for the Darwinian evolution that led from the first cell to all the organisms that now exist, and it explains the fundamental similarity of all living organisms.
- Life originated about 3.5 billion years ago, most likely with the formation of a membrane-enclosed compartment containing a self-replicating RNA molecule. The components for the first cell were produced by the action of lightning and high temperature on simple atmospheric molecules such as CO₂ and NH₃.
- The catalytic and genetic roles of the early RNA genome were separated over time, with DNA becoming the genomic material and proteins the major catalytic species.
- Eukaryotic cells acquired the capacity for photosynthesis and for oxidative phosphorylation from endosymbiotic bacteria. In multicellular organisms, differentiated cell types specialize in one or more of the functions essential to the organism's survival.
- Knowledge of the complete genomic nucleotide sequences of organisms from different branches of the phylogenetic tree provides insights into the evolution and function of extant organisms and offers great opportunities in human medicine.

Key Terms

All terms are defined in the glossary.

metabolite	3	stereoisomers	16	exergonic reaction	23
nucleus	3	configuration	16	equilibrium	24
genome	3	chiral center	17	standard free-energy change, ΔG°	26
eukaryote	4	conformation	19	activation energy, $\Delta G^\ddagger$	26
prokaryote	4	entropy, S	23	catabolism	27
archaeobacteria	4	enthalpy, H	23	anabolism	27
eubacteria	4	free-energy change, ΔG	23	metabolism	27
cytoskeleton	9	endergonic reaction	23	mutation	31

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Problems

Some problems related to the contents of the chapter follow. (In solving end-of-chapter problems, you may wish to refer to the tables on the inside of the back cover.) Each problem has a title for easy reference and discussion.

1. The Size of Cells and Their Components

(a) If you were to magnify a cell 10,000 fold (typical of the magnification achieved using an electron microscope), how big would it appear? Assume you are viewing a “typical” eukaryotic cell with a cellular diameter of 50 μm .

(b) If this cell were a muscle cell (myocyte), how many molecules of actin could it hold? (Assume the cell is spherical and no other cellular components are present; actin molecules are spherical, with a diameter of 3.6 nm. The volume of a sphere is $\frac{4}{3}\pi r^3$.)

(c) If this were a liver cell (hepatocyte) of the same dimensions, how many mitochondria could it hold? (Assume the cell is spherical; no other cellular components are present; and the mitochondria are spherical, with a diameter of 1.5 μm .)

(d) Glucose is the major energy-yielding nutrient for most cells. Assuming a cellular concentration of 1 mM, calculate how many molecules of glucose would be present in our hypothetical (and spherical) eukaryotic cell. (Avogadro’s number, the number of molecules in 1 mol of a nonionized substance, is 6.02×10^{23} .)

(e) Hexokinase is an important enzyme in the metabolism of glucose. If the concentration of hexokinase in our eukaryotic cell is 20 μM , how many glucose molecules are present per hexokinase molecule?

2. Components of *E. coli* *E. coli* cells are rod-shaped, about 2 μm long and 0.8 μm in diameter. The volume of a cylinder is $\pi r^2 h$, where h is the height of the cylinder.

(a) If the average density of *E. coli* (mostly water) is $1.1 \times 10^3 \text{ g/L}$, what is the mass of a single cell?

(b) *E. coli* has a protective cell envelope 10 nm thick. What percentage of the total volume of the bacterium does the cell envelope occupy?

(c) *E. coli* is capable of growing and multiplying rapidly because it contains some 15,000 spherical ribosomes (diameter 18 nm), which carry out protein synthesis. What percentage of the cell volume do the ribosomes occupy?

3. Genetic Information in *E. coli* DNA The genetic information contained in DNA consists of a linear sequence of coding units, known as codons. Each codon is a specific sequence of three deoxyribonucleotides (three deoxyribonucleotide pairs in double-stranded DNA), and each codon codes for a single amino acid unit in a protein. The molecular weight of an *E. coli* DNA molecule is about $3.1 \times 10^9 \text{ g/mol}$. The average molecular weight of a nucleotide pair is 660 g/mol, and each nucleotide pair contributes 0.34 nm to the length of DNA.

(a) Calculate the length of an *E. coli* DNA molecule. Compare the length of the DNA molecule with the cell dimensions (see Problem 2). How does the DNA molecule fit into the cell?

(b) Assume that the average protein in *E. coli* consists of a chain of 400 amino acids. What is the maximum number of proteins that can be coded by an *E. coli* DNA molecule?

4. The High Rate of Bacterial Metabolism Bacterial cells have a much higher rate of metabolism than animal cells. Under ideal conditions some bacteria double in size and divide every 20 min, whereas most animal cells under rapid growth conditions require 24 hours. The high rate of bacterial metabolism requires a high ratio of surface area to cell volume.

(a) Why does surface-to-volume ratio affect the maximum rate of metabolism?

(b) Calculate the surface-to-volume ratio for the spherical bacterium *Neisseria gonorrhoeae* (diameter 0.5 μm), responsible for the disease gonorrhea. Compare it with the surface-to-volume ratio for a globular amoeba, a large eukaryotic cell (diameter 150 μm). The surface area of a sphere is $4\pi r^2$.

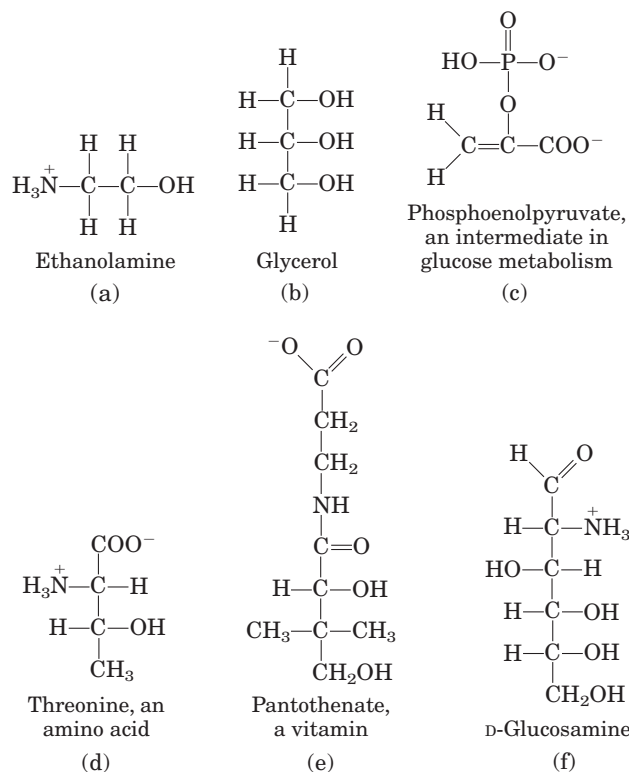
5. Fast Axonal Transport Neurons have long thin processes called axons, structures specialized for conducting signals throughout the organism’s nervous system. Some axonal processes can be as long as 2 m—for example, the axons that originate in your spinal cord and terminate in the muscles of your toes. Small membrane-enclosed vesicles carrying materials essential to axonal function move along microtubules of the cytoskeleton, from the cell body to the tips of the axons.

(a) If the average velocity of a vesicle is 1 $\mu\text{m/s}$, how long does it take a vesicle to move from a cell body in the spinal cord to the axonal tip in the toes?

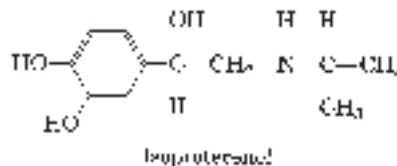
(b) Movement of large molecules by diffusion occurs relatively slowly in cells. (For example, hemoglobin diffuses at a rate of approximately $5 \mu\text{m/s}$.) However, the diffusion of sucrose in an aqueous solution occurs at a rate approaching that of fast cellular transport mechanisms (about $4 \mu\text{m/s}$). What are some advantages to a cell or an organism of fast, directed transport mechanisms, compared with diffusion alone?

6. Vitamin C: Is the Synthetic Vitamin as Good as the Natural One? A claim put forth by some purveyors of health foods is that vitamins obtained from natural sources are more healthful than those obtained by chemical synthesis. For example, pure L-ascorbic acid (vitamin C) extracted from rose hips is better than pure L-ascorbic acid manufactured in a chemical plant. Are the vitamins from the two sources different? Can the body distinguish a vitamin's source?

7. Identification of Functional Groups Figures 1–15 and 1–16 show some common functional groups of biomolecules. Because the properties and biological activities of biomolecules are largely determined by their functional groups, it is important to be able to identify them. In each of the compounds below, circle and identify by name each functional group.



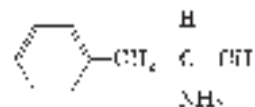
8. Drug Activity and Stereochemistry The quantitative differences in biological activity between the two enantiomers of a compound are sometimes quite large. For example, the D isomer of the drug isoproterenol, used to treat mild asthma, is 50 to 80 times more effective as a bronchodilator than the L isomer. Identify the chiral center in isoproterenol. Why do the two enantiomers have such radically different bioactivity?



9. Separating Biomolecules In studying a particular biomolecule (a protein, nucleic acid, carbohydrate, or lipid) in the laboratory, the biochemist first needs to separate it from other biomolecules in the sample—that is, to *purify* it. Specific purification techniques are described later in the text. However, by looking at the monomeric subunits of a biomolecule, you should have some ideas about the characteristics of the molecule that would allow you to separate it from other molecules. For example, how would you separate (a) amino acids from fatty acids and (b) nucleotides from glucose?

10. Silicon-Based Life? Silicon is in the same group of the periodic table as carbon and, like carbon, can form up to four single bonds. Many science fiction stories have been based on the premise of silicon-based life. Is this realistic? What characteristics of silicon make it *less* well adapted than carbon as the central organizing element for life? To answer this question, consider what you have learned about carbon's bonding versatility, and refer to a beginning inorganic chemistry textbook for silicon's bonding properties.

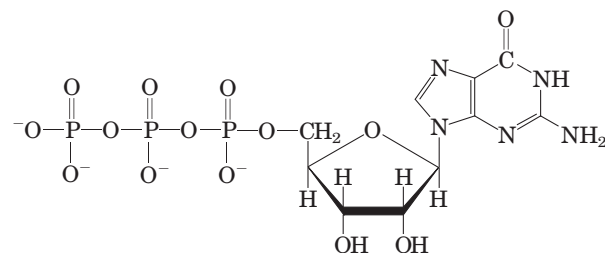
11. Drug Action and Shape of Molecules Some years ago two drug companies marketed a drug under the trade names Dexedrine and Benzedrine. The structure of the drug is shown below.



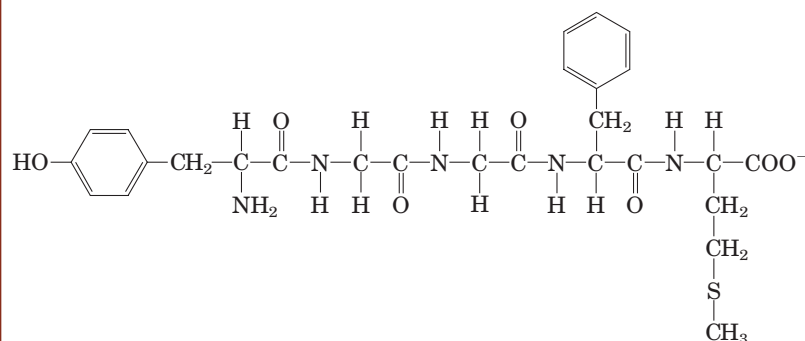
The physical properties (C, H, and N analysis, melting point, solubility, etc.) of Dexedrine and Benzedrine were identical. The recommended oral dosage of Dexedrine (which is still available) was 5 mg/day, but the recommended dosage of Benzedrine (no longer available) was twice that. Apparently it required considerably more Benzedrine than Dexedrine to yield the same physiological response. Explain this apparent contradiction.

12. Components of Complex Biomolecules Figure 1–10 shows the major components of complex biomolecules. For each of the three important biomolecules below (shown in their ionized forms at physiological pH), identify the constituents.

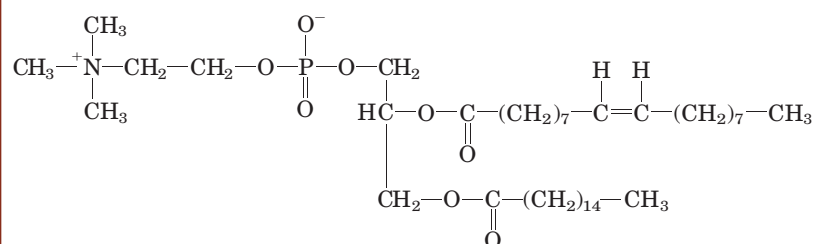
(a) Guanosine triphosphate (GTP), an energy-rich nucleotide that serves as a precursor to RNA:



(b) Phosphatidylcholine, a component of many membranes:



(c) Methionine enkephalin, the brain's own opiate:



13. Determination of the Structure of a Biomolecule

An unknown substance, X, was isolated from rabbit muscle. Its structure was determined from the following observations and experiments. Qualitative analysis showed that X was composed entirely of C, H, and O. A weighed sample of X was completely oxidized, and the H_2O and CO_2 produced were measured; this quantitative analysis revealed that X contained 40.00% C, 6.71% H, and 53.29% O by weight. The molecular mass of X, determined by mass spectrometry, was 90.00 u (atomic mass units; see Box 1-1). Infrared spectroscopy showed that X contained one double bond. X dissolved readily in water to give an acidic solution; the solution demonstrated optical activity when tested in a polarimeter.

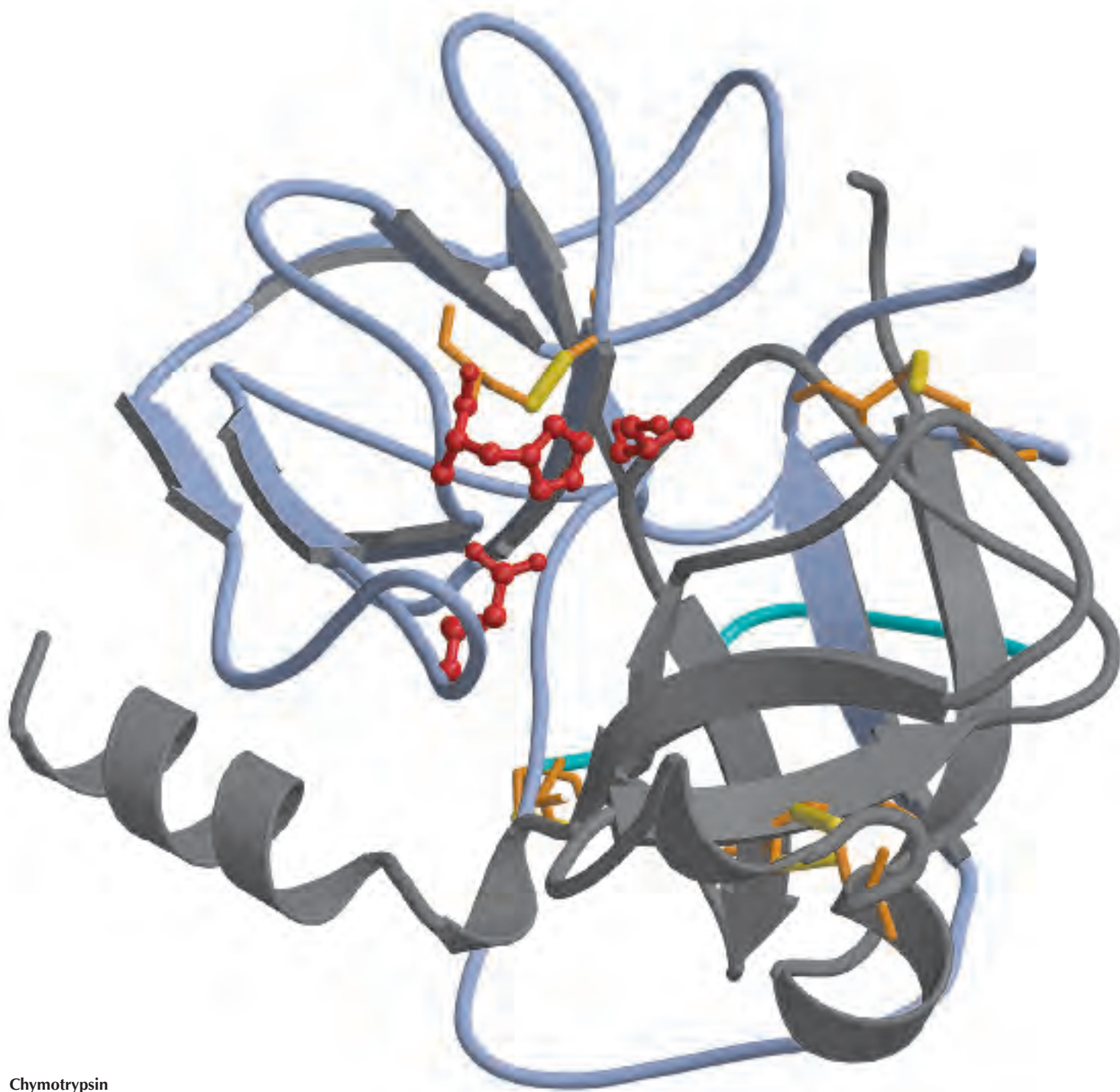
(a) Determine the empirical and molecular formula of X.

(b) Draw the possible structures of X that fit the molecular formula and contain one double bond. Consider *only* linear or branched structures and disregard cyclic structures. Note that oxygen makes very poor bonds to itself.

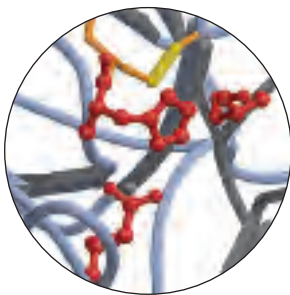
(c) What is the structural significance of the observed optical activity? Which structures in (b) are consistent with the observation?

(d) What is the structural significance of the observation that a solution of X was acidic? Which structures in (b) are consistent with the observation?

(e) What is the structure of X? Is more than one structure consistent with all the data?



Chymotrypsin



STRUCTURE AND CATALYSIS

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In 1897 Eduard Buchner, the German research worker, discovered that sugar can be made to ferment, not only with ordinary yeast, but also with the help of the expressed juices of yeast which contain none of the cells of the *Saccharomyces* . . . Why was this apparently somewhat trivial experiment considered to be of such significance? The answer to this question is self-evident, if the development within the research work directed on the elucidation of the chemical nature of (life) is followed . . . there, more than in most fields, a tendency has showed itself to consider the unexplained as inexplicable . . . Thus ordinary yeast consists of living cells, and fermentation was considered by the majority of research workers—among them Pasteur—to be a manifestation of life, i.e. to be inextricably associated with the vital processes in these cells. Buchner's discovery showed that this was not the case. It may be said that thereby, at a blow, an important class of vital processes

was removed from the cells into the chemists' laboratories, to be studied there by the chemists' methods. It proved, too, that, apart from fermentation, combustion and respiration, the splitting up of protein substances, fats and carbohydrates, and many other similar reactions which characterise the living cell, could be imitated in the test tube without any cooperation at all from the cells, and that on the whole the same laws held for these reactions as for ordinary chemical processes.

—A. Tiselius, in presentation speech for the award of the Nobel Prize in Chemistry to James B. Sumner, John H. Northrop, and Wendell M. Stanley, 1946

The science of biochemistry can be dated to Eduard Buchner's pioneering discovery. His finding opened a world of chemistry that has inspired researchers for well over a century. Biochemistry is nothing less than the chemistry of life, and, yes, life can be investigated, analyzed, and understood. To begin, every student of biochemistry needs both a language and some fundamentals; these are provided in Part I.

The chapters of Part I are devoted to the structure and function of the major classes of cellular constituents: water (Chapter 2), amino acids and proteins (Chapters 3 through 6), sugars and polysaccharides (Chapter 7), nucleotides and nucleic acids (Chapter 8), fatty acids and lipids (Chapter 10), and, finally, membranes and membrane signaling proteins (Chapters 11 and 12). We supplement this discourse on molecules with information about the technologies used to study them. Some of the techniques sections are woven throughout the molecular descriptions, although one entire chapter (Chapter 9) is devoted to an integrated

suite of modern advances in biotechnology that have greatly accelerated the pace of discovery.

The molecules found in a cell are a major part of the language of biochemistry; familiarity with them is a prerequisite for understanding more advanced topics covered in this book and for appreciating the rapidly growing and exciting literature of biochemistry. We begin with water because its properties affect the structure and function of all other cellular constituents. For each class of organic molecules, we first consider the covalent chemistry of the monomeric units (amino acids, monosaccharides, nucleotides, and fatty acids) and then describe the structure of the macromolecules and supramolecular complexes derived from them. An overriding theme is that the polymeric macromolecules in living systems, though large, are highly ordered chemical entities, with specific sequences of monomeric subunits giving rise to discrete structures and functions. This fundamental theme can be broken down into three interrelated principles: (1) the unique structure of each macromolecule determines its function; (2) noncovalent interactions play a critical role in the structure and thus the function of macromolecules; and (3) the monomeric subunits in polymeric macromolecules occur in specific sequences, representing a form of information upon which the ordered living state depends.

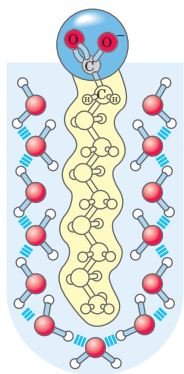
The relationship between structure and function is especially evident in proteins, which exhibit an extraordinary diversity of functions. One particular polymeric sequence of amino acids produces a strong, fibrous structure found in hair and wool; another produces a protein that transports oxygen in the blood; a third binds other proteins and catalyzes the cleavage of the bonds between their amino acids. Similarly, the special functions of polysaccharides, nucleic acids, and lipids can be understood as a direct manifestation of their chemical structure, with their characteristic monomeric subunits linked in precise functional polymers. Sugars linked together become energy stores, structural fibers, and points of specific molecular recognition; nucleotides strung together in DNA or RNA provide the blueprint for an entire organ-

ism; and aggregated lipids form membranes. Chapter 12 unifies the discussion of biomolecule function, describing how specific signaling systems regulate the activities of biomolecules—within a cell, within an organ, and among organs—to keep an organism in homeostasis.

As we move from monomeric units to larger and larger polymers, the chemical focus shifts from covalent bonds to noncovalent interactions. The properties of covalent bonds, both in the monomeric subunits and in the bonds that connect them in polymers, place constraints on the shapes assumed by large molecules. It is the numerous noncovalent interactions, however, that dictate the stable native conformations of large molecules while permitting the flexibility necessary for their biological function. As we shall see, noncovalent interactions are essential to the catalytic power of enzymes, the critical interaction of complementary base pairs in nucleic acids, the arrangement and properties of lipids in membranes, and the interaction of a hormone or growth factor with its membrane receptor.

The principle that sequences of monomeric subunits are rich in information emerges most fully in the discussion of nucleic acids (Chapter 8). However, proteins and some short polymers of sugars (oligosaccharides) are also information-rich molecules. The amino acid sequence is a form of information that directs the folding of the protein into its unique three-dimensional structure, and ultimately determines the function of the protein. Some oligosaccharides also have unique sequences and three-dimensional structures that are recognized by other macromolecules.

Each class of molecules has a similar structural hierarchy: subunits of fixed structure are connected by bonds of limited flexibility to form macromolecules with three-dimensional structures determined by noncovalent interactions. These macromolecules then interact to form the supramolecular structures and organelles that allow a cell to carry out its many metabolic functions. Together, the molecules described in Part I are the stuff of life. We begin with water.



chapter 2

WATER

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- 2.2 Ionization of Water, Weak Acids, and Weak Bases** 60
- 2.3 Buffering against pH Changes in Biological Systems** 65
- 2.4 Water as a Reactant** 69
- 2.5 The Fitness of the Aqueous Environment for Living Organisms** 70

I believe that as the methods of structural chemistry are further applied to physiological problems, it will be found that the significance of the hydrogen bond for physiology is greater than that of any other single structural feature.

—Linus Pauling, *The Nature of the Chemical Bond*, 1939

What in water did Bloom, water lover, drawer of water, water carrier returning to the range, admire? Its universality, its democratic quality.

—James Joyce, *Ulysses*, 1922

Water is the most abundant substance in living systems, making up 70% or more of the weight of most organisms. The first living organisms doubtless arose in an aqueous environment, and the course of evolution has been shaped by the properties of the aqueous medium in which life began.

This chapter begins with descriptions of the physical and chemical properties of water, to which all aspects of cell structure and function are adapted. The attractive forces between water molecules and the slight tendency of water to ionize are of crucial importance to the structure and function of biomolecules. We review the topic of ionization in terms of equilibrium constants, pH,

and titration curves, and consider how aqueous solutions of weak acids or bases and their salts act as buffers against pH changes in biological systems. The water molecule and its ionization products, H^+ and OH^- , profoundly influence the structure, self-assembly, and properties of all cellular components, including proteins, nucleic acids, and lipids. The noncovalent interactions responsible for the strength and specificity of “recognition” among biomolecules are decisively influenced by the solvent properties of water, including its ability to form hydrogen bonds with itself and with solutes.

2.1 Weak Interactions in Aqueous Systems

Hydrogen bonds between water molecules provide the cohesive forces that make water a liquid at room temperature and that favor the extreme ordering of molecules that is typical of crystalline water (ice). Polar biomolecules dissolve readily in water because they can replace water-water interactions with more energetically favorable water-solute interactions. In contrast, nonpolar biomolecules interfere with water-water interactions but are unable to form water-solute interactions—consequently, nonpolar molecules are poorly soluble in water. In aqueous solutions, nonpolar molecules tend to cluster together.

Hydrogen bonds and ionic, hydrophobic (Greek, “water-fearing”), and van der Waals interactions are individually weak, but collectively they have a very significant influence on the three-dimensional structures of proteins, nucleic acids, polysaccharides, and membrane lipids.

Hydrogen Bonding Gives Water Its Unusual Properties

Water has a higher melting point, boiling point, and heat of vaporization than most other common solvents (Table 2–1). These unusual properties are a consequence of

TABLE 2-1 Melting Point, Boiling Point, and Heat of Vaporization of Some Common Solvents

	Melting point ($^{\circ}\text{C}$)	Boiling point ($^{\circ}\text{C}$)	Heat of vaporization (J/g)*
Water	0	100	2,260
Methanol (CH_3OH)	-98	65	1,100
Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)	-117	78	854
Propanol ($\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$)	-127	97	687
Butanol ($\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{OH}$)	-90	117	590
Acetone (CH_3COCH_3)	-95	56	523
Hexane ($\text{CH}_3(\text{CH}_2)_4\text{CH}_3$)	-98	69	423
Benzene (C_6H_6)	6	80	394
Butane ($\text{CH}_3(\text{CH}_2)_2\text{CH}_3$)	-135	-0.5	381
Chloroform (CHCl_3)	-63	61	247

*The heat energy required to convert 1.0 g of a liquid at its boiling point, at atmospheric pressure, into its gaseous state at the same temperature. It is a direct measure of the energy required to overcome attractive forces between molecules in the liquid phase.

attractions between adjacent water molecules that give liquid water great internal cohesion. A look at the electron structure of the H_2O molecule reveals the cause of these intermolecular attractions.

Each hydrogen atom of a water molecule shares an electron pair with the central oxygen atom. The geometry of the molecule is dictated by the shapes of the outer electron orbitals of the oxygen atom, which are similar to the sp_3 bonding orbitals of carbon (see Fig. 1-14). These orbitals describe a rough tetrahedron, with a hydrogen atom at each of two corners and unshared electron pairs at the other two corners (Fig. 2-1a). The $\text{H}-\text{O}-\text{H}$ bond angle is 104.5° , slightly less than the 109.5° of a perfect tetrahedron because of crowding by the nonbonding orbitals of the oxygen atom.

The oxygen nucleus attracts electrons more strongly than does the hydrogen nucleus (a proton); that is, oxygen is more electronegative. The sharing of electrons between H and O is therefore unequal; the electrons are more often in the vicinity of the oxygen atom than of the hydrogen. The result of this unequal electron sharing is two electric dipoles in the water molecule, one along each of the $\text{H}-\text{O}$ bonds; each hydrogen bears a partial positive charge (δ^+) and the oxygen atom bears a partial negative charge equal to the sum of the two partial positives ($2\delta^-$). As a result, there is an electrostatic attraction between the oxygen atom of one water molecule and the hydrogen of another (Fig. 2-1c), called a **hydrogen bond**. Throughout this book, we represent hydrogen bonds with three parallel blue lines, as in Figure 2-1c.

Hydrogen bonds are relatively weak. Those in liquid water have a **bond dissociation energy** (the energy required to break a bond) of about 23 kJ/mol, compared with 470 kJ/mol for the covalent $\text{O}-\text{H}$ bond in

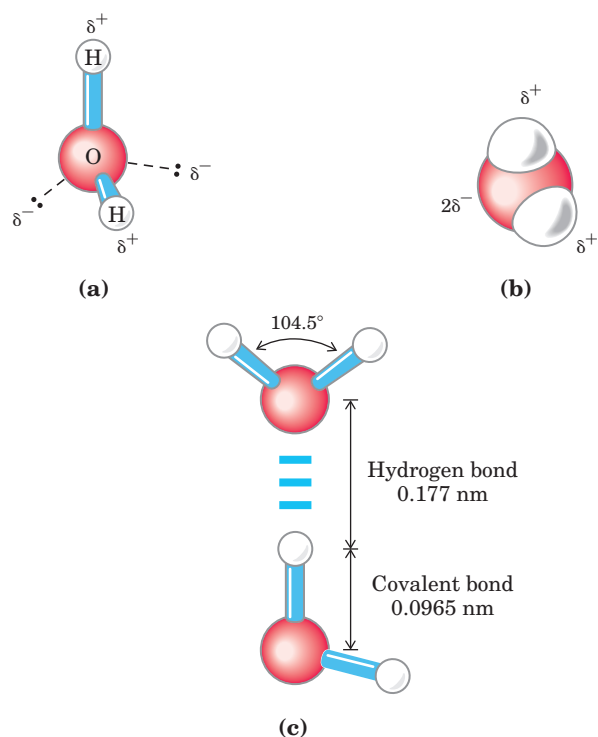
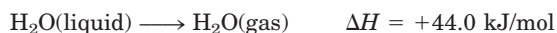


FIGURE 2-1 Structure of the water molecule. The dipolar nature of the H_2O molecule is shown by (a) ball-and-stick and (b) space-filling models. The dashed lines in (a) represent the nonbonding orbitals. There is a nearly tetrahedral arrangement of the outer-shell electron pairs around the oxygen atom; the two hydrogen atoms have localized partial positive charges (δ^+) and the oxygen atom has a partial negative charge ($2\delta^-$). (c) Two H_2O molecules joined by a hydrogen bond (designated here, and throughout this book, by three blue lines) between the oxygen atom of the upper molecule and a hydrogen atom of the lower one. Hydrogen bonds are longer and weaker than covalent $\text{O}-\text{H}$ bonds.

water or 348 kJ/mol for a covalent C—C bond. The hydrogen bond is about 10% covalent, due to overlaps in the bonding orbitals, and about 90% electrostatic. At room temperature, the thermal energy of an aqueous solution (the kinetic energy of motion of the individual atoms and molecules) is of the same order of magnitude as that required to break hydrogen bonds. When water is heated, the increase in temperature reflects the faster motion of individual water molecules. At any given time, most of the molecules in liquid water are engaged in hydrogen bonding, but the lifetime of each hydrogen bond is just 1 to 20 picoseconds ($1 \text{ ps} = 10^{-12} \text{ s}$); upon breakage of one hydrogen bond, another hydrogen bond forms, with the same partner or a new one, within 0.1 ps. The apt phrase “flickering clusters” has been applied to the short-lived groups of water molecules interlinked by hydrogen bonds in liquid water. The sum of all the hydrogen bonds between H_2O molecules confers great internal cohesion on liquid water. Extended networks of hydrogen-bonded water molecules also form bridges between solutes (proteins and nucleic acids, for example) that allow the larger molecules to interact with each other over distances of several nanometers without physically touching.

The nearly tetrahedral arrangement of the orbitals about the oxygen atom (Fig. 2-1a) allows each water molecule to form hydrogen bonds with as many as four neighboring water molecules. In liquid water at room temperature and atmospheric pressure, however, water molecules are disorganized and in continuous motion, so that each molecule forms hydrogen bonds with an average of only 3.4 other molecules. In ice, on the other hand, each water molecule is fixed in space and forms hydrogen bonds with a full complement of four other water molecules to yield a regular lattice structure (Fig. 2-2). Breaking a sufficient proportion of hydrogen bonds to destabilize the crystal lattice of ice requires much thermal energy, which accounts for the relatively high melting point of water (Table 2-1). When ice melts or water evaporates, heat is taken up by the system:



During melting or evaporation, the entropy of the aqueous system increases as more highly ordered arrays of water molecules relax into the less orderly hydrogen-bonded arrays in liquid water or the wholly disordered gaseous state. At room temperature, both the melting of ice and the evaporation of water occur spontaneously; the tendency of the water molecules to associate through hydrogen bonds is outweighed by the energetic push toward randomness. Recall that the free-energy change (ΔG) must have a negative value for a process to occur spontaneously: $\Delta G = \Delta H - T\Delta S$, where ΔG represents the driving force, ΔH the enthalpy change from making

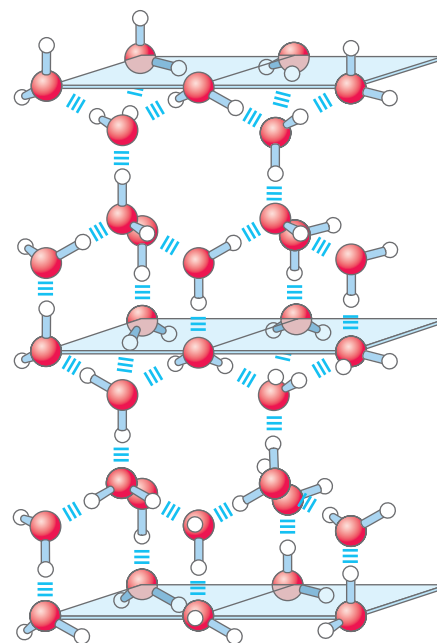


FIGURE 2-2 Hydrogen bonding in ice. In ice, each water molecule forms the maximum of four hydrogen bonds, creating a regular crystal lattice. By contrast, in liquid water at room temperature and atmospheric pressure, each water molecule hydrogen-bonds with an average of 3.4 other water molecules. This crystal lattice of ice makes it less dense than liquid water, and thus ice floats on liquid water.

and breaking bonds, and ΔS the change in randomness. Because ΔH is positive for melting and evaporation, it is clearly the increase in entropy (ΔS) that makes ΔG negative and drives these transformations.

Water Forms Hydrogen Bonds with Polar Solutes

Hydrogen bonds are not unique to water. They readily form between an electronegative atom (the hydrogen acceptor, usually oxygen or nitrogen with a lone pair of electrons) and a hydrogen atom covalently bonded to another electronegative atom (the hydrogen donor) in the same or another molecule (Fig. 2-3). Hydrogen atoms covalently bonded to carbon atoms do not participate in hydrogen bonding, because carbon is only

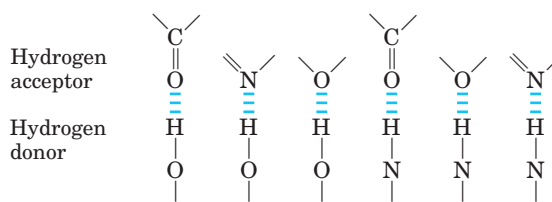


FIGURE 2-3 Common hydrogen bonds in biological systems. The hydrogen acceptor is usually oxygen or nitrogen; the hydrogen donor is another electronegative atom.

slightly more electronegative than hydrogen and thus the C—H bond is only very weakly polar. The distinction explains why butanol ($\text{CH}_3(\text{CH}_2)_2\text{CH}_2\text{OH}$) has a relatively high boiling point of 117°C , whereas butane ($\text{CH}_3(\text{CH}_2)_2\text{CH}_3$) has a boiling point of only -0.5°C . Butanol has a polar hydroxyl group and thus can form intermolecular hydrogen bonds. Uncharged but polar biomolecules such as sugars dissolve readily in water because of the stabilizing effect of hydrogen bonds between the hydroxyl groups or carbonyl oxygen of the sugar and the polar water molecules. Alcohols, aldehydes, ketones, and compounds containing N—H bonds all form hydrogen bonds with water molecules (Fig. 2-4) and tend to be soluble in water.

Hydrogen bonds are strongest when the bonded molecules are oriented to maximize electrostatic interaction, which occurs when the hydrogen atom and the two atoms that share it are in a straight line—that is, when the acceptor atom is in line with the covalent bond between the donor atom and H (Fig. 2-5). Hydrogen bonds are thus highly directional and capable of hold-

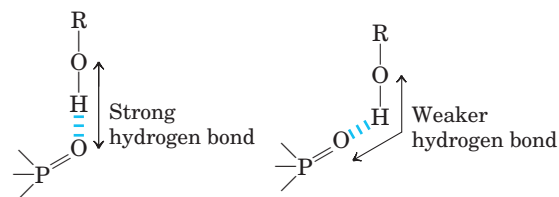


FIGURE 2-5 Directionality of the hydrogen bond. The attraction between the partial electric charges (see Fig. 2-1) is greatest when the three atoms involved (in this case O, H, and O) lie in a straight line. When the hydrogen-bonded moieties are structurally constrained (as when they are parts of a single protein molecule, for example), this ideal geometry may not be possible and the resulting hydrogen bond is weaker.

ing two hydrogen-bonded molecules or groups in a specific geometric arrangement. As we shall see later, this property of hydrogen bonds confers very precise three-dimensional structures on protein and nucleic acid molecules, which have many intramolecular hydrogen bonds.

Water Interacts Electrostatically with Charged Solutes

Water is a polar solvent. It readily dissolves most biomolecules, which are generally charged or polar compounds (Table 2-2); compounds that dissolve easily in water are **hydrophilic** (Greek, “water-loving”). In contrast, nonpolar solvents such as chloroform and benzene are poor solvents for polar biomolecules but easily dissolve those that are **hydrophobic**—nonpolar molecules such as lipids and waxes.

Water dissolves salts such as NaCl by hydrating and stabilizing the Na^+ and Cl^- ions, weakening the electrostatic interactions between them and thus counteracting their tendency to associate in a crystalline lattice (Fig. 2-6). The same factors apply to charged biomolecules, compounds with functional groups such as ionized carboxylic acids ($-\text{COO}^-$), protonated amines ($-\text{NH}_3^+$), and phosphate esters or anhydrides. Water readily dissolves such compounds by replacing solute-solute hydrogen bonds with solute-water hydrogen bonds, thus screening the electrostatic interactions between solute molecules.

Water is especially effective in screening the electrostatic interactions between dissolved ions because it has a high dielectric constant, a physical property reflecting the number of dipoles in a solvent. The strength, or force (F), of ionic interactions in a solution depends upon the magnitude of the charges (Q), the distance between the charged groups (r), and the dielectric constant (ϵ) of the solvent in which the interactions occur:

$$F = \frac{Q_1 Q_2}{\epsilon r^2}$$

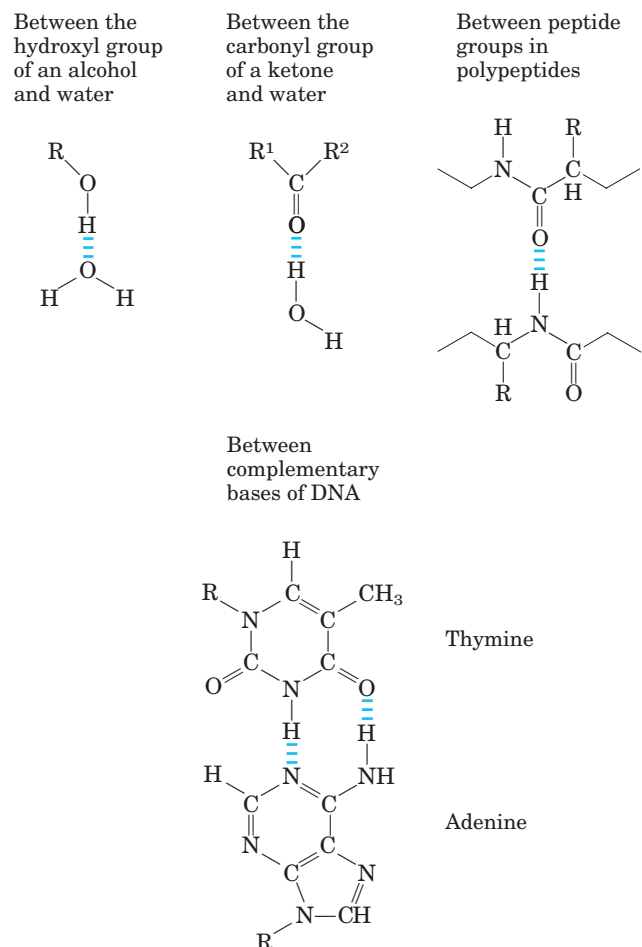
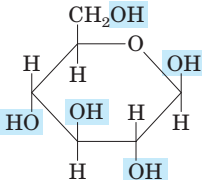
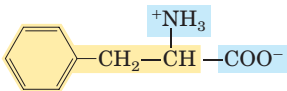
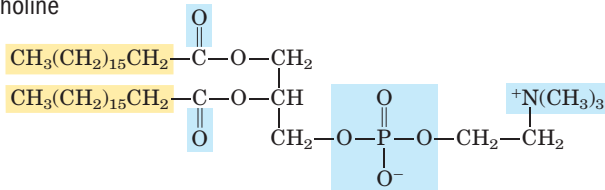


FIGURE 2-4 Some biologically important hydrogen bonds.

TABLE 2-2 Some Examples of Polar, Nonpolar, and Amphipathic Biomolecules (Shown as Ionic Forms at pH 7)

Polar	Nonpolar
Glucose	Typical wax
	$\text{CH}_3(\text{CH}_2)_7-\text{CH}=\text{CH}-(\text{CH}_2)_6-\text{CH}_2-\text{C}(=\text{O})\text{O}-$ $\text{CH}_3(\text{CH}_2)_7-\text{CH}=\text{CH}-(\text{CH}_2)_7-\text{CH}_2-$
Glycine	Amphipathic
$^+\text{NH}_3-\text{CH}_2-\text{COO}^-$	Phenylalanine
Aspartate	
$^-\text{OOC}-\text{CH}_2-\text{CH}(\text{NH}_3^+)-\text{COO}^-$	Phosphatidylcholine
Lactate	
$\text{CH}_3-\text{CH}(\text{OH})-\text{COO}^-$	
Glycerol	
$\text{HOCH}_2-\text{CH}(\text{OH})-\text{CH}_2\text{OH}$	

 Polar groups  Nonpolar groups

For water at 25 °C, ϵ (which is dimensionless) is 78.5, and for the very nonpolar solvent benzene, ϵ is 4.6. Thus, ionic interactions are much stronger in less polar environments. The dependence on r^2 is such that ionic attractions or repulsions operate only over short distances—in the range of 10 to 40 nm (depending on the electrolyte concentration) when the solvent is water.

Entropy Increases as Crystalline Substances Dissolve

As a salt such as NaCl dissolves, the Na^+ and Cl^- ions leaving the crystal lattice acquire far greater freedom of motion (Fig. 2-6). The resulting increase in entropy (randomness) of the system is largely responsible for the ease of dissolving salts such as NaCl in water. In

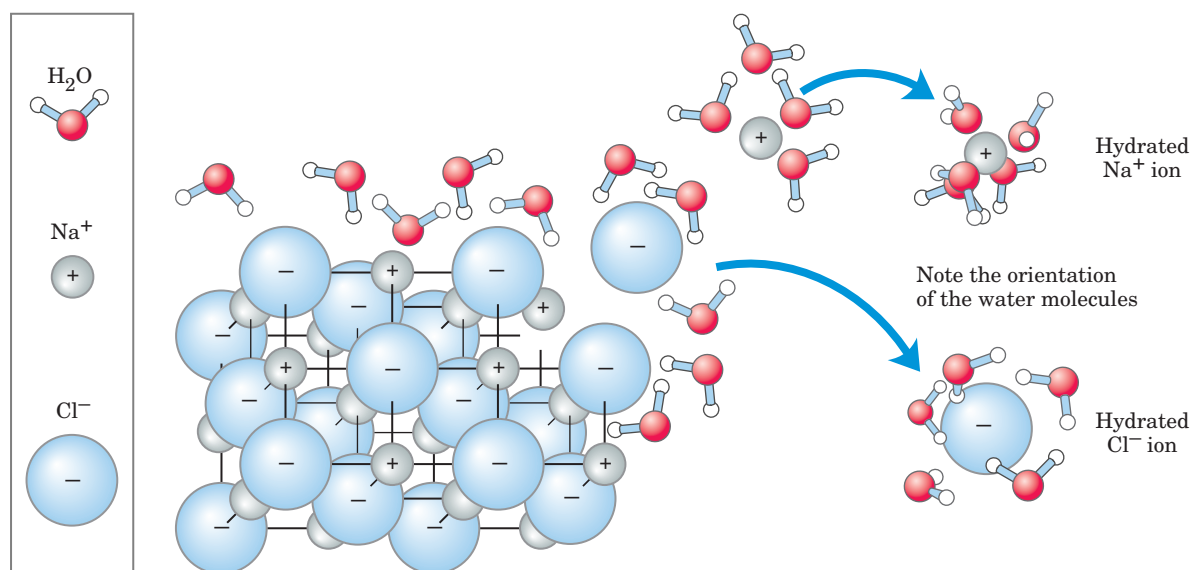


FIGURE 2-6 Water as solvent. Water dissolves many crystalline salts by hydrating their component ions. The NaCl crystal lattice is disrupted as water molecules cluster about the Cl^- and Na^+ ions. The ionic

charges are partially neutralized, and the electrostatic attractions necessary for lattice formation are weakened.

thermodynamic terms, formation of the solution occurs with a favorable free-energy change: $\Delta G = \Delta H - T \Delta S$, where ΔH has a small positive value and $T \Delta S$ a large positive value; thus ΔG is negative.

Nonpolar Gases Are Poorly Soluble in Water

The molecules of the biologically important gases CO_2 , O_2 , and N_2 are nonpolar. In O_2 and N_2 , electrons are shared equally by both atoms. In CO_2 , each $\text{C}=\text{O}$ bond is polar, but the two dipoles are oppositely directed and cancel each other (Table 2–3). The movement of molecules from the disordered gas phase into aqueous solution constrains their motion and the motion of water molecules and therefore represents a decrease in entropy. The nonpolar nature of these gases and the decrease in entropy when they enter solution combine to make them very poorly soluble in water (Table 2–3). Some organisms have water-soluble carrier proteins (hemoglobin and myoglobin, for example) that facilitate the transport of O_2 . Carbon dioxide forms carbonic acid (H_2CO_3) in aqueous solution and is transported as the HCO_3^- (bicarbonate) ion, either free—bicarbonate is very soluble in water (~ 100 g/L at 25°C)—or bound to hemoglobin. Two other gases, NH_3 and H_2S , also have biological roles in some organisms; these gases are polar and dissolve readily in water.

Nonpolar Compounds Force Energetically Unfavorable Changes in the Structure of Water

When water is mixed with benzene or hexane, two phases form; neither liquid is soluble in the other. Nonpolar compounds such as benzene and hexane are

hydrophobic—they are unable to undergo energetically favorable interactions with water molecules, and they interfere with the hydrogen bonding among water molecules. All molecules or ions in aqueous solution interfere with the hydrogen bonding of some water molecules in their immediate vicinity, but polar or charged solutes (such as NaCl) compensate for lost water-water hydrogen bonds by forming new solute-water interactions. The net change in enthalpy (ΔH) for dissolving these solutes is generally small. Hydrophobic solutes, however, offer no such compensation, and their addition to water may therefore result in a small gain of enthalpy; the breaking of hydrogen bonds between water molecules takes up energy from the system. Furthermore, dissolving hydrophobic compounds in water produces a measurable decrease in entropy. Water molecules in the immediate vicinity of a nonpolar solute are constrained in their possible orientations as they form a highly ordered cagelike shell around each solute molecule. These water molecules are not as highly oriented as those in **clathrates**, crystalline compounds of nonpolar solutes and water, but the effect is the same in both cases: the ordering of water molecules reduces entropy. The number of ordered water molecules, and therefore the magnitude of the entropy decrease, is proportional to the surface area of the hydrophobic solute enclosed within the cage of water molecules. The free-energy change for dissolving a nonpolar solute in water is thus unfavorable: $\Delta G = \Delta H - T \Delta S$, where ΔH has a positive value, ΔS has a negative value, and ΔG is positive.

Amphipathic compounds contain regions that are polar (or charged) and regions that are nonpolar (Table 2–2). When an amphipathic compound is mixed with

TABLE 2–3 Solubilities of Some Gases in Water

Gas	Structure*	Polarity	Solubility in water (g/L) [†]
Nitrogen	$\text{N}\equiv\text{N}$	Nonpolar	0.018 (40°C)
Oxygen	$\text{O}=\text{O}$	Nonpolar	0.035 (50°C)
Carbon dioxide	$\begin{array}{c} \delta^- \quad \delta^- \\ \leftarrow \quad \rightarrow \\ \text{O}=\text{C}=\text{O} \end{array}$	Nonpolar	0.97 (45°C)
Ammonia	$\begin{array}{c} \text{H} \quad \text{H} \quad \text{H} \\ \diagdown \quad \quad \diagup \\ \text{N} \\ \downarrow \delta^- \end{array}$	Polar	900 (10°C)
Hydrogen sulfide	$\begin{array}{c} \text{H} \quad \text{H} \\ \diagdown \quad \diagup \\ \text{S} \\ \downarrow \delta^- \end{array}$	Polar	1,860 (40°C)

*The arrows represent electric dipoles; there is a partial negative charge (δ^-) at the head of the arrow, a partial positive charge (δ^+ ; not shown here) at the tail.

[†]Note that polar molecules dissolve far better even at low temperatures than do nonpolar molecules at relatively high temperatures.

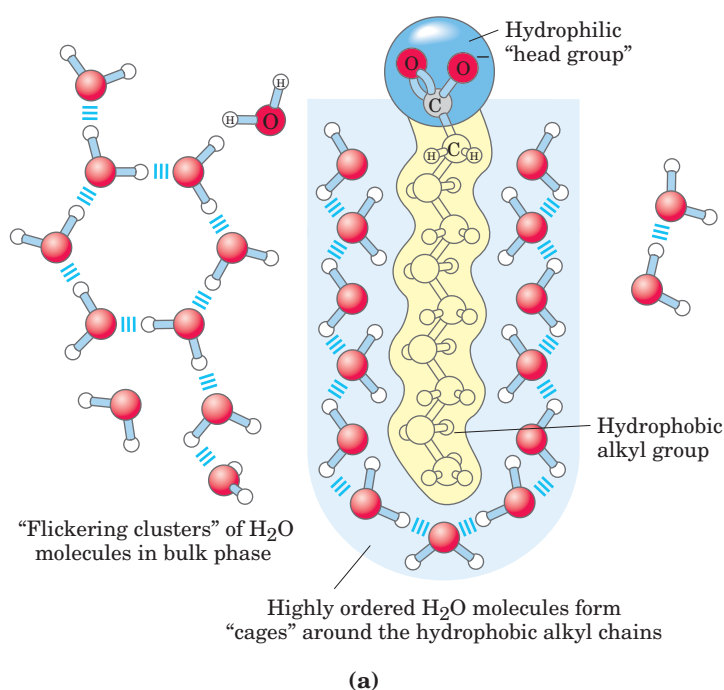
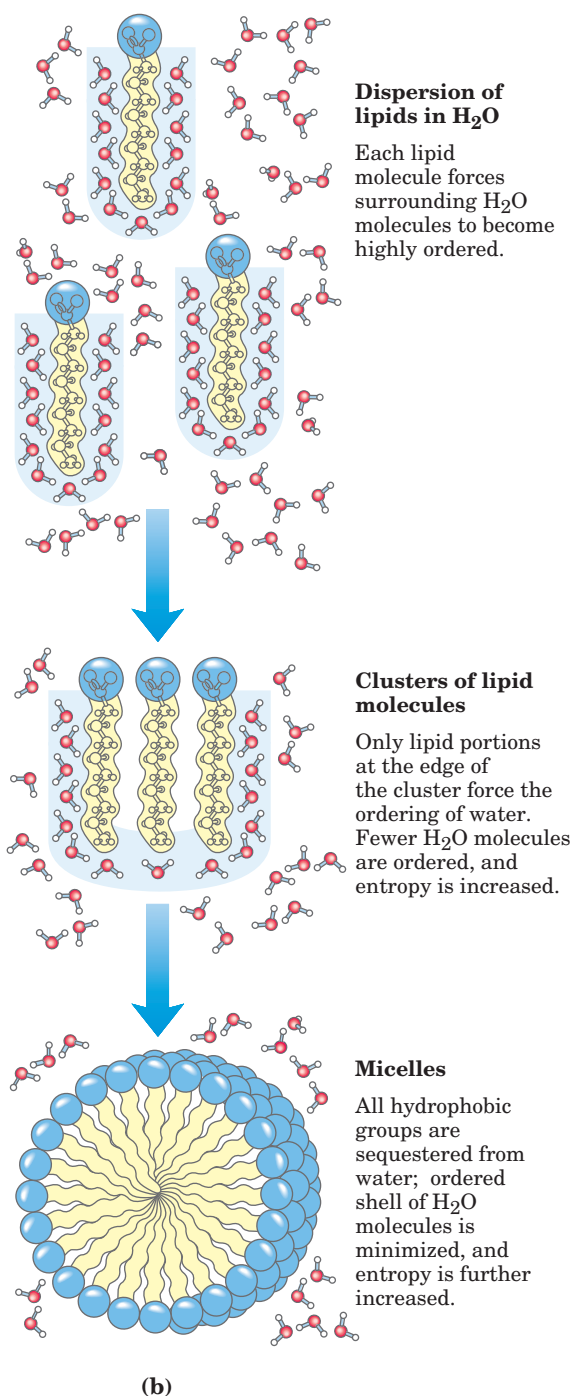


FIGURE 2-7 Amphipathic compounds in aqueous solution. (a) Long-chain fatty acids have very hydrophobic alkyl chains, each of which is surrounded by a layer of highly ordered water molecules. (b) By clustering together in micelles, the fatty acid molecules expose the smallest possible hydrophobic surface area to the water, and fewer water molecules are required in the shell of ordered water. The energy gained by freeing immobilized water molecules stabilizes the micelle.

water, the polar, hydrophilic region interacts favorably with the solvent and tends to dissolve, but the nonpolar, hydrophobic region tends to avoid contact with the water (Fig. 2-7a). The nonpolar regions of the molecules cluster together to present the smallest hydrophobic area to the aqueous solvent, and the polar regions are arranged to maximize their interaction with the solvent (Fig. 2-7b). These stable structures of amphipathic compounds in water, called **micelles**, may contain hundreds or thousands of molecules. The forces that hold the nonpolar regions of the molecules together are called **hydrophobic interactions**. The strength of hydrophobic interactions is not due to any intrinsic attraction between nonpolar moieties. Rather, it results from the system's achieving greatest thermodynamic stability by minimizing the number of ordered water molecules required to surround hydrophobic portions of the solute molecules.

Many biomolecules are amphipathic; proteins, pigments, certain vitamins, and the sterols and phospholipids of membranes all have polar and nonpolar surface regions. Structures composed of these molecules are stabilized by hydrophobic interactions among the non-



polar regions. Hydrophobic interactions among lipids, and between lipids and proteins, are the most important determinants of structure in biological membranes. Hydrophobic interactions between nonpolar amino acids also stabilize the three-dimensional structures of proteins.

Hydrogen bonding between water and polar solutes also causes some ordering of water molecules, but the effect is less significant than with nonpolar solutes. Part

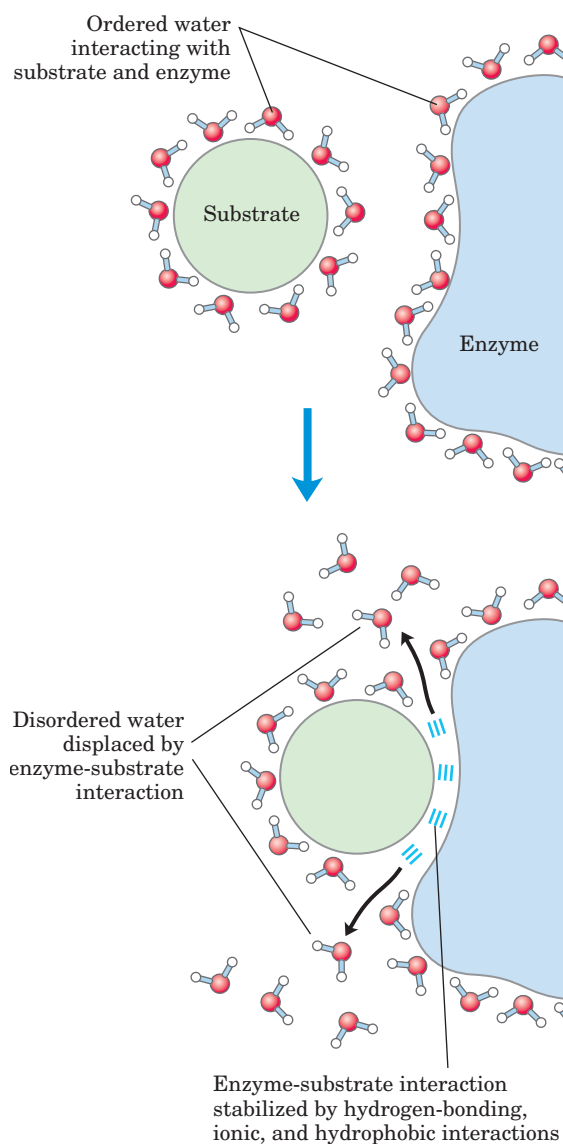


FIGURE 2-8 Release of ordered water favors formation of an enzyme-substrate complex. While separate, both enzyme and substrate force neighboring water molecules into an ordered shell. Binding of substrate to enzyme releases some of the ordered water, and the resulting increase in entropy provides a thermodynamic push toward formation of the enzyme-substrate complex.

of the driving force for binding of a polar substrate (reactant) to the complementary polar surface of an enzyme is the entropy increase as the enzyme displaces ordered water from the substrate (Fig. 2-8).

van der Waals Interactions Are Weak Interatomic Attractions

When two uncharged atoms are brought very close together, their surrounding electron clouds influence each

other. Random variations in the positions of the electrons around one nucleus may create a transient electric dipole, which induces a transient, opposite electric dipole in the nearby atom. The two dipoles weakly attract each other, bringing the two nuclei closer. These weak attractions are called **van der Waals interactions**. As the two nuclei draw closer together, their electron clouds begin to repel each other. At the point where the van der Waals attraction exactly balances this repulsive force, the nuclei are said to be in van der Waals contact. Each atom has a characteristic **van der Waals radius**, a measure of how close that atom will allow another to approach (Table 2-4). In the “space-filling” molecular models shown throughout this book, the atoms are depicted in sizes proportional to their van der Waals radii.

Weak Interactions Are Crucial to Macromolecular Structure and Function

The noncovalent interactions we have described (hydrogen bonds and ionic, hydrophobic, and van der Waals interactions) (Table 2-5) are much weaker than covalent bonds. An input of about 350 kJ of energy is required to break a mole of (6×10^{23}) C—C single bonds, and about 410 kJ to break a mole of C—H bonds, but as little as 4 kJ is sufficient to disrupt a mole of typical van der Waals interactions. Hydrophobic interactions are also much weaker than covalent bonds, although they are substantially strengthened by a highly polar solvent (a concentrated salt solution, for example). Ionic interactions and hydrogen bonds are variable in strength, depending on the polarity of the solvent and

TABLE 2-4 van der Waals Radii and Covalent (Single-Bond) Radii of Some Elements

Element	van der Waals radius (nm)	Covalent radius for single bond (nm)
H	0.11	0.030
O	0.15	0.066
N	0.15	0.070
C	0.17	0.077
S	0.18	0.104
P	0.19	0.110
I	0.21	0.133

Sources: For van der Waals radii, Chauvin, R. (1992) Explicit periodic trend of van der Waals radii. *J. Phys. Chem.* **96**, 9194-9197. For covalent radii, Pauling, L. (1960) *Nature of the Chemical Bond*, 3rd edn, Cornell University Press, Ithaca, NY.

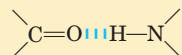
Note: van der Waals radii describe the space-filling dimensions of atoms. When two atoms are joined covalently, the atomic radii at the point of bonding are less than the van der Waals radii, because the joined atoms are pulled together by the shared electron pair. The distance between nuclei in a van der Waals interaction or a covalent bond is about equal to the sum of the van der Waals or covalent radii, respectively, for the two atoms. Thus the length of a carbon-carbon single bond is about $0.077 \text{ nm} + 0.077 \text{ nm} = 0.154 \text{ nm}$.

TABLE 2-5 Four Types of Noncovalent (“Weak”) Interactions among Biomolecules in Aqueous Solvent**Hydrogen bonds**

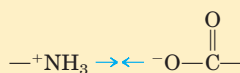
Between neutral groups



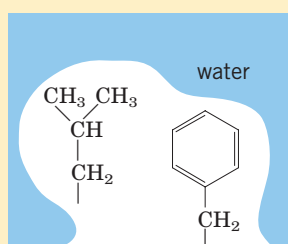
Between peptide bonds

**Ionic interactions**

Attraction



Repulsion

**Hydrophobic interactions****van der Waals interactions**

Any two atoms in close proximity

the alignment of the hydrogen-bonded atoms, but they are always significantly weaker than covalent bonds. In aqueous solvent at 25 °C, the available thermal energy can be of the same order of magnitude as the strength of these weak interactions, and the interaction between solute and solvent (water) molecules is nearly as favorable as solute-solute interactions. Consequently, hydrogen bonds and ionic, hydrophobic, and van der Waals interactions are continually formed and broken.

Although these four types of interactions are individually weak relative to covalent bonds, the cumulative effect of many such interactions can be very significant. For example, the noncovalent binding of an enzyme to its substrate may involve several hydrogen bonds and one or more ionic interactions, as well as hydrophobic and van der Waals interactions. The formation of each of these weak bonds contributes to a net decrease in the free energy of the system. We can calculate the stability of a noncovalent interaction, such as that of a small molecule hydrogen-bonded to its macromolecular partner, from the binding energy. Stability, as measured by the equilibrium constant (see below) of the binding reaction, varies *exponentially* with binding energy. The dissociation of two biomolecules (such as an enzyme and its bound substrate) associated noncovalently

through multiple weak interactions requires all these interactions to be disrupted at the same time. Because the interactions fluctuate randomly, such simultaneous disruptions are very unlikely. The molecular stability bestowed by 5 or 20 weak interactions is therefore much greater than would be expected intuitively from a simple summation of small binding energies.

Macromolecules such as proteins, DNA, and RNA contain so many sites of potential hydrogen bonding or ionic, van der Waals, or hydrophobic interactions that the cumulative effect of the many small binding forces can be enormous. For macromolecules, the most stable (that is, the native) structure is usually that in which weak-bonding possibilities are maximized. The folding of a single polypeptide or polynucleotide chain into its three-dimensional shape is determined by this principle. The binding of an antigen to a specific antibody depends on the cumulative effects of many weak interactions. As noted earlier, the energy released when an enzyme binds noncovalently to its substrate is the main source of the enzyme's catalytic power. The binding of a hormone or a neurotransmitter to its cellular receptor protein is the result of weak interactions. One consequence of the large size of enzymes and receptors is that their extensive surfaces provide many opportunities for weak interactions. At the molecular level, the complementarity between interacting biomolecules reflects the complementarity and weak interactions between polar, charged, and hydrophobic groups on the surfaces of the molecules.

When the structure of a protein such as hemoglobin (Fig. 2-9) is determined by x-ray crystallography (see

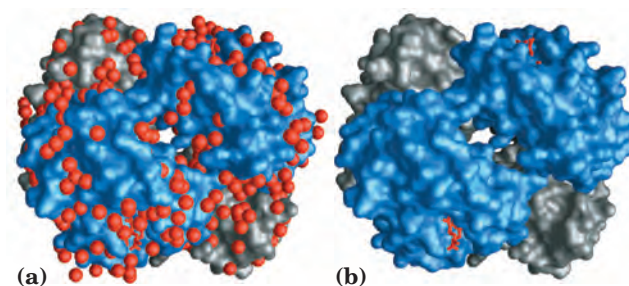


FIGURE 2-9 Water binding in hemoglobin. The crystal structure of hemoglobin, shown (a) with bound water molecules (red spheres) and (b) without the water molecules. These water molecules are so firmly bound to the protein that they affect the x-ray diffraction pattern as though they were fixed parts of the crystal. The gray structures with red and orange atoms are the four hemes of hemoglobin, discussed in detail in Chapter 5.

Box 4–4, p. XX), water molecules are often found to be bound so tightly as to be part of the crystal structure; the same is true for water in crystals of RNA or DNA. These bound water molecules, which can also be detected in aqueous solutions by nuclear magnetic resonance, have distinctly different properties from those of the “bulk” water of the solvent. They are, for example, not osmotically active (see below). For many proteins, tightly bound water molecules are essential to their function. In a reaction central to the process of photosynthesis, for example, light drives protons across a biological membrane as electrons flow through a series of electron-carrying proteins (see Fig. 19–XX). One of these proteins, cytochrome *f*, has a chain of five bound water molecules (Fig. 2–10) that may provide a path for protons to move through the membrane by a process known as “proton hopping” (described below). Another such light-driven proton pump, bacteriorhodopsin, almost certainly uses a chain of precisely oriented bound water molecules in the transmembrane movement of protons (see Fig. 19–XX).

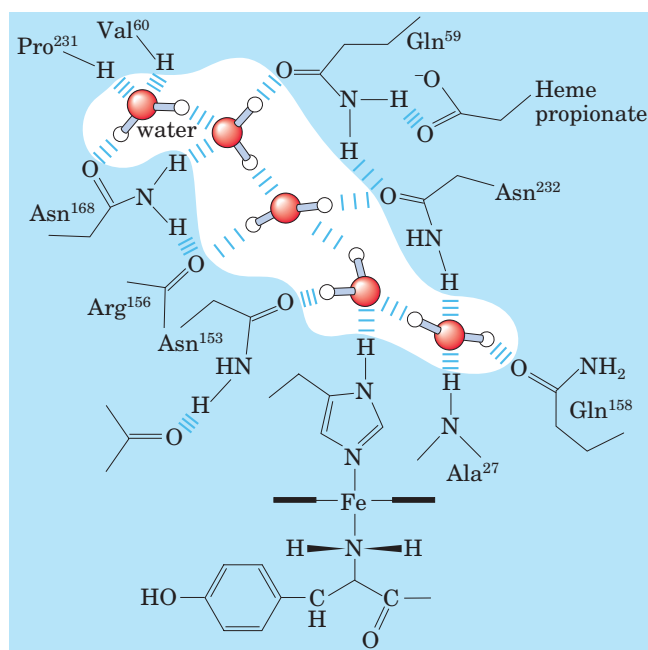


FIGURE 2–10 Water chain in cytochrome *f*. Water is bound in a proton channel of the membrane protein cytochrome *f*, which is part of the energy-trapping machinery of photosynthesis in chloroplasts (see Fig. 19–XX). Five water molecules are hydrogen-bonded to each other and to functional groups of the protein, which include the side chains of valine, proline, arginine, alanine, two asparagine, and two glutamine residues. The protein has a bound heme (see Fig. 5–1), its iron ion facilitating electron flow during photosynthesis. Electron flow is coupled to the movement of protons across the membrane, which probably involves “electron hopping” (see Fig. 2–14) through this chain of bound water molecules.

Solutes Affect the Colligative Properties of Aqueous Solutions

Solutes of all kinds alter certain physical properties of the solvent, water: its vapor pressure, boiling point, melting point (freezing point), and osmotic pressure. These are called **colligative** (“tied together”) **properties**, because the effect of solutes on all four properties has the same basis: the concentration of water is lower in solutions than in pure water. The effect of solute concentration on the colligative properties of water is independent of the chemical properties of the solute; it depends only on the *number* of solute particles (molecules, ions) in a given amount of water. A compound such as NaCl, which dissociates in solution, has twice the effect on osmotic pressure, for example, as does an equal number of moles of a nondissociating solute such as glucose.

Solutes alter the colligative properties of aqueous solutions by lowering the effective concentration of water. For example, when a significant fraction of the molecules at the surface of an aqueous solution are not water but solute, the tendency of water molecules to escape into the vapor phase—that is, the vapor pressure—is lowered (Fig. 2–11). Similarly, the tendency of water molecules to move from the aqueous phase to the surface of a forming ice crystal is reduced when some of the molecules that collide with the crystal are solute, not water. In that case, the solution will freeze more slowly than pure water and at a lower temperature. For a 1.00 molal aqueous solution (1.00 mol of solute per 1,000 g of water) of an ideal, nonvolatile, and nondissociating solute at 101 kPa (1 atm) of pressure, the freezing point is 1.86 °C lower and the boiling point is 0.543 °C higher than for pure water. For a 0.100 molal solution of the same solute, the changes are one-tenth as large.

Water molecules tend to move from a region of higher water concentration to one of lower water concentration. When two different aqueous solutions are separated by a semipermeable membrane (one that allows the passage of water but not solute molecules), water molecules diffusing from the region of higher water concentration to that of lower water concentration produce osmotic pressure (Fig. 2–12). This pressure, Π , measured as the force necessary to resist water movement (Fig. 2–12c), is approximated by the van’t Hoff equation:

$$\Pi = icRT$$

in which R is the gas constant and T is the absolute temperature. The term ic is the **osmolarity** of the solution, the product of the solute’s molar concentration c and the van’t Hoff factor i , which is a measure of the extent to which the solute dissociates into two or more ionic species. In dilute NaCl solutions, the solute completely

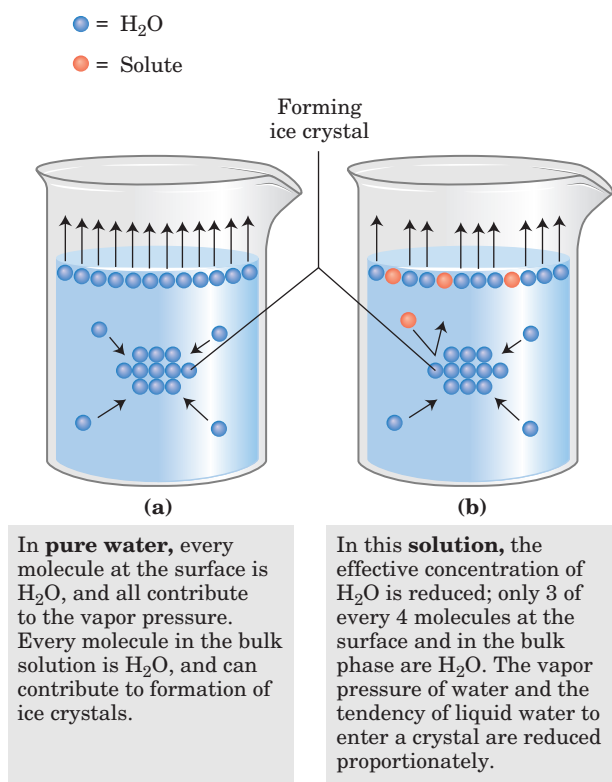


FIGURE 2-11 Solutes alter the colligative properties of aqueous solutions. (a) At 101 kPa (1 atm) pressure, pure water boils at 100 °C and freezes at 0 °C. (b) The presence of solute molecules reduces the probability of a water molecule leaving the solution and entering the gas phase, thereby reducing the vapor pressure of the solution and increasing the boiling point. Similarly, the probability of a water molecule colliding with and joining a forming ice crystal is reduced when some of the molecules colliding with the crystal are solute, not water, molecules. The effect is depression of the freezing point.

dissociates into Na⁺ and Cl⁻, doubling the number of solute particles, and thus $i = 2$. For nonionizing solutes, i is always 1. For solutions of several (n) solutes, Π is the sum of the contributions of each species:

$$\Pi = RT(i_1c_1 + i_2c_2 + \cdots + i_nc_n)$$

Osmosis, water movement across a semipermeable membrane driven by differences in osmotic pressure, is an important factor in the life of most cells. Plasma membranes are more permeable to water than to most other small molecules, ions, and macromolecules. This permeability is due partly to simple diffusion of water through the lipid bilayer and partly to protein channels (aquaporins; see Fig. 11-XX) in the membrane that selectively permit the passage of water. Solutions of equal osmolarity are said to be **isotonic**. Surrounded by an isotonic solution, a cell neither gains nor loses water (Fig. 2-13). In a **hypertonic** solution, one with higher

osmolarity than the cytosol, the cell shrinks as water flows out. In a **hypotonic** solution, with lower osmolarity than the cytosol, the cell swells as water enters. In their natural environments, cells generally contain higher concentrations of biomolecules and ions than their surroundings, so osmotic pressure tends to drive water into cells. If not somehow counterbalanced, this inward movement of water would distend the plasma membrane and eventually cause bursting of the cell (osmotic lysis).

Several mechanisms have evolved to prevent this catastrophe. In bacteria and plants, the plasma membrane is surrounded by a nonexpandable cell wall of sufficient rigidity and strength to resist osmotic pressure and prevent osmotic lysis. Certain freshwater protists that live in a highly hypotonic medium have an organelle (contractile vacuole) that pumps water out of the cell. In multicellular animals, blood plasma and interstitial fluid (the extracellular fluid of tissues) are maintained at an osmolarity close to that of the cytosol. The high concentration of albumin and other proteins in blood plasma contributes to its osmolarity. Cells also actively pump out ions such as Na⁺ into the interstitial fluid to stay in osmotic balance with their surroundings.

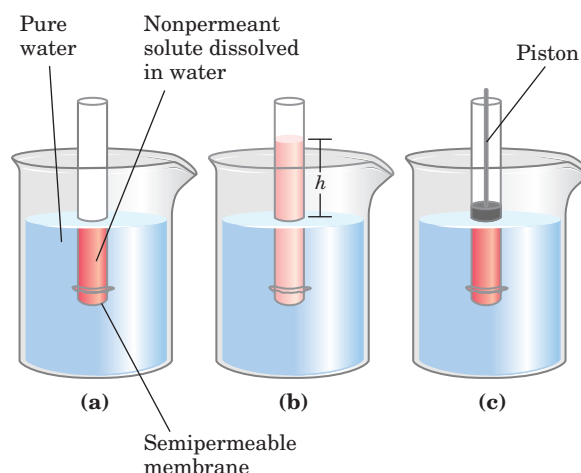


FIGURE 2-12 Osmosis and the measurement of osmotic pressure.

(a) The initial state. The tube contains an aqueous solution, the beaker contains pure water, and the semipermeable membrane allows the passage of water but not solute. Water flows from the beaker into the tube to equalize its concentration across the membrane. (b) The final state. Water has moved into the solution of the nonpermeant compound, diluting it and raising the column of water within the tube. At equilibrium, the force of gravity operating on the solution in the tube exactly balances the tendency of water to move into the tube, where its concentration is lower. (c) Osmotic pressure (Π) is measured as the force that must be applied to return the solution in the tube to the level of that in the beaker. This force is proportional to the height, h , of the column in (b).

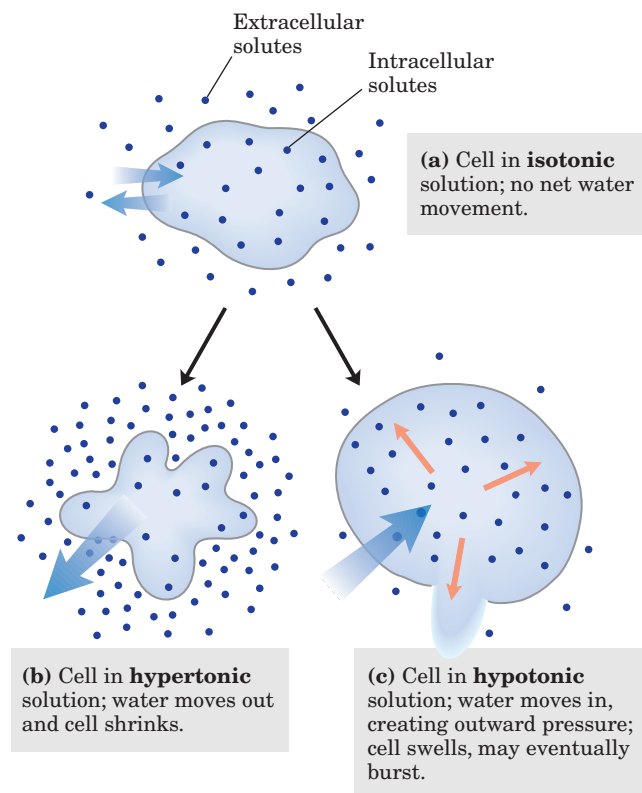


FIGURE 2-13 Effect of extracellular osmolarity on water movement across a plasma membrane. When a cell in osmotic balance with its surrounding medium (that is, in an isotonic medium) (a) is transferred into a hypertonic solution (b) or hypotonic solution (c), water moves across the plasma membrane in the direction that tends to equalize osmolarity outside and inside the cell.

Because the effect of solutes on osmolarity depends on the *number* of dissolved particles, not their *mass*, macromolecules (proteins, nucleic acids, polysaccharides) have far less effect on the osmolarity of a solution than would an equal mass of their monomeric components. For example, a *gram* of a polysaccharide composed of 1,000 glucose units has the same effect on osmolarity as a *milligram* of glucose. One effect of storing fuel as polysaccharides (starch or glycogen) rather than as glucose or other simple sugars is prevention of an enormous increase in osmotic pressure within the storage cell.

Plants use osmotic pressure to achieve mechanical rigidity. The very high solute concentration in the plant cell vacuole draws water into the cell (Fig. 2-13). The resulting osmotic pressure against the cell wall (turgor pressure) stiffens the cell, the tissue, and the plant body. When the lettuce in your salad wilts, it is because loss of water has reduced turgor pressure. Sudden alterations in turgor pressure produce the movement of plant

parts seen in touch-sensitive plants such as the Venus flytrap and mimosa (Box 2-1).

Osmosis also has consequences for laboratory protocols. Mitochondria, chloroplasts, and lysosomes, for example, are bounded by semipermeable membranes. In isolating these organelles from broken cells, biochemists must perform the fractionations in isotonic solutions (see Fig. 1-8). Buffers used in cellular fractionations commonly contain sufficient concentrations (about 0.2 M) of sucrose or some other inert solute to protect the organelles from osmotic lysis.

SUMMARY 2.1 Weak Interactions in Aqueous Systems

- The very different electronegativities of H and O make water a highly polar molecule, capable of forming hydrogen bonds with itself and with solutes. Hydrogen bonds are fleeting, primarily electrostatic, and weaker than covalent bonds. Water is a good solvent for polar (hydrophilic) solutes, with which it forms hydrogen bonds, and for charged solutes, with which it interacts electrostatically.
- Nonpolar (hydrophobic) compounds dissolve poorly in water; they cannot hydrogen-bond with the solvent, and their presence forces an energetically unfavorable ordering of water molecules at their hydrophobic surfaces. To minimize the surface exposed to water, nonpolar compounds such as lipids form aggregates (micelles) in which the hydrophobic moieties are sequestered in the interior, associating through hydrophobic interactions, and only the more polar moieties interact with water.
- Numerous weak, noncovalent interactions decisively influence the folding of macromolecules such as proteins and nucleic acids. The most stable macromolecular conformations are those in which hydrogen bonding is maximized within the molecule and between the molecule and the solvent, and in which hydrophobic moieties cluster in the interior of the molecule away from the aqueous solvent.
- The physical properties of aqueous solutions are strongly influenced by the concentrations of solutes. When two aqueous compartments are separated by a semipermeable membrane (such as the plasma membrane separating a cell from its surroundings), water moves across that membrane to equalize the osmolarity in the two compartments. This tendency for water to move across a semipermeable membrane is the osmotic pressure.

BOX 2-1 THE WORLD OF BIOCHEMISTRY

Touch Response in Plants: An Osmotic Event

The highly specialized leaves of the Venus flytrap (*Dionaea muscipula*) rapidly fold together in response to a light touch by an unsuspecting insect, entrapping the insect for later digestion. Attracted by nectar on the leaf surface, the insect touches three mechanically sensitive hairs, triggering the traplike closing of the leaf (Fig. 1). This leaf movement is produced by sudden (within 0.5 s) changes of turgor pressure in mesophyll cells (the inner cells of the leaf), probably achieved by the release of K^+ ions from the

cells and the resulting efflux, by osmosis, of water. Digestive glands in the leaf's surface release enzymes that extract nutrients from the insect.

The sensitive plant (*Mimosa pudica*) also undergoes a remarkable change in leaf shape triggered by mechanical touch (Fig. 2). A light touch or vibration produces a sudden drooping of the leaves, the result of a dramatic reduction in turgor pressure in cells at the base of each leaflet and leaf. As in the Venus flytrap, the drop in turgor pressure results from K^+ release followed by the efflux of water.



(a)



(b)

FIGURE 1 Touch response in the Venus flytrap. A fly approaching an open leaf (a) is trapped for digestion by the plant (b).



(a)



(b)

FIGURE 2 The feathery leaflets of the sensitive plant (a) close and drop (b) to protect the plant from structural damage by wind.

2.2 Ionization of Water, Weak Acids, and Weak Bases

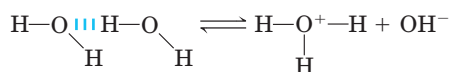
Although many of the solvent properties of water can be explained in terms of the uncharged H_2O molecule, the small degree of ionization of water to hydrogen ions (H^+) and hydroxide ions (OH^-) must also be taken into account. Like all reversible reactions, the ionization of water can be described by an equilibrium constant. When weak acids are dissolved in water, they contribute H^+ by ionizing; weak bases consume H^+ by becoming protonated. These processes are also governed by equilibrium constants. The total hydrogen ion concentration from all sources is experimentally measurable and is expressed as the pH of the solution. To predict the state of ionization of solutes in water, we must take into account the relevant equilibrium constants for each ionization reaction. We therefore turn now to a brief discussion of the ionization of water and of weak acids and bases dissolved in water.

Pure Water Is Slightly Ionized

Water molecules have a slight tendency to undergo reversible ionization to yield a hydrogen ion (a proton) and a hydroxide ion, giving the equilibrium



Although we commonly show the dissociation product of water as H^+ , free protons do not exist in solution; hydrogen ions formed in water are immediately hydrated to **hydronium ions** (H_3O^+). Hydrogen bonding between water molecules makes the hydration of dissociating protons virtually instantaneous:



The ionization of water can be measured by its electrical conductivity; pure water carries electrical current as H^+ migrates toward the cathode and OH^- toward the anode. The movement of hydronium and hydroxide ions in the electric field is anomalously fast compared with that of other ions such as Na^+ , K^+ , and Cl^- . This high ionic mobility results from the kind of “proton hopping” shown in Figure 2-14. No individual proton moves very far through the bulk solution, but a series of proton hops between hydrogen-bonded water molecules causes the net movement of a proton over a long distance in a remarkably short time. As a result of the high ionic mobility of H^+ (and of OH^- , which also moves rapidly by proton hopping, but in the opposite direction), acid-base reactions in aqueous solutions are generally exceptionally fast. As noted above, proton hopping very likely also plays a role in biological proton-transfer reactions (Fig. 2-10; see also Fig. 19-XX).

Because reversible ionization is crucial to the role of water in cellular function, we must have a means of

Hydronium ion gives up a proton

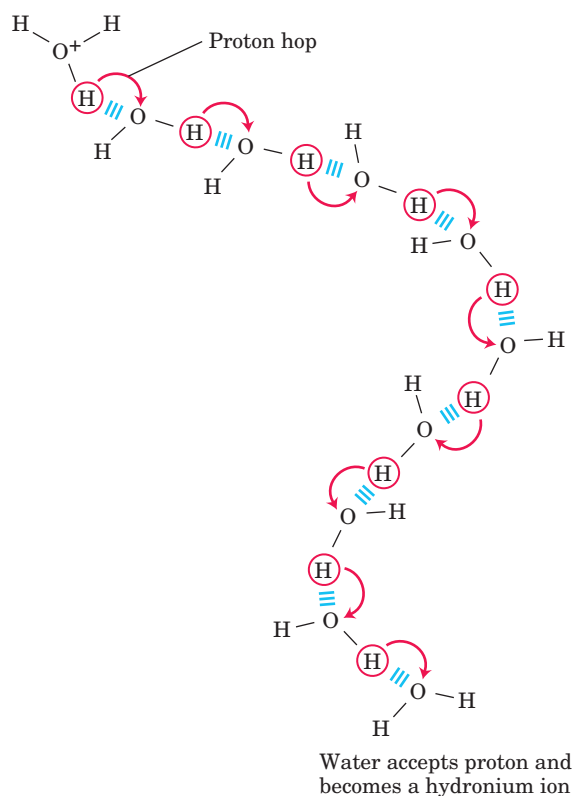


FIGURE 2-14 Proton hopping. Short “hops” of protons between a series of hydrogen-bonded water molecules effect an extremely rapid net movement of a proton over a long distance. As a hydronium ion (upper left) gives up a proton, a water molecule some distance away (lower right) acquires one, becoming a hydronium ion. Proton hopping is much faster than true diffusion and explains the remarkably high ionic mobility of H^+ ions compared with other monovalent cations such as Na^+ or K^+ .

expressing the extent of ionization of water in quantitative terms. A brief review of some properties of reversible chemical reactions shows how this can be done.

The position of equilibrium of any chemical reaction is given by its **equilibrium constant**, K_{eq} (sometimes expressed simply as K). For the generalized reaction



an equilibrium constant can be defined in terms of the concentrations of reactants (A and B) and products (C and D) at equilibrium:

$$K_{\text{eq}} = \frac{[\text{C}][\text{D}]}{[\text{A}][\text{B}]}$$

Strictly speaking, the concentration terms should be the *activities*, or effective concentrations in nonideal solutions, of each species. Except in very accurate work, however, the equilibrium constant may be approxi-

mated by measuring the *concentrations* at equilibrium. For reasons beyond the scope of this discussion, equilibrium constants are dimensionless. Nonetheless, we have generally retained the concentration units (M) in the equilibrium expressions used in this book to remind you that molarity is the unit of concentration used in calculating K_{eq} .

The equilibrium constant is fixed and characteristic for any given chemical reaction at a specified temperature. It defines the composition of the final equilibrium mixture, regardless of the starting amounts of reactants and products. Conversely, we can calculate the equilibrium constant for a given reaction at a given temperature if the equilibrium concentrations of all its reactants and products are known. As we will show in Chapter 13, the standard free-energy change (ΔG°) is directly related to K_{eq} .

The Ionization of Water Is Expressed by an Equilibrium Constant

The degree of ionization of water at equilibrium (Eqn 2-1) is small; at 25 °C only about two of every 10^9 molecules in pure water are ionized at any instant. The equilibrium constant for the reversible ionization of water (Eqn 2-1) is

$$K_{\text{eq}} = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]} \quad (2-3)$$

In pure water at 25 °C, the concentration of water is 55.5 M (grams of H_2O in 1 L divided by its gram molecular weight: (1,000 g/L)/(18.015 g/mol)) and is essentially constant in relation to the very low concentrations of H^+ and OH^- , namely, 1×10^{-7} M. Accordingly, we can substitute 55.5 M in the equilibrium constant expression (Eqn 2-3) to yield

$$K_{\text{eq}} = \frac{[\text{H}^+][\text{OH}^-]}{55.5 \text{ M}},$$

which, on rearranging, becomes

$$(55.5 \text{ M})(K_{\text{eq}}) = [\text{H}^+][\text{OH}^-] = K_{\text{w}} \quad (2-4)$$

where K_{w} designates the product $(55.5 \text{ M})(K_{\text{eq}})$, the **ion product of water** at 25 °C.

The value for K_{eq} , determined by electrical-conductivity measurements of pure water, is 1.8×10^{-16} M at 25 °C. Substituting this value for K_{eq} in Equation 2-4 gives the value of the ion product of water:

$$\begin{aligned} K_{\text{w}} &= [\text{H}^+][\text{OH}^-] = (55.5 \text{ M})(1.8 \times 10^{-16} \text{ M}) \\ &= 1.0 \times 10^{-14} \text{ M}^2 \end{aligned}$$

Thus the product $[\text{H}^+][\text{OH}^-]$ in aqueous solutions at 25 °C always equals $1 \times 10^{-14} \text{ M}^2$. When there are exactly equal concentrations of H^+ and OH^- , as in pure water, the solution is said to be at **neutral pH**. At this pH, the concentration of H^+ and OH^- can be calculated from the ion product of water as follows:

$$K_{\text{w}} = [\text{H}^+][\text{OH}^-] = [\text{H}^+]^2$$

Solving for $[\text{H}^+]$ gives

$$[\text{H}^+] = \sqrt{K_{\text{w}}} = \sqrt{1 \times 10^{-14} \text{ M}^2}$$

$$[\text{H}^+] = [\text{OH}^-] = 10^{-7} \text{ M}$$

As the ion product of water is constant, whenever $[\text{H}^+]$ is greater than $1 \times 10^{-7} \text{ M}$, $[\text{OH}^-]$ must become less than $1 \times 10^{-7} \text{ M}$, and vice versa. When $[\text{H}^+]$ is very high, as in a solution of hydrochloric acid, $[\text{OH}^-]$ must be very low. From the ion product of water we can calculate $[\text{H}^+]$ if we know $[\text{OH}^-]$, and vice versa (Box 2-2).

The pH Scale Designates the H^+ and OH^- Concentrations

The ion product of water, K_{w} , is the basis for the **pH scale** (Table 2-6). It is a convenient means of designating the concentration of H^+ (and thus of OH^-) in any aqueous solution in the range between 1.0 M H^+ and 1.0 M OH^- . The term **pH** is defined by the expression

$$\text{pH} = \log \frac{1}{[\text{H}^+]} = -\log [\text{H}^+]$$

The symbol p denotes “negative logarithm of.” For a precisely neutral solution at 25 °C, in which the concentration of hydrogen ions is $1.0 \times 10^{-7} \text{ M}$, the pH can be calculated as follows:

$$\begin{aligned} \text{pH} &= \log \frac{1}{1.0 \times 10^{-7}} = \log (1.0 \times 10^7) \\ &= \log 1.0 + \log 10^7 = 0 + 7 = 7 \end{aligned}$$

TABLE 2-6 The pH Scale

$[\text{H}^+] \text{ (M)}$	pH	$[\text{OH}^-] \text{ (M)}$	pOH*
10^0 (1)	0	10^{-14}	14
10^{-1}	1	10^{-13}	13
10^{-2}	2	10^{-12}	12
10^{-3}	3	10^{-11}	11
10^{-4}	4	10^{-10}	10
10^{-5}	5	10^{-9}	9
10^{-6}	6	10^{-8}	8
10^{-7}	7	10^{-7}	7
10^{-8}	8	10^{-6}	6
10^{-9}	9	10^{-5}	5
10^{-10}	10	10^{-4}	4
10^{-11}	11	10^{-3}	3
10^{-12}	12	10^{-2}	2
10^{-13}	13	10^{-1}	1
10^{-14}	14	10^0 (1)	0

*The expression pOH is sometimes used to describe the basicity, or OH^- concentration, of a solution; pOH is defined by the expression $\text{pOH} = -\log [\text{OH}^-]$, which is analogous to the expression for pH. Note that in all cases, $\text{pH} + \text{pOH} = 14$.

BOX 2-2 WORKING IN BIOCHEMISTRY

The Ion Product of Water: Two Illustrative Problems

The ion product of water makes it possible to calculate the concentration of H^+ , given the concentration of OH^- , and vice versa; the following problems demonstrate this.

1. What is the concentration of H^+ in a solution of 0.1 M NaOH?

$$K_w = [H^+][OH^-]$$

Solving for $[H^+]$ gives

$$\begin{aligned}[H^+] &= \frac{K_w}{[OH^-]} = \frac{1 \times 10^{-14} \text{ M}^2}{0.1 \text{ M}} = \frac{10^{-14} \text{ M}^2}{10^{-1} \text{ M}} \\ &= 10^{-13} \text{ M} \quad (\text{answer})\end{aligned}$$

2. What is the concentration of OH^- in a solution with an H^+ concentration of $1.3 \times 10^{-4} \text{ M}$?

$$K_w = [H^+][OH^-]$$

Solving for $[OH^-]$ gives

$$\begin{aligned}[OH^-] &= \frac{K_w}{[H^+]} = \frac{1.0 \times 10^{-14} \text{ M}^2}{1.3 \times 10^{-4} \text{ M}} \\ &= 7.7 \times 10^{-11} \text{ M} \quad (\text{answer})\end{aligned}$$

When doing these or any other calculations, be sure to round your answers to the correct number of significant figures.

The value of 7 for the pH of a precisely neutral solution is not an arbitrarily chosen figure; it is derived from the absolute value of the ion product of water at 25 °C, which by convenient coincidence is a round number. Solutions having a pH greater than 7 are alkaline or basic; the concentration of OH^- is greater than that of H^+ . Conversely, solutions having a pH less than 7 are acidic.

Note that the pH scale is logarithmic, not arithmetic. To say that two solutions differ in pH by 1 pH unit means that one solution has ten times the H^+ concentration of the other, but it does not tell us the absolute magnitude of the difference. Figure 2-15 gives the pH of some common aqueous fluids. A cola drink (pH 3.0) or red wine (pH 3.7) has an H^+ concentration approximately 10,000 times that of blood (pH 7.4).

The pH of an aqueous solution can be approximately measured using various indicator dyes, including litmus, phenolphthalein, and phenol red, which undergo color changes as a proton dissociates from the dye molecule. Accurate determinations of pH in the chemical or clinical laboratory are made with a glass electrode that is selectively sensitive to H^+ concentration but insensitive to Na^+ , K^+ , and other cations. In a pH meter the signal from such an electrode is amplified and compared with the signal generated by a solution of accurately known pH.

Measurement of pH is one of the most important and frequently used procedures in biochemistry. The pH affects the structure and activity of biological macromolecules; for example, the catalytic activity of enzymes is strongly dependent on pH (see Fig. 2-21). Measurements of the pH of blood and urine are commonly used in medical diagnoses. The pH of the blood plasma of people

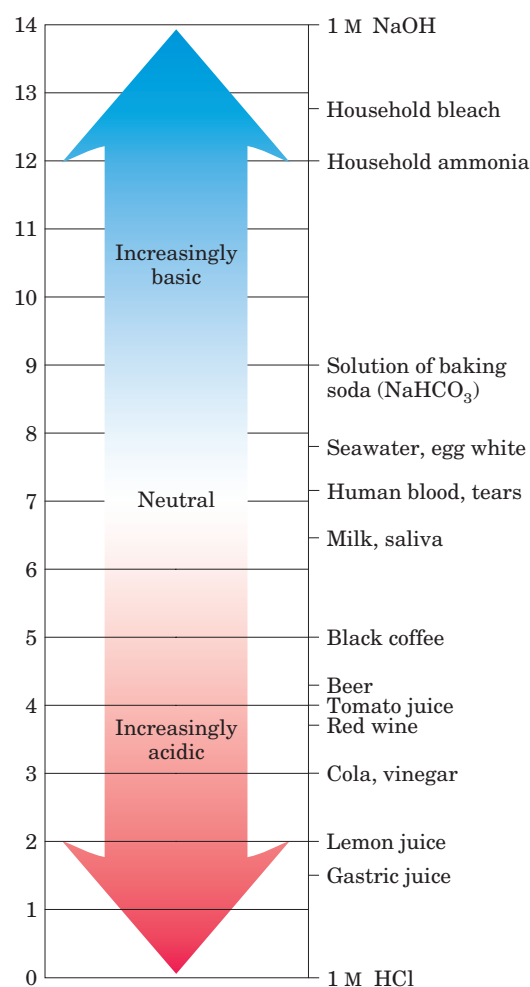


FIGURE 2-15 The pH of some aqueous fluids.

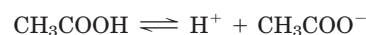
with severe, uncontrolled diabetes, for example, is often below the normal value of 7.4; this condition is called acidosis. In certain other disease states the pH of the blood is higher than normal, the condition of alkalosis.

Weak Acids and Bases Have Characteristic Dissociation Constants

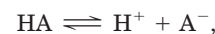
Hydrochloric, sulfuric, and nitric acids, commonly called strong acids, are completely ionized in dilute aqueous solutions; the strong bases NaOH and KOH are also completely ionized. Of more interest to biochemists is the behavior of weak acids and bases—those not completely ionized when dissolved in water. These are common in biological systems and play important roles in metabolism and its regulation. The behavior of aqueous solutions of weak acids and bases is best understood if we first define some terms.

Acids may be defined as proton donors and bases as proton acceptors. A proton donor and its corresponding proton acceptor make up a **conjugate acid-base pair** (Fig. 2–16). Acetic acid (CH_3COOH), a proton donor, and the acetate anion (CH_3COO^-), the corre-

sponding proton acceptor, constitute a conjugate acid-base pair, related by the reversible reaction



Each acid has a characteristic tendency to lose its proton in an aqueous solution. The stronger the acid, the greater its tendency to lose its proton. The tendency of any acid (HA) to lose a proton and form its conjugate base (A^-) is defined by the equilibrium constant (K_{eq}) for the reversible reaction



which is

$$K_{\text{eq}} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = K_a$$

Equilibrium constants for ionization reactions are usually called ionization or **dissociation constants**, often designated K_a . The dissociation constants of some acids are given in Figure 2–16. Stronger acids, such as phosphoric and carbonic acids, have larger dissociation constants; weaker acids, such as monohydrogen phosphate (HPO_4^{2-}), have smaller dissociation constants.

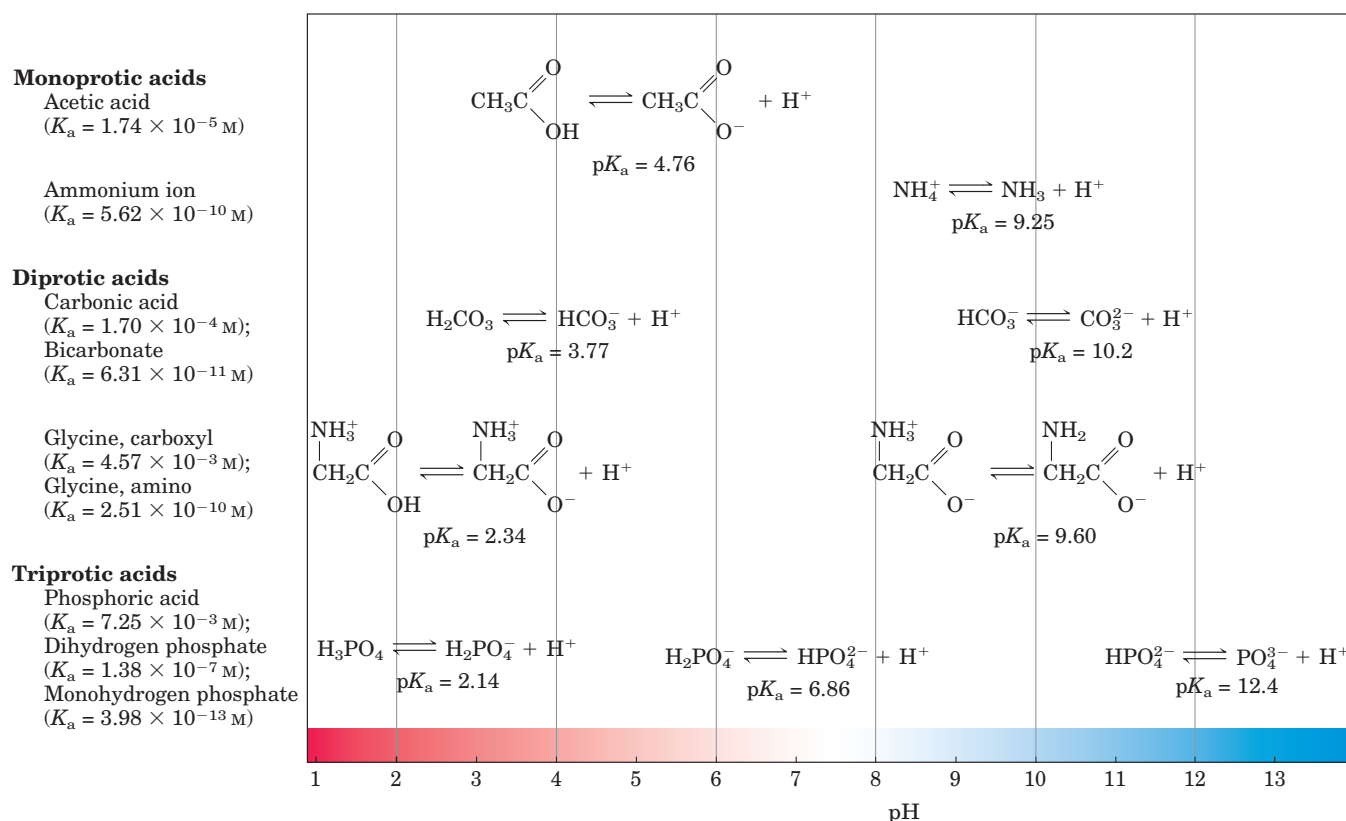


FIGURE 2–16 Conjugate acid-base pairs consist of a proton donor and a proton acceptor. Some compounds, such as acetic acid and ammonium ion, are monoprotic; they can give up only one proton. Others are diprotic (H_2CO_3 (carbonic acid) and glycine) or triprotic

(H_3PO_4 (phosphoric acid)). The dissociation reactions for each pair are shown where they occur along a pH gradient. The equilibrium or dissociation constant (K_a) and its negative logarithm, the $\text{p}K_a$, are shown for each reaction.

Also included in Figure 2-16 are values of pK_a , which is analogous to pH and is defined by the equation

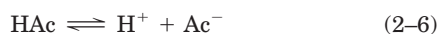
$$pK_a = \log \frac{1}{K_a} = -\log K_a$$

The stronger the tendency to dissociate a proton, the stronger is the acid and the lower its pK_a . As we shall now see, the pK_a of any weak acid can be determined quite easily.

Titration Curves Reveal the pK_a of Weak Acids

Titration is used to determine the amount of an acid in a given solution. A measured volume of the acid is titrated with a solution of a strong base, usually sodium hydroxide (NaOH), of known concentration. The NaOH is added in small increments until the acid is consumed (neutralized), as determined with an indicator dye or a pH meter. The concentration of the acid in the original solution can be calculated from the volume and concentration of NaOH added.

A plot of pH against the amount of NaOH added (a **titration curve**) reveals the pK_a of the weak acid. Consider the titration of a 0.1 M solution of acetic acid (for simplicity denoted as HAc) with 0.1 M NaOH at 25 °C (Fig. 2-17). Two reversible equilibria are involved in the process:



The equilibria must simultaneously conform to their characteristic equilibrium constants, which are, respectively,

$$K_w = [H^+][OH^-] = 1 \times 10^{-14} \text{ M}^2 \quad (2-7)$$

$$K_a = \frac{[H^+][Ac^-]}{[HAc]} = 1.74 \times 10^{-5} \text{ M} \quad (2-8)$$

At the beginning of the titration, before any NaOH is added, the acetic acid is already slightly ionized, to an extent that can be calculated from its dissociation constant (Eqn 2-8).

As NaOH is gradually introduced, the added OH^- combines with the free H^+ in the solution to form H_2O , to an extent that satisfies the equilibrium relationship in Equation 2-7. As free H^+ is removed, HAc dissociates further to satisfy its own equilibrium constant (Eqn 2-8). The net result as the titration proceeds is that more and more HAc ionizes, forming Ac^- , as the NaOH is added. At the midpoint of the titration, at which exactly 0.5 equivalent of NaOH has been added, one-half of the original acetic acid has undergone dissociation, so that the concentration of the proton donor, [HAc], now equals that of the proton acceptor, [Ac^-]. At this midpoint a very important relationship holds: the pH of the equimolar solution of acetic acid and acetate is ex-

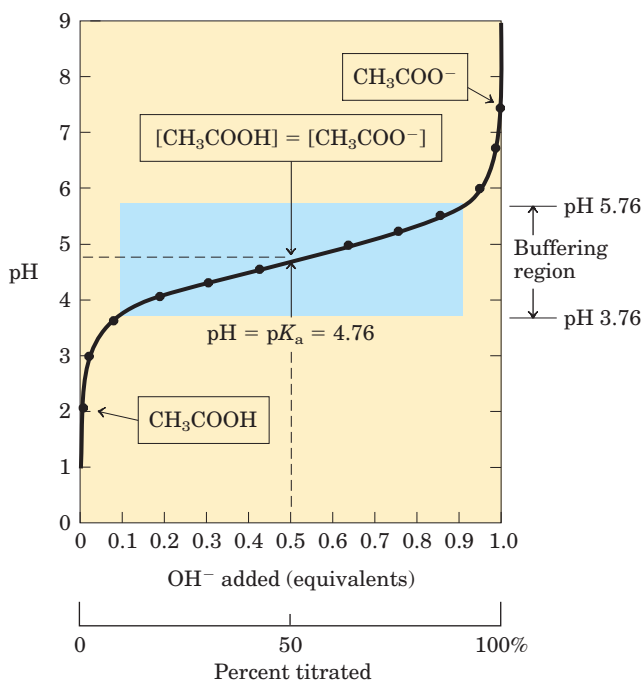


FIGURE 2-17 The titration curve of acetic acid. After addition of each increment of NaOH to the acetic acid solution, the pH of the mixture is measured. This value is plotted against the amount of NaOH expressed as a fraction of the total NaOH required to convert all the acetic acid to its deprotonated form, acetate. The points so obtained yield the titration curve. Shown in the boxes are the predominant ionic forms at the points designated. At the midpoint of the titration, the concentrations of the proton donor and proton acceptor are equal, and the pH is numerically equal to the pK_a . The shaded zone is the useful region of buffering power, generally between 10% and 90% titration of the weak acid.

actly equal to the pK_a of acetic acid ($pK_a = 4.76$; Figs 2-16, 2-17). The basis for this relationship, which holds for all weak acids, will soon become clear.

As the titration is continued by adding further increments of NaOH, the remaining nondissociated acetic acid is gradually converted into acetate. The end point of the titration occurs at about pH 7.0: all the acetic acid has lost its protons to OH^- , to form H_2O and acetate. Throughout the titration the two equilibria (Eqns 2-5, 2-6) coexist, each always conforming to its equilibrium constant.

Figure 2-18 compares the titration curves of three weak acids with very different dissociation constants: acetic acid ($pK_a = 4.76$); dihydrogen phosphate, $H_2PO_4^-$ ($pK_a = 6.86$); and ammonium ion, NH_4^+ ($pK_a = 9.25$). Although the titration curves of these acids have the same shape, they are displaced along the pH axis because the three acids have different strengths. Acetic acid, with the highest K_a (lowest pK_a) of the three, is the strongest (loses its proton most readily); it is al-

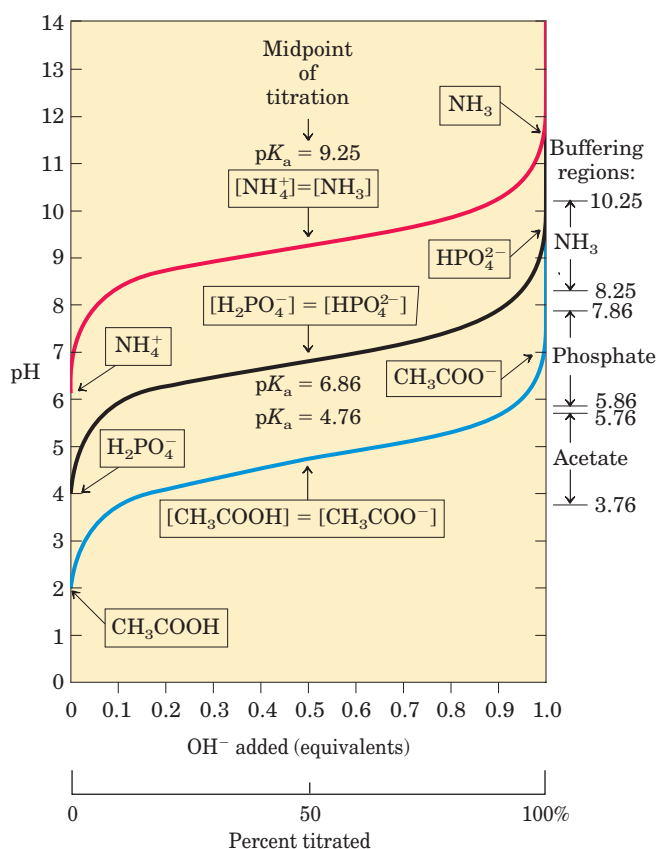


FIGURE 2-18 Comparison of the titration curves of three weak acids. Shown here are the titration curves for CH_3COOH , H_2PO_4^- , and NH_4^+ . The predominant ionic forms at designated points in the titration are given in boxes. The regions of buffering capacity are indicated at the right. Conjugate acid-base pairs are effective buffers between approximately 10% and 90% neutralization of the proton-donor species.

ready half dissociated at pH 4.76. Dihydrogen phosphate loses a proton less readily, being half dissociated at pH 6.86. Ammonium ion is the weakest acid of the three and does not become half dissociated until pH 9.25.

The most important point about the titration curve of a weak acid is that it shows graphically that a weak acid and its anion—a conjugate acid-base pair—can act as a buffer.

SUMMARY 2.2 Ionization of Water, Weak Acids, and Weak Bases

- Pure water ionizes slightly, forming equal numbers of hydrogen ions (hydronium ions, H_3O^+) and hydroxide ions. The extent of ionization is described by an equilibrium constant, $K_{\text{eq}} = \frac{[\text{H}^+][\text{OH}^-]}{[\text{H}_2\text{O}]}$, from which the ion product of

water, K_w , is derived. At 25°C , $K_w = [\text{H}^+][\text{OH}^-] = (55.5 \text{ M})(K_{\text{eq}}) = 10^{-14} \text{ M}^2$.

- The pH of an aqueous solution reflects, on a logarithmic scale, the concentration of hydrogen ions: $\text{pH} = \log \frac{1}{[\text{H}^+]} = -\log [\text{H}^+]$.
- The greater the acidity of a solution, the lower its pH. Weak acids partially ionize to release a hydrogen ion, thus lowering the pH of the aqueous solution. Weak bases accept a hydrogen ion, increasing the pH. The extent of these processes is characteristic of each particular weak acid or base and is expressed as a dissociation constant, K_a : $K_{\text{eq}} = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]} = K_a$.
- The pK_a expresses, on a logarithmic scale, the relative strength of a weak acid or base: $\text{pK}_a = \log \frac{1}{K_a} = -\log K_a$.
- The stronger the acid, the lower its pK_a ; the stronger the base, the higher its pK_a . The pK_a can be determined experimentally; it is the pH at the midpoint of the titration curve for the acid or base.

2.3 Buffering against pH Changes in Biological Systems

Almost every biological process is pH dependent; a small change in pH produces a large change in the rate of the process. This is true not only for the many reactions in which the H^+ ion is a direct participant, but also for those in which there is no apparent role for H^+ ions. The enzymes that catalyze cellular reactions, and many of the molecules on which they act, contain ionizable groups with characteristic pK_a values. The protonated amino and carboxyl groups of amino acids and the phosphate groups of nucleotides, for example, function as weak acids; their ionic state depends on the pH of the surrounding medium. As we noted above, ionic interactions are among the forces that stabilize a protein molecule and allow an enzyme to recognize and bind its substrate.

Cells and organisms maintain a specific and constant cytosolic pH, keeping biomolecules in their optimal ionic state, usually near pH 7. In multicellular organisms, the pH of extracellular fluids is also tightly regulated. Constancy of pH is achieved primarily by biological buffers: mixtures of weak acids and their conjugate bases.

We describe here the ionization equilibria that account for buffering, and we show the quantitative relationship between the pH of a buffered solution and the pK_a of the buffer. Biological buffering is illustrated by the phosphate and carbonate buffering systems of humans.

Buffers Are Mixtures of Weak Acids and Their Conjugate Bases

Buffers are aqueous systems that tend to resist changes in pH when small amounts of acid (H^+) or base (OH^-) are added. A buffer system consists of a weak acid (the proton donor) and its conjugate base (the proton acceptor). As an example, a mixture of equal concentrations of acetic acid and acetate ion, found at the midpoint of the titration curve in Figure 2-17, is a buffer system. The titration curve of acetic acid has a relatively flat zone extending about 1 pH unit on either side of its midpoint pH of 4.76. In this zone, an amount of H^+ or OH^- added to the system has much less effect on pH than the same amount added outside the buffer range. This relatively flat zone is the buffering region of the acetic acid–acetate buffer pair. At the midpoint of the buffering region, where the concentration of the proton donor (acetic acid) exactly equals that of the proton acceptor (acetate), the buffering power of the system is maximal; that is, its pH changes least on addition of H^+ or OH^- . The pH at this point in the titration curve of acetic acid is equal to its $\text{p}K_a$. The pH of the acetate buffer system does change slightly when a small amount of H^+ or OH^- is added, but this change is very small compared with the pH change that would result if the same amount of H^+ or OH^- were added to pure water or to a solution of the salt of a strong acid and strong base, such as NaCl, which has no buffering power.

Buffering results from two reversible reaction equilibria occurring in a solution of nearly equal concentrations of a proton donor and its conjugate proton acceptor. Figure 2-19 explains how a buffer system works. Whenever H^+ or OH^- is added to a buffer, the result is a small change in the ratio of the relative concentrations of the weak acid and its anion and thus a small change in pH. The decrease in concentration of one component of the system is balanced exactly by an increase in the other. The sum of the buffer components does not change, only their ratio.

Each conjugate acid-base pair has a characteristic pH zone in which it is an effective buffer (Fig. 2-18). The $\text{H}_2\text{PO}_4^-/\text{HPO}_4^{2-}$ pair has a $\text{p}K_a$ of 6.86 and thus can serve as an effective buffer system between approximately pH 5.9 and pH 7.9; the $\text{NH}_4^+/\text{NH}_3$ pair, with a $\text{p}K_a$ of 9.25, can act as a buffer between approximately pH 8.3 and pH 10.3.

A Simple Expression Relates pH, $\text{p}K_a$, and Buffer Concentration

The titration curves of acetic acid, H_2PO_4^- , and NH_4^+ (Fig. 2-18) have nearly identical shapes, suggesting that these curves reflect a fundamental law or relationship. This is indeed the case. The shape of the titration curve of any weak acid is described by the **Henderson-**

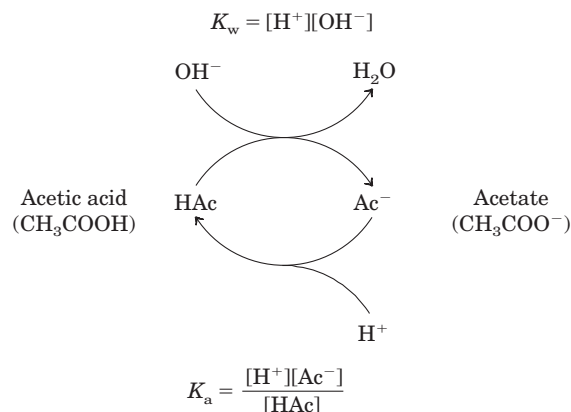


FIGURE 2-19 The acetic acid–acetate pair as a buffer system. The system is capable of absorbing either H^+ or OH^- through the reversibility of the dissociation of acetic acid. The proton donor, acetic acid (HAc), contains a reserve of bound H^+ , which can be released to neutralize an addition of OH^- to the system, forming H_2O . This happens because the product $[\text{H}^+][\text{OH}^-]$ transiently exceeds K_w ($1 \times 10^{-14} \text{ M}^2$). The equilibrium quickly adjusts so that this product equals $1 \times 10^{-14} \text{ M}^2$ (at 25 °C), thus transiently reducing the concentration of H^+ . But now the quotient $[\text{H}^+][\text{Ac}^-] / [\text{HAc}]$ is less than K_a , so HAc dissociates further to restore equilibrium. Similarly, the conjugate base, Ac^- , can react with H^+ ions added to the system; again, the two ionization reactions simultaneously come to equilibrium. Thus a conjugate acid-base pair, such as acetic acid and acetate ion, tends to resist a change in pH when small amounts of acid or base are added. Buffering action is simply the consequence of two reversible reactions taking place simultaneously and reaching their points of equilibrium as governed by their equilibrium constants, K_w and K_a .

Hasselbalch equation, which is important for understanding buffer action and acid-base balance in the blood and tissues of vertebrates. This equation is simply a useful way of restating the expression for the dissociation constant of an acid. For the dissociation of a weak acid HA into H^+ and A^- , the Henderson-Hasselbalch equation can be derived as follows:

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

First solve for $[\text{H}^+]$:

$$[\text{H}^+] = K_a \frac{[\text{HA}]}{[\text{A}^-]}$$

Then take the negative logarithm of both sides:

$$-\log [\text{H}^+] = -\log K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

Substitute pH for $-\log [\text{H}^+]$ and $\text{p}K_a$ for $-\log K_a$:

$$\text{pH} = \text{p}K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$$

Now invert $-\log [HA]/[A^-]$, which involves changing its sign, to obtain the Henderson-Hasselbalch equation:

$$\text{pH} = \text{p}K_a + \log \frac{[A^-]}{[HA]} \quad (2-9)$$

Stated more generally,

$$\text{pH} = \text{p}K_a + \log \frac{[\text{proton acceptor}]}{[\text{proton donor}]}$$

This equation fits the titration curve of all weak acids and enables us to deduce a number of important quantitative relationships. For example, it shows why the $\text{p}K_a$ of a weak acid is equal to the pH of the solution at the midpoint of its titration. At that point, $[HA]$ equals $[A^-]$, and

$$\text{pH} = \text{p}K_a + \log 1 = \text{p}K_a + 0 = \text{p}K_a$$

As shown in Box 2-3, the Henderson-Hasselbalch equation also allows us to (1) calculate $\text{p}K_a$, given pH and the molar ratio of proton donor and acceptor; (2) calculate pH, given $\text{p}K_a$ and the molar ratio of proton donor and acceptor; and (3) calculate the molar ratio of proton donor and acceptor, given pH and $\text{p}K_a$.

Weak Acids or Bases Buffer Cells and Tissues against pH Changes

The intracellular and extracellular fluids of multicellular organisms have a characteristic and nearly constant

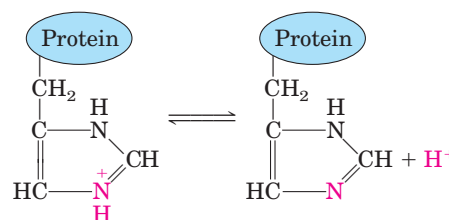


FIGURE 2-20 The amino acid histidine, a component of proteins, is a weak acid. The $\text{p}K_a$ of the protonated nitrogen of the side chain is 6.0.

pH. The organism's first line of defense against changes in internal pH is provided by buffer systems. The cytoplasm of most cells contains high concentrations of proteins, which contain many amino acids with functional groups that are weak acids or weak bases. For example, the side chain of histidine (Fig. 2-20) has a $\text{p}K_a$ of 6.0; proteins containing histidine residues therefore buffer effectively near neutral pH. Nucleotides such as ATP, as well as many low molecular weight metabolites, contain ionizable groups that can contribute buffering power to the cytoplasm. Some highly specialized organelles and extracellular compartments have high concentrations of compounds that contribute buffering capacity: organic acids buffer the vacuoles of plant cells; ammonia buffers urine.

BOX 2-3 WORKING IN BIOCHEMISTRY

Solving Problems Using the Henderson-Hasselbalch Equation

1. Calculate the $\text{p}K_a$ of lactic acid, given that when the concentration of lactic acid is 0.010 M and the concentration of lactate is 0.087 M, the pH is 4.80.

$$\begin{aligned} \text{pH} &= \text{p}K_a + \log \frac{[\text{lactate}]}{[\text{lactic acid}]} \\ \text{p}K_a &= \text{pH} - \log \frac{[\text{lactate}]}{[\text{lactic acid}]} \\ &= 4.80 - \log \frac{0.087}{0.010} = 4.80 - \log 8.7 \\ &= 4.80 - 0.94 = 3.9 \quad (\text{answer}) \end{aligned}$$

2. Calculate the pH of a mixture of 0.10 M acetic acid and 0.20 M sodium acetate. The $\text{p}K_a$ of acetic acid is 4.76.

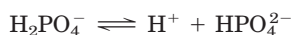
$$\begin{aligned} \text{pH} &= \text{p}K_a + \log \frac{[\text{acetate}]}{[\text{acetic acid}]} \\ &= 4.76 + \log \frac{0.20}{0.10} = 4.76 + 0.30 \\ &= 5.1 \quad (\text{answer}) \end{aligned}$$

3. Calculate the ratio of the concentrations of acetate and acetic acid required in a buffer system of pH 5.30.

$$\begin{aligned} \text{pH} &= \text{p}K_a + \log \frac{[\text{acetate}]}{[\text{acetic acid}]} \\ \log \frac{[\text{acetate}]}{[\text{acetic acid}]} &= \text{pH} - \text{p}K_a \\ &= 5.30 - 4.76 = 0.54 \\ \frac{[\text{acetate}]}{[\text{acetic acid}]} &= \text{antilog } 0.54 = 3.5 \quad (\text{answer}) \end{aligned}$$

To see the effect of pH on the degree of ionization of a weak acid, see the Living Graph for Equation 2-9.

Two especially important biological buffers are the phosphate and bicarbonate systems. The phosphate buffer system, which acts in the cytoplasm of all cells, consists of H_2PO_4^- as proton donor and HPO_4^{2-} as proton acceptor:



The phosphate buffer system is maximally effective at a pH close to its $\text{p}K_a$ of 6.86 (Figs 2–16, 2–18) and thus tends to resist pH changes in the range between about 5.9 and 7.9. It is therefore an effective buffer in biological fluids; in mammals, for example, extracellular fluids and most cytoplasmic compartments have a pH in the range of 6.9 to 7.4.

Blood plasma is buffered in part by the bicarbonate system, consisting of carbonic acid (H_2CO_3) as proton donor and bicarbonate (HCO_3^-) as proton acceptor:



$$K_1 = \frac{[\text{H}^+][\text{HCO}_3^-]}{[\text{H}_2\text{CO}_3]}$$

This buffer system is more complex than other conjugate acid-base pairs because one of its components, carbonic acid (H_2CO_3), is formed from dissolved (d) carbon dioxide and water, in a reversible reaction:



$$K_2 = \frac{[\text{H}_2\text{CO}_3]}{[\text{CO}_2(\text{d})][\text{H}_2\text{O}]}$$

Carbon dioxide is a gas under normal conditions, and the concentration of dissolved CO_2 is the result of equilibration with CO_2 of the gas (g) phase:



$$K_3 = \frac{[\text{CO}_2(\text{d})]}{[\text{CO}_2(\text{g})]}$$

The pH of a bicarbonate buffer system depends on the concentration of H_2CO_3 and HCO_3^- , the proton donor and acceptor components. The concentration of H_2CO_3 in turn depends on the concentration of dissolved CO_2 , which in turn depends on the concentration of CO_2 in the gas phase, called the **partial pressure** of CO_2 . Thus the pH of a bicarbonate buffer exposed to a gas phase is ultimately determined by the concentration of HCO_3^- in the aqueous phase and the partial pressure of CO_2 in the gas phase (Box 2–4).

Human blood plasma normally has a pH close to 7.4. Should the pH-regulating mechanisms fail or be overwhelmed, as may happen in severe uncontrolled diabetes when an overproduction of metabolic acids causes acidosis, the pH of the blood can fall to 6.8 or below, leading to irreparable cell damage and death. In other diseases the pH may rise to lethal levels.

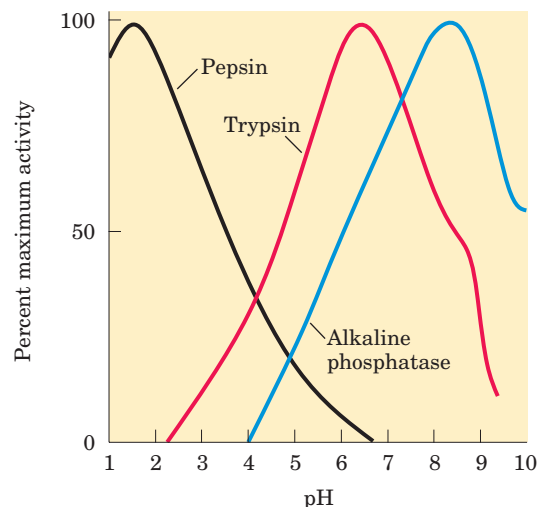


FIGURE 2-21 The pH optima of some enzymes. Pepsin is a digestive enzyme secreted into gastric juice; trypsin, a digestive enzyme that acts in the small intestine; alkaline phosphatase of bone tissue, a hydrolytic enzyme thought to aid in bone mineralization.

Although many aspects of cell structure and function are influenced by pH, it is the catalytic activity of enzymes that is especially sensitive. Enzymes typically show maximal catalytic activity at a characteristic pH, called the **pH optimum** (Fig. 2–21). On either side of the optimum pH their catalytic activity often declines sharply. Thus, a small change in pH can make a large difference in the rate of some crucial enzyme-catalyzed reactions. Biological control of the pH of cells and body fluids is therefore of central importance in all aspects of metabolism and cellular activities.

SUMMARY 2.3 Buffering against pH Changes in Biological Systems

- A mixture of a weak acid (or base) and its salt resists changes in pH caused by the addition of H^+ or OH^- . The mixture thus functions as a buffer.
- The pH of a solution of a weak acid (or base) and its salt is given by the Henderson-Hasselbalch equation: $\text{pH} = \text{p}K_a - \log \frac{[\text{HA}]}{[\text{A}^-]}$.
- In cells and tissues, phosphate and bicarbonate buffer systems maintain intracellular and extracellular fluids at their optimum (physiological) pH, which is usually close to pH 7. Enzymes generally work optimally at this pH.



BOX 2-4 BIOCHEMISTRY IN MEDICINE

Blood, Lungs, and Buffer: The Bicarbonate Buffer System

In animals with lungs, the bicarbonate buffer system is an effective physiological buffer near pH 7.4, because the H_2CO_3 of blood plasma is in equilibrium with a large reserve capacity of $\text{CO}_2(\text{g})$ in the air space of the lungs. This buffer system involves three reversible equilibria between gaseous CO_2 in the lungs and bicarbonate (HCO_3^-) in the blood plasma (Fig. 1).

When H^+ (from lactic acid produced in muscle tissue during vigorous exercise, for example) is added to blood as it passes through the tissues, reaction 1 proceeds toward a new equilibrium, in which the concentration of H_2CO_3 is increased. This increases the concentration of $\text{CO}_2(\text{d})$ in the blood plasma (reaction 2) and thus increases the pressure of $\text{CO}_2(\text{g})$ in the air space of the lungs (reaction 3); the extra CO_2 is exhaled. Conversely, when the pH of blood plasma is raised (by NH_3 production during protein catabolism, for example), the opposite events occur: the H^+ concentration of blood plasma is lowered, causing more H_2CO_3 to dissociate into H^+ and HCO_3^- . This in turn causes more $\text{CO}_2(\text{g})$ from the lungs to dissolve

in the blood plasma. The rate of breathing—that is, the rate of inhaling and exhaling CO_2 —can quickly adjust these equilibria to keep the blood pH nearly constant.

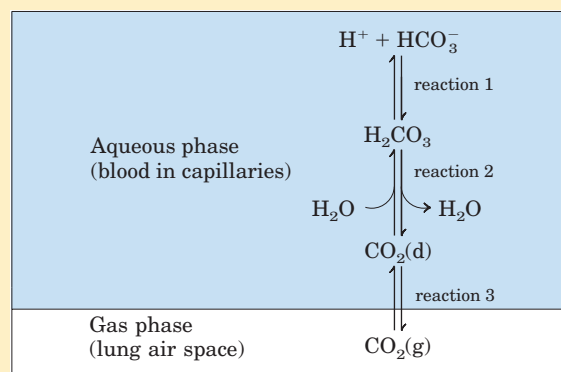


FIGURE 1 The CO_2 in the air space of the lungs is in equilibrium with the bicarbonate buffer in the blood plasma passing through the lung capillaries. Because the concentration of dissolved CO_2 can be adjusted rapidly through changes in the rate of breathing, the bicarbonate buffer system of the blood is in near-equilibrium with a large potential reservoir of CO_2 .

2.4 Water as a Reactant

Water is not just the solvent in which the chemical reactions of living cells occur; it is very often a direct participant in those reactions. The formation of ATP from ADP and inorganic phosphate is an example of a **condensation reaction** in which the elements of water are eliminated (Fig. 2-22a). The reverse of this reaction—cleavage accompanied by the addition of the elements of water—is a **hydrolysis reaction**. Hydrolysis reactions are also responsible for the enzymatic depolymerization of proteins, carbohydrates, and nucleic acids. Hydrolysis reactions, catalyzed by enzymes called

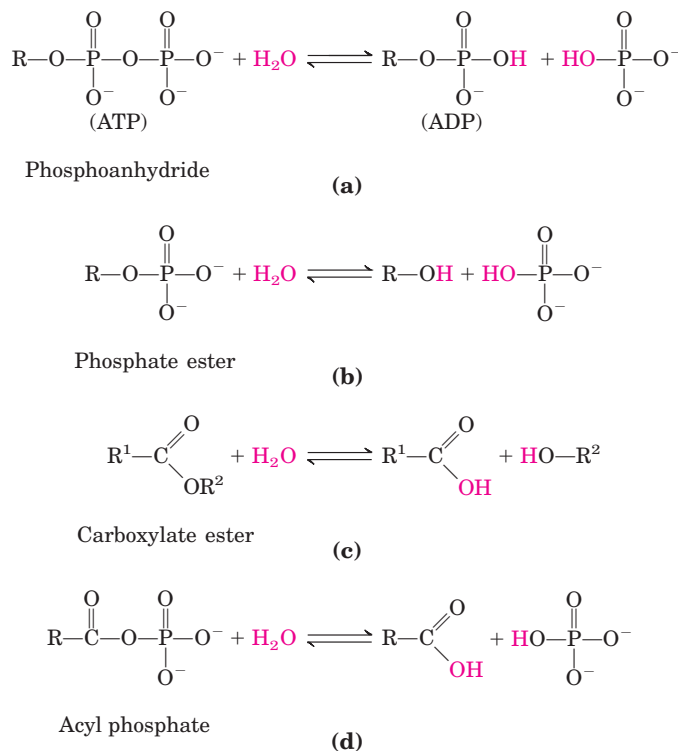
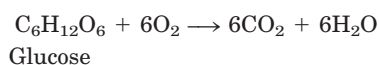


FIGURE 2-22 Participation of water in biological reactions. (a) ATP is a phosphoanhydride formed by a condensation reaction (loss of the elements of water) between ADP and phosphate. R represents adenosine monophosphate (AMP). This condensation reaction requires energy. The hydrolysis of (addition of the elements of water to) ATP to form ADP and phosphate releases an equivalent amount of energy. Also shown are some other condensation and hydrolysis reactions common in biological systems (b), (c), (d).

hydrolases, are almost invariably exergonic. The formation of cellular polymers from their subunits by simple reversal of hydrolysis (that is, by condensation reactions) would be endergonic and therefore does not occur. As we shall see, cells circumvent this thermodynamic obstacle by coupling endergonic condensation reactions to exergonic processes, such as breakage of the anhydride bond in ATP.

You are (we hope!) consuming oxygen as you read. Water and carbon dioxide are the end products of the oxidation of fuels such as glucose. The overall reaction can be summarized as



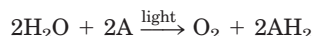
The “metabolic water” formed by oxidation of foods and stored fats is actually enough to allow some animals in very dry habitats (gerbils, kangaroo rats, camels) to survive for extended periods without drinking water.

The CO_2 produced by glucose oxidation is converted in erythrocytes to the more soluble HCO_3^- , in a reaction catalyzed by the enzyme carbonic anhydrase:



In this reaction, water not only is a substrate but also functions in proton transfer by forming a network of hydrogen-bonded water molecules through which proton hopping occurs (Fig. 2–14).

Green plants and algae use the energy of sunlight to split water in the process of photosynthesis:



In this reaction, A is an electron-accepting species, which varies with the type of photosynthetic organism, and water serves as the electron donor in an oxidation-reduction sequence (see Fig. 19–XX) that is fundamental to all life.

SUMMARY 2.4 Water as a Reactant

- Water is both the solvent in which metabolic reactions occur and a reactant in many biochemical processes, including hydrolysis, condensation, and oxidation-reduction reactions.

2.5 The Fitness of the Aqueous Environment for Living Organisms

Organisms have effectively adapted to their aqueous environment and have evolved means of exploiting the unusual properties of water. The high specific heat of water (the heat energy required to raise the temperature of 1 g of water by 1 °C) is useful to cells and or-

ganisms because it allows water to act as a “heat buffer,” keeping the temperature of an organism relatively constant as the temperature of the surroundings fluctuates and as heat is generated as a byproduct of metabolism. Furthermore, some vertebrates exploit the high heat of vaporization of water (Table 2–1) by using (thus losing) excess body heat to evaporate sweat. The high degree of internal cohesion of liquid water, due to hydrogen bonding, is exploited by plants as a means of transporting dissolved nutrients from the roots to the leaves during the process of transpiration. Even the density of ice, lower than that of liquid water, has important biological consequences in the life cycles of aquatic organisms. Ponds freeze from the top down, and the layer of ice at the top insulates the water below from frigid air, preventing the pond (and the organisms in it) from freezing solid. Most fundamental to all living organisms is the fact that many physical and biological properties of cell macromolecules, particularly the proteins and nucleic acids, derive from their interactions with water molecules of the surrounding medium. The influence of water on the course of biological evolution has been profound and determinative. If life forms have evolved elsewhere in the universe, they are unlikely to resemble those of Earth unless their extraterrestrial origin is also a place in which plentiful liquid water is available.



Aqueous environments support countless species. Soft corals, sponges, bryozoans, and algae compete for space on this reef substrate off the Philippine Islands.

Key Terms

Terms in bold are defined in the glossary.

hydrogen bond 48	osmosis 57	pK_a 64
bond energy 48	isotonic 57	titration curve 64
hydrophilic 50	hypertonic 57	buffer 66
hydrophobic 50	hypotonic 57	Henderson-Hasselbalch
amphipathic 52	equilibrium constant (K_{eq}) 60	equation 66
micelle 53	ion product of water (K_w) 61	condensation 69
hydrophobic interactions 53	pH 61	hydrolysis 69
van der Waals interactions 54	conjugate acid-base pair 63	
osmolarity 56	dissociation constant (K_a) 63	

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Weak Acids, Weak Bases, and Buffers: Problems for Practice

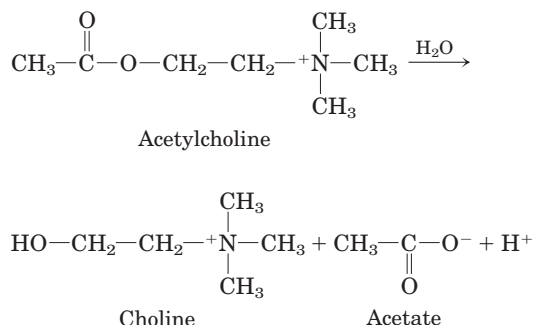
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Problems

1. Simulated Vinegar One way to make vinegar (*not* the preferred way) is to prepare a solution of acetic acid, the sole acid component of vinegar, at the proper pH (see Fig. 2–15) and add appropriate flavoring agents. Acetic acid (M_r 60) is a liquid at 25 °C, with a density of 1.049 g/mL. Calculate the volume that must be added to distilled water to make 1 L of simulated vinegar (see Fig. 2–16).

 **2. Acidity of Gastric HCl** In a hospital laboratory, a 10.0 mL sample of gastric juice, obtained several hours after a meal, was titrated with 0.1 M NaOH to neutrality; 7.2 mL of NaOH was required. The patient's stomach contained no ingested food or drink, thus assume that no buffers were present. What was the pH of the gastric juice?

3. Measurement of Acetylcholine Levels by pH Changes The concentration of acetylcholine (a neurotransmitter) in a sample can be determined from the pH changes that accompany its hydrolysis. When the sample is incubated with the enzyme acetylcholinesterase, acetylcholine is quantitatively converted into choline and acetic acid, which dissociates to yield acetate and a hydrogen ion:



In a typical analysis, 15 mL of an aqueous solution containing an unknown amount of acetylcholine had a pH of 7.65. When incubated with acetylcholinesterase, the pH of the solution decreased to 6.87. Assuming that there was no buffer in the assay mixture, determine the number of moles of acetylcholine in the 15 mL sample.

4. Osmotic Balance in a Marine Frog The crab-eating frog of Southeast Asia, *Rana cancrivora*, develops and matures in fresh water but searches for its food in coastal mangrove swamps (composed of 80% to full-strength seawater). When the frog moves from its freshwater home to seawater it experiences a large change in the osmolarity of its environment (from hypotonic to hypertonic).

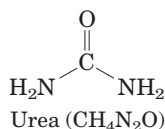
(a) Eighty percent seawater contains 460 mM NaCl, 10 mM KCl, 10 mM CaCl_2 , and 50 mM MgCl_2 . What are the concentrations of the various ionic species in this seawater? Assuming that these salts account for nearly all the solutes in seawater, calculate the osmolarity of the seawater.

(b) The chart below lists the cytoplasmic concentrations of ions in *R. cancrivora*. Ignoring dissolved proteins, amino acids, nucleic acids, and other small metabolites, calculate the osmolarity of the frog's cells based solely on the ionic concentrations given below.

	Na^+ (mM)	K^+ (mM)	Cl^- (mM)	Ca^{2+} (mM)	Mg^{2+} (mM)
<i>R. cancrivora</i>	122	10	100	2	1

(c) Like all frogs, the crab-eating frog can exchange gases through its permeable skin, allowing it to stay underwater for long periods of time without breathing. How does the high permeability of frog skin affect the frog's cells when it moves from fresh water to seawater?

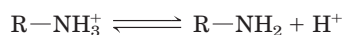
(d) The crab-eating frog uses two mechanisms to maintain its cells in osmotic balance with its environment. First, it allows the Na^+ and Cl^- concentrations in its cells to increase slowly as the ions diffuse down their concentration gradients. Second, like many elasmobranchs (sharks), it retains the waste product urea in its cells. The addition of both NaCl and urea increases the osmolarity of the cytosol to a level nearly equal to that of the surrounding environment.



Assuming the volume of water in a typical frog is 100 mL, calculate how many grams of NaCl (formula weight (FW) 58.44) the frog must take up to make its tissues isotonic with seawater.

(e) How many grams of urea (FW 60) must it retain to accomplish the same thing?

5. Properties of a Buffer The amino acid glycine is often used as the main ingredient of a buffer in biochemical experiments. The amino group of glycine, which has a $\text{p}K_a$ of 9.6, can exist either in the protonated form ($-\text{NH}_3^+$) or as the free base ($-\text{NH}_2$), because of the reversible equilibrium



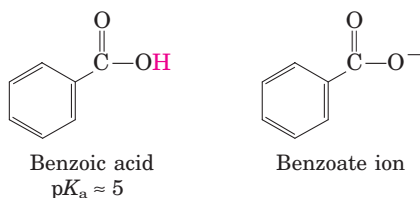
(a) In what pH range can glycine be used as an effective buffer due to its amino group?

(b) In a 0.1 M solution of glycine at pH 9.0, what fraction of glycine has its amino group in the $-\text{NH}_3^+$ form?

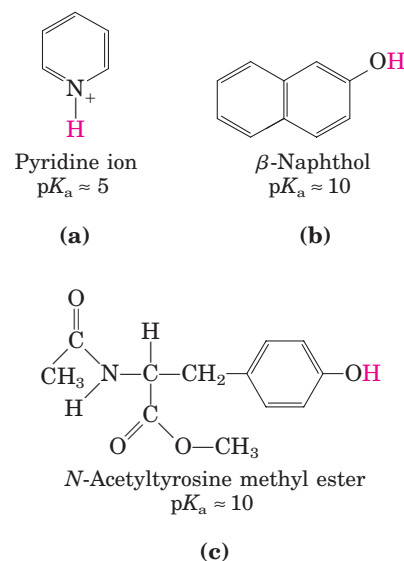
(c) How much 5 M KOH must be added to 1.0 L of 0.1 M glycine at pH 9.0 to bring its pH to exactly 10.0?

(d) When 99% of the glycine is in its $-\text{NH}_3^+$ form, what is the numerical relation between the pH of the solution and the $\text{p}K_a$ of the amino group?

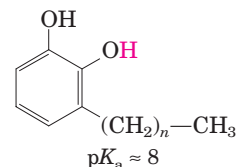
6. The Effect of pH on Solubility The strongly polar, hydrogen-bonding properties of water make it an excellent solvent for ionic (charged) species. By contrast, nonionized, nonpolar organic molecules, such as benzene, are relatively insoluble in water. In principle, the aqueous solubility of any organic acid or base can be increased by converting the molecules to charged species. For example, the solubility of benzoic acid in water is low. The addition of sodium bicarbonate to a mixture of water and benzoic acid raises the pH and deprotonates the benzoic acid to form benzoate ion, which is quite soluble in water.



Are the following compounds more soluble in an aqueous solution of 0.1 M NaOH or 0.1 M HCl? (The dissociable protons are shown in red.)



7. Treatment of Poison Ivy Rash The components of poison ivy and poison oak that produce the characteristic itchy rash are catechols substituted with long-chain alkyl groups.

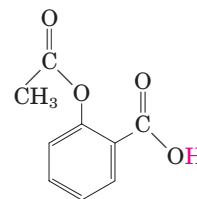


If you were exposed to poison ivy, which of the treatments below would you apply to the affected area? Justify your choice.

- Wash the area with cold water.
- Wash the area with dilute vinegar or lemon juice.
- Wash the area with soap and water.
- Wash the area with soap, water, and baking soda (sodium bicarbonate).



8. pH and Drug Absorption Aspirin is a weak acid with a $\text{p}K_a$ of 3.5.



It is absorbed into the blood through the cells lining the stomach and the small intestine. Absorption requires passage through the plasma membrane, the rate of which is determined by the polarity of the molecule: charged and highly polar molecules pass slowly, whereas neutral hydrophobic ones pass rapidly. The pH of the stomach contents is about 1.5, and the pH of the contents of the small intestine is about 6. Is more aspirin absorbed into the bloodstream from the stomach or from the small intestine? Clearly justify your choice.

9. Preparation of Standard Buffer for Calibration of a pH Meter

The glass electrode used in commercial pH meters gives an electrical response proportional to the concentration of hydrogen ion. To convert these responses into pH, glass electrodes must be calibrated against standard solutions of known H^+ concentration. Determine the weight in grams of sodium dihydrogen phosphate ($NaH_2PO_4 \cdot H_2O$; FW 138.01) and disodium hydrogen phosphate (Na_2HPO_4 ; FW 141.98) needed to prepare 1 L of a standard buffer at pH 7.00 with a total phosphate concentration of 0.100 M (see Fig. 2–16).

10. Calculating pH from Hydrogen Ion Concentration

What is the pH of a solution that has an H^+ concentration of (a) 1.75×10^{-5} mol/L; (b) 6.50×10^{-10} mol/L; (c) 1.0×10^{-4} mol/L; (d) 1.50×10^{-5} mol/L?

11. Calculating Hydrogen Ion Concentration from pH

What is the H^+ concentration of a solution with pH of (a) 3.82; (b) 6.52; (c) 11.11?

12. Calculating pH from Molar Ratios Calculate the pH of a dilute solution that contains a molar ratio of potassium acetate to acetic acid ($pK_a = 4.76$) of (a) 2:1; (b) 1:3; (c) 5:1; (d) 1:1; (e) 1:10.

13. Working with Buffers A buffer contains 0.010 mol of lactic acid ($pK_a = 3.86$) and 0.050 mol of sodium lactate per liter. (a) Calculate the pH of the buffer. (b) Calculate the change in pH when 5 mL of 0.5 M HCl is added to 1 L of the buffer. (c) What pH change would you expect if you added the same quantity of HCl to 1 L of pure water?

14. Calculating pH from Concentrations What is the pH of a solution containing 0.12 mol/L of NH_4Cl and 0.03 mol/L of NaOH (pK_a of NH_4^+/NH_3 is 9.25)?

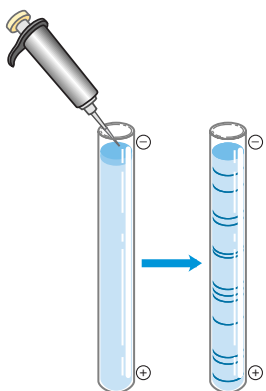
15. Calculating pK_a An unknown compound, X, is thought to have a carboxyl group with a pK_a of 2.0 and another ionizable group with a pK_a between 5 and 8. When 75 mL of 0.1 M NaOH was added to 100 mL of a 0.1 M solution of X at pH 2.0, the pH increased to 6.72. Calculate the pK_a of the second ionizable group of X.

16. Control of Blood pH by Respiration Rate

(a) The partial pressure of CO_2 in the lungs can be varied rapidly by the rate and depth of breathing. For example, a common remedy to alleviate hiccups is to increase the concentration of CO_2 in the lungs. This can be achieved by holding one's breath, by very slow and shallow breathing (hypoventilation), or by breathing in and out of a paper bag. Under such conditions, the partial pressure of CO_2 in the air space of the lungs rises above normal. Qualitatively explain the effect of these procedures on the blood pH.

(b) A common practice of competitive short-distance runners is to breathe rapidly and deeply (hyperventilate) for about half a minute to remove CO_2 from their lungs just before running in, say, a 100 m dash. Blood pH may rise to 7.60. Explain why the blood pH increases.

(c) During a short-distance run the muscles produce a large amount of lactic acid ($CH_3CH(OH)COOH$, $K_a = 1.38 \times 10^{-4}$) from their glucose stores. In view of this fact, why might hyperventilation before a dash be useful?



AMINO ACIDS, PEPTIDES, AND PROTEINS

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- 3.5 Protein Sequences and Evolution 106

The word protein that I propose to you . . . I would wish to derive from *proteios*, because it appears to be the primitive or principal substance of animal nutrition that plants prepare for the herbivores, and which the latter then furnish to the carnivores.

—J. J. Berzelius, letter to G. J. Mulder, 1838

Proteins are the most abundant biological macromolecules, occurring in all cells and all parts of cells. Proteins also occur in great variety; thousands of different kinds, ranging in size from relatively small peptides to huge polymers with molecular weights in the millions, may be found in a single cell. Moreover, proteins exhibit enormous diversity of biological function and are the most important final products of the information pathways discussed in Part III of this book. Proteins are the molecular instruments through which genetic information is expressed.

Relatively simple monomeric subunits provide the key to the structure of the thousands of different proteins. All proteins, whether from the most ancient lines of bacteria or from the most complex forms of life, are constructed from the same ubiquitous set of 20 amino

acids, covalently linked in characteristic linear sequences. Because each of these amino acids has a side chain with distinctive chemical properties, this group of 20 precursor molecules may be regarded as the alphabet in which the language of protein structure is written.

What is most remarkable is that cells can produce proteins with strikingly different properties and activities by joining the same 20 amino acids in many different combinations and sequences. From these building blocks different organisms can make such widely diverse products as enzymes, hormones, antibodies, transporters, muscle fibers, the lens protein of the eye, feathers, spider webs, rhinoceros horn, milk proteins, antibiotics, mushroom poisons, and myriad other substances having distinct biological activities (Fig. 3–1). Among these protein products, the enzymes are the most varied and specialized. Virtually all cellular reactions are catalyzed by enzymes.

Protein structure and function are the topics of this and the next three chapters. We begin with a description of the fundamental chemical properties of amino acids, peptides, and proteins.

3.1 Amino Acids

Protein Architecture—Amino Acids

Proteins are polymers of amino acids, with each **amino acid residue** joined to its neighbor by a specific type of covalent bond. (The term “residue” reflects the loss of the elements of water when one amino acid is joined to another.) Proteins can be broken down (hydrolyzed) to their constituent amino acids by a variety of methods, and the earliest studies of proteins naturally focused on

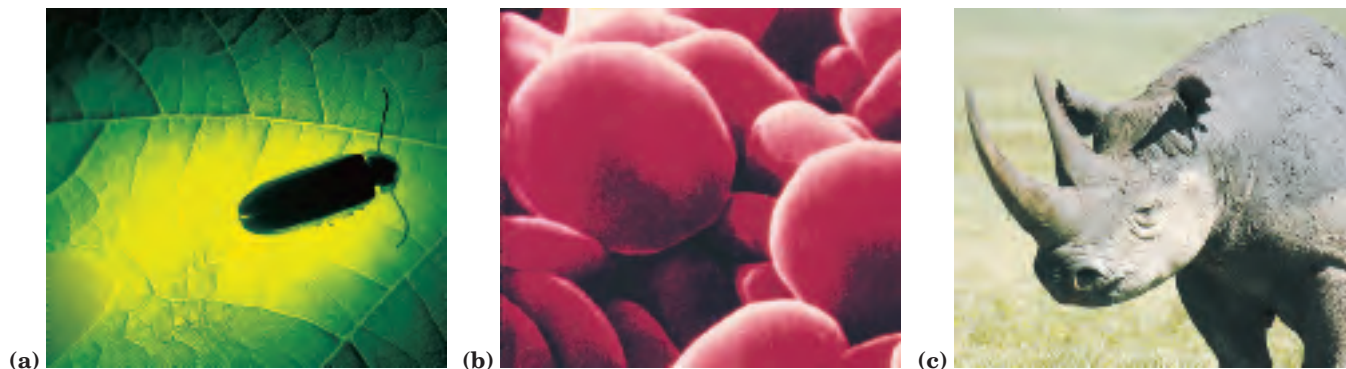


FIGURE 3-1 Some functions of proteins. (a) The light produced by fireflies is the result of a reaction involving the protein luciferin and ATP, catalyzed by the enzyme luciferase (see Box 13-2). (b) Erythrocytes contain large amounts of the oxygen-transporting protein hemoglobin. (c) The protein keratin, formed by all vertebrates, is the chief structural component of hair, scales, horn, wool, nails, and feathers.

the free amino acids derived from them. Twenty different amino acids are commonly found in proteins. The first to be discovered was asparagine, in 1806. The last of the 20 to be found, threonine, was not identified until 1938. All the amino acids have trivial or common names, in some cases derived from the source from which they were first isolated. Asparagine was first found in asparagus, and glutamate in wheat gluten; tyrosine was first isolated from cheese (its name is derived from the Greek *tyros*, “cheese”); and glycine (Greek *glykos*, “sweet”) was so named because of its sweet taste.

Amino Acids Share Common Structural Features

All 20 of the common amino acids are α -amino acids. They have a carboxyl group and an amino group bonded to the same carbon atom (the α carbon) (Fig. 3-2). They differ from each other in their side chains, or **R groups**, which vary in structure, size, and electric charge, and which influence the solubility of the amino acids in water. In addition to these 20 amino acids there are many less common ones. Some are residues modified after a protein has been synthesized; others are amino acids present in living organisms but not as constituents of proteins. The common amino acids of proteins have been assigned three-letter abbreviations and one-letter

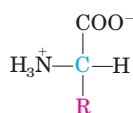
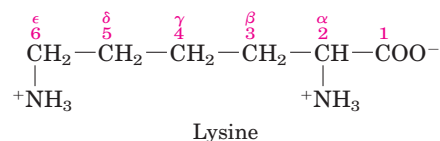


FIGURE 3-2 General structure of an amino acid. This structure is common to all but one of the α -amino acids. (Proline, a cyclic amino acid, is the exception.) The R group or side chain (red) attached to the α carbon (blue) is different in each amino acid.

ers. The black rhinoceros is nearing extinction in the wild because of the belief prevalent in some parts of the world that a powder derived from its horn has aphrodisiac properties. In reality, the chemical properties of powdered rhinoceros horn are no different from those of powdered bovine hooves or human fingernails.

symbols (Table 3-1), which are used as shorthand to indicate the composition and sequence of amino acids polymerized in proteins.

Two conventions are used to identify the carbons in an amino acid—a practice that can be confusing. The additional carbons in an R group are commonly designated β , γ , δ , ϵ , and so forth, proceeding out from the α carbon. For most other organic molecules, carbon atoms are simply numbered from one end, giving highest priority (C-1) to the carbon with the substituent containing the atom of highest atomic number. Within this latter convention, the carboxyl carbon of an amino acid would be C-1 and the α carbon would be C-2. In some cases, such as amino acids with heterocyclic R groups, the Greek lettering system is ambiguous and the numbering convention is therefore used.



For all the common amino acids except glycine, the α carbon is bonded to four different groups: a carboxyl group, an amino group, an R group, and a hydrogen atom (Fig. 3-2; in glycine, the R group is another hydrogen atom). The α -carbon atom is thus a **chiral center** (p. 17). Because of the tetrahedral arrangement of the bonding orbitals around the α -carbon atom, the four different groups can occupy two unique spatial arrangements, and thus amino acids have two possible stereoisomers. Since they are nonsuperimposable mirror images of each other (Fig. 3-3), the two forms represent a class of stereoisomers called **enantiomers** (see Fig. 1-19). All molecules with a chiral center are also **optically active**—that is, they rotate plane-polarized light (see Box 1-2).

Special nomenclature has been developed to specify the **absolute configuration** of the four substituents of asymmetric carbon atoms. The absolute configurations of simple sugars and amino acids are specified by the **D, L system** (Fig. 3–4), based on the absolute configuration of the three-carbon sugar glyceraldehyde, a convention proposed by Emil Fischer in 1891. (Fischer knew what groups surrounded the asymmetric carbon of glyceraldehyde but had to guess at their absolute configuration; his guess was later confirmed by x-ray diffraction analysis.) For all chiral compounds, stereoisomers having a configuration related to that of L-glyceraldehyde are designated L, and stereoisomers related to D-glyceraldehyde are designated D. The functional groups of L-alanine are matched with those of L-glyceraldehyde by aligning those that can be interconverted by simple, one-step chemical reactions. Thus the carboxyl group of L-alanine occupies the same position about the chiral carbon as does the aldehyde group of L-glyceraldehyde, because an aldehyde is readily converted to a carboxyl group via a one-step oxidation. Historically, the similar *l* and *d* designations were used for levorotatory (rotating light to the left) and dextrorotatory (rotating light to the right). However, not all

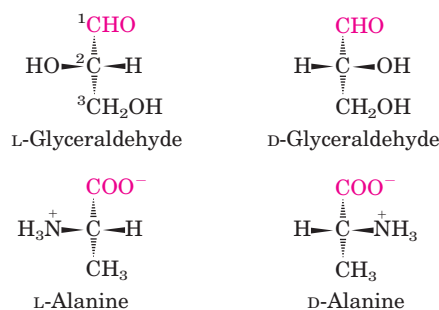


FIGURE 3–4 Steric relationship of the stereoisomers of alanine to the absolute configuration of L- and D-glyceraldehyde. In these perspective formulas, the carbons are lined up vertically, with the chiral atom in the center. The carbons in these molecules are numbered beginning with the terminal aldehyde or carboxyl carbon (red), 1 to 3 from top to bottom as shown. When presented in this way, the R group of the amino acid (in this case the methyl group of alanine) is always below the α carbon. L-Amino acids are those with the α -amino group on the left, and D-amino acids have the α -amino group on the right.

L-amino acids are levorotatory, and the convention shown in Figure 3–4 was needed to avoid potential ambiguities about absolute configuration. By Fischer's convention, L and D refer *only* to the absolute configuration of the four substituents around the chiral carbon, not to optical properties of the molecule.

Another system of specifying configuration around a chiral center is the **RS system**, which is used in the systematic nomenclature of organic chemistry and describes more precisely the configuration of molecules with more than one chiral center (see p. 18).

The Amino Acid Residues in Proteins Are L Stereoisomers

Nearly all biological compounds with a chiral center occur naturally in only one stereoisomeric form, either D or L. The amino acid residues in protein molecules are exclusively L stereoisomers. D-Amino acid residues have been found only in a few, generally small peptides, including some peptides of bacterial cell walls and certain peptide antibiotics.

It is remarkable that virtually all amino acid residues in proteins are L stereoisomers. When chiral compounds are formed by ordinary chemical reactions, the result is a racemic mixture of D and L isomers, which are difficult for a chemist to distinguish and separate. But to a living system, D and L isomers are as different as the right hand and the left. The formation of stable, repeating substructures in proteins (Chapter 4) generally requires that their constituent amino acids be of one stereochemical series. Cells are able to specifically synthesize the L isomers of amino acids because the active sites of enzymes are asymmetric, causing the reactions they catalyze to be stereospecific.

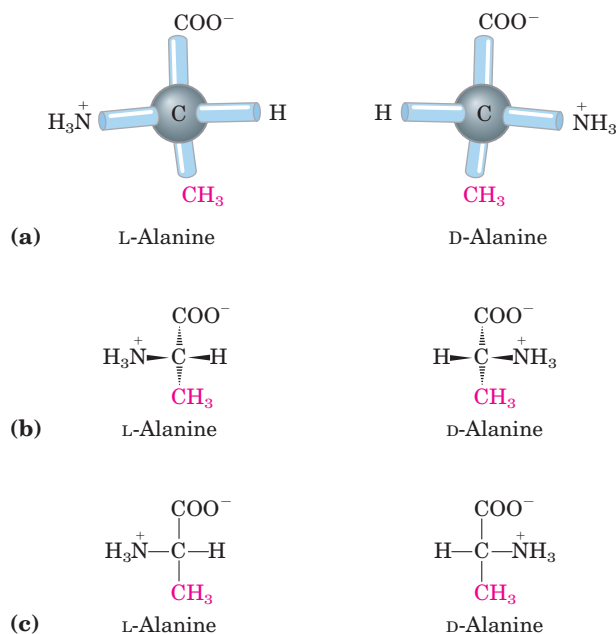


FIGURE 3–3 Stereoisomerism in α -amino acids. (a) The two stereoisomers of alanine, L- and D-alanine, are nonsuperimposable mirror images of each other (enantiomers). (b, c) Two different conventions for showing the configurations in space of stereoisomers. In perspective formulas (b) the solid wedge-shaped bonds project out of the plane of the paper, the dashed bonds behind it. In projection formulas (c) the horizontal bonds are assumed to project out of the plane of the paper, the vertical bonds behind. However, projection formulas are often used casually and are not always intended to portray a specific stereochemical configuration.

TABLE 3-1 Properties and Conventions Associated with the Common Amino Acids Found in Proteins

Amino acid	Abbreviation/ symbol	M_r	pK_a values			pI	Hydropathy index*	Occurrence in proteins (%) [†]
			pK_1 (—COOH)	pK_2 (—NH ₃ ⁺)	pK_R (R group)			
Nonpolar, aliphatic								
R groups								
Glycine	Gly G	75	2.34	9.60		5.97	−0.4	7.2
Alanine	Ala A	89	2.34	9.69		6.01	1.8	7.8
Proline	Pro P	115	1.99	10.96		6.48	1.6	5.2
Valine	Val V	117	2.32	9.62		5.97	4.2	6.6
Leucine	Leu L	131	2.36	9.60		5.98	3.8	9.1
Isoleucine	Ile I	131	2.36	9.68		6.02	4.5	5.3
Methionine	Met M	149	2.28	9.21		5.74	1.9	2.3
Aromatic R groups								
Phenylalanine	Phe F	165	1.83	9.13		5.48	2.8	3.9
Tyrosine	Tyr Y	181	2.20	9.11	10.07	5.66	−1.3	3.2
Tryptophan	Trp W	204	2.38	9.39		5.89	−0.9	1.4
Polar, uncharged								
R groups								
Serine	Ser S	105	2.21	9.15		5.68	−0.8	6.8
Threonine	Thr T	119	2.11	9.62		5.87	−0.7	5.9
Cysteine	Cys C	121	1.96	10.28	8.18	5.07	2.5	1.9
Asparagine	Asn N	132	2.02	8.80		5.41	−3.5	4.3
Glutamine	Gln Q	146	2.17	9.13		5.65	−3.5	4.2
Positively charged								
R groups								
Lysine	Lys K	146	2.18	8.95	10.53	9.74	−3.9	5.9
Histidine	His H	155	1.82	9.17	6.00	7.59	−3.2	2.3
Arginine	Arg R	174	2.17	9.04	12.48	10.76	−4.5	5.1
Negatively charged								
R groups								
Aspartate	Asp D	133	1.88	9.60	3.65	2.77	−3.5	5.3
Glutamate	Glu E	147	2.19	9.67	4.25	3.22	−3.5	6.3

*A scale combining hydrophobicity and hydrophilicity of R groups; it can be used to measure the tendency of an amino acid to seek an aqueous environment (− values) or a hydrophobic environment (+ values). See Chapter 11. From Kyte, J. & Doolittle, R.F. (1982) A simple method for displaying the hydropathic character of a protein. *J. Mol. Biol.* **157**, 105–132.

[†]Average occurrence in more than 1,150 proteins. From Doolittle, R.F. (1989) Redundancies in protein sequences. In *Prediction of Protein Structure and the Principles of Protein Conformation* (Fasman, G.D., ed.), pp. 599–623, Plenum Press, New York.

Amino Acids Can Be Classified by R Group

Knowledge of the chemical properties of the common amino acids is central to an understanding of biochemistry. The topic can be simplified by grouping the amino acids into five main classes based on the properties of their R groups (Table 3–1), in particular, their **polarity**, or tendency to interact with water at biological pH (near pH 7.0). The polarity of the R groups varies widely, from nonpolar and hydrophobic (water-insoluble) to highly polar and hydrophilic (water-soluble).

The structures of the 20 common amino acids are shown in Figure 3–5, and some of their properties are

listed in Table 3–1. Within each class there are gradations of polarity, size, and shape of the R groups.

Nonpolar, Aliphatic R Groups The R groups in this class of amino acids are nonpolar and hydrophobic. The side chains of **alanine**, **valine**, **leucine**, and **isoleucine** tend to cluster together within proteins, stabilizing protein structure by means of hydrophobic interactions. **Glycine** has the simplest structure. Although it is formally nonpolar, its very small side chain makes no real contribution to hydrophobic interactions. **Methionine**, one of the two sulfur-containing amino acids, has a nonpolar thioether group in its side chain. **Proline** has an

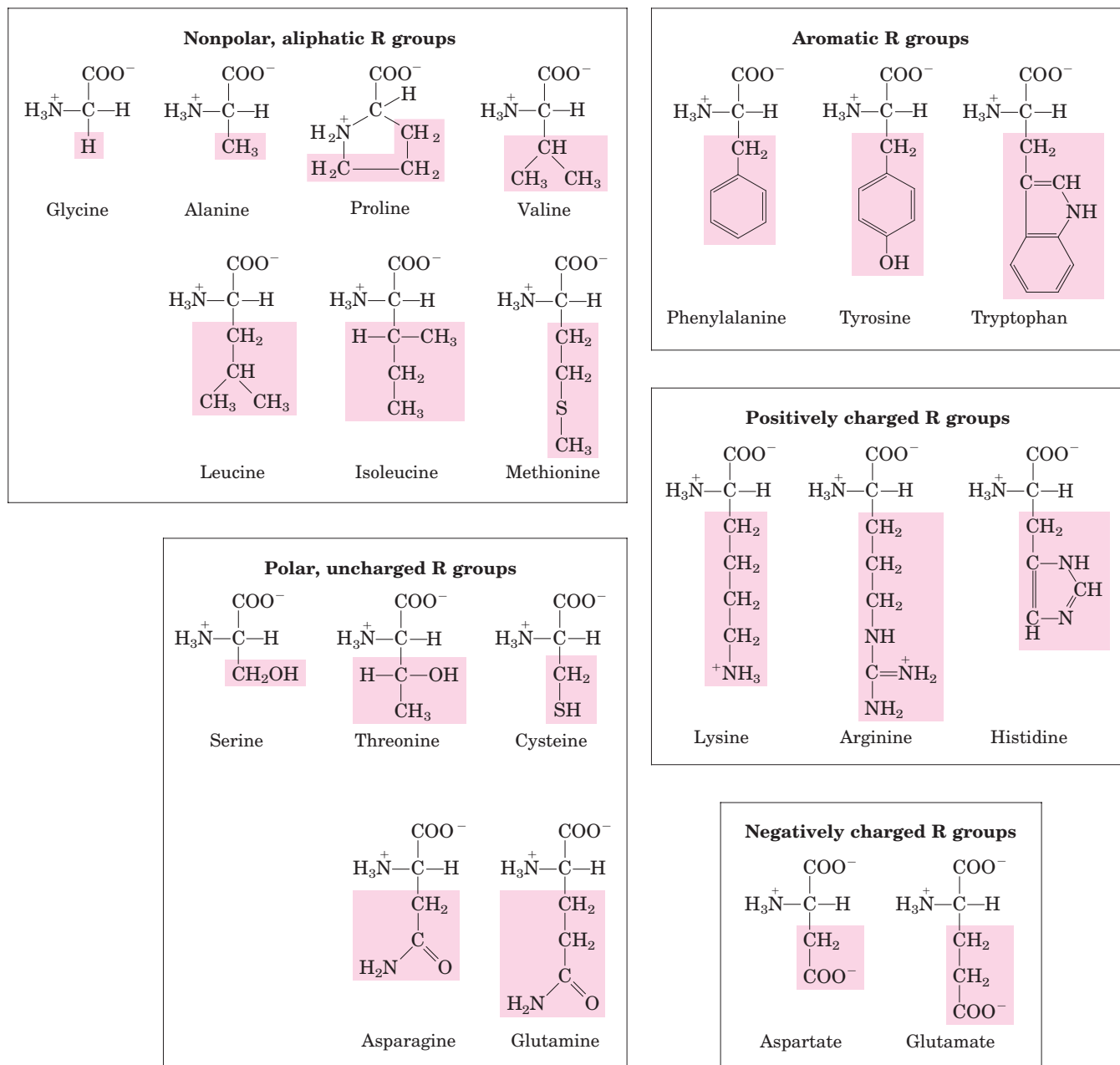


FIGURE 3-5 The 20 common amino acids of proteins. The structural formulas show the state of ionization that would predominate at pH 7.0. The unshaded portions are those common to all the amino acids; the portions shaded in red are the R groups. Although the R group of

histidine is shown uncharged, its pK_a (see Table 3-1) is such that a small but significant fraction of these groups are positively charged at pH 7.0.

aliphatic side chain with a distinctive cyclic structure. The secondary amino (imino) group of proline residues is held in a rigid conformation that reduces the structural flexibility of polypeptide regions containing proline.

Aromatic R Groups Phenylalanine, tyrosine, and tryptophan, with their aromatic side chains, are relatively nonpolar (hydrophobic). All can participate in hydrophobic interactions. The hydroxyl group of tyrosine can form hydrogen bonds, and it is an important func-

tional group in some enzymes. Tyrosine and tryptophan are significantly more polar than phenylalanine, because of the tyrosine hydroxyl group and the nitrogen of the tryptophan indole ring.

Tryptophan and tyrosine, and to a much lesser extent phenylalanine, absorb ultraviolet light (Fig. 3-6; Box 3-1). This accounts for the characteristic strong absorbance of light by most proteins at a wavelength of 280 nm, a property exploited by researchers in the characterization of proteins.

Polar, Uncharged R Groups The R groups of these amino acids are more soluble in water, or more hydrophilic, than those of the nonpolar amino acids, because they contain functional groups that form hydrogen bonds with water. This class of amino acids includes **serine**, **threonine**, **cysteine**, **asparagine**, and **glutamine**. The polarity of serine and threonine is contributed by their hydroxyl groups; that of cysteine by its sulfhydryl group; and that of asparagine and glutamine by their amide groups.

Asparagine and glutamine are the amides of two other amino acids also found in proteins, aspartate and glutamate, respectively, to which asparagine and glutamine are easily hydrolyzed by acid or base. Cysteine is readily oxidized to form a covalently linked dimeric amino acid called **cystine**, in which two cysteine molecules or residues are joined by a disulfide bond (Fig. 3-7). The disulfide-linked residues are strongly hydrophobic (nonpolar). Disulfide bonds play a special role in the structures of many proteins by forming covalent links between parts of a protein molecule or between two different polypeptide chains.

Positively Charged (Basic) R Groups The most hydrophilic R groups are those that are either positively or negatively charged. The amino acids in which the R groups have significant positive charge at pH 7.0 are **lysine**, which has a second primary amino group at the ϵ posi-

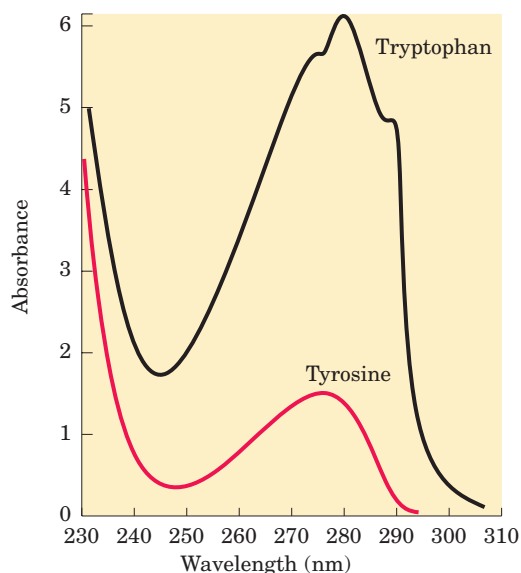


FIGURE 3-6 Absorption of ultraviolet light by aromatic amino acids. Comparison of the light absorption spectra of the aromatic amino acids tryptophan and tyrosine at pH 6.0. The amino acids are present in equimolar amounts (10^{-3} M) under identical conditions. The measured absorbance of tryptophan is as much as four times that of tyrosine. Note that the maximum light absorption for both tryptophan and tyrosine occurs near a wavelength of 280 nm. Light absorption by the third aromatic amino acid, phenylalanine (not shown), generally contributes little to the spectroscopic properties of proteins.

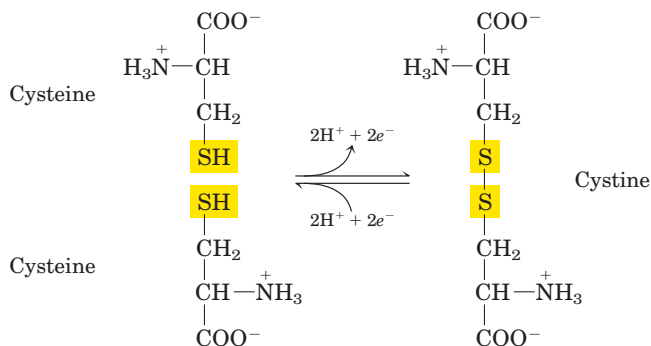


FIGURE 3-7 Reversible formation of a disulfide bond by the oxidation of two molecules of cysteine. Disulfide bonds between Cys residues stabilize the structures of many proteins.

tion on its aliphatic chain; **arginine**, which has a positively charged guanidino group; and **histidine**, which has an imidazole group. Histidine is the only common amino acid having an ionizable side chain with a pK_a near neutrality. In many enzyme-catalyzed reactions, a His residue facilitates the reaction by serving as a proton donor/acceptor.

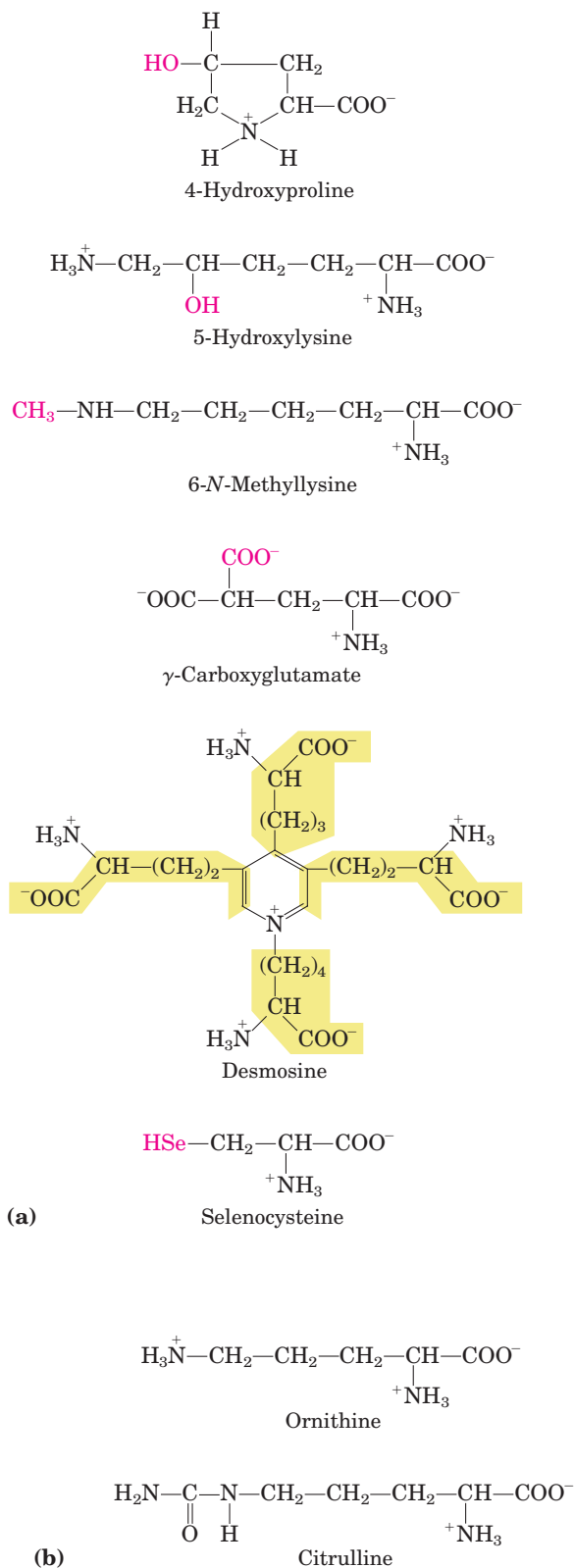
Negatively Charged (Acidic) R Groups The two amino acids having R groups with a net negative charge at pH 7.0 are **aspartate** and **glutamate**, each of which has a second carboxyl group.

Uncommon Amino Acids Also Have Important Functions

In addition to the 20 common amino acids, proteins may contain residues created by modification of common residues already incorporated into a polypeptide (Fig. 3-8a). Among these uncommon amino acids are **4-hydroxyproline**, a derivative of proline, and **5-hydroxylysine**, derived from lysine. The former is found in plant cell wall proteins, and both are found in collagen, a fibrous protein of connective tissues. **6-N-Methyllysine** is a constituent of myosin, a contractile protein of muscle. Another important uncommon amino acid is **γ -carboxyglutamate**, found in the blood-clotting protein prothrombin and in certain other proteins that bind Ca^{2+} as part of their biological function. More complex is **desmosine**, a derivative of four Lys residues, which is found in the fibrous protein elastin.

Selenocysteine is a special case. This rare amino acid residue is introduced during protein synthesis rather than created through a postsynthetic modification. It contains selenium rather than the sulfur of cysteine. Actually derived from serine, selenocysteine is a constituent of just a few known proteins.

Some 300 additional amino acids have been found in cells. They have a variety of functions but are not constituents of proteins. **Ornithine** and **citrulline**



(Fig. 3-8b) deserve special note because they are key intermediates (metabolites) in the biosynthesis of arginine (Chapter 22) and in the urea cycle (Chapter 18).

FIGURE 3-8 Uncommon amino acids. (a) Some uncommon amino acids found in proteins. All are derived from common amino acids. Extra functional groups added by modification reactions are shown in red. Desmosine is formed from four Lys residues (the four carbon backbones are shaded in yellow). Note the use of either numbers or Greek letters to identify the carbon atoms in these structures. (b) Ornithine and citrulline, which are not found in proteins, are intermediates in the biosynthesis of arginine and in the urea cycle.

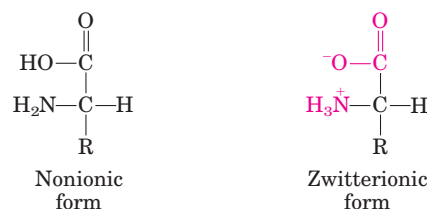
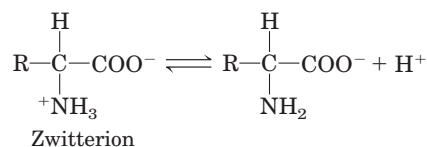


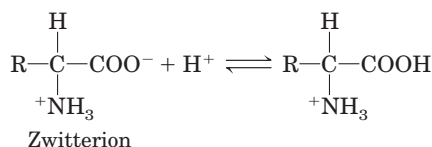
FIGURE 3-9 Nonionic and zwitterionic forms of amino acids. The nonionic form does not occur in significant amounts in aqueous solutions. The zwitterion predominates at neutral pH.

Amino Acids Can Act as Acids and Bases

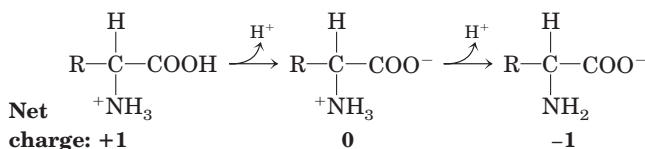
When an amino acid is dissolved in water, it exists in solution as the dipolar ion, or **zwitterion** (German for “hybrid ion”), shown in Figure 3-9. A zwitterion can act as either an acid (proton donor):



or a base (proton acceptor):



Substances having this dual nature are **amphoteric** and are often called **ampholytes** (from “amphoteric electrolytes”). A simple monoamino monocarboxylic α -amino acid, such as alanine, is a diprotic acid when fully protonated—it has two groups, the $-\text{COOH}$ group and the $-\text{NH}_3^+$ group, that can yield protons:



BOX 3-1 WORKING IN BIOCHEMISTRY

**Absorption of Light by Molecules:
The Lambert-Beer Law**

A wide range of biomolecules absorb light at characteristic wavelengths, just as tryptophan absorbs light at 280 nm (see Fig. 3-6). Measurement of light absorption by a spectrophotometer is used to detect and identify molecules and to measure their concentration in solution. The fraction of the incident light absorbed by a solution at a given wavelength is related to the thickness of the absorbing layer (path length) and the concentration of the absorbing species (Fig. 1). These two relationships are combined into the Lambert-Beer law,

$$\log \frac{I_0}{I} = \epsilon cl$$

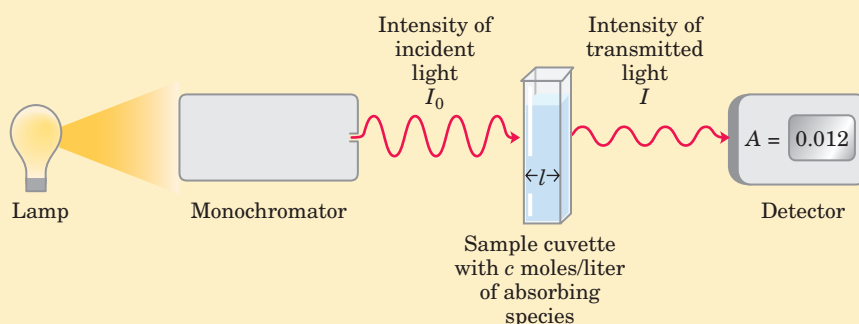
where I_0 is the intensity of the incident light, I is the intensity of the transmitted light, ϵ is the molar extinction coefficient (in units of liters per mole-centimeter), c is the concentration of the absorbing species (in

moles per liter), and l is the path length of the light-absorbing sample (in centimeters). The Lambert-Beer law assumes that the incident light is parallel and monochromatic (of a single wavelength) and that the solvent and solute molecules are randomly oriented. The expression $\log (I_0/I)$ is called the **absorbance**, designated A .

It is important to note that each successive millimeter of path length of absorbing solution in a 1.0 cm cell absorbs not a constant amount but a constant fraction of the light that is incident upon it. However, with an absorbing layer of fixed path length, *the absorbance, A , is directly proportional to the concentration of the absorbing solute.*

The molar extinction coefficient varies with the nature of the absorbing compound, the solvent, and the wavelength, and also with pH if the light-absorbing species is in equilibrium with an ionization state that has different absorbance properties.

FIGURE 1 The principal components of a spectrophotometer. A light source emits light along a broad spectrum, then the monochromator selects and transmits light of a particular wavelength. The monochromatic light passes through the sample in a cuvette of path length l and is absorbed by the sample in proportion to the concentration of the absorbing species. The transmitted light is measured by a detector.

**Amino Acids Have Characteristic Titration Curves**

Acid-base titration involves the gradual addition or removal of protons (Chapter 2). Figure 3-10 shows the titration curve of the diprotic form of glycine. The plot has two distinct stages, corresponding to deprotonation of two different groups on glycine. Each of the two stages resembles in shape the titration curve of a monoprotic acid, such as acetic acid (see Fig. 2-17), and can be analyzed in the same way. At very low pH, the predominant ionic species of glycine is the fully protonated form, $^+\text{H}_3\text{N}-\text{CH}_2-\text{COOH}$. At the midpoint in the first stage of the titration, in which the $-\text{COOH}$ group of glycine loses its proton, equimolar concentrations of the proton-donor ($^+\text{H}_3\text{N}-\text{CH}_2-\text{COOH}$) and proton-acceptor ($^+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^-$) species are present. At the midpoint of any titration, a point of inflection is reached where the pH is equal to the pK_a of the protonated group being titrated (see Fig. 2-18). For glycine, the pH at the midpoint is 2.34, thus its $-\text{COOH}$ group has a pK_a (labeled pK_1 in Fig. 3-10) of 2.34.

(Recall from Chapter 2 that pH and pK_a are simply convenient notations for proton concentration and the equilibrium constant for ionization, respectively. The pK_a is a measure of the tendency of a group to give up a proton, with that tendency decreasing tenfold as the pK_a increases by one unit.) As the titration proceeds, another important point is reached at pH 5.97. Here there is another point of inflection, at which removal of the first proton is essentially complete and removal of the second has just begun. At this pH glycine is present largely as the dipolar ion $^+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^-$. We shall return to the significance of this inflection point in the titration curve (labeled pI in Fig. 3-10) shortly.

The second stage of the titration corresponds to the removal of a proton from the $-\text{NH}_3^+$ group of glycine. The pH at the midpoint of this stage is 9.60, equal to the pK_a (labeled pK_2 in Fig. 3-10) for the $-\text{NH}_3^+$ group. The titration is essentially complete at a pH of about 12, at which point the predominant form of glycine is $\text{H}_2\text{N}-\text{CH}_2-\text{COO}^-$.

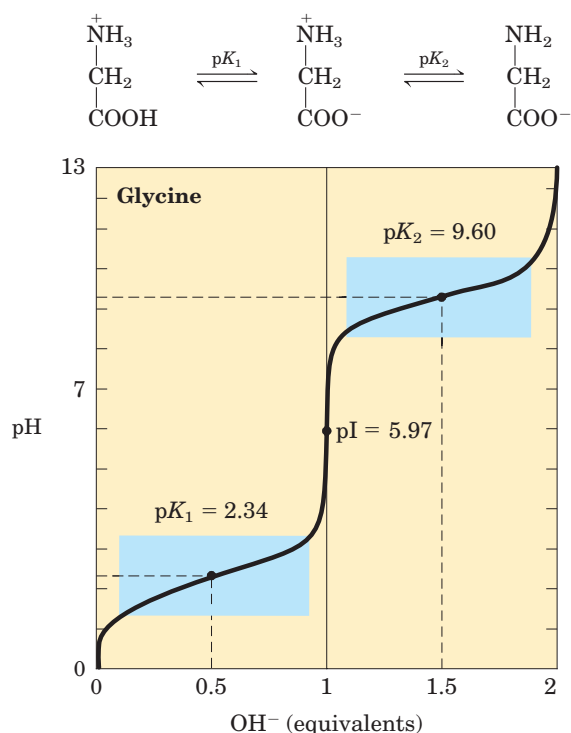


FIGURE 3-10 Titration of an amino acid. Shown here is the titration curve of 0.1 M glycine at 25 °C. The ionic species predominating at key points in the titration are shown above the graph. The shaded boxes, centered at about $pK_1 = 2.34$ and $pK_2 = 9.60$, indicate the regions of greatest buffering power.

From the titration curve of glycine we can derive several important pieces of information. First, it gives a quantitative measure of the pK_a of each of the two ionizing groups: 2.34 for the $-\text{COOH}$ group and 9.60 for the $-\text{NH}_3^+$ group. Note that the carboxyl group of glycine is over 100 times more acidic (more easily ionized) than the carboxyl group of acetic acid, which, as we saw in Chapter 2, has a pK_a of 4.76—about average for a carboxyl group attached to an otherwise unsubstituted aliphatic hydrocarbon. The perturbed pK_a of glycine is caused by repulsion between the departing proton and the nearby positively charged amino group on the α -carbon atom, as described in Figure 3-11. The opposite charges on the resulting zwitterion are stabilizing, nudging the equilibrium farther to the right. Similarly, the pK_a of the amino group in glycine is perturbed downward relative to the average pK_a of an amino group. This effect is due partly to the electronegative oxygen atoms in the carboxyl groups, which tend to pull electrons toward them, increasing the tendency of the amino group to give up a proton. Hence, the α -amino group has a pK_a that is lower than that of an aliphatic amine such as methylamine (Fig. 3-11). In short, the pK_a of any functional group is greatly affected by its chemical environment, a phenomenon sometimes exploited in the active sites of enzymes to promote exquisitely adapted reaction mechanisms that depend on the perturbed pK_a values of proton donor/acceptor groups of specific residues.

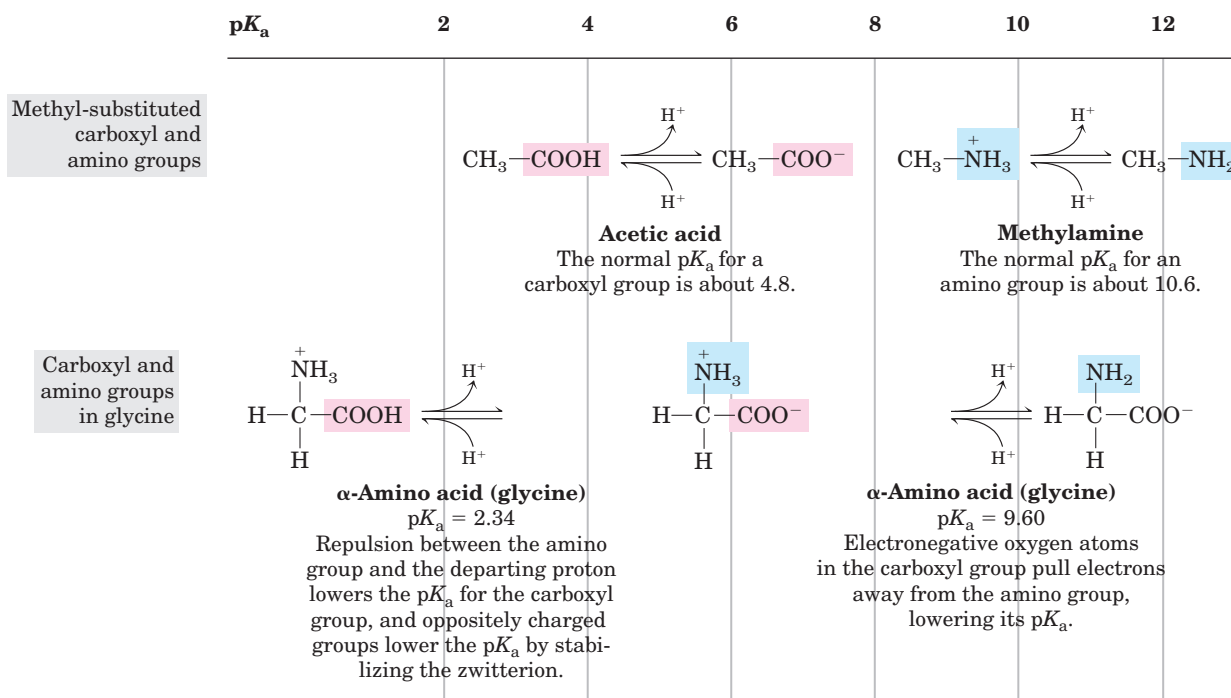


FIGURE 3-11 Effect of the chemical environment on pK_a . The pK_a values for the ionizable groups in glycine are lower than those for simple, methyl-substituted amino and carboxyl groups. These downward

perturbations of pK_a are due to intramolecular interactions. Similar effects can be caused by chemical groups that happen to be positioned nearby—for example, in the active site of an enzyme.

The second piece of information provided by the titration curve of glycine is that this amino acid has *two* regions of buffering power. One of these is the relatively flat portion of the curve, extending for approximately 1 pH unit on either side of the first pK_a of 2.34, indicating that glycine is a good buffer near this pH. The other buffering zone is centered around pH 9.60. (Note that glycine is not a good buffer at the pH of intracellular fluid or blood, about 7.4.) Within the buffering ranges of glycine, the Henderson-Hasselbalch equation (see Box 2–3) can be used to calculate the proportions of proton-donor and proton-acceptor species of glycine required to make a buffer at a given pH.

Titration Curves Predict the Electric Charge of Amino Acids

Another important piece of information derived from the titration curve of an amino acid is the relationship between its net electric charge and the pH of the solution. At pH 5.97, the point of inflection between the two stages in its titration curve, glycine is present predominantly as its dipolar form, fully ionized but with no *net* electric charge (Fig. 3–10). The characteristic pH at which the net electric charge is zero is called the **isoelectric point** or **isoelectric pH**, designated **pI**. For glycine, which has no ionizable group in its side chain, the isoelectric point is simply the arithmetic mean of the two pK_a values:

$$pI = \frac{1}{2} (pK_1 + pK_2) = \frac{1}{2} (2.34 + 9.60) = 5.97$$

As is evident in Figure 3–10, glycine has a net negative charge at any pH above its pI and will thus move toward the positive electrode (the anode) when placed in an electric field. At any pH below its pI, glycine has a net positive charge and will move toward the negative electrode (the cathode). The farther the pH of a glycine solution is from its isoelectric point, the greater the net electric charge of the population of glycine molecules. At pH 1.0, for example, glycine exists almost entirely as the form $^+H_3N-CH_2-COOH$, with a net positive charge of 1.0. At pH 2.34, where there is an equal mixture of $^+H_3N-CH_2-COOH$ and $^+H_3N-CH_2-COO^-$, the average or net positive charge is 0.5. The sign and the magnitude of the net charge of any amino acid at any pH can be predicted in the same way.

Amino Acids Differ in Their Acid-Base Properties

The shared properties of many amino acids permit some simplifying generalizations about their acid-base behaviors. First, all amino acids with a single α -amino group, a single α -carboxyl group, and an R group that does not ionize have titration curves resembling that of glycine (Fig. 3–10). These amino acids have very similar, although not identical, pK_a values: pK_a of the $-COOH$

group in the range of 1.8 to 2.4, and pK_a of the $-NH_3^+$ group in the range of 8.8 to 11.0 (Table 3–1).

Second, amino acids with an ionizable R group have more complex titration curves, with *three* stages corresponding to the three possible ionization steps; thus they have three pK_a values. The additional stage for the titration of the ionizable R group merges to some extent with the other two. The titration curves for two amino acids of this type, glutamate and histidine, are shown in Figure 3–12. The isoelectric points reflect the nature of the ionizing R groups present. For example, glutamate

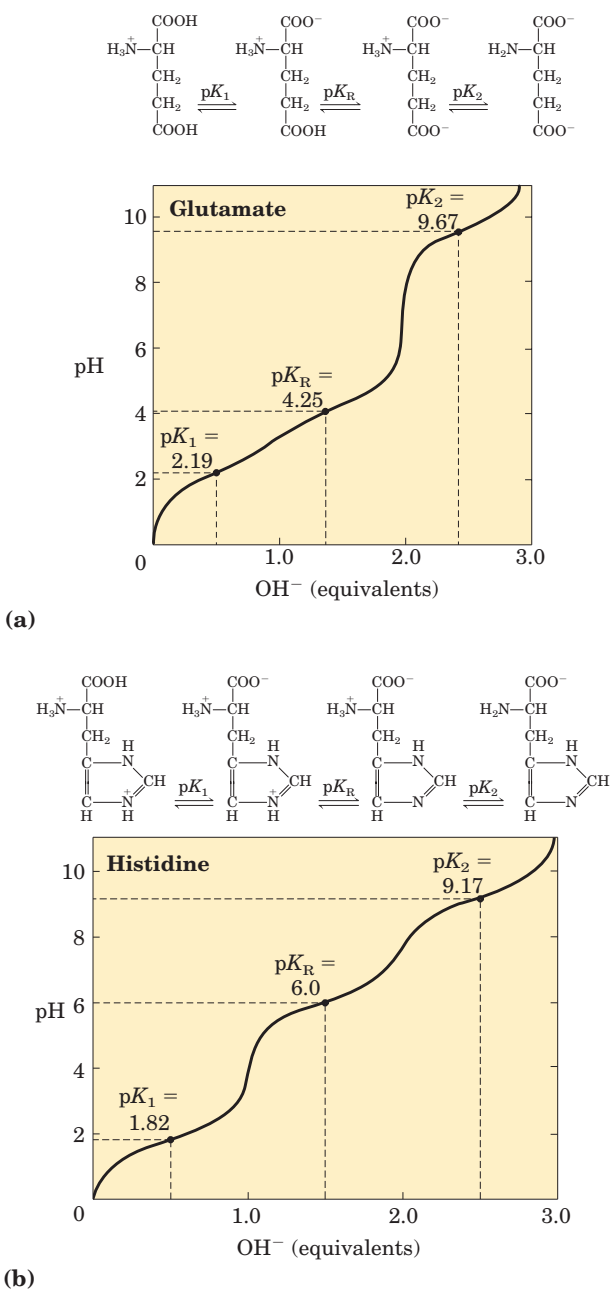


FIGURE 3–12 Titration curves for (a) glutamate and (b) histidine. The pK_a of the R group is designated here as pK_R .

has a pI of 3.22, considerably lower than that of glycine. This is due to the presence of two carboxyl groups, which, at the average of their pK_a values (3.22), contribute a net charge of -1 that balances the $+1$ contributed by the amino group. Similarly, the pI of histidine, with two groups that are positively charged when protonated, is 7.59 (the average of the pK_a values of the amino and imidazole groups), much higher than that of glycine.

Finally, as pointed out earlier, under the general condition of free and open exposure to the aqueous environment, only histidine has an R group ($pK_a = 6.0$) providing significant buffering power near the neutral pH usually found in the intracellular and extracellular fluids of most animals and bacteria (Table 3–1).

SUMMARY 3.1 Amino Acids

- The 20 amino acids commonly found as residues in proteins contain an α -carboxyl group, an α -amino group, and a distinctive R group substituted on the α -carbon atom. The α -carbon atom of all amino acids except glycine is asymmetric, and thus amino acids can exist in at least two stereoisomeric forms. Only the L stereoisomers, with a configuration related to the absolute configuration of the reference molecule L-glyceraldehyde, are found in proteins.
- Other, less common amino acids also occur, either as constituents of proteins (through modification of common amino acid residues after protein synthesis) or as free metabolites.
- Amino acids are classified into five types on the basis of the polarity and charge (at pH 7) of their R groups.
- Amino acids vary in their acid-base properties and have characteristic titration curves. Monoamino monocarboxylic amino acids (with nonionizable R groups) are diprotic acids ($^+H_3NCH(R)COOH$) at low pH and exist in several different ionic forms as the pH is increased. Amino acids with ionizable R groups have additional ionic species, depending on the pH of the medium and the pK_a of the R group.

3.2 Peptides and Proteins

We now turn to polymers of amino acids, the **peptides** and **proteins**. Biologically occurring polypeptides range in size from small to very large, consisting of two or three to thousands of linked amino acid residues. Our focus is on the fundamental chemical properties of these polymers.

Peptides Are Chains of Amino Acids

Two amino acid molecules can be covalently joined through a substituted amide linkage, termed a **peptide bond**, to yield a dipeptide. Such a linkage is formed by removal of the elements of water (dehydration) from the α -carboxyl group of one amino acid and the α -amino group of another (Fig. 3–13). Peptide bond formation is an example of a condensation reaction, a common class of reactions in living cells. Under standard biochemical conditions, the equilibrium for the reaction shown in Figure 3–13 favors the amino acids over the dipeptide. To make the reaction thermodynamically more favorable, the carboxyl group must be chemically modified or activated so that the hydroxyl group can be more readily eliminated. A chemical approach to this problem is outlined later in this chapter. The biological approach to peptide bond formation is a major topic of Chapter 27.

Three amino acids can be joined by two peptide bonds to form a tripeptide; similarly, amino acids can be linked to form tetrapeptides, pentapeptides, and so forth. When a few amino acids are joined in this fashion, the structure is called an **oligopeptide**. When many amino acids are joined, the product is called a **polypeptide**. Proteins may have thousands of amino acid residues. Although the terms “protein” and “polypeptide” are sometimes used interchangeably, molecules referred to as polypeptides generally have molecular weights below 10,000, and those called proteins have higher molecular weights.

Figure 3–14 shows the structure of a pentapeptide. As already noted, an amino acid unit in a peptide is often called a residue (the part left over after losing a hydrogen atom from its amino group and the hydroxyl moiety from its carboxyl group). In a peptide, the amino acid residue at the end with a free α -amino group is the **amino-terminal** (or *N*-terminal) residue; the residue

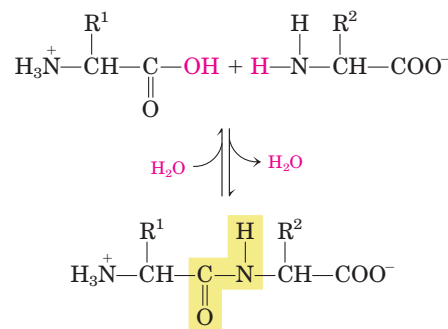


FIGURE 3–13 Formation of a peptide bond by condensation. The α -amino group of one amino acid (with R^2 group) acts as a nucleophile to displace the hydroxyl group of another amino acid (with R^1 group), forming a peptide bond (shaded in yellow). Amino groups are good nucleophiles, but the hydroxyl group is a poor leaving group and is not readily displaced. At physiological pH, the reaction shown does not occur to any appreciable extent.

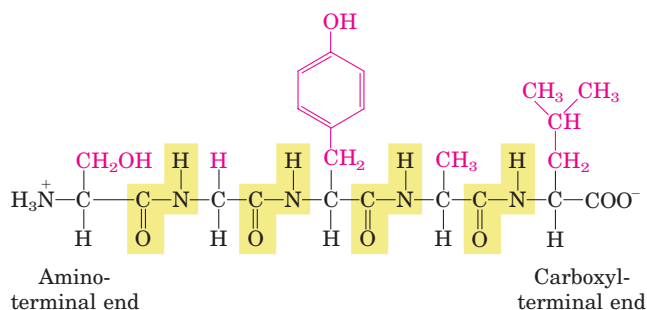


FIGURE 3-14 The pentapeptide serylglycyltyrosylalanylleucine, or Ser-Gly-Tyr-Ala-Leu. Peptides are named beginning with the amino-terminal residue, which by convention is placed at the left. The peptide bonds are shaded in yellow; the R groups are in red.

at the other end, which has a free carboxyl group, is the **carboxyl-terminal** (C-terminal) residue.

Although hydrolysis of a peptide bond is an exergonic reaction, it occurs slowly because of its high activation energy. As a result, the peptide bonds in proteins are quite stable, with an average half-life ($t_{1/2}$) of about 7 years under most intracellular conditions.

Peptides Can Be Distinguished by Their Ionization Behavior

Peptides contain only one free α -amino group and one free α -carboxyl group, at opposite ends of the chain (Fig. 3-15). These groups ionize as they do in free amino acids, although the ionization constants are different because an oppositely charged group is no longer linked to the α carbon. The α -amino and α -carboxyl groups of all nonterminal amino acids are covalently joined in the peptide bonds, which do not ionize and thus do not contribute to the total acid-base behavior of peptides. How-

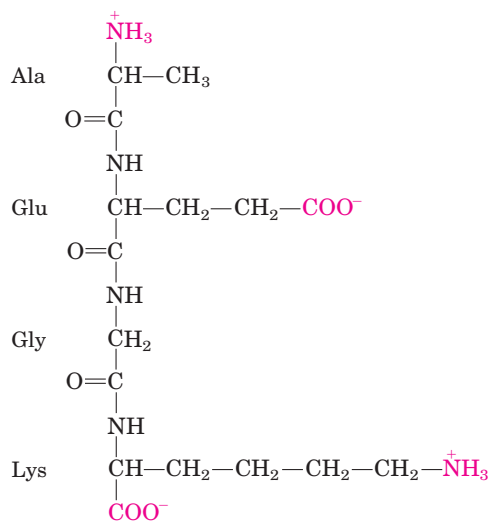


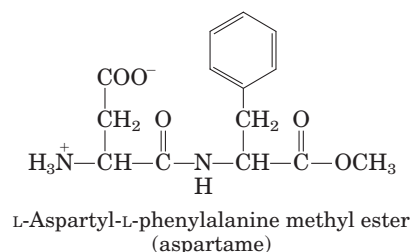
FIGURE 3-15 Alanylglutamylglycyllysine. This tetrapeptide has one free α -amino group, one free α -carboxyl group, and two ionizable R groups. The groups ionized at pH 7.0 are in red.

ever, the R groups of some amino acids can ionize (Table 3-1), and in a peptide these contribute to the overall acid-base properties of the molecule (Fig. 3-15). Thus the acid-base behavior of a peptide can be predicted from its free α -amino and α -carboxyl groups as well as the nature and number of its ionizable R groups.

Like free amino acids, peptides have characteristic titration curves and a characteristic isoelectric pH (pI) at which they do not move in an electric field. These properties are exploited in some of the techniques used to separate peptides and proteins, as we shall see later in the chapter. It should be emphasized that the pK_a value for an ionizable R group can change somewhat when an amino acid becomes a residue in a peptide. The loss of charge in the α -carboxyl and α -amino groups, the interactions with other peptide R groups, and other environmental factors can affect the pK_a . The pK_a values for R groups listed in Table 3-1 can be a useful guide to the pH range in which a given group will ionize, but they cannot be strictly applied to peptides.

Biologically Active Peptides and Polypeptides Occur in a Vast Range of Sizes

No generalizations can be made about the molecular weights of biologically active peptides and proteins in relation to their functions. Naturally occurring peptides range in length from two to many thousands of amino acid residues. Even the smallest peptides can have biologically important effects. Consider the commercially synthesized dipeptide L-aspartyl-L-phenylalanine methyl ester, the artificial sweetener better known as aspartame or NutraSweet.



Many small peptides exert their effects at very low concentrations. For example, a number of vertebrate hormones (Chapter 23) are small peptides. These include oxytocin (nine amino acid residues), which is secreted by the posterior pituitary and stimulates uterine contractions; bradykinin (nine residues), which inhibits inflammation of tissues; and thyrotropin-releasing factor (three residues), which is formed in the hypothalamus and stimulates the release of another hormone, thyrotropin, from the anterior pituitary gland. Some extremely toxic mushroom poisons, such as amanitin, are also small peptides, as are many antibiotics.

Slightly larger are small polypeptides and oligopeptides such as the pancreatic hormone insulin, which contains two polypeptide chains, one having 30 amino acid

residues and the other 21. Glucagon, another pancreatic hormone, has 29 residues; it opposes the action of insulin. Corticotropin is a 39-residue hormone of the anterior pituitary gland that stimulates the adrenal cortex.

How long are the polypeptide chains in proteins? As Table 3-2 shows, lengths vary considerably. Human cytochrome *c* has 104 amino acid residues linked in a single chain; bovine chymotrypsinogen has 245 residues. At the extreme is titin, a constituent of vertebrate muscle, which has nearly 27,000 amino acid residues and a molecular weight of about 3,000,000. The vast majority of naturally occurring proteins are much smaller than this, containing fewer than 2,000 amino acid residues.

Some proteins consist of a single polypeptide chain, but others, called **multisubunit** proteins, have two or more polypeptides associated noncovalently (Table 3-2). The individual polypeptide chains in a multisubunit protein may be identical or different. If at least two are identical the protein is said to be **oligomeric**, and the identical units (consisting of one or more polypeptide chains) are referred to as **protomers**. Hemoglobin, for example, has four polypeptide subunits: two identical α chains and two identical β chains, all four held together by noncovalent interactions. Each α subunit is paired in an identical way with a β subunit within the structure of this multisubunit protein, so that hemoglobin can be considered either a tetramer of four polypeptide subunits or a dimer of $\alpha\beta$ protomers.

A few proteins contain two or more polypeptide chains linked covalently. For example, the two polypeptide chains of insulin are linked by disulfide bonds. In such cases, the individual polypeptides are not considered subunits but are commonly referred to simply as chains.

We can calculate the approximate number of amino acid residues in a simple protein containing no other

chemical constituents by dividing its molecular weight by 110. Although the average molecular weight of the 20 common amino acids is about 138, the smaller amino acids predominate in most proteins. If we take into account the proportions in which the various amino acids occur in proteins (Table 3-1), the average molecular weight of protein amino acids is nearer to 128. Because a molecule of water (M_r 18) is removed to create each peptide bond, the average molecular weight of an amino acid residue in a protein is about $128 - 18 = 110$.

Polypeptides Have Characteristic Amino Acid Compositions

Hydrolysis of peptides or proteins with acid yields a mixture of free α -amino acids. When completely hydrolyzed, each type of protein yields a characteristic proportion or mixture of the different amino acids. The 20 common amino acids almost never occur in equal amounts in a protein. Some amino acids may occur only once or not at all in a given type of protein; others may occur in large numbers. Table 3-3 shows the composition of the amino acid mixtures obtained on complete hydrolysis of bovine cytochrome *c* and chymotrypsinogen, the inactive precursor of the digestive enzyme chymotrypsin. These two proteins, with very different functions, also differ significantly in the relative numbers of each kind of amino acid they contain.

Complete hydrolysis alone is not sufficient for an exact analysis of amino acid composition, however, because some side reactions occur during the procedure. For example, the amide bonds in the side chains of asparagine and glutamine are cleaved by acid treatment, yielding aspartate and glutamate, respectively. The side chain of tryptophan is almost completely degraded by acid hydrolysis, and small amounts of serine, threonine,

TABLE 3-2 Molecular Data on Some Proteins

	Molecular weight	Number of residues	Number of polypeptide chains
Cytochrome <i>c</i> (human)	13,000	104	1
Ribonuclease A (bovine pancreas)	13,700	124	1
Lysozyme (chicken egg white)	13,930	129	1
Myoglobin (equine heart)	16,890	153	1
Chymotrypsin (bovine pancreas)	21,600	241	3
Chymotrypsinogen (bovine)	22,000	245	1
Hemoglobin (human)	64,500	574	4
Serum albumin (human)	68,500	609	1
Hexokinase (yeast)	102,000	972	2
RNA polymerase (<i>E. coli</i>)	450,000	4,158	5
Apolipoprotein B (human)	513,000	4,536	1
Glutamine synthetase (<i>E. coli</i>)	619,000	5,628	12
Titin (human)	2,993,000	26,926	1

TABLE 3-3 Amino Acid Composition of Two Proteins

Amino acid	Number of residues per molecule of protein*	
	Bovine cytochrome c	Bovine chymotrypsinogen
Ala	6	22
Arg	2	4
Asn	5	15
Asp	3	8
Cys	2	10
Gln	3	10
Glu	9	5
Gly	14	23
His	3	2
Ile	6	10
Leu	6	19
Lys	18	14
Met	2	2
Phe	4	6
Pro	4	9
Ser	1	28
Thr	8	23
Trp	1	8
Tyr	4	4
Val	3	23
Total	104	245

*In some common analyses, such as acid hydrolysis, Asp and Asn are not readily distinguished from each other and are together designated Asx (or B). Similarly, when Glu and Gln cannot be distinguished, they are together designated Glx (or Z). In addition, Trp is destroyed. Additional procedures must be employed to obtain an accurate assessment of complete amino acid content.

and tyrosine are also lost. When a precise amino acid composition is required, biochemists use additional procedures to resolve the ambiguities that remain from acid hydrolysis.

Some Proteins Contain Chemical Groups Other Than Amino Acids

Many proteins, for example the enzymes ribonuclease A and chymotrypsinogen, contain only amino acid residues and no other chemical constituents; these are considered simple proteins. However, some proteins contain permanently associated chemical components in addition to amino acids; these are called **conjugated proteins**. The non-amino acid part of a conjugated protein is usually called its **prosthetic group**. Conjugated proteins are classified on the basis of the chemical nature of their prosthetic groups (Table 3-4); for example, **lipoproteins** contain lipids, **glycoproteins** contain sugar groups, and **metalloproteins** contain a specific

TABLE 3-4 Conjugated Proteins

Class	Prosthetic group	Example
Lipoproteins	Lipids	β_1 -Lipoprotein of blood
Glycoproteins	Carbohydrates	Immunoglobulin G
Phosphoproteins	Phosphate groups	Casein of milk
Hemoproteins	Heme (iron porphyrin)	Hemoglobin
Flavoproteins	Flavin nucleotides	Succinate dehydrogenase
Metalloproteins	Iron	Ferritin
	Zinc	Alcohol dehydrogenase
	Calcium	Calmodulin
	Molybdenum	Dinitrogenase
	Copper	Plastocyanin

metal. A number of proteins contain more than one prosthetic group. Usually the prosthetic group plays an important role in the protein's biological function.

There Are Several Levels of Protein Structure

For large macromolecules such as proteins, the tasks of describing and understanding structure are approached at several levels of complexity, arranged in a kind of conceptual hierarchy. Four levels of protein structure are commonly defined (Fig. 3-16). A description of all covalent bonds (mainly peptide bonds and disulfide bonds) linking amino acid residues in a polypeptide chain is its **primary structure**. The most important element of primary structure is the *sequence* of amino acid residues. **Secondary structure** refers to particularly stable arrangements of amino acid residues giving rise to recurring structural patterns. **Tertiary structure** describes all aspects of the three-dimensional folding of a polypeptide. When a protein has two or more polypeptide subunits, their arrangement in space is referred to as **quaternary structure**. Primary structure is the focus of Section 3.4; the higher levels of structure are discussed in Chapter 4.

SUMMARY 3.2 Peptides and Proteins

- Amino acids can be joined covalently through peptide bonds to form peptides and proteins. Cells generally contain thousands of different proteins, each with a different biological activity.
- Proteins can be very long polypeptide chains of 100 to several thousand amino acid residues. However, some naturally occurring peptides have only a few amino acid residues. Some proteins are composed of several noncovalently

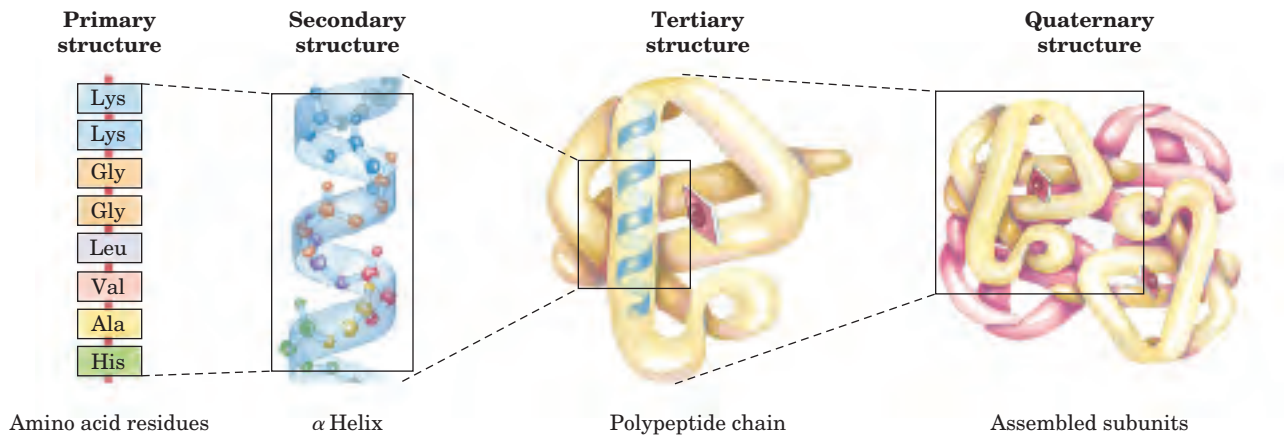


FIGURE 3-16 Levels of structure in proteins. The *primary structure* consists of a sequence of amino acids linked together by peptide bonds and includes any disulfide bonds. The resulting polypeptide can be coiled into units of *secondary structure*, such as an α helix. The he-

lix is a part of the *tertiary structure* of the folded polypeptide, which is itself one of the subunits that make up the *quaternary structure* of the multisubunit protein, in this case hemoglobin.

associated polypeptide chains, called subunits. Simple proteins yield only amino acids on hydrolysis; conjugated proteins contain in addition some other component, such as a metal or organic prosthetic group.

- The sequence of amino acids in a protein is characteristic of that protein and is called its primary structure. This is one of four generally recognized levels of protein structure.

3.3 Working with Proteins

Our understanding of protein structure and function has been derived from the study of many individual proteins. To study a protein in detail, the researcher must be able to separate it from other proteins and must have the techniques to determine its properties. The necessary methods come from protein chemistry, a discipline as old as biochemistry itself and one that retains a central position in biochemical research.

Proteins Can Be Separated and Purified

A pure preparation is essential before a protein's properties and activities can be determined. Given that cells contain thousands of different kinds of proteins, how can one protein be purified? Methods for separating proteins take advantage of properties that vary from one protein to the next, including size, charge, and binding properties.

The source of a protein is generally tissue or microbial cells. The first step in any protein purification procedure is to break open these cells, releasing their proteins into a solution called a **crude extract**. If necessary, differential centrifugation can be used to pre-

pare subcellular fractions or to isolate specific organelles (see Fig. 1-8).

Once the extract or organelle preparation is ready, various methods are available for purifying one or more of the proteins it contains. Commonly, the extract is subjected to treatments that separate the proteins into different fractions based on a property such as size or charge, a process referred to as **fractionation**. Early fractionation steps in a purification utilize differences in protein solubility, which is a complex function of pH, temperature, salt concentration, and other factors. The solubility of proteins is generally lowered at high salt concentrations, an effect called "salting out." The addition of a salt in the right amount can selectively precipitate some proteins, while others remain in solution. Ammonium sulfate $((\text{NH}_4)_2\text{SO}_4)$ is often used for this purpose because of its high solubility in water.

A solution containing the protein of interest often must be further altered before subsequent purification steps are possible. For example, **dialysis** is a procedure that separates proteins from solvents by taking advantage of the proteins' larger size. The partially purified extract is placed in a bag or tube made of a semipermeable membrane. When this is suspended in a much larger volume of buffered solution of appropriate ionic strength, the membrane allows the exchange of salt and buffer but not proteins. Thus dialysis retains large proteins within the membranous bag or tube while allowing the concentration of other solutes in the protein preparation to change until they come into equilibrium with the solution outside the membrane. Dialysis might be used, for example, to remove ammonium sulfate from the protein preparation.

The most powerful methods for fractionating proteins make use of **column chromatography**, which takes advantage of differences in protein charge, size,

binding affinity, and other properties (Fig. 3-17). A porous solid material with appropriate chemical properties (the stationary phase) is held in a column, and a buffered solution (the mobile phase) percolates through it. The protein-containing solution, layered on the top of the column, percolates through the solid matrix as an ever-expanding band within the larger mobile phase (Fig. 3-17). Individual proteins migrate faster or more slowly through the column depending on their properties. For example, in **cation-exchange chromatography** (Fig. 3-18a), the solid matrix has negatively charged groups. In the mobile phase, proteins with a net positive charge migrate through the matrix more slowly than those with a net negative charge, because the migration of the former is retarded more by interaction with the stationary phase. The two types of protein can separate into two distinct bands. The expansion of the protein band in the mobile phase (the protein solution) is caused both by separation of proteins with different properties and by diffusional spreading. As the length of the column increases, the resolution of two types of protein with different net charges generally improves. However, the rate at which the protein solution can flow through the column usually decreases with column

length. And as the length of time spent on the column increases, the resolution can decline as a result of diffusional spreading within each protein band.

Figure 3-18 shows two other variations of column chromatography in addition to ion exchange. **Size-exclusion chromatography** separates proteins according to size. In this method, large proteins emerge from the column sooner than small ones—a somewhat counterintuitive result. The solid phase consists of beads with engineered pores or cavities of a particular size. Large proteins cannot enter the cavities, and so take a short (and rapid) path through the column, around the beads. Small proteins enter the cavities, and migrate through the column more slowly as a result (Fig. 3-18b). **Affinity chromatography** is based on the binding affinity of a protein. The beads in the column have a covalently attached chemical group. A protein with affinity for this particular chemical group will bind to the beads in the column, and its migration will be retarded as a result (Fig. 3-18c).

A modern refinement in chromatographic methods is **HPLC**, or **high-performance liquid chromatography**. HPLC makes use of high-pressure pumps that speed the movement of the protein molecules down the column, as well as higher-quality chromatographic materials that can withstand the crushing force of the pressurized flow. By reducing the transit time on the column, HPLC can limit diffusional spreading of protein bands and thus greatly improve resolution.

The approach to purification of a protein that has not previously been isolated is guided both by established precedents and by common sense. In most cases, several different methods must be used sequentially to purify a protein completely. The choice of method is

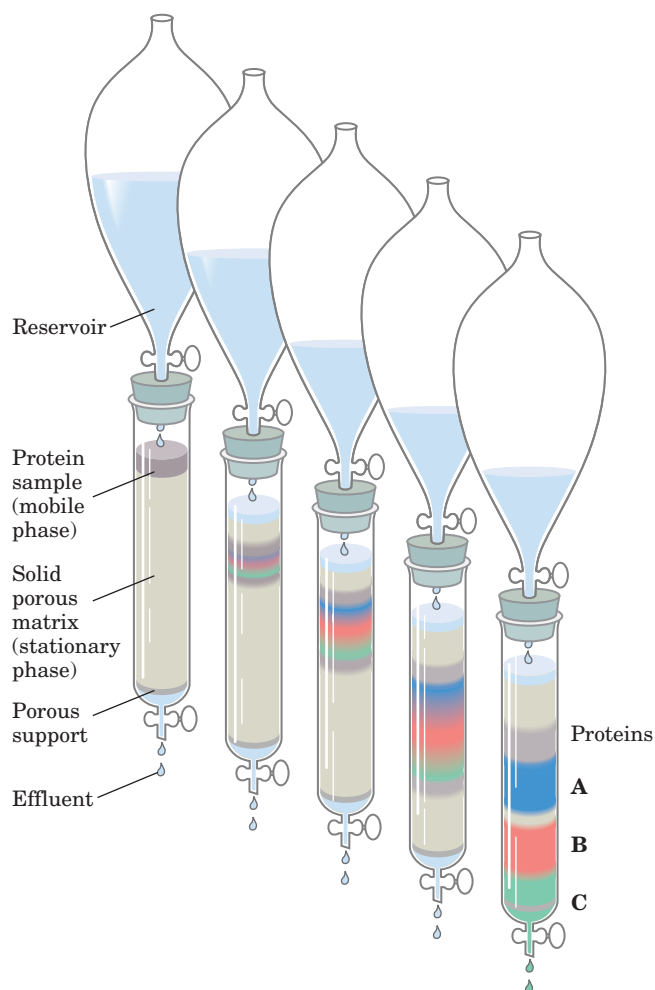


FIGURE 3-17 Column chromatography. The standard elements of a chromatographic column include a solid, porous material supported inside a column, generally made of plastic or glass. The solid material (matrix) makes up the stationary phase through which flows a solution, the mobile phase. The solution that passes out of the column at the bottom (the effluent) is constantly replaced by solution supplied from a reservoir at the top. The protein solution to be separated is layered on top of the column and allowed to percolate into the solid matrix. Additional solution is added on top. The protein solution forms a band within the mobile phase that is initially the depth of the protein solution applied to the column. As proteins migrate through the column, they are retarded to different degrees by their different interactions with the matrix material. The overall protein band thus widens as it moves through the column. Individual types of proteins (such as A, B, and C, shown in blue, red, and green) gradually separate from each other, forming bands within the broader protein band. Separation improves (resolution increases) as the length of the column increases. However, each individual protein band also broadens with time due to diffusional spreading, a process that decreases resolution. In this example, protein A is well separated from B and C, but diffusional spreading prevents complete separation of B and C under these conditions.

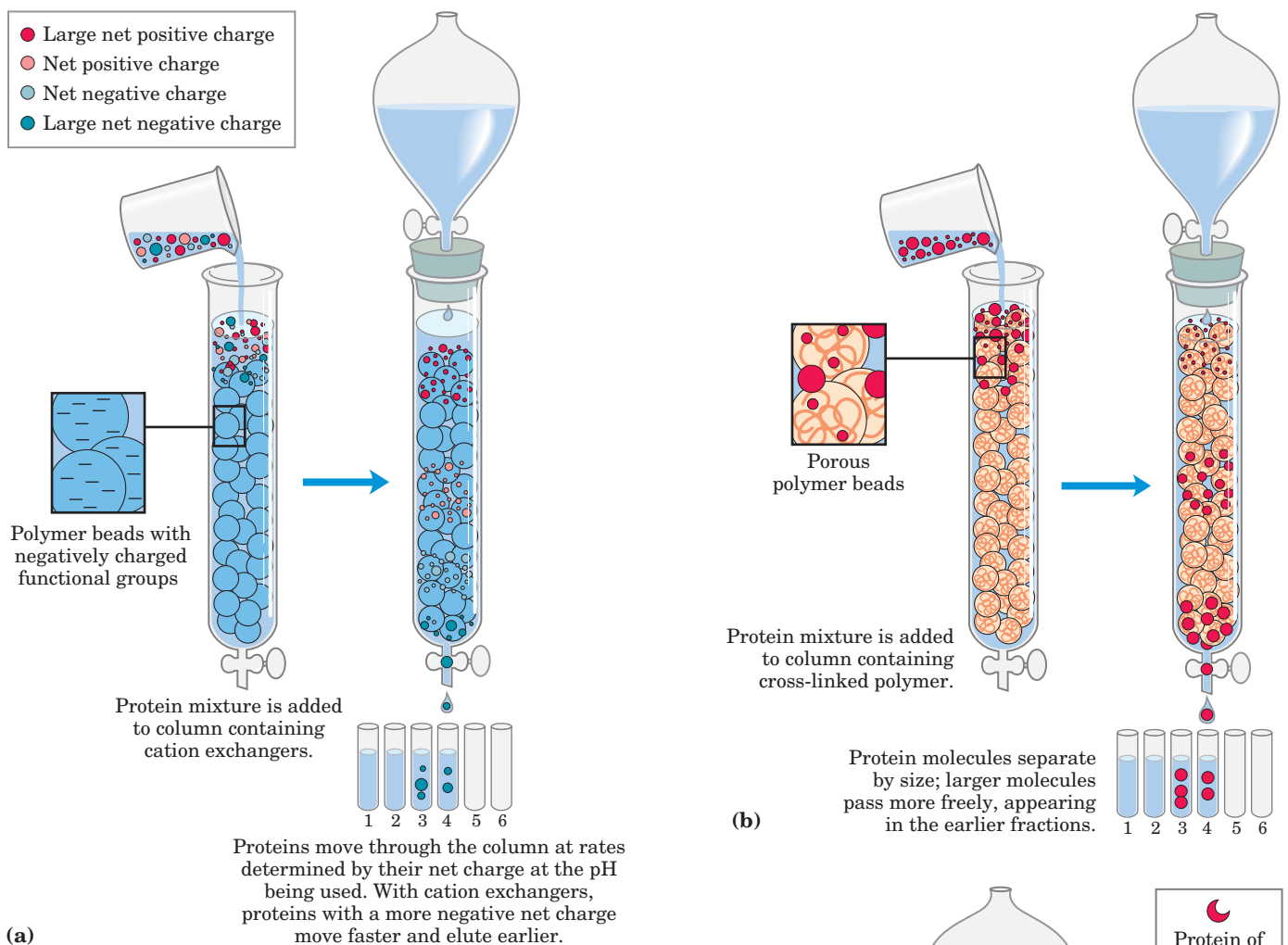


FIGURE 3-18 Three chromatographic methods used in protein purification. (a) **Ion-exchange chromatography** exploits differences in the sign and magnitude of the net electric charges of proteins at a given pH. The column matrix is a synthetic polymer containing bound charged groups; those with bound anionic groups are called **cation exchangers**, and those with bound cationic groups are called **anion exchangers**. Ion-exchange chromatography on a cation exchanger is shown here. The affinity of each protein for the charged groups on the column is affected by the pH (which determines the ionization state of the molecule) and the concentration of competing free salt ions in the surrounding solution. Separation can be optimized by gradually changing the pH and/or salt concentration of the mobile phase so as to create a pH or salt gradient. (b) **Size-exclusion chromatography**, also called gel filtration, separates proteins according to size. The column matrix is a cross-linked polymer with pores of selected size. Larger proteins migrate faster than smaller ones, because they are too large to enter the pores in the beads and hence take a more direct route through the column. The smaller proteins enter the pores and are slowed by their more labyrinthine path through the column. (c) **Affinity chromatography** separates proteins by their binding specificities. The proteins retained on the column are those that bind specifically to a ligand cross-linked to the beads. (In biochemistry, the term “ligand” is used to refer to a group or molecule that binds to a macromolecule such as a protein.) After proteins that do not bind to the ligand are washed through the column, the bound protein of particular interest is eluted (washed out of the column) by a solution containing free ligand.

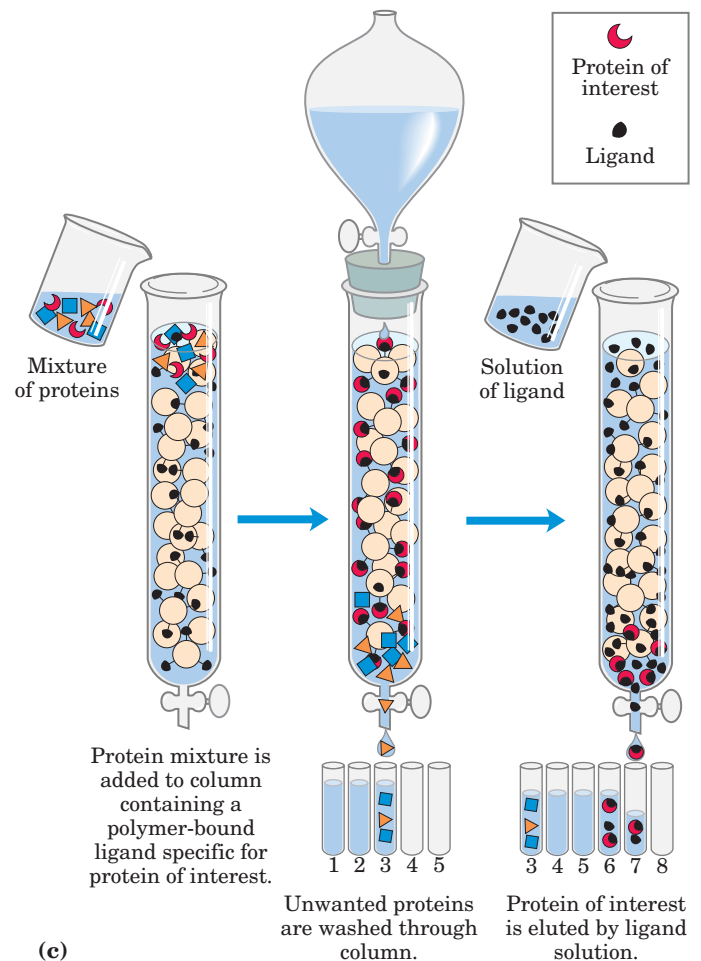


TABLE 3-5 A Purification Table for a Hypothetical Enzyme

Procedure or step	Fraction volume (ml)	Total protein (mg)	Activity (units)	Specific activity (units/mg)
1. Crude cellular extract	1,400	10,000	100,000	10
2. Precipitation with ammonium sulfate	280	3,000	96,000	32
3. Ion-exchange chromatography	90	400	80,000	200
4. Size-exclusion chromatography	80	100	60,000	600
5. Affinity chromatography	6	3	45,000	15,000

Note: All data represent the status of the sample *after* the designated procedure has been carried out. Activity and specific activity are defined on page 94.

somewhat empirical, and many protocols may be tried before the most effective one is found. Trial and error can often be minimized by basing the procedure on purification techniques developed for similar proteins. Published purification protocols are available for many thousands of proteins. Common sense dictates that inexpensive procedures such as salting out be used first, when the total volume and the number of contaminants are greatest. Chromatographic methods are often impractical at early stages, because the amount of chromatographic medium needed increases with sample size. As each purification step is completed, the sample size generally becomes smaller (Table 3-5), making it feasible to use more sophisticated (and expensive) chromatographic procedures at later stages.

Proteins Can Be Separated and Characterized by Electrophoresis

Another important technique for the separation of proteins is based on the migration of charged proteins in an electric field, a process called **electrophoresis**. These procedures are not generally used to purify proteins in large amounts, because simpler alternatives are usually available and electrophoretic methods often adversely affect the structure and thus the function of proteins. Electrophoresis is, however, especially useful as an analytical method. Its advantage is that proteins can be visualized as well as separated, permitting a researcher to estimate quickly the number of different proteins in a mixture or the degree of purity of a particular protein preparation. Also, electrophoresis allows determination of crucial properties of a protein such as its isoelectric point and approximate molecular weight.

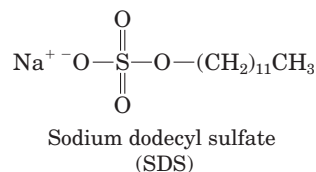
Electrophoresis of proteins is generally carried out in gels made up of the cross-linked polymer polyacrylamide (Fig. 3-19). The polyacrylamide gel acts as a molecular sieve, slowing the migration of proteins approximately in proportion to their charge-to-mass ratio. Migration may also be affected by protein shape. In electrophoresis, the force moving the macromolecule is the electrical potential, E . The electrophoretic mobility of the molecule, μ , is the ratio of the velocity of the par-

ticle molecule, V , to the electrical potential. Electrophoretic mobility is also equal to the net charge of the molecule, Z , divided by the frictional coefficient, f , which reflects in part a protein's shape. Thus:

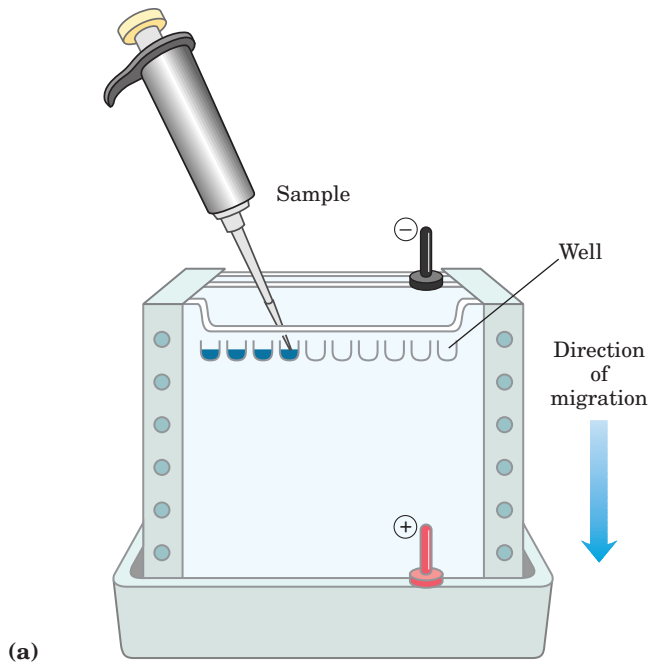
$$\mu = \frac{V}{E} = \frac{Z}{f}$$

The migration of a protein in a gel during electrophoresis is therefore a function of its size and its shape.

An electrophoretic method commonly employed for estimation of purity and molecular weight makes use of the detergent **sodium dodecyl sulfate (SDS)**.



SDS binds to most proteins in amounts roughly proportional to the molecular weight of the protein, about one molecule of SDS for every two amino acid residues. The bound SDS contributes a large net negative charge, rendering the intrinsic charge of the protein insignificant and conferring on each protein a similar charge-to-mass ratio. In addition, the native conformation of a protein is altered when SDS is bound, and most proteins assume a similar shape. Electrophoresis in the presence of SDS therefore separates proteins almost exclusively on the basis of mass (molecular weight), with smaller polypeptides migrating more rapidly. After electrophoresis, the proteins are visualized by adding a dye such as Coomassie blue, which binds to proteins but not to the gel itself (Fig. 3-19b). Thus, a researcher can monitor the progress of a protein purification procedure as the number of protein bands visible on the gel decreases after each new fractionation step. When compared with the positions to which proteins of known molecular weight migrate in the gel, the position of an unidentified protein can provide an excellent measure of its molecular weight (Fig. 3-20). If the protein has two or more different subunits, the subunits will generally be separated by the SDS treatment and a separate band will appear for each.  **SDS Gel Electrophoresis**



(a)

FIGURE 3-19 Electrophoresis. (a) Different samples are loaded in wells or depressions at the top of the polyacrylamide gel. The proteins move into the gel when an electric field is applied. The gel minimizes convection currents caused by small temperature gradients, as well as protein movements other than those induced by the electric field. (b) Proteins can be visualized after electrophoresis by treating the gel with a stain such as Coomassie blue, which binds to the proteins but not to the gel itself. Each band on the gel represents a different pro-

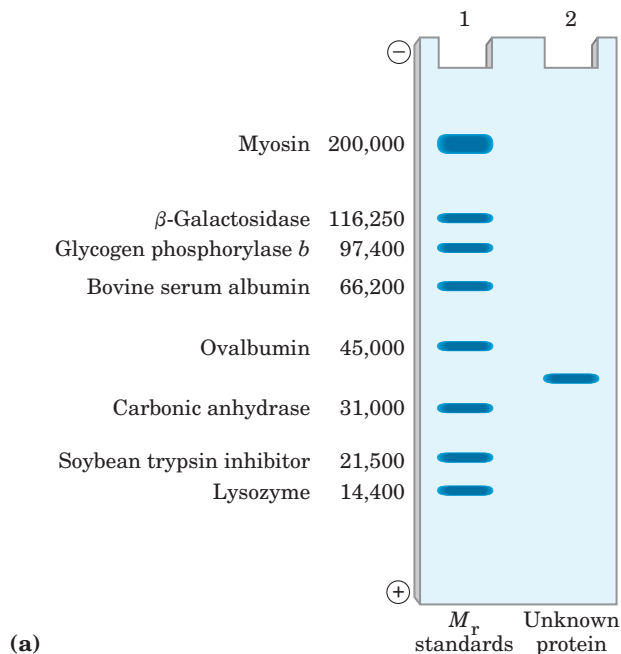
(b)



Isoelectric focusing is a procedure used to determine the isoelectric point (pI) of a protein (Fig. 3-21). A pH gradient is established by allowing a mixture of low molecular weight organic acids and bases (ampholytes; p. 81) to distribute themselves in an electric field generated across the gel. When a protein mix-

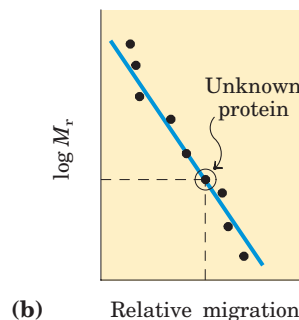
ture is applied, each protein migrates until it reaches the pH that matches its pI (Table 3-6). Proteins with different isoelectric points are thus distributed differently throughout the gel.

Combining isoelectric focusing and SDS electrophoresis sequentially in a process called **two-dimensional**



(a)

FIGURE 3-20 Estimating the molecular weight of a protein. The electrophoretic mobility of a protein on an SDS polyacrylamide gel is related to its molecular weight, M_r . (a) Standard proteins of known molecular weight are subjected to electrophoresis (lane 1). These marker proteins can be used to estimate the molecular weight of an unknown protein (lane 2). (b) A plot of $\log M_r$ of the marker proteins versus relative migration during electrophoresis is linear, which allows the molecular weight of the unknown protein to be read from the graph.



(b)

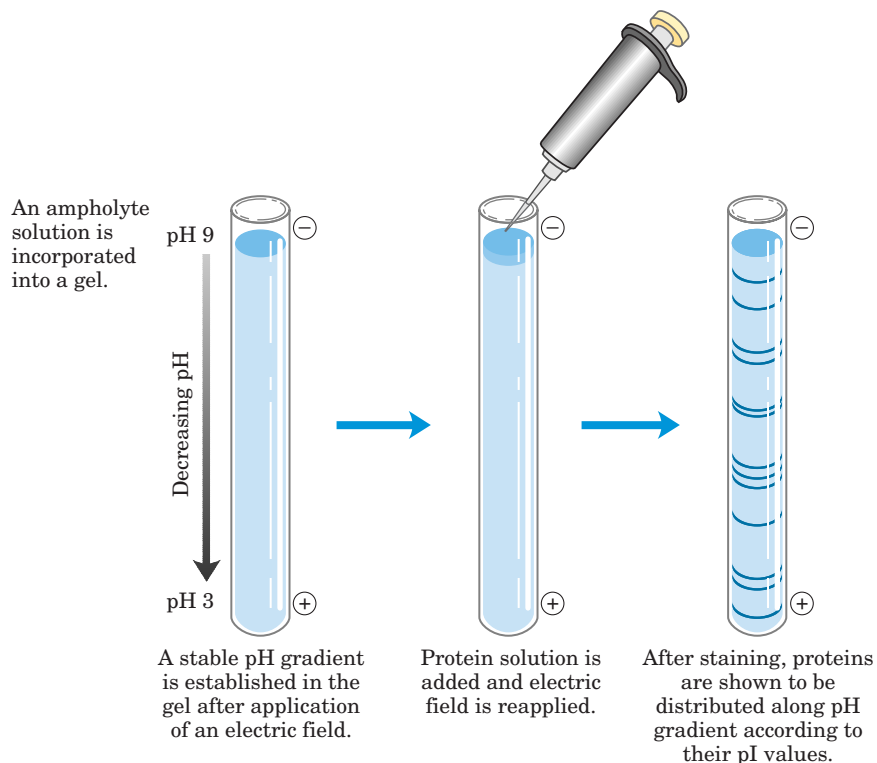


FIGURE 3-21 Isoelectric focusing. This technique separates proteins according to their isoelectric points. A stable pH gradient is established in the gel by the addition of appropriate ampholytes. A protein mixture is placed in a well on the gel. With an applied electric field, proteins enter the gel and migrate until each reaches a pH equivalent to its pI. Remember that when $\text{pH} = \text{pI}$, the net charge of a protein is zero.

electrophoresis permits the resolution of complex mixtures of proteins (Fig. 3-22). This is a more sensitive analytical method than either electrophoretic method alone. Two-dimensional electrophoresis separates proteins of identical molecular weight that differ in pI, or proteins with similar pI values but different molecular weights.

Unseparated Proteins Can Be Quantified

To purify a protein, it is essential to have a way of detecting and quantifying that protein in the presence of many other proteins at each stage of the procedure. Often, purification must proceed in the absence of any information about the size and physical properties of the protein or about the fraction of the total protein mass it represents in the extract. For proteins that are enzymes, the amount in a given solution or tissue extract can be measured, or assayed, in terms of the catalytic effect the enzyme produces—that is, the *increase* in the rate at which its substrate is converted to reaction products when the enzyme is present. For this purpose one must know (1) the overall equation of the reaction catalyzed, (2) an analytical procedure for determining the disappearance of the substrate or the appearance of a reaction product, (3) whether the enzyme requires co-factors such as metal ions or coenzymes, (4) the dependence of the enzyme activity on substrate concentration, (5) the optimum pH, and (6) a temperature zone in which the enzyme is stable and has high activity. Enzymes are usually assayed at their optimum pH and at some convenient temperature within the range

25 to 38 °C. Also, very high substrate concentrations are generally used so that the initial reaction rate, measured experimentally, is proportional to enzyme concentration (Chapter 6).

By international agreement, 1.0 unit of enzyme activity is defined as the amount of enzyme causing transformation of 1.0 μmol of substrate per minute at 25 °C under optimal conditions of measurement. The term **activity** refers to the total units of enzyme in a solution. The **specific activity** is the number of enzyme units per milligram of total protein (Fig. 3-23). The specific activity is a measure of enzyme purity: it increases during purification of an enzyme and becomes maximal and constant when the enzyme is pure (Table 3-5).

TABLE 3-6 The Isoelectric Points of Some Proteins

Protein	pI
Pepsin	<1.0
Egg albumin	4.6
Serum albumin	4.9
Urease	5.0
β -Lactoglobulin	5.2
Hemoglobin	6.8
Myoglobin	7.0
Chymotrypsinogen	9.5
Cytochrome c	10.7
Lysozyme	11.0

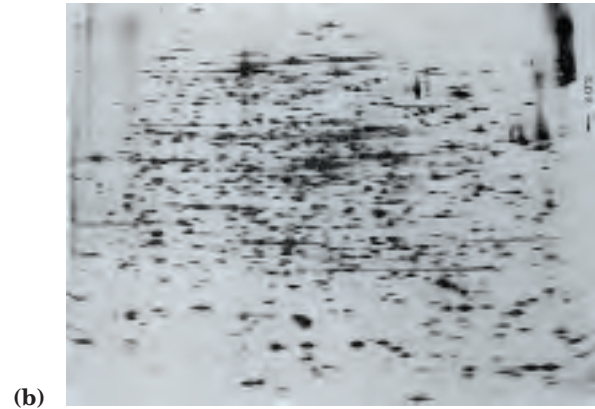
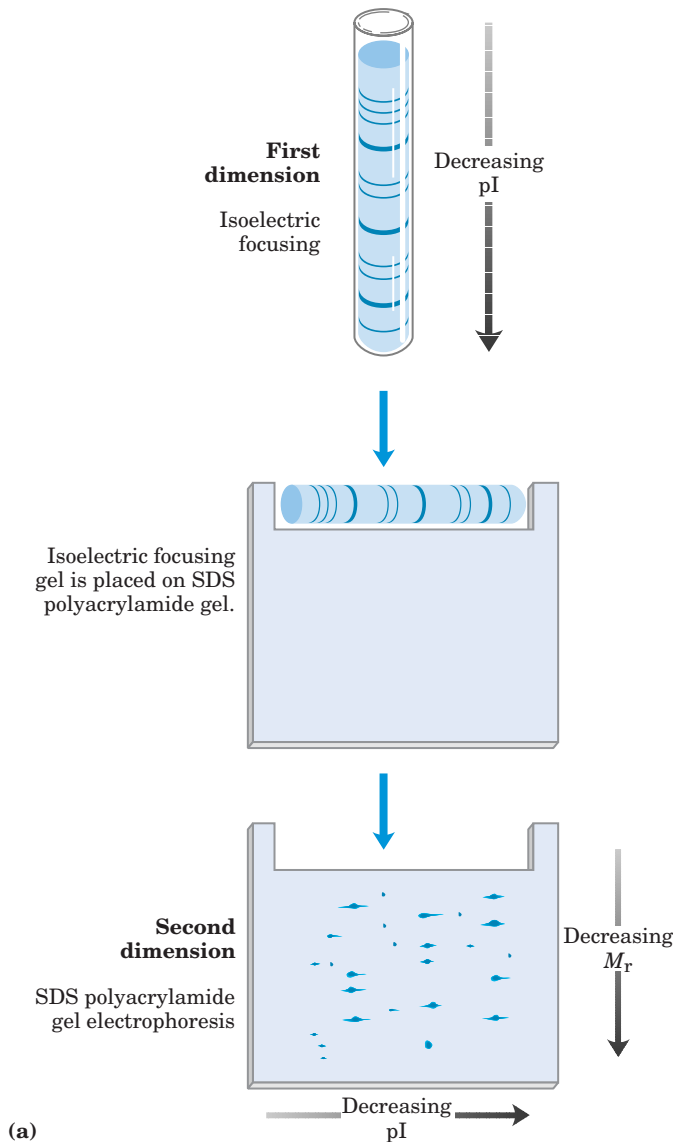


FIGURE 3-22 Two-dimensional electrophoresis. (a) Proteins are first separated by isoelectric focusing in a cylindrical gel. The gel is then laid horizontally on a second, slab-shaped gel, and the proteins are separated by SDS polyacrylamide gel electrophoresis. Horizontal separation reflects differences in pI; vertical separation reflects differences in molecular weight. (b) More than 1,000 different proteins from *E. coli* can be resolved using this technique.

when further purification steps fail to increase specific activity and when only a single protein species can be detected (for example, by electrophoresis).

For proteins that are not enzymes, other quantification methods are required. Transport proteins can be assayed by their binding to the molecule they transport, and hormones and toxins by the biological effect they produce; for example, growth hormones will stimulate the growth of certain cultured cells. Some structural proteins represent such a large fraction of a tissue mass that they can be readily extracted and purified without a functional assay. The approaches are as varied as the proteins themselves.

After each purification step, the activity of the preparation (in units of enzyme activity) is assayed, the total amount of protein is determined independently, and the ratio of the two gives the specific activity. Activity and total protein generally decrease with each step. Activity decreases because some loss always occurs due to inactivation or nonideal interactions with chromatographic materials or other molecules in the solution. Total protein decreases because the objective is to remove as much unwanted or nonspecific protein as possible. In a successful step, the loss of nonspecific protein is much greater than the loss of activity; therefore, specific activity increases even as total activity falls. The data are then assembled in a purification table similar to Table 3-5. A protein is generally considered pure

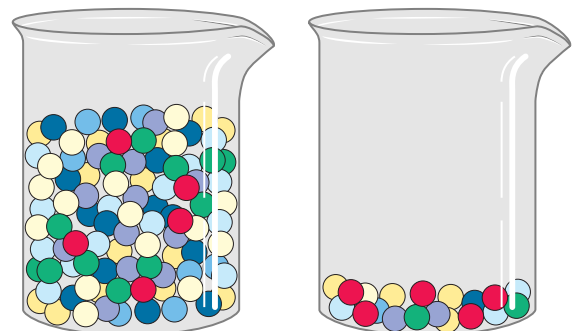


FIGURE 3-23 Activity versus specific activity. The difference between these two terms can be illustrated by considering two beakers of marbles. The beakers contain the same number of red marbles, but different numbers of marbles of other colors. If the marbles represent proteins, both beakers contain the same *activity* of the protein represented by the red marbles. The second beaker, however, has the higher *specific activity* because here the red marbles represent a much higher fraction of the total.

SUMMARY 3.3 Working with Proteins

- Proteins are separated and purified by taking advantage of differences in their properties. Proteins can be selectively precipitated by the addition of certain salts. A wide range of chromatographic procedures makes use of differences in size, binding affinities, charge, and other properties. These include ion-exchange, size-exclusion, affinity, and high-performance liquid chromatography.
- Electrophoresis separates proteins on the basis of mass or charge. SDS gel electrophoresis and isoelectric focusing can be used separately or in combination for higher resolution.
- All purification procedures require a method for quantifying or assaying the protein of interest in the presence of other proteins. Purification can be monitored by assaying specific activity.

3.4 The Covalent Structure of Proteins

Purification of a protein is usually only a prelude to a detailed biochemical dissection of its structure and function. What is it that makes one protein an enzyme, another a hormone, another a structural protein, and still another an antibody? How do they differ chemically? The most obvious distinctions are structural, and these distinctions can be approached at every level of structure defined in Figure 3–16.

The differences in primary structure can be especially informative. Each protein has a distinctive number and sequence of amino acid residues. As we shall see in Chapter 4, the primary structure of a protein determines how it folds up into a unique three-dimensional structure, and this in turn determines the function of the protein. Primary structure is the focus of the remainder of this chapter. We first consider empirical clues that amino acid sequence and protein function are closely linked, then describe how amino acid sequence is determined; finally, we outline the many uses to which this information can be put.

The Function of a Protein Depends on Its Amino Acid Sequence

The bacterium *Escherichia coli* produces more than 3,000 different proteins; a human produces 25,000 to 35,000. In both cases, each type of protein has a unique three-dimensional structure and this structure confers a unique function. Each type of protein also has a unique amino acid sequence. Intuition suggests that the amino acid sequence must play a fundamental role in determining the three-dimensional structure of the protein, and ultimately its function, but is this supposition cor-

rect? A quick survey of proteins and how they vary in amino acid sequence provides a number of empirical clues that help substantiate the important relationship between amino acid sequence and biological function.

First, as we have already noted, proteins with different functions always have different amino acid sequences. Second, thousands of human genetic diseases have been traced to the production of defective proteins. Perhaps one-third of these proteins are defective because of a single change in their amino acid sequence; hence, if the primary structure is altered, the function of the protein may also be changed. Finally, on comparing functionally similar proteins from different species, we find that these proteins often have similar amino acid sequences. An extreme case is ubiquitin, a 76-residue protein involved in regulating the degradation of other proteins. The amino acid sequence of ubiquitin is identical in species as disparate as fruit flies and humans.

Is the amino acid sequence absolutely fixed, or invariant, for a particular protein? No; some flexibility is possible. An estimated 20% to 30% of the proteins in humans are **polymorphic**, having amino acid sequence variants in the human population. Many of these variations in sequence have little or no effect on the function of the protein. Furthermore, proteins that carry out a broadly similar function in distantly related species can differ greatly in overall size and amino acid sequence.

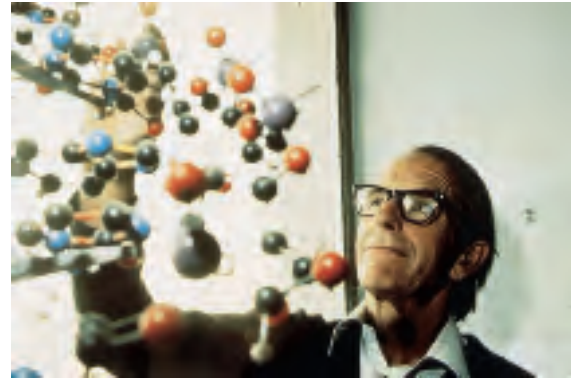
Although the amino acid sequence in some regions of the primary structure might vary considerably without affecting biological function, most proteins contain crucial regions that are essential to their function and whose sequence is therefore conserved. The fraction of the overall sequence that is critical varies from protein to protein, complicating the task of relating sequence to three-dimensional structure, and structure to function. Before we can consider this problem further, however, we must examine how sequence information is obtained.

The Amino Acid Sequences of Millions of Proteins Have Been Determined

Two major discoveries in 1953 were of crucial importance in the history of biochemistry. In that year James D. Watson and Francis Crick deduced the double-helical structure of DNA and proposed a structural basis for its precise replication (Chapter 8). Their proposal illuminated the molecular reality behind the idea of a gene. In that same year, Frederick Sanger worked out the sequence of amino acid residues in the polypeptide chains of the hormone insulin (Fig. 3–24), surprising many researchers who had long thought that elucidation of the amino acid sequence of a polypeptide would be a hopelessly difficult task. It quickly became evident that the nucleotide sequence in DNA and the amino acid sequence in proteins were somehow related. Barely a decade after these discoveries, the role of the nucleotide

sequence of DNA in determining the amino acid sequence of protein molecules was revealed (Chapter 27). An enormous number of protein sequences can now be derived indirectly from the DNA sequences in the rapidly growing genome databases. However, many are still deduced by traditional methods of polypeptide sequencing.

The amino acid sequences of thousands of different proteins from many species have been determined using principles first developed by Sanger. These methods are still in use, although with many variations and improvements in detail. Chemical protein sequencing now



Frederick Sanger

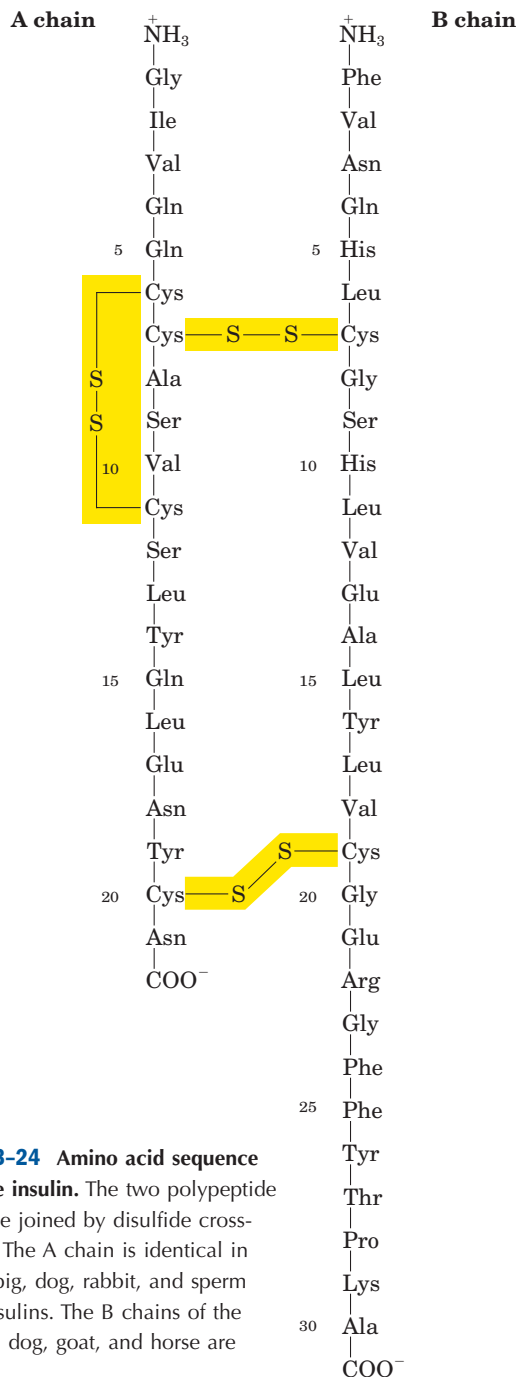
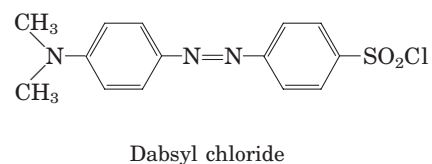
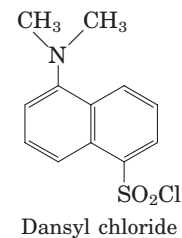


FIGURE 3-24 Amino acid sequence of bovine insulin. The two polypeptide chains are joined by disulfide cross-linkages. The A chain is identical in human, pig, dog, rabbit, and sperm whale insulins. The B chains of the cow, pig, dog, goat, and horse are identical.

complements a growing list of newer methods, providing multiple avenues to obtain amino acid sequence data. Such data are now critical to every area of biochemical investigation.

Short Polypeptides Are Sequenced Using Automated Procedures

Various procedures are used to analyze protein primary structure. Several protocols are available to label and identify the amino-terminal amino acid residue (Fig. 3-25a). Sanger developed the reagent 1-fluoro-2,4-dinitrobenzene (FDNB) for this purpose; other reagents used to label the amino-terminal residue, dansyl chloride and dabsyl chloride, yield derivatives that are more easily detectable than the dinitrophenyl derivatives. After the amino-terminal residue is labeled with one of these reagents, the polypeptide is hydrolyzed to its constituent amino acids and the labeled amino acid is identified. Because the hydrolysis stage destroys the polypeptide, this procedure cannot be used to sequence a polypeptide beyond its amino-terminal residue. However, it can help determine the number of chemically distinct polypeptides in a protein, provided each has a different amino-terminal residue. For example, two residues—Phe and Gly—would be labeled if insulin (Fig. 3-24) were subjected to this procedure.



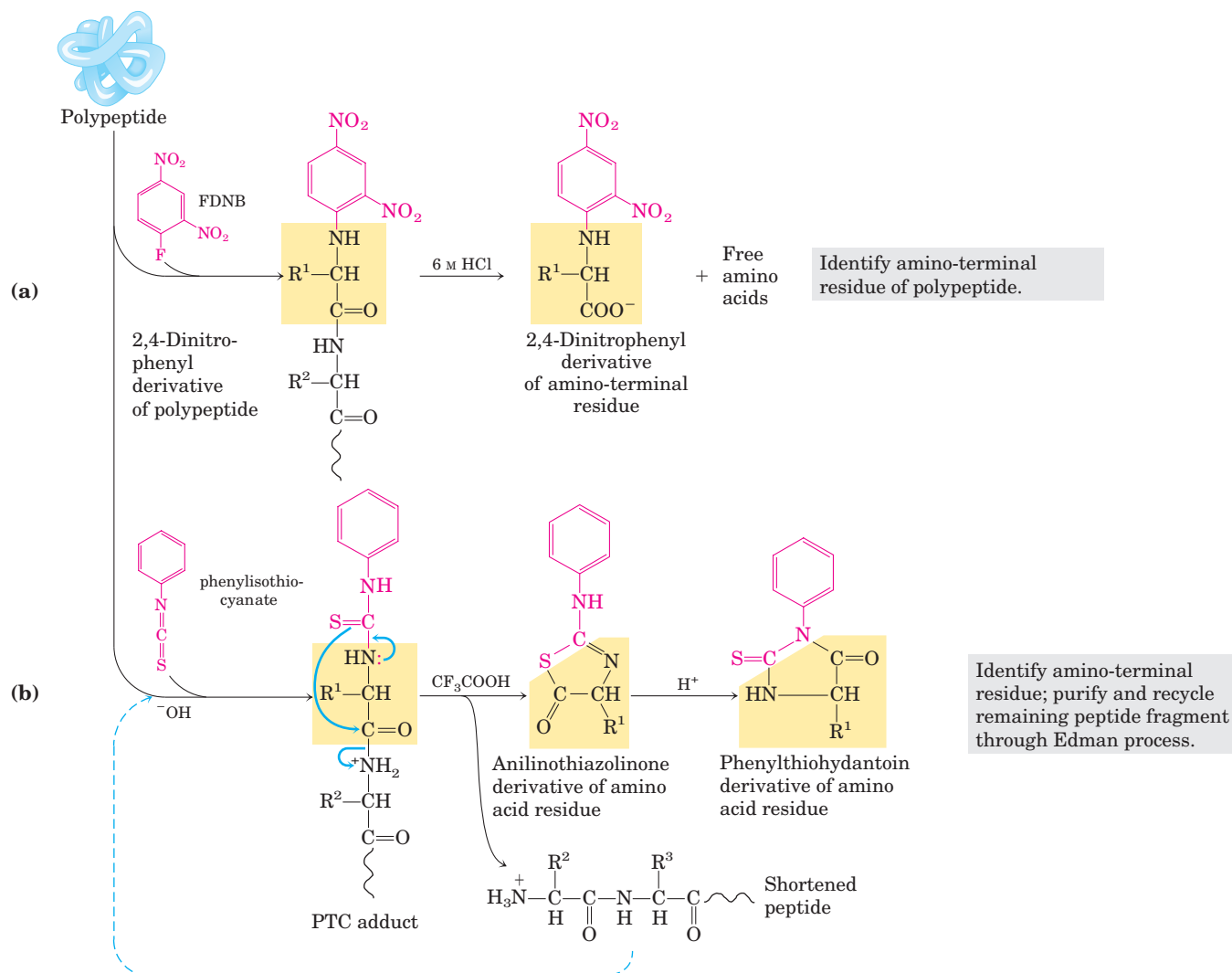


FIGURE 3-25 Steps in sequencing a polypeptide. (a) Identification of the amino-terminal residue can be the first step in sequencing a polypeptide. Sanger's method for identifying the amino-terminal residue is shown here. (b) The Edman degradation procedure reveals

the entire sequence of a peptide. For shorter peptides, this method alone readily yields the entire sequence, and step (a) is often omitted. Step (a) is useful in the case of larger polypeptides, which are often fragmented into smaller peptides for sequencing (see Fig. 3-27).

To sequence an entire polypeptide, a chemical method devised by Pehr Edman is usually employed. The **Edman degradation** procedure labels and removes only the amino-terminal residue from a peptide, leaving all other peptide bonds intact (Fig. 3-25b). The peptide is reacted with phenylisothiocyanate under mildly alkaline conditions, which converts the amino-terminal amino acid to a phenylthiocarbamoyl (PTC) adduct. The peptide bond next to the PTC adduct is then cleaved in a step carried out in anhydrous trifluoroacetic acid, with removal of the amino-terminal amino acid as an anilinothiazolinone derivative. The derivatized amino acid is extracted with organic solvents, converted to the more stable phenylthiohydantoin derivative by treatment with aqueous acid, and then identified. The use of sequential reactions carried out under first basic and then acidic conditions provides control over

the entire process. Each reaction with the amino-terminal amino acid can go essentially to completion without affecting any of the other peptide bonds in the peptide. After removal and identification of the amino-terminal residue, the *new* amino-terminal residue so exposed can be labeled, removed, and identified through the same series of reactions. This procedure is repeated until the entire sequence is determined. The Edman degradation is carried out on a machine, called a **sequenator**, that mixes reagents in the proper proportions, separates the products, identifies them, and records the results. These methods are extremely sensitive. Often, the complete amino acid sequence can be determined starting with only a few micrograms of protein.

The length of polypeptide that can be accurately sequenced by the Edman degradation depends on the

efficiency of the individual chemical steps. Consider a peptide beginning with the sequence Gly–Pro–Lys– at its amino terminus. If glycine were removed with 97% efficiency, 3% of the polypeptide molecules in the solution would retain a Gly residue at their amino terminus. In the second Edman cycle, 97% of the liberated amino acids would be proline, and 3% glycine, while 3% of the polypeptide molecules would retain Gly (0.1%) or Pro (2.9%) residues at their amino terminus. At each cycle, peptides that did not react in earlier cycles would contribute amino acids to an ever-increasing background, eventually making it impossible to determine which amino acid is next in the original peptide sequence. Modern sequencers achieve efficiencies of better than 99% per cycle, permitting the sequencing of more than 50 contiguous amino acid residues in a polypeptide. The primary structure of insulin, worked out by Sanger and colleagues over a period of 10 years, could now be completely determined in a day or two.

Large Proteins Must Be Sequenced in Smaller Segments

The overall accuracy of amino acid sequencing generally declines as the length of the polypeptide increases. The very large polypeptides found in proteins must be broken down into smaller pieces to be sequenced efficiently. There are several steps in this process. First, the protein is cleaved into a set of specific fragments by chemical or enzymatic methods. If any disulfide bonds

are present, they must be broken. Each fragment is purified, then sequenced by the Edman procedure. Finally, the order in which the fragments appear in the original protein is determined and disulfide bonds (if any) are located.

Breaking Disulfide Bonds Disulfide bonds interfere with the sequencing procedure. A cystine residue (Fig. 3–7) that has one of its peptide bonds cleaved by the Edman procedure may remain attached to another polypeptide strand via its disulfide bond. Disulfide bonds also interfere with the enzymatic or chemical cleavage of the polypeptide. Two approaches to irreversible breakage of disulfide bonds are outlined in Figure 3–26.

Cleaving the Polypeptide Chain Several methods can be used for fragmenting the polypeptide chain. Enzymes called **proteases** catalyze the hydrolytic cleavage of peptide bonds. Some proteases cleave only the peptide bond adjacent to particular amino acid residues (Table 3–7) and thus fragment a polypeptide chain in a predictable and reproducible way. A number of chemical reagents also cleave the peptide bond adjacent to specific residues.

Among proteases, the digestive enzyme trypsin catalyzes the hydrolysis of only those peptide bonds in which the carbonyl group is contributed by either a Lys or an Arg residue, regardless of the length or amino acid sequence of the chain. The number of smaller peptides produced by trypsin cleavage can thus be predicted

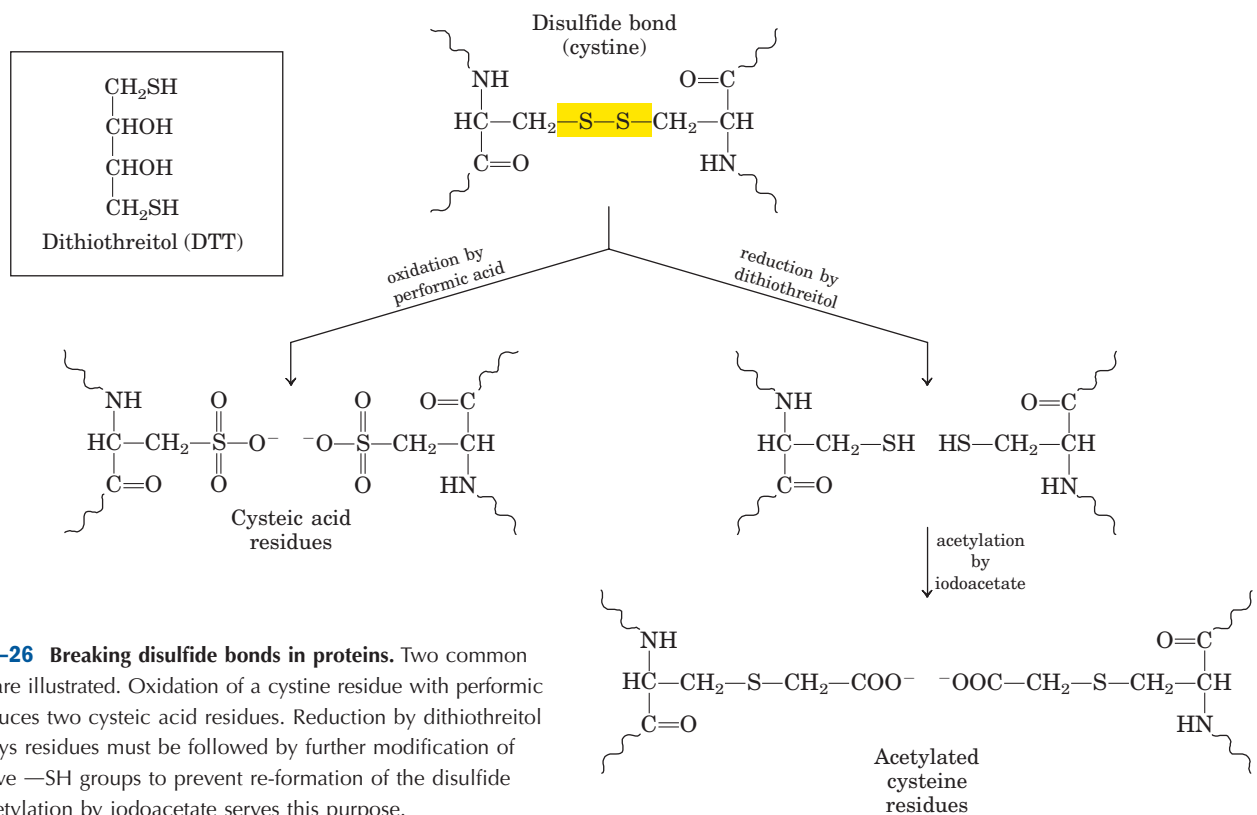


FIGURE 3–26 Breaking disulfide bonds in proteins. Two common methods are illustrated. Oxidation of a cystine residue with performic acid produces two cysteic acid residues. Reduction by dithiothreitol to form Cys residues must be followed by further modification of the reactive –SH groups to prevent re-formation of the disulfide bond. Acetylation by iodoacetate serves this purpose.

TABLE 3-7 The Specificity of Some Common Methods for Fragmenting Polypeptide Chains

Reagent (biological source)*	Cleavage points†
Trypsin (bovine pancreas)	Lys, Arg (C)
Submaxillary protease (mouse submaxillary gland)	Arg (C)
Chymotrypsin (bovine pancreas)	Phe, Trp, Tyr (C)
<i>Staphylococcus aureus</i> V8 protease (bacterium <i>S. aureus</i>)	Asp, Glu (C)
Asp-N-protease (bacterium <i>Pseudomonas fragi</i>)	Asp, Glu (N)
Pepsin (porcine stomach)	Phe, Trp, Tyr (N)
Endoproteinase Lys C (bacterium <i>Lyso bacter</i> <i>enzymogenes</i>)	Lys (C)
Cyanogen bromide	Met (C)

*All reagents except cyanogen bromide are proteases. All are available from commercial sources.

†Residues furnishing the primary recognition point for the protease or reagent; peptide bond cleavage occurs on either the carbonyl (C) or the amino (N) side of the indicated amino acid residues.

from the total number of Lys or Arg residues in the original polypeptide, as determined by hydrolysis of an intact sample (Fig. 3-27). A polypeptide with five Lys and/or Arg residues will usually yield six smaller peptides on cleavage with trypsin. Moreover, all except one of these will have a carboxyl-terminal Lys or Arg. The fragments produced by trypsin (or other enzyme or chemical) action are then separated by chromatographic or electrophoretic methods.

Sequencing of Peptides Each peptide fragment resulting from the action of trypsin is sequenced separately by the Edman procedure.

Ordering Peptide Fragments The order of the “trypsin fragments” in the original polypeptide chain must now be determined. Another sample of the intact polypeptide is cleaved into fragments using a different enzyme or reagent, one that cleaves peptide bonds at points other than those cleaved by trypsin. For example, cyanogen bromide cleaves only those peptide bonds in which the carbonyl group is contributed by Met. The fragments resulting from this second procedure are then separated and sequenced as before.

The amino acid sequences of each fragment obtained by the two cleavage procedures are examined, with the objective of finding peptides from the second procedure whose sequences establish continuity, be-

cause of overlaps, between the fragments obtained by the first cleavage procedure (Fig. 3-27). Overlapping peptides obtained from the second fragmentation yield the correct order of the peptide fragments produced in the first. If the amino-terminal amino acid has been identified before the original cleavage of the protein, this information can be used to establish which fragment is derived from the amino terminus. The two sets of fragments can be compared for possible errors in determining the amino acid sequence of each fragment. If the second cleavage procedure fails to establish continuity between all peptides from the first cleavage, a third or even a fourth cleavage method must be used to obtain a set of peptides that can provide the necessary overlap(s).

Locating Disulfide Bonds If the primary structure includes disulfide bonds, their locations are determined in an additional step after sequencing is completed. A sample of the protein is again cleaved with a reagent such as trypsin, this time without first breaking the disulfide bonds. The resulting peptides are separated by electrophoresis and compared with the original set of peptides generated by trypsin. For each disulfide bond, two of the original peptides will be missing and a new, larger peptide will appear. The two missing peptides represent the regions of the intact polypeptide that are linked by the disulfide bond.

Amino Acid Sequences Can Also Be Deduced by Other Methods

The approach outlined above is not the only way to determine amino acid sequences. New methods based on mass spectrometry permit the sequencing of short polypeptides (20 to 30 amino acid residues) in just a few minutes (Box 3-2). In addition, with the development of rapid DNA sequencing methods (Chapter 8), the elucidation of the genetic code (Chapter 27), and the development of techniques for isolating genes (Chapter 9), researchers can deduce the sequence of a polypeptide by determining the sequence of nucleotides in the gene that codes for it (Fig. 3-28). The techniques used to determine protein and DNA sequences are complementary. When the gene is available, sequencing the DNA can be faster and more accurate than sequencing the protein. Most proteins are now sequenced in this indirect way. If the gene has not been isolated, direct sequencing of peptides is necessary, and this can provide information (the location of disulfide bonds, for example) not available in a DNA sequence. In addition, a knowledge of the amino acid sequence of even a part of a polypeptide can greatly facilitate the isolation of the corresponding gene (Chapter 9).

The array of methods now available to analyze both proteins and nucleic acids is ushering in a new disci-

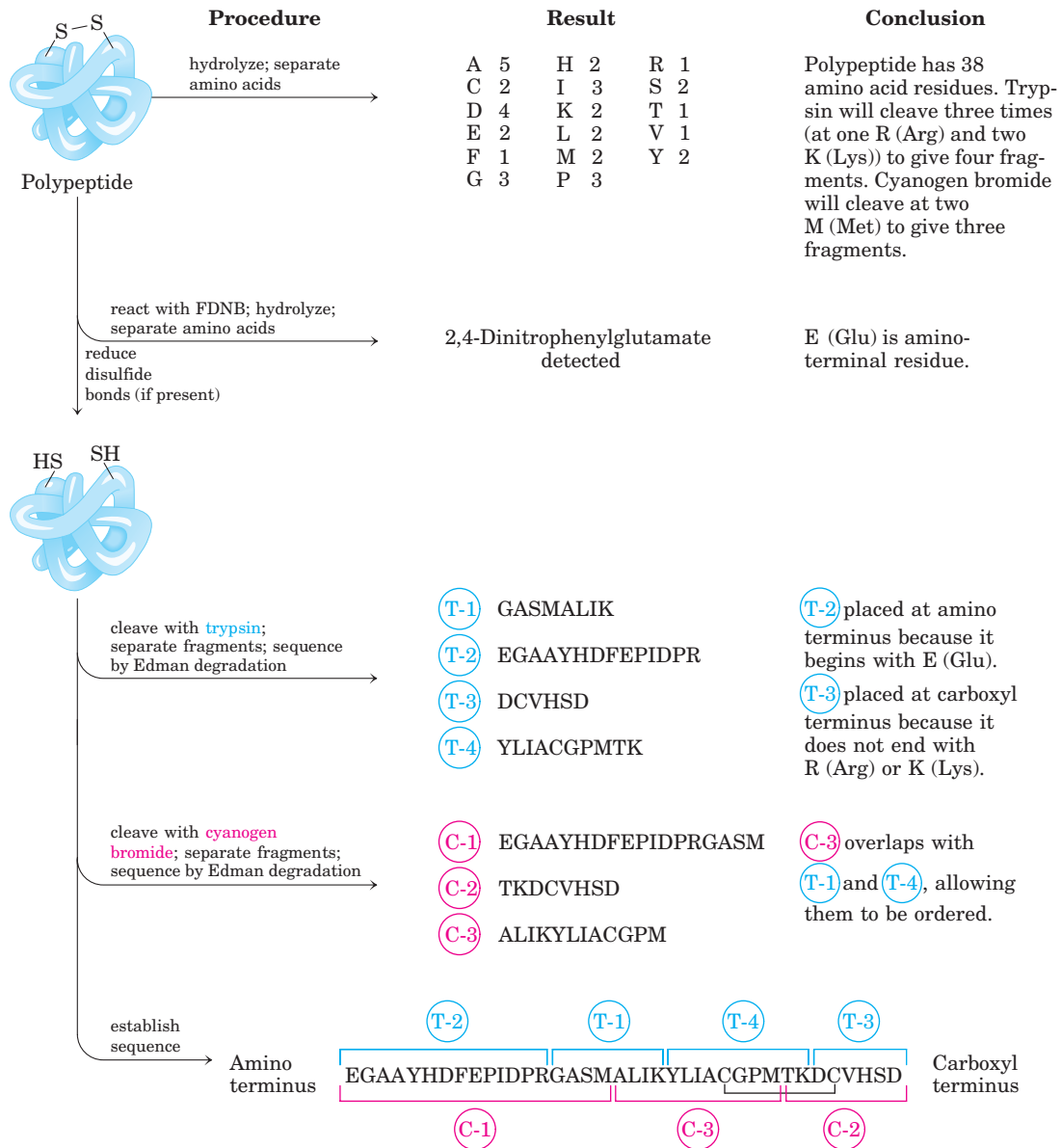


FIGURE 3-27 Cleaving proteins and sequencing and ordering the peptide fragments. First, the amino acid composition and amino-terminal residue of an intact sample are determined. Then any disulfide bonds are broken before fragmenting so that sequencing can proceed efficiently. In this example, there are only two Cys (C) residues and

thus only one possibility for location of the disulfide bond. In polypeptides with three or more Cys residues, the position of disulfide bonds can be determined as described in the text. (The one-letter symbols for amino acids are given in Table 3-1.)

pline of “whole cell biochemistry.” The complete sequence of an organism’s DNA, its genome, is now available for organisms ranging from viruses to bacteria to multicellular eukaryotes (see Table 1-4). Genes are being discovered by the millions, including many that encode proteins with no known function. To describe the entire protein complement encoded by an organism’s DNA, researchers have coined the term **proteome**. As described in Chapter 9, the new disciplines of genomics and proteomics are complementing work carried out on cellular intermediary metabolism and nucleic acid

metabolism to provide a new and increasingly complete picture of biochemistry at the level of cells and even organisms.

Amino acid sequence (protein)	Gln-Tyr-Pro-Thr-Ile-Trp
DNA sequence (gene)	CAGTATCCTACGATTGG

FIGURE 3-28 Correspondence of DNA and amino acid sequences. Each amino acid is encoded by a specific sequence of three nucleotides in DNA. The genetic code is described in detail in Chapter 27.

BOX 3-2 WORKING IN BIOCHEMISTRY

Investigating Proteins with Mass Spectrometry

The mass spectrometer has long been an indispensable tool in chemistry. Molecules to be analyzed, referred to as **analytes**, are first ionized in a vacuum. When the newly charged molecules are introduced into an electric and/or magnetic field, their paths through the field are a function of their mass-to-charge ratio, m/z . This measured property of the ionized species can be used to deduce the mass (M) of the analyte with very high precision.

Although mass spectrometry has been in use for many years, it could not be applied to macromolecules such as proteins and nucleic acids. The m/z measurements are made on molecules in the gas phase, and the heating or other treatment needed to transfer a macromolecule to the gas phase usually caused its rapid decomposition. In 1988, two different techniques were developed to overcome this problem. In one, proteins are placed in a light-absorbing matrix. With a short pulse of laser light, the proteins are ionized and then desorbed from the matrix into the vacuum system. This process, known as **matrix-assisted laser desorption/ionization mass spectrometry**, or **MALDI MS**, has been successfully used to measure the mass of a wide range of macromolecules. In a second and equally successful method, macromolecules in solution are forced directly from the liquid to the gas phase. A solution of analytes is passed through a charged needle that is kept at a high electrical potential, dispersing the solution into a fine mist of charged microdroplets. The solvent surrounding the macromolecules rapidly evaporates, and the resulting multiply charged macromolecular ions are thus introduced nondestructively into the gas phase. This technique is called **electrospray ionization mass spectrometry**, or **ESI MS**. Protons added during passage through the needle give additional charge to the macromolecule. The m/z of the molecule can be analyzed in the vacuum chamber.

Mass spectrometry provides a wealth of information for proteomics research, enzymology, and protein chemistry in general. The techniques require only miniscule amounts of sample, so they can be readily applied to the small amounts of protein that can be extracted from a two-dimensional electrophoretic gel. The accurately measured molecular mass of a protein is one of the critical parameters in its identification. Once the mass of a protein is accurately known, mass spectrometry is a convenient and accurate method for detecting changes in mass due to the presence of bound cofactors, bound metal ions, covalent modifications, and so on.

The process for determining the molecular mass of a protein with ESI MS is illustrated in Figure 1. As it is injected into the gas phase, a protein acquires a variable number of protons, and thus positive charges, from the solvent. This creates a spectrum of species with different mass-to-charge ratios. Each successive peak corresponds to a species that differs from that

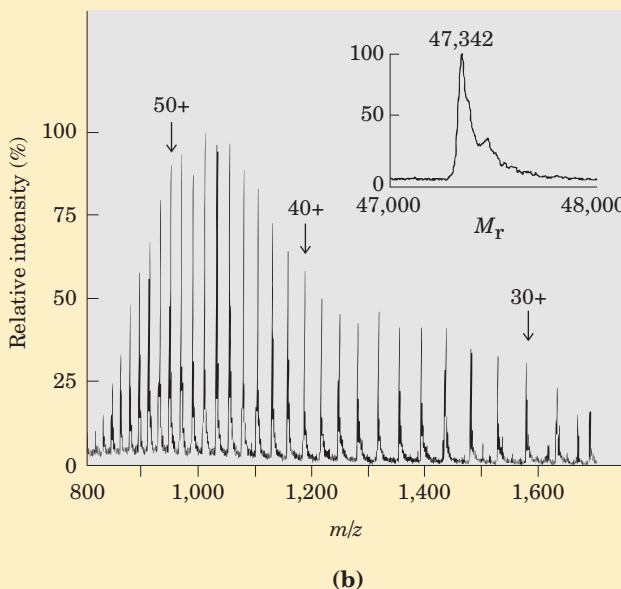
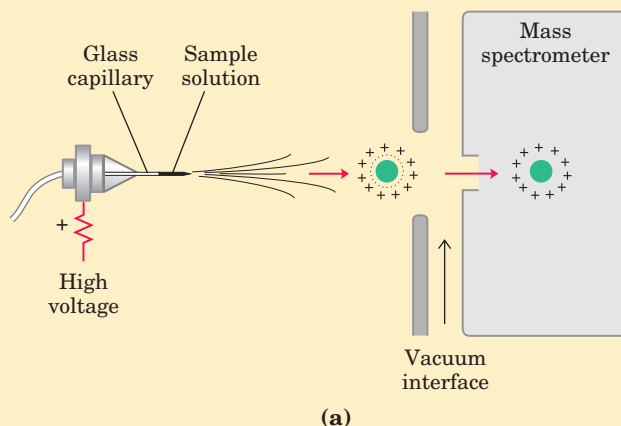


FIGURE 1 Electrospray mass spectrometry of a protein. (a) A protein solution is dispersed into highly charged droplets by passage through a needle under the influence of a high-voltage electric field. The droplets evaporate, and the ions (with added protons in this case) enter the mass spectrometer for m/z measurement. The spectrum generated (b) is a family of peaks, with each successive peak (from right to left) corresponding to a charged species increased by 1 in both mass and charge. A computer-generated transformation of this spectrum is shown in the inset.

of its neighboring peak by a charge difference of 1 and a mass difference of 1 (1 proton). The mass of the protein can be determined from any two neighboring peaks. The measured m/z of one peak is

$$(m/z)_2 = \frac{M + n_2 X}{n_2}$$

where M is the mass of the protein, n_2 is the number of charges, and X is the mass of the added groups (protons in this case). Similarly for the neighboring peak,

$$(m/z)_1 = \frac{M + (n_2 + 1)X}{n_2 + 1}$$

We now have two unknowns (M and n_2) and two equations. We can solve first for n_2 and then for M :

$$n_2 = \frac{(m/z)_2 - X}{(m/z)_2 - (m/z)_1}$$

$$M = n_2 [(m/z)_2 - X]$$

This calculation using the m/z values for any two peaks in a spectrum such as that shown in Figure 1b usually provides the mass of the protein (in this case, aerolysin k; 47,342 Da) with an error of only $\pm 0.01\%$. Generating several sets of peaks, repeating the calculation, and averaging the results generally provides an even more accurate value for M . Computer algorithms can transform the m/z spectrum into a single peak that

also provides a very accurate mass measurement (Fig. 1b, inset).

Mass spectrometry can also be used to sequence short stretches of polypeptide, an application that has emerged as an invaluable tool for quickly identifying unknown proteins. Sequence information is extracted using a technique called **tandem MS**, or **MS/MS**. A solution containing the protein under investigation is first treated with a protease or chemical reagent to hydrolyze it to a mixture of shorter peptides. The mixture is then injected into a device that is essentially two mass spectrometers in tandem (Fig. 2a, top). In the first, the peptide mixture is sorted and the ionized fragments are manipulated so that only one of the several types of peptides produced by cleavage emerges at the other end. The sample of the selected

(continued on next page)

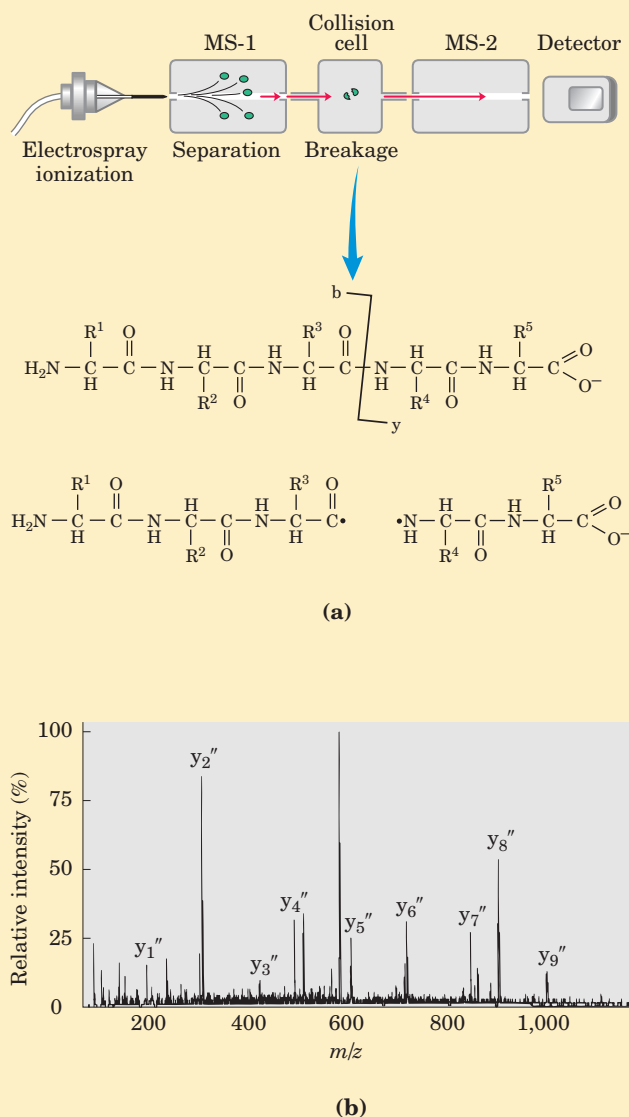


FIGURE 2 Obtaining protein sequence information with tandem MS. **(a)** After proteolytic hydrolysis, a protein solution is injected into a mass spectrometer (MS-1). The different peptides are sorted so that only one type is selected for further analysis. The selected peptide is further fragmented in a chamber between the two mass spectrometers, and m/z for each fragment is measured in the second mass spectrometer (MS-2). Many of the ions generated during this second fragmentation result from breakage of the peptide bond, as shown. These are called b-type or y-type ions, depending on whether the charge is retained on the amino- or carboxyl-terminal side, respectively. **(b)** A typical spectrum with peaks representing the peptide fragments generated from a sample of one small peptide (10 residues). The labeled peaks are y-type ions. The large peak next to y_5'' is a doubly charged ion and is not part of the y set. The successive peaks differ by the mass of a particular amino acid in the original peptide. In this case, the deduced sequence was Phe-Pro-Gly-Gln-(Ile/Leu)-Asn-Ala-Asp-(Ile/Leu)-Arg. Note the ambiguity about Ile and Leu residues, because they have the same molecular mass. In this example, the set of peaks derived from y-type ions predominates, and the spectrum is greatly simplified as a result. This is because an Arg residue occurs at the carboxyl terminus of the peptide, and most of the positive charges are retained on this residue.

BOX 3-2 WORKING IN BIOCHEMISTRY *(continued from previous page)*

peptide, each molecule of which has a charge somewhere along its length, then travels through a vacuum chamber between the two mass spectrometers. In this collision cell, the peptide is further fragmented by high-energy impact with a “collision gas,” a small amount of a noble gas such as helium or argon that is bled into the vacuum chamber. This procedure is designed to fragment many of the peptide molecules in the sample, with each individual peptide broken in only one place, on average. Most breaks occur at peptide bonds. This fragmentation does not involve the addition of water (it is done in a near-vacuum), so the products may include molecular ion radicals such as carbonyl radicals (Fig. 2a, bottom). The charge on the original peptide is retained on one of the fragments generated from it.

The second mass spectrometer then measures the m/z ratios of all the charged fragments (uncharged fragments are not detected). This generates one or more sets of peaks. A given set of peaks (Fig. 2b) consists of all the charged fragments that were generated by breaking the same type of bond (but at different points in the peptide) and are derived from the same side of the bond breakage, either the carboxyl- or amino-terminal side. Each successive peak in a given set has one less amino acid than the peak before. The difference in mass from peak to peak identifies the amino acid that was lost in each case, thus revealing the sequence of the peptide. The only ambiguities involve leucine and isoleucine, which have the same mass.

The charge on the peptide can be retained on either the carboxyl- or amino-terminal fragment, and

bonds other than the peptide bond can be broken in the fragmentation process, with the result that multiple sets of peaks are usually generated. The two most prominent sets generally consist of charged fragments derived from breakage of the peptide bonds. The set consisting of the carboxyl-terminal fragments can be unambiguously distinguished from that consisting of the amino-terminal fragments. Because the bond breaks generated between the spectrometers (in the collision cell) do not yield full carboxyl and amino groups at the sites of the breaks, the only intact α -amino and α -carboxyl groups on the peptide fragments are those at the very ends (Fig. 2a). The two sets of fragments can thereby be identified by the resulting slight differences in mass. The amino acid sequence derived from one set can be confirmed by the other, improving the confidence in the sequence information obtained.

Even a short sequence is often enough to permit unambiguous association of a protein with its gene, if the gene sequence is known. Sequencing by mass spectrometry cannot replace the Edman degradation procedure for the sequencing of long polypeptides, but it is ideal for proteomics research aimed at cataloging the hundreds of cellular proteins that might be separated on a two-dimensional gel. In the coming decades, detailed genomic sequence data will be available from hundreds, eventually thousands, of organisms. The ability to rapidly associate proteins with genes using mass spectrometry will greatly facilitate the exploitation of this extraordinary information resource.

Small Peptides and Proteins Can Be Chemically Synthesized

Many peptides are potentially useful as pharmacologic agents, and their production is of considerable commercial importance. There are three ways to obtain a peptide: (1) purification from tissue, a task often made difficult by the vanishingly low concentrations of some peptides; (2) genetic engineering (Chapter 9); or (3) direct chemical synthesis. Powerful techniques now make direct chemical synthesis an attractive option in many cases. In addition to commercial applications, the synthesis of specific peptide portions of larger proteins is an increasingly important tool for the study of protein structure and function.

The complexity of proteins makes the traditional synthetic approaches of organic chemistry impractical for peptides with more than four or five amino acid

residues. One problem is the difficulty of purifying the product after each step.

The major breakthrough in this technology was provided by R. Bruce Merrifield in 1962. His innovation involved synthesizing a peptide while keeping it attached at one end to a solid support. The support is an insoluble polymer (resin) contained within a column, similar to that used for chromatographic procedures. The peptide is built up on this support one amino acid at a time using a standard set of reactions in a repeating cycle (Fig. 3-29). At each successive step in the cycle, protective chemical groups block unwanted reactions.

The technology for chemical peptide synthesis is now automated. As in the sequencing reactions already considered, the most important limitation of the process is the efficiency of each chemical cycle, as can be seen by calculating the overall yields of peptides of various

lengths when the yield for addition of each new amino acid is 96.0% versus 99.8% (Table 3-8). Incomplete reaction at one stage can lead to formation of an impurity (in the form of a shorter peptide) in the next. The chemistry has been optimized to permit the synthesis

of proteins of 100 amino acid residues in a few days in reasonable yield. A very similar approach is used to synthesize nucleic acids (see Fig. 8-38). It is worth noting that this technology, impressive as it is, still pales when compared with biological processes. The same

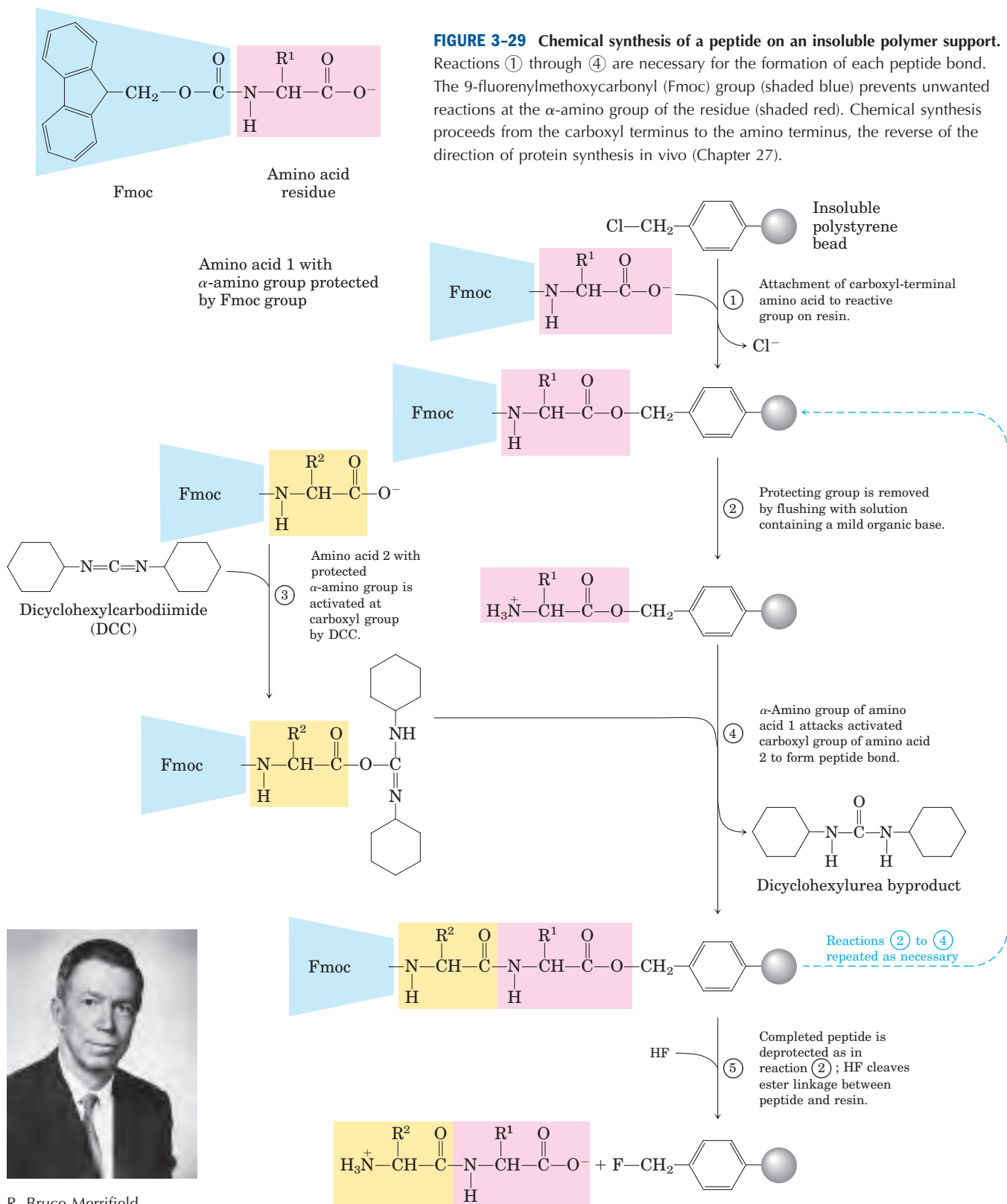


TABLE 3-8 Effect of Stepwise Yield on Overall Yield in Peptide Synthesis

Number of residues in the final polypeptide	Overall yield of final peptide (%) when the yield of each step is:	
	96.0%	99.8%
11	66	98
21	44	96
31	29	94
51	13	90
100	1.7	82

100-amino-acid protein would be synthesized with exquisite fidelity in about 5 seconds in a bacterial cell.

A variety of new methods for the efficient ligation (joining together) of peptides has made possible the assembly of synthetic peptides into larger proteins. With these methods, novel forms of proteins can be created with precisely positioned chemical groups, including those that might not normally be found in a cellular protein. These novel forms provide new ways to test theories of enzyme catalysis, to create proteins with new chemical properties, and to design protein sequences that will fold into particular structures. This last application provides the ultimate test of our increasing ability to relate the primary structure of a peptide to the three-dimensional structure that it takes up in solution.

Amino Acid Sequences Provide Important Biochemical Information

Knowledge of the sequence of amino acids in a protein can offer insights into its three-dimensional structure and its function, cellular location, and evolution. Most of these insights are derived by searching for similarities with other known sequences. Thousands of sequences are known and available in databases accessible through the Internet. A comparison of a newly obtained sequence with this large bank of stored sequences often reveals relationships both surprising and enlightening.

Exactly how the amino acid sequence determines three-dimensional structure is not understood in detail, nor can we always predict function from sequence. However, protein families that have some shared structural or functional features can be readily identified on the basis of amino acid sequence similarities. Individual proteins are assigned to families based on the degree of similarity in amino acid sequence. Members of a family are usually identical across 25% or more of their sequences, and proteins in these families generally share at least some structural and functional characteristics. Some families are defined, however, by identities involving only a few amino acid residues that are critical

to a certain function. A number of similar substructures (to be defined in Chapter 4 as “domains”) occur in many functionally unrelated proteins. These domains often fold into structural configurations that have an unusual degree of stability or that are specialized for a certain environment. Evolutionary relationships can also be inferred from the structural and functional similarities within protein families.

Certain amino acid sequences serve as signals that determine the cellular location, chemical modification, and half-life of a protein. Special signal sequences, usually at the amino terminus, are used to target certain proteins for export from the cell; other proteins are targeted for distribution to the nucleus, the cell surface, the cytosol, and other cellular locations. Other sequences act as attachment sites for prosthetic groups, such as sugar groups in glycoproteins and lipids in lipoproteins. Some of these signals are well characterized and are easily recognized in the sequence of a newly characterized protein (Chapter 27).

SUMMARY 3.4 The Covalent Structure of Proteins

- Differences in protein function result from differences in amino acid composition and sequence. Some variations in sequence are possible for a particular protein, with little or no effect on function.
- Amino acid sequences are deduced by fragmenting polypeptides into smaller peptides using reagents known to cleave specific peptide bonds; determining the amino acid sequence of each fragment by the automated Edman degradation procedure; then ordering the peptide fragments by finding sequence overlaps between fragments generated by different reagents. A protein sequence can also be deduced from the nucleotide sequence of its corresponding gene in DNA.
- Short proteins and peptides (up to about 100 residues) can be chemically synthesized. The peptide is built up, one amino acid residue at a time, while remaining tethered to a solid support.

3.5 Protein Sequences and Evolution

The simple string of letters denoting the amino acid sequence of a given protein belies the wealth of information this sequence holds. As more protein sequences have become available, the development of more powerful methods for extracting information from them has become a major biochemical enterprise. Each protein's function relies on its three-dimensional structure, which

in turn is determined largely by its primary structure. Thus, the biochemical information conveyed by a protein sequence is in principle limited only by our own understanding of structural and functional principles. On a different level of inquiry, protein sequences are beginning to tell us how the proteins evolved and, ultimately, how life evolved on this planet.

Protein Sequences Can Elucidate the History of Life on Earth

The field of molecular evolution is often traced to Emile Zuckerkandl and Linus Pauling, whose work in the mid-1960s advanced the use of nucleotide and protein sequences to explore evolution. The premise is deceptively straightforward. If two organisms are closely related, the sequences of their genes and proteins should be similar. The sequences increasingly diverge as the evolutionary distance between two organisms increases. The promise of this approach began to be realized in the 1970s, when Carl Woese used ribosomal RNA sequences to define archaeobacteria as a group of living organisms distinct from other bacteria and eukaryotes (see Fig. 1–4). Protein sequences offer an opportunity to greatly refine the available information. With the advent of genome projects investigating organisms from bacteria to humans, the number of available sequences is growing at an enormous rate. This information can be used to trace biological history. The challenge is in learning to read the genetic hieroglyphics.

Evolution has not taken a simple linear path. Complexities abound in any attempt to mine the evolutionary information stored in protein sequences. For a given protein, the amino acid residues essential for the activity of the protein are conserved over evolutionary time. The residues that are less important to function may vary over time—that is, one amino acid may substitute for another—and these variable residues can provide the information used to trace evolution. Amino acid substitutions are not always random, however. At some positions in the primary structure, the need to maintain protein function may mean that only particular amino acid substitutions can be tolerated. Some proteins have more variable amino acid residues than others. For these and other reasons, proteins can evolve at different rates.

Another complicating factor in tracing evolutionary history is the rare transfer of a gene or group of genes from one organism to another, a process called **lateral gene transfer**. The transferred genes may be quite sim-

ilar to the genes they were derived from in the original organism, whereas most other genes in the same two organisms may be quite distantly related. An example of lateral gene transfer is the recent rapid spread of antibiotic-resistance genes in bacterial populations. The proteins derived from these transferred genes would not be good candidates for the study of bacterial evolution, because they share only a very limited evolutionary history with their “host” organisms.

The study of molecular evolution generally focuses on families of closely related proteins. In most cases, the families chosen for analysis have essential functions in cellular metabolism that must have been present in the earliest viable cells, thus greatly reducing the chance that they were introduced relatively recently by lateral gene transfer. For example, a protein called EF-1 α (elongation factor 1 α) is involved in the synthesis of proteins in all eukaryotes. A similar protein, EF-Tu, with the same function, is found in bacteria. Similarities in sequence and function indicate that EF-1 α and EF-Tu are members of a family of proteins that share a common ancestor. The members of protein families are called **homologous proteins**, or **homologs**. The concept of a homolog can be further refined. If two proteins within a family (that is, two homologs) are present in the same species, they are referred to as **paralogs**. Homologs from different species are called **orthologs** (see Fig. 1–37). The process of tracing evolution involves first identifying suitable families of homologous proteins and then using them to reconstruct evolutionary paths.

Homologs are identified using increasingly powerful computer programs that can directly compare two or more chosen protein sequences, or can search vast databases to find the evolutionary relatives of one selected protein sequence. The electronic search process can be thought of as sliding one sequence past the other until a section with a good match is found. Within this sequence alignment, a positive score is assigned for each position where the amino acid residues in the two sequences are identical—the value of the score varying from one program to the next—to provide a measure of the quality of the alignment. The process has some complications. Sometimes the proteins being compared match well at, say, two sequence segments, and these segments are connected by less related sequences of different lengths. Thus the two matching segments cannot be aligned at the same time. To handle this, the computer program introduces “gaps” in one of the sequences to bring the matching segments into register (Fig. 3–30).



FIGURE 3–30 Aligning protein sequences with the use of gaps. Shown here is the sequence alignment of a short section of the EF-Tu protein from two well-studied bacterial species, *E. coli* and *Bacillus*

subtilis. Introduction of a gap in the *B. subtilis* sequence allows a better alignment of amino acid residues on either side of the gap. Identical amino acid residues are shaded.

Of course, if a sufficient number of gaps are introduced, almost any two sequences could be brought into some sort of alignment. To avoid uninformative alignments, the programs include penalties for each gap introduced, thus lowering the overall alignment score. With electronic trial and error, the program selects the alignment with the optimal score that maximizes identical amino acid residues while minimizing the introduction of gaps.

Identical amino acids are often inadequate to identify related proteins or, more importantly, to determine how closely related the proteins are on an evolutionary time scale. A more useful analysis includes a consideration of the chemical properties of substituted amino acids. When amino acid substitutions are found within a protein family, many of the differences may be conservative—that is, an amino acid residue is replaced by a residue having similar chemical properties. For example, a Glu residue may substitute in one family member for the Asp residue found in another; both amino acids are negatively charged. Such a conservative substitution should logically garner a higher score in a sequence alignment than does a nonconservative substitution, such as the replacement of the Asp residue with a hydrophobic Phe residue.

To determine what scores to assign to the many different amino acid substitutions, Steven Henikoff and Jorja Henikoff examined the aligned sequences from a variety of different proteins. They did not analyze entire protein sequences, focusing instead on thousands of short conserved blocks where the fraction of identical amino acids was high and the alignments were thus reliable. Looking at the aligned sequence blocks, the Henikoffs analyzed the nonidentical amino acid residues within the blocks. Higher scores were given to nonidentical residues that occurred frequently than to those that appeared rarely. Even the identical residues were given scores based on how often they were replaced, such that amino acids with unique chemical properties (such as Cys and Trp) received higher scores than those more conservatively replaced (such as Asp and Glu). The result of this scoring system is a Blosom (*blocks substitution matrix*) table. The table in Figure 3-31 was generated from sequences that were identical in at least 62% of their amino acid residues, and it is thus referred to as Blosom62. Similar tables have been generated for blocks of homologous sequences that are 50% or 80% identical. When higher levels of identity are required, the most conservative amino acid substitutions can be

	A Ala	C Cys	D Asp	E Glu	F Phe	G Gly	H His	I Ile	K Lys	L Leu	M Met	N Asn	P Pro	Q Gln	R Arg	S Ser	T Thr	V Val	W Trp	Y Tyr
A	4	0	-2	-1	-2	0	-2	-1	-1	-1	-1	-2	-1	-1	-1	1	0	0	-3	-2
C		9	-3	-4	-2	-3	-3	-1	-3	-1	-1	-3	-3	-3	-3	-1	-1	-1	-2	-2
D			6	2	-3	-1	-1	-3	-1	-4	-3	1	-1	0	-2	0	-1	-3	-4	-3
E				5	-3	-2	0	-3	1	-3	-2	0	-1	2	0	0	-1	-2	-3	-2
F					6	-3	-1	0	-3	0	0	-3	-4	-3	-3	-2	-2	-1	1	3
G						6	-2	-4	-2	-4	-3	0	-2	-2	-2	0	-2	-3	-2	-3
H							8	-3	-1	-3	-2	1	-2	0	0	-1	-2	-3	-2	2
I								4	-3	2	1	-3	-3	-3	-3	-2	-1	3	-3	-1
K									5	-2	-1	0	-1	1	2	0	-1	-2	-3	-2
L										4	2	-3	-3	-2	-2	-2	-1	1	-2	-1
M											5	-2	-2	0	-1	-1	-1	1	-1	-1
N												6	-2	0	0	1	0	-3	-4	-2
P													7	-1	-2	-1	-1	-2	-4	-3
Q														5	1	0	-1	-2	-2	-1
R															5	-1	-1	-3	-3	-2
S																4	1	-2	-3	-2
T																	5	0	-2	-2
V																		4	-3	-1
W																			11	2
Y																				7

FIGURE 3-31 The Blosom62 table. This *blocks substitution matrix* was created by comparing thousands of short blocks of aligned sequences that were identical in at least 62% of their amino acid residues. The nonidentical residues were assigned scores based on how frequently they were replaced by each of the other amino acids. Each substitution contributes to the score given to a particular alignment. Positive numbers (shaded yellow) add to the score for a particular alignment; negative numbers subtract from the score. Identical

residues in sequences being compared (the shaded diagonal from top left to bottom right in the matrix) receive scores based on how often they are replaced, such that amino acids with unique chemical properties (e.g., Cys and Trp) receive higher scores (9 and 11, respectively) than those more easily replaced in conservative substitutions (e.g., Asp (6) and Glu (5)). Many computer programs use Blosom62 to assign scores to new sequence alignments.

Archaeobacteria	<i>Halobacterium halobium</i>	IGHVDHGKST	TMVGR	Signature sequence	LYETGSVPEHV	IEQH
	<i>Sulfolobus solfataricus</i>	IGHVDHGKST	TLVGR	LLMDRG	GFIDEKT	VKEA
Eukaryotes	<i>Saccharomyces cerevisiae</i>	IGHVDSGKST	TTTGH	LIYKC	GGIDKRT	IEKF
	<i>Homo sapiens</i>	IGHVDSGKST	TTTGH	LIYKC	GGIDKRT	IEKF
Gram-positive bacterium	<i>Bacillus subtilis</i>	IGHVDHGKST	TMVGR			ITTV
Gram-negative bacterium	<i>Escherichia coli</i>	IGHVDHGKT	TLTAA			ITTV

FIGURE 3-32 A signature sequence in the EF-1 α /EF-Tu protein family. The signature sequence (boxed) is a 12-amino-acid insertion near the amino terminus of the sequence. Residues that align in all species are shaded yellow. Both archaeobacteria and eukaryotes have

the signature, although the sequences of the insertions are quite distinct for the two groups. The variation in the signature sequence reflects the significant evolutionary divergence that has occurred at this site since it first appeared in a common ancestor of both groups.

overrepresented, which limits the usefulness of the matrix in identifying homologs that are somewhat distantly related. Tests have shown that the Blosum62 table provides the most reliable alignments over a wide range of protein families, and it is the default table in many sequence alignment programs.

For most efforts to find homologies and explore evolutionary relationships, protein sequences (derived either directly from protein sequencing or from the sequencing of the DNA encoding the protein) are superior to nongenic nucleic acid sequences (those that do not encode a protein or functional RNA). For a nucleic acid, with its four different types of residues, random alignment of nonhomologous sequences will generally yield matches for at least 25% of the positions. Introduction of a few gaps can often increase the fraction of matched residues to 40% or more, and the probability of chance alignment of unrelated sequences becomes quite high. The 20 different amino acid residues in proteins greatly lower the probability of uninformative chance alignments of this type.

The programs used to generate a sequence alignment are complemented by methods that test the reliability of the alignments. A common computerized test is to shuffle the amino acid sequence of one of the proteins being compared to produce a random sequence, then instruct the program to align the shuffled sequence with the other, unshuffled one. Scores are assigned to the new alignment, and the shuffling and alignment process is repeated many times. The original alignment, before shuffling, should have a score significantly higher than any of those within the distribution of scores generated by the random alignments; this increases the confidence that the sequence alignment has identified a pair of homologs. Note that the *absence* of a significant alignment score does not necessarily mean that no evolutionary relationship exists between two proteins. As we shall see in Chapter 4, three-dimensional structural similarities sometimes reveal evolutionary relationships where sequence homology has been wiped away by time.

Using a protein family to explore evolution requires the identification of family members with similar molecular functions in the widest possible range of organ-

isms. Information from the family can then be used to trace the evolution of those organisms. By analyzing the sequence divergence in selected protein families, investigators can segregate organisms into classes based on their evolutionary relationships. This information must be reconciled with more classical examinations of the physiology and biochemistry of the organisms.

Certain segments of a protein sequence may be found in the organisms of one taxonomic group but not in other groups; these segments can be used as **signature sequences** for the group in which they are found. An example of a signature sequence is an insertion of 12 amino acids near the amino terminus of the EF-1 α /EF-Tu proteins in all archaeobacteria and eukaryotes but not in other types of bacteria (Fig. 3-32). The signature is one of many biochemical clues that can help establish the evolutionary relatedness of eukaryotes and archaeobacteria. For example, the major taxa of bacteria can be distinguished by signature sequences in several different proteins. The β and γ proteobacteria have signature sequences in the Hsp70 and DNA gyrase protein families (families of proteins involved in protein folding and DNA replication, respectively) that are not present in any other bacteria, including the other proteobacteria. The other types of proteobacteria (α , δ , ϵ), along with the β and γ proteobacteria, have a separate Hsp70 signature sequence and a signature in alanyl-tRNA synthetase (an enzyme of protein synthesis) that are not present in other bacteria. The appearance of unique signatures in the β and γ proteobacteria suggests the α , δ , and ϵ proteobacteria arose before their β and γ cousins.

By considering the entire sequence of a protein, researchers can now construct more elaborate evolutionary trees with many species in each taxonomic group. Figure 3-33 presents one such tree for bacteria, based on sequence divergence in the protein GroEL (a protein present in all bacteria that assists in the proper folding of proteins). The tree can be refined by basing it on the sequences of multiple proteins and by supplementing the sequence information with data on the unique biochemical and physiological properties of each species. There are many methods for generating trees, each with its own advantages and shortcomings, and

many ways to represent the resulting evolutionary relationships. In Figure 3-33, the free end points of lines are called “external nodes”; each represents an extant species, and each is so labeled. The points where two lines come together, the “internal nodes,” represent extinct ancestor species. In most representations (including Fig. 3-33), the lengths of the lines connecting the nodes are proportional to the number of amino acid substitutions separating one species from another. If we trace two extant species to a common internal node (representing the common ancestor of the two species), the length of the branch connecting each external node to the internal node represents the number of amino acid substitutions separating one extant species from this ancestor. The sum of the lengths of all the line segments that connect an extant species to another extant species through a common ancestor reflects the number of substitutions separating the two extant species. To determine how much time was needed for the various species to diverge, the tree must be calibrated by comparing it with information from the fossil record and other sources.

As more sequence information is made available in databases, we can generate evolutionary trees based on a variety of different proteins. Some proteins evolve faster than others, or change faster within one group of species than another. A large protein, with many vari-

able amino acid residues, may exhibit a few differences between two closely related species. Another, smaller protein may be identical in the same two species. For many reasons, some details of an evolutionary tree based on the sequences of one protein may differ from those of a tree based on the sequences of another protein. Increasingly sophisticated analyses using the sequences of many different proteins can provide an exquisitely detailed and accurate picture of evolutionary relationships. The story is a work in progress, and the questions being asked and answered are fundamental to how humans view themselves and the world around them. The field of molecular evolution promises to be among the most vibrant of the scientific frontiers in the twenty-first century.

SUMMARY 3.5 Protein Sequences and Evolution

- Protein sequences are a rich source of information about protein structure and function, as well as the evolution of life on this planet. Sophisticated methods are being developed to trace evolution by analyzing the resultant slow changes in the amino acid sequences of homologous proteins.

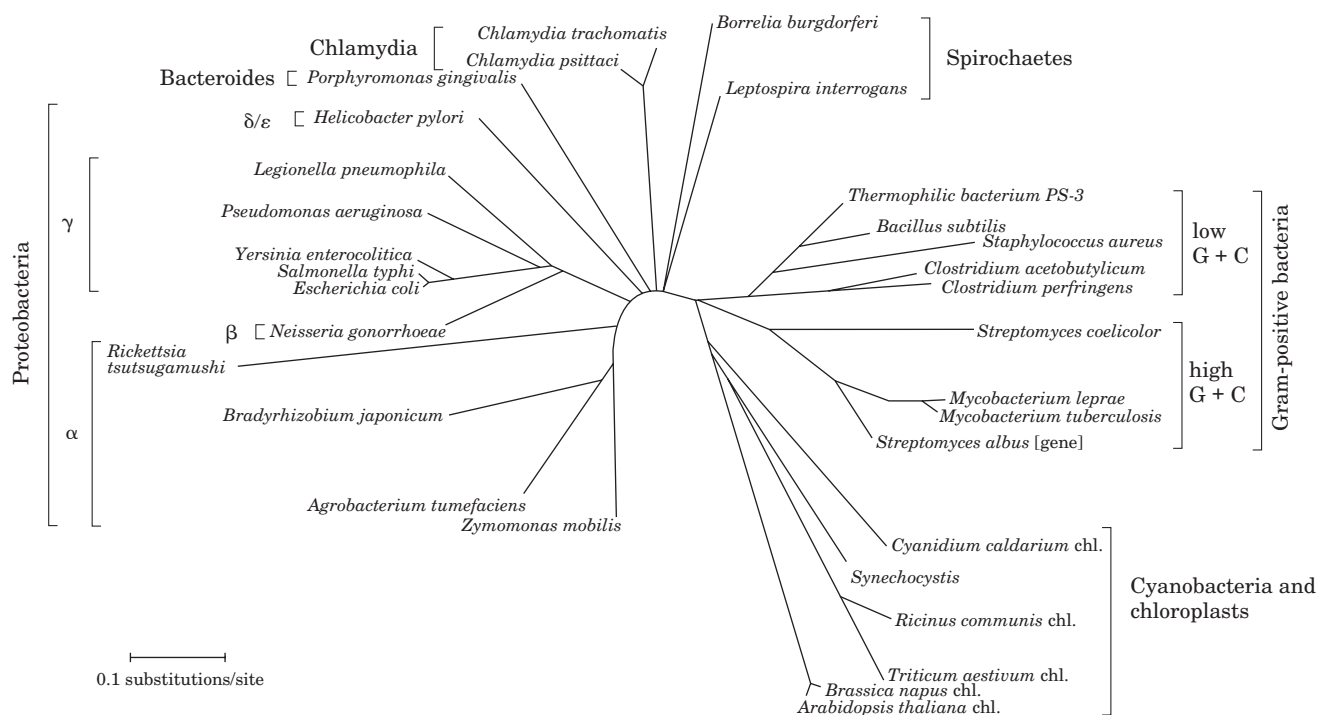


FIGURE 3-33 Evolutionary tree derived from amino acid sequence comparisons. A bacterial evolutionary tree, based on the sequence divergence observed in the GroEL family of proteins. Also included in this tree (lower right) are the chloroplasts (chl.) of some nonbacterial species.

Key Terms

Terms in bold are defined in the glossary.

amino acids 75	protein 85	quaternary structure 88	isoelectric focusing 93
R group 76	peptide bond 85	crude extract 89	Edman degradation 98
chiral center 76	oligopeptide 85	fractionation 89	proteases 99
enantiomers 76	polypeptide 85	dialysis 89	proteome 101
absolute configuration 77	oligomeric protein 87	column chromatography 89	lateral gene transfer 107
D, L system 77	protomer 87	high-performance liquid chromatography (HPLC) 90	homologous proteins 107
polarity 78	conjugated protein 88	electrophoresis 92	homolog 107
zwitterion 81	prosthetic group 88	sodium dodecyl sulfate (SDS) 92	paralog 107
absorbance, A 82	primary structure 88		ortholog 107
isoelectric pH (isoelectric point, pI) 84	secondary structure 88		signature sequence 109
peptide 85	tertiary structure 88		

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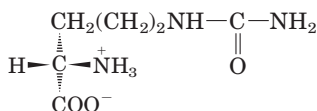
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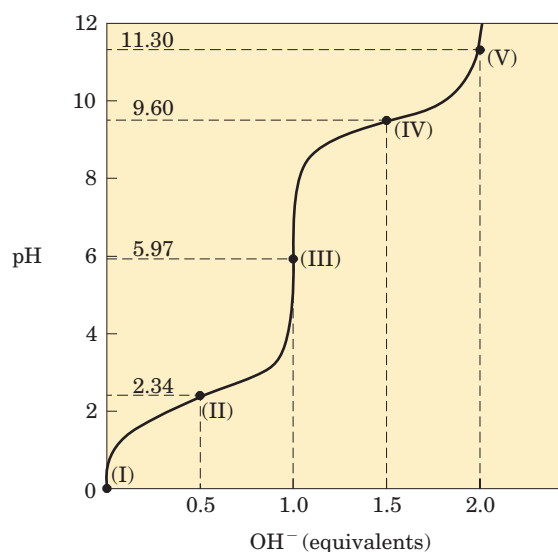
Problems

1. Absolute Configuration of Citrulline The citrulline isolated from watermelons has the structure shown below. Is it a D- or L-amino acid? Explain.

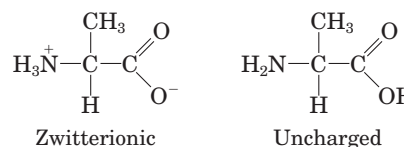


2. Relationship between the Titration Curve and the Acid-Base Properties of Glycine A 100 mL solution of 0.1 M glycine at pH 1.72 was titrated with 2 M NaOH solution. The pH was monitored and the results were plotted on a graph, as shown at right. The key points in the titration are designated I to V. For each of the statements (a) to (o), *identify* the appropriate key point in the titration and *justify* your choice.

- Glycine is present predominantly as the species $^+\text{H}_3\text{N}-\text{CH}_2-\text{COOH}$.
- The *average* net charge of glycine is $+\frac{1}{2}$.
- Half of the amino groups are ionized.
- The pH is equal to the pK_a of the carboxyl group.
- The pH is equal to the pK_a of the protonated amino group.
- Glycine has its maximum buffering capacity.
- The *average* net charge of glycine is zero.
- The carboxyl group has been completely titrated (first equivalence point).
- Glycine is completely titrated (second equivalence point).
- The predominant species is $^+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^-$.
- The *average* net charge of glycine is -1 .
- Glycine is present predominantly as a 50:50 mixture of $^+\text{H}_3\text{N}-\text{CH}_2-\text{COOH}$ and $^+\text{H}_3\text{N}-\text{CH}_2-\text{COO}^-$.
- This is the isoelectric point.
- This is the end of the titration.
- These are the *worst* pH regions for buffering power.



3. How Much Alanine Is Present as the Completely Uncharged Species? At a pH equal to the isoelectric point of alanine, the *net* charge on alanine is zero. Two structures can be drawn that have a net charge of zero, but the predominant form of alanine at its pI is zwitterionic.



- Why is alanine predominantly zwitterionic rather than completely uncharged at its pI?
- What fraction of alanine is in the completely uncharged form at its pI? Justify your assumptions.

4. Ionization State of Amino Acids Each ionizable group of an amino acid can exist in one of two states, charged or neutral. The electric charge on the functional group is determined by the relationship between its pK_a and the pH of the solution. This relationship is described by the Henderson-Hasselbalch equation.

(a) Histidine has three ionizable functional groups. Write the equilibrium equations for its three ionizations and assign the proper pK_a for each ionization. Draw the structure of histidine in each ionization state. What is the net charge on the histidine molecule in each ionization state?

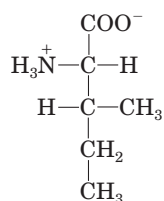
(b) Draw the structures of the predominant ionization state of histidine at pH 1, 4, 8, and 12. Note that the ionization state can be approximated by treating each ionizable group independently.

(c) What is the net charge of histidine at pH 1, 4, 8, and 12? For each pH, will histidine migrate toward the anode (+) or cathode (−) when placed in an electric field?

5. Separation of Amino Acids by Ion-Exchange Chromatography Mixtures of amino acids are analyzed by first separating the mixture into its components through ion-exchange chromatography. Amino acids placed on a cation-exchange resin containing sulfonate groups (see Fig. 3-18a) flow down the column at different rates because of two factors that influence their movement: (1) ionic attraction between the $-\text{SO}_3^-$ residues on the column and positively charged functional groups on the amino acids, and (2) hydrophobic interactions between amino acid side chains and the strongly hydrophobic backbone of the polystyrene resin. For each pair of amino acids listed, determine which will be eluted first from an ion-exchange column using a pH 7.0 buffer.

- Asp and Lys
- Arg and Met
- Glu and Val
- Gly and Leu
- Ser and Ala

6. Naming the Stereoisomers of Isoleucine The structure of the amino acid isoleucine is



- How many chiral centers does it have?
- How many optical isomers?
- Draw perspective formulas for all the optical isomers of isoleucine.

7. Comparing the pK_a Values of Alanine and Polyalanine The titration curve of alanine shows the ionization of two functional groups with pK_a values of 2.34 and 9.69, corresponding to the ionization of the carboxyl and the protonated amino groups, respectively. The titration of di-, tri-, and larger oligopeptides of alanine also shows the ionization of only two functional groups, although the experimental pK_a values are different. The trend in pK_a values is summarized in the table.

Amino acid or peptide	pK_1	pK_2
Ala	2.34	9.69
Ala-Ala	3.12	8.30
Ala-Ala-Ala	3.39	8.03
Ala-(Ala) $_n$ -Ala, $n \geq 4$	3.42	7.94

(a) Draw the structure of Ala-Ala-Ala. Identify the functional groups associated with pK_1 and pK_2 .

(b) Why does the value of pK_1 *increase* with each addition of an Ala residue to the Ala oligopeptide?

(c) Why does the value of pK_2 *decrease* with each addition of an Ala residue to the Ala oligopeptide?

8. The Size of Proteins What is the approximate molecular weight of a protein with 682 amino acid residues in a single polypeptide chain?

9. The Number of Tryptophan Residues in Bovine Serum Albumin A quantitative amino acid analysis reveals that bovine serum albumin (BSA) contains 0.58% tryptophan (M_r 204) by weight.

(a) Calculate the *minimum* molecular weight of BSA (i.e., assuming there is only one tryptophan residue per protein molecule).

(b) Gel filtration of BSA gives a molecular weight estimate of 70,000. How many tryptophan residues are present in a molecule of serum albumin?

10. Net Electric Charge of Peptides A peptide has the sequence



(a) What is the net charge of the molecule at pH 3, 8, and 11? (Use pK_a values for side chains and terminal amino and carboxyl groups as given in Table 3-1.)

(b) Estimate the pI for this peptide.

11. Isoelectric Point of Pepsin Pepsin is the name given to several digestive enzymes secreted (as larger precursor proteins) by glands that line the stomach. These glands also secrete hydrochloric acid, which dissolves the particulate matter in food, allowing pepsin to enzymatically cleave individual protein molecules. The resulting mixture of food, HCl, and digestive enzymes is known as chyme and has a pH near 1.5. What pI would you predict for the pepsin proteins? What functional groups must be present to confer this pI on pepsin? Which amino acids in the proteins would contribute such groups?

12. The Isoelectric Point of Histones Histones are proteins found in eukaryotic cell nuclei, tightly bound to DNA, which has many phosphate groups. The pI of histones is very high, about 10.8. What amino acid residues must be present in relatively large numbers in histones? In what way do these residues contribute to the strong binding of histones to DNA?

13. Solubility of Polypeptides One method for separating polypeptides makes use of their differential solubilities. The solubility of large polypeptides in water depends upon the relative polarity of their R groups, particularly on the number of ionized groups: the more ionized groups there are, the more soluble the polypeptide. Which of each pair of the polypeptides that follow is more soluble at the indicated pH?

- (a) (Gly)₂₀ or (Glu)₂₀ at pH 7.0
 (b) (Lys-Ala)₃ or (Phe-Met)₃ at pH 7.0
 (c) (Ala-Ser-Gly)₅ or (Asn-Ser-His)₅ at pH 6.0
 (d) (Ala-Asp-Gly)₅ or (Asn-Ser-His)₅ at pH 3.0

14. Purification of an Enzyme A biochemist discovers and purifies a new enzyme, generating the purification table below.

Procedure	Total protein (mg)	Activity (units)
1. Crude extract	20,000	4,000,000
2. Precipitation (salt)	5,000	3,000,000
3. Precipitation (pH)	4,000	1,000,000
4. Ion-exchange chromatography	200	800,000
5. Affinity chromatography	50	750,000
6. Size-exclusion chromatography	45	675,000

(a) From the information given in the table, calculate the specific activity of the enzyme solution after each purification procedure.

(b) Which of the purification procedures used for this enzyme is most effective (i.e., gives the greatest relative increase in purity)?

(c) Which of the purification procedures is least effective?

(d) Is there any indication based on the results shown in the table that the enzyme after step 6 is now pure? What else could be done to estimate the purity of the enzyme preparation?

15. Sequence Determination of the Brain Peptide Leucine Enkephalin

A group of peptides that influence nerve transmission in certain parts of the brain has been isolated from normal brain tissue. These peptides are known as opioids, because they bind to specific receptors that also bind opiate drugs, such as morphine and naloxone. Opioids thus mimic some of the properties of opiates. Some researchers consider these peptides to be the brain's own pain killers. Using the information below, determine the amino acid sequence of the opioid leucine enkephalin. Explain how your structure is consistent with each piece of information.

(a) Complete hydrolysis by 6 M HCl at 110 °C followed by amino acid analysis indicated the presence of Gly, Leu, Phe, and Tyr, in a 2:1:1:1 molar ratio.

(b) Treatment of the peptide with 1-fluoro-2,4-dinitrobenzene followed by complete hydrolysis and chromatography indicated the presence of the 2,4-dinitrophenyl derivative of tyrosine. No free tyrosine could be found.

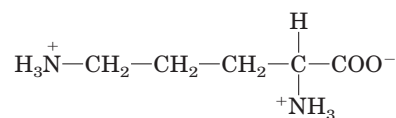
(c) Complete digestion of the peptide with pepsin followed by chromatography yielded a dipeptide containing Phe and Leu, plus a tripeptide containing Tyr and Gly in a 1:2 ratio.

16. Structure of a Peptide Antibiotic from *Bacillus brevis*

Extracts from the bacterium *Bacillus brevis* contain a peptide with antibiotic properties. This peptide forms complexes with metal ions and apparently disrupts ion transport across the cell membranes of other bacterial species, killing them. The structure of the peptide has been determined from the following observations.

(a) Complete acid hydrolysis of the peptide followed by amino acid analysis yielded equimolar amounts of Leu, Orn,

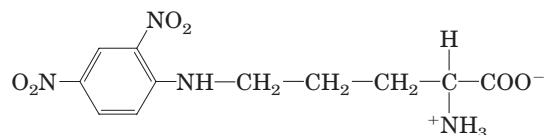
Phe, Pro, and Val. Orn is ornithine, an amino acid not present in proteins but present in some peptides. It has the structure



(b) The molecular weight of the peptide was estimated as about 1,200.

(c) The peptide failed to undergo hydrolysis when treated with the enzyme carboxypeptidase. This enzyme catalyzes the hydrolysis of the carboxyl-terminal residue of a polypeptide unless the residue is Pro or, for some reason, does not contain a free carboxyl group.

(d) Treatment of the intact peptide with 1-fluoro-2,4-dinitrobenzene, followed by complete hydrolysis and chromatography, yielded only free amino acids and the following derivative:



(Hint: Note that the 2,4-dinitrophenyl derivative involves the amino group of a side chain rather than the α-amino group.)

(e) Partial hydrolysis of the peptide followed by chromatographic separation and sequence analysis yielded the following di- and tripeptides (the amino-terminal amino acid is always at the left):

Leu-Phe Phe-Pro Orn-Leu Val-Orn

Val-Orn-Leu Phe-Pro-Val Pro-Val-Orn

Given the above information, deduce the amino acid sequence of the peptide antibiotic. Show your reasoning. When you have arrived at a structure, demonstrate that it is consistent with *each* experimental observation.

17. Efficiency in Peptide Sequencing A peptide with the primary structure Lys-Arg-Pro-Leu-Ile-Asp-Gly-Ala is sequenced by the Edman procedure. If each Edman cycle is 96% efficient, what percentage of the amino acids liberated in the fourth cycle will be leucine? Do the calculation a second time, but assume a 99% efficiency for each cycle.

18. Biochemistry Protocols: Your First Protein Purification

As the newest and least experienced student in a biochemistry research lab, your first few weeks are spent washing glassware and labeling test tubes. You then graduate to making buffers and stock solutions for use in various laboratory procedures. Finally, you are given responsibility for purifying a protein. It is a citric acid cycle enzyme, citrate synthase, located in the mitochondrial matrix. Following a protocol for the purification, you proceed through the steps below. As you work, a more experienced student questions you about the rationale for each procedure. Supply the answers. (Hint: See Chapter 2 for information about osmolarity; see p. 6 for information on separation of organelles from cells.)

(a) You pick up 20 kg of beef hearts from a nearby slaughterhouse. You transport the hearts on ice, and perform

each step of the purification on ice or in a walk-in cold room. You homogenize the beef heart tissue in a high-speed blender in a medium containing 0.2 M sucrose, buffered to a pH of 7.2. *Why do you use beef heart tissue, and in such large quantity? What is the purpose of keeping the tissue cold and suspending it in 0.2 M sucrose, at pH 7.2? What happens to the tissue when it is homogenized?*

(b) You subject the resulting heart homogenate, which is dense and opaque, to a series of differential centrifugation steps. *What does this accomplish?*

(c) You proceed with the purification using the supernatant fraction that contains mostly intact mitochondria. Next you osmotically lyse the mitochondria. The lysate, which is less dense than the homogenate, but still opaque, consists primarily of mitochondrial membranes and internal mitochondrial contents. To this lysate you add ammonium sulfate, a highly soluble salt, to a specific concentration. You centrifuge the solution, decant the supernatant, and discard the pellet. To the supernatant, which is clearer than the lysate, you add *more* ammonium sulfate. Once again, you centrifuge the sample, but this time you save the pellet because it contains the protein of interest. *What is the rationale for the two-step addition of the salt?*

(d) You solubilize the ammonium sulfate pellet containing the mitochondrial proteins and dialyze it overnight against large volumes of buffered (pH 7.2) solution. *Why isn't ammonium sulfate included in the dialysis buffer? Why do you use the buffer solution instead of water?*

(e) You run the dialyzed solution over a size-exclusion chromatographic column. Following the protocol, you collect the *first* protein fraction that exits the column, and discard the rest of the fractions that elute from the column later. You detect the protein by measuring UV absorbance (at 280 nm) in the fractions. *What does the instruction to collect the first fraction tell you about the protein? Why is UV absorbance at 280 nm a good way to monitor for the presence of protein in the eluted fractions?*

(f) You place the fraction collected in (e) on a cation-exchange chromatographic column. After discarding the initial solution that exits the column (the flowthrough), you add a washing solution of higher pH to the column and collect the protein fraction that immediately elutes. *Explain what you are doing.*

(g) You run a small sample of your fraction, now very reduced in volume and quite clear (though tinged pink), on an isoelectric focusing gel. When stained, the gel shows three sharp bands. According to the protocol, the protein of interest is the one with the pI of 5.6, but you decide to do one more assay of the protein's purity. You cut out the pI 5.6 band and subject it to SDS polyacrylamide gel electrophoresis. The protein resolves as a single band. *Why were you unconvinced of the purity of the "single" protein band on your isoelectric focusing gel? What did the results of the SDS gel tell you? Why is it important to do the SDS gel electrophoresis after the isoelectric focusing?*



THE THREE-DIMENSIONAL STRUCTURE OF PROTEINS

- 4.1 Overview of Protein Structure 116**
- 4.2 Protein Secondary Structure 120**
- 4.3 Protein Tertiary and Quaternary Structures 125**
- 4.4 Protein Denaturation and Folding 147**

Perhaps the more remarkable features of [myoglobin] are its complexity and its lack of symmetry. The arrangement seems to be almost totally lacking in the kind of regularities which one instinctively anticipates, and it is more complicated than has been predicted by any theory of protein structure.

—John Kendrew, article in *Nature*, 1958

The covalent backbone of a typical protein contains hundreds of individual bonds. Because free rotation is possible around many of these bonds, the protein can assume an unlimited number of conformations. However, each protein has a specific chemical or structural function, strongly suggesting that each has a unique three-dimensional structure (Fig. 4–1). By the late 1920s, several proteins had been crystallized, including hemoglobin (M_r 64,500) and the enzyme urease (M_r 483,000). Given that the ordered array of molecules in a crystal can generally form only if the molecular units are identical, the simple fact that many proteins can be crystallized provides strong evidence that even very large proteins are discrete chemical entities with unique structures. This conclusion revolutionized thinking about proteins and their functions.

In this chapter, we explore the three-dimensional structure of proteins, emphasizing five themes. First, the three-dimensional structure of a protein is determined by its amino acid sequence. Second, the function

of a protein depends on its structure. Third, an isolated protein usually exists in one or a small number of stable structural forms. Fourth, the most important forces stabilizing the specific structures maintained by a given protein are noncovalent interactions. Finally, amid the huge number of unique protein structures, we can recognize some common structural patterns that help us organize our understanding of protein architecture.

These themes should not be taken to imply that proteins have static, unchanging three-dimensional structures. Protein function often entails an interconversion between two or more structural forms. The dynamic aspects of protein structure will be explored in Chapters 5 and 6.

The relationship between the amino acid sequence of a protein and its three-dimensional structure is an intricate puzzle that is gradually yielding to techniques used in modern biochemistry. An understanding of structure, in turn, is essential to the discussion of function in succeeding chapters. We can find and understand the patterns within the biochemical labyrinth of protein structure by applying fundamental principles of chemistry and physics.

4.1 Overview of Protein Structure

The spatial arrangement of atoms in a protein is called its **conformation**. The possible conformations of a protein include any structural state that can be achieved without breaking covalent bonds. A change in conformation could occur, for example, by rotation about single bonds. Of the numerous conformations that are theoretically possible in a protein containing hundreds of single bonds, one or (more commonly) a few generally predominate under biological conditions. The need for multiple stable conformations reflects the changes that must occur in most proteins as they bind to other

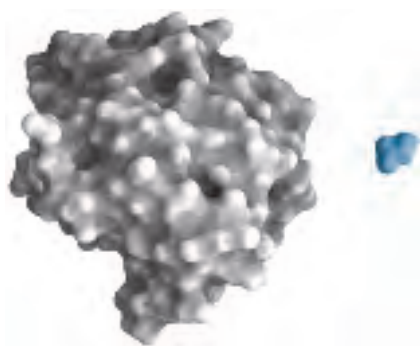


FIGURE 4-1 Structure of the enzyme chymotrypsin, a globular protein. Proteins are large molecules and, as we shall see, each has a unique structure. A molecule of glycine (blue) is shown for size comparison. The known three-dimensional structures of proteins are archived in the Protein Data Bank, or PDB (www.rcsb.org/pdb). Each structure is assigned a unique four-character identifier, or PDB ID. Where appropriate, we will provide the PDB IDs for molecular graphic images in the figure captions. The image shown here was made using data from the PDB file 6GCH. The data from the PDB files provide only a series of coordinates detailing the location of atoms and their connectivity. Viewing the images requires easy-to-use graphics programs such as RasMol and Chime that convert the coordinates into an image and allow the viewer to manipulate the structure in three dimensions. You will find instructions for downloading Chime with the structure tutorials on the textbook website (www.whfreeman.com/lehninger). The PDB website has instructions for downloading other viewers. We encourage all students to take advantage of the resources of the PDB and the free molecular graphics programs.

molecules or catalyze reactions. The conformations existing under a given set of conditions are usually the ones that are thermodynamically the most stable, having the lowest Gibbs free energy (G). Proteins in any of their functional, folded conformations are called **native proteins**.

What principles determine the most stable conformations of a protein? An understanding of protein conformation can be built stepwise from the discussion of primary structure in Chapter 3 through a consideration of secondary, tertiary, and quaternary structures. To this traditional approach must be added a new emphasis on supersecondary structures, a growing set of known and classifiable protein folding patterns that provides an important organizational context to this complex endeavor. We begin by introducing some guiding principles.

A Protein's Conformation Is Stabilized Largely by Weak Interactions

In the context of protein structure, the term **stability** can be defined as the tendency to maintain a native conformation. Native proteins are only marginally stable; the ΔG separating the folded and unfolded states in typical proteins under physiological conditions is in the range of only 20 to 65 kJ/mol. A given polypeptide chain

can theoretically assume countless different conformations, and as a result the unfolded state of a protein is characterized by a high degree of conformational entropy. This entropy, and the hydrogen-bonding interactions of many groups in the polypeptide chain with solvent (water), tend to maintain the unfolded state. The chemical interactions that counteract these effects and stabilize the native conformation include disulfide bonds and the weak (noncovalent) interactions described in Chapter 2: hydrogen bonds, and hydrophobic and ionic interactions. An appreciation of the role of these weak interactions is especially important to our understanding of how polypeptide chains fold into specific secondary and tertiary structures, and how they combine with other polypeptides to form quaternary structures.

About 200 to 460 kJ/mol are required to break a single covalent bond, whereas weak interactions can be disrupted by a mere 4 to 30 kJ/mol. Individual covalent bonds that contribute to the native conformations of proteins, such as disulfide bonds linking separate parts of a single polypeptide chain, are clearly much stronger than individual weak interactions. Yet, because they are so numerous, it is weak interactions that predominate as a stabilizing force in protein structure. In general, the protein conformation with the lowest free energy (that is, the most stable conformation) is the one with the maximum number of weak interactions.

The stability of a protein is not simply the sum of the free energies of formation of the many weak interactions within it. Every hydrogen-bonding group in a folded polypeptide chain was hydrogen-bonded to water prior to folding, and for every hydrogen bond formed in a protein, a hydrogen bond (of similar strength) between the same group and water was broken. The net stability contributed by a given weak interaction, or the *difference* in free energies of the folded and unfolded states, may be close to zero. We must therefore look elsewhere to explain why the native conformation of a protein is favored.

We find that the contribution of weak interactions to protein stability can be understood in terms of the properties of water (Chapter 2). Pure water contains a network of hydrogen-bonded H_2O molecules. No other molecule has the hydrogen-bonding potential of water, and other molecules present in an aqueous solution disrupt the hydrogen bonding of water. When water surrounds a hydrophobic molecule, the optimal arrangement of hydrogen bonds results in a highly structured shell, or **solvation layer**, of water in the immediate vicinity. The increased order of the water molecules in the solvation layer correlates with an unfavorable decrease in the entropy of the water. However, when nonpolar groups are clustered together, there is a decrease in the extent of the solvation layer because each group no longer presents its entire surface to the solution. The result is a favorable increase in entropy. As described in

Chapter 2, this entropy term is the major thermodynamic driving force for the association of hydrophobic groups in aqueous solution. Hydrophobic amino acid side chains therefore tend to be clustered in a protein's interior, away from water.

Under physiological conditions, the formation of hydrogen bonds and ionic interactions in a protein is driven largely by this same entropic effect. Polar groups can generally form hydrogen bonds with water and hence are soluble in water. However, the number of hydrogen bonds per unit mass is generally greater for pure water than for any other liquid or solution, and there are limits to the solubility of even the most polar molecules as their presence causes a net decrease in hydrogen bonding per unit mass. Therefore, a solvation shell of structured water will also form to some extent around polar molecules. Even though the energy of formation of an intramolecular hydrogen bond or ionic interaction between two polar groups in a macromolecule is largely canceled out by the elimination of such interactions between the same groups and water, the release of structured water when the intramolecular interaction is formed provides an entropic driving force for folding. Most of the net change in free energy that occurs when weak interactions are formed within a protein is therefore derived from the increased entropy in the surrounding aqueous solution resulting from the burial of hydrophobic surfaces. This more than counterbalances the large loss of conformational entropy as a polypeptide is constrained into a single folded conformation.

Hydrophobic interactions are clearly important in stabilizing a protein conformation; the interior of a protein is generally a densely packed core of hydrophobic amino acid side chains. It is also important that any polar or charged groups in the protein interior have suitable partners for hydrogen bonding or ionic interactions. One hydrogen bond seems to contribute little to the stability of a native structure, but the presence of hydrogen-bonding or charged groups without partners in the hydrophobic core of a protein can be so *destabilizing* that conformations containing these groups are often thermodynamically untenable. The favorable free-energy change realized by combining such a group with a partner in the surrounding solution can be greater than the difference in free energy between the folded and unfolded states. In addition, hydrogen bonds between groups in proteins form cooperatively. Formation of one hydrogen bond facilitates the formation of additional hydrogen bonds. The overall contribution of hydrogen bonds and other noncovalent interactions to the stabilization of protein conformation is still being evaluated. The interaction of oppositely charged groups that form an ion pair (salt bridge) may also have a stabilizing effect on one or more native conformations of some proteins.

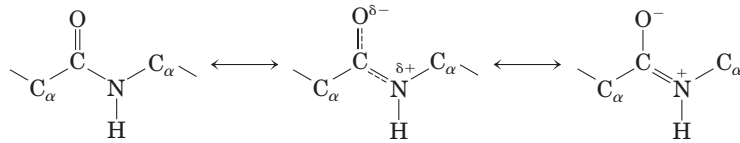
Most of the structural patterns outlined in this chapter reflect two simple rules: (1) hydrophobic residues

are largely buried in the protein interior, away from water; and (2) the number of hydrogen bonds within the protein is maximized. Insoluble proteins and proteins within membranes (which we examine in Chapter 11) follow somewhat different rules because of their function or their environment, but weak interactions are still critical structural elements.

The Peptide Bond Is Rigid and Planar

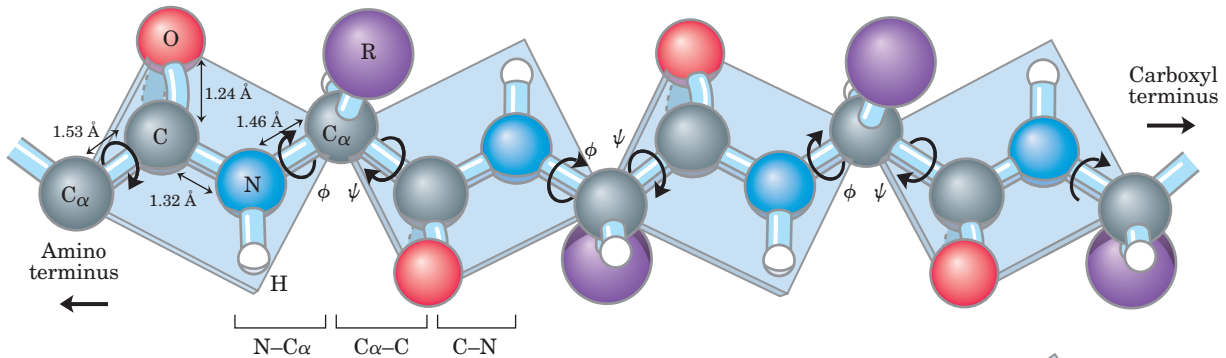
 **Protein Architecture—Primary Structure** Covalent bonds also place important constraints on the conformation of a polypeptide. In the late 1930s, Linus Pauling and Robert Corey embarked on a series of studies that laid the foundation for our present understanding of protein structure. They began with a careful analysis of the peptide bond. The α carbons of adjacent amino acid residues are separated by three covalent bonds, arranged as $C_\alpha-C-N-C_\alpha$. X-ray diffraction studies of crystals of amino acids and of simple dipeptides and tripeptides demonstrated that the peptide $C-N$ bond is somewhat shorter than the $C-N$ bond in a simple amine and that the atoms associated with the peptide bond are coplanar. This indicated a resonance or partial sharing of two pairs of electrons between the carbonyl oxygen and the amide nitrogen (Fig. 4-2a). The oxygen has a partial negative charge and the nitrogen a partial positive charge, setting up a small electric dipole. The six atoms of the **peptide group** lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other. From these findings Pauling and Corey concluded that the peptide $C-N$ bonds are unable to rotate freely because of their partial double-bond character. Rotation is permitted about the $N-C_\alpha$ and the $C_\alpha-C$ bonds. The backbone of a polypeptide chain can thus be pictured as a series of rigid planes with consecutive planes sharing a common point of rotation at C_α (Fig. 4-2b). The rigid peptide bonds limit the range of conformations that can be assumed by a polypeptide chain.

By convention, the bond angles resulting from rotations at C_α are labeled ϕ (phi) for the $N-C_\alpha$ bond and ψ (psi) for the $C_\alpha-C$ bond. Again by convention, both ϕ and ψ are defined as 180° when the polypeptide is in its fully extended conformation and all peptide groups are in the same plane (Fig. 4-2b). In principle, ϕ and ψ can have any value between -180° and $+180^\circ$, but many values are prohibited by steric interference between atoms in the polypeptide backbone and amino acid side chains. The conformation in which both ϕ and ψ are 0° (Fig. 4-2c) is prohibited for this reason; this conformation is used merely as a reference point for describing the angles of rotation. Allowed values for ϕ and ψ are graphically revealed when ψ is plotted versus ϕ in a **Ramachandran plot** (Fig. 4-3), introduced by G. N. Ramachandran.



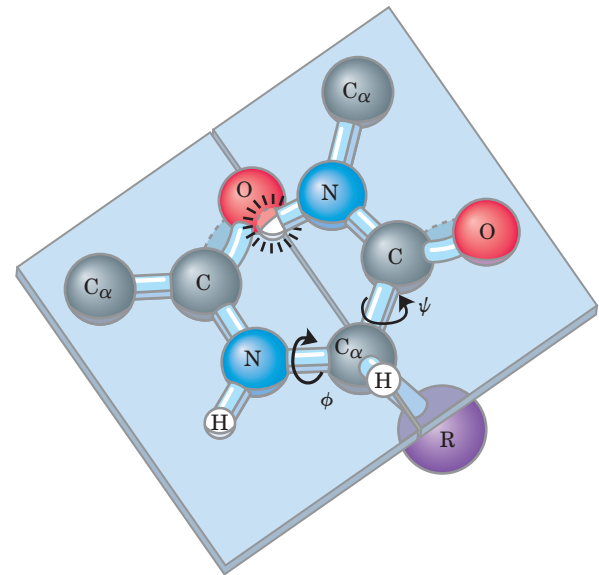
The carbonyl oxygen has a partial negative charge and the amide nitrogen a partial positive charge, setting up a small electric dipole. Virtually all peptide bonds in proteins occur in this trans configuration; an exception is noted in Figure 4–8b.

(a)



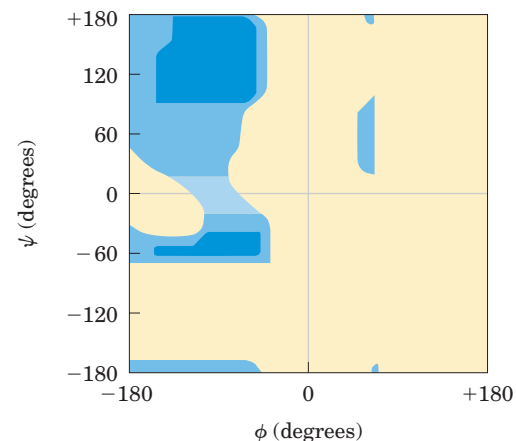
(b)

FIGURE 4–2 The planar peptide group. (a) Each peptide bond has some double-bond character due to resonance and cannot rotate. (b) Three bonds separate sequential α carbons in a polypeptide chain. The $N-C_\alpha$ and $C_\alpha-C$ bonds can rotate, with bond angles designated ϕ and ψ , respectively. The peptide $C-N$ bond is not free to rotate. Other single bonds in the backbone may also be rotationally hindered, depending on the size and charge of the R groups. In the conformation shown, ϕ and ψ are 180° (or -180°). As one looks out from the α carbon, the ψ and ϕ angles increase as the carbonyl or amide nitrogens (respectively) rotate clockwise. (c) By convention, both ϕ and ψ are defined as 0° when the two peptide bonds flanking that α carbon are in the same plane and positioned as shown. In a protein, this conformation is prohibited by steric overlap between an α -carbonyl oxygen and an α -amino hydrogen atom. To illustrate the bonds between atoms, the balls representing each atom are smaller than the van der Waals radii for this scale. $1 \text{ \AA} = 0.1 \text{ nm}$.



(c)

FIGURE 4–3 Ramachandran plot for L-Ala residues. The conformations of peptides are defined by the values of ϕ and ψ . Conformations deemed possible are those that involve little or no steric interference, based on calculations using known van der Waals radii and bond angles. The areas shaded dark blue reflect conformations that involve no steric overlap and thus are fully allowed; medium blue indicates conformations allowed at the extreme limits for unfavorable atomic contacts; the lightest blue area reflects conformations that are permissible if a little flexibility is allowed in the bond angles. The asymmetry of the plot results from the L stereochemistry of the amino acid residues. The plots for other L-amino acid residues with unbranched side chains are nearly identical. The allowed ranges for branched amino acid residues such as Val, Ile, and Thr are somewhat smaller than for Ala. The Gly residue, which is less sterically hindered, exhibits a much broader range of allowed conformations. The range for Pro residues is greatly restricted because ϕ is limited by the cyclic side chain to the range of -35° to -85° .





Linus Pauling, 1901–1994



Robert Corey, 1897–1971

SUMMARY 4.1 Overview of Protein Structure

- Every protein has a three-dimensional structure that reflects its function.
- Protein structure is stabilized by multiple weak interactions. Hydrophobic interactions are the major contributors to stabilizing the globular form of most soluble proteins; hydrogen bonds and ionic interactions are optimized in the specific structures that are thermodynamically most stable.
- The nature of the covalent bonds in the polypeptide backbone places constraints on structure. The peptide bond has a partial double-bond character that keeps the entire six-atom peptide group in a rigid planar configuration. The N—C_α and C_α—C bonds can rotate to assume bond angles of ϕ and ψ , respectively.

4.2 Protein Secondary Structure

The term **secondary structure** refers to the local conformation of some part of a polypeptide. The discussion of secondary structure most usefully focuses on common regular folding patterns of the polypeptide backbone. A few types of secondary structure are particularly stable and occur widely in proteins. The most prominent are the α helix and β conformations described below. Using fundamental chemical principles and a few experimental observations, Pauling and Corey predicted the existence of these secondary structures in 1951, several years before the first complete protein structure was elucidated.

The α Helix Is a Common Protein Secondary Structure

 **Protein Architecture— α Helix** Pauling and Corey were aware of the importance of hydrogen bonds in orient-

ing polar chemical groups such as the C=O and N—H groups of the peptide bond. They also had the experimental results of William Astbury, who in the 1930s had conducted pioneering x-ray studies of proteins. Astbury demonstrated that the protein that makes up hair and porcupine quills (the fibrous protein α -keratin) has a regular structure that repeats every 5.15 to 5.2 Å. (The angstrom, Å, named after the physicist Anders J. Ångström, is equal to 0.1 nm. Although not an SI unit, it is used universally by structural biologists to describe atomic distances.) With this information and their data on the peptide bond, and with the help of precisely constructed models, Pauling and Corey set out to determine the likely conformations of protein molecules.

The simplest arrangement the polypeptide chain could assume with its rigid peptide bonds (but other single bonds free to rotate) is a helical structure, which Pauling and Corey called the **α helix** (Fig. 4–4). In this structure the polypeptide backbone is tightly wound around an imaginary axis drawn longitudinally through the middle of the helix, and the R groups of the amino acid residues protrude outward from the helical backbone. The repeating unit is a single turn of the helix, which extends about 5.4 Å along the long axis, slightly greater than the periodicity Astbury observed on x-ray analysis of hair keratin. The amino acid residues in an α helix have conformations with $\psi = -45^\circ$ to -50° and $\phi = -60^\circ$, and each helical turn includes 3.6 amino acid residues. The helical twist of the α helix found in all proteins is right-handed (Box 4–1). The α helix proved to be the predominant structure in α -keratins. More generally, about one-fourth of all amino acid residues in polypeptides are found in α helices, the exact fraction varying greatly from one protein to the next.

Why does the α helix form more readily than many other possible conformations? The answer is, in part, that an α helix makes optimal use of internal hydrogen bonds. The structure is stabilized by a hydrogen bond between the hydrogen atom attached to the electronegative nitrogen atom of a peptide linkage and the electronegative carbonyl oxygen atom of the fourth amino acid on the amino-terminal side of that peptide bond (Fig. 4–4b). Within the α helix, every peptide bond (except those close to each end of the helix) participates in such hydrogen bonding. Each successive turn of the α helix is held to adjacent turns by three to four hydrogen bonds. All the hydrogen bonds combined give the entire helical structure considerable stability.

Further model-building experiments have shown that an α helix can form in polypeptides consisting of either L- or D-amino acids. However, all residues must be of one stereoisomeric series; a D-amino acid will disrupt a regular structure consisting of L-amino acids, and vice versa. Naturally occurring L-amino acids can form either right- or left-handed α helices, but extended left-handed helices have not been observed in proteins.

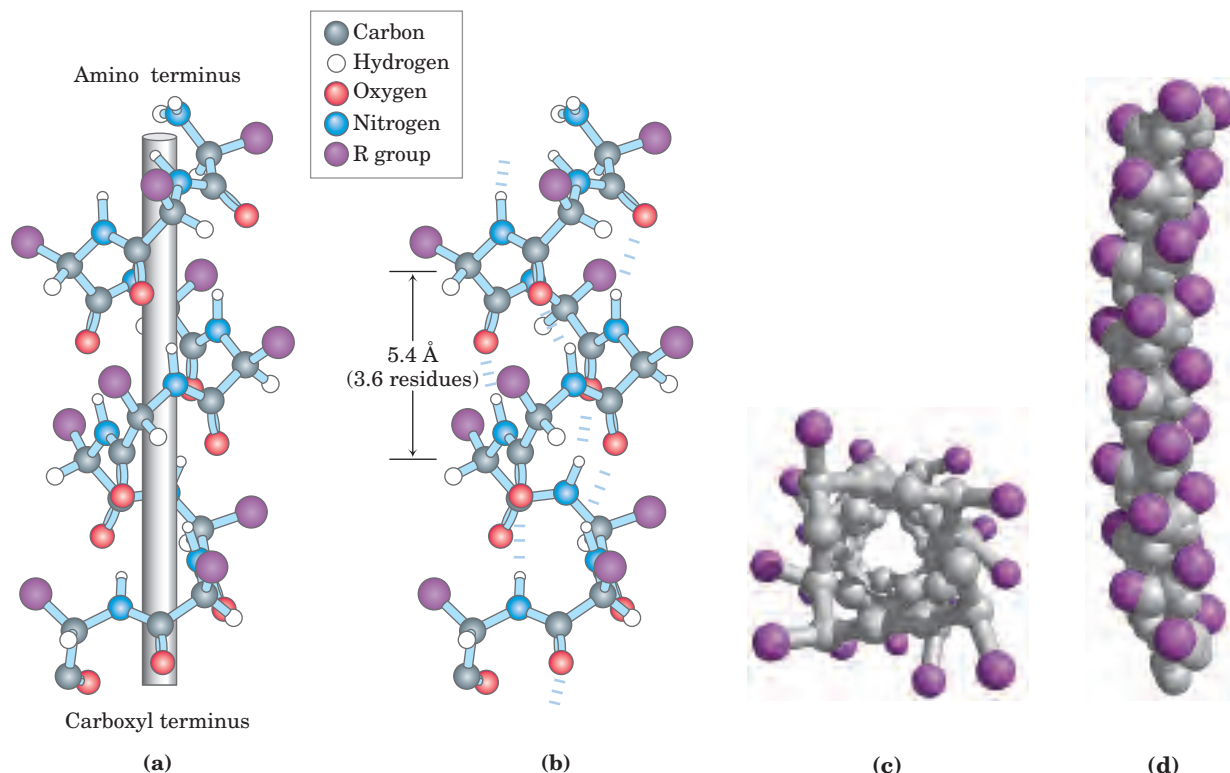


FIGURE 4-4 Four models of the α helix, showing different aspects of its structure. (a) Formation of a right-handed α helix. The planes of the rigid peptide bonds are parallel to the long axis of the helix, depicted here as a vertical rod. (b) Ball-and-stick model of a right-handed α helix, showing the intrachain hydrogen bonds. The repeat unit is a single turn of the helix, 3.6 residues. (c) The α helix as viewed from one end, looking down the longitudinal axis (derived from PDB

ID 4TNC). Note the positions of the R groups, represented by purple spheres. This ball-and-stick model, used to emphasize the helical arrangement, gives the false impression that the helix is hollow, because the balls do not represent the van der Waals radii of the individual atoms. As the space-filling model (d) shows, the atoms in the center of the α helix are in very close contact.

Amino Acid Sequence Affects α Helix Stability

Not all polypeptides can form a stable α helix. Interactions between amino acid side chains can stabilize or destabilize this structure. For example, if a polypeptide chain has a long block of Glu residues, this segment of the chain will not form an α helix at pH 7.0. The negatively charged carboxyl groups of adjacent Glu residues repel each other so strongly that they prevent formation of the α helix. For the same reason, if there are many adjacent Lys and/or Arg residues, which have positively charged R groups at pH 7.0, they will also repel each other and prevent formation of the α helix. The bulk and shape of Asn, Ser, Thr, and Cys residues can also destabilize an α helix if they are close together in the chain.

The twist of an α helix ensures that critical interactions occur between an amino acid side chain and the side chain three (and sometimes four) residues away on either side of it (Fig. 4-5). Positively charged amino acids are often found three residues away from negatively charged amino acids, permitting the formation of an ion pair. Two aromatic amino acid residues are often similarly spaced, resulting in a hydrophobic interaction.

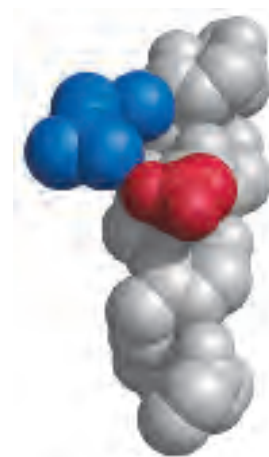


FIGURE 4-5 Interactions between R groups of amino acids three residues apart in an α helix. An ionic interaction between Asp¹⁰⁰ and Arg¹⁰³ in an α -helical region of the protein troponin C, a calcium-binding protein associated with muscle, is shown in this space-filling model (derived from PDB ID 4TNC). The polypeptide backbone (carbons, α -amino nitrogens, and α -carbonyl oxygens) is shown in gray for a helix segment 13 residues long. The only side chains represented here are the interacting Asp (red) and Arg (blue) side chains.

BOX 4-1 WORKING IN BIOCHEMISTRY

Knowing the Right Hand from the Left

There is a simple method for determining whether a helical structure is right-handed or left-handed. Make fists of your two hands with thumbs outstretched and pointing straight up. Looking at your right hand, think of a helix spiraling up your right thumb in the direction in which the other four fingers are curled as shown (counterclockwise). The resulting helix is right-handed. Your left hand will demonstrate a left-handed helix, which rotates in the clockwise direction as it spirals up your thumb.



A constraint on the formation of the α helix is the presence of Pro or Gly residues. In proline, the nitrogen atom is part of a rigid ring (see Fig. 4-8b), and rotation about the N—C $_{\alpha}$ bond is not possible. Thus, a Pro residue introduces a destabilizing kink in an α helix. In addition, the nitrogen atom of a Pro residue in peptide linkage has no substituent hydrogen to participate in hydrogen bonds with other residues. For these reasons, proline is only rarely found within an α helix. Glycine occurs infrequently in α helices for a different reason: it has more conformational flexibility than the other amino acid residues. Polymers of glycine tend to take up coiled structures quite different from an α helix.

A final factor affecting the stability of an α helix in a polypeptide is the identity of the amino acid residues near the ends of the α -helical segment. A small electric dipole exists in each peptide bond (Fig. 4-2a). These dipoles are connected through the hydrogen bonds of the helix, resulting in a net dipole extending along the helix that increases with helix length (Fig. 4-6). The four amino acid residues at each end of the helix do not participate fully in the helix hydrogen bonds. The partial positive and negative charges of the helix dipole actually reside on the peptide amino and carbonyl groups near the amino-terminal and carboxyl-terminal ends of the helix, respectively. For this reason, negatively charged amino acids are often found near the amino terminus of the helical segment, where they have a stabilizing interaction with the positive charge of the helix dipole; a positively charged amino acid at the amino-terminal end is destabilizing. The opposite is true at the carboxyl-terminal end of the helical segment.

Thus, five different kinds of constraints affect the stability of an α helix: (1) the electrostatic repulsion (or attraction) between successive amino acid residues with charged R groups, (2) the bulkiness of adjacent R groups, (3) the interactions between R groups spaced

three (or four) residues apart, (4) the occurrence of Pro and Gly residues, and (5) the interaction between amino acid residues at the ends of the helical segment and the electric dipole inherent to the α helix. The tendency of a given segment of a polypeptide chain to fold up as an α helix therefore depends on the identity and sequence of amino acid residues within the segment.

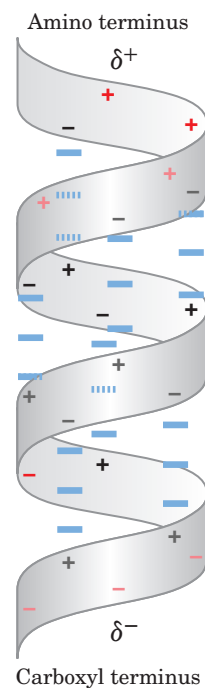


FIGURE 4-6 Helix dipole. The electric dipole of a peptide bond (see Fig. 4-2a) is transmitted along an α -helical segment through the intrachain hydrogen bonds, resulting in an overall helix dipole. In this illustration, the amino and carbonyl constituents of each peptide bond are indicated by + and - symbols, respectively. Non-hydrogen-bonded amino and carbonyl constituents in the peptide bonds near each end of the α -helical region are shown in red.

The β Conformation Organizes Polypeptide Chains into Sheets

Protein Architecture— β Sheet Pauling and Corey predicted a second type of repetitive structure, the **β conformation**. This is a more extended conformation of polypeptide chains, and its structure has been confirmed by x-ray analysis. In the β conformation, the backbone of the polypeptide chain is extended into a zigzag rather than helical structure (Fig. 4–7). The zigzag polypeptide chains can be arranged side by side to form a structure resembling a series of pleats. In this arrangement, called a **β sheet**, hydrogen bonds are formed between adjacent segments of polypeptide chain. The individual segments that form a β sheet are usually nearby on the polypeptide chain, but can also be quite distant from each other in the linear sequence of the polypeptide; they may even be segments in different polypeptide chains. The R groups of adjacent amino acids protrude from the zigzag structure in opposite directions, creating the alternating pattern seen in the side views in Figure 4–7.

The adjacent polypeptide chains in a β sheet can be either parallel or antiparallel (having the same or opposite amino-to-carboxyl orientations, respectively). The structures are somewhat similar, although the repeat period is shorter for the parallel conformation (6.5 Å, versus 7 Å for antiparallel) and the hydrogen-bonding patterns are different.

Some protein structures limit the kinds of amino acids that can occur in the β sheet. When two or more β sheets are layered close together within a protein, the R groups of the amino acid residues on the touching surfaces must be relatively small. β -Keratins such as silk fibroin and the fibroin of spider webs have a very high content of Gly and Ala residues, the two amino acids with the smallest R groups. Indeed, in silk fibroin Gly and Ala alternate over large parts of the sequence.

β Turns Are Common in Proteins

Protein Architecture— β Turn In globular proteins, which have a compact folded structure, nearly one-third of the amino acid residues are in turns or loops where the polypeptide chain reverses direction (Fig. 4–8). These are the connecting elements that link successive runs of α helix or β conformation. Particularly common are **β turns** that connect the ends of two adjacent segments of an antiparallel β sheet. The structure is a 180° turn involving four amino acid residues, with the carbonyl oxygen of the first residue forming a hydrogen bond with the amino-group hydrogen of the fourth. The peptide groups of the central two residues do not participate in any interresidue hydrogen bonding. Gly and Pro residues often occur in β turns, the former because it is small and flexible, the latter because peptide bonds

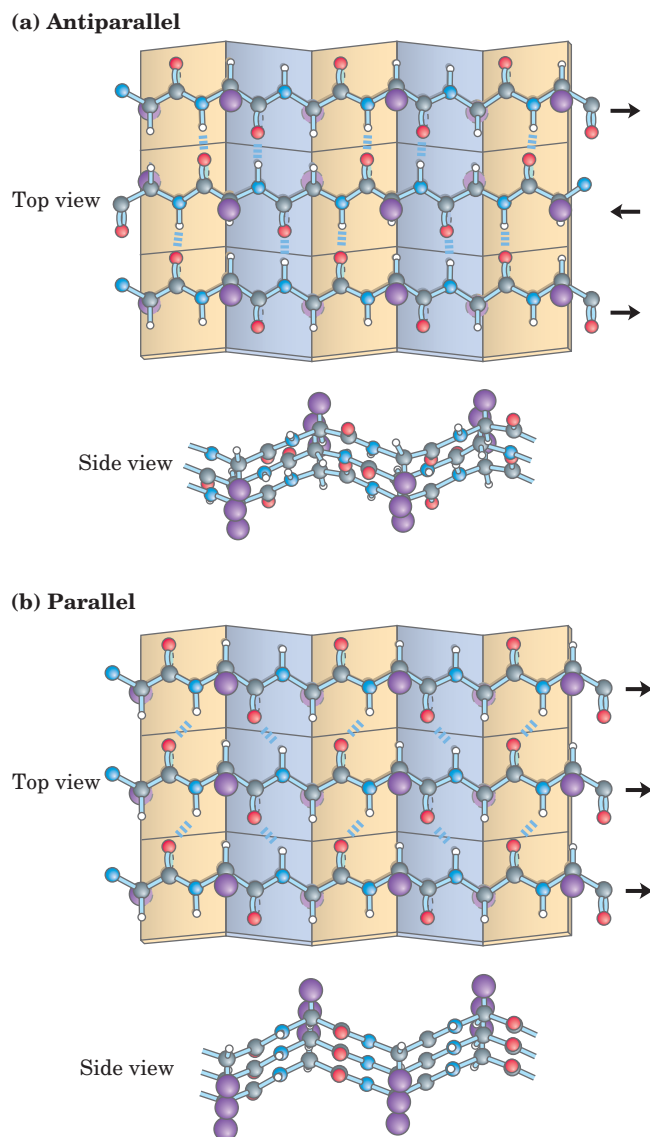


FIGURE 4–7 The β conformation of polypeptide chains. These top and side views reveal the R groups extending out from the β sheet and emphasize the pleated shape described by the planes of the peptide bonds. (An alternative name for this structure is β -pleated sheet.) Hydrogen-bond cross-links between adjacent chains are also shown. (a) Antiparallel β sheet, in which the amino-terminal to carboxyl-terminal orientation of adjacent chains (arrows) is inverse. (b) Parallel β sheet.

involving the imino nitrogen of proline readily assume the cis configuration (Fig. 4–8b), a form that is particularly amenable to a tight turn. Of the several types of β turns, the two shown in Figure 4–8a are the most common. Beta turns are often found near the surface of a protein, where the peptide groups of the central two amino acid residues in the turn can hydrogen-bond with water. Considerably less common is the γ turn, a three-residue turn with a hydrogen bond between the first and third residues.

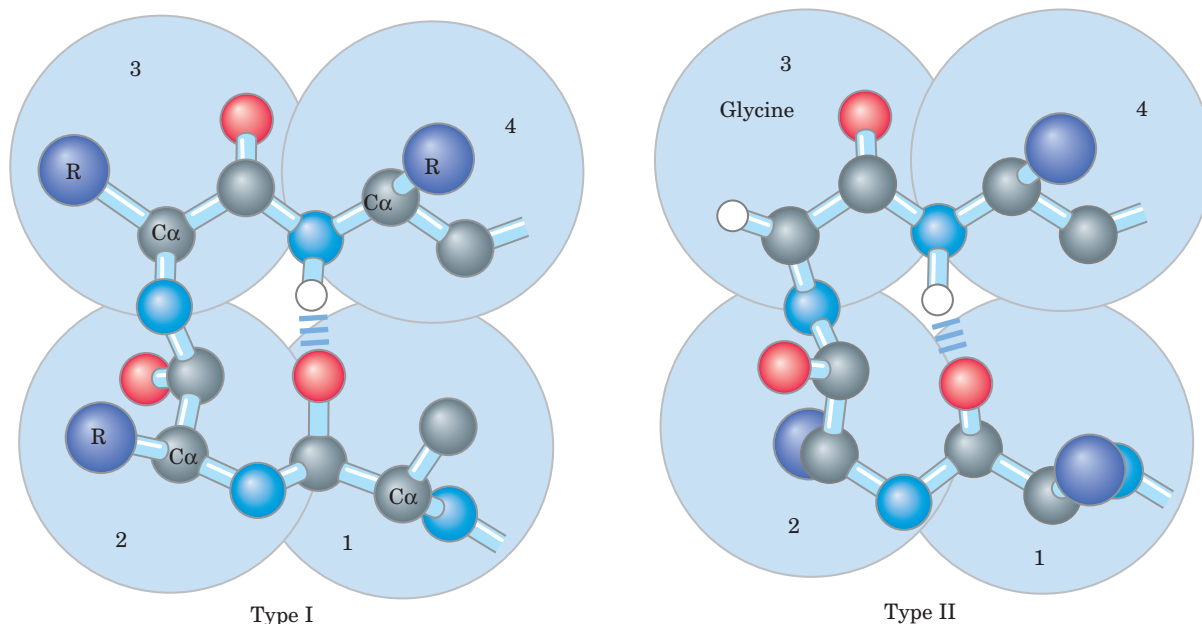
(a) β Turns

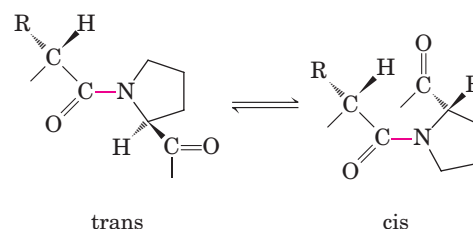
FIGURE 4-8 Structures of β turns. (a) Type I and type II β turns are most common; type I turns occur more than twice as frequently as type II. Type II β turns always have Gly as the third residue. Note the hydrogen bond between the peptide groups of the first and fourth residues of the bends. (Individual amino acid residues are framed by large blue circles.) (b) The trans and cis isomers of a peptide bond involving the imino nitrogen of proline. Of the peptide bonds between amino acid residues other than Pro, over 99.95% are in the trans configuration. For peptide bonds involving the imino nitrogen of proline, however, about 6% are in the cis configuration; many of these occur at β turns.

Common Secondary Structures Have Characteristic Bond Angles and Amino Acid Content

The α helix and the β conformation are the major repetitive secondary structures in a wide variety of proteins, although other repetitive structures do exist in some specialized proteins (an example is collagen; see Fig. 4-13 on page 128). Every type of secondary structure can be completely described by the bond angles ϕ and ψ at each residue. As shown by a Ramachandran plot, the α helix and β conformation fall within a relatively restricted range of sterically allowed structures (Fig. 4-9a). Most values of ϕ and ψ taken from known protein structures fall into the expected regions, with high concentrations near the α helix and β conformation values as predicted (Fig. 4-9b). The only amino acid residue often found in a conformation outside these regions is glycine. Because its side chain, a single hydrogen atom, is small, a Gly residue can take part in many conformations that are sterically forbidden for other amino acids.

Some amino acids are accommodated better than others in the different types of secondary structures. An overall summary is presented in Figure 4-10. Some

(b) Proline isomers



biases, such as the common presence of Pro and Gly residues in β turns and their relative absence in α helices, are readily explained by the known constraints on the different secondary structures. Other evident biases may be explained by taking into account the sizes or charges of side chains, but not all the trends shown in Figure 4-10 are understood.

SUMMARY 4.2 Protein Secondary Structure

- Secondary structure is the regular arrangement of amino acid residues in a segment of a polypeptide chain, in which each residue is spatially related to its neighbors in the same way.
- The most common secondary structures are the α helix, the β conformation, and β turns.
- The secondary structure of a polypeptide segment can be completely defined if the ϕ and ψ angles are known for all amino acid residues in that segment.

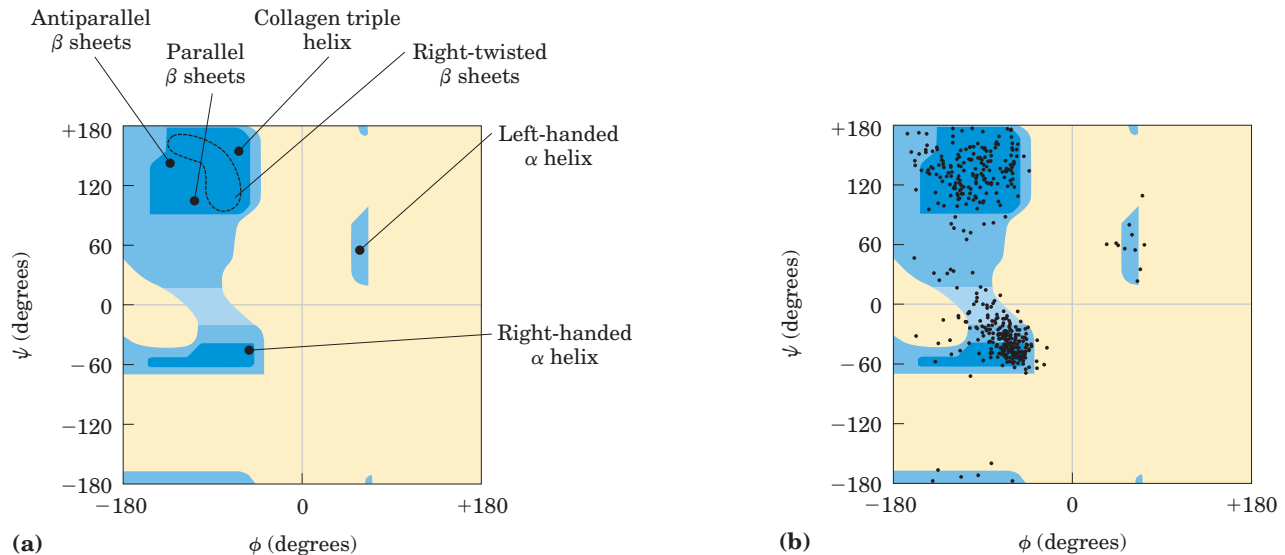


FIGURE 4-9 Ramachandran plots for a variety of structures. (a) The values of ϕ and ψ for various allowed secondary structures are overlaid on the plot from Figure 4-3. Although left-handed α helices extending over several amino acid residues are theoretically possible, they have not been observed in proteins. (b) The values of ϕ and ψ

for all the amino acid residues except Gly in the enzyme pyruvate kinase (isolated from rabbit) are overlaid on the plot of theoretically allowed conformations (Fig. 4-3). The small, flexible Gly residues were excluded because they frequently fall outside the expected ranges (blue).

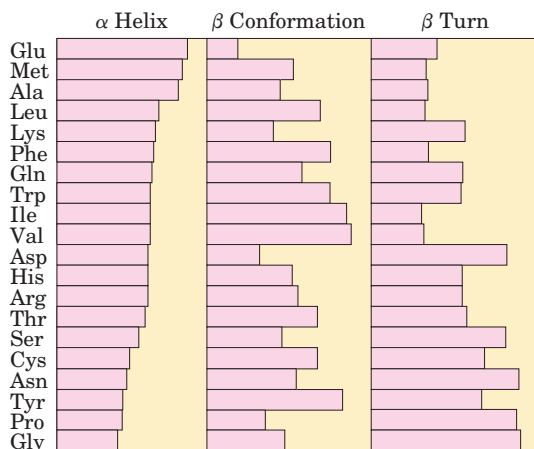


FIGURE 4-10 Relative probabilities that a given amino acid will occur in the three common types of secondary structure.

4.3 Protein Tertiary and Quaternary Structures

Protein Architecture—Introduction to Tertiary Structure The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's **tertiary structure**. Whereas the term “secondary structure” refers to the spatial arrangement of amino acid residues that are adjacent in the primary structure, tertiary structure includes *longer-range* aspects of amino acid sequence. Amino acids that are far apart in the polypeptide sequence and that reside in different types of secondary structure may interact within the completely folded structure of a protein. The location of bends (including

β turns) in the polypeptide chain and the direction and angle of these bends are determined by the number and location of specific bend-producing residues, such as Pro, Thr, Ser, and Gly. Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak bonding interactions (and sometimes by covalent bonds such as disulfide cross-links) between the segments.

Some proteins contain two or more separate polypeptide chains, or subunits, which may be identical or different. The arrangement of these protein subunits in three-dimensional complexes constitutes **quaternary structure**.

In considering these higher levels of structure, it is useful to classify proteins into two major groups: **fibrous proteins**, having polypeptide chains arranged in long strands or sheets, and **globular proteins**, having polypeptide chains folded into a spherical or globular shape. The two groups are structurally distinct: fibrous proteins usually consist largely of a single type of secondary structure; globular proteins often contain several types of secondary structure. The two groups differ functionally in that the structures that provide support, shape, and external protection to vertebrates are made of fibrous proteins, whereas most enzymes and regulatory proteins are globular proteins. Certain fibrous proteins played a key role in the development of our modern understanding of protein structure and provide particularly clear examples of the relationship between structure and function. We begin our discussion with fibrous proteins, before turning to the more complex folding patterns observed in globular proteins.

Fibrous Proteins Are Adapted for a Structural Function

Protein Architecture—Tertiary Structure of Fibrous Proteins α -Keratin, collagen, and silk fibroin nicely illustrate the relationship between protein structure and biological function (Table 4–1). Fibrous proteins share properties that give strength and/or flexibility to the structures in which they occur. In each case, the fundamental structural unit is a simple repeating element of secondary structure. All fibrous proteins are insoluble in water, a property conferred by a high concentration of hydrophobic amino acid residues both in the interior of the protein and on its surface. These hydrophobic surfaces are largely buried by packing many similar polypeptide chains together to form elaborate supramolecular complexes. The underlying structural simplicity of fibrous proteins makes them particularly useful for illustrating some of the fundamental principles of protein structure discussed above.

α -Keratin The α -keratins have evolved for strength. Found in mammals, these proteins constitute almost the entire dry weight of hair, wool, nails, claws, quills, horns, hooves, and much of the outer layer of skin. The α -keratins are part of a broader family of proteins called intermediate filament (IF) proteins. Other IF proteins are found in the cytoskeletons of animal cells. All IF proteins have a structural function and share structural features exemplified by the α -keratins.

The α -keratin helix is a right-handed α helix, the same helix found in many other proteins. Francis Crick

and Linus Pauling in the early 1950s independently suggested that the α helices of keratin were arranged as a coiled coil. Two strands of α -keratin, oriented in parallel (with their amino termini at the same end), are wrapped about each other to form a supertwisted coiled coil. The supertwisting amplifies the strength of the overall structure, just as strands are twisted to make a strong rope (Fig. 4–11). The twisting of the axis of an α helix to form a coiled coil explains the discrepancy between the 5.4 Å per turn predicted for an α helix by Pauling and Corey and the 5.15 to 5.2 Å repeating structure observed in the x-ray diffraction of hair (p. 120). The helical path of the supertwists is left-handed, opposite in sense to the α helix. The surfaces where the two α helices touch are made up of hydrophobic amino acid residues, their R groups meshed together in a regular interlocking pattern. This permits a close packing of the polypeptide chains within the left-handed supertwist. Not surprisingly, α -keratin is rich in the hydrophobic residues Ala, Val, Leu, Ile, Met, and Phe.

An individual polypeptide in the α -keratin coiled coil has a relatively simple tertiary structure, dominated by an α -helical secondary structure with its helical axis twisted in a left-handed superhelix. The intertwining of the two α -helical polypeptides is an example of quaternary structure. Coiled coils of this type are common structural elements in filamentous proteins and in the muscle protein myosin (see Fig. 5–29). The quaternary structure of α -keratin can be quite complex. Many coiled coils can be assembled into large supramolecular complexes, such as the arrangement of α -keratin to form the intermediate filament of hair (Fig. 4–11b).

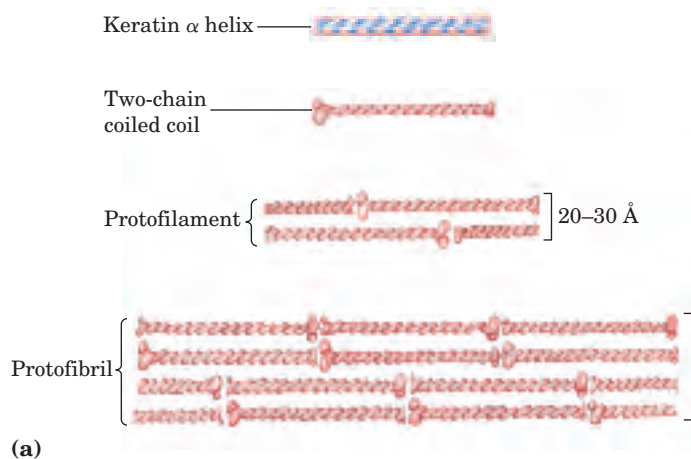
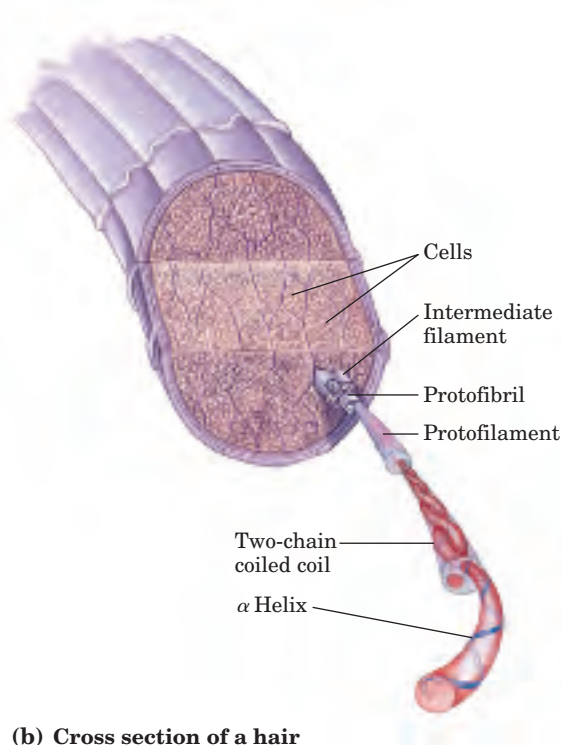


FIGURE 4–11 Structure of hair. (a) Hair α -keratin is an elongated α helix with somewhat thicker elements near the amino and carboxyl termini. Pairs of these helices are interwound in a left-handed sense to form two-chain coiled coils. These then combine in higher-order structures called protofilaments and microfibrils. About four microfibrils—32 strands of α -keratin altogether—combine to form an intermediate filament. The individual two-chain coiled coils in the various substructures also appear to be interwound, but the handedness of the intertwining and other structural details are unknown. (b) A hair is an array of many α -keratin filaments, made up of the substructures shown in (a).



(b) Cross section of a hair

TABLE 4-1 Secondary Structures and Properties of Fibrous Proteins

Structure	Characteristics	Examples of occurrence
α Helix, cross-linked by disulfide bonds	Tough, insoluble protective structures of varying hardness and flexibility	α -Keratin of hair, feathers, and nails
β Conformation	Soft, flexible filaments	Silk fibroin
Collagen triple helix	High tensile strength, without stretch	Collagen of tendons, bone matrix

The strength of fibrous proteins is enhanced by covalent cross-links between polypeptide chains within the multihelical “ropes” and between adjacent chains in a supramolecular assembly. In α -keratins, the cross-links stabilizing quaternary structure are disulfide bonds (Box 4-2). In the hardest and toughest α -keratins, such as those of rhinoceros horn, up to 18% of the residues are cysteines involved in disulfide bonds.

Collagen Like the α -keratins, collagen has evolved to provide strength. It is found in connective tissue such as tendons, cartilage, the organic matrix of bone, and the cornea of the eye. The collagen helix is a unique

secondary structure quite distinct from the α helix. It is left-handed and has three amino acid residues per turn (Fig. 4-12). Collagen is also a coiled coil, but one with distinct tertiary and quaternary structures: three separate polypeptides, called α chains (not to be confused with α helices), are supertwisted about each other (Fig. 4-12c). The superhelical twisting is right-handed in collagen, opposite in sense to the left-handed helix of the α chains.

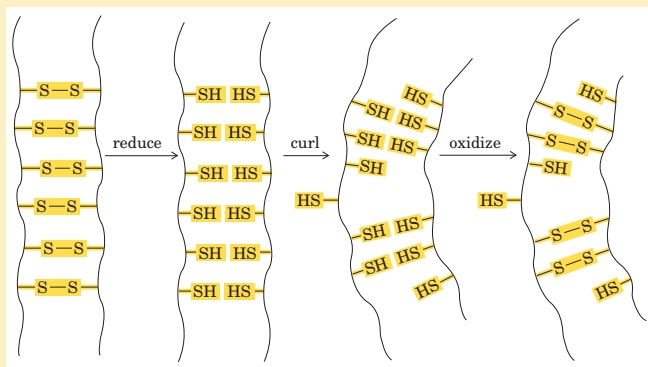
There are many types of vertebrate collagen. Typically they contain about 35% Gly, 11% Ala, and 21% Pro and 4-Hyp (4-hydroxyproline, an uncommon amino acid; see Fig. 3-8a). The food product gelatin is derived

BOX 4-2 THE WORLD OF BIOCHEMISTRY

Permanent Waving Is Biochemical Engineering

When hair is exposed to moist heat, it can be stretched. At the molecular level, the α helices in the α -keratin of hair are stretched out until they arrive at the fully extended β conformation. On cooling they spontaneously revert to the α -helical conformation. The characteristic “stretchability” of α -keratins, and their numerous disulfide cross-linkages, are the basis of permanent waving. The hair to be waved or curled is first bent around a form of appropriate shape. A solution of a reducing agent, usually a compound containing a thiol or sulfhydryl group ($-\text{SH}$), is then applied with heat. The reducing agent cleaves the cross-linkages by reducing each disulfide bond to form two Cys residues. The moist heat breaks hydrogen

bonds and causes the α -helical structure of the polypeptide chains to uncoil. After a time the reducing solution is removed, and an oxidizing agent is added to establish *new* disulfide bonds between pairs of Cys residues of adjacent polypeptide chains, but not the same pairs as before the treatment. After the hair is washed and cooled, the polypeptide chains revert to their α -helical conformation. The hair fibers now curl in the desired fashion because the new disulfide cross-linkages exert some torsion or twist on the bundles of α -helical coils in the hair fibers. A permanent wave is not truly permanent, because the hair grows; in the new hair replacing the old, the α -keratin has the natural, nonwavy pattern of disulfide bonds.



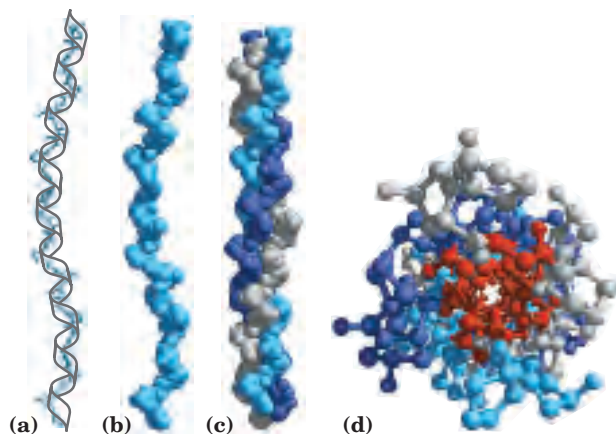
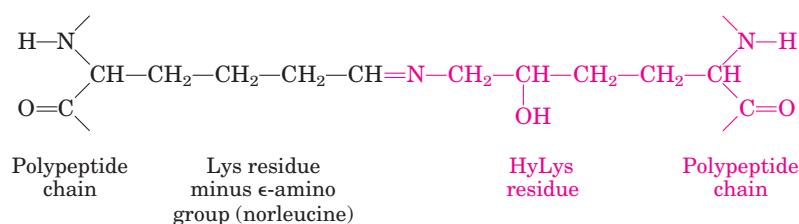


FIGURE 4-12 Structure of collagen. (Derived from PDB ID 1CGD.)

(a) The α chain of collagen has a repeating secondary structure unique to this protein. The repeating tripeptide sequence Gly-X-Pro or Gly-X-4-Hyp adopts a left-handed helical structure with three residues per turn. The repeating sequence used to generate this model is Gly-Pro-4-Hyp. (b) Space-filling model of the same α chain. (c) Three of these helices (shown here in gray, blue, and purple) wrap around one another with a right-handed twist. (d) The three-stranded collagen superhelix shown from one end, in a ball-and-stick representation. Gly residues are shown in red. Glycine, because of its small size, is required at the tight junction where the three chains are in contact. The balls in this illustration do not represent the van der Waals radii of the individual atoms. The center of the three-stranded superhelix is not hollow, as it appears here, but is very tightly packed.

from collagen; it has little nutritional value as a protein, because collagen is extremely low in many amino acids that are essential in the human diet. The unusual amino acid content of collagen is related to structural constraints unique to the collagen helix. The amino acid sequence in collagen is generally a repeating tripeptide unit, Gly–X–Y, where X is often Pro, and Y is often 4-Hyp. Only Gly residues can be accommodated at the very tight junctions between the individual α chains (Fig. 4-12d); The Pro and 4-Hyp residues permit the sharp twisting of the collagen helix. The amino acid sequence and the supertwisted quaternary structure of collagen allow a very close packing of its three polypeptides. 4-Hydroxyproline has a special role in the structure of collagen—and in human history (Box 4-3).

The tight wrapping of the α chains in the collagen triple helix provides tensile strength greater than that



Dehydrohydroxylysinoonorleucine

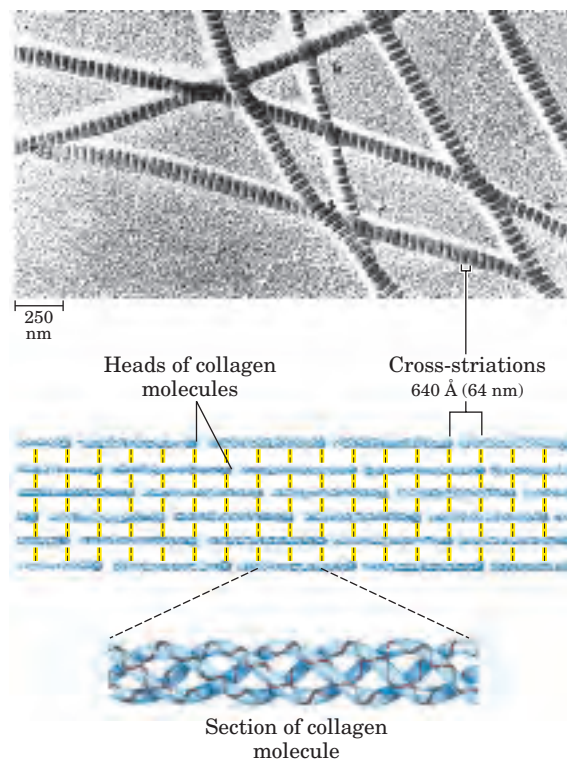


FIGURE 4-13 Structure of collagen fibrils. Collagen (M_r 300,000) is

a rod-shaped molecule, about 3,000 Å long and only 15 Å thick. Its three helically intertwined α chains may have different sequences, but each has about 1,000 amino acid residues. Collagen fibrils are made up of collagen molecules aligned in a staggered fashion and cross-linked for strength. The specific alignment and degree of cross-linking vary with the tissue and produce characteristic cross-striations in an electron micrograph. In the example shown here, alignment of the head groups of every fourth molecule produces striations 640 Å apart.

of a steel wire of equal cross section. Collagen fibrils (Fig. 4-13) are supramolecular assemblies consisting of triple-helical collagen molecules (sometimes referred to as tropocollagen molecules) associated in a variety of ways to provide different degrees of tensile strength. The α chains of collagen molecules and the collagen molecules of fibrils are cross-linked by unusual types of covalent bonds involving Lys, HyLys (5-hydroxylysine; see Fig. 3-8a), or His residues that are present at a few of the X and Y positions in collagens. These links create uncommon amino acid residues such as dehydrohydroxylysinonorleucine. The increasingly rigid and brittle character of aging connective tissue results from accumulated covalent cross-links in collagen fibrils.

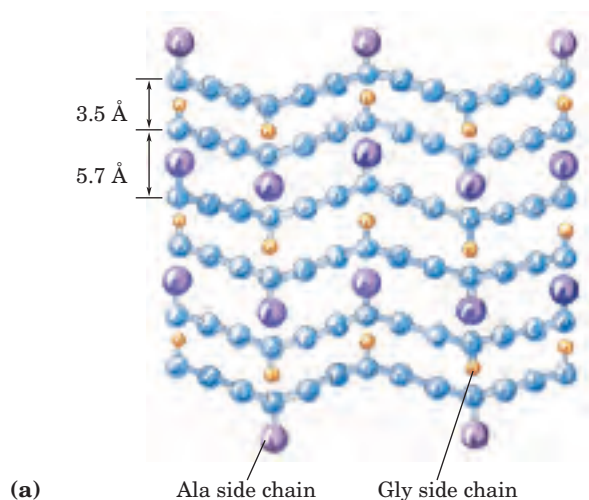
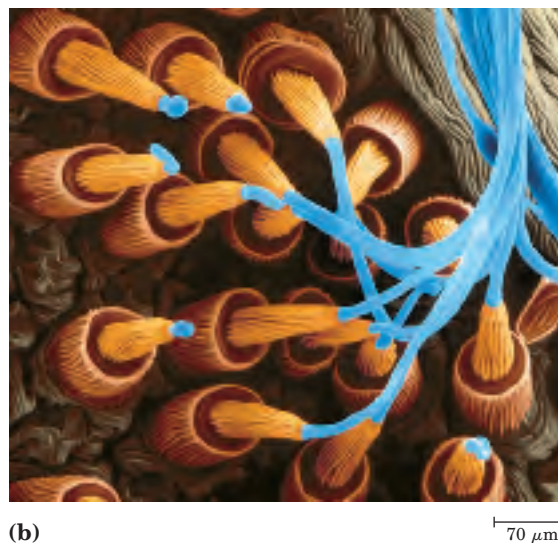


FIGURE 4-14 Structure of silk. The fibers used to make silk cloth or a spider web are made up of the protein fibroin. (a) Fibroin consists of layers of antiparallel β sheets rich in Ala (purple) and Gly (yellow) residues. The small side chains interdigitate and allow close packing

 A typical mammal has more than 30 structural variants of collagen, particular to certain tissues and each somewhat different in sequence and function. Some human genetic defects in collagen structure illustrate the close relationship between amino acid sequence and three-dimensional structure in this protein. Osteogenesis imperfecta is characterized by abnormal bone formation in babies; Ehlers-Danlos syndrome is characterized by loose joints. Both conditions can be lethal, and both result from the substitution of an amino acid residue with a larger R group (such as Cys or Ser) for a single Gly residue in each α chain (a different Gly residue in each disorder). These single-residue substitutions have a catastrophic effect on collagen function because they disrupt the Gly-X-Y repeat that gives collagen its unique helical structure. Given its role in the collagen triple helix (Fig. 4-12d), Gly cannot be replaced by another amino acid residue without substantial deleterious effects on collagen structure. ■

Silk Fibroin Fibroin, the protein of silk, is produced by insects and spiders. Its polypeptide chains are predominantly in the β conformation. Fibroin is rich in Ala and Gly residues, permitting a close packing of β sheets and an interlocking arrangement of R groups (Fig. 4-14). The overall structure is stabilized by extensive hydrogen bonding between all peptide linkages in the polypeptides of each β sheet and by the optimization of van der Waals interactions between sheets. Silk does not stretch, because the β conformation is already highly extended (Fig. 4-7; see also Fig. 4-15). However, the structure is flexible because the sheets are held together by numerous weak interactions rather than by covalent bonds such as the disulfide bonds in α -keratins.



of each layered sheet, as shown in this side view. (b) Strands of fibroin (blue) emerge from the spinnerets of a spider in this colored electron micrograph.

Structural Diversity Reflects Functional Diversity in Globular Proteins

In a globular protein, different segments of a polypeptide chain (or multiple polypeptide chains) fold back on each other. As illustrated in Figure 4-15, this folding generates a compact form relative to polypeptides in a fully extended conformation. The folding also provides the structural diversity necessary for proteins to carry out a wide array of biological functions. Globular proteins include enzymes, transport proteins, motor proteins, regulatory proteins, immunoglobulins, and proteins with many other functions.

As a new millennium begins, the number of known three-dimensional protein structures is in the thousands and more than doubles every two years. This wealth of structural information is revolutionizing our understanding of protein structure, the relation of structure

β Conformation
2,000 $\times$ 5 Å

α Helix
900 $\times$ 11 Å

Native globular form
100 $\times$ 60 Å

FIGURE 4-15 Globular protein structures are compact and varied. Human serum albumin (M_r 64,500) has 585 residues in a single chain. Given here are the approximate dimensions its single polypeptide chain would have if it occurred entirely in extended β conformation or as an α helix. Also shown is the size of the protein in its native globular form, as determined by X-ray crystallography; the polypeptide chain must be very compactly folded to fit into these dimensions.



BOX 4-3 BIOCHEMISTRY IN MEDICINE

Why Sailors, Explorers, and College Students Should Eat Their Fresh Fruits and Vegetables

... from this misfortune, together with the unhealthiness of the country, where there never falls a drop of rain, we were stricken with the “camp-sickness,” which was such that the flesh of our limbs all shrivelled up, and the skin of our legs became all blotched with black, mouldy patches, like an old jack-boot, and proud flesh came upon the gums of those of us who had the sickness, and none escaped from this sickness save through the jaws of death. The signal was this: when the nose began to bleed, then death was at hand . . .

—from *The Memoirs of the Lord of Joinville*, ca. 1300

This excerpt describes the plight of Louis IX’s army toward the end of the Seventh Crusade (1248–1254), immediately preceding the battle of Fariskur, where the scurvy-weakened Crusader army was destroyed by the Egyptians. What was the nature of the malady afflicting these thirteenth-century soldiers?

Scurvy is caused by lack of vitamin C, or ascorbic acid (ascorbate). Vitamin C is required for, among other things, the hydroxylation of proline and lysine in collagen; scurvy is a deficiency disease characterized by general degeneration of connective tissue. Manifestations of advanced scurvy include numerous small hemorrhages caused by fragile blood vessels, tooth loss, poor wound healing and the reopening of old wounds, bone pain and degeneration, and eventually heart failure. Despondency and oversensitivity to stimuli of many kinds are also observed. Milder cases of vitamin

C deficiency are accompanied by fatigue, irritability, and an increased severity of respiratory tract infections. Most animals make large amounts of vitamin C, converting glucose to ascorbate in four enzymatic steps. But in the course of evolution, humans and some other animals—gorillas, guinea pigs, and fruit bats—have lost the last enzyme in this pathway and must obtain ascorbate in their diet. Vitamin C is available in a wide range of fruits and vegetables. Until 1800, however, it was often absent in the dried foods and other food supplies stored for winter or for extended travel.

Scurvy was recorded by the Egyptians in 1500 BCE, and it is described in the fifth century BCE writings of Hippocrates. Although scurvy played a critical role in medieval wars and made regular winter appearances in northern climates, it did not come to wide public notice until the European voyages of discovery from 1500 to 1800. The first circumnavigation of the globe, led by Ferdinand Magellan (1520), was accomplished only with the loss of more than 80% of his crew to scurvy. Vasco da Gama lost two-thirds of his crew to the same fate during his first exploration of trade routes to India (1499). During Jacques Cartier’s second voyage to explore the St. Lawrence River (1535–1536), his band suffered numerous fatalities and was threatened with complete disaster until the native Americans taught the men to make a cedar tea that cured and prevented scurvy (it contained vitamin C) (Fig. 1). It is estimated that a million sailors died of scurvy in the years 1600 to 1800. Winter outbreaks of scurvy in Europe were gradually eliminated in the nineteenth century as the cultivation of the potato, introduced from South America, became widespread.

In 1747, James Lind, a Scottish surgeon in the Royal Navy (Fig. 2), carried out the first controlled clinical study in recorded history. During an extended voyage on the 50-gun warship *HMS Salisbury*, Lind selected 12 sailors suffering from scurvy and separated them into groups of two. All 12 received the same diet, except that each group was given a different remedy for scurvy from among those recommended at the time. The sailors given lemons and oranges recovered and returned to duty. The sailors given boiled apple juice improved slightly. The remainder continued to deteriorate. Lind’s *Treatise on the Scurvy* was published in 1753, but inaction persisted in the Royal Navy for another 40 years.



FIGURE 1 Iroquois showing Jacques Cartier how to make cedar tea as a remedy for scurvy.

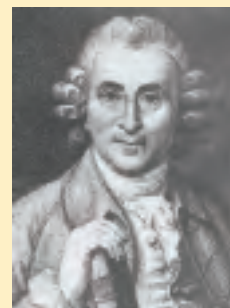


FIGURE 2 James Lind, 1716–1794; naval surgeon, 1739–1748.

In 1795 the British admiralty finally mandated a ration of concentrated lime or lemon juice for all British sailors (hence the name “limeys”). Scurvy continued to be a problem in some other parts of the world until 1932, when Hungarian scientist Albert Szent-Györgyi, and W. A. Waugh and C. G. King at the University of Pittsburgh, isolated and synthesized ascorbic acid.

L-Ascorbic acid (vitamin C) is a white, odorless, crystalline powder. It is freely soluble in water and relatively insoluble in organic solvents. In a dry state, away from light, it is stable for a considerable length of time. The appropriate daily intake of this vitamin is still in dispute. The recommended daily allowance in the United States is 60 mg (Australia and the United Kingdom recommend 30 to 40 mg; Russia recommends 100 mg). Higher doses of vitamin C are sometimes recommended, although the benefit of such a regimen is disputed. Notably, animals that synthesize their own vitamin C maintain levels found in humans only if they consume hundreds of times the recommended daily allowance. Along with citrus fruits and almost all other fresh fruits, other good sources of vitamin C include peppers, tomatoes, potatoes, and broccoli. The vitamin C of fruits and vegetables is destroyed by overcooking or prolonged storage.

So why is ascorbate so necessary to good health? Of particular interest to us here is its role in the formation of collagen. The proline derivative 4(R)-L-hydroxyproline (4-Hyp) plays an essential role in the folding of collagen and in maintaining its structure. As noted in the text, collagen is constructed of the repeating tripeptide unit Gly–X–Y, where X and Y are generally Pro or 4-Hyp. A constructed peptide with 10 Gly–Pro–Pro repeats will fold to form a collagen triple helix, but the structure melts at 41 °C. If the 10 repeats are changed to Gly–Pro–4-Hyp, the melting temperature jumps to 69 °C. The stability of collagen arises from the detailed structure of the collagen helix, determined independently by Helen Berman and Adriana Zagari and their colleagues. The proline ring is normally found as a mixture of two puckered conformations, called C_γ-endo and C_γ-exo (Fig. 3). The collagen helix structure requires the Pro residue in the Y positions to be in the C_γ-exo conformation, and it is this conformation that is enforced by the hydroxyl substitution at C-4 in 4-hydroxyproline. However, the collagen structure requires the Pro residue in the X positions to have the C_γ-endo conformation, and introduction of 4-Hyp here can destabilize the helix. The inability to hydroxylate the Pro at the Y positions when vitamin C is absent leads to collagen instability and the connective tissue problems seen in scurvy.

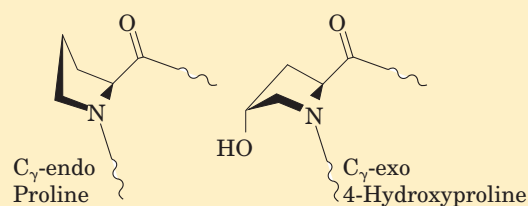


FIGURE 3 The C_γ-endo conformation of proline and the C_γ-exo conformation of 4-hydroxyproline.

The hydroxylation of specific Pro residues in procollagen, the precursor of collagen, requires the action of the enzyme prolyl 4-hydroxylase. This enzyme (*M_r* 240,000) is an α₂β₂ tetramer in all vertebrate sources. The proline-hydroxylating activity is found in the α subunits. (Researchers were surprised to find that the β subunits are identical to the enzyme protein disulfide isomerase (PDI; p. 152); these subunits do not participate in the prolyl hydroxylation activity.) Each α subunit contains one atom of nonheme iron (Fe²⁺), and the enzyme is one of a class of hydroxylases that require α-ketoglutarate in their reactions.

In the normal prolyl 4-hydroxylase reaction (Fig. 4a), one molecule of α-ketoglutarate and one of O₂ bind to the enzyme. The α-ketoglutarate is oxidatively decarboxylated to form CO₂ and succinate. The remaining oxygen atom is then used to hydroxylate an appropriate Pro residue in procollagen. No ascorbate is needed in this reaction. However, prolyl 4-hydroxylase also catalyzes an oxidative decarboxylation of α-ketoglutarate that is not coupled to proline hydroxylation—and this is the reaction that requires ascorbate (Fig. 4b). During this reaction, the heme Fe²⁺ becomes oxidized, and the oxidized form of the enzyme is inactive—unable to hydroxylate proline. The ascorbate consumed in the reaction presumably functions to reduce the heme iron and restore enzyme activity.

But there is more to the vitamin C story than proline hydroxylation. Very similar hydroxylation reactions generate the less abundant 3-hydroxyproline and 5-hydroxylysine residues that also occur in collagen. The enzymes that catalyze these reactions are members of the same α-ketoglutarate-dependent dioxygenase family, and for all these enzymes ascorbate plays the same role. These dioxygenases are just a few of the dozens of closely related enzymes that play a variety of metabolic roles in different classes of organisms. Ascorbate serves other roles too. It is an antioxidant, reacting enzymatically and nonenzymatically with reactive oxygen species, which in mammals play an important role in aging and cancer.

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BOX 4-3 BIOCHEMISTRY IN MEDICINE (continued from previous page)

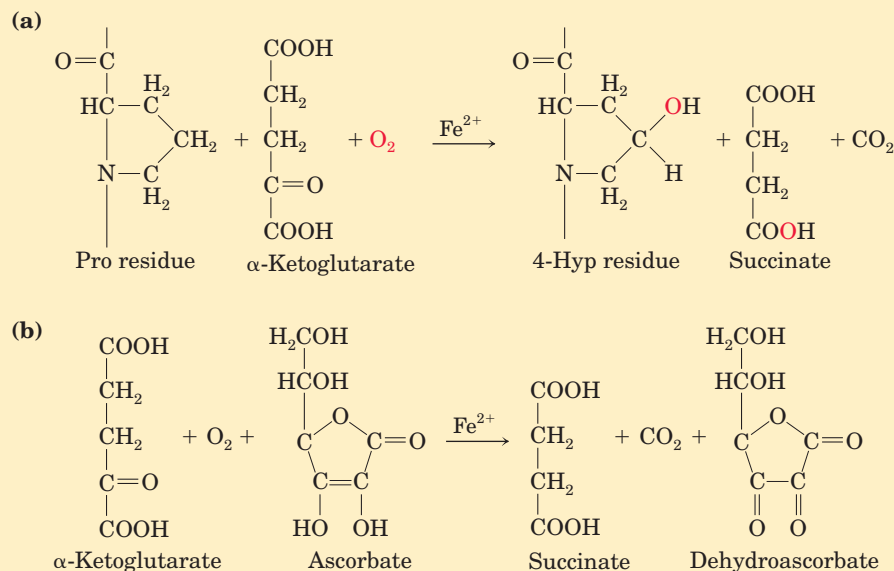


FIGURE 4 The reactions catalyzed by prolyl 4-hydroxylase. (a) The normal reaction, coupled to proline hydroxylation, which does not require ascorbate. The fate of the two oxygen atoms from O_2 is shown in red. (b) The uncoupled reaction, in which α -ketoglutarate is oxidatively decarboxylated without hydroxylation of proline. Ascorbate is consumed stoichiometrically in this process as it is converted to dehydroascorbate.

In plants, ascorbate is required as a substrate for the enzyme ascorbate peroxidase, which converts H_2O_2 to water. The peroxide is generated from the O_2 produced in photosynthesis, an unavoidable consequence of generating O_2 in a compartment laden with powerful oxidation-reduction systems (Chapter 19). Ascorbate is also a precursor of oxalate and tartrate in plants, and is involved in the hydroxylation of Pro residues in cell wall proteins called extensins. Ascorbate is found in all subcellular compartments of plants, at concentrations of 2 to 25 mM—which is why plants are such good sources of vitamin C.

Scurvy remains a problem today. The malady is still encountered not only in remote regions where nutritious food is scarce but, surprisingly, on U.S. college campuses. The only vegetables consumed by some students are those in tossed salads, and days go by without these young adults consuming fruit. A 1998 study of 230 students at Arizona State University revealed that 10% had serious vitamin C deficiencies, and 2 students had vitamin C levels so low that they probably had scurvy. Only half the students in the study consumed the recommended daily allowance of vitamin C. Eat your fresh fruit and vegetables.

to function, and even the evolutionary paths by which proteins arrived at their present state, which can be glimpsed in the family resemblances that are revealed as protein databases are sifted and sorted. The sheer variety of structures can seem daunting. Yet as new protein structures become available it is becoming increasingly clear that they are manifestations of a finite set of recognizable, stable folding patterns.

Our discussion of globular protein structure begins with the principles gleaned from the earliest protein structures to be elucidated. This is followed by a detailed description of protein substructure and comparative categorization. Such discussions are possible only because of the vast amount of information available over the Internet from resources such as the Protein Data Bank (PDB; www.rcsb.org/pdb), an archive of experimentally determined three-dimensional structures of biological macromolecules.

Myoglobin Provided Early Clues about the Complexity of Globular Protein Structure

Protein Architecture—Tertiary Structure of Small Globular Proteins, II. Myoglobin The first breakthrough in understanding the three-dimensional structure of a globular protein came from x-ray diffraction studies of myoglobin carried out by John Kendrew and his colleagues in the 1950s. Myoglobin is a relatively small (M_r 16,700), oxygen-binding protein of muscle cells. It functions both to store oxygen and to facilitate oxygen diffusion in rapidly contracting muscle tissue. Myoglobin contains a single polypeptide chain of 153 amino acid residues of known sequence and a single iron protoporphyrin, or heme, group. The same heme group is found in hemoglobin, the oxygen-binding protein of erythrocytes, and is responsible for the deep red-brown color of both myoglobin and hemoglobin. Myoglobin is particularly abun-

dant in the muscles of diving mammals such as the whale, seal, and porpoise, whose muscles are so rich in this protein that they are brown. Storage and distribution of oxygen by muscle myoglobin permit these animals to remain submerged for long periods of time. The activities of myoglobin and other globin molecules are investigated in greater detail in Chapter 5.

Figure 4–16 shows several structural representations of myoglobin, illustrating how the polypeptide chain is folded in three dimensions—its tertiary structure. The red group surrounded by protein is heme. The backbone of the myoglobin molecule is made up of eight relatively straight segments of α helix interrupted by bends, some of which are β turns. The longest α helix has 23 amino acid residues and the shortest only 7; all helices are right-handed. More than 70% of the residues in myoglobin are in these α -helical regions. X-ray analysis has revealed the precise position of each of the R groups, which occupy nearly all the space within the folded chain.

Many important conclusions were drawn from the structure of myoglobin. The positioning of amino acid side chains reflects a structure that derives much of its stability from hydrophobic interactions. Most of the hydrophobic R groups are in the interior of the myoglobin molecule, hidden from exposure to water. All but two of the polar R groups are located on the outer surface of the molecule, and all are hydrated. The myoglobin molecule is so compact that its interior has room for only four molecules of water. This dense hydrophobic core is typical of globular proteins. The fraction of space occupied by atoms in an organic liquid is 0.4 to 0.6; in a typical crystal the fraction is 0.70 to 0.78, near the theoretical maximum. In a globular protein the fraction is about 0.75, comparable to that in a crystal. In this packed environment, weak interactions strengthen and reinforce each other. For example, the nonpolar side chains in the core are so close together that short-range van der Waals interactions make a significant contribution to stabilizing hydrophobic interactions.

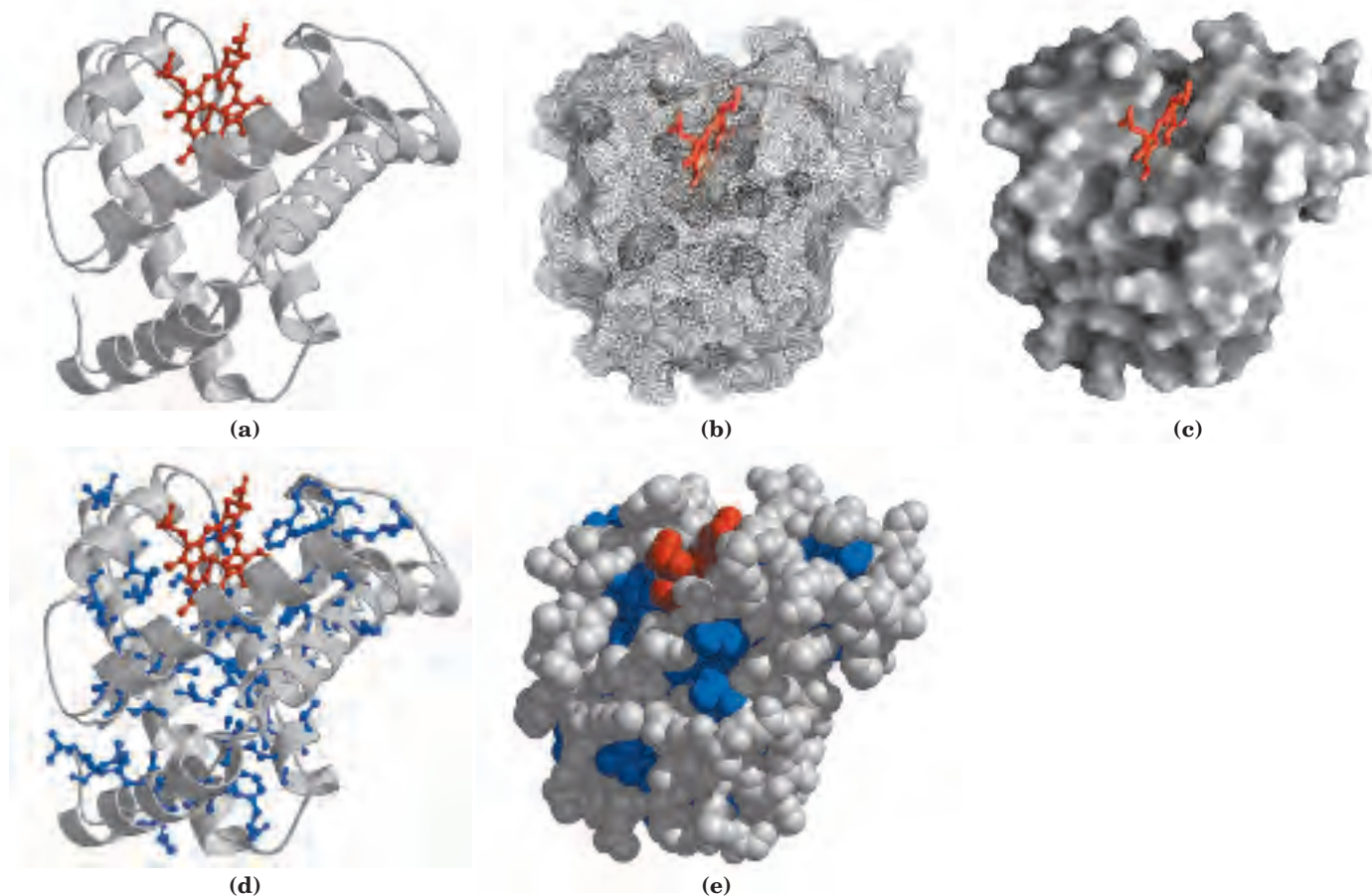


FIGURE 4-16 Tertiary structure of sperm whale myoglobin. (PDB ID 1MBO) The orientation of the protein is similar in all panels; the heme group is shown in red. In addition to illustrating the myoglobin structure, this figure provides examples of several different ways to display protein structure. **(a)** The polypeptide backbone, shown in a ribbon representation of a type introduced by Jane Richardson, which highlights regions of secondary structure. The α -helical regions are evident. **(b)** A “mesh” image emphasizes the protein surface. **(c)** A sur-

face contour image is useful for visualizing pockets in the protein where other molecules might bind. **(d)** A ribbon representation, including side chains (blue) for the hydrophobic residues Leu, Ile, Val, and Phe. **(e)** A space-filling model with all amino acid side chains. Each atom is represented by a sphere encompassing its van der Waals radius. The hydrophobic residues are again shown in blue; most are not visible, because they are buried in the interior of the protein.

Deduction of the structure of myoglobin confirmed some expectations and introduced some new elements of secondary structure. As predicted by Pauling and Corey, all the peptide bonds are in the planar trans configuration. The α helices in myoglobin provided the first direct experimental evidence for the existence of this type of secondary structure. Three of the four Pro residues of myoglobin are found at bends (recall that proline, with its fixed ϕ bond angle and lack of a peptide-bond N—H group for participation in hydrogen bonds, is largely incompatible with α -helical structure). The fourth Pro residue occurs within an α helix, where it creates a kink necessary for tight helix packing. Other bends contain Ser, Thr, and Asn residues, which are among the amino acids whose bulk and shape tend to make them incompatible with α -helical structure if they are in close proximity in the amino acid sequence (p. 121).

The flat heme group rests in a crevice, or pocket, in the myoglobin molecule. The iron atom in the center of the heme group has two bonding (coordination) positions perpendicular to the plane of the heme (Fig. 4-17). One of these is bound to the R group of the His residue at position 93; the other is the site at which an O_2 molecule binds. Within this pocket, the accessibility of the heme group to solvent is highly restricted. This is important for function, because free heme groups in an oxygenated solution are rapidly oxidized from the ferrous (Fe^{2+}) form, which is active in the reversible binding of O_2 , to the ferric (Fe^{3+}) form, which does not bind O_2 .

Knowledge of the structure of myoglobin allowed researchers for the first time to understand in detail the

correlation between the structure and function of a protein. Many different myoglobin structures have been elucidated, allowing investigators to see how the structure changes when oxygen or other molecules bind to it. Hundreds of proteins have been subjected to similar analysis since then. Today, techniques such as NMR spectroscopy supplement x-ray diffraction data, providing more information on a protein's structure (Box 4-4). The ongoing sequencing of genomic DNA from many organisms (Chapter 9) has identified thousands of genes that encode proteins of known sequence but unknown function. Our first insight into what these proteins do often comes from our still-limited understanding of how primary structure determines tertiary structure, and how tertiary structure determines function.

Globular Proteins Have a Variety of Tertiary Structures

With elucidation of the tertiary structures of hundreds of other globular proteins by x-ray analysis, it became clear that myoglobin illustrates only one of many ways in which a polypeptide chain can be folded. In Figure 4-18 the structures of cytochrome *c*, lysozyme, and ribonuclease are compared. These proteins have different amino acid sequences and different tertiary structures, reflecting differences in function. All are relatively small and easy to work with, facilitating structural analysis. Cytochrome *c* is a component of the respiratory chain of mitochondria (Chapter 19). Like myoglobin, cytochrome *c* is a heme protein. It contains a single polypeptide chain of about 100 residues (M_r 12,400) and a single heme group. In this case, the protoporphyrin of the heme group is covalently attached to the polypeptide. Only about 40% of the polypeptide is in α -helical segments, compared with 70% of the myoglobin chain. The rest of the cytochrome *c* chain contains β turns and irregularly coiled and extended segments.

Lysozyme (M_r 14,600) is an enzyme abundant in egg white and human tears that catalyzes the hydrolytic cleavage of polysaccharides in the protective cell walls of some families of bacteria. Lysozyme, because it can lyse, or degrade, bacterial cell walls, serves as a bactericidal agent. As in cytochrome *c*, about 40% of its 129 amino acid residues are in α -helical segments, but the arrangement is different and some β -sheet structure is also present (Fig. 4-18). Four disulfide bonds contribute stability to this structure. The α helices line a long crevice in the side of the molecule, called the active site, which is the site of substrate binding and catalysis. The bacterial polysaccharide that is the substrate for lysozyme fits into this crevice.

Protein Architecture—Tertiary Structure of Small Globular Proteins, III. Lysozyme

Ribonuclease, another small globular protein (M_r 13,700), is an enzyme secreted by the pancreas into the small intestine, where it catalyzes the hydrolysis of certain bonds in the ribonucleic acids present in ingested

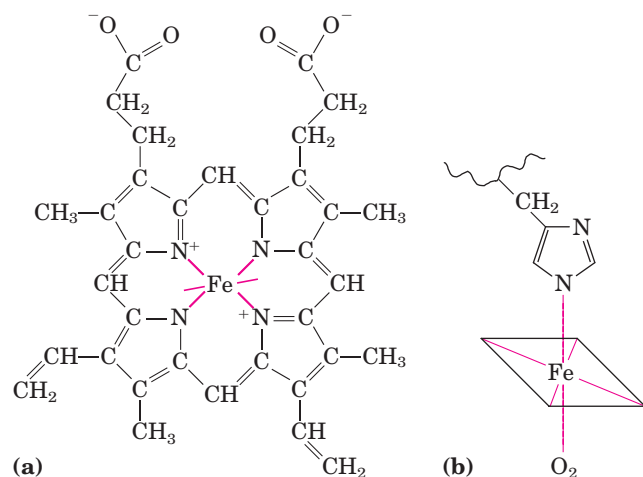


FIGURE 4-17 The heme group. This group is present in myoglobin, hemoglobin, cytochromes, and many other heme proteins. (a) Heme consists of a complex organic ring structure, protoporphyrin, to which is bound an iron atom in its ferrous (Fe^{2+}) state. The iron atom has six coordination bonds, four in the plane of, and bonded to, the flat porphyrin molecule and two perpendicular to it. (b) In myoglobin and hemoglobin, one of the perpendicular coordination bonds is bound to a nitrogen atom of a His residue. The other is “open” and serves as the binding site for an O_2 molecule.

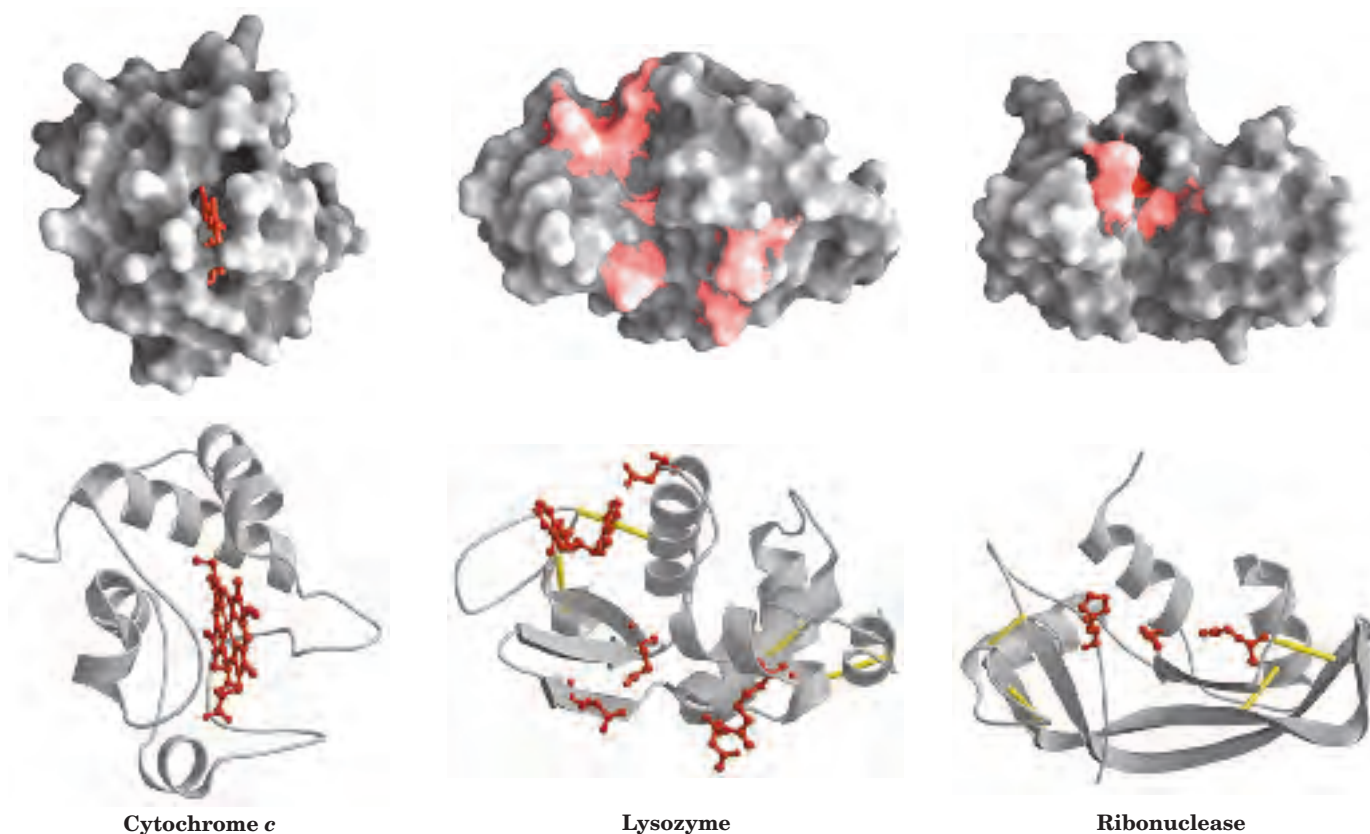


FIGURE 4-18 Three-dimensional structures of some small proteins. Shown here are cytochrome *c* (PDB ID 1CCR), lysozyme (PDB ID 3LYM), and ribonuclease (PDB ID 3RN3). Each protein is shown in surface contour and in a ribbon representation, in the same orientation. In the ribbon depictions, regions in the β conformation are

represented by flat arrows and the α helices are represented by spiral ribbons. Key functional groups (the heme in cytochrome *c*; amino acid side chains in the active site of lysozyme and ribonuclease) are shown in red. Disulfide bonds are shown (in the ribbon representations) in yellow.

food. Its tertiary structure, determined by x-ray analysis, shows that little of its 124 amino acid polypeptide chain is in an α -helical conformation, but it contains many segments in the β conformation (Fig. 4-18). Like lysozyme, ribonuclease has four disulfide bonds between loops of the polypeptide chain.

In small proteins, hydrophobic residues are less likely to be sheltered in a hydrophobic interior—simple geometry dictates that the smaller the protein, the lower the ratio of volume to surface area. Small proteins also have fewer potential weak interactions available to stabilize them. This explains why many smaller proteins such as those in Figure 4-18 are stabilized by a number of covalent bonds. Lysozyme and ribonuclease, for example, have disulfide linkages, and the heme group in cytochrome *c* is covalently linked to the protein on two sides, providing significant stabilization of the entire protein structure.

Table 4-2 shows the proportions of α helix and β conformation (expressed as percentage of residues in each secondary structure) in several small, single-chain, globular proteins. Each of these proteins has a distinct structure, adapted for its particular biological function, but together they share several important properties. Each is folded compactly, and in each case the hydro-

phobic amino acid side chains are oriented toward the interior (away from water) and the hydrophilic side chains are on the surface. The structures are also stabilized by a multitude of hydrogen bonds and some ionic interactions.

TABLE 4-2 Approximate Amounts of α Helix and β Conformation in Some Single-Chain Proteins

Protein (total residues)	Residues (%) [*]	
	α Helix	β Conformation
Chymotrypsin (247)	14	45
Ribonuclease (124)	26	35
Carboxypeptidase (307)	38	17
Cytochrome <i>c</i> (104)	39	0
Lysozyme (129)	40	12
Myoglobin (153)	78	0

Source: Data from Cantor, C.R. & Schimmel, P.R. (1980) *Biophysical Chemistry, Part I: The Conformation of Biological Macromolecules*, p. 100, W. H. Freeman and Company, New York.

^{*}Portions of the polypeptide chains that are not accounted for by α helix or β conformation consist of bends and irregularly coiled or extended stretches. Segments of α helix and β conformation sometimes deviate slightly from their normal dimensions and geometry.

BOX 4-4 WORKING IN BIOCHEMISTRY

Methods for Determining the Three-Dimensional Structure of a Protein**X-Ray Diffraction**

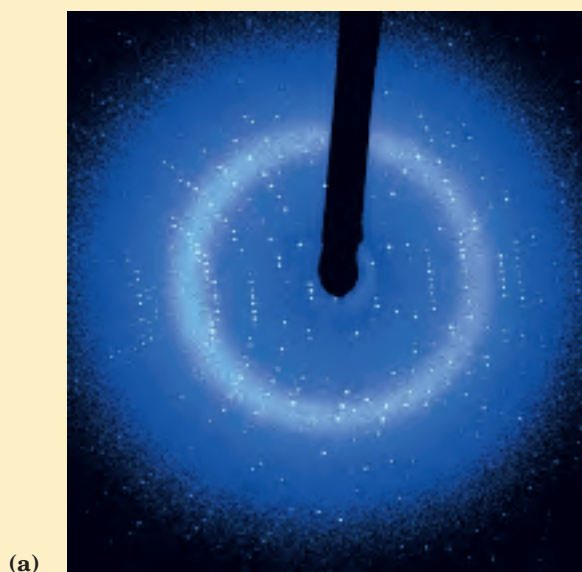
The spacing of atoms in a crystal lattice can be determined by measuring the locations and intensities of spots produced on photographic film by a beam of x rays of given wavelength, after the beam has been diffracted by the electrons of the atoms. For example, x-ray analysis of sodium chloride crystals shows that Na^+ and Cl^- ions are arranged in a simple cubic lattice. The spacing of the different kinds of atoms in complex organic molecules, even very large ones such as proteins, can also be analyzed by x-ray diffraction methods. However, the technique for analyzing crystals of complex molecules is far more laborious than for simple salt crystals. When the repeating pattern of the crystal is a molecule as large as, say, a protein, the numerous atoms in the molecule yield thousands of diffraction spots that must be analyzed by computer.

The process may be understood at an elementary level by considering how images are generated in a light microscope. Light from a point source is focused on an object. The light waves are scattered by the object, and these scattered waves are recombined by a series of lenses to generate an enlarged image of the object. The smallest object whose structure can be determined by such a system—that is, the resolving power of the microscope—is determined by the wavelength of the light, in this case visible light, with

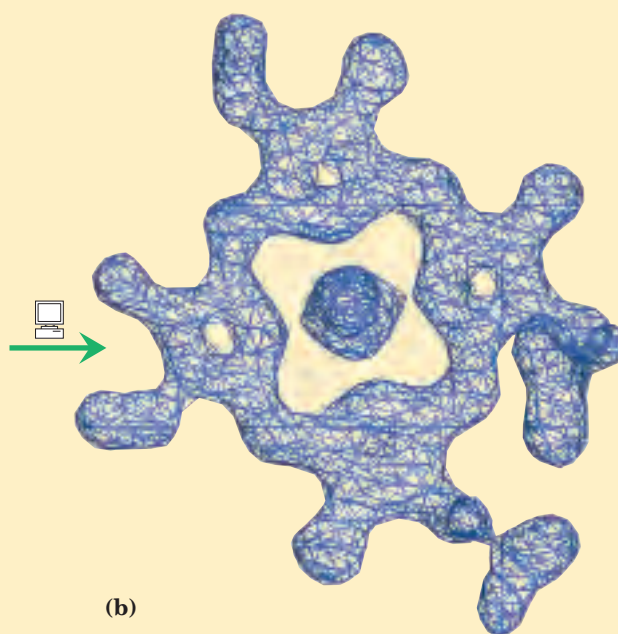
wavelengths in the range of 400 to 700 nm. Objects smaller than half the wavelength of the incident light cannot be resolved. To resolve objects as small as proteins we must use x rays, with wavelengths in the range of 0.7 to 1.5 Å (0.07 to 0.15 nm). However, there are no lenses that can recombine x rays to form an image; instead the pattern of diffracted x rays is collected directly and an image is reconstructed by mathematical techniques.

The amount of information obtained from x-ray crystallography depends on the degree of structural order in the sample. Some important structural parameters were obtained from early studies of the diffraction patterns of the fibrous proteins arranged in fairly regular arrays in hair and wool. However, the orderly bundles formed by fibrous proteins are not crystals—the molecules are aligned side by side, but not all are oriented in the same direction. More detailed three-dimensional structural information about proteins requires a highly ordered protein crystal. Protein crystallization is something of an empirical science, and the structures of many important proteins are not yet known, simply because they have proved difficult to crystallize. Practitioners have compared making protein crystals to holding together a stack of bowling balls with cellophane tape.

Operationally, there are several steps in x-ray structural analysis (Fig. 1). Once a crystal is obtained, it is placed in an x-ray beam between the x-ray source and a detector, and a regular array of spots called re-



(a)



(b)

flections is generated. The spots are created by the diffracted x-ray beam, and each atom in a molecule makes a contribution to each spot. An electron-density map of the protein is reconstructed from the overall diffraction pattern of spots by using a mathematical technique called a Fourier transform. In effect, the computer acts as a “computational lens.” A model for the structure is then built that is consistent with the electron-density map.

John Kendrew found that the x-ray diffraction pattern of crystalline myoglobin (isolated from muscles of the sperm whale) is very complex, with nearly 25,000 reflections. Computer analysis of these reflections took place in stages. The resolution improved at each stage, until in 1959 the positions of virtually all the non-hydrogen atoms in the protein had been determined. The amino acid sequence of the protein, obtained by chemical analysis, was consistent with the molecular model. The structures of thousands of proteins, many of them much more complex than myoglobin, have since been determined to a similar level of resolution.

The physical environment within a crystal, of course, is not identical to that in solution or in a living cell. A crystal imposes a space and time average on the structure deduced from its analysis, and x-ray diffraction studies provide little information about molecular motion within the protein. The conformation of proteins in a crystal could in principle also be affected by nonphysiological factors such as incidental

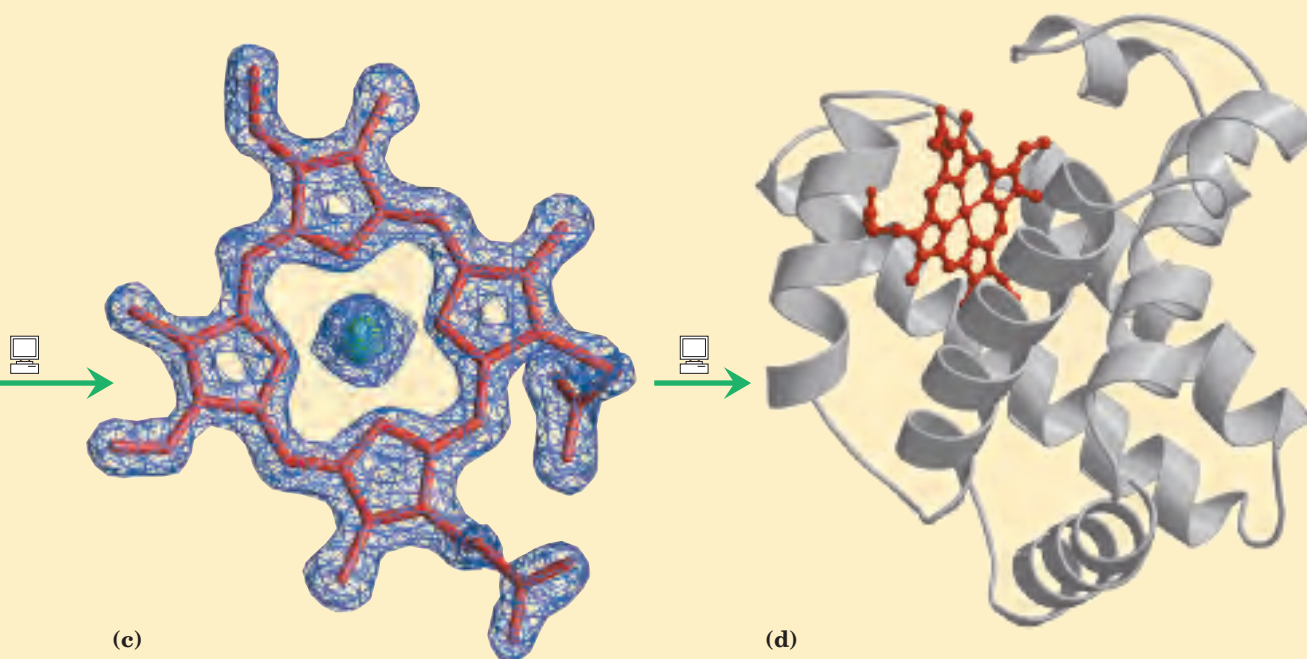
protein-protein contacts within the crystal. However, when structures derived from the analysis of crystals are compared with structural information obtained by other means (such as NMR, as described below), the crystal-derived structure almost always represents a functional conformation of the protein. X-ray crystallography can be applied successfully to proteins too large to be structurally analyzed by NMR.

Nuclear Magnetic Resonance

An important complementary method for determining the three-dimensional structures of macromolecules is nuclear magnetic resonance (NMR). Modern NMR techniques are being used to determine the structures of ever-larger macromolecules, including carbohydrates, nucleic acids, and small to average-sized proteins. An advantage of NMR studies is that they are

(continued on next page)

FIGURE 1 Steps in the determination of the structure of sperm whale myoglobin by x-ray crystallography. **(a)** X-ray diffraction patterns are generated from a crystal of the protein. **(b)** Data extracted from the diffraction patterns are used to calculate a three-dimensional electron-density map of the protein. The electron density of only part of the structure, the heme, is shown. **(c)** Regions of greatest electron density reveal the location of atomic nuclei, and this information is used to piece together the final structure. Here, the heme structure is modeled into its electron-density map. **(d)** The completed structure of sperm whale myoglobin, including the heme (PDB ID 2MBW).



BOX 4-4 WORKING IN BIOCHEMISTRY (continued from previous page)

carried out on macromolecules in solution, whereas x-ray crystallography is limited to molecules that can be crystallized. NMR can also illuminate the dynamic side of protein structure, including conformational changes, protein folding, and interactions with other molecules.

NMR is a manifestation of nuclear spin angular momentum, a quantum mechanical property of atomic nuclei. Only certain atoms, including ^1H , ^{13}C , ^{15}N , ^{19}F , and ^{31}P , possess the kind of nuclear spin that gives rise to an NMR signal. Nuclear spin generates a magnetic dipole. When a strong, static magnetic field is applied to a solution containing a single type of macromolecule, the magnetic dipoles are aligned in the field in one of two orientations, parallel (low energy) or antiparallel (high energy). A short ($\sim 10\ \mu\text{s}$) pulse of electromagnetic energy of suitable frequency (the resonant frequency, which is in the radio frequency range) is applied at right angles to the nuclei aligned in the magnetic field. Some energy is absorbed as nuclei switch to the high-energy state, and the absorption spectrum that results contains information about the identity of the nuclei and their immediate chemical environment. The data from many such experiments performed on a sample are averaged, increasing the signal-to-noise ratio, and an NMR spectrum such as that in Figure 2 is generated.

^1H is particularly important in NMR experiments because of its high sensitivity and natural abundance. For macromolecules, ^1H NMR spectra can become quite complicated. Even a small protein has hundreds of ^1H atoms, typically resulting in a one-dimensional NMR spectrum too complex for analysis. Structural analysis of proteins became possible with the advent of two-dimensional NMR techniques (Fig. 3). These methods allow measurement of distance-dependent coupling of nuclear spins in nearby atoms through space (the nuclear Overhauser effect (NOE), in a method dubbed NOESY) or the coupling of nuclear spins in atoms connected by covalent bonds (total correlation spectroscopy, or TOCSY).

Translating a two-dimensional NMR spectrum into a complete three-dimensional structure can be a laborious process. The NOE signals provide some information about the distances between individual atoms, but

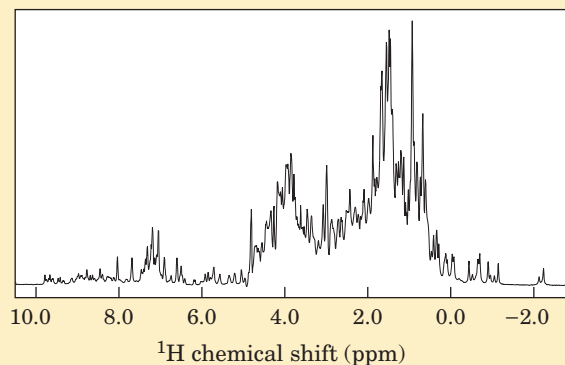


FIGURE 2 A one-dimensional NMR spectrum of a globin from a marine blood worm. This protein and sperm whale myoglobin are very close structural analogs, belonging to the same protein structural family and sharing an oxygen-transport function.

for these distance constraints to be useful, the atoms giving rise to each signal must be identified. Complementary TOCSY experiments can help identify which NOE signals reflect atoms that are linked by covalent bonds. Certain patterns of NOE signals have been associated with secondary structures such as α helices. Modern genetic engineering (Chapter 9) can be used to prepare proteins that contain the rare isotopes ^{13}C or ^{15}N . The new NMR signals produced by these atoms, and the coupling with ^1H signals resulting from these substitutions, help in the assignment of individual ^1H NOE signals. The process is also aided by a knowledge of the amino acid sequence of the polypeptide.

To generate a three-dimensional structure, researchers feed the distance constraints into a computer along with known geometric constraints such as chirality, van der Waals radii, and bond lengths and angles. The computer generates a family of closely related structures that represent the range of conformations consistent with the NOE distance constraints (Fig. 3c). The uncertainty in structures generated by NMR is in part a reflection of the molecular vibrations (breathing) within a protein structure in solution, discussed in more detail in Chapter 5. Normal experimental uncertainty can also play a role.

When a protein structure has been determined by both x-ray crystallography and NMR, the structures

Analysis of Many Globular Proteins Reveals Common Structural Patterns

Protein Architecture—Tertiary Structure of Large Globular Proteins For the beginning student, the very complex tertiary structures of globular proteins much larger than those shown in Figure 4-18 are best approached by fo-

cusing on structural patterns that recur in different and often unrelated proteins. The three-dimensional structure of a typical globular protein can be considered an assemblage of polypeptide segments in the α -helix and β -sheet conformations, linked by connecting segments. The structure can then be described to a first approximation by defining how these segments stack on one

generally agree well. In some cases, the precise locations of particular amino acid side chains on the protein exterior are different, often because of effects related to the packing of adjacent protein molecules in

a crystal. The two techniques together are at the heart of the rapid increase in the availability of structural information about the macromolecules of living cells.

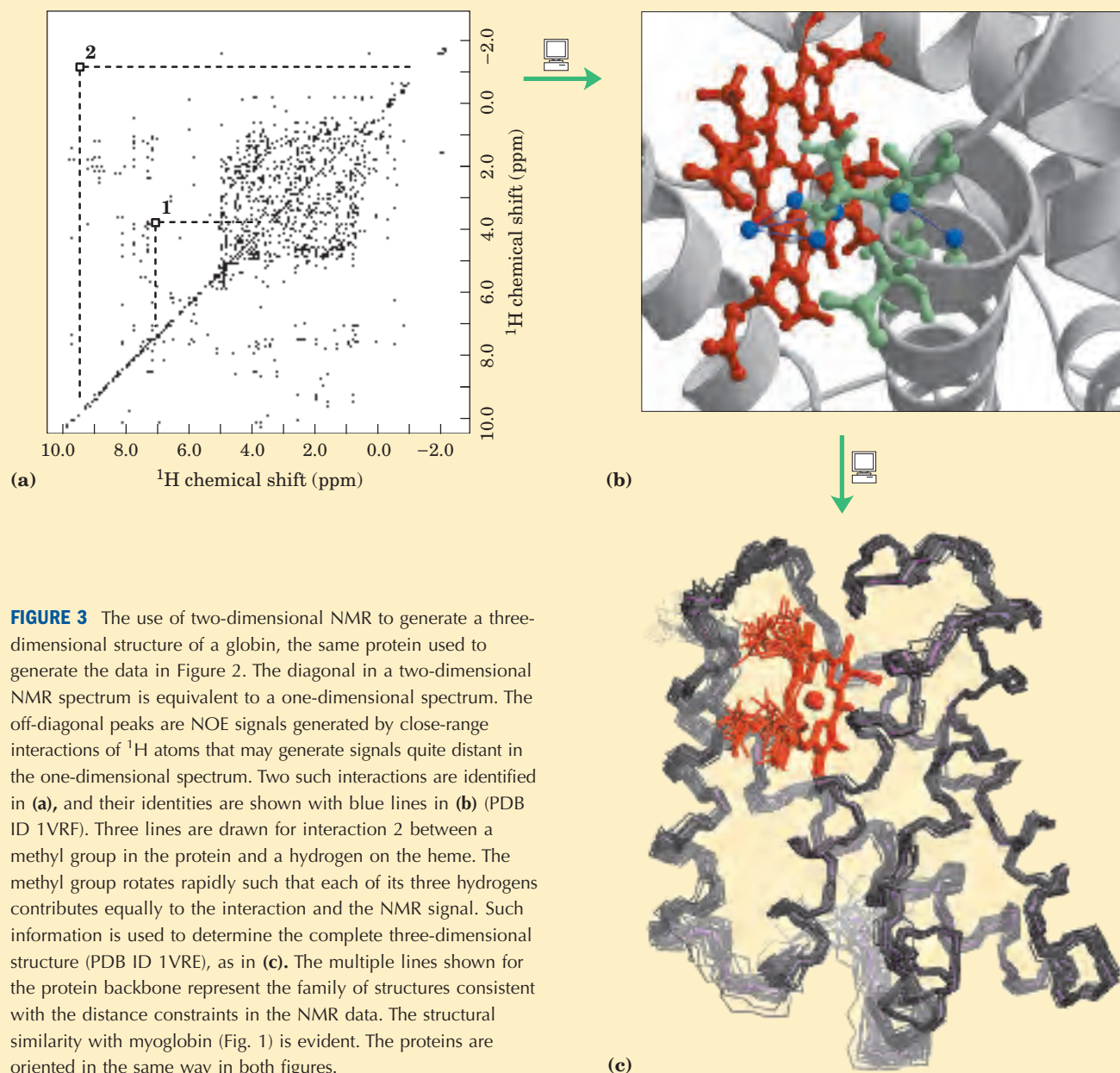


FIGURE 3 The use of two-dimensional NMR to generate a three-dimensional structure of a globin, the same protein used to generate the data in Figure 2. The diagonal in a two-dimensional NMR spectrum is equivalent to a one-dimensional spectrum. The off-diagonal peaks are NOE signals generated by close-range interactions of ^1H atoms that may generate signals quite distant in the one-dimensional spectrum. Two such interactions are identified in (a), and their identities are shown with blue lines in (b) (PDB ID 1VRF). Three lines are drawn for interaction 2 between a methyl group in the protein and a hydrogen on the heme. The methyl group rotates rapidly such that each of its three hydrogens contributes equally to the interaction and the NMR signal. Such information is used to determine the complete three-dimensional structure (PDB ID 1VRE), as in (c). The multiple lines shown for the protein backbone represent the family of structures consistent with the distance constraints in the NMR data. The structural similarity with myoglobin (Fig. 1) is evident. The proteins are oriented in the same way in both figures.

another and how the segments that connect them are arranged. This formalism has led to the development of databases that allow informative comparisons of protein structures, complementing other databases that permit comparisons of protein sequences.

An understanding of a complete three-dimensional structure is built upon an analysis of its parts. We begin

by defining terms used to describe protein substructures, then turn to the folding rules elucidated from analysis of the structures of many proteins.

Supersecondary structures, also called **motifs** or simply **folds**, are particularly stable arrangements of several elements of secondary structure and the connections between them. There is no universal agreement

among biochemists on the application of the three terms, and they are often used interchangeably. The terms are also applied to a wide range of structures. Recognized motifs range from simple to complex, sometimes appearing in repeating units or combinations. A single large motif may comprise the entire protein. We have already encountered one well-studied motif, the coiled coil of α -keratin, also found in a number of other proteins.

Polypeptides with more than a few hundred amino acid residues often fold into two or more stable, globular units called **domains**. In many cases, a domain from a large protein will retain its correct three-dimensional structure even when it is separated (for example, by proteolytic cleavage) from the remainder of the polypeptide chain. A protein with multiple domains may appear to have a distinct globular lobe for each domain (Fig. 4-19), but, more commonly, extensive contacts between domains make individual domains hard to discern. Different domains often have distinct functions, such as the binding of small molecules or interaction with other proteins. Small proteins usually have only one domain (the domain *is* the protein).

Folding of polypeptides is subject to an array of physical and chemical constraints. A sampling of the prominent folding rules that have emerged provides an opportunity to introduce some simple motifs.

1. Hydrophobic interactions make a large contribution to the stability of protein structures. Burial of hydrophobic amino acid R groups so as to exclude water requires at least two layers of secondary structure. Two simple motifs, the **β - α - β loop** and the **α - α corner** (Fig. 4-20a), create two layers.
2. Where they occur together in proteins, α helices and β sheets generally are found in different structural layers. This is because the backbone of a polypeptide segment in the β conformation (Fig. 4-7) cannot readily hydrogen-bond to an α helix aligned with it.

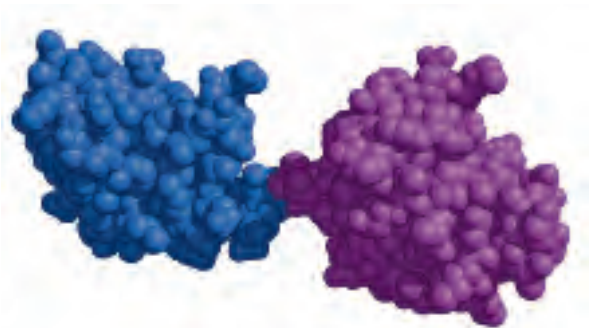


FIGURE 4-19 Structural domains in the polypeptide troponin C. (PDB ID 4TNC) This calcium-binding protein associated with muscle has separate calcium-binding domains, indicated in blue and purple.

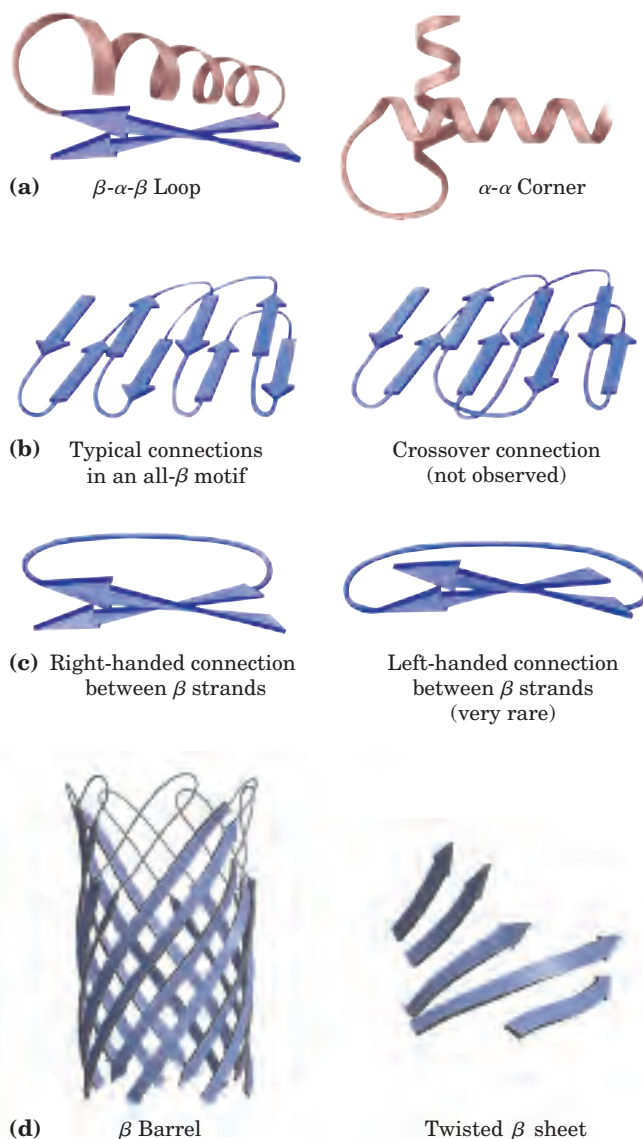


FIGURE 4-20 Stable folding patterns in proteins. (a) Two simple and common motifs that provide two layers of secondary structure. Amino acid side chains at the interface between elements of secondary structure are shielded from water. Note that the β strands in the β - α - β loop tend to twist in a right-handed fashion. (b) Connections between β strands in layered β sheets. The strands are shown from one end, with no twisting included in the schematic. Thick lines represent connections at the ends nearest the viewer; thin lines are connections at the far ends of the β strands. The connections on a given end (e.g., near the viewer) do not cross each other. (c) Because of the twist in β strands, connections between strands are generally right-handed. Left-handed connections must traverse sharper angles and are harder to form. (d) Two arrangements of β strands stabilized by the tendency of the strands to twist. This β barrel is a single domain of α -hemolysin (a pore-forming toxin that kills a cell by creating a hole in its membrane) from the bacterium *Staphylococcus aureus* (derived from PDB ID 7AHL). The twisted β sheet is from a domain of photolyase (a protein that repairs certain types of DNA damage) from *E. coli* (derived from PDB ID 1DNP).

3. Polypeptide segments adjacent to each other in the primary sequence are usually stacked adjacent to each other in the folded structure. Although distant segments of a polypeptide may come together in the tertiary structure, this is not the norm.
4. Connections between elements of secondary structure cannot cross or form knots (Fig. 4–20b).
5. The β conformation is most stable when the individual segments are twisted slightly in a right-handed sense. This influences both the arrangement of β sheets relative to one another and the path of the polypeptide connection between them. Two parallel β strands, for example, must be connected by a crossover strand (Fig. 4–20c). In principle, this crossover could have a right- or left-handed conformation, but in proteins it is almost always right-handed. Right-handed connections tend to be shorter than left-handed connections and tend to bend through smaller angles, making them easier to form. The twisting of β sheets also leads to a characteristic twisting of the structure formed when many segments are put together. Two examples of resulting structures are the β barrel and twisted β sheet (Fig. 4–20d), which form the core of many larger structures.

Following these rules, complex motifs can be built up from simple ones. For example, a series of β - α - β loops, arranged so that the β strands form a barrel, creates a particularly stable and common motif called the **α/β barrel** (Fig. 4–21). In this structure, each parallel β segment is attached to its neighbor by an α -helical segment. All connections are right-handed. The α/β barrel is found in many enzymes, often with a binding site for a

cofactor or substrate in the form of a pocket near one end of the barrel. Note that domains exhibiting similar folding patterns are said to have the same motif even though their constituent α helices and β sheets may differ in length.

Protein Motifs Are the Basis for Protein Structural Classification

Protein Architecture—Tertiary Structure of Large Globular Proteins, IV. Structural Classification of Proteins As we have seen, the complexities of tertiary structure are decreased by considering substructures. Taking this idea further, researchers have organized the complete contents of databases according to hierarchical levels of structure. The Structural Classification of Proteins (SCOP) database offers a good example of this very important trend in biochemistry. At the highest level of classification, the SCOP database (<http://scop.mrc-lmb.cam.ac.uk/scop>) borrows a scheme already in common use, in which protein structures are divided into four classes: all α , all β , α/β (in which the α and β segments are interspersed or alternate), and $\alpha + \beta$ (in which the α and β regions are somewhat segregated) (Fig. 4–22). Within each class are tens to hundreds of different folding arrangements, built up from increasingly identifiable substructures. Some of the substructure arrangements are very common, others have been found in just one protein. Figure 4–22 displays a variety of motifs arrayed among the four classes of protein structure. Those illustrated are just a minute sample of the hundreds of known motifs. The number of folding patterns is not infinite, however. As the rate at which new protein structures are elucidated has increased, the fraction of those structures containing a new motif has steadily declined. Fewer than 1,000 different folds or motifs may exist in all proteins. Figure 4–22 also shows how proteins can be organized based on the presence of the various motifs. The top two levels of organization, **class** and **fold**, are purely structural. Below the fold level, categorization is based on evolutionary relationships.

Many examples of recurring domain or motif structures are available, and these reveal that protein tertiary structure is more reliably conserved than primary sequence. The comparison of protein structures can thus provide much information about evolution. Proteins with significant primary sequence similarity, and/or with demonstrably similar structure and function, are said to be in the same **protein family**. A strong evolutionary relationship is usually evident within a protein family. For example, the globin family has many different proteins with both structural and sequence similarity to myoglobin (as seen in the proteins used as examples in Box 4–4 and again in the next chapter). Two or more families with little primary sequence similarity sometimes make use of the same major structural

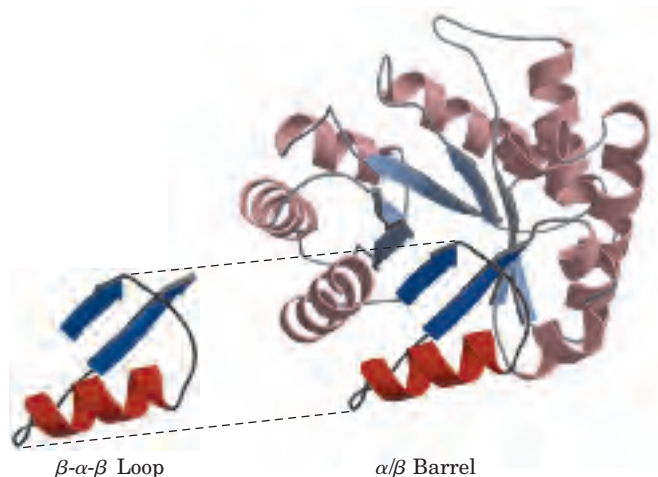


FIGURE 4-21 Constructing large motifs from smaller ones. The α/β barrel is a common motif constructed from repetitions of the simpler β - α - β loop motif. This α/β barrel is a domain of the pyruvate kinase (a glycolytic enzyme) from rabbit (derived from PDB ID 1PKN).

All α 

1AO6
Serum albumin
Serum albumin
Serum albumin
Human (*Homo sapiens*)



1BCF
Ferritin-like
Ferritin-like
Bacterioferritin (cytochrome b_1)
Escherichia coli



1GAI
 α/α toroid
Six-hairpin glycosyltransferase
Glucoamylase
Glucoamylase
Aspergillus awamori,
variant x100



1ENH
DNA/RNA-binding
3-helical bundle
Homeodomain-like
Homeodomain
engrailed Homeodomain
Drosophila melanogaster

All β 

1HOE
 α -Amylase inhibitor tendamistat
 α -Amylase inhibitor tendamistat
 α -Amylase inhibitor tendamistat
 α -Amylase inhibitor tendamistat
Streptomyces tendae



1LXA
Single-stranded left-handed β helix
Trimeric LpxA-like enzymes
UDP *N*-acetylglucosamine acyltransferase
UDP *N*-acetylglucosamine acyltransferase
Escherichia coli



1PEX
Four-bladed β propeller
Hemopexin-like domain
Hemopexin-like domain
Collagenase-3 (MMP-13),
carboxyl-terminal domain
Human (*Homo sapiens*)



1JPC
 β -Prism II
 α -D-Mannose-specific plant lectins
 α -D-Mannose-specific plant lectins
Lectin (agglutinin)
Snowdrop (*Galanthus nivalis*)



1CD8
Immunoglobulin-like β sandwich
Immunoglobulin
V set domains (antibody variable domain-like)
CD8
Human (*Homo sapiens*)

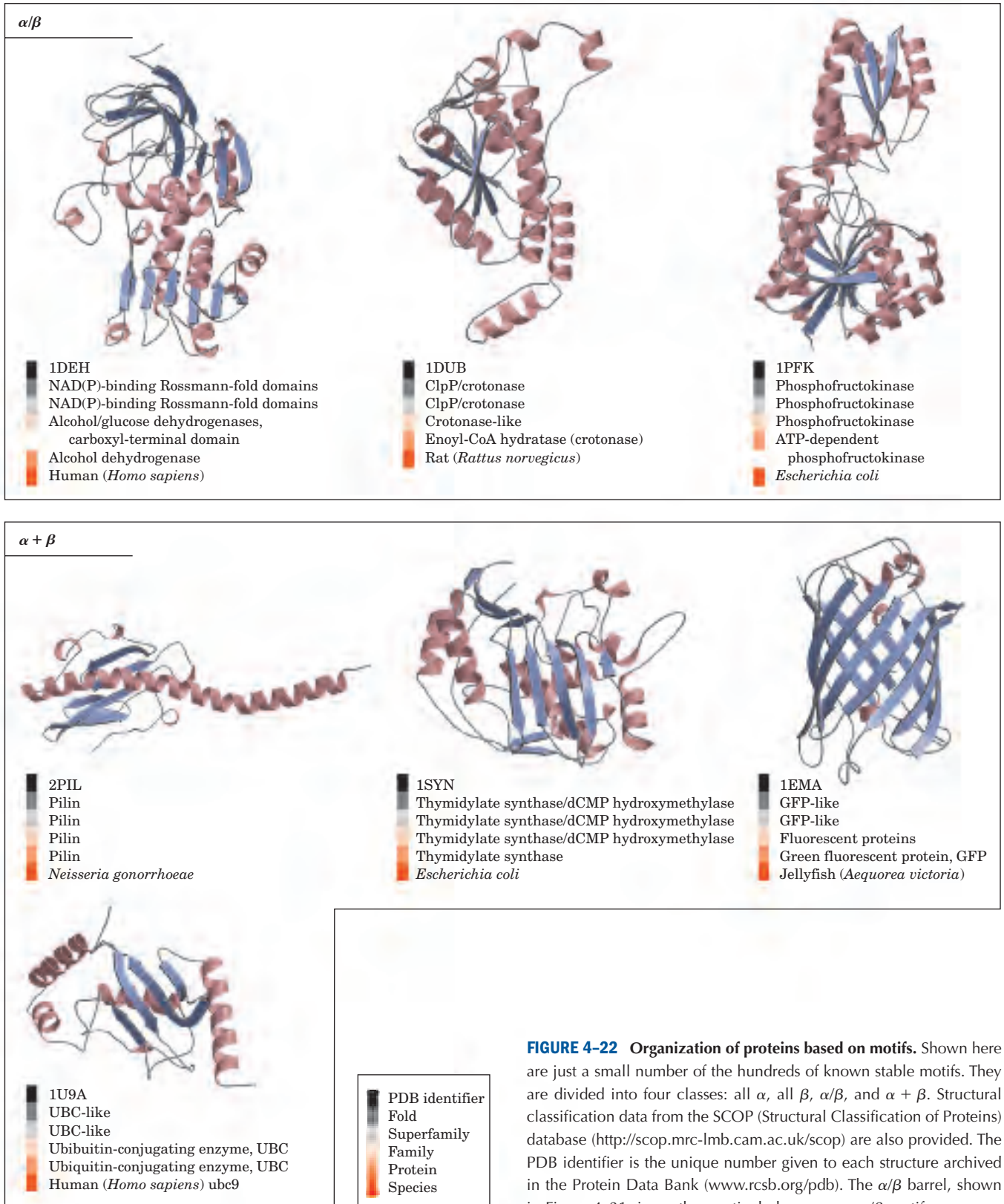


FIGURE 4-22 Organization of proteins based on motifs. Shown here are just a small number of the hundreds of known stable motifs. They are divided into four classes: all α , all β , α/β , and $\alpha + \beta$. Structural classification data from the SCOP (Structural Classification of Proteins) database (<http://scop.mrc-lmb.cam.ac.uk/scop>) are also provided. The PDB identifier is the unique number given to each structure archived in the Protein Data Bank (www.rcsb.org/pdb). The α/β barrel, shown in Figure 4-21, is another particularly common α/β motif.

motif and have functional similarities; these families are grouped as **superfamilies**. An evolutionary relationship between the families in a superfamily is considered probable, even though time and functional distinctions—hence different adaptive pressures—may have erased many of the telltale sequence relationships. A protein family may be widespread in all three domains of cellular life, the Bacteria, Archaea, and Eukarya, suggesting a very ancient origin. Other families may be present in only a small group of organisms, indicating that the structure arose more recently. Tracing the natural history of structural motifs, using structural classifications in databases such as SCOP, provides a powerful complement to sequence analyses in tracing many evolutionary relationships.

The SCOP database is curated manually, with the objective of placing proteins in the correct evolutionary framework based on conserved structural features. Two similar enterprises, the CATH (class, architecture, topology, and homologous superfamily) and FSSP (fold classification based on structure-structure alignment of proteins) databases, make use of more automated methods and can provide additional information.

Structural motifs become especially important in defining protein families and superfamilies. Improved classification and comparison systems for proteins lead inevitably to the elucidation of new functional relationships. Given the central role of proteins in living systems, these structural comparisons can help illuminate every aspect of biochemistry, from the evolution of individual proteins to the evolutionary history of complete metabolic pathways.

Protein Quaternary Structures Range from Simple Dimers to Large Complexes

 **Protein Architecture—Quaternary Structure** Many proteins have multiple polypeptide subunits. The association of polypeptide chains can serve a variety of functions. Many multisubunit proteins have regulatory roles; the binding of small molecules may affect the interaction between subunits, causing large changes in the protein's activity in response to small changes in the concentration of substrate or regulatory molecules (Chapter 6). In other cases, separate subunits can take on separate but related functions, such as catalysis and regulation. Some associations, such as the fibrous proteins considered earlier in this chapter and the coat proteins of viruses, serve primarily structural roles. Some very large protein assemblies are the site of complex, multistep reactions. One example is the ribosome, site of protein synthesis, which incorporates dozens of protein subunits along with a number of RNA molecules.

A multisubunit protein is also referred to as a **multimer**. Multimeric proteins can have from two to hundreds of subunits. A multimer with just a few subunits

is often called an **oligomer**. If a multimer is composed of a number of nonidentical subunits, the overall structure of the protein can be asymmetric and quite complicated. However, most multimers have identical subunits or repeating groups of nonidentical subunits, usually in symmetric arrangements. As noted in Chapter 3, the repeating structural unit in such a multimeric protein, whether it is a single subunit or a group of subunits, is called a **protomer**.

The first oligomeric protein for which the three-dimensional structure was determined was hemoglobin (M_r 64,500), which contains four polypeptide chains and four heme prosthetic groups, in which the iron atoms are in the ferrous (Fe^{2+}) state (Fig. 4-17). The protein portion, called globin, consists of two α chains (141 residues each) and two β chains (146 residues each). Note that in this case α and β do not refer to secondary structures. Because hemoglobin is four times as large as myoglobin, much more time and effort were required to solve its three-dimensional structure by x-ray analysis, finally achieved by Max Perutz, John Kendrew, and their colleagues in 1959. The subunits of hemoglobin are arranged in symmetric pairs (Fig. 4-23), each pair having one α and one β subunit. Hemoglobin can therefore be described either as a tetramer or as a dimer of $\alpha\beta$ protomers.

Identical subunits of multimeric proteins are generally arranged in one or a limited set of symmetric patterns. A description of the structure of these proteins requires an understanding of conventions used to define symmetries. Oligomers can have either **rotational symmetry** or **helical symmetry**; that is, individual subunits can be superimposed on others (brought to coincidence) by rotation about one or more rotational axes, or by a helical rotation. In proteins with rotational symmetry, the subunits pack about the rotational axes to form closed structures. Proteins with helical symme-



Max Perutz, 1914–2002 (left)
John Kendrew, 1917–1997 (right)

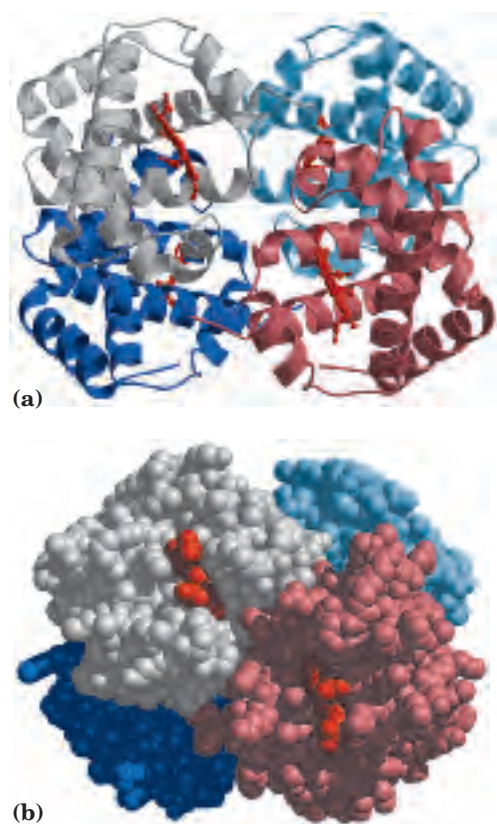


FIGURE 4-23 Quaternary structure of deoxyhemoglobin. (PDB ID 2HHB) X-ray diffraction analysis of deoxyhemoglobin (hemoglobin without oxygen molecules bound to the heme groups) shows how the four polypeptide subunits are packed together. **(a)** A ribbon representation. **(b)** A space-filling model. The α subunits are shown in gray and light blue; the β subunits in pink and dark blue. Note that the heme groups (red) are relatively far apart.

try tend to form structures that are more open-ended, with subunits added in a spiraling array.

There are several forms of rotational symmetry. The simplest is **cyclic symmetry**, involving rotation about a single axis (Fig. 4-24a). If subunits can be superimposed by rotation about a single axis, the protein has a symmetry defined by convention as C_n (C for cyclic, n for the number of subunits related by the axis). The axis itself is described as an n -fold rotational axis. The $\alpha\beta$ protomers of hemoglobin (Fig. 4-23) are related by C_2 symmetry. A somewhat more complicated rotational symmetry is **dihedral symmetry**, in which a twofold rotational axis intersects an n -fold axis at right angles. The symmetry is defined as D_n (Fig. 4-24b). A protein with dihedral symmetry has $2n$ protomers.

Proteins with cyclic or dihedral symmetry are particularly common. More complex rotational symmetries are possible, but only a few are regularly encountered. One example is **icosahedral symmetry**. An icosahedron is a regular 12-cornered polyhedron having 20 equilateral triangular faces (Fig. 4-24c). Each face can

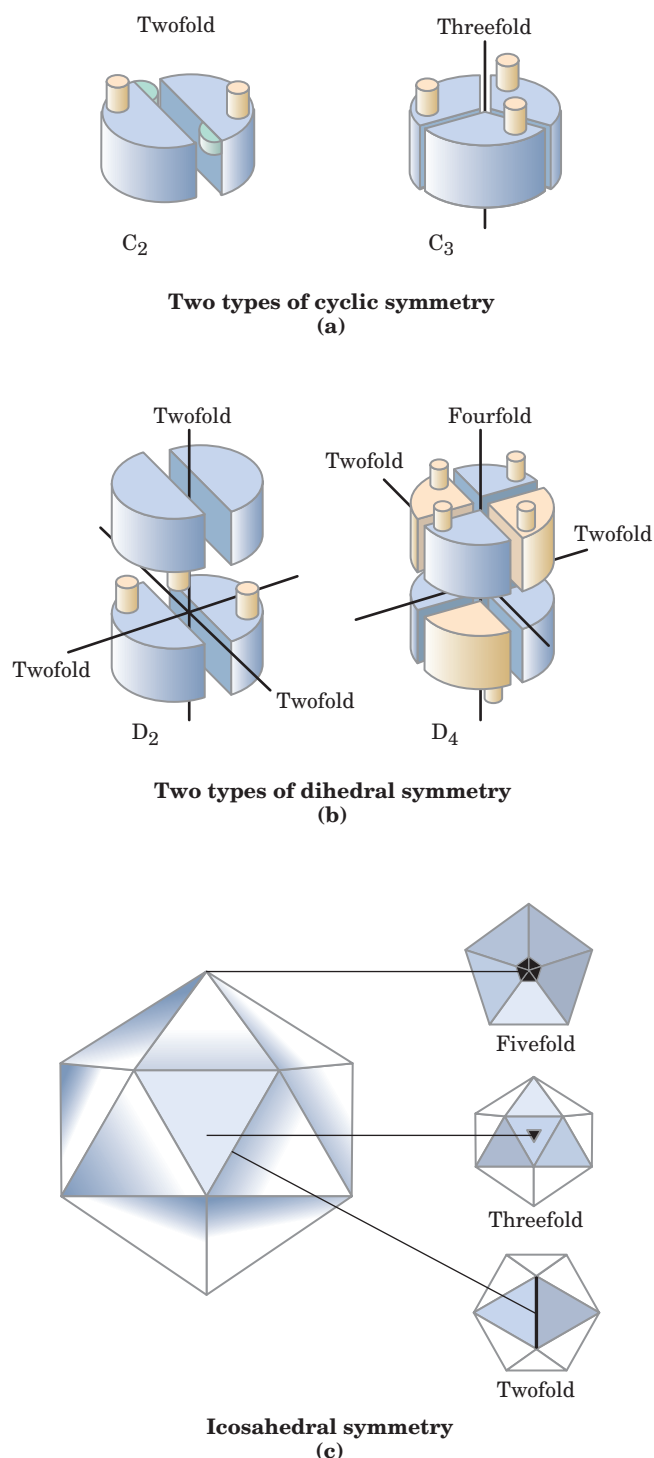
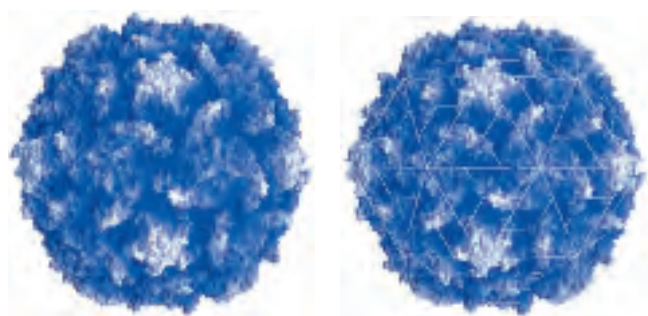


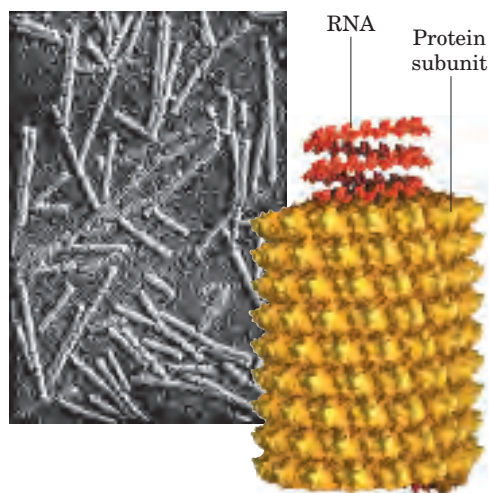
FIGURE 4-24 Rotational symmetry in proteins. **(a)** In cyclic symmetry, subunits are related by rotation about a single n -fold axis, where n is the number of subunits so related. The axes are shown as black lines; the numbers are values of n . Only two of many possible C_n arrangements are shown. **(b)** In dihedral symmetry, all subunits can be related by rotation about one or both of two axes, one of which is twofold. D_2 symmetry is most common. **(c)** Icosahedral symmetry. Relating all 20 triangular faces of an icosahedron requires rotation about one or more of three separate rotational axes: twofold, threefold, and fivefold. An end-on view of each of these axes is shown at the right.

be brought to coincidence with another by rotation about one or more of three rotational axes. This is a common structure in virus coats, or capsids. The human poliovirus has an icosahedral capsid (Fig. 4–25a). Each triangular face is made up of three protomers, each protomer containing single copies of four different polypeptide chains, three of which are accessible at the outer surface. Sixty protomers form the 20 faces of the icosahedral shell enclosing the genetic material (RNA).

The other major type of symmetry found in oligomers, helical symmetry, also occurs in capsids. Tobacco mosaic virus is a right-handed helical filament made up of 2,130 identical subunits (Fig. 4–25b). This cylindrical structure encloses the viral RNA. Proteins with subunits arranged in helical filaments can also form long, fibrous structures such as the actin filaments of muscle (see Fig. 5–30).



(a)



(b)

FIGURE 4–25 Viral capsids. (a) Poliovirus (derived from PDB ID 2PLV). The coat proteins of poliovirus assemble into an icosahedron 300 Å in diameter. Icosahedral symmetry is a type of rotational symmetry (see Fig. 4–24c). On the left is a surface contour image of the poliovirus capsid. In the image on the right, lines have been superimposed to show the axes of symmetry. (b) Tobacco mosaic virus (derived from PDB ID 1VTM). This rod-shaped virus (as shown in the electron micrograph) is 3,000 Å long and 180 Å in diameter; it has helical symmetry.

There Are Limits to the Size of Proteins

The relatively large size of proteins reflects their functions. The function of an enzyme, for example, requires a stable structure containing a pocket large enough to bind its substrate and catalyze a reaction. Protein size has limits, however, imposed by two factors: the genetic coding capacity of nucleic acids and the accuracy of the protein biosynthetic process. The use of many copies of one or a few proteins to make a large enclosing structure (capsid) is important for viruses because this strategy conserves genetic material. Remember that there is a linear correspondence between the sequence of a gene in the nucleic acid and the amino acid sequence of the protein for which it codes (see Fig. 1–31). The nucleic acids of viruses are much too small to encode the information required for a protein shell made of a single polypeptide. By using many copies of much smaller polypeptides, a much shorter nucleic acid is needed for coding the capsid subunits, and this nucleic acid can be efficiently used over and over again. Cells also use large complexes of polypeptides in muscle, cilia, the cytoskeleton, and other structures. It is simply more efficient to make many copies of a small polypeptide than one copy of a very large protein. In fact, most proteins with a molecular weight greater than 100,000 have multiple subunits, identical or different.

The second factor limiting the size of proteins is the error frequency during protein biosynthesis. The error frequency is low (about 1 mistake per 10,000 amino acid residues added), but even this low rate results in a high probability of a damaged protein if the protein is very large. Simply put, the potential for incorporating a “wrong” amino acid in a protein is greater for a large protein than for a small one.

SUMMARY 4.3 Protein Tertiary and Quaternary Structures

- Tertiary structure is the complete three-dimensional structure of a polypeptide chain. There are two general classes of proteins based on tertiary structure: fibrous and globular.
- Fibrous proteins, which serve mainly structural roles, have simple repeating elements of secondary structure.
- Globular proteins have more complicated tertiary structures, often containing several types of secondary structure in the same polypeptide chain. The first globular protein structure to be determined, using x-ray diffraction methods, was that of myoglobin.
- The complex structures of globular proteins can be analyzed by examining stable substructures called supersecondary structures,

motifs, or folds. The thousands of known protein structures are generally assembled from a repertoire of only a few hundred motifs. Regions of a polypeptide chain that can fold stably and independently are called domains.

- Quaternary structure results from interactions between the subunits of multisubunit (multimeric) proteins or large protein assemblies. Some multimeric proteins have a repeated unit consisting of a single subunit or a group of subunits referred to as a protomer. Protomers are usually related by rotational or helical symmetry.

4.4 Protein Denaturation and Folding

All proteins begin their existence on a ribosome as a linear sequence of amino acid residues (Chapter 27). This polypeptide must fold during and following synthesis to take up its native conformation. We have seen that a native protein conformation is only marginally stable. Modest changes in the protein's environment can bring about structural changes that can affect function. We now explore the transition that occurs between the folded and unfolded states.

Loss of Protein Structure Results in Loss of Function

Protein structures have evolved to function in particular cellular environments. Conditions different from those in the cell can result in protein structural changes, large and small. A loss of three-dimensional structure sufficient to cause loss of function is called **denaturation**. The denatured state does not necessarily equate with complete unfolding of the protein and randomization of conformation. Under most conditions, denatured proteins exist in a set of partially folded states that are poorly understood.

Most proteins can be denatured by heat, which affects the weak interactions in a protein (primarily hydrogen bonds) in a complex manner. If the temperature is increased slowly, a protein's conformation generally remains intact until an abrupt loss of structure (and function) occurs over a narrow temperature range (Fig. 4–26). The abruptness of the change suggests that unfolding is a cooperative process: loss of structure in one part of the protein destabilizes other parts. The effects of heat on proteins are not readily predictable. The very heat-stable proteins of thermophilic bacteria have evolved to function at the temperature of hot springs (~100 °C). Yet the structures of these proteins often differ only slightly from those of homologous proteins derived from bacteria such as *Escherichia coli*. How these small differences promote structural stability at high temperatures is not yet understood.

Proteins can be denatured not only by heat but by extremes of pH, by certain miscible organic solvents

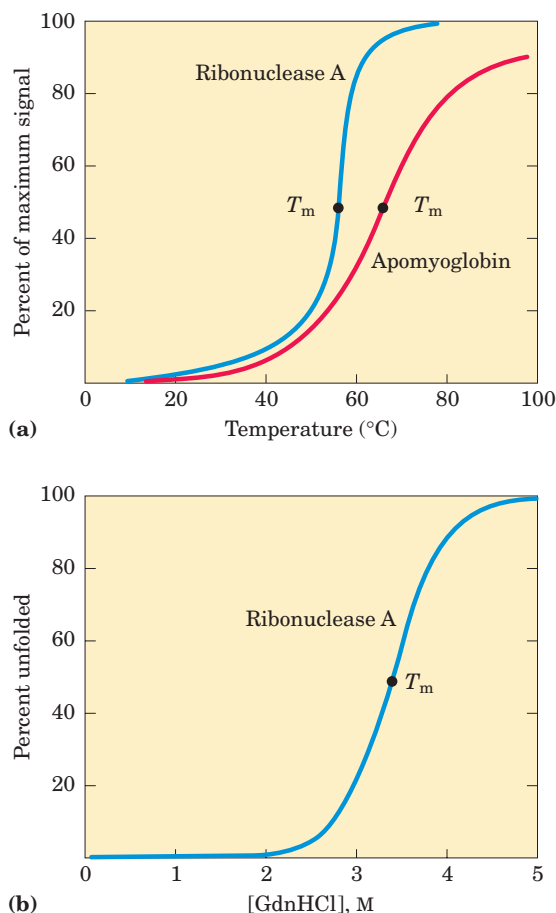


FIGURE 4–26 Protein denaturation. Results are shown for proteins denatured by two different environmental changes. In each case, the transition from the folded to unfolded state is fairly abrupt, suggesting cooperativity in the unfolding process. **(a)** Thermal denaturation of horse apomyoglobin (myoglobin without the heme prosthetic group) and ribonuclease A (with its disulfide bonds intact; see Fig. 4–27). The midpoint of the temperature range over which denaturation occurs is called the melting temperature, or T_m . The denaturation of apomyoglobin was monitored by circular dichroism, a technique that measures the amount of helical structure in a macromolecule. Denaturation of ribonuclease A was tracked by monitoring changes in the intrinsic fluorescence of the protein, which is affected by changes in the environment of Trp residues. **(b)** Denaturation of disulfide-intact ribonuclease A by guanidine hydrochloride (GdnHCl), monitored by circular dichroism.

such as alcohol or acetone, by certain solutes such as urea and guanidine hydrochloride, or by detergents. Each of these denaturing agents represents a relatively mild treatment in the sense that no covalent bonds in the polypeptide chain are broken. Organic solvents, urea, and detergents act primarily by disrupting the hydrophobic interactions that make up the stable core of globular proteins; extremes of pH alter the net charge on the protein, causing electrostatic repulsion and the disruption of some hydrogen bonding. The denatured states obtained with these various treatments need not be equivalent.

Amino Acid Sequence Determines Tertiary Structure

The tertiary structure of a globular protein is determined by its amino acid sequence. The most important proof of this came from experiments showing that denaturation of some proteins is reversible. Certain globular proteins denatured by heat, extremes of pH, or denaturing reagents will regain their native structure and their biological activity if returned to conditions in which the native conformation is stable. This process is called **renaturation**.

A classic example is the denaturation and renaturation of ribonuclease. Purified ribonuclease can be completely denatured by exposure to a concentrated urea solution in the presence of a reducing agent. The reducing agent cleaves the four disulfide bonds to yield eight Cys residues, and the urea disrupts the stabilizing hydrophobic interactions, thus freeing the entire polypeptide from its folded conformation. Denaturation of ribonuclease is accompanied by a complete loss of catalytic activity. When the urea and the reducing agent are removed, the randomly coiled, denatured ribonuclease spontaneously refolds into its correct tertiary structure, with full restoration of its catalytic activity (Fig. 4-27). The refolding of ribonuclease is so accurate that the four intrachain disulfide bonds are re-formed in the same positions in the renatured molecule as in the native ribonuclease. As calculated mathematically, the eight Cys residues could recombine at random to form up to four disulfide bonds in 105 different ways. In fact, an essentially random distribution of disulfide bonds is obtained when the disulfides are allowed to re-form in the presence of denaturant, indicating that weak bonding interactions are required for correct positioning of disulfide bonds and assumption of the native conformation.

This classic experiment, carried out by Christian Anfinsen in the 1950s, provided the first evidence that the amino acid sequence of a polypeptide chain contains all the information required to fold the chain into its native, three-dimensional structure. Later, similar results were obtained using chemically synthesized, catalytically active ribonuclease. This eliminated the possibility that some minor contaminant in Anfinsen's purified ribonuclease preparation might have contributed to the renaturation of the enzyme, thus dispelling any remaining doubt that this enzyme folds spontaneously.

Polypeptides Fold Rapidly by a Stepwise Process

In living cells, proteins are assembled from amino acids at a very high rate. For example, *E. coli* cells can make a complete, biologically active protein molecule containing 100 amino acid residues in about 5 seconds at 37 °C. How does such a polypeptide chain arrive at its native conformation? Let's assume conservatively that each of the amino acid residues could take up 10 dif-

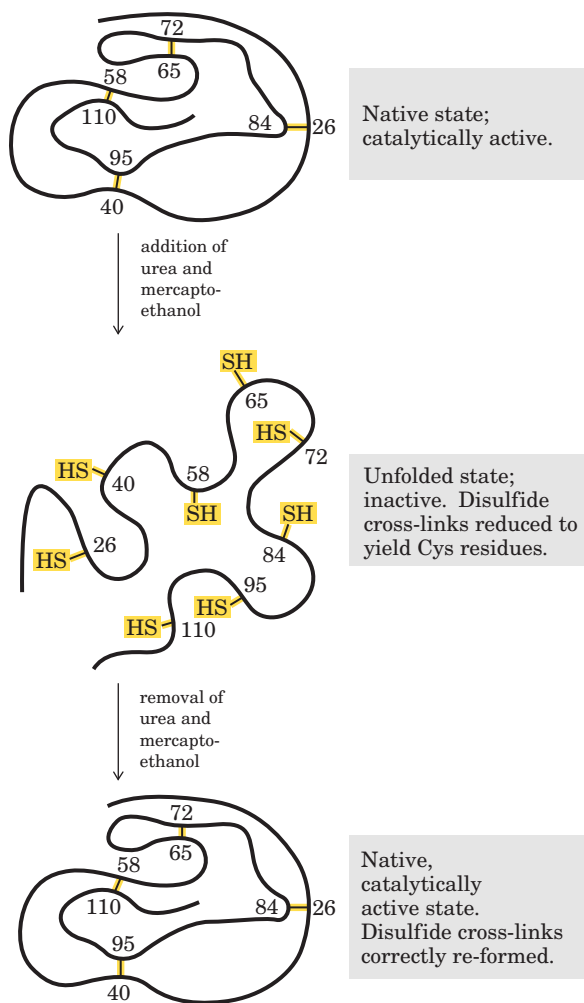


FIGURE 4-27 Renaturation of unfolded, denatured ribonuclease. Urea is used to denature ribonuclease, and mercaptoethanol ($\text{HOCH}_2\text{CH}_2\text{SH}$) to reduce and thus cleave the disulfide bonds to yield eight Cys residues. Renaturation involves reestablishment of the correct disulfide cross-links.

ferent conformations on average, giving 10^{100} different conformations for the polypeptide. Let's also assume that the protein folds itself spontaneously by a random process in which it tries out all possible conformations around every single bond in its backbone until it finds its native, biologically active form. If each conformation were sampled in the shortest possible time ($\sim 10^{-13}$ second, or the time required for a single molecular vibration), it would take about 10^{77} years to sample all possible conformations. Thus protein folding cannot be a completely random, trial-and-error process. There must be shortcuts. This problem was first pointed out by Cyrus Levinthal in 1968 and is sometimes called Levinthal's paradox.

The folding pathway of a large polypeptide chain is unquestionably complicated, and not all the principles that guide the process have been worked out. However, extensive study has led to the development of several

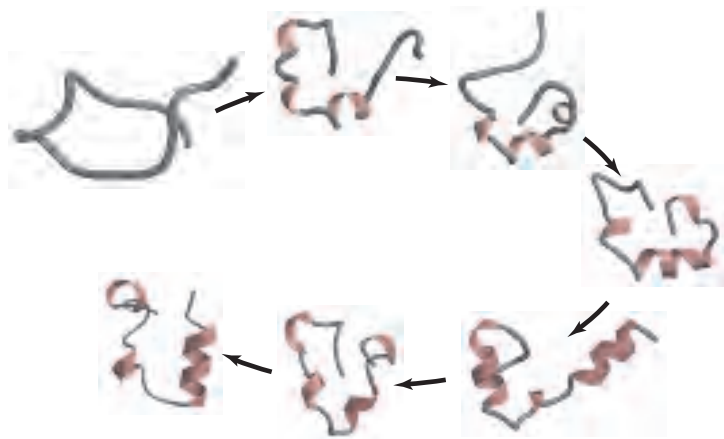


FIGURE 4-28 A simulated folding pathway. The folding pathway of a 36-residue segment of the protein villin (an actin-binding protein found principally in the microvilli lining the intestine) was simulated by computer. The process started with the randomly coiled peptide and 3,000 surrounding water molecules in a virtual “water box.” The molecular motions of the peptide and the effects of the water molecules were taken into account in mapping the most likely paths to the final structure among the countless alternatives. The simulated folding took place in a theoretical time span of 1 ms; however, the calculation required half a billion integration steps on two Cray supercomputers, each running for two months.

plausible models. In one, the folding process is envisioned as hierarchical. Local secondary structures form first. Certain amino acid sequences fold readily into α helices or β sheets, guided by constraints we have reviewed in our discussion of secondary structure. This is followed by longer-range interactions between, say, two α helices that come together to form stable supersecondary structures. The process continues until complete domains form and the entire polypeptide is folded (Fig. 4-28). In an alternative model, folding is initiated by a spontaneous collapse of the polypeptide into a compact state, mediated by hydrophobic interactions among nonpolar residues. The state resulting from this “hydrophobic collapse” may have a high content of secondary structure, but many amino acid side chains are not entirely fixed. The collapsed state is often referred to as a **molten globule**. Most proteins probably fold by a process that incorporates features of both models. Instead of following a single pathway, a population of peptide molecules may take a variety of routes to the same end point, with the number of different partly folded conformational species decreasing as folding nears completion.

Thermodynamically, the folding process can be viewed as a kind of free-energy funnel (Fig. 4-29). The unfolded states are characterized by a high degree of conformational entropy and relatively high free energy. As folding proceeds, the narrowing of the funnel repre-

sents a decrease in the number of conformational species present. Small depressions along the sides of the free-energy funnel represent semistable intermediates that can briefly slow the folding process. At the bottom of the funnel, an ensemble of folding intermediates has been reduced to a single native conformation (or one of a small set of native conformations).

Defects in protein folding may be the molecular basis for a wide range of human genetic disorders. For example, cystic fibrosis is caused by defects in a membrane-bound protein called *cystic fibrosis transmembrane conductance regulator* (CFTR), which acts as a channel for chloride ions. The most common cystic

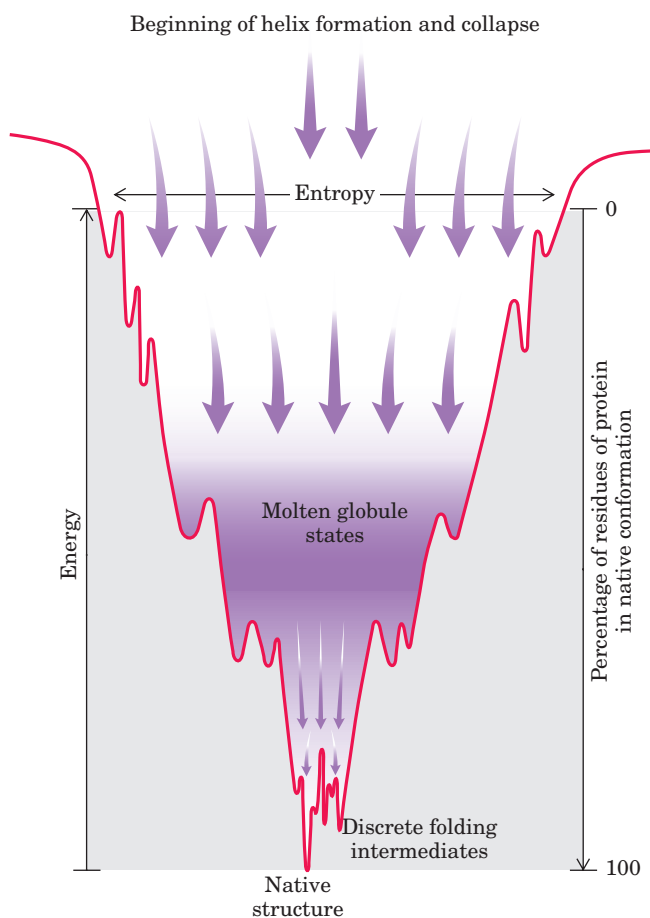


FIGURE 4-29 The thermodynamics of protein folding depicted as a free-energy funnel. At the top, the number of conformations, and hence the conformational entropy, is large. Only a small fraction of the intramolecular interactions that will exist in the native conformation are present. As folding progresses, the thermodynamic path down the funnel reduces the number of states present (decreases entropy), increases the amount of protein in the native conformation, and decreases the free energy. Depressions on the sides of the funnel represent semistable folding intermediates, which may, in some cases, slow the folding process.

fibrosis—causing mutation is the deletion of a Phe residue at position 508 in CFTR, which causes improper protein folding (see Box 11–3). Many of the disease-related mutations in collagen (p. 129) also cause defective folding. An improved understanding of protein

folding may lead to new therapies for these and many other diseases (Box 4–5). ■

Thermodynamic stability is not evenly distributed over the structure of a protein—the molecule has regions of high and low stability. For example, a protein



BOX 4–5 BIOCHEMISTRY IN MEDICINE

Death by Misfolding: The Prion Diseases

A misfolded protein appears to be the causative agent of a number of rare degenerative brain diseases in mammals. Perhaps the best known of these is mad cow disease (bovine spongiform encephalopathy, BSE), an outbreak of which made international headlines in the spring of 1996. Related diseases include kuru and Creutzfeldt-Jakob disease in humans, scrapie in sheep, and chronic wasting disease in deer and elk. These diseases are also referred to as spongiform encephalopathies, because the diseased brain frequently becomes riddled with holes (Fig. 1). Typical symptoms include dementia and loss of coordination. The diseases are fatal.

In the 1960s, investigators found that preparations of the disease-causing agents appeared to lack nucleic acids. At this time, Tikvah Alper suggested that the agent was a protein. Initially, the idea seemed heretical. All disease-causing agents known up to that time—viruses, bacteria, fungi, and so on—contained nucleic acids, and their virulence was related to genetic reproduction and propagation. However, four decades of investigations, pursued most notably by Stanley Prusiner, have provided evidence that spongiform encephalopathies are different.

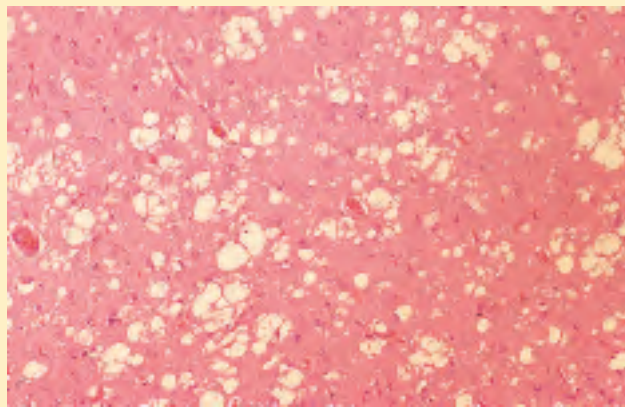


FIGURE 1 A stained section of the cerebral cortex from a patient with Creutzfeldt-Jakob disease shows spongiform (vacuolar) degeneration, the most characteristic neurohistological feature. The yellowish vacuoles are intracellular and occur mostly in pre- and post-synaptic processes of neurons. The vacuoles in this section vary in diameter from 20 to 100 μm .

The infectious agent has been traced to a single protein (M_r 28,000), which Prusiner dubbed **prion** (from *proteinaceous infectious only*) protein (PrP). Prion protein is a normal constituent of brain tissue in all mammals. Its role in the mammalian brain is not known in detail, but it appears to have a molecular signaling function. Strains of mice lacking the gene for PrP (and thus the protein itself) suffer no obvious ill effects. Illness occurs only when the normal cellular PrP, or PrP^C, occurs in an altered conformation called PrP^{Sc} (Sc denotes scrapie). The interaction of PrP^{Sc} with PrP^C converts the latter to PrP^{Sc}, initiating a domino effect in which more and more of the brain protein converts to the disease-causing form. The mechanism by which the presence of PrP^{Sc} leads to spongiform encephalopathy is not understood.

In inherited forms of prion diseases, a mutation in the gene encoding PrP produces a change in one amino acid residue that is believed to make the conversion of PrP^C to PrP^{Sc} more likely. A complete understanding of prion diseases awaits new information about how prion protein affects brain function. Structural information about PrP is beginning to provide insights into the molecular process that allows the prion proteins to interact so as to alter their conformation (Fig. 2).

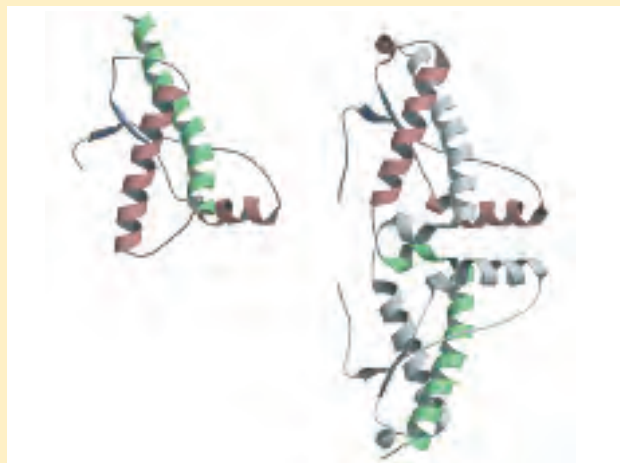


FIGURE 2 The structure of the globular domain of human PrP in monomeric (left) and dimeric (right) forms. The second subunit is gray to highlight the dramatic conformational change in the green α helix when the dimer is formed.

may have two stable domains joined by a segment with lower structural stability, or one small part of a domain may have a lower stability than the remainder. The regions of low stability allow a protein to alter its conformation between two or more states. As we shall see in the next two chapters, variations in the stability of regions within a given protein are often essential to protein function.

Some Proteins Undergo Assisted Folding

Not all proteins fold spontaneously as they are synthesized in the cell. Folding for many proteins is facilitated by the action of specialized proteins. **Molecular chaperones** are proteins that interact with partially folded or improperly folded polypeptides, facilitating correct folding pathways or providing microenvironments in which folding can occur. Two classes of molecular chaperones have been well studied. Both are found in organisms ranging from bacteria to humans. The first class, a family of proteins called **Hsp70**, generally have

a molecular weight near 70,000 and are more abundant in cells stressed by elevated temperatures (hence, *heat shock proteins* of M_r 70,000, or Hsp70). Hsp70 proteins bind to regions of unfolded polypeptides that are rich in hydrophobic residues, preventing inappropriate aggregation. These chaperones thus “protect” proteins that have been denatured by heat and peptides that are being synthesized (and are not yet folded). Hsp70 proteins also block the folding of certain proteins that must remain unfolded until they have been translocated across membranes (as described in Chapter 27). Some chaperones also facilitate the quaternary assembly of oligomeric proteins. The Hsp70 proteins bind to and release polypeptides in a cycle that also involves several other proteins (including a class called Hsp40) and ATP hydrolysis. Figure 4–30 illustrates chaperone-assisted folding as elucidated for the chaperones DnaK and DnaJ in *E. coli*, homologs of the eukaryotic Hsp70 and Hsp40. DnaK and DnaJ were first identified as proteins required for *in vitro* replication of certain viral DNA molecules (hence the “Dna” designation).

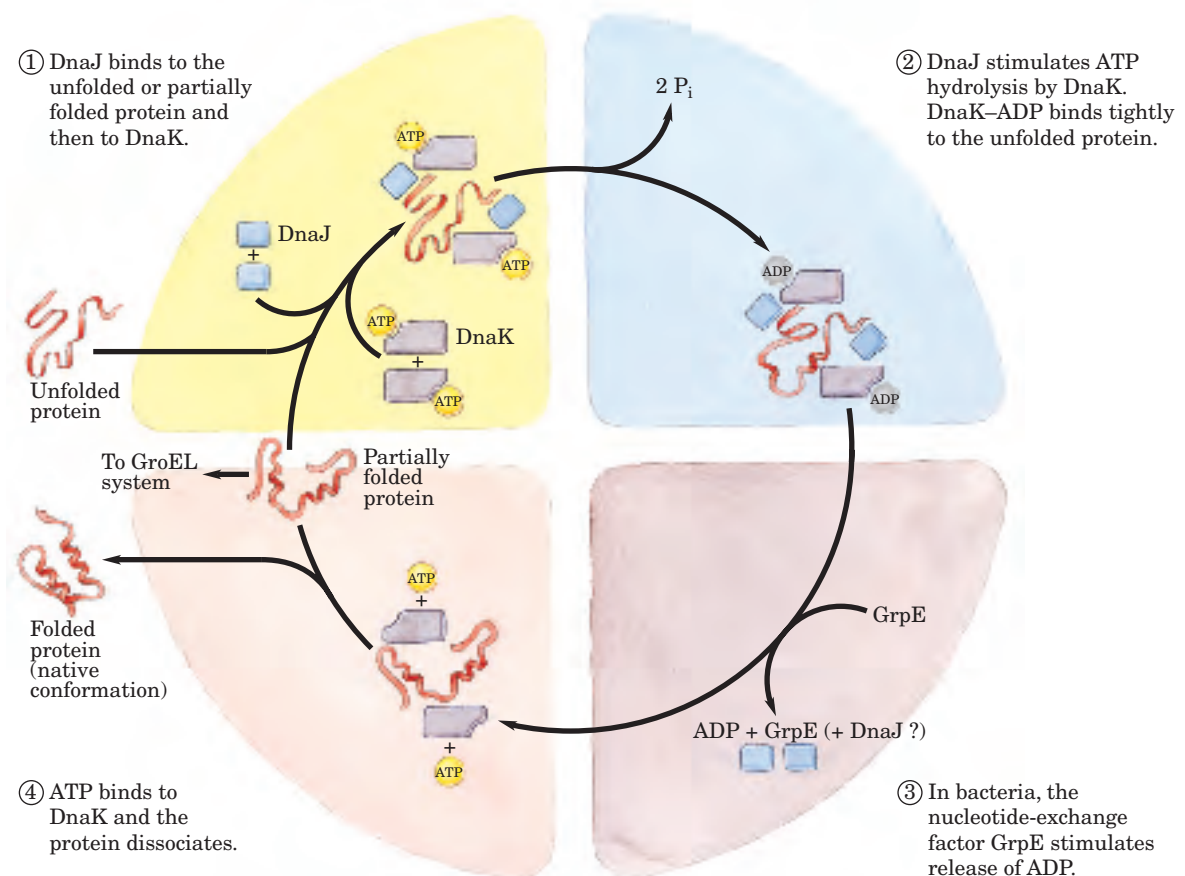


FIGURE 4–30 Chaperones in protein folding. The cyclic pathway by which chaperones bind and release polypeptides is illustrated for the *E. coli* chaperone proteins DnaK and DnaJ, homologs of the eukaryotic chaperones Hsp70 and Hsp40. The chaperones do not actively promote the folding of the substrate protein, but instead prevent aggregation of unfolded peptides. For a population of polypeptides, some

fraction of the polypeptides released at the end of the cycle are in the native conformation. The remainder are rebound by DnaK or are diverted to the chaperonin system (GroEL; see Fig. 4–31). In bacteria, a protein called GrpE interacts transiently with DnaK late in the cycle (step ③), promoting dissociation of ADP and possibly DnaJ. No eukaryotic analog of GrpE is known.

The second class of chaperones is called **chaperonins**. These are elaborate protein complexes required for the folding of a number of cellular proteins that do not fold spontaneously. In *E. coli* an estimated 10% to 15% of cellular proteins require the resident chaperonin system, called GroEL/GroES, for folding under normal conditions (up to 30% require this assistance when the cells are heat stressed). These proteins first became known when they were found to be necessary for the growth of certain bacterial viruses (hence the designation “Gro”). Unfolded proteins are bound within pockets in the GroEL complex, and the pockets are capped transiently by the GroES “lid” (Fig. 4–31). GroEL un-

dergoes substantial conformational changes, coupled to ATP hydrolysis and the binding and release of GroES, which promote folding of the bound polypeptide. Although the structure of the GroEL/GroES chaperonin is known, many details of its mechanism of action remain unresolved.

Finally, the folding pathways of a number of proteins require two enzymes that catalyze isomerization reactions. **Protein disulfide isomerase (PDI)** is a widely distributed enzyme that catalyzes the interchange or shuffling of disulfide bonds until the bonds of the native conformation are formed. Among its functions, PDI catalyzes the elimination of folding interme-

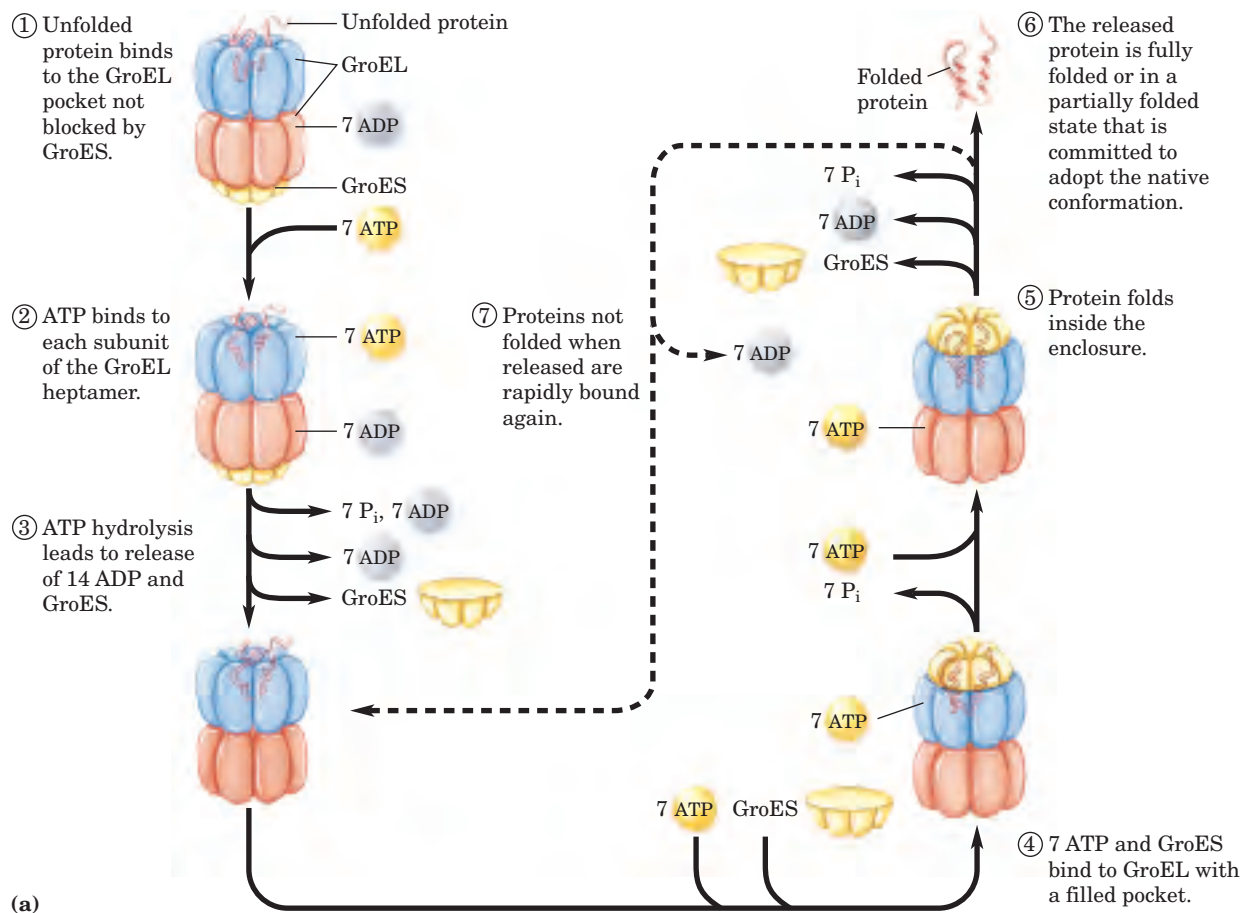
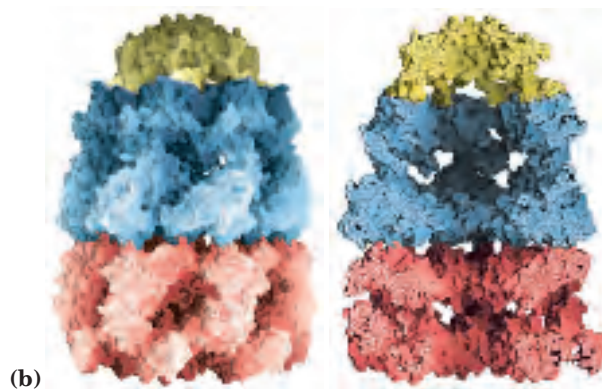


FIGURE 4–31 Chaperonins in protein folding. (a) A proposed pathway for the action of the *E. coli* chaperonins GroEL (a member of the Hsp60 protein family) and GroES. Each GroEL complex consists of two large pockets formed by two heptameric rings (each subunit M_r 57,000). GroES, also a heptamer (subunits M_r 10,000), blocks one of the GroEL pockets. (b) Surface and cut-away images of the GroEL/GroES complex (PDB ID 1AON). The cut-away (right) illustrates the large interior space within which other proteins are bound.



diates with inappropriate disulfide cross-links. **Peptide prolyl cis-trans isomerase (PPI)** catalyzes the interconversion of the cis and trans isomers of Pro peptide bonds (Fig. 4–8b), which can be a slow step in the folding of proteins that contain some Pro residue peptide bonds in the cis conformation.

Protein folding is likely to be a more complex process in the densely packed cellular environment than in the test tube. More classes of proteins that facilitate protein folding may be discovered as the biochemical dissection of the folding process continues.

SUMMARY 4.4 Protein Denaturation and Folding

- The three-dimensional structure and the function of proteins can be destroyed by denaturation, demonstrating a relationship

between structure and function. Some denatured proteins can renature spontaneously to form biologically active protein, showing that protein tertiary structure is determined by amino acid sequence.

- Protein folding in cells probably involves multiple pathways. Initially, regions of secondary structure may form, followed by folding into supersecondary structures. Large ensembles of folding intermediates are rapidly brought to a single native conformation.
- For many proteins, folding is facilitated by Hsp70 chaperones and by chaperonins. Disulfide bond formation and the cis-trans isomerization of Pro peptide bonds are catalyzed by specific enzymes.

Key Terms

Terms in bold are defined in the glossary.

conformation 116	β conformation 123	collagen 127	protomer 144
native conformation 117	β sheet 123	silk fibroin 129	symmetry 144
solvation layer 117	β turn 123	supersecondary structures 139	denaturation 147
peptide group 118	tertiary structure 125	motif 139	molten globule 149
Ramachandran plot 118	quaternary structure 125	fold 139	prion 150
secondary structure 120	fibrous proteins 125	domain 140	molecular chaperone 151
α helix 120	globular proteins 125	protein family 141	Hsp70 151
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		oligomer 144	

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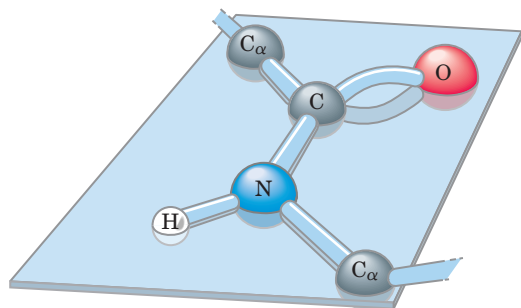
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Problems

1. Properties of the Peptide Bond In x-ray studies of crystalline peptides, Linus Pauling and Robert Corey found that the C—N bond in the peptide link is intermediate in length (1.32 Å) between a typical C—N single bond (1.49 Å) and a C=N double bond (1.27 Å). They also found that the peptide bond is planar (all four atoms attached to the C—N group are located in the same plane) and that the two α -carbon atoms attached to the C—N are always trans to each other (on opposite sides of the peptide bond):



(a) What does the length of the C—N bond in the peptide linkage indicate about its strength and its bond order (i.e., whether it is single, double, or triple)?

(b) What do the observations of Pauling and Corey tell us about the ease of rotation about the C—N peptide bond?

2. Structural and Functional Relationships in Fibrous Proteins William Astbury discovered that the x-ray pattern of wool shows a repeating structural unit spaced about 5.2 Å along the length of the wool fiber. When he steamed and

stretched the wool, the x-ray pattern showed a new repeating structural unit at a spacing of 7.0 Å. Steaming and stretching the wool and then letting it shrink gave an x-ray pattern consistent with the original spacing of about 5.2 Å. Although these observations provided important clues to the molecular structure of wool, Astbury was unable to interpret them at the time.

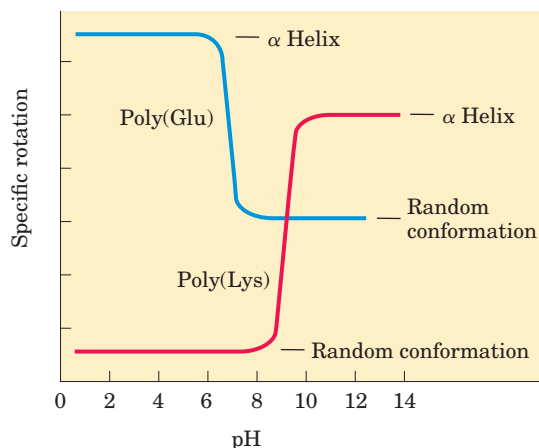
(a) Given our current understanding of the structure of wool, interpret Astbury's observations.

(b) When wool sweaters or socks are washed in hot water or heated in a dryer, they shrink. Silk, on the other hand, does not shrink under the same conditions. Explain.

3. Rate of Synthesis of Hair α -Keratin Hair grows at a rate of 15 to 20 cm/yr. All this growth is concentrated at the base of the hair fiber, where α -keratin filaments are synthesized inside living epidermal cells and assembled into ropelike structures (see Fig. 4–11). The fundamental structural element of α -keratin is the α helix, which has 3.6 amino acid residues per turn and a rise of 5.4 Å per turn (see Fig. 4–4b). Assuming that the biosynthesis of α -helical keratin chains is the rate-limiting factor in the growth of hair, calculate the rate at which peptide bonds of α -keratin chains must be synthesized (peptide bonds per second) to account for the observed yearly growth of hair.

4. Effect of pH on the Conformation of α -Helical Secondary Structures The unfolding of the α helix of a polypeptide to a randomly coiled conformation is accompanied by a large decrease in a property called its specific rotation, a measure of a solution's capacity to rotate plane-polarized light. Polyglutamate, a polypeptide made up of only L-Glu residues,

has the α -helical conformation at pH 3. When the pH is raised to 7, there is a large decrease in the specific rotation of the solution. Similarly, polylysine (L-Lys residues) is an α helix at pH 10, but when the pH is lowered to 7 the specific rotation also decreases, as shown by the following graph.



What is the explanation for the effect of the pH changes on the conformations of poly(Glu) and poly(Lys)? Why does the transition occur over such a narrow range of pH?

5. Disulfide Bonds Determine the Properties of Many Proteins A number of natural proteins are very rich in disulfide bonds, and their mechanical properties (tensile strength, viscosity, hardness, etc.) are correlated with the degree of disulfide bonding. For example, glutenin, a wheat protein rich in disulfide bonds, is responsible for the cohesive and elastic character of dough made from wheat flour. Similarly, the hard, tough nature of tortoise shell is due to the extensive disulfide bonding in its α -keratin.

(a) What is the molecular basis for the correlation between disulfide-bond content and mechanical properties of the protein?

(b) Most globular proteins are denatured and lose their activity when briefly heated to 65 °C. However, globular proteins that contain multiple disulfide bonds often must be heated longer at higher temperatures to denature them. One such protein is bovine pancreatic trypsin inhibitor (BPTI), which has 58 amino acid residues in a single chain and contains three disulfide bonds. On cooling a solution of denatured BPTI, the activity of the protein is restored. What is the molecular basis for this property?

6. Amino Acid Sequence and Protein Structure Our growing understanding of how proteins fold allows researchers to make predictions about protein structure based on primary amino acid sequence data.

1	2	3	4	5	6	7	8	9	10
Ile	Ala	His	Thr	Tyr	Gly	Pro	Phe	Glu	Ala
11	12	13	14	15	16	17	18	19	20
Ala	Met	Cys	Lys	Trp	Glu	Ala	Gln	Pro	Asp
21	22	23	24	25	26	27	28		
Gly	Met	Glu	Cys	Ala	Phe	His	Arg		

(a) In the amino acid sequence above, where would you predict that bends or β turns would occur?

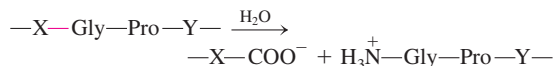
(b) Where might intrachain disulfide cross-linkages be formed?

(c) Assuming that this sequence is part of a larger globular protein, indicate the probable location (the external surface or interior of the protein) of the following amino acid residues: Asp, Ile, Thr, Ala, Gln, Lys. Explain your reasoning. (Hint: See the hydropathy index in Table 3-1.)

7. Bacteriorhodopsin in Purple Membrane Proteins

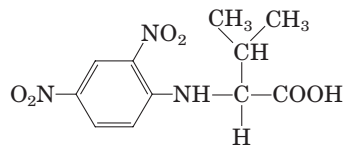
Under the proper environmental conditions, the salt-loving bacterium *Halobacterium halobium* synthesizes a membrane protein (M_r 26,000) known as bacteriorhodopsin, which is purple because it contains retinal (see Fig. 10-21). Molecules of this protein aggregate into "purple patches" in the cell membrane. Bacteriorhodopsin acts as a light-activated proton pump that provides energy for cell functions. X-ray analysis of this protein reveals that it consists of seven parallel α -helical segments, each of which traverses the bacterial cell membrane (thickness 45 Å). Calculate the minimum number of amino acid residues necessary for one segment of α helix to traverse the membrane completely. Estimate the fraction of the bacteriorhodopsin protein that is involved in membrane-spanning helices. (Use an average amino acid residue weight of 110.)

8. Pathogenic Action of Bacteria That Cause Gas Gangrene The highly pathogenic anaerobic bacterium *Clostridium perfringens* is responsible for gas gangrene, a condition in which animal tissue structure is destroyed. This bacterium secretes an enzyme that efficiently catalyzes the hydrolysis of the peptide bond indicated in red:



where X and Y are any of the 20 common amino acids. How does the secretion of this enzyme contribute to the invasiveness of this bacterium in human tissues? Why does this enzyme not affect the bacterium itself?

9. Number of Polypeptide Chains in a Multisubunit Protein A sample (660 mg) of an oligomeric protein of M_r 132,000 was treated with an excess of 1-fluoro-2,4-dinitrobenzene (Sanger's reagent) under slightly alkaline conditions until the chemical reaction was complete. The peptide bonds of the protein were then completely hydrolyzed by heating it with concentrated HCl. The hydrolysate was found to contain 5.5 mg of the following compound:



2,4-Dinitrophenyl derivatives of the α -amino groups of other amino acids could not be found.

(a) Explain how this information can be used to determine the number of polypeptide chains in an oligomeric protein.

(b) Calculate the number of polypeptide chains in this protein.

(c) What other protein analysis technique could you employ to determine whether the polypeptide chains in this protein are similar or different?

Biochemistry on the Internet

10. Protein Modeling on the Internet A group of patients suffering from Crohn's disease (an inflammatory bowel disease) underwent biopsies of their intestinal mucosa in an attempt to identify the causative agent. A protein was identified that was expressed at higher levels in patients with Crohn's disease than in patients with an unrelated inflammatory bowel disease or in unaffected controls. The protein was isolated and the following *partial* amino acid sequence was obtained (reads left to right):

EAELCPDRCI	HSFQNLGIQC	VKKRDLEQAI
SQRIQTNNNP	FQVPIEEQRG	DYDLNAVRLC
FQVTVRDPSG	RPLRLPPVLP	HPIFDNRAPN
TAEIKICRVN	RNSGSLGSD	EIFLLCDKVQ
KEDIEVYFTG	PGWEARGSFS	QADVHRQVAI
VFRTPPYADP	SLQAPVRVSM	QLRRPSDREL
SEPMEFQYLP	DTDDRHRIEE	KRKRTYETFK
SIMKKSPFSG	PTDPRPPRR	IAPSRSSAS
VPKPAPQPYP		

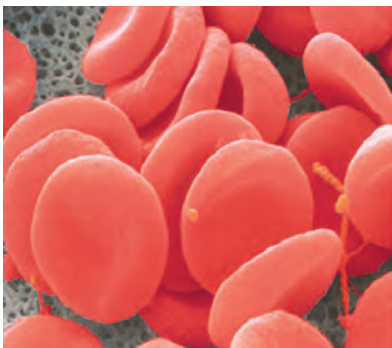
(a) You can identify this protein using a protein database on the Internet. Some good places to start include Protein Information Resource (PIR; pir.georgetown.edu/pirwww), Structural Classification of Proteins (SCOP; <http://scop.berkeley.edu>), and Prosite (<http://us.expasy.org/prosite>).

At your selected database site, follow links to locate the sequence comparison engine. Enter about 30 residues from the sequence of the protein in the appropriate search field and submit it for analysis. What does this analysis tell you about the identity of the protein?

(b) Try using different portions of the protein amino acid sequence. Do you always get the same result?

(c) A variety of websites provide information about the three-dimensional structure of proteins. Find information about the protein's secondary, tertiary, and quaternary structure using database sites such as the Protein Data Bank (PDB; www.rcsb.org/pdb) or SCOP.

(d) In the course of your Web searches try to find information about the cellular function of the protein.



PROTEIN FUNCTION

- 5.1 Reversible Binding of a Protein to a Ligand: Oxygen-Binding Proteins 158
- 5.2 Complementary Interactions between Proteins and Ligands: The Immune System and Immunoglobulins 174
- 5.3 Protein Interactions Modulated by Chemical Energy: Actin, Myosin, and Molecular Motors 182

I have occasionally seen in almost dried blood, placed between glass plates in a desiccator, rectangular crystalline structures, which under the microscope had sharp edges and were bright red.

—Friedrich Ludwig Hünefeld, *Der Chemismus in der thierischen Organisation*, 1840
(one of the first observations of hemoglobin)

Since the proteins participate in one way or another in all chemical processes in the living organism, one may expect highly significant information for biological chemistry from the elucidation of their structure and their transformations.

—Emil Fischer, *article in Berichte der deutschen chemischen Gesellschaft zu Berlin*, 1906

Knowing the three-dimensional structure of a protein is an important part of understanding how the protein functions. However, the structure shown in two dimensions on a page is deceptively static. Proteins are dynamic molecules whose functions almost invariably depend on interactions with other molecules, and these interactions are affected in physiologically important ways by sometimes subtle, sometimes striking changes in protein conformation. In this chapter, we explore how

proteins interact with other molecules and how their interactions are related to dynamic protein structure. The importance of molecular interactions to a protein's function can hardly be overemphasized. In Chapter 4, we saw that the function of fibrous proteins as structural elements of cells and tissues depends on stable, long-term quaternary interactions between identical polypeptide chains. As we shall see in this chapter, the functions of many other proteins involve interactions with a variety of different molecules. Most of these interactions are fleeting, though they may be the basis of complex physiological processes such as oxygen transport, immune function, and muscle contraction—the topics we examine in detail in this chapter. The proteins that carry out these processes illustrate the following key principles of protein function, some of which will be familiar from the previous chapter:

The functions of many proteins involve the reversible binding of other molecules. A molecule bound reversibly by a protein is called a **ligand**. A ligand may be any kind of molecule, including another protein. The transient nature of protein-ligand interactions is critical to life, allowing an organism to respond rapidly and reversibly to changing environmental and metabolic circumstances.

A ligand binds at a site on the protein called the **binding site**, which is complementary to the ligand in size, shape, charge, and hydrophobic or hydrophilic character. Furthermore, the interaction is specific: the protein can discriminate among the thousands of different molecules in its environment and selectively bind only one or a few. A given protein may have separate binding sites for several different ligands. These specific molecular interactions are crucial in maintaining the high degree of order in a living system. (This discussion

excludes the binding of water, which may interact weakly and nonspecifically with many parts of a protein. In Chapter 6, we consider water as a specific ligand for many enzymes.)

Proteins are flexible. Changes in conformation may be subtle, reflecting molecular vibrations and small movements of amino acid residues throughout the protein. A protein flexing in this way is sometimes said to “breathe.” Changes in conformation may also be quite dramatic, with major segments of the protein structure moving as much as several nanometers. Specific conformational changes are frequently essential to a protein’s function.

The binding of a protein and ligand is often coupled to a conformational change in the protein that makes the binding site more complementary to the ligand, permitting tighter binding. The structural adaptation that occurs between protein and ligand is called **induced fit**.

In a multisubunit protein, a conformational change in one subunit often affects the conformation of other subunits.

Interactions between ligands and proteins may be regulated, usually through specific interactions with one or more additional ligands. These other ligands may cause conformational changes in the protein that affect the binding of the first ligand.

Enzymes represent a special case of protein function. Enzymes bind and chemically transform other molecules—they catalyze reactions. The molecules acted upon by enzymes are called reaction **substrates** rather than ligands, and the ligand-binding site is called the **catalytic site** or **active site**. In this chapter we emphasize the noncatalytic functions of proteins. In Chapter 6 we consider catalysis by enzymes, a central topic in biochemistry. You will see that the themes of this chapter—binding, specificity, and conformational change—are continued in the next chapter, with the added element of proteins acting as reactants in chemical transformations.

5.1 Reversible Binding of a Protein to a Ligand: Oxygen-Binding Proteins

Myoglobin and hemoglobin may be the most-studied and best-understood proteins. They were the first proteins for which three-dimensional structures were determined, and our current understanding of myoglobin and hemoglobin is garnered from the work of thousands of biochemists over several decades. Most important, these

molecules illustrate almost every aspect of that most central of biochemical processes: the reversible binding of a ligand to a protein. This classic model of protein function tells us a great deal about how proteins work.

 **Oxygen-Binding Proteins—Myoglobin: Oxygen Storage**

Oxygen Can Be Bound to a Heme Prosthetic Group

Oxygen is poorly soluble in aqueous solutions (see Table 2–3) and cannot be carried to tissues in sufficient quantity if it is simply dissolved in blood serum. Diffusion of oxygen through tissues is also ineffective over distances greater than a few millimeters. The evolution of larger, multicellular animals depended on the evolution of proteins that could transport and store oxygen. However, none of the amino acid side chains in proteins is suited for the reversible binding of oxygen molecules. This role is filled by certain transition metals, among them iron and copper, that have a strong tendency to bind oxygen. Multicellular organisms exploit the properties of metals, most commonly iron, for oxygen transport. However, free iron promotes the formation of highly reactive oxygen species such as hydroxyl radicals that can damage DNA and other macromolecules. Iron used in cells is therefore bound in forms that sequester it and/or make it less reactive. In multicellular organisms—especially those in which iron, in its oxygen-carrying capacity, must be transported over large distances—iron is often incorporated into a protein-bound prosthetic group called **heme**. (Recall from Chapter 3 that a prosthetic group is a compound permanently associated with a protein that contributes to the protein’s function.)

Heme (or haem) consists of a complex organic ring structure, **protoporphyrin**, to which is bound a single iron atom in its ferrous (Fe^{2+}) state (Fig. 5–1). The iron atom has six coordination bonds, four to nitrogen atoms that are part of the flat **porphyrin ring** system and two perpendicular to the porphyrin. The coordinated nitrogen atoms (which have an electron-donating character) help prevent conversion of the heme iron to the ferric (Fe^{3+}) state. Iron in the Fe^{2+} state binds oxygen reversibly; in the Fe^{3+} state it does not bind oxygen. Heme is found in a number of oxygen-transporting proteins, as well as in some proteins, such as the cytochromes, that participate in oxidation-reduction (electron-transfer) reactions (Chapter 19).

In free heme molecules (heme not bound to protein), reaction of oxygen at one of the two “open” coordination bonds of iron (perpendicular to the plane of the porphyrin molecule, above and below) can result in irreversible conversion of Fe^{2+} to Fe^{3+} . In heme-containing proteins, this reaction is prevented by sequestering of the heme deep within the protein structure where access to the two open coordination bonds is restricted. One of these two coordination bonds is occupied by a side-chain nitrogen of a His residue. The

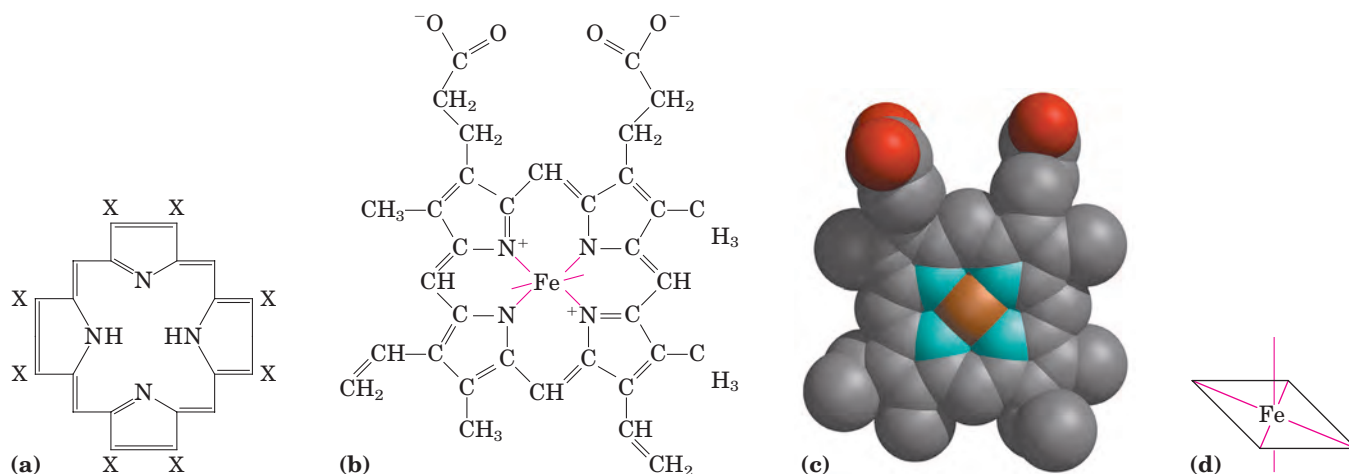


FIGURE 5-1 Heme. The heme group is present in myoglobin, hemoglobin, and many other proteins, designated heme proteins. Heme consists of a complex organic ring structure, protoporphyrin IX, to which is bound an iron atom in its ferrous (Fe^{2+}) state. **(a)** Porphyrins, of which protoporphyrin IX is only one example, consist of four pyr-

role rings linked by methene bridges, with substitutions at one or more of the positions denoted X. **(b, c)** Two representations of heme. (Derived from PDB ID 1CCR.) The iron atom of heme has six coordination bonds: four in the plane of, and bonded to, the flat porphyrin ring system, and **(d)** two perpendicular to it.

other is the binding site for molecular oxygen (O_2) (Fig. 5-2). When oxygen binds, the electronic properties of heme iron change; this accounts for the change in color from the dark purple of oxygen-depleted venous blood to the bright red of oxygen-rich arterial blood. Some small molecules, such as carbon monoxide (CO) and nitric oxide (NO), coordinate to heme iron with greater affinity than does O_2 . When a molecule of CO is bound to heme, O_2 is excluded, which is why CO is highly toxic to aerobic organisms (a topic explored later, in Box 5-1). By surrounding and sequestering heme, oxygen-binding proteins regulate the access of CO and other small molecules to the heme iron.

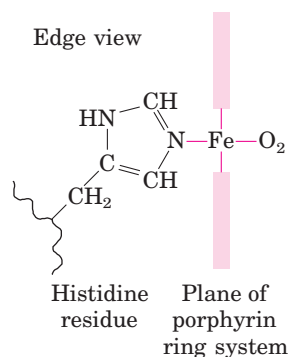


FIGURE 5-2 The heme group viewed from the side. This view shows the two coordination bonds to Fe^{2+} perpendicular to the porphyrin ring system. One of these two bonds is occupied by a His residue, sometimes called the proximal His. The other bond is the binding site for oxygen. The remaining four coordination bonds are in the plane of, and bonded to, the flat porphyrin ring system.

Myoglobin Has a Single Binding Site for Oxygen

Myoglobin (M_r 16,700; abbreviated Mb) is a relatively simple oxygen-binding protein found in almost all mammals, primarily in muscle tissue. As a transport protein, it facilitates oxygen diffusion in muscle. Myoglobin is particularly abundant in the muscles of diving mammals such as seals and whales, where it also has an oxygen-storage function for prolonged excursions undersea. Proteins very similar to myoglobin are widely distributed, occurring even in some single-celled organisms.

Myoglobin is a single polypeptide of 153 amino acid residues with one molecule of heme. It is typical of the family of proteins called **globins**, all of which have similar primary and tertiary structures. The polypeptide is made up of eight α -helical segments connected by bends (Fig. 5-3). About 78% of the amino acid residues in the protein are found in these α helices.

Any detailed discussion of protein function inevitably involves protein structure. To facilitate our treatment of myoglobin, we first introduce some structural conventions peculiar to globins. As seen in Figure 5-3, the helical segments are named A through H. An individual amino acid residue is designated either by its position in the amino acid sequence or by its location within the sequence of a particular α -helical segment. For example, the His residue coordinated to the heme in myoglobin, His⁹³ (the 93rd amino acid residue from the amino-terminal end of the myoglobin polypeptide sequence), is also called His F8 (the 8th residue in α helix F). The bends in the structure are designated AB, CD, EF, FG, and so forth, reflecting the α -helical segments they connect.

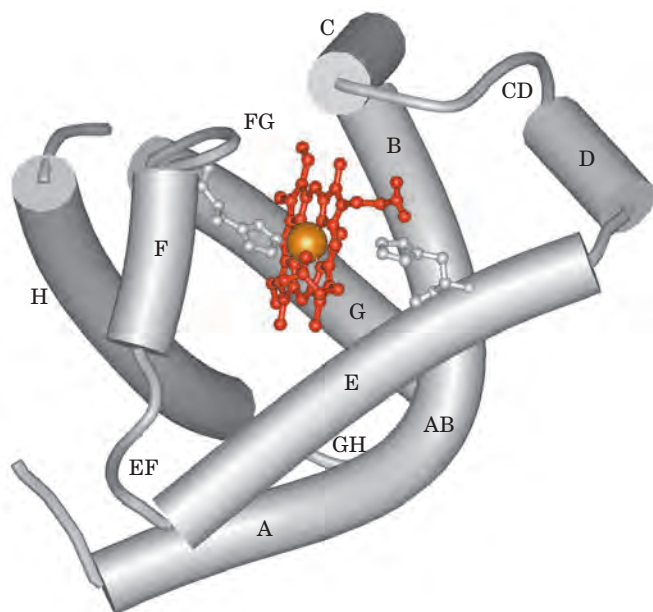


FIGURE 5-3 The structure of myoglobin. (PDB ID 1MBO) The eight α -helical segments (shown here as cylinders) are labeled A through H. Nonhelical residues in the bends that connect them are labeled AB, CD, EF, and so forth, indicating the segments they interconnect. A few bends, including BC and DE, are abrupt and do not contain any residues; these are not normally labeled. (The short segment visible between D and E is an artifact of the computer representation.) The heme is bound in a pocket made up largely of the E and F helices, although amino acid residues from other segments of the protein also participate.

Protein-Ligand Interactions Can Be Described Quantitatively

The function of myoglobin depends on the protein's ability not only to bind oxygen but also to release it when and where it is needed. Function in biochemistry often revolves around a reversible protein-ligand interaction of this type. A quantitative description of this interaction is therefore a central part of many biochemical investigations.

In general, the reversible binding of a protein (P) to a ligand (L) can be described by a simple **equilibrium expression**:



The reaction is characterized by an equilibrium constant, K_a , such that

$$K_a = \frac{[PL]}{[P][L]} \quad (5-2)$$

The term K_a is an **association constant** (not to be confused with the K_a that denotes an acid dissociation constant; p. 63). The association constant provides a measure of the affinity of the ligand L for the protein. K_a has units of M^{-1} ; a higher value of K_a corresponds to

a higher affinity of the ligand for the protein. A rearrangement of Equation 5-2 shows that the ratio of bound to free protein is directly proportional to the concentration of free ligand:

$$K_a[L] = \frac{[PL]}{[P]} \quad (5-3)$$

When the concentration of the ligand is much greater than the concentration of ligand-binding sites, the binding of the ligand by the protein does not appreciably change the concentration of free (unbound) ligand—that is, [L] remains constant. This condition is broadly applicable to most ligands that bind to proteins in cells and simplifies our description of the binding equilibrium.

We can now consider the binding equilibrium from the standpoint of the fraction, θ (theta), of ligand-binding sites on the protein that are occupied by ligand:

$$\theta = \frac{\text{binding sites occupied}}{\text{total binding sites}} = \frac{[PL]}{[PL] + [P]} \quad (5-4)$$

Substituting $K_a[L][P]$ for [PL] (see Eqn 5-3) and rearranging terms gives

$$\theta = \frac{K_a[L][P]}{K_a[L][P] + [P]} = \frac{K_a[L]}{K_a[L] + 1} = \frac{[L]}{[L] + \frac{1}{K_a}} \quad (5-5)$$

The value of K_a can be determined from a plot of θ versus the concentration of free ligand, [L] (Fig. 5-4a). Any equation of the form $x = y/(y + z)$ describes a hyperbola, and θ is thus found to be a hyperbolic function of [L]. The fraction of ligand-binding sites occupied approaches saturation asymptotically as [L] increases. The [L] at which half of the available ligand-binding sites are occupied (at $\theta = 0.5$) corresponds to $1/K_a$.

It is more common (and intuitively simpler), however, to consider the **dissociation constant, K_d** , which is the reciprocal of K_a ($K_d = 1/K_a$) and is given in units of molar concentration (M). K_d is the equilibrium constant for the release of ligand. The relevant expressions change to

$$K_d = \frac{[P][L]}{[PL]} \quad (5-6)$$

$$[PL] = \frac{[P][L]}{K_d} \quad (5-7)$$

$$\theta = \frac{[L]}{[L] + K_d} \quad (5-8)$$

When [L] is equal to K_d , half of the ligand-binding sites are occupied. As [L] falls below K_d , progressively less of the protein has ligand bound to it. In order for 90% of the available ligand-binding sites to be occupied, [L] must be nine times greater than K_d .

In practice, K_d is used much more often than K_a to express the affinity of a protein for a ligand. Note that

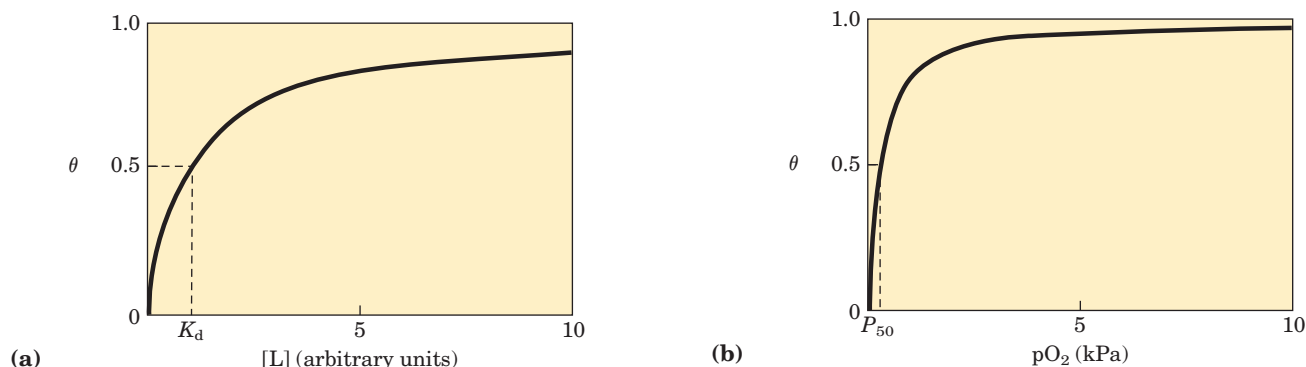


FIGURE 5-4 Graphical representations of ligand binding. The fraction of ligand-binding sites occupied, θ , is plotted against the concentration of free ligand. Both curves are rectangular hyperbolas. (a) A hypothetical binding curve for a ligand L. The $[L]$ at which half of the available ligand-binding sites are occupied is equivalent to $1/K_d$,

a lower value of K_d corresponds to a higher affinity of ligand for the protein. The mathematics can be reduced to simple statements: K_d is equivalent to the molar concentration of ligand at which half of the available ligand-binding sites are occupied. At this point, the protein is said to have reached half-saturation with respect to ligand binding. The more tightly a protein binds a ligand, the lower the concentration of ligand required for half the binding sites to be occupied, and thus the lower the value of K_d . Some representative dissociation constants are given in Table 5-1.

The binding of oxygen to myoglobin follows the patterns discussed above. However, because oxygen is a gas, we must make some minor adjustments to the equations so that laboratory experiments can be carried out more conveniently. We first substitute the concentration of dissolved oxygen for $[L]$ in Equation 5-8 to give

$$\theta = \frac{[O_2]}{[O_2] + K_d} \quad (5-9)$$

or K_d . The curve has a horizontal asymptote at $\theta = 1$ and a vertical asymptote (not shown) at $[L] = -1/K_d$. (b) A curve describing the binding of oxygen to myoglobin. The partial pressure of O_2 in the air above the solution is expressed in kilopascals (kPa). Oxygen binds tightly to myoglobin, with a P_{50} of only 0.26 kPa.

As for any ligand, K_d is equal to the $[O_2]$ at which half of the available ligand-binding sites are occupied, or $[O_2]_{0.5}$. Equation 5-9 thus becomes

$$\theta = \frac{[O_2]}{[O_2] + [O_2]_{0.5}} \quad (5-10)$$

In experiments using oxygen as a ligand, it is the partial pressure of oxygen in the gas phase above the solution, pO_2 , that is varied, because this is easier to measure than the concentration of oxygen dissolved in the solution. The concentration of a volatile substance in solution is always proportional to the local partial pressure of the gas. So, if we define the partial pressure of oxygen at $[O_2]_{0.5}$ as P_{50} , substitution in Equation 5-10 gives

$$\theta = \frac{pO_2}{pO_2 + P_{50}} \quad (5-11)$$

A binding curve for myoglobin that relates θ to pO_2 is shown in Figure 5-4b.

TABLE 5-1 Some Protein Dissociation Constants

Protein	Ligand	K_d (M)*
Avidin (egg white) [†]	Biotin	1×10^{-15}
Insulin receptor (human)	Insulin	1×10^{-10}
Anti-HIV immunoglobulin (human) [‡]	gp41 (HIV-1 surface protein)	4×10^{-10}
Nickel-binding protein (<i>E. coli</i>)	Ni^{2+}	1×10^{-7}
Calmodulin (rat) [§]	Ca^{2+}	3×10^{-6}
		2×10^{-5}

*A reported dissociation constant is valid only for the particular solution conditions under which it was measured. K_d values for a protein-ligand interaction can be altered, sometimes by several orders of magnitude, by changes in the solution's salt concentration, pH, or other variables.

[†]Interaction of avidin with biotin, an enzyme cofactor, is among the strongest noncovalent biochemical interactions known.

[‡]This immunoglobulin was isolated as part of an effort to develop a vaccine against HIV. Immunoglobulins (described later in the chapter) are highly variable, and the K_d reported here should not be considered characteristic of all immunoglobulins.

[§]Calmodulin has four binding sites for calcium. The values shown reflect the highest- and lowest-affinity binding sites observed in one set of measurements.

Protein Structure Affects How Ligands Bind

The binding of a ligand to a protein is rarely as simple as the above equations would suggest. The interaction is greatly affected by protein structure and is often accompanied by conformational changes. For example, the specificity with which heme binds its various ligands is altered when the heme is a component of myoglobin. Carbon monoxide binds to free heme molecules more than 20,000 times better than does O_2 (that is, the K_d or P_{50} for CO binding to free heme is more than 20,000 times lower than that for O_2), but it binds only about 200 times better when the heme is bound in myoglobin. The difference may be partly explained by steric hindrance. When O_2 binds to free heme, the axis of the oxygen molecule is positioned at an angle to the Fe—O bond (Fig. 5–5a). In contrast, when CO binds to free heme, the Fe, C, and O atoms lie in a straight line (Fig. 5–5b). In both cases, the binding reflects the geometry of hybrid orbitals in each ligand. In myoglobin, His⁶⁴ (His E7), on the O_2 -binding side of the heme, is too far away to coordinate with the heme iron, but it does interact with a ligand bound to heme. This residue, called the *distal His*, does not affect the binding of O_2 (Fig. 5–5c) but may preclude the linear binding of CO, providing one explanation for the diminished binding of CO to heme in myoglobin (and hemoglobin). A reduction in CO binding is physiologically important, because CO is a low-level byproduct of cellular metabolism. Other factors, not yet well-defined, also seem to modulate the interaction of heme with CO in these proteins.

The binding of O_2 to the heme in myoglobin also depends on molecular motions, or “breathing,” in the protein structure. The heme molecule is deeply buried in the folded polypeptide, with no direct path for oxygen to move from the surrounding solution to the ligand-binding site. If the protein were rigid, O_2 could not enter or leave the heme pocket at a measurable rate. However, rapid molecular flexing of the amino acid side chains produces transient cavities in the protein structure, and O_2 evidently makes its way in and out by moving through these cavities. Computer simulations of rapid structural fluctuations in myoglobin suggest that there are many such pathways. One major route is provided by rotation of the side chain of the distal His (His⁶⁴), which occurs on a nanosecond (10^{-9} s) time scale. Even subtle conformational changes can be critical for protein activity.

Oxygen Is Transported in Blood by Hemoglobin

Oxygen-Binding Proteins—Hemoglobin: Oxygen Transport

Nearly all the oxygen carried by whole blood in animals is bound and transported by hemoglobin in erythrocytes (red blood cells). Normal human erythrocytes are small (6 to 9 μm in diameter), biconcave disks. They are formed from precursor stem cells called **hemocytoblasts**. In

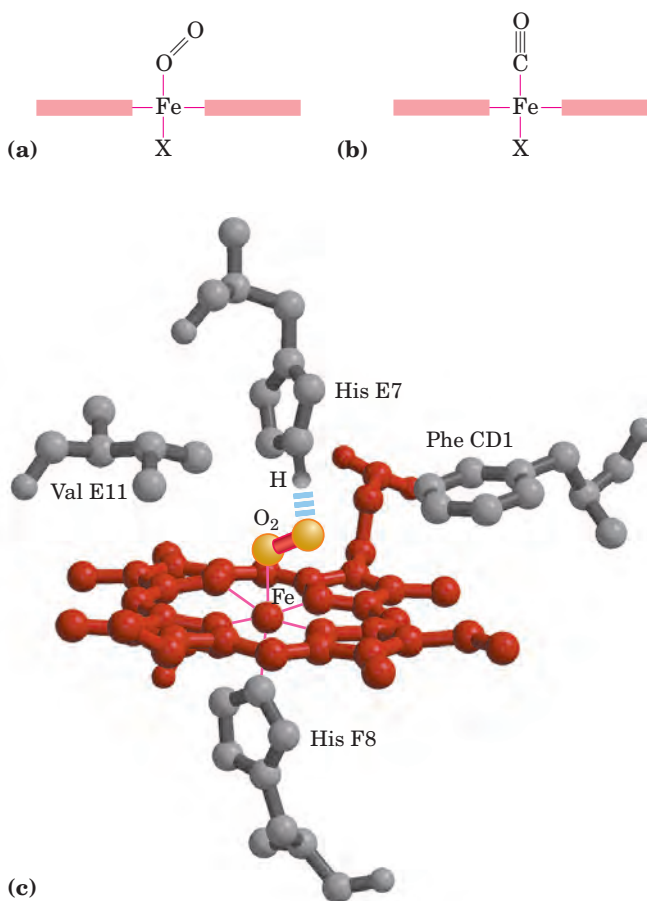


FIGURE 5-5 Steric effects on the binding of ligands to the heme of myoglobin. (a) Oxygen binds to heme with the O_2 axis at an angle, a binding conformation readily accommodated by myoglobin. (b) Carbon monoxide binds to free heme with the CO axis perpendicular to the plane of the porphyrin ring. When binding to the heme in myoglobin, CO is forced to adopt a slight angle because the perpendicular arrangement is sterically blocked by His E7, the distal His. This effect weakens the binding of CO to myoglobin. (c) Another view (derived from PDB ID 1MBO), showing the arrangement of key amino acid residues around the heme of myoglobin. The bound O_2 is hydrogen-bonded to the distal His, His E7 (His⁶⁴), further facilitating the binding of O_2 .

the maturation process, the stem cell produces daughter cells that form large amounts of hemoglobin and then lose their intracellular organelles—nucleus, mitochondria, and endoplasmic reticulum. Erythrocytes are thus incomplete, vestigial cells, unable to reproduce and, in humans, destined to survive for only about 120 days. Their main function is to carry hemoglobin, which is dissolved in the cytosol at a very high concentration (~34% by weight).

In arterial blood passing from the lungs through the heart to the peripheral tissues, hemoglobin is about 96% saturated with oxygen. In the venous blood returning to the heart, hemoglobin is only about 64% saturated. Thus, each 100 mL of blood passing through a tissue releases

about one-third of the oxygen it carries, or 6.5 mL of O₂ gas at atmospheric pressure and body temperature.

Myoglobin, with its hyperbolic binding curve for oxygen (Fig. 5-4b), is relatively insensitive to small changes in the concentration of dissolved oxygen and so functions well as an oxygen-storage protein. Hemoglobin, with its multiple subunits and O₂-binding sites, is better suited to oxygen transport. As we shall see, interactions between the subunits of a multimeric protein can permit a highly sensitive response to small changes in ligand concentration. Interactions among the subunits in hemoglobin cause conformational changes that alter the affinity of the protein for oxygen. The modulation of oxygen binding allows the O₂-transport protein to respond to changes in oxygen demand by tissues.

Hemoglobin Subunits Are Structurally Similar to Myoglobin

Hemoglobin (*M_r* 64,500; abbreviated Hb) is roughly spherical, with a diameter of nearly 5.5 nm. It is a tetrameric protein containing four heme prosthetic groups, one associated with each polypeptide chain. Adult hemoglobin contains two types of globin, two α chains (141 residues each) and two β chains (146 residues each). Although fewer than half of the amino acid residues in the polypeptide sequences of the α and β subunits are identical, the three-dimensional structures of the two types of subunits are very similar. Furthermore, their structures are very similar to that of myoglobin (Fig. 5-6), even though the amino acid sequences of the three polypeptides are identical at only 27 positions (Fig. 5-7). All three polypeptides are members of the globin family of proteins. The helix-naming convention described for myoglobin is also applied to the hemoglobin polypeptides, except that the α subunit lacks the short D helix. The heme-binding pocket is made up largely of the E and F helices.

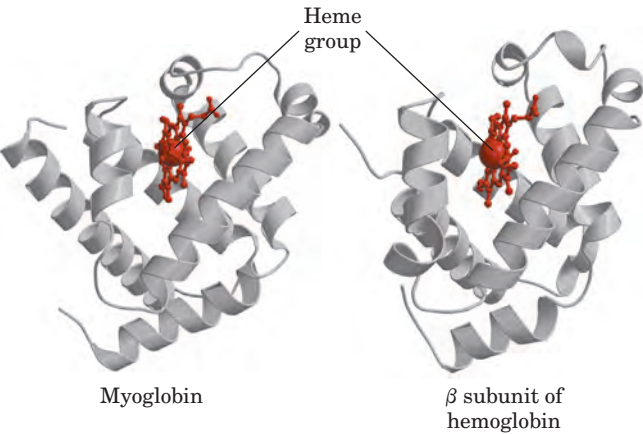


FIGURE 5-6 A comparison of the structures of myoglobin (PDB ID 1MBO) and the β subunit of hemoglobin (derived from PDB ID 1HGA).

	Mb	Hb α	Hb β		Mb	Hb α	Hb β
NA1	1 V	1 V	1 V		K	A	A
	—	—	H		K	H	H
	L	L	L		K	V	L
A1	S	S	T	80	G	D	D
	E	P	P		H	D	80 N
	G	A	E		H	M	L
	E	D	E		E	P	K
	W	K	K		A	N	G
	Q	T	S		E	A	T
	L	N	A	F1	L	80 L	F
	V	V	V		K	S	A
	L	K	T		P	A	T
	H	A	A		L	L	L
	V	A	L		A	S	S
	W	W	W		Q	D	E
	A	G	G		S	L	L
	K	K	K	Proximal His F8	H	H	H
	V	V	V	F9	A	A	C
A16	E	G	—		T	H	D
	A	A	—		K	K	K
B1	20 D	20 H	N		H	L	L
	V	A	20 V		K	R	H
	A	G	D		I	V	V
	G	E	E	G1	100 P	D	D
	H	Y	V		I	P	100 P
	G	G	G		K	V	E
	Q	A	G		Y	N	N
	D	E	E		L	F	F
	I	A	A		E	K	R
	L	L	L		F	100 L	L
	I	E	G		I	L	L
	R	R	R		S	S	G
	L	M	L		E	H	N
	F	F	L		A	C	V
	K	L	V		I	L	L
B16	S	S	V		I	L	V
C1	H	F	Y		H	V	C
	P	P	P		V	T	V
	E	T	W		L	L	L
	T	T	T		H	A	A
40 L	40 K	Q			S	A	H
	E	T	40 R	G19	H	H	H
	K	Y	F		H	L	F
	F	F	F		120 P	P	G
	D	P	E		G	A	120 K
	R	H	S		D	E	E
	F	F	F		F	F	F
	K	—	G		H1	T	T
	H	D	D		A	P	P
	L	L	L		D	120 A	P
	K	S	S		A	V	V
D1	T	H	T		Q	H	Q
	E	—	P		G	A	A
	A	—	D		A	S	A
	E	—	A		M	L	Y
	M	—	V		N	D	Q
	K	—	M		K	K	K
D7	A	G	G		A	F	V
E1	S	S	N		L	L	V
	E	A	P		E	A	A
60 D	Q	K			L	S	G
	L	V	60 V		F	V	V
	K	K	K		R	S	A
	K	G	A		140 K	T	N
Distal His E7	H	H	H		D	V	140 A
	G	G	G		I	L	L
	V	60 K	K		A	T	A
	T	K	K		H21	S	H
	V	V	V		K	K	K
	L	A	L		Y	140 Y	Y
	T	D	G		K	141 R	146 H
	A	A	A		E	—	—
	L	L	F		H26	L	—
	G	T	S		G	—	—
	A	N	D		Y	—	—
	I	A	G		Q	—	—
E19	L	V	L		153 G	—	—

FIGURE 5-7 The amino acid sequences of whale myoglobin and the α and β chains of human hemoglobin. Dashed lines mark helix boundaries. To align the sequences optimally, short gaps must be introduced into both Hb sequences where a few amino acids are present in the compared sequences. With the exception of the missing D helix in Hb α , this alignment permits the use of the helix lettering convention that emphasizes the common positioning of amino acid residues that are identical in all three structures (shaded). Residues shaded in pink are conserved in all known globins. Note that the common helix-letter-and-number designation for amino acids does not necessarily correspond to a common position in the linear sequence of amino acids in the polypeptides. For example, the distal His residue is His E7 in all three structures, but corresponds to His⁶⁴, His⁵⁸, and His⁶³ in the linear sequences of Mb, Hb α , and Hb β , respectively. Nonhelical residues at the amino and carboxyl termini, beyond the first (A) and last (H) α -helical segments, are labeled NA and HC, respectively.

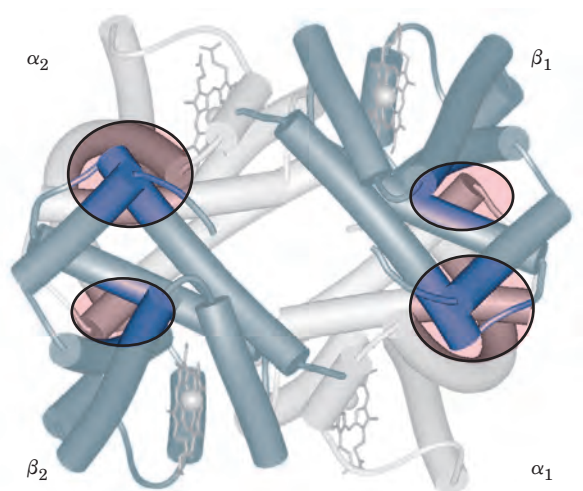


FIGURE 5-8 Dominant interactions between hemoglobin subunits. In this representation, α subunits are light and β subunits are dark. The strongest subunit interactions (highlighted) occur between unlike subunits. When oxygen binds, the $\alpha_1\beta_1$ contact changes little, but there is a large change at the $\alpha_1\beta_2$ contact, with several ion pairs broken (PDB ID 1HGA).

The quaternary structure of hemoglobin features strong interactions between unlike subunits. The $\alpha_1\beta_1$ interface (and its $\alpha_2\beta_2$ counterpart) involves more than 30 residues, and its interaction is sufficiently strong that although mild treatment of hemoglobin with urea tends to cause the tetramer to disassemble into $\alpha\beta$ dimers, these dimers remain intact. The $\alpha_1\beta_2$ (and $\alpha_2\beta_1$) interface involves 19 residues (Fig. 5–8). Hydrophobic interactions predominate at the interfaces, but there are also many hydrogen bonds and a few ion pairs (sometimes referred to as salt bridges), whose importance is discussed below.

Hemoglobin Undergoes a Structural Change on Binding Oxygen

X-ray analysis has revealed two major conformations of hemoglobin: the **R state** and the **T state**. Although oxygen binds to hemoglobin in either state, it has a significantly higher affinity for hemoglobin in the R state. Oxygen binding stabilizes the R state. When oxygen is absent experimentally, the T state is more stable and is thus the predominant conformation of **deoxyhemoglobin**. T and R originally denoted “tense” and “relaxed,” respectively, because the T state is stabilized by a greater number of ion pairs, many of which lie at the $\alpha_1\beta_2$ (and $\alpha_2\beta_1$) interface (Fig. 5–9). The binding of O_2 to a hemoglobin subunit in the T state triggers a change in conformation to the R state. When the entire protein undergoes this transition, the structures of the individual subunits change little, but the $\alpha\beta$ subunit pairs slide past each other and rotate, narrowing the pocket between the β subunits (Fig. 5–10). In this process, some

of the ion pairs that stabilize the T state are broken and some new ones are formed.

Max Perutz proposed that the $T \rightarrow R$ transition is triggered by changes in the positions of key amino acid side chains surrounding the heme. In the T state, the porphyrin is slightly puckered, causing the heme iron to protrude somewhat on the proximal His (His F8) side. The binding of O_2 causes the heme to assume a more planar conformation, shifting the position of the proximal His and the attached F helix (Fig. 5–11). These changes lead to adjustments in the ion pairs at the $\alpha_1\beta_2$ interface.

Hemoglobin Binds Oxygen Cooperatively

Hemoglobin must bind oxygen efficiently in the lungs, where the pO_2 is about 13.3 kPa, and release oxygen in the tissues, where the pO_2 is about 4 kPa. Myoglobin, or any protein that binds oxygen with a hyperbolic binding curve, would be ill-suited to this function, for the reason illustrated in Figure 5–12. A protein that bound

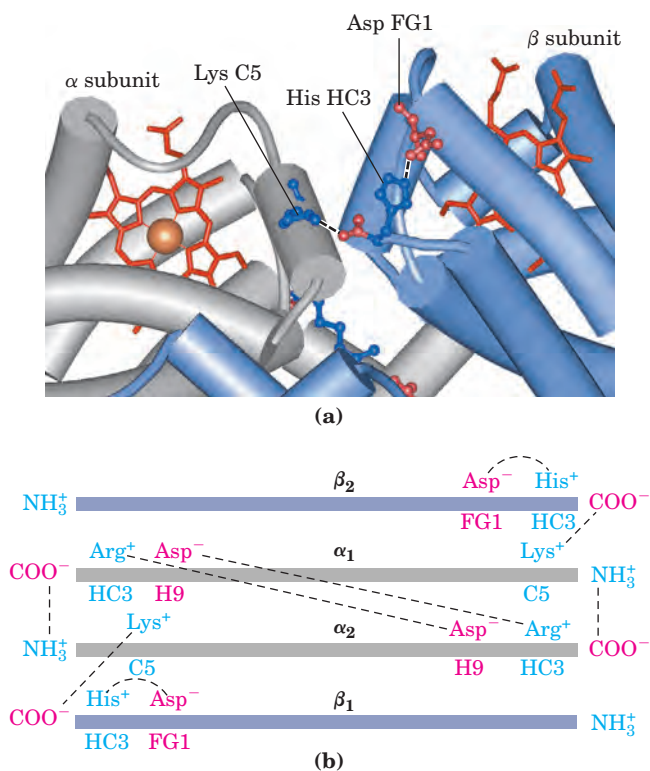


FIGURE 5-9 Some ion pairs that stabilize the T state of deoxyhemoglobin. (a) A close-up view of a portion of a deoxyhemoglobin molecule in the T state (PDB ID 1HGA). Interactions between the ion pairs His HC3 and Asp FG1 of the β subunit (blue) and between Lys C5 of the α subunit (gray) and His HC3 (its α -carboxyl group) of the β subunit are shown with dashed lines. (Recall that HC3 is the carboxyl-terminal residue of the β subunit.) (b) The interactions between these ion pairs, and between others not shown in (a), are schematized in this representation of the extended polypeptide chains of hemoglobin.

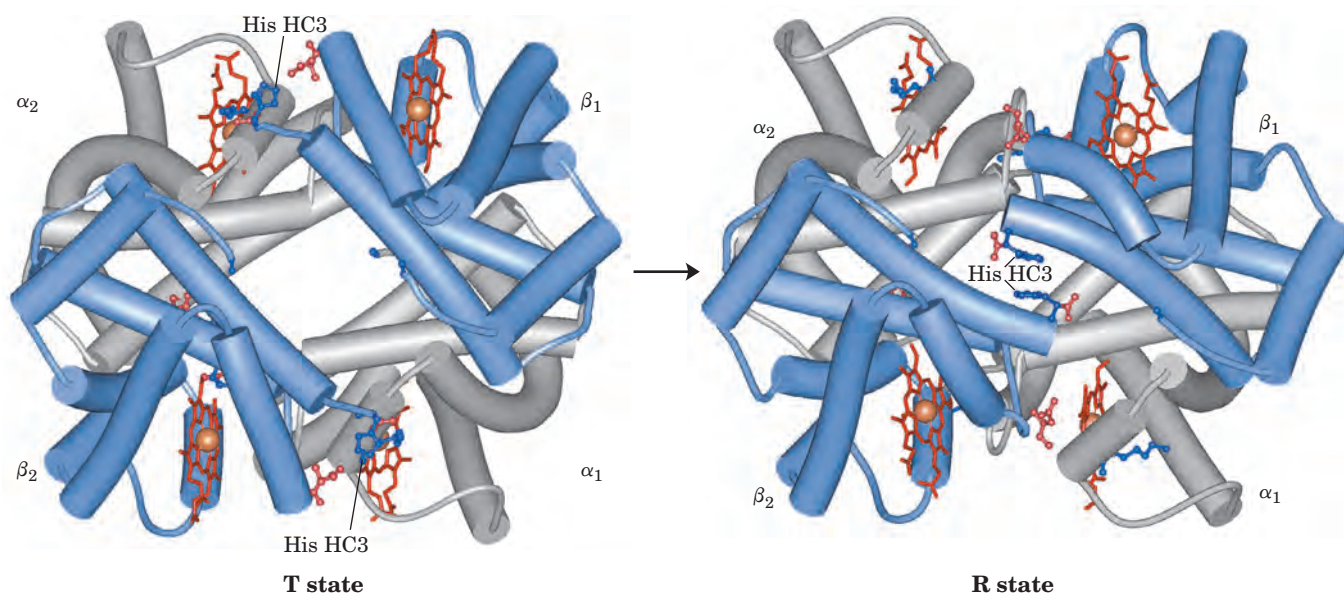


FIGURE 5-10 The T $\rightarrow$ R transition. (PDB ID 1HGA and 1BBB) In these depictions of deoxyhemoglobin, as in Figure 5-9, the β subunits are blue and the α subunits are gray. Positively charged side chains and chain termini involved in ion pairs are shown in blue, their negatively charged partners in red. The Lys C5 of each α subunit and Asp FG1 of each β subunit are visible but not labeled (compare Fig. 5-9a). Note that the molecule is oriented slightly differently than in Figure

5-9. The transition from the T state to the R state shifts the subunit pairs substantially, affecting certain ion pairs. Most noticeably, the His HC3 residues at the carboxyl termini of the β subunits, which are involved in ion pairs in the T state, rotate in the R state toward the center of the molecule, where they are no longer in ion pairs. Another dramatic result of the T $\rightarrow$ R transition is a narrowing of the pocket between the β subunits.

O₂ with high affinity would bind it efficiently in the lungs but would not release much of it in the tissues. If the protein bound oxygen with a sufficiently low affinity to release it in the tissues, it would not pick up much oxygen in the lungs.

Hemoglobin solves the problem by undergoing a transition from a low-affinity state (the T state) to a high-affinity state (the R state) as more O₂ molecules are bound. As a result, hemoglobin has a hybrid S-shaped, or sigmoid, binding curve for oxygen (Fig. 5-12). A single-subunit protein with a single ligand-binding site cannot produce a sigmoid binding curve—even if binding elicits a conformational change—because each molecule of ligand binds independently and cannot affect the binding of another molecule. In contrast, O₂ binding to individual subunits of hemoglobin can alter the affinity for O₂ in adjacent subunits. The first molecule of O₂ that interacts with deoxyhemoglobin binds weakly, because it binds to a subunit in the T state. Its binding, however, leads to conformational changes that are communicated to adjacent subunits, making it easier for additional molecules of O₂ to bind. In effect, the T $\rightarrow$ R transition occurs more readily in the second subunit once O₂ is bound to the first subunit. The last (fourth) O₂ molecule binds to a heme in a subunit that is already in the R state, and hence it binds with much higher affinity than the first molecule.

An **allosteric protein** is one in which the binding of a ligand to one site affects the binding properties of another site on the same protein. The term “allosteric” derives from the Greek *allos*, “other,” and *stereos*, “solid” or “shape.” Allosteric proteins are those having “other shapes,” or conformations, induced by the binding of ligands referred to as modulators. The conformational changes induced by the modulator(s) interconvert more-active and less-active forms of the protein. The modulators for allosteric proteins may be either inhibitors or activators. When the normal ligand and

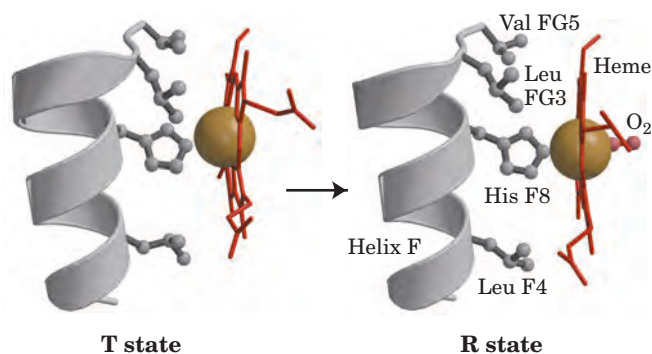


FIGURE 5-11 Changes in conformation near heme on O₂ binding to deoxyhemoglobin. (Derived from PDB ID 1HGA and 1BBB.) The shift in the position of the F helix when heme binds O₂ is thought to be one of the adjustments that triggers the T $\rightarrow$ R transition.

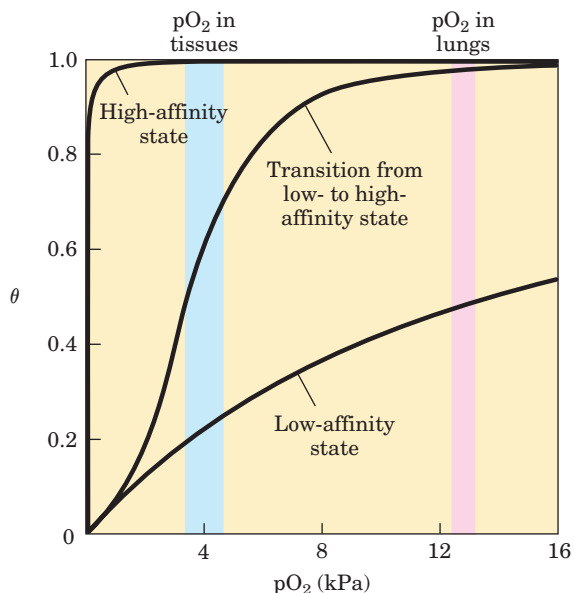


FIGURE 5-12 A sigmoid (cooperative) binding curve. A sigmoid binding curve can be viewed as a hybrid curve reflecting a transition from a low-affinity to a high-affinity state. Cooperative binding, as manifested by a sigmoid binding curve, renders hemoglobin more sensitive to the small differences in O_2 concentration between the tissues and the lungs, allowing hemoglobin to bind oxygen in the lungs (where pO_2 is high) and release it in the tissues (where pO_2 is low).

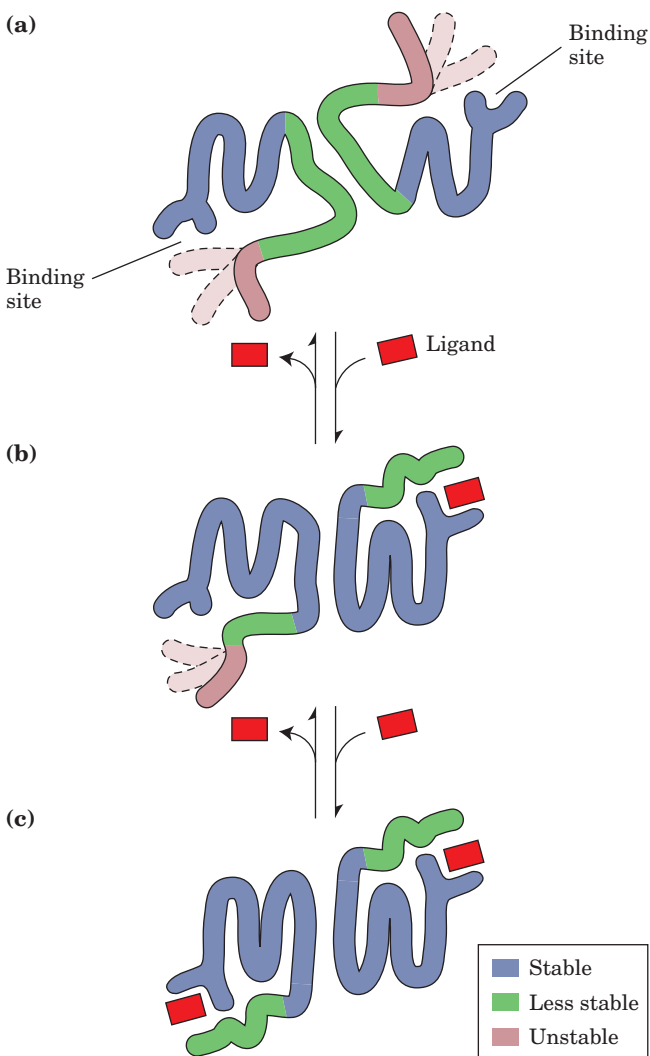
modulator are identical, the interaction is termed **homotropic**. When the modulator is a molecule other than the normal ligand the interaction is **heterotropic**. Some proteins have two or more modulators and therefore can have both homotropic and heterotropic interactions.

Cooperative binding of a ligand to a multimeric protein, such as we observe with the binding of O_2 to hemoglobin, is a form of allosteric binding often observed in multimeric proteins. The binding of one ligand affects

FIGURE 5-13 Structural changes in a multisubunit protein undergoing cooperative binding to ligand. Structural stability is not uniform throughout a protein molecule. Shown here is a hypothetical dimeric protein, with regions of high (blue), medium (green), and low (red) stability. The ligand-binding sites are composed of both high- and low-stability segments, so affinity for ligand is relatively low. (a) In the absence of ligand, the red segments are quite flexible and take up a variety of conformations, few of which facilitate ligand binding. The green segments are most stable in the low-affinity state. (b) The binding of ligand to one subunit stabilizes a high-affinity conformation of the nearby red segment (now shown in green), inducing a conformational change in the rest of the polypeptide. This is a form of induced fit. The conformational change is transmitted to the other subunit through protein-protein interactions, such that a higher-affinity conformation of the binding site is stabilized in the other subunit. (c) A second ligand molecule can now bind to the second subunit, with a higher affinity than the binding of the first, giving rise to the observed positive cooperativity.

the affinities of any remaining unfilled binding sites, and O_2 can be considered as both a ligand and an activating homotropic modulator. There is only one binding site for O_2 on each subunit, so the allosteric effects giving rise to cooperativity are mediated by conformational changes transmitted from one subunit to another by subunit-subunit interactions. A sigmoid binding curve is diagnostic of cooperative binding. It permits a much more sensitive response to ligand concentration and is important to the function of many multisubunit proteins. The principle of allostery extends readily to regulatory enzymes, as we shall see in Chapter 6.

Cooperative conformational changes depend on variations in the structural stability of different parts of a protein, as described in Chapter 4. The binding sites of an allosteric protein typically consist of stable segments in proximity to relatively unstable segments, with the latter capable of frequent changes in conformation or disorganized motion (Fig. 5-13). When a ligand binds, the moving parts of the protein's binding site may be stabilized in a particular conformation, affecting the conformation of adjacent polypeptide subunits. If the



entire binding site were highly stable, then few structural changes could occur in this site or be propagated to other parts of the protein when a ligand binds.

As is the case with myoglobin, ligands other than oxygen can bind to hemoglobin. An important example is carbon monoxide, which binds to hemoglobin about 250 times better than does oxygen. Human exposure to CO can have tragic consequences (Box 5-1).

Cooperative Ligand Binding Can Be Described Quantitatively

Cooperative binding of oxygen by hemoglobin was first analyzed by Archibald Hill in 1910. From this work came a general approach to the study of cooperative ligand binding to multisubunit proteins.

For a protein with n binding sites, the equilibrium of Equation 5-1 becomes



and the expression for the association constant becomes

$$K_a = \frac{[PL_n]}{[P][L]^n} \quad (5-13)$$

The expression for θ (see Eqn 5-8) is

$$\theta = \frac{[L]^n}{[L]^n + K_d} \quad (5-14)$$

Rearranging, then taking the log of both sides, yields

$$\frac{\theta}{1 - \theta} = \frac{[L]^n}{K_d} \quad (5-15)$$

$$\log \left(\frac{\theta}{1 - \theta} \right) = n \log [L] - \log K_d \quad (5-16)$$

where $K_d = [L]_{0.5}^n$.

Equation 5-16 is the **Hill equation**, and a plot of $\log [\theta/(1 - \theta)]$ versus $\log [L]$ is called a **Hill plot**. Based on the equation, the Hill plot should have a slope of n . However, the experimentally determined slope actually reflects not the number of binding sites but the degree of interaction between them. The slope of a Hill plot is therefore denoted by n_H , the **Hill coefficient**, which is a measure of the degree of cooperativity. If n_H equals 1, ligand binding is not cooperative, a situation that can arise even in a multisubunit protein if the subunits do not communicate. An n_H of greater than 1 indicates positive cooperativity in ligand binding. This is the situation observed in hemoglobin, in which the binding of one molecule of ligand facilitates the binding of others. The theoretical upper limit for n_H is reached when $n_H = n$. In this case the binding would be completely cooperative: all binding sites on the protein would bind ligand simultaneously, and no protein molecules partially saturated with ligand would be present under any conditions. This limit is never reached in

practice, and the measured value of n_H is always less than the actual number of ligand-binding sites in the protein.

An n_H of less than 1 indicates negative cooperativity, in which the binding of one molecule of ligand *impedes* the binding of others. Well-documented cases of negative cooperativity are rare.

To adapt the Hill equation to the binding of oxygen to hemoglobin we must again substitute pO_2 for $[L]$ and P_{50}^n for K_d :

$$\log \left(\frac{\theta}{1 - \theta} \right) = n \log pO_2 - n \log P_{50}^n \quad (5-17)$$

Hill plots for myoglobin and hemoglobin are given in Figure 5-14.

Two Models Suggest Mechanisms for Cooperative Binding

Biochemists now know a great deal about the T and R states of hemoglobin, but much remains to be learned about how the $T \rightarrow R$ transition occurs. Two models for the cooperative binding of ligands to proteins with multiple binding sites have greatly influenced thinking about this problem.

The first model was proposed by Jacques Monod, Jeffries Wyman, and Jean-Pierre Changeux in 1965, and is called the **MWC model** or the **concerted model** (Fig. 5-15a). The concerted model assumes that the subunits of a cooperatively binding protein are functionally identical, that each subunit can exist in (at

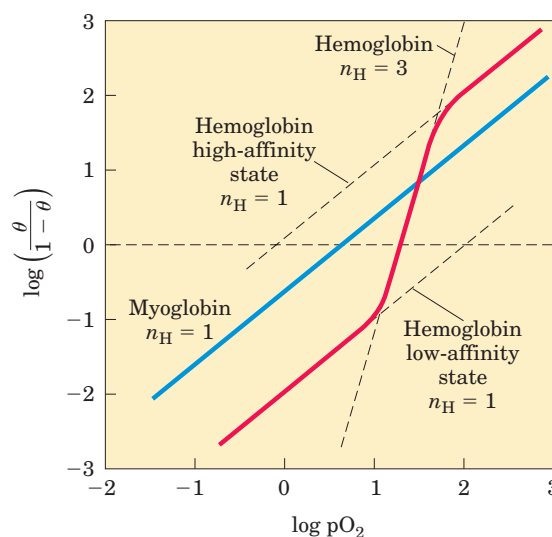


FIGURE 5-14 Hill plots for the binding of oxygen to myoglobin and hemoglobin. When $n_H = 1$, there is no evident cooperativity. The maximum degree of cooperativity observed for hemoglobin corresponds approximately to $n_H = 3$. Note that while this indicates a high level of cooperativity, n_H is less than n , the number of O_2 -binding sites in hemoglobin. This is normal for a protein that exhibits allosteric binding behavior.



BOX 5-1 BIOCHEMISTRY IN MEDICINE

Carbon Monoxide: A Stealthy Killer

Lake Powell, Arizona, August 2000. A family was vacationing in a rented houseboat. They turned on the electrical generator to power an air conditioner and a television. About 15 minutes later, two brothers, aged 8 and 11, jumped off the swim deck at the stern. Situated immediately below the deck was the exhaust port for the generator. Within two minutes, both boys were overcome by the carbon monoxide in the exhaust, which had become concentrated in the space under the deck. Both drowned. These deaths, along with a series of deaths in the 1990s linked to houseboats of similar design, eventually led to the recall and redesign of the generator exhaust assembly.

Carbon monoxide (CO), a colorless, odorless gas, is responsible for more than half of yearly deaths due to poisoning worldwide. CO has an approximately 250-fold greater affinity for hemoglobin than does oxygen. Consequently, relatively low levels of CO can have substantial and tragic effects. When CO combines with hemoglobin, the complex is referred to as carboxyhemoglobin, or COHb.

Some CO is produced by natural processes, but locally high levels generally result only from human activities. Engine and furnace exhausts are important sources, as CO is a byproduct of the incomplete combustion of fossil fuels. In the United States alone, nearly 4,000 people succumb to CO poisoning each year, both accidentally and intentionally. Many of the accidental deaths involve undetected CO buildup in enclosed spaces, such as when a household furnace malfunctions or leaks, venting CO into a home. However, CO poisoning can also occur in open spaces, as unsuspecting people at work or play inhale the exhaust from generators, outboard motors, tractor engines, recreational vehicles, or lawn mowers.

Carbon monoxide levels in the atmosphere are rarely dangerous, ranging from less than 0.05 parts per million (ppm) in remote and uninhabited areas to 3 to 4 ppm in some cities of the northern hemisphere. In the United States, the government-mandated (Occupational Safety and Health Administration, OSHA) limit for CO at worksites is 50 ppm for people working an eight-hour shift. The tight binding of CO to hemoglobin means that COHb can accumulate over

time as people are exposed to a constant low-level source of CO.

In an average, healthy individual, 1% or less of the total hemoglobin is complexed as COHb. Since CO is a product of tobacco smoke, many smokers have COHb levels in the range of 3% to 8% of total hemoglobin, and the levels can rise to 15% for chain-smokers. COHb levels equilibrate at 50% in people who breathe air containing 570 ppm of CO for several hours. Reliable methods have been developed that relate CO content in the atmosphere to COHb levels in the blood (Fig. 1). In tests of houseboats with a generator exhaust like the one responsible for the Lake Powell deaths, CO levels reached 6,000 to 30,000 ppm under the swim deck, and atmospheric O₂ levels under the deck declined from 21% to 12%. Even above the swim deck, CO levels of up to 7,200 ppm were detected, high enough to cause death within a few minutes.

How is a human affected by COHb? At levels of less than 10% of total hemoglobin, symptoms are rarely observed. At 15%, the individual experiences mild headaches. At 20% to 30%, the headache is severe and

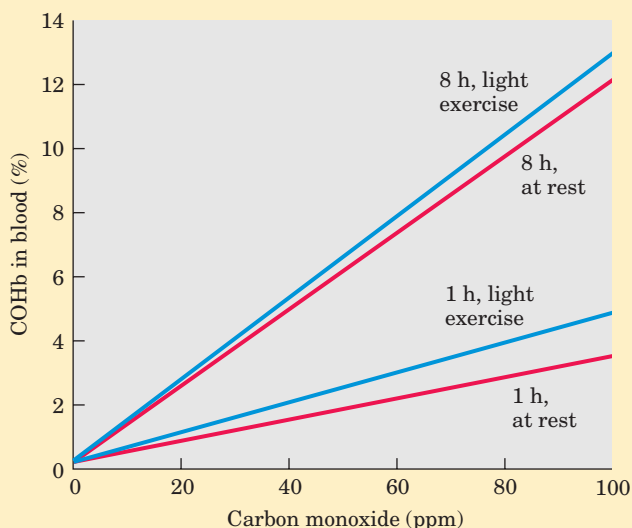


FIGURE 1 Relationship between the levels of COHb in blood and the concentration of CO in the surrounding air. Four different conditions of exposure are shown, comparing the effects of short versus extended exposure, and exposure at rest versus exposure during light exercise.

least) two conformations, and that all subunits undergo the transition from one conformation to the other simultaneously. In this model, no protein has individual subunits in different conformations. The two conformations are in equilibrium. The ligand can bind to either

conformation, but binds each with different affinity. Successive binding of ligand molecules to the low-affinity conformation (which is more stable in the absence of ligand) makes a transition to the high-affinity conformation more likely.

is generally accompanied by nausea, dizziness, confusion, disorientation, and some visual disturbances; these symptoms are generally reversed rapidly if the individual is treated with oxygen. At COHb levels of 30% to 50%, the neurological symptoms become more severe, and at levels near 50%, the individual loses consciousness and can sink into coma. Respiratory failure may follow. With prolonged exposure, some damage becomes permanent. Death normally occurs when COHb levels rise above 60%. Autopsy on the boys who died at Lake Powell revealed COHb levels of 59% and 52%.

Binding of CO to hemoglobin is affected by many factors, including exercise (Fig. 1) and changes in air pressure related to altitude. Because of their higher base levels of COHb, smokers exposed to a source of CO often develop symptoms faster than nonsmokers. Individuals with heart and lung conditions or blood diseases that reduce the availability of oxygen to tissues may also experience symptoms at lower levels of CO exposure. Fetuses are at particular risk for CO poisoning, because fetal hemoglobin has a somewhat higher affinity for CO than adult hemoglobin. Cases of CO exposure have been recorded in which the fetus died but the mother recovered.

It may seem surprising that the loss of half of one's hemoglobin to COHb can prove fatal—we know that people with any of several anemic conditions manage to function reasonably well with half the usual complement of active hemoglobin. However, the binding of CO to hemoglobin does more than remove protein from the pool available to bind oxygen. It also affects the affinity of the remaining hemoglobin subunits for oxygen. As CO binds to one or two subunits of a hemoglobin tetramer, the affinity for O_2 is increased substantially in the remaining subunits (Fig. 2). Thus, a hemoglobin tetramer with two bound CO molecules can efficiently bind O_2 in the lungs—but it releases very little of it in the tissues. Oxygen deprivation in the tissues rapidly becomes severe. To add to the problem, the effects of CO are not limited to interference with hemoglobin function. CO binds to other heme proteins and a variety of metalloproteins. The effects of these interactions are not yet well understood, but they may be responsible for some of the longer-term effects of acute but nonfatal CO poisoning.

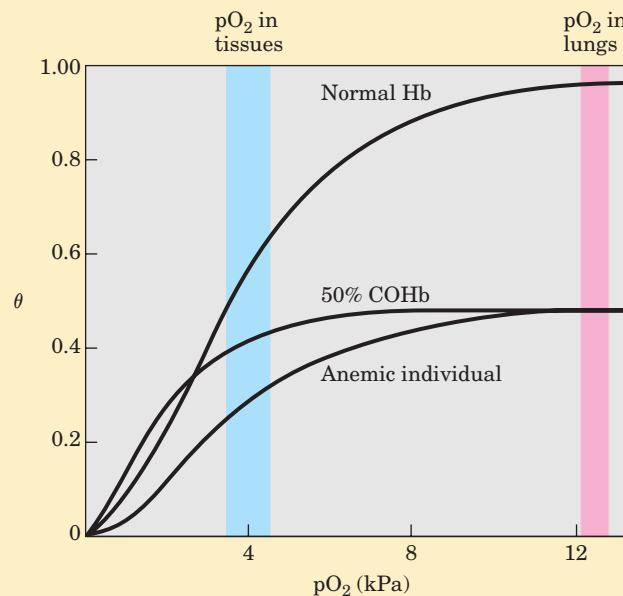


FIGURE 2 Several oxygen-binding curves: for normal hemoglobin, hemoglobin from an anemic individual with only 50% of her hemoglobin functional, and hemoglobin from an individual with 50% of his hemoglobin subunits complexed with CO. The pO_2 in human lungs and tissues is indicated.

When CO poisoning is suspected, rapid evacuation of the person away from the CO source is essential, but this does not always result in rapid recovery. When an individual is moved from the CO-polluted site to a normal, outdoor atmosphere, O_2 begins to replace the CO in hemoglobin. The COHb levels drop rather slowly, however; the half-time is 2 to 6.5 hours, depending on individual and environmental factors. If 100% oxygen is administered with a mask, the rate of exchange can be increased about fourfold; the half-time for O_2 -CO exchange can be reduced to tens of minutes if 100% oxygen at a pressure of 3 atm (303 kPa) is supplied. Thus, rapid treatment by a properly equipped medical team is critical.

Carbon monoxide detectors in all homes are highly recommended. This is a simple and inexpensive measure to avoid possible tragedy. After completing the research for this box, we immediately purchased several new CO detectors for our homes.

In the second model, the **sequential model** (Fig. 5–15b), proposed in 1966 by Daniel Koshland and colleagues, ligand binding can induce a change of conformation in an individual subunit. A conformational change in one subunit makes a similar change in an

adjacent subunit, as well as the binding of a second ligand molecule, more likely. There are more potential intermediate states in this model than in the concerted model. The two models are not mutually exclusive; the concerted model may be viewed as the “all-or-none” limiting

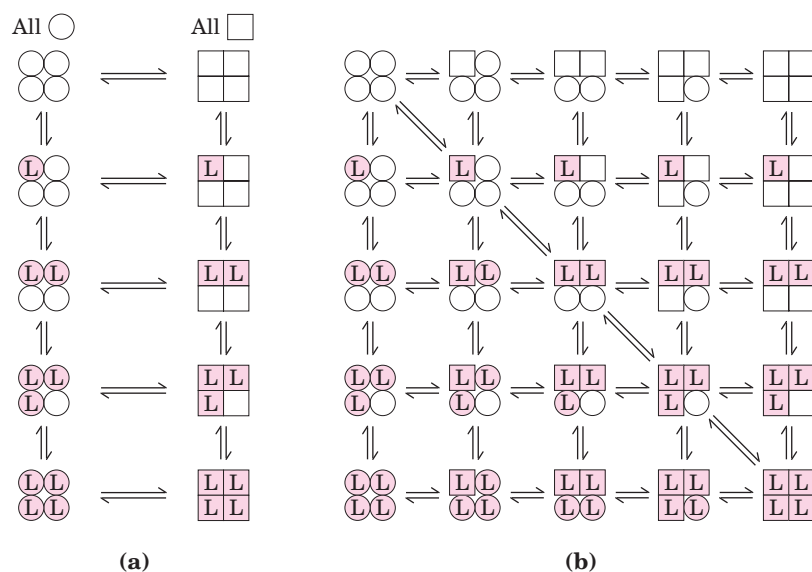


FIGURE 5-15 Two general models for the interconversion of inactive and active forms of cooperative ligand-binding proteins. Although the models may be applied to any protein—including any enzyme (Chapter 6)—that exhibits cooperative binding, we show here four subunits because the model was originally proposed for hemoglobin. In the concerted, or all-or-none, model (MWC model) (a) all the subunits are postulated to be in the same conformation, either all ○ (low affinity or inactive) or all □ (high affinity or active). Depending on the equilibrium, K_1 , between ○ and □ forms, the binding of one or more ligand molecules (L) will pull the equilibrium toward the □ form. Subunits with bound L are shaded. In the sequential model (b), each individual subunit can be in either the ○ or □ form. A very large number of conformations is thus possible.

case of the sequential model. In Chapter 6 we use these models to investigate allosteric enzymes.

Hemoglobin Also Transports H^+ and CO_2

In addition to carrying nearly all the oxygen required by cells from the lungs to the tissues, hemoglobin carries two end products of cellular respiration— H^+ and CO_2 —from the tissues to the lungs and the kidneys, where they are excreted. The CO_2 , produced by oxidation of organic fuels in mitochondria, is hydrated to form bicarbonate:



This reaction is catalyzed by **carbonic anhydrase**, an enzyme particularly abundant in erythrocytes. Carbon dioxide is not very soluble in aqueous solution, and bubbles of CO_2 would form in the tissues and blood if it were not converted to bicarbonate. As you can see from the equation, the hydration of CO_2 results in an increase in the H^+ concentration (a decrease in pH) in the tissues. The binding of oxygen by hemoglobin is profoundly influenced by pH and CO_2 concentration, so the interconversion of CO_2 and bicarbonate is of great importance to the regulation of oxygen binding and release in the blood.

Hemoglobin transports about 40% of the total H^+ and 15% to 20% of the CO_2 formed in the tissues to the lungs and the kidneys. (The remainder of the H^+ is absorbed by the plasma's bicarbonate buffer; the remainder of the CO_2 is transported as dissolved HCO_3^- and CO_2 .) The binding of H^+ and CO_2 is inversely related to the binding of oxygen. At the relatively low pH and high CO_2 concentration of peripheral tissues, the affinity of hemoglobin for oxygen decreases as H^+ and CO_2 are bound, and O_2 is released to the tissues. Conversely, in the capillaries of the lung, as CO_2 is excreted

and the blood pH consequently rises, the affinity of hemoglobin for oxygen increases and the protein binds more O_2 for transport to the peripheral tissues. This effect of pH and CO_2 concentration on the binding and release of oxygen by hemoglobin is called the **Bohr effect**, after Christian Bohr, the Danish physiologist (and father of physicist Niels Bohr) who discovered it in 1904.

The binding equilibrium for hemoglobin and one molecule of oxygen can be designated by the reaction



but this is not a complete statement. To account for the effect of H^+ concentration on this binding equilibrium, we rewrite the reaction as



where HHb^+ denotes a protonated form of hemoglobin. This equation tells us that the O_2 -saturation curve of hemoglobin is influenced by the H^+ concentration (Fig. 5-16). Both O_2 and H^+ are bound by hemoglobin, but with inverse affinity. When the oxygen concentration is high, as in the lungs, hemoglobin binds O_2 and releases protons. When the oxygen concentration is low, as in the peripheral tissues, H^+ is bound and O_2 is released.

Oxygen and H^+ are not bound at the same sites in hemoglobin. Oxygen binds to the iron atoms of the hemes, whereas H^+ binds to any of several amino acid residues in the protein. A major contribution to the Bohr effect is made by His^{146} (His HC3) of the β subunits. When protonated, this residue forms one of the ion pairs—to Asp^{94} (Asp FG1)—that helps stabilize deoxyhemoglobin in the T state (Fig. 5-9). The ion pair stabilizes the protonated form of His HC3, giving this residue an abnormally high pK_a in the T state. The pK_a falls to its normal value of 6.0 in the R state because the ion pair cannot form, and this residue is largely unpro-

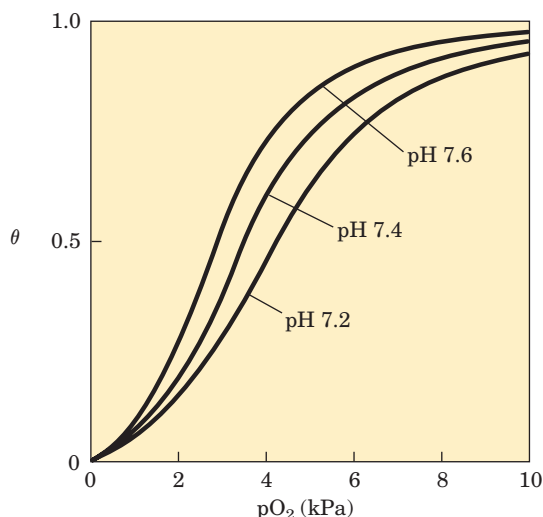
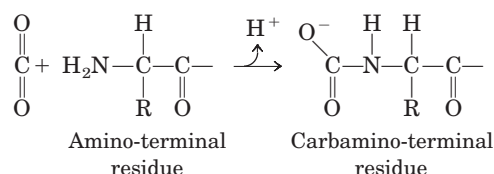


FIGURE 5-16 Effect of pH on the binding of oxygen to hemoglobin. The pH of blood is 7.6 in the lungs and 7.2 in the tissues. Experimental measurements on hemoglobin binding are often performed at pH 7.4.

tonated in oxyhemoglobin at pH 7.6, the blood pH in the lungs. As the concentration of H^+ rises, protonation of His HC3 promotes release of oxygen by favoring a transition to the T state. Protonation of the amino-terminal residues of the α subunits, certain other His residues, and perhaps other groups has a similar effect.

Thus we see that the four polypeptide chains of hemoglobin communicate with each other about not only O_2 binding to their heme groups but also H^+ binding to specific amino acid residues. And there is still more to the story. Hemoglobin also binds CO_2 , again in a manner inversely related to the binding of oxygen. Carbon dioxide binds as a carbamate group to the α -amino group at the amino-terminal end of each globin chain, forming carbaminohemoglobin:

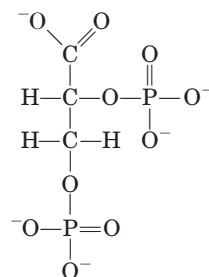


This reaction produces H^+ , contributing to the Bohr effect. The bound carbamates also form additional salt bridges (not shown in Fig. 5-9) that help to stabilize the T state and promote the release of oxygen.

When the concentration of carbon dioxide is high, as in peripheral tissues, some CO_2 binds to hemoglobin and the affinity for O_2 decreases, causing its release. Conversely, when hemoglobin reaches the lungs, the high oxygen concentration promotes binding of O_2 and release of CO_2 . It is the capacity to communicate ligand-binding information from one polypeptide subunit to the others that makes the hemoglobin molecule so beautifully adapted to integrating the transport of O_2 , CO_2 , and H^+ by erythrocytes.

Oxygen Binding to Hemoglobin Is Regulated by 2,3-Bisphosphoglycerate

The interaction of **2,3-bisphosphoglycerate (BPG)** with hemoglobin provides an example of heterotropic allosteric modulation.



2,3-Bisphosphoglycerate

BPG is present in relatively high concentrations in erythrocytes. When hemoglobin is isolated, it contains substantial amounts of bound BPG, which can be difficult to remove completely. In fact, the O_2 -binding curves for hemoglobin that we have examined to this point were obtained in the presence of bound BPG. 2,3-Bisphosphoglycerate is known to greatly reduce the affinity of hemoglobin for oxygen—there is an inverse relationship between the binding of O_2 and the binding of BPG. We can therefore describe another binding process for hemoglobin:



BPG binds at a site distant from the oxygen-binding site and regulates the O_2 -binding affinity of hemoglobin in relation to the pO_2 in the lungs. BPG plays an important role in the physiological adaptation to the lower pO_2 available at high altitudes. For a healthy human strolling by the ocean, the binding of O_2 to hemoglobin is regulated such that the amount of O_2 delivered to the tissues is equivalent to nearly 40% of the maximum that could be carried by the blood (Fig. 5-17). Imagine that this person is quickly transported to a mountainside at an altitude of 4,500 meters, where the pO_2 is considerably lower. The delivery of O_2 to the tissues is now reduced. However, after just a few hours at the higher altitude, the BPG concentration in the blood has begun to rise, leading to a decrease in the affinity of hemoglobin for oxygen. This adjustment in the BPG level has only a small effect on the binding of O_2 in the lungs but a considerable effect on the release of O_2 in the tissues. As a result, the delivery of oxygen to the tissues is restored to nearly 40% of that which can be transported by the blood. The situation is reversed when the person returns to sea level. The BPG concentration in erythrocytes also increases in people suffering from **hypoxia**, lowered oxygenation of peripheral tissues due to inadequate functioning of the lungs or circulatory system.

The site of BPG binding to hemoglobin is the cavity between the β subunits in the T state (Fig. 5-18).

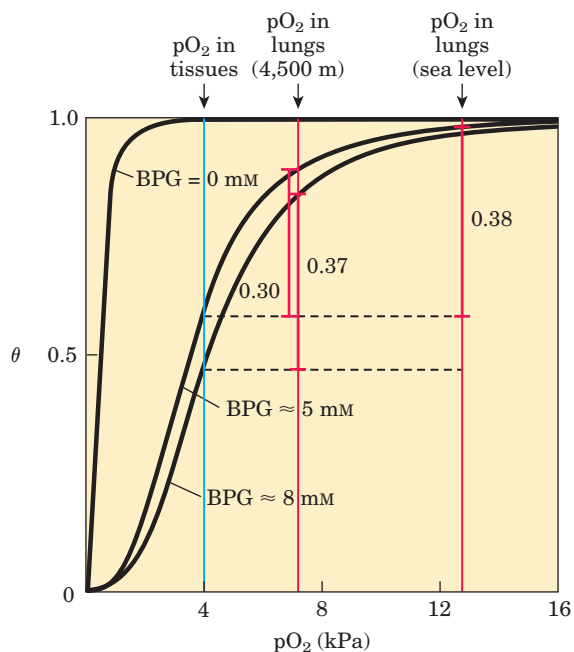


FIGURE 5-17 Effect of BPG on the binding of oxygen to hemoglobin. The BPG concentration in normal human blood is about 5 mM at sea level and about 8 mM at high altitudes. Note that hemoglobin binds to oxygen quite tightly when BPG is entirely absent, and the binding curve appears to be hyperbolic. In reality, the measured Hill coefficient for O_2 -binding cooperativity decreases only slightly (from 3 to about 2.5) when BPG is removed from hemoglobin, but the rising part of the sigmoid curve is confined to a very small region close to the origin. At sea level, hemoglobin is nearly saturated with O_2 in the lungs, but only 60% saturated in the tissues, so the amount of oxygen released in the tissues is close to 40% of the maximum that can be carried in the blood. At high altitudes, O_2 delivery declines by about one-fourth, to 30% of maximum. An increase in BPG concentration, however, decreases the affinity of hemoglobin for O_2 , so nearly 40% of what can be carried is again delivered to the tissues.

This cavity is lined with positively charged amino acid residues that interact with the negatively charged groups of BPG. Unlike O_2 , only one molecule of BPG is bound to each hemoglobin tetramer. BPG lowers hemoglobin's affinity for oxygen by stabilizing the T state. The transition to the R state narrows the binding pocket for BPG, precluding BPG binding. In the absence of BPG, hemoglobin is converted to the R state more easily.

Regulation of oxygen binding to hemoglobin by BPG has an important role in fetal development. Because a fetus must extract oxygen from its mother's blood, fetal hemoglobin must have greater affinity than the maternal hemoglobin for O_2 . The fetus synthesizes γ subunits rather than β subunits, forming $\alpha_2\gamma_2$ hemoglobin. This tetramer has a much lower affinity for BPG than normal adult hemoglobin, and a correspondingly higher affinity for O_2 . **Oxygen-Binding Proteins—Hemoglobin Is Susceptible to Allosteric Regulation**

Sickle-Cell Anemia Is a Molecular Disease of Hemoglobin



The great importance of the amino acid sequence in determining the secondary, tertiary, and quaternary structures of globular proteins, and thus their biological functions, is strikingly demonstrated by the hereditary human disease sickle-cell anemia. Almost 500 genetic variants of hemoglobin are known to occur in the human population; all but a few are quite rare. Most variations consist of differences in a single amino acid residue. The effects on hemoglobin structure and function are often minor but can sometimes be extraordinary. Each hemoglobin variation is the product of an altered gene. The variant genes are called alleles. Because humans generally have two copies of each gene, an in-

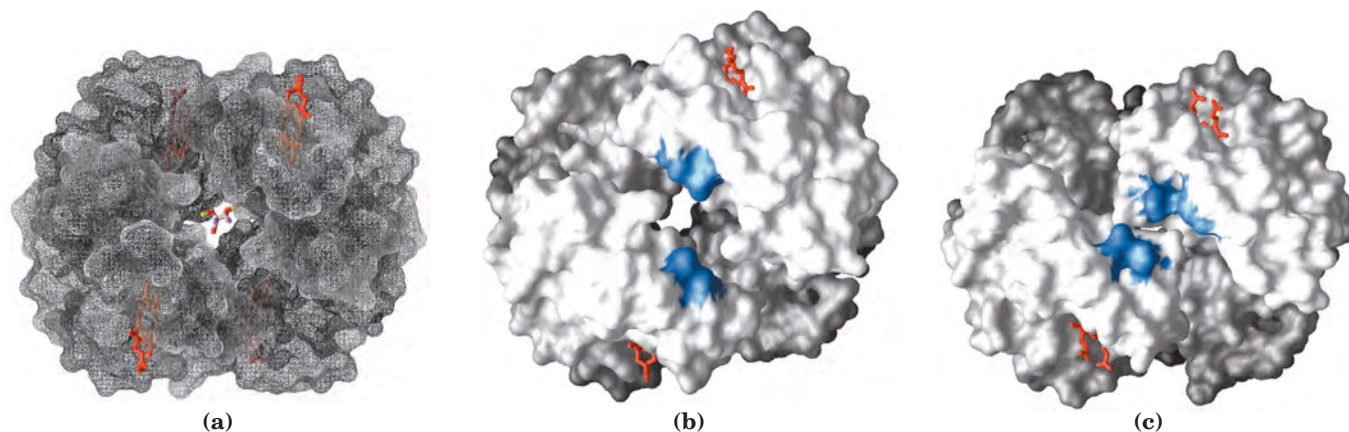


FIGURE 5-18 Binding of BPG to deoxyhemoglobin. (a) BPG binding stabilizes the T state of deoxyhemoglobin (PDB ID 1HGA), shown here as a mesh surface image. (b) The negative charges of BPG interact with several positively charged groups (shown in blue in this surface

contour image) that surround the pocket between the β subunits in the T state. (c) The binding pocket for BPG disappears on oxygenation, following transition to the R state (PDB ID 1BBB). (Compare (b) and (c) with Fig. 5-10.)

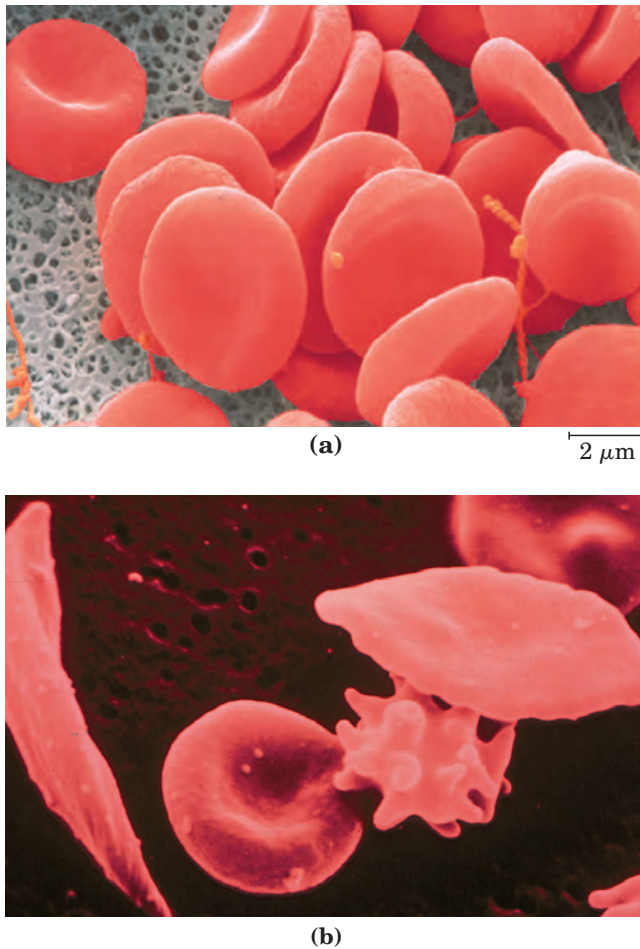


FIGURE 5-19 A comparison of uniform, cup-shaped, normal erythrocytes (a) with the variably shaped erythrocytes seen in sickle-cell anemia (b), which range from normal to spiny or sickle-shaped.

dividual may have two copies of one allele (thus being homozygous for that gene) or one copy of each of two different alleles (thus heterozygous).

Sickle-cell anemia is a genetic disease in which an individual has inherited the allele for sickle-cell hemoglobin from both parents. The erythrocytes of these individuals are fewer and also abnormal. In addition to an unusually large number of immature cells, the blood contains many long, thin, crescent-shaped erythrocytes that look like the blade of a sickle (Fig. 5-19). When hemoglobin from sickle cells (called hemoglobin S) is deoxygenated, it becomes insoluble and forms polymers that aggregate into tubular fibers (Fig. 5-20). Normal hemoglobin (hemoglobin A) remains soluble on deoxygenation. The insoluble fibers of deoxygenated hemoglobin S are responsible for the deformed sickle shape of the erythrocytes, and the proportion of sickled cells increases greatly as blood is deoxygenated.

The altered properties of hemoglobin S result from a single amino acid substitution, a Val instead of a Glu residue at position 6 in the two β chains. The R group

of valine has no electric charge, whereas glutamate has a negative charge at pH 7.4. Hemoglobin S therefore has two fewer negative charges than hemoglobin A, one for each of the two β chains. Replacement of the Glu residue by Val creates a “sticky” hydrophobic contact point at position 6 of the β chain, which is on the outer surface

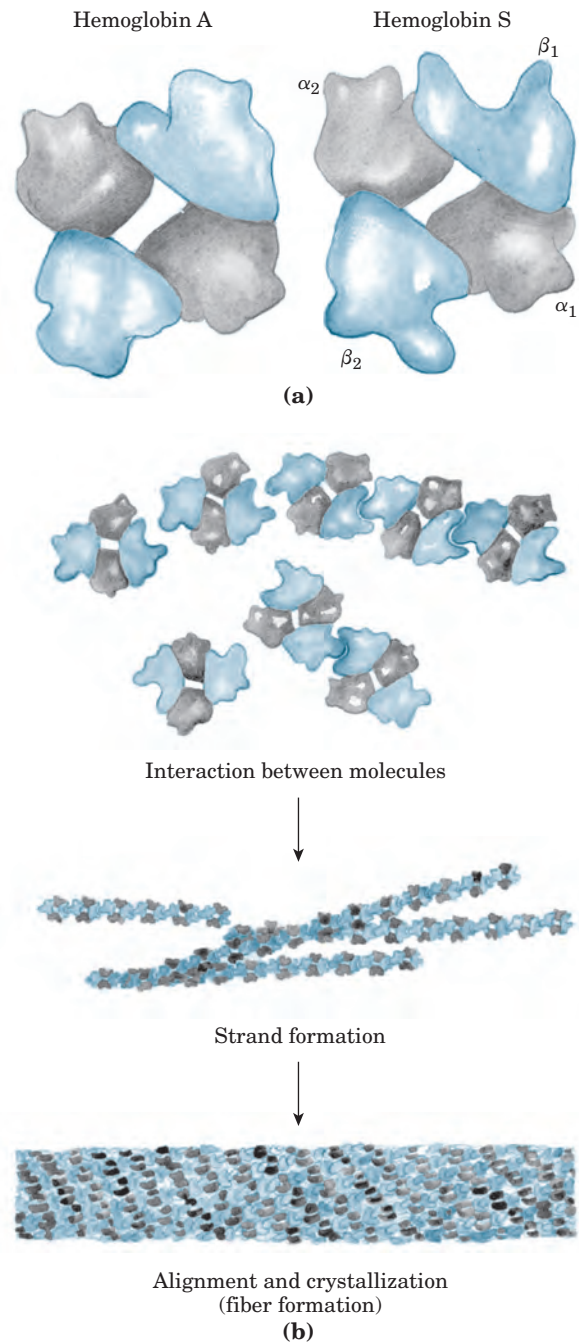


FIGURE 5-20 Normal and sickle-cell hemoglobin. (a) Subtle differences between the conformations of hemoglobin A and hemoglobin S result from a single amino acid change in the β chains. (b) As a result of this change, deoxyhemoglobin S has a hydrophobic patch on its surface, which causes the molecules to aggregate into strands that align into insoluble fibers.

of the molecule. These sticky spots cause deoxyhemoglobin S molecules to associate abnormally with each other, forming the long, fibrous aggregates characteristic of this disorder.  **Oxygen-Binding Proteins—Defects in Hb Lead to Serious Genetic Disease**

Sickle-cell anemia, as we have noted, occurs in individuals homozygous for the sickle-cell allele of the gene encoding the β subunit of hemoglobin. Individuals who receive the sickle-cell allele from only one parent and are thus heterozygous experience a milder condition called sickle-cell trait; only about 1% of their erythrocytes become sickled on deoxygenation. These individuals may live completely normal lives if they avoid vigorous exercise or other stresses on the circulatory system.

Sickle-cell anemia is a life-threatening and painful disease. People with sickle-cell anemia suffer from repeated crises brought on by physical exertion. They become weak, dizzy, and short of breath, and they also experience heart murmurs and an increased pulse rate. The hemoglobin content of their blood is only about half the normal value of 15 to 16 g/100 mL, because sickled cells are very fragile and rupture easily; this results in anemia (“lack of blood”). An even more serious consequence is that capillaries become blocked by the long, abnormally shaped cells, causing severe pain and interfering with normal organ function—a major factor in the early death of many people with the disease.

Without medical treatment, people with sickle-cell anemia usually die in childhood. Nevertheless, the sickle-cell allele is surprisingly common in certain parts of Africa. Investigation into the persistence of an allele that is so obviously deleterious in homozygous individuals led to the finding that in heterozygous individuals, the allele confers a small but significant resistance to lethal forms of malaria. Natural selection has resulted in an allele population that balances the deleterious effects of the homozygous condition against the resistance to malaria afforded by the heterozygous condition. ■

SUMMARY 5.1 Reversible Binding of a Protein to a Ligand: Oxygen-Binding Proteins

- Protein function often entails interactions with other molecules. A molecule bound by a protein is called a ligand, and the site to which it binds is called the binding site. Proteins may undergo conformational changes when a ligand binds, a process called induced fit. In a multi-subunit protein, the binding of a ligand to one subunit may affect ligand binding to other subunits. Ligand binding can be regulated.
- Myoglobin contains a heme prosthetic group, which binds oxygen. Heme consists of a single atom of Fe^{2+} coordinated within a porphyrin. Oxygen binds to myoglobin reversibly; this

simple reversible binding can be described by an association constant K_a or a dissociation constant K_d . For a monomeric protein such as myoglobin, the fraction of binding sites occupied by a ligand is a hyperbolic function of ligand concentration.

- Normal adult hemoglobin has four heme-containing subunits, two α and two β , similar in structure to each other and to myoglobin. Hemoglobin exists in two interchangeable structural states, T and R. The T state is most stable when oxygen is not bound. Oxygen binding promotes transition to the R state.
- Oxygen binding to hemoglobin is both allosteric and cooperative. As O_2 binds to one binding site, the hemoglobin undergoes conformational changes that affect the other binding sites—an example of allosteric behavior. Conformational changes between the T and R states, mediated by subunit-subunit interactions, result in cooperative binding; this is described by a sigmoid binding curve and can be analyzed by a Hill plot.
- Two major models have been proposed to explain the cooperative binding of ligands to multisubunit proteins: the concerted model and the sequential model.
- Hemoglobin also binds H^+ and CO_2 , resulting in the formation of ion pairs that stabilize the T state and lessen the protein's affinity for O_2 (the Bohr effect). Oxygen binding to hemoglobin is also modulated by 2,3-bisphosphoglycerate, which binds to and stabilizes the T state.
- Sickle-cell anemia is a genetic disease caused by a single amino acid substitution (Glu^6 to Val^6) in each β chain of hemoglobin. The change produces a hydrophobic patch on the surface of the hemoglobin that causes the molecules to aggregate into bundles of fibers. This homozygous condition results in serious medical complications.

5.2 Complementary Interactions between Proteins and Ligands: The Immune System and Immunoglobulins

Our discussion of oxygen-binding proteins showed how the conformations of these proteins affect and are affected by the binding of small ligands (O_2 or CO) to the heme group. However, most protein-ligand interactions do not involve a prosthetic group. Instead, the binding site for a ligand is more often like the hemoglobin binding site for BPG—a cleft in the protein lined with amino

acid residues, arranged to render the binding interaction highly specific. Effective discrimination between ligands is the norm at binding sites, even when the ligands have only minor structural differences.

All vertebrates have an immune system capable of distinguishing molecular “self” from “nonself” and then destroying those entities identified as nonself. In this way, the immune system eliminates viruses, bacteria, and other pathogens and molecules that may pose a threat to the organism. On a physiological level, the response of the immune system to an invader is an intricate and coordinated set of interactions among many classes of proteins, molecules, and cell types. However, at the level of individual proteins, the immune response demonstrates how an acutely sensitive and specific biochemical system is built upon the reversible binding of ligands to proteins.

The Immune Response Features a Specialized Array of Cells and Proteins

Immunity is brought about by a variety of **leukocytes** (white blood cells), including **macrophages** and **lymphocytes**, all developing from undifferentiated stem cells in the bone marrow. Leukocytes can leave the bloodstream and patrol the tissues, each cell producing one or more proteins capable of recognizing and binding to molecules that might signal an infection.

The immune response consists of two complementary systems, the humoral and cellular immune systems. The **humoral immune system** (Latin *humor*, “fluid”) is directed at bacterial infections and extracellular viruses (those found in the body fluids), but can also respond to individual proteins introduced into the organism. The **cellular immune system** destroys host cells infected by viruses and also destroys some parasites and foreign tissues.

The proteins at the heart of the humoral immune response are soluble proteins called **antibodies** or **immunoglobulins**, often abbreviated **Ig**. Immunoglobulins bind bacteria, viruses, or large molecules identified as foreign and target them for destruction. Making up 20% of blood protein, the immunoglobulins are produced by **B lymphocytes**, or **B cells**, so named because they complete their development in the bone marrow.

The agents at the heart of the cellular immune response are a class of **T lymphocytes**, or **T cells** (so called because the latter stages of their development occur in the thymus), known as **cytotoxic T cells** (**T_C cells**, also called killer T cells). Recognition of infected cells or parasites involves proteins called **T-cell receptors** on the surface of T_C cells. Receptors are proteins, usually found on the outer surface of cells and extending through the plasma membrane; they recognize and bind extracellular ligands, triggering changes inside the cell.

In addition to cytotoxic T cells, there are **helper T cells** (**T_H cells**), whose function it is to produce soluble signaling proteins called cytokines, which include the interleukins. T_H cells interact with macrophages. Table 5–2 summarizes the functions of the various leukocytes of the immune system.

Each recognition protein of the immune system, either an antibody produced by a B cell or a receptor on the surface of a T cell, specifically binds some particular chemical structure, distinguishing it from virtually all others. Humans are capable of producing more than 10⁸ different antibodies with distinct binding specificities. This extraordinary diversity makes it likely that any chemical structure on the surface of a virus or invading cell will be recognized and bound by one or more antibodies. Antibody diversity is derived from random reassembly of a set of immunoglobulin gene segments through genetic recombination mechanisms that are discussed in Chapter 25 (see Fig. 25–44).

Some properties of the interactions between antibodies or T-cell receptors and the molecules they bind are unique to the immune system, and a specialized lexicon is used to describe them. Any molecule or pathogen capable of eliciting an immune response is called an **antigen**. An antigen may be a virus, a bacterial cell wall, or an individual protein or other macromolecule. A complex antigen may be bound by a number of different antibodies. An individual antibody or T-cell receptor binds only a particular molecular structure within the antigen, called its **antigenic determinant** or **epitope**.

It would be unproductive for the immune system to respond to small molecules that are common intermediates and products of cellular metabolism. Molecules of $M_r < 5,000$ are generally not antigenic. However, small

TABLE 5–2 Some Types of Leukocytes Associated with the Immune System

Cell type	Function
Macrophages	Ingest large particles and cells by phagocytosis
B lymphocytes (B cells)	Produce and secrete antibodies
T lymphocytes (T cells)	
Cytotoxic (killer) T cells (T _C)	Interact with infected host cells through receptors on T-cell surface
Helper T cells (T _H)	Interact with macrophages and secrete cytokines (interleukins) that stimulate T _C , T _H , and B cells to proliferate.

molecules can be covalently attached to large proteins in the laboratory, and in this form they may elicit an immune response. These small molecules are called **hap- tens**. The antibodies produced in response to protein-linked haptens will then bind to the same small molecules when they are free. Such antibodies are sometimes used in the development of analytical tests described later in this chapter or as catalytic antibodies (see Box 6–3).

The interactions of antibody and antigen are much better understood than are the binding properties of T-cell receptors. However, before focusing on antibodies, we need to look at the humoral and cellular immune systems in more detail to put the fundamental biochemical interactions into their proper context.

Self Is Distinguished from Nonself by the Display of Peptides on Cell Surfaces

The immune system must identify and destroy pathogens, but it must also recognize and *not* destroy the normal proteins and cells of the host organism—the “self.” Detection of protein antigens in the host is mediated by **MHC (major histocompatibility complex) proteins**. MHC proteins bind peptide fragments of proteins digested in the cell and present them on the outside surface of the cell. These peptides normally come from the digestion of typical cellular proteins, but during a viral infection viral proteins are also digested and presented on the cell surface by MHC proteins. Peptide

fragments from foreign proteins that are displayed by MHC proteins are the antigens the immune system recognizes as nonself. T-cell receptors bind these fragments and launch the subsequent steps of the immune response. There are two classes of MHC proteins (Fig. 5–21), which differ in their distribution among cell types and in the source of digested proteins whose peptides they display.

Class I MHC proteins (Fig. 5–22) are found on the surface of virtually all vertebrate cells. There are countless variants in the human population, placing them among the most polymorphic of proteins. Because each individual produces up to six class I MHC protein variants, any two individuals are unlikely to have the same set. Class I MHC proteins bind and display peptides derived from the proteolytic degradation and turnover of proteins that occurs randomly within the cell. These complexes of peptides and class I MHC proteins are the recognition targets of the T-cell receptors of the T_C cells in the cellular immune system. The general pattern of immune system recognition was first described by Rolf Zinkernagel and Peter Doherty in 1974.

Each T_C cell has many copies of only one T-cell receptor that is specific for a particular class I MHC protein–peptide complex. To avoid creating a legion of T_C cells that would set upon and destroy normal cells, the maturation of T_C cells in the thymus includes a stringent selection process that eliminates more than 95% of the developing T_C cells, including those that might recognize and bind class I MHC proteins displaying pep-

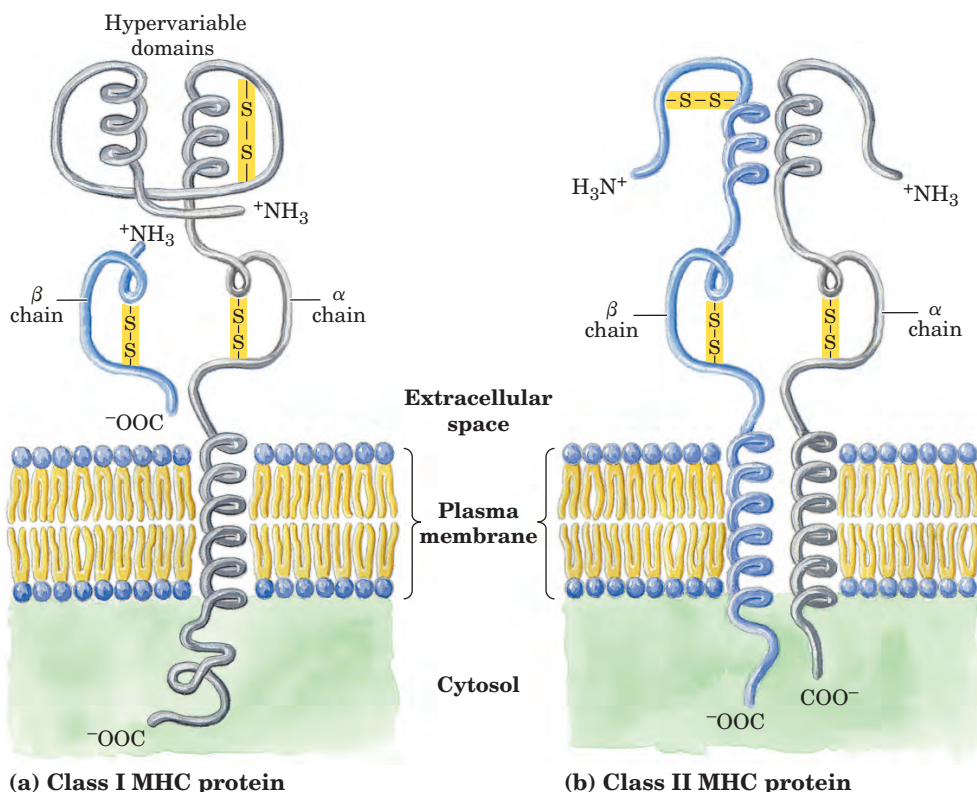
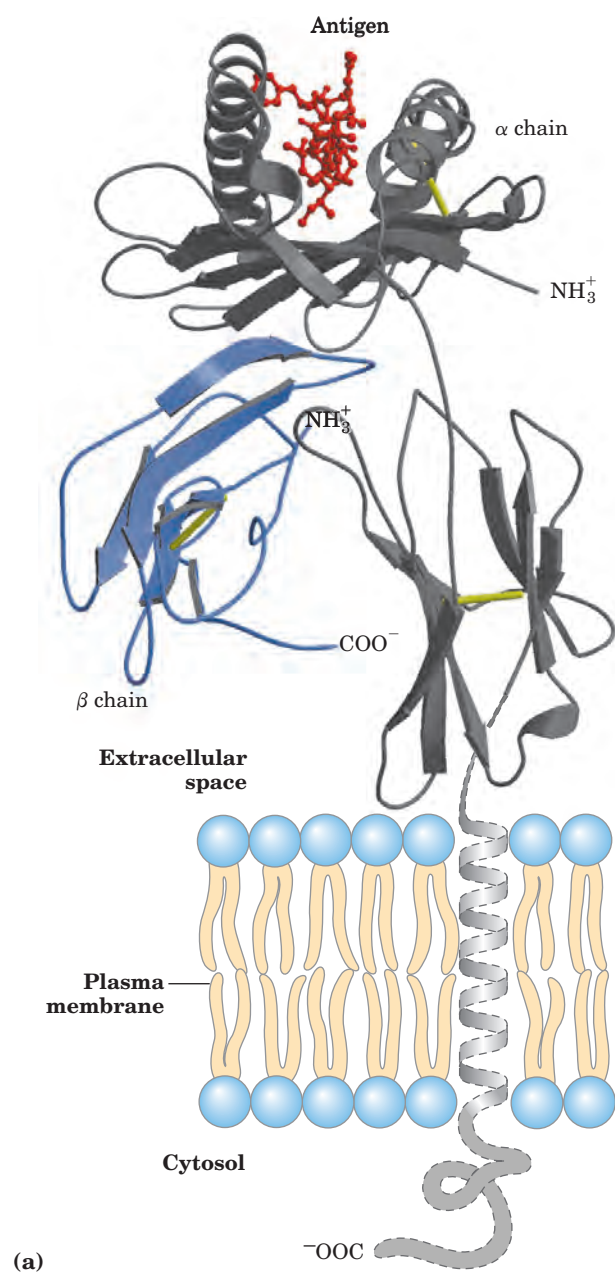


FIGURE 5–21 MHC proteins.

These proteins consist of α and β chains. (a) In class I MHC proteins, the small β chain is invariant but the amino acid sequence of the α chain exhibits a high degree of variability, localized in specific domains of the protein that appear on the outside of the cell. Each human produces up to six different α chains for class I MHC proteins. (b) In class II MHC proteins, both the α and β chains have regions of relatively high variability near their amino-terminal ends.



ptides from cellular proteins of the organism itself. The T_C cells that survive and mature are those with T-cell receptors that do not bind to the organism's own proteins. The result is a population of cells that bind foreign peptides bound to class I MHC proteins of the host cell. These binding interactions lead to the destruction of parasites and virus-infected cells. Following organ transplantation, the donor's class I MHC proteins, recognized as foreign, are bound by the recipient's T_C cells, leading to tissue rejection.

Class II MHC proteins occur on the surfaces of a few types of specialized cells, including macrophages and B lymphocytes that take up foreign antigens. Like class I MHC proteins, the class II proteins are highly polymorphic, with many variants in the human popula-



(b)

FIGURE 5-22 Structure of a human class I MHC protein. (a) This model is derived in part from the known structure of the extracellular portion of the protein (PDB ID 1DDH). The α chain of MHC is shown in gray; the small β chain is blue; the disulfide bonds are yellow. A bound ligand, a peptide derived from HIV, is shown in red. (b) Top view of the protein, showing a surface contour image of the site where peptides are bound and displayed. The HIV peptide (red) occupies the site. This part of the class I MHC protein interacts with T-cell receptors.

tion. Each human is capable of producing up to 12 variants, and thus it is unlikely that any two individuals have an identical set. The class II MHC proteins bind and display peptides derived not from cellular proteins but from external proteins ingested by the cells. The resulting class II MHC protein-peptide complexes are the binding targets of the T-cell receptors of the various helper T cells. T_H cells, like T_C cells, undergo a stringent selection process in the thymus, eliminating those that recognize the individual's own cellular proteins. **MHC Molecules**

Despite the elimination of most T_C and T_H cells during the selection process in the thymus, a very large number survive, and these provide the immune response. Each survivor has a single type of T-cell receptor that can bind to one particular chemical structure. The T cells patrolling the bloodstream and the tissues carry millions of different binding specificities in the T-cell receptors. Within the highly varied T-cell population there is almost always a contingent of cells that can specifically bind any antigen that might appear. The vast majority of these cells never encounter a foreign antigen to which they can bind, and they typically die within a few days, replaced by new generations of T cells endlessly patrolling in search of the interaction that will launch the full immune response.

The T_H cells participate only indirectly in the destruction of infected cells and pathogens, stimulating the selective proliferation of those T_C and B cells that

can bind to a particular antigen. This process, called **clonal selection**, increases the number of immune system cells that can respond to a particular pathogen. The importance of T_H cells is dramatically illustrated by the epidemic produced by HIV (human immunodeficiency virus), the virus that causes AIDS (acquired immune deficiency syndrome). The primary targets of HIV infection are T_H cells. Elimination of these cells progressively incapacitates the entire immune system.

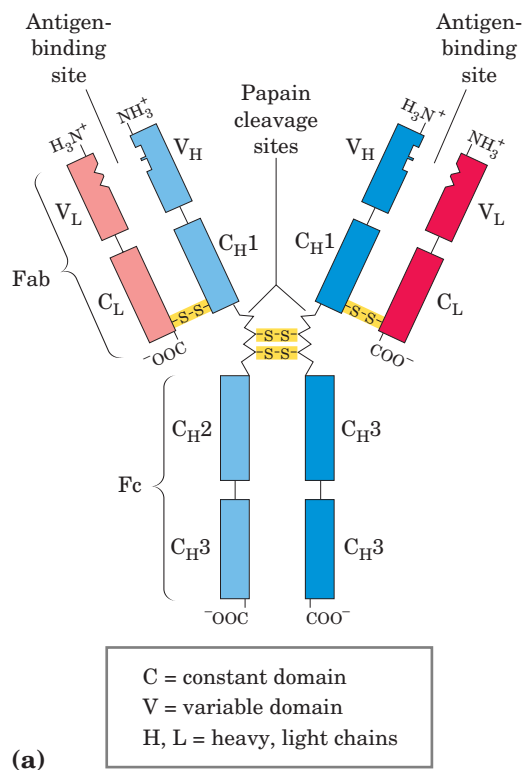
Antibodies Have Two Identical Antigen-Binding Sites

Immunoglobulin G (IgG) is the major class of antibody molecule and one of the most abundant proteins in the blood serum. IgG has four polypeptide chains: two large ones, called heavy chains, and two light chains, linked by noncovalent and disulfide bonds into a complex of M_r 150,000. The heavy chains of an IgG molecule interact at one end, then branch to interact separately with the light chains, forming a Y-shaped molecule (Fig. 5–23). At the “hinges” separating the base of an IgG molecule from its branches, the immunoglobulin can be cleaved with proteases. Cleavage with the protease papain liberates the basal fragment, called **Fc** because it usually crystallizes readily, and the two branches, called

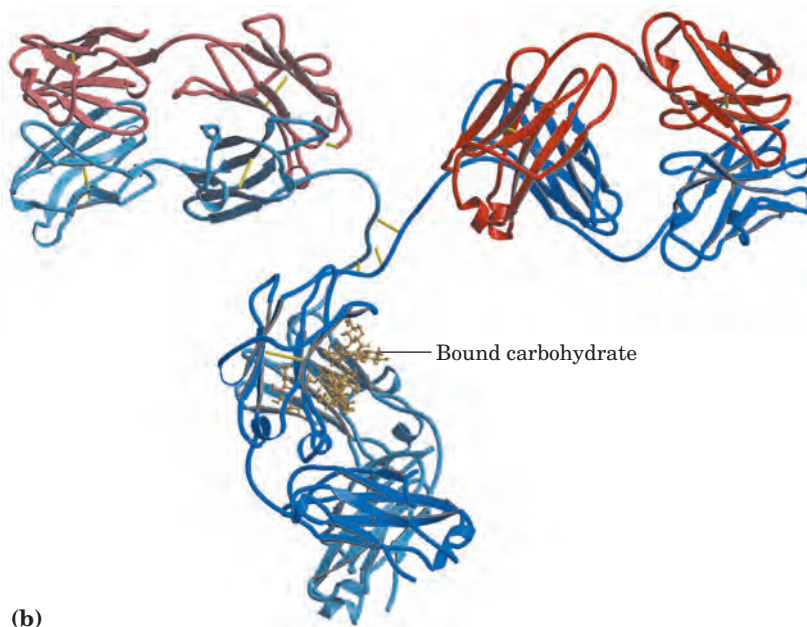
Fab, the antigen-binding fragments. Each branch has a single antigen-binding site.

The fundamental structure of immunoglobulins was first established by Gerald Edelman and Rodney Porter. Each chain is made up of identifiable domains; some are constant in sequence and structure from one IgG to the next, others are variable. The constant domains have a characteristic structure known as the **immunoglobulin fold**, a well-conserved structural motif in the all β class of proteins (Chapter 4). There are three of these constant domains in each heavy chain and one in each light chain. The heavy and light chains also have one variable domain each, in which most of the variability in amino acid residue sequence is found. The variable domains associate to create the antigen-binding site (Fig. 5–24).

In many vertebrates, IgG is but one of five classes of immunoglobulins. Each class has a characteristic type of heavy chain, denoted α , δ , ϵ , γ , and μ for IgA, IgD, IgE, IgG, and IgM, respectively. Two types of light chain, κ and λ , occur in all classes of immunoglobulins. The overall structures of **IgD** and **IgE** are similar to that of IgG. **IgM** occurs either in a monomeric, membrane-bound form or in a secreted form that is a cross-linked pentamer of this basic structure (Fig. 5–25). **IgA**, found



(a)



(b)

FIGURE 5–23 The structure of immunoglobulin G. (a) Pairs of heavy and light chains combine to form a Y-shaped molecule. Two antigen-binding sites are formed by the combination of variable domains from one light (V_L) and one heavy (V_H) chain. Cleavage with papain separates the Fab and Fc portions of the protein in the hinge region. The Fc portion of the molecule also contains bound carbohydrate.

(b) A ribbon model of the first complete IgG molecule to be crystallized and structurally analyzed (PDB ID 1IGT). Although the molecule contains two identical heavy chains (two shades of blue) and two identical light chains (two shades of red), it crystallized in the asymmetric conformation shown. Conformational flexibility may be important to the function of immunoglobulins.

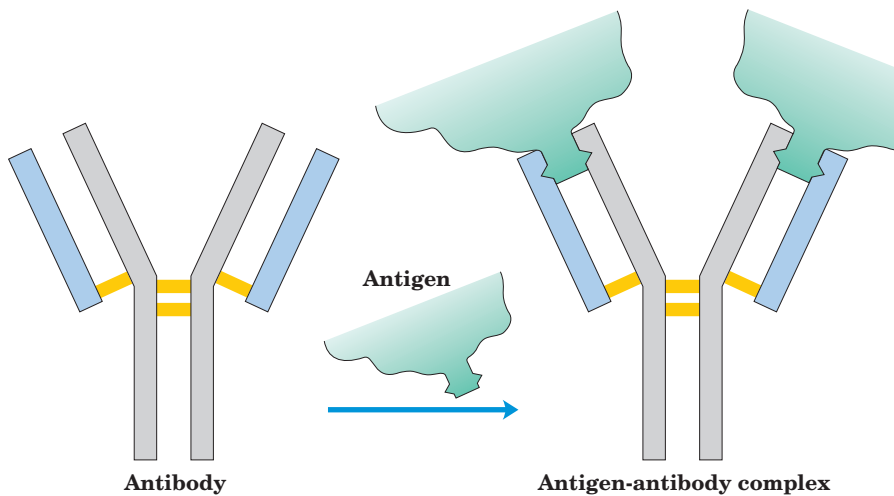


FIGURE 5-24 Binding of IgG to an antigen. To generate an optimal fit for the antigen, the binding sites of IgG often undergo slight conformational changes. Such induced fit is common to many protein-ligand interactions.

principally in secretions such as saliva, tears, and milk, can be a monomer, dimer, or trimer. IgM is the first antibody to be made by B lymphocytes and is the major antibody in the early stages of a primary immune response. Some B cells soon begin to produce IgD (with the same antigen-binding site as the IgM produced by the same cell), but the unique function of IgD is less clear.

The IgG described above is the major antibody in secondary immune responses, which are initiated by memory B cells. As part of the organism's ongoing immunity to antigens already encountered and dealt with, IgG is the most abundant immunoglobulin in the blood. When IgG binds to an invading bacterium or virus, it

activates certain leukocytes such as macrophages to engulf and destroy the invader, and also activates some other parts of the immune response. Yet another class of receptors on the cell surface of macrophages recognizes and binds the Fc region of IgG. When these Fc receptors bind an antibody-pathogen complex, the macrophage engulfs the complex by phagocytosis (Fig. 5-26).

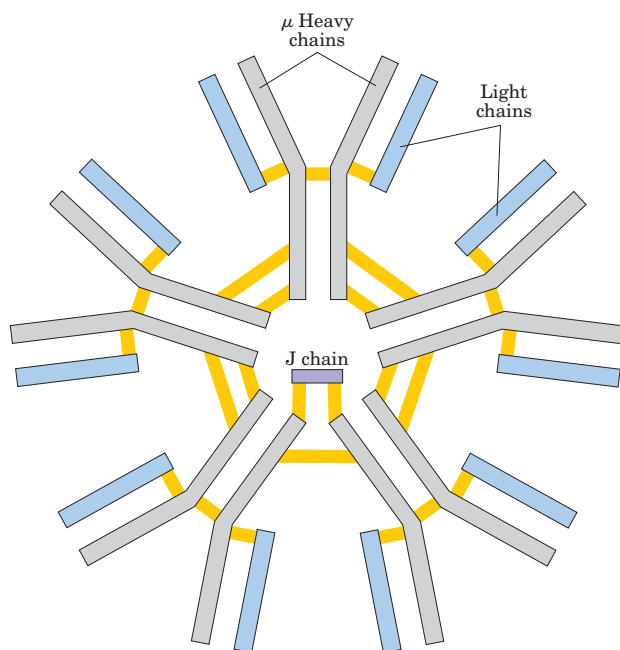


FIGURE 5-25 IgM pentamer of immunoglobulin units. The pentamer is cross-linked with disulfide bonds (yellow). The J chain is a polypeptide of M_r 20,000 found in both IgA and IgM.

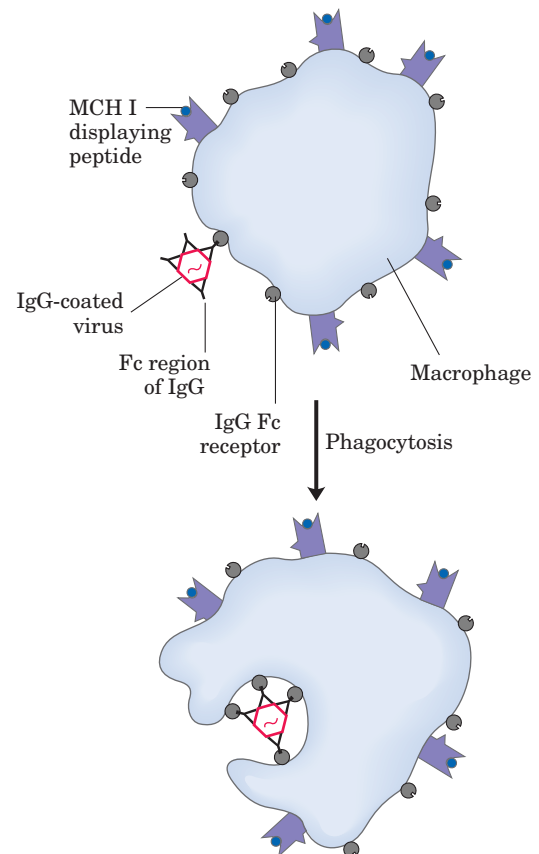


FIGURE 5-26 Phagocytosis of an antibody-bound virus by a macrophage. The Fc regions of the antibodies bind to Fc receptors on the surface of the macrophage, triggering the macrophage to engulf and destroy the virus.

IgE plays an important role in the allergic response, interacting with basophils (phagocytic leukocytes) in the blood and histamine-secreting cells called mast cells that are widely distributed in tissues. This immunoglobulin binds, through its Fc region, to special Fc receptors on the basophils or mast cells. In this form, IgE serves as a kind of receptor for antigen. If antigen is bound, the cells are induced to secrete histamine and other biologically active amines that cause dilation and increased permeability of blood vessels. These effects on the blood vessels are thought to facilitate the movement of immune system cells and proteins to sites of inflammation. They also produce the symptoms normally associated with allergies. Pollen or other allergens are recognized as foreign, triggering an immune response normally reserved for pathogens.

Antibodies Bind Tightly and Specifically to Antigen

The binding specificity of an antibody is determined by the amino acid residues in the variable domains of its heavy and light chains. Many residues in these domains are variable, but not equally so. Some, particularly those lining the antigen-binding site, are hypervariable—especially likely to differ. Specificity is conferred by chemical complementarity between the antigen and its specific binding site, in terms of shape and the location of charged, nonpolar, and hydrogen-bonding groups. For example, a binding site with a negatively charged group may bind an antigen with a positive charge in the complementary position. In many instances, complementarity is achieved interactively as the structures of antigen and binding site are influenced by each other during the approach of the ligand. Conformational changes in the antibody and/or the antigen then occur that allow the complementary groups to interact fully. This is an example of induced fit (Fig. 5–27).

A typical antibody-antigen interaction is quite strong, characterized by K_d values as low as 10^{-10} M (recall that a lower K_d corresponds to a stronger binding interaction). The K_d reflects the energy derived from the various ionic, hydrogen-bonding, hydrophobic, and van der Waals interactions that stabilize the binding. The binding energy required to produce a K_d of 10^{-10} M is about 65 kJ/mol.

The complex of a peptide derived from HIV (a model antigen) and an Fab molecule, shown in Figure 5–27, illustrates some of these properties. The changes in structure observed on antigen binding are particularly striking in this example.

The Antibody-Antigen Interaction Is the Basis for a Variety of Important Analytical Procedures

The extraordinary binding affinity and specificity of antibodies make them valuable analytical reagents. Two types of antibody preparations are in use: polyclonal and monoclonal. **Polyclonal antibodies** are those produced by many different B lymphocytes responding to one antigen, such as a protein injected into an animal. Cells in the population of B lymphocytes produce antibodies that bind specific, different epitopes within the antigen. Thus, polyclonal preparations contain a mixture of antibodies that recognize different parts of the protein. **Monoclonal antibodies**, in contrast, are synthesized by a population of identical B cells (a **clone**) grown in cell culture. These antibodies are homogeneous, all recognizing the same epitope. The techniques for producing monoclonal antibodies were developed by Georges Köhler and Cesar Milstein.

The specificity of antibodies has practical uses. A selected antibody can be covalently attached to a resin and used in a chromatography column of the type shown in Figure 3–18c. When a mixture of proteins is added to

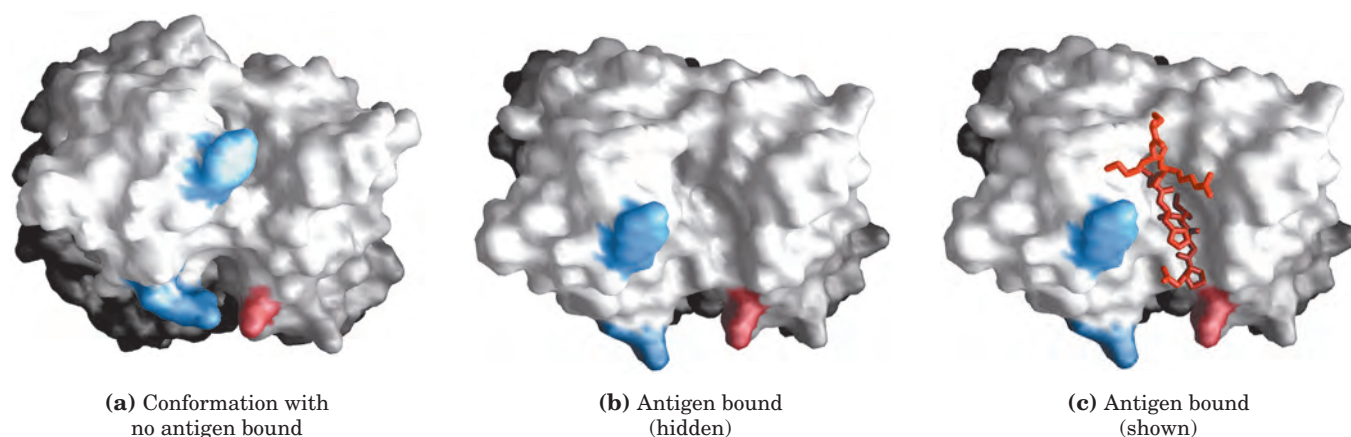


FIGURE 5–27 Induced fit in the binding of an antigen to IgG. The molecule, shown in surface contour, is the Fab fragment of an IgG. The antigen bound by this IgG is a small peptide derived from HIV. Two residues from the heavy chain (blue) and one from the light chain (pink) are colored to provide visual points of reference. (a) View of the Fab fragment, looking down on the antigen-binding site (PDB ID

1GCC). (b) The same view, but here the Fab fragment is in the “bound” conformation (PDB ID 1GGI); the antigen has been omitted from the image to provide an unobstructed view of the altered binding site. Note how the binding cavity has enlarged and several groups have shifted position. (c) The same view as in (b), but with the antigen in the binding site, pictured as a red stick structure.



Georges Köhler, 1946–1995



Cesar Milstein, 1927–2002

the column, the antibody specifically binds its target protein and retains it on the column while other proteins are washed through. The target protein can then be eluted from the resin by a salt solution or some other agent. This is a powerful tool for protein purification.

In another versatile analytical technique, an antibody is attached to a radioactive label or some other reagent that makes it easy to detect. When the antibody binds the target protein, the label reveals the presence of the protein in a solution or its location in a gel or even a living cell. Several variations of this procedure are illustrated in Figure 5–28.

An **ELISA** (enzyme-linked immunosorbent assay) allows for rapid screening and quantification of the presence of an antigen in a sample (Fig. 5–28b). Proteins in a sample are adsorbed to an inert surface, usually a 96-well polystyrene plate. The surface is washed with a solution of an inexpensive nonspecific protein (often casein from nonfat dry milk powder) to block proteins introduced in subsequent steps from also adsorbing to these surfaces. The surface is then treated with a solution containing the primary antibody—an antibody against the protein of interest. Unbound antibody is washed away and the surface is treated with a solution containing antibodies against the primary antibody. These secondary antibodies have been linked to an enzyme that catalyzes a reaction that forms a colored product. After unbound secondary antibody is washed away, the substrate of the antibody-linked enzyme is added. Product formation (monitored as color intensity) is proportional to the concentration of the protein of interest in the sample.

In an **immunoblot assay** (Fig. 5–28c), proteins that have been separated by gel electrophoresis are transferred electrophoretically to a nitrocellulose membrane. The membrane is blocked (as described above for ELISA), then treated successively with primary

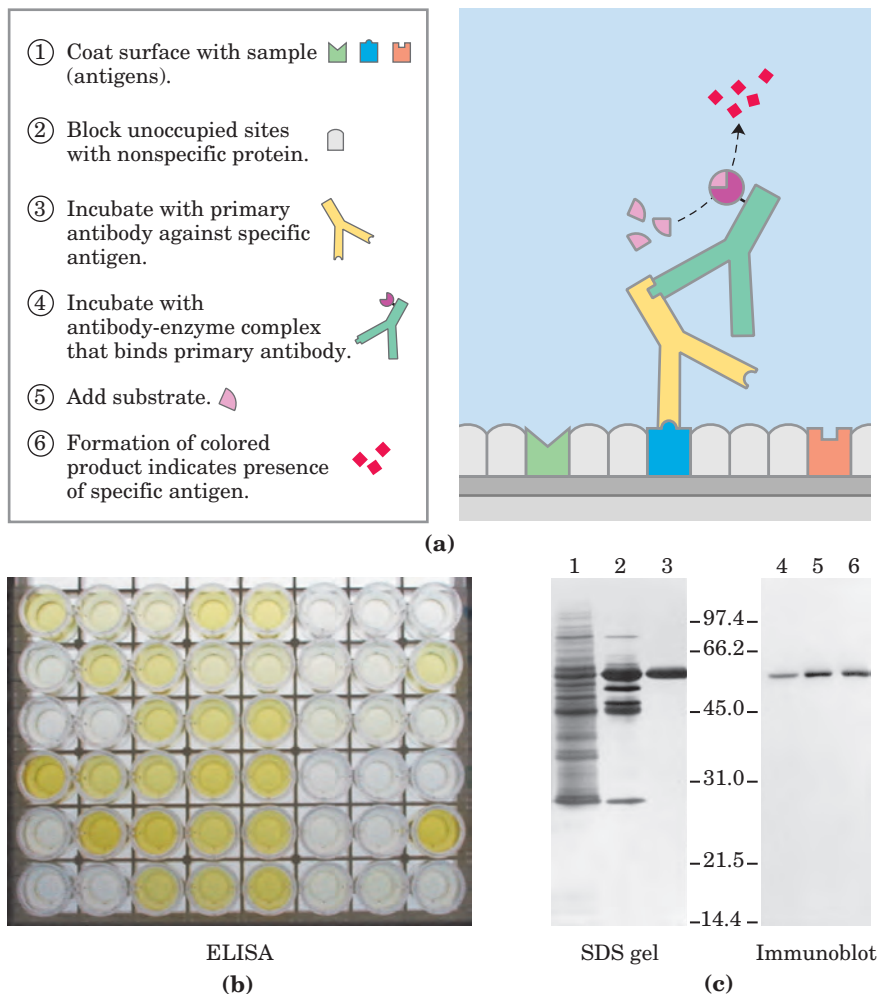


FIGURE 5–28 Antibody techniques. The specific reaction of an antibody with its antigen is the basis of several techniques that identify and quantify a specific protein in a complex sample. **(a)** A schematic representation of the general method. **(b)** An ELISA to test for the presence of herpes simplex virus (HSV) antibodies in blood samples. Wells were coated with an HSV antigen, to which antibodies against HSV will bind. The second antibody is anti-human IgG linked to horseradish peroxidase. Blood samples with greater amounts of HSV antibody turn brighter yellow. **(c)** An immunoblot. Lanes 1 to 3 are from an SDS gel; samples from successive stages in the purification of a protein kinase have been separated and stained with Coomassie blue. Lanes 4 to 6 show the same samples, but these were electrophoretically transferred to a nitrocellulose membrane after separation on an SDS gel. The membrane was then “probed” with antibody against the protein kinase. The numbers between the SDS gel and the immunoblot indicate M_r in thousands.

antibody, secondary antibody linked to enzyme, and substrate. A colored precipitate forms only along the band containing the protein of interest. The immunoblot allows the detection of a minor component in a sample and provides an approximation of its molecular weight.

Immunoblotting

We will encounter other aspects of antibodies in later chapters. They are extremely important in medicine and can tell us much about the structure of proteins and the action of genes.

SUMMARY 5.2 Complementary Interactions between Proteins and Ligands: The Immune System and Immunoglobulins

- The immune response is mediated by interactions among an array of specialized leukocytes and their associated proteins. T lymphocytes produce T-cell receptors. B lymphocytes produce immunoglobulins. All cells produce MHC proteins, which display host (self) or antigenic (nonself) peptides on the cell surface. In a process called clonal selection, helper T cells induce the proliferation of B cells and cytotoxic T cells that produce immunoglobulins or of T-cell receptors that bind to a specific antigen.
- Humans have five classes of immunoglobulins, each with different biological functions. The most abundant class is IgG, a Y-shaped protein with two heavy and two light chains. The domains near the upper ends of the Y are hypervariable within the broad population of IgGs and form two antigen-binding sites.
- A given immunoglobulin generally binds to only a part, called the epitope, of a large antigen. Binding often involves a conformational change in the IgG, an induced fit to the antigen.

5.3 Protein Interactions Modulated by Chemical Energy: Actin, Myosin, and Molecular Motors

Organisms move. Cells move. Organelles and macromolecules within cells move. Most of these movements arise from the activity of a fascinating class of protein-based molecular motors. Fueled by chemical energy, usually derived from ATP, large aggregates of motor proteins undergo cyclic conformational changes that accumulate into a unified, directional force—the tiny force that pulls apart chromosomes in a dividing cell, and the immense force that levers a pouncing, quarter-ton jungle cat into the air.

The interactions among motor proteins, as you might predict, feature complementary arrangements of ionic, hydrogen-bonding, hydrophobic, and van der Waals interactions at protein binding sites. In motor proteins, however, these interactions achieve exceptionally high levels of spatial and temporal organization.

Motor proteins underlie the contraction of muscles, the migration of organelles along microtubules, the rotation of bacterial flagella, and the movement of some proteins along DNA. Proteins called kinesins and dyneins move along microtubules in cells, pulling along organelles or reorganizing chromosomes during cell division. An interaction of dynein with microtubules brings about the motion of eukaryotic flagella and cilia. Flagellar motion in bacteria involves a complex rotational motor at the base of the flagellum (see Fig. 19–35). Helicases, polymerases, and other proteins move along DNA as they carry out their functions in DNA metabolism (Chapter 25). Here, we focus on the well-studied example of the contractile proteins of vertebrate skeletal muscle as a paradigm for how proteins translate chemical energy into motion.

The Major Proteins of Muscle Are Myosin and Actin

The contractile force of muscle is generated by the interaction of two proteins, myosin and actin. These proteins are arranged in filaments that undergo transient interactions and slide past each other to bring about contraction. Together, actin and myosin make up more than 80% of the protein mass of muscle.

Myosin (M_r 540,000) has six subunits: two heavy chains (each of M_r 220,000) and four light chains (each of M_r 20,000). The heavy chains account for much of the overall structure. At their carboxyl termini, they are arranged as extended α helices, wrapped around each other in a fibrous, left-handed coiled coil similar to that of α -keratin (Fig. 5–29a). At its amino terminus, each heavy chain has a large globular domain containing a site where ATP is hydrolyzed. The light chains are associated with the globular domains. When myosin is treated briefly with the protease trypsin, much of the fibrous tail is cleaved off, dividing the protein into components called light and heavy meromyosin (Fig. 5–29b). The globular domain, called myosin subfragment 1, or S1, or simply the myosin head group, is liberated from heavy meromyosin by cleavage with papain. The S1 fragment produced by this procedure is the motor domain that makes muscle contraction possible. S1 fragments can be crystallized and their structure has been determined. The overall structure of the S1 fragment as determined by Ivan Rayment and Hazel Holden is shown in Figure 5–29c.

In muscle cells, molecules of myosin aggregate to form structures called **thick filaments** (Fig. 5–30a). These rodlike structures serve as the core of the con-

tractile unit. Within a thick filament, several hundred myosin molecules are arranged with their fibrous “tails” associated to form a long bipolar structure. The globular domains project from either end of this structure, in regular stacked arrays.

The second major muscle protein, **actin**, is abundant in almost all eukaryotic cells. In muscle, molecules

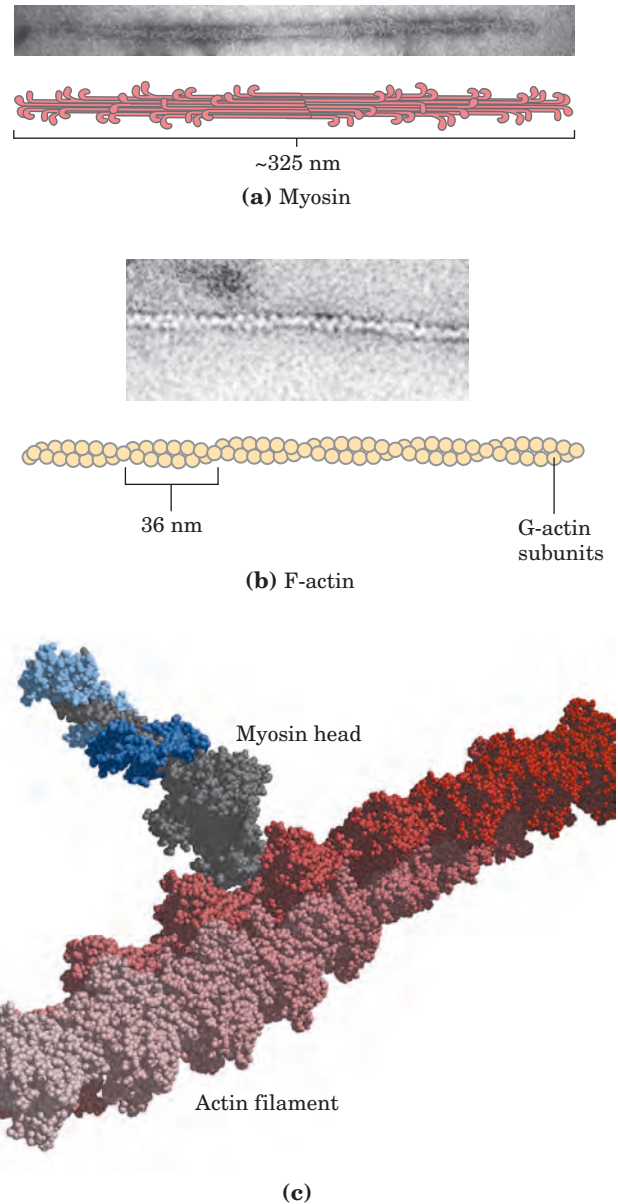
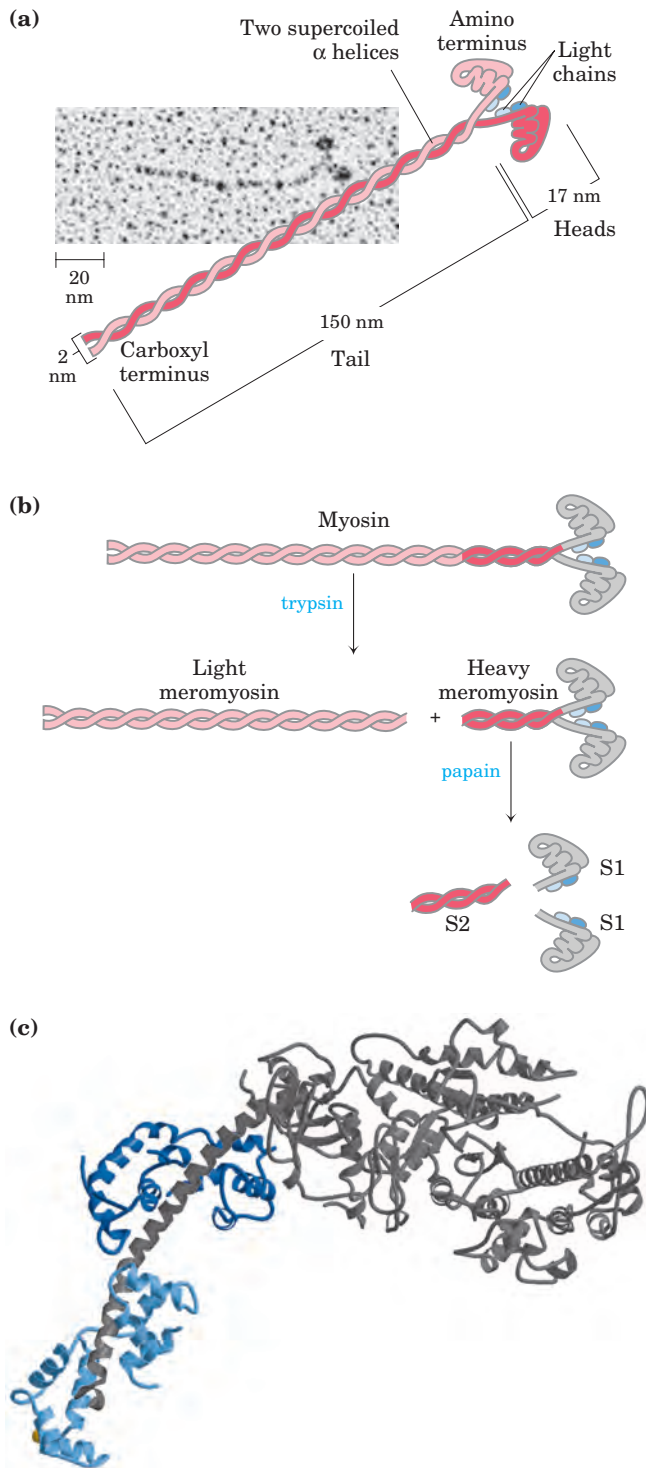


FIGURE 5-30 The major components of muscle. (a) Myosin aggregates to form a bipolar structure called a thick filament. (b) F-actin is a filamentous assemblage of G-actin monomers that polymerize two by two, giving the appearance of two filaments spiraling about one another in a right-handed fashion. An electron micrograph and a model of the myosin thick filament and F-actin are shown. (c) Space-filling model of an actin filament (shades of red) with one myosin head (gray and two shades of blue) bound to an actin monomer within the filament. (From coordinates supplied by Ivan Rayment.)

◀ **FIGURE 5-29 (at left) Myosin.** (a) Myosin has two heavy chains (in two shades of pink), the carboxyl termini forming an extended coiled coil (tail) and the amino termini having globular domains (heads). Two light chains (blue) are associated with each myosin head. (b) Cleavage with trypsin and papain separates the myosin heads (S1 fragments) from the tails. (c) Ribbon representation of the myosin S1 fragment. The heavy chain is in gray, the two light chains in two shades of blue. (From coordinates supplied by Ivan Rayment.)

of monomeric actin, called G-actin (*globular actin*; M_r 42,000), associate to form a long polymer called F-actin (*filamentous actin*). The **thin filament** (Fig. 5–30b) consists of F-actin, along with the proteins troponin and tropomyosin. The filamentous parts of thin filaments assemble as successive monomeric actin molecules add to one end. On addition, each monomer binds ATP, then hydrolyzes it to ADP, so every actin molecule in the filament is complexed to ADP. This ATP hydrolysis by actin functions only in the assembly of the filaments; it does not contribute directly to the energy expended in muscle contraction. Each actin monomer in the thin filament can bind tightly and specifically to one myosin head group (Fig. 5–30c).

Additional Proteins Organize the Thin and Thick Filaments into Ordered Structures

Skeletal muscle consists of parallel bundles of **muscle fibers**, each fiber a single, very large, multinucleated cell, 20 to 100 μm in diameter, formed from many cells fused together and often spanning the length of the muscle. Each fiber, in turn, contains about 1,000 **myofibrils**, 2 μm in diameter, each consisting of a vast number of regularly arrayed thick and thin filaments complexed to other proteins (Fig. 5–31). A system of flat membranous vesicles called the **sarcoplasmic reticulum** surrounds each myofibril. Examined under the electron microscope, muscle fibers reveal alternating regions of high and low electron density, called the **A bands** and **I bands** (Fig. 5–31b, c). The A and I bands arise from the arrangement of thick and thin filaments,

which are aligned and partially overlapping. The I band is the region of the bundle that in cross section would contain only thin filaments. The darker A band stretches the length of the thick filament and includes the region where parallel thick and thin filaments overlap. Bisecting the I band is a thin structure called the **Z disk**, perpendicular to the thin filaments and serving as an anchor to which the thin filaments are attached. The A band too is bisected by a thin line, the **M line** or M disk, a region of high electron density in the middle of the thick filaments. The entire contractile unit, consisting of bundles of thick filaments interleaved at either end with bundles of thin filaments, is called the **sarcomere**. The arrangement of interleaved bundles allows the thick and thin filaments to slide past each other (by a mechanism discussed below), causing a progressive shortening of each sarcomere (Fig. 5–32).

The thin actin filaments are attached at one end to the Z disk in a regular pattern. The assembly includes the minor muscle proteins **α -actinin**, **desmin**, and **vimentin**. Thin filaments also contain a large protein called **nebulin** (~7,000 amino acid residues), thought to be structured as an α helix that is long enough to span the length of the filament. The M line similarly organizes the thick filaments. It contains the proteins **paramyosin**, **C-protein**, and **M-protein**. Another class of proteins called **titins**, the largest single polypeptide chains discovered thus far (the titin of human cardiac muscle has 26,926 amino acid residues), link the thick filaments to the Z disk, providing additional organization to the overall structure. Among their structural functions, the proteins nebulin and titin are believed to act as “molecular

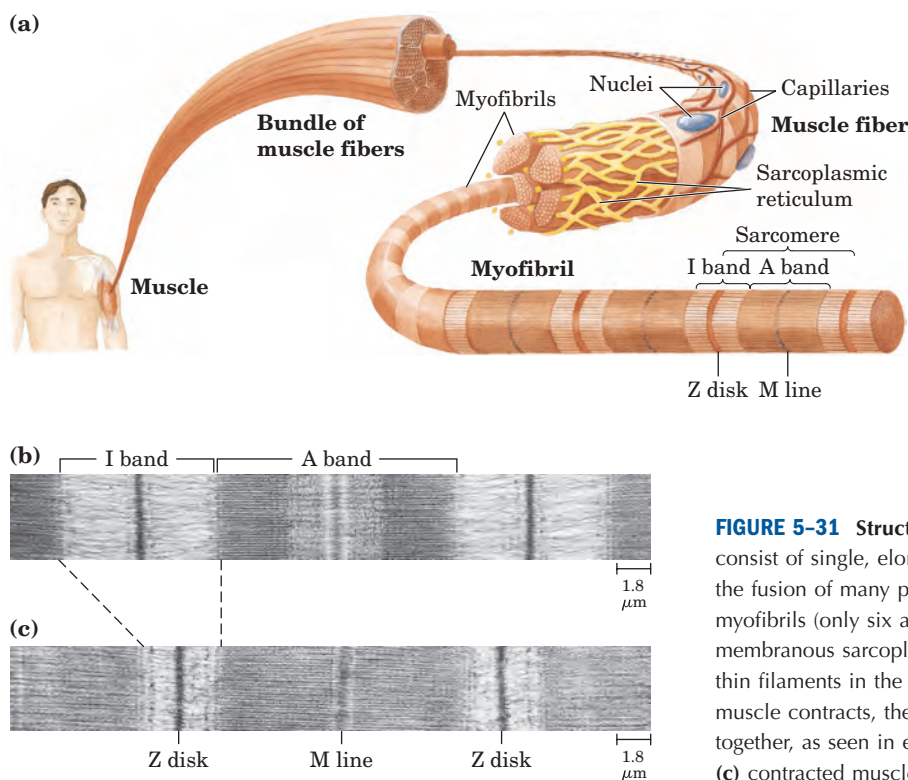


FIGURE 5–31 Structure of skeletal muscle. (a) Muscle fibers consist of single, elongated, multinucleated cells that arise from the fusion of many precursor cells. Within the fibers are many myofibrils (only six are shown here for simplicity) surrounded by the membranous sarcoplasmic reticulum. The organization of thick and thin filaments in the myofibril gives it a striated appearance. When muscle contracts, the I bands narrow and the Z disks come closer together, as seen in electron micrographs of (b) relaxed and (c) contracted muscle.

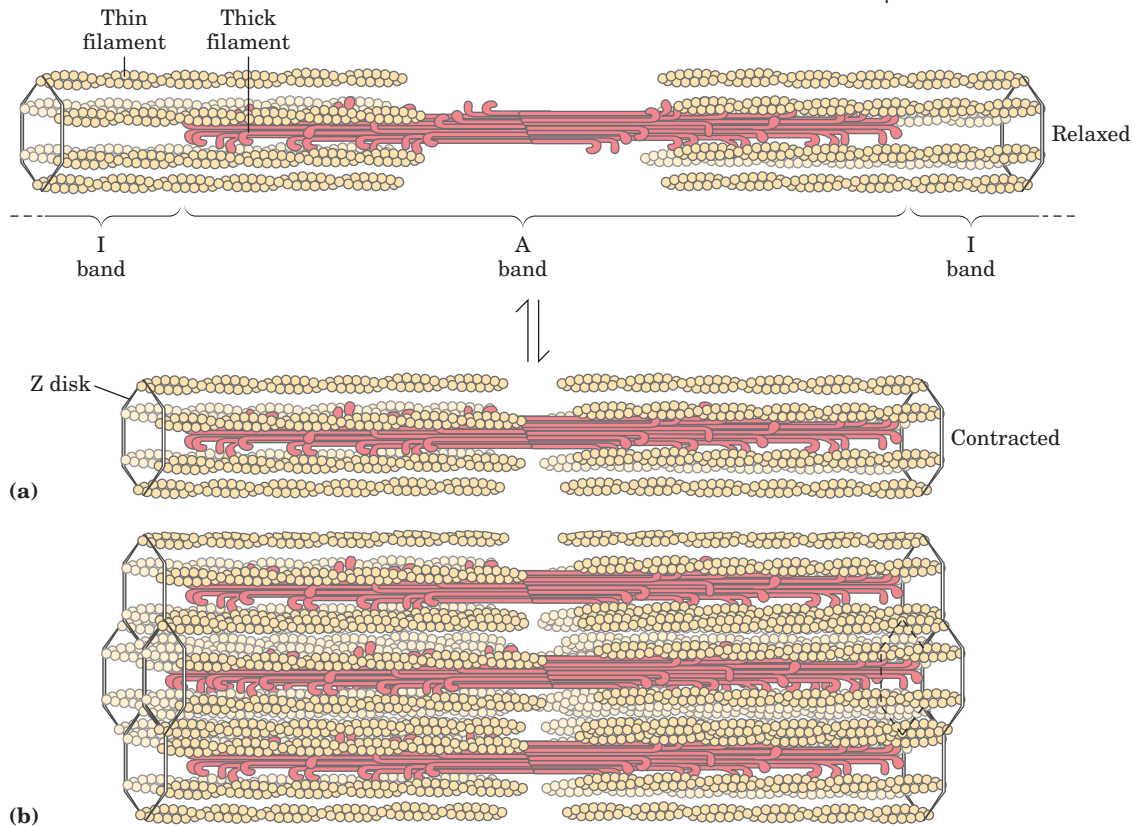


FIGURE 5-32 Muscle contraction. Thick filaments are bipolar structures created by the association of many myosin molecules. **(a)** Muscle contraction occurs by the sliding of the thick and thin filaments

past each other so that the Z disks in neighboring I bands approach each other. **(b)** The thick and thin filaments are interleaved such that each thick filament is surrounded by six thin filaments.

rulers,” regulating the length of the thin and thick filaments, respectively. Titin extends from the Z disk to the M line, regulating the length of the sarcomere itself and preventing overextension of the muscle. The characteristic sarcomere length varies from one muscle tissue to the next in a vertebrate organism, a finding attributed in large part to the different titin variants in the tissues.

Myosin Thick Filaments Slide along Actin Thin Filaments

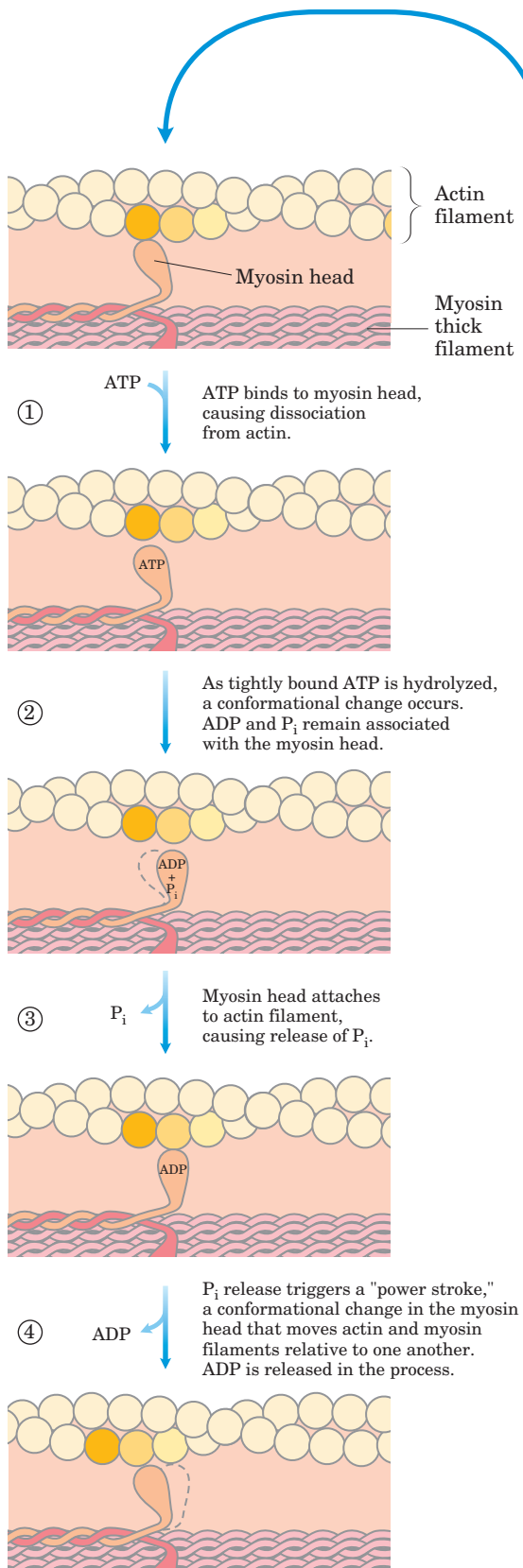
The interaction between actin and myosin, like that between all proteins and ligands, involves weak bonds. When ATP is not bound to myosin, a face on the myosin head group binds tightly to actin (Fig. 5-33). When ATP binds to myosin and is hydrolyzed to ADP and phosphate, a coordinated and cyclic series of conformational changes occurs in which myosin releases the F-actin subunit and binds another subunit farther along the thin filament.

The cycle has four major steps (Fig. 5-33). In step ①, ATP binds to myosin and a cleft in the myosin molecule opens, disrupting the actin-myosin interaction so that the bound actin is released. ATP is then hydrolyzed in step ②, causing a conformational change in the protein to a “high-energy” state that moves the myosin head and changes its orientation in relation to the actin thin filament. Myosin then binds weakly to an F-actin subunit

closer to the Z disk than the one just released. As the phosphate product of ATP hydrolysis is released from myosin in step ③, another conformational change occurs in which the myosin cleft closes, strengthening the myosin-actin binding. This is followed quickly by step ④, a “power stroke” during which the conformation of the myosin head returns to the original resting state, its orientation relative to the bound actin changing so as to pull the tail of the myosin toward the Z disk. ADP is then released to complete the cycle. Each cycle generates about 3 to 4 pN (piconewtons) of force and moves the thick filament 5 to 10 nm relative to the thin filament.

Because there are many myosin heads in a thick filament, at any given moment some (probably 1% to 3%) are bound to the thin filaments. This prevents the thick filaments from slipping backward when an individual myosin head releases the actin subunit to which it was bound. The thick filament thus actively slides forward past the adjacent thin filaments. This process, coordinated among the many sarcomeres in a muscle fiber, brings about muscle contraction.

The interaction between actin and myosin must be regulated so that contraction occurs only in response to appropriate signals from the nervous system. The regulation is mediated by a complex of two proteins, **tropomyosin** and **troponin**. Tropomyosin binds to the thin filament, blocking the attachment sites for the myosin head groups. Troponin is a Ca^{2+} -binding protein.



A nerve impulse causes release of Ca^{2+} from the sarcoplasmic reticulum. The released Ca^{2+} binds to troponin (another protein-ligand interaction) and causes a conformational change in the tropomyosin-troponin complexes, exposing the myosin-binding sites on the thin filaments. Contraction follows.

Working skeletal muscle requires two types of molecular functions that are common in proteins—binding and catalysis. The actin-myosin interaction, a protein-ligand interaction like that of immunoglobulins with antigens, is reversible and leaves the participants unchanged. When ATP binds myosin, however, it is hydrolyzed to ADP and P_i . Myosin is not only an actin-binding protein, it is also an ATPase—an enzyme. The function of enzymes in catalyzing chemical transformations is the topic of the next chapter.

SUMMARY 5.3 Protein Interactions Modulated by Chemical Energy: Actin, Myosin, and Molecular Motors

- Protein-ligand interactions achieve a special degree of spatial and temporal organization in motor proteins. Muscle contraction results from choreographed interactions between myosin and actin, coupled to the hydrolysis of ATP by myosin.
- Myosin consists of two heavy and four light chains, forming a fibrous coiled coil (tail) domain and a globular (head) domain. Myosin molecules are organized into thick filaments, which slide past thin filaments composed largely of actin. ATP hydrolysis in myosin is coupled to a series of conformational changes in the myosin head, leading to dissociation of myosin from one F-actin subunit and its eventual reassociation with another, farther along the thin filament. The myosin thus slides along the actin filaments.
- Muscle contraction is stimulated by the release of Ca^{2+} from the sarcoplasmic reticulum. The Ca^{2+} binds to the protein troponin, leading to a conformational change in a troponin-tropomyosin complex that triggers the cycle of actin-myosin interactions.

FIGURE 5-33 Molecular mechanism of muscle contraction. Conformational changes in the myosin head that are coupled to stages in the ATP hydrolytic cycle cause myosin to successively dissociate from one actin subunit, then associate with another farther along the actin filament. In this way the myosin heads slide along the thin filaments, drawing the thick filament array into the thin filament array (see Fig. 5–32).

Key Terms

Terms in bold are defined in the glossary.

ligand 157	allosteric protein 165	epitope 175
binding site 157	Hill equation 167	hapten 176
induced fit 158	Bohr effect 170	immunoglobulin fold 178
heme 158	lymphocytes 175	polyclonal antibodies 180
porphyrin 158	antibody 175	monoclonal antibodies 180
globins 159	immunoglobulin 175	ELISA 181
equilibrium expression 160	B lymphocytes or B cells 175	myosin 182
association constant, K_a 160	T lymphocytes or T cells 175	actin 183
dissociation constant, K_d 160	antigen 175	sarcomere 184

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Problems

1. Relationship between Affinity and Dissociation Constant Protein A has a binding site for ligand X with a K_d of 10^{-6} M. Protein B has a binding site for ligand X with a K_d of 10^{-9} M. Which protein has a higher affinity for ligand X? Explain your reasoning. Convert the K_d to K_a for both proteins.

2. Negative Cooperativity Which of the following situations would produce a Hill plot with $n_H < 1.0$? Explain your reasoning in each case.

(a) The protein has multiple subunits, each with a single ligand-binding site. Binding of ligand to one site decreases the binding affinity of other sites for the ligand.

(b) The protein is a single polypeptide with two ligand-binding sites, each having a different affinity for the ligand.

(c) The protein is a single polypeptide with a single ligand-binding site. As purified, the protein preparation is heterogeneous, containing some protein molecules that are partially denatured and thus have a lower binding affinity for the ligand.

3. Affinity for Oxygen in Myoglobin and Hemoglobin What is the effect of the following changes on the O_2 affinity of myoglobin and hemoglobin? (a) A drop in the pH of blood plasma from 7.4 to 7.2. (b) A decrease in the partial pressure of CO_2 in the lungs from 6 kPa (holding one's breath) to 2 kPa (normal). (c) An increase in the BPG level from 5 mM (normal altitudes) to 8 mM (high altitudes).

4. Cooperativity in Hemoglobin Under appropriate conditions, hemoglobin dissociates into its four subunits. The isolated α subunit binds oxygen, but the O_2 -saturation curve is hyperbolic rather than sigmoid. In addition, the binding of oxygen to the isolated α subunit is not affected by the presence of H^+ , CO_2 , or BPG. What do these observations indicate about the source of the cooperativity in hemoglobin?

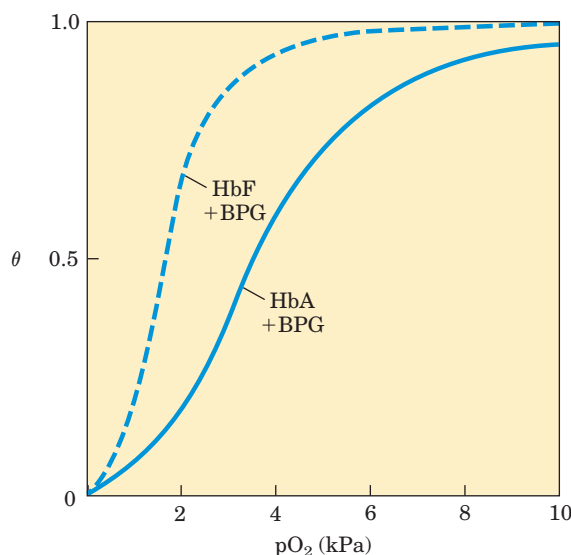
5. Comparison of Fetal and Maternal Hemoglobins Studies of oxygen transport in pregnant mammals have shown that the O_2 -saturation curves of fetal and maternal blood are markedly different when measured under the same conditions. Fetal erythrocytes contain a structural variant of hemoglobin, HbF, consisting of two α and two γ subunits ($\alpha_2\gamma_2$), whereas maternal erythrocytes contain HbA ($\alpha_2\beta_2$).

(a) Which hemoglobin has a higher affinity for oxygen under physiological conditions, HbA or HbF? Explain.

(b) What is the physiological significance of the different O_2 affinities?

(c) When all the BPG is carefully removed from samples of HbA and HbF, the measured O_2 -saturation curves (and consequently the O_2 affinities) are displaced to the left. However,

HbA now has a greater affinity for oxygen than does HbF. When BPG is reintroduced, the O_2 -saturation curves return to normal, as shown in the graph. What is the effect of BPG on the O_2 affinity of hemoglobin? How can the above information be used to explain the different O_2 affinities of fetal and maternal hemoglobin?



6. Hemoglobin Variants There are almost 500 naturally occurring variants of hemoglobin. Most are the result of a single amino acid substitution in a globin polypeptide chain. Some variants produce clinical illness, though not all variants have deleterious effects. A brief sample is presented below.

HbS (sickle-cell Hb): substitutes a Val for a Glu on the surface
Hb Cowtown: eliminates an ion pair involved in T-state stabilization

Hb Memphis: substitutes one uncharged polar residue for another of similar size on the surface

Hb Bibba: substitutes a Pro for a Leu involved in an α helix
Hb Milwaukee: substitutes a Glu for a Val

Hb Providence: substitutes an Asn for a Lys that normally projects into the central cavity of the tetramer

Hb Philly: substitutes a Phe for a Tyr, disrupting hydrogen bonding at the $\alpha_1\beta_1$ interface

Explain your choices for each of the following:

(a) The Hb variant *least* likely to cause pathological symptoms.

(b) The variant(s) most likely to show pI values different from that of HbA when run on an isoelectric focusing gel.

(c) The variant(s) most likely to show a decrease in BPG binding and an increase in the overall affinity of the hemoglobin for oxygen.

7. Reversible (but Tight) Binding to an Antibody An antibody binds to an antigen with a K_d of 5×10^{-8} M. At what concentration of antigen will θ be (a) 0.2, (b) 0.5, (c) 0.6, (d) 0.8?

8. Using Antibodies to Probe Structure-Function Relationships in Proteins A monoclonal antibody binds to G-actin but not to F-actin. What does this tell you about the epitope recognized by the antibody?



9. The Immune System and Vaccines A host organism needs time, often days, to mount an immune response against a new antigen, but memory cells permit a rapid response to pathogens previously encountered. A vaccine to protect against a particular viral infection often consists of weakened or killed virus or isolated proteins from a viral protein coat. When injected into a human patient, the vaccine generally does not cause an infection and illness, but it effectively “teaches” the immune system what the viral particles look like, stimulating the production of memory cells. On subsequent infection, these cells can bind to the virus and trigger a rapid immune response. Some pathogens, including HIV, have developed mechanisms to evade the immune system, making it difficult or impossible to develop effective vaccines against them. What strategy could a pathogen use to evade the immune system? Assume that antibodies and/or T-cell receptors are available to bind to any structure that might appear on the surface of a pathogen and that, once bound, the pathogen is destroyed.

10. How We Become a “Stiff” When a higher vertebrate dies, its muscles stiffen as they are deprived of ATP, a state called rigor mortis. Explain the molecular basis of the rigor state.

11. Sarcomeres from Another Point of View The symmetry of thick and thin filaments in a sarcomere is such that six thin filaments ordinarily surround each thick filament in a hexagonal array. Draw a cross section (transverse cut) of a myofibril at the following points: (a) at the M line; (b) through the I band; (c) through the dense region of the A band; (d) through the less dense region of the A band, adjacent to the M line (see Fig. 5–31b, c).

Biochemistry on the Internet

12. Lysozyme and Antibodies To fully appreciate how proteins function in a cell, it is helpful to have a three-dimensional view of how proteins interact with other cellular components. Fortunately, this is possible using on-line protein databases and the three-dimensional molecular viewing utilities Chime and Protein Explorer. If you have not yet installed the Chime plug-in on your computer, go to www.mdlchime.com/chime and follow the instructions for your operating system and browser. Once chime is installed, go to the Protein Data Bank (www.rcsb.org/pdb).

In this exercise you will examine the interactions between the enzyme lysozyme (Chapter 4) and the Fab portion of the anti-lysozyme antibody. Use the PDB identifier 1FDL to explore the structure of the IgG1 Fab fragment–lysozyme complex (antibody-antigen complex). View the structure using Protein Explorer, and also use the information in the PDBsum summary of the structure to answer the following questions.

(a) Which chains in the three-dimensional model correspond to the antibody fragment and which correspond to the antigen, lysozyme?

(b) What secondary structure predominates in this Fab fragment?

(c) How many amino acid residues are in the heavy and light chains of the Fab fragment? In lysozyme? Estimate the percentage of the lysozyme that interacts with the antigen-binding site of the antibody fragment.

(d) Identify the specific amino acid residues in lysozyme and in the variable regions of the Fab heavy and light chains that appear to be situated at the antigen-antibody interface. Are the residues contiguous in the primary sequence of the polypeptide chains?

13. Exploring Reversible Interactions of Proteins and Ligands with Living Graphs  Use the living graphs for Equations 5–8, 5–11, 5–14, and 5–16 to work through the following exercises.

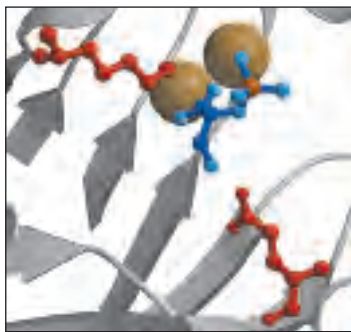
(a) Reversible binding of a ligand to a simple protein, without cooperativity. For Equation 5–8, set up a plot of θ versus $[L]$ (vertical and horizontal axes, respectively). Examine the plots generated when K_d is set at 5 μM , 10 μM , 20 μM , and 100 μM . Higher affinity of the protein for the ligand means more binding at lower ligand concentrations. Suppose that four different proteins exhibit these four different K_d values for ligand L. Which protein would have the highest affinity for L?

Examine the plot generated when $K_d = 10 \mu\text{M}$. How much does θ increase when $[L]$ increases from 0.2 μM to 0.4 μM ? How much does θ increase when $[L]$ increases from 40 μM to 80 μM ?

You can do the same exercise for Equation 5–11. Convert $[L]$ to $p\text{O}_2$ and K_d to P_{50} . Examine the curves generated when P_{50} is set at 0.5 kPa, 1 kPa, 2 kPa, and 10 kPa. For the curve generated when $P_{50} = 1$ kPa, how much does θ change when the $p\text{O}_2$ increases from 0.02 kPa to 0.04 kPa? From 4 kPa to 8 kPa?

(b) Cooperative binding of a ligand to a multisubunit protein. Using Equation 5–14, generate a binding curve for a protein and ligand with $K_d = 10 \mu\text{M}$ and $n = 3$. Note the altered definition of K_d in Equation 5–16. On the same plot, add a curve for a protein with $K_d = 20 \mu\text{M}$ and $n = 3$. Now see how both curves change when you change to $n = 4$. Generate Hill plots (Eqn 5–16) for each of these cases. For $K_d = 10 \mu\text{M}$ and $n = 3$, what is θ when $[L] = 20 \mu\text{M}$?

(c) Explore these equations further by varying all the parameters used above.



chapter 6

ENZYMES

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- 6.2 How Enzymes Work 193**
- 6.3 Enzyme Kinetics as an Approach to Understanding Mechanism 202**
- 6.4 Examples of Enzymatic Reactions 213**
- 6.5 Regulatory Enzymes 225**

One way in which this condition might be fulfilled would be if the molecules when combined with the enzyme, lay slightly further apart than their equilibrium distance when [covalently joined], but nearer than their equilibrium distance when free. . . . Using Fischer's lock and key simile, the key does not fit the lock quite perfectly but exercises a certain strain on it.

—J. B. S. Haldane, *Enzymes*, 1930

Catalysis can be described formally in terms of a stabilization of the transition state through tight binding to the catalyst.

—William P. Jencks, *article in Advances in Enzymology*, 1975

There are two fundamental conditions for life. First, the living entity must be able to self-replicate (a topic considered in Part III); second, the organism must be able to catalyze chemical reactions efficiently and selectively. The central importance of catalysis may surprise some beginning students of biochemistry, but it is easy to demonstrate. As described in Chapter 1, living systems make use of energy from the environment. Many of us, for example, consume substantial amounts of sucrose—common table sugar—as a kind of fuel, whether in the form of sweetened foods and drinks or as sugar itself. The conversion of sucrose to CO_2 and

H_2O in the presence of oxygen is a highly exergonic process, releasing free energy that we can use to think, move, taste, and see. However, a bag of sugar can remain on the shelf for years without any obvious conversion to CO_2 and H_2O . Although this chemical process is thermodynamically favorable, it is very slow! Yet when sucrose is consumed by a human (or almost any other organism), it releases its chemical energy in seconds. The difference is catalysis. Without catalysis, chemical reactions such as sucrose oxidation could not occur on a useful time scale, and thus could not sustain life.

In this chapter, then, we turn our attention to the reaction catalysts of biological systems: the enzymes, the most remarkable and highly specialized proteins. Enzymes have extraordinary catalytic power, often far greater than that of synthetic or inorganic catalysts. They have a high degree of specificity for their substrates, they accelerate chemical reactions tremendously, and they function in aqueous solutions under very mild conditions of temperature and pH. Few non-biological catalysts have all these properties.

Enzymes are central to every biochemical process. Acting in organized sequences, they catalyze the hundreds of stepwise reactions that degrade nutrient molecules, conserve and transform chemical energy, and make biological macromolecules from simple precursors. Through the action of regulatory enzymes, metabolic pathways are highly coordinated to yield a harmonious interplay among the many activities necessary to sustain life.

The study of enzymes has immense practical importance. In some diseases, especially inheritable genetic disorders, there may be a deficiency or even a total absence of one or more enzymes. For other disease conditions, excessive activity of an enzyme may be the cause. Measurements of the activities of enzymes in blood plasma, erythrocytes, or tissue samples are important in diagnosing certain illnesses. Many drugs exert their biological effects through interactions with enzymes. And enzymes are important practical tools,

not only in medicine but in the chemical industry, food processing, and agriculture.

We begin with descriptions of the properties of enzymes and the principles underlying their catalytic power, then introduce enzyme kinetics, a discipline that provides much of the framework for any discussion of enzymes. Specific examples of enzyme mechanisms are then provided, illustrating principles introduced earlier in the chapter. We end with a discussion of how enzyme activity is regulated.

6.1 An Introduction to Enzymes

Much of the history of biochemistry is the history of enzyme research. Biological catalysis was first recognized and described in the late 1700s, in studies on the digestion of meat by secretions of the stomach, and research continued in the 1800s with examinations of the conversion of starch to sugar by saliva and various plant extracts. In the 1850s, Louis Pasteur concluded that fermentation of sugar into alcohol by yeast is catalyzed by “ferments.” He postulated that these ferments were inseparable from the structure of living yeast cells; this view, called *vitalism*, prevailed for decades. Then in 1897 Eduard Buchner discovered that yeast extracts could ferment sugar to alcohol, proving that fermentation was promoted by molecules that continued to function when removed from cells. Frederick W. Kühne called these molecules **enzymes**. As vitalistic notions of life were disproved, the isolation of new enzymes and the investigation of their properties advanced the science of biochemistry.

The isolation and crystallization of urease by James Sumner in 1926 provided a breakthrough in early enzyme studies. Sumner found that urease crystals consisted entirely of protein, and he postulated that all enzymes are proteins. In the absence of other examples, this idea remained controversial for some time. Only in the 1930s was Sumner’s conclusion widely accepted, after John Northrop and Moses Kunitz crystallized pepsin, trypsin, and other digestive enzymes and found them also to be proteins. During this period, J. B. S. Haldane wrote a treatise entitled *Enzymes*. Although the molecular nature of enzymes was not yet fully appreciated, Haldane made the remarkable suggestion that weak bonding interactions between an enzyme and its substrate might be used to catalyze a reaction. This insight lies at the heart of our current understanding of enzymatic catalysis.

Since the latter part of the twentieth century, research on enzymes has been intensive. It has led to the purification of thousands of enzymes, elucidation of the

structure and chemical mechanism of many of them, and a general understanding of how enzymes work.

Most Enzymes Are Proteins

With the exception of a small group of catalytic RNA molecules (Chapter 26), all enzymes are proteins. Their catalytic activity depends on the integrity of their native protein conformation. If an enzyme is denatured or dissociated into its subunits, catalytic activity is usually lost. If an enzyme is broken down into its component amino acids, its catalytic activity is always destroyed. Thus the primary, secondary, tertiary, and quaternary structures of protein enzymes are essential to their catalytic activity.

Enzymes, like other proteins, have molecular weights ranging from about 12,000 to more than 1 million. Some enzymes require no chemical groups for activity other than their amino acid residues. Others require an additional chemical component called a **cofactor**—either one or more inorganic ions, such as Fe^{2+} , Mg^{2+} , Mn^{2+} , or Zn^{2+} (Table 6-1), or a complex organic or metalloorganic molecule called a **coenzyme** (Table 6-2). Some enzymes require *both* a coenzyme

TABLE 6-1 Some Inorganic Elements That Serve as Cofactors for Enzymes

Cu^{2+}	Cytochrome oxidase
Fe^{2+} or Fe^{3+}	Cytochrome oxidase, catalase, peroxidase
K^{+}	Pyruvate kinase
Mg^{2+}	Hexokinase, glucose 6-phosphatase, pyruvate kinase
Mn^{2+}	Arginase, ribonucleotide reductase
Mo	Dinitrogenase
Ni^{2+}	Urease
Se	Glutathione peroxidase
Zn^{2+}	Carbonic anhydrase, alcohol dehydrogenase, carboxypeptidases A and B



Eduard Buchner,
1860–1917



James Sumner,
1887–1955



J. B. S. Haldane,
1892–1964

TABLE 6–2 Some Coenzymes That Serve as Transient Carriers of Specific Atoms or Functional Groups

Coenzyme	Examples of chemical groups transferred	Dietary precursor in mammals
Biocytin	CO ₂	Biotin
Coenzyme A	Acyl groups	Pantothenic acid and other compounds
5'-Deoxyadenosylcobalamin (coenzyme B ₁₂)	H atoms and alkyl groups	Vitamin B ₁₂
Flavin adenine dinucleotide	Electrons	Riboflavin (vitamin B ₂)
Lipoate	Electrons and acyl groups	Not required in diet
Nicotinamide adenine dinucleotide	Hydride ion (:H [−])	Nicotinic acid (niacin)
Pyridoxal phosphate	Amino groups	Pyridoxine (vitamin B ₆)
Tetrahydrofolate	One-carbon groups	Folate
Thiamine pyrophosphate	Aldehydes	Thiamine (vitamin B ₁)

Note: The structures and modes of action of these coenzymes are described in Part II.

and one or more metal ions for activity. A coenzyme or metal ion that is very tightly or even covalently bound to the enzyme protein is called a **prosthetic group**. A complete, catalytically active enzyme together with its bound coenzyme and/or metal ions is called a **holoenzyme**. The protein part of such an enzyme is called the **apoenzyme** or **apoprotein**. Coenzymes act as transient carriers of specific functional groups. Most are derived from vitamins, organic nutrients required in small amounts in the diet. We consider coenzymes in more detail as we encounter them in the metabolic pathways discussed in Part II. Finally, some enzyme proteins are modified covalently by phosphorylation, glycosylation, and other processes. Many of these alterations are involved in the regulation of enzyme activity.

Enzymes Are Classified by the Reactions They Catalyze

Many enzymes have been named by adding the suffix “-ase” to the name of their substrate or to a word or phrase describing their activity. Thus urease catalyzes hydrolysis of urea, and DNA polymerase catalyzes the polymerization of nucleotides to form DNA. Other enzymes were named by their discoverers for a broad func-

tion, before the specific reaction catalyzed was known. For example, an enzyme known to act in the digestion of foods was named pepsin, from the Greek *pepsis*, “digestion,” and lysozyme was named for its ability to lyse bacterial cell walls. Still others were named for their source: trypsin, named in part from the Greek *tryein*, “to wear down,” was obtained by rubbing pancreatic tissue with glycerin. Sometimes the same enzyme has two or more names, or two different enzymes have the same name. Because of such ambiguities, and the ever-increasing number of newly discovered enzymes, biochemists, by international agreement, have adopted a system for naming and classifying enzymes. This system divides enzymes into six classes, each with subclasses, based on the type of reaction catalyzed (Table 6–3). Each enzyme is assigned a four-part classification number and a systematic name, which identifies the reaction it catalyzes. As an example, the formal systematic name of the enzyme catalyzing the reaction



is ATP:glucose phosphotransferase, which indicates that it catalyzes the transfer of a phosphoryl group from ATP to glucose. Its Enzyme Commission number (E.C. number) is 2.7.1.1. The first number (2) denotes the

TABLE 6–3 International Classification of Enzymes

No.	Class	Type of reaction catalyzed
1	Oxidoreductases	Transfer of electrons (hydride ions or H atoms)
2	Transferases	Group transfer reactions
3	Hydrolases	Hydrolysis reactions (transfer of functional groups to water)
4	Lyases	Addition of groups to double bonds, or formation of double bonds by removal of groups
5	Isomerases	Transfer of groups within molecules to yield isomeric forms
6	Ligases	Formation of C—C, C—S, C—O, and C—N bonds by condensation reactions coupled to ATP cleavage

Note: Most enzymes catalyze the transfer of electrons, atoms, or functional groups. They are therefore classified, given code numbers, and assigned names according to the type of transfer reaction, the group donor, and the group acceptor.

class name (transferase); the second number (7), the subclass (phosphotransferase); the third number (1), a phosphotransferase with a hydroxyl group as acceptor; and the fourth number (1), D-glucose as the phosphoryl group acceptor. For many enzymes, a trivial name is more commonly used—in this case hexokinase. A complete list and description of the thousands of known enzymes is maintained by the Nomenclature Committee of the International Union of Biochemistry and Molecular Biology (www.chem.qmul.ac.uk/iubmb/enzyme). This chapter is devoted primarily to principles and properties common to all enzymes.

SUMMARY 6.1 An Introduction to Enzymes

- Life depends on the existence of powerful and specific catalysts: the enzymes. Almost every biochemical reaction is catalyzed by an enzyme.
- With the exception of a few catalytic RNAs, all known enzymes are proteins. Many require nonprotein coenzymes or cofactors for their catalytic function.
- Enzymes are classified according to the type of reaction they catalyze. All enzymes have formal E.C. numbers and names, and most have trivial names.

6.2 How Enzymes Work

The enzymatic catalysis of reactions is essential to living systems. Under biologically relevant conditions, uncatalyzed reactions tend to be slow—most biological molecules are quite stable in the neutral-pH, mild-temperature, aqueous environment inside cells. Furthermore, many common reactions in biochemistry entail chemical events that are unfavorable or unlikely in the cellular environment, such as the transient formation of unstable charged intermediates or the collision of two or more molecules in the precise orientation required for reaction. Reactions required to digest food, send nerve signals, or contract a muscle simply do not occur at a useful rate without catalysis.

An enzyme circumvents these problems by providing a specific environment within which a given reaction can occur more rapidly. The distinguishing feature of an enzyme-catalyzed reaction is that it takes place within the confines of a pocket on the enzyme called the **active site** (Fig. 6–1). The molecule that is bound in the active site and acted upon by the enzyme is called the **substrate**. The surface of the active site is lined with amino acid residues with substituent groups that bind the substrate and catalyze its chemical transformation. Often, the active site encloses a substrate, sequestering it completely from solution. The enzyme-

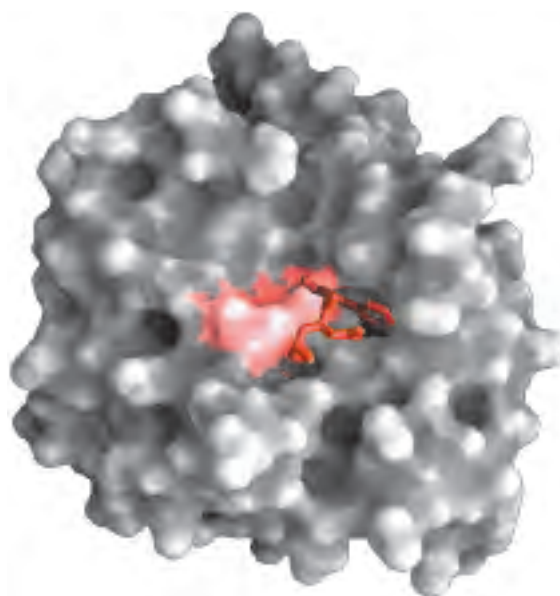


FIGURE 6–1 Binding of a substrate to an enzyme at the active site. The enzyme chymotrypsin, with bound substrate in red (PDB ID 7GCH). Some key active-site amino acid residues appear as a red splotch on the enzyme surface.

substrate complex, whose existence was first proposed by Charles-Adolphe Wurtz in 1880, is central to the action of enzymes. It is also the starting point for mathematical treatments that define the kinetic behavior of enzyme-catalyzed reactions and for theoretical descriptions of enzyme mechanisms.

Enzymes Affect Reaction Rates, Not Equilibria

A simple enzymatic reaction might be written



where E, S, and P represent the enzyme, substrate, and product; ES and EP are transient complexes of the enzyme with the substrate and with the product.

To understand catalysis, we must first appreciate the important distinction between reaction equilibria and reaction rates. The function of a catalyst is to increase the *rate* of a reaction. Catalysts do not affect reaction *equilibria*. Any reaction, such as $S \rightleftharpoons P$, can be described by a reaction coordinate diagram (Fig. 6–2), a picture of the energy changes during the reaction. As discussed in Chapter 1, energy in biological systems is described in terms of free energy, G . In the coordinate diagram, the free energy of the system is plotted against the progress of the reaction (the reaction coordinate). The starting point for either the forward or the reverse reaction is called the **ground state**, the contribution to the free energy of the system by an average molecule (S or P) under a given set of conditions. To describe the free-energy changes for reactions, chemists define a standard set of conditions (temperature 298 K; partial pressure of each gas 1 atm, or 101.3 kPa; concentration

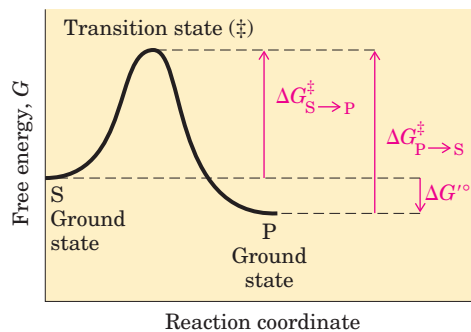


FIGURE 6-2 Reaction coordinate diagram for a chemical reaction.

The free energy of the system is plotted against the progress of the reaction $S \rightarrow P$. A diagram of this kind is a description of the energy changes during the reaction, and the horizontal axis (reaction coordinate) reflects the progressive chemical changes (e.g., bond breakage or formation) as S is converted to P. The activation energies, $\Delta G^\ddagger$, for the $S \rightarrow P$ and $P \rightarrow S$ reactions are indicated. $\Delta G'^\circ$ is the overall standard free-energy change in the direction $S \rightarrow P$.

of each solute 1 M) and express the free-energy change for this reacting system as ΔG° , the **standard free-energy change**. Because biochemical systems commonly involve H^+ concentrations far below 1 M, biochemists define a **biochemical standard free-energy change**, $\Delta G'^\circ$, the standard free-energy change at pH 7.0; we employ this definition throughout the book. A more complete definition of $\Delta G'^\circ$ is given in Chapter 13.

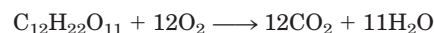
The equilibrium between S and P reflects the difference in the free energies of their ground states. In the example shown in Figure 6-2, the free energy of the ground state of P is lower than that of S, so $\Delta G'^\circ$ for the reaction is negative and the equilibrium favors P. The position and direction of equilibrium are *not* affected by any catalyst.

A favorable equilibrium does not mean that the $S \rightarrow P$ conversion will occur at a detectable rate. The *rate* of a reaction is dependent on an entirely different parameter. There is an energy barrier between S and P: the energy required for alignment of reacting groups, formation of transient unstable charges, bond rearrangements, and other transformations required for the reaction to proceed in either direction. This is illustrated by the energy “hill” in Figures 6-2 and 6-3. To undergo reaction, the molecules must overcome this barrier and therefore must be raised to a higher energy level. At the top of the energy hill is a point at which decay to the S or P state is equally probable (it is downhill either way). This is called the **transition state**. The transition state is not a chemical species with any significant stability and should not be confused with a reaction intermediate (such as ES or EP). It is simply a fleeting molecular moment in which events such as bond breakage, bond formation, and charge development have proceeded to the precise point at which decay to

either substrate or product is equally likely. The difference between the energy levels of the ground state and the transition state is the **activation energy**, $\Delta G^\ddagger$. The rate of a reaction reflects this activation energy: a higher activation energy corresponds to a slower reaction. Reaction rates can be increased by raising the temperature, thereby increasing the number of molecules with sufficient energy to overcome the energy barrier. Alternatively, the activation energy can be lowered by adding a catalyst (Fig. 6-3). *Catalysts enhance reaction rates by lowering activation energies.*

Enzymes are no exception to the rule that catalysts do not affect reaction equilibria. The bidirectional arrows in Equation 6-1 make this point: any enzyme that catalyzes the reaction $S \rightarrow P$ also catalyzes the reaction $P \rightarrow S$. The role of enzymes is to *accelerate* the interconversion of S and P. The enzyme is not used up in the process, and the equilibrium point is unaffected. However, the reaction reaches equilibrium much faster when the appropriate enzyme is present, because the rate of the reaction is increased.

This general principle can be illustrated by considering the conversion of sucrose and oxygen to carbon dioxide and water:



This conversion, which takes place through a series of separate reactions, has a very large and negative $\Delta G'^\circ$, and at equilibrium the amount of sucrose present is negligible. Yet sucrose is a stable compound, because the activation energy barrier that must be overcome before sucrose reacts with oxygen is quite high. Sucrose can be stored in a container with oxygen almost indefinitely without reacting. In cells, however, sucrose is readily broken down to CO_2 and H_2O in a series of reactions catalyzed by enzymes. These enzymes not only accel-

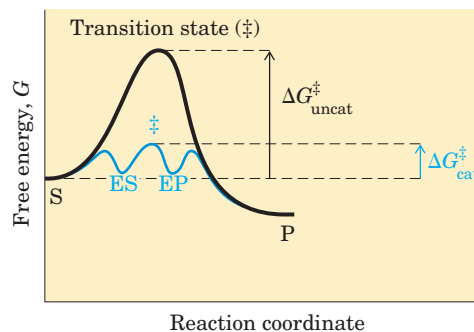


FIGURE 6-3 Reaction coordinate diagram comparing enzyme-catalyzed and uncatalyzed reactions. In the reaction $S \rightarrow P$, the ES and EP intermediates occupy minima in the energy progress curve of the enzyme-catalyzed reaction. The terms $\Delta G_{\text{uncat}}^\ddagger$ and $\Delta G_{\text{cat}}^\ddagger$ correspond to the activation energy for the uncatalyzed reaction and the overall activation energy for the catalyzed reaction, respectively. The activation energy is lower when the enzyme catalyzes the reaction.

erate the reactions, they organize and control them so that much of the energy released is recovered in other chemical forms and made available to the cell for other tasks. The reaction pathway by which sucrose (and other sugars) is broken down is the primary energy-yielding pathway for cells, and the enzymes of this pathway allow the reaction sequence to proceed on a biologically useful time scale.

Any reaction may have several steps, involving the formation and decay of transient chemical species called **reaction intermediates**.^{*} A reaction intermediate is any species on the reaction pathway that has a finite chemical lifetime (longer than a molecular vibration, $\sim 10^{-13}$ seconds). When the $S \rightleftharpoons P$ reaction is catalyzed by an enzyme, the ES and EP complexes can be considered intermediates, even though S and P are stable chemical species (Eqn 6-1); the ES and EP complexes occupy valleys in the reaction coordinate diagram (Fig. 6-3). Additional, less stable chemical intermediates often exist in the course of an enzyme-catalyzed reaction. The interconversion of two sequential reaction intermediates thus constitutes a reaction step. When several steps occur in a reaction, the overall rate is determined by the step (or steps) with the highest activation energy; this is called the **rate-limiting step**. In a simple case, the rate-limiting step is the highest-energy point in the diagram for interconversion of S and P. In practice, the rate-limiting step can vary with reaction conditions, and for many enzymes several steps may have similar activation energies, which means they are all partially rate-limiting.

Activation energies are energy barriers to chemical reactions. These barriers are crucial to life itself. The rate at which a molecule undergoes a particular reaction decreases as the activation barrier for that reaction increases. Without such energy barriers, complex macromolecules would revert spontaneously to much simpler molecular forms, and the complex and highly ordered structures and metabolic processes of cells could not exist. Over the course of evolution, enzymes have developed lower activation energies *selectively* for reactions that are needed for cell survival.

Reaction Rates and Equilibria Have Precise Thermodynamic Definitions

Reaction *equilibria* are inextricably linked to the standard free-energy change for the reaction, $\Delta G'^{\circ}$, and re-

action *rates* are linked to the activation energy, $\Delta G^{\ddagger}$. A basic introduction to these thermodynamic relationships is the next step in understanding how enzymes work.

An equilibrium such as $S \rightleftharpoons P$ is described by an **equilibrium constant**, K'_{eq} , or simply K (p. 26). Under the standard conditions used to compare biochemical processes, an equilibrium constant is denoted K'_{eq} (or K'):

$$K'_{eq} = \frac{[P]}{[S]} \quad (6-2)$$

From thermodynamics, the relationship between K'_{eq} and $\Delta G'^{\circ}$ can be described by the expression

$$\Delta G'^{\circ} = -RT \ln K'_{eq} \quad (6-3)$$

where R is the gas constant, $8.315 \text{ J/mol} \cdot \text{K}$, and T is the absolute temperature, 298 K (25°C). Equation 6-3 is developed and discussed in more detail in Chapter 13. The important point here is that the equilibrium constant is directly related to the overall standard free-energy change for the reaction (Table 6-4). A large negative value for $\Delta G'^{\circ}$ reflects a favorable reaction equilibrium—but as already noted, this does not mean the reaction will proceed at a rapid rate.

The rate of any reaction is determined by the concentration of the reactant (or reactants) and by a **rate constant**, usually denoted by k . For the unimolecular reaction $S \rightarrow P$, the rate (or velocity) of the reaction, V —representing the amount of S that reacts per unit time—is expressed by a **rate equation**:

$$V = k[S] \quad (6-4)$$

In this reaction, the rate depends only on the concentration of S. This is called a first-order reaction. The factor k is a proportionality constant that reflects the probability of reaction under a given set of conditions (pH, temperature, and so forth). Here, k is a first-order rate constant and has units of reciprocal time, such as s^{-1} . If a first-order reaction has a rate constant k of 0.03 s^{-1} ,

TABLE 6-4 Relationship between K'_{eq} and $\Delta G'^{\circ}$

K'_{eq}	$\Delta G'^{\circ}$ (kJ/mol)
10^{-6}	34.2
10^{-5}	28.5
10^{-4}	22.8
10^{-3}	17.1
10^{-2}	11.4
10^{-1}	5.7
1	0.0
10^1	-5.7
10^2	-11.4
10^3	-17.1

Note: The relationship is calculated from $\Delta G'^{\circ} = -RT \ln K'_{eq}$ (Eqn 6-3).

^{*}In this chapter, *step* and *intermediate* refer to chemical species in the reaction pathway of a single enzyme-catalyzed reaction. In the context of metabolic pathways involving many enzymes (discussed in Part II), these terms are used somewhat differently. An entire enzymatic reaction is often referred to as a “step” in a pathway, and the product of one enzymatic reaction (which is the substrate for the next enzyme in the pathway) is referred to as an “intermediate.”

this may be interpreted (qualitatively) to mean that 3% of the available S will be converted to P in 1 s. A reaction with a rate constant of $2,000 \text{ s}^{-1}$ will be over in a small fraction of a second. If a reaction rate depends on the concentration of two different compounds, or if the reaction is between two molecules of the same compound, the reaction is second order and k is a second-order rate constant, with units of $\text{M}^{-1}\text{s}^{-1}$. The rate equation then becomes

$$V = k[\text{S}_1][\text{S}_2] \quad (6-5)$$

From transition-state theory we can derive an expression that relates the magnitude of a rate constant to the activation energy:

$$k = \frac{kT}{h} e^{-\Delta G^\ddagger/RT} \quad (6-6)$$

where k is the Boltzmann constant and h is Planck's constant. The important point here is that the relationship between the rate constant k and the activation energy $\Delta G^\ddagger$ is inverse and exponential. In simplified terms, this is the basis for the statement that a lower activation energy means a faster reaction rate.

Now we turn from *what* enzymes do to *how* they do it.

A Few Principles Explain the Catalytic Power and Specificity of Enzymes

Enzymes are extraordinary catalysts. The rate enhancements they bring about are in the range of 5 to 17 orders of magnitude (Table 6-5). Enzymes are also very specific, readily discriminating between substrates with quite similar structures. How can these enormous and highly selective rate enhancements be explained? What is the source of the energy for the dramatic lowering of the activation energies for specific reactions?

The answer to these questions has two distinct but interwoven parts. The first lies in the rearrangements of covalent bonds during an enzyme-catalyzed reaction. Chemical reactions of many types take place between substrates and enzymes' functional groups (specific

amino acid side chains, metal ions, and coenzymes). Catalytic functional groups on an enzyme may form a transient covalent bond with a substrate and activate it for reaction, or a group may be transiently transferred from the substrate to the enzyme. In many cases, these reactions occur only in the enzyme active site. Covalent interactions between enzymes and substrates lower the activation energy (and thereby accelerate the reaction) by providing an alternative, lower-energy reaction path. The specific types of rearrangements that occur are described in Section 6.4.

The second part of the explanation lies in the *non-covalent* interactions between enzyme and substrate. Much of the energy required to lower activation energies is derived from weak, noncovalent interactions between substrate and enzyme. What really sets enzymes apart from most other catalysts is the formation of a specific ES complex. The interaction between substrate and enzyme in this complex is mediated by the same forces that stabilize protein structure, including hydrogen bonds and hydrophobic and ionic interactions (Chapter 4). Formation of each weak interaction in the ES complex is accompanied by release of a small amount of free energy that provides a degree of stability to the interaction. The energy derived from enzyme-substrate interaction is called **binding energy, ΔG_B** . Its significance extends beyond a simple stabilization of the enzyme-substrate interaction. *Binding energy is a major source of free energy used by enzymes to lower the activation energies of reactions.*

Two fundamental and interrelated principles provide a general explanation for how enzymes use noncovalent binding energy:

1. Much of the catalytic power of enzymes is ultimately derived from the free energy released in forming many weak bonds and interactions between an enzyme and its substrate. This binding energy contributes to specificity as well as to catalysis.
2. Weak interactions are optimized in the reaction transition state; enzyme active sites are complementary not to the substrates per se but to the transition states through which substrates pass as they are converted to products during an enzymatic reaction.

These themes are critical to an understanding of enzymes, and they now become our primary focus.

Weak Interactions between Enzyme and Substrate Are Optimized in the Transition State

How does an enzyme use binding energy to lower the activation energy for a reaction? Formation of the ES complex is not the explanation in itself, although some

TABLE 6-5 Some Rate Enhancements Produced by Enzymes

Cyclophilin	10^5
Carbonic anhydrase	10^7
Triose phosphate isomerase	10^9
Carboxypeptidase A	10^{11}
Phosphoglucosmutase	10^{12}
Succinyl-CoA transferase	10^{13}
Urease	10^{14}
Orotidine monophosphate decarboxylase	10^{17}

of the earliest considerations of enzyme mechanisms began with this idea. Studies on enzyme specificity carried out by Emil Fischer led him to propose, in 1894, that enzymes were structurally complementary to their substrates, so that they fit together like a lock and key (Fig. 6-4). This elegant idea, that a specific (exclusive) interaction between two biological molecules is mediated by molecular surfaces with complementary shapes, has greatly influenced the development of biochemistry, and such interactions lie at the heart of many biochemical processes. However, the “lock and key” hypothesis can be misleading when applied to enzymatic catalysis. An enzyme completely complementary to its substrate would be a very poor enzyme, as we can demonstrate.

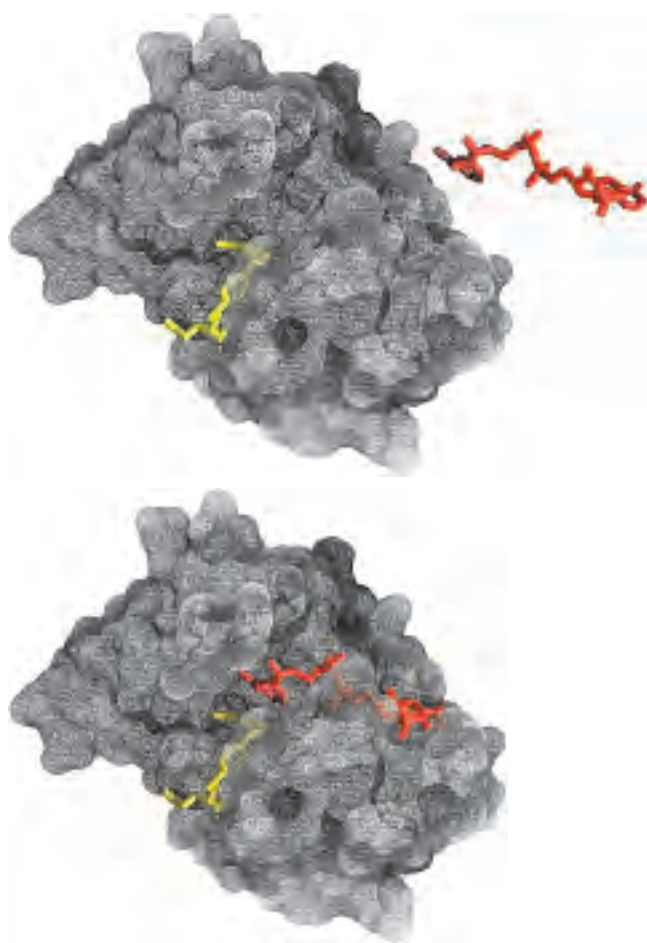


FIGURE 6-4 Complementary shapes of a substrate and its binding site on an enzyme. The enzyme dihydrofolate reductase with its substrate NADP^+ (red), unbound (top) and bound (bottom). Another bound substrate, tetrahydrofolate (yellow), is also visible (PDB ID 1RA2). The NADP^+ binds to a pocket that is complementary to it in shape and ionic properties. In reality, the complementarity between protein and ligand (in this case substrate) is rarely perfect, as we saw in Chapter 5. The interaction of a protein with a ligand often involves changes in the conformation of one or both molecules, a process called induced fit. This *lack* of perfect complementarity between enzyme and substrate (not evident in this figure) is important to enzymatic catalysis.

Consider an imaginary reaction, the breaking of a magnetized metal stick. The uncatalyzed reaction is shown in Figure 6-5a. Let’s examine two imaginary enzymes—two “stickases”—that could catalyze this reaction, both of which employ magnetic forces as a paradigm for the binding energy used by real enzymes. We first design an enzyme perfectly complementary to the substrate (Fig. 6-5b). The active site of this stickase is a pocket lined with magnets. To react (break), the stick must reach the transition state of the reaction, but the stick fits so tightly in the active site that it cannot bend, because bending would eliminate some of the magnetic interactions between stick and enzyme. Such an enzyme *impedes* the reaction, stabilizing the substrate instead. In a reaction coordinate diagram (Fig. 6-5b), this kind of ES complex would correspond to an energy trough from which the substrate would have difficulty escaping. Such an enzyme would be useless.

The modern notion of enzymatic catalysis, first proposed by Michael Polanyi (1921) and Haldane (1930), was elaborated by Linus Pauling in 1946: in order to catalyze reactions, an enzyme must be complementary to the *reaction transition state*. This means that optimal interactions between substrate and enzyme occur only in the transition state. Figure 6-5c demonstrates how such an enzyme can work. The metal stick binds to the stickase, but only a subset of the possible magnetic interactions are used in forming the ES complex. The bound substrate must still undergo the increase in free energy needed to reach the transition state. Now, however, the increase in free energy required to draw the stick into a bent and partially broken conformation is offset, or “paid for,” by the magnetic interactions (binding energy) that form between the enzyme and substrate in the transition state. Many of these interactions involve parts of the stick that are distant from the point of breakage; thus interactions between the stickase and nonreacting parts of the stick provide some of the energy needed to catalyze stick breakage. This “energy payment” translates into a lower net activation energy and a faster reaction rate.

Real enzymes work on an analogous principle. Some weak interactions are formed in the ES complex, but the full complement of such interactions between substrate and enzyme is formed only when the substrate reaches the transition state. The free energy (binding energy) released by the formation of these interactions partially offsets the energy required to reach the top of the energy hill. The summation of the unfavorable (positive) activation energy $\Delta G^\ddagger$ and the favorable (negative) binding energy ΔG_B results in a lower *net* activation energy (Fig. 6-6). Even on the enzyme, the transition state is not a stable species but a brief point in time that the substrate spends atop an energy hill. The enzyme-catalyzed reaction is much faster than the uncatalyzed process, however, because the hill is much smaller. The

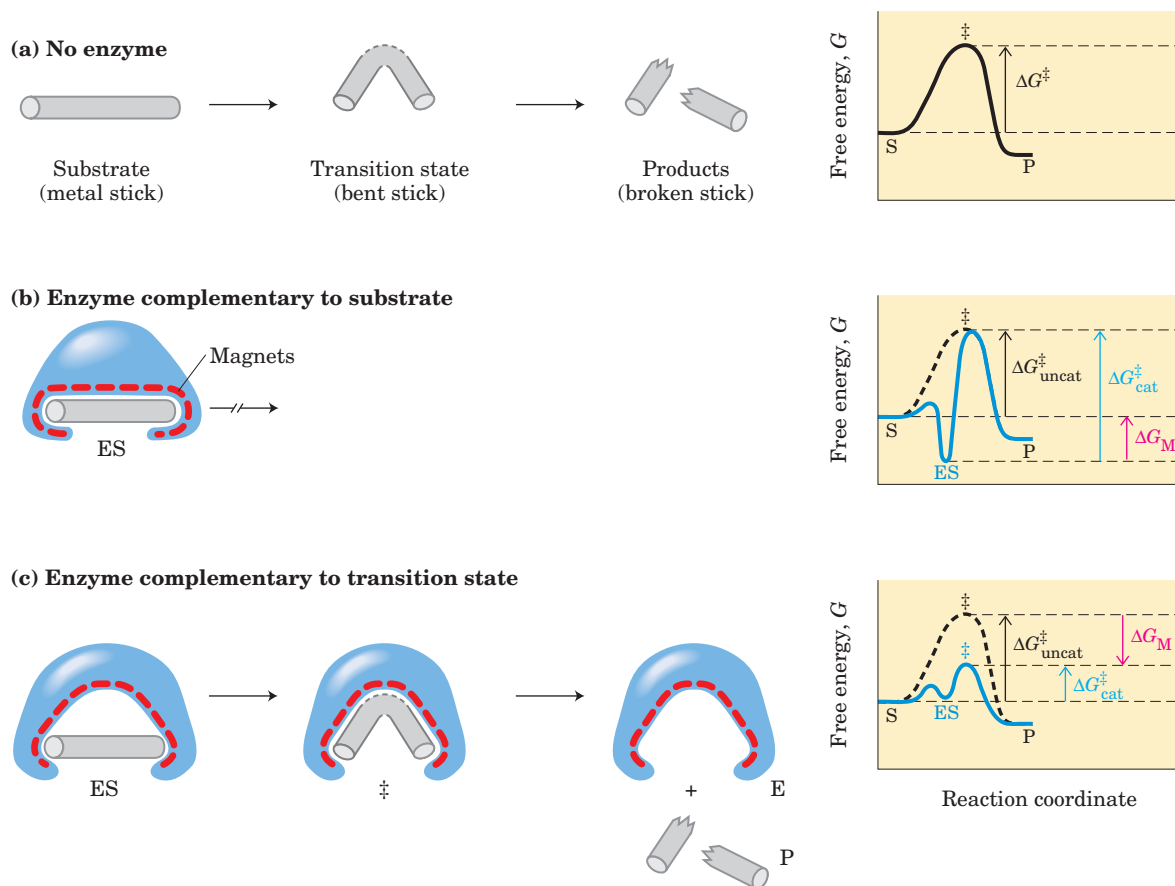


FIGURE 6-5 An imaginary enzyme (stickase) designed to catalyze breakage of a metal stick. (a) Before the stick is broken, it must first be bent (the transition state). In both stickase examples, magnetic interactions take the place of weak bonding interactions between enzyme and substrate. (b) A stickase with a magnet-lined pocket complementary in structure to the stick (the substrate) stabilizes the substrate. Bending is impeded by the magnetic attraction between stick and stickase. (c) An enzyme with a pocket complementary to the reaction transition state helps to destabilize the stick, contributing to catalysis of the reaction. The binding energy of the magnetic interactions

compensates for the increase in free energy required to bend the stick. Reaction coordinate diagrams (right) show the energy consequences of complementarity to substrate versus complementarity to transition state (EP complexes are omitted). ΔG_M , the difference between the transition-state energies of the uncatalyzed and catalyzed reactions, is contributed by the magnetic interactions between the stick and stickase. When the enzyme is complementary to the substrate (b), the ES complex is more stable and has less free energy in the ground state than substrate alone. The result is an *increase* in the activation energy.

important principle is that *weak binding interactions between the enzyme and the substrate provide a substantial driving force for enzymatic catalysis*. The groups on the substrate that are involved in these weak interactions can be at some distance from the bonds that are broken or changed. The weak interactions formed only in the transition state are those that make the primary contribution to catalysis.

The requirement for multiple weak interactions to drive catalysis is one reason why enzymes (and some coenzymes) are so large. An enzyme must provide functional groups for ionic, hydrogen-bond, and other interactions, and also must precisely position these groups so that binding energy is optimized in the transition state. Adequate binding is accomplished most readily by positioning a substrate in a cavity (the active site) where it is effectively removed from water. The size of proteins

reflects the need for superstructure to keep interacting groups properly positioned and to keep the cavity from collapsing.

Binding Energy Contributes to Reaction Specificity and Catalysis

Can we demonstrate quantitatively that binding energy accounts for the huge rate accelerations brought about by enzymes? Yes. As a point of reference, Equation 6-6 allows us to calculate that $\Delta G^\ddagger$ must be lowered by about 5.7 kJ/mol to accelerate a first-order reaction by a factor of ten, under conditions commonly found in cells. The energy available from formation of a single weak interaction is generally estimated to be 4 to 30 kJ/mol. The overall energy available from a number of such interactions is therefore sufficient to lower activation en-

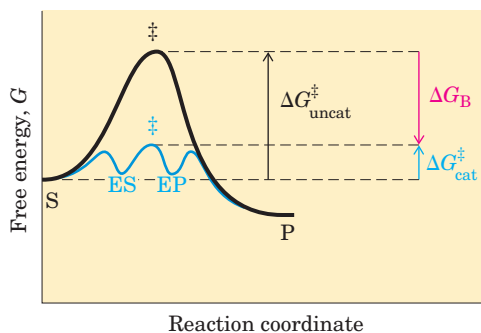
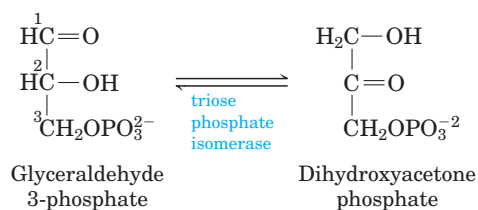


FIGURE 6-6 Role of binding energy in catalysis. To lower the activation energy for a reaction, the system must acquire an amount of energy equivalent to the amount by which $\Delta G^\ddagger$ is lowered. Much of this energy comes from binding energy (ΔG_B) contributed by formation of weak noncovalent interactions between substrate and enzyme in the transition state. The role of ΔG_B is analogous to that of ΔG_M in Figure 6-5.

ergies by the 60 to 100 kJ/mol required to explain the large rate enhancements observed for many enzymes.

The same binding energy that provides energy for catalysis also gives an enzyme its **specificity**, the ability to discriminate between a substrate and a competing molecule. Conceptually, specificity is easy to distinguish from catalysis, but this distinction is much more difficult to make experimentally, because catalysis and specificity arise from the same phenomenon. If an enzyme active site has functional groups arranged optimally to form a variety of weak interactions with a particular substrate in the transition state, the enzyme will not be able to interact to the same degree with any other molecule. For example, if the substrate has a hydroxyl group that forms a hydrogen bond with a specific Glu residue on the enzyme, any molecule lacking a hydroxyl group at that particular position will be a poorer substrate for the enzyme. In addition, any molecule with an extra functional group for which the enzyme has no pocket or binding site is likely to be excluded from the enzyme. In general, *specificity* is derived from the formation of many weak interactions between the enzyme and its specific substrate molecule.

The importance of binding energy to catalysis can be readily demonstrated. For example, the glycolytic enzyme triose phosphate isomerase catalyzes the interconversion of glyceraldehyde 3-phosphate and dihydroxyacetone phosphate:



This reaction rearranges the carbonyl and hydroxyl groups on carbons 1 and 2. However, more than 80% of the enzymatic rate acceleration has been traced to enzyme-substrate interactions involving the phosphate group on carbon 3 of the substrate. This was determined by a careful comparison of the enzyme-catalyzed reactions with glyceraldehyde 3-phosphate and with glyceraldehyde (no phosphate group at position 3) as substrate.

The general principles outlined above can be illustrated by a variety of recognized catalytic mechanisms. These mechanisms are not mutually exclusive, and a given enzyme might incorporate several types in its overall mechanism of action. For most enzymes, it is difficult to quantify the contribution of any one catalytic mechanism to the rate and/or specificity of a particular enzyme-catalyzed reaction.

As we have noted, binding energy makes an important, and sometimes the dominant, contribution to catalysis. Consider what needs to occur for a reaction to take place. Prominent physical and thermodynamic factors contributing to $\Delta G^\ddagger$, the barrier to reaction, might include (1) a reduction in entropy, in the form of decreased freedom of motion of two molecules in solution; (2) the solvation shell of hydrogen-bonded water that surrounds and helps to stabilize most biomolecules in aqueous solution; (3) the distortion of substrates that must occur in many reactions; and (4) the need for proper alignment of catalytic functional groups on the enzyme. Binding energy can be used to overcome all these barriers.

First, a large restriction in the relative motions of two substrates that are to react, or **entropy reduction**, is one obvious benefit of binding them to an enzyme. Binding energy holds the substrates in the proper orientation to react—a substantial contribution to catalysis, because productive collisions between molecules in solution can be exceedingly rare. Substrates can be precisely aligned on the enzyme, with many weak interactions between each substrate and strategically located groups on the enzyme clamping the substrate molecules into the proper positions. Studies have shown that constraining the motion of two reactants can produce rate enhancements of many orders of magnitude (Fig. 6-7).

Second, formation of weak bonds between substrate and enzyme also results in **desolvation** of the substrate. Enzyme-substrate interactions replace most or all of the hydrogen bonds between the substrate and water. Third, binding energy involving weak interactions formed only in the reaction transition state helps to compensate thermodynamically for any distortion, primarily electron redistribution, that the substrate must undergo to react.

Finally, the enzyme itself usually undergoes a change in conformation when the substrate binds, induced by multiple weak interactions with the substrate.

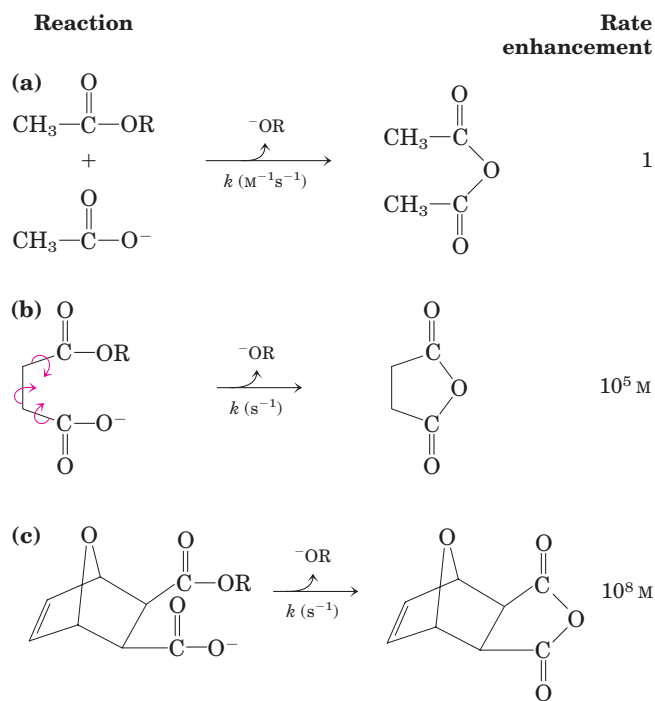


FIGURE 6-7 Rate enhancement by entropy reduction. Shown here are reactions of an ester with a carboxylate group to form an anhydride. The R group is the same in each case. (a) For this bimolecular reaction, the rate constant k is second order, with units of $\text{M}^{-1}\text{s}^{-1}$. (b) When the two reacting groups are in a single molecule, the reaction is much faster. For this unimolecular reaction, k has units of s^{-1} . Dividing the rate constant for (b) by the rate constant for (a) gives a rate enhancement of about 10^5 M . (The enhancement has units of molarity because we are comparing a unimolecular and a bimolecular reaction.) Put another way, if the reactant in (b) were present at a concentration of 1 M , the reacting groups would behave as though they were present at a concentration of 10^5 M . Note that the reactant in (b) has freedom of rotation about three bonds (shown with curved arrows), but this still represents a substantial reduction of entropy over (a). If the bonds that rotate in (b) are constrained as in (c), the entropy is reduced further and the reaction exhibits a rate enhancement of 10^8 M relative to (a).

This is referred to as **induced fit**, a mechanism postulated by Daniel Koshland in 1958. Induced fit serves to bring specific functional groups on the enzyme into the proper position to catalyze the reaction. The conformational change also permits formation of additional weak bonding interactions in the transition state. In either case, the new enzyme conformation has enhanced catalytic properties. As we have seen, induced fit is a common feature of the reversible binding of ligands to proteins (Chapter 5). Induced fit is also important in the interaction of almost every enzyme with its substrate.

Specific Catalytic Groups Contribute to Catalysis

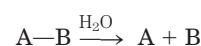
In most enzymes, the binding energy used to form the ES complex is just one of several contributors to the

overall catalytic mechanism. Once a substrate is bound to an enzyme, properly positioned catalytic functional groups aid in the cleavage and formation of bonds by a variety of mechanisms, including general acid-base catalysis, covalent catalysis, and metal ion catalysis. These are distinct from mechanisms based on binding energy, because they generally involve transient *covalent* interaction with a substrate or group transfer to or from a substrate.

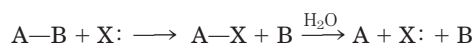
General Acid-Base Catalysis Many biochemical reactions involve the formation of unstable charged intermediates that tend to break down rapidly to their constituent reactant species, thus impeding the reaction (Fig. 6–8). Charged intermediates can often be stabilized by the transfer of protons to or from the substrate or intermediate to form a species that breaks down more readily to products. For nonenzymatic reactions, the proton transfers can involve either the constituents of water alone or other weak proton donors or acceptors. Catalysis of this type that uses only the H^+ (H_3O^+) or OH^- ions present in water is referred to as **specific acid-base catalysis**. If protons are transferred between the intermediate and water faster than the intermediate breaks down to reactants, the intermediate is effectively stabilized every time it forms. No additional catalysis mediated by other proton acceptors or donors will occur. In many cases, however, water is not enough. The term **general acid-base catalysis** refers to proton transfers mediated by other classes of molecules. For nonenzymatic reactions in aqueous solutions, this occurs only when the unstable reaction intermediate breaks down to reactants faster than protons can be transferred to or from water. Many weak organic acids can supplement water as proton donors in this situation, or weak organic bases can serve as proton acceptors.

In the active site of an enzyme, a number of amino acid side chains can similarly act as proton donors and acceptors (Fig. 6–9). These groups can be precisely positioned in an enzyme active site to allow proton transfers, providing rate enhancements of the order of 10^2 to 10^5 . This type of catalysis occurs on the vast majority of enzymes. In fact, proton transfers are the most common biochemical reactions.

Covalent Catalysis In covalent catalysis, a transient covalent bond is formed between the enzyme and the substrate. Consider the hydrolysis of a bond between groups A and B:



In the presence of a covalent catalyst (an enzyme with a nucleophilic group X:) the reaction becomes



This alters the pathway of the reaction, and it results in catalysis only when the new pathway has a lower activation energy than the uncatalyzed pathway. Both of the new steps must be faster than the uncatalyzed

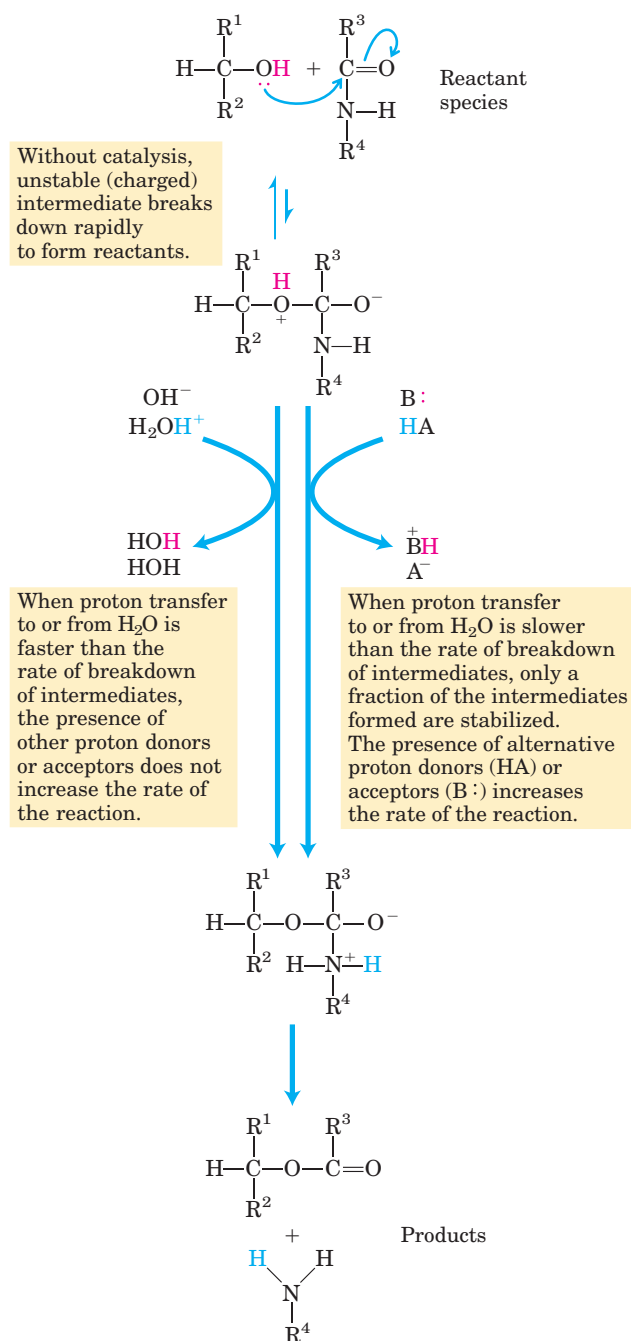


FIGURE 6-8 How a catalyst circumvents unfavorable charge development during cleavage of an amide. The hydrolysis of an amide bond, shown here, is the same reaction as that catalyzed by chymotrypsin and other proteases. Charge development is unfavorable and can be circumvented by donation of a proton by H_3O^+ (specific acid catalysis) or HA (general acid catalysis), where HA represents any acid. Similarly, charge can be neutralized by proton abstraction by OH^- (specific base catalysis) or $\text{B}:$ (general base catalysis), where $\text{B}:$ represents any base.

reaction. A number of amino acid side chains, including all those in Figure 6-9, and the functional groups of some enzyme cofactors can serve as nucleophiles in the formation of covalent bonds with substrates. These covalent complexes always undergo further reaction to regenerate the free enzyme. The covalent bond formed between the enzyme and the substrate can activate a substrate for further reaction in a manner that is usually specific to the particular group or coenzyme.

Metal Ion Catalysis Metals, whether tightly bound to the enzyme or taken up from solution along with the substrate, can participate in catalysis in several ways. Ionic interactions between an enzyme-bound metal and a substrate can help orient the substrate for reaction or stabilize charged reaction transition states. This use of weak bonding interactions between metal and substrate is similar to some of the uses of enzyme-substrate binding energy described earlier. Metals can also mediate oxidation-reduction reactions by reversible changes in the metal ion's oxidation state. Nearly a third of all known enzymes require one or more metal ions for catalytic activity.

Most enzymes employ a combination of several catalytic strategies to bring about a rate enhancement. A good example of the use of both covalent catalysis and general acid-base catalysis is the reaction catalyzed by chymotrypsin. The first step is cleavage of a peptide bond, which is accompanied by formation of a covalent linkage between a Ser residue on the enzyme and part

Amino acid residues	General acid form (proton donor)	General base form (proton acceptor)
Glu, Asp	$\text{R}-\text{COOH}$	$\text{R}-\text{COO}^-$
Lys, Arg	$\text{R}-\text{NH}_3^+$	$\text{R}-\text{NH}_2$
Cys	$\text{R}-\text{SH}$	$\text{R}-\text{S}^-$
His	$\text{R}-\text{C}(\text{NH}_2)=\text{CH}-\text{NH}^+$	$\text{R}-\text{C}(\text{NH}_2)=\text{CH}-\text{N}:$
Ser	$\text{R}-\text{OH}$	$\text{R}-\text{O}^-$
Tyr	$\text{R}-\text{C}_6\text{H}_4-\text{OH}$	$\text{R}-\text{C}_6\text{H}_4-\text{O}^-$

FIGURE 6-9 Amino acids in general acid-base catalysis. Many organic reactions are promoted by proton donors (general acids) or proton acceptors (general bases). The active sites of some enzymes contain amino acid functional groups, such as those shown here, that can participate in the catalytic process as proton donors or proton acceptors.

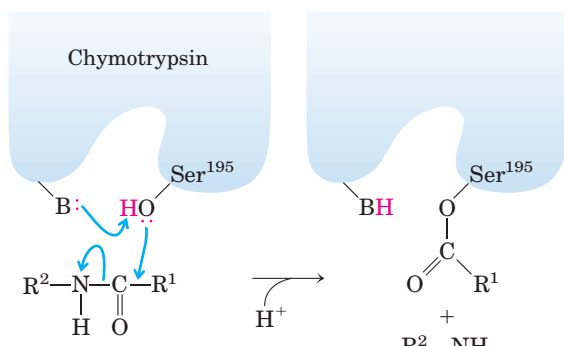


FIGURE 6-10 Covalent and general acid-base catalysis. The first step in the reaction catalyzed by chymotrypsin is the acylation step. The hydroxyl group of Ser¹⁹⁵ is the nucleophile in a reaction aided by general base catalysis (the base is the side chain of His⁵⁷). This provides a new pathway for the hydrolytic cleavage of a peptide bond. Catalysis occurs only if each step in the new pathway is faster than the uncatalyzed reaction. The chymotrypsin reaction is described in more detail in Figure 6-21.

of the substrate; the reaction is enhanced by general base catalysis by other groups on the enzyme (Fig. 6-10). The chymotrypsin reaction is described in more detail in Section 6.4.

SUMMARY 6.2 How Enzymes Work

- Enzymes are highly effective catalysts, commonly enhancing reaction rates by a factor of 10^5 to 10^{17} .
- Enzyme-catalyzed reactions are characterized by the formation of a complex between substrate and enzyme (an ES complex). Substrate binding occurs in a pocket on the enzyme called the active site.
- The function of enzymes and other catalysts is to lower the activation energy, $\Delta G^\ddagger$, for a reaction and thereby enhance the reaction rate. The equilibrium of a reaction is unaffected by the enzyme.
- A significant part of the energy used for enzymatic rate enhancements is derived from weak interactions (hydrogen bonds and hydrophobic and ionic interactions) between substrate and enzyme. The enzyme active site is structured so that some of these weak interactions occur preferentially in the reaction transition state, thus stabilizing the transition state. The need for multiple interactions is one reason for the large size of enzymes. The binding energy, ΔG_B , can be used to lower substrate entropy or to cause a conformational change in the enzyme (induced fit). Binding energy also accounts for the exquisite specificity of enzymes for their substrates.

- Additional catalytic mechanisms employed by enzymes include general acid-base catalysis, covalent catalysis, and metal ion catalysis. Catalysis often involves transient covalent interactions between the substrate and the enzyme, or group transfers to and from the enzyme, so as to provide a new, lower-energy reaction path.

6.3 Enzyme Kinetics as an Approach to Understanding Mechanism

Biochemists commonly use several approaches to study the mechanism of action of purified enzymes. A knowledge of the three-dimensional structure of the protein provides important information, and the value of structural information is greatly enhanced by classical protein chemistry and modern methods of site-directed mutagenesis (changing the amino acid sequence of a protein by genetic engineering; see Fig. 9-12). These technologies permit enzymologists to examine the role of individual amino acids in enzyme structure and action. However, the central approach to studying the mechanism of an enzyme-catalyzed reaction is to determine the *rate* of the reaction and how it changes in response to changes in experimental parameters, a discipline known as **enzyme kinetics**. This is the oldest approach to understanding enzyme mechanisms and remains the most important. We provide here a basic introduction to the kinetics of enzyme-catalyzed reactions. More advanced treatments are available in the sources cited at the end of the chapter.

Substrate Concentration Affects the Rate of Enzyme-Catalyzed Reactions

A key factor affecting the rate of a reaction catalyzed by an enzyme is the concentration of substrate, [S]. However, studying the effects of substrate concentration is complicated by the fact that [S] changes during the course of an *in vitro* reaction as substrate is converted to product. One simplifying approach in kinetics experiments is to measure the **initial rate** (or **initial velocity**), designated V_0 , when [S] is much greater than the concentration of enzyme, [E]. In a typical reaction, the enzyme may be present in nanomolar quantities, whereas [S] may be five or six orders of magnitude higher. If only the beginning of the reaction is monitored (often the first 60 seconds or less), changes in [S] can be limited to a few percent, and [S] can be regarded as constant. V_0 can then be explored as a function of [S], which is adjusted by the investigator. The effect on V_0 of varying [S] when the enzyme concentration is held constant is shown in Figure 6-11. At relatively low concentrations of substrate, V_0 increases almost linearly

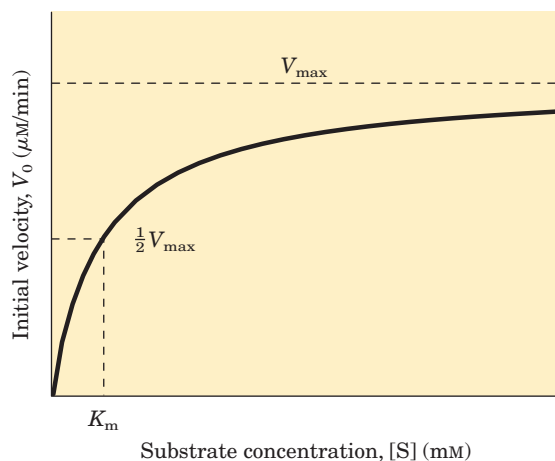


FIGURE 6-11 Effect of substrate concentration on the initial velocity of an enzyme-catalyzed reaction. $V_{\max}$ is extrapolated from the plot, because V_0 approaches but never quite reaches $V_{\max}$. The substrate concentration at which V_0 is half maximal is K_m , the Michaelis constant. The concentration of enzyme in an experiment such as this is generally so low that $[S] \gg [E]$ even when $[S]$ is described as low or relatively low. The units shown are typical for enzyme-catalyzed reactions and are given only to help illustrate the meaning of V_0 and $[S]$. (Note that the curve describes *part* of a rectangular hyperbola, with one asymptote at $V_{\max}$. If the curve were continued below $[S] = 0$, it would approach a vertical asymptote at $[S] = -K_m$.)

with an increase in $[S]$. At higher substrate concentrations, V_0 increases by smaller and smaller amounts in response to increases in $[S]$. Finally, a point is reached beyond which increases in V_0 are vanishingly small as $[S]$ increases. This plateau-like V_0 region is close to the **maximum velocity, $V_{\max}$** .

The ES complex is the key to understanding this kinetic behavior, just as it was a starting point for our discussion of catalysis. The kinetic pattern in Figure 6-11 led Victor Henri, following the lead of Wurtz, to propose in 1903 that the combination of an enzyme with its substrate molecule to form an ES complex is a necessary step in enzymatic catalysis. This idea was expanded into a general theory of enzyme action, particularly by Leonor Michaelis and Maud Menten in 1913. They postulated that the enzyme first combines reversibly with

its substrate to form an enzyme-substrate complex in a relatively fast reversible step:



The ES complex then breaks down in a slower second step to yield the free enzyme and the reaction product P:



Because the slower second reaction (Eqn 6-8) must limit the rate of the overall reaction, the overall rate must be proportional to the concentration of the species that reacts in the second step, that is, ES.

At any given instant in an enzyme-catalyzed reaction, the enzyme exists in two forms, the free or uncombined form E and the combined form ES. At low $[S]$, most of the enzyme is in the uncombined form E. Here, the rate is proportional to $[S]$ because the equilibrium of Equation 6-7 is pushed toward formation of more ES as $[S]$ increases. The maximum initial rate of the catalyzed reaction ($V_{\max}$) is observed when virtually all the enzyme is present as the ES complex and $[E]$ is vanishingly small. Under these conditions, the enzyme is “saturated” with its substrate, so that further increases in $[S]$ have no effect on rate. This condition exists when $[S]$ is sufficiently high that essentially all the free enzyme has been converted to the ES form. After the ES complex breaks down to yield the product P, the enzyme is free to catalyze reaction of another molecule of substrate. The saturation effect is a distinguishing characteristic of enzymatic catalysts and is responsible for the plateau observed in Figure 6-11. The pattern seen in Figure 6-11 is sometimes referred to as **saturation kinetics**.

When the enzyme is first mixed with a large excess of substrate, there is an initial period, the **pre-steady state**, during which the concentration of ES builds up. This period is usually too short to be easily observed, lasting just microseconds. The reaction quickly achieves a **steady state** in which $[ES]$ (and the concentrations of any other intermediates) remains approximately constant over time. The concept of a steady state was introduced by G. E. Briggs and Haldane in 1925. The measured V_0 generally reflects the steady state, even though V_0 is limited to the early part of the reaction, and analysis of these initial rates is referred to as **steady-state kinetics**.

The Relationship between Substrate Concentration and Reaction Rate Can Be Expressed Quantitatively

The curve expressing the relationship between $[S]$ and V_0 (Fig. 6-11) has the same general shape for most enzymes (it approaches a rectangular hyperbola), which can be expressed algebraically by the Michaelis-Menten



Leonor Michaelis,
1875–1949



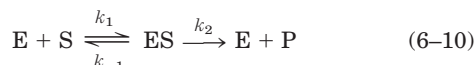
Maud Menten,
1879–1960

equation. Michaelis and Menten derived this equation starting from their basic hypothesis that the rate-limiting step in enzymatic reactions is the breakdown of the ES complex to product and free enzyme. The equation is

$$V_0 = \frac{V_{\max} [S]}{K_m + [S]} \quad (6-9)$$

The important terms are $[S]$, V_0 , $V_{\max}$, and a constant called the Michaelis constant, K_m . All these terms are readily measured experimentally.

Here we develop the basic logic and the algebraic steps in a modern derivation of the Michaelis-Menten equation, which includes the steady-state assumption introduced by Briggs and Haldane. The derivation starts with the two basic steps of the formation and breakdown of ES (Eqns 6-7 and 6-8). Early in the reaction, the concentration of the product, $[P]$, is negligible, and we make the simplifying assumption that the reverse reaction, $P \rightarrow S$ (described by k_{-2}), can be ignored. This assumption is not critical but it simplifies our task. The overall reaction then reduces to



V_0 is determined by the breakdown of ES to form product, which is determined by $[ES]$:

$$V_0 = k_2[ES] \quad (6-11)$$

Because $[ES]$ in Equation 6-11 is not easily measured experimentally, we must begin by finding an alternative expression for this term. First, we introduce the term $[E_t]$, representing the total enzyme concentration (the sum of free and substrate-bound enzyme). Free or unbound enzyme can then be represented by $[E_t] - [ES]$. Also, because $[S]$ is ordinarily far greater than $[E_t]$, the amount of substrate bound by the enzyme at any given time is negligible compared with the total $[S]$. With these conditions in mind, the following steps lead us to an expression for V_0 in terms of easily measurable parameters.

Step 1 The rates of formation and breakdown of ES are determined by the steps governed by the rate constants k_1 (formation) and $k_{-1} + k_2$ (breakdown), according to the expressions

$$\text{Rate of ES formation} = k_1([E_t] - [ES])[S] \quad (6-12)$$

$$\text{Rate of ES breakdown} = k_{-1}[ES] + k_2[ES] \quad (6-13)$$

Step 2 We now make an important assumption: that the initial rate of reaction reflects a steady state in which $[ES]$ is constant—that is, the rate of formation of ES is equal to the rate of its breakdown. This is called the **steady-state assumption**. The expressions in Equations 6-12 and 6-13 can be equated for the steady state, giving

$$k_1([E_t] - [ES])[S] = k_{-1}[ES] + k_2[ES] \quad (6-14)$$

Step 3 In a series of algebraic steps, we now solve Equation 6-14 for $[ES]$. First, the left side is multiplied out and the right side simplified to give

$$k_1[E_t][S] - k_1[ES][S] = (k_{-1} + k_2)[ES] \quad (6-15)$$

Adding the term $k_1[ES][S]$ to both sides of the equation and simplifying gives

$$k_1[E_t][S] = (k_1[S] + k_{-1} + k_2)[ES] \quad (6-16)$$

We then solve this equation for $[ES]$:

$$[ES] = \frac{k_1[E_t][S]}{k_1[S] + k_{-1} + k_2} \quad (6-17)$$

This can now be simplified further, combining the rate constants into one expression:

$$[ES] = \frac{[E_t][S]}{[S] + (k_2 + k_{-1})/k_1} \quad (6-18)$$

The term $(k_2 + k_{-1})/k_1$ is defined as the **Michaelis constant**, K_m . Substituting this into Equation 6-18 simplifies the expression to

$$[ES] = \frac{[E_t][S]}{K_m + [S]} \quad (6-19)$$

Step 4 We can now express V_0 in terms of $[ES]$. Substituting the right side of Equation 6-19 for $[ES]$ in Equation 6-11 gives

$$V_0 = \frac{k_2[E_t][S]}{K_m + [S]} \quad (6-20)$$

This equation can be further simplified. Because the maximum velocity occurs when the enzyme is saturated (that is, with $[ES] = [E_t]$) $V_{\max}$ can be defined as $k_2[E_t]$. Substituting this in Equation 6-20 gives Equation 6-9:

$$V_0 = \frac{V_{\max} [S]}{K_m + [S]}$$

This is the **Michaelis-Menten equation**, the **rate equation** for a one-substrate enzyme-catalyzed reaction. It is a statement of the quantitative relationship between the initial velocity V_0 , the maximum velocity $V_{\max}$, and the initial substrate concentration $[S]$, all related through the Michaelis constant K_m . Note that K_m has units of concentration. Does the equation fit experimental observations? Yes; we can confirm this by considering the limiting situations where $[S]$ is very high or very low, as shown in Figure 6-12.

An important numerical relationship emerges from the Michaelis-Menten equation in the special case when V_0 is exactly one-half $V_{\max}$ (Fig. 6-12). Then

$$\frac{V_{\max}}{2} = \frac{V_{\max} [S]}{K_m + [S]} \quad (6-21)$$

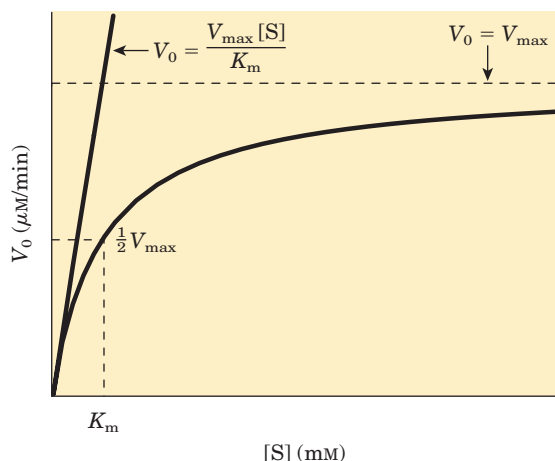


FIGURE 6-12 Dependence of initial velocity on substrate concentration. This graph shows the kinetic parameters that define the limits of the curve at high and low $[S]$. At low $[S]$, $K_m \gg [S]$ and the $[S]$ term in the denominator of the Michaelis-Menten equation (Eqn 6-9) becomes insignificant. The equation simplifies to $V_0 = V_{\max}[S]/K_m$ and V_0 exhibits a linear dependence on $[S]$, as observed here. At high $[S]$, where $[S] \gg K_m$, the K_m term in the denominator of the Michaelis-Menten equation becomes insignificant and the equation simplifies to $V_0 = V_{\max}$; this is consistent with the plateau observed at high $[S]$. The Michaelis-Menten equation is therefore consistent with the observed dependence of V_0 on $[S]$, and the shape of the curve is defined by the terms $V_{\max}/K_m$ at low $[S]$ and $V_{\max}$ at high $[S]$.

On dividing by $V_{\max}$, we obtain

$$\frac{1}{2} = \frac{[S]}{K_m + [S]} \quad (6-22)$$

Solving for K_m , we get $K_m + [S] = 2[S]$, or

$$K_m = [S], \text{ when } V_0 = \frac{1}{2}V_{\max} \quad (6-23)$$

This is a very useful, practical definition of K_m : K_m is equivalent to the substrate concentration at which V_0 is one-half $V_{\max}$.

The Michaelis-Menten equation (Eqn 6-9) can be algebraically transformed into versions that are useful in the practical determination of K_m and $V_{\max}$ (Box 6-1) and, as we describe later, in the analysis of inhibitor action (see Box 6-2 on page 210).

Kinetic Parameters Are Used to Compare Enzyme Activities

It is important to distinguish between the Michaelis-Menten equation and the specific kinetic mechanism on which it was originally based. The equation describes the kinetic behavior of a great many enzymes, and all enzymes that exhibit a hyperbolic dependence of V_0 on $[S]$ are said to follow **Michaelis-Menten kinetics**. The practical rule that

$K_m = [S]$ when $V_0 = \frac{1}{2}V_{\max}$ (Eqn 6-23) holds for all enzymes that follow Michaelis-Menten kinetics. (The most important exceptions to Michaelis-Menten kinetics are the regulatory enzymes, discussed in Section 6.5.) However, the Michaelis-Menten equation does not depend on the relatively simple two-step reaction mechanism proposed by Michaelis and Menten (Eqn 6-10). Many enzymes that follow Michaelis-Menten kinetics have quite different reaction mechanisms, and enzymes that catalyze reactions with six or eight identifiable steps often exhibit the same steady-state kinetic behavior. Even though Equation 6-23 holds true for many enzymes, both the magnitude and the real meaning of $V_{\max}$ and K_m can differ from one enzyme to the next. This is an important limitation of the steady-state approach to enzyme kinetics. The parameters $V_{\max}$ and K_m can be obtained experimentally for any given enzyme, but by themselves they provide little information about the number, rates, or chemical nature of discrete steps in the reaction. Steady-state kinetics nevertheless is the standard language by which biochemists compare and characterize the catalytic efficiencies of enzymes.

Interpreting $V_{\max}$ and K_m Figure 6-12 shows a simple graphical method for obtaining an approximate value for K_m . A more convenient procedure, using a **double-reciprocal plot**, is presented in Box 6-1. The K_m can vary greatly from enzyme to enzyme, and even for different substrates of the same enzyme (Table 6-6). The term is sometimes used (often inappropriately) as an indicator of the affinity of an enzyme for its substrate. The actual meaning of K_m depends on specific aspects of the reaction mechanism such as the number and relative rates of the individual steps. For reactions with two steps,

$$K_m = \frac{k_2 + k_{-1}}{k_1} \quad (6-24)$$

When k_2 is rate-limiting, $k_2 \ll k_{-1}$ and K_m reduces to k_{-1}/k_1 , which is defined as the **dissociation constant**, K_d , of the ES complex. Where these conditions hold, K_m does represent a measure of the affinity of the enzyme

TABLE 6-6 K_m for Some Enzymes and Substrates

Enzyme	Substrate	K_m (mM)
Hexokinase (brain)	ATP	0.4
	D-Glucose	0.05
	D-Fructose	1.5
Carbonic anhydrase	HCO_3^-	26
Chymotrypsin	Glycyltyrosinylglycine	108
	N-Benzoyltyrosinamide	2.5
β -Galactosidase	D-Lactose	4.0
Threonine dehydratase	L-Threonine	5.0

BOX 6-1 WORKING IN BIOCHEMISTRY

Transformations of the Michaelis-Menten Equation: The Double-Reciprocal Plot

The Michaelis-Menten equation

$$V_0 = \frac{V_{\max} [S]}{K_m + [S]}$$

can be algebraically transformed into equations that are more useful in plotting experimental data. One common transformation is derived simply by taking the reciprocal of both sides of the Michaelis-Menten equation:

$$\frac{1}{V_0} = \frac{K_m + [S]}{V_{\max} [S]}$$

Separating the components of the numerator on the right side of the equation gives

$$\frac{1}{V_0} = \frac{K_m}{V_{\max} [S]} + \frac{[S]}{V_{\max} [S]}$$

which simplifies to

$$\frac{1}{V_0} = \frac{K_m}{V_{\max} [S]} + \frac{1}{V_{\max}}$$

This form of the Michaelis-Menten equation is called the **Lineweaver-Burk equation**. For enzymes obeying the Michaelis-Menten relationship, a plot of $1/V_0$ versus $1/[S]$ (the “double reciprocal” of the V_0 versus $[S]$ plot we have been using to this point) yields a straight line (Fig. 1). This line has a slope of $K_m/V_{\max}$, an intercept of $1/V_{\max}$ on the $1/V_0$ axis, and an intercept

of $-1/K_m$ on the $1/[S]$ axis. The double-reciprocal presentation, also called a Lineweaver-Burk plot, has the great advantage of allowing a more accurate determination of $V_{\max}$, which can only be *approximated* from a simple plot of V_0 versus $[S]$ (see Fig. 6-12).

Other transformations of the Michaelis-Menten equation have been derived, each with some particular advantage in analyzing enzyme kinetic data. (See Problem 11 at the end of this chapter.)

The double-reciprocal plot of enzyme reaction rates is very useful in distinguishing between certain types of enzymatic reaction mechanisms (see Fig. 6-14) and in analyzing enzyme inhibition (see Box 6-2).

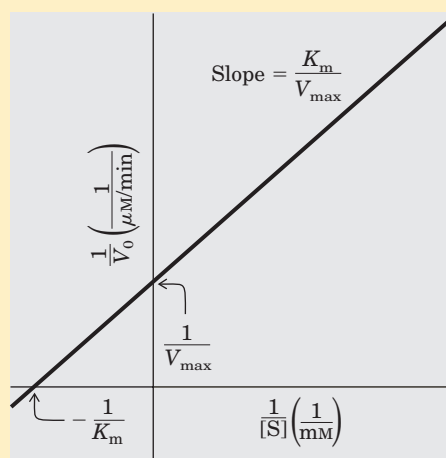
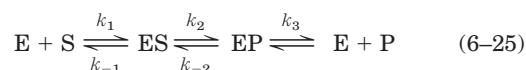


FIGURE 1 A double-reciprocal or Lineweaver-Burk plot.

for its substrate in the ES complex. However, this scenario does not apply for most enzymes. Sometimes $k_2 \gg k_{-1}$, and then $K_m = k_2/k_1$. In other cases, k_2 and k_{-1} are comparable and K_m remains a more complex function of all three rate constants (Eqn 6-24). The Michaelis-Menten equation and the characteristic saturation behavior of the enzyme still apply, but K_m cannot be considered a simple measure of substrate affinity. Even more common are cases in which the reaction goes through several steps after formation of ES; K_m can then become a very complex function of many rate constants.

The quantity $V_{\max}$ also varies greatly from one enzyme to the next. If an enzyme reacts by the two-step Michaelis-Menten mechanism, $V_{\max} = k_2[E_t]$, where k_2 is rate-limiting. However, the number of reaction steps and the identity of the rate-limiting step(s) can vary from enzyme to enzyme. For example, consider the quite common situation where product release, $EP \rightarrow E + P$, is rate-limiting. Early in the reaction (when $[P]$ is low), the overall reaction can be described by the scheme



In this case, most of the enzyme is in the EP form at saturation, and $V_{\max} = k_3[E_t]$. It is useful to define a more general rate constant, k_{cat} , to describe the limiting rate of any enzyme-catalyzed reaction at saturation. If the reaction has several steps and one is clearly rate-limiting, k_{cat} is equivalent to the rate constant for that limiting step. For the simple reaction of Equation 6-10, $k_{\text{cat}} = k_2$. For the reaction of Equation 6-25, $k_{\text{cat}} = k_3$. When several steps are partially rate-limiting, k_{cat} can become a complex function of several of the rate constants that define each individual reaction step. In the Michaelis-Menten equation, $k_{\text{cat}} = V_{\max}/[E_t]$, and Equation 6-9 becomes

$$V_0 = \frac{k_{\text{cat}} [E_t][S]}{K_m + [S]} \quad (6-26)$$

The constant k_{cat} is a first-order rate constant and hence has units of reciprocal time. It is also called the

turnover number. It is equivalent to the number of substrate molecules converted to product in a given unit of time on a single enzyme molecule when the enzyme is saturated with substrate. The turnover numbers of several enzymes are given in Table 6-7.

Comparing Catalytic Mechanisms and Efficiencies

The kinetic parameters k_{cat} and K_{m} are generally useful for the study and comparison of different enzymes, whether their reaction mechanisms are simple or complex. Each enzyme has values of k_{cat} and K_{m} that reflect the cellular environment, the concentration of substrate normally encountered in vivo by the enzyme, and the chemistry of the reaction being catalyzed.

The parameters k_{cat} and K_{m} also allow us to evaluate the kinetic efficiency of enzymes, but either parameter alone is insufficient for this task. Two enzymes catalyzing different reactions may have the same k_{cat} (turnover number), yet the rates of the uncatalyzed reactions may be different and thus the rate enhancements brought about by the enzymes may differ greatly. Experimentally, the K_{m} for an enzyme tends to be similar to the cellular concentration of its substrate. An enzyme that acts on a substrate present at a very low concentration in the cell usually has a lower K_{m} than an enzyme that acts on a substrate that is more abundant.

The best way to compare the catalytic efficiencies of different enzymes or the turnover of different substrates by the same enzyme is to compare the ratio $k_{\text{cat}}/K_{\text{m}}$ for the two reactions. This parameter, sometimes called the **specificity constant**, is the rate constant for the conversion of $E + S$ to $E + P$. When $[S] \ll K_{\text{m}}$, Equation 6-26 reduces to the form

$$V_0 = \frac{k_{\text{cat}}}{K_{\text{m}}} [E]_t [S] \quad (6-27)$$

V_0 in this case depends on the concentration of two reactants, $[E]_t$ and $[S]$; therefore this is a second-order rate equation and the constant $k_{\text{cat}}/K_{\text{m}}$ is a second-order rate

TABLE 6-7 Turnover Numbers, k_{cat} , of Some Enzymes

Enzyme	Substrate	k_{cat} (s^{-1})
Catalase	H_2O_2	40,000,000
Carbonic anhydrase	HCO_3^-	400,000
Acetylcholinesterase	Acetylcholine	14,000
β -Lactamase	Benzylpenicillin	2,000
Fumarase	Fumarate	800
RecA protein (an ATPase)	ATP	0.4

constant with units of $\text{M}^{-1}\text{s}^{-1}$. There is an upper limit to $k_{\text{cat}}/K_{\text{m}}$, imposed by the rate at which E and S can diffuse together in an aqueous solution. This diffusion-controlled limit is 10^8 to $10^9 \text{ M}^{-1}\text{s}^{-1}$, and many enzymes have a $k_{\text{cat}}/K_{\text{m}}$ near this range (Table 6-8). Such enzymes are said to have achieved catalytic perfection. Note that different values of k_{cat} and K_{m} can produce the maximum ratio.

Many Enzymes Catalyze Reactions with Two or More Substrates

We have seen how $[S]$ affects the rate of a simple enzymatic reaction ($S \rightarrow P$) with only one substrate molecule. In most enzymatic reactions, however, two (and sometimes more) different substrate molecules bind to the enzyme and participate in the reaction. For example, in the reaction catalyzed by hexokinase, ATP and glucose are the substrate molecules, and ADP and glucose 6-phosphate are the products:



The rates of such bisubstrate reactions can also be analyzed by the Michaelis-Menten approach. Hexokinase has a characteristic K_{m} for each of its substrates (Table 6-6).

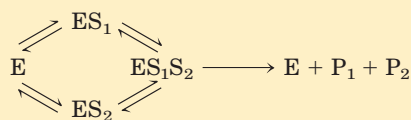
Enzymatic reactions with two substrates usually involve transfer of an atom or a functional group from one substrate to the other. These reactions proceed by one

TABLE 6-8 Enzymes for Which $k_{\text{cat}}/K_{\text{m}}$ Is Close to the Diffusion-Controlled Limit (10^8 to $10^9 \text{ M}^{-1}\text{s}^{-1}$)

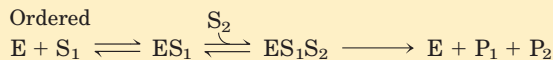
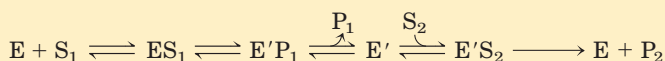
Enzyme	Substrate	k_{cat} (s^{-1})	K_{m} (M)	$k_{\text{cat}}/K_{\text{m}}$ ($\text{M}^{-1}\text{s}^{-1}$)
Acetylcholinesterase	Acetylcholine	1.4×10^4	9×10^{-5}	1.6×10^8
Carbonic anhydrase	CO_2	1.1×10^6	1.2×10^{-2}	8.3×10^7
	HCO_3^-	1.4×10^5	2.6×10^{-2}	1.5×10^7
Catalase	H_2O_2	1.4×10^7	1.1×10^0	4×10^7
Crotonase	Crotonyl-CoA	5.7×10^3	2×10^{-5}	2.8×10^8
Fumarase	Fumarate	1.8×10^2	5×10^{-6}	1.6×10^8
	Malate	1.9×10^2	2.5×10^{-5}	3.6×10^7
β -Lactamase	Benzylpenicillin	2.0×10^3	2×10^{-5}	1×10^8

(a) Enzyme reaction involving a ternary complex

Random order



Ordered

**(b) Enzyme reaction in which no ternary complex is formed**

of several different pathways. In some cases, both substrates are bound to the enzyme concurrently at some point in the course of the reaction, forming a noncovalent ternary complex (Fig. 6-13a); the substrates bind in a random sequence or in a specific order. In other cases, the first substrate is converted to product and dissociates before the second substrate binds, so no ternary complex is formed. An example of this is the Ping-Pong, or double-displacement, mechanism (Fig. 6-13b). Steady-state kinetics can often help distinguish among these possibilities (Fig. 6-14).

Pre-Steady State Kinetics Can Provide Evidence for Specific Reaction Steps

We have introduced kinetics as the primary method for studying the steps in an enzymatic reaction, and we have also outlined the limitations of the most common kinetic parameters in providing such information. The two most important experimental parameters obtained from steady-state kinetics are k_{cat} and $k_{\text{cat}}/K_{\text{m}}$. Variation in k_{cat} and $k_{\text{cat}}/K_{\text{m}}$ with changes in pH or temperature can provide additional information about steps in a reaction pathway. In the case of bisubstrate reactions, steady-state kinetics can help determine whether a ternary complex is formed during the reaction (Fig. 6-14). A more complete picture generally requires more sophisticated kinetic methods that go beyond the scope of an introductory text. Here, we briefly introduce one of the most important kinetic approaches for studying reaction mechanisms, pre-steady state kinetics.

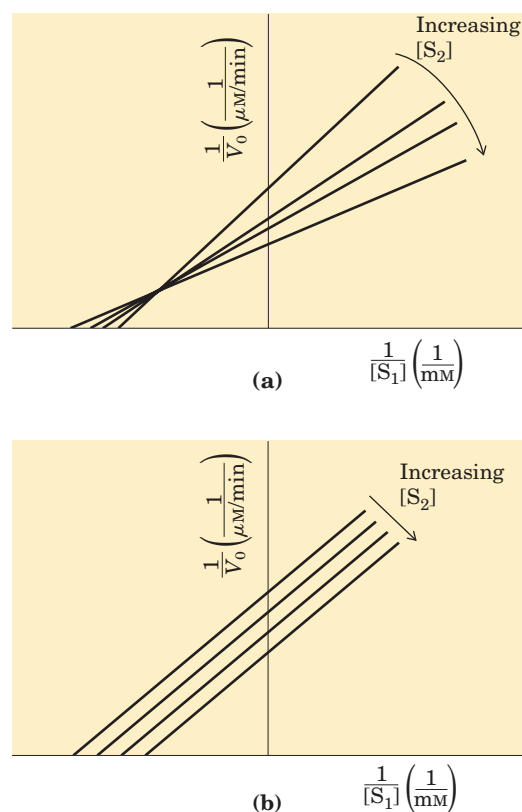
A complete description of an enzyme-catalyzed reaction requires direct measurement of the rates of individual reaction steps—for example, measurement of the association of enzyme and substrate to form the ES complex. It is during the pre-steady state that the rates of many reaction steps can be measured independently. Experimenters adjust reaction conditions so that they can observe events during reaction of a single substrate molecule. Because the pre-steady state phase is gener-

FIGURE 6-13 Common mechanisms for enzyme-catalyzed bisubstrate reactions.

(a) The enzyme and both substrates come together to form a ternary complex. In ordered binding, substrate 1 must bind before substrate 2 can bind productively. In random binding, the substrates can bind in either order.

(b) An enzyme-substrate complex forms, a product leaves the complex, the altered enzyme forms a second complex with another substrate molecule, and the second product leaves, regenerating the enzyme. Substrate 1 may transfer a functional group to the enzyme (to form the covalently modified E'), which is subsequently transferred to substrate 2. This is called a Ping-Pong or double-displacement mechanism.

ally very short, the experiments often require specialized techniques for very rapid mixing and sampling. One objective is to gain a complete and quantitative picture of the energy changes during the reaction. As we have already noted, reaction rates and equilibria are related to the free-energy changes during a reaction. Measur-

**FIGURE 6-14 Steady-state kinetic analysis of bisubstrate reactions.**

In these double-reciprocal plots (see Box 6-1), the concentration of substrate 1 is varied while the concentration of substrate 2 is held constant. This is repeated for several values of $[S_2]$, generating several separate lines. **(a)** Intersecting lines indicate that a ternary complex is formed in the reaction; **(b)** parallel lines indicate a Ping-Pong (double-displacement) pathway.

ing the rate of individual reaction steps reveals how energy is used by a specific enzyme, which is an important component of the overall reaction mechanism. In a number of cases investigators have been able to record the rates of every individual step in a multistep enzymatic reaction. Some examples of the application of pre-steady state kinetics are included in the descriptions of specific enzymes in Section 6.4.

Enzymes Are Subject to Reversible or Irreversible Inhibition

Enzyme inhibitors are molecular agents that interfere with catalysis, slowing or halting enzymatic reactions. Enzymes catalyze virtually all cellular processes, so it should not be surprising that enzyme inhibitors are among the most important pharmaceutical agents known. For example, aspirin (acetylsalicylate) inhibits the enzyme that catalyzes the first step in the synthesis of prostaglandins, compounds involved in many processes, including some that produce pain. The study of enzyme inhibitors also has provided valuable information about enzyme mechanisms and has helped define some metabolic pathways. There are two broad classes of enzyme inhibitors: reversible and irreversible.

Reversible Inhibition One common type of **reversible inhibition** is called competitive (Fig. 6-15a). A **competitive inhibitor** competes with the substrate for the active site of an enzyme. While the inhibitor (I) occupies the active site it prevents binding of the substrate to the enzyme. Many competitive inhibitors are compounds that resemble the substrate and combine with the enzyme to form an EI complex, but without leading to catalysis. Even fleeting combinations of this type will reduce the efficiency of the enzyme. By taking into account the molecular geometry of inhibitors that resemble the substrate, we can reach conclusions about which parts of the normal substrate bind to the enzyme. Competitive inhibition can be analyzed quantitatively by steady-state kinetics. In the presence of a competitive inhibitor, the Michaelis-Menten equation (Eqn 6-9) becomes

$$V_0 = \frac{V_{\max} [S]}{\alpha K_m + [S]} \quad (6-28)$$

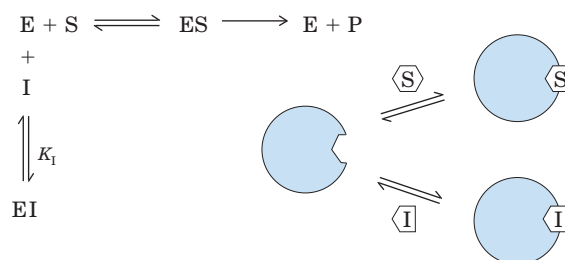
where

$$\alpha = 1 + \frac{[I]}{K_I} \quad \text{and} \quad K_I = \frac{[E][I]}{[EI]}$$

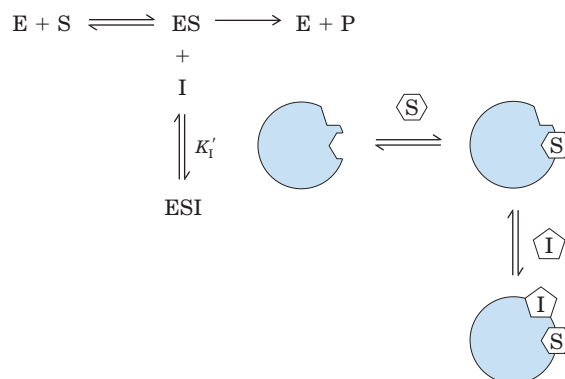
Equation 6-28 describes the important features of competitive inhibition. The experimentally determined variable αK_m , the K_m observed in the presence of the inhibitor, is often called the “apparent” K_m .

Because the inhibitor binds reversibly to the enzyme, the competition can be biased to favor the substrate sim-

(a) Competitive inhibition



(b) Uncompetitive inhibition



(c) Mixed inhibition

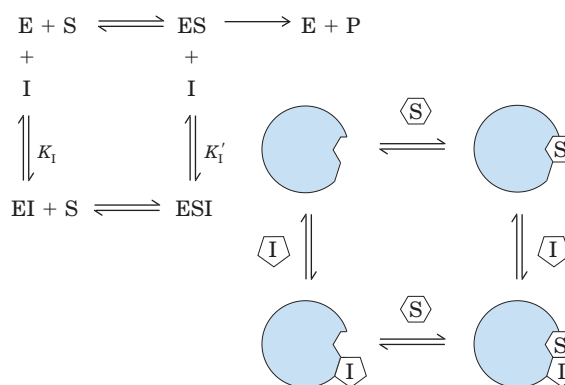


FIGURE 6-15 Three types of reversible inhibition. (a) Competitive inhibitors bind to the enzyme’s active site. (b) Uncompetitive inhibitors bind at a separate site, but bind only to the ES complex. K_I is the equilibrium constant for inhibitor binding to E; K'_I is the equilibrium constant for inhibitor binding to ES. (c) Mixed inhibitors bind at a separate site, but may bind to either E or ES.

ply by adding more substrate. When $[S]$ far exceeds $[I]$, the probability that an inhibitor molecule will bind to the enzyme is minimized and the reaction exhibits a normal $V_{\max}$. However, the $[S]$ at which $V_0 = \frac{1}{2} V_{\max}$, the apparent K_m , increases in the presence of inhibitor by the factor α . This effect on apparent K_m , combined with the absence of an effect on $V_{\max}$, is diagnostic of competitive inhibition and is readily revealed in a double-reciprocal plot (Box 6-2). The equilibrium constant for inhibitor binding, K_I , can be obtained from the same plot.

BOX 6-2 WORKING IN BIOCHEMISTRY

Kinetic Tests for Determining Inhibition Mechanisms

The double-reciprocal plot (see Box 6-1) offers an easy way of determining whether an enzyme inhibitor is competitive, uncompetitive, or mixed. Two sets of rate experiments are carried out, with the enzyme concentration held constant in each set. In the first set, [S] is also held constant, permitting measurement of the effect of increasing inhibitor concentration [I] on the initial rate V_0 (not shown). In the second set, [I] is held constant but [S] is varied. The results are plotted as $1/V_0$ versus $1/[S]$.

Figure 1 shows a set of double-reciprocal plots, one obtained in the absence of inhibitor and two at different concentrations of a competitive inhibitor. Increasing [I] results in a family of lines with a common intercept on the $1/V_0$ axis but with different slopes. Because the intercept on the $1/V_0$ axis equals $1/V_{\max}$, we know that $V_{\max}$ is unchanged by the presence of a competitive inhibitor. That is, regardless of the concentration of a competitive inhibitor, a sufficiently high substrate concentration will always displace the inhibitor from the enzyme's active site. Above the graph is the rearrangement of Equation 6-28 on which the plot is based. The value of α can be calculated

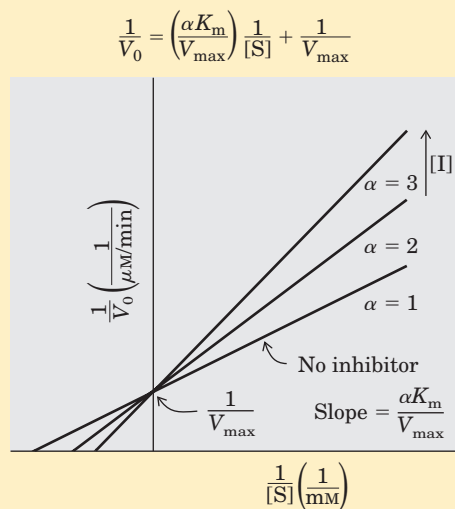


FIGURE 1 Competitive inhibition.

from the change in slope at any given [I]. Knowing [I] and α , we can calculate K_I from the expression

$$\alpha = 1 + \frac{[I]}{K_I}$$

For uncompetitive and mixed inhibition, similar plots of rate data give the families of lines shown in Figures 2 and 3. Changes in axis intercepts signal changes in $V_{\max}$ and K_m .

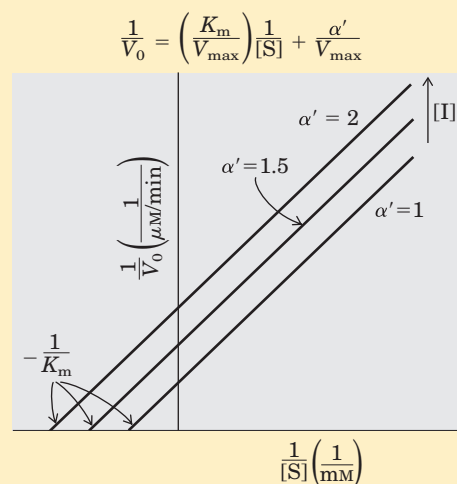


FIGURE 2 Uncompetitive inhibition.

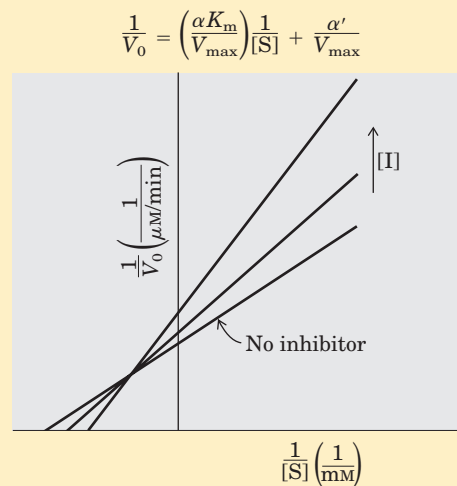


FIGURE 3 Mixed inhibition.



A medical therapy based on competition at the active site is used to treat patients who have ingested methanol, a solvent found in gas-line antifreeze. The liver enzyme alcohol dehydrogenase converts methanol to formaldehyde, which is damaging to many tissues. Blindness is a common result of methanol ingestion, because

the eyes are particularly sensitive to formaldehyde. Ethanol competes effectively with methanol as an alternative substrate for alcohol dehydrogenase. The effect of ethanol is much like that of a competitive inhibitor, with the distinction that ethanol is also a substrate for alcohol dehydrogenase and its concentration will decrease over

time as the enzyme converts it to acetaldehyde. The therapy for methanol poisoning is slow intravenous infusion of ethanol, at a rate that maintains a controlled concentration in the bloodstream for several hours. This slows the formation of formaldehyde, lessening the danger while the kidneys filter out the methanol to be excreted harmlessly in the urine. ■

Two other types of reversible inhibition, uncompetitive and mixed, though often defined in terms of one-substrate enzymes, are in practice observed only with enzymes having two or more substrates. An **uncompetitive inhibitor** (Fig. 6-15b) binds at a site distinct from the substrate active site and, unlike a competitive inhibitor, binds only to the ES complex. In the presence of an uncompetitive inhibitor, the Michaelis-Menten equation is altered to

$$V_0 = \frac{V_{\max} [S]}{K_m + \alpha' [S]} \quad (6-29)$$

where

$$\alpha' = 1 + \frac{[I]}{K_I} \quad \text{and} \quad K_I = \frac{[ES][I]}{[ESI]}$$

As described by Equation 6-29, at high concentrations of substrate, V_0 approaches $V_{\max}/\alpha'$. Thus, an uncompetitive inhibitor lowers the measured $V_{\max}$. Apparent K_m also decreases, because the $[S]$ required to reach one-half $V_{\max}$ decreases by the factor α' .

A **mixed inhibitor** (Fig. 6-15c) also binds at a site distinct from the substrate active site, but it binds to either E or ES. The rate equation describing mixed inhibition is

$$V_0 = \frac{V_{\max} [S]}{\alpha K_m + \alpha' [S]} \quad (6-30)$$

where α and α' are defined as above. A mixed inhibitor usually affects both K_m and $V_{\max}$. The special case of $\alpha = \alpha'$, rarely encountered in experiments, classically has been defined as **noncompetitive inhibition**. Examine Equation 6-30 to see why a noncompetitive inhibitor would affect the $V_{\max}$ but not the K_m .

Equation 6-30 serves as a general expression for the effects of reversible inhibitors, simplifying to the expressions for competitive and uncompetitive inhibition when $\alpha' = 1.0$ or $\alpha = 1.0$, respectively. From this expression we can summarize the effects of inhibitors on individual kinetic parameters. For all reversible inhibitors, apparent $V_{\max} = V_{\max}/\alpha'$, because the right side of Equation 6-30 always simplifies to $V_{\max}/\alpha'$ at sufficiently high substrate concentrations. For competitive inhibitors, $\alpha' = 1.0$ and can thus be ignored. Taking this expression for apparent $V_{\max}$, we can also derive a general expression for apparent K_m to show how this parameter changes in the presence of reversible inhibitors. Apparent K_m , as always, equals the $[S]$ at which V_0 is one-half apparent $V_{\max}$ or, more generally, when $V_0 = V_{\max}/2\alpha'$. This condition is

TABLE 6-9 Effects of Reversible Inhibitors on Apparent $V_{\max}$ and Apparent K_m

Inhibitor type	Apparent $V_{\max}$	Apparent K_m
None	$V_{\max}$	K_m
Competitive	$V_{\max}$	αK_m
Uncompetitive	$V_{\max}/\alpha'$	K_m/α'
Mixed	$V_{\max}/\alpha'$	$\alpha K_m/\alpha'$

met when $[S] = \alpha K_m/\alpha'$. Thus, apparent $K_m = \alpha K_m/\alpha'$. This expression is simpler when either α or α' is 1.0 (for uncompetitive or competitive inhibitors), as summarized in Table 6-9.

In practice, uncompetitive and mixed inhibition are observed only for enzymes with two or more substrates—say, S_1 and S_2 —and are very important in the experimental analysis of such enzymes. If an inhibitor binds to the site normally occupied by S_1 , it may act as a competitive inhibitor in experiments in which $[S_1]$ is varied. If an inhibitor binds to the site normally occupied by S_2 , it may act as a mixed or uncompetitive inhibitor of S_1 . The actual inhibition patterns observed depend on whether the S_1 - and S_2 -binding events are ordered or random, and thus the order in which substrates bind and products leave the active site can be determined. Use of one of the reaction products as an inhibitor is often particularly informative. If only one of two reaction products is present, no reverse reaction can take place. However, a product generally binds to some part of the active site, thus serving as an inhibitor. Enzymologists can use elaborate kinetic studies involving different combinations and amounts of products and inhibitors to develop a detailed picture of the mechanism of a bisubstrate reaction.

Irreversible Inhibition The **irreversible inhibitors** are those that bind covalently with or destroy a functional group on an enzyme that is essential for the enzyme's activity, or those that form a particularly stable noncovalent association. Formation of a covalent link between an irreversible inhibitor and an enzyme is common. Irreversible inhibitors are another useful tool for studying reaction mechanisms. Amino acids with key catalytic functions in the active site can sometimes be identified by determining which residue is covalently linked to an inhibitor after the enzyme is inactivated. An example is shown in Figure 6-16.

A special class of irreversible inhibitors is the **suicide inactivators**. These compounds are relatively unreactive until they bind to the active site of a specific enzyme. A suicide inactivator undergoes the first few chemical steps of the normal enzymatic reaction, but instead of being transformed into the normal product, the

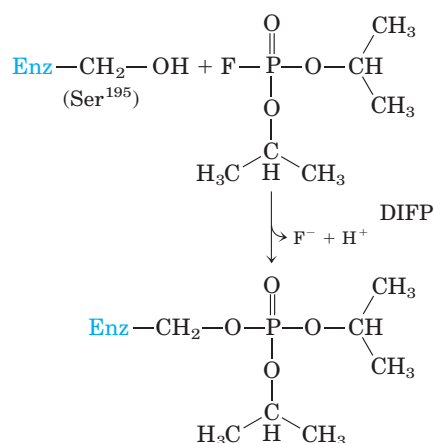


FIGURE 6-16 Irreversible inhibition. Reaction of chymotrypsin with diisopropylfluorophosphate (DIFP) irreversibly inhibits the enzyme. This has led to the conclusion that Ser¹⁹⁵ is the key active-site Ser residue in chymotrypsin.

inactivator is converted to a very reactive compound that combines irreversibly with the enzyme. These compounds are also called **mechanism-based inactivators**, because they hijack the normal enzyme reaction mechanism to inactivate the enzyme. Suicide inactivators play a significant role in *rational drug design*, a modern approach to obtaining new pharmaceutical agents in which chemists synthesize novel substrates based on knowledge of substrates and reaction mechanisms. A well-designed suicide inactivator is specific for a single enzyme and is unreactive until within that enzyme's active site, so drugs based on this approach can offer the important advantage of few side effects (see Box 22-2).

Enzyme Activity Depends on pH

Enzymes have an optimum pH (or pH range) at which their activity is maximal (Fig. 6-17); at higher or lower pH, activity decreases. This is not surprising. Amino acid side chains in the active site may act as weak acids and bases with critical functions that depend on their maintaining a certain state of ionization, and elsewhere in the protein ionized side chains may play an essential role in the interactions that maintain protein structure. Removing a proton from a His residue, for example, might eliminate an ionic interaction essential for stabilizing the active conformation of the enzyme. A less common cause of pH sensitivity is titration of a group on the substrate.

The pH range over which an enzyme undergoes changes in activity can provide a clue to the type of amino acid residue involved (see Table 3-1). A change in activity near pH 7.0, for example, often reflects titration of a His residue. The effects of pH must be interpreted with some caution, however. In the closely packed environment of a protein, the pK_a of amino acid

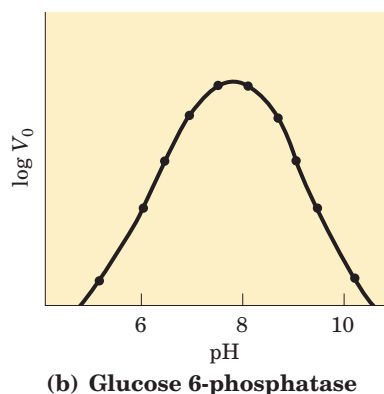
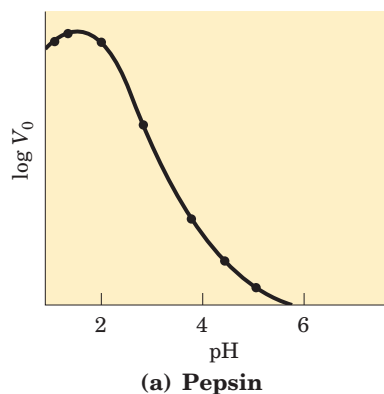


FIGURE 6-17 The pH-activity profiles of two enzymes. These curves are constructed from measurements of initial velocities when the reaction is carried out in buffers of different pH. Because pH is a logarithmic scale reflecting tenfold changes in $[\text{H}^+]$, the changes in V_0 are also plotted on a logarithmic scale. The pH optimum for the activity of an enzyme is generally close to the pH of the environment in which the enzyme is normally found. (a) Pepsin, which hydrolyzes certain peptide bonds of proteins during digestion in the stomach, has a pH optimum of about 1.6. The pH of gastric juice is between 1 and 2. (b) Glucose 6-phosphatase of hepatocytes (liver cells), with a pH optimum of about 7.8, is responsible for releasing glucose into the blood. The normal pH of the cytosol of hepatocytes is about 7.2.

side chains can be significantly altered. For example, a nearby positive charge can lower the pK_a of a Lys residue, and a nearby negative charge can increase it. Such effects sometimes result in a pK_a that is shifted by several pH units from its value in the free amino acid. In the enzyme acetoacetate decarboxylase, for example, one Lys residue has a pK_a of 6.6 (compared with 10.5 in free lysine) due to electrostatic effects of nearby positive charges.

SUMMARY 6.3 Enzyme Kinetics As an Approach to Understanding Mechanism

- Most enzymes have certain kinetic properties in common. When substrate is added to an enzyme, the reaction rapidly achieves a steady state in which the rate at which the ES

complex forms balances the rate at which it reacts. As $[S]$ increases, the steady-state activity of a fixed concentration of enzyme increases in a hyperbolic fashion to approach a characteristic maximum rate, $V_{\max}$, at which essentially all the enzyme has formed a complex with substrate.

- The substrate concentration that results in a reaction rate equal to one-half $V_{\max}$ is the Michaelis constant K_m , which is characteristic for each enzyme acting on a given substrate. The Michaelis-Menten equation

$$V_0 = \frac{V_{\max} [S]}{K_m + [S]}$$

relates initial velocity to $[S]$ and $V_{\max}$ through the constant K_m . Michaelis-Menten kinetics is also called steady-state kinetics.

- K_m and $V_{\max}$ have different meanings for different enzymes. The limiting rate of an enzyme-catalyzed reaction at saturation is described by the constant k_{cat} , the turnover number. The ratio k_{cat}/K_m provides a good measure of catalytic efficiency. The Michaelis-Menten equation is also applicable to bisubstrate reactions, which occur by ternary-complex or Ping-Pong (double-displacement) pathways.
- Reversible inhibition of an enzyme is competitive, uncompetitive, or mixed. Competitive inhibitors compete with substrate by binding reversibly to the active site, but they are not transformed by the enzyme. Uncompetitive inhibitors bind only to the ES complex, at a site distinct from the active site. Mixed inhibitors bind to either E or ES, again at a site distinct from the active site. In irreversible inhibition an inhibitor binds permanently to an active site by forming a covalent bond or a very stable noncovalent interaction.
- Every enzyme has an optimum pH (or pH range) at which it has maximal activity.

6.4 Examples of Enzymatic Reactions

Thus far we have focused on the general principles of catalysis and on introducing some of the kinetic parameters used to describe enzyme action. We now turn to several examples of specific enzyme reaction mechanisms.

An understanding of the complete mechanism of action of a purified enzyme requires identification of all substrates, cofactors, products, and regulators. Moreover, it requires a knowledge of (1) the temporal sequence in which enzyme-bound reaction intermediates

form, (2) the structure of each intermediate and each transition state, (3) the rates of interconversion between intermediates, (4) the structural relationship of the enzyme to each intermediate, and (5) the energy contributed by all reacting and interacting groups to intermediate complexes and transition states. As yet, there is probably no enzyme for which we have an understanding that meets all these requirements. Many decades of research, however, have produced mechanistic information about hundreds of enzymes, and in some cases this information is highly detailed.

We present here the mechanisms for four enzymes: chymotrypsin, hexokinase, enolase, and lysozyme. These examples are not intended to cover all possible classes of enzyme chemistry. They are chosen in part because they are among the best understood enzymes, and in part because they clearly illustrate some general principles outlined in this chapter. The discussion concentrates on selected principles, along with some key experiments that have helped to bring these principles into focus. We use the chymotrypsin example to review some of the conventions used to depict enzyme mechanisms. Much mechanistic detail and experimental evidence is necessarily omitted; no one book could completely document the rich experimental history of these enzymes. Also absent from these discussions is the special contribution of coenzymes to the catalytic activity of many enzymes. The function of coenzymes is chemically varied, and we describe each as it is encountered in Part II.

The Chymotrypsin Mechanism Involves Acylation and Deacylation of a Ser Residue

Bovine pancreatic chymotrypsin (M_r 25,191) is a protease, an enzyme that catalyzes the hydrolytic cleavage of peptide bonds. This protease is specific for peptide bonds adjacent to aromatic amino acid residues (Trp, Phe, Tyr). The three-dimensional structure of chymotrypsin is shown in Figure 6-18, with functional groups in the active site emphasized. The reaction catalyzed by this enzyme illustrates the principle of transition-state stabilization and also provides a classic example of general acid-base catalysis and covalent catalysis.

Chymotrypsin enhances the rate of peptide bond hydrolysis by a factor of at least 10^9 . It does not catalyze a direct attack of water on the peptide bond; instead, a transient covalent acyl-enzyme intermediate is formed. The reaction thus has two distinct phases. In the acylation phase, the peptide bond is cleaved and an ester linkage is formed between the peptide carbonyl carbon and the enzyme. In the deacylation phase, the ester linkage is hydrolyzed and the nonacylated enzyme regenerated.

The first evidence for a covalent acyl-enzyme intermediate came from a classic application of pre-steady state kinetics. In addition to its action on polypeptides,

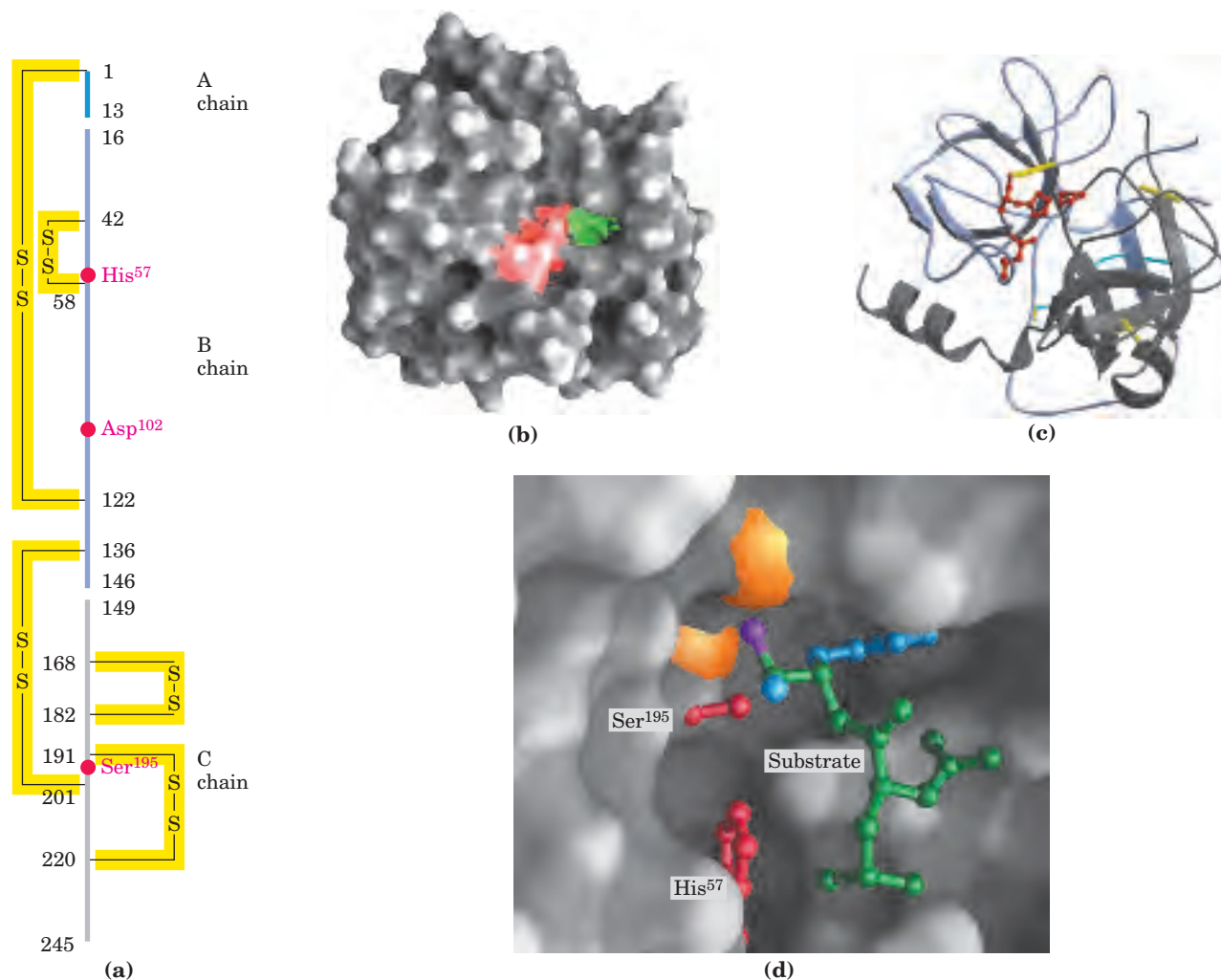


FIGURE 6-18 Structure of chymotrypsin. (PDB ID 7GCH) (a) A representation of primary structure, showing disulfide bonds and the amino acid residues crucial to catalysis. The protein consists of three polypeptide chains linked by disulfide bonds. (The numbering of residues in chymotrypsin, with “missing” residues 14, 15, 147, and 148, is explained in Fig. 6–33.) The active-site amino acid residues are grouped together in the three-dimensional structure. (b) A depiction of the enzyme emphasizing its surface. The pocket in which the aromatic amino acid side chain of the substrate is bound is shown in green. Key active-site residues, including Ser¹⁹⁵, His⁵⁷, and Asp¹⁰², are red. The roles of these residues in catalysis are illustrated in Fig-

ure 6–21. (c) The polypeptide backbone as a ribbon structure. Disulfide bonds are yellow; the three chains are colored as in part (a). (d) A close-up of the active site with a substrate (mostly green) bound. Two of the active-site residues, Ser¹⁹⁵ and His⁵⁷ (both red), are partly visible. Ser¹⁹⁵ attacks the carbonyl group of the substrate (the oxygen is purple); the developing negative charge on the oxygen is stabilized by the oxyanion hole (amide nitrogens in orange), as explained in Figure 6–21. In the substrate, the aromatic amino acid side chain and the amide nitrogen of the peptide bond to be cleaved (protruding toward the viewer and projecting the path of the rest of the substrate polypeptide chain) are in blue.

chymotrypsin also catalyzes the hydrolysis of small esters and amides. These reactions are much slower than hydrolysis of peptides because less binding energy is available with smaller substrates, and they are therefore easier to study. Investigations by B. S. Hartley and B. A. Kilby in 1954 found that chymotrypsin hydrolysis of the ester *p*-nitrophenylacetate, as measured by release of *p*-nitrophenol, proceeded with a rapid burst before leveling off to a slower rate (Fig. 6–19). By extrapolating back to zero time, they concluded that the

burst phase corresponded to just under one molecule of *p*-nitrophenol released for every enzyme molecule present. Hartley and Kilby suggested that this reflected a rapid acylation of all the enzyme molecules (with release of *p*-nitrophenol), with the rate for subsequent turnover of the enzyme limited by a slow deacylation step. Similar results have since been obtained with many other enzymes. The observation of a burst phase provides yet another example of the use of kinetics to break down a reaction into its constituent steps.

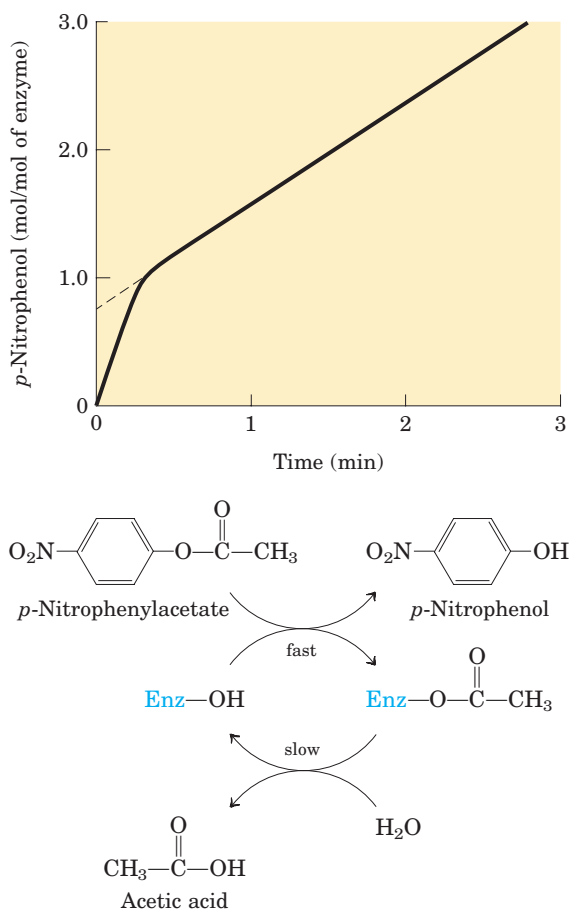


FIGURE 6-19 Pre-steady state kinetic evidence for an acyl-enzyme intermediate. The hydrolysis of p -nitrophenylacetate by chymotrypsin is measured by release of p -nitrophenol (a colored product). Initially, the reaction releases a rapid burst of p -nitrophenol nearly stoichiometric with the amount of enzyme present. This reflects the fast acylation phase of the reaction. The subsequent rate is slower, because enzyme turnover is limited by the rate of the slower deacylation phase.

Additional features of the chymotrypsin mechanism have been elucidated by analyzing the dependence of the reaction on pH. The rate of chymotrypsin-catalyzed cleavage generally exhibits a bell-shaped pH-rate profile (Fig. 6-20). The rates plotted in Figure 6-20a are obtained at low (subsaturating) substrate concentrations and therefore represent $k_{\text{cat}}/K_{\text{m}}$. The plot can be dissected further by first obtaining the maximum rates at each pH, and then plotting k_{cat} alone versus pH (Fig. 6-20b); after obtaining the K_{m} at each pH, researchers can then plot $1/K_{\text{m}}$ (Fig. 6-20c). Kinetic and structural analyses have revealed that the change in k_{cat} reflects the ionization state of His⁵⁷. The decline in k_{cat} at low pH results from protonation of His⁵⁷ (so that it cannot extract a proton from Ser¹⁹⁵ in step ① of the reaction; see Fig. 6-21). This rate reduction illustrates the importance of general acid and general base catalysis in the mechanism for chymotrypsin. The changes in the

$1/K_{\text{m}}$ term reflect the ionization of the α -amino group of Ile¹⁶ (at the amino-terminal end of one of chymotrypsin's three polypeptide chains). This group forms a salt bridge to Asp¹⁹⁴, stabilizing the active conformation of the enzyme. When this group loses its proton at high pH, the salt bridge is eliminated and a conformational change closes the hydrophobic pocket where the

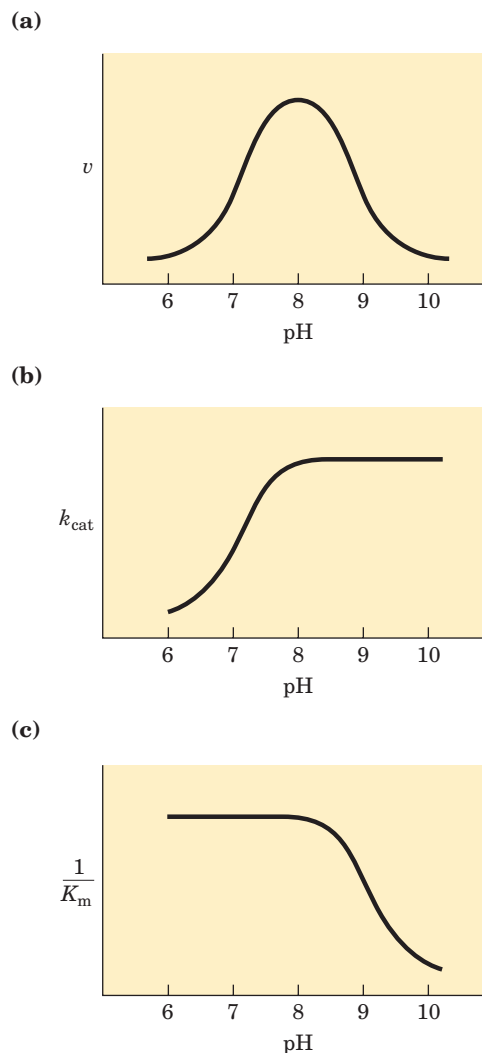


FIGURE 6-20 The pH dependence of chymotrypsin-catalyzed reactions. (a) The rates of chymotrypsin-mediated cleavage produce a bell-shaped pH-rate profile with an optimum at pH 8.0. The rate (v) being plotted is that at low substrate concentrations and thus reflects the term $k_{\text{cat}}/K_{\text{m}}$. The plot can be broken down to its components by using kinetic methods to determine the terms k_{cat} and K_{m} separately at each pH. When this is done (b and c), it becomes clear that the transition just above pH 7 is due to changes in k_{cat} , whereas the transition above pH 8.5 is due to changes in $1/K_{\text{m}}$. Kinetic and structural studies have shown that the transitions illustrated in (b) and (c) reflect the ionization states of the His⁵⁷ side chain (when substrate is not bound) and the α -amino group of Ile¹⁶ (at the amino terminus of the B chain), respectively. For optimal activity, His⁵⁷ must be unprotonated and Ile¹⁶ must be protonated.

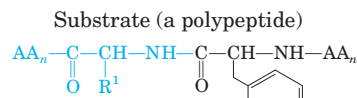
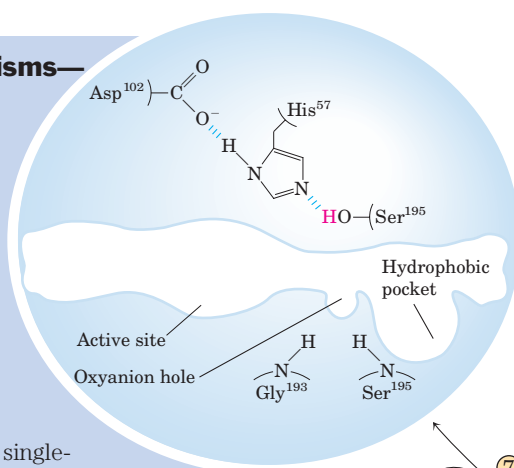
How to Read Reaction Mechanisms—A Refresher

Chemical reaction mechanisms, which trace the formation and breakage of covalent bonds, are communicated with dots and curved arrows, a convention known informally as “electron pushing.” A covalent bond consists of a shared pair of electrons. Nonbonded electrons important to the reaction mechanism are designated by dots ($-\ddot{\text{O}}\text{H}$). Curved arrows ($\curvearrowright$) represent the movement of electron pairs. For movement of a single electron (as in a free radical reaction), a single-headed (fishhook-type) arrow is used ($\frown$). Most reaction steps involve an unshared electron pair (as in the chymotrypsin mechanism).

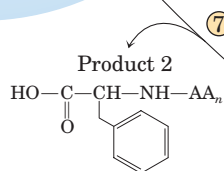
Some atoms are more electronegative than others; that is, they more strongly attract electrons. The relative electronegativities of atoms encountered in this text are $\text{F} > \text{O} > \text{N} > \text{C} \approx \text{S} > \text{P} \approx \text{H}$. For example, the two electron pairs making up a $\text{C}=\text{O}$ (carbonyl) bond are not shared equally; the carbon is relatively electron-deficient as the oxygen draws away the electrons. Many reactions involve an electron-rich atom (a nucleophile) reacting with an electron-deficient atom (an electrophile). Some common nucleophiles and electrophiles in biochemistry are shown at right.

In general, a reaction mechanism is initiated at an unshared electron pair of a nucleophile. In mechanism diagrams, the base of the electron-pushing arrow originates near the electron-pair dots, and the head of the arrow points directly at the electrophilic center being attacked. Where the unshared electron pair confers a formal negative charge on the nucleophile, the negative charge symbol itself can represent the unshared electron pair, and serves as the base of the arrow. In the chymotrypsin mechanism, the nucleophilic electron pair in the ES complex between steps ① and ② is provided by the oxygen of the Ser^{195} hydroxyl group. This electron pair (2 of the 8 valence electrons of the hydroxyl oxygen) provides the base of the curved arrow. The electrophilic center under attack is the carbonyl carbon of the peptide bond to be cleaved. The C, O, and N atoms have a maximum of 8 valence electrons, and H has a maximum of 2. These atoms are occasionally found in unstable states with less than their maximum allotment of electrons, but C, O, and N cannot have more than 8. Thus, when the electron pair from chymotrypsin's Ser^{195} attacks the substrate's carbonyl carbon, an electron pair is displaced from the carbon valence shell (you cannot have 5 bonds to carbon!). These electrons move toward the more electronegative carbonyl oxygen. The oxygen has 8 valence electrons both before and after this chemical process, but the number shared with the carbon is reduced from 4 to 2, and the carbonyl oxygen acquires a negative charge. In the next step, the electron pair conferring the negative charge on the oxygen moves back to re-form a bond with carbon and reestablish the carbonyl linkage. Again, an electron pair must be displaced from the carbon, and this time it is the electron pair shared with the amino group of the peptide linkage. This breaks the peptide bond. The remaining steps follow a similar pattern.

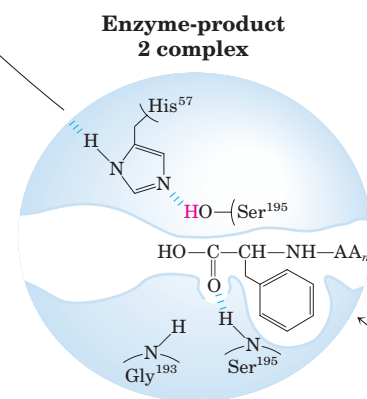
Chymotrypsin (free enzyme)



When substrate binds, the side chain of the residue adjacent to the peptide bond to be cleaved nestles in a hydrophobic pocket on the enzyme, positioning the peptide bond for attack.



Diffusion of the second product from the active site regenerates free enzyme.



Nucleophiles

$-\ddot{\text{O}}^-$
Negatively charged oxygen (as in an unprotonated hydroxyl group or an ionized carboxylic acid)

$-\text{S}^-$
Negatively charged sulfhydryl

$-\text{C}^-$
Carbanion

$-\ddot{\text{N}}^-$
Uncharged amine group

$\text{HN} \cdots \text{N}:$
Imidazole

$\text{H}-\ddot{\text{O}}^-$
Hydroxide ion

Electrophiles

$\text{C}=\ddot{\text{O}}$
Carbon atom of a carbonyl group (the more electronegative oxygen of the carbonyl group pulls electrons away from the carbon)

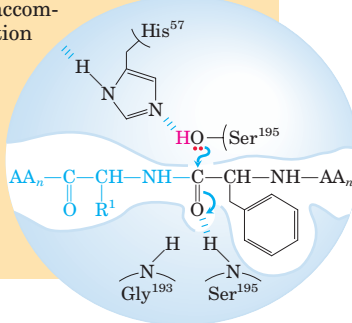
$\text{C}=\text{N}^+\text{H}$
Protonated imine group (activated for nucleophilic attack at the carbon by protonation of the imine)

$\text{O}=\text{P}(=\text{O})(\text{O}^-)_2$
Phosphorus of a phosphate group

H^+
Proton

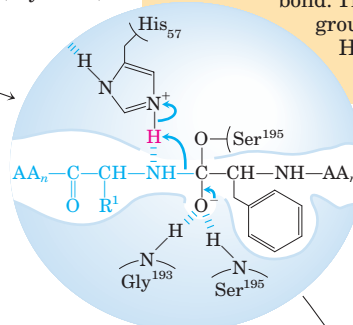
Interaction of Ser¹⁹⁵ and His⁵⁷ generates a strongly nucleophilic alkoxide ion on Ser¹⁹⁵; the ion attacks the peptide carbonyl group, forming a tetrahedral acyl-enzyme. This is accompanied by formation of a short-lived negative charge on the carbonyl oxygen of the substrate, which is stabilized by hydrogen bonding in the oxyanion hole.

ES complex



②

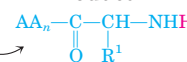
Short-lived intermediate (acylation)



Instability of the negative charge on the substrate carbonyl oxygen leads to collapse of the tetrahedral intermediate; re-formation of a double bond with carbon displaces the bond between carbon and the amino group of the peptide linkage, breaking the peptide bond. The amino leaving group is protonated by His⁵⁷, facilitating its displacement.

③

Product 1

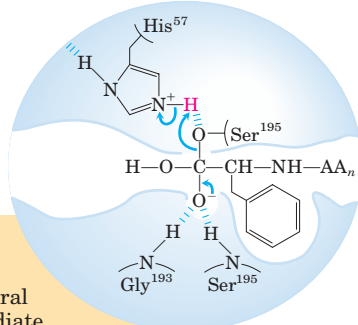


MECHANISM FIGURE 6-21 Hydrolytic cleavage of a peptide bond by chymotrypsin. The reaction has two phases. In the acylation phase (steps ① to ③), formation of a covalent acyl-enzyme intermediate is coupled to cleavage of the peptide bond. In the deacylation phase (steps ④ to ⑦), deacylation regenerates the free enzyme; this is essentially the reverse of the acylation phase, with water mirroring, in reverse, the role of the amine component of the substrate. **Chymotrypsin Mechanism**

Short-lived intermediate (deacylation)

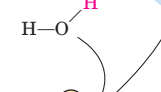
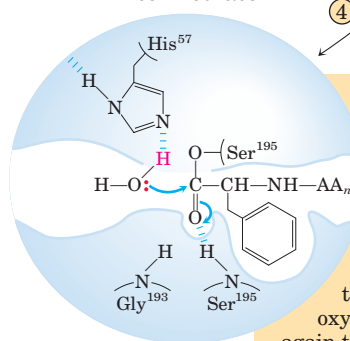
⑥

Collapse of the tetrahedral intermediate forms the second product, a carboxylate anion, and displaces Ser¹⁹⁵.



⑤

Acyl-enzyme intermediate



Acyl-enzyme intermediate

An incoming water molecule is deprotonated by general base catalysis, generating a strongly nucleophilic hydroxide ion. Attack of hydroxide on the ester linkage of the acyl-enzyme generates a second tetrahedral intermediate, with oxygen in the oxyanion hole again taking on a negative charge.

*The tetrahedral intermediate in the chymotrypsin reaction pathway, and the second tetrahedral intermediate that forms later, are sometimes referred to as transition states, which can lead to confusion. An *intermediate* is any chemical species with a finite lifetime, “finite” being defined as longer than the time required for a molecular vibration ($\sim 10^{-13}$ seconds). A *transition state* is simply the maximum-energy species formed on the reaction coordinate and does not have a finite lifetime. The tetrahedral intermediates formed in the chymotrypsin reaction closely resemble, both energetically and structurally, the transition states leading to their formation and breakdown. However, the intermediate represents a committed stage of completed

bond formation, whereas the transition state is part of the process of reaction. In the case of chymotrypsin, given the close relationship between the intermediate and the actual transition state the distinction between them is routinely glossed over. Furthermore, the interaction of the negatively charged oxygen with the amide nitrogens in the oxyanion hole, often referred to as transition-state stabilization, also serves to stabilize the intermediate in this case. Not all intermediates are so short-lived that they resemble transition states. The chymotrypsin acyl-enzyme intermediate is much more stable and more readily detected and studied, and it is never confused with a transition state.

aromatic amino acid side chain of the substrate inserts (Fig. 6-18). Substrates can no longer bind properly, which is measured kinetically as an increase in K_m .

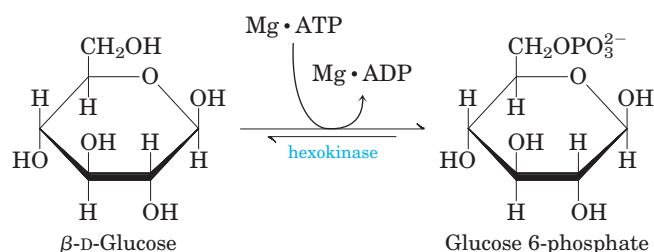
The nucleophile in the acylation phase is the oxygen of Ser¹⁹⁵. (Proteases with a Ser residue that plays this role in reaction mechanisms are called serine proteases.) The pK_a of a Ser hydroxyl group is generally too high for the unprotonated form to be present in significant concentrations at physiological pH. However, in chymotrypsin, Ser¹⁹⁵ is linked to His⁵⁷ and Asp¹⁰² in a hydrogen-bonding network referred to as the **catalytic triad**. When a peptide substrate binds to chymotrypsin, a subtle change in conformation compresses the hydrogen bond between His⁵⁷ and Asp¹⁰², resulting in a stronger interaction, called a low-barrier hydrogen bond. This enhanced interaction increases the pK_a of His⁵⁷ from ~ 7 (for free histidine) to >12 , allowing the His residue to act as an enhanced general base that can remove the proton from the Ser¹⁹⁵ hydroxyl group. Deprotonation prevents development of a very unstable positive charge on the Ser¹⁹⁵ hydroxyl and makes the Ser side chain a stronger nucleophile. At later reaction stages, His⁵⁷ also acts as a proton donor, protonating the amino group in the displaced portion of the substrate (the leaving group).

As the Ser¹⁹⁵ oxygen attacks the carbonyl group of the substrate, a very short-lived tetrahedral intermediate is formed in which the carbonyl oxygen acquires a negative charge (Fig 6-21). This charge, forming within a pocket on the enzyme called the oxyanion hole, is stabilized by hydrogen bonds contributed by the amide groups of two peptide bonds in the chymotrypsin backbone. One of these hydrogen bonds (contributed by Gly¹⁹³) is present only in this intermediate and in the transition states for its formation and breakdown; it reduces the energy required to reach these states. This is an example of the use of binding energy in catalysis.

The role of transition state complementarity in enzyme catalysis is further explored in Box 6-3.

Hexokinase Undergoes Induced Fit on Substrate Binding

Yeast hexokinase (M_r 107,862) is a bisubstrate enzyme that catalyzes the reversible reaction



ATP and ADP always bind to enzymes as a complex with the metal ion Mg^{2+} .

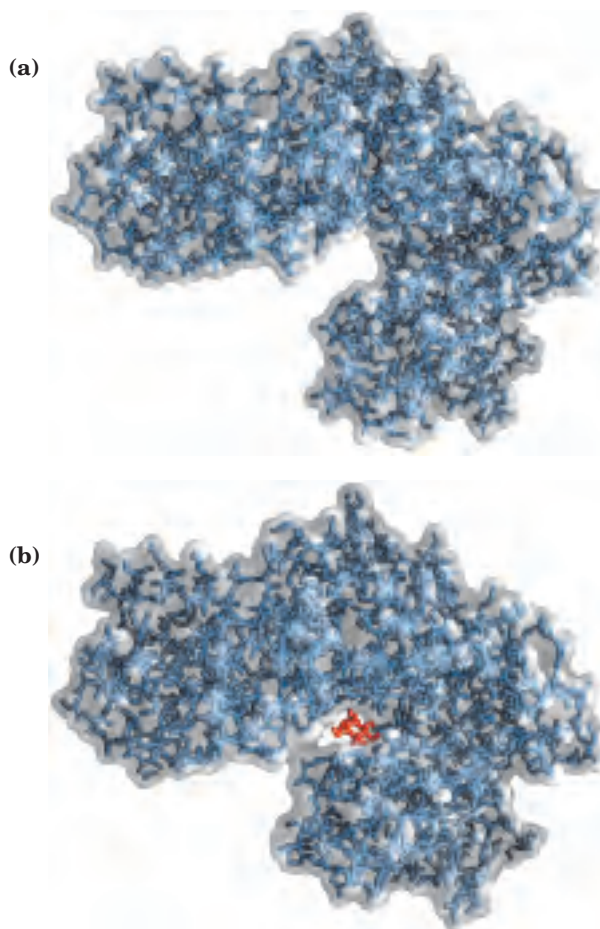
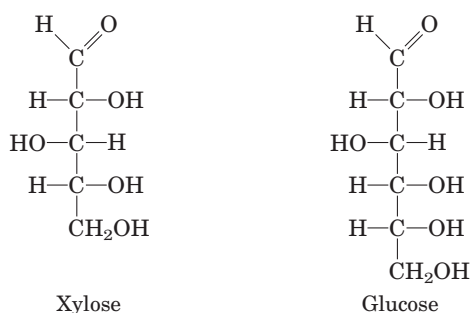


FIGURE 6-22 Induced fit in hexokinase. (a) Hexokinase has a U-shaped structure (PDB ID 2YHX). (b) The ends pinch toward each other in a conformational change induced by binding of D-glucose (red) (derived from PDB ID 1HKG and PDB ID 1GLK).

The hydroxyl at C-6 of glucose (to which the γ -phosphoryl of ATP is transferred in the hexokinase reaction) is similar in chemical reactivity to water, and water freely enters the enzyme active site. Yet hexokinase favors the reaction with glucose by a factor of 10^6 . The enzyme can discriminate between glucose and water because of a conformational change in the enzyme when the correct substrates bind (Fig. 6-22). Hexokinase thus provides a good example of induced fit. When glucose is not present, the enzyme is in an inactive conformation with the active-site amino acid side chains out of position for reaction. When glucose (but not water) and $\text{Mg} \cdot \text{ATP}$ bind, the binding energy derived from this interaction induces a conformational change in hexokinase to the catalytically active form.

This model has been reinforced by kinetic studies. The five-carbon sugar xylose, stereochemically similar to glucose but one carbon shorter, binds to hexokinase but in a position where it cannot be phosphorylated. Nevertheless, addition of xylose to the reaction mixture increases the rate of ATP hydrolysis. Evidently, the binding of xylose is sufficient to induce a change in

hexokinase to its active conformation, and the enzyme is thereby “tricked” into phosphorylating water. The hexokinase reaction also illustrates that enzyme specificity is not always a simple matter of binding one compound but not another. In the case of hexokinase, specificity is observed not in the formation of the ES complex but in the relative rates of subsequent catalytic steps. Water is not excluded from the active site, but reaction rates increase greatly in the presence of the functional phosphoryl group acceptor (glucose).

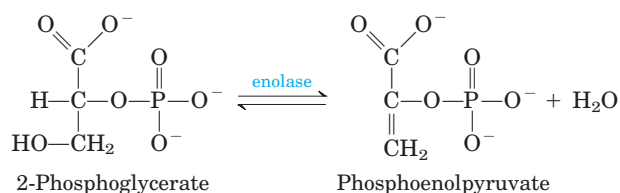


Induced fit is only one aspect of the catalytic mechanism of hexokinase—like chymotrypsin, hexokinase uses several catalytic strategies. For example, the active-

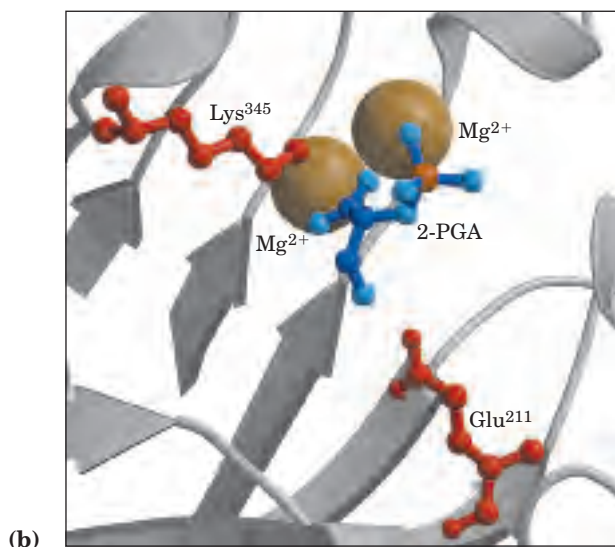
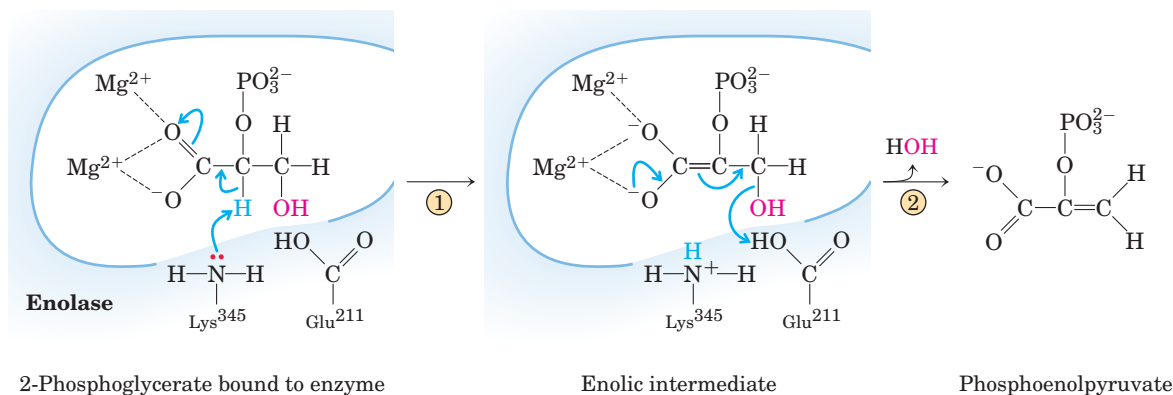
site amino acid residues (those brought into position by the conformational change that follows substrate binding) participate in general acid-base catalysis and transition-state stabilization.

The Enolase Reaction Mechanism Requires Metal Ions

Another glycolytic enzyme, enolase, catalyzes the reversible dehydration of 2-phosphoglycerate to phosphoenolpyruvate:



Yeast enolase (M_r 93,316) is a dimer with 436 amino acid residues per subunit. The enolase reaction illustrates one type of metal ion catalysis and provides an additional example of general acid-base catalysis and transition-state stabilization. The reaction occurs in two steps (Fig. 6-23a). First, Lys³⁴⁵ acts as a general base catalyst,



MECHANISM FIGURE 6-23 Two-step reaction catalyzed by enolase. (a) The mechanism by which enolase converts 2-phosphoglycerate (2-PGA) to phosphoenolpyruvate. The carboxyl group of 2-PGA is coordinated by two magnesium ions at the active site. A proton is abstracted in step ① by general base catalysis (Lys³⁴⁵), and the resulting enolic intermediate is stabilized by the two Mg^{2+} ions. Elimination of the $-OH$ in step ② is facilitated by general acid catalysis (Glu²¹¹). (b) The substrate, 2-PGA, in relation to the Mg^{2+} ions, Lys³⁴⁵, and Glu²¹¹ in the enolase active site. Hydrogen atoms are not shown. All the oxygen atoms of 2-PGA are light blue; phosphorus is orange (PDB ID 1ONE).

Evidence for Enzyme–Transition State Complementarity

The transition state of a reaction is difficult to study because it is so short-lived. To understand enzymatic catalysis, however, we must dissect the interaction between the enzyme and this ephemeral moment in the course of a reaction. Complementarity between an enzyme and the transition state is virtually a requirement for catalysis, because the energy hill upon which the transition state sits is what the enzyme must lower if catalysis is to occur. How can we obtain evidence for enzyme–transition state complementarity? Fortunately, we have a variety of approaches, old and new, to address this problem, each providing compelling evidence in support of this general principle of enzyme action.

Structure–Activity Correlations

If enzymes are complementary to reaction transition states, then some functional groups in both the substrate and the enzyme must interact preferentially in the transition state rather than in the ES complex. Changing these groups should have little effect on formation of the ES complex and hence should not affect kinetic parameters (the dissociation constant, K_d ; or sometimes K_m , if $K_d = K_m$) that reflect the $E + S \rightleftharpoons ES$ equilibrium. Changing these same groups should have a large effect on the overall rate (k_{cat} or k_{cat}/K_m) of the reaction, however, because the bound substrate lacks potential binding interactions needed to lower the activation energy.

An excellent example of this effect is seen in the kinetics associated with a series of related substrates for the enzyme chymotrypsin (Fig. 1). Chymotrypsin

normally catalyzes the hydrolysis of peptide bonds next to aromatic amino acids. The substrates shown in Figure 1 are convenient smaller models for the natural substrates (long polypeptides and proteins). The additional chemical groups added in each substrate (A to B to C) are shaded. As the table shows, the interaction between the enzyme and these added functional groups has a minimal effect on K_m (taken here as a reflection of K_d) but a large, positive effect on k_{cat} and k_{cat}/K_m . This is what we would expect if the interaction contributed largely to stabilization of the transition state. The results also demonstrate that the rate of a reaction can be affected greatly by enzyme–substrate interactions that are physically remote from the covalent bonds that are altered in the enzyme-catalyzed reaction. Chymotrypsin is described in more detail in the text.

A complementary experimental approach is to modify the enzyme, eliminating certain enzyme–substrate interactions by replacing specific amino acid residues through site-directed mutagenesis (see Fig. 9–12). Results from such experiments again demonstrate the importance of binding energy in stabilizing the transition state.

Transition-State Analogs

Even though transition states cannot be observed directly, chemists can often predict the approximate structure of a transition state based on accumulated knowledge about reaction mechanisms. The transition state is by definition transient and so unstable that direct measurement of the binding interaction between this species and the enzyme is impossible. In some

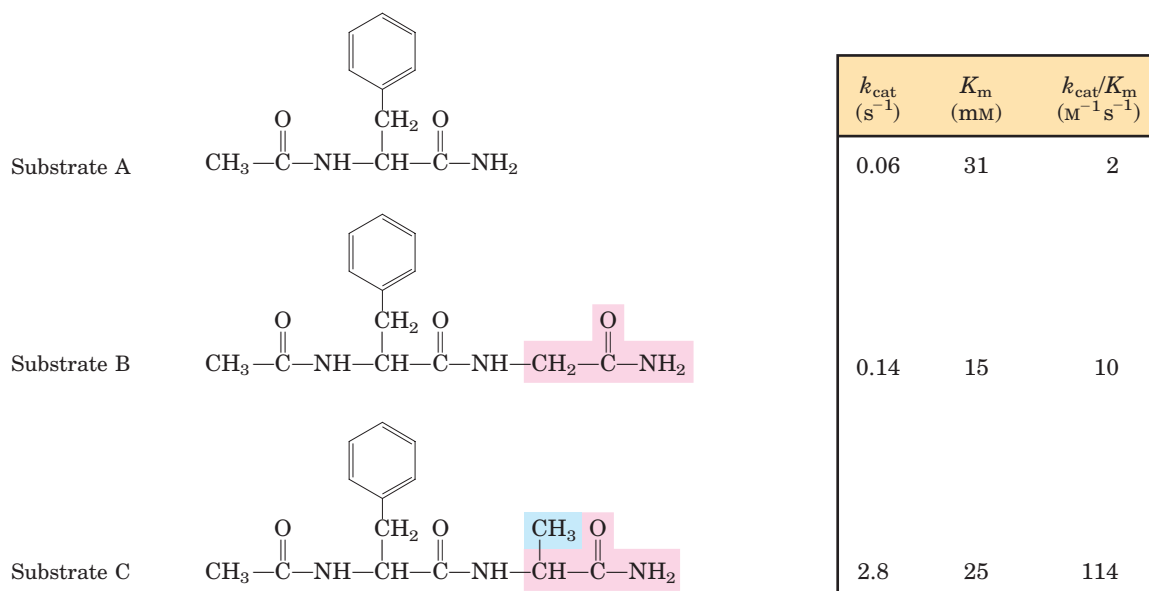


FIGURE 1 Effects of small structural changes in the substrate on kinetic parameters for chymotrypsin-catalyzed amide hydrolysis.

cases, however, stable molecules can be designed that resemble transition states. These are called **transition-state analogs**. In principle, they should bind to an enzyme more tightly than does the substrate in the ES complex, because they should fit the active site better (that is, form a greater number of weak interactions) than the substrate itself. The idea of transition-state analogs was suggested by Pauling in the 1940s, and it has been explored using a number of enzymes. These experiments have the limitation that a transition-state analog cannot perfectly mimic a transition state. Some analogs, however, bind an enzyme 10^2 to 10^6 times more tightly than does the normal substrate, providing good evidence that enzyme active sites are indeed complementary to transition states. The same principle is now used in the pharmaceutical industry to design new drugs. The powerful anti-HIV drugs called protease inhibitors were designed in part as tight-binding transition-state analogs directed at the active site of HIV protease.

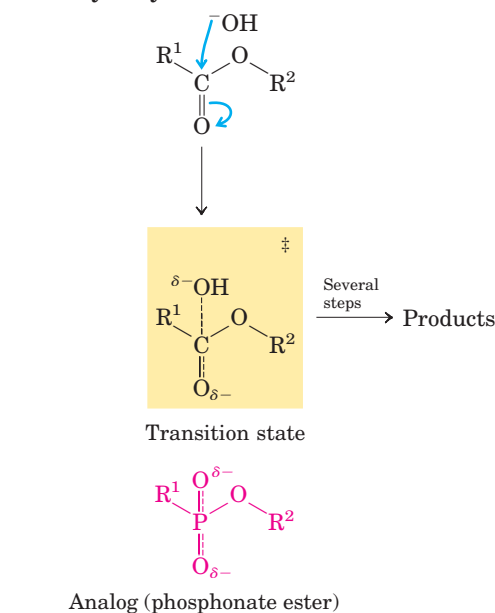
Catalytic Antibodies

If a transition-state analog can be designed for the reaction $S \rightarrow P$, then an antibody that binds tightly to this analog might be expected to catalyze $S \rightarrow P$. Antibodies (immunoglobulins; see Fig. 5-23) are key components of the immune response. When a transition-state analog is used as a protein-bound epitope to stimulate the production of antibodies, the antibodies that bind it are potential catalysts of the corresponding reaction. This use of “catalytic antibodies,” first suggested by William P. Jencks in 1969, has become practical with the development of laboratory techniques to produce quantities of identical antibodies that bind one specific antigen (monoclonal antibodies; see Chapter 5).

Pioneering work in the laboratories of Richard Lerner and Peter Schultz has resulted in the isolation of a number of monoclonal antibodies that catalyze the hydrolysis of esters or carbonates (Fig. 2). In these reactions, the attack by water (OH^-) on the carbonyl carbon produces a tetrahedral transition state in which a partial negative charge has developed on the carbonyl oxygen. Phosphonate ester compounds mimic the structure and charge distribution of this transition state in ester hydrolysis, making them good transition-state analogs; phosphate ester compounds are used for carbonate hydrolysis reactions. Antibodies that bind the phosphonate or phosphate compound tightly have been found to accelerate the corresponding ester or carbonate hydrolysis reaction by factors of 10^3 to 10^4 . Structural analyses of a few of these catalytic antibodies have shown that some catalytic amino acid side chains are arranged such that they could interact with the substrate in the transition state.

Catalytic antibodies generally do not approach the catalytic efficiency of enzymes, but medical and industrial uses for them are nevertheless emerging. For example, catalytic antibodies designed to degrade cocaine are being investigated as a potential aid in the treatment of cocaine addiction.

Ester hydrolysis



Carbonate hydrolysis

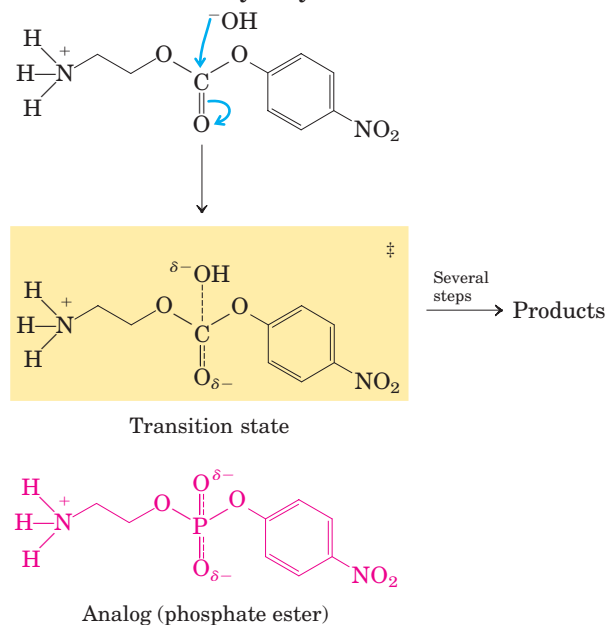


FIGURE 2 The expected transition states for ester or carbonate hydrolysis reactions. Phosphonate ester and phosphate ester compounds, respectively, make good transition-state analogs for these reactions.

abstracting a proton from C-2 of 2-phosphoglycerate; then Glu²¹¹ acts as a general acid catalyst, donating a proton to the —OH leaving group. The proton at C-2 of 2-phosphoglycerate is not very acidic and thus is not readily removed. However, in the enzyme active site, 2-phosphoglycerate undergoes strong ionic interactions with two bound Mg²⁺ ions (Fig. 6-23b), making the C-2 proton more acidic (lowering the pK_a) and easier to abstract. Hydrogen bonding to other active-site amino acid residues also contributes to the overall mechanism. The various interactions effectively stabilize both the enolate intermediate and the transition state preceding its formation.

Lysozyme Uses Two Successive Nucleophilic Displacement Reactions

Lysozyme is a natural antibacterial agent found in tears and egg whites. The hen egg white lysozyme (M_r 14,296) is a monomer with 129 amino acid residues. This was the first enzyme to have its three-dimensional structure determined, by David Phillips and colleagues in 1965. The structure revealed four stabilizing disulfide bonds and a cleft containing the active site (Fig. 6-24a; see also Fig. 4-18). More than five decades of lysozyme investigations have provided a detailed picture of the structure and activity of the enzyme, and an interesting story of how biochemical science progresses.

The substrate of lysozyme is peptidoglycan, a carbohydrate found in many bacterial cell walls (see Fig. 7-22). Lysozyme cleaves the (β 1→4) glycosidic C—O bond between the two types of sugar residue in the molecule, *N*-acetylmuramic acid (Mur2Ac) and *N*-acetylglucosamine (GlcNAc), often referred to as NAM and NAG, respectively, in the research literature on enzymology (Fig. 6-24b). Six residues of the alternating Mur2Ac and GlcNAc in peptidoglycan bind in the active site, in binding sites labeled A through F. Model building has shown that the lactyl side chain of Mur2Ac cannot be accommodated in sites C and E, restricting Mur2Ac binding to sites B, D, and F. Only one of the bound glycosidic bonds is cleaved, that between a Mur2Ac residue in site D and a GlcNAc residue in site E. The key catalytic amino acid residues in the active site are Glu³⁵ and Asp⁵² (Fig. 6-25a). The reaction is a nucleophilic substitution, with —OH from water replacing the GlcNAc at C-1 of Mur2Ac.

With the active site residues identified and a detailed structure of the enzyme available, the path to understanding the reaction mechanism seemed open in the 1960s. However, definitive evidence for a particular mechanism eluded investigators for nearly four decades. There are two chemically reasonable mechanisms that could generate the observed product of lysozyme-mediated cleavage of the glycosidic bond. Phillips and

colleagues proposed a dissociative (S_N1 -type) mechanism (Fig. 6-25a, left), in which the GlcNAc initially dissociates in step ① to leave behind a glycosyl cation (a carbocation) intermediate. In this mechanism, the departing GlcNAc is protonated by general acid catalysis by Glu³⁵, located in a hydrophobic pocket that gives its carboxyl group an unusually high pK_a . The carbocation is stabilized by resonance involving the adjacent ring oxygen, as well as by electrostatic interaction with the negative charge on the nearby Asp⁵². In step ②, water attacks at C-1 of Mur2Ac to yield the product. The alternative mechanism (Fig. 6-25a, right) involves two consecutive direct-displacement (S_N2 -type) steps. In step ①, Asp⁵² attacks C-1 of Mur2Ac to displace the GlcNAc. As in the first mechanism, Glu³⁵ acts as a general acid to protonate the departing GlcNAc. In step ②, water attacks at C-1 of Mur2Ac to displace the Asp⁵² and generate product.

The Phillips mechanism (S_N1), based on structural considerations and bolstered by a variety of binding studies with artificial substrates, was widely accepted for more than three decades. However, some controversy persisted and tests continued. The scientific method sometimes advances an issue slowly, and a truly insightful experiment can be difficult to design. Some early arguments against the Phillips mechanism were suggestive but not completely persuasive. For example, the half-life of the proposed glycosyl cation was estimated to be 10^{-12} seconds, just longer than a molecular vibration and not long enough for the needed diffusion of other molecules. More important, lysozyme is a member of a family of enzymes called “retaining glycosidases,” all of which catalyze reactions in which the product has the same anomeric configuration as the substrate (anomeric configurations of carbohydrates are examined in Chapter 7), and all of which are known to have reactive covalent intermediates like that envisioned in the alternative (S_N2) pathway. Hence, the Phillips mechanism ran counter to experimental findings for closely related enzymes.

A compelling experiment tipped the scales decidedly in favor of the S_N2 pathway, as reported by Stephen Withers and colleagues in 2001. Making use of a mutant enzyme (with residue 35 changed from Glu to Gln) and artificial substrates, which combined to slow the rate of key steps in the reaction, these workers were able to stabilize the elusive covalent intermediate. This in turn allowed them to observe the intermediate directly, using both mass spectrometry and x-ray crystallography (Fig. 6-25b).

Is the lysozyme mechanism now proven? No. A key feature of the scientific method, as Albert Einstein once summarized it, is “No amount of experimentation can ever prove me right; a single experiment can prove me wrong.” In the case of the lysozyme mechanism,

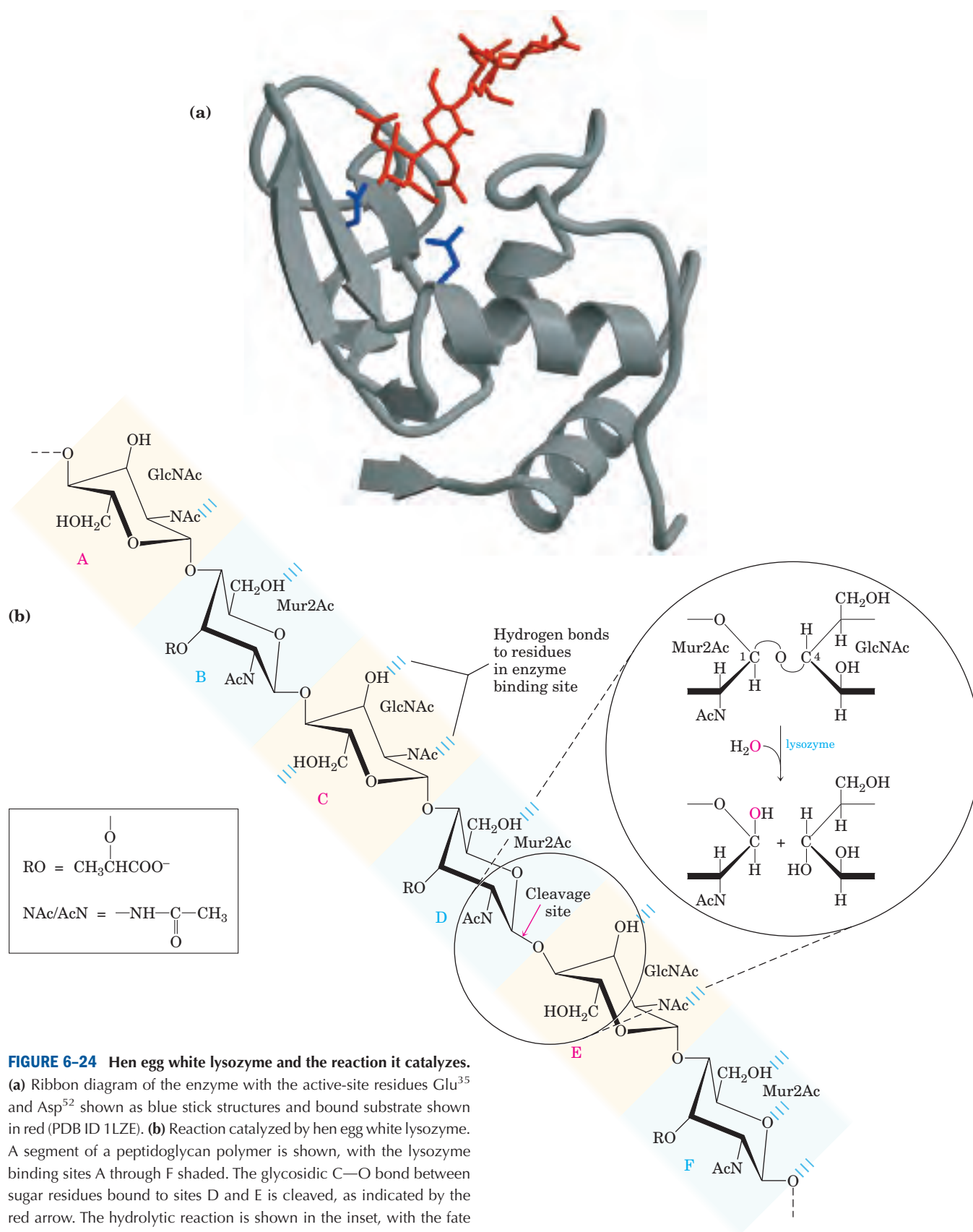
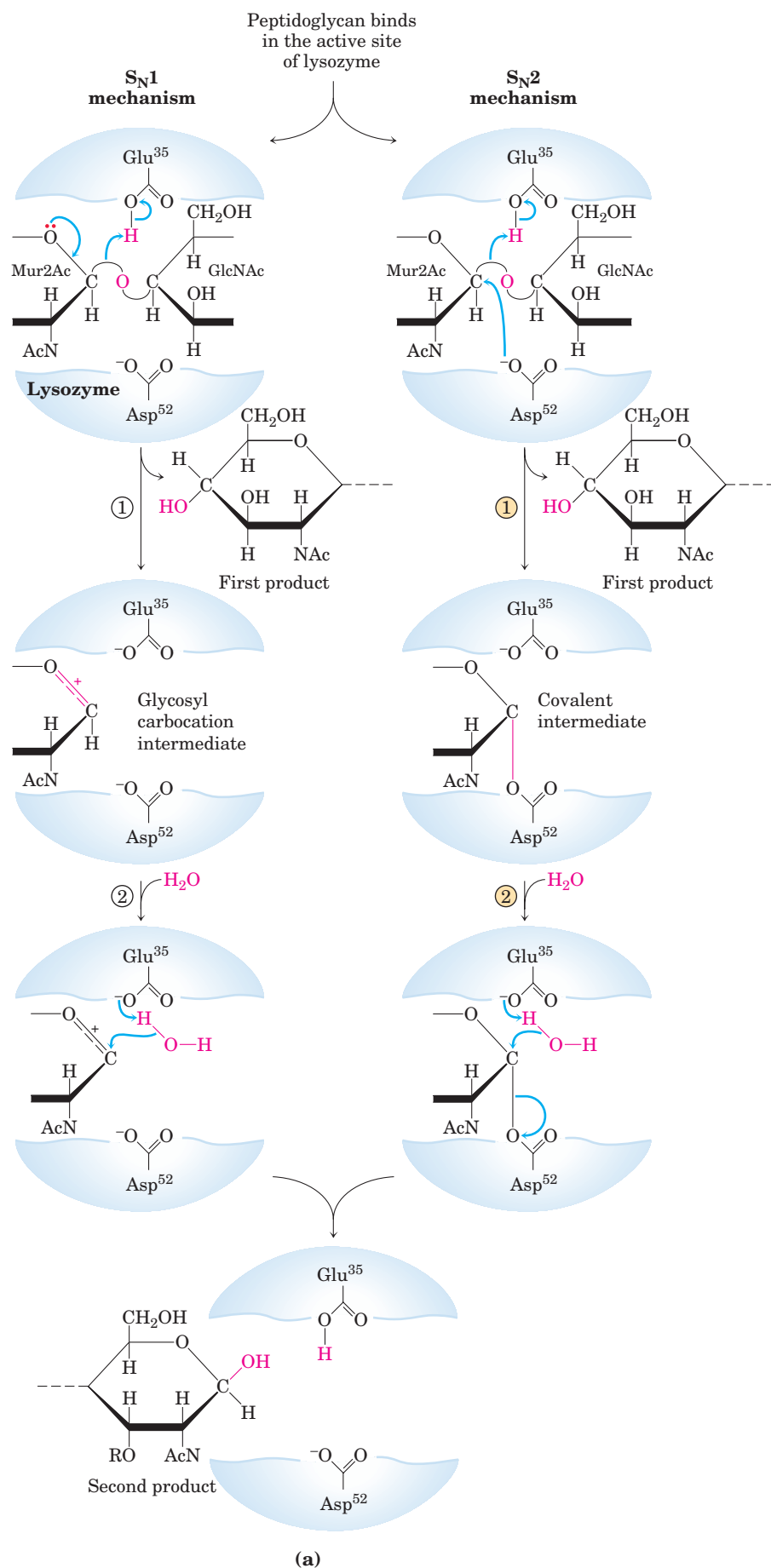
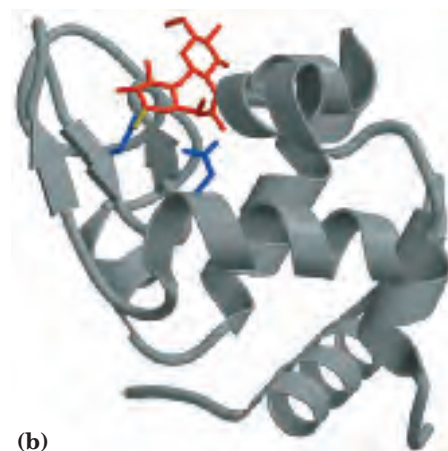


FIGURE 6-24 Hen egg white lysozyme and the reaction it catalyzes.

(a) Ribbon diagram of the enzyme with the active-site residues Glu³⁵ and Asp⁵² shown as blue stick structures and bound substrate shown in red (PDB ID 1LZE). (b) Reaction catalyzed by hen egg white lysozyme. A segment of a peptidoglycan polymer is shown, with the lysozyme binding sites A through F shaded. The glycosidic C—O bond between sugar residues bound to sites D and E is cleaved, as indicated by the red arrow. The hydrolytic reaction is shown in the inset, with the fate of the oxygen in the H₂O traced in red. Mur2Ac is *N*-acetylmuramic acid; GlcNAc, *N*-acetylglucosamine. RO— represents a lactyl (lactic acid) group; —NAc and AcN—, an *N*-acetyl group (see key).

**MECHANISM FIGURE 6-25** Lysozyme

reaction. In this reaction (described on p. 222), the water introduced into the product at C-1 of Mur2Ac is in the same configuration as the original glycosidic bond. The reaction is thus a molecular substitution with retention of configuration. **(a)** Two proposed pathways potentially explain the overall reaction and its properties. The S_N1 pathway (left) is the original Phillips mechanism. The S_N2 pathway (right) is the mechanism most consistent with current data. **(b)** A ribbon diagram of the covalent enzyme-substrate intermediate with the active-site residues (blue) and bound substrate (red) shown as stick structures (PDB ID 1H6M).



one might argue (and some have) that the artificial substrates, with fluorine substitutions at C-1 and C-2, that were used to stabilize the covalent intermediate might have altered the reaction pathway. The highly electronegative fluorine could destabilize an already electron-deficient oxocarbenium ion in the glycosyl cation intermediate that might occur in an S_N1 pathway. However, the S_N2 pathway is now the mechanism most in concert with available data.

SUMMARY 6.4 Examples of Enzymatic Reactions

- Chymotrypsin is a serine protease with a well-understood mechanism, featuring general acid-base catalysis, covalent catalysis, and transition-state stabilization.
- Hexokinase provides an excellent example of induced fit as a means of using substrate binding energy.
- The enolase reaction proceeds via metal ion catalysis.
- Lysozyme makes use of covalent catalysis and general acid catalysis as it promotes two successive nucleophilic displacement reactions.

6.5 Regulatory Enzymes

In cellular metabolism, groups of enzymes work together in sequential pathways to carry out a given metabolic process, such as the multireaction breakdown of glucose to lactate or the multireaction synthesis of an amino acid from simpler precursors. In such enzyme systems, the reaction product of one enzyme becomes the substrate of the next.

Most of the enzymes in each metabolic pathway follow the kinetic patterns we have already described. Each pathway, however, includes one or more enzymes that have a greater effect on the rate of the overall sequence. These **regulatory enzymes** exhibit increased or decreased catalytic activity in response to certain signals. Adjustments in the rate of reactions catalyzed by regulatory enzymes, and therefore in the rate of entire metabolic sequences, allow the cell to meet changing needs for energy and for biomolecules required in growth and repair.

In most multienzyme systems, the first enzyme of the sequence is a regulatory enzyme. This is an excellent place to regulate a pathway, because catalysis of even the first few reactions of a sequence that leads to an unneeded product diverts energy and metabolites from more important processes. Other enzymes in the sequence are usually present at levels that provide an excess of catalytic activity; they can generally promote

their reactions as fast as their substrates are made available from preceding reactions.

The activities of regulatory enzymes are modulated in a variety of ways. **Allosteric enzymes** function through reversible, noncovalent binding of regulatory compounds called **allosteric modulators** or **allosteric effectors**, which are generally small metabolites or cofactors. Other enzymes are regulated by reversible **covalent modification**. Both classes of regulatory enzymes tend to be multisubunit proteins, and in some cases the regulatory site(s) and the active site are on separate subunits. Metabolic systems have at least two other mechanisms of enzyme regulation. Some enzymes are stimulated or inhibited when they are bound by separate regulatory proteins. Others are activated when peptide segments are removed by proteolytic cleavage; unlike effector-mediated regulation, regulation by proteolytic cleavage is irreversible. Important examples of both mechanisms are found in physiological processes such as digestion, blood clotting, hormone action, and vision.

Cell growth and survival depend on efficient use of resources, and this efficiency is made possible by regulatory enzymes. No single rule governs the occurrence of different types of regulation in different systems. To a degree, allosteric (noncovalent) regulation may permit fine-tuning of metabolic pathways that are required continuously but at different levels of activity as cellular conditions change. Regulation by covalent modification may be all or none—usually the case with proteolytic cleavage—or it may allow for subtle changes in activity. Several types of regulation may occur in a single regulatory enzyme. The remainder of this chapter is devoted to a discussion of these methods of enzyme regulation.

Allosteric Enzymes Undergo Conformational Changes in Response to Modulator Binding

As we saw in Chapter 5, allosteric proteins are those having “other shapes” or conformations induced by the binding of modulators. The same concept applies to certain regulatory enzymes, as conformational changes induced by one or more modulators interconvert more-active and less-active forms of the enzyme. The modulators for allosteric enzymes may be inhibitory or stimulatory. Often the modulator is the substrate itself; regulatory enzymes for which substrate and modulator are identical are called homotropic. The effect is similar to that of O_2 binding to hemoglobin (Chapter 5): binding of the ligand—or substrate, in the case of enzymes—causes conformational changes that affect the subsequent activity of other sites on the protein. When the modulator is a molecule other than the substrate, the enzyme is said to be heterotropic. Note that allosteric modulators should not be confused with uncompetitive

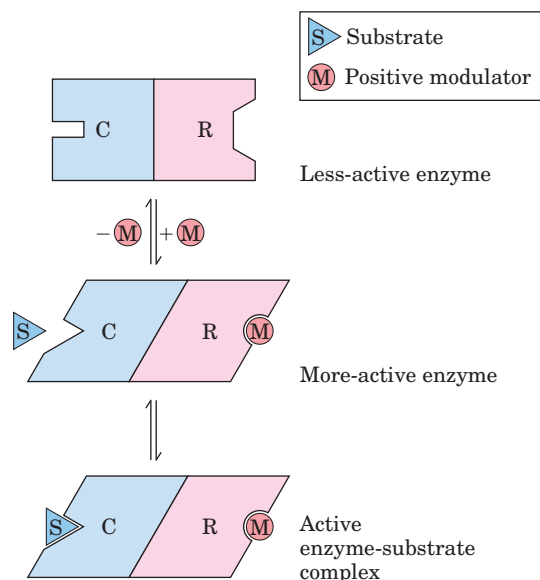


FIGURE 6-26 Subunit interactions in an allosteric enzyme, and interactions with inhibitors and activators. In many allosteric enzymes the substrate binding site and the modulator binding site(s) are on different subunits, the catalytic (C) and regulatory (R) subunits, respectively. Binding of the positive (stimulatory) modulator (M) to its specific site on the regulatory subunit is communicated to the catalytic subunit through a conformational change. This change renders the catalytic subunit active and capable of binding the substrate (S) with higher affinity. On dissociation of the modulator from the regulatory subunit, the enzyme reverts to its inactive or less active form.

and mixed inhibitors. Although the latter bind at a second site on the enzyme, they do not necessarily mediate conformational changes between active and inactive forms, and the kinetic effects are distinct.

The properties of allosteric enzymes are significantly different from those of simple nonregulatory enzymes. Some of the differences are structural. In addition to active sites, allosteric enzymes generally have one or more regulatory, or allosteric, sites for binding the modulator (Fig. 6-26). Just as an enzyme's active site is specific for its substrate, each regulatory site is specific for its modulator. Enzymes with several modulators generally have different specific binding sites for each. In homotropic enzymes, the active site and regulatory site are the same.

Allosteric enzymes are generally larger and more complex than nonallosteric enzymes. Most have two or more subunits. Aspartate transcarbamoylase, which catalyzes an early reaction in the biosynthesis of pyrimidine nucleotides (see Fig. 22-36), has 12 polypeptide chains organized into catalytic and regulatory subunits. Figure 6-27 shows the quaternary structure of this enzyme, deduced from x-ray analysis.

In Many Pathways a Regulated Step Is Catalyzed by an Allosteric Enzyme

In some multienzyme systems, the regulatory enzyme is specifically inhibited by the end product of the pathway whenever the concentration of the end product exceeds the cell's requirements. When the regulatory enzyme reaction is slowed, all subsequent enzymes operate at reduced rates as their substrates are depleted. The rate

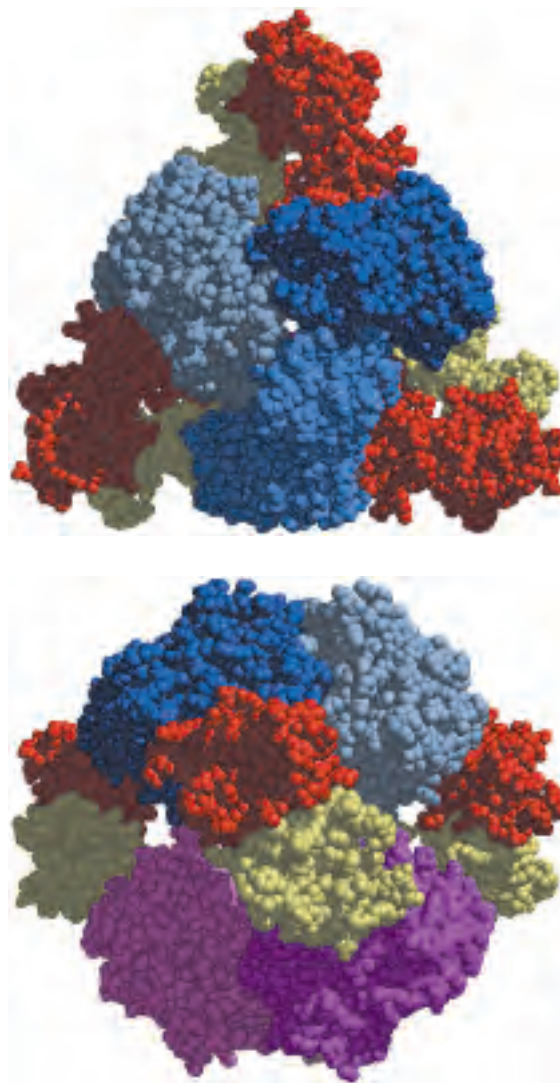


FIGURE 6-27 Two views of the regulatory enzyme aspartate transcarbamoylase. (Derived from PDB ID 2AT2.) This allosteric regulatory enzyme has two stacked catalytic clusters, each with three catalytic polypeptide chains (in shades of blue and purple), and three regulatory clusters, each with two regulatory polypeptide chains (in red and yellow). The regulatory clusters form the points of a triangle surrounding the catalytic subunits. Binding sites for allosteric modulators are on the regulatory subunits. Modulator binding produces large changes in enzyme conformation and activity. The role of this enzyme in nucleotide synthesis, and details of its regulation, are discussed in Chapter 22.

of production of the pathway's end product is thereby brought into balance with the cell's needs. This type of regulation is called **feedback inhibition**. Buildup of the end product ultimately slows the entire pathway.

One of the first known examples of allosteric feedback inhibition was the bacterial enzyme system that catalyzes the conversion of L-threonine to L-isoleucine in five steps (Fig. 6–28). In this system, the first enzyme, threonine dehydratase, is inhibited by isoleucine, the product of the last reaction of the series. This is an example of heterotropic allosteric inhibition. Isoleucine is quite specific as an inhibitor. No other intermediate in this sequence inhibits threonine dehydratase, nor is any other enzyme in the sequence inhibited by isoleucine. Isoleucine binds not to the active site but to another specific site on the enzyme molecule, the regulatory site. This binding is noncovalent and readily reversible; if the isoleucine concentration decreases, the rate of threonine dehydration increases. Thus threonine dehydratase activity responds rapidly and reversibly to fluctuations in the cellular concentration of isoleucine.

The Kinetic Properties of Allosteric Enzymes Diverge from Michaelis-Menten Behavior

Allosteric enzymes show relationships between V_0 and $[S]$ that differ from Michaelis-Menten kinetics. They do exhibit saturation with the substrate when $[S]$ is sufficiently high, but for some allosteric enzymes, plots of V_0 versus $[S]$ (Fig. 6–29) produce a sigmoid saturation curve, rather than the hyperbolic curve typical of non-regulatory enzymes. On the sigmoid saturation curve we can find a value of $[S]$ at which V_0 is half-maximal, but we cannot refer to it with the designation K_m , because the enzyme does not follow the hyperbolic Michaelis-Menten relationship. Instead, the symbol $[S]_{0.5}$ or $K_{0.5}$ is often used to represent the substrate concentration giving half-maximal velocity of the reaction catalyzed by an allosteric enzyme (Fig. 6–29).

Sigmoid kinetic behavior generally reflects cooperative interactions between protein subunits. In other words, changes in the structure of one subunit are translated into structural changes in adjacent subunits, an effect mediated by noncovalent interactions at the interface between subunits. The principles are particularly well illustrated by a nonenzyme: O_2 binding to hemoglobin. Sigmoid kinetic behavior is explained by the concerted and sequential models for subunit interactions (see Fig. 5–15).

Homotropic allosteric enzymes generally are multi-subunit proteins and, as noted earlier, the same binding site on each subunit functions as both the active site and the regulatory site. Most commonly, the substrate acts as a positive modulator (an activator), because the subunits act cooperatively: the binding of one molecule

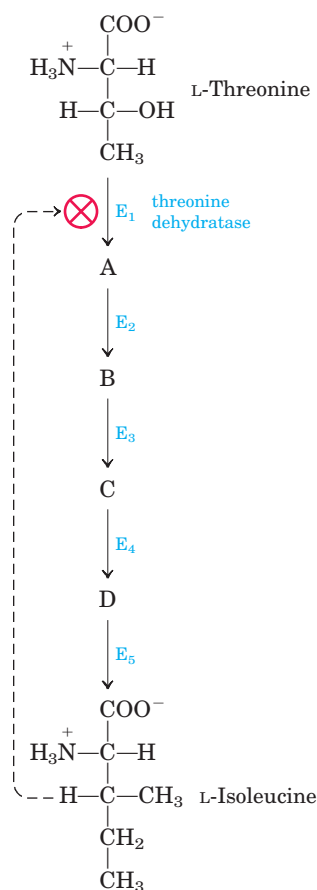


FIGURE 6–28 Feedback inhibition. The conversion of L-threonine to L-isoleucine is catalyzed by a sequence of five enzymes (E_1 to E_5). Threonine dehydratase (E_1) is specifically inhibited allosterically by L-isoleucine, the end product of the sequence, but not by any of the four intermediates (A to D). Feedback inhibition is indicated by the dashed feedback line and the $\otimes$ symbol at the threonine dehydratase reaction arrow, a device used throughout this book.

of substrate to one binding site alters the enzyme's conformation and enhances the binding of subsequent substrate molecules. This accounts for the sigmoid rather than hyperbolic change in V_0 with increasing $[S]$. One characteristic of sigmoid kinetics is that small changes in the concentration of a modulator can be associated with large changes in activity. As is evident in Figure 6–29a, a relatively small increase in $[S]$ in the steep part of the curve causes a comparatively large increase in V_0 .

For heterotropic allosteric enzymes, those whose modulators are metabolites other than the normal substrate, it is difficult to generalize about the shape of the substrate-saturation curve. An activator may cause the curve to become more nearly hyperbolic, with a decrease in $K_{0.5}$ but no change in $V_{\max}$, resulting in an increased reaction velocity at a fixed substrate concentration (V_0 is higher for any value of $[S]$; Fig. 6–29b, upper curve).

Other heterotropic allosteric enzymes respond to an activator by an increase in $V_{\max}$ with little change in $K_{0.5}$ (Fig. 6–29c). A negative modulator (an inhibitor) may produce a *more* sigmoid substrate-saturation curve,

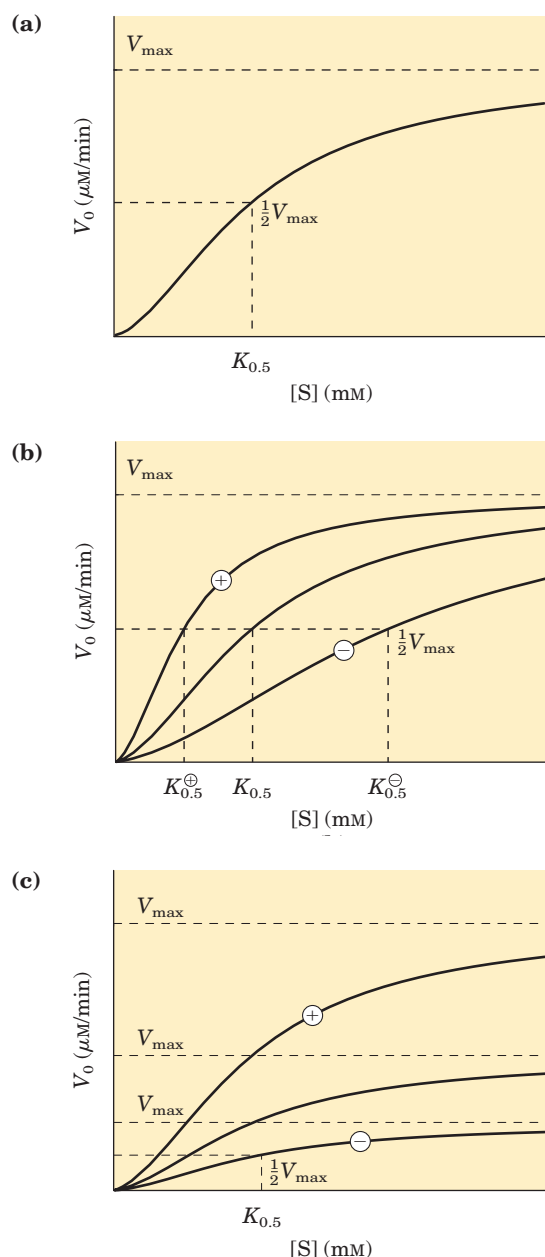


FIGURE 6–29 Substrate-activity curves for representative allosteric enzymes. Three examples of complex responses of allosteric enzymes to their modulators. **(a)** The sigmoid curve of a homotropic enzyme, in which the substrate also serves as a positive (stimulatory) modulator, or activator. Note the resemblance to the oxygen-saturation curve of hemoglobin (see Fig. 5–12). **(b)** The effects of a positive modulator (+) and a negative modulator (–) on an allosteric enzyme in which $K_{0.5}$ is altered without a change in $V_{\max}$. The central curve shows the substrate-activity relationship without a modulator. **(c)** A less common type of modulation, in which $V_{\max}$ is altered and $K_{0.5}$ is nearly constant.

with an increase in $K_{0.5}$ (Fig. 6–29b, lower curve). Heterotropic allosteric enzymes therefore show different kinds of responses in their substrate-activity curves, because some have inhibitory modulators, some have activating modulators, and some have both.

Some Regulatory Enzymes Undergo Reversible Covalent Modification

In another important class of regulatory enzymes, activity is modulated by covalent modification of the enzyme molecule. Modifying groups include phosphoryl, adenylyl, uridylyl, methyl, and adenosine diphosphate ribosyl groups (Fig. 6–30). These groups are generally linked to and removed from the regulatory enzyme by separate enzymes.

An example of an enzyme regulated by methylation is the methyl-accepting chemotaxis protein of bacteria. This protein is part of a system that permits a bacterium to swim toward an attractant (such as a sugar) in solution and away from repellent chemicals. The methylating agent is *S*-adenosylmethionine (adoMet) (see Fig. 18–18b). ADP-ribosylation is an especially interesting reaction, observed in only a few proteins; the ADP-ribose is derived from nicotinamide adenine dinucleotide (NAD) (see Fig. 8–41). This type of modification occurs for the bacterial enzyme dinitrogenase reductase, resulting in regulation of the important process of biological nitrogen fixation. Diphtheria toxin and cholera toxin are enzymes that catalyze the ADP-ribosylation (and inactivation) of key cellular enzymes or proteins. Diphtheria toxin acts on and inhibits elongation factor 2, a protein involved in protein biosynthesis. Cholera toxin acts on a G protein that is part of a signaling pathway (see Fig. 12–39), leading to several physiological responses including a massive loss of body fluids and, sometimes, death.

Phosphorylation is the most common type of regulatory modification; one-third to one-half of all proteins in a eukaryotic cell are phosphorylated. Some proteins have only one phosphorylated residue, others have several, and a few have dozens of sites for phosphorylation. This mode of covalent modification is central to a large number of regulatory pathways, and we therefore discuss it in considerable detail.

Phosphoryl Groups Affect the Structure and Catalytic Activity of Proteins

The attachment of phosphoryl groups to specific amino acid residues of a protein is catalyzed by **protein kinases**; removal of phosphoryl groups is catalyzed by **protein phosphatases**. The addition of a phosphoryl group to a Ser, Thr, or Tyr residue introduces a bulky, charged group into a region that was only moderately polar. The oxygen atoms of a phosphoryl group can hydrogen-bond with one or several groups in a protein, commonly the

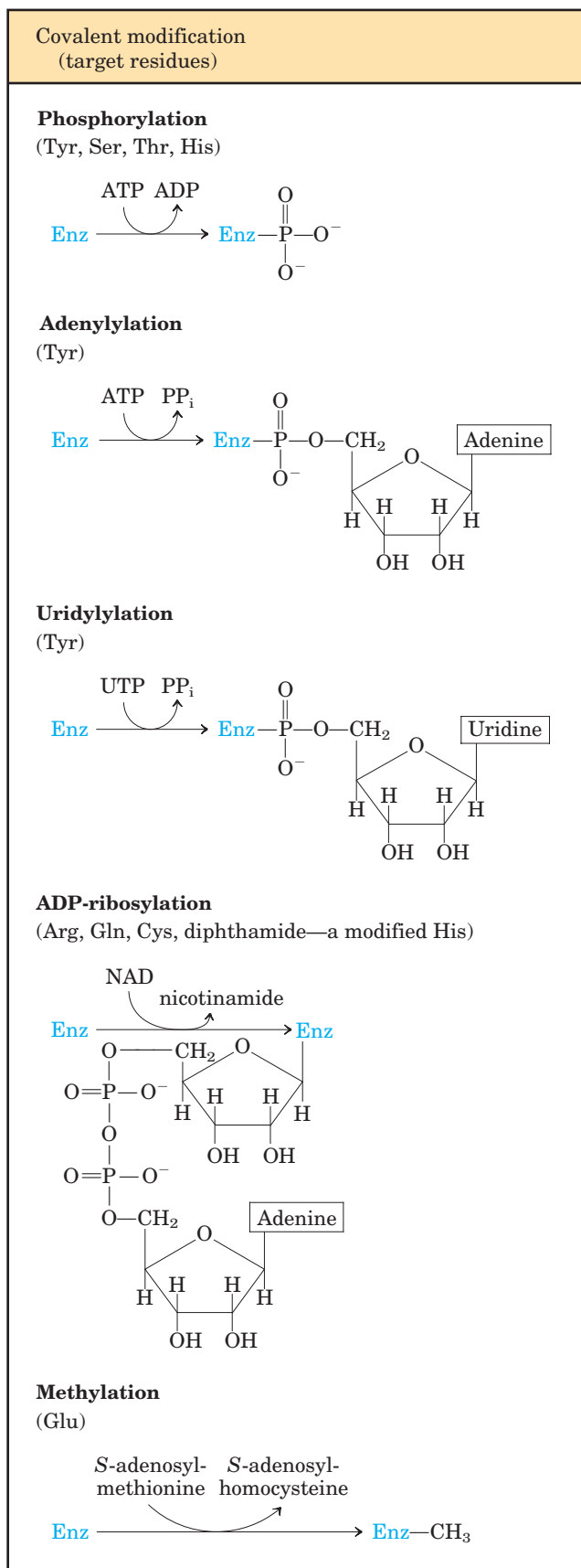
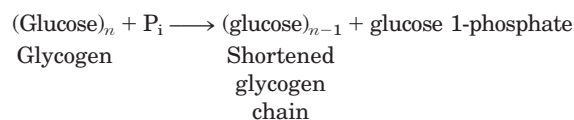


FIGURE 6-30 Some enzyme modification reactions.

amide groups of the peptide backbone at the start of an α helix or the charged guanidinium group of an Arg residue. The two negative charges on a phosphorylated side chain can also repel neighboring negatively charged (Asp or Glu) residues. When the modified side chain is located in a region of the protein critical to its three-dimensional structure, phosphorylation can have dramatic effects on protein conformation and thus on substrate binding and catalysis.

An important example of regulation by phosphorylation is seen in glycogen phosphorylase (M_r 94,500) of muscle and liver (Chapter 15), which catalyzes the reaction



The glucose 1-phosphate so formed can be used for ATP synthesis in muscle or converted to free glucose in the liver. Glycogen phosphorylase occurs in two forms: the more active phosphorylase *a* and the less active phosphorylase *b* (Fig. 6-31). Phosphorylase *a* has two subunits, each with a specific Ser residue that is phosphorylated at its hydroxyl group. These serine phosphate residues are required for maximal activity of the enzyme.

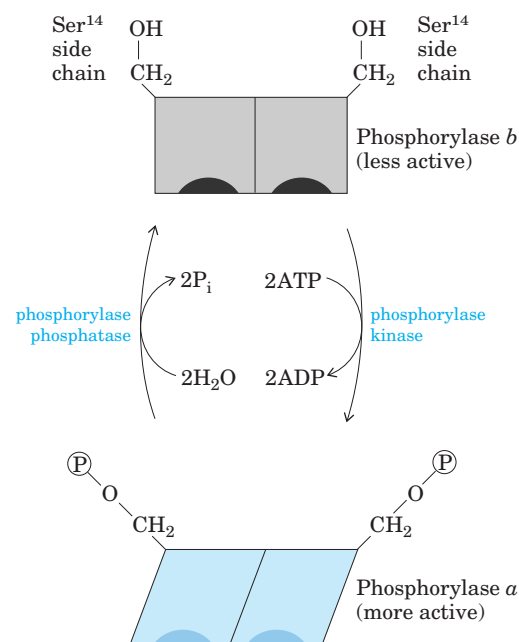
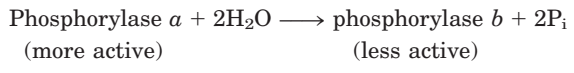


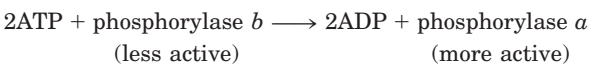
FIGURE 6-31 Regulation of glycogen phosphorylase activity by covalent modification. In the more active form of the enzyme, phosphorylase *a*, specific Ser residues, one on each subunit, are phosphorylated. Phosphorylase *a* is converted to the less active phosphorylase *b* by enzymatic loss of these phosphoryl groups, promoted by phosphorylase phosphatase. Phosphorylase *b* can be reconverted (reactivated) to phosphorylase *a* by the action of phosphorylase kinase.

The phosphoryl groups can be hydrolytically removed by a separate enzyme called phosphorylase phosphatase:



In this reaction, phosphorylase *a* is converted to phosphorylase *b* by the cleavage of two serine phosphate covalent bonds, one on each subunit of glycogen phosphorylase.

Phosphorylase *b* can in turn be reactivated—covalently transformed back into active phosphorylase *a*—by another enzyme, phosphorylase kinase, which catalyzes the transfer of phosphoryl groups from ATP to the hydroxyl groups of the two specific Ser residues in phosphorylase *b*:



The breakdown of glycogen in skeletal muscles and the liver is regulated by variations in the ratio of the two forms of glycogen phosphorylase. The *a* and *b* forms differ in their secondary, tertiary, and quaternary structures; the active site undergoes changes in structure and, consequently, changes in catalytic activity as the two forms are interconverted.

The regulation of glycogen phosphorylase by phosphorylation illustrates the effects on both structure and catalytic activity of adding a phosphoryl group. In the unphosphorylated state, each subunit of this protein is folded so as to bring the 20 residues at its amino terminus, including a number of basic residues, into a region containing several acidic amino acids; this produces an electrostatic interaction that stabilizes the conformation. Phosphorylation of Ser¹⁴ interferes with this interaction, forcing the amino-terminal domain out of the acidic environment and into a conformation that allows interaction between the P^- -Ser and several Arg side chains. In this conformation, the enzyme is much more active.

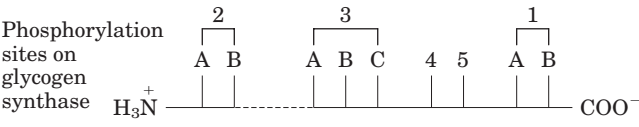
Phosphorylation of an enzyme can affect catalysis in another way: by altering substrate-binding affinity. For example, when isocitrate dehydrogenase (an enzyme of the citric acid cycle; Chapter 16) is phosphorylated, electrostatic repulsion by the phosphoryl group inhibits the binding of citrate (a tricarboxylic acid) at the active site.

Multiple Phosphorylations Allow Exquisite Regulatory Control

The Ser, Thr, or Tyr residues that are phosphorylated in regulated proteins occur within common structural motifs, called consensus sequences, that are recognized by specific protein kinases (Table 6-10). Some kinases are basophilic, preferring to phosphorylate a residue having basic neighbors; others have different substrate preferences, such as for a residue near a Pro residue.

Primary sequence is not the only important factor in determining whether a given residue will be phosphorylated, however. Protein folding brings together residues that are distant in the primary sequence; the resulting three-dimensional structure can determine whether a protein kinase has access to a given residue and can recognize it as a substrate. Another factor influencing the substrate specificity of certain protein kinases is the proximity of other phosphorylated residues.

Regulation by phosphorylation is often complicated. Some proteins have consensus sequences recognized by several different protein kinases, each of which can phosphorylate the protein and alter its enzymatic activity. In some cases, phosphorylation is hierarchical: a certain residue can be phosphorylated only if a neighboring residue has already been phosphorylated. For example, glycogen synthase, the enzyme that catalyzes the condensation of glucose monomers to form glycogen (Chapter 15), is inactivated by phosphorylation of specific Ser residues and is also modulated by at least four other protein kinases that phosphorylate four other sites in the protein (Fig. 6-32). The protein is not a substrate for glycogen synthase kinase 3, for example, until one site has been phosphorylated by casein kinase II. Some phosphorylations inhibit glycogen synthase more than



Kinase	Phosphorylation sites	Degree of synthase inactivation
Protein kinase A	1A, 1B, 2, 4	+
Protein kinase G	1A, 1B, 2	+
Protein kinase C	1A	+
Ca ²⁺ /calmodulin kinase	1B, 2	+
Phosphorylase <i>b</i> kinase	2	+
Casein kinase I	At least nine	+ + + +
Casein kinase II	5	0
Glycogen synthase kinase 3	3A, 3B, 3C	+ + +
Glycogen synthase kinase 4	2	+

FIGURE 6-32 Multiple regulatory phosphorylations. The enzyme glycogen synthase has at least nine separate sites in five designated regions susceptible to phosphorylation by one of the cellular protein kinases. Thus, regulation of this enzyme is a matter not of binary (on/off) switching but of finely tuned modulation of activity over a wide range in response to a variety of signals.

TABLE 6–10 Consensus Sequences for Protein Kinases

Protein kinase	Consensus sequence and phosphorylated residue*
Protein kinase A	-X-R-(R/K)-X-(S/T)-B-
Protein kinase G	-X-R-(R/K)-X-(S/T)-X-
Protein kinase C	-(R/K)-(R/K)-X-(S/T)-B-(R/K)-(R/K)-
Protein kinase B	-X-R-X-(S/T)-X-K-
Ca ²⁺ /calmodulin kinase I	-B-X-R-X-X-(S/T)-X-X-X-B-
Ca ²⁺ /calmodulin kinase II	-B-X-(R/K)-X-X-(S/T)-X-X-
Myosin light chain kinase (smooth muscle)	-K-K-R-X-X- S -X-B-B-
Phosphorylase b kinase	-K-R-K-Q-I- S -V-R-
Extracellular signal-regulated kinase (ERK)	-P-X-(S/T)-P-P-
Cyclin-dependent protein kinase (cdc2)	-X-(S/T)-P-X-(K/R)-
Casein kinase I	-(Sp/Tp)-X-X-(X)-(S/T)-B
Casein kinase II	-X-(S/T)-X-X-(E/D/Sp/Yp)-X-
β -Adrenergic receptor kinase	-(D/E) _n -(S/T)-X-X-X-
Rhodopsin kinase	-X-X-(S/T)-(E) _n -
Insulin receptor kinase	-X-E-E-E- Y -M-M-M-M-K-K-S-R-G-D- Y -M-T-M-Q-I-G-K-K-K- L-P-A-T-G-D- Y -M-N-M-S-P-V-G-D-
Epidermal growth factor (EGF) receptor kinase	-E-E-E-E- Y -F-E-L-V-

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*Shown here are deduced consensus sequences (in roman type) and actual sequences from known substrates (italic). The Ser (S), Thr (T), or Tyr (Y) residue that undergoes phosphorylation is in red; all amino acid residues are shown as their one-letter abbreviations (see Table 3–1). X represents any amino acid; B, any hydrophobic amino acid; Sp, Tp, and Yp, already phosphorylated Ser, Thr, and Tyr residues.

others, and some combinations of phosphorylations are cumulative. These multiple regulatory phosphorylations provide the potential for extremely subtle modulation of enzyme activity.

To serve as an effective regulatory mechanism, phosphorylation must be reversible. In general, phosphoryl groups are added and removed by different enzymes, and the processes can therefore be separately regulated. Cells contain a family of phosphoprotein phosphatases that hydrolyze specific P-Ser , P-Thr , and P-Tyr esters, releasing P_i . The phosphoprotein phosphatases we know of thus far act only on a subset of phosphoproteins, but they show less substrate specificity than protein kinases.

Some Enzymes and Other Proteins Are Regulated by Proteolytic Cleavage of an Enzyme Precursor

For some enzymes, an inactive precursor called a **zymogen** is cleaved to form the active enzyme. Many proteolytic enzymes (proteases) of the stomach and pancreas are regulated in this way. Chymotrypsin and trypsin are initially synthesized as chymotrypsinogen and trypsinogen (Fig. 6–33). Specific cleavage causes conformational changes that expose the enzyme active site. Because this type of activation is irreversible, other

mechanisms are needed to inactivate these enzymes. Proteases are inactivated by inhibitor proteins that bind very tightly to the enzyme active site. For example, pancreatic trypsin inhibitor (M_r 6,000) binds to and inhibits trypsin; α_1 -antiprotease (M_r 53,000) primarily inhibits neutrophil elastase (neutrophils are a type of leukocyte, or white blood cell; elastase is a protease acting on elastin, a component of some connective tissues). An insufficiency of α_1 -antiprotease, which can be caused by exposure to cigarette smoke, has been associated with lung damage, including emphysema.

Proteases are not the only proteins activated by proteolysis. In other cases, however, the precursors are called not zymogens but, more generally, **proproteins** or **proenzymes**, as appropriate. For example, the connective tissue protein collagen is initially synthesized as the soluble precursor procollagen. The blood clotting system provides many examples of the proteolytic activation of proteins. Fibrin, the protein of blood clots, is produced by proteolysis of fibrinogen, its inactive proprotein. The protease responsible for this activation is thrombin (similar in many respects to chymotrypsin), which itself is produced by proteolysis of a proprotein (in this case a zymogen), prothrombin. Blood clotting is mediated by a complicated cascade of proteolytic activations.

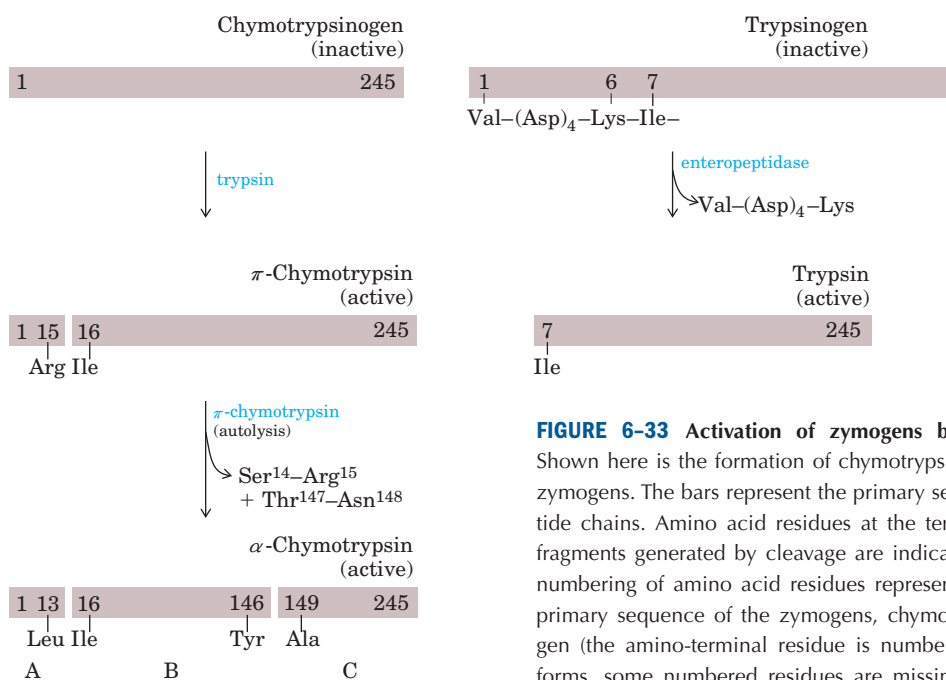


FIGURE 6-33 Activation of zymogens by proteolytic cleavage.

Shown here is the formation of chymotrypsin and trypsin from their zymogens. The bars represent the primary sequences of the polypeptide chains. Amino acid residues at the termini of the polypeptide fragments generated by cleavage are indicated below the bars. The numbering of amino acid residues represents their positions in the primary sequence of the zymogens, chymotrypsinogen or trypsinogen (the amino-terminal residue is number 1). Thus, in the active forms, some numbered residues are missing. Recall that the three polypeptide chains (A, B, and C) of chymotrypsin are linked by disulfide bonds (see Fig. 6-18).

Some Regulatory Enzymes Use Several Regulatory Mechanisms

Glycogen phosphorylase catalyzes the first reaction in a pathway that feeds stored glucose into energy-yielding carbohydrate metabolism (Chapters 14 and 15). This is an important metabolic step, and its regulation is correspondingly complex. Although its primary regulation is through covalent modification, as outlined in Figure 6-31, glycogen phosphorylase is also modulated allosterically by AMP, which is an activator of phosphorylase *b*, and by several other molecules that are inhibitors.

Other complex regulatory enzymes are found at key metabolic crossroads. Bacterial glutamine synthetase, which catalyzes a reaction that introduces reduced nitrogen into cellular metabolism (Chapter 22), is among the most complex regulatory enzymes known. It is regulated allosterically (with at least eight different modulators); by reversible covalent modification; and by the association of other regulatory proteins, a mechanism examined in detail when we consider the regulation of specific metabolic pathways.

What is the advantage of such complexity in the regulation of enzymatic activity? We began this chapter by stressing the central importance of catalysis to the very existence of life. The *control* of catalysis is also critical to life. If all possible reactions in a cell were catalyzed simultaneously, macromolecules and metabolites would quickly be broken down to much simpler chem-

ical forms. Instead, cells catalyze only the reactions they need at a given moment. When chemical resources are plentiful, cells synthesize and store glucose and other metabolites. When chemical resources are scarce, cells use these stores to fuel cellular metabolism. Chemical energy is used economically, parceled out to various metabolic pathways as cellular needs dictate. The availability of powerful catalysts, each specific for a given reaction, makes the regulation of these reactions possible. This in turn gives rise to the complex, highly regulated symphony we call life.

SUMMARY 6.5 Regulatory Enzymes

- The activities of metabolic pathways in cells are regulated by control of the activities of certain enzymes.
- In feedback inhibition, the end product of a pathway inhibits the first enzyme of that pathway.
- The activity of allosteric enzymes is adjusted by reversible binding of a specific modulator to a regulatory site. Modulators may be the substrate itself or some other metabolite, and the effect of the modulator may be inhibitory or stimulatory. The kinetic behavior of allosteric enzymes reflects cooperative interactions among enzyme subunits.

- Other regulatory enzymes are modulated by covalent modification of a specific functional group necessary for activity. The phosphorylation of specific amino acid residues is a particularly common way to regulate enzyme activity.
- Many proteolytic enzymes are synthesized as inactive precursors called zymogens, which are

activated by cleavage of small peptide fragments.

- Enzymes at important metabolic intersections may be regulated by complex combinations of effectors, allowing coordination of the activities of interconnected pathways.

Key Terms

Terms in bold are defined in the glossary.

enzyme 191	rate constant 195	k_{cat} 206
cofactor 191	binding energy (ΔG_{B}) 196	turnover number 207
coenzyme 191	specificity 199	reversible inhibition 209
prosthetic group 192	induced fit 200	competitive inhibition 209
holoenzyme 192	specific acid-base catalysis 200	uncompetitive inhibition 211
apoenzyme 192	general acid-base catalysis 200	mixed inhibition 211
apoprotein 192	covalent catalysis 200	noncompetitive inhibition 211
active site 193	enzyme kinetics 202	irreversible inhibitors 211
substrate 193	initial rate (initial velocity), V_0 202	suicide inactivator 211
ground state 193	V_{max} 203	transition state analogs 220
standard free-energy change (ΔG°) 194	pre-steady state 203	regulatory enzyme 225
transition state 194	steady state 203	allosteric enzyme 225
activation energy ($\Delta G^\ddagger$) 194	steady-state kinetics 203	allosteric modulator 225
reaction intermediate 195	Michaelis constant (K_{m}) 204	feedback inhibition 227
rate-limiting step 195	Michaelis-Menten equation 204	protein kinases 228
equilibrium constant (K_{eq}) 195	dissociation constant (K_{d}) 205	zymogen 231
	Lineweaver-Burk equation 206	

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A classic paper introducing the concept of allosteric regulation.

Problems

1. Keeping the Sweet Taste of Corn The sweet taste of freshly picked corn (maize) is due to the high level of sugar in the kernels. Store-bought corn (several days after picking) is not as sweet, because about 50% of the free sugar is converted to starch within one day of picking. To preserve the sweetness of fresh corn, the husked ears can be immersed in boiling water for a few minutes (“blanched”) then cooled in cold water. Corn processed in this way and stored in a freezer maintains its sweetness. What is the biochemical basis for this procedure?

2. Intracellular Concentration of Enzymes To approximate the actual concentration of enzymes in a bacterial cell, assume that the cell contains equal concentrations of 1,000 different enzymes in solution in the cytosol and that each protein has a molecular weight of 100,000. Assume also that the bacterial cell is a cylinder (diameter 1.0 μm , height 2.0 μm), that the cytosol (specific gravity 1.20) is 20% soluble protein by weight, and that the soluble protein consists entirely of enzymes. Calculate the *average* molar concentration of each enzyme in this hypothetical cell.

3. Rate Enhancement by Urease The enzyme urease enhances the rate of urea hydrolysis at pH 8.0 and 20 °C by a factor of 10^{14} . If a given quantity of urease can completely hydrolyze a given quantity of urea in 5.0 min at 20 °C and pH 8.0, how long would it take for this amount of urea to be hydrolyzed under the same conditions in the absence of urease? Assume that both reactions take place in sterile systems so that bacteria cannot attack the urea.

4. Protection of an Enzyme against Denaturation by Heat When enzyme solutions are heated, there is a progressive loss of catalytic activity over time due to denaturation of the enzyme. A solution of the enzyme hexokinase incubated at 45 °C lost 50% of its activity in 12 min, but when incubated at 45 °C in the presence of a very large concentration of one of its substrates, it lost only 3% of its activity in 12 min. Suggest why thermal denaturation of hexokinase was retarded in the presence of one of its substrates.

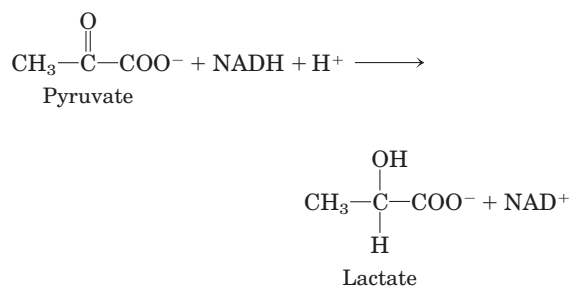
5. Requirements of Active Sites in Enzymes Carboxypeptidase, which sequentially removes carboxyl-terminal

amino acid residues from its peptide substrates, is a single polypeptide of 307 amino acids. The two essential catalytic groups in the active site are furnished by Arg¹⁴⁵ and Glu²⁷⁰.

(a) If the carboxypeptidase chain were a perfect α helix, how far apart (in Å) would Arg¹⁴⁵ and Glu²⁷⁰ be? (Hint: See Fig. 4-4b.)

(b) Explain how the two amino acid residues can catalyze a reaction occurring in the space of a few angstroms.

6. Quantitative Assay for Lactate Dehydrogenase The muscle enzyme lactate dehydrogenase catalyzes the reaction



NADH and NAD⁺ are the reduced and oxidized forms, respectively, of the coenzyme NAD. Solutions of NADH, but *not* NAD⁺, absorb light at 340 nm. This property is used to determine the concentration of NADH in solution by measuring spectrophotometrically the amount of light absorbed at 340 nm by the solution. Explain how these properties of NADH can be used to design a quantitative assay for lactate dehydrogenase.

7. Relation between Reaction Velocity and Substrate Concentration: Michaelis-Menten Equation (a) At what substrate concentration would an enzyme with a k_{cat} of 30.0 s⁻¹ and a K_m of 0.0050 M operate at one-quarter of its maximum rate? (b) Determine the fraction of V_{max} that would be obtained at the following substrate concentrations: $[S] = \frac{1}{2}K_m$, $2K_m$, and $10K_m$.

8. Estimation of V_{max} and K_m by Inspection Although graphical methods are available for accurate determination of the V_{max} and K_m of an enzyme-catalyzed reaction (see Box 6-1), sometimes these quantities can be quickly estimated by inspecting values of V_0 at increasing $[S]$. Estimate the V_{max} and K_m of the enzyme-catalyzed reaction for which the following data were obtained.

[S] (M)	V_0 ($\mu\text{M}/\text{min}$)
2.5×10^{-6}	28
4.0×10^{-6}	40
1×10^{-5}	70
2×10^{-5}	95
4×10^{-5}	112
1×10^{-4}	128
2×10^{-3}	139
1×10^{-2}	140

9. Properties of an Enzyme of Prostaglandin Synthesis Prostaglandins are a class of eicosanoids, fatty acid derivatives with a variety of extremely potent actions on

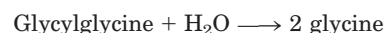
vertebrate tissues. They are responsible for producing fever and inflammation and its associated pain. Prostaglandins are derived from the 20-carbon fatty acid arachidonic acid in a reaction catalyzed by the enzyme prostaglandin endoperoxide synthase. This enzyme, a cyclooxygenase, uses oxygen to convert arachidonic acid to PGG₂, the immediate precursor of many different prostaglandins (prostaglandin synthesis is described in Chapter 21).

(a) The kinetic data given below are for the reaction catalyzed by prostaglandin endoperoxide synthase. Focusing here on the first two columns, determine the V_{max} and K_m of the enzyme.

[Arachidonic acid] (mM)	Rate of formation of PGG ₂ (mM/min)	Rate of formation of PGG ₂ with 10 mg/mL ibuprofen (mM/min)
0.5	23.5	16.67
1.0	32.2	25.25
1.5	36.9	30.49
2.5	41.8	37.04
3.5	44.0	38.91

(b) Ibuprofen is an inhibitor of prostaglandin endoperoxide synthase. By inhibiting the synthesis of prostaglandins, ibuprofen reduces inflammation and pain. Using the data in the first and third columns of the table, determine the type of inhibition that ibuprofen exerts on prostaglandin endoperoxide synthase.

 **10. Graphical Analysis of V_{max} and K_m** The following experimental data were collected during a study of the catalytic activity of an intestinal peptidase with the substrate glycylglycine:



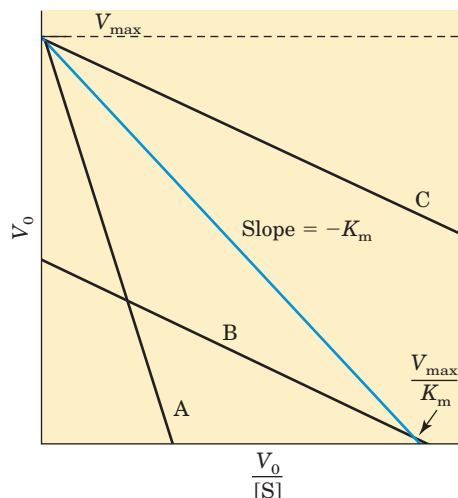
[S] (mM)	Product formed ($\mu\text{mol}/\text{min}$)
1.5	0.21
2.0	0.24
3.0	0.28
4.0	0.33
8.0	0.40
16.0	0.45

Use graphical analysis (see Box 6-1 and its associated Living Graph) to determine the K_m and V_{max} for this enzyme preparation and substrate.

11. The Eadie-Hofstee Equation One transformation of the Michaelis-Menten equation is the Lineweaver-Burk, or double-reciprocal, equation. Multiplying both sides of the Lineweaver-Burk equation by V_{max} and rearranging gives the Eadie-Hofstee equation:

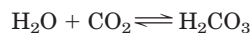
$$V_0 = (-K_m) \frac{V_0}{[S]} + V_{\text{max}}$$

A plot of V_0 vs. $V_0/[S]$ for an enzyme-catalyzed reaction is shown below. The blue curve was obtained in the absence of inhibitor. Which of the other curves (A, B, or C) shows the enzyme activity when a competitive inhibitor is added to the reaction mixture? Hint: See Equation 6–30.



12. The Turnover Number of Carbonic Anhydrase

Carbonic anhydrase of erythrocytes (M_r 30,000) has one of the highest turnover numbers we know of. It catalyzes the reversible hydration of CO_2 :



This is an important process in the transport of CO_2 from the tissues to the lungs. If $10.0 \mu\text{g}$ of pure carbonic anhydrase catalyzes the hydration of 0.30 g of CO_2 in 1 min at 37°C at $V_{\max}$, what is the turnover number (k_{cat}) of carbonic anhydrase (in units of min^{-1})?

13. Deriving a Rate Equation for Competitive Inhibition The rate equation for an enzyme subject to competitive inhibition is

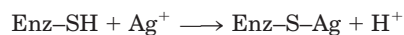
$$V_0 = \frac{V_{\max} [S]}{\alpha K_m + [S]}$$

Beginning with a new definition of total enzyme as

$$[E]_t = [E] + [EI] + [ES]$$

and the definitions of α and K_1 provided in the text, derive the rate equation above. Use the derivation of the Michaelis-Menten equation as a guide.

14. Irreversible Inhibition of an Enzyme Many enzymes are inhibited irreversibly by heavy metal ions such as Hg^{2+} , Cu^{2+} , or Ag^+ , which can react with essential sulfhydryl groups to form mercaptides:



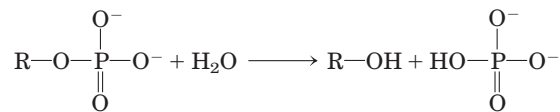
The affinity of Ag^+ for sulfhydryl groups is so great that Ag^+ can be used to titrate $-\text{SH}$ groups quantitatively. To 10.0 mL of a solution containing 1.0 mg/mL of a pure enzyme, an investigator added just enough AgNO_3 to completely inactivate the enzyme. A total of $0.342 \mu\text{mol}$ of AgNO_3 was required.

Calculate the minimum molecular weight of the enzyme. Why does the value obtained in this way give only the *minimum* molecular weight?



15. Clinical Application of Differential Enzyme Inhibition

Human blood serum contains a class of enzymes known as acid phosphatases, which hydrolyze biological phosphate esters under slightly acidic conditions (pH 5.0):

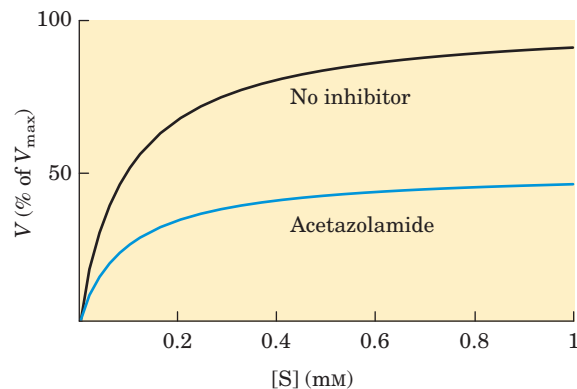


Acid phosphatases are produced by erythrocytes, the liver, kidney, spleen, and prostate gland. The enzyme of the prostate gland is clinically important, because its increased activity in the blood can be an indication of prostate cancer. The phosphatase from the prostate gland is strongly inhibited by tartrate ion, but acid phosphatases from other tissues are not. How can this information be used to develop a specific procedure for measuring the activity of the acid phosphatase of the prostate gland in human blood serum?



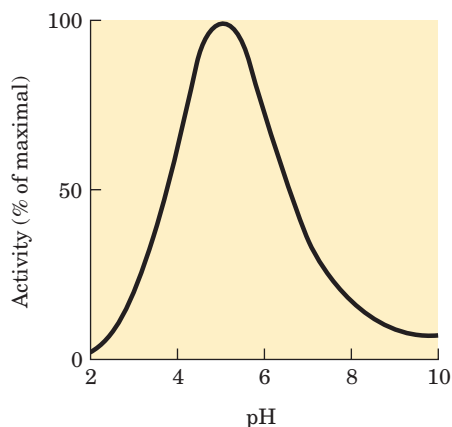
16. Inhibition of Carbonic Anhydrase by Acetazolamide

Carbonic anhydrase is strongly inhibited by the drug acetazolamide, which is used as a diuretic (i.e., to increase the production of urine) and to lower excessively high pressure in the eye (due to accumulation of intraocular fluid) in glaucoma. Carbonic anhydrase plays an important role in these and other secretory processes, because it participates in regulating the pH and bicarbonate content of several body fluids. The experimental curve of initial reaction velocity (as percentage of $V_{\max}$) versus $[S]$ for the carbonic anhydrase reaction is illustrated below (upper curve). When the experiment is repeated in the presence of acetazolamide, the lower curve is obtained. From an inspection of the curves and your knowledge of the kinetic properties of competitive and mixed enzyme inhibitors, determine the nature of the inhibition by acetazolamide. Explain your reasoning.



17. The Effects of Reversible Inhibitors Derive the expression for the effect of a reversible inhibitor on observed K_m (apparent $K_m = \alpha K_m / \alpha'$). Start with Equation 6–30 and the statement that apparent K_m is equivalent to the $[S]$ at which $V_0 = V_{\max} / 2\alpha'$.

18. pH Optimum of Lysozyme The active site of lysozyme contains two amino acid residues essential for catalysis: Glu³⁵ and Asp⁵². The pK_a values of the carboxyl side chains of these residues are 5.9 and 4.5, respectively. What is the ionization state (protonated or deprotonated) of each residue at pH 5.2, the pH optimum of lysozyme? How can the ionization states of these residues explain the pH-activity profile of lysozyme shown below?

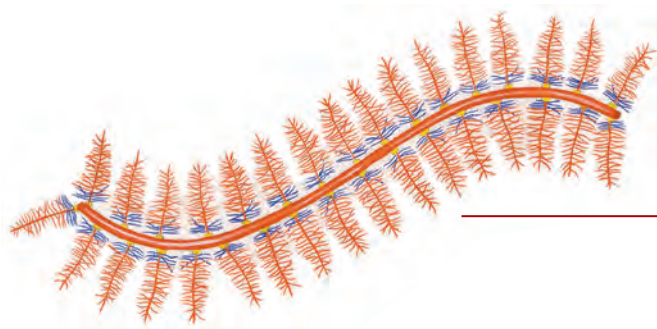


19. Working with Kinetics Go to the Living Graphs for Chapter 6.

(a) Using the Living Graph for Equation 6-9, create a V versus $[S]$ plot. Use $V_{\max} = 100 \mu\text{M s}^{-1}$, and $K_m = 10 \mu\text{M}$. How much does V_0 increase when $[S]$ is doubled, from 0.2 to 0.4 μM ? What is V_0 when $[S] = 10 \mu\text{M}$? How much does the V_0 increase when $[S]$ increases from 100 to 200 μM ? Observe how the graph changes when the values for $V_{\max}$ or K_m are halved or doubled.

(b) Using the Living Graph for Equation 6-30 and the kinetic parameters in (a), create a plot in which both α and α' are 1.0. Now observe how the plot changes when $\alpha = 2.0$; when $\alpha' = 3.0$; and when $\alpha = 2.0$ and $\alpha' = 3.0$.

(c) Using the Living Graphs for Equation 6-30 and the Lineweaver-Burk equation in Box 6-1, create Lineweaver-Burk (double-reciprocal) plots for all the cases in (a) and (b). When $\alpha = 2.0$, does the x intercept move to the right or to the left? If $\alpha = 2.0$ and $\alpha' = 3.0$, does the x intercept move to the right or to the left?



CARBOHYDRATES AND GLYCOBIOLOGY

- 7.1 Monosaccharides and Disaccharides** 239
- 7.2 Polysaccharides** 247
- 7.3 Glycoconjugates: Proteoglycans, Glycoproteins, and Glycolipids** 255
- 7.4 Carbohydrates as Informational Molecules: The Sugar Code** 261
- 7.5 Working with Carbohydrates** 267

Ah! sweet mystery of life . . .

—Rida Johnson Young (lyrics) and Victor Herbert (music),
“Ah! Sweet Mystery of Life,” 1910

I would feel more optimistic about a bright future for man
if he spent less time proving that he can outwit Nature
and more time tasting her sweetness and respecting her
seniority.

—E. B. White, “Coon Tree,” 1977

Carbohydrates are the most abundant biomolecules on Earth. Each year, photosynthesis converts more than 100 billion metric tons of CO₂ and H₂O into cellulose and other plant products. Certain carbohydrates (sugar and starch) are a dietary staple in most parts of the world, and the oxidation of carbohydrates is the central energy-yielding pathway in most nonphotosynthetic cells. Insoluble carbohydrate polymers serve as structural and protective elements in the cell walls of bacteria and plants and in the connective tissues of animals. Other carbohydrate polymers lubricate skeletal joints and participate in recognition and adhesion between cells. More complex carbohydrate polymers covalently

attached to proteins or lipids act as signals that determine the intracellular location or metabolic fate of these hybrid molecules, called **glycoconjugates**. This chapter introduces the major classes of carbohydrates and glycoconjugates and provides a few examples of their many structural and functional roles.

Carbohydrates are polyhydroxy aldehydes or ketones, or substances that yield such compounds on hydrolysis. Many, but not all, carbohydrates have the empirical formula (CH₂O)_n; some also contain nitrogen, phosphorus, or sulfur.

There are three major size classes of carbohydrates: monosaccharides, oligosaccharides, and polysaccharides (the word “saccharide” is derived from the Greek *sakcharon*, meaning “sugar”). **Monosaccharides**, or simple sugars, consist of a single polyhydroxy aldehyde or ketone unit. The most abundant monosaccharide in nature is the six-carbon sugar D-glucose, sometimes referred to as dextrose. Monosaccharides of more than four carbons tend to have cyclic structures.

Oligosaccharides consist of short chains of monosaccharide units, or residues, joined by characteristic linkages called glycosidic bonds. The most abundant are the **disaccharides**, with two monosaccharide units. Typical is sucrose (cane sugar), which consists of the six-carbon sugars D-glucose and D-fructose. All common monosaccharides and disaccharides have names ending with the suffix “-ose.” In cells, most oligosaccharides consisting of three or more units do not occur as free entities but are joined to nonsugar molecules (lipids or proteins) in glycoconjugates.

The **polysaccharides** are sugar polymers containing more than 20 or so monosaccharide units, and some have hundreds or thousands of units. Some polysaccharides, such as cellulose, are linear chains; others,

such as glycogen, are branched. Both glycogen and cellulose consist of recurring units of D-glucose, but they differ in the type of glycosidic linkage and consequently have strikingly different properties and biological roles.

7.1 Monosaccharides and Disaccharides

The simplest of the carbohydrates, the monosaccharides, are either aldehydes or ketones with two or more hydroxyl groups; the six-carbon monosaccharides glucose and fructose have five hydroxyl groups. Many of the carbon atoms to which hydroxyl groups are attached are chiral centers, which give rise to the many sugar stereoisomers found in nature. We begin by describing the families of monosaccharides with backbones of three to seven carbons—their structure and stereoisomeric forms, and the means of representing their three-dimensional structures on paper. We then discuss several chemical reactions of the carbonyl groups of monosaccharides. One such reaction, the addition of a hydroxyl group from within the same molecule, generates the cyclic forms of five- and six-carbon sugars (the forms that predominate in aqueous solution) and creates a new chiral center, adding further stereochemical complexity to this class of compounds. The nomenclature for unambiguously specifying the configuration about each carbon atom in a cyclic form and the means of representing these structures on paper are therefore described in some detail; this information will be useful as we discuss the metabolism of monosaccharides in Part II. We also introduce here some important monosaccharide derivatives encountered in later chapters.

The Two Families of Monosaccharides Are Aldoses and Ketoses

Monosaccharides are colorless, crystalline solids that are freely soluble in water but insoluble in nonpolar solvents. Most have a sweet taste. The backbones of common monosaccharide molecules are unbranched carbon chains in which all the carbon atoms are linked by single bonds. In the open-chain form, one of the carbon atoms is double-bonded to an oxygen atom to form a carbonyl group; each of the other carbon atoms has a hydroxyl group. If the carbonyl group is at an end of the carbon chain (that is, in an aldehyde group) the monosaccharide is an **aldose**; if the carbonyl group is at any other position (in a ketone group) the monosaccharide is a **ketose**. The simplest monosaccharides are the two three-carbon trioses: glyceraldehyde, an aldotriose, and dihydroxyacetone, a ketotriose (Fig. 7-1a).

Monosaccharides with four, five, six, and seven carbon atoms in their backbones are called, respectively, tetroses, pentoses, hexoses, and heptoses. There are aldoses and ketoses of each of these chain lengths:

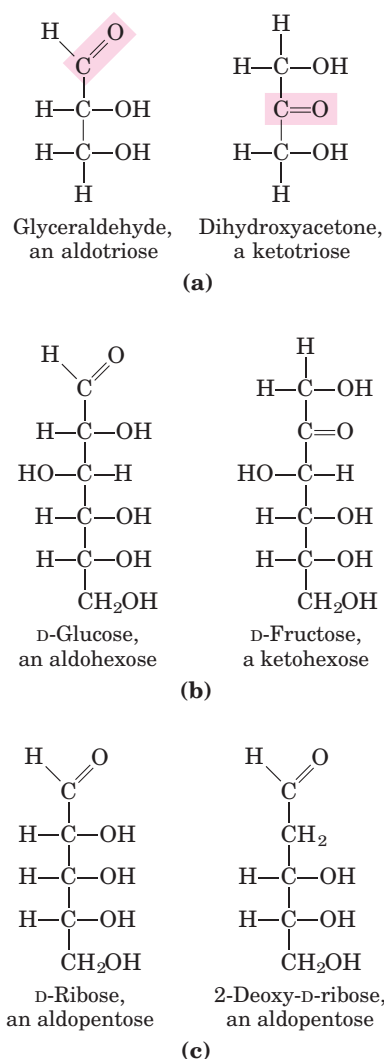
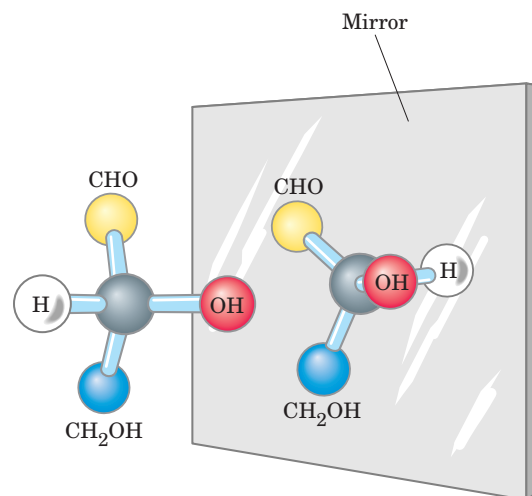


FIGURE 7-1 Representative monosaccharides. (a) Two trioses, an aldose and a ketose. The carbonyl group in each is shaded. (b) Two common hexoses. (c) The pentose components of nucleic acids. D-Ribose is a component of ribonucleic acid (RNA), and 2-deoxy-D-ribose is a component of deoxyribonucleic acid (DNA).

aldotetroses and ketotetroses, aldopentoses and ketopentoses, and so on. The hexoses, which include the aldohexose D-glucose and the ketohexose D-fructose (Fig. 7-1b), are the most common monosaccharides in nature. The aldopentoses D-ribose and 2-deoxy-D-ribose (Fig. 7-1c) are components of nucleotides and nucleic acids (Chapter 8).

Monosaccharides Have Asymmetric Centers

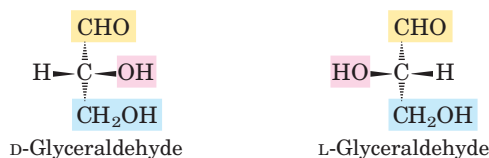
All the monosaccharides except dihydroxyacetone contain one or more asymmetric (chiral) carbon atoms and thus occur in optically active isomeric forms (pp. 17–19). The simplest aldose, glyceraldehyde, contains one chiral center (the middle carbon atom) and therefore has two different optical isomers, or enantiomers (Fig. 7-2).



Ball-and-stick models



Fischer projection formulas



Perspective formulas

FIGURE 7-2 Three ways to represent the two stereoisomers of glyceraldehyde. The stereoisomers are mirror images of each other. Ball-and-stick models show the actual configuration of molecules. By convention, in Fischer projection formulas, horizontal bonds project out of the plane of the paper, toward the reader; vertical bonds project behind the plane of the paper, away from the reader. Recall (see Fig. 1-17) that in perspective formulas, solid wedge-shaped bonds point toward the reader, dashed wedges point away.

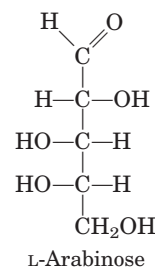
By convention, one of these two forms is designated the *D* isomer, the other the *L* isomer. As for other biomolecules with chiral centers, the absolute configurations of sugars are known from x-ray crystallography. To represent three-dimensional sugar structures on paper, we often use **Fischer projection formulas** (Fig. 7-2).

In general, a molecule with n chiral centers can have 2^n stereoisomers. Glyceraldehyde has $2^1 = 2$; the aldohexoses, with four chiral centers, have $2^4 = 16$ stereoisomers. The stereoisomers of monosaccharides

of each carbon-chain length can be divided into two groups that differ in the configuration about the chiral center *most distant* from the carbonyl carbon. Those in which the configuration at this reference carbon is the same as that of *D*-glyceraldehyde are designated *D* isomers, and those with the same configuration as *L*-glyceraldehyde are *L* isomers. When the hydroxyl group on the reference carbon is on the right in the projection formula, the sugar is the *D* isomer; when on the left, it is the *L* isomer. Of the 16 possible aldohexoses, eight are *D* forms and eight are *L*. Most of the hexoses of living organisms are *D* isomers.

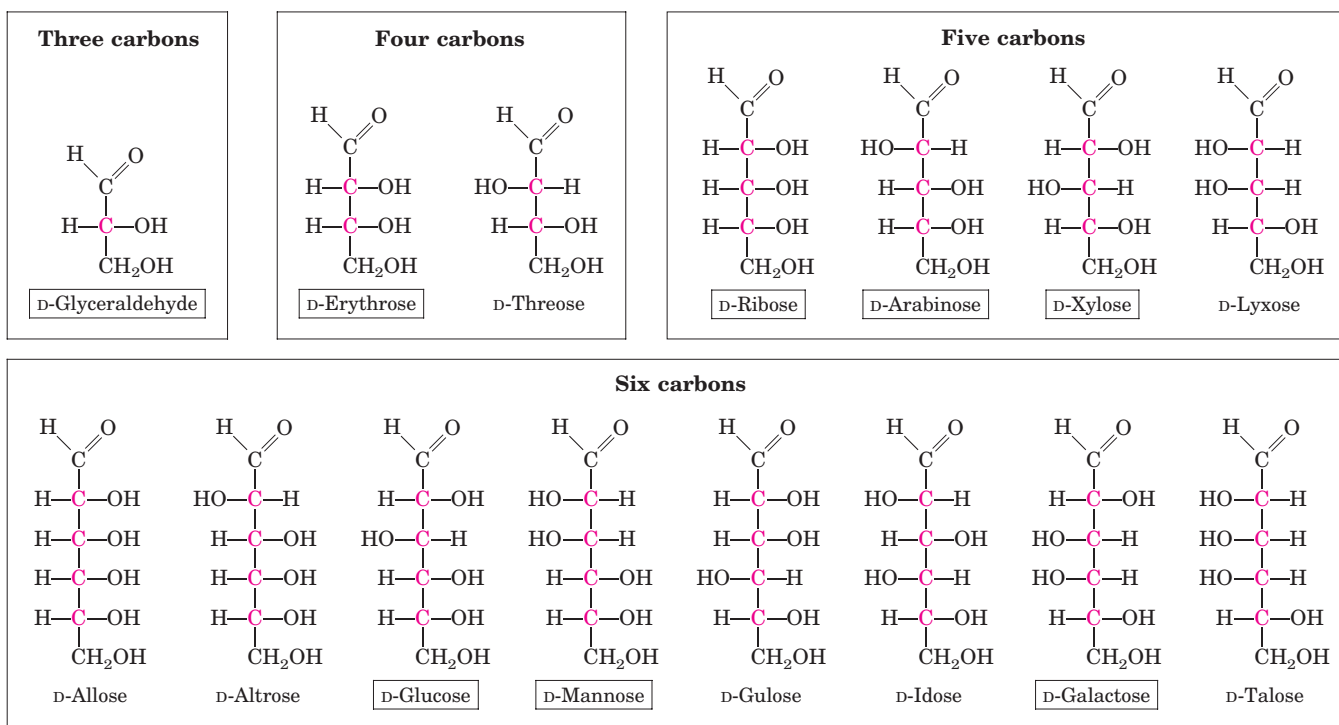
Figure 7-3 shows the structures of the *D* stereoisomers of all the aldoses and ketoses having three to six carbon atoms. The carbons of a sugar are numbered beginning at the end of the chain nearest the carbonyl group. Each of the eight *D*-aldohexoses, which differ in the stereochemistry at C-2, C-3, or C-4, has its own name: *D*-glucose, *D*-galactose, *D*-mannose, and so forth (Fig. 7-3a). The four- and five-carbon ketoses are designated by inserting “ul” into the name of a corresponding aldose; for example, *D*-ribulose is the ketopentose corresponding to the aldopentose *D*-ribose. The ketohexoses are named otherwise: for example, fructose (from the Latin *fructus*, “fruit”; fruits are rich in this sugar) and sorbose (from *Sorbus*, the genus of mountain ash, which has berries rich in the related sugar alcohol sorbitol). Two sugars that differ only in the configuration around one carbon atom are called **epimers**; *D*-glucose and *D*-mannose, which differ only in the stereochemistry at C-2, are epimers, as are *D*-glucose and *D*-galactose (which differ at C-4) (Fig. 7-4).

Some sugars occur naturally in their *L* form; examples are *L*-arabinose and the *L* isomers of some sugar derivatives that are common components of glycoconjugates (Section 7.3).



The Common Monosaccharides Have Cyclic Structures

For simplicity, we have thus far represented the structures of aldoses and ketoses as straight-chain molecules (Figs 7-3, 7-4). In fact, in aqueous solution, aldohexoses and all monosaccharides with five or more carbon atoms in the backbone occur predominantly as cyclic (ring) structures in which the carbonyl group has formed a covalent bond with the oxygen of a hydroxyl

**D-Aldoses**

(a)

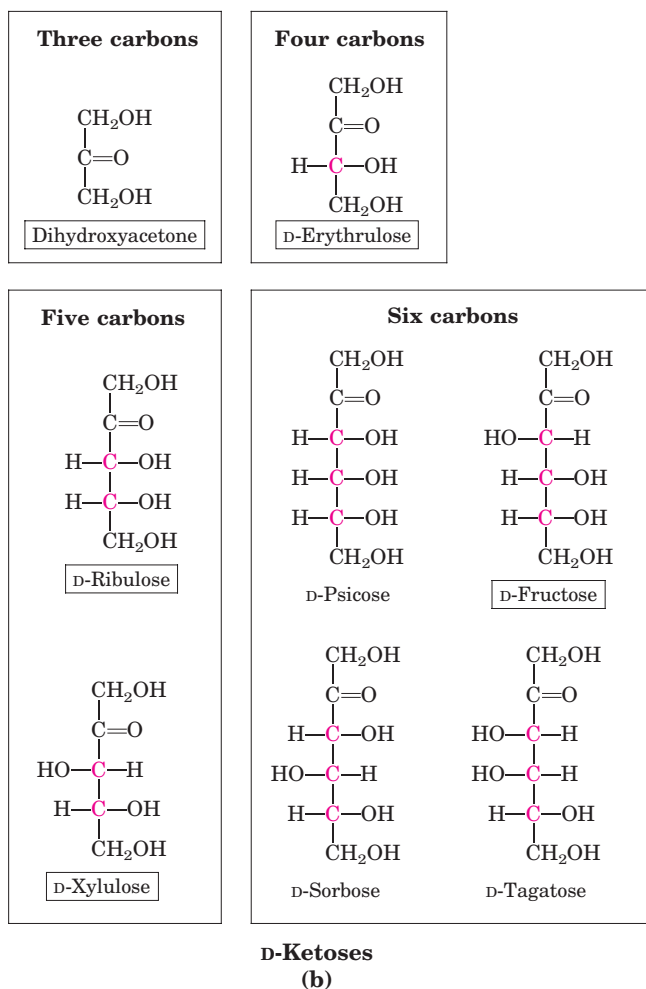


FIGURE 7-3 Aldoses and ketoses. The series of (a) D-aldoses and (b) D-ketoses having from three to six carbon atoms, shown as projection formulas. The carbon atoms in red are chiral centers. In all these D isomers, the chiral carbon *most distant from the carbonyl carbon* has the same configuration as the chiral carbon in D-glyceraldehyde. The sugars named in boxes are the most common in nature; you will encounter these again in this and later chapters.

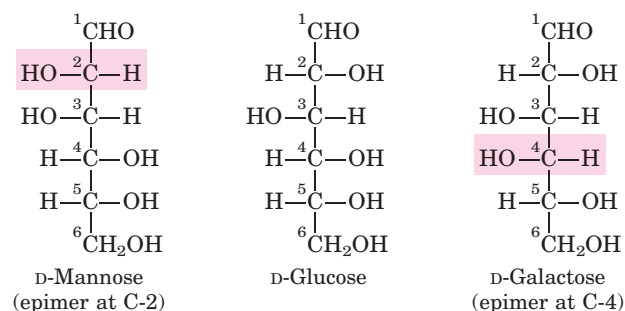
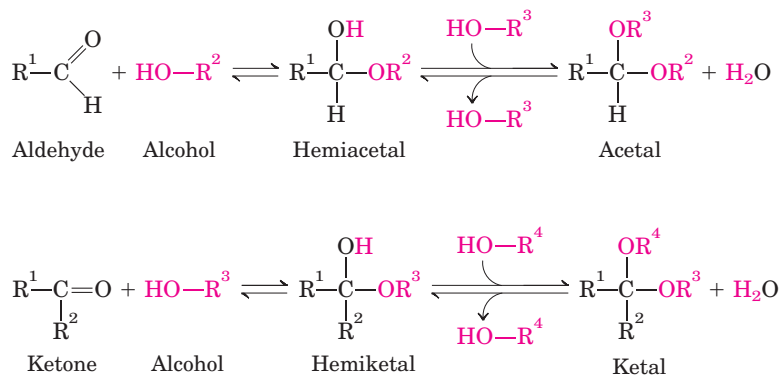


FIGURE 7-4 Epimers. D-Glucose and two of its epimers are shown as projection formulas. Each epimer differs from D-glucose in the configuration at one chiral center (shaded red).

**FIGURE 7-5** Formation of hemiacetals and hemiketals.

An aldehyde or ketone can react with an alcohol in a 1:1 ratio to yield a hemiacetal or hemiketal, respectively, creating a new chiral center at the carbonyl carbon. Substitution of a second alcohol molecule produces an acetal or ketal. When the second alcohol is part of another sugar molecule, the bond produced is a glycosidic bond (p. 245).

group along the chain. The formation of these ring structures is the result of a general reaction between alcohols and aldehydes or ketones to form derivatives called **hemiacetals** or **hemiketals** (Fig. 7-5), which contain an additional asymmetric carbon atom and thus can exist in two stereoisomeric forms. For example, *D*-glucose exists in solution as an intramolecular hemiacetal in which the free hydroxyl group at C-5 has reacted with the aldehydic C-1, rendering the latter carbon asymmetric and producing two stereoisomers, designated α and β (Fig. 7-6). These six-membered ring compounds are called **pyranoses** because they resemble the six-membered ring compound pyran (Fig. 7-7). The systematic names for the two ring forms of *D*-glucose are α -*D*-glucopyranose and β -*D*-glucopyranose.

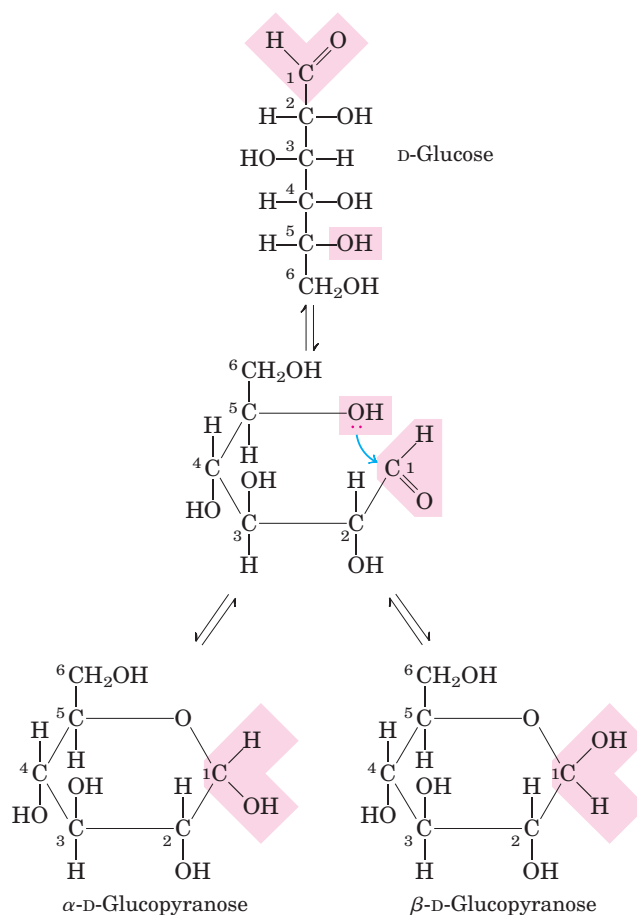
Aldohexoses also exist in cyclic forms having five-membered rings, which, because they resemble the five-membered ring compound furan, are called **furanoses**. However, the six-membered aldopyranose ring is much more stable than the aldofuranose ring and predominates in aldohexose solutions. Only aldoses having five or more carbon atoms can form pyranose rings.

Isomeric forms of monosaccharides that differ only in their configuration about the hemiacetal or hemiketal carbon atom are called **anomers**. The hemiacetal (or carbonyl) carbon atom is called the **anomeric carbon**. The α and β anomers of *D*-glucose interconvert in aqueous solution by a process called **mutarotation**. Thus, a solution of α -*D*-glucose and a solution of β -*D*-glucose eventually form identical equilibrium mixtures having identical optical properties. This mixture consists of about one-third α -*D*-glucose, two-thirds β -*D*-glucose, and very small amounts of the linear and five-membered ring (glucofuranose) forms.

Ketohexoses also occur in α and β anomeric forms. In these compounds the hydroxyl group at C-5 (or C-6) reacts with the keto group at C-2, forming a furanose (or pyranose) ring containing a hemiketal linkage (Fig. 7-5). *D*-Fructose readily forms the furanose ring (Fig. 7-7); the more common anomer of this sugar in combined forms or in derivatives is β -*D*-fructofuranose.

Haworth perspective formulas like those in Figure 7-7 are commonly used to show the stereochem-

istry of ring forms of monosaccharides. However, the six-membered pyranose ring is not planar, as Haworth perspectives suggest, but tends to assume either of two “chair” conformations (Fig. 7-8). Recall from Chapter 1 (p. 19) that two *conformations* of a molecule are interconvertible without the breakage of covalent bonds,

**FIGURE 7-6** Formation of the two cyclic forms of *D*-glucose. Reaction between the aldehyde group at C-1 and the hydroxyl group at C-5 forms a hemiacetal linkage, producing either of two stereoisomers, the α and β anomers, which differ only in the stereochemistry around the hemiacetal carbon. The interconversion of α and β anomers is called mutarotation.

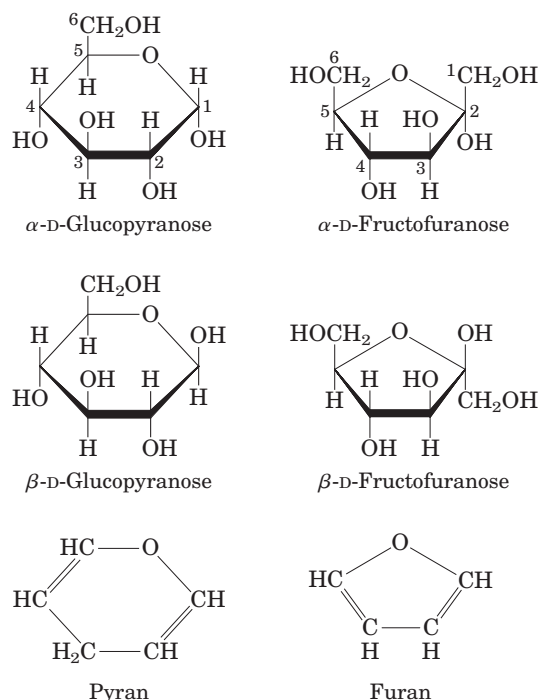


FIGURE 7-7 Pyranoses and furanoses. The pyranose forms of D-glucose and the furanose forms of D-fructose are shown here as Haworth perspective formulas. The edges of the ring nearest the reader are represented by bold lines. Hydroxyl groups below the plane of the ring in these Haworth perspectives would appear at the right side of a Fischer projection (compare with Fig. 7-6). Pyran and furan are shown for comparison.

whereas two *configurations* can be interconverted only by breaking a covalent bond—for example, in the case of α and β configurations, the bond involving the ring oxygen atom. The specific three-dimensional conformations of the monosaccharide units are important in determining the biological properties and functions of some polysaccharides, as we shall see.

Organisms Contain a Variety of Hexose Derivatives

In addition to simple hexoses such as glucose, galactose, and mannose, there are a number of sugar derivatives in which a hydroxyl group in the parent compound is replaced with another substituent, or a carbon atom is oxidized to a carboxyl group (Fig. 7-9). In glucosamine, galactosamine, and mannosamine, the hydroxyl at C-2 of the parent compound is replaced with an amino group. The amino group is nearly always condensed with acetic acid, as in *N*-acetylglucosamine. This glucosamine derivative is part of many structural polymers, including those of the bacterial cell wall. Bacterial cell walls also contain a derivative of glucosamine, *N*-acetylmuramic acid, in which lactic acid (a three-carbon carboxylic acid) is ether-linked to the oxygen at C-3 of *N*-acetylglucosamine. The substitution of a hydrogen for

the hydroxyl group at C-6 of L-galactose or L-mannose produces L-fucose or L-rhamnose, respectively; these deoxy sugars are found in plant polysaccharides and in the complex oligosaccharide components of glycoproteins and glycolipids.

Oxidation of the carbonyl (aldehyde) carbon of glucose to the carboxyl level produces gluconic acid; other aldoses yield other **aldonic acids**. Oxidation of the carbon at the other end of the carbon chain—C-6 of glucose, galactose, or mannose—forms the corresponding **uronic acid**: glucuronic, galacturonic, or mannuronic acid. Both aldonic and uronic acids form stable intramolecular esters called lactones (Fig. 7-9, lower left). In addition to these acidic hexose derivatives, one nine-carbon acidic sugar deserves mention: *N*-acetylneuraminic acid (a sialic acid, but often referred to simply as “sialic acid”), a derivative of *N*-acetylmannosamine, is a component of many glycoproteins and glycolipids in animals. The carboxylic acid groups of the acidic sugar derivatives are ionized at pH 7, and the compounds are therefore correctly named as the carboxylates—glucuronate, galacturonate, and so forth.

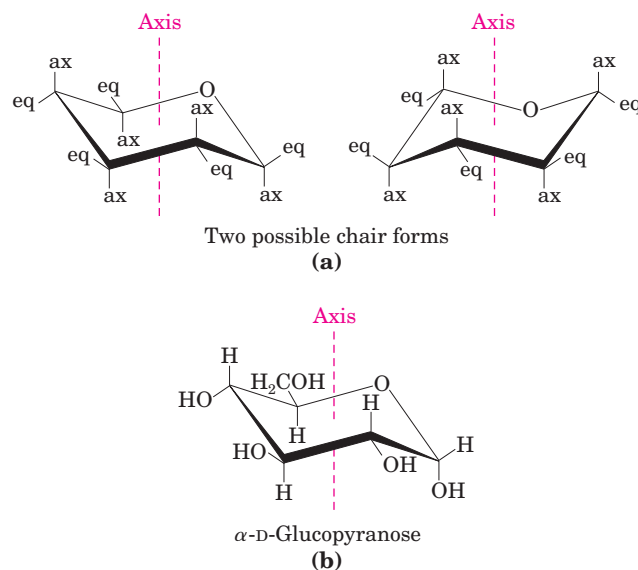


FIGURE 7-8 Conformational formulas of pyranoses. (a) Two chair forms of the pyranose ring. Substituents on the ring carbons may be either axial (ax), projecting parallel to the vertical axis through the ring, or equatorial (eq), projecting roughly perpendicular to this axis. Two conformers such as these are not readily interconvertible without breaking the ring. However, when the molecule is “stretched” (by atomic force microscopy), an input of about 46 kJ of energy per mole of sugar can force the interconversion of chair forms. Generally, substituents in the equatorial positions are less sterically hindered by neighboring substituents, and conformers with bulky substituents in equatorial positions are favored. Another conformation, the “boat” (not shown), is seen only in derivatives with very bulky substituents. (b) A chair conformation of α -D-glucopyranose.

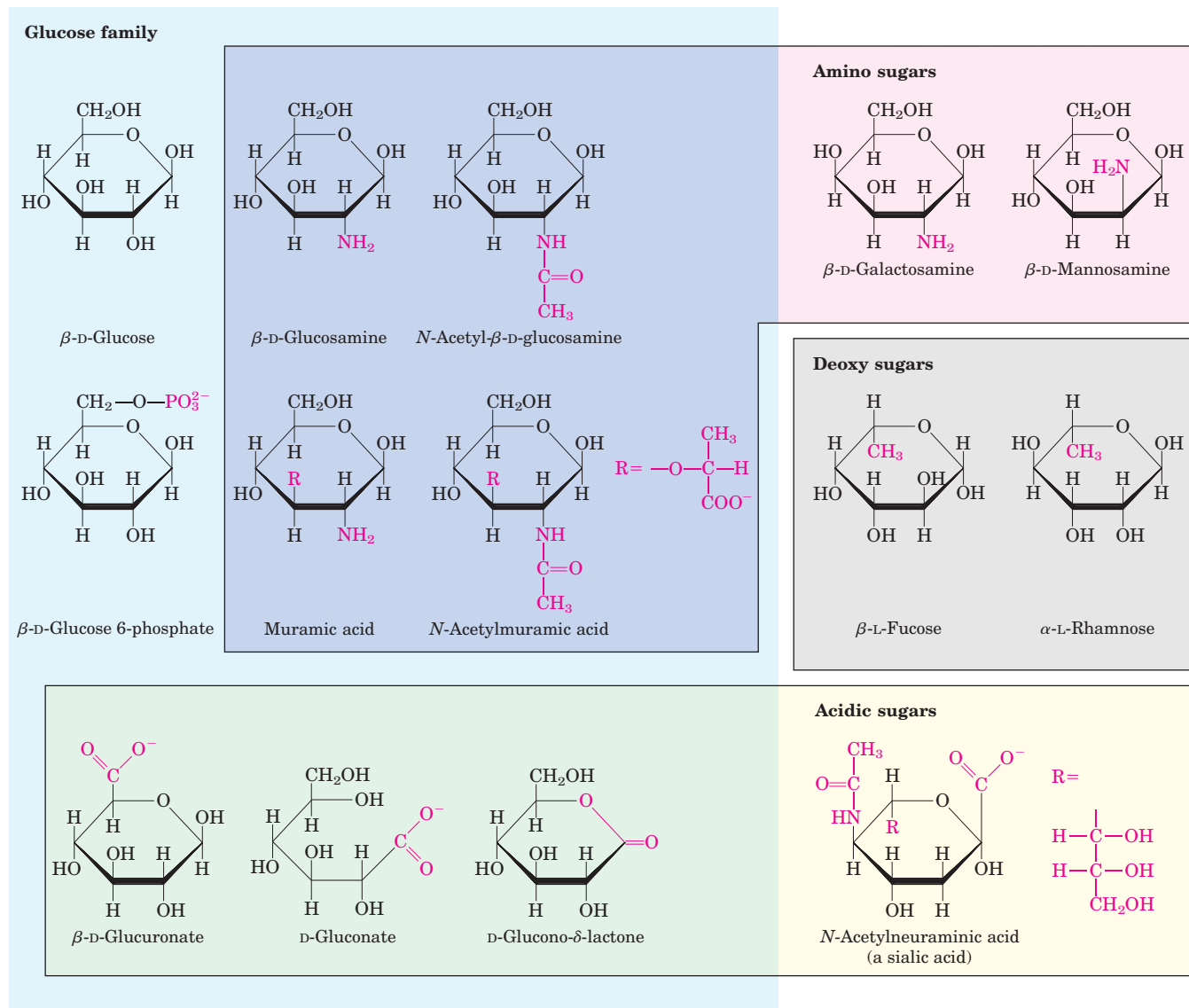


FIGURE 7-9 Some hexose derivatives important in biology. In amino sugars, an —NH_2 group replaces one of the —OH groups in the parent hexose. Substitution of —H for —OH produces a deoxy sugar; note that the deoxy sugars shown here occur in nature as the L iso-

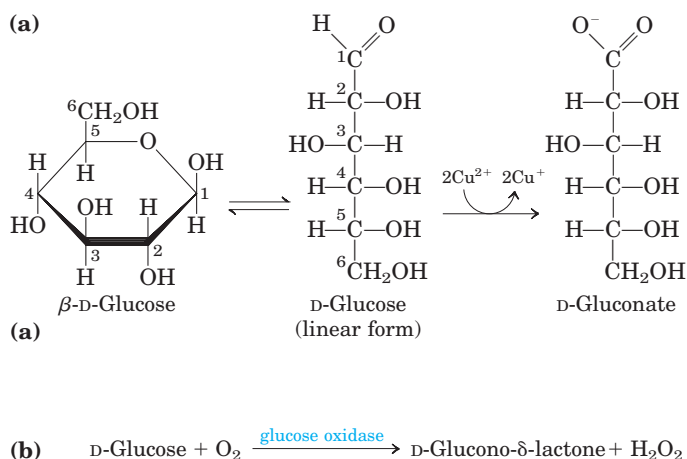
mers. The acidic sugars contain a carboxylate group, which confers a negative charge at neutral pH. D-Glucono- δ -lactone results from formation of an ester linkage between the C-1 carboxylate group and the C-5 (also known as the δ carbon) hydroxyl group of D-gluconate.

In the synthesis and metabolism of carbohydrates, the intermediates are very often not the sugars themselves but their phosphorylated derivatives. Condensation of phosphoric acid with one of the hydroxyl groups of a sugar forms a phosphate ester, as in glucose 6-phosphate (Fig. 7-9). Sugar phosphates are relatively stable at neutral pH and bear a negative charge. One effect of sugar phosphorylation within cells is to trap the sugar inside the cell; most cells do not have plasma membrane transporters for phosphorylated sugars. Phosphorylation also activates sugars for subsequent chemical transformation. Several important phosphorylated derivatives of sugars are components of nucleotides (discussed in the next chapter).

Monosaccharides Are Reducing Agents



Monosaccharides can be oxidized by relatively mild oxidizing agents such as ferric (Fe^{3+}) or cupric (Cu^{2+}) ion (Fig. 7-10a). The carbonyl carbon is oxidized to a carboxyl group. Glucose and other sugars capable of reducing ferric or cupric ion are called **reducing sugars**. This property is the basis of Fehling's reaction, a qualitative test for the presence of reducing sugar. By measuring the amount of oxidizing agent reduced by a solution of a sugar, it is also possible to estimate the concentration of that sugar. For many years this test was used to detect and measure elevated glucose levels in blood and urine in the diagnosis of dia-



betes mellitus. Now, more sensitive methods for measuring blood glucose employ an enzyme, glucose oxidase (Fig. 7–10b). ■

Disaccharides Contain a Glycosidic Bond

Disaccharides (such as maltose, lactose, and sucrose) consist of two monosaccharides joined covalently by an **O-glycosidic bond**, which is formed when a hydroxyl group of one sugar reacts with the anomeric carbon of the other (Fig. 7–11). This reaction represents the formation of an acetal from a hemiacetal (such as glucopyranose) and an alcohol (a hydroxyl group of the second sugar molecule) (Fig. 7–5). Glycosidic bonds are readily hydrolyzed by acid but resist cleavage by base. Thus disaccharides can be hydrolyzed to yield their free monosaccharide components by boiling with dilute acid. *N*-glycosyl bonds join the anomeric carbon of a sugar to a nitrogen atom in glycoproteins (see Fig. 7–31) and nucleotides (see Fig. 8–1).

The oxidation of a sugar's anomeric carbon by cupric or ferric ion (the reaction that defines a reducing sugar) occurs only with the linear form, which exists in equilibrium with the cyclic form(s). When the anomeric carbon is involved in a glycosidic bond, that sugar residue cannot take the linear form and therefore becomes a nonreducing sugar. In describing disaccharides or polysaccharides, the end of a chain with a free anomeric carbon (one not involved in a glycosidic bond) is commonly called the **reducing end**.

The disaccharide maltose (Fig. 7–11) contains two D-glucose residues joined by a glycosidic linkage between C-1 (the anomeric carbon) of one glucose residue and C-4 of the other. Because the disaccharide retains a free anomeric carbon (C-1 of the glucose residue on the right in Fig. 7–11), maltose is a reducing sugar. The configuration of the anomeric carbon atom in the glycosidic linkage is α . The glucose residue with the free anomeric carbon is capable of existing in α - and β -pyranose forms.

FIGURE 7–10 Sugars as reducing agents. (a) Oxidation of the anomeric carbon of glucose and other sugars is the basis for Fehling's reaction. The cuprous ion (Cu^+) produced under alkaline conditions forms a red cuprous oxide precipitate. In the hemiacetal (ring) form, C-1 of glucose cannot be oxidized by Cu^{2+} . However, the open-chain form is in equilibrium with the ring form, and eventually the oxidation reaction goes to completion. The reaction with Cu^{2+} is not as simple as the equation here implies; in addition to D-gluconate, a number of shorter-chain acids are produced by the fragmentation of glucose. (b) Blood glucose concentration is commonly determined by measuring the amount of H_2O_2 produced in the reaction catalyzed by glucose oxidase. In the reaction mixture, a second enzyme, peroxidase, catalyzes reaction of the H_2O_2 with a colorless compound to produce a colored compound, the amount of which is then measured spectrophotometrically.

To name reducing disaccharides such as maltose unambiguously, and especially to name more complex oligosaccharides, several rules are followed. By convention, the name describes the compound with its nonreducing end to the left, and we can “build up” the name in the following order. (1) Give the configuration (α or β) at the anomeric carbon joining the first monosaccharide unit (on the left) to the second. (2) Name the

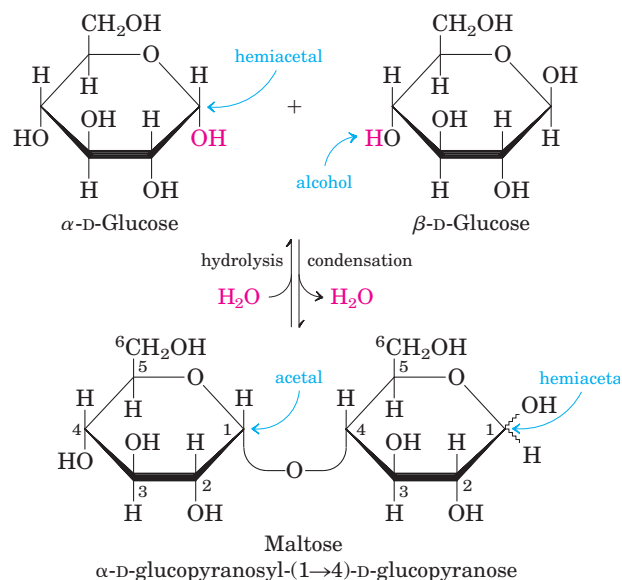


FIGURE 7–11 Formation of maltose. A disaccharide is formed from two monosaccharides (here, two molecules of D-glucose) when an —OH (alcohol) of one glucose molecule (right) condenses with the intramolecular hemiacetal of the other glucose molecule (left), with elimination of H_2O and formation of an O-glycosidic bond. The reversal of this reaction is hydrolysis—attack by H_2O on the glycosidic bond. The maltose molecule retains a reducing hemiacetal at the C-1 not involved in the glycosidic bond. Because mutarotation interconverts the α and β forms of the hemiacetal, the bonds at this position are sometimes depicted with wavy lines, as shown here, to indicate that the structure may be either α or β .

nonreducing residue; to distinguish five- and six-membered ring structures, insert “furano” or “pyrano” into the name. (3) Indicate in parentheses the two carbon atoms joined by the glycosidic bond, with an arrow connecting the two numbers; for example, (1→4) shows that C-1 of the first-named sugar residue is joined to C-4 of the second. (4) Name the second residue. If there is a third residue, describe the second glycosidic bond by the same conventions. (To shorten the description of complex polysaccharides, three-letter abbreviations for the monosaccharides are often used, as given in Table 7-1.) Following this convention for naming oligosaccharides, maltose is α -D-glucopyranosyl-(1→4)-D-glucopyranose. Because most sugars encountered in this book are the D enantiomers and the pyranose form of hexoses predominates, we generally use a shortened version of the formal name of such compounds, giving the configuration of the anomeric carbon and naming the carbons joined by the glycosidic bond. In this abbreviated nomenclature, maltose is Glc(α 1→4)Glc.

The disaccharide lactose (Fig. 7-12), which yields D-galactose and D-glucose on hydrolysis, occurs naturally only in milk. The anomeric carbon of the glucose residue is available for oxidation, and thus lactose is a reducing disaccharide. Its abbreviated name is Gal(β 1→4)Glc. Sucrose (table sugar) is a disaccharide of glucose and fructose. It is formed by plants but not by animals. In contrast to maltose and lactose, sucrose contains no free anomeric carbon atom; the anomeric carbons of both monosaccharide units are involved in the glycosidic bond (Fig. 7-12). Sucrose is therefore a nonreducing sugar. Nonreducing disaccharides are named as glycosides; in this case, the positions joined are the anomeric carbons. In the abbreviated nomenclature, a double-headed arrow connects the symbols specifying the anomeric carbons and their configurations. For example, the abbreviated name of sucrose is either Glc(α 1↔2 β)Fru or Fru(β 2↔1 α)Glc. Sucrose is a major intermediate product of photosynthesis; in

many plants it is the principal form in which sugar is transported from the leaves to other parts of the plant body. Trehalose, Glc(α 1↔1 α)Glc (Fig. 7-12)—a disaccharide of D-glucose that, like sucrose, is a nonreducing sugar—is a major constituent of the circulating fluid (hemolymph) of insects, serving as an energy-storage compound.

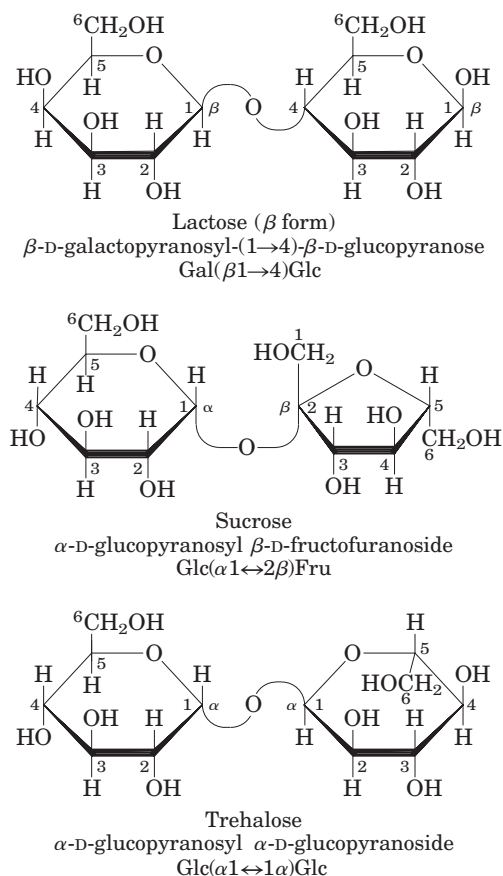


FIGURE 7-12 Some common disaccharides. Like maltose in Figure 7-11, these are shown as Haworth perspectives. The common name, full systematic name, and abbreviation are given for each disaccharide.

TABLE 7-1 Abbreviations for Common Monosaccharides and Some of Their Derivatives

Abequose	Abe	Glucuronic acid	GlcA
Arabinose	Ara	Galactosamine	GalN
Fructose	Fru	Glucosamine	GlcN
Fucose	Fuc	N-Acetylgalactosamine	GalNAc
Galactose	Gal	N-Acetylglucosamine	GlcNAc
Glucose	Glc	Iduronic acid	IdoA
Mannose	Man	Muramic acid	Mur
Rhamnose	Rha	N-Acetylmuramic acid	Mur2Ac
Ribose	Rib	N-Acetylneuraminic acid	Neu5Ac
Xylose	Xyl	(a sialic acid)	

SUMMARY 7.1 Monosaccharides and Disaccharides

- Sugars (also called saccharides) are compounds containing an aldehyde or ketone group and two or more hydroxyl groups.
- Monosaccharides generally contain several chiral carbons and therefore exist in a variety of stereochemical forms, which may be represented on paper as Fischer projections. Epimers are sugars that differ in configuration at only one carbon atom.
- Monosaccharides commonly form internal hemiacetals or hemiketals, in which the aldehyde or ketone group joins with a hydroxyl group of the same molecule, creating a cyclic structure; this can be represented as a Haworth perspective formula. The carbon atom originally found in the aldehyde or ketone group (the anomeric carbon) can assume either of two configurations, α and β , which are interconvertible by mutarotation. In the linear form, which is in equilibrium with the cyclized forms, the anomeric carbon is easily oxidized.
- A hydroxyl group of one monosaccharide can add to the anomeric carbon of a second monosaccharide to form an acetal. In this disaccharide, the glycosidic bond protects the anomeric carbon from oxidation.
- Oligosaccharides are short polymers of several monosaccharides joined by glycosidic bonds. At one end of the chain, the reducing end, is a monosaccharide unit whose anomeric carbon is not involved in a glycosidic bond.
- The common nomenclature for di- or oligosaccharides specifies the order of monosaccharide units, the configuration at each anomeric carbon, and the carbon atoms involved in the glycosidic linkage(s).

7.2 Polysaccharides

Most carbohydrates found in nature occur as polysaccharides, polymers of medium to high molecular weight. Polysaccharides, also called **glycans**, differ from each other in the identity of their recurring monosaccharide units, in the length of their chains, in the types of bonds linking the units, and in the degree of branching. **Homopolysaccharides** contain only a single type of monomer; **heteropolysaccharides** contain two or more different kinds (Fig. 7–13). Some homopolysaccharides serve as storage forms of monosaccharides that are used as fuels; starch and glycogen are homopolysaccharides of this type. Other homopolysaccharides (cellulose and chitin,

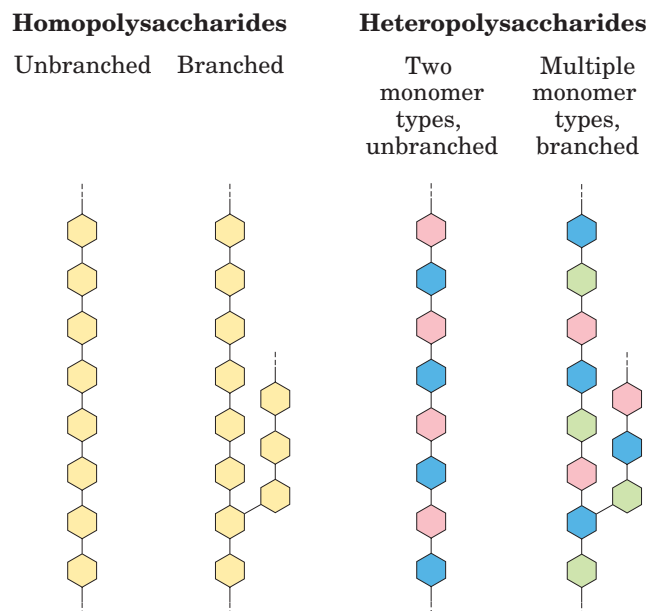


FIGURE 7-13 Homo- and heteropolysaccharides. Polysaccharides may be composed of one, two, or several different monosaccharides, in straight or branched chains of varying length.

for example) serve as structural elements in plant cell walls and animal exoskeletons. Heteropolysaccharides provide extracellular support for organisms of all kingdoms. For example, the rigid layer of the bacterial cell envelope (the peptidoglycan) is composed in part of a heteropolysaccharide built from two alternating monosaccharide units. In animal tissues, the extracellular space is occupied by several types of heteropolysaccharides, which form a matrix that holds individual cells together and provides protection, shape, and support to cells, tissues, and organs.

Unlike proteins, polysaccharides generally do not have definite molecular weights. This difference is a consequence of the mechanisms of assembly of the two types of polymers. As we shall see in Chapter 27, proteins are synthesized on a template (messenger RNA) of defined sequence and length, by enzymes that follow the template exactly. For polysaccharide synthesis there is no template; rather, the program for polysaccharide synthesis is intrinsic to the enzymes that catalyze the polymerization of the monomeric units, and there is no specific stopping point in the synthetic process.

Some Homopolysaccharides Are Stored Forms of Fuel

The most important storage polysaccharides are starch in plant cells and glycogen in animal cells. Both polysaccharides occur intracellularly as large clusters or granules (Fig. 7–14). Starch and glycogen molecules are heavily hydrated, because they have many exposed hydroxyl groups available to hydrogen-bond with water. Most plant cells have the ability to form starch, but it is

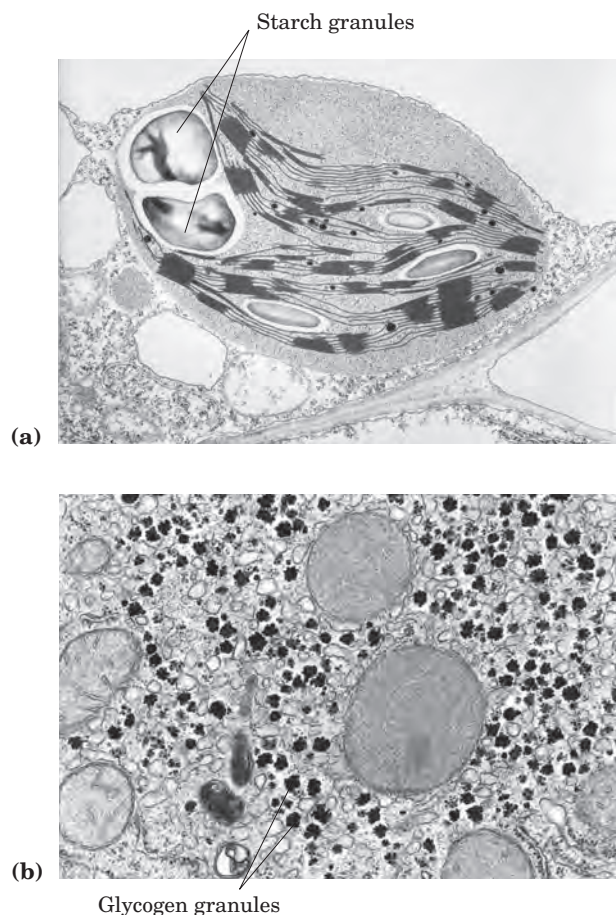


FIGURE 7-14 Electron micrographs of starch and glycogen granules. (a) Large starch granules in a single chloroplast. Starch is made in the chloroplast from D-glucose formed photosynthetically. (b) Glycogen granules in a hepatocyte. These granules form in the cytosol and are much smaller ($\sim 0.1 \mu\text{m}$) than starch granules ($\sim 1.0 \mu\text{m}$).

especially abundant in tubers, such as potatoes, and in seeds.

Starch contains two types of glucose polymer, amylose and amylopectin (Fig. 7-15). The former consists of long, unbranched chains of D-glucose residues connected by ($\alpha 1 \rightarrow 4$) linkages. Such chains vary in molecular weight from a few thousand to more than a million. Amylopectin also has a high molecular weight (up to 100 million) but unlike amylose is highly branched. The glycosidic linkages joining successive glucose residues in amylopectin chains are ($\alpha 1 \rightarrow 4$); the branch points (occurring every 24 to 30 residues) are ($\alpha 1 \rightarrow 6$) linkages.

Glycogen is the main storage polysaccharide of animal cells. Like amylopectin, glycogen is a polymer of ($\alpha 1 \rightarrow 4$)-linked subunits of glucose, with ($\alpha 1 \rightarrow 6$)-linked branches, but glycogen is more extensively branched (on average, every 8 to 12 residues) and more compact than starch. Glycogen is especially abundant in the liver,

where it may constitute as much as 7% of the wet weight; it is also present in skeletal muscle. In hepatocytes glycogen is found in large granules (Fig. 7-14b), which are themselves clusters of smaller granules composed of single, highly branched glycogen molecules with an average molecular weight of several million. Such glycogen granules also contain, in tightly bound form, the enzymes responsible for the synthesis and degradation of glycogen.

Because each branch in glycogen ends with a nonreducing sugar unit, a glycogen molecule has as many nonreducing ends as it has branches, but only one reducing end. When glycogen is used as an energy source, glucose units are removed one at a time from the nonreducing ends. Degradative enzymes that act only at nonreducing ends can work simultaneously on the many branches, speeding the conversion of the polymer to monosaccharides.

Why not store glucose in its monomeric form? It has been calculated that hepatocytes store glycogen equivalent to a glucose concentration of 0.4 M. The actual concentration of glycogen, which is insoluble and contributes little to the osmolarity of the cytosol, is about $0.01 \mu\text{M}$. If the cytosol contained 0.4 M glucose, the osmolarity would be threateningly elevated, leading to osmotic entry of water that might rupture the cell (see Fig. 2-13). Furthermore, with an intracellular glucose concentration of 0.4 M and an external concentration of about 5 mM (the concentration in the blood of a mammal), the free-energy change for glucose uptake into cells against this very high concentration gradient would be prohibitively large.

Dextrans are bacterial and yeast polysaccharides made up of ($\alpha 1 \rightarrow 6$)-linked poly-D-glucose; all have ($\alpha 1 \rightarrow 3$) branches, and some also have ($\alpha 1 \rightarrow 2$) or ($\alpha 1 \rightarrow 4$) branches. Dental plaque, formed by bacteria growing on the surface of teeth, is rich in dextrans. Synthetic dextrans are used in several commercial products (for example, Sephadex) that serve in the fractionation of proteins by size-exclusion chromatography (see Fig. 3-18b). The dextrans in these products are chemically cross-linked to form insoluble materials of various porosities, admitting macromolecules of various sizes.

Some Homopolysaccharides Serve Structural Roles

Cellulose, a fibrous, tough, water-insoluble substance, is found in the cell walls of plants, particularly in stalks, stems, trunks, and all the woody portions of the plant body. Cellulose constitutes much of the mass of wood, and cotton is almost pure cellulose. Like amylose and the main chains of amylopectin and glycogen, the cellulose molecule is a linear, unbranched homopolysaccharide, consisting of 10,000 to 15,000 D-glucose units. But there is a very important difference: in cellulose the glucose residues have the β configuration (Fig. 7-16),

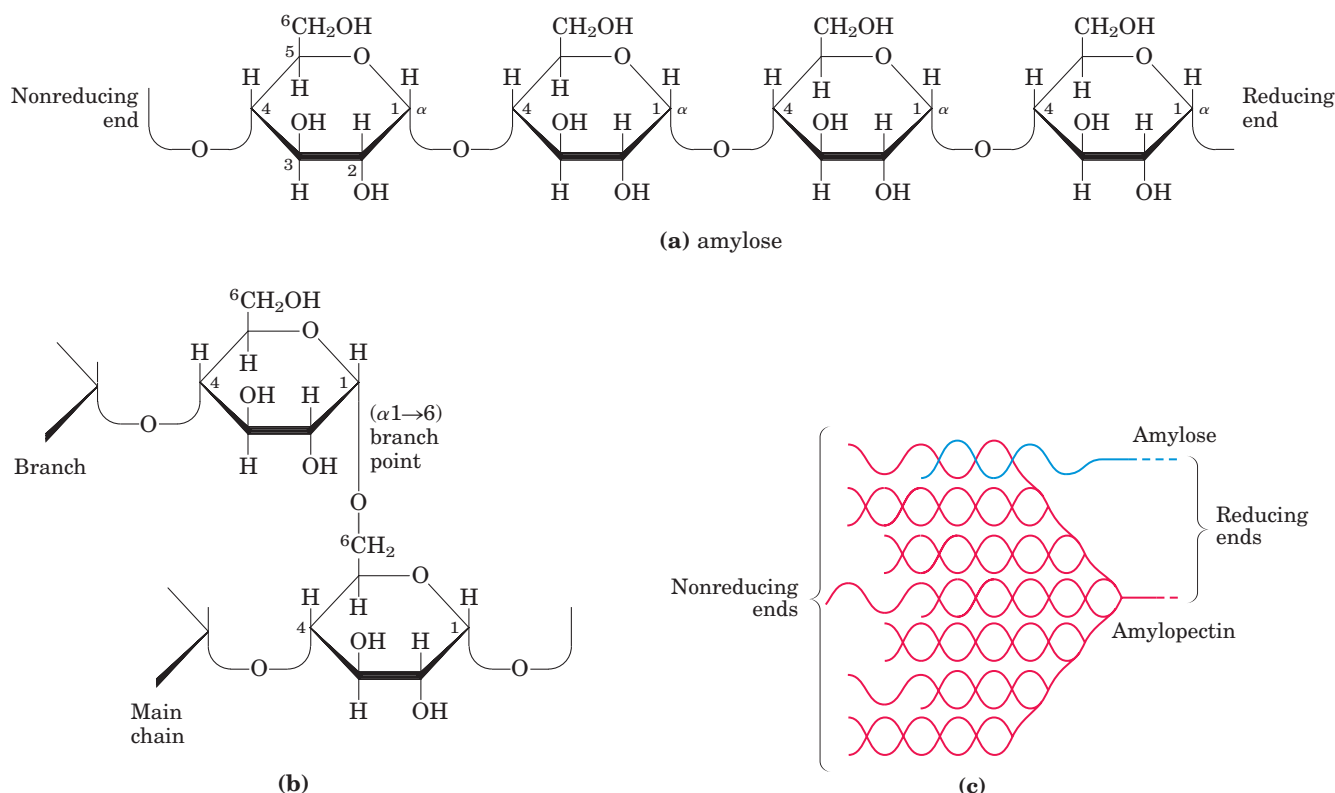


FIGURE 7-15 Amylose and amylopectin, the polysaccharides of starch. (a) A short segment of amylose, a linear polymer of D-glucose residues in ($\alpha 1 \rightarrow 4$) linkage. A single chain can contain several thousand glucose residues. Amylopectin has stretches of similarly linked residues between branch points. (b) An ($\alpha 1 \rightarrow 6$) branch point of amylopectin. (c) A cluster of amylose and amylopectin like that believed

to occur in starch granules. Strands of amylopectin (red) form double-helical structures with each other or with amylose strands (blue). Glucose residues at the nonreducing ends of the outer branches are removed enzymatically during the mobilization of starch for energy production. Glycogen has a similar structure but is more highly branched and more compact.

whereas in amylose, amylopectin, and glycogen the glucose is in the α configuration. The glucose residues in cellulose are linked by ($\beta 1 \rightarrow 4$) glycosidic bonds, in contrast to the ($\alpha 1 \rightarrow 4$) bonds of amylose, starch, and glycogen. This difference gives cellulose and amylose very different structures and physical properties.

Glycogen and starch ingested in the diet are hydrolyzed by α -amylases, enzymes in saliva and intestinal secretions that break ($\alpha 1 \rightarrow 4$) glycosidic bonds between glucose units. Most animals cannot use cellulose as a fuel source, because they lack an enzyme to hydrolyze the ($\beta 1 \rightarrow 4$) linkages. Termites readily digest cellulose

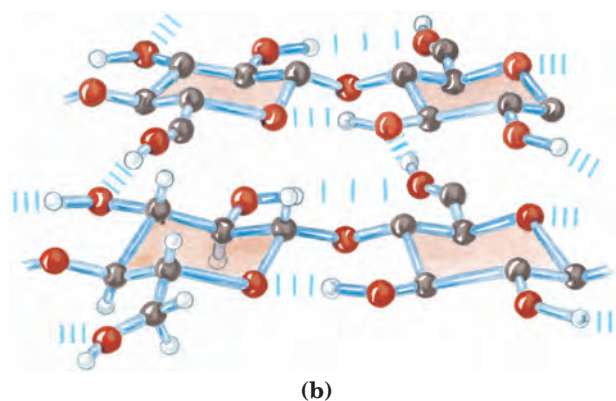
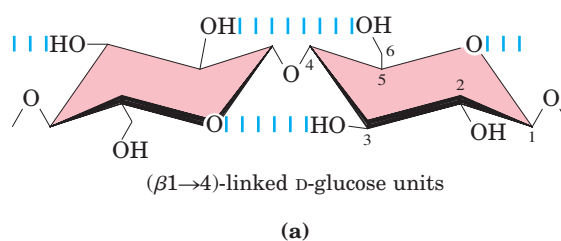


FIGURE 7-16 The structure of cellulose. (a) Two units of a cellulose chain; the D-glucose residues are in ($\beta 1 \rightarrow 4$) linkage. The rigid chair structures can rotate relative to one another. (b) Scale drawing of segments of two parallel cellulose chains, showing the conformation of the D-glucose residues and the hydrogen-bond cross-links. In the hexose unit at the lower left, all hydrogen atoms are shown; in the other three hexose units, the hydrogens attached to carbon have been omitted for clarity as they do not participate in hydrogen bonding.



FIGURE 7-17 Cellulose breakdown by wood fungi. A wood fungus growing on an oak log. All wood fungi have the enzyme cellulase, which breaks the ($\beta 1 \rightarrow 4$) glycosidic bonds in cellulose, such that wood is a source of metabolizable sugar (glucose) for the fungus. The only vertebrates able to use cellulose as food are cattle and other ruminants (sheep, goats, camels, giraffes). The extra stomach compartment (rumen) of a ruminant teems with bacteria and protists that secrete cellulase.

(and therefore wood), but only because their intestinal tract harbors a symbiotic microorganism, *Trichonympha*, that secretes cellulase, which hydrolyzes the ($\beta 1 \rightarrow 4$) linkages. Wood-rot fungi and bacteria also produce cellulase (Fig. 7-17).

Chitin is a linear homopolysaccharide composed of *N*-acetylglucosamine residues in β linkage (Fig. 7-18). The only chemical difference from cellulose is the replacement of the hydroxyl group at C-2 with an acetylated amino group. Chitin forms extended fibers similar to those of cellulose, and like cellulose cannot be digested by vertebrates. Chitin is the principal component of the hard exoskeletons of nearly a million species of arthropods—insects, lobsters, and crabs, for example—and is probably the second most abundant polysaccharide, next to cellulose, in nature.

Steric Factors and Hydrogen Bonding Influence Homopolysaccharide Folding

The folding of polysaccharides in three dimensions follows the same principles as those governing polypeptide structure: subunits with a more-or-less rigid structure dictated by covalent bonds form three-dimensional macromolecular structures that are stabilized by weak interactions within or between molecules: hydrogen-bond, hydrophobic, and van der Waals interactions, and, for polymers with charged subunits, electrostatic interactions. Because polysaccharides have so many hydroxyl groups, hydrogen bonding has an especially important influence on their structure. Glycogen, starch, and cellulose are composed of pyranoside subunits (having six-membered rings), as are the oligosaccharides of glycoproteins and glycolipids to be discussed later. Such molecules can be represented as a series of rigid pyranose rings connected by an oxygen atom bridging two carbon atoms (the glycosidic bond). There is, in princi-

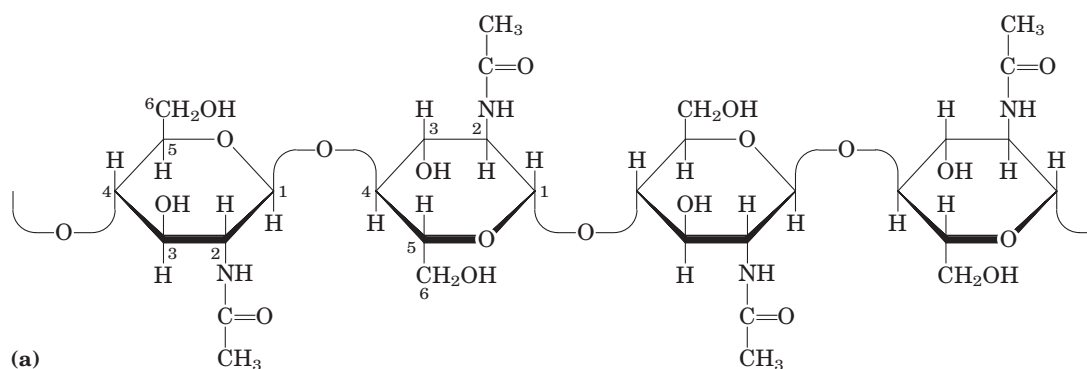


FIGURE 7-18 Chitin. (a) A short segment of chitin, a homopolymer of *N*-acetyl-D-glucosamine units in ($\beta 1 \rightarrow 4$) linkage. (b) A spotted June beetle (*Pellidnota punetata*), showing its surface armor (exoskeleton) of chitin.



(b)

ple, free rotation about both C—O bonds linking the residues (Fig. 7-16a), but as in polypeptides (see Figs 4-2, 4-9), rotation about each bond is limited by steric hindrance by substituents. The three-dimensional structures of these molecules can be described in terms of the dihedral angles, ϕ and ψ , made with the glycosidic bond (Fig. 7-19), analogous to angles ϕ and ψ made by the peptide bond (see Fig. 4-2). Because of the bulkiness of the pyranose ring and its substituents, their size and shape place constraints on the angles ϕ and ψ ; certain conformations are much more stable than others, as can be shown on a map of energy as a function of ϕ and ψ (Fig. 7-20).

The most stable three-dimensional structure for starch and glycogen is a tightly coiled helix (Fig. 7-21), stabilized by interchain hydrogen bonds. In amylose (with no branches) this structure is regular enough to allow crystallization and thus determination of the structure by x-ray diffraction. Each residue along the amylose chain forms a 60° angle with the preceding residue, so the helical structure has six residues per turn. For amylose, the core of the helix is of precisely the right dimensions to accommodate iodine in the form I^{3-} or I^{5-} (iodide ions), and this interaction with iodine is a common qualitative test for amylose.

For cellulose, the most stable conformation is that in which each chair is turned 180° relative to its neighbors, yielding a straight, extended chain. All —OH groups are available for hydrogen bonding with neighboring chains. With several chains lying side by side, a stabilizing network of interchain and intrachain hydrogen bonds produces straight, stable supramolecular

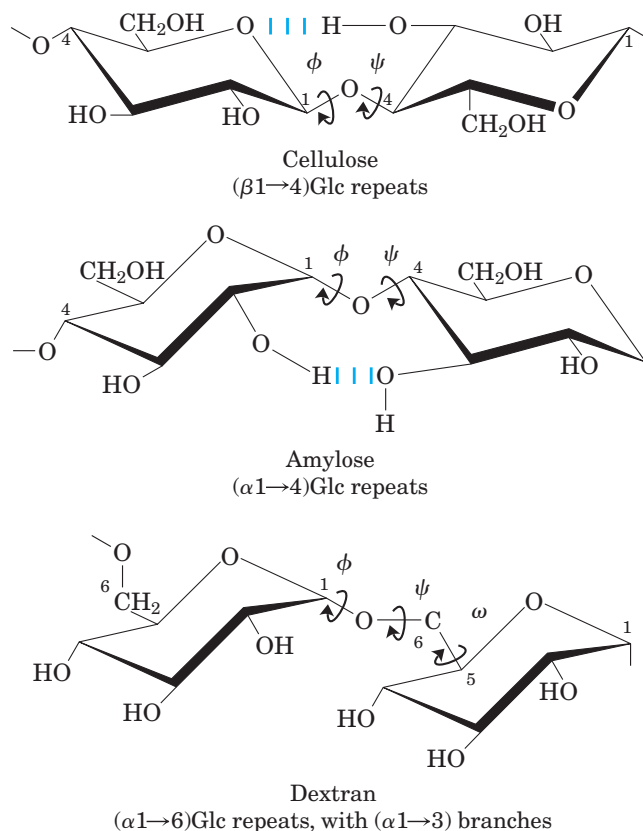


FIGURE 7-19 Conformation at the glycosidic bonds of cellulose, amylose, and dextran. The polymers are depicted as rigid pyranose rings joined by glycosidic bonds, with free rotation about these bonds. Note that in dextran there is also free rotation about the bond between C-5 and C-6 (torsion angle ω (omega)).

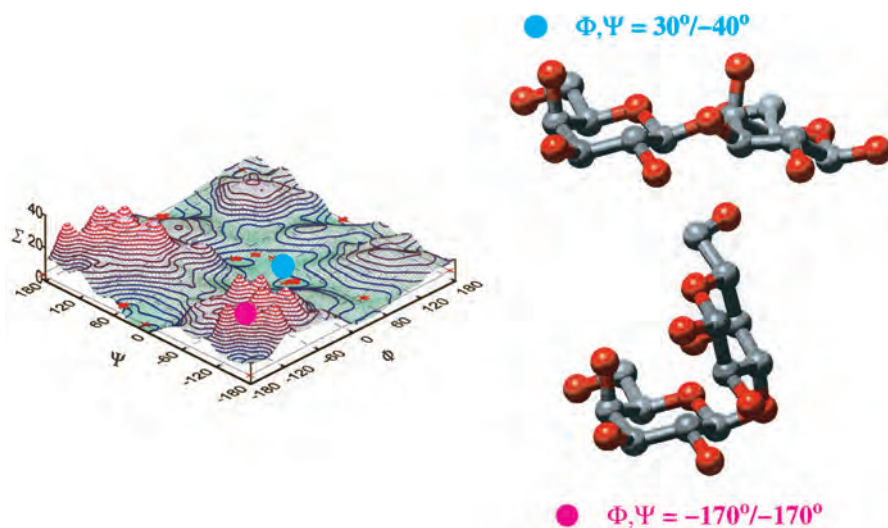


FIGURE 7-20 A map of favored conformations for oligosaccharides and polysaccharides. The torsion angles ψ and ϕ (see Fig. 7-19), which define the spatial relationship between adjacent rings, can in principle have any value from 0° to 360° . In fact, some of the torsion angles would give conformations that are sterically hindered, whereas others give conformations that maximize hydrogen bonding. When the relative energy is plotted for each value of ϕ and ψ , with iso-

ergy ("same energy") contours drawn at intervals of 1 kcal/mol above the minimum energy state, the result is a map of preferred conformations. This is analogous to the Ramachandran plot for peptides (see Figs 4-3, 4-9). The known conformations of the three polysaccharides shown in Figure 7-19 have been determined by x-ray crystallography, and all fall within the lowest-energy regions of the map.

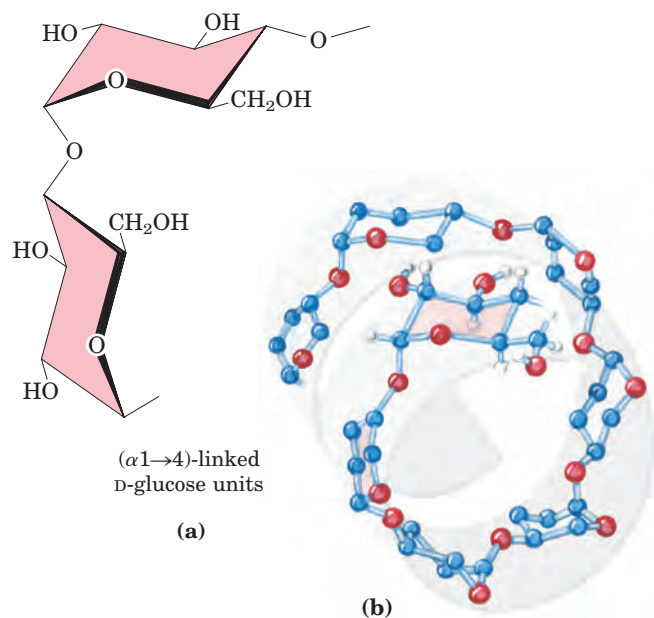


FIGURE 7-21 The structure of starch (amylose). (a) In the most stable conformation, with adjacent rigid chairs, the polysaccharide chain is curved, rather than linear as in cellulose (see Fig. 7-16). (b) Scale drawing of a segment of amylose. The conformation of $(\alpha 1 \rightarrow 4)$ linkages in amylose, amylopectin, and glycogen causes these polymers to assume tightly coiled helical structures. These compact structures produce the dense granules of stored starch or glycogen seen in many cells (see Fig. 7-14).

fibers of great tensile strength (Fig. 7-16b). This property of cellulose has made it a useful substance to civilizations for millennia. Many manufactured products, including papyrus, paper, cardboard, rayon, insulating tiles, and a variety of other useful materials, are derived from cellulose. The water content of these materials is low because extensive interchain hydrogen bonding between cellulose molecules satisfies their capacity for hydrogen-bond formation.

Bacterial and Algal Cell Walls Contain Structural Heteropolysaccharides

The rigid component of bacterial cell walls is a heteropolymer of alternating $(\beta 1 \rightarrow 4)$ -linked *N*-acetylglucosamine and *N*-acetylmuramic acid residues (Fig. 7-22). The linear polymers lie side by side in the cell wall, cross-linked by short peptides, the exact structure of which depends on the bacterial species. The peptide cross-links weld the polysaccharide chains into a strong sheath that envelops the entire cell and prevents cellular swelling and lysis due to the osmotic entry of water. The enzyme lysozyme kills bacteria by hydrolyzing the $(\beta 1 \rightarrow 4)$ glycosidic bond between *N*-acetylglucosamine and *N*-acetylmuramic acid (see Fig. 6-24). Lysozyme is notably present in tears, presumably as a defense against bacterial infections of the eye. It is also produced by certain bacterial viruses to ensure their release from the host

bacterial cell, an essential step of the viral infection cycle. Penicillin and related antibiotics kill bacteria by preventing synthesis of the cross-links, leaving the cell wall too weak to resist osmotic lysis (see Box 20-1).

Certain marine red algae, including some of the seaweeds, have cell walls that contain **agar**, a mixture of sulfated heteropolysaccharides made up of D-galactose and an L-galactose derivative ether-linked between C-3 and C-6 (Fig. 7-23). The two major components of agar are the unbranched polymer **agarose** ($M_r \sim 120,000$) and a branched component, agarpectin. The remarkable gel-forming property of agarose makes it useful in the biochemistry laboratory. When a suspension of agarose in water is heated and cooled, the agarose forms a double helix: two molecules in parallel orientation twist together with a helix repeat of three residues; water molecules are trapped in the central cavity. These struc-

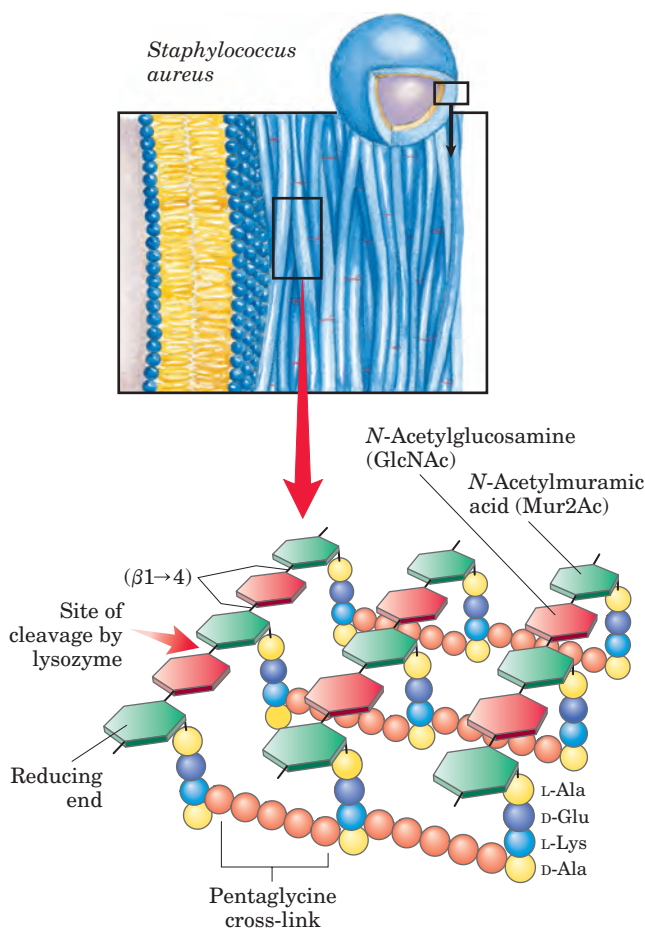


FIGURE 7-22 Peptidoglycan. Shown here is the peptidoglycan of the cell wall of *Staphylococcus aureus*, a gram-positive bacterium. Peptides (strings of colored spheres) covalently link *N*-acetylmuramic acid residues in neighboring polysaccharide chains. Note the mixture of L and D amino acids in the peptides. Gram-positive bacteria have a pentaglycine chain in the cross-link. Gram-negative bacteria, such as *E. coli*, lack the pentaglycine; instead, the terminal D-Ala residue of one tetrapeptide is attached directly to a neighboring tetrapeptide through either L-Lys or a lysine-like amino acid, diaminopimelic acid.

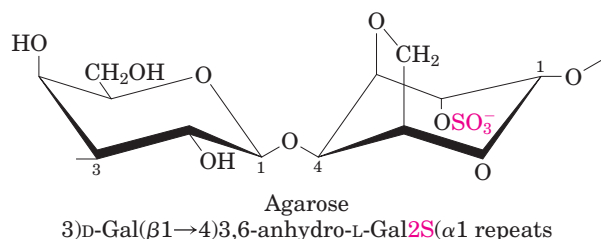


FIGURE 7-23 The structure of agarose. The repeating unit consists of D-galactose (β1→4)-linked to 3,6-anhydro-L-galactose (in which an ether ring connects C-3 and C-6). These units are joined by (α1→3) glycosidic links to form a polymer 600 to 700 residues long. A small fraction of the 3,6-anhydrogalactose residues have a sulfate ester at C-2 (as shown here).

tures in turn associate with each other to form a gel—a three-dimensional matrix that traps large amounts of water. Agarose gels are used as inert supports for the electrophoretic separation of nucleic acids, an essential part of the DNA sequencing process (p. 8-24). Agar is also used to form a surface for the growth of bacterial colonies. Another commercial use of agar is for the capsules in which some vitamins and drugs are packaged; the dried agar material dissolves readily in the stomach and is metabolically inert.

Glycosaminoglycans Are Heteropolysaccharides of the Extracellular Matrix

The extracellular space in the tissues of multicellular animals is filled with a gel-like material, the **extracellular matrix**, also called ground substance, which holds the cells together and provides a porous pathway for the diffusion of nutrients and oxygen to individual cells. The extracellular matrix is composed of an interlocking meshwork of heteropolysaccharides and fibrous proteins such as collagen, elastin, fibronectin, and laminin. These heteropolysaccharides, the **glycosaminoglycans**, are a family of linear polymers composed of repeating disaccharide units (Fig. 7-24). One of the two monosaccharides is always either *N*-acetylglucosamine or *N*-acetylgalactosamine; the other is in most cases a uronic acid, usually D-glucuronic or L-iduronic acid. In some glycosaminoglycans, one or more of the hydroxyls of the amino sugar are esterified with sulfate. The combination

of sulfate groups and the carboxylate groups of the uronic acid residues gives glycosaminoglycans a very high density of negative charge. To minimize the repulsive forces among neighboring charged groups, these molecules assume an extended conformation in solution. The specific patterns of sulfated and nonsulfated sugar residues in glycosaminoglycans provide for specific recognition by a

Glycosaminoglycan

Number of
disaccharides
per chain

Repeating disaccharide

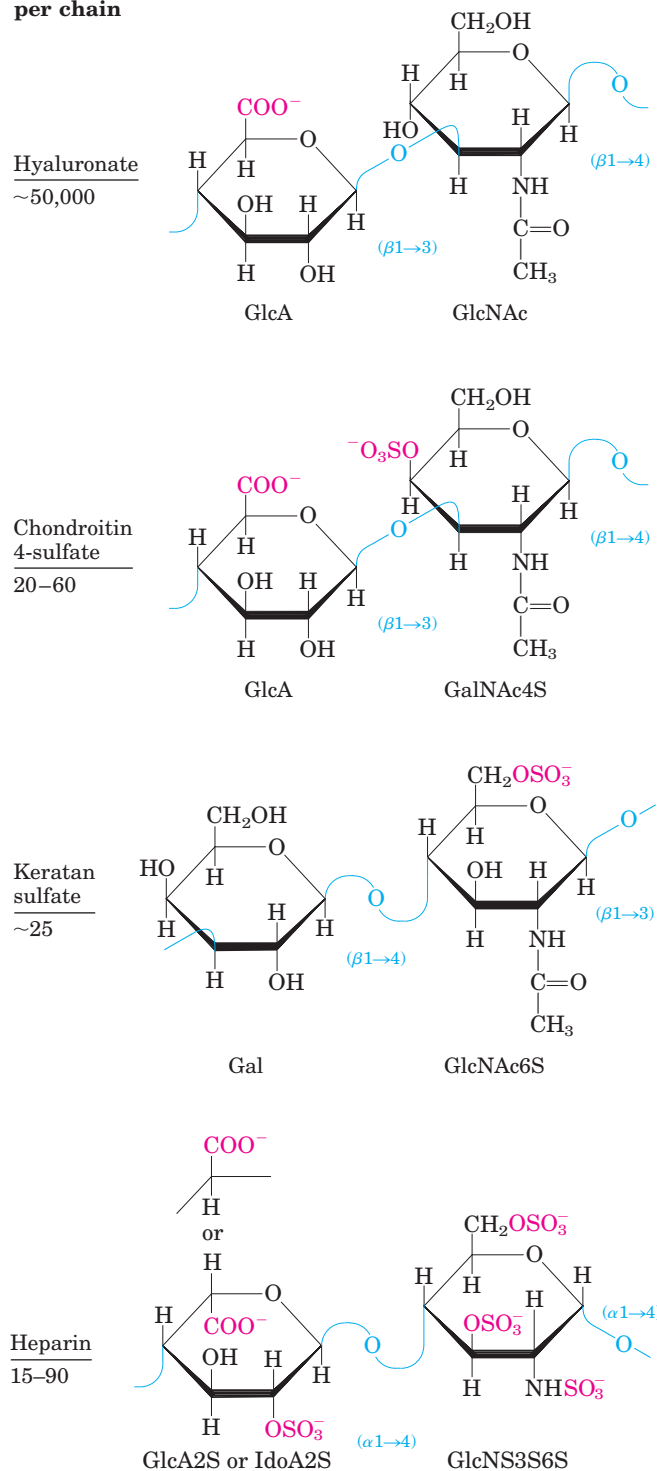
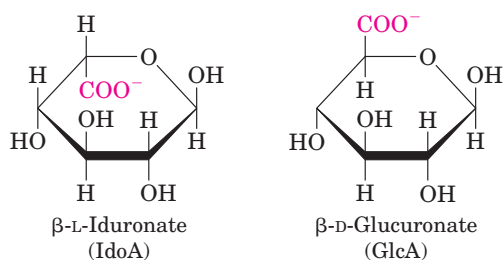


FIGURE 7-24 Repeating units of some common glycosaminoglycans of extracellular matrix. The molecules are copolymers of alternating uronic acid and amino sugar residues, with sulfate esters in any of several positions. The ionized carboxylate and sulfate groups (red) give these polymers their characteristic high negative charge. Heparin contains primarily iduronic acid (IdoA) and a smaller proportion of glucuronic acid (GlcA), and is generally highly sulfated and heterogeneous in length. Heparan sulfate (not shown) is similar to heparin but has a higher proportion of GlcA and fewer sulfate groups, arranged in a less regular pattern.

variety of protein ligands that bind electrostatically to these molecules. Glycosaminoglycans are attached to extracellular proteins to form proteoglycans (Section 7.3).

The glycosaminoglycan **hyaluronic acid** (hyaluronate at physiological pH) contains alternating residues of D-glucuronic acid and N-acetylglucosamine (Fig. 7–24). With up to 50,000 repeats of the basic disaccharide unit, hyaluronates have molecular weights greater than 1 million; they form clear, highly viscous solutions that serve as lubricants in the synovial fluid of joints and give the vitreous humor of the vertebrate eye its jelly-like consistency (the Greek *hyalos* means “glass”; hyaluronates can have a glassy or translucent appearance). Hyaluronate is also an essential component of the extracellular matrix of cartilage and tendons, to which it contributes tensile strength and elasticity as a result of its strong interactions with other components of the matrix. Hyaluronidase, an enzyme secreted by some pathogenic bacteria, can hydrolyze the glycosidic linkages of hyaluronate, rendering tissues more susceptible to bacterial invasion. In many organisms, a similar enzyme in sperm hydrolyzes an outer glycosaminoglycan coat around the ovum, allowing sperm penetration.

Other glycosaminoglycans differ from hyaluronate in two respects: they are generally much shorter polymers and they are covalently linked to specific proteins (proteoglycans). Chondroitin sulfate (Greek *chondros*, “cartilage”) contributes to the tensile strength of cartilage, tendons, ligaments, and the walls of the aorta. Dermatan sulfate (Greek *derma*, “skin”) contributes to the pliability of skin and is also present in blood vessels and heart valves. In this polymer, many of the glucuronate (GlcA) residues present in chondroitin sulfate are replaced by their epimer, iduronate (IdoA).



Keratan sulfates (Greek *keras*, “horn”) have no uronic acid and their sulfate content is variable. They are present in cornea, cartilage, bone, and a variety of horny structures formed of dead cells: horn, hair, hoofs, nails, and claws. Heparin (Greek *hēpar*, “liver”) is a natural anticoagulant made in mast cells (a type of leukocyte) and released into the blood, where it inhibits blood coagulation by binding to the protein antithrombin. Heparin binding causes antithrombin to bind to and inhibit thrombin, a protease essential to blood clotting. The interaction is strongly electrostatic; heparin has the highest negative charge density of any known biological macromolecule (Fig. 7–25). Purified heparin is routinely

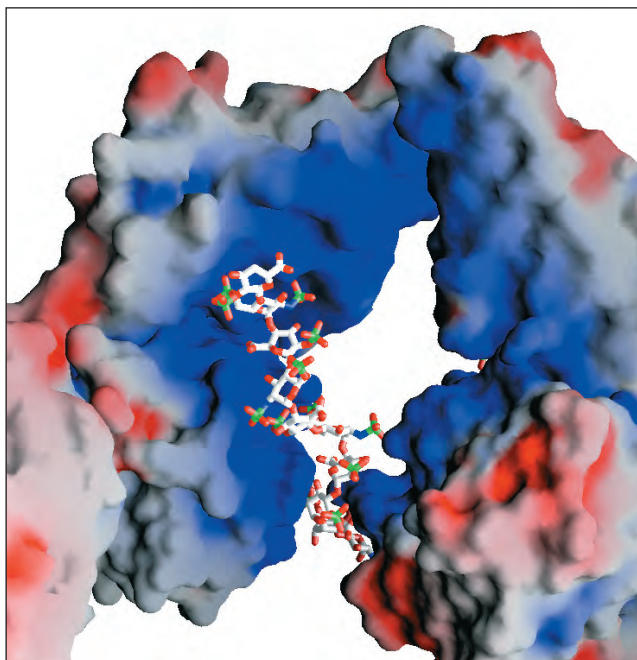


FIGURE 7-25 Interaction between a glycosaminoglycan and its binding protein. Fibroblast growth factor (FGF1), its cell surface receptor (FGFR), and a short segment of a glycosaminoglycan (heparin) were co-crystallized to yield the structure shown here (PDB ID 1E0O). The proteins are represented as surface contour images, with color to represent surface electrostatic potential: red, predominantly negative charge; blue, predominantly positive charge. Heparin is shown in a ball-and-stick representation, with the negative charges ($-\text{SO}_3^-$ and $-\text{COO}^-$) attracted to the positive (blue) surface of the FGF protein. Heparin was used in this experiment, but, in vivo, the glycosaminoglycan that binds FGF is heparan sulfate on the cell surface.

added to blood samples obtained for clinical analysis, and to blood donated for transfusion, to prevent clotting.

Table 7–2 summarizes the composition, properties, roles, and occurrence of the polysaccharides described in Section 7.2.

SUMMARY 7.2 Polysaccharides

- Polysaccharides (glycans) serve as stored fuel and as structural components of cell walls and extracellular matrix.
- The homopolysaccharides starch and glycogen are stored fuels in plant, animal, and bacterial cells. They consist of D-glucose with linkages, and all three contain some branches.
- The homopolysaccharides cellulose, chitin, and dextran serve structural roles. Cellulose, composed of ($\beta 1 \rightarrow 4$)-linked D-glucose residues, lends strength and rigidity to plant cell walls. Chitin, a polymer of ($\beta 1 \rightarrow 4$)-linked N-acetylglucosamine, strengthens the

TABLE 7-2 Structures and Roles of Some Polysaccharides

Polymer	Type*	Repeating unit [†]	Size (number of monosaccharide units)	Roles/significance
Starch				Energy storage: in plants
Amylose	Homo-	(α 1 $\rightarrow$ 4)Glc, linear	50–5,000	
Amylopectin	Homo-	(α 1 $\rightarrow$ 4)Glc, with (α 1 $\rightarrow$ 6)Glc branches every 24–30 residues	Up to 10 ⁶	
Glycogen	Homo-	(α 1 $\rightarrow$ 4)Glc, with (α 1 $\rightarrow$ 6)Glc branches every 8–12 residues	Up to 50,000	Energy storage: in bacteria and animal cells
Cellulose	Homo-	(β 1 $\rightarrow$ 4)Glc	Up to 15,000	Structural: in plants, gives rigidity and strength to cell walls
Chitin	Homo-	(β 1 $\rightarrow$ 4)GlcNAc	Very large	Structural: in insects, spiders, crustaceans, gives rigidity and strength to exoskeletons
Dextran	Homo-	(α 1 $\rightarrow$ 6)Glc, with (α 1 $\rightarrow$ 3) branches	Wide range	Structural: in bacteria, extracellular adhesive
Peptidoglycan	Hetero-; peptides attached	4)Mur2Ac(β 1 $\rightarrow$ 4)GlcNAc(β 1	Very large	Structural: in bacteria, gives rigidity and strength to cell envelope
Agarose	Hetero-	3) α -Gal(β 1 $\rightarrow$ 4)3,6-anhydro-L-Gal(α 1	1,000	Structural: in algae, cell wall material
Hyaluronate (a glycosaminoglycan)	Hetero-; acidic	4)GlcA(β 1 $\rightarrow$ 3)GlcNAc(β 1	Up to 100,000	Structural: in vertebrates, extracellular matrix of skin and connective tissue; viscosity and lubrication in joints

*Each polymer is classified as a homopolysaccharide (homo-) or heteropolysaccharide (hetero-).

[†]The abbreviated names for the peptidoglycan, agarose, and hyaluronate repeating units indicate that the polymer contains repeats of this disaccharide unit. For example, in peptidoglycan, the GlcNAc of one disaccharide unit is (β 1 $\rightarrow$ 4)-linked to the first residue of the next disaccharide unit.

exoskeletons of arthropods. Dextran forms an adhesive coat around certain bacteria.

- Homopolysaccharides fold in three dimensions. The chair form of the pyranose ring is essentially rigid, so the conformation of the polymers is determined by rotation about the bonds to the oxygen on the anomeric carbon. Starch and glycogen form helical structures with intrachain hydrogen bonding; cellulose and chitin form long, straight strands that interact with neighboring strands.
- Bacterial and algal cell walls are strengthened by heteropolysaccharides—peptidoglycan in bacteria, agar in red algae. The repeating disaccharide in peptidoglycan is GlcNAc(β 1 $\rightarrow$ 4)Mur2Ac; in agarose, it is α -Gal(β 1 $\rightarrow$ 4)3,6-anhydro-L-Gal.
- Glycosaminoglycans are extracellular heteropolysaccharides in which one of the two monosaccharide units is a uronic acid and the

other an *N*-acetylated amino sugar. Sulfate esters on some of the hydroxyl groups give these polymers a high density of negative charge, forcing them to assume extended conformations. These polymers (hyaluronate, chondroitin sulfate, dermatan sulfate, keratan sulfate, and heparin) provide viscosity, adhesiveness, and tensile strength to the extracellular matrix.

7.3 Glycoconjugates: Proteoglycans, Glycoproteins, and Glycolipids

In addition to their important roles as stored fuels (starch, glycogen, dextran) and as structural materials (cellulose, chitin, peptidoglycans), polysaccharides and oligosaccharides are information carriers: they serve as destination labels for some proteins and as mediators of specific cell-cell interactions and interactions between cells and the extracellular matrix. Specific carbohydrate-containing molecules act in cell-cell recognition and

adhesion, cell migration during development, blood clotting, the immune response, and wound healing, to name but a few of their many roles. In most of these cases, the informational carbohydrate is covalently joined to a protein or a lipid to form a **glycoconjugate**, which is the biologically active molecule.

Proteoglycans are macromolecules of the cell surface or extracellular matrix in which one or more glycosaminoglycan chains are joined covalently to a membrane protein or a secreted protein. The glycosaminoglycan moiety commonly forms the greater fraction (by mass) of the proteoglycan molecule, dominates the structure, and is often the main site of biological activity. In many cases the biological activity is the provision of multiple binding sites, rich in opportunities for hydrogen bonding and electrostatic interactions with other proteins of the cell surface or the extracellular matrix. Proteoglycans are major components of connective tissue such as cartilage, in which their many noncovalent interactions with other proteoglycans, proteins, and glycosaminoglycans provide strength and resilience.

Glycoproteins have one or several oligosaccharides of varying complexity joined covalently to a protein. They are found on the outer face of the plasma membrane, in the extracellular matrix, and in the blood. Inside cells they are found in specific organelles such as Golgi complexes, secretory granules, and lysosomes. The oligosaccharide portions of glycoproteins are less monotonous than the glycosaminoglycan chains of proteoglycans; they are rich in information, forming highly specific sites for recognition and high-affinity binding by other proteins.

Glycolipids are membrane lipids in which the hydrophilic head groups are oligosaccharides, which, as in glycoproteins, act as specific sites for recognition by carbohydrate-binding proteins.

Proteoglycans Are Glycosaminoglycan-Containing Macromolecules of the Cell Surface and Extracellular Matrix

Mammalian cells can produce at least 30 types of molecules that are members of the proteoglycan superfamily. These molecules act as tissue organizers, influence the development of specialized tissues, mediate the activities of various growth factors, and regulate the extracellular assembly of collagen fibrils. The basic proteoglycan unit consists of a “core protein” with covalently attached glycosaminoglycan(s). For example, the sheet-like extracellular matrix (basal lamina) that separates organized groups of cells contains a family of core proteins (M_r 20,000 to 40,000), each with several covalently attached heparan sulfate chains. (Heparan sulfate is structurally similar to heparin but has a lower density of sulfate esters.) The point of attachment is commonly a

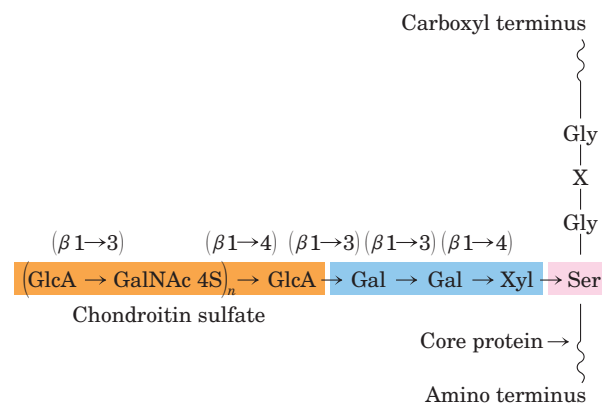


FIGURE 7-26 Proteoglycan structure, showing the trisaccharide bridge. A typical trisaccharide linker (blue) connects a glycosaminoglycan—in this case chondroitin sulfate (orange)—to a Ser residue (red) in the core protein. The xylose residue at the reducing end of the linker is joined by its anomeric carbon to the hydroxyl of the Ser residue.

Ser residue, to which the glycosaminoglycan is joined through a trisaccharide bridge (Fig. 7-26). The Ser residue is generally in the sequence –Ser–Gly–X–Gly– (where X is any amino acid residue), although not every protein with this sequence has an attached glycosaminoglycan. Many proteoglycans are secreted into the extracellular matrix, but some are integral membrane proteins (see Fig. 11-7). For example, syndecan core protein (M_r 56,000) has a single transmembrane domain and an extracellular domain bearing three chains of heparan sulfate and two of chondroitin sulfate, each attached to a Ser residue (Fig. 7-27a). There are at least four members of the syndecan family in mammals. Another family of core proteins is the glypicans, with six members. These proteins are attached to the membrane by a lipid anchor, a derivative of the membrane lipid phosphatidylinositol (Chapter 11).

The heparan sulfate moieties in proteoglycans bind a variety of extracellular ligands and thereby modulate the ligands' interaction with specific receptors of the cell surface. Detailed examination of the glycan moiety of proteoglycans has revealed a sequence heterogeneity that is not random; some domains (typically 3 to 8 disaccharide units long) differ from neighboring domains in sequence and in ability to bind to specific proteins. Heparan sulfate, for example, is initially synthesized as a long polymer (50 to 200 disaccharide units) of alternating *N*-acetylglucosamine (GlcNAc) and glucuronic acid (GlcA) residues. This simple chain is acted on by a series of enzymes that introduce alterations in specific regions. First, an *N*-deacetylase: *N*-sulfotransferase replaces some acetyl groups of GlcNAc residues with sulfates, creating clusters of *N*-sulfated glucosamine (GlcN) residues. These clusters then attract enzymes that carry out further modifications: an epimerase con-

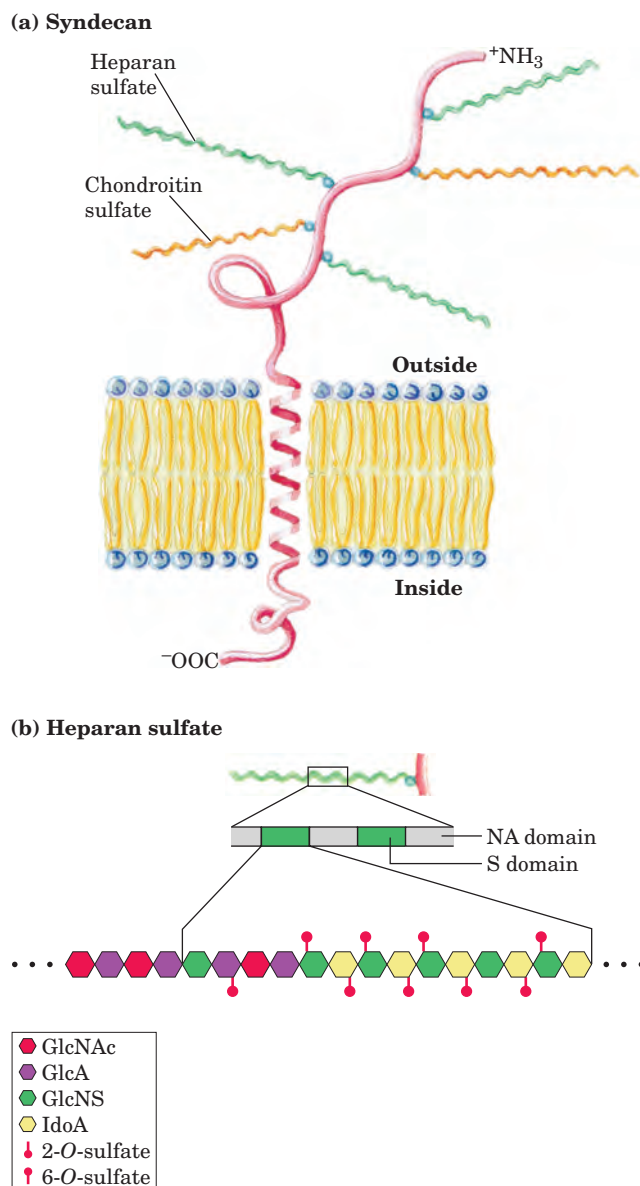
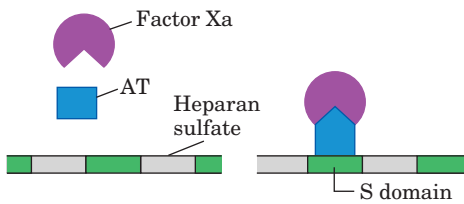


FIGURE 7-27 Proteoglycan structure of an integral membrane protein. **(a)** Schematic diagram of syndecan, a core protein of the plasma membrane. The amino-terminal domain on the extracellular surface of the membrane is covalently attached (by trisaccharide linkers such as those in Fig. 7-26) to three heparan sulfate chains and two chondroitin sulfate chains. Some core proteins (syndecans, as here) are anchored by a single transmembrane helix; others (glypicans), by a covalently attached membrane glycolipid. In a third class of core proteins, the protein is released into the extracellular space, where it forms part of the basement membrane. **(b)** Along a heparan sulfate chain, regions rich in sulfated sugars, the S domains (green), alternate with regions with chiefly unmodified residues of GlcNAc and GlcA, the NA domains (gray). One of the S domains is shown in more detail, revealing a high density of modified residues: GlcA, with a sulfate ester at C-6; and IdoA, with a sulfate ester at C-2. The exact pattern of sulfation in the S domain differs among proteoglycans. Given all the possible modifications of the GlcNAc–IdoA dimer, at least 32 different disaccharide units are possible.

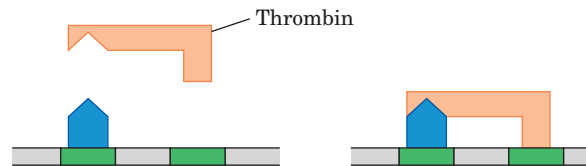
verts GlcA to IdoA; sulfotransferases then create sulfate esters at the C-2 hydroxyl of IdoA and the C-6 hydroxyl of *N*-sulfated GlcN, but only in regions that already have *N*-sulfated GlcN residues. The result is a polymer in which highly sulfated domains (S domains) alternate with domains having unmodified GlcNAc and GlcA residues (*N*-acetylated, or NA, domains) (Fig. 7-27b). The exact pattern of sulfation in the S domain differs in different proteoglycans; given the number of possible modifications of the GlcNAc–IdoA dimer, at least 32 different disaccharide units are possible. Furthermore, the same core protein can display different heparan sulfate structures when synthesized in different cell types.

The S domains bind specifically to extracellular proteins and signaling molecules to alter their activities. The change in activity may result from a conformational change in the protein that is induced by the binding (Fig. 7-28a), or it may be due to the ability of adjacent domains of heparan sulfate to bind to two different proteins, bringing them into close proximity and enhancing protein-protein interactions (Fig. 7-28b). A third general mechanism of action is the binding of extracellular signal molecules (growth factors, for example) to heparan sulfate, which increases their local concentrations and enhances their interaction with growth factor receptors in the cell surface; in this case, the heparan sulfate acts as a coreceptor (Fig. 7-28c). For example, fibroblast growth factor (FGF), an extracellular protein signal that stimulates cell division, first binds to heparan sulfate moieties of syndecan molecules in the target cell's plasma membrane. Syndecan presents FGF to the FGF plasma membrane receptor, and only then can FGF interact productively with its receptor to trigger cell division. Finally, the S domains interact—electrostatically and otherwise—with a variety of soluble molecules outside the cell, maintaining high local concentrations at the cell surface (Fig. 7-28d). The importance of correctly synthesizing sulfated domains in heparan sulfate is demonstrated in “knockout” mice that lack the enzyme that places sulfates at the C-2 hydroxyl of IdoA. Such animals are born without kidneys and with very severe abnormalities in development of the skeleton and eyes.

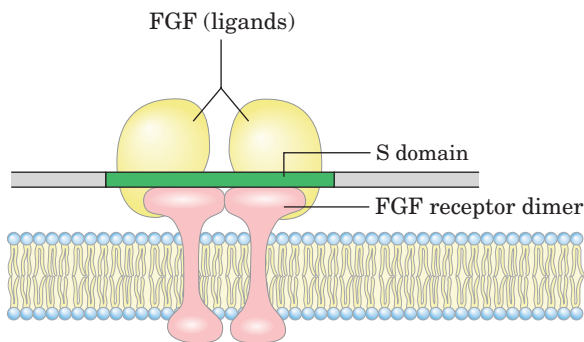
Some proteoglycans can form **proteoglycan aggregates**, enormous supramolecular assemblies of many core proteins all bound to a single molecule of hyaluronate. Aggrecan core protein ($M_r \sim 250,000$) has multiple chains of chondroitin sulfate and keratan sulfate, joined to Ser residues in the core protein through trisaccharide linkers, to give an aggrecan monomer of $M_r \sim 2 \times 10^6$. When a hundred or more of these “decorated” core proteins bind a single, extended molecule of hyaluronate (Fig. 7-29), the resulting proteoglycan aggregate ($M_r > 2 \times 10^8$) and its associated water of hydration occupy a volume about equal to that of a bacterial cell! Aggrecan interacts strongly with collagen in the

(a) Conformational activation

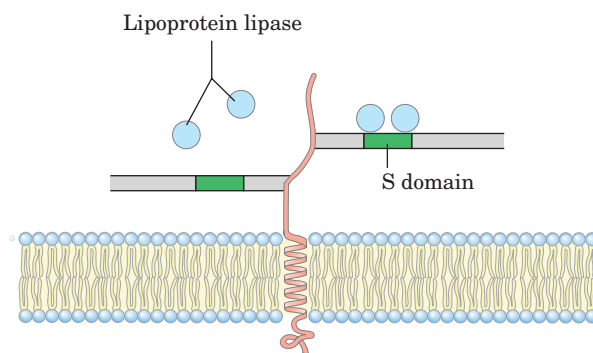
A conformational change induced in the protein antithrombin (AT) on binding a specific pentasaccharide S domain allows its interaction with Factor Xa, a blood clotting factor, preventing clotting.

(b) Enhanced protein-protein interaction

Binding of AT and thrombin to two adjacent S domains brings the two proteins into close proximity, favoring their interaction, which inhibits blood clotting.

(c) Coreceptor for extracellular ligands

S domains interact with both the fibroblast growth factor (FGF) and its receptor, bringing the oligomeric complex together and increasing the effectiveness of a low concentration of FGF.

(d) Cell surface localization/concentration

The high density of negative charges in heparan sulfate brings positively charged molecules of lipoprotein lipase into the vicinity and holds them by electrostatic interactions as well as by sequence-specific interactions with S domains. Such interactions are also central in the first step in the entry of certain viruses (such as herpes simplex viruses HSV-1 and HSV-2) into cells.

FIGURE 7-28 Four types of protein interactions with S domains of heparan sulfate.

extracellular matrix of cartilage, contributing to the development and tensile strength of this connective tissue.

Interwoven with these enormous extracellular proteoglycans are fibrous matrix proteins such as collagen, elastin, and fibronectin, forming a cross-linked meshwork that gives the whole extracellular matrix strength and resilience. Some of these proteins are multiadhesive, a single protein having binding sites for several different matrix molecules. Fibronectin, for example, has separate domains that bind fibrin, heparan sulfate, collagen, and a family of plasma membrane proteins called integrins that mediate signaling between the cell interior and the extracellular matrix (see Fig. 11-24). Integrins, in turn, have binding sites for a number of other extracellular macromolecules. The overall picture of cell-matrix interactions that emerges (Fig. 7-30) shows an array of interactions between cellular and extracellular molecules. These interactions serve not merely to

anchor cells to the extracellular matrix but also to provide paths that direct the migration of cells in developing tissue and, through integrins, to convey information in both directions across the plasma membrane.

Glycoproteins Have Covalently Attached Oligosaccharides

Glycoproteins are carbohydrate-protein conjugates in which the carbohydrate moieties are smaller and more structurally diverse than the glycosaminoglycans of proteoglycans. The carbohydrate is attached at its anomeric carbon through a glycosidic link to the —OH of a Ser or Thr residue (*O*-linked), or through an *N*-glycosyl link to the amide nitrogen of an Asn residue (*N*-linked) (Fig. 7-31). Some glycoproteins have a single oligosaccharide chain, but many have more than one; the carbohydrate may constitute from 1% to 70% or more of the glyco-

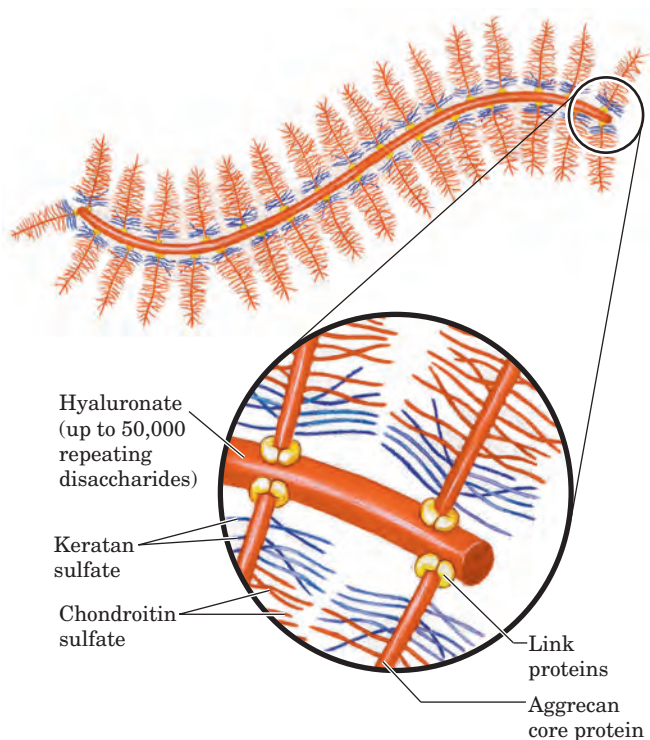


FIGURE 7-29 Proteoglycan aggregate of the extracellular matrix.

One very long molecule of hyaluronate is associated noncovalently with about 100 molecules of the core protein aggrecan. Each aggrecan molecule contains many covalently bound chondroitin sulfate and keratan sulfate chains. Link proteins situated at the junction between each core protein and the hyaluronate backbone mediate the core protein–hyaluronate interaction.

protein by mass. The structures of a large number of *O*- and *N*-linked oligosaccharides from a variety of glycoproteins are known; Figure 7-31 shows a few typical examples.

As we shall see in Chapter 11, the external surface of the plasma membrane has many membrane glycoproteins with arrays of covalently attached oligosaccharides of varying complexity. One of the best-characterized membrane glycoproteins is glycophorin A of the erythrocyte membrane (see Fig. 11-8). It contains 60% carbohydrate by mass, in the form of 16 oligosaccharide chains (totaling 60 to 70 monosaccharide residues) covalently attached to amino acid residues near the amino terminus of the polypeptide chain. Fifteen of the oligosaccharide chains are *O*-linked to Ser or Thr residues, and one is *N*-linked to an Asn residue.

Many of the proteins secreted by eukaryotic cells are glycoproteins, including most of the proteins of blood. For example, immunoglobulins (antibodies) and certain hormones, such as follicle-stimulating hormone, luteinizing hormone, and thyroid-stimulating hormone, are glycoproteins. Many milk proteins, including lactalbumin, and some of the proteins secreted by the pancreas (such as ribonuclease) are glycosylated, as are most of the proteins contained in lysosomes.

A number of cases are known in which the same protein produced in two types of tissues has different glycosylation patterns. For example, the human protein interferon IFN- β 1 has one set of oligosaccharide chains when produced in ovarian cells and a different set when produced in breast epithelial cells. The biological significance of these **tissue glycoforms** is not understood, but in some way the oligosaccharide chains represent a tissue-specific marker.

The biological advantages of adding oligosaccharides to proteins are not fully understood. The very hydrophilic clusters of carbohydrate alter the polarity and solubility of the proteins with which they are conjugated. Oligosaccharide chains that are attached to newly synthesized proteins in the endoplasmic reticulum and elaborated in the Golgi complex may also influence the sequence of polypeptide-folding events that determine the tertiary structure of the protein (see Fig. 27-34). Steric interactions between peptide and oligosaccharide may preclude one folding route and favor another. When numerous negatively charged oligosaccharide chains are clustered in a single region of a protein, the charge repulsion among them favors the formation of an extended, rodlike structure in that region. The bulkiness and negative charge of oligosaccharide chains also

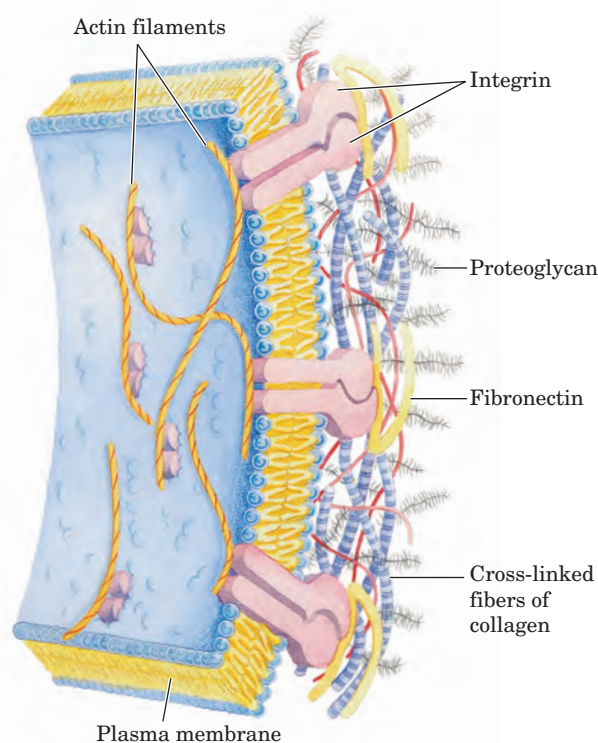


FIGURE 7-30 Interactions between cells and the extracellular matrix.

The association between cells and the proteoglycan of the extracellular matrix is mediated by a membrane protein (integrin) and by an extracellular protein (fibronectin in this example) with binding sites for both integrin and the proteoglycan. Note the close association of collagen fibers with the fibronectin and proteoglycan.

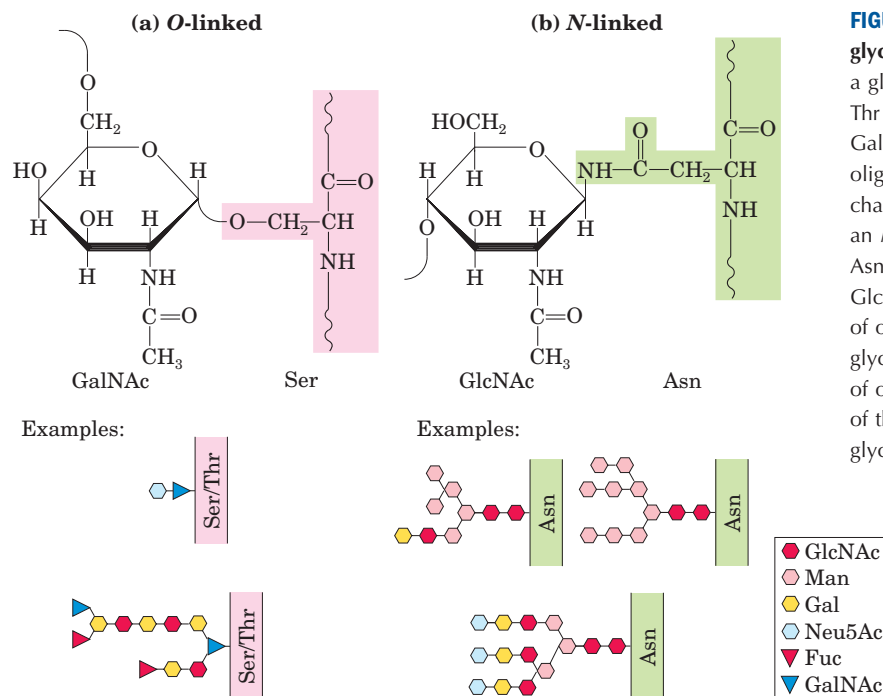


FIGURE 7-31 Oligosaccharide linkages in glycoproteins. **(a)** O-linked oligosaccharides have a glycosidic bond to the hydroxyl group of Ser or Thr residues (shaded pink), illustrated here with GalNAc as the sugar at the reducing end of the oligosaccharide. One simple chain and one complex chain are shown. **(b)** N-linked oligosaccharides have an N-glycosyl bond to the amide nitrogen of an Asn residue (shaded green), illustrated here with GlcNAc as the terminal sugar. Three common types of oligosaccharide chains that are N-linked in glycoproteins are shown. A complete description of oligosaccharide structure requires specification of the position and stereochemistry (α or β) of each glycosidic linkage.

protect some proteins from attack by proteolytic enzymes. Beyond these global physical effects on protein structure, there are also more specific biological effects of oligosaccharide chains in glycoproteins (Section 7.4).

Glycolipids and Lipopolysaccharides Are Membrane Components

Glycoproteins are not the only cellular components that bear complex oligosaccharide chains; some lipids, too, have covalently bound oligosaccharides. **Gangliosides** are membrane lipids of eukaryotic cells in which the polar head group, the part of the lipid that forms the outer surface of the membrane, is a complex oligosaccharide containing sialic acid (Fig. 7-9) and other monosaccharide residues. Some of the oligosaccharide moieties of gangliosides, such as those that determine human blood groups (see Fig. 10-14), are identical with those found in certain glycoproteins, which therefore also contribute to blood group type determination. Like the oligosaccharide moieties of glycoproteins, those of membrane lipids are generally, perhaps always, found on the outer face of the plasma membrane.

Lipopolysaccharides are the dominant surface feature of the outer membrane of gram-negative bacteria such as *Escherichia coli* and *Salmonella typhimurium*. These molecules are prime targets of the antibodies produced by the vertebrate immune system in response to bacterial infection and are therefore important determinants of the serotype of bacterial strains (serotypes are strains that are distinguished on the basis of antigenic properties). The lipopolysaccharides of *S. typhimurium* contain six fatty acids bound to two

glucosamine residues, one of which is the point of attachment for a complex oligosaccharide (Fig. 7-32). *E. coli* has similar but unique lipopolysaccharides. The lipopolysaccharides of some bacteria are toxic to humans and other animals; for example, they are responsible for the dangerously lowered blood pressure that occurs in toxic shock syndrome resulting from gram-negative bacterial infections. ■

SUMMARY 7.3 Glycoconjugates: Proteoglycans, Glycoproteins, and Glycolipids

- Proteoglycans are glycoconjugates in which a core protein is attached covalently to one or more large glycans, such as heparan sulfate, chondroitin sulfate, or keratan sulfate. The glycan is the greater portion (by mass) of the molecule. Bound to the outside of the plasma membrane by a transmembrane peptide or a covalently attached lipid, proteoglycans provide points of adhesion, recognition, and information transfer between cells, or between the cell and the extracellular matrix.
- Glycoproteins contain covalently linked oligosaccharides that are smaller but more structurally complex, and therefore more information-rich, than glycosaminoglycans. Many cell surface or extracellular proteins are glycoproteins, as are most secreted proteins. The covalently attached oligosaccharides influence the folding and stability of the proteins, provide critical information about the

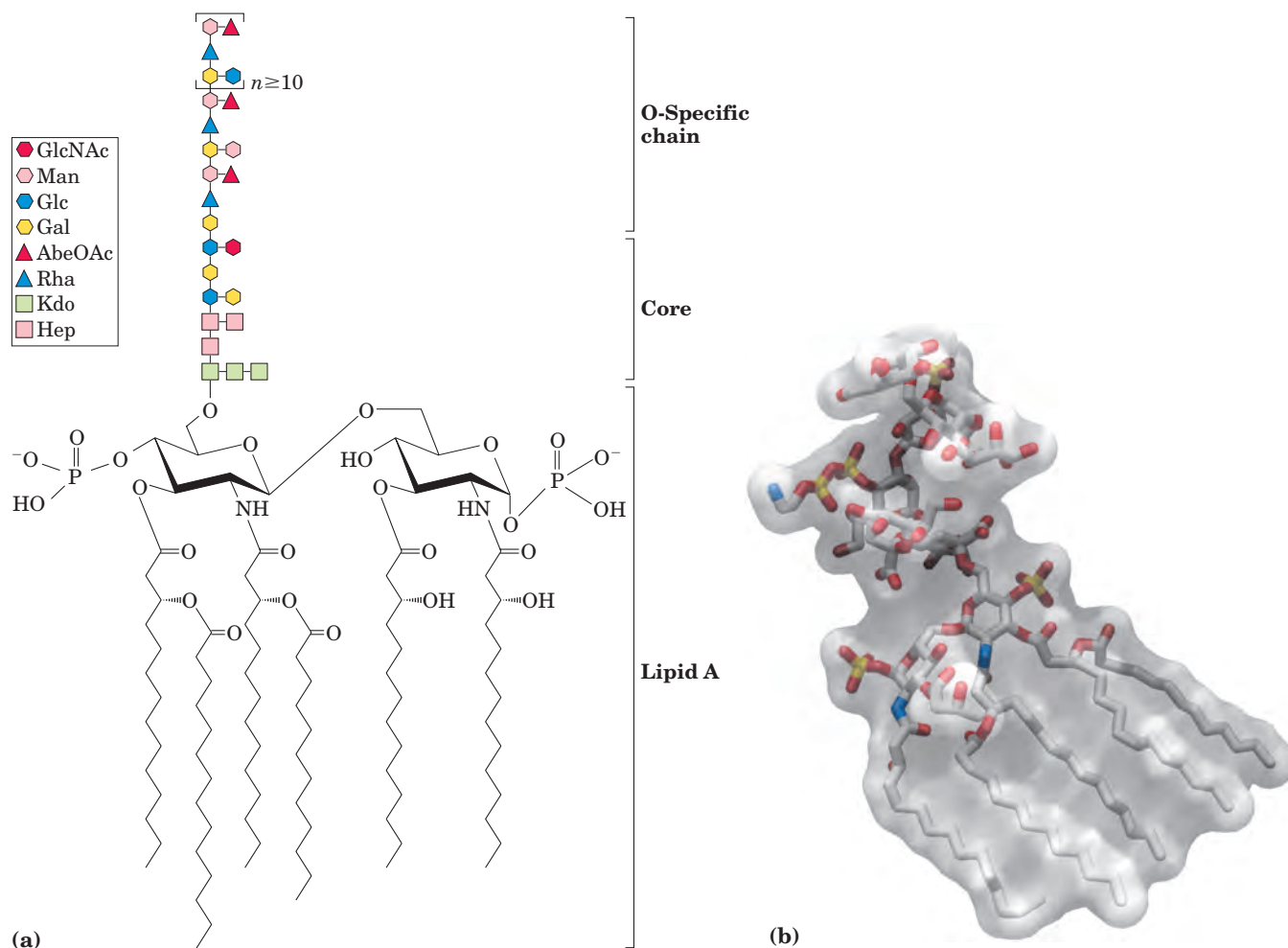


FIGURE 7-32 Bacterial lipopolysaccharides. (a) Schematic diagram of the lipopolysaccharide of the outer membrane of *Salmonella typhimurium*. Kdo is 3-deoxy-D-manno-octulosonic acid, previously called ketodeoxyoctonic acid; Hep is L-glycero-D-mannoheptose; AbeOAc is abequose (a 3,6-dideoxyhexose) acetylated on one of its hydroxyls. There are six fatty acids in the lipid A portion of the molecule. Different bacterial species have subtly different lipopolysaccharide structures, but they have in common a lipid region (lipid A), a core oligosaccharide, and an “O-specific” chain, which is the prin-

cipal determinant of the serotype (immunological reactivity) of the bacterium. The outer membranes of the gram-negative bacteria *S. typhimurium* and *E. coli* contain so many lipopolysaccharide molecules that the cell surface is virtually covered with O-specific chains. (b) The stick structure of the lipopolysaccharide of *E. coli* is visible through a transparent surface contour model of the molecule. The position of the sixth fatty acyl chain was not defined in the crystallographic study, so it is not shown.

targeting of newly synthesized proteins, and allow for specific recognition by other proteins.

- Glycolipids and lipopolysaccharides are components of the plasma membrane with covalently attached oligosaccharide chains exposed on the cell's outer surface.

7.4 Carbohydrates as Informational Molecules: The Sugar Code

Glycobiology, the study of the structure and function of glycoconjugates, is one of the most active and exciting areas of biochemistry and cell biology. As is becoming

increasingly clear, cells use specific oligosaccharides to encode important information about intracellular targeting of proteins, cell-cell interaction, tissue development, and extracellular signals. Our discussion uses just a few examples to illustrate the diversity of structure and the range of biological activity of the glycoconjugates. In Chapter 20 we discuss the biosynthesis of polysaccharides, including the peptidoglycans; and in Chapter 27, the assembly of oligosaccharide chains on glycoproteins.

Improved methods for the analysis of oligosaccharide and polysaccharide structure have revealed remarkable complexity and diversity in the oligosaccharides of glycoproteins and glycolipids. Consider the oligosaccharide chains in Figure 7-31, typical of those

found in many glycoproteins. The most complex of those shown contains 14 monosaccharide residues of four different kinds, variously linked as (1→2), (1→3), (1→4), (1→6), (2→3), and (2→6), some with the α and some with the β configuration. Branched structures, not found in nucleic acids or proteins, are common in oligosaccharides. With the reasonable assumption that 20 different monosaccharide subunits are available for construction of oligosaccharides, we can calculate that 1.44×10^{15} different hexameric oligosaccharides are possible; this compares with 6.4×10^7 (20^6) different hexapeptides possible with the 20 common amino acids, and 4,096 (4^6) different hexanucleotides with the four nucleotide subunits. If we also allow for variations in oligosaccharides resulting from sulfation of one or more residues, the number of possible oligosaccharides increases by two orders of magnitude. Oligosaccharides are enormously rich in structural information, not merely rivaling but far surpassing nucleic acids in the density of information contained in a molecule of modest size. Each of the oligosaccharides represented in Figure 7–31 presents a unique, three-dimensional face—a word in the sugar code—readable by the proteins that interact with it.

Lectins Are Proteins That Read the Sugar Code and Mediate Many Biological Processes

Lectins, found in all organisms, are proteins that bind carbohydrates with high affinity and specificity (Table 7–3). Lectins serve in a wide variety of cell-cell recognition, signaling, and adhesion processes and in intra-

cellular targeting of newly synthesized proteins. In the laboratory, purified lectins are useful reagents for detecting and separating glycoproteins with different oligosaccharide moieties. Here we discuss just a few examples of the roles of lectins in cells.

Some peptide hormones that circulate in the blood have oligosaccharide moieties that strongly influence their circulatory half-life. Luteinizing hormone and thyrotropin (polypeptide hormones produced in the adrenal cortex) have *N*-linked oligosaccharides that end with the disaccharide GalNAc4S(β 1→4)GlcNAc, which is recognized by a lectin (receptor) of hepatocytes. (GalNAc4S is *N*-acetylgalactosamine sulfated on the —OH group of C-4.) Receptor-hormone interaction mediates the uptake and destruction of luteinizing hormone and thyrotropin, reducing their concentration in the blood. Thus the blood levels of these hormones undergo a periodic rise (due to secretion by the adrenal cortex) and fall (due to destruction by hepatocytes).



The importance of the oligosaccharide moiety of these hormones is apparent from studies of individuals with a defective enzyme in the pathway that produces this oligosaccharide. Females with this congenital defect often fail to undergo the sexual changes of puberty (although males with the same defect develop normally). ■

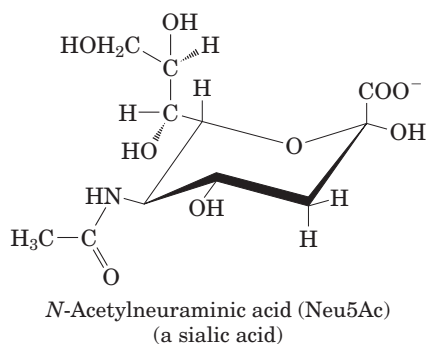
The residues of Neu5Ac (a sialic acid) situated at the ends of the oligosaccharide chains of many plasma glycoproteins (Fig. 7–31) protect those proteins from uptake and degradation in the liver. For example, ceruloplasmin, a copper-containing serum glycoprotein, has several oligosaccharide chains ending in Neu5Ac. Re-

TABLE 7–3 Some Lectins and the Oligosaccharide Ligands They Bind

Lectin source and lectin	Abbreviation	Ligand(s)
Plant		
Concanavalin A	ConA	Man α 1—OCH ₃
<i>Griffonia simplicifolia</i> lectin 4	GS4	Lewis b (Le ^b) tetrasaccharide
Wheat germ agglutinin	WGA	Neu5Ac(α 2→3)Gal(β 1→4)Glc GlcNAc(β 1→4)GlcNAc
Ricin		Gal(β 1→4)Glc
Animal		
Galectin-1		Gal(β 1→4)Glc
Mannose-binding protein A	MBP-A	High-mannose octasaccharide
Viral		
Influenza virus hemagglutinin	HA	Neu5Ac(α 2→6)Gal(β 1→4)Glc
Polyoma virus protein 1	VP1	Neu5Ac(α 2→3)Gal(β 1→4)Glc
Bacterial		
Enterotoxin	LT	Gal
Cholera toxin	CT	GM1 pentasaccharide

Source: Weiss, W.I. & Drickamer, K. (1996) Structural basis of lectin-carbohydrate recognition. *Annu. Rev. Biochem.* **65**, 441–473.

removal of the sialic acid residues by the enzyme sialidase (also called neuraminidase) is one way in which the body marks “old” proteins for destruction and replacement. The plasma membrane of hepatocytes has lectin molecules (asialoglycoprotein receptors; “asialo-” indicating “without sialic acid”) that specifically bind oligosaccharide chains with galactose residues no longer “protected” by a terminal Neu5Ac residue. Receptor-ceruloplasmin interaction triggers endocytosis and destruction of the ceruloplasmin.



A similar mechanism is apparently responsible for removing old erythrocytes from the mammalian bloodstream. Newly synthesized erythrocytes have several membrane glycoproteins with oligosaccharide chains that end in Neu5Ac. When the sialic acid residues are removed by withdrawing a sample of blood, treating it with sialidase *in vitro*, and reintroducing it into the circulation, the treated erythrocytes disappear from the bloodstream within a few hours; those with intact oligosaccharides (erythrocytes withdrawn and reintroduced without sialidase treatment) continue to circulate for days.

Several animal viruses, including the influenza virus, attach to their host cells through interactions with oligosaccharides displayed on the host cell surface. The lectin of the influenza virus, the HA protein, is essential for viral entry and infection (see Fig. 11–25). After initial binding of the virus to a sialic acid-containing oligosaccharide on the host surface, a viral sialidase removes the terminal sialic acid residue, triggering the entry of the virus into the cell. Inhibitors of this enzyme are used clinically in the treatment of influenza. Lectins on the surface of the herpes simplex viruses HS-1 and HS-2 (the causative agents of oral and genital herpes, respectively) bind specifically to heparan sulfate on the cell surface as a first step in their infection cycle; infection requires precisely the right pattern of sulfation on this polymer.

Selectins are a family of plasma membrane lectins that mediate cell-cell recognition and adhesion in a wide range of cellular processes. One such process is the movement of immune cells (T lymphocytes) through the capillary wall, from blood to tissues, at sites of infection or inflammation (Fig. 7–33). At an infection site, P-selectin on the surface of capillary endothelial cells interacts with a specific oligosaccharide of the glycoproteins of circu-

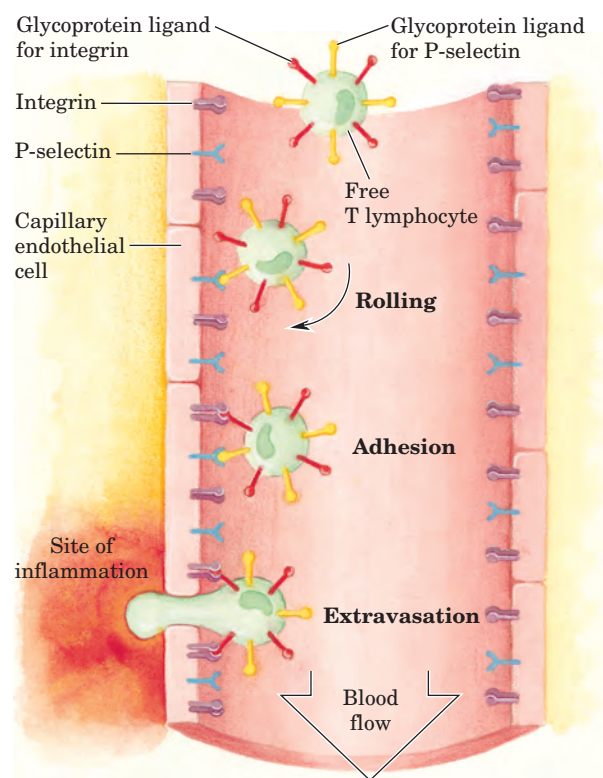


FIGURE 7–33 Role of lectin-ligand interactions in lymphocyte movement to the site of an infection or injury. A T lymphocyte circulating through a capillary is slowed by transient interactions between P-selectin molecules in the plasma membrane of the capillary endothelial cells and glycoprotein ligands for P-selectin on the T-cell surface. As it interacts with successive P-selectin molecules, the T cell rolls along the capillary surface. Near a site of inflammation, stronger interactions between integrin in the capillary surface and its ligand in the T-cell surface lead to tight adhesion. The T cell stops rolling and, under the influence of signals sent out from the site of inflammation, begins extravasation—escape through the capillary wall—as it moves toward the site of inflammation.

lating T cells. This interaction slows the T cells as they adhere to and roll along the endothelial lining of the capillaries. A second interaction, between integrin molecules (see p. XXX) in the T-cell plasma membrane and an adhesion protein on the endothelial cell surface, now stops the T cell and allows it to move through the capillary wall into the infected tissues to initiate the immune attack. Two other selectins participate in this “lymphocyte homing”: E-selectin on the endothelial cell and L-selectin on the T cell bind their cognate oligosaccharides on the T cell and endothelial cell, respectively.



Some microbial pathogens have lectins that mediate bacterial adhesion to host cells or toxin entry into cells. The bacterium believed responsible for most gastric ulcers, *Helicobacter pylori*, adheres to the inner surface of the stomach by interactions between bacterial membrane lectins and specific oligosaccharides of membrane glycoproteins of the gastric epithelial cells

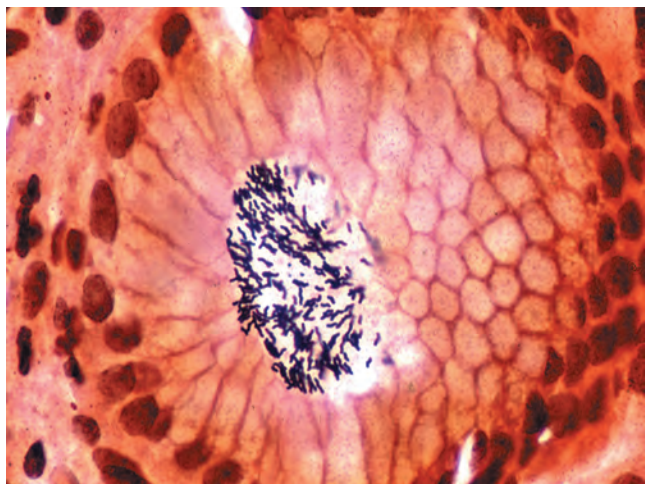


FIGURE 7-34 An ulcer in the making. *Helicobacter pylori* cells adhering to the gastric surface. This bacterium causes ulcers by interactions between a bacterial surface lectin and the Le^b oligosaccharide (a blood group antigen) of the gastric epithelium.

(Fig. 7-34). Among the binding sites recognized by *H. pylori* is the oligosaccharide Le^b when it is part of the type O blood group determinant. This observation helps to explain the severalfold greater incidence of gastric ulcers in people of blood type O than in those of type A or B. Chemically synthesized analogs of the Le^b oligosaccharide may prove useful in treating this type of ulcer. Administered orally, they could prevent bacterial adhesion (and thus infection) by competing with the gastric glycoproteins for binding to the bacterial lectin.

The cholera toxin molecule (produced by *Vibrio cholerae*) triggers diarrhea after entering intestinal cells responsible for water absorption from the intestine. The toxin attaches to its target cell through the oligosaccharide of ganglioside GM1, a membrane phospholipid (for the structure of GM1 see Box 10-2, Fig. 1), on the surface of intestinal epithelial cells. Similarly, the pertussis toxin produced by *Bordetella pertussis*, the bacterium that causes whooping cough, enters target cells only after interacting with an oligosaccharide (or perhaps several oligosaccharides) with a terminal sialic acid residue. Understanding the details of the oligosaccharide-binding sites of these toxins (lectins) may allow the development of genetically engineered toxin analogs for use in vaccines. Toxin analogs engineered to lack the carbohydrate binding site would be harmless because they could not bind to and enter cells, but they might elicit an immune response that would protect the recipient if later exposed to the natural toxin. It is also possible to imagine drugs that would act by mimicking the oligosaccharides of the cell surface, binding to the lectins of bacteria or toxins and preventing their productive binding to cell surfaces. ■

Lectins also act intracellularly. An oligosaccharide containing mannose 6-phosphate marks newly synthe-

sized proteins in the Golgi complex for transfer to the lysosome (see Fig. 27-36). A common structural feature on the surface of these glycoproteins, the signal patch, causes them to be recognized by an enzyme that phosphorylates a mannose residue at the terminus of an oligosaccharide chain. This mannose phosphate residue is recognized by the cation-dependent mannose 6-phosphate receptor, a membrane-associated lectin with its mannose phosphate binding site on the luminal side of the Golgi complex. When a section of the Golgi complex containing this receptor buds off to form a transport vesicle, proteins containing mannose phosphate residues are dragged into the forming bud by interaction of their mannose phosphates with the receptor; the vesicle then moves to and fuses with a lysosome, depositing its cargo therein. Many, perhaps all, of the degradative enzymes (hydrolases) of the lysosome are targeted and delivered by this mechanism.

Lectin-Carbohydrate Interactions Are Very Strong and Highly Specific

In all the functions of lectins described above, and in many more known to involve lectin-oligosaccharide interactions, it is essential that the oligosaccharide have a unique structure, so that recognition by the lectin is highly specific. The high density of information in oligosaccharides provides a sugar code with an essentially unlimited number of unique “words” small enough to be read by a single protein. In their carbohydrate-binding sites, lectins have a subtle molecular complementarity that allows interaction only with their correct carbohydrate cognates. The result is extraordinarily high specificity in these interactions.

X-ray crystallographic studies of the structures of several lectin-carbohydrate complexes have provided rich details of the lectin-sugar interaction. Sialoadhesin (also called siglec-1) is a membrane-bound lectin on the surface of mouse macrophages that recognizes certain sialic acid-containing oligosaccharides. This protein has a β sandwich domain (see this motif in the CD8 protein in Fig. 4-22) that contains the sialic acid binding site (Fig. 7-35a). Each of the ring substituents unique to Neu5Ac is involved in the interaction between sugar and lectin; the acetyl group at C-5 undergoes both hydrogen-bond and van der Waals interactions with the protein; the carboxyl group makes a salt bridge with Arg⁹⁷; and the hydroxyls of the glycerol moiety hydrogen-bond with the protein (Fig. 7-35b).

The structure of the mannose 6-phosphate receptor/lectin has also been resolved crystallographically, revealing details of its interaction with mannose 6-phosphate that explain the specificity of the binding and the necessity for a divalent cation in the lectin-sugar interaction (Fig. 7-35c). Arg¹¹¹ of the receptor is hydrogen-bonded to the C-2 hydroxyl of mannose and coordinated with Mn²⁺. His¹⁰⁵ is hydrogen-bonded to

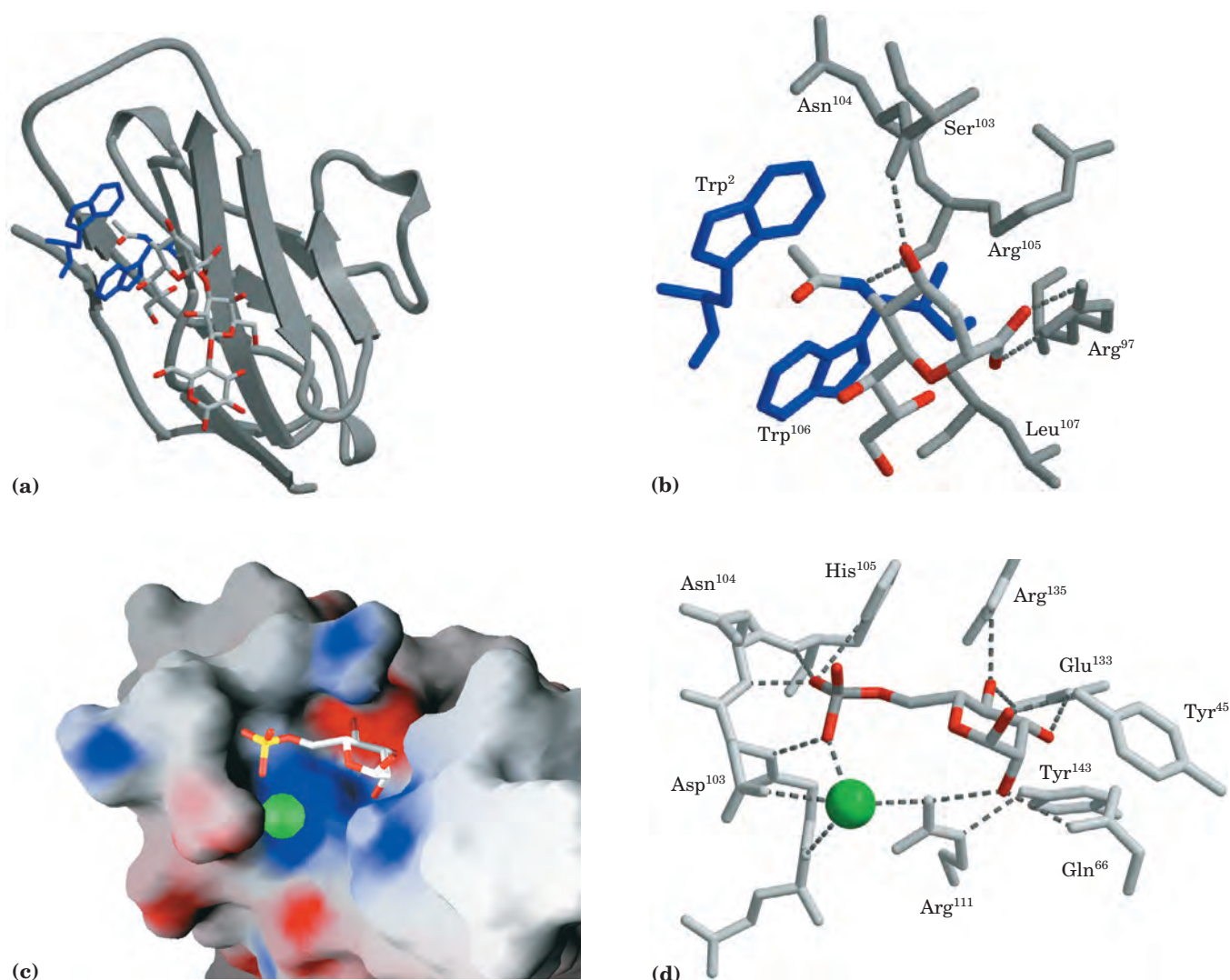


FIGURE 7-35 Details of lectin-carbohydrate interaction. (a) X-ray crystallographic studies of a sialic acid-specific lectin (derived from PDB ID 1QFO) show how a protein can recognize and bind to a sialic acid (Neu5Ac) residue. Sialoadhesin (also called siglec-1), a membrane-bound lectin of the surface of mouse macrophages, has a β sandwich domain (gray) that contains the Neu5Ac binding site (dark blue). Neu5Ac is shown as a stick structure. (b) Each ring substituent unique to Neu5Ac is involved in the interaction between sugar and lectin: the acetyl group at C-5 has both hydrogen-bond and van der Waals interactions with the protein; the carboxyl group makes a salt bridge with Arg⁹⁷; and the hydroxyls of the glycerol moiety hydrogen-bond with the

protein. (c) Structure of the bovine mannose 6-phosphate receptor complexed with mannose 6-phosphate (PDB ID 1M6P). The protein is represented here as a surface contour image, with color to indicate the surface electrostatic potential: red, predominantly negative charge; blue, predominantly positive charge. Mannose 6-phosphate is shown as a stick structure; a manganese ion is shown in green. (d) In this complex, mannose 6-phosphate is hydrogen-bonded to Arg¹¹¹ and coordinated with the manganese ion (green). The His¹⁰⁵ hydrogen-bonded to a phosphate oxygen of mannose 6-phosphate may be the residue that, when protonated at low pH, causes the receptor to release mannose 6-phosphate into the lysosome.

one of the oxygen atoms of the phosphate (Fig. 7-35d). When the protein tagged with mannose 6-phosphate reaches the lysosome (which has a lower internal pH than the Golgi complex), the receptor apparently loses its affinity for mannose 6-phosphate. Protonation of His¹⁰⁵ may be responsible for this change in binding.

In addition to these very specific interactions, there are more general interactions that contribute to the binding of many carbohydrates to their lectins. For example, many sugars have a more polar and a less polar

side (Fig. 7-36); the more polar side hydrogen-bonds with the lectin, while the less polar undergoes hydrophobic interactions with nonpolar amino acid residues. The sum of all these interactions produces high-affinity binding (K_d often 10^{-8} M or less) and high specificity of lectins for their carbohydrates. This represents a kind of information transfer that is clearly central in many processes within and between cells. Figure 7-37 summarizes some of the biological interactions mediated by the sugar code.

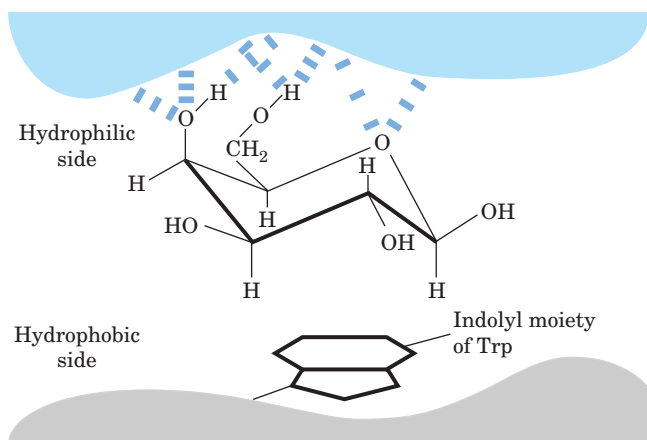


FIGURE 7-36 Hydrophobic interactions of sugar residues. Sugar units such as galactose have a more polar side (the top of the chair, with the ring oxygen and several hydroxyls), available to hydrogen-bond with the lectin, and a less polar side that can have hydrophobic interactions with nonpolar side chains in the protein, such as the indole ring of tryptophan.

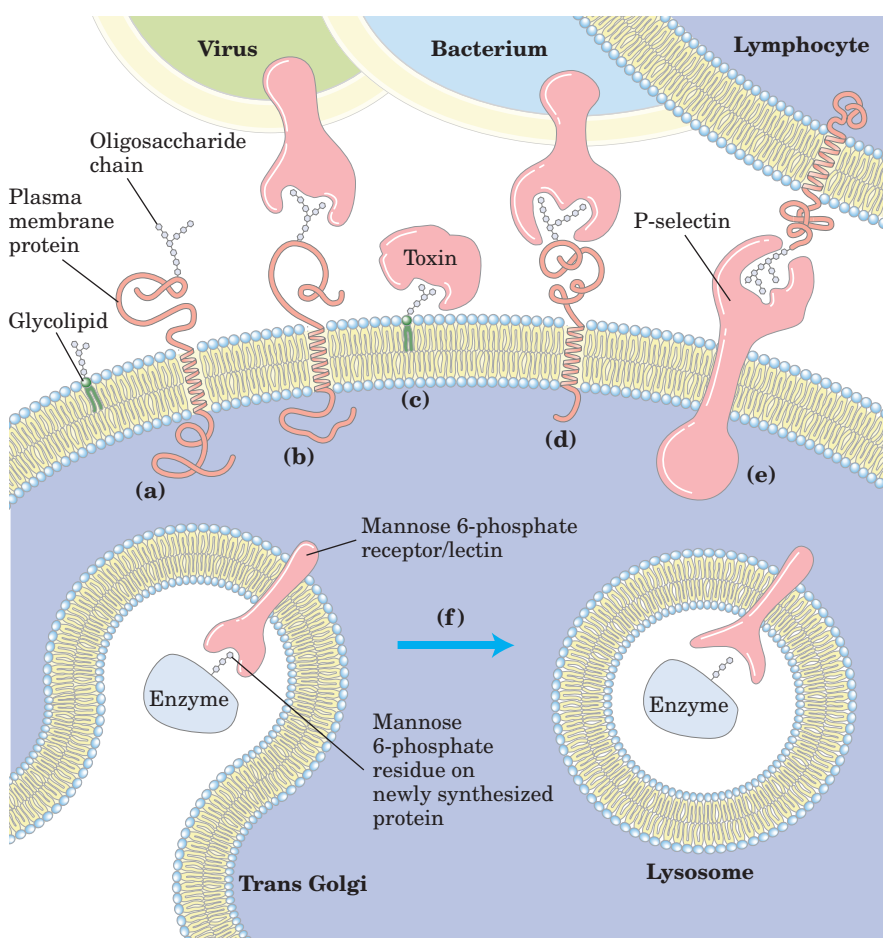


FIGURE 7-37 Roles of oligosaccharides in recognition and adhesion at the cell surface.

(a) Oligosaccharides with unique structures (represented as strings of hexagons), components of a variety of glycoproteins or glycolipids on the outer surface of plasma membranes, interact with high specificity and affinity with lectins in the extracellular milieu. (b) Viruses that infect animal cells, such as the influenza virus, bind to cell surface glycoproteins as the first step in infection. (c) Bacterial toxins, such as the cholera and pertussis toxins, bind to a surface glycolipid before entering a cell. (d) Some bacteria, such as *H. pylori*, adhere to and then colonize or infect animal cells. (e) Selectins (lectins) in the plasma membrane of certain cells mediate cell-cell interactions, such as those of T lymphocytes with the endothelial cells of the capillary wall at an infection site. (f) The mannose 6-phosphate receptor/lectin of the trans Golgi complex binds to the oligosaccharide of lysosomal enzymes, targeting them for transfer into the lysosome.

SUMMARY 7.4 Carbohydrates as Informational Molecules: The Sugar Code

- Monosaccharides can be assembled into an almost limitless variety of oligosaccharides, which differ in the stereochemistry and position of glycosidic bonds, the type and orientation of substituent groups, and the number and type of branches. Oligosaccharides are far more information-dense than nucleic acids or proteins.

- Lectins, proteins with highly specific carbohydrate-binding domains, are commonly found on the outer surface of cells, where they initiate interaction with other cells. In vertebrates, oligosaccharide tags “read” by lectins govern the rate of degradation of certain peptide hormones, circulating proteins, and blood cells.
- The adhesion of bacterial and viral pathogens to their animal-cell targets occurs through binding of lectins in the pathogens to

oligosaccharides in the target cell surface. Lectins are also present inside cells, where they mediate intracellular protein targeting.

- X-ray crystallography of lectin-sugar complexes shows the detailed complementarity between the two molecules, which accounts for the strength and specificity of their interactions with carbohydrates.
- Selectins are plasma membrane lectins that bind carbohydrate chains in the extracellular matrix or on the surfaces of other cells, thereby mediating the flow of information between cell and matrix or between cells.

7.5 Working with Carbohydrates

The growing appreciation of the importance of oligosaccharide structure in biological recognition has been the driving force behind the development of methods for analyzing the structure and stereochemistry of complex oligosaccharides. Oligosaccharide analysis is complicated by the fact that, unlike nucleic acids and proteins, oligosaccharides can be branched and are joined by a variety of linkages. Oligosaccharides are generally removed from their protein or lipid conjugates before analysis, then subjected to stepwise degradation with specific reagents that reveal bond position or stereochemistry. Mass spectrometry and NMR spectroscopy have also become invaluable in deciphering oligosaccharide structure.

The oligosaccharide moieties of glycoproteins or glycolipids can be released by purified enzymes—glycosidases that specifically cleave *O*- or *N*-linked oligosaccharides or lipases that remove lipid head groups. Mixtures of carbohydrates are resolved into their individual components (Fig. 7–38) by some of the same techniques useful in protein and amino acid separation: fractional precipitation by solvents, and ion-exchange and size-exclusion chromatography (see Fig. 3–18). Highly purified lectins, attached covalently to an insoluble support, are commonly used in affinity chromatography of carbohydrates (see Fig. 3–18c). Hydrolysis of oligosaccharides and polysaccharides in strong acid yields a mixture of monosaccharides, which, after conversion to suitable volatile derivatives, may be separated, identified, and quantified by gas-liquid chromatography (p. XXX) to yield the overall composition of the polymer.

For simple, linear polymers such as amylose, the positions of the glycosidic bonds are determined by treating the intact polysaccharide with methyl iodide in a strongly basic medium to convert all free hydroxyls to acid-stable methyl ethers, then hydrolyzing the methylated polysaccharide in acid. The only free hydroxyls present in the monosaccharide derivatives so produced are those that were involved in glycosidic bonds. To de-

termine the sequence of monosaccharide residues, including branches if they are present, exoglycosidases of known specificity are used to remove residues one at a time from the nonreducing end(s). The specificity of these exoglycosidases often allows deduction of the position and stereochemistry of the linkages. Polysaccharides and large oligosaccharides can be treated chemically or with endoglycosidases to split specific internal glycosidic bonds, producing several smaller, more easily analyzable oligosaccharides.

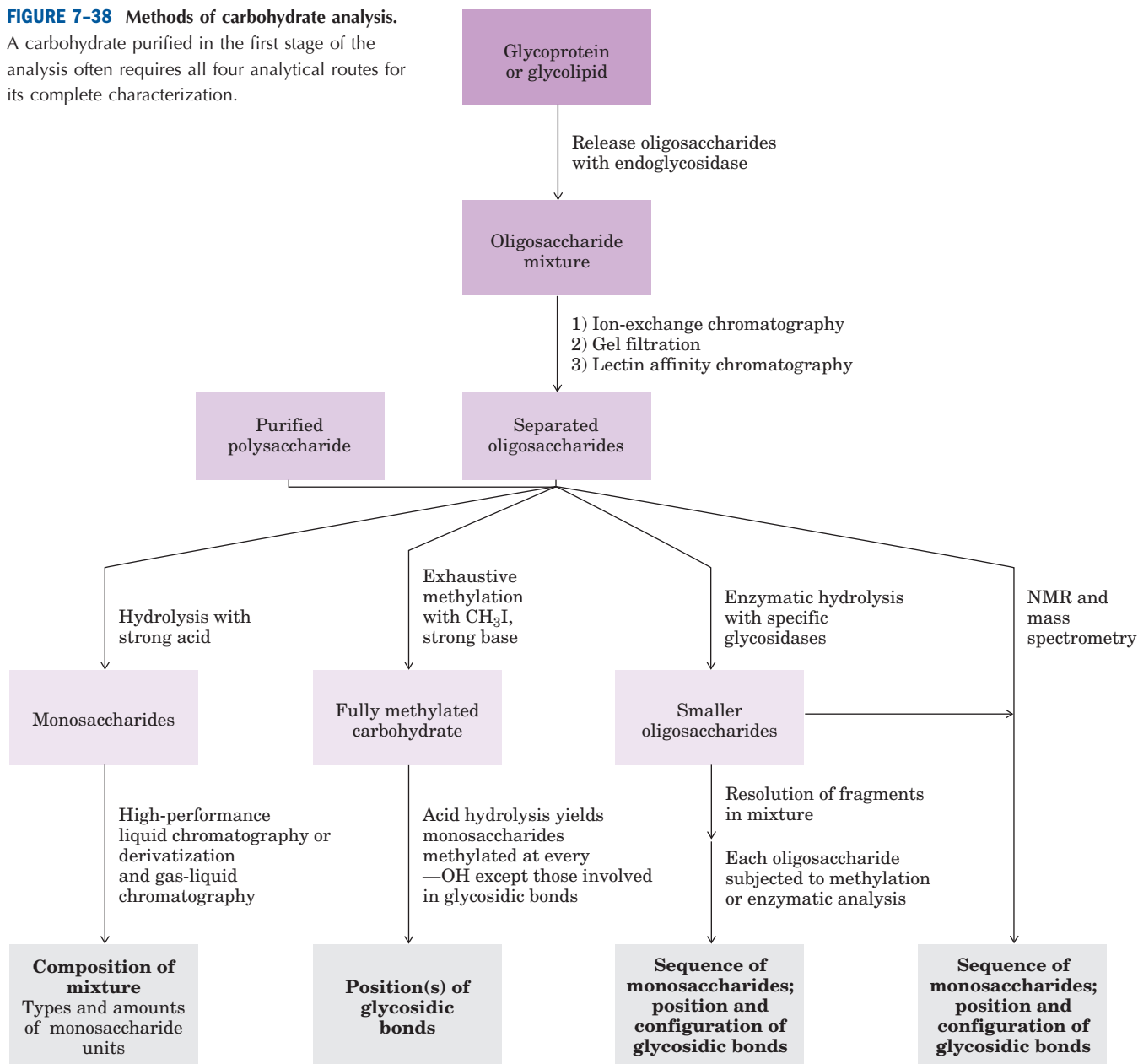
Oligosaccharide analysis relies increasingly on mass spectrometry and high-resolution NMR spectroscopy (see Box 4–4). Matrix-assisted laser desorption/ionization mass spectrometry (MALDI MS) and tandem mass spectrometry (MS/MS) (described in Box 3–2), are readily applicable to polar compounds like oligosaccharides. MALDI MS is a very sensitive method for determining the mass of the molecular ion (the entire oligosaccharide chain). Tandem MS reveals the mass of the molecular ion and many of its fragments, which are usually the result of breakage of the glycosidic bonds. A comparison of the masses of each fragment therefore gives information about the sequence of monosaccharide units. NMR analysis alone, especially for oligosaccharides of moderate size, can yield much information about sequence, linkage position, and anomeric carbon configuration. Automated procedures and commercial instruments are used for the routine determination of oligosaccharide structure, but the sequencing of branched oligosaccharides joined by more than one type of bond remains a far more formidable task than determining the linear sequences of proteins and nucleic acids, with monomers joined by a single bond type.

SUMMARY 7.5 Working with Carbohydrates

- Establishing the complete structure of oligosaccharides and polysaccharides requires determination of branching positions, the sequence in each branch, the configuration of each monosaccharide unit, and the positions of the glycosidic links—a more complex problem than protein and nucleic acid analysis.
- The structures of oligosaccharides and polysaccharides are usually determined by a combination of methods: specific enzymatic hydrolysis to determine stereochemistry and produce smaller fragments for further analysis; methylation analysis to locate glycosidic bonds; and stepwise degradation to determine sequence and configuration of anomeric carbons.
- Mass spectrometry and high-resolution NMR spectroscopy, applicable to small samples of carbohydrate, yield essential information about sequence, configuration at anomeric and other carbons, and positions of glycosidic bonds.

FIGURE 7-38 Methods of carbohydrate analysis.

A carbohydrate purified in the first stage of the analysis often requires all four analytical routes for its complete characterization.



Key Terms

Terms in bold are defined in the glossary.

glycoconjugate	238	hemiacetal	242	glycosidic bonds	245	glycoprotein	256
monosaccharide	238	hemiketal	242	reducing end	245	glycolipid	256
oligosaccharide	238	pyranose	242	glycan	247	lectin	262
disaccharide	238	furanose	242	starch	248	selectins	263
polysaccharide	238	anomers	242	glycogen	248		
aldose	239	anomeric carbon	242	extracellular matrix	253		
ketose	239	mutarotation	242	glycosaminoglycan	253		
Fischer projection		Haworth perspective					
formulas	240	formulas	242	hyaluronic acid	254		
epimers	240	reducing sugar	244	proteoglycan	256		

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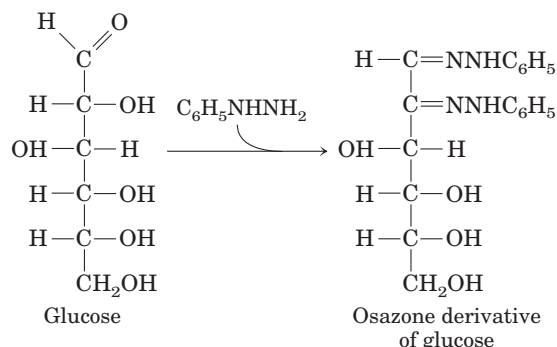
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Problems

1. Determination of an Empirical Formula An unknown substance containing only C, H, and O was isolated from goose liver. A 0.423 g sample produced 0.620 g of CO₂ and 0.254 g of H₂O after complete combustion in excess oxygen. Is the empirical formula of this substance consistent with its being a carbohydrate? Explain.

2. Sugar Alcohols In the monosaccharide derivatives known as sugar alcohols, the carbonyl oxygen is reduced to a hydroxyl group. For example, D-glyceraldehyde can be reduced to glycerol. However, this sugar alcohol is no longer designated D or L. Why?

3. Melting Points of Monosaccharide Osazone Derivatives Many carbohydrates react with phenylhydrazine ($\text{C}_6\text{H}_5\text{NHNH}_2$) to form bright yellow crystalline derivatives known as osazones:



The melting temperatures of these derivatives are easily determined and are characteristic for each osazone. This information was used to help identify monosaccharides before the development of HPLC or gas-liquid chromatography. Listed below are the melting points (MPs) of some aldose-osazone derivatives:

Monosaccharide	MP of anhydrous monosaccharide ($^{\circ}\text{C}$)	MP of osazone derivative ($^{\circ}\text{C}$)
Glucose	146	205
Mannose	132	205
Galactose	165–168	201
Talose	128–130	201

As the table shows, certain pairs of derivatives have the same melting points, although the underivatized monosaccharides do not. Why do glucose and mannose, and galactose and talose, form osazone derivatives with the same melting points?

4. Interconversion of D-Glucose Forms A solution of one stereoisomer of a given monosaccharide rotates plane-polarized light to the left (counterclockwise) and is called the levorotatory isomer, designated (–); the other stereoisomer rotates plane-polarized light to the same extent but to the right (clockwise) and is called the dextrorotatory isomer, designated (+). An equimolar mixture of the (+) and (–) forms does not rotate plane-polarized light.

The optical activity of a stereoisomer is expressed quantitatively by its *optical rotation*, the number of degrees by which plane-polarized light is rotated on passage through a given path length of a solution of the compound at a given concentration. The *specific rotation* $[\alpha]_{\text{D}}^{25^{\circ}\text{C}}$ of an optically active compound is defined thus:

$$[\alpha]_{\text{D}}^{25^{\circ}\text{C}} = \frac{\text{observed optical rotation } (^{\circ})}{\text{optical path length (dm)} \times \text{concentration (g/mL)}}$$

The temperature and the wavelength of the light employed (usually the D line of sodium, 589 nm) must be specified in the definition.

A freshly prepared solution of α -D-glucose shows a specific rotation of $+112^{\circ}$. Over time, the rotation of the solution gradually decreases and reaches an equilibrium value corresponding to $[\alpha]_{\text{D}}^{25^{\circ}\text{C}} = +52.5^{\circ}$. In contrast, a freshly pre-

pared solution of β -D-glucose has a specific rotation of $+19^{\circ}$. The rotation of this solution increases over time to the same equilibrium value as that shown by the α anomer.

(a) Draw the Haworth perspective formulas of the α and β forms of D-glucose. What feature distinguishes the two forms?

(b) Why does the specific rotation of a freshly prepared solution of the α form gradually decrease with time? Why do solutions of the α and β forms reach the same specific rotation at equilibrium?

(c) Calculate the percentage of each of the two forms of D-glucose present at equilibrium.

5. A Taste of Honey The fructose in honey is mainly in the β -D-pyranose form. This is one of the sweetest carbohydrates known, about twice as sweet as glucose. The β -D-furanose form of fructose is much less sweet. The sweetness of honey gradually decreases at a high temperature. Also, high-fructose corn syrup (a commercial product in which much of the glucose in corn syrup is converted to fructose) is used for sweetening *cold* but not *hot* drinks. What chemical property of fructose could account for both these observations?



6. Glucose Oxidase in Determination of Blood Glucose

The enzyme glucose oxidase isolated from the mold *Penicillium notatum* catalyzes the oxidation of β -D-glucose to D-glucono- δ -lactone. This enzyme is highly specific for the β anomer of glucose and does not affect the α anomer. In spite of this specificity, the reaction catalyzed by glucose oxidase is commonly used in a clinical assay for total blood glucose—that is, for solutions consisting of a mixture of β - and α -D-glucose. How is this possible? Aside from allowing the detection of smaller quantities of glucose, what advantage does glucose oxidase offer over Fehling's reagent for the determination of blood glucose?

7. Invertase “Inverts” Sucrose The hydrolysis of sucrose (specific rotation $+66.5^{\circ}$) yields an equimolar mixture of D-glucose (specific rotation $+52.5^{\circ}$) and D-fructose (specific rotation -92°). (See Problem 4 for details of specific rotation.)

(a) Suggest a convenient way to determine the rate of hydrolysis of sucrose by an enzyme preparation extracted from the lining of the small intestine.

(b) Explain why an equimolar mixture of D-glucose and D-fructose formed by hydrolysis of sucrose is called invert sugar in the food industry.

(c) The enzyme invertase (now commonly called sucrase) is allowed to act on a 10% (0.1 g/mL) solution of sucrose until hydrolysis is complete. What will be the observed optical rotation of the solution in a 10 cm cell? (Ignore a possible small contribution from the enzyme.)

8. Manufacture of Liquid-Filled Chocolates The manufacture of chocolates containing a liquid center is an interesting application of enzyme engineering. The flavored liquid center consists largely of an aqueous solution of sugars rich in fructose to provide sweetness. The technical dilemma is the following: the chocolate coating must be prepared by pouring hot melted chocolate over a solid (or almost solid) core, yet the final product must have a liquid, fructose-rich center. Suggest a way to solve this problem. (Hint: Sucrose is much less soluble than a mixture of glucose and fructose.)

9. Anomers of Sucrose? Although lactose exists in two anomeric forms, no anomeric forms of sucrose have been reported. Why?

10. Physical Properties of Cellulose and Glycogen The almost pure cellulose obtained from the seed threads of *Gossypium* (cotton) is tough, fibrous, and completely insoluble in water. In contrast, glycogen obtained from muscle or liver disperses readily in hot water to make a turbid solution. Although they have markedly different physical properties, both substances are composed of (1→4)-linked D-glucose polymers of comparable molecular weight. What structural features of these two polysaccharides underlie their different physical properties? Explain the biological advantages of their respective properties.

11. Growth Rate of Bamboo The stems of bamboo, a tropical grass, can grow at the phenomenal rate of 0.3 m/day under optimal conditions. Given that the stems are composed almost entirely of cellulose fibers oriented in the direction of growth, calculate the number of sugar residues per second that must be added enzymatically to growing cellulose chains to account for the growth rate. Each D-glucose unit contributes ~0.5 nm to the length of a cellulose molecule.

12. Glycogen as Energy Storage: How Long Can a Game Bird Fly? Since ancient times it has been observed that certain game birds, such as grouse, quail, and pheasants, are easily fatigued. The Greek historian Xenophon wrote, “The bustards . . . can be caught if one is quick in starting them up, for they will fly only a short distance, like partridges, and soon tire; and their flesh is delicious.” The flight muscles of game birds rely almost entirely on the use of glucose 1-phosphate for energy, in the form of ATP (Chapter 14). In game birds, glucose 1-phosphate is formed by the breakdown of stored muscle glycogen, catalyzed by the enzyme glycogen phosphorylase. The rate of ATP production is limited by the rate at which glycogen can be broken down. During a “panic flight,” the game bird’s rate of glycogen breakdown is quite high, approximately 120 $\mu\text{mol}/\text{min}$ of glucose 1-phosphate produced per gram of fresh tissue. Given that the flight muscles usually contain about 0.35% glycogen by weight, calculate how long a game bird can fly. (Assume the average molecular weight of a glucose residue in glycogen is 160 g/mol.)

13. Volume of Chondroitin Sulfate in Solution One critical function of chondroitin sulfate is to act as a lubricant in skeletal joints by creating a gel-like medium that is resilient to friction and shock. This function appears to be related to a distinctive property of chondroitin sulfate: the volume occupied by the molecule is much greater in solution than in the dehydrated solid. Why is the volume occupied by the molecule so much larger in solution?



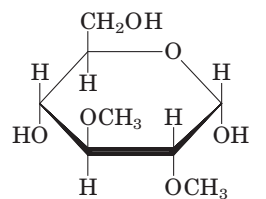
14. Heparin Interactions Heparin, a highly negatively charged glycosaminoglycan, is used clinically as an anticoagulant. It acts by binding several plasma proteins,

including antithrombin III, an inhibitor of blood clotting. The 1:1 binding of heparin to antithrombin III appears to cause a conformational change in the protein that greatly increases its ability to inhibit clotting. What amino acid residues of antithrombin III are likely to interact with heparin?

15. Information Content of Oligosaccharides The carbohydrate portion of some glycoproteins may serve as a cellular recognition site. In order to perform this function, the oligosaccharide moiety of glycoproteins must have the potential to exist in a large variety of forms. Which can produce a greater variety of structures: oligopeptides composed of five different amino acid residues or oligosaccharides composed of five different monosaccharide residues? Explain.

16. Determination of the Extent of Branching in Amylopectin The extent of branching (number of (α 1→6) glycosidic bonds) in amylopectin can be determined by the following procedure. A sample of amylopectin is exhaustively methylated—treated with a methylating agent (methyl iodide) that replaces all the hydrogens of the sugar hydroxyls with methyl groups, converting —OH to —OCH₃. All the glycosidic bonds in the treated sample are then hydrolyzed in aqueous acid. The amount of 2,3-di-O-methylglucose in the hydrolyzed sample is determined.

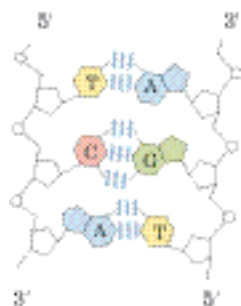
(a) Explain the basis of this procedure for determining the number of (α 1→6) branch points in amylopectin. What happens to the unbranched glucose residues in amylopectin during the methylation and hydrolysis procedure?



2,3-Di-O-methylglucose

(b) A 258 mg sample of amylopectin treated as described above yielded 12.4 mg of 2,3-di-O-methylglucose. Determine what percentage of the glucose residues in amylopectin contain an (α 1→6) branch. (Assume that the average molecular weight of a glucose residue in amylopectin is 162 g/mol.)

17. Structural Analysis of a Polysaccharide A polysaccharide of unknown structure was isolated, subjected to exhaustive methylation, and hydrolyzed. Analysis of the products revealed three methylated sugars in the ratio 20:1:1. The sugars were 2,3,4-tri-O-methyl-D-glucose; 2,4-di-O-methyl-D-glucose; and 2,3,4,6-tetra-O-methyl-D-glucose. What is the structure of the polysaccharide?



NUCLEOTIDES AND NUCLEIC ACIDS

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- 8.4 Other Functions of Nucleotides 300

A structure this pretty just had to exist.

—James Watson, *The Double Helix*, 1968

Nucleotides have a variety of roles in cellular metabolism. They are the energy currency in metabolic transactions, the essential chemical links in the response of cells to hormones and other extracellular stimuli, and the structural components of an array of enzyme cofactors and metabolic intermediates. And, last but certainly not least, they are the constituents of nucleic acids: deoxyribonucleic acid (DNA) and ribonucleic acid (RNA), the molecular repositories of genetic information. The structure of every protein, and ultimately of every biomolecule and cellular component, is a product of information programmed into the nucleotide sequence of a cell's nucleic acids. The ability to store and transmit genetic information from one generation to the next is a fundamental condition for life.

This chapter provides an overview of the chemical nature of the nucleotides and nucleic acids found in most cells; a more detailed examination of the function of nucleic acids is the focus of Part III of this text.

8.1 Some Basics

 **Nucleotides, Building Blocks of Nucleic Acids** The amino acid sequence of every protein in a cell, and the nucleotide sequence of every RNA, is specified by a nucleotide se-

quence in the cell's DNA. A segment of a DNA molecule that contains the information required for the synthesis of a functional biological product, whether protein or RNA, is referred to as a **gene**. A cell typically has many thousands of genes, and DNA molecules, not surprisingly, tend to be very large. The storage and transmission of biological information are the only known functions of DNA.

RNAs have a broader range of functions, and several classes are found in cells. **Ribosomal RNAs (rRNAs)** are components of ribosomes, the complexes that carry out the synthesis of proteins. **Messenger RNAs (mRNAs)** are intermediaries, carrying genetic information from one or a few genes to a ribosome, where the corresponding proteins can be synthesized. **Transfer RNAs (tRNAs)** are adapter molecules that faithfully translate the information in mRNA into a specific sequence of amino acids. In addition to these major classes there is a wide variety of RNAs with special functions, described in depth in Part III.

Nucleotides and Nucleic Acids Have Characteristic Bases and Pentoses

Nucleotides have three characteristic components: (1) a nitrogenous (nitrogen-containing) base, (2) a pentose, and (3) a phosphate (Fig. 8–1). The molecule without the phosphate group is called a **nucleoside**. The nitrogenous bases are derivatives of two parent compounds, **pyrimidine** and **purine**. The bases and pentoses of the common nucleotides are heterocyclic compounds. The carbon and nitrogen atoms in the parent structures are conventionally numbered to facilitate the naming and identification of the many derivative compounds. The convention for the pentose ring follows rules outlined in Chapter 7, but in the pentoses of nucleotides

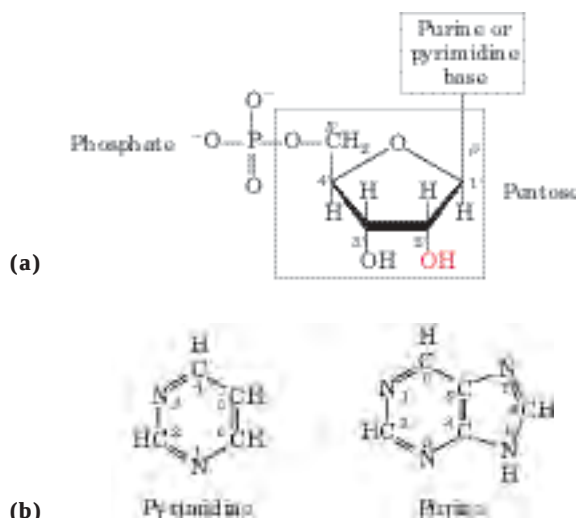


FIGURE 8-1 Structure of nucleotides. (a) General structure showing the numbering convention for the pentose ring. This is a ribonucleotide. In deoxyribonucleotides the —OH group on the 2' carbon (in red) is replaced with —H. (b) The parent compounds of the pyrimidine and purine bases of nucleotides and nucleic acids, showing the numbering conventions.

and nucleosides the carbon numbers are given a prime (') designation to distinguish them from the numbered atoms of the nitrogenous bases.

The base of a nucleotide is joined covalently (at N-1 of pyrimidines and N-9 of purines) in an *N*- β -glycosyl bond to the 1' carbon of the pentose, and the phosphate is esterified to the 5' carbon. The *N*- β -glycosyl bond is formed by removal of the elements of water (a hydroxyl group from the pentose and hydrogen from the base), as in *O*-glycosidic bond formation (see Fig. 7-31).

Both DNA and RNA contain two major purine bases, **adenine** (A) and **guanine** (G), and two major pyrimidines. In both DNA and RNA one of the pyrimidines is **cytosine** (C), but the second major pyrimidine is not the same in both: it is **thymine** (T) in DNA and **uracil** (U) in RNA. Only rarely does thymine occur in RNA or uracil in DNA. The structures of the five major bases are shown in Figure 8-2, and the nomenclature of their corresponding nucleotides and nucleosides is summarized in Table 8-1.

Nucleic acids have two kinds of pentoses. The recurring deoxyribonucleotide units of DNA contain 2'-deoxy-D-ribose, and the ribonucleotide units of RNA contain D-ribose. In nucleotides, both types of pentoses are in their β -furanose (closed five-membered ring) form. As Figure 8-3 shows, the pentose ring is not planar but occurs in one of a variety of conformations generally described as "puckered."

Figure 8-4 gives the structures and names of the four major **deoxyribonucleotides** (deoxyribonucleoside 5'-monophosphates), the structural units of DNAs, and the four major **ribonucleotides** (ribonucleoside 5'-monophosphates), the structural units of RNAs. Specific

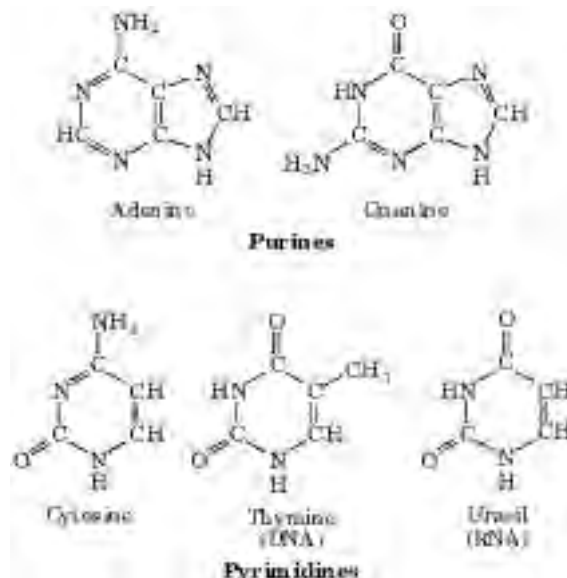


FIGURE 8-2 Major purine and pyrimidine bases of nucleic acids.

Some of the common names of these bases reflect the circumstances of their discovery. Guanine, for example, was first isolated from guano (bird manure), and thymine was first isolated from thymus tissue.

long sequences of A, T, G, and C nucleotides in DNA are the repository of genetic information.

Although nucleotides bearing the major purines and pyrimidines are most common, both DNA and RNA also

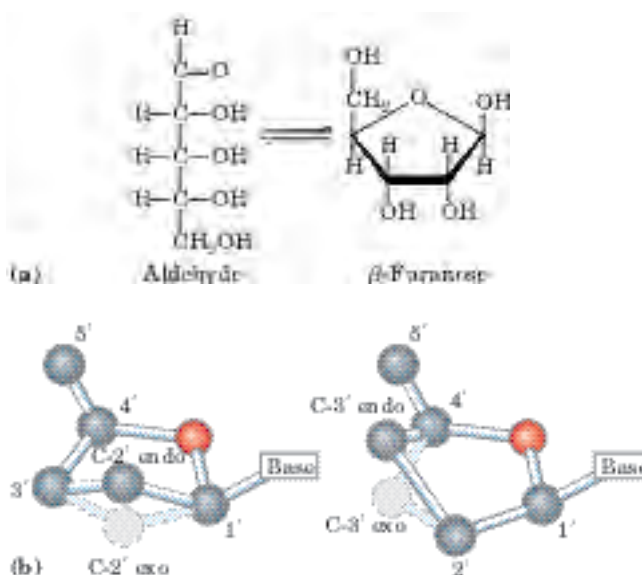


FIGURE 8-3 Conformations of ribose. (a) In solution, the straight-chain (aldehyde) and ring (β -furanose) forms of free ribose are in equilibrium. RNA contains only the ring form, β -D-ribofuranose. Deoxyribose undergoes a similar interconversion in solution, but in DNA exists solely as β -2'-deoxy-D-ribofuranose. (b) Ribofuranose rings in nucleotides can exist in four different puckered conformations. In all cases, four of the five atoms are in a single plane. The fifth atom (C-2' or C-3') is on either the same (endo) or the opposite (exo) side of the plane relative to the C-5' atom.

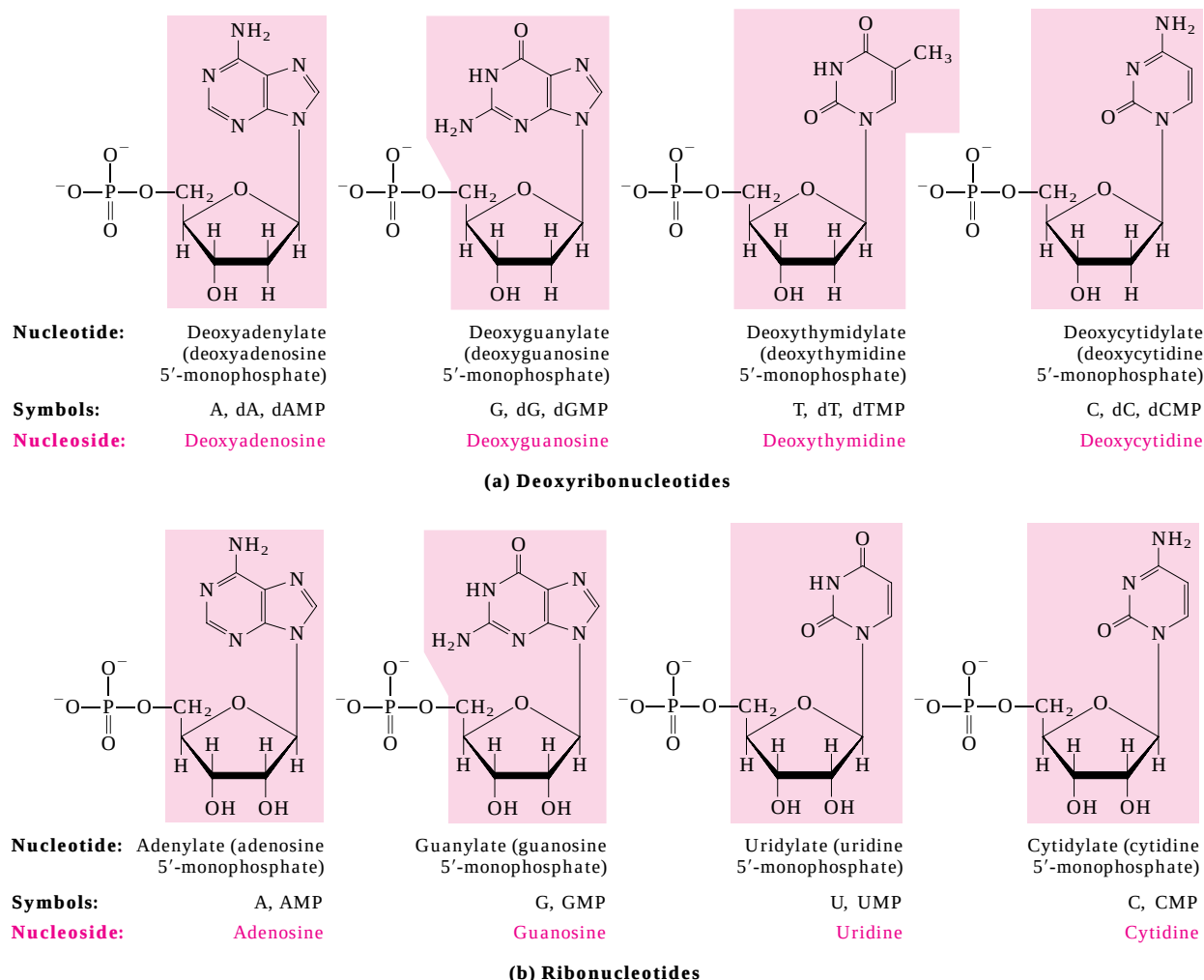


FIGURE 8-4 Deoxyribonucleotides and ribonucleotides of nucleic acids. All nucleotides are shown in their free form at pH 7.0. The nucleotide units of DNA **(a)** are usually symbolized as A, G, T, and C, sometimes as dA, dG, dT, and dC; those of RNA **(b)** as A, G, U, and C. In their free form the deoxyribonucleotides are commonly abbreviated dAMP, dGMP, dTMP, and dCMP; the ribonucleotides, AMP,

GMP, UMP, and CMP. For each nucleotide, the more common name is followed by the complete name in parentheses. All abbreviations assume that the phosphate group is at the 5' position. The nucleoside portion of each molecule is shaded in light red. In this and the following illustrations, the ring carbons are not shown.

TABLE 8-1 Nucleotide and Nucleic Acid Nomenclature

Base	Nucleoside	Nucleotide	Nucleic acid
Purines			
Adenine	Adenosine	Adenylate	RNA
	Deoxyadenosine	Deoxyadenylate	DNA
Guanine	Guanosine	Guanylate	RNA
	Deoxyguanosine	Deoxyguanylate	DNA
Pyrimidines			
Cytosine	Cytidine	Cytidylate	RNA
	Deoxycytidine	Deoxycytidylate	DNA
Thymine	Thymidine or deoxythymidine	Thymidylate or deoxythymidylate	DNA
Uracil	Uridine	Uridylate	RNA

Note: "Nucleoside" and "nucleotide" are generic terms that include both ribo- and deoxyribo- forms. Also, ribonucleosides and ribonucleotides are here designated simply as nucleosides and nucleotides (e.g., ribo-adenosine as adenosine), and deoxyribo-nucleosides and deoxyribo-nucleotides as deoxynucleosides and deoxynucleotides (e.g., deoxyriboadenosine as deoxyadenosine). Both forms of naming are acceptable, but the shortened names are more commonly used. Thymine is an exception; "ribthymidine" is used to describe its unusual occurrence in RNA.

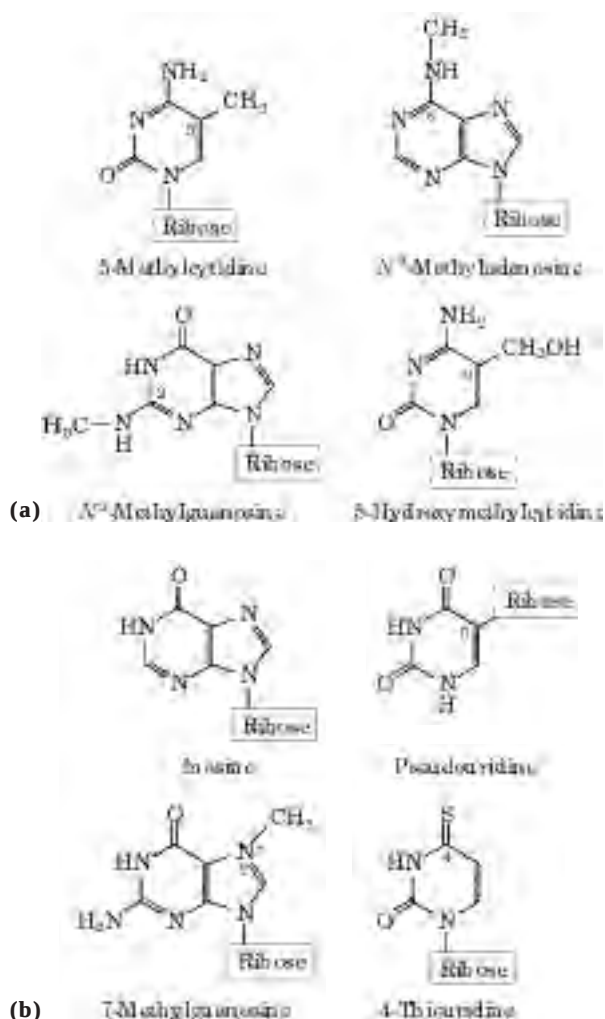


FIGURE 8-5 Some minor purine and pyrimidine bases, shown as the nucleosides. (a) Minor bases of DNA. 5-Methylcytidine occurs in the DNA of animals and higher plants, N^6 -methyladenosine in bacterial DNA, and 5-hydroxymethylcytidine in the DNA of bacteria infected with certain bacteriophages. (b) Some minor bases of tRNAs. Inosine contains the base hypoxanthine. Note that pseudouridine, like uridine, contains uracil; they are distinct in the point of attachment to the ribose—in uridine, uracil is attached through N-1, the usual attachment point for pyrimidines; in pseudouridine, through C-5.

contain some minor bases (Fig. 8-5). In DNA the most common of these are methylated forms of the major bases; in some viral DNAs, certain bases may be hydroxymethylated or glucosylated. Altered or unusual bases in DNA molecules often have roles in regulating or protecting the genetic information. Minor bases of many types are also found in RNAs, especially in tRNAs (see Fig. 26-24).

The nomenclature for the minor bases can be confusing. Like the major bases, many have common names—hypoxanthine, for example, shown as its nucleoside inosine in Figure 8-5. When an atom in the purine or pyrimidine ring is substituted, the usual convention (used

here) is simply to indicate the ring position of the substituent by its number—for example, 5-methylcytosine, 7-methylguanine, and 5-hydroxymethylcytosine (shown as the nucleosides in Fig. 8-5). The element to which the substituent is attached (N, C, O) is not identified. The convention changes when the substituted atom is exocyclic (not within the ring structure), in which case the type of atom is identified and the ring position to which it is attached is denoted with a superscript. The amino nitrogen attached to C-6 of adenine is N^6 ; similarly, the carbonyl oxygen and amino nitrogen at C-6 and C-2 of guanine are O^6 and N^2 , respectively. Examples of this nomenclature are N^6 -methyladenosine and N^2 -methylguanosine (Fig. 8-5).

Cells also contain nucleotides with phosphate groups in positions other than on the 5' carbon (Fig. 8-6). **Ribonucleoside 2',3'-cyclic monophosphates** are isolatable intermediates, and **ribonucleoside 3'-monophosphates** are end products of the hydrolysis of RNA by certain ribonucleases. Other variations are adenosine 3',5'-cyclic monophosphate (cAMP) and guanosine 3',5'-cyclic monophosphate (cGMP), considered at the end of this chapter.

Phosphodiester Bonds Link Successive Nucleotides in Nucleic Acids

The successive nucleotides of both DNA and RNA are covalently linked through phosphate-group “bridges,” in which the 5'-phosphate group of one nucleotide unit is

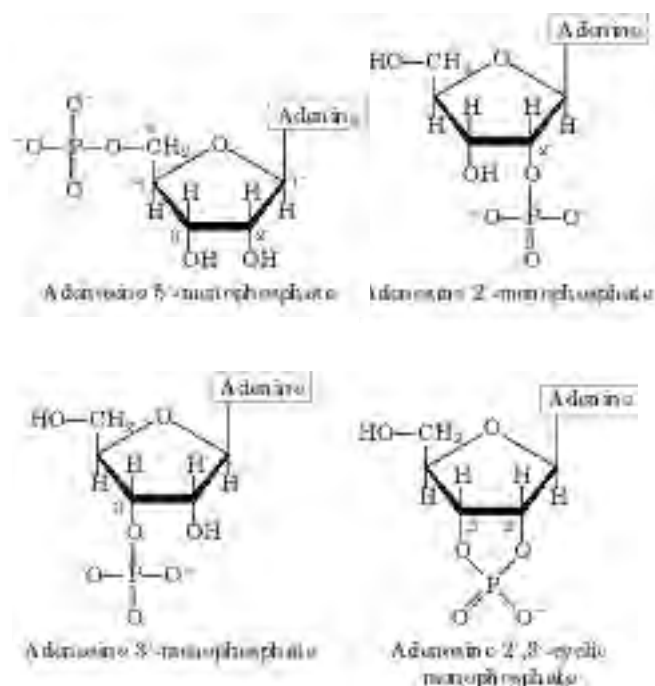


FIGURE 8-6 Some adenosine monophosphates. Adenosine 2'-monophosphate, 3'-monophosphate, and 2',3'-cyclic monophosphate are formed by enzymatic and alkaline hydrolysis of RNA.

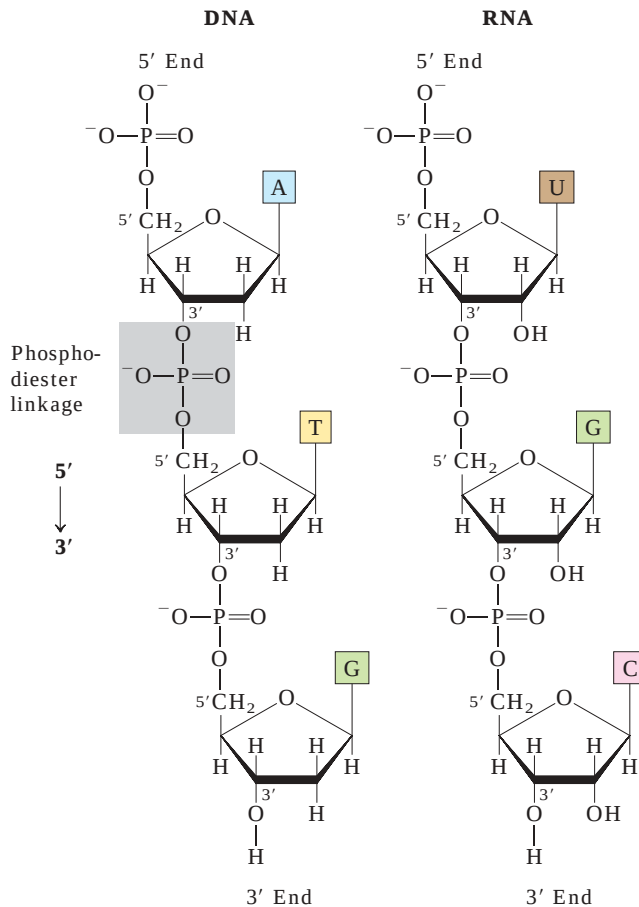


FIGURE 8-7 Phosphodiester linkages in the covalent backbone of DNA and RNA. The phosphodiester bonds (one of which is shaded in the DNA) link successive nucleotide units. The backbone of alternating pentose and phosphate groups in both types of nucleic acid is highly polar. The 5' end of the macromolecule lacks a nucleotide at the 5' position, and the 3' end lacks a nucleotide at the 3' position.

joined to the 3'-hydroxyl group of the next nucleotide, creating a **phosphodiester linkage** (Fig. 8-7). Thus the covalent backbones of nucleic acids consist of alternating phosphate and pentose residues, and the nitrogenous bases may be regarded as side groups joined to the backbone at regular intervals. The backbones of both DNA and RNA are hydrophilic. The hydroxyl groups of the sugar residues form hydrogen bonds with water. The phosphate groups, with a pK_a near 0, are completely ionized and negatively charged at pH 7, and the negative charges are generally neutralized by ionic interactions with positive charges on proteins, metal ions, and polyamines.

All the phosphodiester linkages have the same orientation along the chain (Fig. 8-7), giving each linear nucleic acid strand a specific polarity and distinct 5' and 3' ends. By definition, the **5' end** lacks a nucleotide at the 5' position and the **3' end** lacks a nucleotide at the 3' position. Other groups (most often one or more phosphates) may be present on one or both ends.

The covalent backbone of DNA and RNA is subject to slow, nonenzymatic hydrolysis of the phosphodiester bonds. In the test tube, RNA is hydrolyzed rapidly under alkaline conditions, but DNA is not; the 2'-hydroxyl groups in RNA (absent in DNA) are directly involved in the process. Cyclic 2',3'-monophosphate nucleotides are the first products of the action of alkali on RNA and are rapidly hydrolyzed further to yield a mixture of 2'- and 3'-nucleoside monophosphates (Fig. 8-8).

The nucleotide sequences of nucleic acids can be represented schematically, as illustrated on the following page by a segment of DNA with five nucleotide units. The phosphate groups are symbolized by $\textcircled{P}$, and each deoxyribose is symbolized by a vertical line, from C-1' at the top to C-5' at the bottom (but keep in mind that

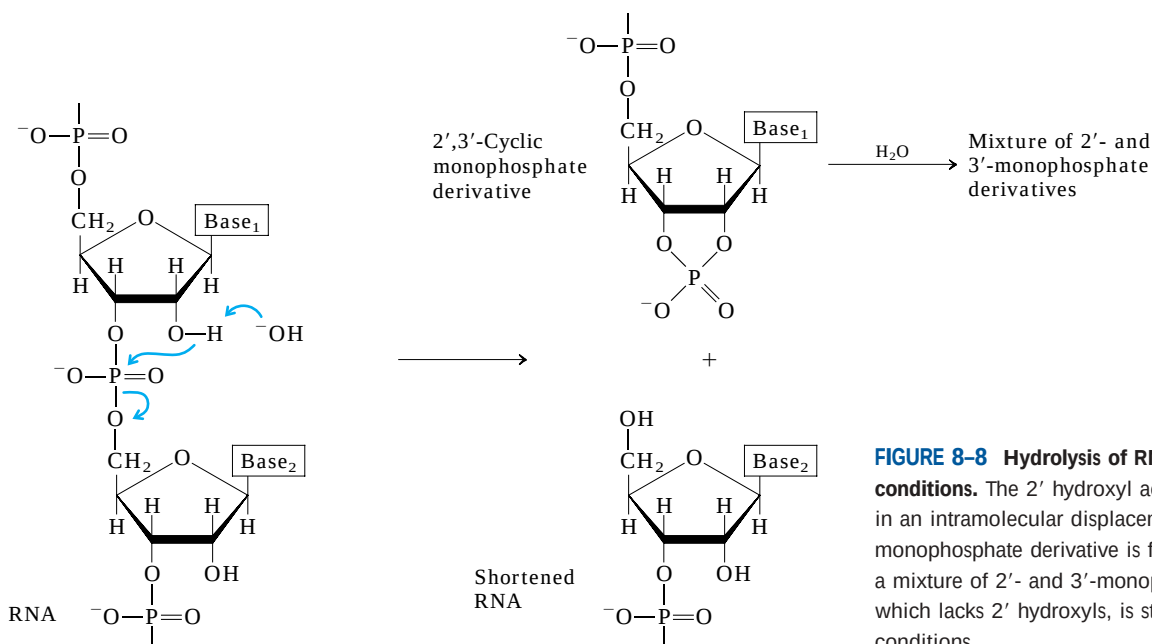
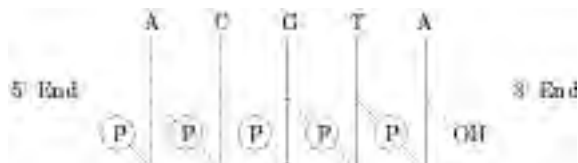


FIGURE 8-8 Hydrolysis of RNA under alkaline conditions. The 2' hydroxyl acts as a nucleophile in an intramolecular displacement. The 2',3'-cyclic monophosphate derivative is further hydrolyzed to a mixture of 2'- and 3'-monophosphates. DNA, which lacks 2' hydroxyls, is stable under similar conditions.

the sugar is always in its closed-ring β -furanose form in nucleic acids). The connecting lines between nucleotides (which pass through $\textcircled{\text{P}}$) are drawn diagonally from the middle (C-3') of the deoxyribose of one nucleotide to the bottom (C-5') of the next.



By convention, the structure of a single strand of nucleic acid is always written with the 5' end at the left and the 3' end at the right—that is, in the 5' $\rightarrow$ 3' direction. Some simpler representations of this pentadeoxyribonucleotide are pA-C-G-T-A_{OH}, pApCpGpTpA, and pACGTA.

A short nucleic acid is referred to as an **oligonucleotide**. The definition of “short” is somewhat arbitrary, but polymers containing 50 or fewer nucleotides are generally called oligonucleotides. A longer nucleic acid is called a **polynucleotide**.

The Properties of Nucleotide Bases Affect the Three-Dimensional Structure of Nucleic Acids

Free pyrimidines and purines are weakly basic compounds and are thus called bases. They have a variety of chemical properties that affect the structure, and ultimately the function, of nucleic acids. The purines and pyrimidines common in DNA and RNA are highly conjugated molecules (Fig. 8–2), a property with important consequences for the structure, electron distribution, and light absorption of nucleic acids. Resonance among atoms in the ring gives most of the bonds partial double-bond character. One result is that pyrimidines are planar molecules; purines are very nearly

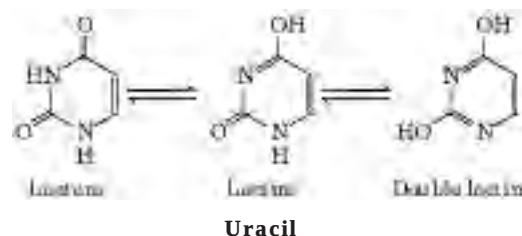


FIGURE 8–9 Tautomeric forms of uracil. The lactam form predominates at pH 7.0; the other forms become more prominent as pH decreases. The other free pyrimidines and the free purines also have tautomeric forms, but they are more rarely encountered.

planar, with a slight pucker. Free pyrimidine and purine bases may exist in two or more tautomeric forms depending on the pH. Uracil, for example, occurs in lactam, lactim, and double lactim forms (Fig. 8–9). The structures shown in Figure 8–2 are the tautomers that predominate at pH 7.0. As a result of resonance, all nucleotide bases absorb UV light, and nucleic acids are characterized by a strong absorption at wavelengths near 260 nm (Fig. 8–10).

The purine and pyrimidine bases are hydrophobic and relatively insoluble in water at the near-neutral pH of the cell. At acidic or alkaline pH the bases become charged and their solubility in water increases. Hydrophobic stacking interactions in which two or more bases are positioned with the planes of their rings parallel (like a stack of coins) are one of two important modes of interaction between bases in nucleic acids. The stacking also involves a combination of van der Waals and dipole-dipole interactions between the bases. Base stacking helps to minimize contact of the bases with water, and base-stacking interactions are very important in stabilizing the three-dimensional structure of nucleic acids, as described later.

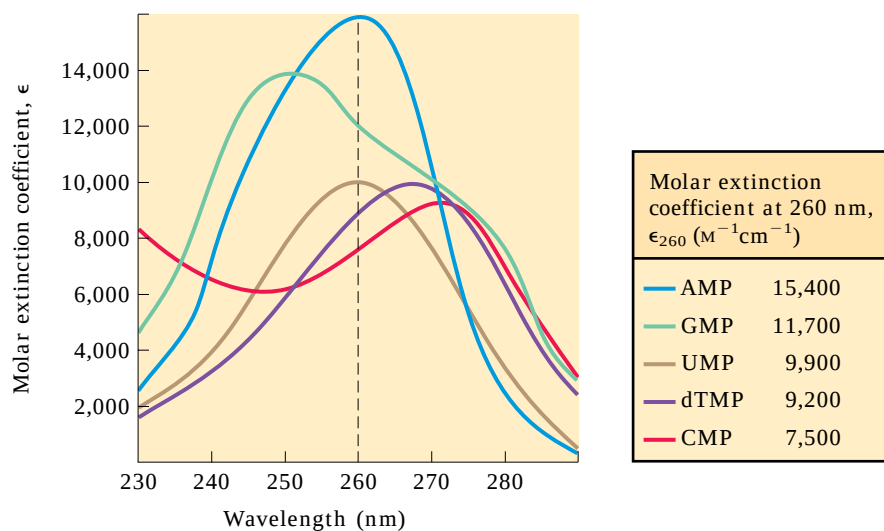


FIGURE 8–10 Absorption spectra of the common nucleotides. The spectra are shown as the variation in molar extinction coefficient with wavelength. The molar extinction coefficients at 260 nm and pH 7.0 (ϵ_{260}) are listed in the table. The spectra of corresponding ribonucleotides and deoxyribonucleotides, as well as the nucleosides, are essentially identical. For mixtures of nucleotides, a wavelength of 260 nm (dashed vertical line) is used for absorption measurements.

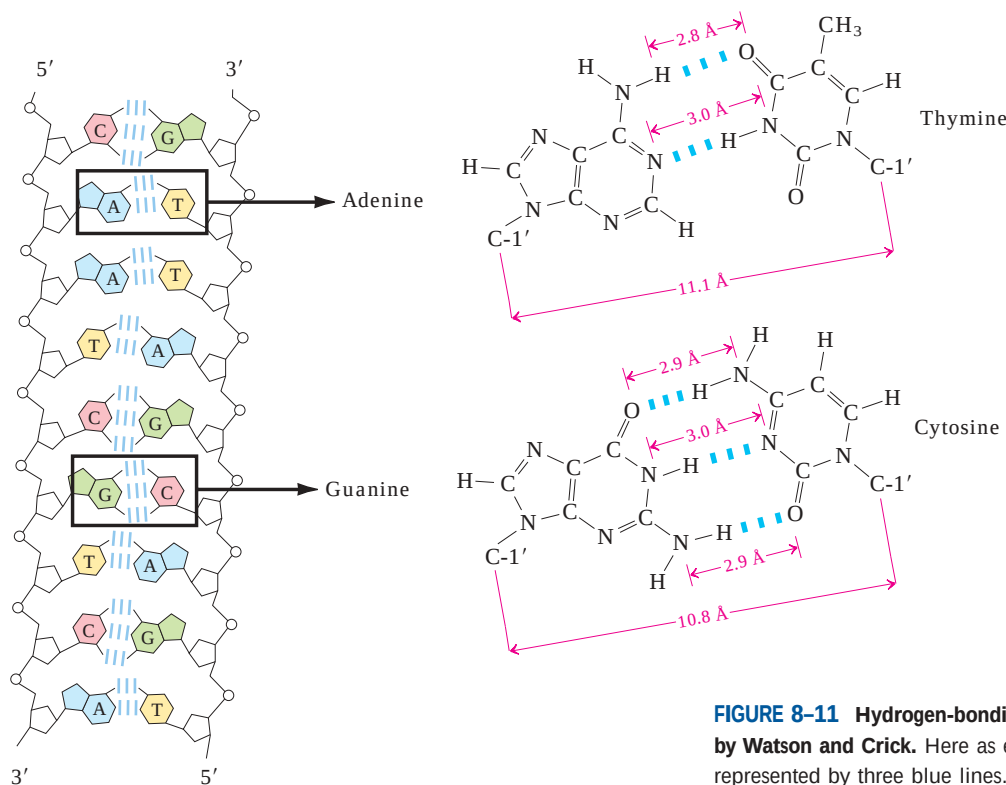


FIGURE 8-11 Hydrogen-bonding patterns in the base pairs defined by Watson and Crick. Here as elsewhere, hydrogen bonds are represented by three blue lines.

The most important functional groups of pyrimidines and purines are ring nitrogens, carbonyl groups, and exocyclic amino groups. Hydrogen bonds involving the amino and carbonyl groups are the second important mode of interaction between bases in nucleic acid molecules. Hydrogen bonds between bases permit a complementary association of two (and occasionally three or four) strands of nucleic acid. The most important hydrogen-bonding patterns are those defined by James D. Watson and Francis Crick in 1953, in which A bonds specifically to T (or U) and G bonds to C (Fig. 8-11). These two types of **base pairs** predominate in double-stranded DNA and RNA, and the tautomers shown in Figure 8-2 are responsible for these patterns. It is this specific pairing of bases that permits the duplication of genetic information, as we shall discuss later in this chapter.

SUMMARY 8.1 Some Basics

- A nucleotide consists of a nitrogenous base (purine or pyrimidine), a pentose sugar, and one or more phosphate groups. Nucleic acids are polymers of nucleotides, joined together by phosphodiester linkages between the 5'-hydroxyl group of one pentose and the 3'-hydroxyl group of the next.
- There are two types of nucleic acid: RNA and DNA. The nucleotides in RNA contain ribose, and the common pyrimidine bases are uracil and cytosine. In DNA, the nucleotides contain 2'-deoxyribose, and the common pyrimidine bases are thymine and cytosine. The primary purines are adenine and guanine in both RNA and DNA.



James Watson



Francis Crick

8.2 Nucleic Acid Structure

The discovery of the structure of DNA by Watson and Crick in 1953 was a momentous event in science, an event that gave rise to entirely new disciplines and influenced the course of many established ones. Our present understanding of the storage and utilization of a cell's genetic information is based on work made possible by this discovery, and an outline of how genetic information is processed by the cell is now a prerequisite for the discussion of any area of biochemistry. Here, we concern ourselves with DNA structure itself, the events

that led to its discovery, and more recent refinements in our understanding. RNA structure is also introduced.

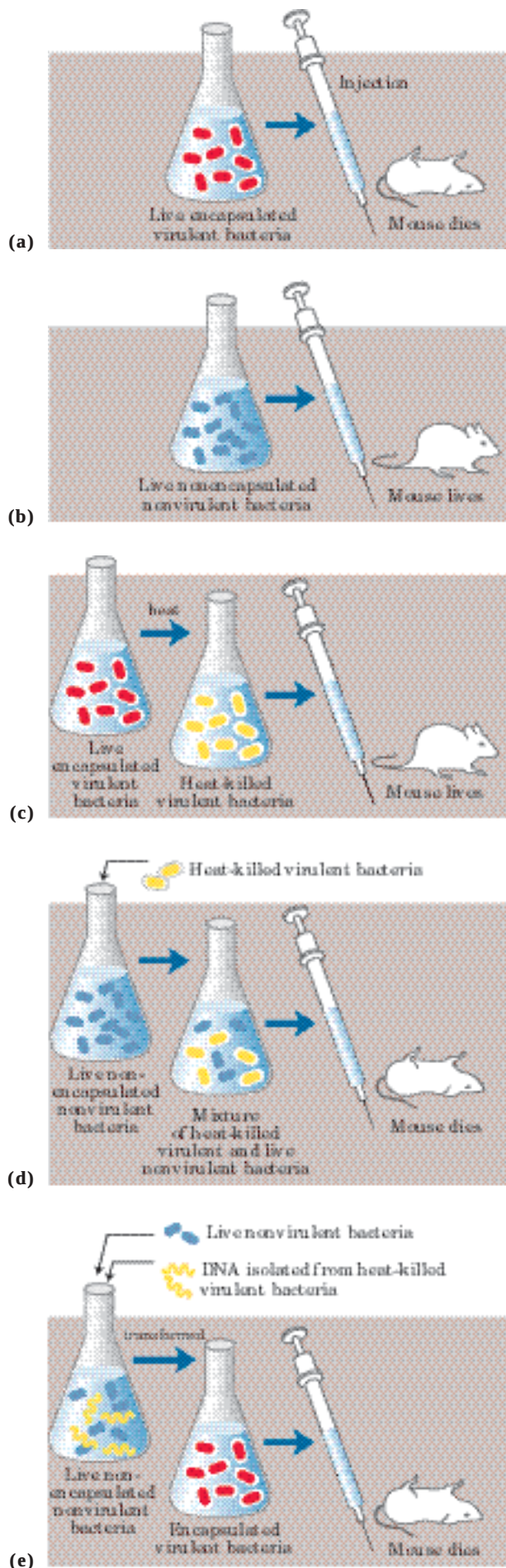
As in the case of protein structure (Chapter 4), it is sometimes useful to describe nucleic acid structure in terms of hierarchical levels of complexity (primary, secondary, tertiary). The primary structure of a nucleic acid is its covalent structure and nucleotide sequence. Any regular, stable structure taken up by some or all of the nucleotides in a nucleic acid can be referred to as secondary structure. All structures considered in the remainder of this chapter fall under the heading of secondary structure. The complex folding of large chromosomes within eukaryotic chromatin and bacterial nucleoids is generally considered tertiary structure; this is discussed in Chapter 24.

DNA Stores Genetic Information

The biochemical investigation of DNA began with Friedrich Miescher, who carried out the first systematic chemical studies of cell nuclei. In 1868 Miescher isolated a phosphorus-containing substance, which he called “nuclein,” from the nuclei of pus cells (leukocytes) obtained from discarded surgical bandages. He found nuclein to consist of an acidic portion, which we know today as DNA, and a basic portion, protein. Miescher later found a similar acidic substance in the heads of sperm cells from salmon. Although he partially purified nuclein and studied its properties, the covalent (primary) structure of DNA (as shown in Fig. 8–7) was not known with certainty until the late 1940s.

Miescher and many others suspected that nuclein (nucleic acid) was associated in some way with cell inheritance, but the first direct evidence that DNA is the bearer of genetic information came in 1944 through a discovery made by Oswald T. Avery, Colin MacLeod, and Maclyn McCarty. These investigators found that DNA extracted from a virulent (disease-causing) strain of the bacterium *Streptococcus pneumoniae*, also known as pneumococcus, genetically transformed a nonvirulent strain of this organism into a virulent form (Fig. 8–12).

FIGURE 8-12 The Avery-MacLeod-McCarty experiment. (a) When injected into mice, the encapsulated strain of pneumococcus is lethal, (b) whereas the nonencapsulated strain, (c) like the heat-killed encapsulated strain, is harmless. (d) Earlier research by the bacteriologist Frederick Griffith had shown that adding heat-killed virulent bacteria (harmless to mice) to a live nonvirulent strain permanently transformed the latter into lethal, virulent, encapsulated bacteria. (e) Avery and his colleagues extracted the DNA from heat-killed virulent pneumococci, removing the protein as completely as possible, and added this DNA to nonvirulent bacteria. The DNA gained entrance into the nonvirulent bacteria, which were permanently transformed into a virulent strain.



Avery and his colleagues concluded that the DNA extracted from the virulent strain carried the inheritable genetic message for virulence. Not everyone accepted these conclusions, because protein impurities present in the DNA could have been the carrier of the genetic information. This possibility was soon eliminated by the finding that treatment of the DNA with proteolytic enzymes did not destroy the transforming activity, but treatment with deoxyribonucleases (DNA-hydrolyzing enzymes) did.

A second important experiment provided independent evidence that DNA carries genetic information. In 1952 Alfred D. Hershey and Martha Chase used radioactive phosphorus (^{32}P) and radioactive sulfur (^{35}S) tracers to show that when the bacterial virus (bacteriophage) T2 infects its host cell, *Escherichia coli*, it is the phosphorus-containing DNA of the viral particle, not the sulfur-containing protein of the viral coat, that enters the host cell and furnishes the genetic information for viral replication (Fig. 8–13). These important early experiments and many other lines of evidence have shown that DNA is the exclusive chromosomal component bearing the genetic information of living cells.

DNA Molecules Have Distinctive Base Compositions

A most important clue to the structure of DNA came from the work of Erwin Chargaff and his colleagues in the late 1940s. They found that the four nucleotide bases of DNA occur in different ratios in the DNAs of different organisms and that the amounts of certain bases are closely related. These data, collected from DNAs of a great many different species, led Chargaff to the following conclusions:

1. The base composition of DNA generally varies from one species to another.
2. DNA specimens isolated from different tissues of the same species have the same base composition.
3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.
4. In *all* cellular DNAs, regardless of the species, the number of adenosine residues is equal to the number of thymidine residues (that is, $A = T$), and the number of guanosine residues is equal to the number of cytidine residues ($G = C$). From these relationships it follows that the sum of the purine residues equals the sum of the pyrimidine residues; that is, $A + G = T + C$.

These quantitative relationships, sometimes called “Chargaff’s rules,” were confirmed by many subsequent researchers. They were a key to establishing the three-dimensional structure of DNA and yielded clues to how genetic information is encoded in DNA and passed from one generation to the next.

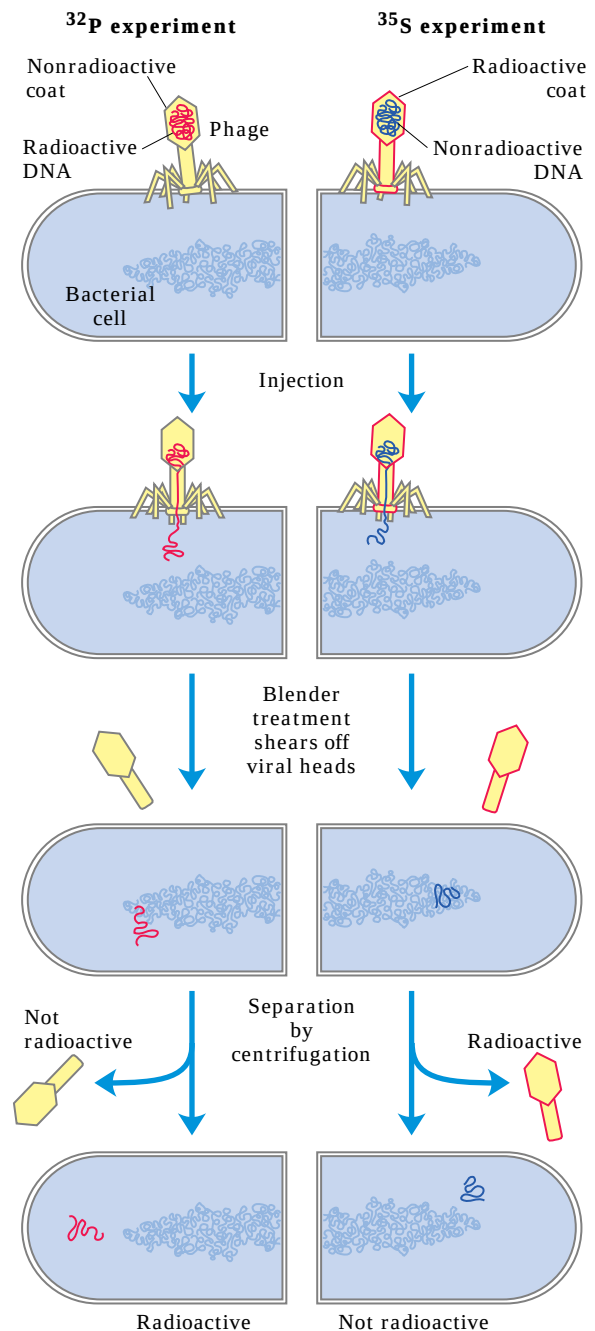


FIGURE 8–13 The Hershey-Chase experiment. Two batches of isotopically labeled bacteriophage T2 particles were prepared. One was labeled with ^{32}P in the phosphate groups of the DNA, the other with ^{35}S in the sulfur-containing amino acids of the protein coats (capsids). (Note that DNA contains no sulfur and viral protein contains no phosphorus.) The two batches of labeled phage were then allowed to infect separate suspensions of unlabeled bacteria. Each suspension of phage-infected cells was agitated in a blender to shear the viral capsids from the bacteria. The bacteria and empty viral coats (called “ghosts”) were then separated by centrifugation. The cells infected with the ^{32}P -labeled phage were found to contain ^{32}P , indicating that the labeled viral DNA had entered the cells; the viral ghosts contained no radioactivity. The cells infected with ^{35}S -labeled phage were found to have no radioactivity after blender treatment, but the viral ghosts contained ^{35}S . Progeny virus particles (not shown) were produced in both batches of bacteria some time after the viral coats were removed, indicating that the genetic message for their replication had been introduced by viral DNA, not by viral protein.

DNA Is a Double Helix

To shed more light on the structure of DNA, Rosalind Franklin and Maurice Wilkins used the powerful method of x-ray diffraction (see Box 4–4) to analyze DNA fibers. They showed in the early 1950s that DNA produces a characteristic x-ray diffraction pattern (Fig. 8–14). From this pattern it was deduced that DNA molecules are helical with two periodicities along their long axis, a primary one of 3.4 Å and a secondary one of 34 Å. The problem then was to formulate a three-dimensional model of the DNA molecule that could account not only for the x-ray diffraction data but also for the specific $A = T$ and $G = C$ base equivalences discovered by Chargaff and for the other chemical properties of DNA.

In 1953 Watson and Crick postulated a three-dimensional model of DNA structure that accounted for all the available data. It consists of two helical DNA chains wound around the same axis to form a right-handed double helix (see Box 4–1 for an explanation of the right- or left-handed sense of a helical structure). The hydrophilic backbones of alternating deoxyribose and phosphate groups are on the outside of the double helix, facing the surrounding water. The furanose ring of each deoxyribose is in the C-2' endo conformation. The purine and pyrimidine bases of both strands are stacked inside the double helix, with their hydrophobic and nearly planar ring structures very close together and perpendicular to the long axis. The offset pairing of the two strands creates a **major groove** and **minor groove** on the surface of the duplex (Fig. 8–15). Each nucleotide base of one strand is paired in the same plane with a base of the other strand. Watson and Crick found that the hydrogen-bonded base pairs illustrated in Figure 8–11, G with C and A with T, are those that fit best within the structure, providing a rationale for Chargaff's rule that in any DNA, $G = C$ and $A = T$. It is important to note that three hydrogen bonds can form between G and C, symbolized $G \equiv C$, but only two can form between A and T, symbolized $A = T$. This is one reason for the

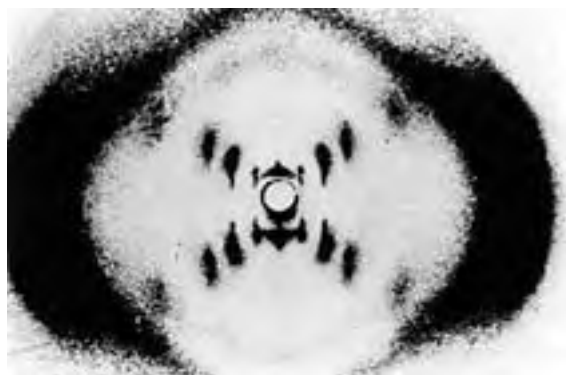


FIGURE 8–14 X-ray diffraction pattern of DNA. The spots forming a cross in the center denote a helical structure. The heavy bands at the left and right arise from the recurring bases.



Rosalind Franklin,
1920–1958



Maurice Wilkins

finding that separation of paired DNA strands is more difficult the higher the ratio of $G \equiv C$ to $A = T$ base pairs. Other pairings of bases tend (to varying degrees) to destabilize the double-helical structure.

When Watson and Crick constructed their model, they had to decide at the outset whether the strands of DNA should be **parallel** or **antiparallel**—whether their 5',3'-phosphodiester bonds should run in the same or opposite directions. An antiparallel orientation produced the most convincing model, and later work with DNA polymerases (Chapter 25) provided experimental evidence that the strands are indeed antiparallel, a finding ultimately confirmed by x-ray analysis.

To account for the periodicities observed in the x-ray diffraction patterns of DNA fibers, Watson and Crick manipulated molecular models to arrive at a structure

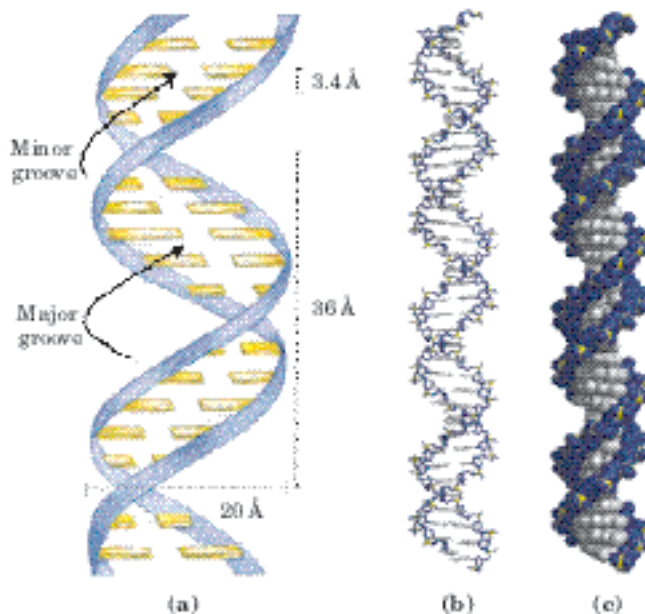


FIGURE 8–15 Watson-Crick model for the structure of DNA. The original model proposed by Watson and Crick had 10 base pairs, or 34 Å (3.4 nm), per turn of the helix; subsequent measurements revealed 10.5 base pairs, or 36 Å (3.6 nm), per turn. **(a)** Schematic representation, showing dimensions of the helix. **(b)** Stick representation showing the backbone and stacking of the bases. **(c)** Space-filling model.

in which the vertically stacked bases inside the double helix would be 3.4 Å apart; the secondary repeat distance of about 34 Å was accounted for by the presence of 10 base pairs in each complete turn of the double helix. In aqueous solution the structure differs slightly from that in fibers, having 10.5 base pairs per helical turn (Fig. 8–15).

As Figure 8–16 shows, the two antiparallel polynucleotide chains of double-helical DNA are not identical in either base sequence or composition. Instead they are **complementary** to each other. Wherever adenine occurs in one chain, thymine is found in the other; similarly, wherever guanine occurs in one chain, cytosine is found in the other.

The DNA double helix, or duplex, is held together by two forces, as described earlier: hydrogen bonding between complementary base pairs (Fig. 8–11) and base-stacking interactions. The complementarity between the DNA strands is attributable to the hydrogen bonding between base pairs. The base-stacking interactions, which are largely nonspecific with respect to the identity of the stacked bases, make the major contribution to the stability of the double helix.

The important features of the double-helical model of DNA structure are supported by much chemical and

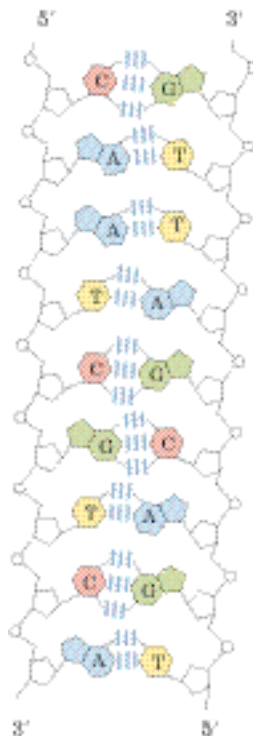


FIGURE 8–16 Complementarity of strands in the DNA double helix.

The complementary antiparallel strands of DNA follow the pairing rules proposed by Watson and Crick. The base-paired antiparallel strands differ in base composition: the left strand has the composition $A_3 T_2 G_1 C_3$; the right, $A_2 T_3 G_3 C_1$. They also differ in sequence when each chain is read in the $5' \rightarrow 3'$ direction. Note the base equivalences: $A = T$ and $G = C$ in the duplex.

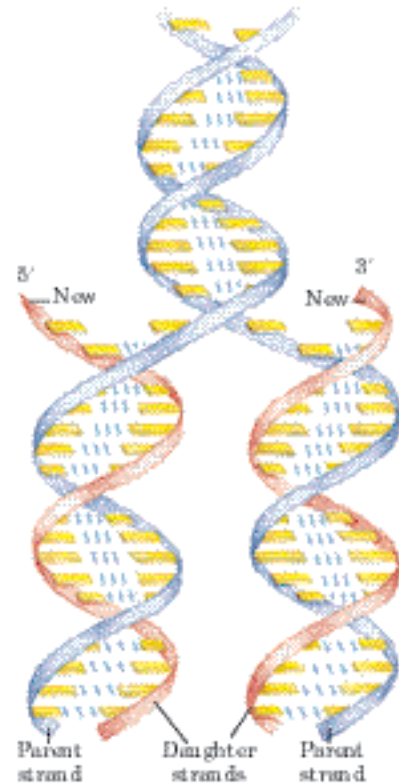


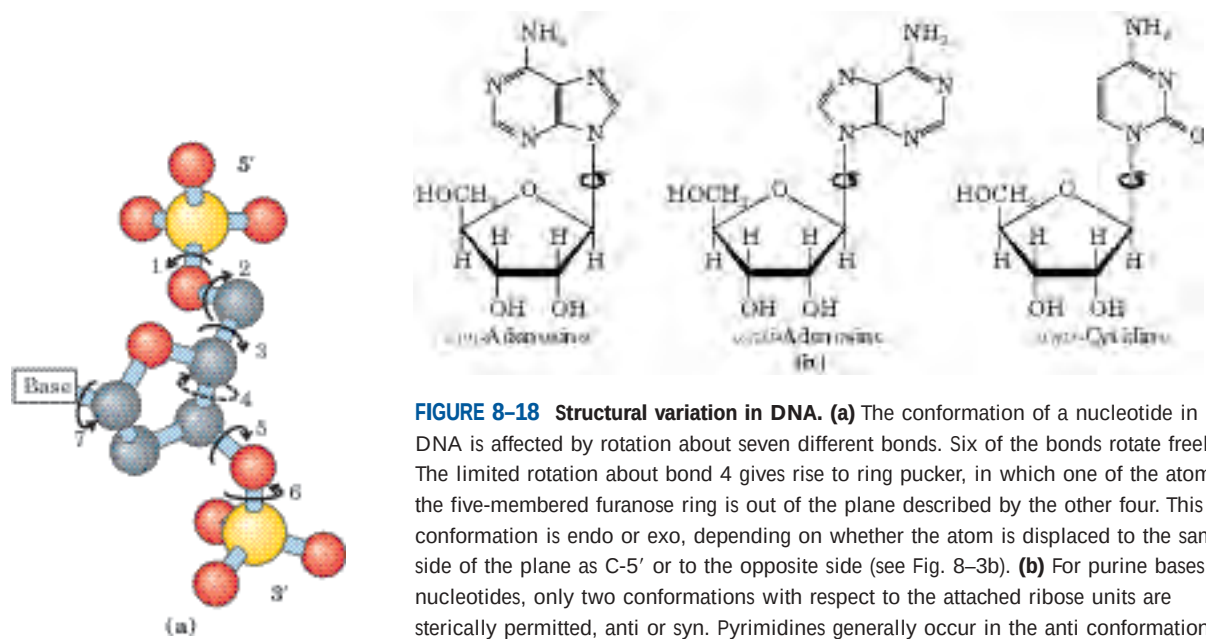
FIGURE 8–17 Replication of DNA as suggested by Watson and Crick.

The preexisting or “parent” strands become separated, and each is the template for biosynthesis of a complementary “daughter” strand (in red).

biological evidence. Moreover, the model immediately suggested a mechanism for the transmission of genetic information. The essential feature of the model is the complementarity of the two DNA strands. As Watson and Crick were able to see, well before confirmatory data became available, this structure could logically be replicated by (1) separating the two strands and (2) synthesizing a complementary strand for each. Because nucleotides in each new strand are joined in a sequence specified by the base-pairing rules stated above, each preexisting strand functions as a template to guide the synthesis of one complementary strand (Fig. 8–17). These expectations were experimentally confirmed, inaugurating a revolution in our understanding of biological inheritance.

DNA Can Occur in Different Three-Dimensional Forms

DNA is a remarkably flexible molecule. Considerable rotation is possible around a number of bonds in the sugar–phosphate (phosphodeoxyribose) backbone, and thermal fluctuation can produce bending, stretching, and unpairing (melting) of the strands. Many significant deviations from the Watson-Crick DNA structure are found in cellular DNA, some or all of which may play important roles in DNA metabolism. These structural variations generally do not affect the key properties of DNA defined by Watson and Crick: strand complementarity,



antiparallel strands, and the requirement for A=T and G=C base pairs.

Structural variation in DNA reflects three things: the different possible conformations of the deoxyribose, rotation about the contiguous bonds that make up the phosphodeoxyribose backbone (Fig. 8-18a), and free rotation about the C-1'-N-glycosyl bond (Fig. 8-18b). Because of steric constraints, purines in purine nucleotides are restricted to two stable conformations with respect to deoxyribose, called syn and anti (Fig. 8-18b). Pyrimidines are generally restricted to the anti conformation because of steric interference between the sugar and the carbonyl oxygen at C-2 of the pyrimidine.

The Watson-Crick structure is also referred to as **B-form DNA**, or B-DNA. The B form is the most stable structure for a random-sequence DNA molecule under physiological conditions and is therefore the standard point of reference in any study of the properties of DNA. Two structural variants that have been well characterized in crystal structures are the **A** and **Z** forms. These three DNA conformations are shown in Figure 8-19, with a summary of their properties. The A form is favored in many solutions that are relatively devoid of water. The DNA is still arranged in a right-handed double helix, but the helix is wider and the number of base pairs per helical turn is 11, rather than 10.5 as in B-DNA. The

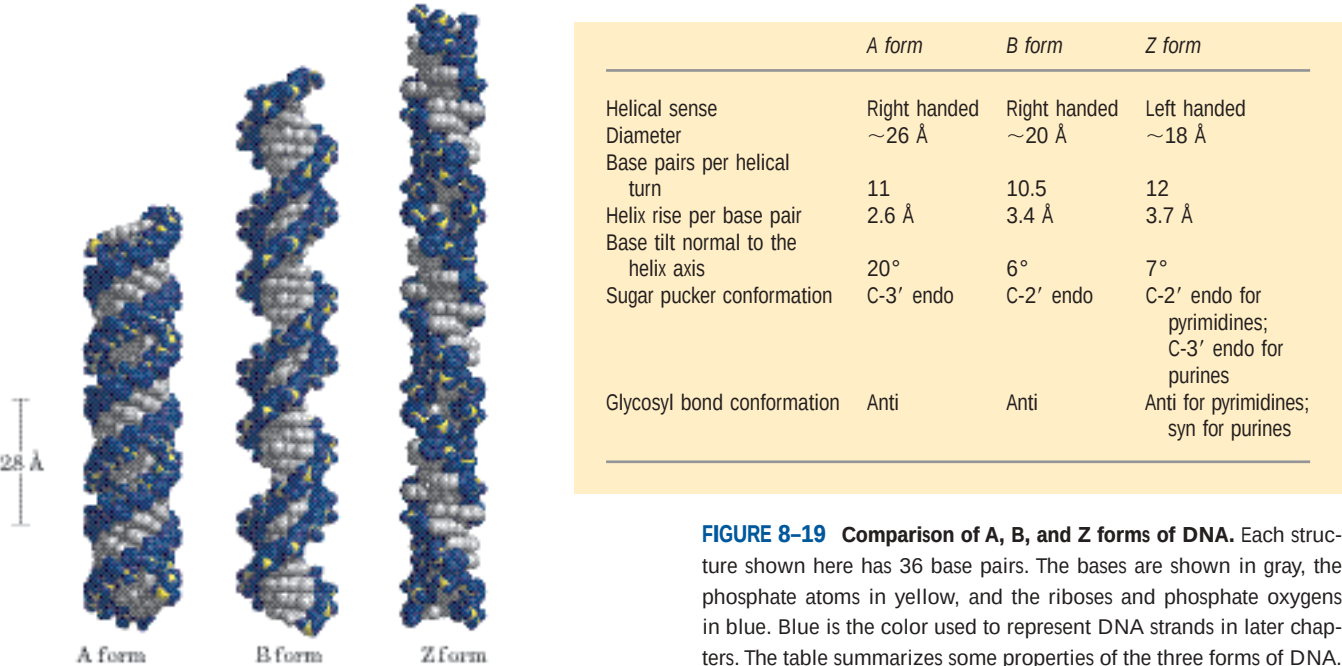


FIGURE 8-19 Comparison of A, B, and Z forms of DNA. Each structure shown here has 36 base pairs. The bases are shown in gray, the phosphate atoms in yellow, and the riboses and phosphate oxygens in blue. Blue is the color used to represent DNA strands in later chapters. The table summarizes some properties of the three forms of DNA.

plane of the base pairs in A-DNA is tilted about 20° with respect to the helix axis. These structural changes deepen the major groove while making the minor groove shallower. The reagents used to promote crystallization of DNA tend to dehydrate it, and thus most short DNA molecules tend to crystallize in the A form.

Z-form DNA is a more radical departure from the B structure; the most obvious distinction is the left-handed helical rotation. There are 12 base pairs per helical turn, and the structure appears more slender and elongated. The DNA backbone takes on a zigzag appearance. Certain nucleotide sequences fold into left-handed Z helices much more readily than others. Prominent examples are sequences in which pyrimidines alternate with purines, especially alternating C and G or 5-methyl-C and G residues. To form the left-handed helix in Z-DNA, the purine residues flip to the syn conformation, alternating with pyrimidines in the anti conformation. The major groove is barely apparent in Z-DNA, and the minor groove is narrow and deep.

Whether A-DNA occurs in cells is uncertain, but there is evidence for some short stretches (tracts) of Z-DNA in both prokaryotes and eukaryotes. These Z-DNA tracts may play a role (as yet undefined) in regulating the expression of some genes or in genetic recombination.

Certain DNA Sequences Adopt Unusual Structures

A number of other sequence-dependent structural variations have been detected within larger chromosomes that may affect the function and metabolism of the DNA segments in their immediate vicinity. For example, bends occur in the DNA helix wherever four or more adenosine residues appear sequentially in one strand. Six adenosines in a row produce a bend of about 18° .

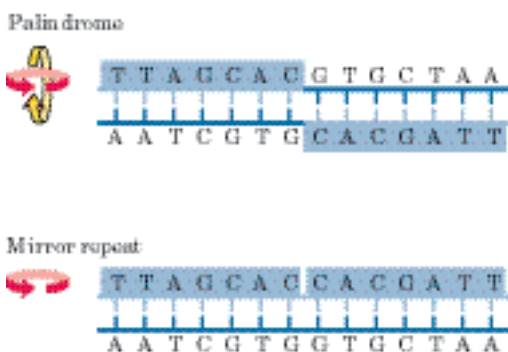


FIGURE 8-20 Palindromes and mirror repeats. Palindromes are sequences of double-stranded nucleic acids with twofold symmetry. In order to superimpose one repeat (shaded sequence) on the other, it must be rotated 180° about the horizontal axis then 180° about the vertical axis, as shown by the colored arrows. A mirror repeat, on the other hand, has a symmetric sequence within each strand. Superimposing one repeat on the other requires only a single 180° rotation about the vertical axis.

The bending observed with this and other sequences may be important in the binding of some proteins to DNA.

A rather common type of DNA sequence is a **palindrome**. A palindrome is a word, phrase, or sentence that is spelled identically read either forward or backward; two examples are ROTATOR and NURSES RUN. The term is applied to regions of DNA with **inverted repeats** of base sequence having twofold symmetry over two strands of DNA (Fig. 8-20). Such sequences are self-complementary within each strand and therefore have the potential to form **hairpin** or **cruciform** (cross-shaped) structures (Fig. 8-21). When the inverted repeat occurs within each individual strand of the DNA, the sequence is called a **mirror repeat**. Mirror repeats do not have complementary sequences within the same strand and cannot form hairpin or cruciform structures. Sequences of these types are found

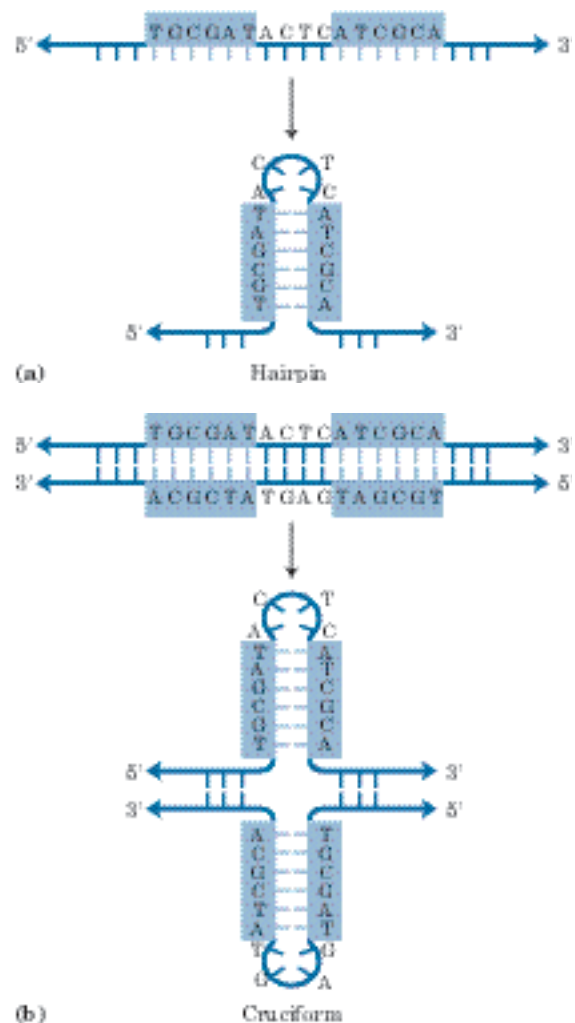


FIGURE 8-21 Hairpins and cruciforms. Palindromic DNA (or RNA) sequences can form alternative structures with intrastrand base pairing. **(a)** When only a single DNA (or RNA) strand is involved, the structure is called a hairpin. **(b)** When both strands of a duplex DNA are involved, it is called a cruciform. Blue shading highlights asymmetric sequences that can pair with the complementary sequence either in the same strand or in the complementary strand.

in virtually every large DNA molecule and can encompass a few base pairs or thousands. The extent to which palindromes occur as cruciforms in cells is not known, although some cruciform structures have been demonstrated *in vivo* in *E. coli*. Self-complementary sequences cause isolated single strands of DNA (or RNA) in solution to fold into complex structures containing multiple hairpins.

Several unusual DNA structures involve three or even four DNA strands. These structural variations merit investigation because there is a tendency for many of them to appear at sites where important events in DNA metabolism (replication, recombination, transcription) are initiated or regulated. Nucleotides participating in a

Watson-Crick base pair (Fig. 8–11) can form a number of additional hydrogen bonds, particularly with functional groups arrayed in the major groove. For example, a cytidine residue (if protonated) can pair with the guanosine residue of a $G \equiv C$ nucleotide pair, and a thymidine can pair with the adenosine of an $A = T$ pair (Fig. 8–22). The N-7, O^6 , and N^6 of purines, the atoms that participate in the hydrogen bonding of triplex DNA, are often referred to as **Hoogsteen positions**, and the non-Watson-Crick pairing is called **Hoogsteen pairing**, after Karst Hoogsteen, who in 1963 first recognized the potential for these unusual pairings. Hoogsteen pairing allows the formation of **triplex DNAs**. The triplexes shown in Figure 8–22 (a, b) are most stable at low pH

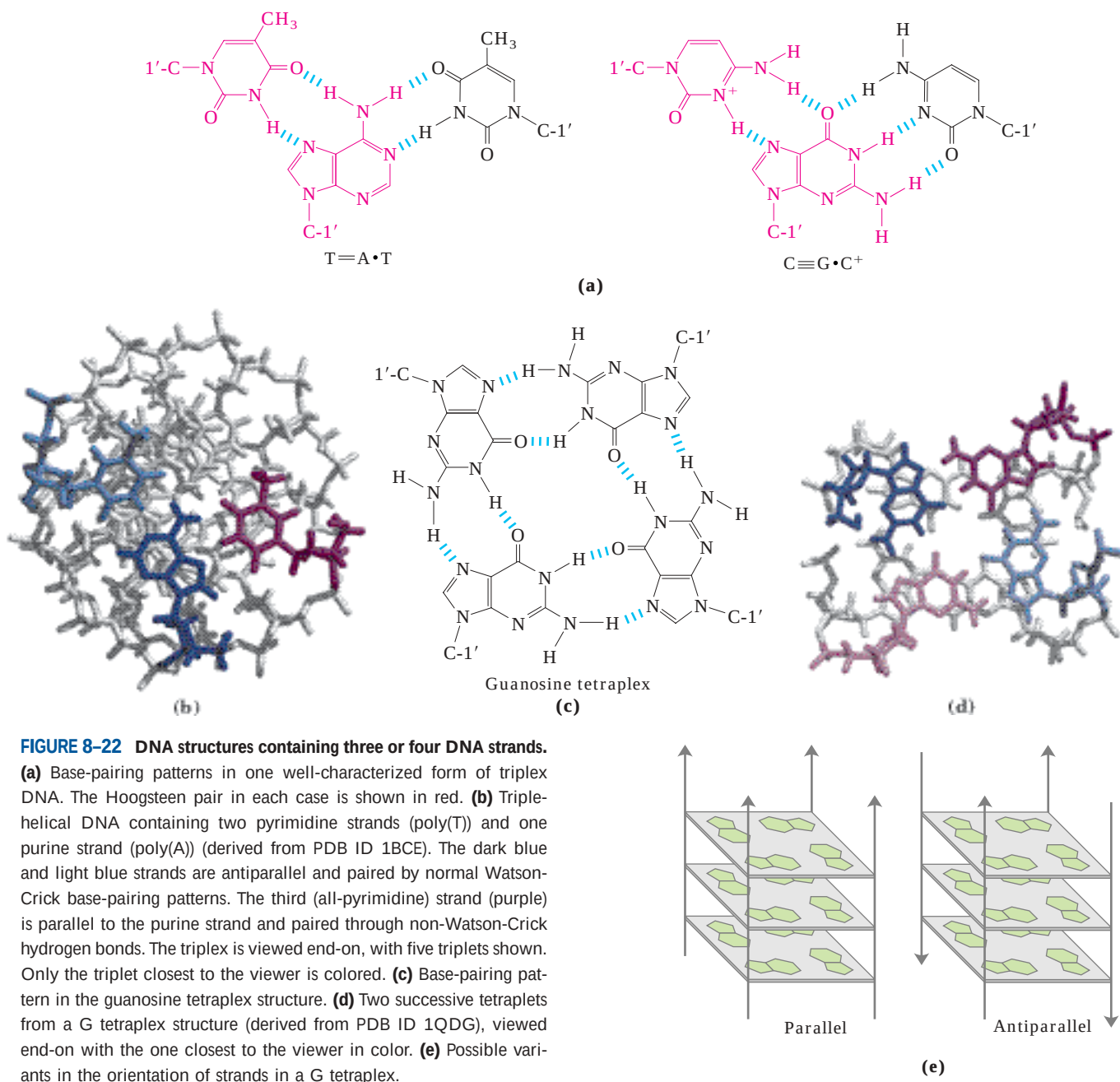


FIGURE 8–22 DNA structures containing three or four DNA strands. (a) Base-pairing patterns in one well-characterized form of triplex DNA. The Hoogsteen pair in each case is shown in red. (b) Triple-helical DNA containing two pyrimidine strands (poly(T)) and one purine strand (poly(A)) (derived from PDB ID 1BCE). The dark blue and light blue strands are antiparallel and paired by normal Watson-Crick base-pairing patterns. The third (all-pyrimidine) strand (purple) is parallel to the purine strand and paired through non-Watson-Crick hydrogen bonds. The triplex is viewed end-on, with five triplets shown. Only the triplet closest to the viewer is colored. (c) Base-pairing pattern in the guanosine tetraplex structure. (d) Two successive tetraplets from a G tetraplex structure (derived from PDB ID 1QDG), viewed end-on with the one closest to the viewer in color. (e) Possible variants in the orientation of strands in a G tetraplex.

because the $C\equiv G \cdot C^+$ triplet requires a protonated cytosine. In the triplex, the pK_a of this cytosine is >7.5 , altered from its normal value of 4.2. The triplexes also form most readily within long sequences containing only pyrimidines or only purines in a given strand. Some triplex DNAs contain two pyrimidine strands and one purine strand; others contain two purine strands and one pyrimidine strand.

Four DNA strands can also pair to form a tetraplex (quadruplex), but this occurs readily only for DNA sequences with a very high proportion of guanosine residues (Fig. 8–22c, d). The guanosine tetraplex, or **G tetraplex**, is quite stable over a wide range of conditions. The orientation of strands in the tetraplex can vary as shown in Figure 8–22e.

A particularly exotic DNA structure, known as **H-DNA**, is found in polypyrimidine or polypurine tracts that also incorporate a mirror repeat. A simple example is a long stretch of alternating T and C residues (Fig. 8–23). The H-DNA structure features the triple-stranded form illustrated in Figure 8–22 (a, b). Two of the three strands in the H-DNA triple helix contain pyrimidines and the third contains purines.

In the DNA of living cells, sites recognized by many sequence-specific DNA-binding proteins (Chapter 28) are arranged as palindromes, and polypyrimidine or polypurine sequences that can form triple helices or even H-DNA are found within regions involved in the regulation of expression of some eukaryotic genes. In principle, synthetic DNA strands designed to pair with these sequences to form triplex DNA could disrupt gene expression. This approach to controlling cellular metabolism is of growing commercial interest for its potential application in medicine and agriculture.

Messenger RNAs Code for Polypeptide Chains

We now turn our attention briefly from DNA structure to the expression of the genetic information that it contains. RNA, the second major form of nucleic acid in cells, has many functions. In gene expression, RNA acts as an intermediary by using the information encoded in DNA to specify the amino acid sequence of a functional protein.

Given that the DNA of eukaryotes is largely confined to the nucleus whereas protein synthesis occurs on ribosomes in the cytoplasm, some molecule other than DNA must carry the genetic message from the nucleus to the cytoplasm. As early as the 1950s, RNA was considered the logical candidate: RNA is found in both the nucleus and the cytoplasm, and an increase in protein synthesis is accompanied by an increase in the amount of cytoplasmic RNA and an increase in its rate of turnover. These and other observations led several researchers to suggest that RNA carries genetic information from DNA to the protein biosynthetic machin-

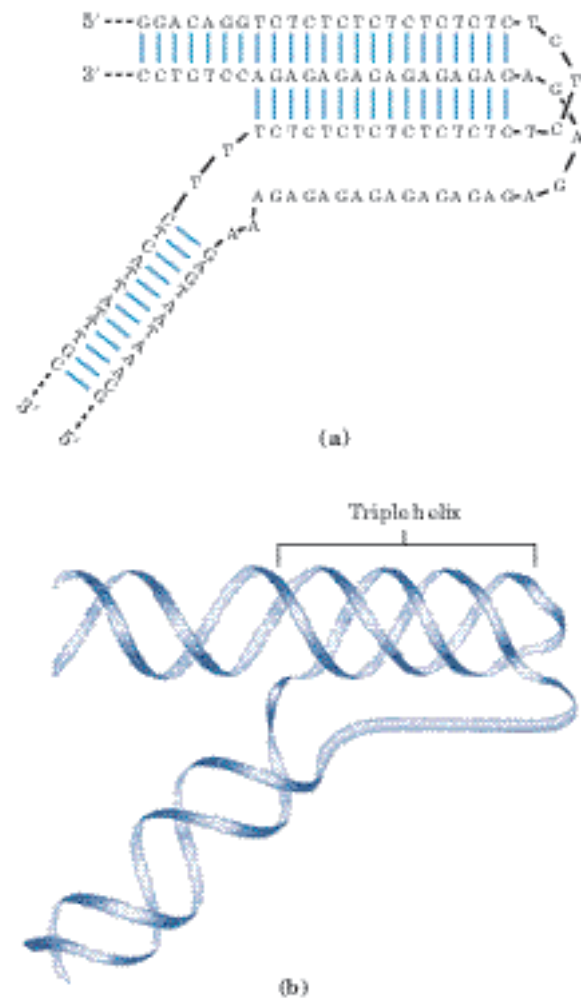


FIGURE 8–23 H-DNA. (a) A sequence of alternating T and C residues can be considered a mirror repeat centered about a central T or C. (b) These sequences form an unusual structure in which the strands in one half of the mirror repeat are separated and the pyrimidine-containing strand (alternating T and C residues) folds back on the other half of the repeat to form a triple helix. The purine strand (alternating A and G residues) is left unpaired. This structure produces a sharp bend in the DNA.

ery of the ribosome. In 1961 François Jacob and Jacques Monod presented a unified (and essentially correct) picture of many aspects of this process. They proposed the name “messenger RNA” (mRNA) for that portion of the total cellular RNA carrying the genetic information from DNA to the ribosomes, where the messengers provide the templates that specify amino acid sequences in polypeptide chains. Although mRNAs from different genes can vary greatly in length, the mRNAs from a particular gene generally have a defined size. The process of forming mRNA on a DNA template is known as transcription.

In prokaryotes, a single mRNA molecule may code for one or several polypeptide chains. If it carries the code for only one polypeptide, the mRNA is **monocistronic**;

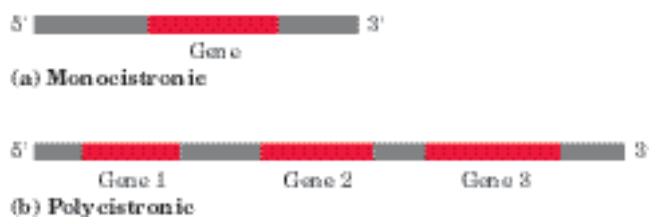


FIGURE 8-24 Prokaryotic mRNA. Schematic diagrams show (a) monocistronic and (b) polycistronic mRNAs of prokaryotes. Red segments represent RNA coding for a gene product; gray segments represent noncoding RNA. In the polycistronic transcript, noncoding RNA separates the three genes.

if it codes for two or more different polypeptides, the mRNA is **polycistronic**. In eukaryotes, most mRNAs are monocistronic. (For the purposes of this discussion, “cistron” refers to a gene. The term itself has historical roots in the science of genetics, and its formal genetic definition is beyond the scope of this text.) The minimum length of an mRNA is set by the length of the polypeptide chain for which it codes. For example, a polypeptide chain of 100 amino acid residues requires an RNA coding sequence of at least 300 nucleotides, because each amino acid is coded by a nucleotide triplet (this and other details of protein synthesis are discussed in Chapter 27). However, mRNAs transcribed from DNA are always somewhat longer than the length needed simply to code for a polypeptide sequence (or sequences). The additional, noncoding RNA includes sequences that regulate protein synthesis. Figure 8-24 summarizes the general structure of prokaryotic mRNAs.

Many RNAs Have More Complex Three-Dimensional Structures

Messenger RNA is only one of several classes of cellular RNA. Transfer RNAs serve as adapter molecules in protein synthesis; covalently linked to an amino acid at one end, they pair with the mRNA in such a way that amino acids are joined to a growing polypeptide in the correct sequence. Ribosomal RNAs are components of ribosomes. There is also a wide variety of special-function RNAs, including some (called ribozymes) that have enzymatic activity. All the RNAs are considered in detail in Chapter 26. The diverse and often complex functions of these RNAs reflect a diversity of structure much richer than that observed in DNA molecules.

The product of transcription of DNA is always single-stranded RNA. The single strand tends to assume a right-handed helical conformation dominated by base-stacking interactions (Fig. 8-25), which are stronger between two purines than between a purine and pyrimidine or between two pyrimidines. The purine-purine interaction is so strong that a pyrimidine separating two purines is often displaced from the stacking pattern so

that the purines can interact. Any self-complementary sequences in the molecule produce more complex structures. RNA can base-pair with complementary regions of either RNA or DNA. Base pairing matches the pattern for DNA: G pairs with C and A pairs with U (or with the occasional T residue in some RNAs). One difference is that base pairing between G and U residues—unusual in DNA—is fairly common in RNA (see Fig. 8-27). The paired strands in RNA or RNA-DNA duplexes are antiparallel, as in DNA.

RNA has no simple, regular secondary structure that serves as a reference point, as does the double helix for DNA. The three-dimensional structures of many RNAs, like those of proteins, are complex and unique. Weak interactions, especially base-stacking interactions, play a major role in stabilizing RNA structures, just as they do in DNA. Where complementary sequences are present, the predominant double-stranded structure is an A-form right-handed double helix. Z-form helices have been made in the laboratory (under very high-salt or high-temperature conditions). The B form of RNA has not been observed. Breaks in the regular A-form helix caused by mismatched or unmatched bases in one or both strands are common and result in bulges or internal loops (Fig. 8-26). Hairpin loops form between nearby self-complementary sequences. The potential for base-paired helical structures in many RNAs is extensive (Fig. 8-27), and the resulting hairpins are the most common type of secondary structure in RNA. Specific



FIGURE 8-25 Typical right-handed stacking pattern of single-stranded RNA. The bases are shown in gray, the phosphate atoms in yellow, and the riboses and phosphate oxygens in green. Green is used to represent RNA strands in succeeding chapters, just as blue is used for DNA.

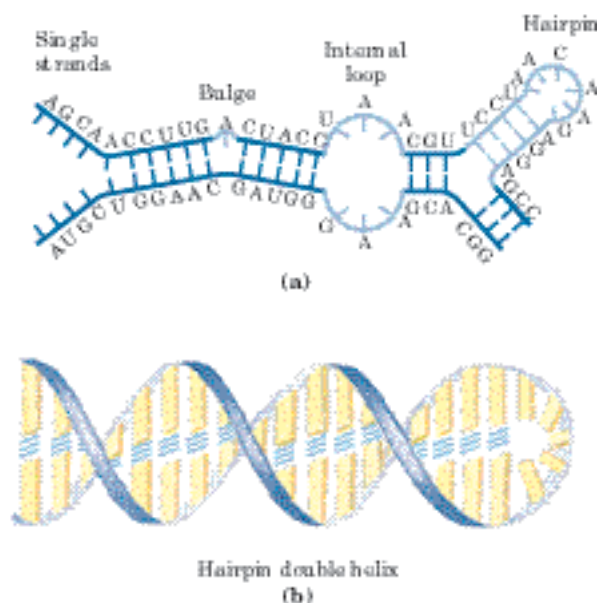
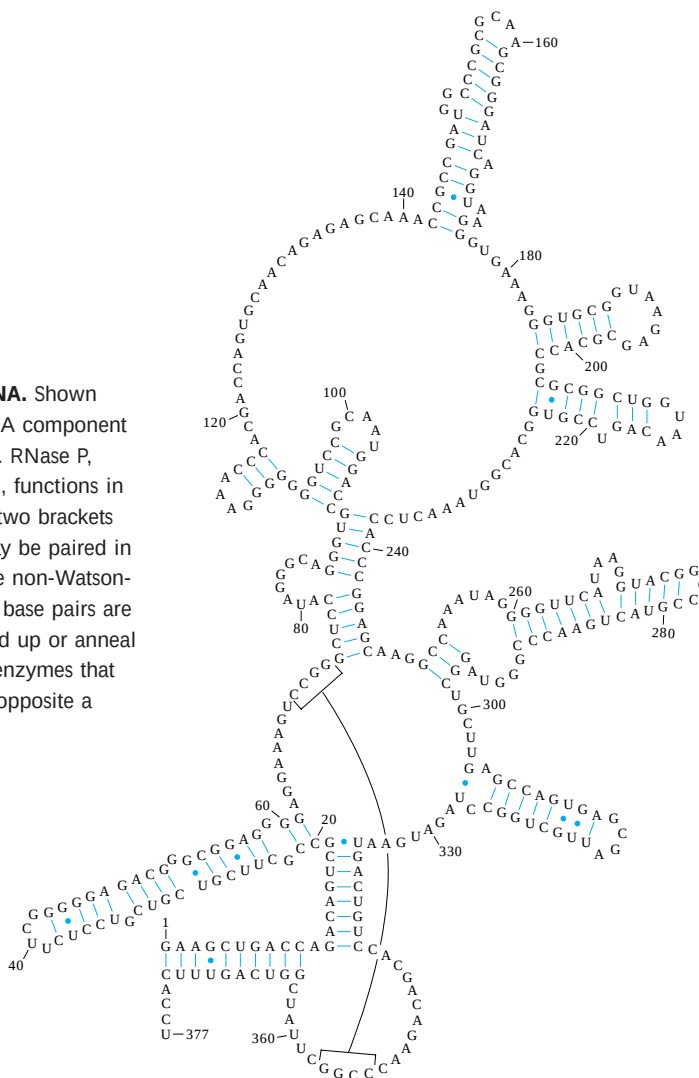
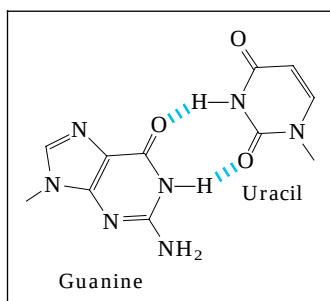


FIGURE 8-26 Secondary structure of RNAs. **(a)** Bulge, internal loop, and hairpin loop. **(b)** The paired regions generally have an A-form right-handed helix, as shown for a hairpin.

short base sequences (such as UUCG) are often found at the ends of RNA hairpins and are known to form particularly tight and stable loops. Such sequences may act as starting points for the folding of an RNA molecule into its precise three-dimensional structure. Important additional structural contributions are made by hydrogen bonds that are not part of standard Watson-Crick base pairs. For example, the 2'-hydroxyl group of ribose can hydrogen-bond with other groups. Some of these properties are evident in the structure of the phenylalanine transfer RNA of yeast—the tRNA responsible for inserting Phe residues into polypeptides—and in two RNA enzymes, or ribozymes, whose functions, like those of protein enzymes, depend on their three-dimensional structures (Fig. 8-28).

The analysis of RNA structure and the relationship between structure and function is an emerging field of inquiry that has many of the same complexities as the analysis of protein structure. The importance of understanding RNA structure grows as we become increasingly aware of the large number of functional roles for RNA molecules.

FIGURE 8-27 Base-paired helical structures in an RNA. Shown here is the possible secondary structure of the M1 RNA component of the enzyme RNase P of *E. coli*, with many hairpins. RNase P, which also contains a protein component (not shown), functions in the processing of transfer RNAs (see Fig. 26-23). The two brackets indicate additional complementary sequences that may be paired in the three-dimensional structure. The blue dots indicate non-Watson-Crick G=U base pairs (boxed inset). Note that G=U base pairs are allowed only when presynthesized strands of RNA fold up or anneal with each other. There are no RNA polymerases (the enzymes that synthesize RNAs on a DNA template) that insert a U opposite a template G, or vice versa, during RNA synthesis.



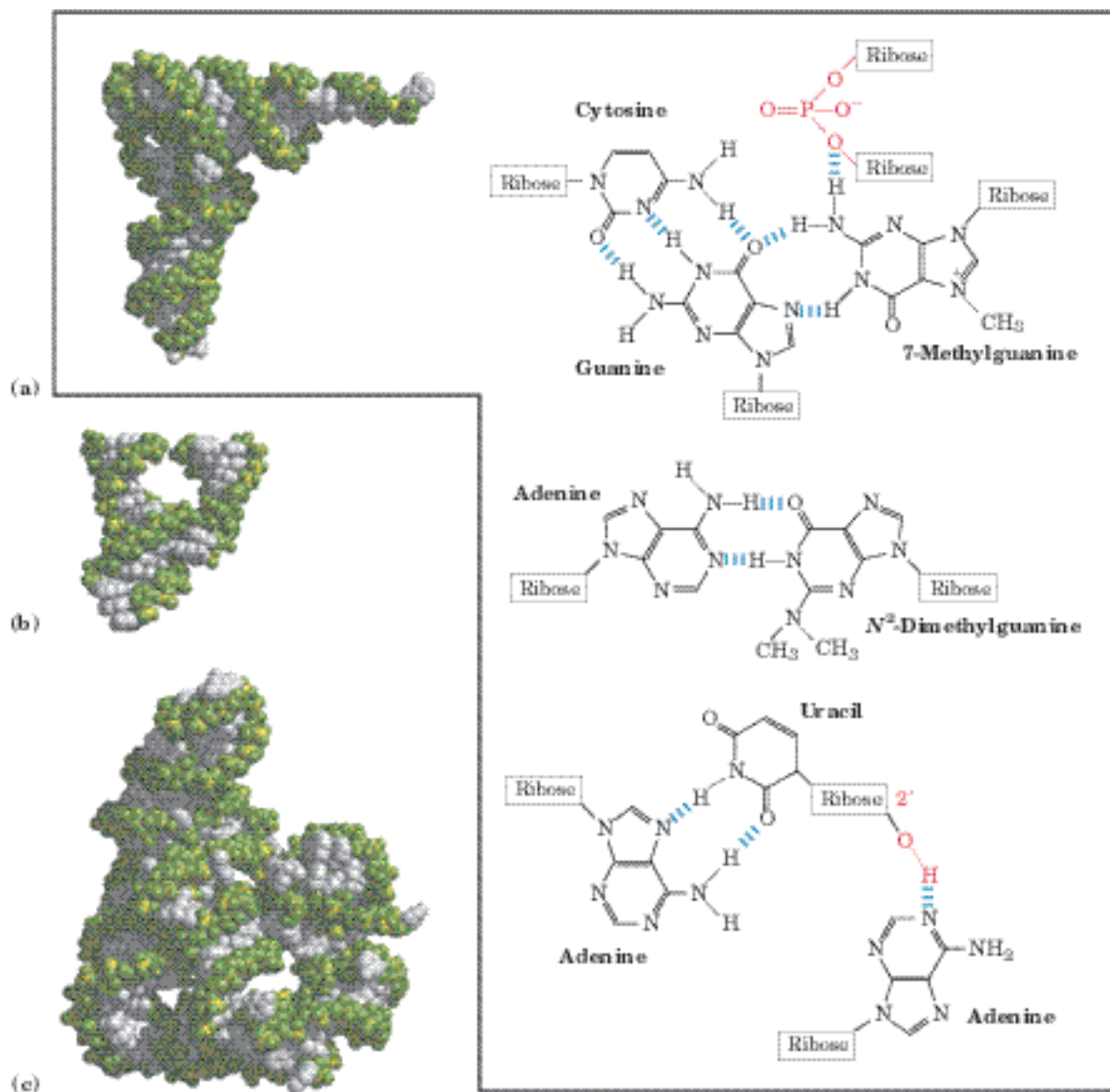


FIGURE 8-28 Three-dimensional structure in RNA. (a) Three-dimensional structure of phenylalanine tRNA of yeast (PDB ID 1TRA). Some unusual base-pairing patterns found in this tRNA are shown. Note also the involvement of the oxygen of a ribose phosphodiester bond in one hydrogen-bonding arrangement, and a ribose 2'-hydroxyl group in another (both in red). (b) A hammerhead ribozyme (so named because the secondary structure at the active site looks like the head of a hammer), derived from certain plant viruses (derived from PDB

ID 1MME). Ribozymes, or RNA enzymes, catalyze a variety of reactions, primarily in RNA metabolism and protein synthesis. The complex three-dimensional structures of these RNAs reflect the complexity inherent in catalysis, as described for protein enzymes in Chapter 6. (c) A segment of mRNA known as an intron, from the ciliated protozoan *Tetrahymena thermophila* (derived from PDB ID 1GRZ). This intron (a ribozyme) catalyzes its own excision from between exons in an mRNA strand (discussed in Chapter 26).

SUMMARY 8.2 Nucleic Acid Structure

- Many lines of evidence show that DNA bears genetic information. In particular, the Avery-MacLeod-McCarty experiment showed that DNA isolated from one bacterial strain can enter and transform the cells of another strain, endowing it with some of the inheritable characteristics of the donor. The Hershey-Chase experiment

showed that the DNA of a bacterial virus, but not its protein coat, carries the genetic message for replication of the virus in a host cell.

- Putting together much published data, Watson and Crick postulated that native DNA consists of two antiparallel chains in a right-handed double-helical arrangement. Complementary base pairs, A=T and G=C, are formed by hydrogen bonding within the helix. The base

pairs are stacked perpendicular to the long axis of the double helix, 3.4 Å apart, with 10.5 base pairs per turn.

- DNA can exist in several structural forms. Two variations of the Watson-Crick form, or B-DNA, are A- and Z-DNA. Some sequence-dependent structural variations cause bends in the DNA molecule. DNA strands with appropriate sequences can form hairpin/cruciform structures or triplex or tetraplex DNA.
- Messenger RNA transfers genetic information from DNA to ribosomes for protein synthesis. Transfer RNA and ribosomal RNA are also involved in protein synthesis. RNA can be structurally complex; single RNA strands can be folded into hairpins, double-stranded regions, or complex loops.

8.3 Nucleic Acid Chemistry

To understand how nucleic acids function, we must understand their chemical properties as well as their structures. The role of DNA as a repository of genetic information depends in part on its inherent stability. The chemical transformations that do occur are generally very slow in the absence of an enzyme catalyst. The long-term storage of information without alteration is so important to a cell, however, that even very slow reactions that alter DNA structure can be physiologically significant. Processes such as carcinogenesis and aging may be intimately linked to slowly accumulating, irreversible alterations of DNA. Other, nondestructive alterations also occur and are essential to function, such as the strand separation that must precede DNA replication or transcription. In addition to providing insights into physiological processes, our understanding of nucleic acid chemistry has given us a powerful array of technologies that have applications in molecular biology, medicine, and forensic science. We now examine the chemical properties of DNA and some of these technologies.

Double-Helical DNA and RNA Can Be Denatured

Solutions of carefully isolated, native DNA are highly viscous at pH 7.0 and room temperature (25 °C). When such a solution is subjected to extremes of pH or to temperatures above 80 °C, its viscosity decreases sharply, indicating that the DNA has undergone a physical change. Just as heat and extremes of pH denature globular proteins, they also cause denaturation, or melting, of double-helical DNA. Disruption of the hydrogen bonds between paired bases and of base stacking causes unwinding of the double helix to form two single strands, completely separate from each other along the entire

length or part of the length (partial denaturation) of the molecule. No covalent bonds in the DNA are broken (Fig. 8–29).

Renaturation of a DNA molecule is a rapid one-step process, as long as a double-helical segment of a dozen or more residues still unites the two strands. When the temperature or pH is returned to the range in which most organisms live, the unwound segments of the two strands spontaneously rewind, or **anneal**, to yield the intact duplex (Fig. 8–29). However, if the two strands are completely separated, renaturation occurs in two steps. In the first, relatively slow step, the two strands “find” each other by random collisions and form a short segment of complementary double helix. The second step is much faster: the remaining unpaired bases successively come into register as base pairs, and the two strands “zipper” themselves together to form the double helix.

The close interaction between stacked bases in a nucleic acid has the effect of decreasing its absorption of UV light relative to that of a solution with the same concentration of free nucleotides, and the absorption is decreased further when two complementary nucleic acids strands are paired. This is called the hypochromic effect. Denaturation of a double-stranded nucleic acid produces the opposite result: an increase in absorption

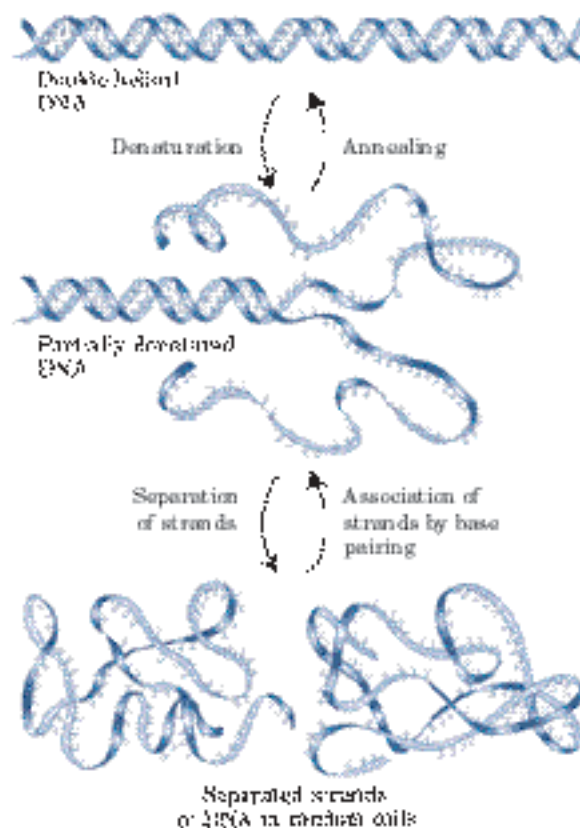


FIGURE 8–29 Reversible denaturation and annealing (renaturation) of DNA.

called the hyperchromic effect. The transition from double-stranded DNA to the single-stranded, denatured form can thus be detected by monitoring the absorption of UV light.

Viral or bacterial DNA molecules in solution denature when they are heated slowly (Fig. 8–30). Each species of DNA has a characteristic denaturation temperature, or melting point (t_m): the higher its content of G≡C base pairs, the higher the melting point of the DNA. This is because G≡C base pairs, with three hydrogen bonds, require more heat energy to dissociate than A=T base pairs. Careful determination of the melting point of a DNA specimen, under fixed conditions of pH and ionic strength, can yield an estimate of its base composition. If denaturation conditions are carefully controlled, regions that are rich in A=T base pairs will specifically denature while most of the DNA remains

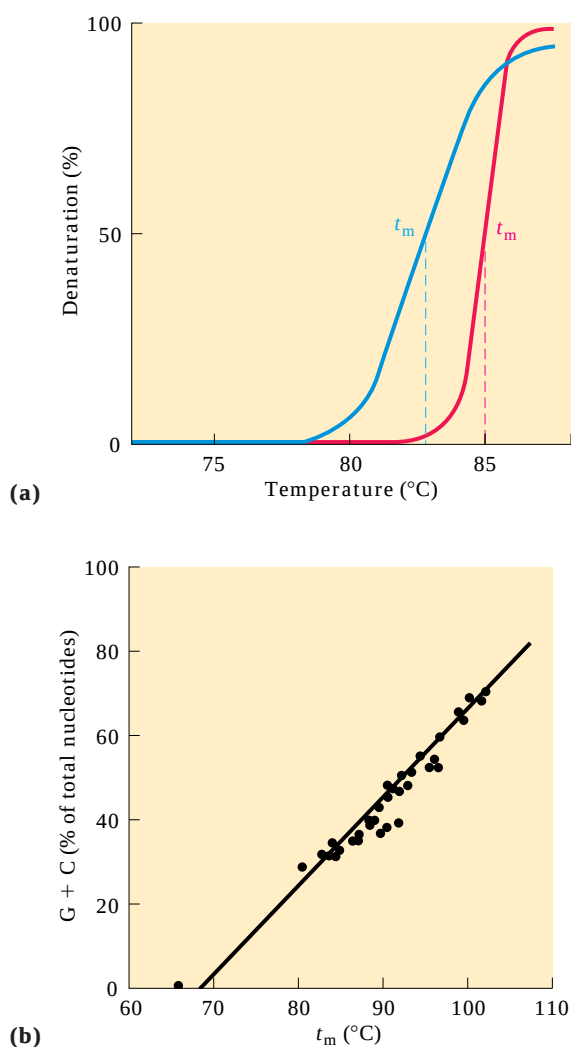


FIGURE 8–30 Heat denaturation of DNA. (a) The denaturation, or melting, curves of two DNA specimens. The temperature at the midpoint of the transition (t_m) is the melting point; it depends on pH and ionic strength and on the size and base composition of the DNA. (b) Relationship between t_m and the G≡C content of a DNA.

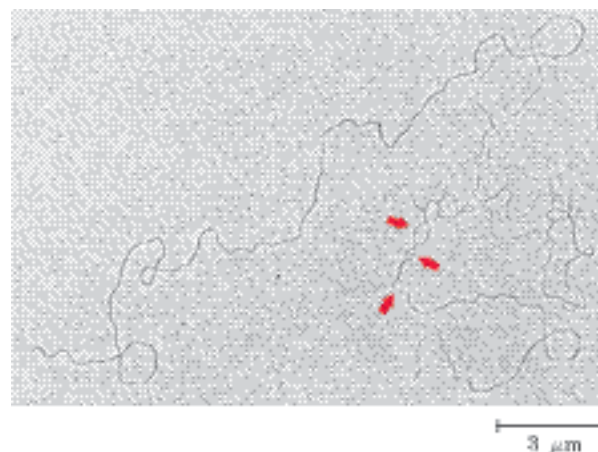


FIGURE 8–31 Partially denatured DNA. This DNA was partially denatured, then fixed to prevent renaturation during sample preparation. The shadowing method used to visualize the DNA in this electron micrograph increases its diameter approximately fivefold and obliterates most details of the helix. However, length measurements can be obtained, and single-stranded regions are readily distinguishable from double-stranded regions. The arrows point to some single-stranded bubbles where denaturation has occurred. The regions that denature are highly reproducible and are rich in A=T base pairs.

double-stranded. Such denatured regions (called bubbles) can be visualized with electron microscopy (Fig. 8–31). Strand separation of DNA *must* occur *in vivo* during processes such as DNA replication and transcription. As we shall see, the DNA sites where these processes are initiated are often rich in A=T base pairs.

Duplexes of two RNA strands or of one RNA strand and one DNA strand (RNA-DNA hybrids) can also be denatured. Notably, RNA duplexes are more stable than DNA duplexes. At neutral pH, denaturation of a double-helical RNA often requires temperatures 20 °C or more higher than those required for denaturation of a DNA molecule with a comparable sequence. The stability of an RNA-DNA hybrid is generally intermediate between that of RNA and that of DNA. The physical basis for these differences in thermal stability is not known.

Nucleic Acids from Different Species Can Form Hybrids

The ability of two complementary DNA strands to pair with one another can be used to detect similar DNA sequences in two different species or within the genome of a single species. If duplex DNAs isolated from human cells and from mouse cells are completely denatured by heating, then mixed and kept at 65 °C for many hours, much of the DNA will anneal. Most of the mouse DNA strands anneal with complementary mouse DNA strands to form mouse duplex DNA; similarly, most human DNA strands anneal with complementary human DNA strands. However, some strands of the mouse DNA will associate with human DNA strands to yield **hybrid**

duplexes, in which segments of a mouse DNA strand form base-paired regions with segments of a human DNA strand (Fig. 8–32). This reflects a common evolutionary heritage; different organisms generally have some proteins and RNAs with similar functions and, often, similar structures. In many cases, the DNAs encoding these proteins and RNAs have similar sequences. The closer the evolutionary relationship between two species, the more extensively their DNAs will hybridize. For example, human DNA hybridizes much more extensively with mouse DNA than with DNA from yeast.

The hybridization of DNA strands from different sources forms the basis for a powerful set of techniques essential to the practice of modern molecular genetics. A specific DNA sequence or gene can be detected in the presence of many other sequences, if one already has an appropriate complementary DNA strand (usually labeled in some way) to hybridize with it (Chapter 9). The complementary DNA can be from a different species or from the same species, or it can be synthesized chemically in the laboratory using techniques described later

in this chapter. Hybridization techniques can be varied to detect a specific RNA rather than DNA. The isolation and identification of specific genes and RNAs rely on these hybridization techniques. Applications of this technology make possible the identification of an individual on the basis of a single hair left at the scene of a crime or the prediction of the onset of a disease decades before symptoms appear (see Box 9–1).

Nucleotides and Nucleic Acids Undergo Nonenzymatic Transformations

Purines and pyrimidines, along with the nucleotides of which they are a part, undergo a number of spontaneous alterations in their covalent structure. The rate of these reactions is generally *very slow*, but they are physiologically significant because of the cell's very low tolerance for alterations in its genetic information. Alterations in DNA structure that produce permanent changes in the genetic information encoded therein are called **mutations**, and much evidence suggests an intimate link between the accumulation of mutations in an individual organism and the processes of aging and carcinogenesis.

Several nucleotide bases undergo spontaneous loss of their exocyclic amino groups (deamination) (Fig. 8–33a). For example, under typical cellular conditions, deamination of cytosine (in DNA) to uracil occurs in about one of every 10^7 cytidine residues in 24 hours. This corresponds to about 100 spontaneous events per day, on average, in a mammalian cell. Deamination of adenine and guanine occurs at about 1/100th this rate.

The slow cytosine deamination reaction seems innocuous enough, but is almost certainly the reason why DNA contains thymine rather than uracil. The product of cytosine deamination (uracil) is readily recognized as foreign in DNA and is removed by a repair system (Chapter 25). If DNA normally contained uracil, recognition of uracils resulting from cytosine deamination would be more difficult, and unrepaired uracils would lead to permanent sequence changes as they were paired with adenines during replication. Cytosine deamination would gradually lead to a decrease in G≡C base pairs and an increase in A=U base pairs in the DNA of all cells. Over the millennia, cytosine deamination could eliminate G≡C base pairs and the genetic code that depends on them. Establishing thymine as one of the four bases in DNA may well have been one of the crucial turning points in evolution, making the long-term storage of genetic information possible.

Another important reaction in deoxyribonucleotides is the hydrolysis of the *N*-β-glycosyl bond between the base and the pentose (Fig. 8–33b). This occurs at a higher rate for purines than for pyrimidines. As many as one in 10^5 purines (10,000 per mammalian cell) are lost from DNA every 24 hours under typical

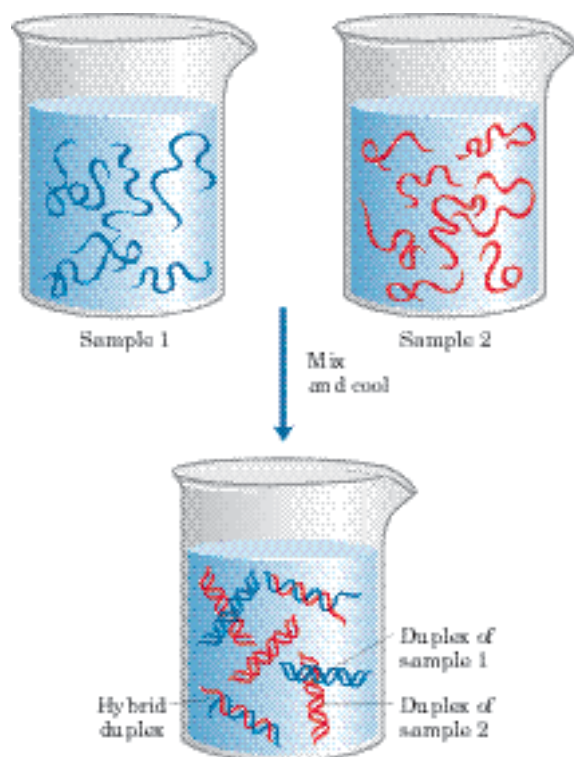
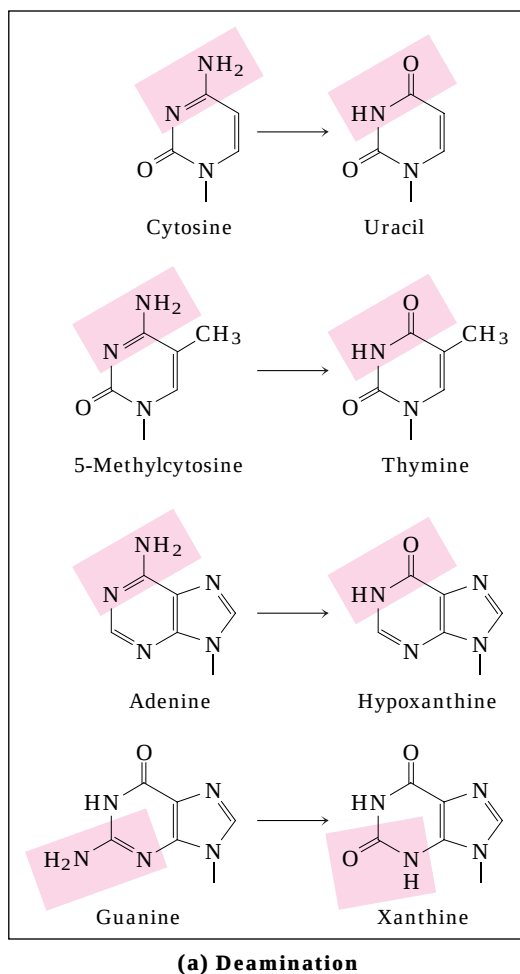


FIGURE 8–32 DNA hybridization. Two DNA samples to be compared are completely denatured by heating. When the two solutions are mixed and slowly cooled, DNA strands of each sample associate with their normal complementary partner and anneal to form duplexes. If the two DNAs have significant sequence similarity, they also tend to form partial duplexes or hybrids with each other: the greater the sequence similarity between the two DNAs, the greater the number of hybrids formed. Hybrid formation can be measured in several ways. One of the DNAs is usually labeled with a radioactive isotope to simplify the measurements.



cellular conditions. Depurination of ribonucleotides and RNA is much slower and generally is not considered physiologically significant. In the test tube, loss of purines can be accelerated by dilute acid. Incubation of DNA at pH 3 causes selective removal of the purine bases, resulting in a derivative called apurinic acid.

Other reactions are promoted by radiation. UV light induces the condensation of two ethylene groups to form a cyclobutane ring. In the cell, the same reaction between adjacent pyrimidine bases in nucleic acids forms cyclobutane pyrimidine dimers. This happens most frequently between adjacent thymidine residues on the same DNA strand (Fig. 8-34). A second type of pyrimidine dimer, called a 6-4 photoproduct, is also formed during UV irradiation. Ionizing radiation (x rays and gamma rays) can cause ring opening and fragmentation of bases as well as breaks in the covalent backbone of nucleic acids.

Virtually all forms of life are exposed to energy-rich radiation capable of causing chemical changes in DNA. Near-UV radiation (with wavelengths of 200 to 400 nm), which makes up a significant portion of the solar spectrum, is known to cause pyrimidine dimer formation and other chemical changes in the DNA of bacteria and of

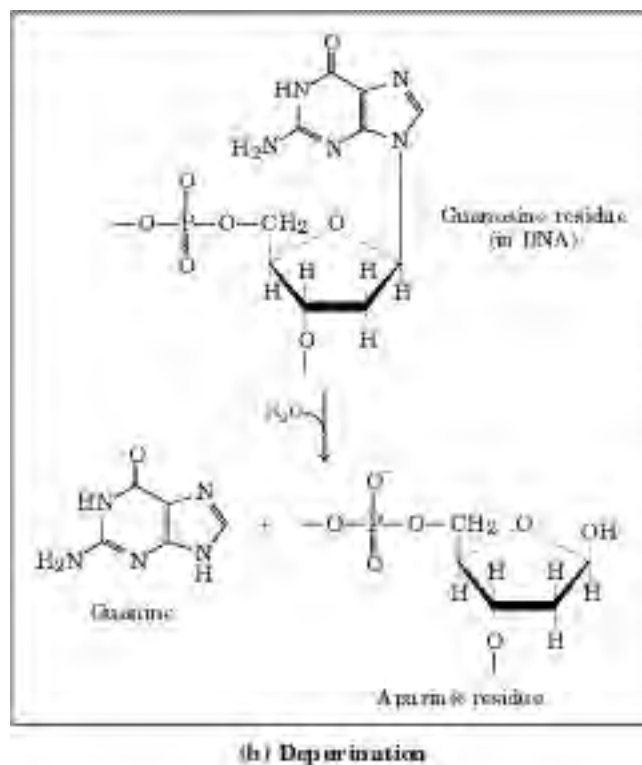


FIGURE 8-33 Some well-characterized nonenzymatic reactions of nucleotides. **(a)** Deamination reactions. Only the base is shown. **(b)** Depurination, in which a purine is lost by hydrolysis of the *N*- β -glycosyl bond. The deoxyribose remaining after depurination is readily converted from the β -furanose to the aldehyde form (see Fig. 8-3). Further nonenzymatic reactions are illustrated in Figures 8-34 and 8-35.

human skin cells. We are subject to a constant field of ionizing radiation in the form of cosmic rays, which can penetrate deep into the earth, as well as radiation emitted from radioactive elements, such as radium, plutonium, uranium, radon, ^{14}C , and ^3H . X rays used in medical and dental examinations and in radiation therapy of cancer and other diseases are another form of ionizing radiation. It is estimated that UV and ionizing radiations are responsible for about 10% of all DNA damage caused by environmental agents.

DNA also may be damaged by reactive chemicals introduced into the environment as products of industrial activity. Such products may not be injurious per se but may be metabolized by cells into forms that are. Two prominent classes of such agents (Fig. 8-35) are (1) deaminating agents, particularly nitrous acid (HNO_2) or compounds that can be metabolized to nitrous acid or nitrites, and (2) alkylating agents.

Nitrous acid, formed from organic precursors such as nitrosamines and from nitrite and nitrate salts, is a potent accelerator of the deamination of bases. Bisulfite has similar effects. Both agents are used as preservatives in processed foods to prevent the growth of toxic bacteria. They do not appear to increase cancer risks

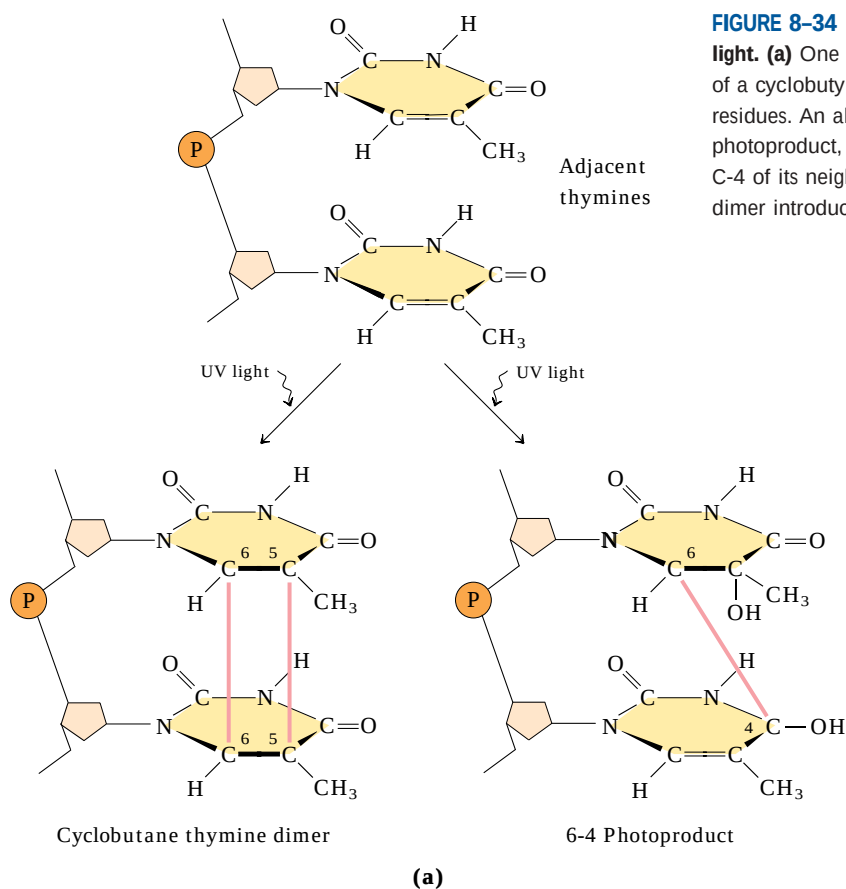
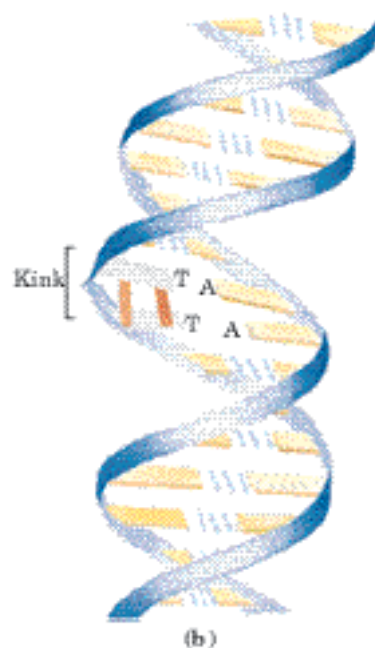
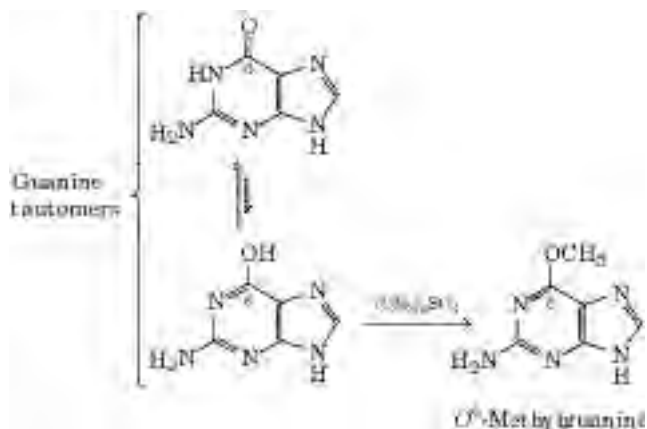


FIGURE 8-34 Formation of pyrimidine dimers induced by UV light. (a) One type of reaction (on the left) results in the formation of a cyclobutyl ring involving C-5 and C-6 of adjacent pyrimidine residues. An alternative reaction (on the right) results in a 6-4 photoproduct, with a linkage between C-6 of one pyrimidine and C-4 of its neighbor. (b) Formation of a cyclobutane pyrimidine dimer introduces a bend or kink into the DNA.



significantly when used in this way, perhaps because they are used in small amounts and make only a minor contribution to the overall levels of DNA damage. (The potential health risk from food spoilage if these preservatives were not used is much greater.)

Alkylating agents can alter certain bases of DNA. For example, the highly reactive chemical dimethylsulfate (Fig. 8-35b) can methylate a guanine to yield O^6 -methylguanine, which cannot base-pair with cytosine.



Many similar reactions are brought about by alkylating agents normally present in cells, such as *S*-adenosyl-methionine.

Possibly the most important source of mutagenic alterations in DNA is oxidative damage. Excited-oxygen species such as hydrogen peroxide, hydroxyl radicals, and superoxide radicals arise during irradiation or as a byproduct of aerobic metabolism. Of these species, the hydroxyl radicals are responsible for most oxidative DNA damage. Cells have an elaborate defense system to destroy reactive oxygen species, including enzymes such as catalase and superoxide dismutase that convert reactive oxygen species to harmless products. A fraction of these oxidants inevitably escape cellular defenses, however, and damage to DNA occurs through any of a large, complex group of reactions ranging from oxidation of deoxyribose and base moieties to strand breaks. Accurate estimates for the extent of this damage are not yet available, but every day the DNA of each human cell is subjected to thousands of damaging oxidative reactions.

This is merely a sampling of the best-understood reactions that damage DNA. Many carcinogenic compounds in food, water, or air exert their cancer-causing effects by modifying bases in DNA. Nevertheless, the integrity of DNA as a polymer is better maintained than that of either RNA or protein, because DNA is the only macromolecule that has the benefit of biochemical repair systems. These repair processes (described in Chapter 25) greatly lessen the impact of damage to DNA.

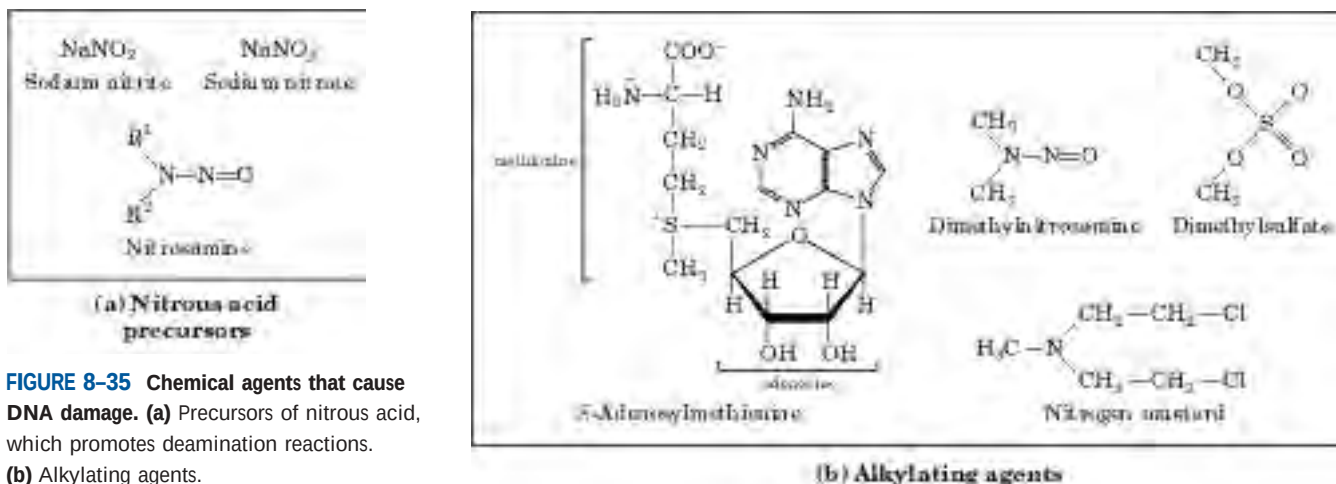


FIGURE 8-35 Chemical agents that cause DNA damage. **(a)** Precursors of nitrous acid, which promotes deamination reactions. **(b)** Alkylating agents.

Some Bases of DNA Are Methylated

Certain nucleotide bases in DNA molecules are enzymatically methylated. Adenine and cytosine are methylated more often than guanine and thymine. Methylation is generally confined to certain sequences or regions of a DNA molecule. In some cases the function of methylation is well understood; in others the function remains unclear. All known DNA methylases use S-adenosylmethionine as a methyl group donor. *E. coli* has two prominent methylation systems. One serves as part of a defense mechanism that helps the cell to distinguish its DNA from foreign DNA by marking its own DNA with methyl groups and destroying (foreign) DNA without the methyl groups (this is known as a restriction-modification system; see Chapter 9). The other system methylates adenosine residues within the sequence (5')GATC(3') to *N*⁶-methyladenosine (Fig. 8-5a). This is mediated by the Dam (DNA adenine methylation) methylase, a component of a system that repairs mismatched base pairs formed occasionally during DNA replication (see Fig. 25-20).

In eukaryotic cells, about 5% of cytidine residues in DNA are methylated to 5-methylcytidine (Fig. 8-5a). Methylation is most common at CpG sequences, producing methyl-CpG symmetrically on both strands of the DNA. The extent of methylation of CpG sequences varies by molecular region in large eukaryotic DNA molecules. Methylation suppresses the migration of segments of DNA called transposons, described in Chapter 25. These methylations of cytosine also have structural significance. The presence of 5-methylcytosine in an alternating CpG sequence markedly increases the tendency for that segment of DNA to assume the Z form.

The Sequences of Long DNA Strands Can Be Determined

In its capacity as a repository of information, a DNA molecule's most important property is its nucleotide se-

quence. Until the late 1970s, determining the sequence of a nucleic acid containing even five or ten nucleotides was difficult and very laborious. The development of two new techniques in 1977, one by Alan Maxam and Walter Gilbert and the other by Frederick Sanger, has made possible the sequencing of ever larger DNA molecules with an ease unimagined just a few decades ago. The techniques depend on an improved understanding of nucleotide chemistry and DNA metabolism, and on electrophoretic methods for separating DNA strands differing in size by only one nucleotide. Electrophoresis of DNA is similar to that of proteins (see Fig. 3-19). Polyacrylamide is often used as the gel matrix in work with short DNA molecules (up to a few hundred nucleotides); agarose is generally used for longer pieces of DNA.

In both Sanger and Maxam-Gilbert sequencing, the general principle is to reduce the DNA to four sets of labeled fragments. The reaction producing each set is base-specific, so the lengths of the fragments correspond to positions in the DNA sequence where a certain base occurs. For example, for an oligonucleotide with the sequence pAATCGACT, labeled at the 5' end (the left end), a reaction that breaks the DNA after each C residue will generate two labeled fragments: a four-nucleotide and a seven-nucleotide fragment; a reaction that breaks the DNA after each G will produce only one labeled, five-nucleotide fragment. Because the fragments are radioactively labeled at their 5' ends, only the fragment to the 5' side of the break is visualized. The fragment sizes correspond to the relative positions of C and G residues in the sequence. When the sets of fragments corresponding to each of the four bases are electrophoretically separated side by side, they produce a ladder of bands from which the sequence can be read directly (Fig. 8-36). We illustrate only the Sanger method, because it has proven to be technically easier and is in more widespread use. It requires the enzymatic synthesis of a DNA strand complementary to the strand under analysis, using a radioactively labeled "primer" and dideoxynucleotides.

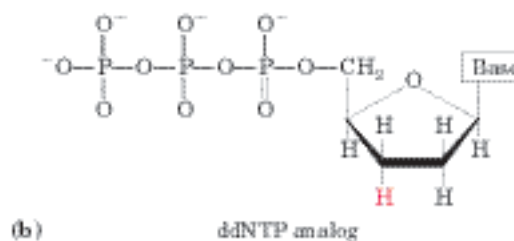
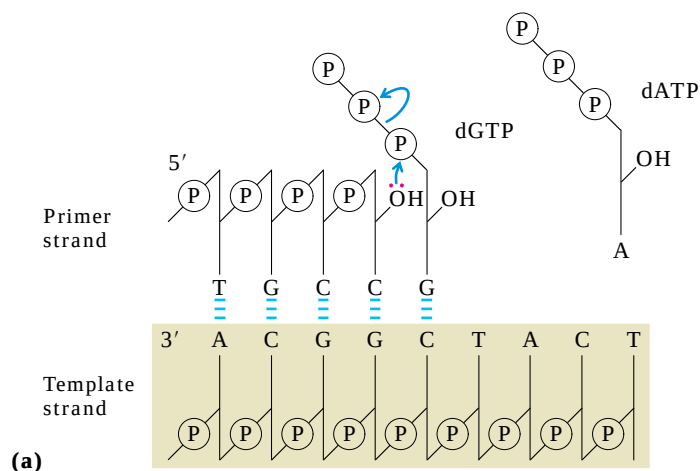
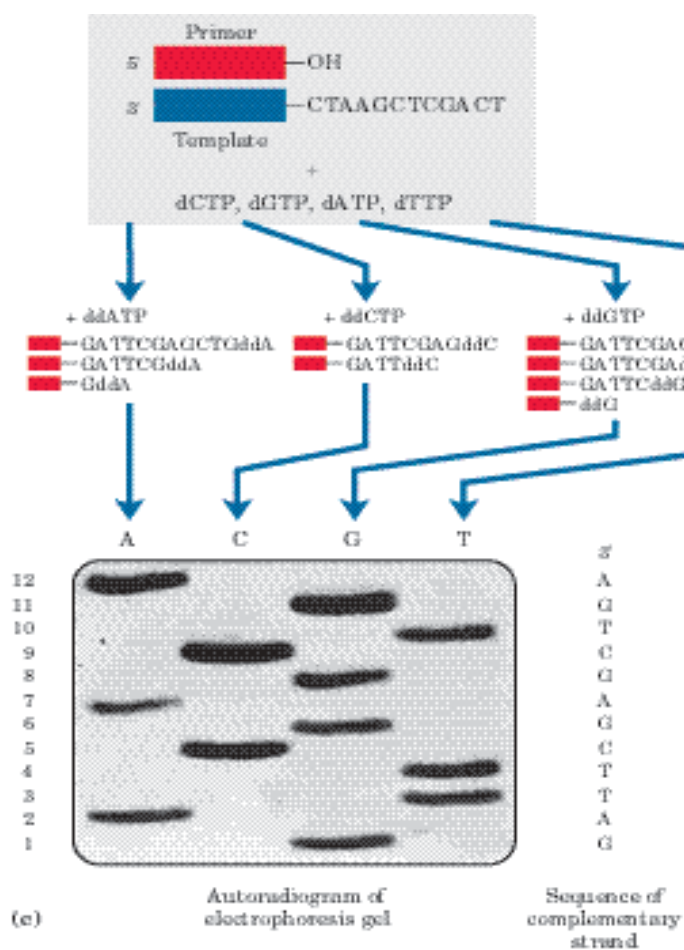


FIGURE 8-36 DNA sequencing by the Sanger method. This method makes use of the mechanism of DNA synthesis by DNA polymerases (Chapter 25). **(a)** DNA polymerases require both a primer (a short oligonucleotide strand), to which nucleotides are added, and a template strand to guide selection of each new nucleotide. In cells, the 3'-hy-

is with an incoming deoxynucleoside *v* phosphodiester bond. **(b)** The Sanger dideoxynucleoside triphosphate (ddNTP) is. (The Sanger method is also known as the dideoxy method. ddNTP is inserted in place of a dNTP, the analog is added, because it lacks the 3' hydroxyl group needed for the next step.

to be sequenced is used as the template, and a short primer, radioactively or fluorescently labeled, is annealed to it. By addition of small amounts of a single ddNTP, for example ddCTP, to an otherwise reaction system, the synthesized strand is prematurely terminated at some location. This normally occurs. Given the excess of template, the analog will be incorporated at a low frequency. However, ddCTP is present in such a concentration that each new strand has a high probability of being terminated at some point during synthesis. The mixture of labeled fragments, each ending at a different position in the sequence, generates a set of bands, such that the different-sized fragments, when separated by electrophoresis, reveal the location of C residues. By repeating the experiment separately for each of the four ddNTPs, and exposing the autoradiogram of the gel, the sequence can be determined. Fragments that migrate faster, the fragments near the 5' end, correspond to cleavage positions closest to the primer. The sequence read (in the 5' → 3' direction) from the bottom of the gel to the top is the sequence obtained is that of the strand being analyzed.



DNA sequencing is readily automated by a variation of Sanger's sequencing method in which the dideoxynucleotides used for each reaction are labeled with a differently colored fluorescent tag (Fig. 8–37). This technology allows DNA sequences containing thousands of nucleotides to be determined in a few hours. Entire genomes of many organisms have now been sequenced (see Table 1–4), and many very large DNA-sequencing projects are in progress. Perhaps the most ambitious of these is the Human Genome Project, in which researchers have sequenced all 3.2 billion base pairs of the DNA in a human cell (Chapter 9).

Dideoxy Sequencing of DNA

The Chemical Synthesis of DNA Has Been Automated

Another technology that has paved the way for many biochemical advances is the chemical synthesis of oligonucleotides with any chosen sequence. The chemical methods for synthesizing nucleic acids were developed primarily by H. Gobind Khorana and his colleagues in the 1970s. Refinement and automation of these methods have made possible the rapid and accurate synthesis of DNA strands. The synthesis is carried out with the growing strand attached to a solid support (Fig. 8–38), using principles similar to those used by Merrifield in peptide synthesis (see Fig. 3–29). The efficiency of each addition step is very high, allowing the routine laboratory synthesis of polymers containing 70 or 80 nucleotides and, in some laboratories, much longer strands. The availability of relatively inexpensive DNA polymers with predesigned sequences is having a powerful impact on all areas of biochemistry (Chapter 9).

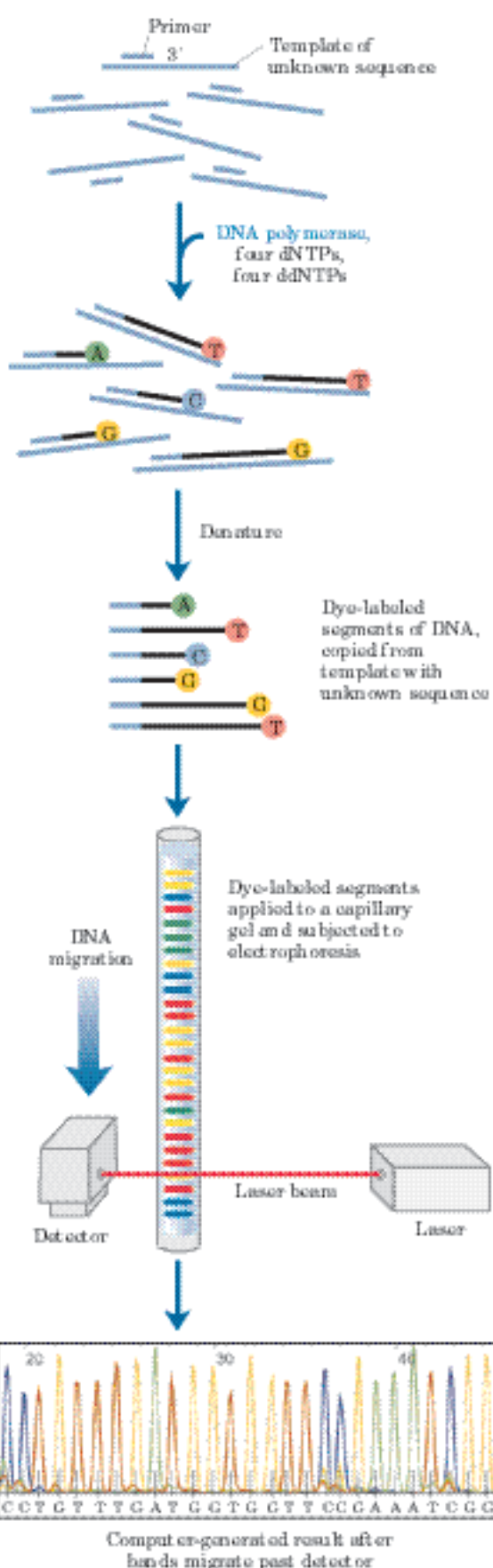


FIGURE 8–37 Strategy for automating DNA sequencing reactions.

Each dideoxynucleotide used in the Sanger method can be linked to a fluorescent molecule that gives all the fragments terminating in that nucleotide a particular color. All four labeled ddNTPs are added to a single tube. The resulting colored DNA fragments are then separated by size in a single electrophoretic gel contained in a capillary tube (a refinement of gel electrophoresis that allows for faster separations). All fragments of a given length migrate through the capillary gel in a single peak, and the color associated with each peak is detected using a laser beam. The DNA sequence is read by determining the sequence of colors in the peaks as they pass the detector. This information is fed directly to a computer, which determines the sequence.

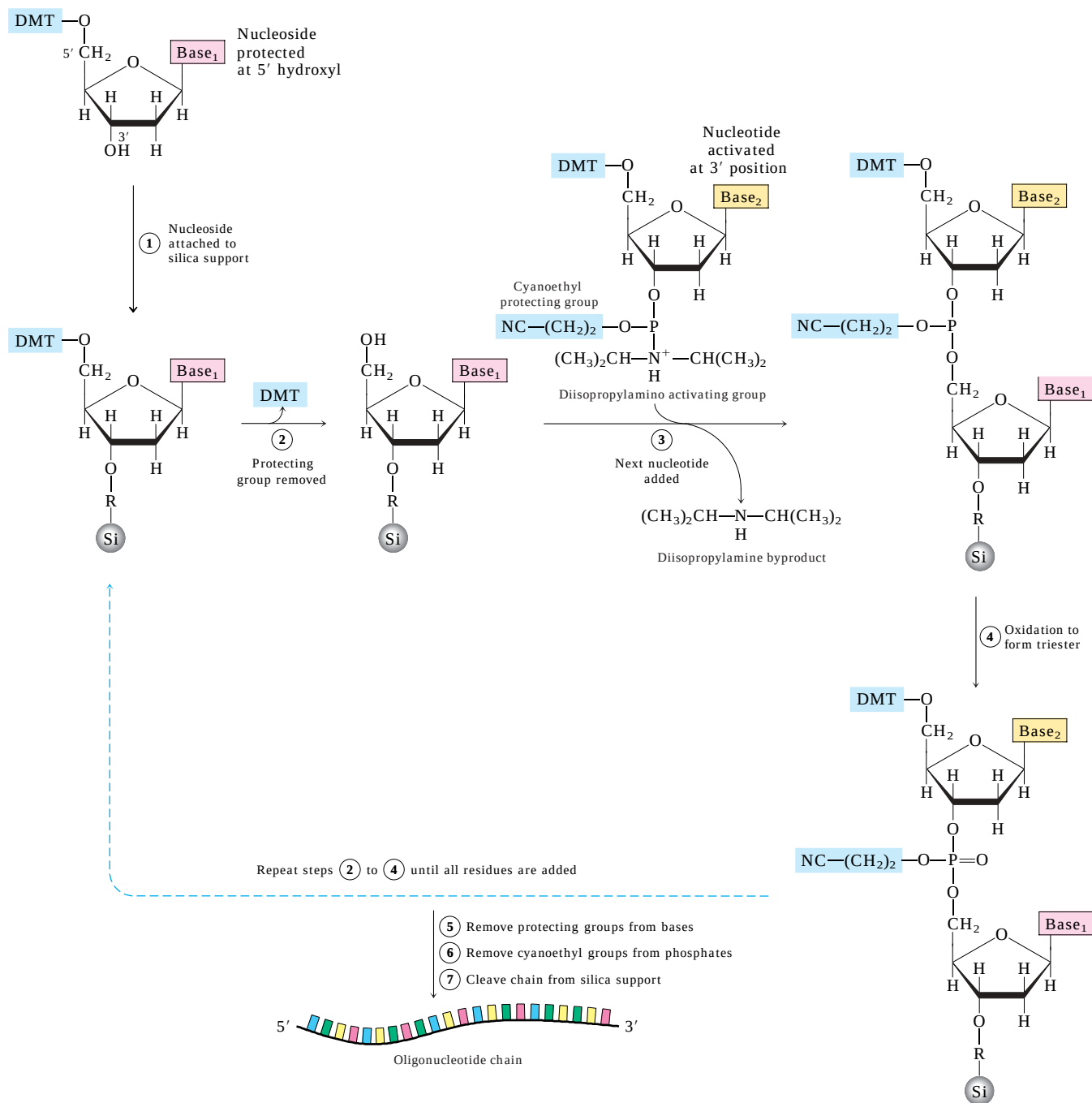


FIGURE 8–38 Chemical synthesis of DNA. Automated DNA synthesis is conceptually similar to the synthesis of polypeptides on a solid support. The oligonucleotide is built up on the solid support (silica), one nucleotide at a time, in a repeated series of chemical reactions with suitably protected nucleotide precursors. ① The first nucleoside (which will be the 3' end) is attached to the silica support at the 3' hydroxyl (through a linking group, R) and is protected at the 5' hydroxyl with an acid-labile dimethoxytrityl group (DMT). The reactive groups on all bases are also chemically protected. ② The protecting DMT group is removed by washing the column with acid (the DMT group is colored, so this reaction can be followed spectrophotometrically). ③ The next nucleotide is activated with a diisopropylamino

group and reacted with the bound nucleotide to form a 5',3' linkage, which in step ④ is oxidized with iodine to produce a phosphotriester linkage. (One of the phosphate oxygens carries a cyanoethyl protecting group.) Reactions ② through ④ are repeated until all nucleotides are added. At each step, excess nucleotide is removed before addition of the next nucleotide. In steps ⑤ and ⑥ the remaining protecting groups on the bases and the phosphates are removed, and in ⑦ the oligonucleotide is separated from the solid support and purified. The chemical synthesis of RNA is somewhat more complicated because of the need to protect the 2' hydroxyl of ribose without adversely affecting the reactivity of the 3' hydroxyl.

SUMMARY 8.3 Nucleic Acid Chemistry

- Native DNA undergoes reversible unwinding and separation of strands (melting) on heating or at extremes of pH. DNAs rich in G≡C pairs have higher melting points than DNAs rich in A=T pairs.
- Denatured single-stranded DNAs from two species can form a hybrid duplex, the degree of hybridization depending on the extent of sequence similarity. Hybridization is the basis for important techniques used to study and isolate specific genes and RNAs.
- DNA is a relatively stable polymer. Spontaneous reactions such as deamination of certain bases, hydrolysis of base-sugar *N*-glycosyl bonds, radiation-induced formation of pyrimidine dimers, and oxidative damage occur at very low rates, yet are important because of cells' very low tolerance for changes in genetic material.
- DNA sequences can be determined and DNA polymers synthesized with simple, automated protocols involving chemical and enzymatic methods.

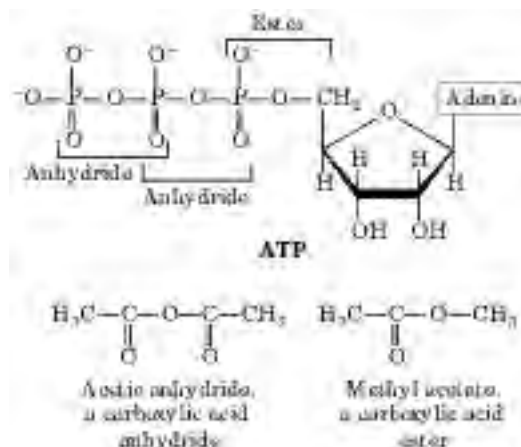


FIGURE 8-40 The phosphate ester and phosphoanhydride bonds of ATP. Hydrolysis of an anhydride bond yields more energy than hydrolysis of the ester. A carboxylic acid anhydride and carboxylic acid ester are shown for comparison.

8.4 Other Functions of Nucleotides

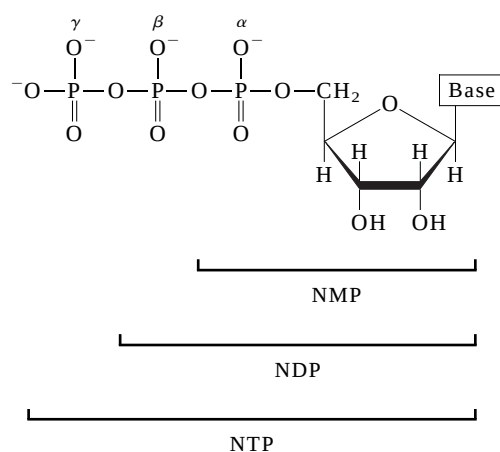
In addition to their roles as the subunits of nucleic acids, nucleotides have a variety of other functions in every cell: as energy carriers, components of enzyme cofactors, and chemical messengers.

Nucleotides Carry Chemical Energy in Cells

The phosphate group covalently linked at the 5' hydroxyl of a ribonucleotide may have one or two additional phosphates attached. The resulting molecules are referred to as nucleoside mono-, di-, and triphosphates (Fig. 8-39). Starting from the ribose, the three phos-

phates are generally labeled α , β , and γ . Hydrolysis of nucleoside triphosphates provides the chemical energy to drive a wide variety of cellular reactions. Adenosine 5'-triphosphate, ATP, is by far the most widely used for this purpose, but UTP, GTP, and CTP are also used in some reactions. Nucleoside triphosphates also serve as the activated precursors of DNA and RNA synthesis, as described in Chapters 25 and 26.

The energy released by hydrolysis of ATP and the other nucleoside triphosphates is accounted for by the structure of the triphosphate group. The bond between the ribose and the α phosphate is an ester linkage. The α,β and β,γ linkages are phosphoanhydrides (Fig. 8-40). Hydrolysis of the ester linkage yields about 14 kJ/mol under standard conditions, whereas hydrolysis of each anhydride bond yields about 30 kJ/mol. ATP hydrolysis often plays an important thermodynamic role in biosynthesis. When coupled to a reaction with a positive free-energy change, ATP hydrolysis shifts the equilibrium of the overall process to favor product forma-



Abbreviations of ribonucleoside 5'-phosphates			
Base	Mono-	Di-	Tri-
Adenine	AMP	ADP	ATP
Guanine	GMP	GDP	GTP
Cytosine	CMP	CDP	CTP
Uracil	UMP	UDP	UTP

Abbreviations of deoxyribonucleoside 5'-phosphates			
Base	Mono-	Di-	Tri-
Adenine	dAMP	dADP	dATP
Guanine	dGMP	dGDP	dGTP
Cytosine	dCMP	dCDP	dCTP
Thymine	dTMP	dTDP	dTTP

FIGURE 8-39 Nucleoside phosphates. General structure of the nucleoside 5'-mono-, di-, and triphosphates (NMPs, NDPs, and NTPs) and their standard abbreviations. In the deoxyribonucleoside phosphates (dNMPs, dNDPs, and dNTPs), the pentose is 2'-deoxy-D-ribose.

tion (recall the relationship between equilibrium constant and free-energy change described by Eqn 6–3 on p. 195).

Adenine Nucleotides Are Components of Many Enzyme Cofactors

A variety of enzyme cofactors serving a wide range of chemical functions include adenosine as part of their structure (Fig. 8–41). They are unrelated structurally except for the presence of adenosine. In none of these cofactors does the adenosine portion participate directly in the primary function, but removal of adenosine generally results in a drastic reduction of cofactor activities. For example, removal of the adenine nucleotide (3'-phosphoadenosine diphosphate) from acetoacetyl-

CoA, the coenzyme A derivative of acetoacetate, reduces its reactivity as a substrate for β -ketoacyl-CoA transferase (an enzyme of lipid metabolism) by a factor of 10^6 . Although this requirement for adenosine has not been investigated in detail, it must involve the binding energy between enzyme and substrate (or cofactor) that is used both in catalysis and in stabilizing the initial enzyme-substrate complex (Chapter 6). In the case of β -ketoacyl-CoA transferase, the nucleotide moiety of coenzyme A appears to be a binding "handle" that helps to pull the substrate (acetoacetyl-CoA) into the active site. Similar roles may be found for the nucleoside portion of other nucleotide cofactors.

Why is adenosine, rather than some other large molecule, used in these structures? The answer here may involve a form of evolutionary economy. Adenosine is

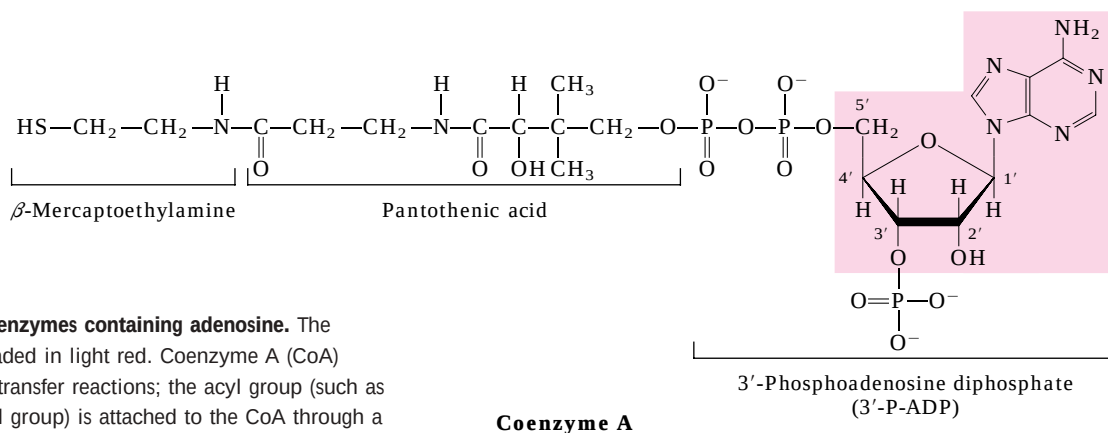
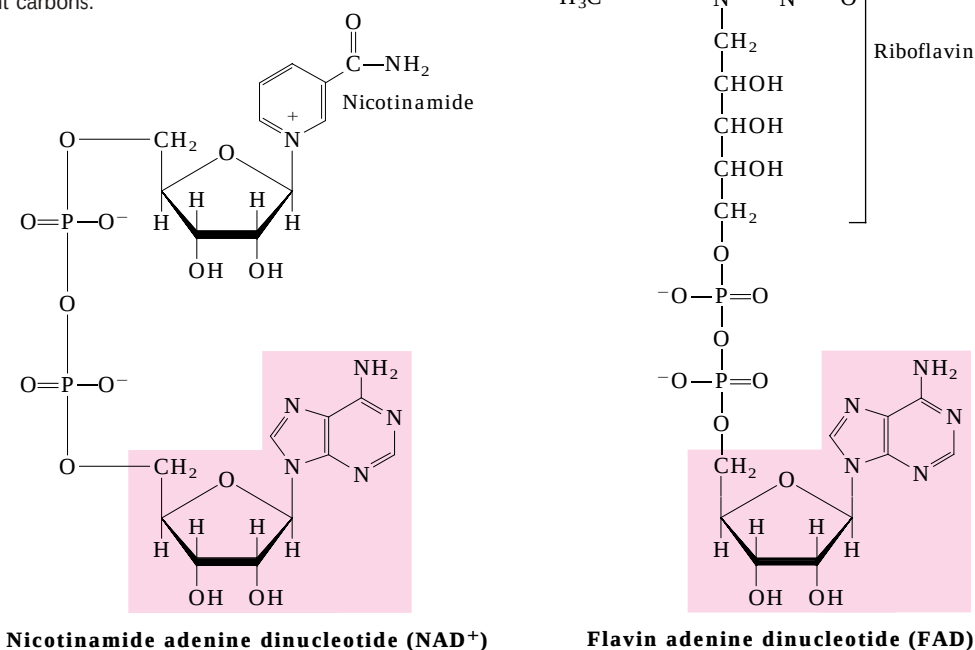


FIGURE 8–41 Some coenzymes containing adenosine. The adenosine portion is shaded in light red. Coenzyme A (CoA) functions in acyl group transfer reactions; the acyl group (such as the acetyl or acetoacetyl group) is attached to the CoA through a thioester linkage to the β -mercaptoethylamine moiety. NAD⁺ functions in hydride transfers, and FAD, the active form of vitamin B₂ (riboflavin), in electron transfers. Another coenzyme incorporating adenosine is 5'-deoxyadenosylcobalamin, the active form of vitamin B₁₂ (see Box 17-2), which participates in intramolecular group transfers between adjacent carbons.



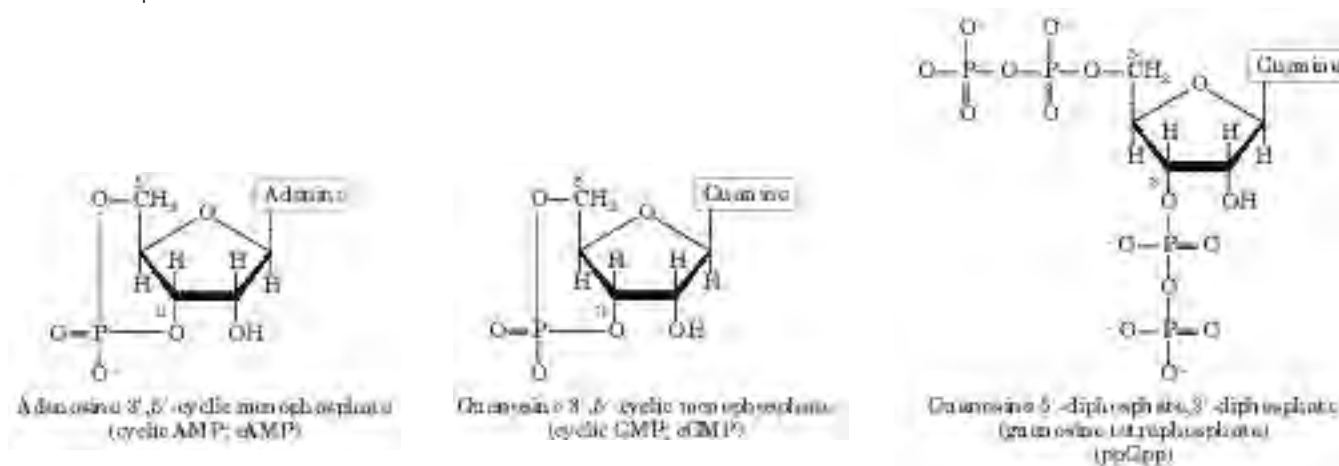


FIGURE 8-42 Three regulatory nucleotides.

certainly not unique in the amount of potential binding energy it can contribute. The importance of adenosine probably lies not so much in some special chemical characteristic as in the evolutionary advantage of using one compound for multiple roles. Once ATP became the universal source of chemical energy, systems developed to synthesize ATP in greater abundance than the other nucleotides; because it is abundant, it becomes the logical choice for incorporation into a wide variety of structures. The economy extends to protein structure. A single protein domain that binds adenosine can be used in a wide variety of enzymes. Such a domain, called a **nucleotide-binding fold**, is found in many enzymes that bind ATP and nucleotide cofactors.

Some Nucleotides Are Regulatory Molecules

Cells respond to their environment by taking cues from hormones or other external chemical signals. The interaction of these extracellular chemical signals (“first messengers”) with receptors on the cell surface often leads to the production of **second messengers** inside the cell, which in turn leads to adaptive changes in the cell interior (Chapter 12). Often, the second messenger is a nucleotide (Fig. 8-42). One of the most common is **adenosine 3',5'-cyclic monophosphate**

(**cyclic AMP**, or **cAMP**), formed from ATP in a reaction catalyzed by adenylyl cyclase, an enzyme associated with the inner face of the plasma membrane. Cyclic AMP serves regulatory functions in virtually every cell outside the plant kingdom. Guanosine 3',5'-cyclic monophosphate (cGMP) occurs in many cells and also has regulatory functions.

Another regulatory nucleotide, ppGpp (Fig. 8-42), is produced in bacteria in response to a slowdown in protein synthesis during amino acid starvation. This nucleotide inhibits the synthesis of the rRNA and tRNA molecules (see Fig. 28-24) needed for protein synthesis, preventing the unnecessary production of nucleic acids.

SUMMARY 8.4 Other Functions of Nucleotides

- ATP is the central carrier of chemical energy in cells. The presence of an adenosine moiety in a variety of enzyme cofactors may be related to binding-energy requirements.
- Cyclic AMP, formed from ATP in a reaction catalyzed by adenylyl cyclase, is a common second messenger produced in response to hormones and other chemical signals.

Key Terms

Terms in bold are defined in the glossary.

gene 273
ribosomal RNA (rRNA) 273
messenger RNA (mRNA) 273
transfer RNA (tRNA) 273
nucleotide 273
nucleoside 273
pyrimidine 273
purine 273
deoxyribonucleotides 274
ribonucleotide 274
phosphodiester linkage 277

5' end 277
3' end 277
oligonucleotide 278
polynucleotide 278
base pair 279
 major groove 282
 minor groove 282
 B-form DNA 284
 A-form DNA 284
 Z-form DNA 284
palindrome 285

hairpin 285
cruciform 285
 triplex DNA 286
 G tetraplex 287
 H-DNA 287
monocistronic mRNA 287
polycistronic mRNA 288
mutation 293
second messenger 302
adenosine 3',5'-cyclic monophosphate (cyclic AMP, cAMP) 302

Further Reading

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ATP As Energy Carrier

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A relatively short article, full of insights.

Problems

1. Nucleotide Structure Which positions in a purine ring of a purine nucleotide in DNA have the potential to form hydrogen bonds but are not involved in Watson-Crick base pairing?

2. Base Sequence of Complementary DNA Strands One strand of a double-helical DNA has the sequence (5')GCGCAATATTTCTCAAATATTGCGC(3'). Write the base sequence of the complementary strand. What special type of sequence is contained in this DNA segment? Does the double-stranded DNA have the potential to form any alternative structures?

3. DNA of the Human Body Calculate the weight in grams of a double-helical DNA molecule stretching from the earth to the moon (~320,000 km). The DNA double helix weighs about 1×10^{-18} g per 1,000 nucleotide pairs; each base pair extends 3.4 Å. For an interesting comparison, your body contains about 0.5 g of DNA!

4. DNA Bending Assume that a poly(A) tract five base pairs long produces a 20° bend in a DNA strand. Calculate

the total (net) bend produced in a DNA if the center base pairs (the third of five) of two successive (dA)₅ tracts are located (a) 10 base pairs apart; (b) 15 base pairs apart. Assume 10 base pairs per turn in the DNA double helix.

5. Distinction between DNA Structure and RNA Structure Hairpins may form at palindromic sequences in single strands of either RNA or DNA. How is the helical structure of a long and fully base-paired (except at the end) hairpin in RNA different from that of a similar hairpin in DNA?

6. Nucleotide Chemistry The cells of many eukaryotic organisms have highly specialized systems that specifically repair G–T mismatches in DNA. The mismatch is repaired to form a G=C (not A=T) base pair. This G–T mismatch repair mechanism occurs in addition to a more general system that repairs virtually all mismatches. Can you suggest why cells might require a specialized system to repair G–T mismatches?

7. Nucleic Acid Structure Explain why the absorption of UV light by double-stranded DNA increases (hyperchromic effect) when the DNA is denatured.

8. Determination of Protein Concentration in a Solution Containing Proteins and Nucleic Acids

The concentration of protein or nucleic acid in a solution containing both can be estimated by using their different light absorption properties: proteins absorb most strongly at 280 nm and nucleic acids at 260 nm. Their respective concentrations in a mixture can be estimated by measuring the absorbance (*A*) of the solution at 280 nm and 260 nm and using the table below, which gives *R*_{280/260}, the ratio of absorbances at 280 and 260 nm; the percentage of total mass that is nucleic acid; and a factor, *F*, that corrects the *A*₂₈₀ reading and gives a more accurate protein estimate. The protein concentration (in mg/ml) = *F* × *A*₂₈₀ (assuming the cuvette is 1 cm wide). Calculate the protein concentration in a solution of *A*₂₈₀ = 0.69 and *A*₂₆₀ = 0.94.

<i>R</i> _{280/260}	Proportion of nucleic acid (%)	<i>F</i>
1.75	0.00	1.116
1.63	0.25	1.081
1.52	0.50	1.054
1.40	0.75	1.023
1.36	1.00	0.994
1.30	1.25	0.970
1.25	1.50	0.944
1.16	2.00	0.899
1.09	2.50	0.852
1.03	3.00	0.814
0.979	3.50	0.776
0.939	4.00	0.743
0.874	5.00	0.682
0.846	5.50	0.656
0.822	6.00	0.632
0.804	6.50	0.607
0.784	7.00	0.585
0.767	7.50	0.565
0.753	8.00	0.545
0.730	9.00	0.508
0.705	10.00	0.478
0.671	12.00	0.422
0.644	14.00	0.377
0.615	17.00	0.322
0.595	20.00	0.278

9. Base Pairing in DNA In samples of DNA isolated from two unidentified species of bacteria, X and Y, adenine makes up 32% and 17%, respectively, of the total bases. What relative proportions of adenine, guanine, thymine, and cytosine would you expect to find in the two DNA samples? What assumptions have you made? One of these species was isolated from a hot spring (64 °C). Suggest which species is the thermophilic bacterium. What is the basis for your answer?

10. Solubility of the Components of DNA Draw the following structures and rate their relative solubilities in water (most soluble to least soluble): deoxyribose, guanine, phosphate. How are these solubilities consistent with the three-dimensional structure of double-stranded DNA?

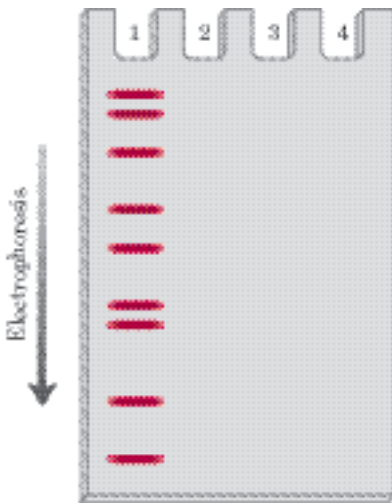
11. DNA Sequencing The following DNA fragment was sequenced by the Sanger method. The red asterisk indicates a fluorescent label.



A sample of the DNA was reacted with DNA polymerase and each of the nucleotide mixtures (in an appropriate buffer) listed below. Dideoxynucleotides (ddNTPs) were added in relatively small amounts.

- 1. dATP, dTTP, dCTP, dGTP, ddTTP
- 2. dATP, dTTP, dCTP, dGTP, ddGTP
- 3. dATP, dCTP, dGTP, ddTTP
- 4. dATP, dTTP, dCTP, dGTP

The resulting DNA was separated by electrophoresis on an agarose gel, and the fluorescent bands on the gel were located. The band pattern resulting from nucleotide mixture 1 is shown below. Assuming that all mixtures were run on the same gel, what did the remaining lanes of the gel look like?



12. Snake Venom Phosphodiesterase An exonuclease is an enzyme that sequentially cleaves nucleotides from the end of a polynucleotide strand. Snake venom phosphodiesterase, which hydrolyzes nucleotides from the 3' end of any oligonucleotide with a free 3'-hydroxyl group, cleaves between the 3' hydroxyl of the ribose or deoxyribose and the phosphoryl group of the next nucleotide. It acts on single-stranded DNA or RNA and has no base specificity. This enzyme was used in sequence determination experiments before the development of modern nucleic acid sequencing techniques. What are the products of partial digestion by snake venom phosphodiesterase of an oligonucleotide with the following sequence?



13. Preserving DNA in Bacterial Endospores Bacterial endospores form when the environment is no longer conducive to active cell metabolism. The soil bacterium *Bacillus subtilis*, for example, begins the process of sporulation when one or more nutrients are depleted. The end product is a

small, metabolically dormant structure that can survive almost indefinitely with no detectable metabolism. Spores have mechanisms to prevent accumulation of potentially lethal mutations in their DNA over periods of dormancy that can exceed 1,000 years. *B. subtilis* spores are much more resistant than the organism's growing cells to heat, UV radiation, and oxidizing agents, all of which promote mutations.

(a) One factor that prevents potential DNA damage in spores is their greatly decreased water content. How would this affect some types of mutations?

(b) Endospores have a category of proteins called small acid-soluble proteins (SASPs) that bind to their DNA, preventing formation of cyclobutane-type dimers. What causes cyclobutane dimers, and why do bacterial endospores need mechanisms to prevent their formation?

Biochemistry on the Internet

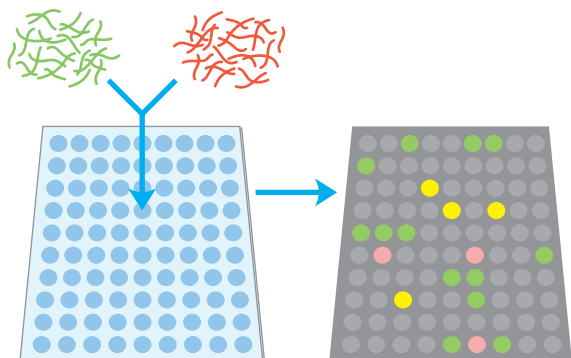
14. The Structure of DNA Elucidation of the three-dimensional structure of DNA helped researchers understand how this molecule conveys information that can be faithfully replicated from one generation to the next. To see the secondary structure of double-stranded DNA, go to the Protein Data Bank website (www.rcsb.org/pdb). Use the PDB identifiers listed below to retrieve the data pages for the two forms of DNA.

Open the structures using RasMol or Chime, and use the different viewing options to complete the following exercises.

(a) Obtain the file for 141D, a highly conserved, repeated DNA sequence from the end of the HIV-1 (the virus that causes AIDS) genome. Display the molecule as a stick or ball-and-stick structure. Identify the sugar-phosphate backbone for each strand of the DNA duplex. Locate and identify individual bases. Which is the 5' end of this molecule? Locate the major and minor grooves. Is this a right- or left-handed helix?

(b) Obtain the file for 145D, a DNA with the Z conformation. Display the molecule as a stick or ball-and-stick structure. Identify the sugar-phosphate backbone for each strand of the DNA duplex. Is this a right- or left-handed helix?

(c) To fully appreciate the secondary structure of DNA, select "Stereo" in the Options menu in the viewer. You will see two images of the DNA molecule. Sit with your nose approximately 10 inches from the monitor and focus on the tip of your nose. In the background you should see three images of the DNA helix. Shift your focus from the tip of your nose to the middle image, which should appear three-dimensional. (Note that only one of the two authors can make this work.) For additional tips, see the Study Guide or the textbook website (www.whfreeman.com/lehninger).



DNA-BASED INFORMATION TECHNOLOGIES

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Of all the natural systems, living matter is the one which, in the face of great transformations, preserves inscribed in its organization the largest amount of its own past history.

—Emile Zuckerkandl and Linus Pauling, article in *Journal of Theoretical Biology*, 1965

We now turn to a technology that is fundamental to the advance of modern biological sciences, defining present and future biochemical frontiers and illustrating many important principles of biochemistry. Elucidation of the laws governing enzymatic catalysis, macromolecular structure, cellular metabolism, and information pathways allows research to be directed at increasingly complex biochemical processes. Cell division, immunity, embryogenesis, vision, taste, oncogenesis, cognition—all are orchestrated in an elaborate symphony of molecular and macromolecular interactions that we are now beginning to understand with increasing clarity. The real implications of the biochemical journey begun in the nineteenth century are found in the ever-increasing power to analyze and alter living systems.

To understand a complex biological process, a biochemist isolates and studies the individual components *in vitro*, then pieces together the parts to get a coherent picture of the overall process. A major source of molecular insights is the cell's own information archive, its DNA. The sheer size of chromosomes, however, pres-

ents an enormous challenge: how does one find and study a particular gene among the tens of thousands of genes nested in the billions of base pairs of a mammalian genome? Solutions began to emerge in the 1970s.

Decades of advances by thousands of scientists working in genetics, biochemistry, cell biology, and physical chemistry came together in the laboratories of Paul Berg, Herbert Boyer, and Stanley Cohen to yield techniques for locating, isolating, preparing, and studying small segments of DNA derived from much larger chromosomes. Techniques for DNA cloning paved the way to the modern fields of **genomics** and **proteomics**, the study of genes and proteins on the scale of whole cells and organisms. These new methods are transforming basic research, agriculture, medicine, ecology, forensics, and many other fields, while occasionally presenting society with difficult choices and ethical dilemmas.

We begin this chapter with an outline of the fundamental biochemical principles of the now-classic discipline of DNA cloning. Next, after laying the groundwork for a discussion of genomics, we illustrate the range of applications and the potential of these technologies, with a broad emphasis on modern advances in genomics and proteomics.

9.1 DNA Cloning: The Basics

A *clone* is an identical copy. This term originally applied to cells of a single type, isolated and allowed to reproduce to create a population of identical cells. **DNA cloning** involves separating a specific gene or DNA segment from a larger chromosome, attaching it to a small molecule of carrier DNA, and then replicating this modified DNA thousands or millions of times through both an increase in cell number and the creation of multiple

copies of the cloned DNA in each cell. The result is selective amplification of a particular gene or DNA segment. Cloning of DNA from any organism entails five general procedures:

1. *Cutting DNA at precise locations.* Sequence-specific endonucleases (restriction endonucleases) provide the necessary molecular scissors.
2. *Selecting a small molecule of DNA capable of self-replication.* These DNAs are called **cloning vectors** (a vector is a delivery agent). They are typically plasmids or viral DNAs.
3. *Joining two DNA fragments covalently.* The enzyme DNA ligase links the cloning vector and DNA to be cloned. Composite DNA molecules comprising covalently linked segments from two or more sources are called **recombinant DNAs**.
4. *Moving recombinant DNA from the test tube to a host cell* that will provide the enzymatic machinery for DNA replication.
5. *Selecting or identifying host cells that contain recombinant DNA.*

The methods used to accomplish these and related tasks are collectively referred to as **recombinant DNA technology** or, more informally, **genetic engineering**.

Much of our initial discussion will focus on DNA cloning in the bacterium *Escherichia coli*, the first organism used for recombinant DNA work and still the most common host cell. *E. coli* has many advantages: its DNA metabolism (like many other of its biochemical processes) is well understood; many naturally occurring cloning vectors associated with *E. coli*, such as plasmids and bacteriophages (bacterial viruses; also called phages), are well characterized; and techniques are available for moving DNA expeditiously from one bac-



Paul Berg



Herbert Boyer



Stanley N. Cohen

terial cell to another. We also address DNA cloning in other organisms, a topic discussed more fully later in the chapter.

Restriction Endonucleases and DNA Ligase Yield Recombinant DNA

Particularly important to recombinant DNA technology is a set of enzymes (Table 9–1) made available through decades of research on nucleic acid metabolism. Two classes of enzymes lie at the heart of the general approach to generating and propagating a recombinant DNA molecule (Fig. 9–1). First, **restriction endonucleases** (also called restriction enzymes) recognize and cleave DNA at specific DNA sequences (recognition sequences or restriction sites) to generate a set of smaller fragments. Second, the DNA fragment to be cloned can be joined to a suitable cloning vector by using **DNA ligases** to link the DNA molecules together. The recombinant vector is then introduced into a host cell, which amplifies the fragment in the course of many generations of cell division.

Restriction endonucleases are found in a wide range of bacterial species. Werner Arber discovered in the early 1960s that their biological function is to recognize and cleave foreign DNA (the DNA of an infecting virus, for example); such DNA is said to be *restricted*. In the host cell's DNA, the sequence that would be recognized

TABLE 9–1 Some Enzymes Used in Recombinant DNA Technology

Enzyme(s)	Function
Type II restriction endonucleases	Cleave DNAs at specific base sequences
DNA ligase	Joins two DNA molecules or fragments
DNA polymerase I (<i>E. coli</i>)	Fills gaps in duplexes by stepwise addition of nucleotides to 3' ends
Reverse transcriptase	Makes a DNA copy of an RNA molecule
Polynucleotide kinase	Adds a phosphate to the 5'-OH end of a polynucleotide to label it or permit ligation
Terminal transferase	Adds homopolymer tails to the 3'-OH ends of a linear duplex
Exonuclease III	Removes nucleotide residues from the 3' ends of a DNA strand
Bacteriophage λ exonuclease	Removes nucleotides from the 5' ends of a duplex to expose single-stranded 3' ends
Alkaline phosphatase	Removes terminal phosphates from either the 5' or 3' end (or both)

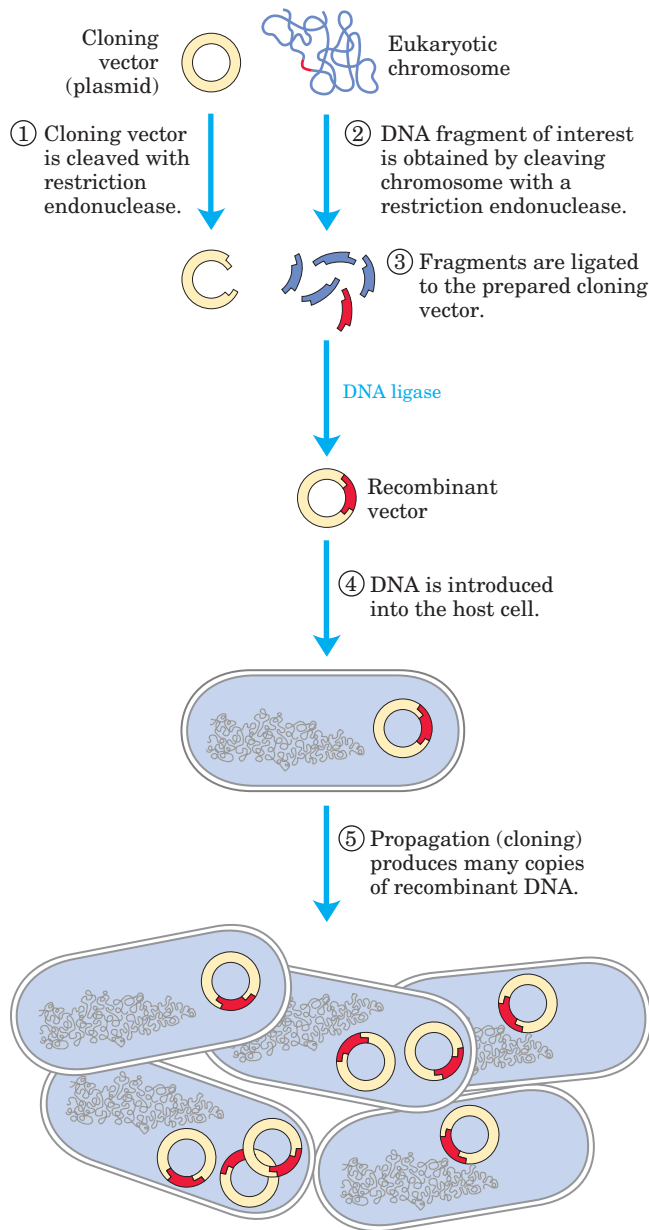


FIGURE 9-1 Schematic illustration of DNA cloning. A cloning vector and eukaryotic chromosomes are separately cleaved with the same restriction endonuclease. The fragments to be cloned are then ligated to the cloning vector. The resulting recombinant DNA (only one recombinant vector is shown here) is introduced into a host cell where it can be propagated (cloned). Note that this drawing is not to scale: the size of the *E. coli* chromosome relative to that of a typical cloning vector (such as a plasmid) is much greater than depicted here.

by its own restriction endonuclease is protected from digestion by methylation of the DNA, catalyzed by a specific DNA methylase. The restriction endonuclease and the corresponding methylase are sometimes referred to as a **restriction-modification system**.

There are three types of restriction endonucleases, designated I, II, and III. Types I and III are generally large, multisubunit complexes containing both the endonuclease

and methylase activities. Type I restriction endonucleases cleave DNA at random sites that can be more than 1,000 base pairs (bp) from the recognition sequence. Type III restriction endonucleases cleave the DNA about 25 bp from the recognition sequence. Both types move along the DNA in a reaction that requires the energy of ATP. **Type II restriction endonucleases**, first isolated by Hamilton Smith in 1970, are simpler, require no ATP, and cleave the DNA within the recognition sequence itself. The extraordinary utility of this group of restriction endonucleases was demonstrated by Daniel Nathans, who first used them to develop novel methods for mapping and analyzing genes and genomes.

Thousands of restriction endonucleases have been discovered in different bacterial species, and more than 100 different DNA sequences are recognized by one or more of these enzymes. The recognition sequences are usually 4 to 6 bp long and palindromic (see Fig. 8-20). Table 9-2 lists sequences recognized by a few type II restriction endonucleases. In some cases, the interaction between a restriction endonuclease and its target sequence has been elucidated in exquisite molecular detail; for example, Figure 9-2 shows the complex of the type II restriction endonuclease *EcoRV* and its target sequence.

Some restriction endonucleases make staggered cuts on the two DNA strands, leaving two to four nucleotides of one strand unpaired at each resulting end. These unpaired strands are referred to as **sticky ends** (Fig. 9-3a), because they can base-pair with each other or with complementary sticky ends of other DNA fragments. Other restriction endonucleases cleave both strands of DNA at the opposing phosphodiester bonds, leaving no unpaired bases on the ends, often called **blunt ends** (Fig. 9-3b).

The average size of the DNA fragments produced by cleaving genomic DNA with a restriction endonuclease depends on the frequency with which a particular restriction site occurs in the DNA molecule; this in turn depends largely on the size of the recognition sequence. In a DNA molecule with a random sequence in which all four nucleotides were equally abundant, a 6 bp sequence recognized by a restriction endonuclease such as *Bam*HI would occur on average once every 4^6 (4,096) bp, assuming the DNA had a 50% G≡C content. Enzymes that recognize a 4 bp sequence would produce smaller DNA fragments from a random-sequence DNA molecule; a recognition sequence of this size would be expected to occur about once every 4^4 (256) bp. In natural DNA molecules, particular recognition sequences tend to occur less frequently than this because nucleotide sequences in DNA are not random and the four nucleotides are not equally abundant. In laboratory experiments, the average size of the fragments produced by restriction endonuclease cleavage of a large DNA can be increased by simply terminating the reaction before completion; the result is called a partial digest. Fragment size can also

TABLE 9–2 Recognition Sequences for Some Type II Restriction Endonucleases

<i>Bam</i> HI	$\begin{array}{c} \downarrow \\ (5') \text{ G G A T C C } (3') \\ \text{C C T A G G} \\ \uparrow \end{array}$	<i>Hind</i> III	$\begin{array}{c} \downarrow \\ (5') \text{ A A G C T T } (3') \\ \text{T T C G A A} \\ \uparrow \end{array}$
<i>Cl</i> aI	$\begin{array}{c} \downarrow \\ (5') \text{ A T C G A T } (3') \\ \text{T A G C T A} \\ \uparrow \end{array}$	<i>Not</i> I	$\begin{array}{c} \downarrow \\ (5') \text{ G C G G C C G C } (3') \\ \text{C G C C G G C G} \\ \uparrow \end{array}$
<i>Eco</i> RI	$\begin{array}{c} \downarrow \\ (5') \text{ G A A T T C } (3') \\ \text{C T T A A G} \\ \uparrow \end{array}$	<i>Pst</i> I	$\begin{array}{c} \downarrow \\ (5') \text{ C T G C A G } (3') \\ \text{G A C G T C} \\ \uparrow \end{array}$
<i>Eco</i> RV	$\begin{array}{c} \downarrow \\ (5') \text{ G A T A T C } (3') \\ \text{C T A T A G} \\ \uparrow \end{array}$	<i>Pvu</i> II	$\begin{array}{c} \downarrow \\ (5') \text{ C A G C T G } (3') \\ \text{G T C G A C} \\ \uparrow \end{array}$
<i>Hae</i> III	$\begin{array}{c} \downarrow \\ (5') \text{ G G C C } (3') \\ \text{C C G G} \\ \uparrow \end{array}$	<i>Tth</i> 111I	$\begin{array}{c} \downarrow \\ (5') \text{ G A C N N N G T C } (3') \\ \text{C T G N N N C A G} \\ \uparrow \end{array}$

Arrows indicate the phosphodiester bonds cleaved by each restriction endonuclease. Asterisks indicate bases that are methylated by the corresponding methylase (where known). N denotes any base. Note that the name of each enzyme consists of a three-letter abbreviation (in italics) of the bacterial species from which it is derived, sometimes followed by a strain designation and Roman numerals to distinguish different restriction endonucleases isolated from the same bacterial species. Thus *Bam*HI is the first (I) restriction endonuclease characterized from *Bacillus amyloliquefaciens*, strain H.

be increased by using a special class of endonucleases called homing endonucleases (see Fig. 26–34). These recognize and cleave much longer DNA sequences (14 to 20 bp).

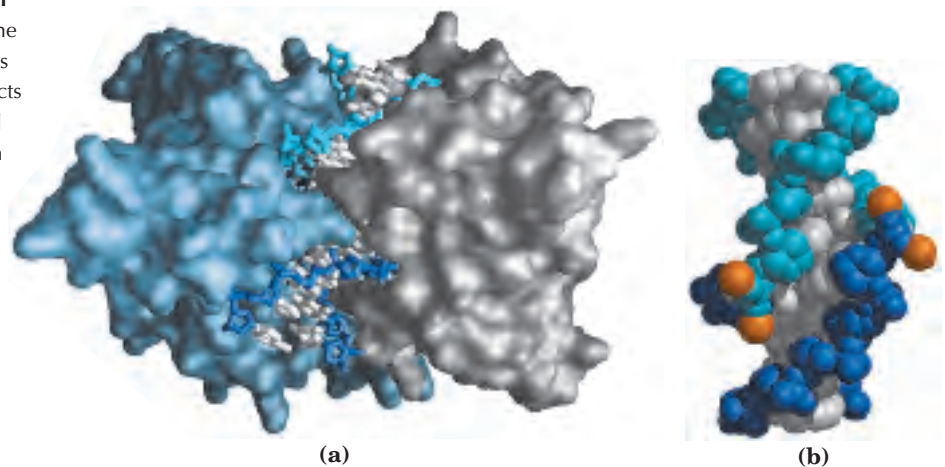
Once a DNA molecule has been cleaved into fragments, a particular fragment of known size can be enriched by agarose or acrylamide gel electrophoresis or by HPLC (pp. 92, 90). For a typical mammalian genome, however, cleavage by a restriction endonuclease usually yields too many different DNA fragments to permit isolation of a particular fragment by electrophoresis or

HPLC. A common intermediate step in the cloning of a specific gene or DNA segment is the construction of a DNA library (as described in Section 9.2).

After the target DNA fragment is isolated, DNA ligase can be used to join it to a similarly digested cloning vector—that is, a vector digested by the *same* restriction endonuclease; a fragment generated by *Eco*RI, for example, generally will not link to a fragment generated by *Bam*HI. As described in more detail in Chapter 25 (see Fig. 25–16), DNA ligase catalyzes the formation of new phosphodiester bonds in a reaction that uses ATP

FIGURE 9–2 Interaction of *Eco*RV restriction endonuclease with its target sequence. (a) The dimeric *Eco*RV endonuclease (its two subunits in light blue and gray) is bound to the products of DNA cleavage at the sequence recognized by the enzyme. The DNA backbone is shown in two shades of blue to distinguish the segments separated by cleavage (PDB ID 1RVC). (b) In this view, showing just the DNA, the DNA segment has been turned 180°. The enzyme creates blunt ends; the cleavage points appear staggered on the two DNA strands because the DNA is kinked. Bound magnesium ions (orange) play a role in catalysis of the cleavage reaction.

 **Restriction Endonucleases**



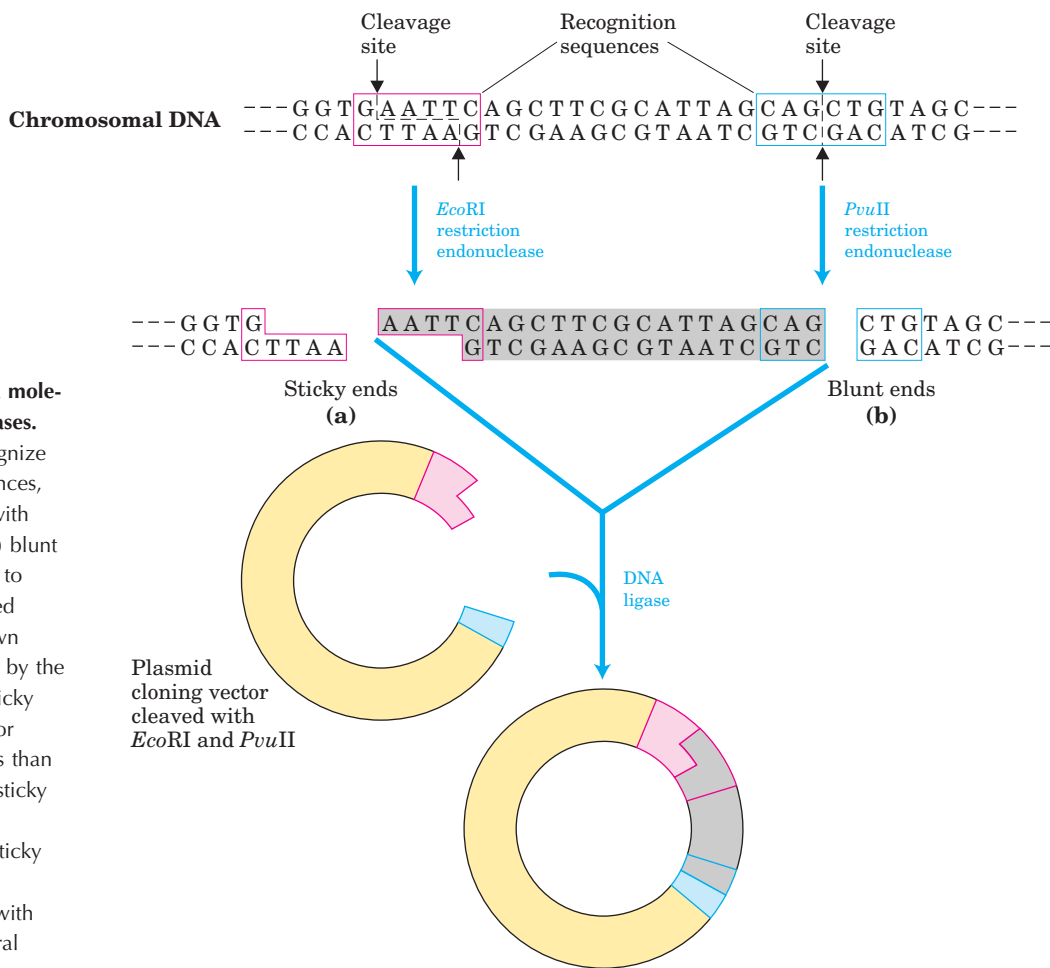
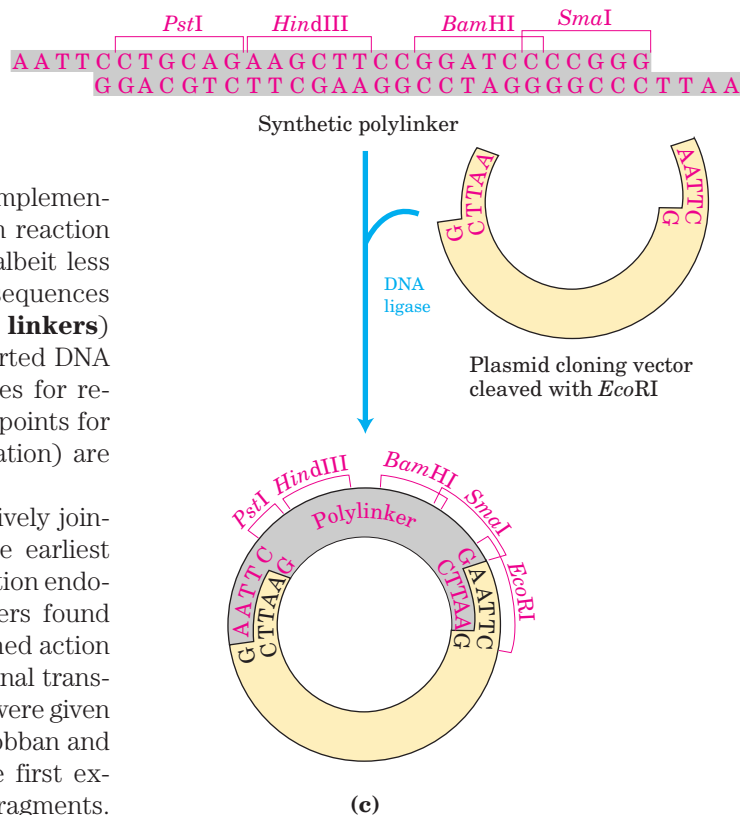


FIGURE 9-3 Cleavage of DNA molecules by restriction endonucleases.

Restriction endonucleases recognize and cleave only specific sequences, leaving either (a) sticky ends (with protruding single strands) or (b) blunt ends. Fragments can be ligated to other DNAs, such as the cleaved cloning vector (a plasmid) shown here. This reaction is facilitated by the annealing of complementary sticky ends. Ligation is less efficient for DNA fragments with blunt ends than for those with complementary sticky ends, and DNA fragments with different (noncomplementary) sticky ends generally are not ligated. (c) A synthetic DNA fragment with recognition sequences for several restriction endonucleases can be inserted into a plasmid that has been cleaved by a restriction endonuclease. The insert is called a linker; an insert with multiple restriction sites is called a polylinker.

or a similar cofactor. The base-pairing of complementary sticky ends greatly facilitates the ligation reaction (Fig. 9-3a). Blunt ends can also be ligated, albeit less efficiently. Researchers can create new DNA sequences by inserting synthetic DNA fragments (called **linkers**) between the ends that are being ligated. Inserted DNA fragments with multiple recognition sequences for restriction endonucleases (often useful later as points for inserting additional DNA by cleavage and ligation) are called **polylinkers** (Fig. 9-3c).

The effectiveness of sticky ends in selectively joining two DNA fragments was apparent in the earliest recombinant DNA experiments. Before restriction endonucleases were widely available, some workers found they could generate sticky ends by the combined action of the bacteriophage λ exonuclease and terminal transferase (Table 9-1). The fragments to be joined were given complementary homopolymeric tails. Peter Lobban and Dale Kaiser used this method in 1971 in the first experiments to join naturally occurring DNA fragments.



Similar methods were used soon after in the laboratory of Paul Berg to join DNA segments from simian virus 40 (SV40) to DNA derived from bacteriophage λ , thereby creating the first recombinant DNA molecule with DNA segments from different species.

Cloning Vectors Allow Amplification of Inserted DNA Segments

The principles that govern the delivery of recombinant DNA in clonable form to a host cell, and its subsequent amplification in the host, are well illustrated by considering three popular cloning vectors commonly used in experiments with *E. coli*—plasmids, bacteriophages, and bacterial artificial chromosomes—and a vector used to clone large DNA segments in yeast.

Plasmids Plasmids are circular DNA molecules that replicate separately from the host chromosome. Naturally occurring bacterial plasmids range in size from 5,000 to 400,000 bp. They can be introduced into bacterial cells by a process called **transformation**. The cells (generally *E. coli*) and plasmid DNA are incubated together at 0 °C in a calcium chloride solution, then subjected to a shock by rapidly shifting the temperature to 37 to 43 °C. For reasons not well understood, some of the cells treated in this way take up the plasmid DNA. Some species of bacteria are naturally competent for DNA uptake and do not require the calcium chloride treatment. In an alternative method, cells incubated with the plasmid DNA are subjected to a high-voltage pulse. This approach, called **electroporation**, transiently renders the bacterial membrane permeable to large molecules.

Regardless of the approach, few cells actually take up the plasmid DNA, so a method is needed to select those that do. The usual strategy is to use a plasmid that includes a gene that the host cell requires for growth under specific conditions, such as a gene that confers resistance to an antibiotic. Only cells transformed by the recombinant plasmid can grow in the presence of that antibiotic, making any cell that contains the plasmid “selectable” under those growth conditions. Such a gene is called a selectable marker.

Investigators have developed many different plasmid vectors suitable for cloning by modifying naturally occurring plasmids. The *E. coli* plasmid pBR322 offers a good example of the features useful in a cloning vector (Fig. 9–4):

1. pBR322 has an origin of replication, *ori*, a sequence where replication is initiated by cellular enzymes (Chapter 25). This sequence is required to propagate the plasmid and maintain it at a level of 10 to 20 copies per cell.
2. The plasmid contains two genes that confer resistance to different antibiotics (*tet^R*, *amp^R*),

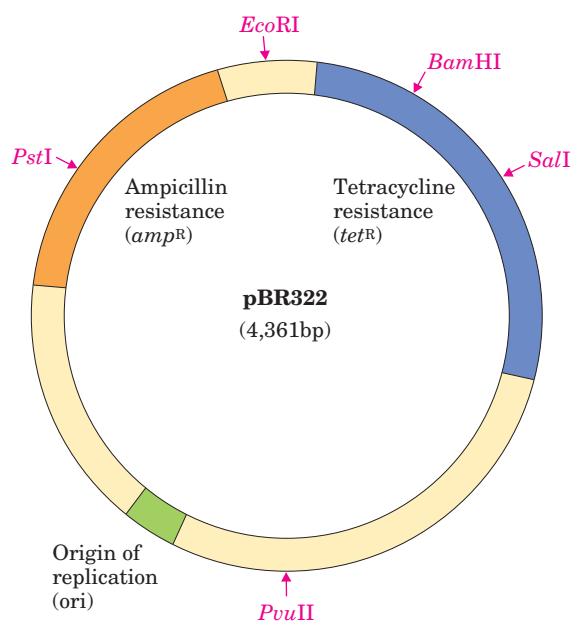


FIGURE 9–4 The constructed *E. coli* plasmid pBR322. Note the location of some important restriction sites—for *Pst*I, *Eco*RI, *Bam*HI, *Sal*I, and *Pvu*II; ampicillin- and tetracycline-resistance genes; and the replication origin (*ori*). Constructed in 1977, this was one of the early plasmids designed expressly for cloning in *E. coli*.

allowing the identification of cells that contain the intact plasmid or a recombinant version of the plasmid (Fig. 9–5).

3. Several unique recognition sequences in pBR322 (*Pst*I, *Eco*RI, *Bam*HI, *Sal*I, *Pvu*II) are targets for different restriction endonucleases, providing sites where the plasmid can later be cut to insert foreign DNA.
4. The small size of the plasmid (4,361 bp) facilitates its entry into cells and the biochemical manipulation of the DNA.

Transformation of typical bacterial cells with purified DNA (never a very efficient process) becomes less successful as plasmid size increases, and it is difficult to clone DNA segments longer than about 15,000 bp when plasmids are used as the vector.

Bacteriophages Bacteriophage λ has a very efficient mechanism for delivering its 48,502 bp of DNA into a bacterium, and it can be used as a vector to clone somewhat larger DNA segments (Fig. 9–6). Two key features contribute to its utility:

1. About one-third of the λ genome is nonessential and can be replaced with foreign DNA.
2. DNA is packaged into infectious phage particles only if it is between 40,000 and 53,000 bp long, a constraint that can be used to ensure packaging of recombinant DNA only.

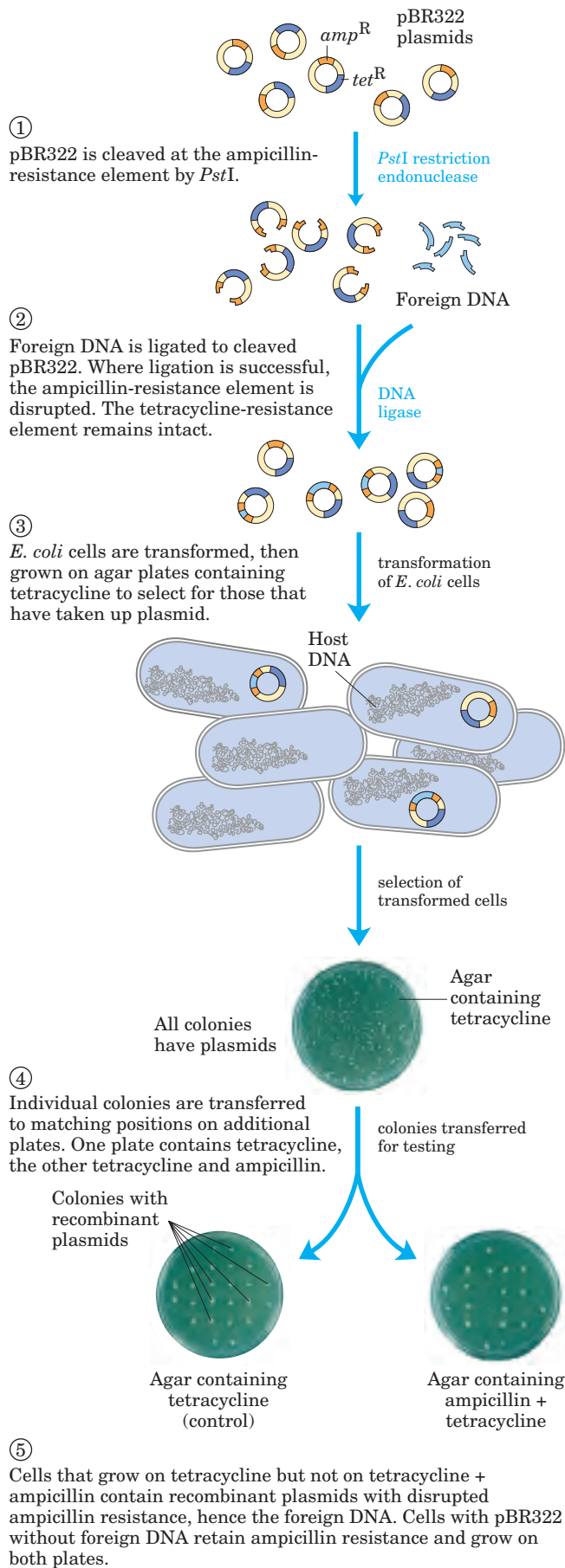


FIGURE 9-5 Use of pBR322 to clone and identify foreign DNA in *E. coli*. **Plasmid Cloning**

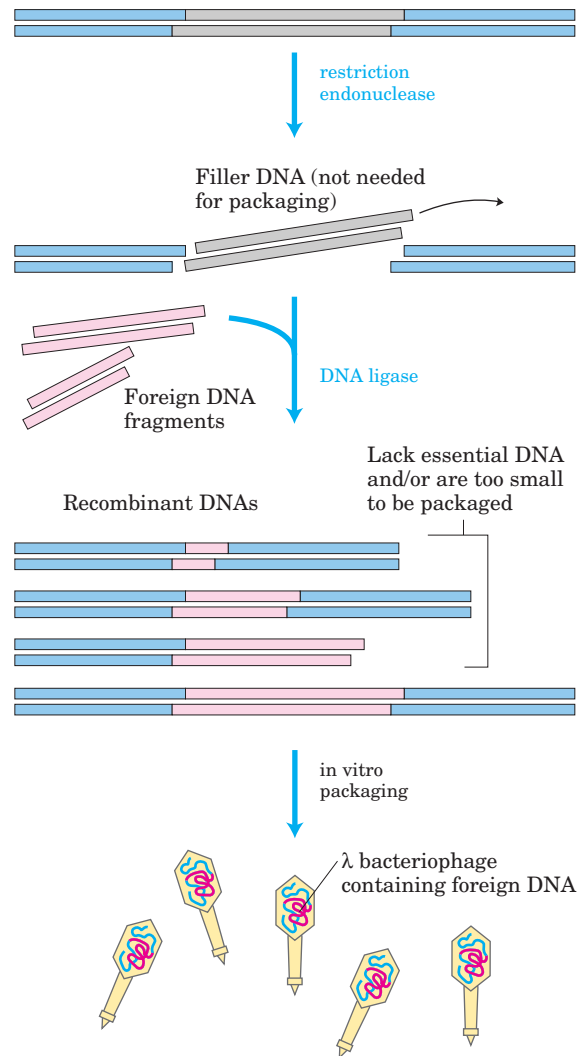


FIGURE 9-6 Bacteriophage λ cloning vectors. Recombinant DNA methods are used to modify the bacteriophage λ genome, removing the genes not needed for phage production and replacing them with “filler” DNA to make the phage DNA large enough for packaging into phage particles. As shown here, the filler is replaced with foreign DNA in cloning experiments. Recombinants are packaged into viable phage particles in vitro only if they include an appropriately sized foreign DNA fragment as well as both of the essential λ DNA end fragments.

Researchers have developed bacteriophage λ vectors that can be readily cleaved into three pieces, two of which contain essential genes but which together are only about 30,000 bp long. The third piece, “filler” DNA, is discarded when the vector is to be used for cloning, and additional DNA is inserted between the two essential segments to generate ligated DNA molecules long enough to produce viable phage particles. In effect, the packaging mechanism *selects for* recombinant viral DNAs.

Bacteriophage λ vectors permit the cloning of DNA fragments of up to 23,000 bp. Once the bacteriophage λ fragments are ligated to foreign DNA fragments of suitable size, the resulting recombinant DNAs can be pack-

aged into phage particles by adding them to crude bacterial cell extracts that contain all the proteins needed to assemble a complete phage. This is called **in vitro packaging** (Fig. 9-6). All viable phage particles will contain a foreign DNA fragment. The subsequent transmission of the recombinant DNA into *E. coli* cells is highly efficient.

Bacterial Artificial Chromosomes (BACs) Bacterial artificial chromosomes are simply plasmids designed for the cloning of very long segments (typically 100,000 to 300,000 bp) of DNA (Fig. 9-7). They generally include selectable markers such as resistance to the antibiotic chloramphenicol (Cm^R), as well as a very stable origin of replication (*ori*) that maintains the plasmid at one or two copies per cell. DNA fragments of several hundred thousand base pairs are cloned into the BAC vector. The large circular DNAs are then introduced into host bacteria by electroporation. These procedures use host bacteria with mutations that compromise the structure of their cell wall, permitting the uptake of the large DNA molecules.

Yeast Artificial Chromosomes (YACs) *E. coli* cells are by no means the only hosts for genetic engineering. Yeasts are particularly convenient eukaryotic organisms for this work. As with *E. coli*, yeast genetics is a well-developed discipline. The genome of the most commonly used yeast, *Saccharomyces cerevisiae*, contains only 14×10^6 bp (a simple genome by eukaryotic standards, less than four times the size of the *E. coli* chromosome), and its entire sequence is known. Yeast is also very easy to maintain and grow on a large scale in the laboratory. Plasmid vectors have been constructed for yeast, employing the same principles that govern the use of *E. coli* vectors described above. Convenient methods are now available for moving DNA into and out of yeast cells, facilitating the study of many aspects of eukaryotic cell biochemistry. Some recombinant plasmids incorporate multiple replication origins and other elements that allow them to be used in more than one species (for example, yeast or *E. coli*). Plasmids that can be propagated in cells of two or more different species are called **shuttle vectors**.

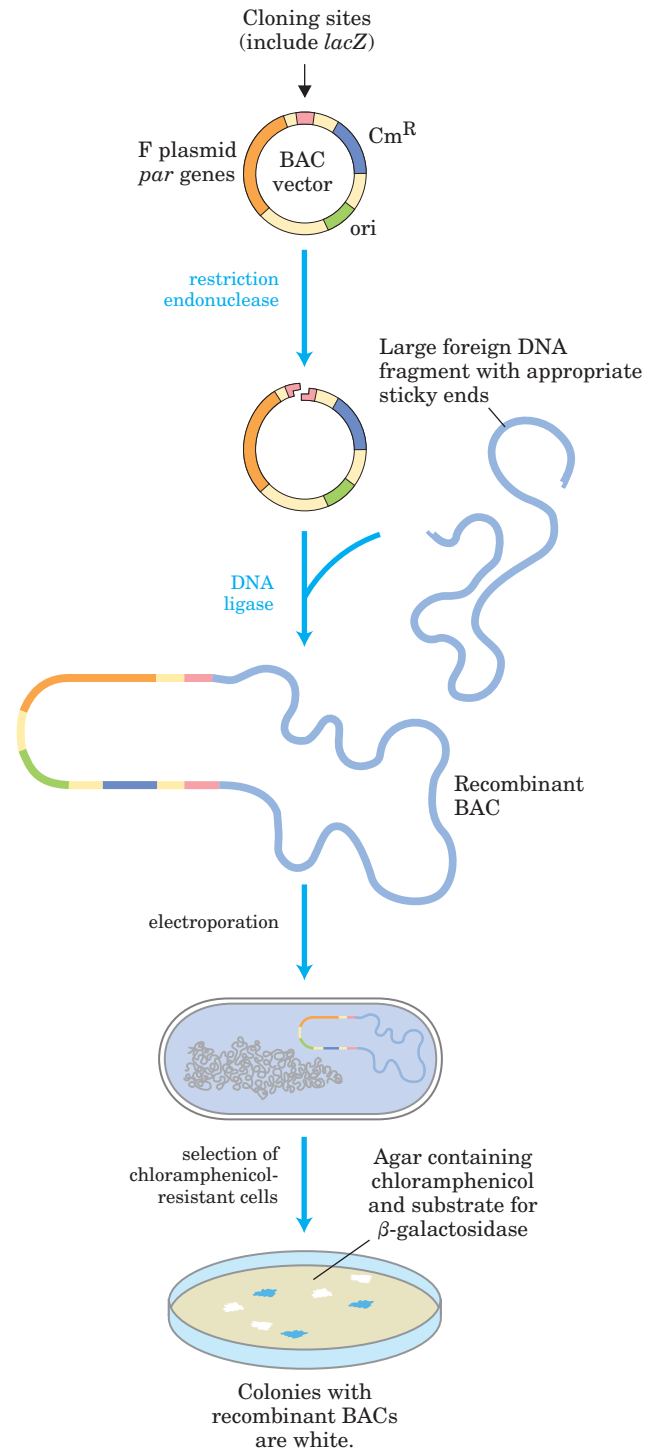


FIGURE 9-7 (above right) Bacterial artificial chromosomes (BACs) as cloning vectors. The vector is a relatively simple plasmid, with a replication origin (*ori*) that directs replication. The *par* genes, derived from a type of plasmid called an F plasmid, assist in the even distribution of plasmids to daughter cells at cell division. This increases the likelihood of each daughter cell carrying one copy of the plasmid, even when few copies are present. The low number of copies is useful in cloning large segments of DNA because it limits the opportunities for unwanted recombination reactions that can unpredictably alter large cloned DNAs over time. The BAC includes selectable markers. A *lacZ*

gene (required for the production of the enzyme β -galactosidase) is situated in the cloning region such that it is inactivated by cloned DNA inserts. Introduction of recombinant BACs into cells by electroporation is promoted by the use of cells with an altered (more porous) cell wall. Recombinant DNAs are screened for resistance to the antibiotic chloramphenicol (Cm^R). Plates also contain a substrate for β -galactosidase that yields a colored product. Colonies with active β -galactosidase and hence no DNA insert in the BAC vector turn blue; colonies without β -galactosidase activity—and thus with the desired DNA inserts—are white.

Research work with large genomes and the associated need for high-capacity cloning vectors led to the development of **yeast artificial chromosomes (YACs)** (Fig. 9–8). YAC vectors contain all the elements needed to maintain a eukaryotic chromosome in the yeast nucleus: a yeast origin of replication, two selectable markers, and specialized sequences (derived from the telomeres and centromere, regions of the chromosome discussed in Chapter 24) needed for stability and

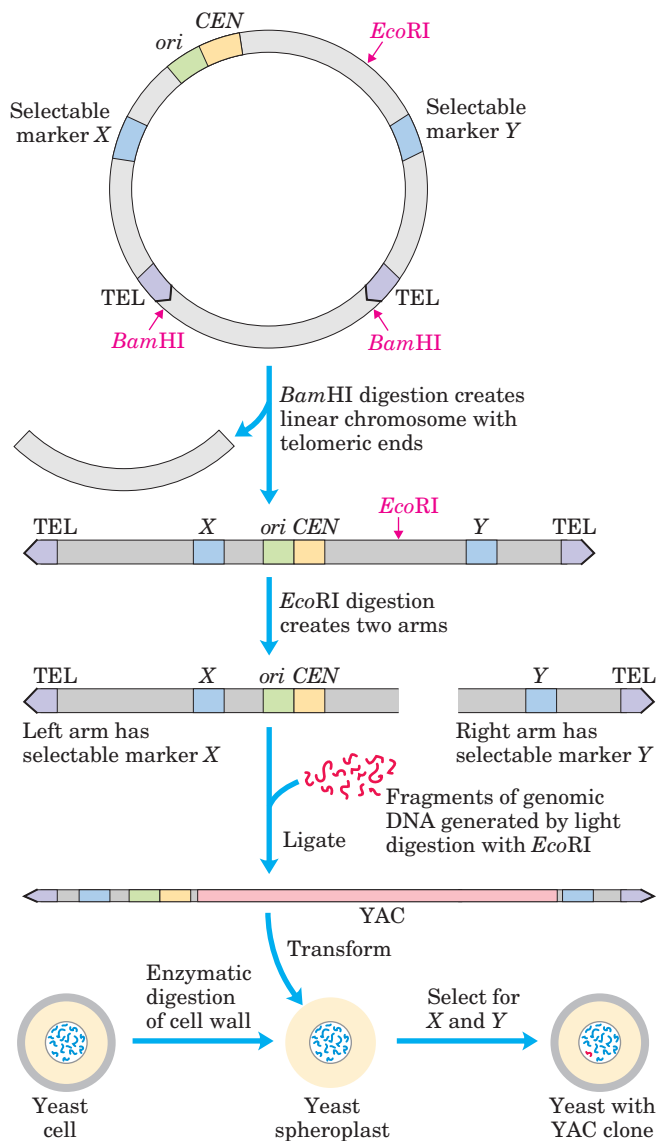


FIGURE 9–8 Construction of a yeast artificial chromosome (YAC). A YAC vector includes an origin of replication (*ori*), a centromere (*CEN*), two telomeres (*TEL*), and selectable markers (*X* and *Y*). Digestion with *Bam*HI and *Eco*RI generates two separate DNA arms, each with a telomeric end and one selectable marker. A large segment of DNA (e.g., up to 2×10^6 bp from the human genome) is ligated to the two arms to create a yeast artificial chromosome. The YAC transforms yeast cells (prepared by removal of the cell wall to form spheroplasts), and the cells are selected for *X* and *Y*; the surviving cells propagate the DNA insert.

proper segregation of the chromosomes at cell division. Before being used in cloning, the vector is propagated as a circular bacterial plasmid. Cleavage with a restriction endonuclease (*Bam*HI in Fig. 9–8) removes a length of DNA between two telomere sequences (*TEL*), leaving the telomeres at the ends of the linearized DNA. Cleavage at another internal site (*Eco*RI in Fig. 9–8) divides the vector into two DNA segments, referred to as vector arms, each with a different selectable marker.

The genomic DNA is prepared by partial digestion with restriction endonucleases (*Eco*RI in Fig. 9–8) to obtain a suitable fragment size. Genomic fragments are then separated by **pulsed field gel electrophoresis**, a variation of gel electrophoresis (see Fig. 3–19) that allows the separation of very large DNA segments. The DNA fragments of appropriate size (up to about 2×10^6 bp) are mixed with the prepared vector arms and ligated. The ligation mixture is then used to transform treated yeast cells with very large DNA molecules. Culture on a medium that requires the presence of both selectable marker genes ensures the growth of only those yeast cells that contain an artificial chromosome with a large insert sandwiched between the two vector arms (Fig. 9–8). The stability of YAC clones increases with size (up to a point). Those with inserts of more than 150,000 bp are nearly as stable as normal cellular chromosomes, whereas those with inserts of less than 100,000 bp are gradually lost during mitosis (so generally there are no yeast cell clones carrying only the two vector ends ligated together or with only short inserts). YACs that lack a telomere at either end are rapidly degraded.

Specific DNA Sequences Are Detectable by Hybridization

DNA hybridization, a process outlined in Chapter 8 (see Fig. 8–32), is the most common sequence-based process for detecting a particular gene or segment of nucleic acid. There are many variations of the basic method, most making use of a labeled (such as radioactive) DNA or RNA fragment, known as a **probe**, complementary to the DNA being sought. In one classic approach to detect a particular DNA sequence within a DNA library (a collection of DNA clones), nitrocellulose paper is pressed onto an agar plate containing many individual bacterial colonies from the library, each colony with a different recombinant DNA. Some cells from each colony adhere to the paper, forming a replica of the plate. The paper is treated with alkali to disrupt the cells and denature the DNA within, which remains bound to the region of the paper around the colony from which it came. Added radioactive DNA probe anneals only to its complementary DNA. After any unannealed probe DNA is washed away, the hybridized DNA can be detected by autoradiography (Fig. 9–9).

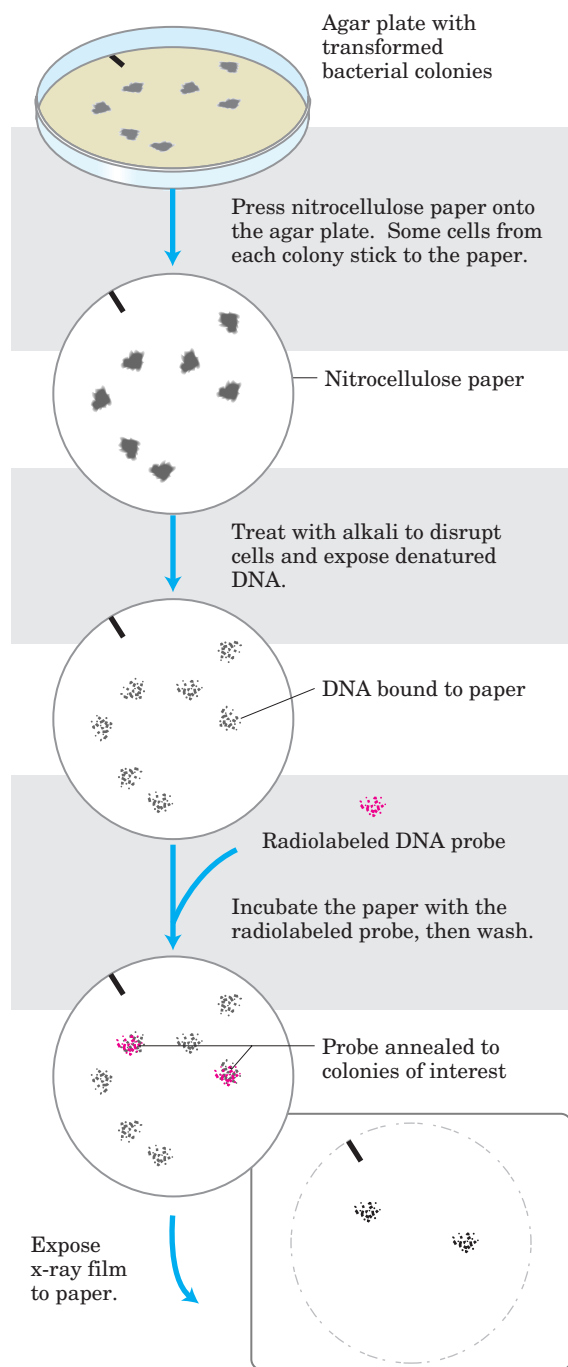


FIGURE 9-9 Use of hybridization to identify a clone with a particular DNA segment. The radioactive DNA probe hybridizes to complementary DNA and is revealed by autoradiography. Once the labeled colonies have been identified, the corresponding colonies on the original agar plate can be used as a source of cloned DNA for further study.

A common limiting step in detecting and cloning a gene is the generation of a complementary strand of nucleic acid to use as a probe. The origin of a probe depends on what is known about the gene under investigation. Sometimes a homologous gene cloned from

another species makes a suitable probe. Or, if the protein product of a gene has been purified, probes can be designed and synthesized by working backward from the amino acid sequence, deducing the DNA sequence that would code for it (Fig. 9–10). Now, researchers typically obtain the necessary DNA sequence information from sequence databases that detail the structure of millions of genes from a wide range of organisms.

Expression of Cloned Genes Produces Large Quantities of Protein

Frequently it is the product of the cloned gene, rather than the gene itself, that is of primary interest—particularly when the protein has commercial, therapeutic, or research value. With an increased understanding of the fundamentals of DNA, RNA, and protein metabolism and their regulation in *E. coli*, investigators can now manipulate cells to express cloned genes in order to study their protein products.

Most eukaryotic genes lack the DNA sequence elements—such as promoters, sequences that instruct RNA polymerase where to bind—required for their expression in *E. coli* cells, so bacterial regulatory sequences for transcription and translation must be inserted at appropriate positions relative to the eukaryotic gene in the vector DNA. (Promoters, regulatory sequences, and other aspects of the regulation of gene expression are discussed in Chapter 28.) In some cases cloned genes are so efficiently expressed that their protein product represents 10% or more of the cellular protein; they are said to be overexpressed. At these concentrations some foreign proteins can kill an *E. coli* cell, so gene expression must be limited to the few hours before the planned harvest of the cells.

Cloning vectors with the transcription and translation signals needed for the regulated expression of a cloned gene are often called **expression vectors**. The rate of expression of the cloned gene is controlled by replacing the gene's own promoter and regulatory sequences with more efficient and convenient versions supplied by the vector. Generally, a well-characterized promoter and its regulatory elements are positioned near several unique restriction sites for cloning, so that genes inserted at the restriction sites will be expressed from the regulated promoter element (Fig. 9–11). Some of these vectors incorporate other features, such as a bacterial ribosome binding site to enhance translation of the mRNA derived from the gene, or a transcription termination sequence.

Genes can similarly be cloned and expressed in eukaryotic cells, with various species of yeast as the usual hosts. A eukaryotic host can sometimes promote post-translational modifications (changes in protein structure made after synthesis on the ribosomes) that might be required for the function of a cloned eukaryotic protein.

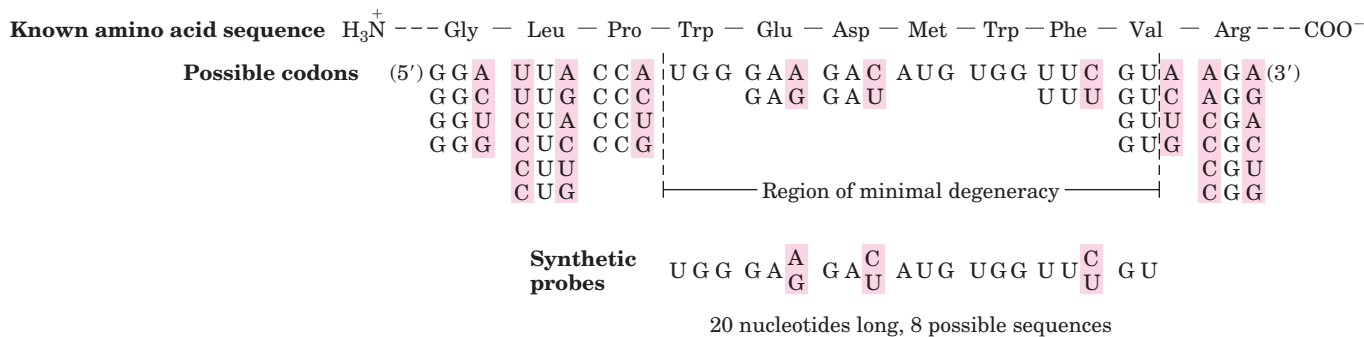


FIGURE 9-10 Probe to detect the gene for a protein of known amino acid sequence. Because more than one DNA sequence can code for any given amino acid sequence, the genetic code is said to be “degenerate.” (As described in Chapter 27, an amino acid is coded for by a set of three nucleotides called a *codon*. Most amino acids have two or more codons; see Fig. 27–7.) Thus the correct DNA sequence for a known amino acid sequence cannot be known in advance. The probe is designed to be complementary to a region of the gene with

minimal degeneracy, that is, a region with the fewest possible codons for the amino acids—two codons at most in the example shown here. Oligonucleotides are synthesized with selectively randomized sequences, so that they contain either of the two possible nucleotides at each position of potential degeneracy (shaded in pink). The oligonucleotide shown here represents a mixture of eight different sequences: one of the eight will complement the gene perfectly, and all eight will match at least 17 of the 20 positions.

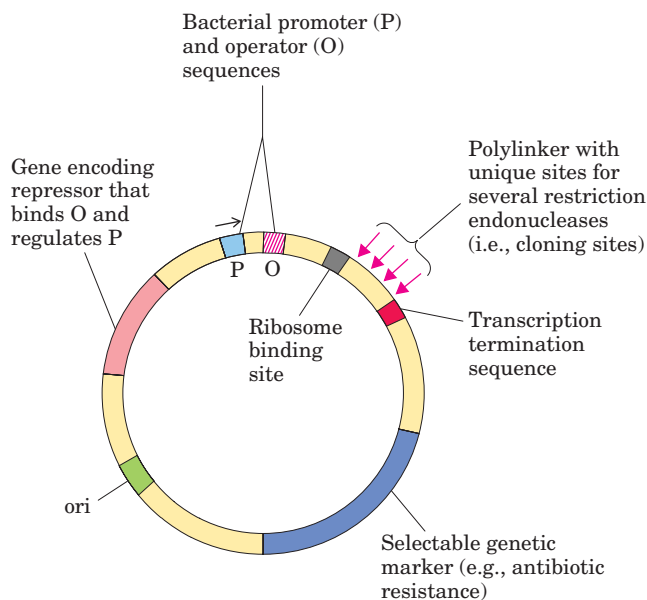


FIGURE 9-11 DNA sequences in a typical *E. coli* expression vector. The gene to be expressed is inserted into one of the restriction sites in the polylinker, near the promoter (P), with the end encoding the amino terminus proximal to the promoter. The promoter allows efficient transcription of the inserted gene, and the transcription termination sequence sometimes improves the amount and stability of the mRNA produced. The operator (O) permits regulation by means of a repressor that binds to it (Chapter 28). The ribosome binding site provides sequence signals needed for efficient translation of the mRNA derived from the gene. The selectable marker allows the selection of cells containing the recombinant DNA.

Alterations in Cloned Genes Produce Modified Proteins

Cloning techniques can be used not only to overproduce proteins but to produce protein products subtly altered from their native forms. Specific amino acids may be replaced individually by **site-directed mutagenesis**. This powerful approach to studying protein structure and function changes the amino acid sequence of a protein by altering the DNA sequence of the cloned gene. If appropriate restriction sites flank the sequence to be altered, researchers can simply remove a DNA segment and replace it with a synthetic one that is identical to the original except for the desired change (Fig. 9–12a). When suitably located restriction sites are not present, an approach called **oligonucleotide-directed mutagenesis** (Fig. 9–12b) can create a specific DNA sequence change. A short synthetic DNA strand with a specific base change is annealed to a single-stranded copy of the cloned gene within a suitable vector. The mismatch of a single base pair in 15 to 20 bp does not prevent annealing if it is done at an appropriate temperature. The annealed strand serves as a primer for the synthesis of a strand complementary to the plasmid vector. This slightly mismatched duplex recombinant plasmid is then used to transform bacteria, where the mismatch is repaired by cellular DNA repair enzymes (Chapter 25). About half of the repair events will remove and replace the altered base and restore the gene to its original sequence; the other half will remove and

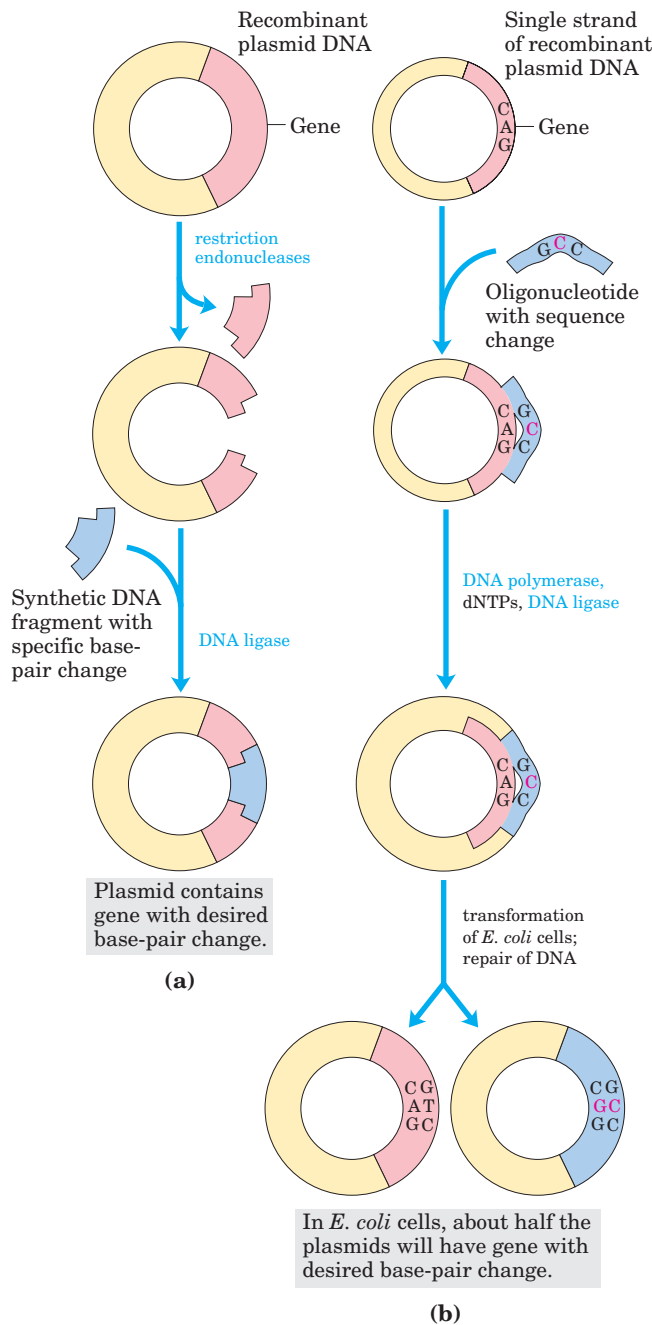


FIGURE 9-12 Two approaches to site-directed mutagenesis. (a) A synthetic DNA segment replaces a DNA fragment that has been removed by cleavage with a restriction endonuclease. (b) A synthetic oligonucleotide with a desired sequence change at one position is hybridized to a single-stranded copy of the gene to be altered. This acts as primer for synthesis of a duplex DNA (with one mismatch), which is then used to transform cells. Cellular DNA repair systems will convert about 50% of the mismatches to reflect the desired sequence change.

replace the *normal* base, retaining the desired mutation. Transformants are screened (often by sequencing their plasmid DNA) until a bacterial colony containing a plasmid with the altered sequence is found.

Changes can also be introduced that involve more than one base pair. Large parts of a gene can be deleted by cutting out a segment with restriction endonucleases and ligating the remaining portions to form a smaller gene. Parts of two different genes can be ligated to create new combinations. The product of such a fused gene is called a **fusion protein**.

Researchers now have ingenious methods to bring about virtually any genetic alteration in vitro. Reintroduction of the altered DNA into the cell permits investigation of the consequences of the alteration. Site-directed mutagenesis has greatly facilitated research on proteins by allowing investigators to make specific changes in the primary structure of a protein and to examine the effects of these changes on the folding, three-dimensional structure, and activity of the protein.

SUMMARY 9.1 DNA Cloning: The Basics

- DNA cloning and genetic engineering involve the cleavage of DNA and assembly of DNA segments in new combinations—recombinant DNA.
- Cloning entails cutting DNA into fragments with enzymes; selecting and possibly modifying a fragment of interest; inserting the DNA fragment into a suitable cloning vector; transferring the vector with the DNA insert into a host cell for replication; and identifying and selecting cells that contain the DNA fragment.
- Key enzymes in gene cloning include restriction endonucleases (especially the type II enzymes) and DNA ligase.
- Cloning vectors include plasmids, bacteriophages, and, for the longest DNA inserts, bacterial artificial chromosomes (BACs) and yeast artificial chromosomes (YACs).
- Cells containing particular DNA sequences can be identified by DNA hybridization methods.
- Genetic engineering techniques manipulate cells to express and/or alter cloned genes.

9.2 From Genes to Genomes

The modern science of **genomics** now permits the study of DNA on a cellular scale, from individual genes to the entire genetic complement of an organism—its genome. Genomic databases are growing rapidly, as one sequencing milestone is superseded by the next. Biology in the twenty-first century will move forward with the aid of informational resources undreamed of only a few years ago. We now turn to a consideration of some of the technologies fueling these advances.

DNA Libraries Provide Specialized Catalogs of Genetic Information

A DNA library is a collection of DNA clones, gathered together as a source of DNA for sequencing, gene discovery, or gene function studies. The library can take a variety of forms, depending on the source of the DNA. Among the largest types of DNA library is a **genomic library**, produced when the complete genome of a particular organism is cleaved into thousands of fragments, and *all* the fragments are cloned by insertion into a cloning vector.

The first step in preparing a genomic library is partial digestion of the DNA by restriction endonucleases, such that any given sequence will appear in fragments of a range of sizes—a range that is compatible with the cloning vector and ensures that virtually all sequences are represented among the clones in the library. Fragments that are too large or too small for cloning are removed by centrifugation or electrophoresis. The cloning vector, such as a BAC or YAC plasmid, is cleaved with the same restriction endonuclease and ligated to the genomic DNA fragments. The ligated DNA mixture is then used to transform bacterial or yeast cells to produce a library of cell types, each type harboring a different recombinant DNA molecule. Ideally, all the DNA in the genome under study will be represented in the library. Each transformed bacterium or yeast cell grows into a colony, or “clone,” of identical cells, each cell bearing the same recombinant plasmid.

Using hybridization methods, researchers can order individual clones in a library by identifying clones with overlapping sequences. A set of overlapping clones represents a catalog for a long contiguous segment of a genome, often referred to as a **contig** (Fig. 9–13). Previously studied sequences or entire genes can be located within the library using hybridization methods to determine which library clones harbor the known sequence. If the sequence has already been mapped on a chromosome, investigators can determine the location (in the genome) of the cloned DNA and any contig of which it is a part. A well-characterized library may contain thousands of long contigs, all assigned to and ordered on particular chromosomes to form a detailed physical map. The known sequences within the library (each called a **sequence-tagged site**, or **STS**) can provide landmarks for genomic sequencing projects.

As more and more genome sequences become available, the utility of genomic libraries is diminishing and investigators are constructing more specialized libraries designed to study gene function. An example is a library that includes only those genes that are *expressed*—that is, are transcribed into RNA—in a given organism or even in certain cells or tissues. Such a library lacks the noncoding DNA that makes up a large portion of many eukaryotic genomes. The researcher first extracts mRNA from an organism or from specific cells of an or-

Segment of chromosome from organism X

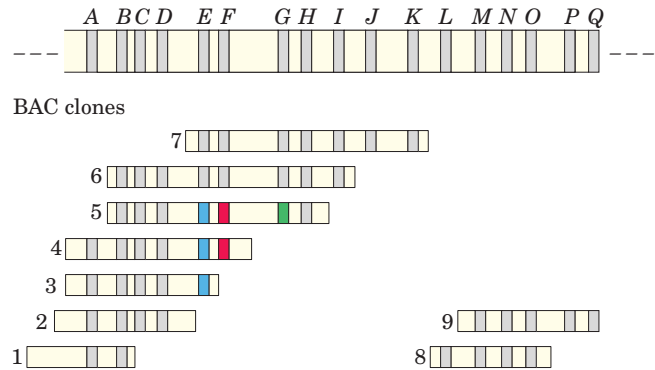


FIGURE 9–13 Ordering of the clones in a DNA library. Shown here is a segment of a chromosome from a hypothetical organism X, with markers A through Q representing sequence-tagged sites (STSs—DNA segments of known sequence, including known genes). Below the chromosome is an array of ordered BAC clones, numbered 1 to 9. Ordering the clones on the genetic map is a many-stage process. The presence or absence of an STS on an individual clone can be determined by hybridization—for example, by probing each clone with PCR-amplified DNA from the STS. Once the STSs on each BAC clone are identified, the clones (and the STSs themselves, if their location is not yet known) can be ordered on the map. For example, compare clones 3, 4, and 5. Marker E (blue) is found on all three clones; F (red) on clones 4 and 5, but not on 3; and G (green) only on clone 5. This indicates that the order of the sites is E, F, G. The clones partially overlap and their order must be 3, 4, 5. The resulting ordered series of clones is called a contig.

ganism and then prepares **complementary DNAs (cDNAs)** from the RNA in a multistep reaction catalyzed by the enzyme reverse transcriptase (Fig. 9–14). The resulting double-stranded DNA fragments are then inserted into a suitable vector and cloned, creating a population of clones called a **cDNA library**. The search for a particular gene is made easier by focusing on a cDNA library generated from the mRNAs of a cell known to express that gene. For example, if we wished to clone globin genes, we could first generate a cDNA library from erythrocyte precursor cells, in which about half the mRNAs code for globins. To aid in the mapping of large genomes, cDNAs in a library can be partially sequenced at random to produce a useful type of STS called an **expressed sequence tag (EST)**. ESTs, ranging in size from a few dozen to several hundred base pairs, can be positioned within the larger genome map, providing markers for expressed genes. Hundreds of thousands of ESTs were included in the detailed physical maps used as a guide to sequencing the human genome.

A cDNA library can be made even more specialized by cloning a cDNA or cDNA fragment into a vector that fuses the cDNA sequence with the sequence for a marker, or reporter gene; the fused genes form a “reporter construct.” Two useful markers are the genes for green fluorescent protein and epitope tags. A target

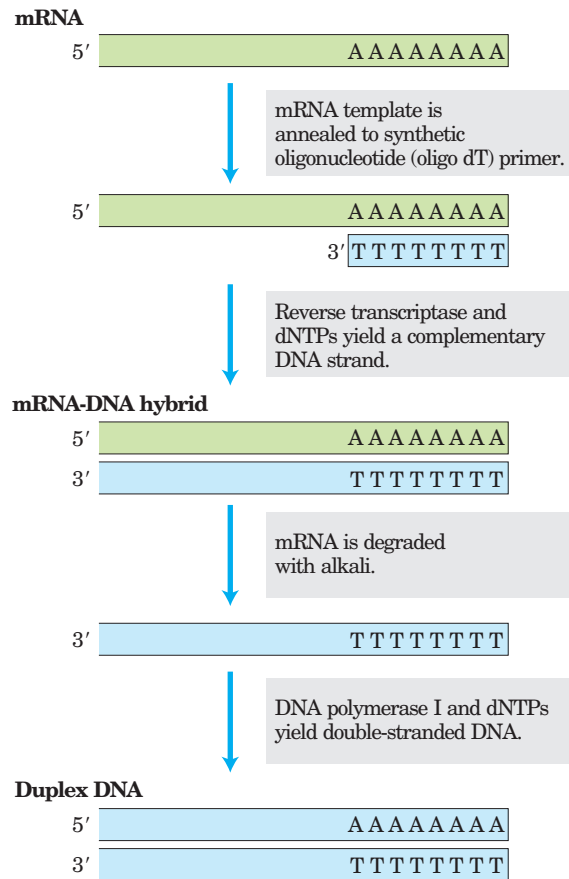


FIGURE 9-14 Construction of a cDNA library from mRNA. A cell's mRNA includes transcripts from thousands of genes, and the cDNAs generated are correspondingly heterogeneous. The duplex DNA produced by this method is inserted into an appropriate cloning vector. Reverse transcriptase can synthesize DNA on an RNA or a DNA template (see Fig. 26–29).

gene fused with a gene for **green fluorescent protein (GFP)** generates a fusion protein that is highly fluorescent—it literally lights up (Fig. 9–15a). Just a few molecules of this protein can be observed microscopically, allowing the study of its location and movements in a cell. An **epitope tag** is a short protein sequence that is bound tightly by a well-characterized monoclonal antibody (Chapter 5). The tagged protein can be specifically precipitated from a crude protein extract by interaction with the antibody (Fig. 9–15b). If any other proteins bind to the tagged protein, those will precipitate as well, providing information about protein-protein interactions in a cell. The diversity and utility of specialized DNA libraries are growing every year.

The Polymerase Chain Reaction Amplifies Specific DNA Sequences

The Human Genome Project, along with the many associated efforts to sequence the genomes of organisms of every type, is providing unprecedented access to gene sequence information. This in turn is simplifying the

process of cloning individual genes for more detailed biochemical analysis. If we know the sequence of at least the flanking parts of a DNA segment to be cloned, we can hugely amplify the number of copies of that DNA segment, using the **polymerase chain reaction (PCR)**, a process conceived by Kary Mullis in 1983. The amplified DNA can be cloned directly or used in a variety of analytical procedures.

The PCR procedure has an elegant simplicity. Two synthetic oligonucleotides are prepared, complementary to sequences on opposite strands of the target DNA at positions just beyond the ends of the segment to be amplified. The oligonucleotides serve as replication primers that can be extended by DNA polymerase. The 3' ends of the hybridized probes are oriented toward each other and positioned to prime DNA synthesis across the desired DNA segment (Fig. 9–16). (DNA polymerases

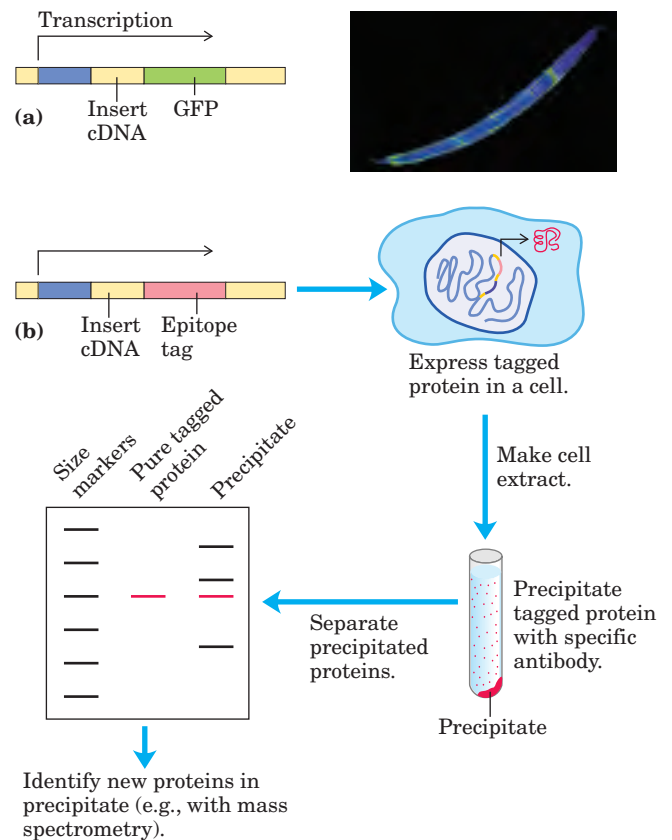


FIGURE 9-15 Specialized DNA libraries. (a) Cloning of cDNA next to a gene for green fluorescent protein (GFP) creates a reporter construct. RNA transcription proceeds through the gene of interest (insert DNA) and the reporter gene, and the mRNA transcript is then expressed as a fusion protein. The GFP part of the protein is visible in the fluorescence microscope. The photograph shows a nematode worm containing a GFP fusion protein expressed only in the four “touch” neurons that run the length of its body. **Reporter Constructs** (b) If the cDNA is cloned next to a gene for an epitope tag, the resulting fusion protein can be precipitated by antibodies to the epitope. Any other proteins that interact with the tagged protein also precipitate, helping to elucidate protein-protein interactions.

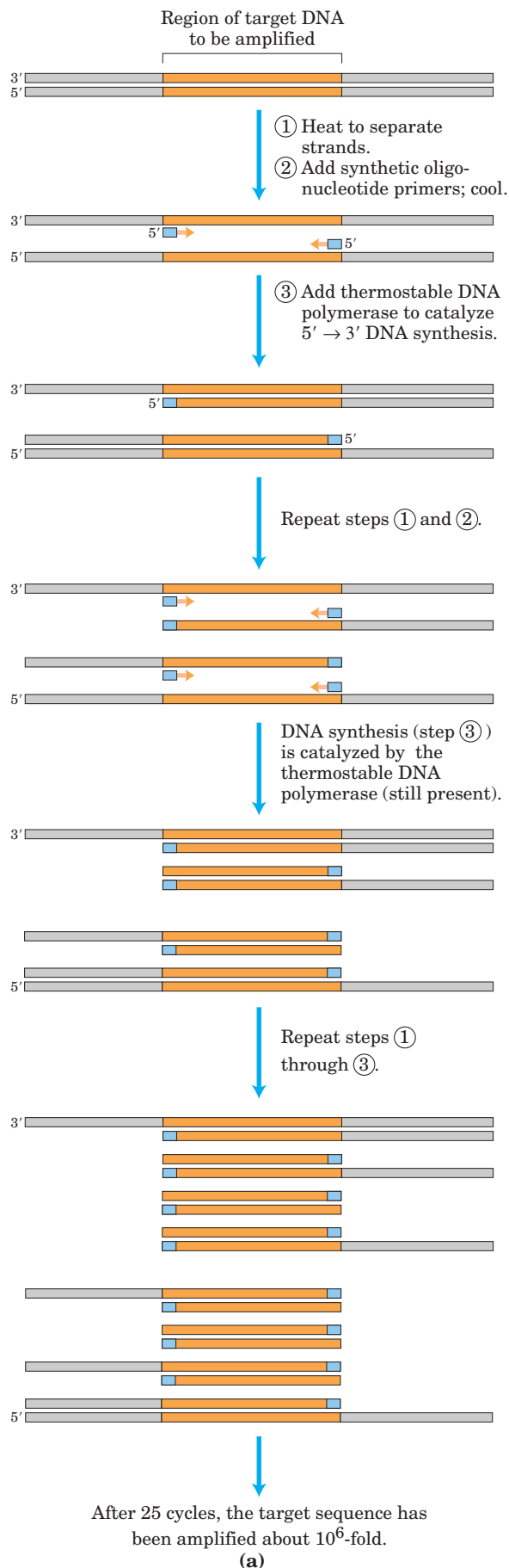
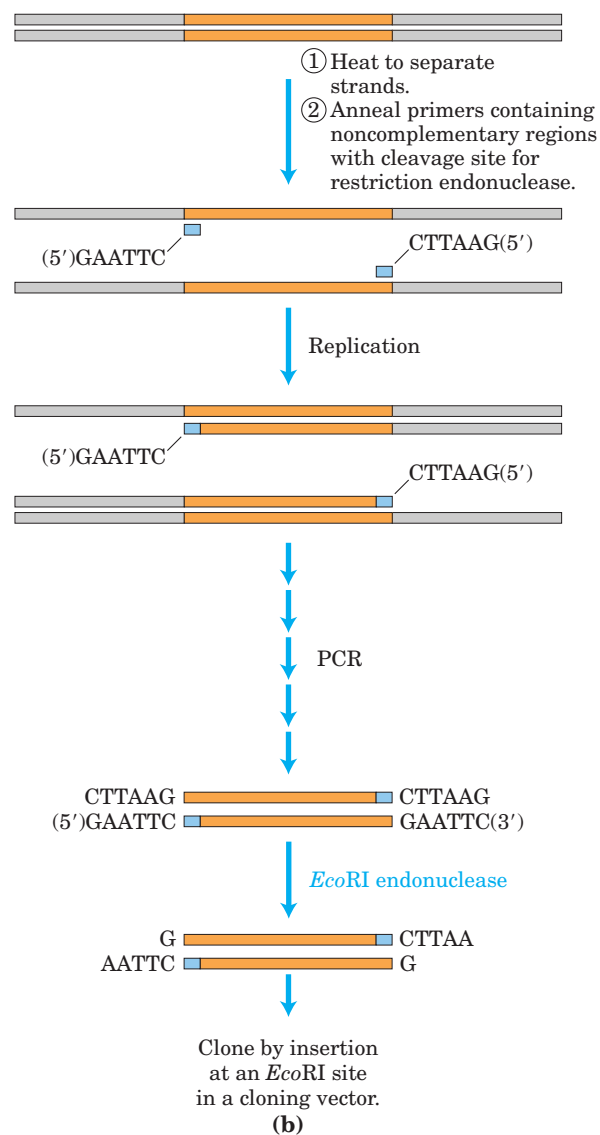


FIGURE 9-16 Amplification of a DNA segment by the polymerase chain reaction. (a) The PCR procedure has three steps. DNA strands are ① separated by heating, then ② annealed to an excess of short synthetic DNA primers (blue) that flank the region to be amplified; ③ new DNA is synthesized by polymerization. The three steps are repeated for 25 or 30 cycles. The thermostable DNA polymerase *TaqI* (from *Thermus aquaticus*, a bacterial species that grows in hot springs) is not denatured by the heating steps. (b) DNA amplified by PCR can be cloned. The primers can include noncomplementary ends that have a site for cleavage by a restriction endonuclease. Although these parts of the primers do not anneal to the target DNA, the PCR process incorporates them into the DNA that is amplified. Cleavage of the amplified fragments at these sites creates sticky ends, used in ligation of the amplified DNA to a cloning vector.

Polymerase Chain Reaction



synthesize DNA strands from deoxyribonucleotides, using a DNA template, as described in Chapter 25.) Isolated DNA containing the segment to be amplified is heated briefly to denature it, and then cooled in the presence of a large excess of the synthetic oligonucleotide primers. The four deoxynucleoside triphosphates are then added, and the primed DNA segment is replicated selectively. The cycle of heating, cooling, and replication is repeated 25 or 30 times over a few hours in an automated process, amplifying the DNA segment flanked by the primers until it can be readily analyzed or cloned. PCR uses a heat-stable DNA polymerase, such as the *Taq* polymerase (derived from a bacterium that lives at 90 °C), which remains active after every heating step and does not have to be replenished. Careful design of the primers used for PCR, such as including restriction endonuclease cleavage sites, can facilitate the subsequent cloning of the amplified DNA (Fig. 9-16b).

This technology is highly sensitive: PCR can detect and amplify as little as one DNA molecule in almost any type of sample. Although DNA degrades over time (p. 293), PCR has allowed successful cloning of DNA from samples more than 40,000 years old. Investigators have used the technique to clone DNA fragments from the mummified remains of humans and extinct animals such as the woolly mammoth, creating the new fields of molecular archaeology and molecular paleontology. DNA from burial sites has been amplified by PCR and used to trace ancient human migrations. Epidemiologists can use PCR-enhanced DNA samples from human remains to trace the evolution of human pathogenic viruses. Thus, in addition to its usefulness for cloning DNA, PCR is a potent tool in forensic medicine (Box 9-1). It is also being used for detection of viral infections before they cause symptoms and for prenatal diagnosis of a wide array of genetic diseases.

The PCR method is also important in advancing the goal of whole genome sequencing. For example, the mapping of expressed sequence tags to particular chromosomes often involves amplification of the EST by PCR, followed by hybridization of the amplified DNA to clones in an ordered library. Investigators found many other applications of PCR in the Human Genome Project, to which we now turn.

Genome Sequences Provide the Ultimate Genetic Libraries

The genome is the ultimate source of information about an organism, and there is no genome we are more interested in than our own. Less than 10 years after the development of practical DNA sequencing methods, serious discussions began about the prospects for sequencing the entire 3 billion base pairs of the human genome. The international Human Genome Project got underway with substantial funding in the late 1980s. The effort eventually included significant contributions from

20 sequencing centers distributed among six nations: the United States, Great Britain, Japan, France, China, and Germany. General coordination was provided by the Office of Genome Research at the National Institutes of Health, led first by James Watson and after 1992 by Francis Collins. At the outset, the task of sequencing a 3×10^9 bp genome seemed to be a titanic job, but it gradually yielded to advances in technology. The completed sequence of the human genome was published in April 2003, several years ahead of schedule.

This advance was the product of a carefully planned international effort spanning 14 years. Research teams first generated a detailed physical map of the human genome, with clones derived from each chromosome organized into a series of long contigs (Fig. 9-17). Each contig contained landmarks in the form of STSs at a distance of every 100,000 bp or less. The genome thus mapped could be divided up between the international sequencing centers, each center sequencing the mapped BAC or YAC clones corresponding to its particular segments of the genome. Because many of the

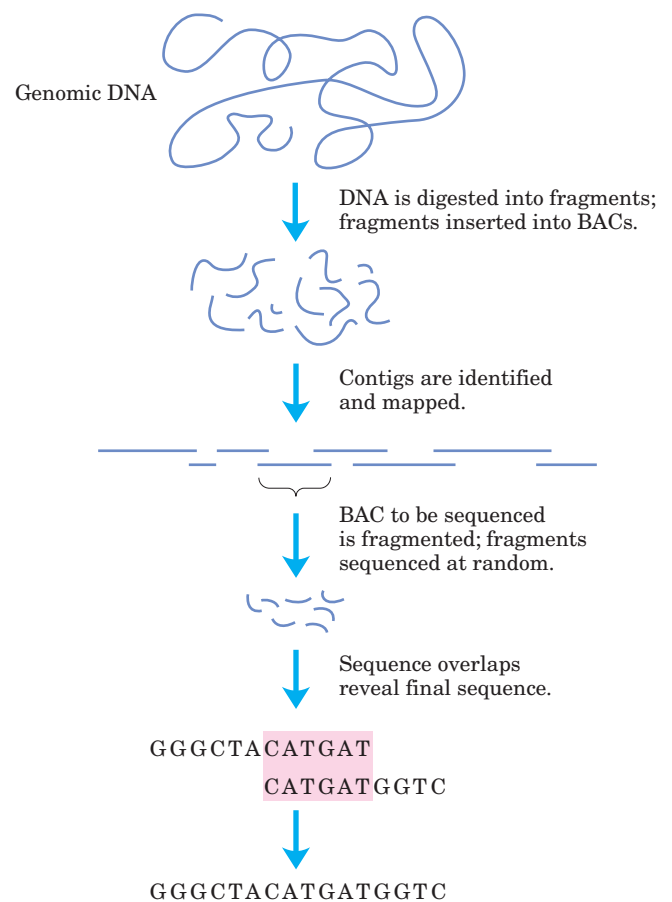


FIGURE 9-17 The Human Genome Project strategy. Clones isolated from a genomic library were ordered into a detailed physical map, then individual clones were sequenced by shotgun sequencing protocols. The strategy used by the commercial sequencing effort eliminated the step of creating the physical map and sequenced the entire genome by shotgun cloning.

BOX 9-1 WORKING IN BIOCHEMISTRY

A Potent Weapon in Forensic Medicine

Traditionally, one of the most accurate methods for placing an individual at the scene of a crime has been a fingerprint. With the advent of recombinant DNA technology, a more powerful tool is now available: **DNA fingerprinting** (also called DNA typing or DNA profiling).

DNA fingerprinting is based on **sequence polymorphisms**, slight sequence differences (usually single base-pair changes) between individuals, 1 bp in every 1,000 bp, on average. Each difference from the prototype human genome sequence (the first one obtained) occurs in some fraction of the human population; every individual has some differences. Some of the sequence changes affect recognition sites for restriction enzymes, resulting in variation in the size of DNA fragments produced by digestion with a particular restriction enzyme. These variations are **restriction fragment length polymorphisms (RFLPs)**.

The detection of RFLPs relies on a specialized hybridization procedure called **Southern blotting** (Fig. 1). DNA fragments from digestion of genomic DNA by restriction endonucleases are separated by size electrophoretically, denatured by soaking the agarose gel in alkali, and then blotted onto a nylon membrane to reproduce the distribution of fragments in the gel. The membrane is immersed in a solution containing a radioactively labeled DNA probe. A probe for a sequence that is repeated several times in the human genome generally identifies a few of the thousands of DNA fragments generated when the human genome is digested with a restriction endonuclease. Autoradiography reveals the fragments to which the probe hybridizes, as in Figure 9-9.

The genomic DNA sequences used in these tests are generally regions containing repetitive DNA

(short sequences repeated thousands of times in tandem), which are common in the genomes of higher eukaryotes (see Fig. 24-8). The number of repeated units in these DNA regions varies among individuals (except between identical twins). With a suitable probe, the pattern of bands produced by DNA fingerprinting is distinctive for each individual. Combining the use of several probes makes the test so selective that it can positively identify a single individual in the entire human population. However, the Southern blot procedure requires relatively fresh DNA samples and larger amounts of DNA than are generally present at a crime scene. RFLP analysis sensitivity is augmented by using PCR (see Fig. 9-16a) to amplify vanishingly small amounts of DNA. This allows investigators to obtain DNA fingerprints from a single hair follicle, a drop of blood, a small semen sample from a rape victim, or samples that might be months or even many years old.

These methods are proving decisive in court cases worldwide. In the example in Figure 1, the DNA from a semen sample obtained from a rape and murder victim was compared with DNA samples from the victim and two suspects. Each sample was cleaved into fragments and separated by gel electrophoresis. Radioactive DNA probes were used to identify a small subset of fragments that contained sequences complementary to the probe. The sizes of the identified fragments varied from one individual to the next, as seen here in the different patterns for the three individuals (victim and two suspects) tested. One suspect's DNA exhibits a banding pattern identical to that of a semen sample taken from the victim. This test used a single probe, but three or four different probes would be used (in separate experiments) to achieve an unambiguous positive identification.

clones were more than 100,000 bp long, and modern sequencing techniques can resolve only 600 to 750 bp of sequence at a time, each clone had to be sequenced in pieces. The sequencing strategy used a shotgun approach, in which researchers used powerful new automated sequencers to sequence random segments of a given clone, then assembled the sequence of the entire clone by computerized identification of overlaps. The number of clone pieces sequenced was determined statistically so that the entire length of the clone was sequenced four to six times on average. The sequenced DNA was then made available in a database covering the entire genome. Construction of the physical map was a

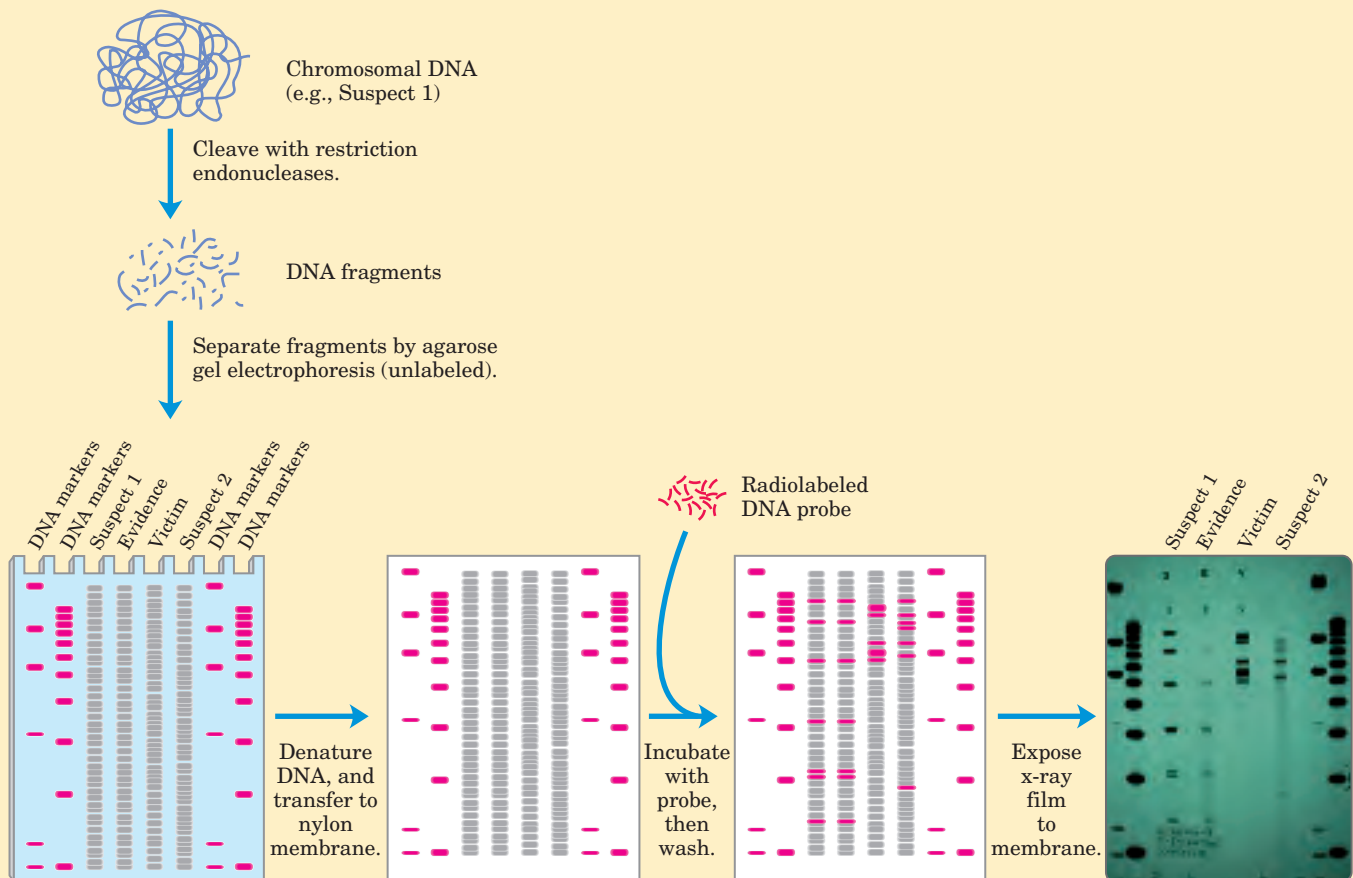
time-consuming task, and its progress was followed in annual reports in major journals throughout the 1990s—by the end of which the map was largely in place. Completion of the entire sequencing project was initially projected for the year 2005, but circumstances and technology intervened to accelerate the process.

A competing commercial effort to sequence the human genome was initiated by the newly established Celera Corporation in 1997. Led by J. Craig Venter, the Celera group made use of a different strategy called “whole genome shotgun sequencing,” which eliminates the step of assembling a physical map of the genome. Instead, teams sequenced DNA segments from through-

Such results have been used to both convict and acquit suspects and, in other cases, to establish paternity with an extraordinary degree of certainty. The impact of these procedures on court cases will continue to grow as societies agree on the standards and as formal methods become widely established in forensic laboratories. Even decades-old murder mysteries

can be solved: in 1996, DNA fingerprinting helped to confirm the identification of the bones of the last Russian czar and his family, who were assassinated in 1918.

FIGURE 1 The Southern blot procedure, as applied to DNA fingerprinting. This procedure was named after Jeremy Southern, who developed the technique.



Francis S. Collins



J. Craig Venter

out the genome at random. The sequenced segments were ordered by the computerized identification of sequence overlaps (with some reference to the public project's detailed physical map). At the outset of the Human Genome Project, shotgun sequencing on this scale had been deemed impractical. However, advances in computer software and sequencing automation had made the approach feasible by 1997. The ensuing race between the private and public sequencing efforts substantially advanced the timeline for completion of the project. Publication of the draft human genome sequence in 2001 was followed by two years of follow-up work to eliminate nearly a thousand discontinuities and

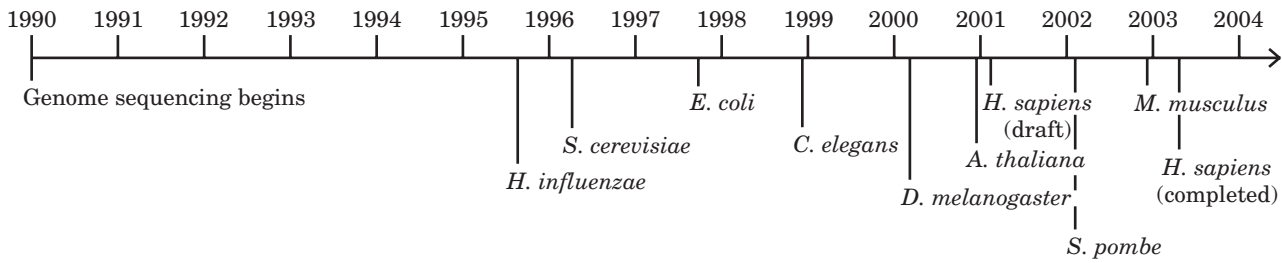


FIGURE 9-18 Genomic sequencing timeline. Discussions in the mid-1980s led to initiation of the project in 1989. Preparatory work, including extensive mapping to provide genome landmarks, occupied much of the 1990s. Separate projects were launched to sequence the genomes of other organisms important to research. The first sequencing efforts to be completed included many bacterial species (such as

Haemophilus influenzae), yeast (*S. cerevisiae*), a nematode worm (*C. elegans*), the fruit fly (*D. melanogaster*), and a plant (*A. thaliana*). Completed sequences for mammalian genomes, including the human genome, began to emerge in 2000. Each genome project has a web-site that serves as a central repository for the latest data.

to provide high-quality sequence data that are contiguous throughout the genome.

The Human Genome Project marks the culmination of twentieth-century biology and promises a vastly changed scientific landscape for the new century. The human genome is only part of the story, as the genomes of many other species are also being (or have been) sequenced, including the yeasts *Saccharomyces cerevisiae* (completed in 1996) and *Schizosaccharomyces pombe* (2002), the nematode *Caenorhabditis elegans* (1998), the fruit fly *Drosophila melanogaster* (2000), the plant *Arabidopsis thaliana* (2000), the mouse *Mus musculus* (2002), zebrafish, and dozens of bacterial and archaeobacterial species (Fig. 9-18). Most of the early efforts have been focused on species commonly used in laboratories. However, genome sequencing is destined to branch out to many other species as experience grows and technology improves. Broad efforts to map genes, attempts to identify new proteins and disease genes, and many other initiatives are currently under way.

The result is a database with the potential not only to fuel rapid advances in biology but to change the way that humans think about themselves. Early insights provided by the human genome sequence range from the intriguing to the profound. We are not as complicated as we thought. Decades-old estimates that humans possessed about 100,000 genes within the approximately 3.2×10^9 bp in the human genome have been supplanted by the discovery that we have only 30,000 to 35,000 genes. This is perhaps three times more genes than a fruit fly (with 13,000) and twice as many as a nematode worm (18,000). Although humans evolved relatively recently, the human genome is very old. Of 1,278 protein families identified in one early screen, only 94 were unique to vertebrates. However, while we share many protein domain types with plants, worms, and flies, we use these domains in more complex arrangements. Alternative modes of gene expression (Chapter 26) allow the production of more than one protein from a single gene—a process that humans and other vertebrates engage in more than do bacteria, worms, or any

other forms of life. This allows for greater complexity in the proteins generated from our gene complement.

We now know that only 1.1% to 1.4% of our DNA actually encodes proteins (Fig. 9-19). More than 50% of our genome consists of short, repeated sequences, the vast majority of which—about 45% of our genome in all—come from transposons, short movable DNA sequences that are molecular parasites (Chapter 25). Many of the transposons have been there a long time, now altered so that they can no longer move to new genomic locations. Others are still actively moving at low frequencies, helping to make the genome an ever-dynamic and evolving entity. At least a few transposons have been co-opted by their host and appear to serve useful cellular functions.

What does all this information tell us about how much one human differs from another? Within the human population are millions of single-base differences, called **single nucleotide polymorphisms**, or **SNPs** (pronounced “snips”). Each human differs from the next

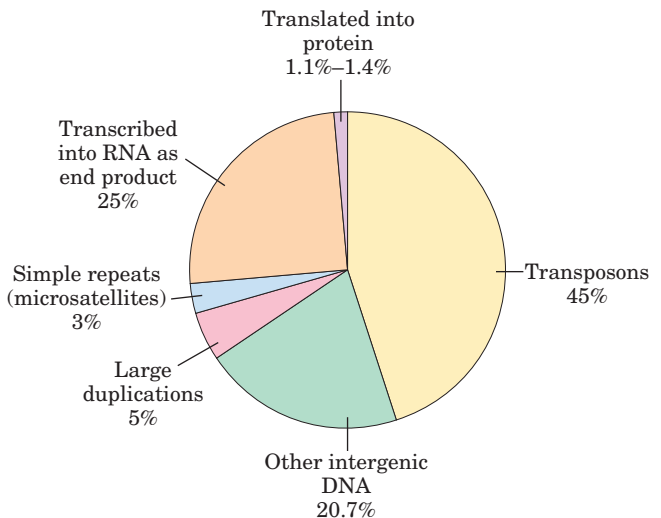


FIGURE 9-19 Snapshot of the human genome. The chart shows the proportions of our genome made up of various types of sequences.

by about 1 bp in every 1,000 bp. From these small genetic differences arises the human variety we are all aware of—differences in hair color, eyesight, allergies to medication, foot size, and even (to some unknown degree) behavior. Some of the SNPs are linked to particular human populations and can provide important information about human migrations that occurred thousands of years ago and about our more distant evolutionary past.

As spectacular as this advance is, the sequencing of the human genome is easy compared with what comes next—the effort to understand all the information in each genome. The genome sequences being added monthly to international databases are roadmaps, parts of which are written in a language we do not yet understand. However, they have great utility in catalyzing the discovery of new proteins and processes affecting every aspect of biochemistry, as will become apparent in chapters to come.

SUMMARY 9.2 From Genes to Genomes

- The science of genomics broadly encompasses the study of genomes and their gene content.
- Genomic DNA segments can be organized in libraries—such as genomic libraries and cDNA libraries—with a wide range of designs and purposes.
- The polymerase chain reaction (PCR) can be used to amplify selected DNA segments from a DNA library or an entire genome.
- In an international cooperative research effort, the genomes of many organisms, including that of humans, have been sequenced in their entirety and are now available in public databases.

9.3 From Genomes to Proteomes

A gene is not simply a DNA sequence; it is information that is converted to a useful product—a protein or functional RNA molecule—when and if needed by the cell. The first and most obvious step in exploring a large sequenced genome is to catalog the products of the genes within that genome. Genes that encode RNA as their final product are somewhat harder to identify than are protein-encoding genes, and even the latter can be very difficult to spot in a vertebrate genome. The explosion of DNA sequence information has also revealed a sobering truth. Despite many years of biochemical advances, there are still thousands of proteins in every eukaryotic cell (and quite a few in bacteria) that we know nothing about. These proteins may have functions in processes not yet discovered, or may contribute in unexpected

ways to processes we think we understand. In addition, the genomic sequences tell us nothing about the three-dimensional structure of proteins or how proteins are modified after they are synthesized. The proteins, with their myriad critical functions in every cell, are now becoming the focus of new strategies for whole cell biochemistry.

The complement of proteins expressed by a genome is called its **proteome**, a term that first appeared in the research literature in 1995. This concept rapidly evolved into a separate field of investigation, called **proteomics**. The problem addressed by proteomics research is straightforward, although the solution is not. Each genome presents us with thousands of genes encoding proteins, and ideally we want to know the structure and function of all those proteins. Given that many proteins offer surprises even after years of study, the investigation of an entire proteome is a daunting enterprise. Simply discovering the function of new proteins requires intensive work. Biochemists can now apply shortcuts in the form of a broad array of new and updated technologies.

Protein function can be described on three levels. **Phenotypic function** describes the effects of a protein on the entire organism. For example, the loss of the protein may lead to slower growth of the organism, an altered development pattern, or even death. **Cellular function** is a description of the network of interactions engaged in by a protein at the cellular level. Interactions with other proteins in the cell can help define the kinds of metabolic processes in which the protein participates. Finally, **molecular function** refers to the precise biochemical activity of a protein, including details such as the reactions an enzyme catalyzes or the ligands a receptor binds.

For several genomes, such as those of the yeast *Saccharomyces cerevisiae* and the plant *Arabidopsis*, a massive effort is underway to inactivate each gene by genetic engineering and to investigate the effect on the organism. If the growth patterns or other properties of the organism change (or if it does not grow at all), this provides information on the phenotypic function of the protein product of the gene.

There are three other main paths to investigating protein function: (1) sequence and structural comparisons with genes and proteins of known function, (2) determination of when and where a gene is expressed, and (3) investigation of the interactions of the protein with other proteins. We discuss each of these approaches in turn.

Sequence or Structural Relationships Provide Information on Protein Function

One of the important reasons to sequence many genomes is to provide a database that can be used to assign gene functions by genome comparisons, an enterprise referred

to as **comparative genomics**. Sometimes a newly discovered gene is related by sequence homologies to a gene previously studied in another or the same species, and its function can be entirely or partly defined by that relationship. Such genes—of different species but possessing a clear sequence and functional relationship to each other—are called **orthologs**. Genes similarly related to each other within a single species are called **paralogs** (see Fig. 1–37). If the function of a gene has been characterized for one species, this information can be used to assign gene function to the ortholog found in the second species. The identity is easiest to make when comparing genomes from relatively closely related species, such as mouse and human, although many clearly orthologous genes have been identified in species as distant as bacteria and humans. Sometimes even the order of genes on a chromosome is conserved over large segments of the genomes of closely related species (Fig. 9–20). Conserved gene order, called **synteny**, provides additional evidence for an orthologous relationship between genes at identical locations within the related segments.

Alternatively, certain sequences associated with particular structural motifs (Chapter 4) may be identified within a protein. The presence of a structural motif may suggest that it, say, catalyzes ATP hydrolysis, binds to DNA, or forms a complex with zinc ions, helping to define molecular function. These relationships are determined with the aid of increasingly sophisticated computer programs, limited only by the current information on gene and protein structure and our capacity to associate sequences with particular structural motifs.

Human 9		Mouse 2
<i>EPB72</i>		<i>Epb7.2</i>
<i>PSMB7</i>		<i>Psmb7</i>
<i>DNM1</i>		<i>Dnm</i>
<i>LMX1B</i>		<i>Lmx1b</i>
<i>CDK9</i>		<i>Cdk9</i>
<i>STXBP1</i>		<i>Stxbp1</i>
<i>AK1</i>		<i>Ak1</i>
<i>LCN2</i>		<i>Lcn2</i>

FIGURE 9–20 Synteny in the mouse and human genomes. Large segments of the mouse and human genomes have closely related genes aligned in the same order on chromosomes, a relationship called synteny. This diagram shows segments of human chromosome 9 and mouse chromosome 2. The genes in these segments exhibit a very high degree of homology as well as the same gene order. The different lettering schemes for the gene names reflect different naming conventions in the two organisms.

To further the assignment of function based on structural relationships, a large-scale structural proteomics project has been initiated. The goal is to crystallize and determine the structure of as many proteins and protein domains as possible, in many cases with little or no existing information about protein function. The project has been assisted by the automation of some of the tedious steps of protein crystallization (see Box 4–4). As these structures are revealed, they will be made available in the structural databases described in Chapter 4. The effort should help define the extent of variation in structural motifs. When a newly discovered protein is found to have structural folds that are clearly related to motifs with known functions in the databases, this information can suggest a molecular function for the protein.

Cellular Expression Patterns Can Reveal the Cellular Function of a Gene

In every newly sequenced genome, researchers find genes that encode proteins with no evident structural relationships to known genes or proteins. In these cases, other approaches must be used to generate information about gene function. Determining which tissues a gene is expressed in, or what circumstances trigger the appearance of the gene product, can provide valuable clues. Many different approaches have been developed to study these patterns.

Two-Dimensional Gel Electrophoresis As shown in Figure 3–22, two-dimensional gel electrophoresis allows the separation and display of up to 1,000 different proteins on a single gel. Mass spectrometry (see Box 3–2) can then be used to partially sequence individual protein spots and assign each to a gene. The appearance and nonappearance (or disappearance) of particular protein spots in samples from different tissues, from similar tissues at different stages of development, or from tissues treated in ways that simulate a variety of biological conditions can help define cellular function.

DNA Microarrays Major refinements of the technology underlying DNA libraries, PCR, and hybridization have come together in the development of **DNA microarrays** (sometimes called **DNA chips**), which allow the rapid and simultaneous screening of many thousands of genes. DNA segments from known genes, a few dozen to hundreds of nucleotides long, are amplified by PCR and placed on a solid surface, using robotic devices that accurately deposit nanoliter quantities of DNA solution. Many thousands of such spots are deposited in a pre-designed array on a surface area of just a few square centimeters. An alternative strategy is to synthesize DNA directly on the solid surface, using photolithography (Fig. 9–21). Once the chip is constructed, it can be probed with mRNAs or cDNAs from a particular cell type

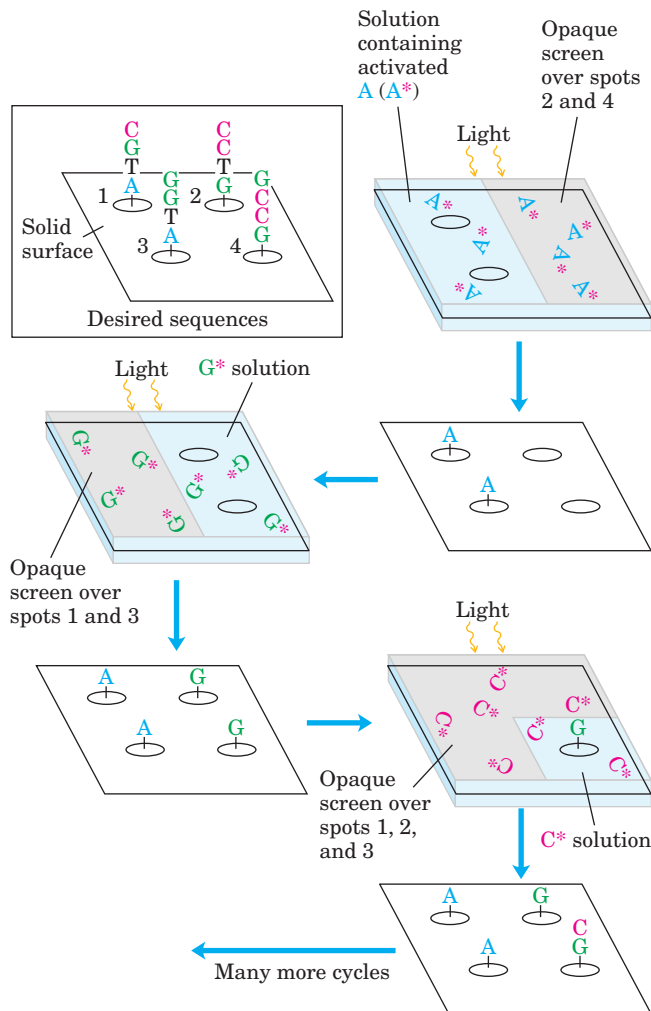


FIGURE 9-21 Photolithography. This technique for preparing a DNA microarray makes use of nucleotide precursors that are activated by light, joining one nucleotide to the next in a photoreaction (as opposed to the chemical process illustrated in Fig. 8–38). A computer is programmed with the oligonucleotide sequences to be synthesized at each point on a solid surface. The surface is washed successively with solutions containing one type of activated nucleotide (A*, G*, etc.). As in the chemical synthesis of DNA, the activated nucleotides are blocked so that only one can be added to a chain in each cycle. A screen covering the surface is opened over the areas programmed to receive a particular nucleotide, and a flash of light joins the nucleotide to the polymers in the uncovered areas. This continues until the required sequences are built up on each spot on the surface. Many polymers with the same sequence are generated on each spot, not just the single polymer shown. Also, the surfaces have thousands of spots with different sequences (see Fig. 9–22); this array shows just four spots, to illustrate the strategy.

or cell culture to identify the genes being expressed in those cells.

A microarray can answer such questions as which genes are expressed at a given stage in the development of an organism. The total complement of mRNA is isolated from cells at two different stages of development

and converted to cDNA, using reverse transcriptase and fluorescently labeled deoxynucleotides. The fluorescent cDNAs are then mixed and used as probes, each hybridizing to complementary sequences on the microarray. In Figure 9–22, for example, the labeled nucleotides used to make the cDNA for each sample fluoresce in two different colors. The cDNA from the two samples is mixed and used to probe the microarray. Spots that fluoresce green represent mRNAs more abundant at the single-cell stage; those that fluoresce red represent sequences more abundant later in development. The mRNAs that are equally abundant at both stages of development fluoresce yellow. By using a mixture of two samples to measure relative rather than absolute abundance of sequences, the method corrects for variations in the amounts of DNA originally deposited in each spot on the grid and other possible inconsistencies among spots in the microarray. The spots that fluoresce provide a snapshot of all the genes being expressed in the cells at the moment they were harvested—gene expression examined on a genome-wide scale. For a gene of unknown function, the time and circumstances of its expression can provide important clues about its role in the cell.

An example of this technique is illustrated in Figure 9–23, showing the dramatic results this technique can produce. Segments from each of the more than 6,000 genes in the completely sequenced yeast genome were separately amplified by PCR, and each segment was deposited in a defined pattern to create the illustrated microarray. In a sense, this array provides a snapshot of the entire yeast genome.

Protein Chips Proteins, too, can be immobilized on a solid surface and used to help define the presence or absence of other proteins in a sample. For example, researchers prepare an array of antibodies to particular proteins by immobilizing them as individual spots on a solid surface. A sample of proteins is added, and if the protein that binds any of the antibodies is present in the sample, it can be detected by a solid-state form of the ELISA assay (see Fig. 5–28). Many other types and applications of protein chips are being developed.

Detection of Protein-Protein Interactions Helps to Define Cellular and Molecular Function

A key to defining the function of any protein is to determine what it binds to. In the case of protein-protein interactions, the association of a protein of unknown function with one whose function is known can provide a useful and compelling “guilt by association.” The techniques used in this effort are quite varied.

Comparisons of Genome Composition Although not evidence of direct association, the mere presence of combinations of genes in particular genomes can hint at

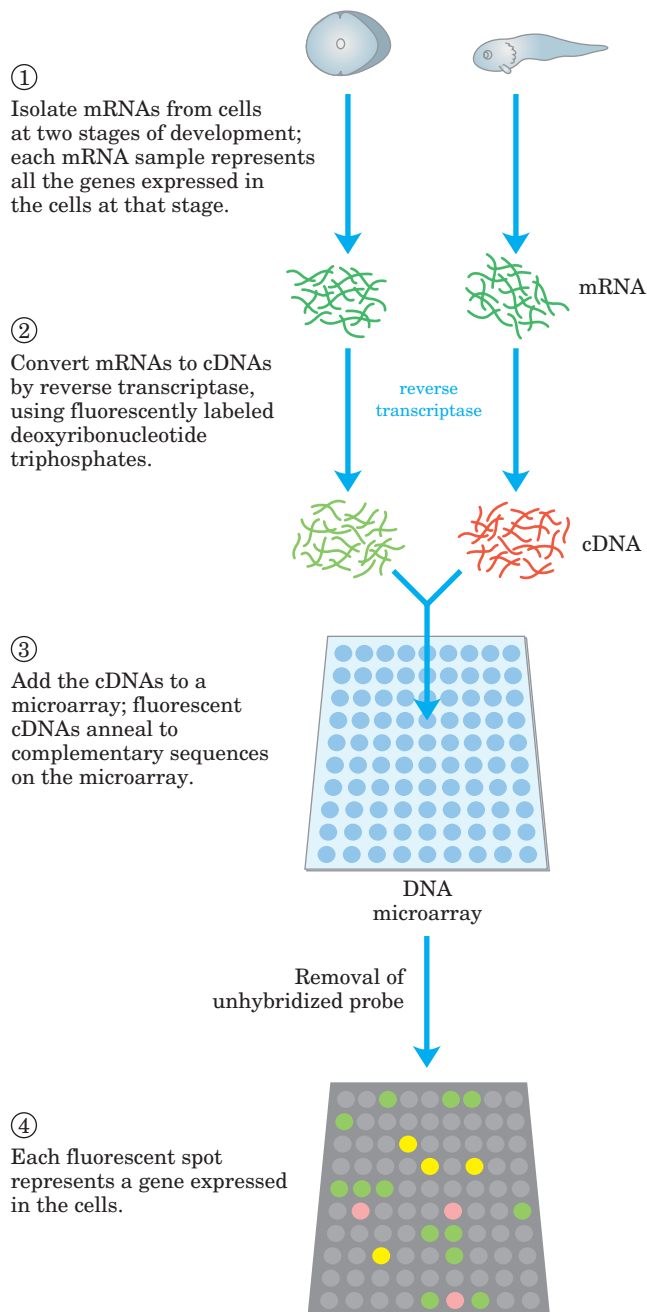


FIGURE 9-22 DNA microarray. A microarray can be prepared from any known DNA sequence, from any source, generated by chemical synthesis or by PCR. The DNA is positioned on a solid surface (usually specially treated glass slides) with the aid of a robotic device capable of depositing very small (nanoliter) drops in precise patterns. UV light cross-links the DNA to the glass slides. Once the DNA is attached to the surface, the microarray can be probed with other fluorescently labeled nucleic acids. Here, mRNA samples are collected from cells at two different stages in the development of a frog. The cDNA probes are made with nucleotides that fluoresce in different colors for each sample; a mixture of the cDNAs is used to probe the microarray. Green spots represent mRNAs more abundant at the single-cell stage; red spots, sequences more abundant later in development. The yellow spots indicate approximately equal abundance at both stages.  **Synthesizing an Oligonucleotide Array**

protein function. We can simply search the genomic databases for particular genes, then determine what other genes are present in the same genomes (Fig. 9-24). When two genes always appear together in a genome, it suggests that the proteins they encode may be functionally related. Such correlations are most useful if the function of at least one of the proteins is known.

Purification of Protein Complexes With the construction of cDNA libraries in which each gene is contiguous with (fused to) an epitope tag, workers can immunoprecipitate the protein product of a gene by using the antibody that binds to the epitope (Fig. 9-15b). If the tagged protein is expressed in cells, other proteins that bind to it may also be precipitated with it. Identification of the associated proteins reveals some of the protein-protein interactions of the tagged protein. There are many variations of this process. For example, a crude extract of cells that express a similarly tagged protein is added to a column containing immobilized antibody. The tagged protein binds to the antibody, and proteins that interact with the tagged protein are sometimes also retained

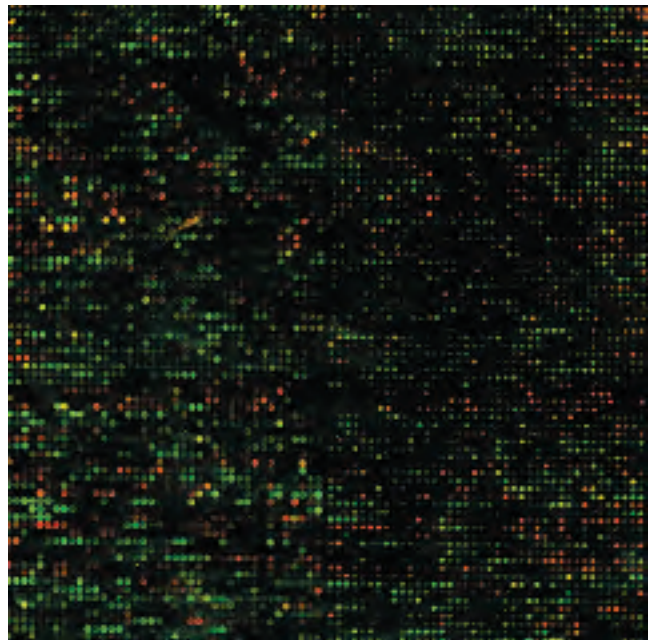


FIGURE 9-23 Enlarged image of a DNA microarray. Each glowing spot in this microarray contains DNA from one of the 6,200 genes of the yeast (*S. cerevisiae*) genome, with every gene represented in the array. The microarray has been probed with fluorescently labeled nucleic acid derived from the mRNAs obtained (1) when the cells were growing normally in culture and (2) five hours after the cells began to form spores. The green spots represent genes expressed at higher levels during normal growth; the red spots, genes expressed at higher levels during sporulation. The yellow spots represent genes that do not change their levels of expression during sporulation. This image is enlarged; the microarray actually measures only 1.8×1.8 cm.

 **Screening Oligonucleotide Array for Patterns of Gene Expression**

Protein	Species			
	1	2	3	4
P1	+	−	+	+
P2	−	−	+	−
P3	+	+	−	+
P4	+	−	+	−
P5	+	−	−	−
P6	+	+	−	+
P7	+	+	+	−

FIGURE 9-24 Use of comparative genomics to identify functionally related genes. One use of comparative genomics is to prepare phylogenetic profiles in order to identify genes that always appear together in a genome. This example shows a comparison of genes from four organisms, but in practice, computer searches can look at dozens of species. The designations P1, P2, and so forth refer to proteins encoded by each species. This technique does not require homologous proteins. In this example, because proteins P3 and P6 always appear together in a genome they may be functionally related. Further testing would be needed to confirm this inference.

on the column. The connection between the protein and the tag is cleaved with a specific protease, and the protein complexes are eluted from the column and analyzed. Researchers can use these methods to define complex networks of interactions within a cell.

A variety of useful protein tags are available. A common one is a histidine tag, often just a string of six His residues. A poly-His sequence binds quite tightly to metals such as nickel. If a protein is cloned so that its sequence is contiguous with a His tag, it will have the extra His residues at its carboxyl terminus. The protein can then be purified by chromatography on columns with immobilized nickel. These procedures are convenient but require caution, because the additional amino acid residues in an epitope or His tag can affect protein activity.

Yeast Two-Hybrid Analysis A sophisticated genetic approach to defining protein-protein interactions is based on the properties of the Gal4 protein (Gal4p), which activates transcription of certain genes in yeast (see Fig. 28–28). Gal4p has two domains, one that binds to a specific DNA sequence and another that activates the RNA polymerase that synthesizes mRNA from an adjacent reporter gene. The domains are stable when separated, but activation of the RNA polymerase requires interaction with the activation domain, which in turn requires positioning by the DNA-binding domain. Hence, the domains must be brought together to function correctly (Fig. 9–25a).

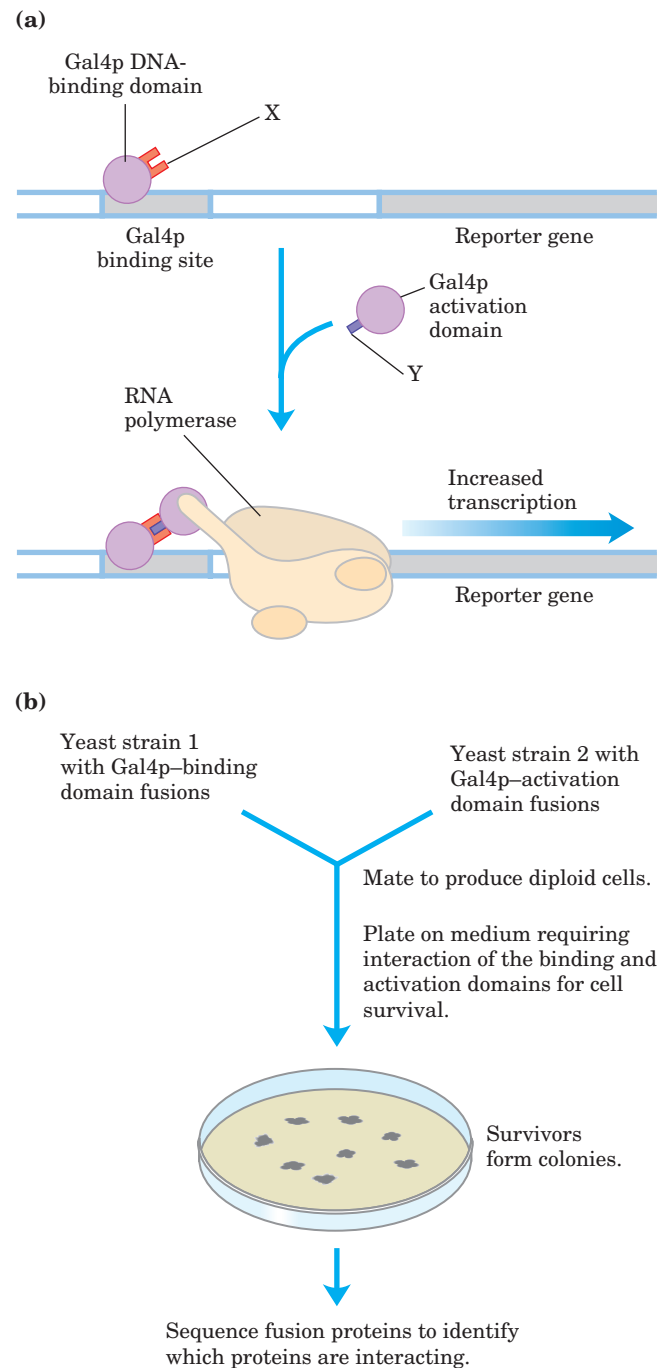


FIGURE 9-25 The yeast two-hybrid system. (a) In this system for detecting protein-protein interactions, the aim is to bring together the DNA-binding domain and the activation domain of the yeast Gal4 protein through the interaction of two proteins, X and Y, to which each domain is fused. This interaction is accompanied by the expression of a reporter gene. (b) The two fusions are created in separate yeast strains, which are then mated. The mated mixture is plated on a medium on which the yeast cannot survive unless the reporter gene is expressed. Thus, all surviving colonies have interacting protein fusion pairs. Sequencing of the fusion proteins in the survivors reveals which proteins are interacting. Yeast Two-Hybrid Systems

In this method, the protein-coding regions of genes to be analyzed are fused to the coding sequences of either the DNA-binding domain or the activation domain of Gal4p, and the resulting genes express a series of fusion proteins. If a protein fused to the DNA-binding domain interacts with a protein fused to the activation domain, transcription is activated. The reporter gene transcribed by this activation is generally one that yields a protein required for growth, or is an enzyme that catalyzes a reaction with a colored product. Thus, when grown on the proper medium, cells that contain such a pair of interacting proteins are easily distinguished from those that do not. Typically, many genes are fused to the Gal4p DNA-binding domain gene in one yeast strain, and many other genes are fused to the Gal4p activation domain gene in another yeast strain, then the yeast strains are mated and individual diploid cells grown into colonies (Fig. 9–25b). This allows for large-scale screening for proteins that interact in the cell.

All these techniques provide important clues to protein function. However, they do not replace classical biochemistry. They simply provide researchers with an expedited entrée into important new biological problems. In the end, a detailed functional understanding of any new protein requires traditional biochemical analyses—such as were used for the many well-studied proteins described in this text. When paired with the simultaneously evolving tools of biochemistry and molecular biology, genomics and proteomics are speeding the discovery not only of new proteins but of new biological processes and mechanisms.

SUMMARY 9.3 From Genomes to Proteomes

- A proteome is the complement of proteins produced by a cell's genome. The new field of proteomics encompasses an effort to catalog and determine the functions of all the proteins in a cell.
- One of the most effective ways to determine the function of a new gene is by comparative genomics, the search of databases for genes with similar sequences. Paralogs and orthologs are proteins (and their genes) with clear functional and sequence relationships in the same or in different species. In some cases, the presence of a gene in combination with certain other genes, observed as a pattern in several genomes, can point toward a possible function.
- Cellular proteomes can be displayed by two-dimensional gel electrophoresis and explored with the aid of mass spectrometry.
- The cellular function of a protein can sometimes be inferred by determining when

and where its gene is expressed. Researchers use DNA microarrays (chips) and protein chips to explore gene expression at the cellular level.

- Several new techniques, including comparative genomics, immunoprecipitation, and yeast two-hybrid analysis, can identify protein-protein interactions. These interactions provide important clues to protein function.

9.4 Genome Alterations and New Products of Biotechnology

We don't need to look far to find practical applications for the new biotechnologies or to find new opportunities for breakthroughs in basic research. Herein lie both the promise and the challenge of genomics. As our knowledge of the genome increases, we will improve our understanding of every aspect of biological function. We will enhance our capacity to engineer organisms and produce new pharmaceutical agents and, as a consequence, will improve human nutrition and health. This promise can be realized only if practical safeguards are in place to ensure responsible application of these techniques.

A Bacterial Plant Parasite Aids Cloning in Plants

We not only can understand genomes, we can change them. This is perhaps the ultimate manifestation of the new technologies. The introduction of recombinant DNA into plants has enormous implications for agriculture, making possible the alteration of the nutritional profile or yield of crops or their resistance to environmental stresses, such as insect pests, diseases, cold, salinity, and drought. Fertile plants of some species may be generated from a single transformed cell, so that an introduced gene passes to progeny through the seeds.

As yet, researchers have not found any naturally occurring plant cell plasmids to facilitate cloning in plants, so the biggest technical challenge is getting DNA into plant cells. An important and adaptable ally in this effort is the soil bacterium *Agrobacterium tumefaciens*. This bacterium can invade plants at the site of a wound, transform nearby cells, and induce them to form a tumor called a crown gall. *Agrobacterium* contains the large (200,000 bp) **Ti plasmid** (Fig. 9–26a). When the bacterium is in contact with a damaged plant cell, a 23,000 bp segment of the Ti plasmid called T DNA is transferred from the plasmid and integrated at a random position in one of the plant cell chromosomes (Fig. 9–26b). The transfer of T DNA from *Agrobacterium* to the plant cell chromosome depends on two 25 bp repeats that flank the T DNA and on the products of the virulence (*vir*) genes on the Ti plasmid (Fig. 9–26a).

The T DNA encodes enzymes that convert plant metabolites to two classes of compounds that benefit

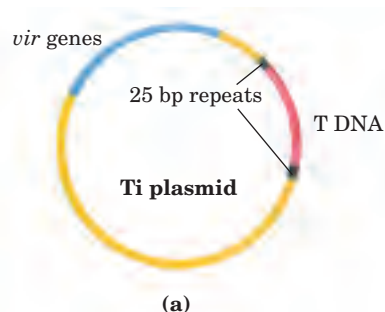
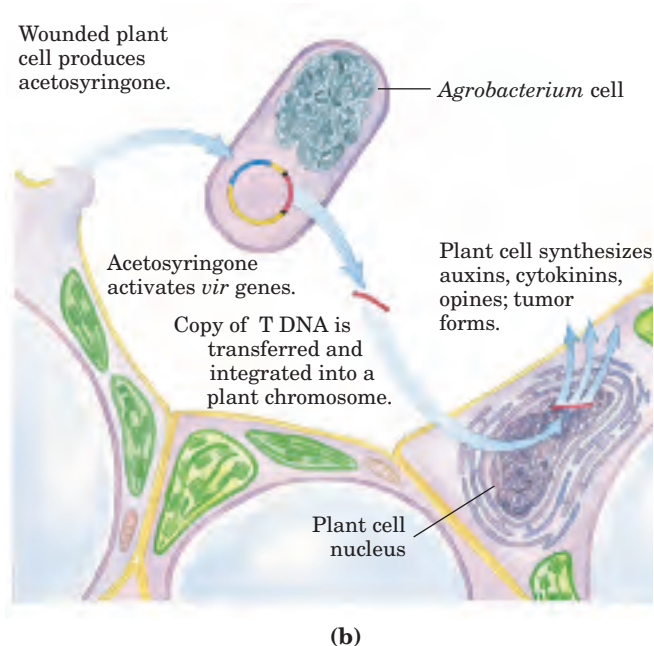


FIGURE 9-26 Transfer of DNA to plant cells by a bacterial parasite.

(a) The Ti (tumor-inducing) plasmid of *Agrobacterium tumefaciens*. (b) Wounded plant cells produce and release the phenolic compound acetosyringone. When *Agrobacterium* detects this compound, the virulence (*vir*) genes on the Ti plasmid are expressed. The *vir* genes encode enzymes needed to introduce the T DNA segment of the Ti plasmid into the genome of nearby plant cells. A single-stranded copy of the T DNA is synthesized and transferred to the plant cell, where it is converted to duplex DNA and integrated into a plant cell chromosome. The T DNA encodes enzymes that synthesize both plant growth hormones and opines (see Fig. 9-27); the latter compounds are metabolized (as a nutrient source) only by *Agrobacterium*. Expression of the T DNA genes by transformed plant cells thus leads to both aberrant plant cell growth (tumor formation) and the diversion of plant cell nutrients to the invading bacteria.



the bacterium (Fig. 9-27). The first group consists of plant growth hormones (auxins and cytokinins) that stimulate growth of the transformed plant cells to form the crown gall tumor. The second constitutes a series of unusual amino acids called opines, which serve as a food source for the bacterium. The opines are produced

in high concentrations in the tumor cells and secreted to the surroundings, where they can be metabolized only by *Agrobacterium*, using enzymes encoded elsewhere on the Ti plasmid. The bacterium thereby diverts plant resources by converting them to a form that benefits only itself.

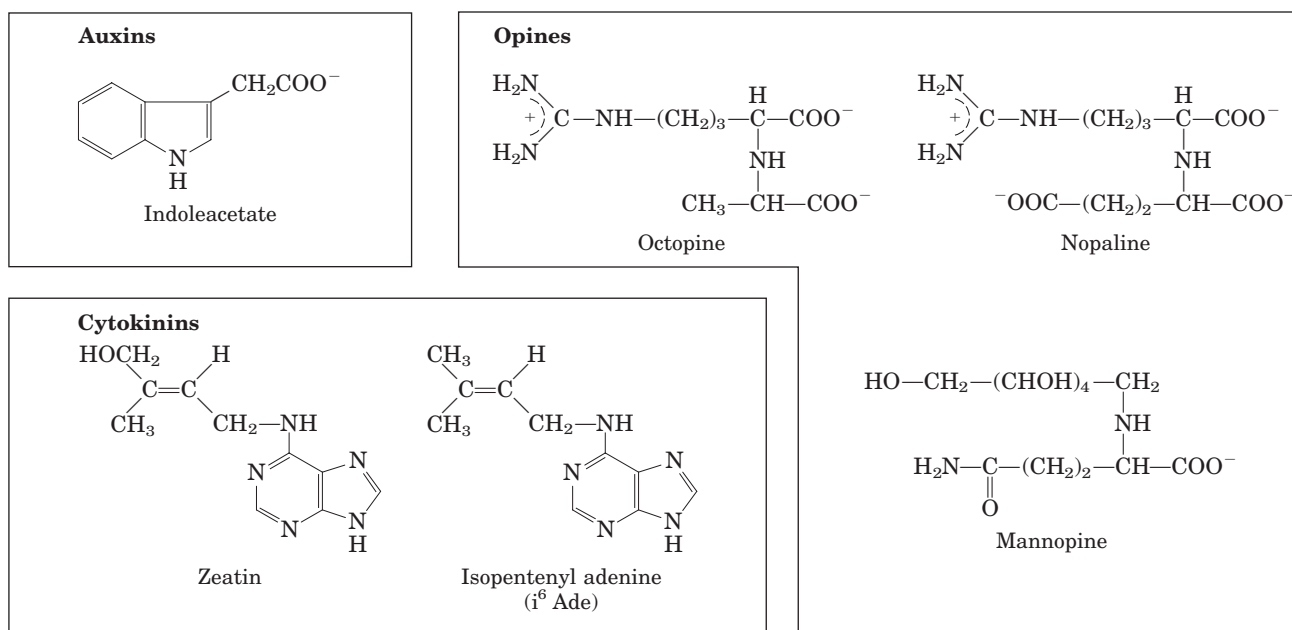


FIGURE 9-27 Metabolites produced in *Agrobacterium*-infected plant cells. Auxins and cytokinins are plant growth hormones. The most common auxin, indoleacetate, is derived from tryptophan. Cytokinins

are adenine derivatives. Opines generally are derived from amino acid precursors; at least 14 different opines are produced by enzymes encoded by the Ti plasmids of different *Agrobacterium* species.

This rare example of DNA transfer from a prokaryote to a eukaryotic cell is a natural genetic engineering process—one that researchers can harness to transfer recombinant DNA (instead of T DNA) to the plant genome. A common cloning strategy employs an *Agrobacterium* with two different recombinant plasmids. The first is a Ti plasmid from which the T DNA segment has been removed in the laboratory (Fig. 9–28a). The second is an *Agrobacterium*–*E. coli* shuttle vector in which the 25 bp repeats of the T DNA flank a foreign gene that the researcher wants to introduce into the plant cell, along with a selectable marker such as resistance to the antibiotic kanamycin (Fig. 9–28b). The engineered *Agrobacterium* is used to infect a leaf, but crown galls are not formed because the T DNA genes for the auxin, cytokinin, and opine biosynthetic enzymes are absent from both plasmids. Instead, the *vir* gene

products from the altered Ti plasmid direct the transformation of the plant cells by the foreign gene—the gene flanked by the T DNA 25 bp repeats in the second plasmid. The transformed plant cells can be selected by growth on agar plates that contain kanamycin, and addition of growth hormones induces the formation of new plants that contain the foreign gene in every cell.

The successful transfer of recombinant DNA into plants was vividly illustrated by an experiment in which the luciferase gene from fireflies was introduced into the cells of a tobacco plant (Fig. 9–29)—a favorite plant for transformation experiments because its cells are particularly easy to transform with *Agrobacterium*. The potential of this technology is not limited to the production of glow-in-the-dark plants, of course. The same approach has been used to produce crop plants that are resistant to herbicides, plant viruses, and insect pests (Fig. 9–30). Potential benefits include increased yields and less need for environmentally harmful agricultural chemicals.

Biotechnology can introduce new traits into a plant much faster than traditional methods of plant breeding. A prominent example is the development of soybeans that are resistant to the general herbicide glyphosate (the active ingredient in the product RoundUp). Glyphosate breaks down rapidly in the environment (glyphosate-sensitive plants can be planted in a treated area after as little as 48 hours), and its use does not generally lead to contamination of groundwater or carryover from one year to the next. A field of glyphosate-resistant soybeans can be treated once with glyphosate during a summer growing season to eliminate essentially all weeds in the field, while leaving the soybeans unaffected (Fig. 9–31). Potential pitfalls of the technology, such as the evolution of glyphosate-resistant weeds or the escape of difficult-to-control recombinant plants, remain a concern of researchers and the public.

FIGURE 9–28 A two-plasmid strategy to create a recombinant plant.

(a) One plasmid is a modified Ti plasmid that contains the *vir* genes but lacks T DNA. (b) The other plasmid contains a segment of DNA that bears both a foreign gene (the gene of interest, e.g., the gene for the insecticidal protein described in Fig. 9–30) and an antibiotic-resistance element (here, kanamycin resistance), flanked by the two 25 bp repeats of T DNA that are required for transfer of the plasmid genes to the plant chromosome. This plasmid also contains the replication origin needed for propagation in *Agrobacterium*.

When bacteria invade at the site of a wound (the edge of the cut leaf), the *vir* genes on the first plasmid mediate transfer into the plant genome of the segment of the second plasmid that is flanked by the 25 bp repeats. Leaf segments are placed on an agar dish that contains both kanamycin and appropriate levels of plant growth hormones, and new plants are generated from segments with the transformed cells. Nontransformed cells are killed by the kanamycin. The foreign gene and the antibiotic-resistance element are normally transferred together, so plant cells that grow in this medium generally contain the foreign gene.

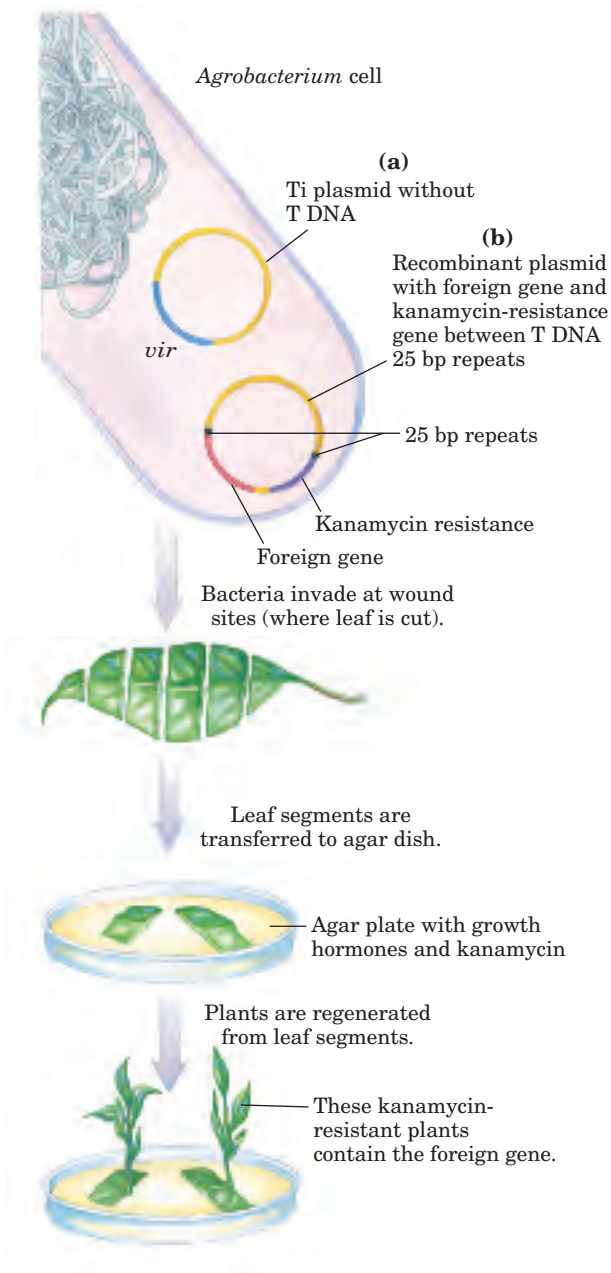




FIGURE 9-29 A tobacco plant expressing the gene for firefly luciferase.

Light was produced after the plant was watered with a solution containing luciferin, the substrate for the light-producing luciferase enzyme (see Box 13-2). Don't expect glow-in-the-dark ornamental plants at your local plant nursery anytime soon. The light is actually quite weak; this photograph required a 24-hour exposure. The real point—that this technology allows the introduction of new traits into plants—is nevertheless elegantly made.



FIGURE 9-30 Tomato plants engineered to be resistant to insect larvae.

Two tomato plants were exposed to equal numbers of moth larvae. The plant on the left has not been genetically altered. The plant on the right expresses a gene for a protein toxin derived from the bacterium *Bacillus thuringiensis*. This protein, introduced by a protocol similar to that depicted in Figure 9-28, is toxic to the larvae of some moth species while being harmless to humans and other organisms. Insect resistance has also been genetically engineered in cotton and other plants.

Manipulation of Animal Cell Genomes Provides Information on Chromosome Structure and Gene Expression

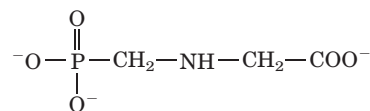
The transformation of animal cells by foreign genetic material offers an important mechanism for expanding our knowledge of the structure and function of animal genomes, as well as for the generation of animals with



(a)



(b)



Glyphosate

FIGURE 9-31 Glyphosate-resistant soybean plants. The photographs

show two areas of a soybean field in Wisconsin. (a) Without glyphosate treatment, this part of the field is overrun with weeds. (b) Glyphosate-resistant soybean plants thrive in the glyphosate-treated section of the field. Glyphosate breaks down rapidly in the environment. Agricultural use of engineered plants such as these proceeds only after considerable deliberation, balancing the extraordinary promise of the technology with the need to select new traits with care. Both science and society as a whole have a stake in ensuring that the use of the resultant plants has no adverse impact on the environment or on human health.

new traits. This potential has stimulated intensive research into more-sophisticated means of cloning animals.

Most work of this kind requires a source of cells into which DNA can be introduced. Although intact tissues are often difficult to maintain and manipulate *in vitro*, many types of animal cells can be isolated and grown in the laboratory if their growth requirements are carefully met. Cells derived from a particular animal tissue and

grown under appropriate **tissue culture** conditions can maintain their differentiated properties (for example, a hepatocyte (liver cell) remains a hepatocyte) for weeks or even months.

No suitable plasmidlike vector is available for introducing DNA into an animal cell, so transformation usually requires the integration of the DNA into a host-cell chromosome. The efficient delivery of DNA to a cell nucleus and integration of this DNA into a chromosome without disrupting any critical genes remain the major technical problems in the genetic engineering of animal cells.

Available methods for carrying DNA into an animal cell vary in efficiency and convenience. Some success has been achieved with spontaneous uptake of DNA or electroporation, techniques roughly comparable to the common methods used to transform bacteria. They are inefficient in animal cells, however, transforming only 1 in 100 to 10,000 cells. **Microinjection**—the injection of DNA directly into a nucleus, using a very fine needle—has a high success rate for skilled practitioners, but the total number of cells that can be treated is small, because each must be injected individually.

The most efficient and widely used methods for transforming animal cells rely on liposomes or viral vectors. Liposomes are small vesicles consisting of a lipid bilayer that encloses an aqueous compartment (see Fig. 11–4). Liposomes that enclose a recombinant DNA molecule can be fused with the membranes of target cells to deliver DNA into the cell. The DNA sometimes reaches the nucleus, where it can integrate into a chromosome (mostly at random locations). **Viral vectors** are even more efficient at delivering DNA. Animal viruses have effective mechanisms for introducing their nucleic acids into cells, and several types also have mechanisms to integrate their DNA into a host-cell chromosome. Some of these, such as retroviruses (see Fig. 26–30) and adenoviruses, have been modified to serve as viral vectors to introduce foreign DNA into mammalian cells.

The work on retroviral vectors illustrates some of the strategies being used (Fig. 9–32). When an engineered retrovirus enters a cell, its RNA genome is transcribed to DNA by reverse transcriptase and then integrated into the host genome by the enzyme viral integrase. Special regions of DNA are required for this

procedure: long terminal repeat (LTR) sequences to integrate retroviral DNA into the host chromosome and the ψ (psi) sequence to package the viral RNA in viral particles (see Fig. 26–30).

The *gag*, *pol*, and *env* genes of the retroviral genome, required for retroviral replication and assembly of viral particles, can be replaced with foreign DNA. To assemble viruses that contain the recombinant genetic information, researchers must introduce the DNA into cultured cells that are simultaneously infected with a “helper virus” that has the genes to produce viral particles but lacks the ψ sequence required for packaging. Thus the recombinant DNA can be transcribed and its

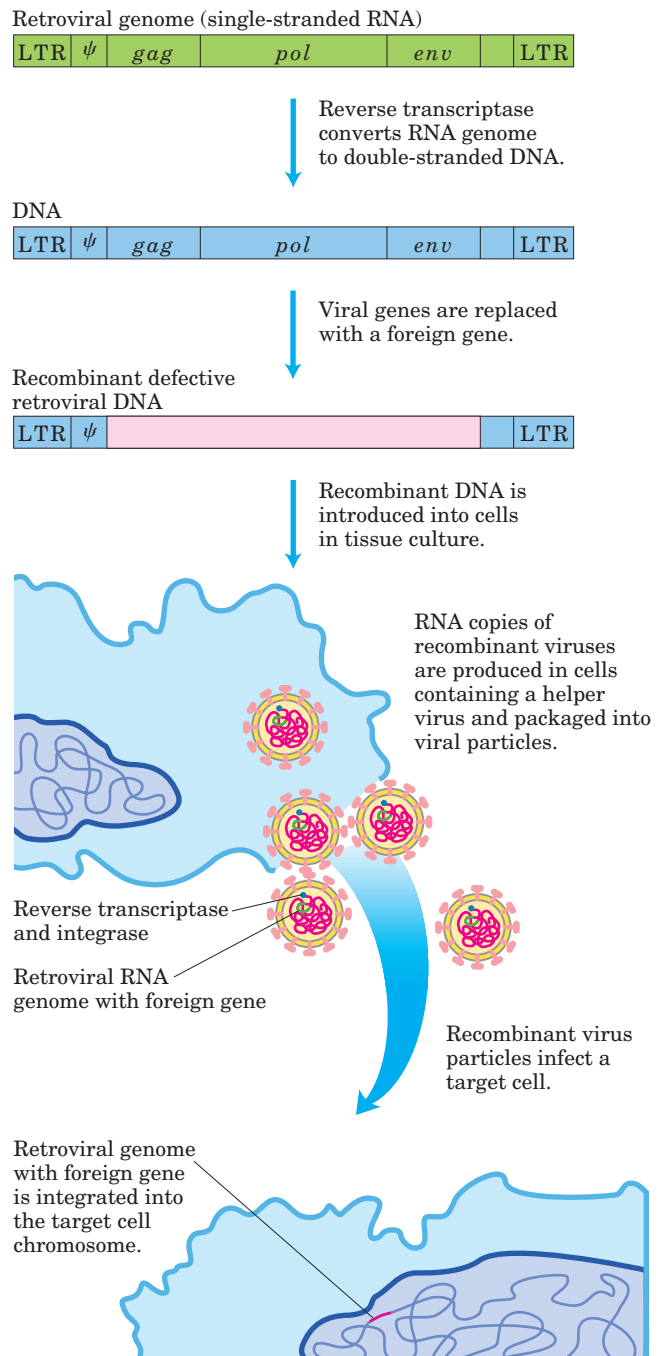


FIGURE 9–32 Use of retroviral vectors in mammalian cell cloning. A typical retroviral genome (somewhat simplified here), engineered to carry a foreign gene (pink), is added to a host-cell tissue culture. The helper virus (not shown) lacks the packaging sequence, ψ , so its RNA transcripts cannot be packaged into viral particles, but it provides the *gag*, *pol*, and *env* gene products needed to package the engineered retrovirus into functional viral particles. This enables the foreign gene in the recombinant retroviral genome to be introduced efficiently into the target cells.

RNA packaged into viral particles. These particles can act as vectors to introduce the recombinant RNA into target cells. Viral reverse transcriptase and integrase enzymes (produced by the helper virus) are also packaged in the viral particle and introduced into the target cells. Once the engineered viral genome is inside a cell, these enzymes create a DNA copy of the recombinant viral RNA genome and integrate it into a host chromosome. The integrated recombinant DNA then becomes a permanent part of the target cell's chromosome and is replicated with the chromosome at every cell division. The cell itself is not endangered by the integrated viral DNA, because the recombinant virus lacks the genes needed to produce RNA copies of its genome and package them into new viral particles. The use of recombinant retroviruses is often the best method for introducing DNA into large numbers of mammalian cells.

Each type of virus has different attributes, so several classes of animal viruses are being engineered as vectors to transform mammalian cells. Adenoviruses, for example, lack a mechanism for integrating DNA into a chromosome. Recombinant DNA introduced via an adenoviral vector is therefore expressed for only a short time and then destroyed. This can be useful if the objective is transient expression of a gene.

Transformation of animal cells by any of the above techniques has its problems. Introduced DNA is generally integrated into chromosomes at random locations. Even when the foreign DNA contains a sequence similar to a sequence in a host chromosome, allowing targeting to that position, nonhomologous integrants still outnumber the targeted ones by several orders of magnitude. If these integration events disrupt essential genes, they can sometimes alter cellular functions (most cells are diploid or polyploid, however, so an integration usually leaves at least one unaffected copy of any given gene). A particularly poor outcome would involve an integration event that inadvertently activated a gene that stimulated cell division, potentially creating a cancer cell. Although such an event was once thought to be rare, recent trials suggest it is a significant hazard (Box 9-2). Finally, the site of an integration can determine the level of expression of the integrated gene, because integrants are not transcribed equally well everywhere in the genome.

Despite these challenges, the transformation of animal cells has been used extensively to study chromosome structure and the function, regulation, and expression of genes. The successful introduction of recombinant DNA into an animal can be illustrated by an experiment that permanently altered an easily observable inheritable physical trait. Microinjection of DNA into the nuclei of fertilized mouse eggs can produce efficient transformation (chromosomal integration). When the injected eggs are introduced into a female mouse and allowed to develop, the new gene is often expressed in some of the newborn mice. Those in which the germ line has been



FIGURE 9-33 Cloning in mice. The gene for human growth hormone was introduced into the genome of the mouse on the right. Expression of the gene resulted in the unusually large size of this mouse.

altered can be identified by testing *their* offspring. By careful breeding of these mice, researchers can establish a **transgenic** mouse line in which all the mice are homozygous for the new gene or genes. This technology was used to introduce into mice the gene for human growth hormone, under the control of an inducible promoter. When the mice were fed a diet that included the inducer, some of the mice that developed from injected embryos grew to an unusually large size (Fig. 9-33). Transgenic mice have now been produced with a wide range of genetic variations, including many relevant to human diseases and their control, pointing the way to human gene therapy (Box 9-2). A very similar approach is used to generate mice in which a particular gene has been inactivated (“knockout mice”), a way of establishing the function of the inactivated gene.  [Creating a Transgenic Mouse](#)

New Technologies Promise to Expedite the Discovery of New Pharmaceuticals



It is difficult to summarize all the ways in which genomics and proteomics might affect the development of pharmaceutical agents, but a few examples illustrate the potential. Hypertension, congestive heart failure, hypercholesterolemia, and obesity are treated by pharmaceutical drugs that alter human physiology. Therapies are arrived at by identifying an enzyme or receptor involved in the process and discovering an inhibitor that interferes with its action. Proteomics will play an increasing role in identifying such potential drug targets. For example, the most potent vasoconstrictor known is the peptide hormone urotensin II. First discovered in fish spinal fluid, urotensin II is a small cyclic peptide, with 11 amino acid residues in humans and 12 or 13 in some other organisms. The vasoconstriction it induces can cause or exacerbate hypertension, congestive heart failure, and coronary artery disease. Some of the methods described in Section 9.3 for elucidating



The Human Genome and Human Gene Therapy

As biotechnology gained momentum in the 1980s, a rational approach to the treatment of genetic diseases became increasingly attractive. In principle, DNA can be introduced into human cells to correct inherited genetic deficiencies. Genetic correction may even be targeted to a specific tissue by inoculating an individual with a genetically engineered, tissue-specific virus carrying a payload of DNA to be incorporated into deficient cells. The goal is entrancing, but the research path is strewn with impediments.

Altering chromosomal DNA entails substantial risk—a risk that cannot be quantified in the early stages of discovery. Consequently, early efforts at human gene therapy were directed at only a small subset of genetic diseases. Panels of scientists and ethicists developed a list of several conditions that should be satisfied to justify the risk involved, including the following. (1) The genetic defect must be a well-characterized, single-gene disorder. (2) Both the mutant and the normal gene must be cloned and sequenced. (3) In the absence of a technique for eliminating the existing mutant gene, the functional gene must function well in the presence of the mutant gene. (4) Finally, and most important, the risks inherent in a new technology must be outweighed by the seriousness of the disease. Protocols for human clinical trials were submitted by scientists in several nations and reviewed for scientific rigor and ethical compliance by carefully selected advisory panels in each country; then human trials commenced.

Early targets of gene therapy included cancer and genetic diseases affecting the immune system. Immunity is mediated by leukocytes (white blood cells) of several different types, all arising from undifferentiated stem cells in the bone marrow. These cells divide quickly and have special metabolic requirements. Differentiation can become blocked in several ways, resulting in a condition called severe combined immune deficiency (SCID). One form of SCID results from genetically inherited defects in the gene encoding adenosine deaminase (ADA), an enzyme involved in nucleotide biosynthesis (discussed in Chapter 22). Another form of SCID arises from a defect in a cell-surface receptor protein that binds chemical signals called cytokines, which trigger differentiation. In both cases, the progenitor stem cells cannot differentiate

into the mature immune system cells, such as T and B lymphocytes (Chapter 5). Children with these rare human diseases are highly susceptible to bacterial and viral infections, and often suffer from a range of related physiological and neurological problems. In the absence of an effective therapy, the children must be confined in a sterile environment. About 20% of these children have a human leukocyte antigen (HLA)-identical sibling who can serve as a bone marrow transplant donor, a procedure that can cure the disease. The remaining children need a different approach.

The first human gene therapy trial was carried out at the National Institutes of Health in Bethesda, Maryland, in 1990. The patient was a four-year-old girl crippled by ADA deficiency. Bone marrow cells from the child were transformed with an engineered retrovirus containing a functional ADA gene; when the alteration of cells is done in this way—in the laboratory rather than in the living patient—the procedure is said to be done *ex vivo*. The treated cells were reintroduced into the patient's marrow. Four years later, the child was leading a normal life, going to school, and even testifying about her experiences before Congress. However, her recovery cannot be uniquely attributed to gene therapy. Before the gene therapy clinical trials began, researchers had developed a new treatment for ADA deficiency, in which synthetic ADA was administered in a complex with polyethylene glycol (PEG). For many ADA-SCID patients, injection of the ADA-PEG complex allowed some immune system development, with weight gain and reduced infection, although not full immune reconstitution. The new gene therapy was risky, and withdrawing the inoculation treatment from patients in the gene therapy trial was judged unethical. So trial participants received both treatments at once, making it unclear which treatment was primarily responsible for the positive clinical outcome. Nevertheless, the clinical trial provided important information: it was feasible to transfer genes *ex vivo* to large numbers of leukocytes, and cells bearing the transferred gene were still detectable years after treatment, suggesting that long-term correction was possible. In addition, the risk associated with use of the retroviral vectors appeared to be low.

Through the 1990s, hundreds of human gene therapy clinical trials were carried out, targeting a variety of genetic diseases, but the results in most cases were

protein-protein interactions have been used to demonstrate that urotensin II is bound by a G-protein-coupled receptor called GPR14. As we shall see in Chapter 12, G proteins play an important role in many signaling pathways. However, GPR14 was an “orphan” receptor,

in that human genome sequencing had identified it as a G-protein-coupled receptor, but with no known function. The association of urotensin II with GPR14 now makes the latter protein a key target for drug therapies aimed at interfering with the action of urotensin II.

discouraging. One major impediment proved to be the inefficiency of introducing new genes into cells. Transformation failed in many cells, and the number of transformed cells often proved insufficient to reverse the disorder. In the ADA trials, achieving a sufficient population of transformed cells was particularly difficult, because of the ongoing ADA-PEG therapy. Normally, stem cells with the correct ADA gene would have a growth advantage over the untreated cells, expanding their population and gradually predominating in the bone marrow. However, the injections of ADA-PEG in the same patients allowed the untransformed (ADA-deficient) cells to live and develop, and the transformed cells did not have the needed growth advantage to expand their population at the expense of the others.

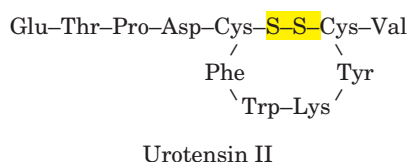
A gene therapy trial initiated in 1999 was successful in correcting a form of SCID caused by defective cytokine receptors (in particular a subunit called γ_c), as reported in 2000 by physician researchers in France, Italy, and Britain. These researchers introduced the corrected gene for the γ_c cytokine–receptor subunit into CD34⁺ cells. (The stem cells that give rise to immune system cells have a protein called CD34 on their surface; these cells can be separated from other bone marrow cells by antibodies to CD34.) The transformed cells were placed back into the patients' bone marrow. In this trial, introduction of the corrected gene clearly conferred a growth advantage over the untreated cells. A functioning immune system was detected in four of the first five patients within 6 to 12 weeks, and levels of mature immune system T lymphocytes reached the levels found in age-matched control subjects (who did not have SCID) within 6 to 8 months. Immune system function was restored, and nearly 4 years later (mid-2003) most of the children are leading normal lives. Similar results have been obtained with four additional patients. This provided dramatic confirmation that human gene therapy could cure a serious genetic disease.

In early 2003 came a setback. One of the original four patients who had received cells with the correct cytokine receptor gene developed a severe form of leukemia. During the gene therapy treatment, one of the introduced retroviruses had by chance inserted itself into a chromosome of one CD34⁺ cell, resulting in abnormally high expression of a gene called LMO-

2. The affected cell differentiated into an immune system T cell, and the elevated expression of LMO-2 led to uncontrolled growth of the cell, giving rise to the leukemia. As of mid-2003 the patient had responded well to chemotherapy, but there may be more chapters to write. The incident shows that early worries about the risk associated with retroviral vectors were well founded. After a review of the gene therapy trial protocols, including consultations with ethicists and parents of children affected by these diseases, further gene therapy trials are still planned for children who are not candidates for bone marrow transplants. The reason is simple enough. The potential benefit to the children with these debilitating conditions has been judged to outweigh the demonstrated risk.

Human gene therapy is not limited to genetic diseases. Cancer cells are being targeted by delivering genes for proteins that might destroy the cell or restore the normal control of cell division. Immune system cells associated with tumors, called tumor-infiltrating lymphocytes, can be genetically modified to produce tumor necrosis factor (TNF; see Fig. 12–50). When these lymphocytes are taken from a cancer patient, modified, and reintroduced, the engineered cells target the tumor, and the TNF they produce causes tumor shrinkage. AIDS may also be treatable with gene therapy; DNA that encodes an RNA molecule complementary to a vital HIV mRNA could be introduced into immune system cells (the targets of HIV). The RNA transcribed from the introduced DNA would pair with the HIV mRNA, preventing its translation and interfering with the virus's life cycle. Alternatively, a gene could be introduced that encodes an inactive form of one subunit of a multisubunit HIV enzyme; with one nonfunctional subunit, the entire enzyme might be inactivated.

Our growing understanding of the human genome and the genetic basis for some diseases brings the promise of early diagnosis and constructive intervention. As the early results demonstrate, however, the road to effective therapies will be a long one, with many detours. We need to learn more about cellular metabolism, more about how genes interact, and more about how to manage the dangers. The prospect of vanquishing life-destroying genetic defects and other debilitating diseases provides the motivation to press on.



Another objective of medical research is to identify new agents that can treat the diseases caused by human pathogens. This now means identifying enzymatic targets in microbial pathogens that can be inactivated with a new drug. The ideal microbial target enzyme

should be (1) essential to the pathogen cell's survival, (2) well-conserved among a wide range of pathogens, and (3) absent or significantly different in humans. The task of identifying metabolic processes that are critical to microorganisms but absent in humans is made much easier by comparative genomics, augmented by the functional information available from genomics and proteomics. ■

Recombinant DNA Technology Yields New Products and Challenges

The products of recombinant DNA technology range from proteins to engineered organisms. The technology can produce large amounts of commercially useful proteins, can design microorganisms for special tasks, and can engineer plants or animals with traits that are useful in agriculture or medicine. Some products of this technology have been approved for consumer or professional use, and many more are in development. Genetic engineering has been transformed over a few years from a promising new technology to a multibillion-dollar industry, with much of the growth occurring in the pharmaceutical industry. Some major classes of new products are listed in Table 9–3.



Erythropoietin is typical of the newer products. This protein hormone (M_r 51,000) stimulates erythrocyte production. People with diseases that com-

promise kidney function often have a deficiency of this protein, resulting in anemia. Erythropoietin produced by recombinant DNA technology can be used to treat these individuals, reducing the need for repeated blood transfusions. ■

Other applications of this technology continue to emerge. Enzymes produced by recombinant DNA technology are already used in the production of detergents, sugars, and cheese. Engineered proteins are being used as food additives to supplement nutrition, flavor, and fragrance. Microorganisms are being engineered with altered or entirely novel metabolic pathways to extract oil and minerals from ground deposits, to digest oil spills, and to detoxify hazardous waste dumps and sewage. Engineered plants with improved resistance to drought, frost, pests, and disease are increasing crop yields and reducing the need for agricultural chemicals. Complete animals can be cloned by moving an entire nucleus and all of its genetic material to a prepared egg from which the nucleus has been removed.

The extraordinary promise of modern biotechnology does not come without controversy. The cloning of mammals challenges societal mores and may be accompanied by serious deficiencies in the health and longevity of the cloned animal. If useful pharmaceutical agents can be produced, so can toxins suitable for biological warfare. The potential for hazards posed by the release of engineered plants and other organisms into

TABLE 9–3 Some Recombinant DNA Products in Medicine

<i>Product category</i>	<i>Examples/uses</i>
Anticoagulants	Tissue plasminogen activator (TPA); activates plasmin, an enzyme involved in dissolving clots; effective in treating heart attack patients.
Blood factors	Factor VIII; promotes clotting; it is deficient in hemophiliacs; treatment with factor VIII produced by recombinant DNA technology eliminates infection risks associated with blood transfusions.
Colony-stimulating factors	Immune system growth factors that stimulate leukocyte production; treatment of immune deficiencies and infections.
Erythropoietin	Stimulates erythrocyte production; treatment of anemia in patients with kidney disease.
Growth factors	Stimulate differentiation and growth of various cell types; promote wound healing.
Human growth hormone	Treatment of dwarfism.
Human insulin	Treatment of diabetes.
Interferons	Interfere with viral reproduction; used to treat some cancers.
Interleukins	Activate and stimulate different classes of leukocytes; possible uses in treatment of wounds, HIV infection, cancer, and immune deficiencies.
Monoclonal antibodies	Extraordinary binding specificity is used in: diagnostic tests; targeted transport of drugs, toxins, or radioactive compounds to tumors as a cancer therapy; many other applications.
Superoxide dismutase	Prevents tissue damage from reactive oxygen species when tissues briefly deprived of O_2 during surgery suddenly have blood flow restored.
Vaccines	Proteins derived from viral coats are as effective in “priming” an immune system as is the killed virus more traditionally used for vaccines, and are safer; first developed was the vaccine for hepatitis B.

the biosphere continues to be monitored carefully. The full range of the long-term consequences of this technology for our species and for the global environment is impossible to foresee, but will certainly demand our increasing understanding of both cellular metabolism and ecology.

SUMMARY 9.4 Genome Alterations and New Products of Biotechnology

- Advances in whole genome sequencing and genetic engineering methods are enhancing our ability to modify genomes in all species.

- Cloning in plants, which makes use of the Ti plasmid vector from *Agrobacterium*, allows the introduction of new plant traits.
- In animal cloning, researchers introduce foreign DNA primarily with the use of viral vectors or microinjection. These techniques can produce transgenic animals and provide new methods for human gene therapy.
- The use of genomics and proteomics in basic and pharmaceutical research is greatly advancing the discovery of new drugs. Biotechnology is also generating an ever-expanding range of other products and technologies.

Key Terms

Terms in bold are defined in the glossary.

cloning 306	genomics 317	restriction fragment length
vector 307	genomic library 318	polymorphisms (RFLPs) 322
recombinant DNA 307	contig 318	Southern blot 322
restriction endonucleases 307	sequence-tagged site (STS) 318	single nucleotide polymorphisms (SNPs) 324
DNA ligase 307	complementary DNA (cDNA) 318	proteome 325
plasmid 311	cDNA library 318	proteomics 325
bacterial artificial chromosome (BAC) 313	expressed sequence tag (EST) 318	orthologs 326
yeast artificial chromosome (YAC) 314	epitope tag 319	synteny 326
site-directed mutagenesis 316	polymerase chain reaction (PCR) 319	DNA microarray 326
fusion protein 317	DNA fingerprinting 322	Ti plasmid 330
		transgenic 335

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Problems

1. Cloning When joining two or more DNA fragments, a researcher can adjust the sequence at the junction in a variety of subtle ways, as seen in the following exercises.

(a) Draw the structure of each end of a linear DNA fragment produced by an *EcoRI* restriction digest (include those sequences remaining from the *EcoRI* recognition sequence).

(b) Draw the structure resulting from the reaction of this end sequence with DNA polymerase I and the four deoxynucleoside triphosphates (see Fig. 8–36).

(c) Draw the sequence produced at the junction that arises if two ends with the structure derived in (b) are ligated (see Fig. 25–16).

(d) Draw the structure produced if the structure derived in (a) is treated with a nuclease that degrades only single-stranded DNA.

(e) Draw the sequence of the junction produced if an

end with structure (b) is ligated to an end with structure (d).

(f) Draw the structure of the end of a linear DNA fragment that was produced by a *PvuII* restriction digest (include those sequences remaining from the *PvuII* recognition sequence).

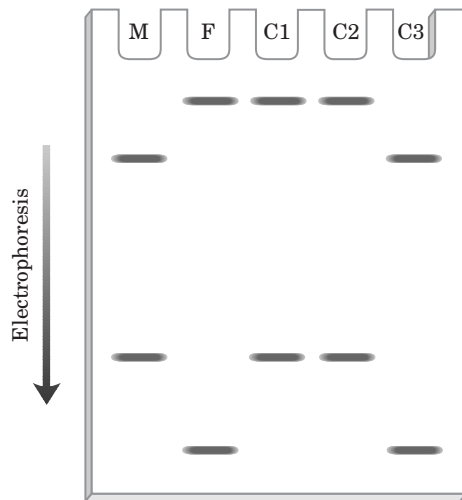
(g) Draw the sequence of the junction produced if an end with structure (b) is ligated to an end with structure (f).

(h) Suppose you can synthesize a short duplex DNA fragment with any sequence you desire. With this synthetic fragment and the procedures described in (a) through (g), design a protocol that would remove an *EcoRI* restriction site from a DNA molecule and incorporate a new *BamHI* restriction site at approximately the same location. (See Fig. 9–3.)

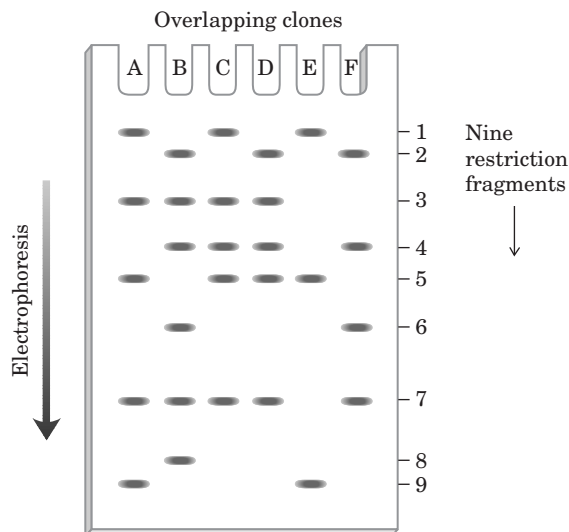
(i) Design four different short synthetic double-stranded DNA fragments that would permit ligation of structure (a) with a DNA fragment produced by a *PstI* restriction

Suggest how the researcher might use the polymerase chain reaction (PCR) to detect the presence of the circular form of the deleted DNA in an extract of the protist.

7. RFLP Analysis for Paternity Testing DNA fingerprinting and RFLP analysis are often used to test for paternity. A child inherits chromosomes from both mother and father, so DNA from a child displays restriction fragments derived from each parent. In the gel shown here, which child, if any, can be excluded as being the biological offspring of the putative father? Explain your reasoning. Lane M is the sample from the mother, F from the putative father, and C1, C2, and C3 from the children.

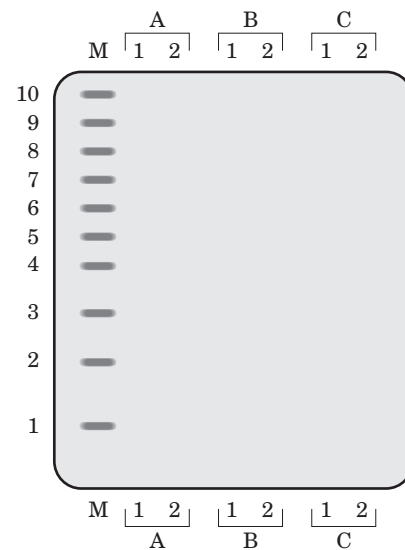
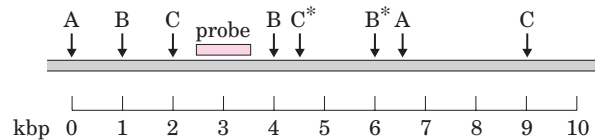


8. Mapping a Chromosome Segment A group of overlapping clones, designated A through F, is isolated from one region of a chromosome. Each of the clones is separately cleaved by a restriction enzyme and the pieces resolved by agarose gel electrophoresis, with the results shown in the figure below. There are nine different restriction fragments in this chromosomal region, with a subset appearing in each clone. Using this information, deduce the order of the restriction fragments in the chromosome.



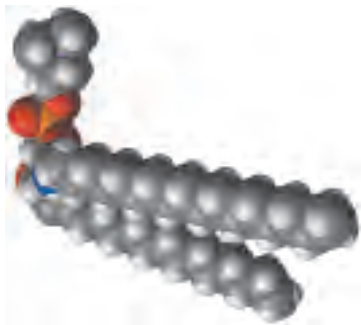
9. Cloning in Plants The strategy outlined in Figure 9–28 employs *Agrobacterium* cells that contain two separate plasmids. Suggest why the sequences on the two plasmids are not combined on one plasmid.

10. DNA Fingerprinting and RFLP Analysis DNA is extracted from the blood cells of two different humans, individuals 1 and 2. In separate experiments, the DNA from each individual is cleaved by restriction endonucleases A, B, and C, and the fragments separated by electrophoresis. A hypothetical map of a 10,000 bp segment of a human chromosome is shown (1 kbp = 1,000 bp). Individual 2 has point mutations that eliminate restriction recognition sites B* and C*. You probe the gel with a radioactive oligonucleotide complementary to the indicated sequence and expose a piece of x-ray film to the gel. Indicate where you would expect to see bands on the film. The lanes of the gel are marked in the accompanying diagram.



11. Use of Photolithography to Make a DNA Microarray Figure 9–21 shows the first steps in the process of making a DNA microarray, or DNA chip, using photolithography. Describe the remaining steps needed to obtain the desired sequences (a different four-nucleotide sequence on each of the four spots) shown in the first panel of the figure. After each step, give the resulting nucleotide sequence attached at each spot.

12. Cloning in Mammals The retroviral vectors described in Figure 9–32 make possible the efficient integration of foreign DNA into a mammalian genome. Explain how these vectors, which lack genes for replication and viral packaging (*gag*, *pol*, *env*), are assembled into infectious viral particles. Suggest why it is important that these vectors lack the replication and packaging genes.



chapter 10

LIPIDS

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The fatty substance, separated from the salifiable bases, was dissolved in boiling alcohol. On cooling, it was obtained crystallized and very pure, and in this state it was examined. As it has not been hitherto described . . . I purpose to call it margarine, from the Greek word signifying pearl, because one of its characters is to have the appearance of mother of pearl, which it communicates to several of the combinations of which it forms with the salifiable bases.

—Michel-Eugène Chevreul,
article in *Philosophical Magazine*, 1814

Biological lipids are a chemically diverse group of compounds, the common and defining feature of which is their insolubility in water. The biological functions of the lipids are as diverse as their chemistry. Fats and oils are the principal stored forms of energy in many organisms. Phospholipids and sterols are major structural elements of biological membranes. Other lipids, although present in relatively small quantities, play crucial roles as enzyme cofactors, electron carriers, light-absorbing pigments, hydrophobic anchors for proteins, “chaperones” to help membrane proteins fold, emulsifying agents in the digestive tract, hormones, and intracellular messengers. This chapter introduces representative lipids of each type, with emphasis on their chemical structure and physical properties. We discuss the energy-yielding oxidation of lipids in Chapter 17 and their synthesis in Chapter 21.

10.1 Storage Lipids

The fats and oils used almost universally as stored forms of energy in living organisms are derivatives of **fatty acids**. The fatty acids are hydrocarbon derivatives, at about the same low oxidation state (that is, as highly reduced) as the hydrocarbons in fossil fuels. The cellular oxidation of fatty acids (to CO_2 and H_2O), like the controlled, rapid burning of fossil fuels in internal combustion engines, is highly exergonic.

We introduce here the structures and nomenclature of the fatty acids most commonly found in living organisms. Two types of fatty acid-containing compounds, triacylglycerols and waxes, are described to illustrate the diversity of structure and physical properties in this family of compounds.

Fatty Acids Are Hydrocarbon Derivatives

Fatty acids are carboxylic acids with hydrocarbon chains ranging from 4 to 36 carbons long (C_4 to C_{36}). In some fatty acids, this chain is unbranched and fully saturated (contains no double bonds); in others the chain contains one or more double bonds (Table 10–1). A few contain three-carbon rings, hydroxyl groups, or methyl-group branches. A simplified nomenclature for these compounds specifies the chain length and number of double bonds, separated by a colon; for example, the 16-carbon saturated palmitic acid is abbreviated 16:0, and the 18-carbon oleic acid, with one double bond, is 18:1. The positions of any double bonds are specified by superscript numbers following Δ (delta); a 20-carbon fatty acid with one double bond between C-9 and C-10 (C-1 being the carboxyl carbon) and another between C-12 and C-13 is designated 20:2($\Delta^{9,12}$). The most commonly occurring fatty acids have even numbers of carbon atoms in an unbranched chain of 12 to 24 carbons (Table 10–1). As we shall see in Chapter 21, the even number of carbons results from the mode of

synthesis of these compounds, which involves condensation of two-carbon (acetate) units.

There is also a common pattern in the location of double bonds; in most monounsaturated fatty acids the double bond is between C-9 and C-10 (Δ^9), and the other double bonds of polyunsaturated fatty acids are generally Δ^{12} and Δ^{15} . (Arachidonic acid is an exception to this generalization.) The double bonds of polyunsaturated fatty acids are almost never conjugated (alternating single and double bonds, as in $-\text{CH}=\text{CH}-\text{CH}=\text{CH}-$), but are separated by a methylene group: $-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}=\text{CH}-$. In nearly all naturally occurring unsaturated fatty acids, the double bonds are in the *cis* configuration. Trans fatty acids are produced by fermentation in the rumen of dairy animals and are obtained from dairy products and meat.

They are also produced during hydrogenation of fish or vegetable oils. Because diets high in trans fatty acids correlate with increased blood levels of LDL (bad cholesterol) and decreased HDL (good cholesterol), it is generally recommended that one avoid large amounts of these fatty acids. Unfortunately, French fries, doughnuts, and cookies tend to be high in trans fatty acids.

The physical properties of the fatty acids, and of compounds that contain them, are largely determined by the length and degree of unsaturation of the hydrocarbon chain. The nonpolar hydrocarbon chain accounts for the poor solubility of fatty acids in water. Lauric acid (12:0, M_r 200), for example, has a solubility in water of 0.063 mg/g—much less than that of glucose (M_r 180), which is 1,100 mg/g. The longer the fatty acyl chain and the fewer the double bonds, the lower is the solubility

TABLE 10-1 Some Naturally Occurring Fatty Acids: Structure, Properties, and Nomenclature

Carbon skeleton	Structure*	Systematic name†	Common name (derivation)	Melting point (°C)	Solubility at 30 °C (mg/g solvent)	
					Water	Benzene
12:0	$\text{CH}_3(\text{CH}_2)_{10}\text{COOH}$	<i>n</i> -Dodecanoic acid	Lauric acid (Latin <i>laurus</i> , “laurel plant”)	44.2	0.063	2,600
14:0	$\text{CH}_3(\text{CH}_2)_{12}\text{COOH}$	<i>n</i> -Tetradecanoic acid	Myristic acid (Latin <i>Myristica</i> , nutmeg genus)	53.9	0.024	874
16:0	$\text{CH}_3(\text{CH}_2)_{14}\text{COOH}$	<i>n</i> -Hexadecanoic acid	Palmitic acid (Latin <i>palma</i> , “palm tree”)	63.1	0.0083	348
18:0	$\text{CH}_3(\text{CH}_2)_{16}\text{COOH}$	<i>n</i> -Octadecanoic acid	Stearic acid (Greek <i>stear</i> , “hard fat”)	69.6	0.0034	124
20:0	$\text{CH}_3(\text{CH}_2)_{18}\text{COOH}$	<i>n</i> -Eicosanoic acid	Arachidic acid (Latin <i>Arachis</i> , legume genus)	76.5		
24:0	$\text{CH}_3(\text{CH}_2)_{22}\text{COOH}$	<i>n</i> -Tetracosanoic acid	Lignoceric acid (Latin <i>lignum</i> , “wood” + <i>cera</i> , “wax”)	86.0		
16:1(Δ^9)	$\text{CH}_3(\text{CH}_2)_5\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -9-Hexadecenoic acid	Palmitoleic acid	1–0.5		
18:1(Δ^9)	$\text{CH}_3(\text{CH}_2)_7\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -9-Octadecenoic acid	Oleic acid (Latin <i>oleum</i> , “oil”)	13.4		
18:2($\Delta^{9,12}$)	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -9,12-Octadecadienoic acid	Linoleic acid (Greek <i>linon</i> , “flax”)	1–5		
18:3($\Delta^{9,12,15}$)	$\text{CH}_3\text{CH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_7\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -9,12,15-Octadecatrienoic acid	α -Linolenic acid	–11		
20:4($\Delta^{5,8,11,14}$)	$\text{CH}_3(\text{CH}_2)_4\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CHCH}_2\text{CH}=\text{CH}(\text{CH}_2)_3\text{COOH}$	<i>cis</i> -, <i>cis</i> -, <i>cis</i> -, <i>cis</i> -5,8,11,14-Icosatetraenoic acid	Arachidonic acid	–49.5		

*All acids are shown in their nonionized form. At pH 7, all free fatty acids have an ionized carboxylate. Note that numbering of carbon atoms begins at the carboxyl carbon.

†The prefix *n*- indicates the “normal” unbranched structure. For instance, “dodecanoic” simply indicates 12 carbon atoms, which could be arranged in a variety of branched forms; “*n*-dodecanoic” specifies the linear, unbranched form. For unsaturated fatty acids, the configuration of each double bond is indicated; in biological fatty acids the configuration is almost always *cis*.

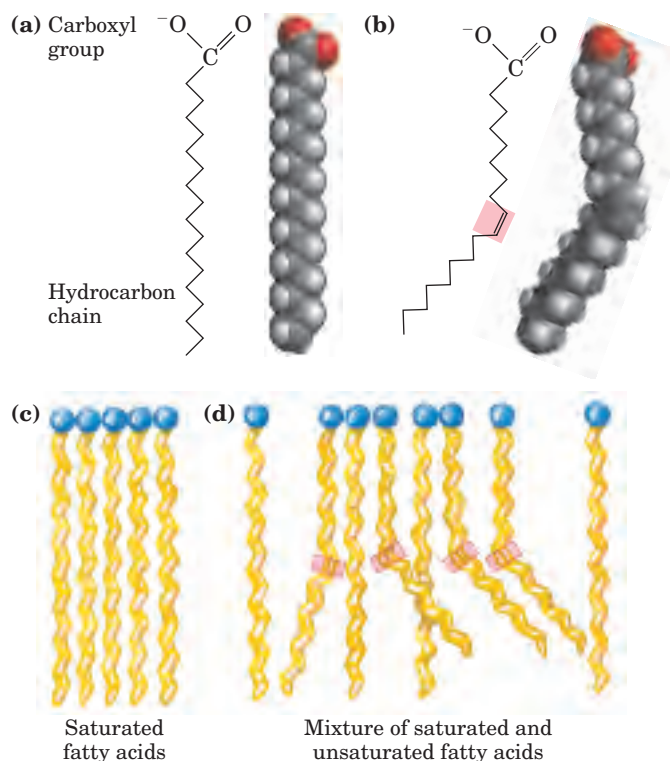


FIGURE 10-1 The packing of fatty acids into stable aggregates. The extent of packing depends on the degree of saturation. (a) Two representations of the fully saturated acid stearic acid (stearate at pH 7) in its usual extended conformation. Each line segment of the zigzag represents a single bond between adjacent carbons. (b) The cis double bond (shaded) in oleic acid (oleate) does not permit rotation and introduces a rigid bend in the hydrocarbon tail. All other bonds in the chain are free to rotate. (c) Fully saturated fatty acids in the extended form pack into nearly crystalline arrays, stabilized by many hydrophobic interactions. (d) The presence of one or more cis double bonds interferes with this tight packing and results in less stable aggregates.

in water. The carboxylic acid group is polar (and ionized at neutral pH) and accounts for the slight solubility of short-chain fatty acids in water.

Melting points are also strongly influenced by the length and degree of unsaturation of the hydrocarbon chain. At room temperature (25°C), the saturated fatty acids from 12:0 to 24:0 have a waxy consistency, whereas unsaturated fatty acids of these lengths are oily liquids. This difference in melting points is due to different degrees of packing of the fatty acid molecules (Fig. 10-1). In the fully saturated compounds, free rotation around each carbon-carbon bond gives the hydrocarbon chain great flexibility; the most stable conformation is the fully extended form, in which the steric hindrance of neighboring atoms is minimized. These molecules can pack together tightly in nearly crystalline arrays, with atoms all along their lengths in van der Waals contact with the atoms of neighboring molecules. In unsaturated fatty acids, a cis double bond forces a kink in the hydrocarbon chain. Fatty acids with one or several such kinks

cannot pack together as tightly as fully saturated fatty acids, and their interactions with each other are therefore weaker. Because it takes less thermal energy to disorder these poorly ordered arrays of unsaturated fatty acids, they have markedly lower melting points than saturated fatty acids of the same chain length (Table 10-1).

In vertebrates, free fatty acids (unesterified fatty acids, with a free carboxylate group) circulate in the blood bound noncovalently to a protein carrier, serum albumin. However, fatty acids are present in blood plasma mostly as carboxylic acid derivatives such as esters or amides. Lacking the charged carboxylate group, these fatty acid derivatives are generally even less soluble in water than are the free fatty acids.

Triacylglycerols Are Fatty Acid Esters of Glycerol

The simplest lipids constructed from fatty acids are the **triacylglycerols**, also referred to as triglycerides, fats, or neutral fats. Triacylglycerols are composed of three fatty acids each in ester linkage with a single glycerol (Fig. 10-2). Those containing the same kind of fatty acid

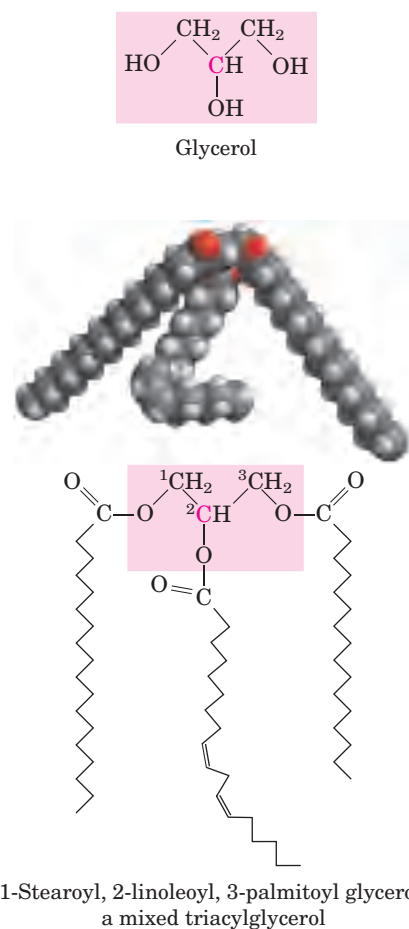


FIGURE 10-2 Glycerol and a triacylglycerol. The mixed triacylglycerol shown here has three different fatty acids attached to the glycerol backbone. When glycerol has two different fatty acids at C-1 and C-3, the C-2 is a chiral center (p. 76).

in all three positions are called simple triacylglycerols and are named after the fatty acid they contain. Simple triacylglycerols of 16:0, 18:0, and 18:1, for example, are tristearin, tripalmitin, and triolein, respectively. Most naturally occurring triacylglycerols are mixed; they contain two or more different fatty acids. To name these compounds unambiguously, the name and position of each fatty acid must be specified.

Because the polar hydroxyls of glycerol and the polar carboxylates of the fatty acids are bound in ester linkages, triacylglycerols are nonpolar, hydrophobic molecules, essentially insoluble in water. Lipids have lower specific gravities than water, which explains why mixtures of oil and water (oil-and-vinegar salad dressing, for example) have two phases: oil, with the lower specific gravity, floats on the aqueous phase.

Triacylglycerols Provide Stored Energy and Insulation

In most eukaryotic cells, triacylglycerols form a separate phase of microscopic, oily droplets in the aqueous cytosol, serving as depots of metabolic fuel. In vertebrates, specialized cells called adipocytes, or fat cells, store large amounts of triacylglycerols as fat droplets that nearly fill the cell (Fig. 10–3a). Triacylglycerols are also stored as oils in the seeds of many types of plants, providing energy and biosynthetic precursors during seed germination (Fig. 10–3b). Adipocytes and germinating seeds contain **lipases**, enzymes that catalyze the hydrolysis of stored triacylglycerols, releasing fatty acids for export to sites where they are required as fuel.

There are two significant advantages to using triacylglycerols as stored fuels, rather than polysaccharides such as glycogen and starch. First, because the carbon atoms of fatty acids are more reduced than those of sugars, oxidation of triacylglycerols yields more than twice as much energy, gram for gram, as the oxidation of carbohydrates. Second, because triacylglycerols are hydrophobic and therefore unhydrated, the organism that carries fat as fuel does not have to carry the extra weight of water of hydration that is associated with stored polysaccharides (2 g per gram of polysaccharide). Humans have fat tissue (composed primarily of adipocytes) under the skin, in the abdominal cavity, and in the mammary glands. Moderately obese people with 15 to 20 kg of triacylglycerols deposited in their adipocytes could meet their energy needs for months by drawing on their fat stores. In contrast, the human body can store less than a day's energy supply in the form of glycogen. Carbohydrates such as glucose and glycogen do offer certain advantages as quick sources of metabolic energy, one of which is their ready solubility in water.

In some animals, triacylglycerols stored under the skin serve not only as energy stores but as insulation against low temperatures. Seals, walruses, penguins, and other warm-blooded polar animals are amply padded with triacylglycerols. In hibernating animals (bears, for

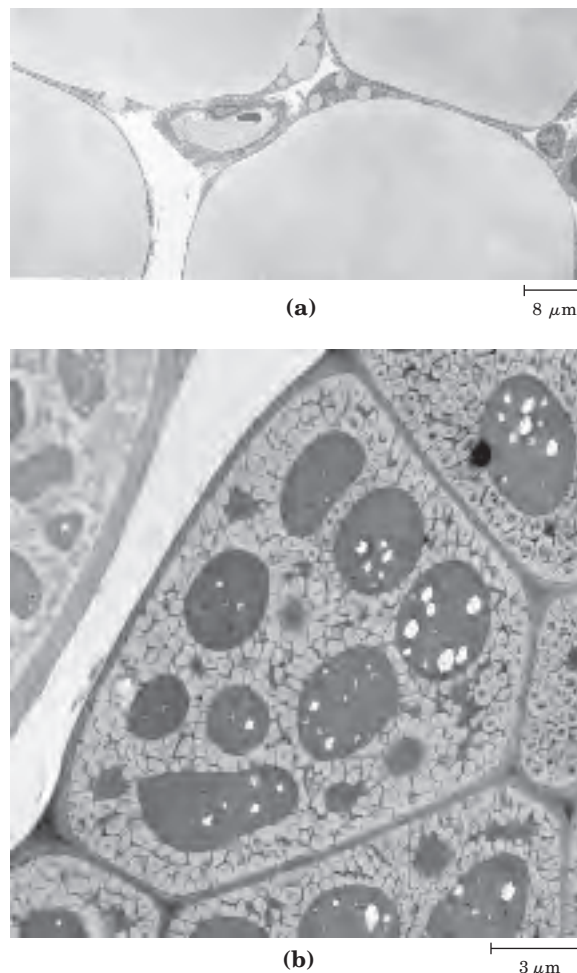


FIGURE 10-3 Fat stores in cells. (a) Cross section of four guinea pig adipocytes, showing huge fat droplets that virtually fill the cells. Also visible are several capillaries in cross section. (b) Cross section of a cotyledon cell from a seed of the plant *Arabidopsis*. The large dark structures are protein bodies, which are surrounded by stored oils in the light-colored oil bodies.

example), the huge fat reserves accumulated before hibernation serve the dual purposes of insulation and energy storage (see Box 17–1). The low density of triacylglycerols is the basis for another remarkable function of these compounds. In sperm whales, a store of triacylglycerols and waxes allows the animals to match the buoyancy of their bodies to that of their surroundings during deep dives in cold water (Box 10–1).

Many Foods Contain Triacylglycerols

Most natural fats, such as those in vegetable oils, dairy products, and animal fat, are complex mixtures of simple and mixed triacylglycerols. These contain a variety of fatty acids differing in chain length and degree of saturation (Fig. 10–4). Vegetable oils such as corn (maize) and olive oil are composed largely of triacylglycerols with unsaturated fatty acids and thus are liquids at room temperature. They are converted industrially into solid

BOX 10-1 THE WORLD OF BIOCHEMISTRY

Sperm Whales: Fatheads of the Deep

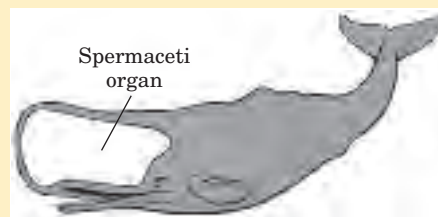
Studies of sperm whales have uncovered another way in which triacylglycerols are biologically useful. The sperm whale's head is very large, accounting for over one-third of its total body weight. About 90% of the weight of the head is made up of the spermaceti organ, a blubbery mass that contains up to 3,600 kg (about 4 tons) of spermaceti oil, a mixture of triacylglycerols and waxes containing an abundance of unsaturated fatty acids. This mixture is liquid at the normal resting body temperature of the whale, about 37°C, but it begins to crystallize at about 31°C and becomes solid when the temperature drops several more degrees.

The probable biological function of spermaceti oil has been deduced from research on the anatomy and feeding behavior of the sperm whale. These mammals feed almost exclusively on squid in very deep water. In their feeding dives they descend 1,000 m or more; the deepest recorded dive is 3,000 m (almost 2 miles). At these depths, there are no competitors for the very plentiful squid; the sperm whale rests quietly, waiting for schools of squid to pass.

For a marine animal to remain at a given depth without a constant swimming effort, it must have the same density as the surrounding water. The sperm whale undergoes changes in buoyancy to match the density of its surroundings—from the tropical ocean surface to great depths where the water is much

colder and thus denser. The key is the freezing point of spermaceti oil. When the temperature of the oil is lowered several degrees during a deep dive, it congeals or crystallizes and becomes denser. Thus the buoyancy of the whale changes to match the density of seawater. Various physiological mechanisms promote rapid cooling of the oil during a dive. During the return to the surface, the congealed spermaceti oil warms and melts, decreasing its density to match that of the surface water. Thus we see in the sperm whale a remarkable anatomical and biochemical adaptation. The triacylglycerols and waxes synthesized by the sperm whale contain fatty acids of the necessary chain length and degree of unsaturation to give the spermaceti oil the proper melting point for the animal's diving habits.

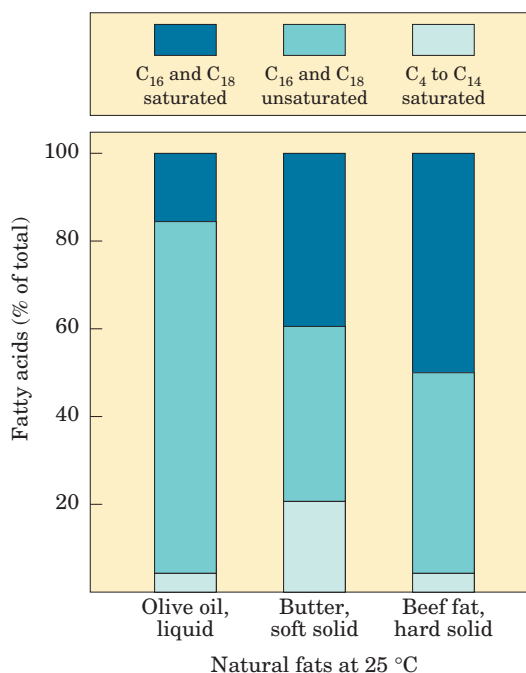
Unfortunately for the sperm whale population, spermaceti oil was at one time considered the finest lamp oil and continues to be commercially valuable as a lubricant. Several centuries of intensive hunting of these mammals have driven sperm whales onto the endangered species list.



fats by catalytic hydrogenation, which reduces some of their double bonds to single bonds and converts others to trans double bonds. Triacylglycerols containing only saturated fatty acids, such as tristearin, the major component of beef fat, are white, greasy solids at room temperature.

When lipid-rich foods are exposed too long to the oxygen in air, they may spoil and become rancid. The unpleasant taste and smell associated with rancidity result from the oxidative cleavage of the double bonds in

FIGURE 10-4 Fatty acid composition of three food fats. Olive oil, butter, and beef fat consist of mixtures of triacylglycerols, differing in their fatty acid composition. The melting points of these fats—and hence their physical state at room temperature (25°C)—are a direct function of their fatty acid composition. Olive oil has a high proportion of long-chain (C_{16} and C_{18}) unsaturated fatty acids, which accounts for its liquid state at 25°C. The higher proportion of long-chain (C_{16} and C_{18}) saturated fatty acids in butter increases its melting point, so butter is a soft solid at room temperature. Beef fat, with an even higher proportion of long-chain saturated fatty acids, is a hard solid.



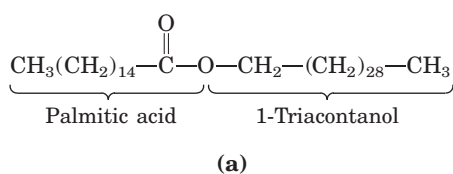
unsaturated fatty acids, which produces aldehydes and carboxylic acids of shorter chain length and therefore higher volatility.

Waxes Serve as Energy Stores and Water Repellents

Biological waxes are esters of long-chain (C_{14} to C_{36}) saturated and unsaturated fatty acids with long-chain (C_{16} to C_{30}) alcohols (Fig. 10–5). Their melting points (60 to 100°C) are generally higher than those of triacylglycerols. In plankton, the free-floating microorganisms at the bottom of the food chain for marine animals, waxes are the chief storage form of metabolic fuel.

Waxes also serve a diversity of other functions related to their water-repellent properties and their firm consistency. Certain skin glands of vertebrates secrete waxes to protect hair and skin and keep it pliable, lubricated, and waterproof. Birds, particularly waterfowl, secrete waxes from their preen glands to keep their feathers water-repellent. The shiny leaves of holly, rhododendrons, poison ivy, and many tropical plants are coated with a thick layer of waxes, which prevents excessive evaporation of water and protects against parasites.

Biological waxes find a variety of applications in the pharmaceutical, cosmetic, and other industries. Lanolin (from lamb's wool), beeswax (Fig. 10–5), carnauba wax (from a Brazilian palm tree), and wax extracted from spermaceti oil (from whales; see Box 10–1) are widely used in the manufacture of lotions, ointments, and polishes.



(b)

FIGURE 10–5 Biological wax. (a) Triacontanoylpalmitate, the major component of beeswax, is an ester of palmitic acid with the alcohol triacontanol. (b) A honeycomb, constructed of beeswax, is firm at 25°C and completely impervious to water. The term “wax” originates in the Old English *weax*, meaning “the material of the honeycomb.”

SUMMARY 10.1 Storage Lipids

- Lipids are water-insoluble cellular components of diverse structure that can be extracted by nonpolar solvents.
- Almost all fatty acids, the hydrocarbon components of many lipids, have an even number of carbon atoms (usually 12 to 24); they are either saturated or unsaturated, with double bonds almost always in the *cis* configuration.
- Triacylglycerols contain three fatty acid molecules esterified to the three hydroxyl groups of glycerol. Simple triacylglycerols contain only one type of fatty acid; mixed triacylglycerols, two or three types. Triacylglycerols are primarily storage fats; they are present in many foods.

10.2 Structural Lipids in Membranes

The central architectural feature of biological membranes is a double layer of lipids, which acts as a barrier to the passage of polar molecules and ions. Membrane lipids are amphipathic: one end of the molecule is hydrophobic, the other hydrophilic. Their hydrophobic interactions with each other and their hydrophilic interactions with water direct their packing into sheets called membrane bilayers. In this section we describe five general types of membrane lipids: glycerophospholipids, in which the hydrophobic regions are composed of two fatty acids joined to glycerol; galactolipids and sulfolipids, which also contain two fatty acids esterified to glycerol, but lack the characteristic phosphate of phospholipids; archaeobacterial tetraether lipids, in which two very long alkyl chains are ether-linked to glycerol at both ends; sphingolipids, in which a single fatty acid is joined to a fatty amine, sphingosine; and sterols, compounds characterized by a rigid system of four fused hydrocarbon rings.

The hydrophilic moieties in these amphipathic compounds may be as simple as a single $-\text{OH}$ group at one end of the sterol ring system, or they may be much more complex. In glycerophospholipids and some sphingolipids, a polar head group is joined to the hydrophobic moiety by a phosphodiester linkage; these are the **phospholipids**. Other sphingolipids lack phosphate but have a simple sugar or complex oligosaccharide at their polar ends; these are the **glycolipids** (Fig. 10–6). Within these groups of membrane lipids, enormous diversity results from various combinations of fatty acid “tails” and polar “heads.” The arrangement of these lipids in membranes, and their structural and functional roles therein, are considered in the next chapter.

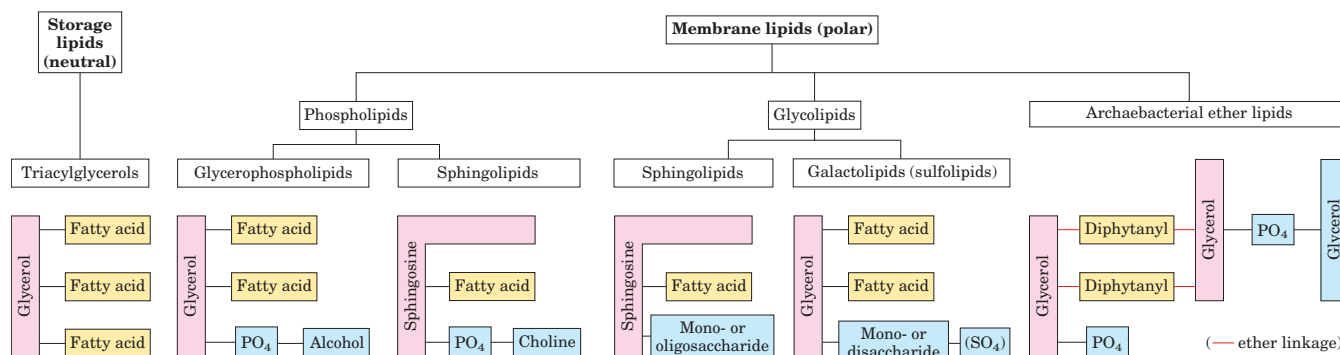


FIGURE 10-6 Some common types of storage and membrane lipids.

All the lipid types shown here have either glycerol or sphingosine as the backbone (pink screen), to which are attached one or more long-chain alkyl groups (yellow) and a polar head group (blue). In triacylglycerols, glycerophospholipids, galactolipids, and sulfolipids, the alkyl groups are fatty acids in ester linkage. Sphingolipids contain a

single fatty acid, in amide linkage to the sphingosine backbone. The membrane lipids of archaeobacteria are variable; that shown here has two very long, branched alkyl chains, each end in ether linkage with a glycerol moiety. In phospholipids the polar head group is joined through a phosphodiester, whereas glycolipids have a direct glycosidic linkage between the head-group sugar and the backbone glycerol.

Glycerophospholipids Are Derivatives of Phosphatidic Acid

Glycerophospholipids, also called phosphoglycerides, are membrane lipids in which two fatty acids are attached in ester linkage to the first and second carbons of glycerol, and a highly polar or charged group is attached through a phosphodiester linkage to the third carbon. Glycerol is prochiral; it has no asymmetric carbons, but attachment of phosphate at one end converts it into a chiral compound, which can be correctly named either *L*-glycerol 3-phosphate, *D*-glycerol 1-phosphate, or *sn*-glycerol 3-phosphate (Fig. 10-7). Glycerophospholipids are named as derivatives of the parent compound, phos-

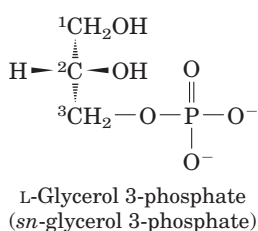


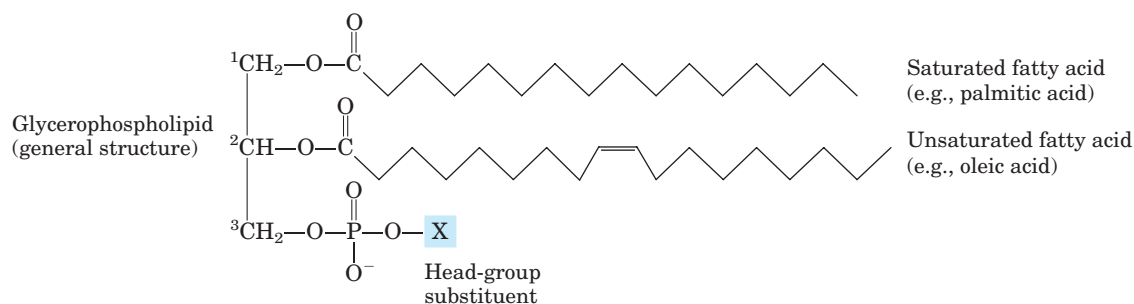
FIGURE 10-7 *L*-Glycerol 3-phosphate, the backbone of phospholipids. Glycerol itself is not chiral, as it has a plane of symmetry through C-2. However, glycerol can be converted to a chiral compound by adding a substituent such as phosphate to either of the $-\text{CH}_2\text{OH}$ groups; that is, glycerol is prochiral. One unambiguous nomenclature for glycerol phosphate is the DL system (described on p. 77), in which the isomers are named according to their stereochemical relationships to glyceraldehyde isomers. By this system, the stereoisomer of glycerol phosphate found in most lipids is correctly named either *L*-glycerol 3-phosphate or *D*-glycerol 1-phosphate. Another way to specify stereoisomers is the stereospecific numbering (*sn*) system, in which C-1 is, by definition, that group of the prochiral compound that occupies the pro-S position. The common form of glycerol phosphate in phospholipids is, by this system, *sn*-glycerol 3-phosphate.

phatidic acid (Fig. 10-8), according to the polar alcohol in the head group. Phosphatidylcholine and phosphatidylethanolamine have choline and ethanolamine in their polar head groups, for example. In all these compounds, the head group is joined to glycerol through a phosphodiester bond, in which the phosphate group bears a negative charge at neutral pH. The polar alcohol may be negatively charged (as in phosphatidylinositol 4,5-bisphosphate), neutral (phosphatidylserine), or positively charged (phosphatidylcholine, phosphatidylethanolamine). As we shall see in Chapter 11, these charges contribute greatly to the surface properties of membranes.

The fatty acids in glycerophospholipids can be any of a wide variety, so a given phospholipid (phosphatidylcholine, for example) may consist of a number of molecular species, each with its unique complement of fatty acids. The distribution of molecular species is specific for different organisms, different tissues of the same organism, and different glycerophospholipids in the same cell or tissue. In general, glycerophospholipids contain a C_{16} or C_{18} saturated fatty acid at C-1 and a C_{18} to C_{20} unsaturated fatty acid at C-2. With few exceptions, the biological significance of the variation in fatty acids and head groups is not yet understood.

Some Phospholipids Have Ether-Linked Fatty Acids

Some animal tissues and some unicellular organisms are rich in **ether lipids**, in which one of the two acyl chains is attached to glycerol in ether, rather than ester, linkage. The ether-linked chain may be saturated, as in the alkyl ether lipids, or may contain a double bond between C-1 and C-2, as in **plasmalogens** (Fig. 10-9). Vertebrate heart tissue is uniquely enriched in ether lipids; about half of the heart phospholipids are plasmalogens. The membranes of halophilic bacteria, ciliated protists, and certain invertebrates also contain high proportions of



Name of glycerophospholipid	Name of X	Formula of X	Net charge (at pH 7)
Phosphatidic acid	—	— H	−1
Phosphatidylethanolamine	Ethanolamine	— CH ₂ -CH ₂ -NH ₃ ⁺	0
Phosphatidylcholine	Choline	— CH ₂ -CH ₂ -N ⁺ (CH ₃) ₃	0
Phosphatidylserine	Serine	— CH ₂ -CH(NH ₃ ⁺)COO ⁻	−1
Phosphatidylglycerol	Glycerol	— CH ₂ -CH(OH)-CH ₂ -OH	−1
Phosphatidylinositol 4,5-bisphosphate	<i>myo</i> -Inositol 4,5-bisphosphate		−4
Cardiolipin	Phosphatidylglycerol		−2

FIGURE 10-8 Glycerophospholipids. The common glycerophospholipids are diacylglycerols linked to head-group alcohols through a phosphodiester bond. Phosphatidic acid, a phosphomonoester, is the

parent compound. Each derivative is named for the head-group alcohol (X), with the prefix “phosphatidyl-.” In cardiolipin, two phosphatidic acids share a single glycerol.

ether lipids. The functional significance of ether lipids in these membranes is unknown; perhaps their resistance to the phospholipases that cleave ester-linked fatty acids from membrane lipids is important in some roles.



At least one ether lipid, **platelet-activating factor**, is a potent molecular signal. It is released

from leukocytes called basophils and stimulates platelet aggregation and the release of serotonin (a vasoconstrictor) from platelets. It also exerts a variety of effects on liver, smooth muscle, heart, uterine, and lung tissues and plays an important role in inflammation and the allergic response. ■

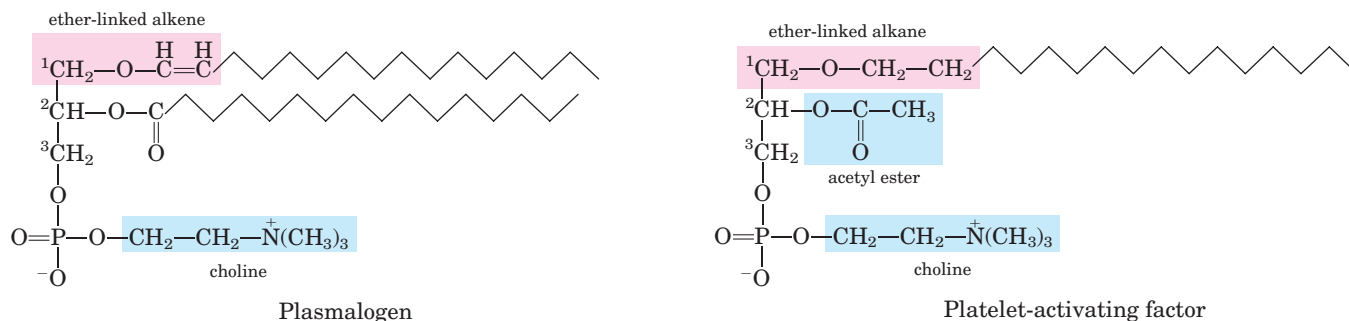


FIGURE 10-9 Ether lipids. Plasmalogens have an ether-linked alkenyl chain where most glycerophospholipids have an ester-linked fatty acid (compare Fig. 10-8). Platelet-activating factor has a long ether-linked alkyl chain at C-1 of glycerol, but C-2 is ester-linked to acetic acid,

which makes the compound much more water-soluble than most glycerophospholipids and plasmalogens. The head-group alcohol is choline in plasmalogens and in platelet-activating factor.

Chloroplasts Contain Galactolipids and Sulfolipids

The second group of membrane lipids are those that predominate in plant cells: the **galactolipids**, in which one or two galactose residues are connected by a glycosidic linkage to C-3 of a 1,2-diacylglycerol (Fig. 10-10; see also Fig. 10-6). Galactolipids are localized in the thylakoid membranes (internal membranes) of chloroplasts; they make up 70% to 80% of the total membrane lipids of a vascular plant. They are probably the most

abundant membrane lipids in the biosphere. Phosphate is often the limiting plant nutrient in soil, and perhaps the evolutionary pressure to conserve phosphate for more critical roles favored plants that made phosphate-free lipids. Plant membranes also contain sulfolipids, in which a sulfonated glucose residue is joined to a diacylglycerol in glycosidic linkage. In sulfolipids, the sulfonate on the head group bears a fixed negative charge like that of the phosphate group in phospholipids (Fig. 10-10).

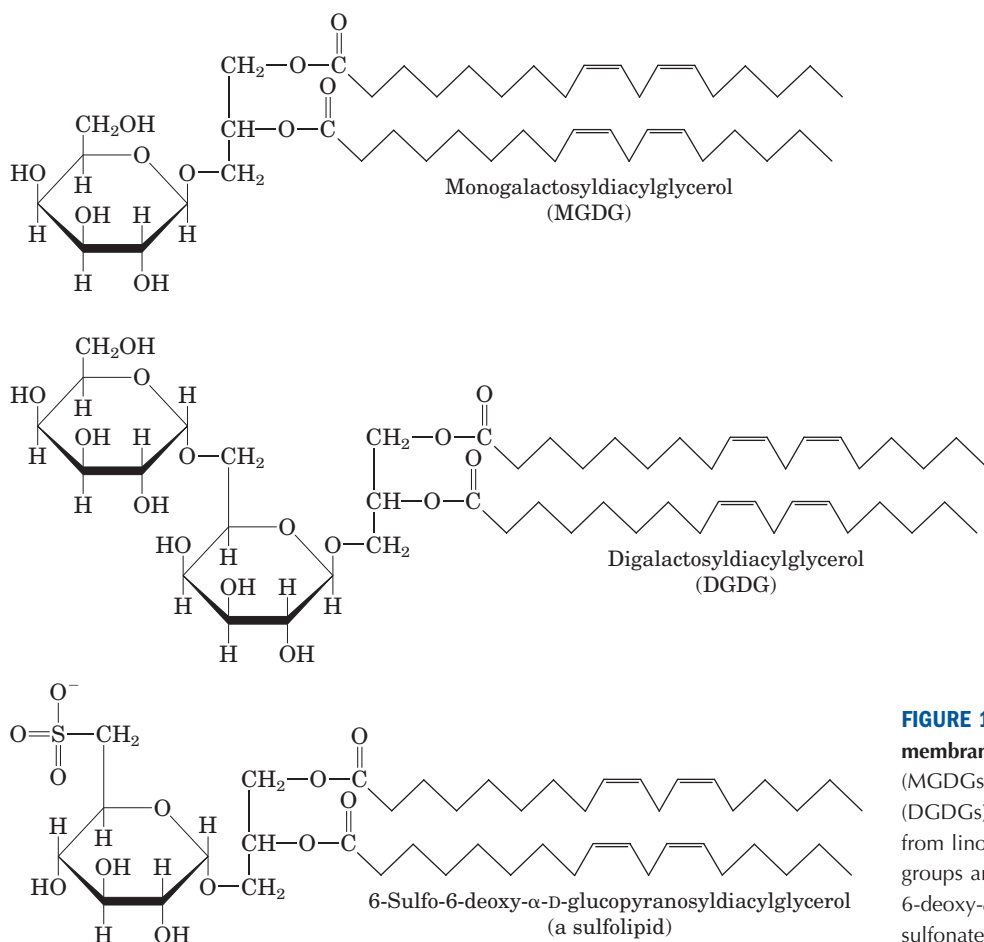


FIGURE 10-10 Three glycolipids of chloroplast membranes. In monogalactosyldiacylglycerols (MGDGs) and digalactosyldiacylglycerols (DGDGs), almost all the acyl groups are derived from linoleic acid (18:2(Δ^{9,12})) and the head groups are uncharged. In the sulfolipid 6-sulfo-6-deoxy-α-D-glucopyranosyldiacylglycerol, the sulfonate carries a fixed negative charge.

Archaeobacteria Contain Unique Membrane Lipids

The archaeobacteria, most of which live in ecological niches with extreme conditions—high temperatures (boiling water), low pH, high ionic strength, for example—have membrane lipids containing long-chain (32 carbons) branched hydrocarbons linked at each end to glycerol (Fig. 10–11). These linkages are through ether bonds, which are much more stable to hydrolysis at low pH and high temperature than are the ester bonds found in the lipids of eubacteria and eukaryotes. In their fully extended form, these archaeobacterial lipids are twice the length of phospholipids and sphingolipids and span the width of the surface membrane. At each end of the extended molecule is a polar head consisting of glycerol linked to either phosphate or sugar residues. The general name for these compounds, glycerol dialkyl glycerol tetraethers (GDGTs), reflects their unique structure. The glycerol moiety of the archaeobacterial lipids is not the same stereoisomer as that in the lipids of eubacteria and eukaryotes; the central carbon is in the R configuration in archaeobacteria, in the S configuration in the other kingdoms (Fig. 10–7).

Sphingolipids Are Derivatives of Sphingosine

Sphingolipids, the fourth large class of membrane lipids, also have a polar head group and two nonpolar tails, but unlike glycerophospholipids and galactolipids they contain no glycerol. Sphingolipids are composed of one molecule of the long-chain amino alcohol sphingosine (also called 4-sphingenine) or one of its derivatives, one molecule of a long-chain fatty acid, and a polar head group that is joined by a glycosidic linkage in some cases and by a phosphodiester in others (Fig. 10–12).

Carbons C-1, C-2, and C-3 of the sphingosine molecule are structurally analogous to the three carbons of

glycerol in glycerophospholipids. When a fatty acid is attached in amide linkage to the —NH_2 on C-2, the resulting compound is a **ceramide**, which is structurally similar to a diacylglycerol. Ceramide is the structural parent of all sphingolipids.

There are three subclasses of sphingolipids, all derivatives of ceramide but differing in their head groups: sphingomyelins, neutral (uncharged) glycolipids, and gangliosides. **Sphingomyelins** contain phosphocholine or phosphoethanolamine as their polar head group and are therefore classified along with glycerophospholipids as phospholipids (Fig. 10–6). Indeed, sphingomyelins resemble phosphatidylcholines in their general properties and three-dimensional structure, and in having no net charge on their head groups (Fig. 10–13). Sphingomyelins are present in the plasma membranes of animal cells and are especially prominent in myelin, a membranous sheath that surrounds and insulates the axons of some neurons—thus the name “sphingomyelins.”

Glycosphingolipids, which occur largely in the outer face of plasma membranes, have head groups with one or more sugars connected directly to the —OH at C-1 of the ceramide moiety; they do not contain phosphate. **Cerebrosides** have a single sugar linked to ceramide; those with galactose are characteristically found in the plasma membranes of cells in neural tissue, and those with glucose in the plasma membranes of cells in nonneural tissues. **Globosides** are neutral (uncharged) glycosphingolipids with two or more sugars, usually D-glucose, D-galactose, or N-acetyl-D-galactosamine. Cerebrosides and globosides are sometimes called **neutral glycolipids**, as they have no charge at pH 7.

Gangliosides, the most complex sphingolipids, have oligosaccharides as their polar head groups and one or more residues of N-acetylneuraminic acid (Neu5Ac), a sialic acid (often simply called “sialic acid”), at the

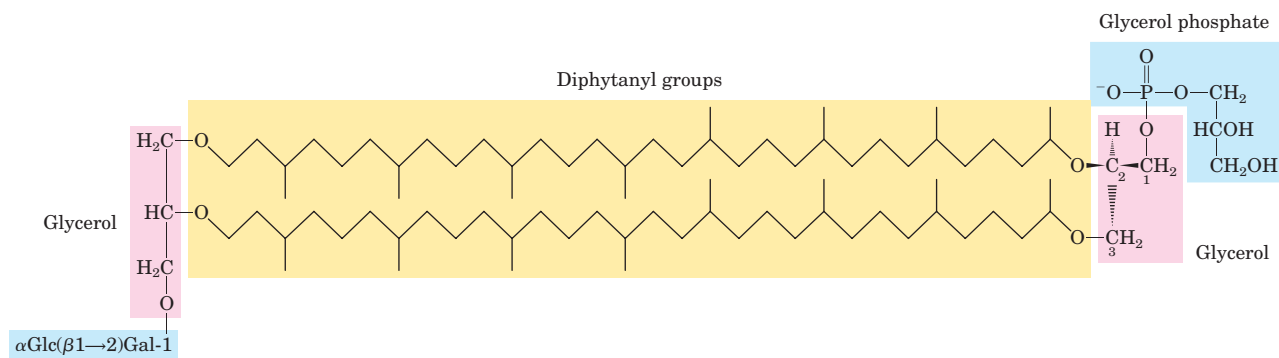
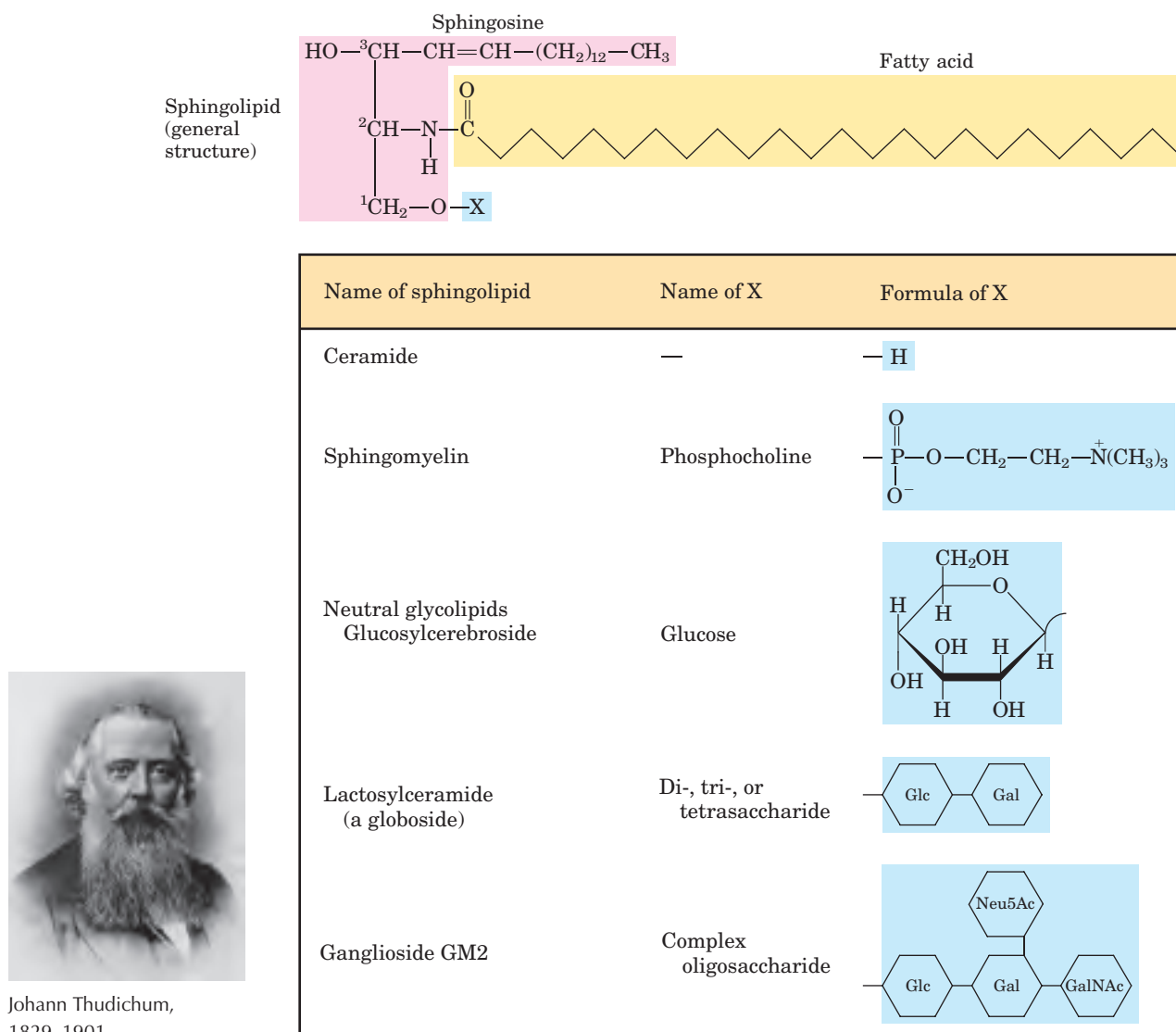


FIGURE 10–11 A typical membrane lipid of archaeobacteria. In this diphytanyl tetraether lipid, the diphytanyl moieties (yellow) are long hydrocarbons composed of eight five-carbon isoprene groups condensed end-to-end (on the condensation of isoprene units, see Fig. 21–36; also, compare the diphytanyl groups with the 20-carbon phytol side chain of chlorophylls in Fig. 19–40a). In this extended form, the diphytanyl groups are about twice the length of a 16-carbon fatty

acid typically found in the membrane lipids of eubacteria and eukaryotes. The glycerol moieties in the archaeobacterial lipids are in the R configuration, in contrast to those of eubacteria and eukaryotes, which have the S configuration. Archaeobacterial lipids differ in the substituents on the glycerols. In the molecule shown here, one glycerol is linked to the disaccharide α -glucopyranosyl-(1 $\rightarrow$ 2)- β -galactofuranose; the other glycerol is linked to a glycerol phosphate head group.

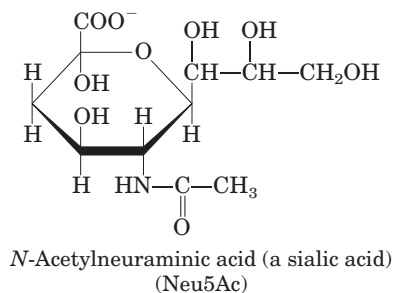


Johann Thudichum,
1829–1901

FIGURE 10-12 Sphingolipids. The first three carbons at the polar end of sphingosine are analogous to the three carbons of glycerol in glycerophospholipids. The amino group at C-2 bears a fatty acid in amide linkage. The fatty acid is usually saturated or monounsaturated, with 16, 18, 22, or 24 carbon atoms. Ceramide is the parent compound

for this group. Other sphingolipids differ in the polar head group (X) attached at C-1. Gangliosides have very complex oligosaccharide head groups. Standard abbreviations for sugars are used in this figure: Glc, D-glucose; Gal, D-galactose; GalNAc, N-acetyl-D-galactosamine; Neu5Ac, N-acetylneuraminic acid (sialic acid).

termini. Sialic acid gives gangliosides the negative charge at pH 7 that distinguishes them from globosides. Gangliosides with one sialic acid residue are in the GM (M for mono-) series, those with two are in the GD (D for di-) series, and so on (GT, three sialic acid residues; GQ, four).



Sphingolipids at Cell Surfaces Are Sites of Biological Recognition

When sphingolipids were discovered a century ago by the physician-chemist Johann Thudichum, their biological role seemed as enigmatic as the Sphinx, for which he therefore named them. In humans, at least 60 different sphingolipids have been identified in cellular membranes. Many of these are especially prominent in the plasma membranes of neurons, and some are clearly recognition sites on the cell surface, but a specific function for only a few sphingolipids has been discovered thus far. The carbohydrate moieties of certain sphingolipids define the human blood groups and therefore determine the type of blood that individuals can safely receive in blood transfusions (Fig. 10-14).

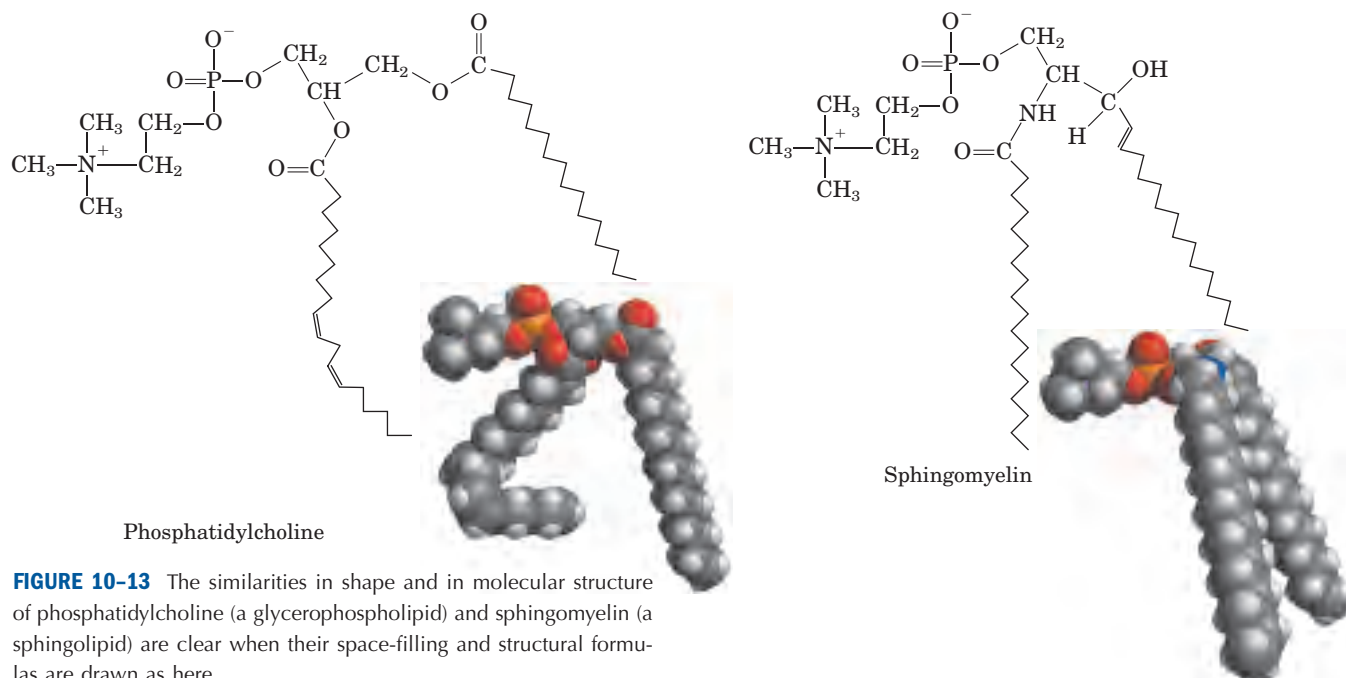


FIGURE 10-13 The similarities in shape and in molecular structure of phosphatidylcholine (a glycerophospholipid) and sphingomyelin (a sphingolipid) are clear when their space-filling and structural formulas are drawn as here.

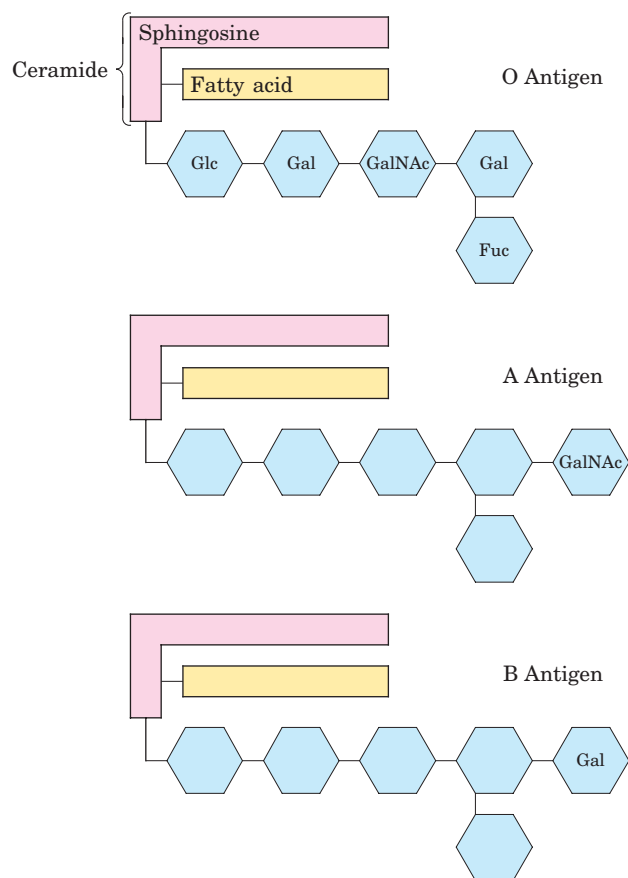


FIGURE 10-14 Glycosphingolipids as determinants of blood groups. The human blood groups (O, A, B) are determined in part by the oligosaccharide head groups (blue) of these glycosphingolipids. The same three oligosaccharides are also found attached to certain blood proteins of individuals of blood types O, A, and B, respectively. (Fuc represents the sugar fucose.)

Gangliosides are concentrated in the outer surface of cells, where they present points of recognition for extracellular molecules or surfaces of neighboring cells. The kinds and amounts of gangliosides in the plasma membrane change dramatically during embryonic development. Tumor formation induces the synthesis of a new complement of gangliosides, and very low concentrations of a specific ganglioside have been found to induce differentiation of cultured neuronal tumor cells. Investigation of the biological roles of diverse gangliosides remains fertile ground for future research.

Phospholipids and Sphingolipids Are Degraded in Lysosomes

Most cells continually degrade and replace their membrane lipids. For each hydrolyzable bond in a glycerophospholipid, there is a specific hydrolytic enzyme in the lysosome (Fig. 10-15). Phospholipases of the A type remove one of the two fatty acids, producing a lysophospholipid. (These esterases do not attack the ether link of plasmalogens.) Lysophospholipases remove the remaining fatty acid.

Gangliosides are degraded by a set of lysosomal enzymes that catalyze the stepwise removal of sugar units, finally yielding a ceramide. A genetic defect in any of these hydrolytic enzymes leads to the accumulation of gangliosides in the cell, with severe medical consequences (Box 10-2).

Sterols Have Four Fused Carbon Rings

Sterols are structural lipids present in the membranes of most eukaryotic cells. The characteristic structure of this fifth group of membrane lipids is the steroid nu-

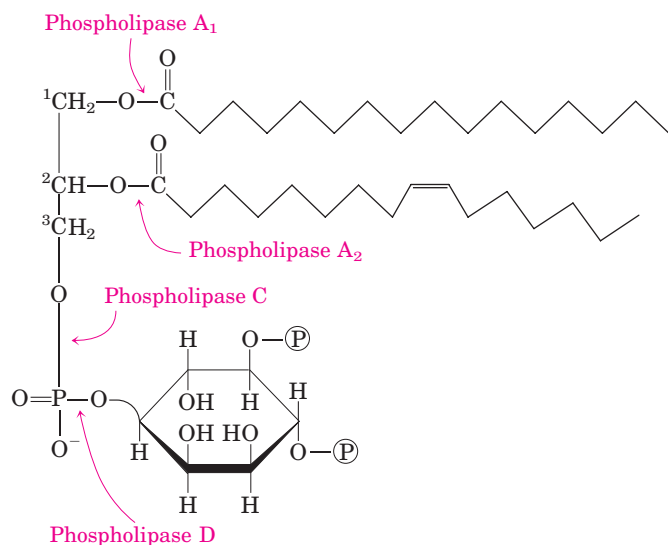


FIGURE 10-15 The specificities of phospholipases. Phospholipases A_1 and A_2 hydrolyze the ester bonds of intact glycerophospholipids at C-1 and C-2 of glycerol, respectively. Phospholipases C and D each split one of the phosphodiester bonds in the head group. Some phospholipases act on only one type of glycerophospholipid, such as phosphatidylinositol 4,5-bisphosphate (shown here) or phosphatidylcholine; others are less specific. When one of the fatty acids has been removed by a type A phospholipase, the second fatty acid is cleaved from the molecule by a lysophospholipase (not shown).

cleus, consisting of four fused rings, three with six carbons and one with five (Fig. 10-16). The steroid nucleus is almost planar and is relatively rigid; the fused rings do not allow rotation about C—C bonds. **Cholesterol**, the major sterol in animal tissues, is amphipathic, with a polar head group (the hydroxyl group at C-3) and a nonpolar hydrocarbon body (the steroid nucleus and the hydrocarbon side chain at C-17), about as long as a 16-carbon fatty acid in its extended form. Similar sterols are found in other eukaryotes: stigmasterol in plants and ergosterol in fungi, for example. Bacteria cannot synthesize sterols; a few bacterial species, however, can incorporate exogenous sterols into their membranes. The sterols of all eukaryotes are synthesized from simple five-carbon isoprene subunits, as are the fat-soluble vitamins, quinones, and dolichols described in Section 10.3.

In addition to their roles as membrane constituents, the sterols serve as precursors for a variety of products with specific biological activities. Steroid hormones, for example, are potent biological signals that regulate gene expression. Bile acids are polar derivatives of cholesterol that act as detergents in the intestine, emulsifying dietary fats to make them more readily accessible to digestive lipases. We return to cholesterol and other sterols in later chapters, to consider the structural role of cholesterol in biological membranes (Chapter 11), signaling by steroid hormones (Chapter 12), the remarkable biosynthetic pathway to cholesterol, and the transport of cholesterol by lipoprotein carriers (Chapter 21).

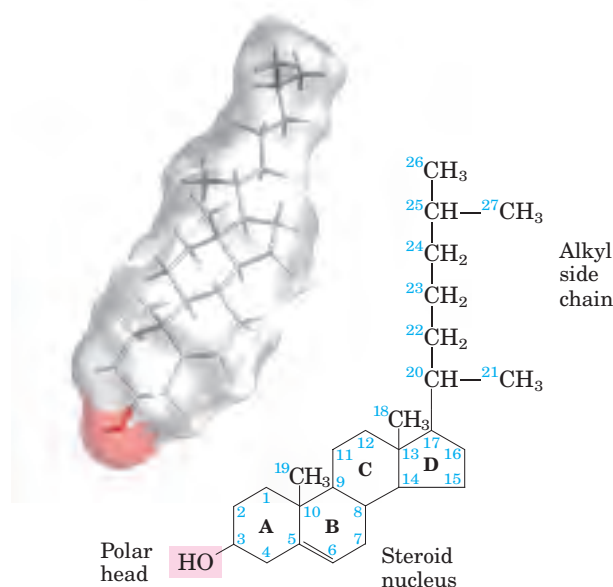
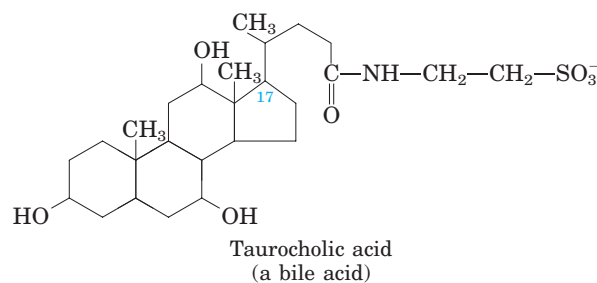


FIGURE 10-16 Cholesterol. The stick structure of cholesterol is visible through a transparent surface contour model of the molecule (from coordinates supplied by Dave Woodcock). In the chemical structure, the rings are labeled A through D to simplify reference to derivatives of the steroid nucleus, and the carbon atoms are numbered in blue. The C-3 hydroxyl group (pink in both representations) is the polar head group. For storage and transport of the sterol, this hydroxyl group condenses with a fatty acid to form a sterol ester.



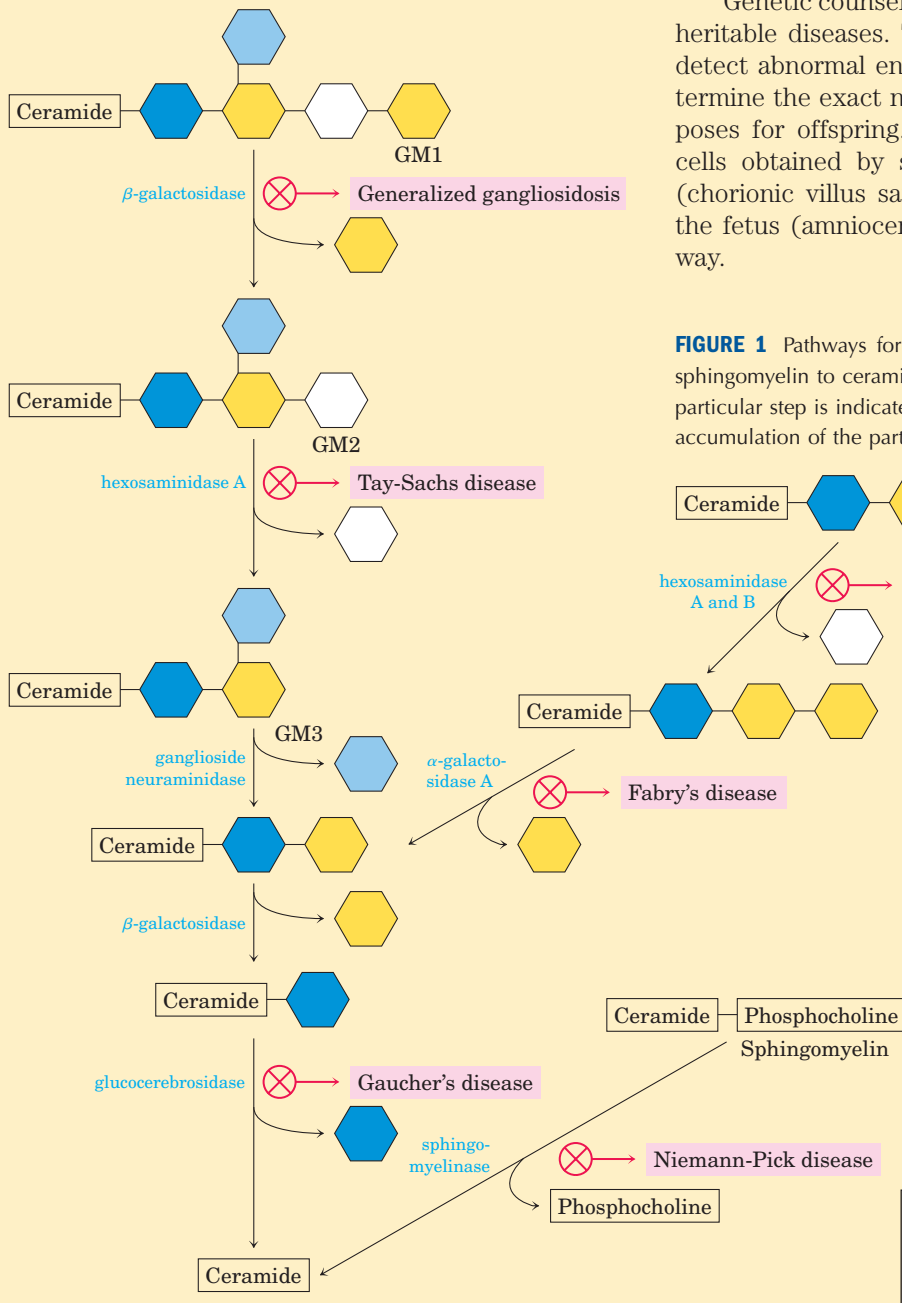
SUMMARY 10.2 Structural Lipids in Membranes

- The polar lipids, with polar heads and nonpolar tails, are major components of membranes. The most abundant are the glycerophospholipids, which contain fatty acids esterified to two of the hydroxyl groups of glycerol, and a second alcohol, the head group, esterified to the third hydroxyl of glycerol via a phosphodiester bond. Other polar lipids are the sterols.
- Glycerophospholipids differ in the structure of their head group; common glycerophospholipids are phosphatidylethanolamine and phosphatidylcholine. The polar heads of the glycerophospholipids carry electric charges at pH near 7.
- Chloroplast membranes are remarkably rich in galactolipids, composed of a diacylglycerol with



Inherited Human Diseases Resulting from Abnormal Accumulations of Membrane Lipids

The polar lipids of membranes undergo constant metabolic turnover, the rate of their synthesis normally counterbalanced by the rate of breakdown. The breakdown of lipids is promoted by hydrolytic enzymes in lysosomes, each enzyme capable of hydrolyzing a specific bond. When sphingolipid degradation is impaired by a defect in one of these enzymes (Fig. 1), partial breakdown products accumulate in the tissues, causing serious disease.



For example, Niemann-Pick disease is caused by a rare genetic defect in the enzyme sphingomyelinase, which cleaves phosphocholine from sphingomyelin. Sphingomyelin accumulates in the brain, spleen, and liver. The disease becomes evident in infants, and causes mental retardation and early death. More common is Tay-Sachs disease, in which ganglioside GM2 accumulates in the brain and spleen (Fig. 2) owing to lack of the enzyme hexosaminidase A. The symptoms of Tay-Sachs disease are progressive retardation in development, paralysis, blindness, and death by the age of 3 or 4 years.

Genetic counseling can predict and avert many inheritable diseases. Tests on prospective parents can detect abnormal enzymes, then DNA testing can determine the exact nature of the defect and the risk it poses for offspring. Once a pregnancy occurs, fetal cells obtained by sampling a part of the placenta (chorionic villus sampling) or the fluid surrounding the fetus (amniocentesis) can be tested in the same way.

FIGURE 1 Pathways for the breakdown of GM1, globoside, and sphingomyelin to ceramide. A defect in the enzyme hydrolyzing a particular step is indicated by $\otimes$, and the disease that results from accumulation of the partial breakdown product is noted.

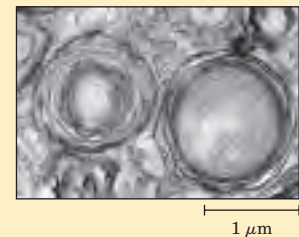


FIGURE 2 Electron micrograph of a portion of a brain cell from an infant with Tay-Sachs disease, showing abnormal ganglioside deposits in the lysosomes.

one or two linked galactose residues, and sulfolipids, diacylglycerols with a linked sulfonated sugar residue and thus a negatively charged head group.

- Archaeobacteria have unique membrane lipids, with long-chain alkyl groups ether-linked to glycerol at both ends and with sugar residues and/or phosphate joined to the glycerol to provide a polar or charged head group. These lipids are stable under the harsh conditions in which archaeobacteria live.
- The sphingolipids contain sphingosine, a long-chain aliphatic amino alcohol, but no glycerol. Sphingomyelin has, in addition to phosphoric acid and choline, two long hydrocarbon chains, one contributed by a fatty acid and the other by sphingosine. Three other classes of sphingolipids are cerebrosides, globosides, and gangliosides, which contain sugar components.
- Sterols have four fused rings and a hydroxyl group. Cholesterol, the major sterol in animals, is both a structural component of membranes and precursor to a wide variety of steroids.

10.3 Lipids as Signals, Cofactors, and Pigments

The two functional classes of lipids considered thus far (storage lipids and structural lipids) are major cellular components; membrane lipids make up 5% to 10% of the dry mass of most cells, and storage lipids more than 80% of the mass of an adipocyte. With some important exceptions, these lipids play a *passive* role in the cell; lipid fuels are stored until oxidized by enzymes, and membrane lipids form impermeable barriers around cells and cellular compartments. Another group of lipids, present in much smaller amounts, have *active* roles in the metabolic traffic as metabolites and messengers. Some serve as potent signals—as hormones, carried in the blood from one tissue to another, or as intracellular messengers generated in response to an extracellular signal (hormone or growth factor). Others function as enzyme cofactors in electron-transfer reactions in chloroplasts and mitochondria, or in the transfer of sugar moieties in a variety of glycosylation (addition of sugar) reactions. A third group consists of lipids with a system of conjugated double bonds: pigment molecules that absorb visible light. Some of these act as light-capturing pigments in vision and photosynthesis; others produce natural colorations, such as the orange of pumpkins and carrots and the yellow of canary feathers. Specialized lipids such as these are derived from lipids of the plasma membrane or from the fat-soluble vitamins A, D, E, and K. We describe in this

section a few of these biologically active lipids. In later chapters, their synthesis and biological roles are considered in more detail.

Phosphatidylinositols and Sphingosine Derivatives Act as Intracellular Signals

Phosphatidylinositol and its phosphorylated derivatives act at several levels to regulate cell structure and metabolism (Fig. 10–17). Phosphatidylinositol 4,5-bisphosphate (Fig. 10–8) in the cytoplasmic (inner) face of plasma membranes serves as a specific binding site for certain cytoskeletal proteins and for some soluble proteins involved in membrane fusion during exocytosis. It also serves as a reservoir of messenger molecules that are released inside the cell in response to extracellular signals interacting with specific receptors on the outer surface of the plasma membrane. The signals act through a series of steps (Fig. 10–17) that begins with enzymatic removal of a phospholipid head group and ends with activation of an enzyme (protein kinase C). For example, when the hormone vasopressin binds to plasma membrane receptors on the epithelial cells of the renal collecting duct, a specific phospholipase C is activated.

Phospholipase C hydrolyzes the bond between glycerol and phosphate in phosphatidylinositol 4,5-bisphosphate, releasing two products: inositol 1,4,5-trisphosphate (IP₃), which is water-soluble, and diacylglycerol, which remains associated with the plasma membrane. IP₃ triggers release of Ca²⁺ from the endoplasmic reticulum, and the combination of diacylglycerol and elevated cytosolic Ca²⁺ activates the enzyme protein kinase C.

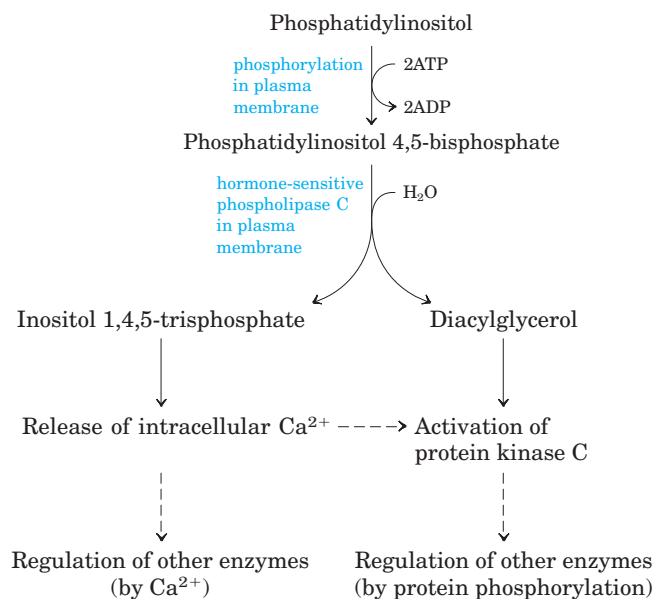


FIGURE 10–17 Phosphatidylinositols in cellular regulation. Phosphatidylinositol 4,5-bisphosphate in the plasma membrane is hydrolyzed by a specific phospholipase C in response to hormonal signals. Both products of hydrolysis act as intracellular messengers.

This enzyme catalyzes the transfer of a phosphoryl group from ATP to a specific residue in one or more target proteins, thereby altering their activity and consequently the cell's metabolism. This signaling mechanism is described more fully in Chapter 12 (see Fig. 12–19).

Inositol phospholipids also serve as points of nucleation for certain supramolecular complexes involved in signaling or in exocytosis. Proteins that contain certain structural motifs, called PH and PX domains (for *pleckstrin* *homology* and *Phox* *homology*, respectively), bind phosphatidylinositols in the membrane with high specificity and affinity, initiating the formation of multienzyme complexes at the membrane's cytosolic surface. A number of proteins bind specifically to phosphatidylinositol 3,4,5-trisphosphate, and the formation of this phospholipid in response to extracellular signals brings the proteins together at the surface of the plasma membrane (see Fig. 12–8).

Membrane sphingolipids also can serve as sources of intracellular messengers. Both ceramide and sphingomyelin (Fig. 10–12) are potent regulators of protein

kinases, and ceramide or its derivatives are known to be involved in the regulation of cell division, differentiation, migration, and programmed cell death (also called apoptosis; see Chapter 12).

Eicosanoids Carry Messages to Nearby Cells



Eicosanoids are paracrine hormones, substances that act only on cells near the point of hormone synthesis instead of being transported in the blood to act on cells in other tissues or organs. These fatty acid derivatives have a variety of dramatic effects on vertebrate tissues. They are known to be involved in reproductive function; in the inflammation, fever, and pain associated with injury or disease; in the formation of blood clots and the regulation of blood pressure; in gastric acid secretion; and in a variety of other processes important in human health or disease.

All eicosanoids are derived from arachidonic acid (20:4($\Delta^{5,8,11,14}$)) (Fig. 10–18), the 20-carbon polyunsaturated fatty acid from which they take their gen-

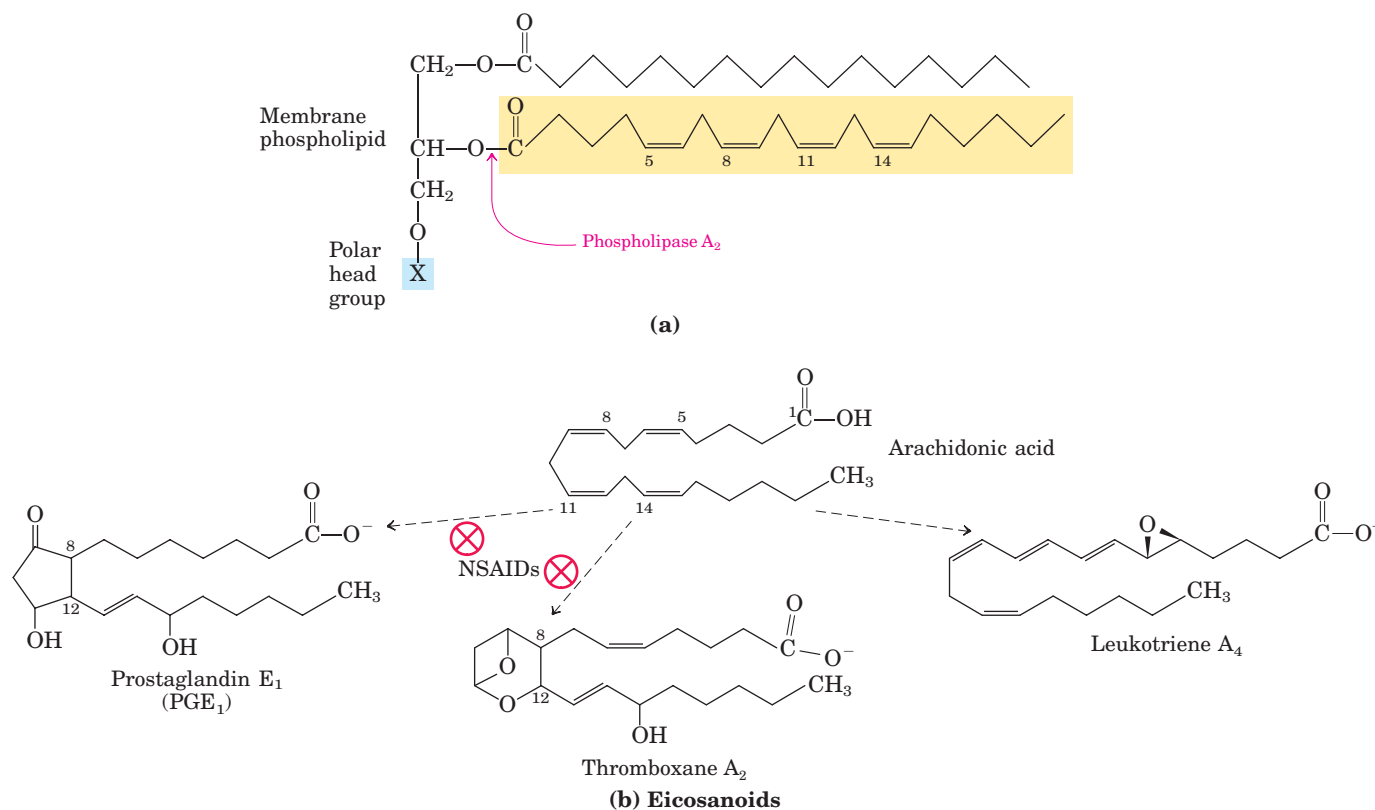


FIGURE 10-18 Arachidonic acid and some eicosanoid derivatives.

(a) In response to hormonal signals, phospholipase A₂ cleaves arachidonic acid-containing membrane phospholipids to release arachidonic acid (arachidonate at pH 7), the precursor to various eicosanoids. (b) These compounds include prostaglandins such as PGE₁, in which C-8 and C-12 of arachidonate are joined to form the characteristic five-membered ring. In thromboxane A₂, the C-8 and

C-12 are joined and an oxygen atom is added to form the six-membered ring. Leukotriene A₄ has a series of three conjugated double bonds. Nonsteroidal antiinflammatory drugs (NSAIDs) such as aspirin and ibuprofen block the formation of prostaglandins and thromboxanes from arachidonate by inhibiting the enzyme cyclooxygenase (prostaglandin H₂ synthase).



John Vane, Sune Bergström, and Bengt Samuelsson

eral name (Greek *eikosi*, “twenty”). There are three classes of eicosanoids: prostaglandins, thromboxanes, and leukotrienes.

Prostaglandins (PG) contain a five-carbon ring originating from the chain of arachidonic acid. Their name derives from the prostate gland, the tissue from which they were first isolated by Bengt Samuelsson and Sune Bergström. Two groups of prostaglandins were originally defined: PGE, for *ether*-soluble, and PGF, for phosphate (*fosfat* in Swedish) buffer-soluble. Each group contains numerous subtypes, named PGE₁, PGE₂, and so forth. Prostaglandins act in many tissues by regulating the synthesis of the intracellular messenger 3',5'-cyclic AMP (cAMP). Because cAMP mediates the action of diverse hormones, the prostaglandins affect a wide range of cellular and tissue functions. Some prostaglandins stimulate contraction of the smooth muscle of the uterus during menstruation and labor. Others affect blood flow to specific organs, the wake-sleep cycle, and the responsiveness of certain tissues to hormones such as epinephrine and glucagon. Prostaglandins in a third group elevate body temperature (producing fever) and cause inflammation and pain.

The **thromboxanes** have a six-membered ring containing an ether. They are produced by platelets (also called thrombocytes) and act in the formation of blood clots and the reduction of blood flow to the site of a clot. The nonsteroidal antiinflammatory drugs (NSAIDs)—aspirin, ibuprofen, and meclofenamate, for example—were shown by John Vane to inhibit the enzyme prostaglandin H₂ synthase (also called cyclooxygenase or COX), which catalyzes an early step in the pathway from arachidonate to prostaglandins and thromboxanes (Fig. 10-18; see also Box 21-2).

Leukotrienes, first found in leukocytes, contain three conjugated double bonds. They are powerful biological signals. For example, leukotriene D₄, derived from leukotriene A₄, induces contraction of the muscle lining the airways to the lung. Overproduction of leukotrienes causes asthmatic attacks, and leukotriene synthesis is one target of antiasthmatic drugs such as prednisone. The strong contraction of the smooth muscles of the lung that occurs during anaphylactic shock is part of the potentially fatal allergic reaction in individuals hypersensitive to bee stings, penicillin, or other agents. ■

Steroid Hormones Carry Messages between Tissues



Steroids are oxidized derivatives of sterols; they have the sterol nucleus but lack the alkyl chain attached to ring D of cholesterol, and they are more polar than cholesterol. Steroid hormones move through the bloodstream (on protein carriers) from their site of production to target tissues, where they enter cells, bind to highly specific receptor proteins in the nucleus, and trigger changes in gene expression and metabolism. Because hormones have very high affinity for their receptors, very low concentrations of hormones (nanomolar or less) are sufficient to produce responses in target tissues. The major groups of steroid hormones are the male and female sex hormones and the hormones produced by the adrenal cortex, cortisol and aldosterone (Fig. 10-19). Prednisone and prednisolone are steroid drugs with potent antiinflammatory activities, mediated in part by the inhibition of arachidonate release by phospholipase A₂ (Fig. 10-18) and consequent inhibition of the

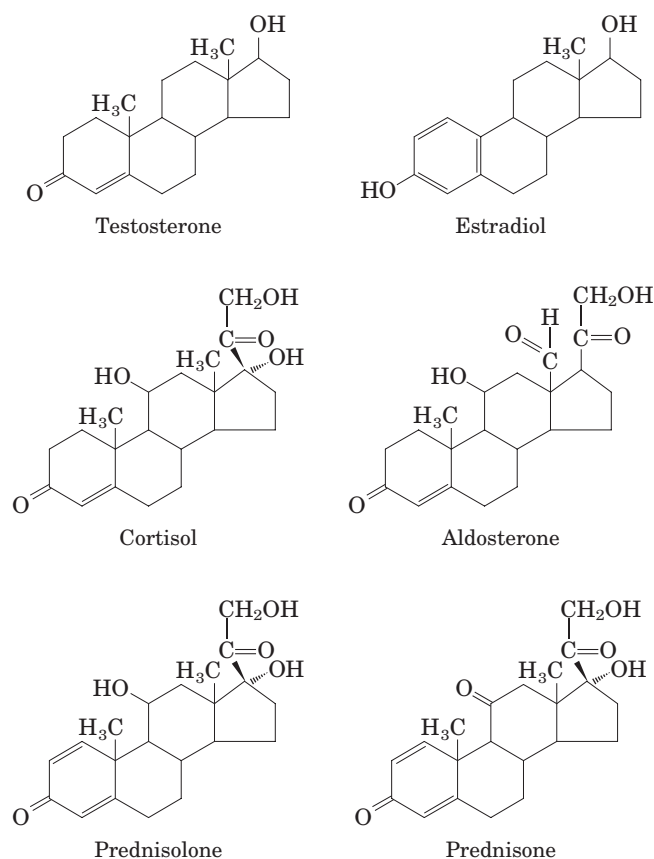
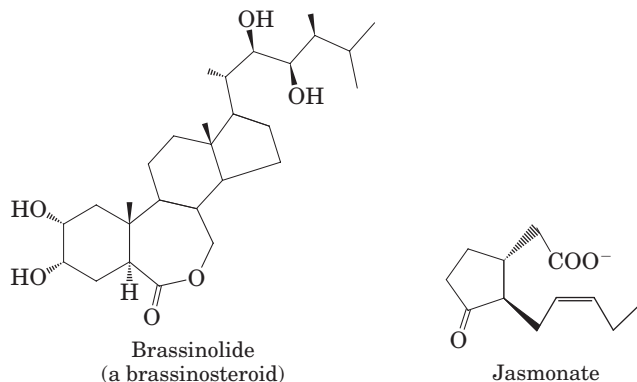


FIGURE 10-19 Steroids derived from cholesterol. Testosterone, the male sex hormone, is produced in the testes. Estradiol, one of the female sex hormones, is produced in the ovaries and placenta. Cortisol and aldosterone are hormones synthesized in the cortex of the adrenal gland; they regulate glucose metabolism and salt excretion, respectively. Prednisolone and prednisone are synthetic steroids used as antiinflammatory agents.

synthesis of leukotrienes, prostaglandins, and thromboxanes. They have a variety of medical applications, including the treatment of asthma and rheumatoid arthritis. ■

Plants Use Phosphatidylinositols, Steroids, and Eicosanoidlike Compounds in Signaling

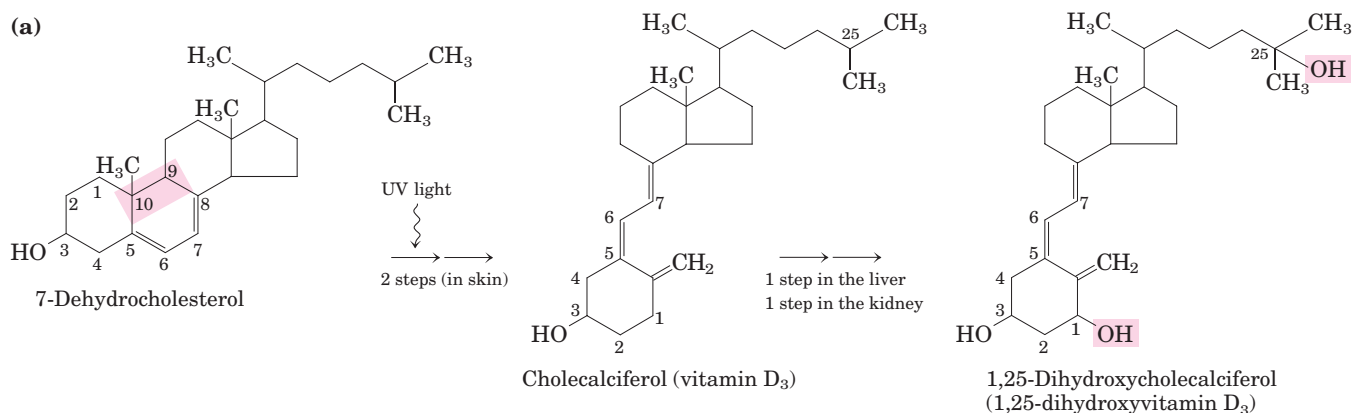
Vascular plants contain phosphatidylinositol 4,5-bisphosphate, as well as the phospholipase that releases IP_3 , and they use IP_3 to regulate the intracellular concentration of Ca^{2+} . Brassinolide and the related group of brassinosteroids are potent growth regulators in plants, increasing the rate of stem elongation and influencing the orientation of cellulose microfibrils in the cell wall during growth. Jasmonate, derived from the fatty acid 18:3($\Delta^{9,12,15}$) in membrane lipids, is chemically similar to the eicosanoids of animal tissues and also serves as a powerful signal, triggering the plant's defenses in response to insect-inflicted damage. The methyl ester of jasmonate gives the characteristic fragrance of jasmine oil, which is widely used in the perfume industry.



Vitamins A and D Are Hormone Precursors



During the first third of the twentieth century, a major focus of research in physiological chemistry was the identification of **vitamins**, compounds that are essential to the health of humans and other vertebrates but cannot be synthesized by these animals and must therefore be obtained in the diet. Early nutritional



Before vitamin D treatment



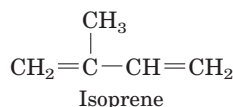
After 14 months of vitamin D treatment
(b)



FIGURE 10-20 Vitamin D_3 production and metabolism. (a) Cholecalciferol

(vitamin D_3) is produced in the skin by UV irradiation of 7-dehydrocholesterol, which breaks the bond shaded pink. In the liver, a hydroxyl group is added at C-25 (pink); in the kidney, a second hydroxylation at C-1 (pink) produces the active hormone, 1,25-dihydroxycholecalciferol. This hormone regulates the metabolism of Ca^{2+} in kidney, intestine, and bone. (b) Dietary vitamin D prevents rickets, a disease once common in cold climates where heavy clothing blocks the UV component of sunlight necessary for the production of vitamin D_3 in skin. On the left is a 2½-year-old boy with severe rickets; on the right, the same boy at age 5, after 14 months of vitamin D therapy.

studies identified two general classes of such compounds: those soluble in nonpolar organic solvents (fat-soluble vitamins) and those that could be extracted from foods with aqueous solvents (water-soluble vitamins). Eventually the fat-soluble group was resolved into the four vitamin groups A, D, E, and K, all of which are isoprenoid compounds synthesized by the condensation of multiple isoprene units. Two of these (D and A) serve as hormone precursors.



Vitamin D₃, also called **cholecalciferol**, is normally formed in the skin from 7-dehydrocholesterol in a photochemical reaction driven by the UV component of sunlight (Fig. 10–20). Vitamin D₃ is not itself biologically active, but it is converted by enzymes in the liver and kidney to 1,25-dihydroxycholecalciferol, a hormone that regulates calcium uptake in the intestine and calcium levels in kidney and bone. Deficiency of vitamin D

leads to defective bone formation and the disease rickets, for which administration of vitamin D produces a dramatic cure. Vitamin D₂ (ergocalciferol) is a commercial product formed by UV irradiation of the ergosterol of yeast. Vitamin D₂ is structurally similar to D₃, with slight modification to the side chain attached to the sterol D ring. Both have the same biological effects, and D₂ is commonly added to milk and butter as a dietary supplement. Like steroid hormones, the product of vitamin D metabolism, 1,25-dihydroxycholecalciferol, regulates gene expression—for example, turning on the synthesis of an intestinal Ca²⁺-binding protein.

Vitamin A (retinol) in its various forms functions as a hormone and as the visual pigment of the vertebrate eye (Fig. 10–21). Acting through receptor proteins in the cell nucleus, the vitamin A derivative retinoic acid regulates gene expression in the development of epithelial tissue, including skin. Retinoic acid is the active ingredient in the drug tretinoin (Retin-A), used in the treatment of severe acne and wrinkled skin. The vitamin A derivative retinal is the pigment that initiates the

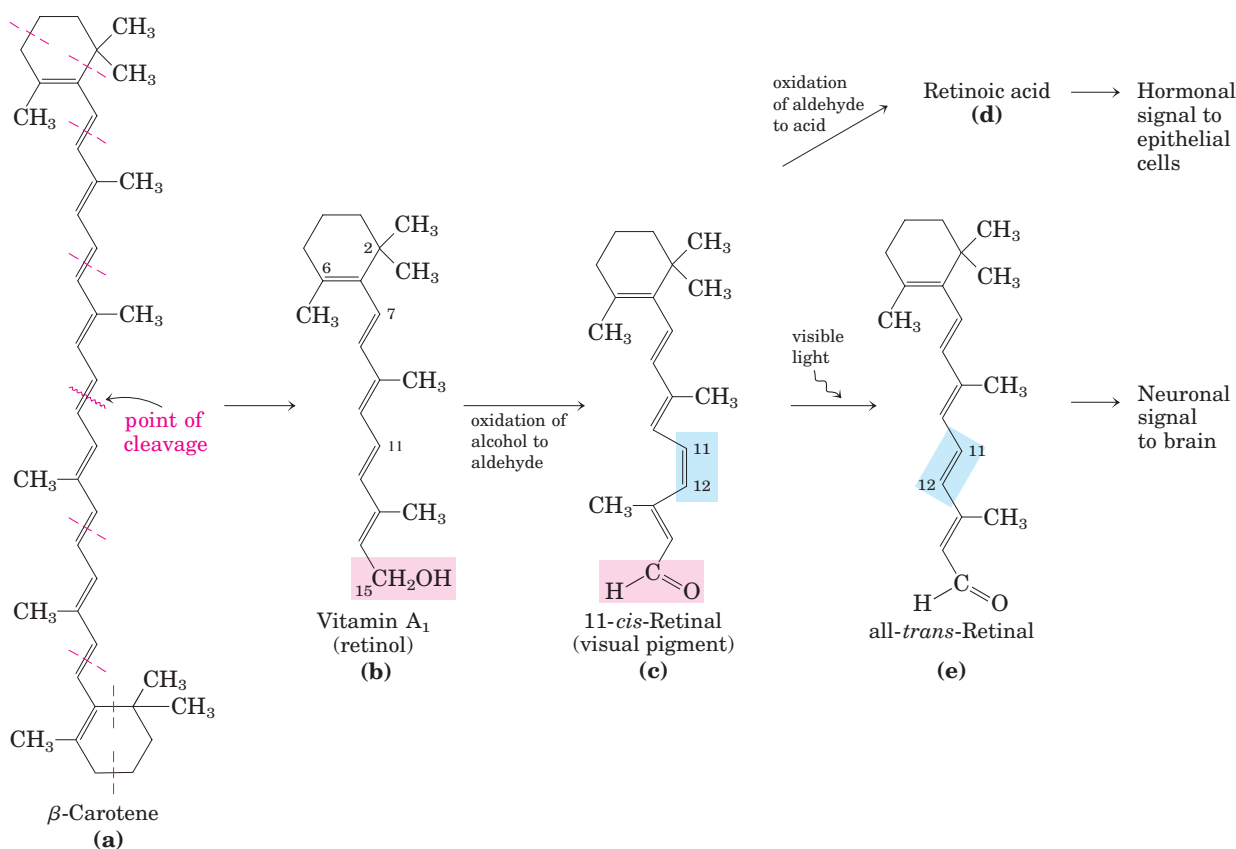


FIGURE 10–21 Vitamin A₁ and its precursor and derivatives.

(a) β-Carotene is the precursor of vitamin A₁. Isoprene structural units are set off by dashed red lines. Cleavage of β-carotene yields two molecules of vitamin A₁ (retinol) **(b)**. Oxidation at C-15 converts retinol to the aldehyde, retinal **(c)**, and further oxidation produces retinoic acid **(d)**, a hormone that regulates gene expression. Retinal combines with the protein opsin to form rhodopsin (not shown), a visual pigment widespread in nature. In the dark, retinal of rhodopsin

is in the 11-*cis* form **(c)**. When a rhodopsin molecule is excited by visible light, the 11-*cis*-retinal undergoes a series of photochemical reactions that convert it to all-*trans*-retinal **(e)**, forcing a change in the shape of the entire rhodopsin molecule. This transformation in the rod cell of the vertebrate retina sends an electrical signal to the brain that is the basis of visual transduction, a topic we address in more detail in Chapter 12.

response of rod and cone cells of the retina to light, producing a neuronal signal to the brain. This role of retinal is described in detail in Chapter 12.

Vitamin A was first isolated from fish liver oils; liver, eggs, whole milk, and butter are good dietary sources. In vertebrates, β -carotene, the pigment that gives carrots, sweet potatoes, and other yellow vegetables their characteristic color, can be enzymatically converted to vitamin A. Deficiency of vitamin A leads to a variety of symptoms in humans, including dryness of the skin, eyes, and mucous membranes; retarded development and growth; and night blindness, an early symptom commonly used in diagnosing vitamin A deficiency. ■

Vitamins E and K and the Lipid Quinones Are Oxidation-Reduction Cofactors



Vitamin E is the collective name for a group of closely related lipids called **tocopherols**, all of which contain a substituted aromatic ring and a long isoprenoid side chain (Fig. 10–22a). Because they are hydrophobic, tocopherols associate with cell membranes, lipid deposits, and lipoproteins in the blood. Tocopherols are biological antioxidants. The aromatic ring reacts with and destroys the most reactive forms of oxygen radicals and other free radicals, protecting unsaturated fatty acids from oxidation and preventing oxidative

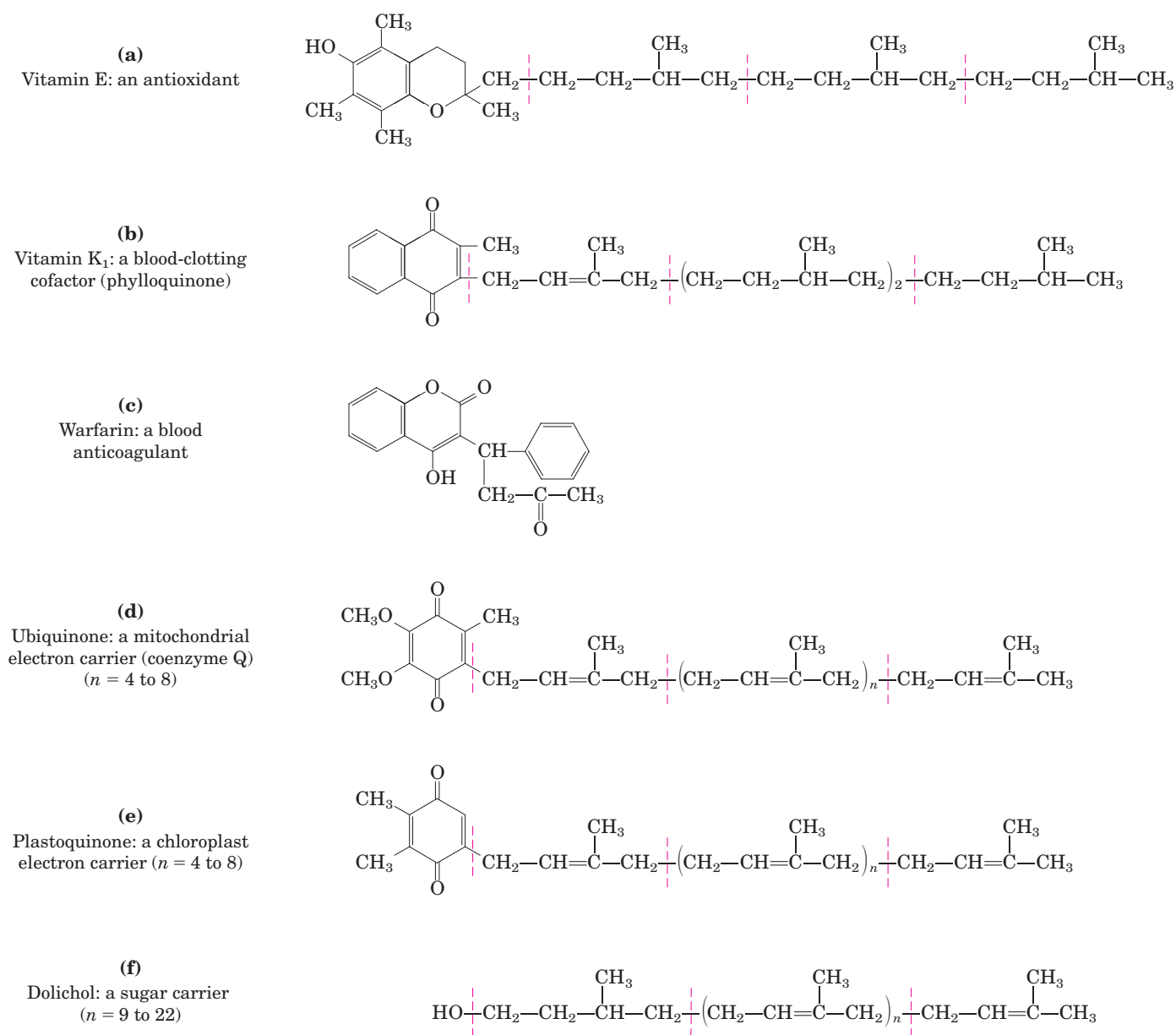


FIGURE 10–22 Some other biologically active isoprenoid compounds or derivatives. Isoprene structural units are set off by dashed red lines. In most mammalian tissues, ubiquinone (also called coen-

zyme Q) has 10 isoprene units. Dolichols of animals have 17 to 21 isoprene units (85 to 105 carbon atoms), bacterial dolichols have 11, and those of plants and fungi have 14 to 24.



Edward A. Doisy,
1893–1986



Henrik Dam,
1895–1976

damage to membrane lipids, which can cause cell fragility. Tocopherols are found in eggs and vegetable oils and are especially abundant in wheat germ. Laboratory animals fed diets depleted of vitamin E develop scaly skin, muscular weakness and wasting, and sterility. Vitamin E deficiency in humans is very rare; the principal symptom is fragile erythrocytes.

The aromatic ring of **vitamin K** (Fig. 10–22b) undergoes a cycle of oxidation and reduction during the formation of active prothrombin, a blood plasma protein essential in blood clot formation. Prothrombin is a proteolytic enzyme that splits peptide bonds in the blood protein fibrinogen to convert it to fibrin, the insoluble fibrous protein that holds blood clots together. Henrik Dam and Edward A. Doisy independently discovered that vitamin K deficiency slows blood clotting, which can be fatal. Vitamin K deficiency is very uncommon in humans, aside from a small percentage of infants who suffer from hemorrhagic disease of the newborn, a potentially fatal disorder. In the United States, newborns are routinely given a 1 mg injection of vitamin K. Vitamin K₁ (phyloquinone) is found in green plant leaves; a related form, vitamin K₂ (menaquinone), is formed by bacteria residing in the vertebrate intestine.

Warfarin (Fig. 10–22c) is a synthetic compound that inhibits the formation of active prothrombin. It is particularly poisonous to rats, causing death by internal bleeding. Ironically, this potent rodenticide is also an invaluable anticoagulant drug for treating humans at risk for excessive blood clotting, such as surgical patients and those with coronary thrombosis. ■

Ubiquinone (also called coenzyme Q) and plastoquinone (Fig. 10–22d, e) are isoprenoids that function as lipophilic electron carriers in the oxidation-reduction reactions that drive ATP synthesis in mitochondria and chloroplasts, respectively. Both ubiquinone and plastoquinone can accept either one or two electrons and either one or two protons (see Fig. 19–54).

Dolichols Activate Sugar Precursors for Biosynthesis

During assembly of the complex carbohydrates of bacterial cell walls, and during the addition of polysaccha-

ride units to certain proteins (glycoproteins) and lipids (glycolipids) in eukaryotes, the sugar units to be added are chemically activated by attachment to isoprenoid alcohols called **dolichols** (Fig. 10–22f). These compounds have strong hydrophobic interactions with membrane lipids, anchoring the attached sugars to the membrane, where they participate in sugar-transfer reactions.

SUMMARY 10.3 Lipids as Signals, Cofactors, and Pigments

- Some types of lipids, although present in relatively small quantities, play critical roles as cofactors or signals.
- Phosphatidylinositol bisphosphate is hydrolyzed to yield two intracellular messengers, diacylglycerol and inositol 1,4,5-trisphosphate. Phosphatidylinositol 3,4,5-trisphosphate is a nucleation point for supramolecular protein complexes involved in biological signaling.
- Prostaglandins, thromboxanes, and leukotrienes (the eicosanoids), derived from arachidonate, are extremely potent hormones.
- Steroid hormones, derived from sterols, serve as powerful biological signals, such as the sex hormones.
- Vitamins D, A, E, and K are fat-soluble compounds made up of isoprene units. All play essential roles in the metabolism or physiology of animals. Vitamin D is precursor to a hormone that regulates calcium metabolism. Vitamin A furnishes the visual pigment of the vertebrate eye and is a regulator of gene expression during epithelial cell growth. Vitamin E functions in the protection of membrane lipids from oxidative damage, and vitamin K is essential in the blood-clotting process.
- Ubiquinones and plastoquinones, also isoprenoid derivatives, function as electron carriers in mitochondria and chloroplasts, respectively.
- Dolichols activate and anchor sugars on cellular membranes for use in the synthesis of certain complex carbohydrates, glycolipids, and glycoproteins.

10.4 Working with Lipids

In exploring the biological role of lipids in cells and tissues, it is essential to know which lipids are present and in what proportions. Because lipids are insoluble in water, their extraction and subsequent fractionation require the use of organic solvents and some techniques

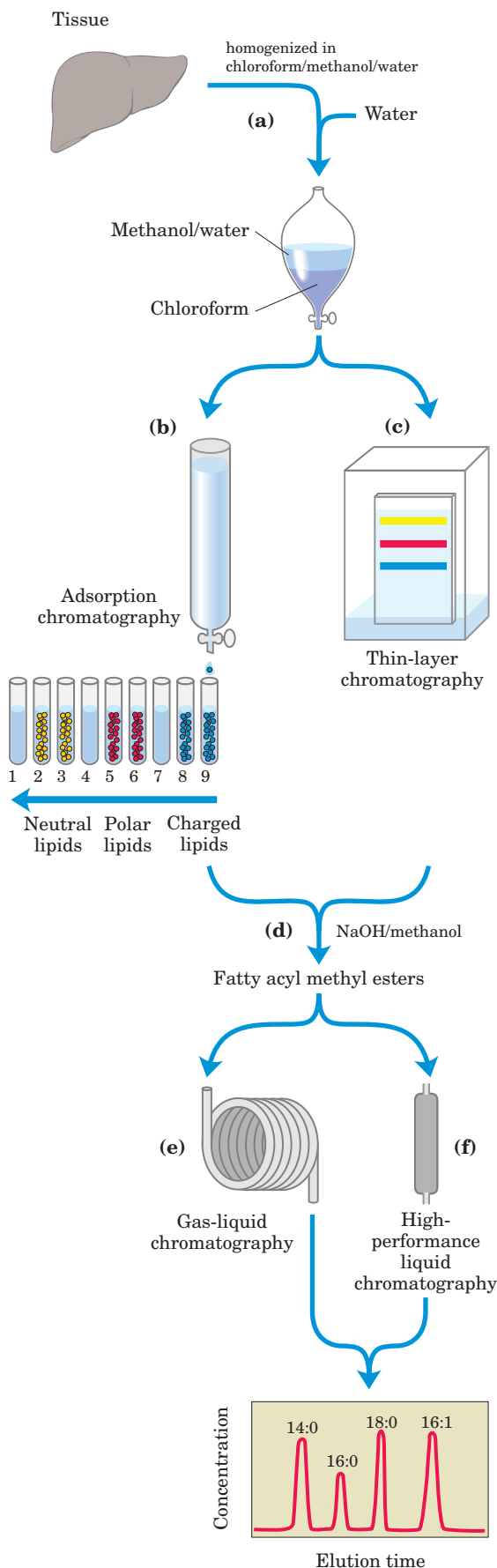
not commonly used in the purification of water-soluble molecules such as proteins and carbohydrates. In general, complex mixtures of lipids are separated by differences in the polarity or solubility of the components in nonpolar solvents. Lipids that contain ester- or amide-linked fatty acids can be hydrolyzed by treatment with acid or alkali or with highly specific hydrolytic enzymes (phospholipases, glycosidases) to yield their component parts for analysis. Some methods commonly used in lipid analysis are shown in Figure 10–23 and discussed below.

Lipid Extraction Requires Organic Solvents

Neutral lipids (triacylglycerols, waxes, pigments, and so forth) are readily extracted from tissues with ethyl ether, chloroform, or benzene, solvents that do not permit lipid clustering driven by hydrophobic interactions. Membrane lipids are more effectively extracted by more polar organic solvents, such as ethanol or methanol, which reduce the hydrophobic interactions among lipid molecules while also weakening the hydrogen bonds and electrostatic interactions that bind membrane lipids to membrane proteins. A commonly used extractant is a mixture of chloroform, methanol, and water, initially in volume proportions (1:2:0.8) that are miscible, producing a single phase. After tissue is homogenized in this solvent to extract all lipids, more water is added to the resulting extract and the mixture separates into two phases, methanol/water (top phase) and chloroform (bottom phase). The lipids remain in the chloroform layer, and more polar molecules such as proteins and sugars partition into the methanol/water layer.

FIGURE 10–23 Common procedures in the extraction, separation, and identification of cellular lipids. (a) Tissue is homogenized in a chloroform/methanol/water mixture, which on addition of water and removal of unextractable sediment by centrifugation yields two phases. Different types of extracted lipids in the chloroform phase may be separated by (b) adsorption chromatography on a column of silica gel, through which solvents of increasing polarity are passed, or (c) thin-layer chromatography (TLC), in which lipids are carried up a silica gel-coated plate by a rising solvent front, less polar lipids traveling farther than more polar or charged lipids. TLC with appropriate solvents can also be used to separate closely related lipid species; for example, the charged lipids phosphatidylserine, phosphatidylglycerol, and phosphatidylinositol are easily separated by TLC.

For the determination of fatty acid composition, a lipid fraction containing ester-linked fatty acids is transesterified in a warm aqueous solution of NaOH and methanol (d), producing a mixture of fatty acyl methyl esters. These methyl esters are then separated on the basis of chain length and degree of saturation by (e) gas-liquid chromatography (GLC) or (f) high-performance liquid chromatography (HPLC). Precise determination of molecular mass by mass spectrometry allows unambiguous identification of individual lipids.



Adsorption Chromatography Separates Lipids of Different Polarity

Complex mixtures of tissue lipids can be fractionated by chromatographic procedures based on the different polarities of each class of lipid. In adsorption chromatography (Fig. 10–23b), an insoluble, polar material such as silica gel (a form of silicic acid, $\text{Si}(\text{OH})_4$) is packed into a glass column, and the lipid mixture (in chloroform solution) is applied to the top of the column. (In high-performance liquid chromatography, the column is of smaller diameter and solvents are forced through the column under high pressure.) The polar lipids bind tightly to the polar silicic acid, but the neutral lipids pass directly through the column and emerge in the first chloroform wash. The polar lipids are then eluted, in order of increasing polarity, by washing the column with solvents of progressively higher polarity. Uncharged but polar lipids (cerebrosides, for example) are eluted with acetone, and very polar or charged lipids (such as glycerophospholipids) are eluted with methanol.

Thin-layer chromatography on silicic acid employs the same principle (Fig. 10–23c). A thin layer of silica gel is spread onto a glass plate, to which it adheres. A small sample of lipids dissolved in chloroform is applied near one edge of the plate, which is dipped in a shallow container of an organic solvent or solvent mixture—all of which is enclosed within a chamber saturated with the solvent vapor. As the solvent rises on the plate by capillary action, it carries lipids with it. The less polar lipids move farthest, as they have less tendency to bind to the silicic acid. The separated lipids can be detected by spraying the plate with a dye (rhodamine) that fluoresces when associated with lipids or by exposing the plate to iodine fumes. Iodine reacts reversibly with the double bonds in fatty acids, such that lipids containing unsaturated fatty acids develop a yellow or brown color. A number of other spray reagents are also useful in detecting specific lipids. For subsequent analysis, regions containing separated lipids can be scraped from the plate and the lipids recovered by extraction with an organic solvent.

Gas-Liquid Chromatography Resolves Mixtures of Volatile Lipid Derivatives

Gas-liquid chromatography separates volatile components of a mixture according to their relative tendencies to dissolve in the inert material packed in the chromatography column and to volatilize and move through the column, carried by a current of an inert gas such as helium. Some lipids are naturally volatile, but most must first be derivatized to increase their volatility (that is, lower their boiling point). For an analysis of the fatty acids in a sample of phospholipids, the lipids are first

heated in a methanol/HCl or methanol/NaOH mixture, which converts fatty acids esterified to glycerol into their methyl esters (in a process of transesterification; Fig. 10–23d). These fatty acyl methyl esters are then loaded onto the gas-liquid chromatography column, and the column is heated to volatilize the compounds. Those fatty acyl esters most soluble in the column material partition into (dissolve in) that material; the less soluble lipids are carried by the stream of inert gas and emerge first from the column. The order of elution depends on the nature of the solid adsorbant in the column and on the boiling point of the components of the lipid mixture. Using these techniques, mixtures of fatty acids of various chain lengths and various degrees of unsaturation can be completely resolved (Fig. 10–23e).

Specific Hydrolysis Aids in Determination of Lipid Structure

Certain classes of lipids are susceptible to degradation under specific conditions. For example, all ester-linked fatty acids in triacylglycerols, phospholipids, and sterol esters are released by mild acid or alkaline treatment, and somewhat harsher hydrolysis conditions release amide-bound fatty acids from sphingolipids. Enzymes that specifically hydrolyze certain lipids are also useful in the determination of lipid structure. Phospholipases A, C, and D (Fig. 10–15) each split particular bonds in phospholipids and yield products with characteristic solubilities and chromatographic behaviors. Phospholipase C, for example, releases a water-soluble phosphoryl alcohol (such as phosphocholine from phosphatidylcholine) and a chloroform-soluble diacylglycerol, each of which can be characterized separately to determine the structure of the intact phospholipid. The combination of specific hydrolysis with characterization of the products by thin-layer, gas-liquid, or high-performance liquid chromatography often allows determination of a lipid structure.

Mass Spectrometry Reveals Complete Lipid Structure

To establish unambiguously the length of a hydrocarbon chain or the position of double bonds, mass spectral analysis of lipids or their volatile derivatives is invaluable. The chemical properties of similar lipids (for example, two fatty acids of similar length unsaturated at different positions, or two isoprenoids with different numbers of isoprene units) are very much alike, and their positions of elution from the various chromatographic procedures often do not distinguish between them. When the effluent from a chromatography column is sampled by mass spectrometry, however, the components of a lipid mixture can be simultaneously separated and identified by their unique pattern of fragmentation (Fig. 10–24).